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ZHENJIANG SHENZHOU SHIPBUILDING CO.,LTD.

DAIHATSU DIESEL ENGINE

DOCUMENT LIST

☐	Approval plans	Please confirm documents in this list and return each one of this list & documents with your signature for approval as soon as possible.
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☑	Finish plans	Please receive documents in this list and return one of this list with your signature.

DAIHATSU DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT
JAPAN

DAIHATSU DIESEL MFG. CO.,LTD.
MAY.10.2017
JAPAN
TECHNICAL DEPT

S.NO.	SZ15045/46/47/48	DRAWN BY	M. Fukumoto
TYPE	6DKM-20eF(L)	CHECKED BY	N. Fujimoto
LIST No.	AQP10008488F		
DATE	2016/6/9	APPROVED BY	K. Tanaka
REVISION	B: Rev.UP 2016/8/9 M.T F Rev.UP 2017/5/9 M.T C: Rev.UP 2016/11/10 M.T D: Rev.UP 2016/11/16 M.T E: Rev.UP 2017/3/1 M.T		

PDK20-0174-5

DOCUMENTS LIST

LIST No. AQP10008488 F 2/3

SHIPYARD S.NO. ZHENJIANG SHENZHOU SHIPBUILDING CO.,LTD. SZ15045/46/47/48

ENGINE MODEL 6DKM-20eF(L)

No	NAME	DRAWING NO.	REMARK	FOR APPROV. COPY	FOR APPROV. APPLY	FOR APPROV. REV	FOR APPROV. DATE	SETTLED APPLY	SETTLED REV	SETTLED DATE	ORDER APPLY	ORDER REV	ORDER DATE	FOR WORK. COPY	FOR WORK. APPLY	FOR WORK. REV	FOR WORK. DATE	AS BUILT COPY	AS BUILT APPLY	AS BUILT REV	AS BUILT DATE
1	DOCUMENTS LIST	AQP10008488 F		6	○	A			B					7	○	E		4	○	F	
2	SPECIFICATIONS (JAP.)	AQP10008489 E		6	○	A								7	○	D		4	○	E	
3	SPECIFICATIONS (ENG.)	BQP10000031 D												7	○	D		4	○		
4	M/E+R/G OUTLINE	AQP10008490 D		6	○	A			B					7	○	D		4	○		
5																					
6	M/E DISASSEMBLE SPACE	E200050540Z A	806847	6	○	A								7	○			4	○		
7	M/E ARRANGE.OF FITTINGS	AQP10008491 D		6	○	A			B					7	○	D		4	○		
8	R/G OUTLINE	AQP10008492 B		6	○	A								7	○	B		4	○		
9	R/G ARRANGE. OF FITTINGS	AQP10008513 C		6	○	A			B					7	○	C		4	○		
10	POINT FIGURE :HOISTING	AG89903R017 A	FOR R/G											7	○	A		4	○		
11	R/G SECTIONAL VIEW	AG32903R020 A		6	○	A								7	○			4	○		
12	F.O. & EXH. GAS PIPING SYSTEM	AQP10008493 C		6	○	A			B					7	○	C		4	○		
13	S.A. PIPING SYSTEM	AQP10008494 A		6	○	A								7	○			4	○		
14	L.O. PIPING SYSTEM	AQP10008495 D		6	○	A			B					7	○	C		4	○	D	
15	C.W. PIPING SYSTEM	AQP10008496 A		6	○	A								7	○			4	○		
16	L.O. PIPING SYSTEM(R/G)	AQP10008497 D		6	○	A			C					7	○	D		4	○		
17	WIRING DIAGRAM	AQP10008498 C		6	○	A			B					7	○	C		4	○		
18	FLEXIBLE COUPLING ASSY	AG24005K013 A		6	○	A								7	○			4	○		
19	FLEXIBLE COUPLING ASSY	AG24005K014 A		6	○	A								7	○			4	○		
20	FRONT PTO SHAFT	AQP10008255 A		6	○	A								7	○			4	○		
21	REAMER BOLT:POWER TRANSMISSON	G253220100A Z	FOR P/G OUTPUT	6	○	Z								7	○			4	○		
22	PULSE SENSOR	NN00736K006 A	FOR R/G	6	○	A								7	○			4	○		
23	PULSE SESOR ASSY.		DRA-25JS															4	○		
24	FLYWHEEL TURN. DEVICE	A590201170Z A	RATCHET TYPE	6	○	A								7	○			4	○		
25	F/W TURN.BAR HOLDER	NN00022K001 C		6	○	C								7	○			4	○		
26	CHOCKFAST SET. ASSY.	AQP10008848 A	FOR ENG.	6	○	A								7	○	A		4	○		
27	CHOCKLINER SET. ASSY.	AQP10002495 B	FOR R/G	6	○	B								7	○			4	○		
28	EXPANSION JOINT	B259905080Z A	300A × 400A FOR T/C	6	○	A								7	○			4	○		
29	EXPANSION JOINT	B259705130Z A	400A X 450A	6	○	A								7	○			4	○		
30	INSTALLATION OF EXPANSION JOINT	AQP10003709 B		6	○	B								7	○			4	○		
31	EXHAUST GAS SILENCER		14091518											7	○			4	○		
32																					
33	MERCURY FILTER	AQP20007600 A	MYC-2000MS	6	○	A								7	○			4	○		
34	MERCURY FILTER	AQP20007601 A	MYC-2000MSL	6	○	A								7	○			4	○		
35	Y STRAINER	AQA20012139 B	FOR S.A. INLET	6	○	B								7	○			4	○		
36	STARTING AIR RECEIVER	AQP20008429 A	200L×2.94MPa	6	○	A								7	○			4	○		
37	M/E STANDBY LO PUMP	AQP10008526 B		6	○	A								7	○	B		4	○		
38	STANDBY LO PUMP STARTER PANEL	AQP20008443 A		6	○	A								7	○			4	○		
39	L.O. COOLER	AQP10008527 A	PLATE TYPE	6	○	A								7	○			4	○		
40	L.O. FILTER FOR ENGINE INLET	AQP10008539 B	AUTO BACKWASH TYPE	6	○	A								7	○	B		4	○		
41	L.O. BYPASS FILTER	AQA20013140 B		6	○	B								7	○	B		4	○		
42	L.O. TEMP CONTROL VALVE	AQP20008498 A	WAX TYPE	6	○	A								7	○	A		4	○		
43	L.O. PRESS.CONTROL VALVE	AQP10008538 A		6	○	A								7	○			4	○		
44	R/G STANDBY LO PUMP	AQP10008528 A		6	○	A								7	○			4	○		
45	R/G STANDBY LO PUMP STARTER PANEL	AQP20008445 A		6	○	A								7	○			4	○		
46	LT-FW PUMP	AQP10008529 A		6	○	A								7	○			4	○		
47	LT-FW COOLER	AQP10008530 A	PLATE TYPE	6	○	A								7	○			4	○		
48	HT-FW TEMP CONTROL VALVE	AQP20003993 A	WAX TYPE	6	○	A								7	○	A		4	○		
49	LT-FW TEMP. CONTROL VALVE	AQP10008537 B	E-P TYPE	6	○	A								7	○	B		4	○		
50																					

DOCUMENTS LIST

LIST No. AQP10008488 F 3/3

SHIPYARD S.NO. ZHENJIANG SHENZHOU SHIPBUILDING CO.,LTD. SZ15045/46/47/48

ENGINE MODEL 6DKM-20eF(L)

No	NAME	DRAWING NO.		REMARK	FOR APPROV. COPY	FOR APPROV. APPLY	FOR APPROV. REV	FOR APPROV. DATE	FOR APPROV. APPLY	FOR APPROV. REV	FOR APPROV. DATE	SETTLED APPLY	SETTLED REV	SETTLED DATE	ORDER APPLY	ORDER REV	ORDER DATE	FOR WORK. COPY	FOR WORK. APPLY	FOR WORK. REV	FOR WORK. DATE	AS BUILT COPY	AS BUILT APPLY	AS BUILT REV	AS BUILT DATE
51	M/E CONTROL SYSTEM		-	91-5228/9	6	○	-											7	○			4	○		
52	SPEED SWITCH UNIT	NN00262001B	G		6	○	F											7	○			4	○	G	
53	NOZZLE TEST PUMP ASSY.	C0913-0986Z	D	476616	6	○	D											7	○			4	○		
54																									
55																									
56																									
57																									
58																									
59	***TOOL LIST***																								
60	·DIESEL ENGINE	AQP10008522	A		7	○	A											7	○			4	○		
61	·REDUCTION GEAR	AQP10008523	A		7	○	A											7	○			4	○		
62	·TURBO CHAGER	B684300140F	C	TPS52F	7	○	C											7	○			4	○		
63	·GOVERNOR	B682500250F	A	RHD-6PC	7	○	A											7	○			4	○		
64	·F.W. COOLER	AQP20008437	B		7	○	A											7	○	B		4	○		
65	·L.O. COOLER	AQP20008438	B		7	○	A											7	○	B		4	○		
66																									
67	***SPARE PARTS LIST***																								
68	·DIESEL ENGINE	AQP10008524	B		7	○	A											7	○	B		4	○		
69	·REDUCTION GEAR	AQP10008525	C		7	○	A											7	○	C		4	○		
70	·TURBO CHAGER	B684305100Z	Z	TPS52F	7	○	Z											7	○			4	○		
71	·INTER COOLER	AQP10007605	A		7	○	A											7	○			4	○		
72	·GOVERNOR	B682500140F	Z	RHD-6PC	7	○	Z											7	○			4	○		
73	·MOTOR DRIVEN PUMP	AQP20008439	A		7	○	A											7	○			4	○		
74	·MOTOR DRIVEN PUMP STARTER	AQP20008444	A		7	○	A											7	○			4	○		
75	·LO FILTER FOR ENGINE INLET	AQP20008440	A		7	○	A											7	○			4	○		
76																									
77	***OTHER***																								
78	TESTING METHOD	AQP10001649	A	BV	7	○	A											7	○			4	○		
79																									
80	POINT FIGURE, FLUSHING	G89146-981A	Z		7	○	Z											7	○			4	○		
81	TORSIONAL VIBRATION	CAL001084	A															7	○			4	○		
82	FLUSHING PROCEDURE	AQP10007680	A		7	○	A											7	○			4	○		
83																									
84	***INSTRUCTION MANUAL***																								
85	·MAIN ENGINE			DKM-20e																		4	○		
86	·REDUCTION GEAR	INB000112	A	DRA-25JS																		4	○	A	
87	·REDUCTION GEAR PARTS LIST	PAL000071	A	DRA-25JS																		4	○	A	
88	·PROP.SOL.VALVE CONT,MANUAL	INB000126	A																			4	○	A	
89	·PROP.SOL.VALVE CONT,SPEC			Ver.1.02																		4	○		
90	·DM LIST																					4	○		
91	·TURBO CHARGER		A	TPS57F																		4	○	A	
92	·GOVERNOR			RHD6																		4	○		
93	·FLEXIBLE COUPLING	INB000081	A	GEISLINGER														7	○	A		4	○		
94	·SPEED SWITCH UNIT	INB000104	A															7	○			4	○	A	
95	·MERCURY FILTER																					4	○		
96	·ELECTRIC CENTRIFUGAL DRIVEN PUMP																					4	○		
97	·ELECTRIC GEAR DRIVEN PUMP																					4	○		
98	·HT-FW ,LO TEMP. CONTROL VALVE																					4	○		
99	·LT-FW TEMP. CONTROL VALVE																					4	○		
100																									

SPECIFICATIONS

FOR

MARINE PROPULSION SYSTEM

1,028 kWm　　2 sets/ship

C

DAIHATSU DIESEL MFG. CO., LTD.
TECHNICAL DEPARTMENT
OSAKA JAPAN

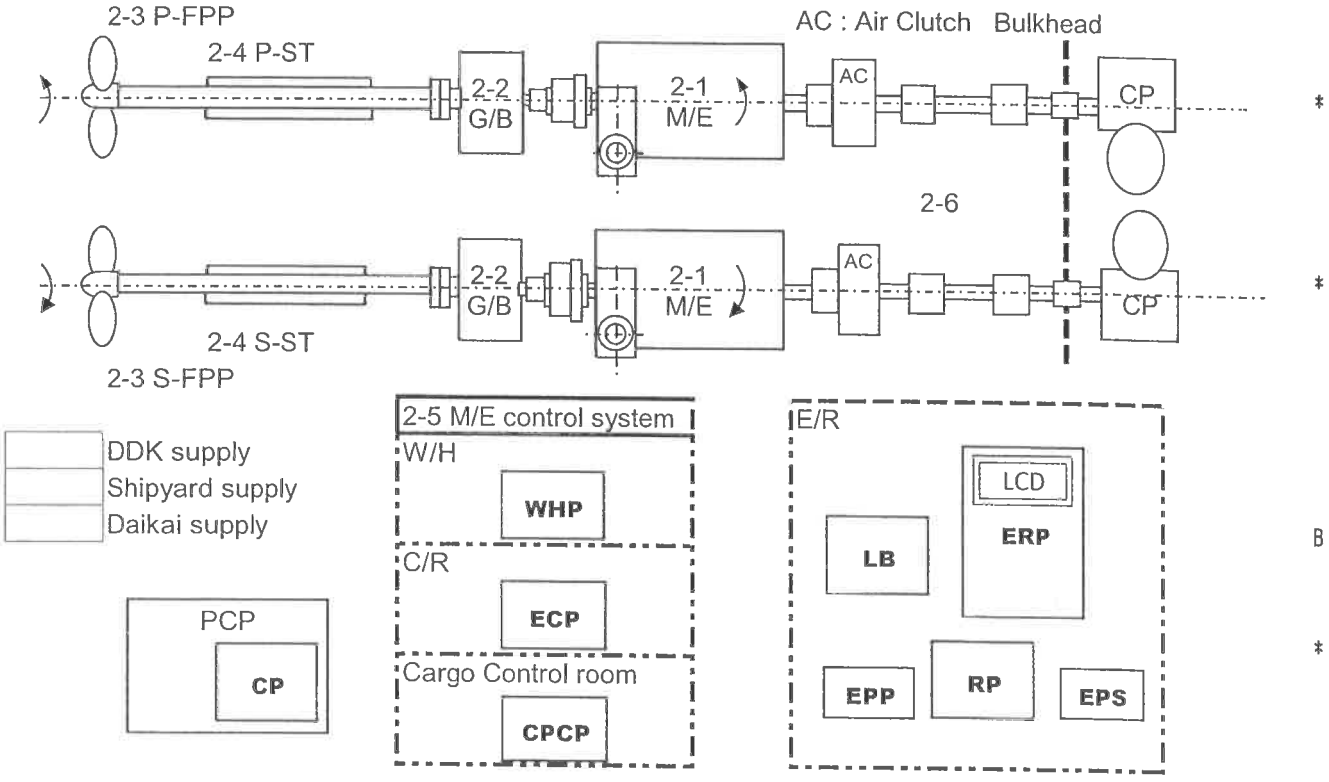
SHIP NO.	SZ15045 / SZ15046	DRAWN BY	*M. Tokumoto*
ENG. MODEL	**6DKM-20eFL　6DKM-20eF**	CHECKED BY	
LIST NO.	BQP10000031　D		
DATE	2016/5/11	APPROVED BY	*Matsushita*
REVISION	DDAP-P-1520 * Revised point from signed specification(BQP90000097A) B : Add sensor of Reduction Gear R.M. 2016.08.29 C : Revised engine output to R/G output shaft end R.M. 2016.09.26 D : Revised P.7,8,9,13 M.T. 2016.11.13		

B
C
D

1	General		
1-1	Design No.		BQP10000031 D
1-2	Shipyard		Zhejiang Shenzhou Shipbuilding Co.,Ltd.
	Ship No.		SZ15045 / SZ15046
1-3	Kind of ship		6,000 T DWT OIL TANKER 3,879 GT
1-4	Ship size	Loa × Lpp × B×D×d	88.90m×83.50m×17.30m×9.60m×6.50m
		Ship speed	11.0 knots (100% MCR)
1-5	Flag		Singapore
1-6	Owner		SINANJU TANKERS PTE LTD
1-7	Guarantee		12 months from the date of ship's delivery or 18 month after engine delivery, whichever comes earlier .
1-8	Classification Society		BV
1-9	Other rules and regulations		SOLAS, IMO, ISO
1-10	Applied industrial standard		JIS JEM
1-11	Measuring Unit		SI
1-12	Nox emissions	**Tier II**	MARPOL 73/78 Annex 4 Regulations on Nox emission EIAPP certificates issued by BV

2	Supply scope of main propulsion system		MARINE PROPULSION SYSTEM
2-1	Main engine	(M/E)	2 sets DDK supply
2-2	Gear box	(G/B)	2 sets DDK supply
2-3	Propeller	(FPP)	2 sets Shipyard supply
2-4	Shaft system	(ST)	2 sets Shipyard supply
2-5	M/E control system		1 set DDK supply
2-6	Cargo oil pump & shafting system		2 sets Daikai supply

Remarks



1-2 Basic properties

1 Figures applied in calculations/designs

			Specific gravity	Specific heat	Viscosity
1)	Diesel oil	MDO	0.890	1.88 kJ/kg℃	3~8 $mm^2/s/40℃$
2)	Gas oil	MGO	0.850	1.88 kJ/kg℃	2~6 $mm^2/s/40℃$
3)	Lubricating oil		0.930	1.93 kJ/kg℃	
4)	Fresh water		1.000	4.19 kJ/kg℃	
5)	Sea water		1.020	3.94 kJ/kg℃	

Remarks : Low Sulphur Fuel Oil (**LSFO**) also can be used. The viscosity at the engine inlet should be kept 2.0 mm^2/s or more. When the fuel viscosity at the engine inlet becomes lower than the 2.0 mm^2/s, due to a rise in fuel temperature, etc., it is recommended to maintain adquate viscosity by installing fuel cooler, chiller, etc. at the engine inlet.

2 Available utility

1) Electrical source			
1	Main circuit	AC 415 V × 50 Hz × 3 φ	
2	Auxiliary circuit	AC 220 V × 50 Hz × 1 φ	
3	Control and buck up circuit	DC 24 V	by battery
2) Air source			
1	Control	0.69 MPa	
2	Starting	3.00 MPa	

3 Engine can operate as bellow condition

1) Statically	Trimming	up to	5	degree
	Heeling	up to	15	degree
2) Dynamically	Pitching	up to	7.5	degree
	Rolling	up to	22.5	degree

2 Principal specifications

2-1 Principal specifications of engine			Specification		
1	Type		Non-reversible, single action,4-cycle, direct injection, trunk piston, with turbo-charger and air cooler, In-line		
2	Model				
2-1		Port	6DKM-20eFL	+ DRA-25JL	C
2-2		Starboard	6DKM-20eF	+ DRA-25J	C
3	Quantity		each 1 set		
4	Rated output (MCR)		1,028 kW	R/G output shaft end	C
			1,060 kW	Crank shaft end	C
	Front PTO rating	abt.	610 kW	(at 900 min^{-1}) max. 961 kWm	
				Please do not use propeller and front end at the same time.	D
5	Rated speed		900 min^{-1}		
6	Idle speed		480 min^{-1}		
7	Number of cylinder		6		
8	Cylinder bore		200 mm		
9	Cylinder stroke		300 mm		
10	Brake mean effective pressure		2.42 MPa		
11	Max. cylinder pressure		17.7 MPa		
12	Mean piston speed		9.00 m/s		
13	Over load		1131 kW	929 min^{-1}	
			10% overload: 60 minutes (every 12 hours)		
14	100% constant toruqe range		765 - 900 min^{-1}	85% - 100%	
15	Propeller design point		85 % at	900 min^{-1}	
16	Direction of rotation	engine			
16-1		Port	Anti-Clockwise (as viewing from the propeller side)		
16-2		Starboard	Clockwise (as viewing from the propeller side)		
17	Engine starting method		Compressed air		

18	Specific fuel oil consumption		198	g/kWh at 4/4 load, max. +5% torerance and lower calorific value of fuel oil 42,700kJ/kg under ISO3046/1 standard reference condition.		
19	Specific lubricating oil consumption		0.8	g/kWh at 4/4 load (Reference value)		
20	Dimensions					
20-1	Length		3,380	mm	T/C at the rear side(STD)	‡
20-2	Height		2,605	mm		
20-3	Width		1,822	mm	960mm Installation	
20-4	Piston overhauling height		1,575	mm	From crank shaft center	
21	Total dry mass		11.3	ton	Without flywheel weight	
22	Conditions					
22-1	Design conditions					
1)	Ambient temperature		5-45	℃		
2)	Relative humidity		~95	%		
3)	Sea water temperature		32	℃		
4)	Ambient atmospheric pressure		100	kPa		
22-2	Usage conditions					
1)	HT-FW temperature		75	℃	at engine outlet	
2)	HT-FW pressure		0.25-0.35	MPa	at engine inlet	
3)	Max exhaust gas pressure		2.94	kPa	at full load	
4)	FO pressure		0.2-0.3	MPa	at engine inlet	
5)	FO viscosity		min. 2.0	mm^2/s	at engine inlet	
23	Combustion air consumption		7,714	m3/h	at full load (ambient temp. 25℃)	‡
24	Exhaust gas					
24-1	Exhaust gas volume		15,115	m3/h	at full load 324 ℃	
24-2	Exhaust gas temperature	abt.	324	℃	at full load	
25	LT-FW capacity		40	m3/h/1eng.	38 ℃	
26	Recommendable lubricating oil					
1)	System oil		API service grade CD SAE30 TBN 10 to 15			
2)	LO for turbo charger		Supplied from system oil			
27	Fluid holdings in engine					
1)	LO	Engine	86	L		
		LO cooler	15.5	L		‡
		Oil pan	0	L	Dry	
		Sump tank	1,800	L		
		Total	1,902	L		
2)	Cooling HT-FW	Engine	85	L		
3)	Cooling LT-FW	Air cooler	23	L		
		LO cooler	15.5	L		‡
		LT-FW cooler	28.9	L		‡
4)	Cooling FW	Expansion Tank	500	L	/1eng.	

2-2 Principal specifications of RG			Specification	RG= Reduction Gear	
1	Maker		DAIHATSU		
2	Type		Single stage reduction gear, turning outwards seeing from top, with hydraulically operated multi-disc clutch		
3	Model		DAIHATSU		‡
3-1		Port	DRA-25JL		
3-2		Starboard	DRA-25J		
4	Quantity		each 1 set		
5	Gear ratio	Propeller	AH : 3.752	AS: 4.612	‡
6	Shaft speed				

6-1		Input	Engine	900 min^{-1}		
6-2		Output	Propeller	240 min^{-1}		
7	Shaft power					
7-1		Input	Engine	1,060 kW	R/G efficiency: 0.97	C
7-2		Output	Propeller	1,028 kW		C
8	Direction of rotation					
8-1	Input	Engine	Port	Anti-Clockwise (as viewing from the propeller side)		
			Starboard	Clockwise (as viewing from the propeller side)		
8-2	Output	Propeller	Port	Anti-Clockwise (as viewing from the propeller side)		
			Starboard	Clockwise (as viewing from the propeller side)		
9	Total dry mass			4.1 ton		

2-3 Principal specifications of Propeller			Shipyard supply	
1	Maker			
2	Type		FPP Propeller	
3	Model			
4	Propeller diameter / Number of blades		2,600 mm / 5 blade	C
5	Direction of rotation	Port	Anti-Clockwise (as viewing from the propeller side)	
		Starboard	Clockwise (as viewing from the propeller side)	
6	Material blades		NiAL Bronze	
7	Hub and blades mass		kg	

2-4 Principal specifications of Shafting system		Shipyard supply				Q/ship
1	Propeller shaft	Dia.	mm	Length	mm	
2	Intermediate shaft	Dia.	mm	Length	mm	
3	Bearing for intermediate shaft					
4	Stern tube bearing					
5	Stern tube					
6	Miscellaneous					

3 General requirements				Specification	
1	Start/stop system			Start Manual at engine side only	
				Stop Automatic, emergency manual	
2	Fuel oil			MDO or MGO	
3	Kind of fuel oil at start			MDO or MGO	
4	Fuel supply system			Open system	
5	Lubricating oil system			Continuous purification	
6	Cooling system				
	1)	Jacket	HT-FW	Fresh water	75℃ at engine outlet
	2)	Cooler	LT-FW	Fresh water	38℃ at R/G LO cooler inlet
7	Starting method			Engine side start only	
8	Turning method			Ratchet type	
9	Installation method			Rigid mounting	
10	Coupling method				
	1)	Engine - R/G		Flexible coupling	
	2)	Front PTO - Cargo pump		Flexible coupling - Air clutch - Cargo pump	

Remarks

4 Accessories

Q/E=Quantity /1 Eng. Sup=supplier Ins=Installation place
Q/G=Quantity /1 R/G D (D)=To be supplied by Daihatsu(Daikai) E=Attached to engine
Q/S=Quantity /1 Ship S=To be supplied by shipyard G=Attached to RG

4-1 Cooling water system		Q/E	Sup	Ins	Specification
1	HT-FW pump (Cooling fresh water pump for jacket)	1	D	E	Engine driven Centrifugal type 36 m3/h × 25 mTH
2	CW piping internal treatment	1	D	E	FW piping Parkerizing
3	Internal treatment of the water cover part of air cooler	1	D	E	Nontar-epoxy coating

Remarks

1	HT- FW temp. control valve	1	D	S	80A 75℃ at engine outlet Wax type
2	Stand-by HT-FW pump	1	S	S	Motor driven Centrifugal type 36 m3/h × 25 m 7.5 kW
3	LT-FW pump	1	D	S	Motor driven Centrifugal type 40 m3/h × 20 m 5.5 kW
4	Stand-by LT-FW pump	1	S	S	Motor driven Centrifugal type 40 m3/h × 20 m 5.5 kW
5	LT-FW cooler	1	D	S	Plate type 20.15 m^2
6	LT- FW temp. control valve	1	D	S	80A 38℃ at LT-FW pump outlet E-P type
7	Cooling SW pump	2	S	S	Motor driven Centrifugal type 50 m3/h × 18 m 5.5 kW
8	CFW expansion tank	1	S	S	500 L

4-2 Fuel oil system		Q/E	Sup	Ins	Specification
1	Fuel oil feed pump	1	D	E	Engine driven Gear type 0.6 m3/h × 0.4 MPa
2	Fuel oil No.2 filter	1	D	E	75 μm 200mesh duplex type
3	FO high-pressure pipe	6	D	E	Block type
4	FO injection pump	6	D	E	Bosch type 150MPa
5	FO injection valve	6	D	E	Non-cool type
6	FO level switch box	1	D	E	Leaked oil detect
7	FO relief valve at engine outlet	1	D	E	Setting press. 0.2-0.3 MPa
8	FO damper	2	D	E	Engine inlet & outlet

Remarks

1	FO marcury filter	1	D	S	10 micron single type installed on to suction side
2	Fuel oil No.1 filter	1	S	S	100 mesh Duplex type installed on to suction side
3	Stand-by Fuel oil pump	1	S	S	Motor driven Gear type 0.5 m3/h × 0.4 MPa 0.75 kW

4-3 Lubricating oil system		Q/E	Sup	Ins	Specification
1	Lubrication oil pump	1	D	E	Engine driven Gear type 32.0 m3/h × 0.64 MPa
2	LO oil pan	1	D	E	Dry type
3	LO by-pass filter	1	D	E	Centrifugal type
4	LO safety filter	1	D	E	75 μm Notch-wire duplex type

Remarks

1	Stand-by Lub.oil pump	1	D	S	Motor driven 20.5 m3/h× 0.6 MPa 7.5 kW	*
2	Stand-by Lub.oil pump starter	1	D	S	AC 415 V × 50 Hz × 3 φ 7.5 kW	
3	LO filter for engine inlet	1	D	S	30 μm Auto-backwash , diff.press. SW 63QD	*
4	LO by-pass filter	1	D	S	35 μm 400 mesh single type	*
5	LO cooler	1	D	S	Plate type 10.46 m^2	*
6	LO temperature control valve	1	D	S	80A 55 ℃ Engine inlet	*
7	LO sump tank	1	S	S	1,800 L	
8	LO filter for pump inlet	1	S	S	150 μm 100mesh duplex type	
9	LO filter for S/B LO pump inlet	1	S	S	150 μm 100mesh simplex type	

4-4 Governing device		Q/E	Sup	Ins	Specification	
1	Governor	1	D	E	Bosch Automobile made All speed type operate by control air Model: RHD-6PC air-less: 450 min^{-1}	*

Remarks

at propeller clutch off :480min-1 ,cancel at front PTO use *
If HT-CW temp, below 60℃:700min-1, for engine warming

1	EP valve panel	**EP P/S**	1	D	S	Wall hanging type with cover E-P converter x 2, magnetic valve for clutch, press.SW for alarm 63AC	*

4-5 Intake/exhaust system		Q/E	Sup	Ins	Specification	
1	Air cooler	1	D	E	Fin tube type LT-FW cooled	
2	Exhaust gas turbo charger	1	D	E	ABB made TPS52F Radial turbine, fit to rear side	
3	T/C cleaning units	1	D	E	Blower side cleaning device	
4	Exhaust gas pipe cover	1	D	E	Full sealed cover for fire protect	
5	Y strainer	1	D	S	300mesh	D

Remarks

1	Exhaust gas expansion joint	1	D	S	For T/C outlet 300A x 400 A	
2	Exhaust gas expansion joint	1	D	S	For Silencer 400A × 450 A	D
3	Exhaust gas silencer	1	(D)	S	35 dB(A) reduce 450 A	*
					Back press. 70 mmAq	*

4-6 Starting air system		Q/E	Sup	Ins	Specification
1	Main air starting valve	1	D	E	
2	Starting rotary valve	1	D	E	
3	Bursting disc	1	D	E	
4	SA piping internal treatment	1	D	E	Parkerizing

Remarks

1	Starting air tank		1/ship	D	S	200 L x 2, 2.94 MPa press.SW for alarm 63A	*
2	Reducing panel	**RP**	1/ship	D	S	Reducing valve, safety valve, pressure gauge	*
3	Drainage valve			S	S	Installed the bottom of piping for drainage	
4	Air compressor			S	S	min. charge capacity 15.5m^3/h	

4-7 Installed & coupling parts		Q/E	Sup	Ins	Specification
1	Fitting bolts & nuts	16	D	E	Both end screw type
2	Reamer bolts & nuts	2	D	E	Both end screw type 24H7 Reamer
3	Jack bolts	4	D	E	M20 x 2.5
4	Flywheel with bolts & nuts	1set	D	E	
5	Flexible coupling with bolts & nuts	1set	D	E	Steel spring type

Remarks

1	Flywheel turning device	1	D	S	Ratchet type
2	Chock fast	1 set	S	S	Material
3	Flywheel cover	1	S	S	Steel plate fabrication
					DDK to be submitted the drawing for reference

4-8 Etceteras		Q/E	Sup	Ins	Specification
1	Free end PTO	1	D	E	Tapered shaft end with key
2	Cam shaft	1	D	E	Mono-block type
3	Rack indicator	1	D	E	
4	Engine frame safety valve	1	D	E	Quantity by rule requirement
5	Oil pan for filter	1	D	E	Welded steel plate with drain hole
6	Name plate	1	D	E	Engine, panel, valves, caution
					Measuring unit SI
					Name English
7	Special parts for Rule	1	D	E	Thermal switch with pocket
		1	D	E	Connection plug for testing
		1	D	E	Fire protection of SOLAS'98
		1	D	E	PL caution plate
8	Counter flange	1	D	E	
9	Connection plug for testing	1	D	E	G1/4 with plug

4-9 Instruments		Q/E	Sup	Ins	Specification			
1	Gauge board	1	D	E	Elastic mounting type			
2	Pressure gauge (Bourbon tube type)				Gauge scale		Blue mark	
	1) LO engine inlet	1	D	E	1.0	MPa	0.4 - 0.55	MPa
	2) LO for T/C inlet	1	D	E	1.0	MPa	0.20 - 0.50	MPa
	3) Boost air engine inlet	1	D	E	0.4	MPa	With damper	
	4) HT-FW engine inlet	1	D	E	0.6	MPa	0.25 - 0.35	MPa
	5) FO engine inlet	1	D	E	0.6	MPa	0.2 - 0.3	MPa With damper
	6) LT-FW engine inlet	1	D	E	0.4	MPa		D
	7) Governor control air	1	D	E	1.0	MPa		*
3	Tachometer	1	D	E	1,500	min-1		
4	Thermometer							
	1) LO							
	1 Engine inlet	1	D	E	Alcohol	type	100 ℃	G 1/2
	2 Oil cooler inlet & outlet	2	D	C	Alcohol	type	100 ℃	G 1/2
	2) Boost air							
	1 Air cooler outlet	1	D	E	Alcohol	type	100 ℃	G 1/2
	3) HT-FW (Jacket CW)							
	1 Engine inlet & outlet	2	D	E	Alcohol	type	100 ℃	G 1/2
	4) FO							
	Engine inlet	1	D	E	Alcohol	type	100 ℃	G 1/2
	5) LT-FW (Cooler CW)							
	1 Air cooler inlet & outlet	2	D	E	Alcohol	type	100 ℃	G 1/2
	2 LT-FW cooler inlet & outlet	2	D	C	Alcohol	type	100 ℃	G 1/2
	3 LO cooler inlet & outlet	2	D	C	Alcohol	type	100 ℃	G 1/2
	6) Exhaust gas							
	1 Each cylinder outlet	6	D	E	Dial	type	650 ℃	G 1/2
	2 Turbo charger inlet	2	D	E	Dial	type	650 ℃	G 1/2
	3 Turbo charger outlet	1	D	E	Dial	type	650 ℃	G 1/2

Remarks

Ins. Marked "C" is Cooler

4-10 Remote sensor and switch for alarm and indicate			Q/E	Sup	Ins	Specification			
1	Pressure transmitter wit root valve 4-20 mA					Scale	Alarm setting		
PTQ	1)	LO engine inlet		D	E	1.0 MPa	0.25 MPa	63Q1	
PTWH	2)	HT-FW engine inlet		D	E	0.6 MPa	0.15 MPa	bellow	
PTF	3)	FO engine inlet		D	E	0.6 MPa	0.15 MPa	bellow	
PTB	4)	Boost air engine inlet		D	E	0.4 MPa	only indicate		
PTQT	5)	LO T/C inlet		D	E	1.0 MPa	0.20 MPa	63QT	
PTAG	6)	Governor control air engine inlet		D	E	1.0 MPa	only indicate		‡
PTQG	7)	LO R/G inlet		D	E	1.0 MPa	0.12 MPa	bellow	D
2	Thermo sensor Pt 100 ohm					Scale	Alarm setting		
TRQ	1)	LO engine inlet		D	E	100 ℃	65 ℃	over	
TRWH	2)	HT-FW engine outlet		D	E	100 ℃	85 ℃	over	
TRF	3)	FO engine inlet		D	E	100 ℃	60 ℃	over	
TRB	4)	Boost air engine inlet		D	E	100 ℃	only indicate		
TRC1-6	5)	Exh. gas each cyl outlet		D	E	700 ℃	480 ℃	over	
TRTI1,2	6)	Exh. gas T/C inlet		D	E	700 ℃	600 ℃	over	
TRTO	7)	Exh. gas T/C outlet		D	E	700 ℃	only indicate		

4-11 Protective device			Q/E	Sup	Ins	Specification			
1	Switch for shut-down (abnormal contact ON)					Setting value			
	1)	LO pressure engine inlet	1	D	E	0.2 MPa	bellow	63Q2	
	2)	Over speed	1	D	E	1,020 min-1	over	12	
	3)	Gear box LO pressure	1	D	G	0.08 MPa	bellow	63QG2	
2	Switch for slow-down request (abnormal contact ON)					Setting value			‡
	1)	Jacket CW temperature engine outlet				90 ℃	over	26WH3	‡
	2)	JW engine inlet press.				0.1 MPa	bellow	63WH3	‡
3	Switch for alarm (abnormal contact OFF)					Setting value			
	1)	FO injection pipe leakage	1	D	E	High level		33F	
	2)	LO filter high deff.press.	1	D	S	1.0 MPa	0.09 MPa	63QD	
	3)	Control air press. Low	1	D	S	1.5 MPa	0.49 MPa	63AC	
4	Switch for control (contact ON)					Setting value			‡
	1)	LO pressure engine inlet	1	D	E	0.27 MPa	over	63QS	‡
	2)	LO pressure engine inlet	1	D	E	0.27 MPa	bellow	63QP	‡
	3)	LO pressure R/G inlet	1	G	E	0.15 MPa	bellow	63QGS	D
	4)	LO pressure R/G inlet	1	G	E	0.15 MPa	bellow	63QGP	D
	5)	HT-FW engine outlet	1	D	E	60 ℃	bellow	26WHC	D
5	Junction box		1 set	D	E				
6	Wiring of electrical equipment		1	D	E	Between each sensors and junction box			
7-1	Shutdown device		1	D	E	DDK made G2 type			
7-2	Shutdown magnetic valve (5V)		1	D	E	Electric source DC24V±20%,15W Control air 0.7-0.9 MPa			
8-1	Shutdown device for each cylinder		6	D	E	Bosh made Installed to FO pump			
8-2	Shutdown magnetic valve for each cylinder shutdown (5S)		1	D	E	Electric source DC24V±20%,15W Control air 0.7-0.9 MPa			
9	Handle switch (HS)		1	D	E	DC30V,6A for trip reset			
10	Turning safety switch (TS)		1	D	E	Limit switch for engine start ready			‡
11	Start interlock magnetic valve (88S)		1	D	E	Electric source DC24V±20%,15W Control air 0.7-0.9 MPa			

All alarm, state signal to be send (**RS422**) to AMS(Alarm monitoring system) which supply by Shipyard.

4-12 Accessories for Reduction gear		Q/G	Sup	Ins	Specification	
1	Clutch	1	D	G	Wet & multi-disc type	
2	Clutch operating valve	1	D	G	Control air operated	
3	Thrust bearing	1set	D	G	Mitchell bearing type	
4	Lubrication oil pump each	1	D	G	Gear driven Gear type	*
					12 m3/h × 1.90~2.10 MPa	*
					19.5 m3/h × 0.25~0.4 MPa	*
5-1	LO relief valve for high-pressure	1	D	G	Set 1.90~2.10 MPa	
5-2	LO relief valve for low-pressure	1	D	G	Set 0.25~0.40 MPa	*
6	Gear box LO cooler	1	D	G	Low-fin type 8.28 m^2	*
					LT-FW capacity 35 m^3/h	
7	LO filter	1	D	G	Notch-wire duplex type	
					125 μm 100 mesh	*
8	Oil pan	1	D	G	120 L initial : 150 L	*
					with level switch for alarm(33QGL), oil gauge	B
9	Gauge board	1	D	G	Elastic mounting type	
	Pressure gauge (Bourbon tube type)				Gauge scale Blue mark	
	1) Clutch LO	2	D	G	3.5 MPa 1.90 - 2.10 MPa	
	2) LO gear box inlet	1	D	G	0.6 MPa 0.25 - 0.40 MPa	
10	Thermometer					
	LO					
	1) Oil cooler inlet & outlet	2	D	G	Alcohol type 100 ℃ G 1/2	
	2) Thrust bearing	1	D	G	Alcohol type 100 ℃ G 1/2	D
	LT-FW					
	1) Oil cooler inlet & outlet	2	D	G	Alcohol type 100 ℃ G 1/2	B
11	Proportional magnetic valve unit	1set	D	G		
12	Installed parts					
	1 Chock liner	8	D	S	t = 55 mm one side finishing	*
	2 Fitting bolts & nuts	4	D	G	Both end screw type	
	3 Reamer bolts & nuts	4	D	G	Both end screw type 36H7 Reamer	
					M33 x 3.5 spare : 1 pc/ship	*
	4 Jack bolts	4	D	G	M24 x 3.0	
	5 Reamer bolts & nuts	8	D	G	Both end screw type 32H7 Reamer	*
	for output flange				M30 x 2.0	*
13	Sensor for slow-down request (abnormal contact ON)					
	1 Clutch LO temperature TRQG	1	D	G	85 ℃ over 26QG3	B
	2 Thrust bearing temperature TRGT	1	D	G	95 ℃ over 26GT3	B

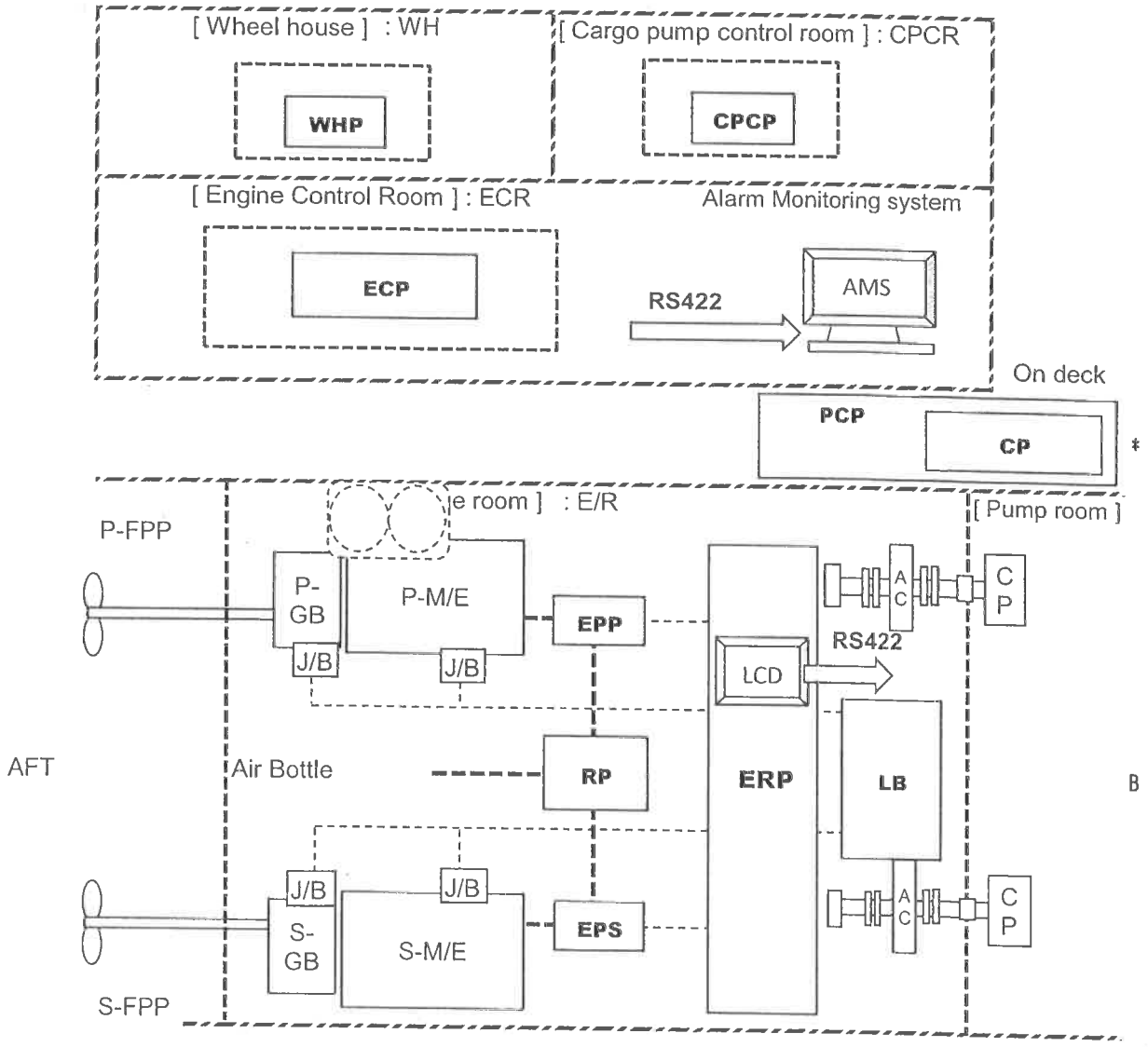
Remarks

1 Stand-by Lub.oil pump	1	D	S	Motor driven	
				10 m3/h × 2.4 MPa	*
2 Stand-by Lub.oil pump starter	1	D	S	AC 415 V × 50 Hz × 3 φ 11 kW	
3 Spacer pipe	1	D	S	for LO Flushing filter	*
4 LO Flushing filter	1	D	S	Gauge-wire simplex type 150 mesh	*

The Main engine and Propulsion syatem to be accordance with BV .

Application list of control and Instrument

WHP	: W/H CONTROL PANEL	Insert type with maneuverng handle Engine / Propeller tachometer	
ECP	: C/R CONTROL PANEL	Insert type with maneuverng handle Engine / Turbochager tachometer	
ERP	: M/E RELAY PANEL	Self-standing type LCD	IP22
LB	: LOCAL BOX	Wall-hanging type	
EP P/S	: E-P VALVE PANEL	Wall-hanging type	63AC
RP	: REDUCING PANEL	Wall-hanging type	
CPCP	: CARGO PUMP CONTROL PANEL	Insert type with control dial Engine tachometer	
PCP	: PUMP CONTROL PANEL	Daikai supply	
CP	: CARGO PANEL	Wall-hanging type	



: air pipe

: electric cable

7 Spare parts and tools

7-1 Tools for one ship		Q/S	Unit	Sup	Specification
1	Tools for M/E overhauling	1	set	D	
1-1					One set of Daihatsu standard tools including oil press. pump for overhaul such as piston, cylinder-liner, etc.
1-2	Indicator	1	set	D	Max pressure gauge, dial type MT31-200
1-3	Fuel injection nozzle tester	1	set	D	
1-4	Deflection gauge	1	set	D	
1-5	Cylinder bore gauge	1	set	D	
1-6	Measuring position guide	1	set	D	For cylinder bore gauge
1-7	Thickness gauge	1	set	D	
1-8	Torque wrench	1	set	D	For cylinder head, main bearing, connecting rod
2	Blower syringe & washing	1	set	D	For turbo-charger cleaning
3	Tools for turbo-charger	1	set	D	Maker standard
4	Tools for governor	1	set	D	Maker standard
5	Sample board of colored	2	pcs	D	Munsel No. 2.5G7/2

7-2 Spare parts for one ship		Q/S	Unit	Sup	Specification
1	For main engine				
1-1	Piston pin metal	1	cyl	D	
1-2	Crank pin metal	1	cyl	D	
1-3	Bolt and nuts for crank pin metal	1	cyl	D	
1-4	Fuel high press. pipe	1	cyl	D	
1-5	Fuel oil injection pump	1	cyl	D	Complete
1-6	Piston ring	1	cyl	D	
1-7	Oil ring	1	cyl	D	
1-8	Air intake valve	1	cyl	D	Complete
1-9	Exhaust valve	2	cyl	D	Complete
1-10	Fuel nozzle holder	6	cyl	D	Complete
1-11	Starting air valve	1	cyl	D	Complete
1-12	Cylinder relief valve	1	cyl	D	Complete
1-13	Cylinder cover	1	cyl	D	Complete with valves
1-14	Cylinder cover blot & nuts	1/2	cyl	D	
1-15	Cylinder liner	1	cyl	D	Complete
1-16	Piston	1	cyl	D	Complete
1-17	Connecting rod	1	cyl	D	Complete
1-18	Main bearing	1	cyl	D	With bolts and nuts
1-19	Gasket & packing	1	cyl	D	Each size
1-20	Thermometer	1	set	D	Each kind
1-21	Bearing and sealing of pumps	1	set	D	Each size
2	For R/G				
2-1	Bearing	1	set	D	Complete, each size
2-2	Thrust bearing pad	1	set	D	With adjusting liners
2-3	Reamer bolts	1	set	D	For input and output shaft coupling
3	For turbo-charger				
3-1	Sealing bush	1	set	D	
3-2	Bearing	1	set	D	
4	Governor spare parts	1	set	D	Maker standard
5	Control device spare parts	1	set	D	Maker standard

8 Shop trials

Details will be described in "SHOP TRIAL PROCEDURE : AQP10007761A BV " for apploval plan.

9 Etceteras

9-1 Supplied parts by shipyard

9-2 Painting color

1	Engine & Panel	Munsel 2.5G7/2
2	Etc	Munsel 2.5G7/2

9-3 Drawings for engine

Kind of DWG.	COPY	SCHEDULE
Approval	6 1	21days after contract come into force For return
Working	6	15days after approval(received comment from shipyard)
Finished	4	On derivery

10. Points of Notice for Installation Work

10-1. Exhaust Gas Piping Work

We strongly recommends that exhaust gas pipe near the engine should be fixed on the hull firmly, in case the exhaust gas pipe moves on the engine. If too much pipe weight is put on the engine, it may cause damage to the turbo charger and expansion joint.

Please understand that the expansion joint and the turbo charger are for the absorption of vibration and thermal deformation of D/G, and not for the weight and thermal deformation of the pipe in the

Kindly refer to the approval drawing "BQP80000021" for details of the installation points of the expansion joint.

Heat insulation for exhaust gas piping of hull side including the expansion joint should be carried out by shipyard.

10-2. Service Air Piping Work

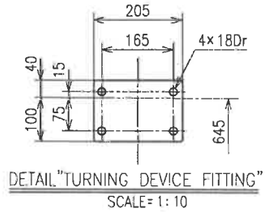
1 Please set up Drain Valve at the bottom on the way of the piping.

2 After finishing the piping work, please carry out cleaning the pipe sufficiently. It is essential work.

10-3. Service LO Piping Work

1 Since the engine may be damaged, the piping from the LO auto backwash filter to the engine inlet should carry out acid washing and flashing carefully.

REDUCTION GEAR		DIESEL ENGINE	
TYPE	DRA-25J(L)	TYPE	6DKM-20eF(L)
REDUCTION RATIO(AH)	3.752	NO. OF CYL	6
REDUCTION RATIO(AS)	4.612	BORE	200 mm
PROPELLER REVO.	240 min⁻¹	STROKE	300 mm
TORQUE INPUT	11247 Nm	REVOLUTION	900 min-1
		B.H.P	1060 kW
CLUTCH	WET & MULTI-PLATE	B.M.E.P	2.50 MPa
THRUST BEARING	MITCHELL TYPE	MAX.PRESS	17.7 MPa
		PISTON SPEED	9.00 m/s
CONTROL SYSTEM	LOCAL & REMOTE	STARTING	COMP.AIR
		COOLING	JACKET: F.W COOLER: F.W
WEIGHT	4100 kg	WEIGHT	11300 kg
DAIHATSU DIESEL MFG.CO.LTD.		DAIHATSU DIESEL MFG.CO.LTD.	



DETAIL"TURNING DEVICE FITTING"
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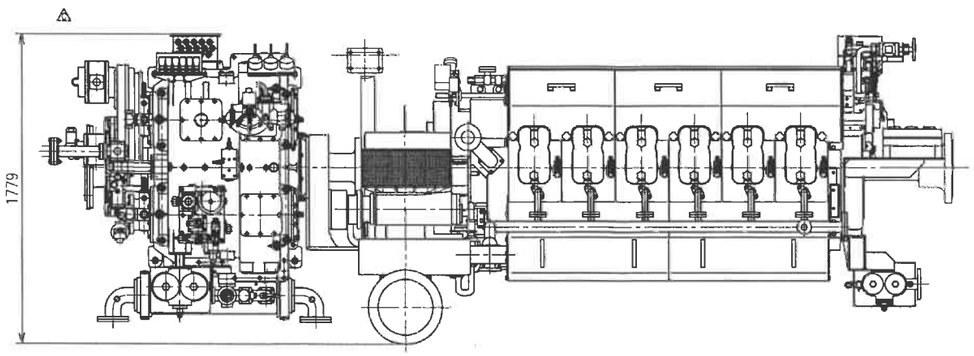
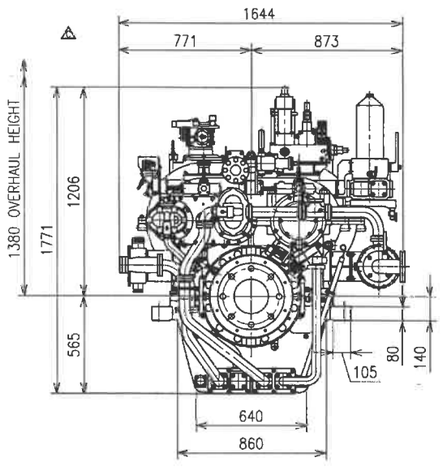
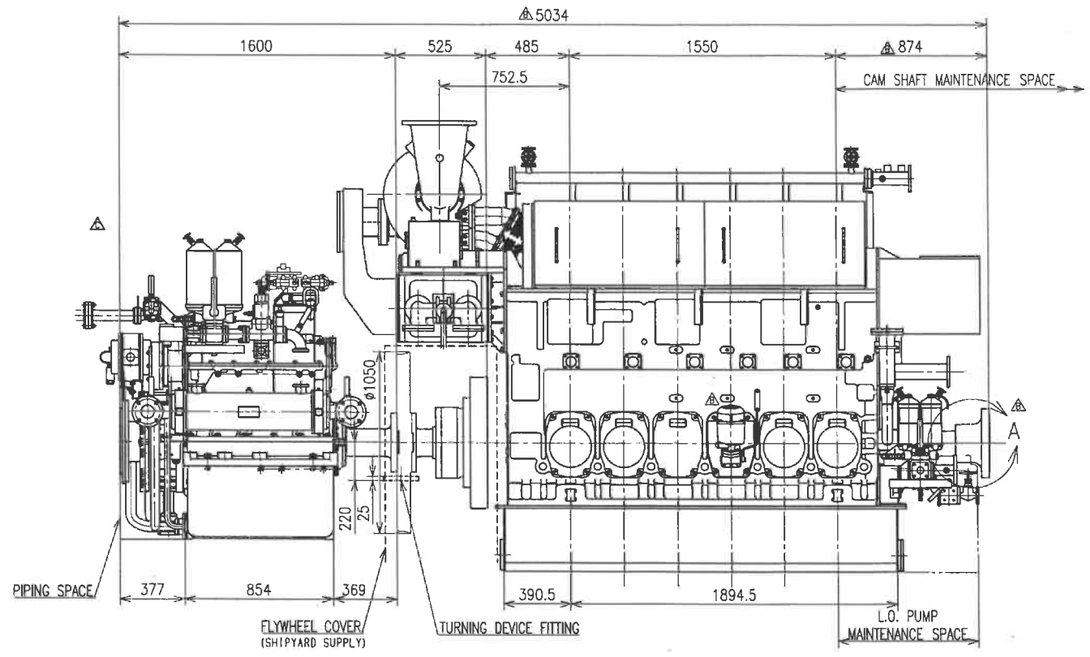
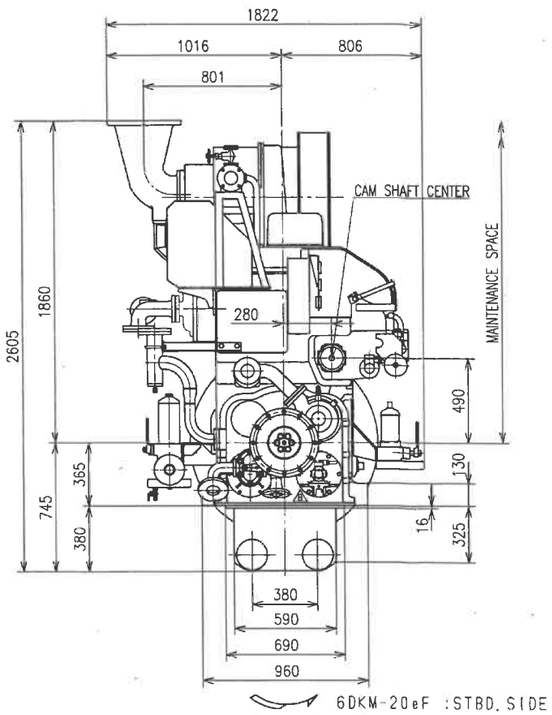
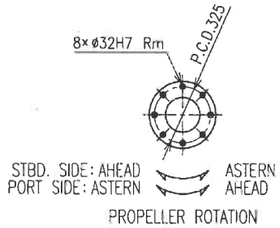




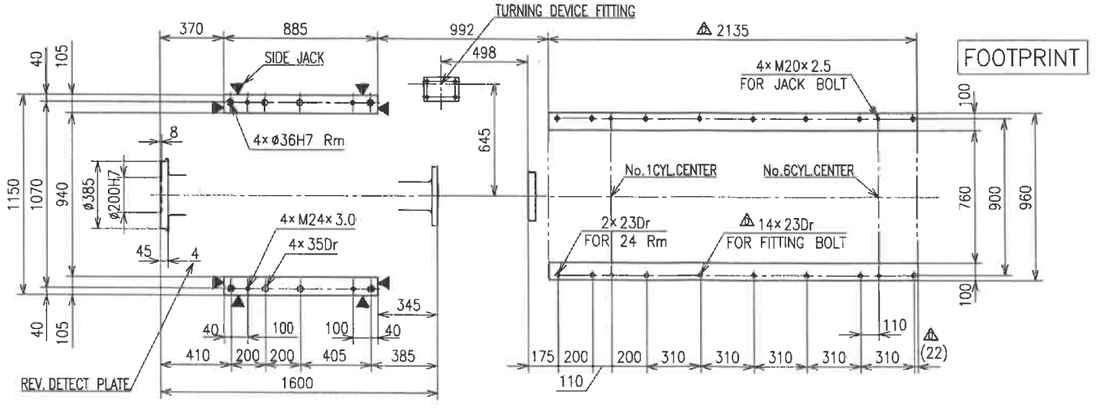


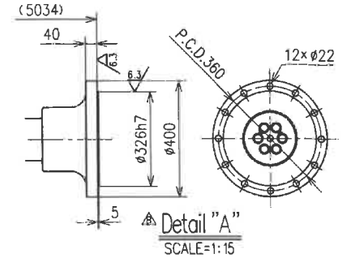


6DKM-20eF :STBD. SIDE
6DKM-20eFL:PORT SIDE
ENGINE ROTATION



PROPELLER ROTATION





REMARKS:
1. THE SPACE OCCUPIED BY ENGINE ARE SHOWN BY ——·——LINE.
2. SUITABLE PROTECTOR UNDER THE FLY-WHEEL AGAINST BILDGE WATER SHOULD BE APPLIED.
3. PLEASE REFER TO THE "DISASSEMBLING SPACE PLAN" WHEN YOU CONSIDER THE LAYOUT OF THE ENGINE ROOM.

⚠x3	(5-B,6-A) REV:DIMENSION	2016/12/20	M.T
⚠x3	(1-D,3-D/G) REV:REDUCTION GEAR	2016/09/20	M.T
⚠x6	(5-D/E,6-E,7-D,8-A,9-C) REV:DIMENSION, BYPASS FILTER POSITION, L.O PUMP INLET ADD:COUPLING (Detail"A")	2016/07/05	M.T
⚠	FIRST DRAWING	2016/05/27	M.T

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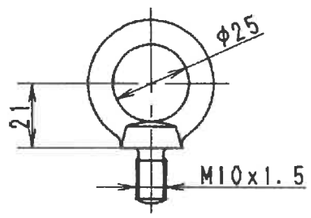
APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5	MATERIAL APRV.	GROUP 0
		BEFORE		MASS ⚠ QTY.	SCALE: 1:25
CHECKED	FUJIMOTO SYUHEI	REF.	BQP10000038	TYPE DK-20e	A2
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN		NAME ガイケイズ: :6DKM-20eF(L) x DRA-25J(L)	
DRAWN	(H)MORICHI YASUKO			OUT LINE: :6DKM-20eF(L) x DRA-25J(L)	
DATE	2016/12/20			PARTS NO. AQP10008490	D x

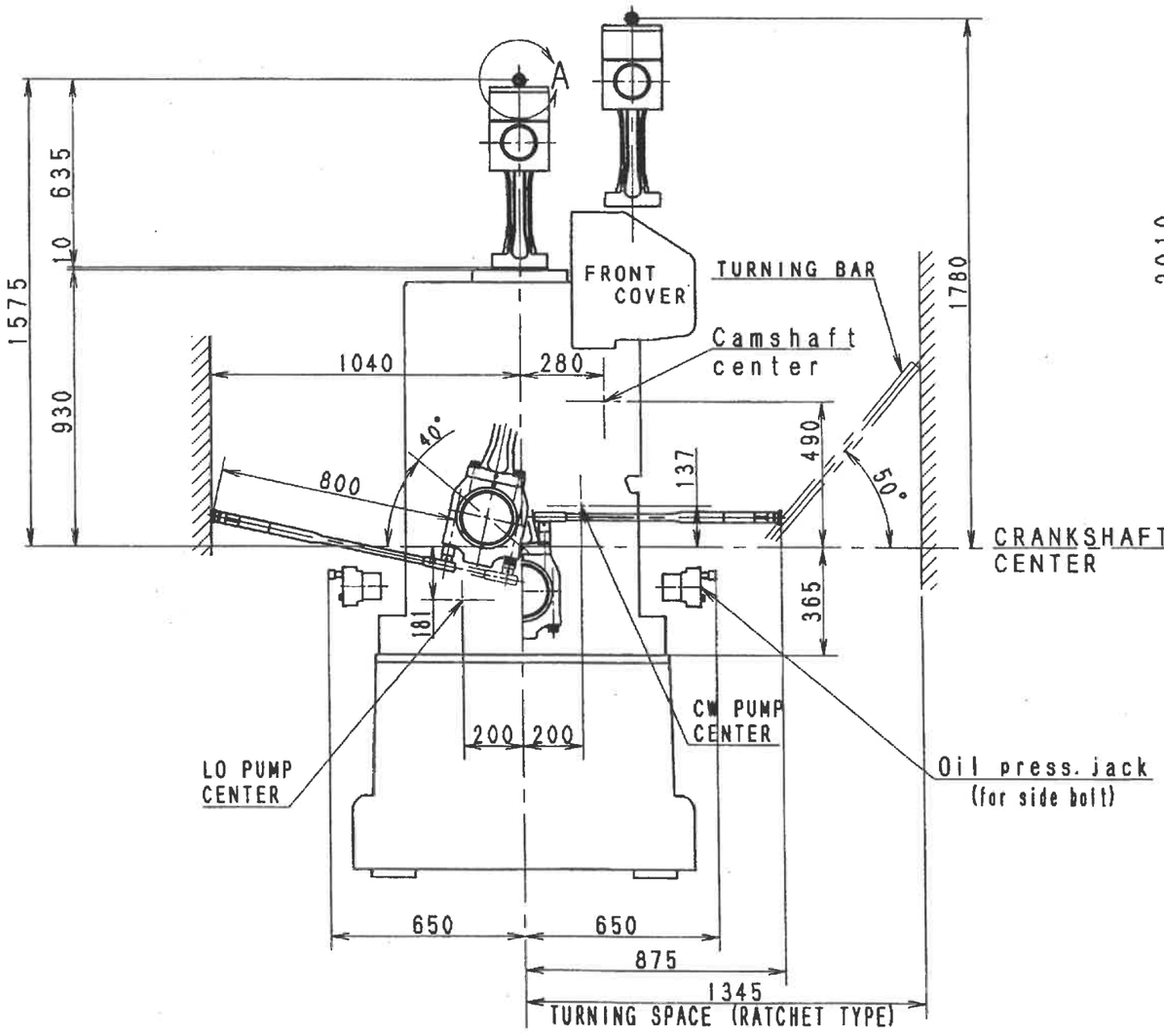
2017/05/09 TECHNICAL DEPARTMENT

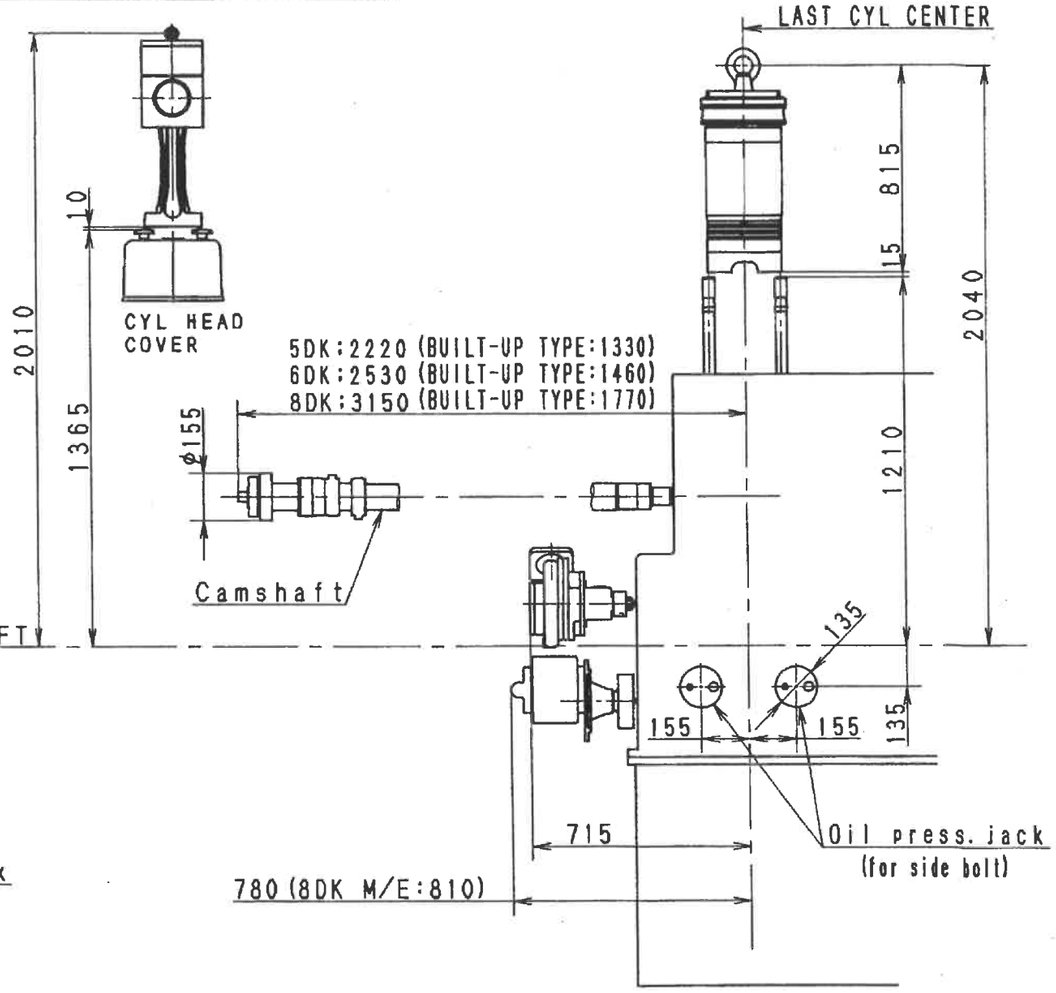
HISTORY OF DWG.		DATE
A	(1-B) Change: 1170 (930) → 1040 (800) (2-A) : 745 → 650 (2-B) Add: Rod&Torquewrench=875	Feb. 15' 99 T. Aiko (DK-20-1431)

NAME OF PARTS	MASS OF PARTS
PISTON&CONNECTING ROD ASSY.	97 Kg
CYLINDER LINER ASSY.	66 Kg
CYLINDER HEAD ASSY.	126 Kg
CAMSHAFT ASSY. (5DK)	140 Kg
CAMSHAFT ASSY. (6DK)	160 Kg
CAMSHAFT ASSY. (8DK)	200 Kg
L. O. PUMP ASSY.	37 Kg
C. W. PUMP ASSY.	35 Kg



Detail "A"





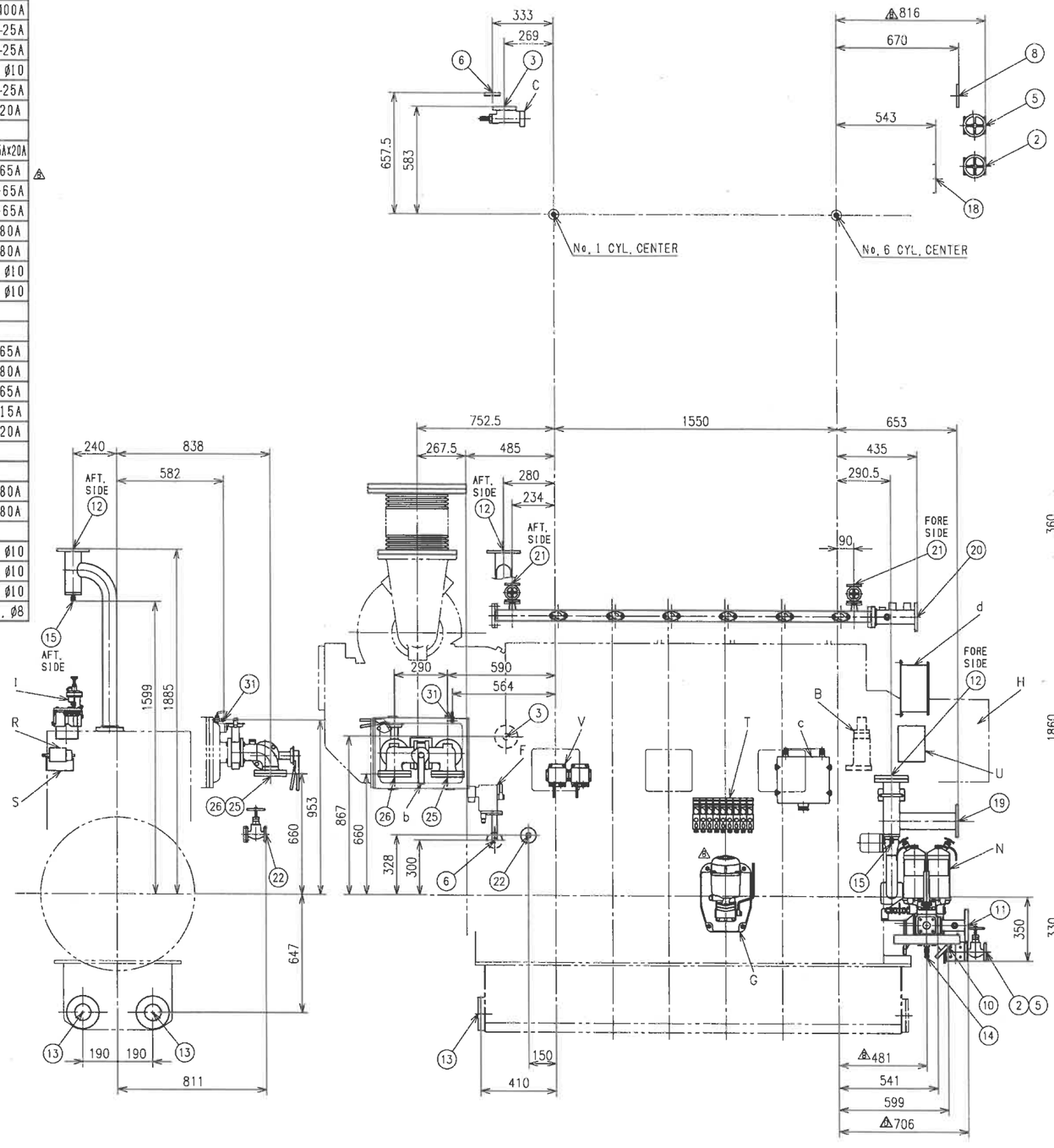
NO.	NAME OF PARTS	PARTS NO.	MATERIAL	MASS.	QTY.
			GROUP 0	KUMI	

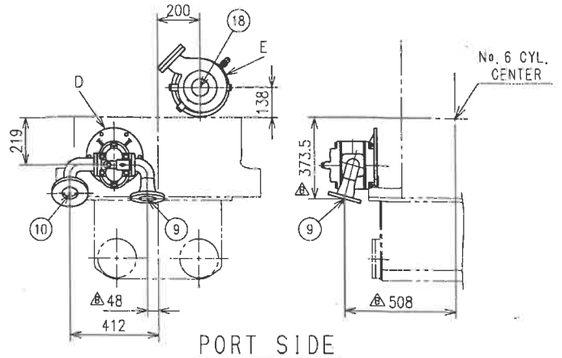
APPROVED BY	LIST NO. E20005	SCALE 1:20	TYPE DK-20
CHECKED BY	BEFORE / AFTER / REF.		NAME ENGINE DISASSEMBLING SPACE-RT
DRAWN BY K. GOI	DAIHATSU DIESEL MFG. CO., LTD. OSAKA JAPAN		PARTS NO. E200050540 Z
DATE APR. 14. '97			DWG. NO. 806847 A

2017/05/09 TECHNICAL DEPARTMENT

NO.	NAME	SIZE
1	EXH. GAS OUTLET	5K-400A
2	F. O. INLET	16K-25A
3	F. O. OUTLET	16K-25A
4	F. O. FILTER DRAIN	O. D. ø10
5	F. O. INLET FROM SPARE PUMP	16K-25A
6	LEAK OIL OUTLET	5K-20A
7		
8	STARTING AIR INLET	30K-25Ax20A
9	L. O. PUMP INLET	5K-65A ⚠
10	L. O. PUMP OUTLET	10K-65A
11	L. O. INLET	10K-65A
12	MIST GAS OULET (x2)	5K-80A
13	L. O. OUTLET (AFT. SIDE x2)	5K-80A
14	L. O. FILTER DRAIN	O. D. ø10
15	MIST GAS DRAIN	O. D. ø10
16		
17		
18	F. W. INLET	5K-65A
19	F. W. INLET (STANDBY F. W. PUMP)	5K-80A
20	F. W. OUTLET	5K-65A
21	F. W. AIR VENT (x2)	5K-15A
22	F. W. DRAIN	5K-20A
23		
24		
25	L. F. W. INLET FOR I. C.	5K-80A
26	L. F. W. OUTLET FOR I. C.	5K-80A
28	GOVERNOR CONT. AIR INLET	O. D. ø10
29	ENGINE STOP CONT. AIR INLET	O. D. ø10
30	RACK LIMIT CONT. AIR INLET	O. D. ø10
31	AIR COOLER AIR VENT	O. D. ø8

I	
h	
g	STARTING OPERATION VALVE
f	EXPANSION JOINT (300Ax400A)
e	
d	JUNCTION BOX-P2
c	JUNCTION BOX-P1
b	BUTTERFLY VALVE
a	STARTING ROTARY VALVE
Z	
Y	
X	
W	
V	THERMAL SWITCH
U	MAGNETIC VALVE BOX
T	PRESS. SWITCH
S	FUEL SHUT DOWN DEVICE
R	FUEL CONTROL CYLINDER
Q	
P	STARTING VALVE
O	
N	L. O. FILTER
M	
L	F. O. NO. 2 FILTER
K	
J	
I	GOVERNOR
H	GAUGE BOARD
G	LUBRICATION OIL FILTER (GLACIER)
F	LEAKED OIL TANK
E	F. W. PUMP
D	L. O. PUMP
C	F. O. PRESS RELIEF VALVE
B	F. O. INJECTION PUMP
A	F. O. PUMP
SIMBOL	NAME





⚠x5	(6-B) REV: DIMENSION	2016/11/14	M. T
⚠x5	(8-E, 9-C/D) REV: DIMENSION	2016/09/16	M. T
⚠x12	(2-F, 5-C, 6-B/G, 8-B/F, 9-B/C/F) REV: BYPASS FILTER POSITION, ⑨, DIMENSION	2016/07/05	M. T
⚠	FIRST DRAWING BY DDAP	2016/05/09	M. T

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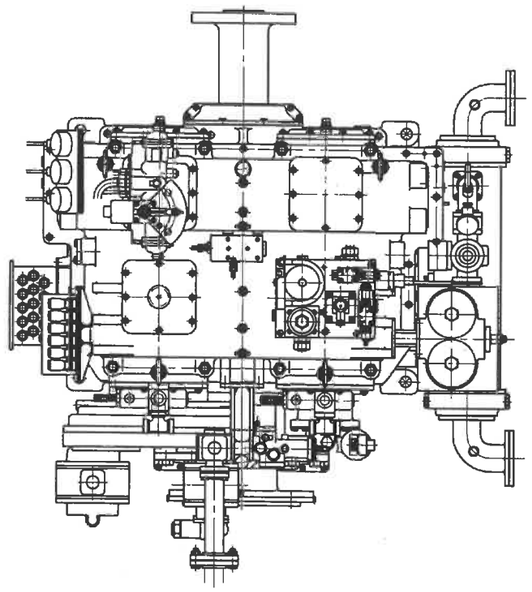
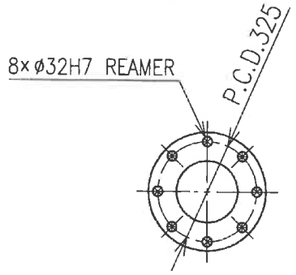
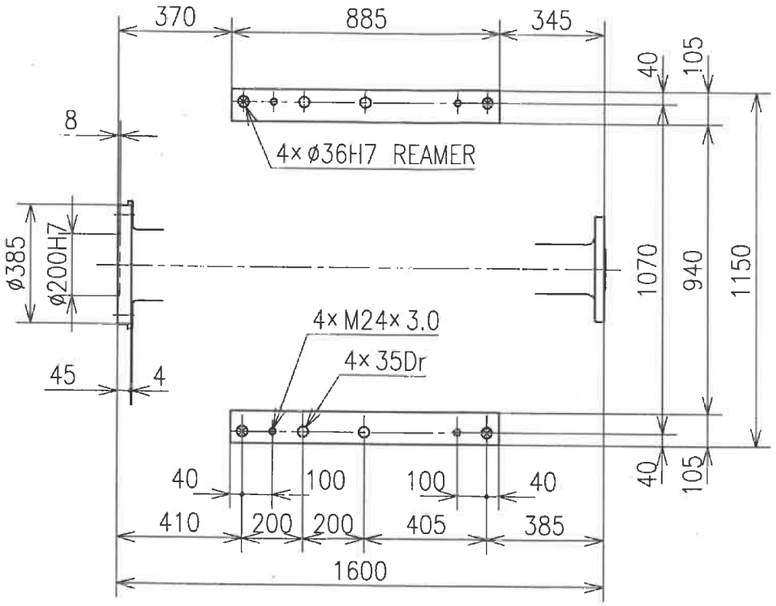
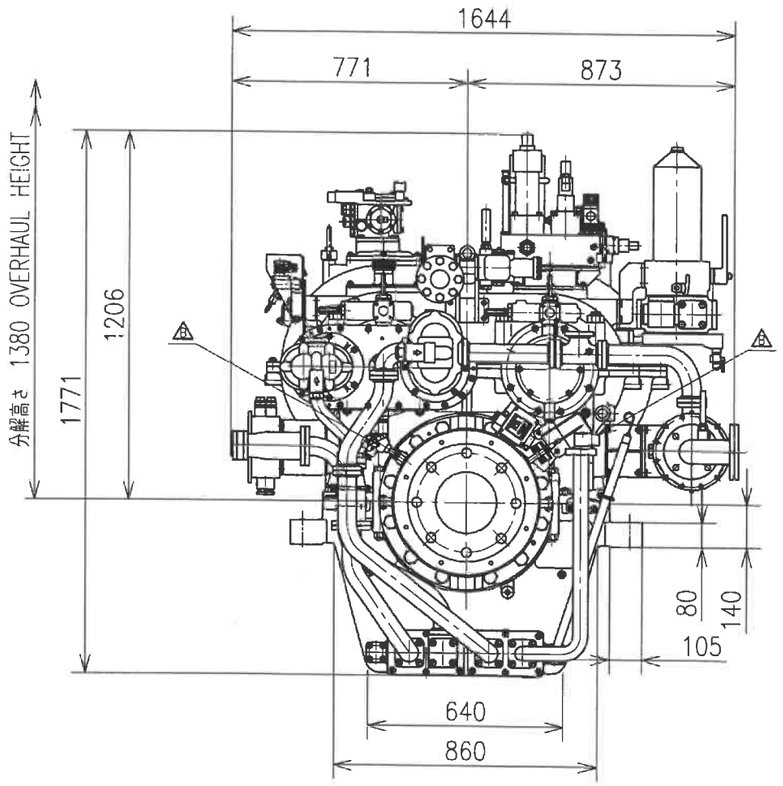
APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5	MATERIAL APRV.	GROUP 0	SCALE: 1:20
CHECKED	NARITA SHIROU	BEFORE		MASS.	QTY.	A2
	FUJIMOTO SYUHEI	REF.	BQP10000040	TYPE	DK-20e	
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG.CO., LTD. OSAKA JAPAN		NAME	ギソウズ: :6DKM-20eF (L)	
DRAWN	(H) MORICHI YASUKO				ARRANGEMENT OF FITTING: :6DKM-20eF (L)	
DATE	2016/11/14			PARTS NO.	AQP10008491	D	*

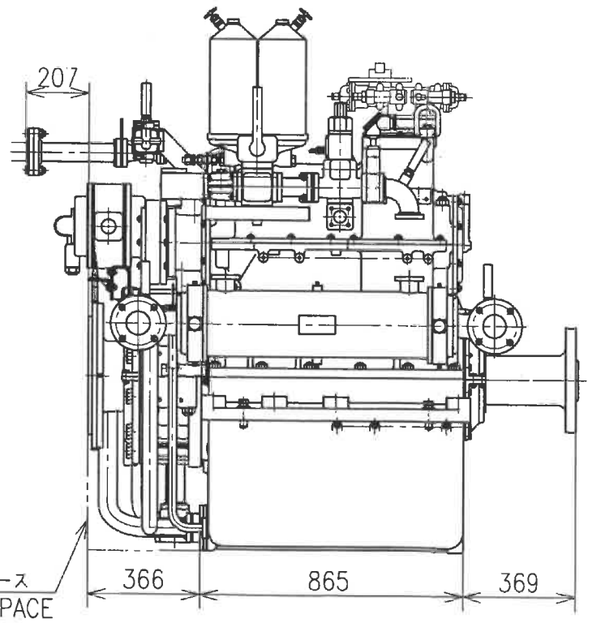
2017/06/09 TECHNICAL DEPARTMENT











⚠ x2	(2-/3-B) ADD:L.O. SUMP TANK LEVEL SWITCH, THERMO SENSOR (THRUST BEARING TEMP.)	2016/09/20	M. T
⚠	FIRST DRAWING	2016/05/27	M. T

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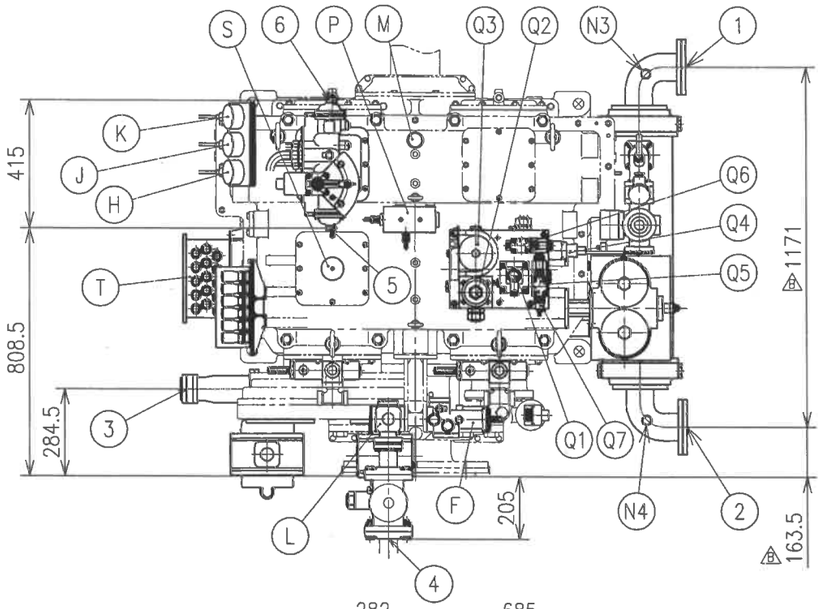
APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5
CHECKED	NARITA SHIROU	BEFORE	
	FUJIMOTO SYUHEI	REF.	BQP10000041
	TOKUMOTO MANABU		
DRAWN	(H)MORICHI YASUKO		
DATE	2016/09/20		

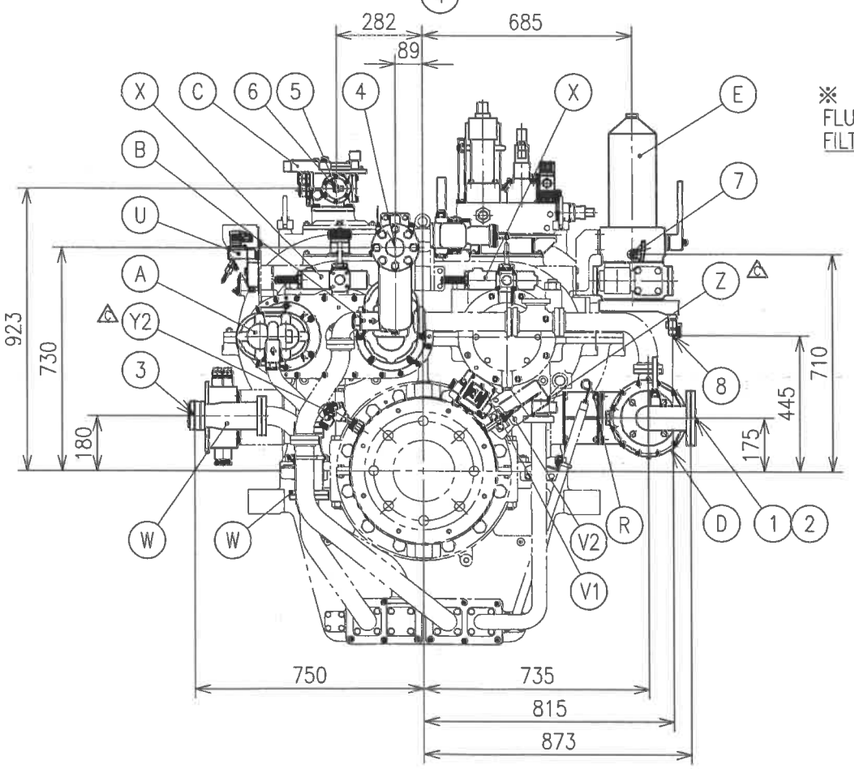
DAIHATSU DIESEL MFG. CO., LTD.
OSAKA JAPAN

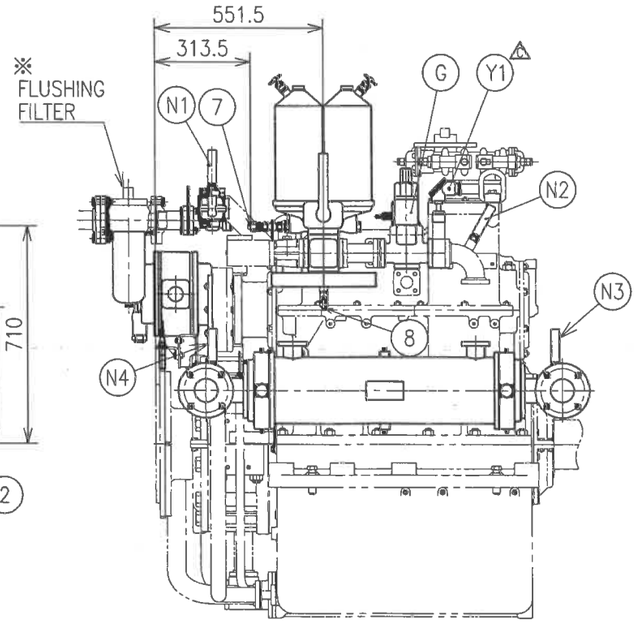
MATERIAL	APRV.	GROUP	0
MASS	A / C	QTY.	
TYPE	DRA-25J		
NAME	ガイケイズ：：DRA-25J(L)		
	OUT LINE：：DRA-25J(L)		
PARTS NO.	AQP10008492	B	*

SCALE: 1:20
A3
LEVEL

2017/05/09 TECHNICAL DEPARTMENT







No.	NAME	名 称	SIZE 管径
1	COOLING WATER INLET,OIL COOLER	冷却水 入口	JIS-5Kx80A/65A
2	COOLING WATER OUTLET,OIL COOLER	冷却水 出口	JIS-5Kx80A/65A
3	LUB.OIL OUTLET TO EMERGENCY PUMP	潤滑油出口(非常用ポンプへ)	□65x50A
4	LUB.OIL OUTLET FROM EMERGENCY PUMP	潤滑油入口(非常用ポンプより)	JIS-16Kx5A
5	AIR INLET,CLUTCH OPERATING VALVE(AHEAD)	操作空気入口(前進用)	φ10ツギワ BANJO
6	AIR INLET,CLUTCH OPERATING VALVE(ASTRN)	操作空気入口(後進用)	φ10ツギワ BANJO
7	FILTER BLOW OFF AND DRAIN	濾器ブローオフ及びドレン	10Aクイコミ BITING UNION
8	DRAIN OF FILTER SAUCER	濾器受け皿ドレン	φ10ユニオン UNION

※)AFTER SEA TRIAL, PLEASE EXCHANGE THE FLUSHING FILTER KIT FOR THE SHORT PIPE.

SYMBOL	NAME	名 称
⚠ Z	L.O. SUMP TANK LEVEL SWITCH	L.O.タンクレベルスイッチ
⚠ Y2	THERMO SENSOR(THRUST BEARING TEMP.)	温度センサー(ｽﾗｽﾄﾍﾞｱﾘﾝｸﾞ)
⚠ Y1	THERMO SENSOR(L.O. COOLER OUTLET)	温度センサー(潤滑油ｸｰﾗｰ出口)
X	OIL FLOW CHANGE INSTRUCTION VALVE	油量切替指令弁
W	SUCTION FILTER	サクションフィルター
V2	PULSE SENSOR(FOR CONTROL)	パルスセンサー(制御用)
V1	PULSE SENSOR(FOR DISPLAY)	パルスセンサー(表示用)
U	PRESS.SWITCH or TRANSMITTER	圧力スイッチ又は圧力発信器
T	JUNCTION BOX	端子箱
S	L.O. REPLENISHMENT ENTRANCE	L.O.補給口
R	OIL LEVEL GAUGE	検油管
Q7	THROTTLE VALVE	絞り弁
Q6	MAGNETIC VALVE SOL2 (FOR CLUTCH OPERATING OIL DRAIN)	電磁弁(クラッチ作動油ドレン用)
Q5	MAGNETIC VALVE SOL3 (FOR CLUTCH OPERATING OIL BYPASS)	電磁弁(クラッチ作動油バイパス用)
Q4	PRECISION THROTTLE VALVE	精密絞り弁
Q3	ACCUMULATOR	アキュムレータ
Q2	LINE FILTER	ラインフィルタ
Q1	MAGNETIC PROPORTIONAL FLOW CONTROL VALVE SOL1	比例電磁弁
P	OIL FLOW CHANGE VALVE	油量切替弁
N4	THERMOMETER(COOLING WATER OUTLET)	温度計(冷却水クーラー出口)
N3	THERMOMETER(COOLING WATER INLET)	温度計(冷却水クーラー入口)
N2	THERMOMETER(LUB.OIL COOLER OUTLET)	温度計(潤滑油クーラー出口)
N1	THERMOMETER(LUB.OIL COOLER INLET)	温度計(潤滑油クーラー入口)
M	BREATHER	ブリーザ
L	CHECK VALVE	逆止弁
K	PRESS.GAUGE (AHEAD CLUTCH OPERATING OIL)	前進クラッチ作動油圧力計
J	PRESS.GAUGE (ASTERN CLUTCH OPERATING OIL)	後進クラッチ作動油圧力計
H	PRESS.GAUGE(LUB.OIL)	潤滑油圧力計
G	RELIEF VALVE(LOW PRESS.)	低圧調圧弁
F	RELIEF VALVE(HIGH PRESS.)	高圧調圧弁
E	FILTER	潤滑油濾器
D	OIL COOLER	潤滑油冷却器
C	CLUTCH OPERATING VALVE	クラッチ操作弁
B	L.O. PUMP(LOW PRESS.)	潤滑油ポンプ(低圧)
A	L.O. PUMP(HIGH PRESS.)	潤滑油ポンプ(高圧)

⚠ x2	(3-B, 5-E) ADD:Y2, Z REV:Y→Y1	2016/09/14	M. T
⚠ x2	(3-C/D) REV:DIMENSION	2016/07/14	M. T
⚠	FIRST DRAWING	2016/05/27	M. T

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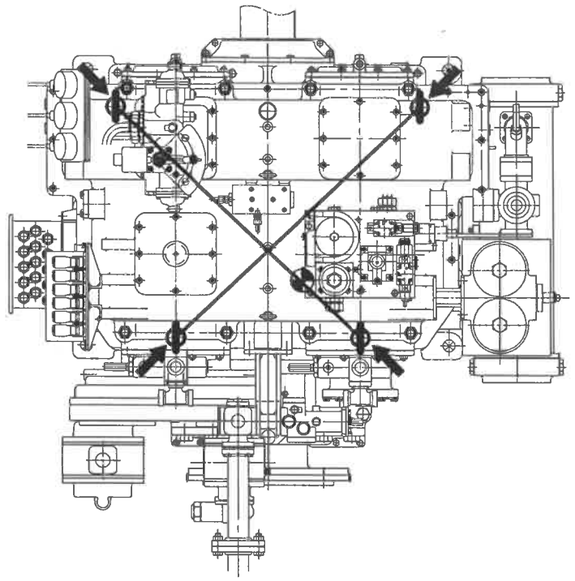
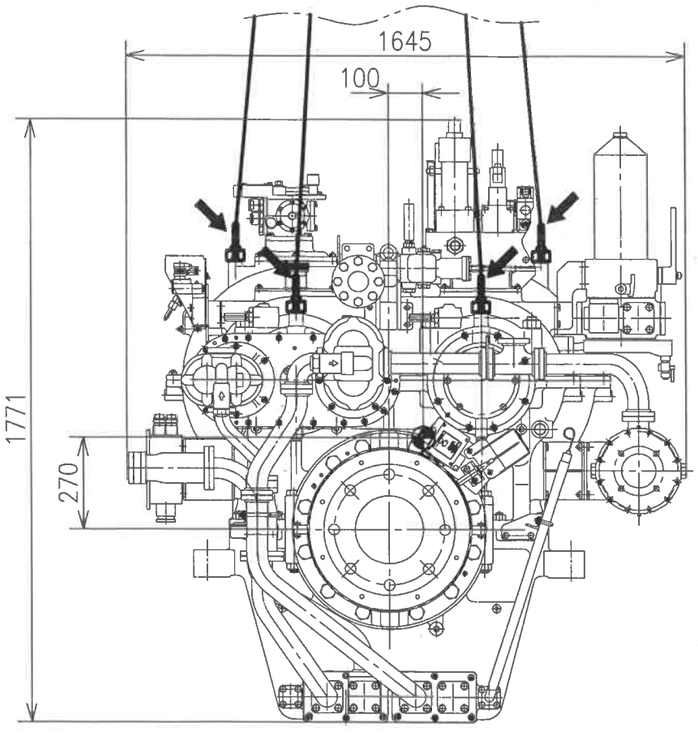
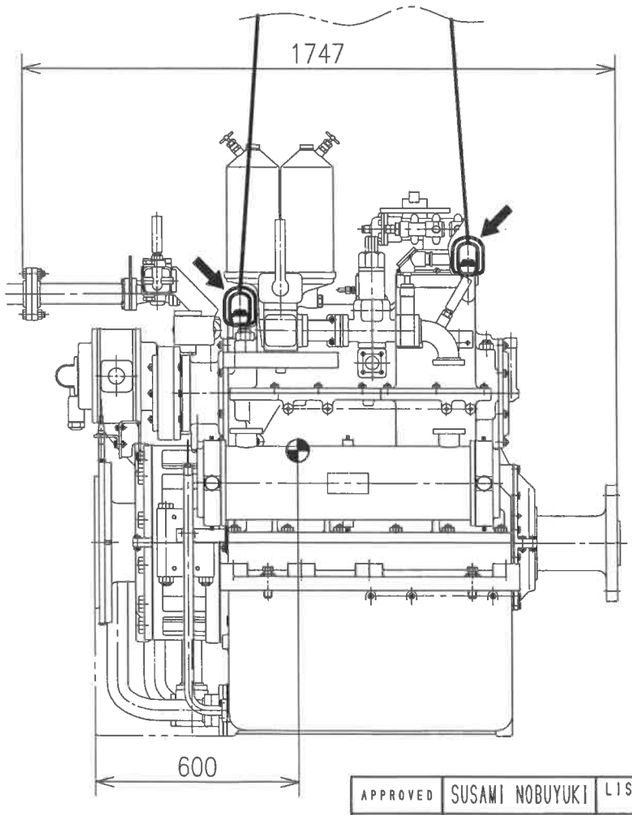
APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5	MATERIAL APRV.	GROUP 0	SCALE: 1:20
CHECKED	NARITA SHIROU	BEFORE		MASS A C	QTY.	A3
	FUJIMOTO SYUHEI	REF.	AG89903R007	TYPE DRA-25J		LEVEL
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN		NAME ギソウズ::DRA-25J(L)		
DRAWN	(H)MORICHI YASUKO			ARRANGEMENT OF FITTING::DRA-25J(L)		
DATE	2016/09/14			PARTS NO. AQP10008513	C	*

2017/05/09 TECHNICAL DEPARTMENT







吊り上げ要領
Procedure for hoisting

1.本減速機の質量は、約4900kgです。吊り上げに使用するワイヤーロープは、
この質量に充分耐えるものを選定願います。
The mass of this reduction gear is about 4900kg.
Select the suitable wire rope.

2.矢印部(➡)吊り上げ金具4箇所にシャックル等を取付け、ワイヤーロープを通します。
Fix the shackle at the arrowed part(➡) on the drawing.

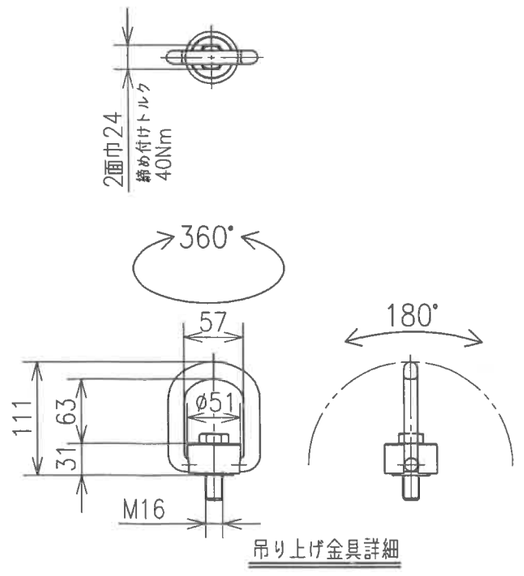
3.ワイヤーロープは、配管や付属機器類に直接掛からない様に配慮願います。
Prevent the wire rope from contacting the pipe and auxiliary equipment.

4.ワイヤーロープが直接触れる箇所には、木片等を当てて保護願います。
また、状況に応じて吊り天秤の使用について検討を願います。
Place a buffer, e.g.,wooden blocks, if the wire rope contacts the reduction gear body.
(Use the hanging tool according to the situation.)

5.吊り上げにより変形の恐れがある配管や機器類は取り外し願います。
Remove temporarily pipes and auxiliary equipments if hoisting might damage them.

6.吊り上げ終了後は、取り外した配管や機器類は元に戻してください。
Put removed parts back after hoisting.

◒:重心位置



吊り上げ金具詳細

⚠	FIRST DRAWING	(E021666)	2016/9/26 H)YOSHIDA

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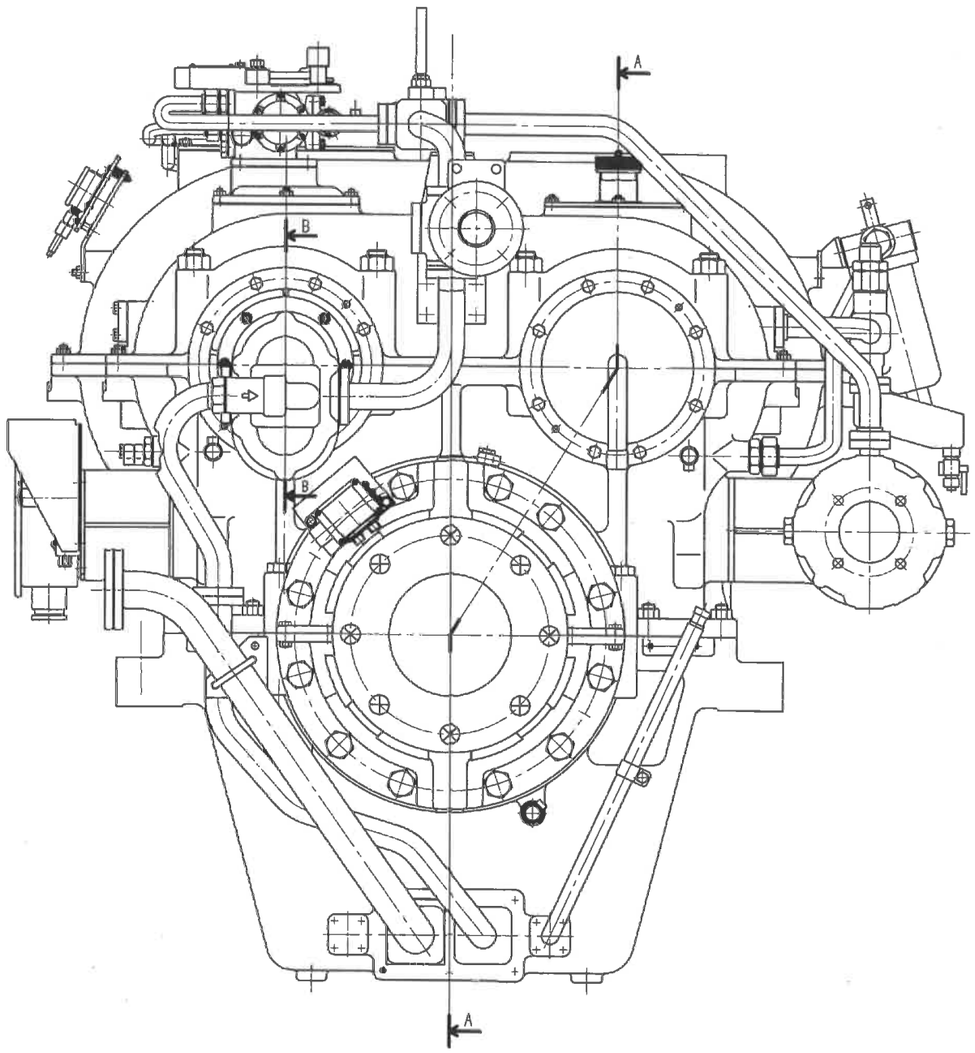
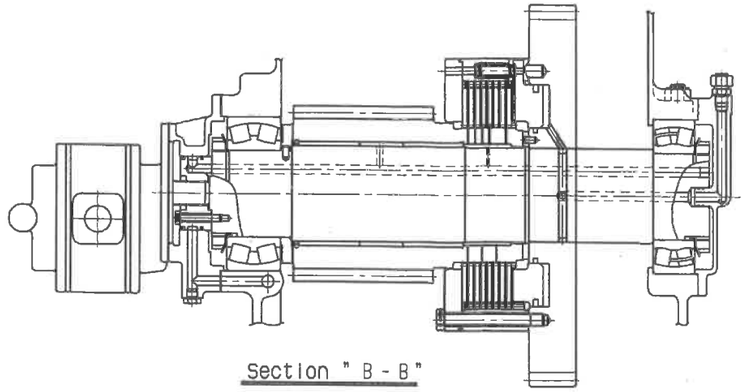
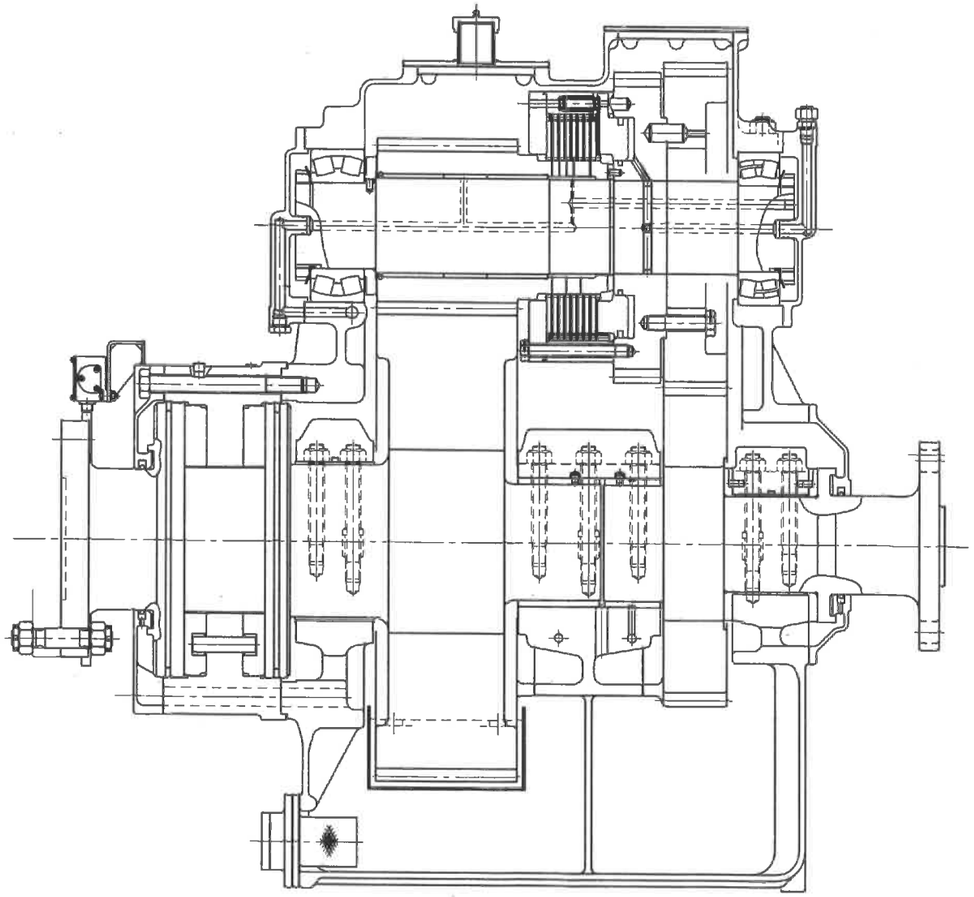
APPROVED	SUSAMI NOBUYUKI	LIST NO. AG89903	MATERIAL APRV	GROUP 0	SCALE: 1:18
CHECKED		BEFORE	MASS 4900Kg Ⓐ Ⓒ	QTY.	A3
	TAKASE YOSHIKUNI	REF.	TYPE DRA-25J		LEVEL
	SAKATA SOHEI	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME ヨウリョウズ：ツリアゲ：DRA-25J		
DRAWN	(H)YOSHIDA KAZUAKI		POINT FIGURE:FOR LIFTING:DRA-25J		
DATE	2016/09/26		PARTS NO. AG89903R017	A	*

2017/05/09 TECHNICAL DEPARTMENT





Section " B - B"



Section " A - A"

⚠	FIRST DRAWING (E016977)	2014/04/01	T.Y

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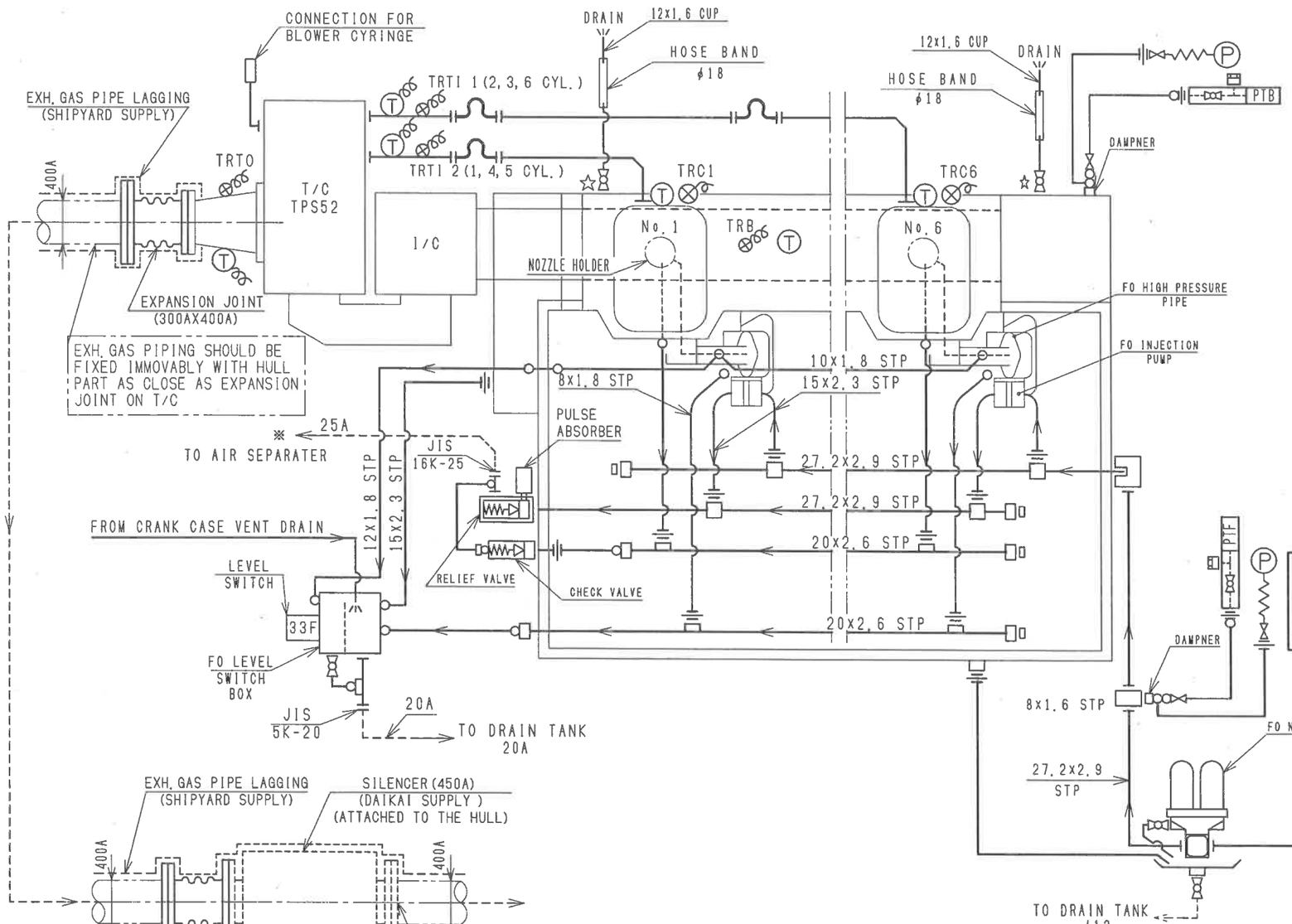
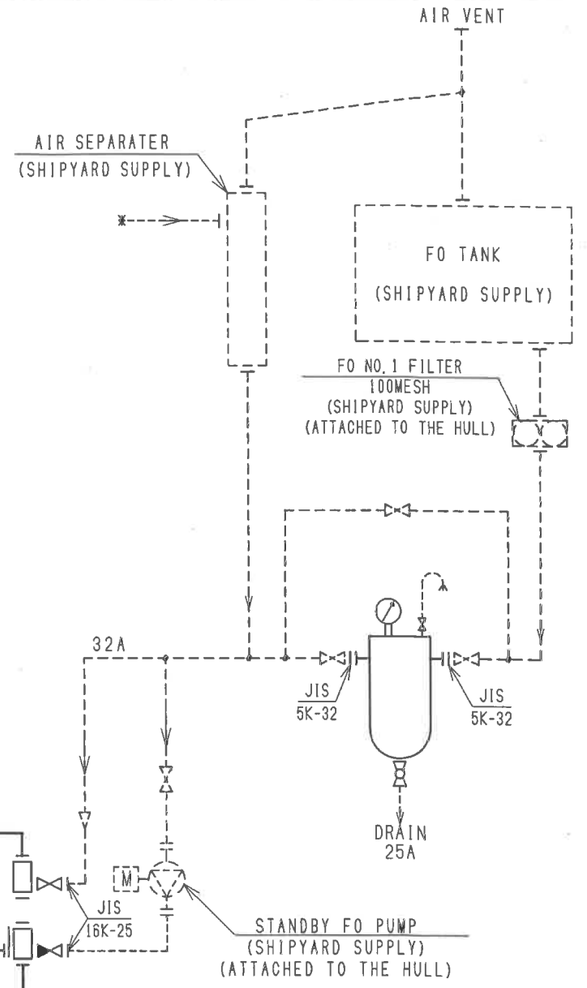
APPROVED	SUSAMI NOBUYUKI	LIST NO. AG32903	MATERIAL APRV	GROUP 0	SCALE: 1:10
CHECKED	TAKASE YOSHIKUNI	BEFORE	MASS A C	QTY.	A3
	UDA YOSHIKI	REF.	TYPE DRA-25F		LEVEL
DRAWN	YAMAGUCHI TAKANOBU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME ショウニンズ：ダンメンズ：DRA-25F		
DATE	2014/03/31		APPROVAL:SECTIONAL VIEW:DRA-25F		
			PARTS NO. AG32903R020	A	*

2017/05/09 TECHNICAL DEPARTMENT

REMARKS
1. OUTLET OF D.O.TANK TO BE ARRANGED HIGHER THEN 1.6m FROM THE CENTER OF CRANKSHAFT.
2. AIR OUTLET FROM THE VENTILATER TO BE ARRANGED NEAR THE TURBO-CHARGER AIR INLET.
3. EXH.GAS BACK PRESS.AT RATED POWER SHOULD BE BELOW 2.45kPa AT TURBO-CHARGER OUTLET. (MAX.2.94kPa)
4. HEAT INSULATION FOR EXHAUST GAS PIPING OF HULL SIDE INCLUDING THE EXPANSOPN JOINT SHOULD BE CARRIED OUT BY SHIPYARD.
5. CHARGE AIR CONDENCATE DRAIN VALVES MARKED "☆" ARE TO BE NORMAL 60° OPEN.
6. ALL COUNTER FLANGE SHOULD BE SUPPLY BY DAIHATSU





REMARKS			
⊣⊢	FLANGE	⊳◁	VALVE
⫴	UNION JOINT	⊳◁	CHECK VALVE
⫴	EYE FLANGE	⋈	COCK
○	JOINT	⋈	3 WAY COCK
▷	REDUCER	CUP	COPPER TUBE
～～	FLEX. TUBE	STP	STEEL TUBE
———	DAIHATSU SUPPLY	SGP	GAS PIPE
- - -	SHIPYARD	Ⓟ	PRESS. GAUGE
═══	LAGGING	Ⓣ	THERMOMETER
⊗	RESISTANCE BULB	⊘	PRESS. GAUGE

⚠1	(6-B) REV:TRF POSITION	2016/11/09 M.T
⚠1	(2-A) REV:NOTE	2016/07/05 M.T
⚠	FIRST DRAWING	2016/05/27 M.T

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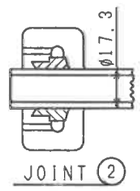
APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5
CHECKED	NARITA SHIROU	BEFORE	
	FUJIMOTO SYUHEI	REF.	BQP10000033A
	TOKUMOTO MANABU		
DRAWN	(H)MORICHI YASUKO		
DATE	2016/11/09		

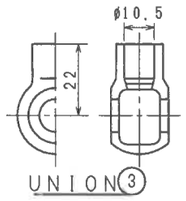
DAIHATSU DIESEL MFG.CO.,LTD.
OSAKA JAPAN

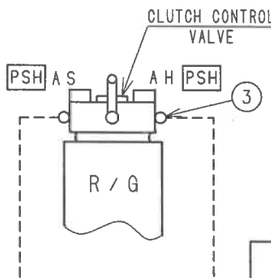
MATERIAL	APRV.	GROUP	0
SCALE:	—:—		
MASS	A C	QTY.	
			A3
TYPE	DK-20e	LEVEL	
NAME	FO&EXH.ガス ケイトウズ::6DKM-20eF(L)		
	F.O.&EXH.GAS PIPING SYSTEM::6DKM-20eF(L)		
PARTS NO.	AQP10008493	C	*

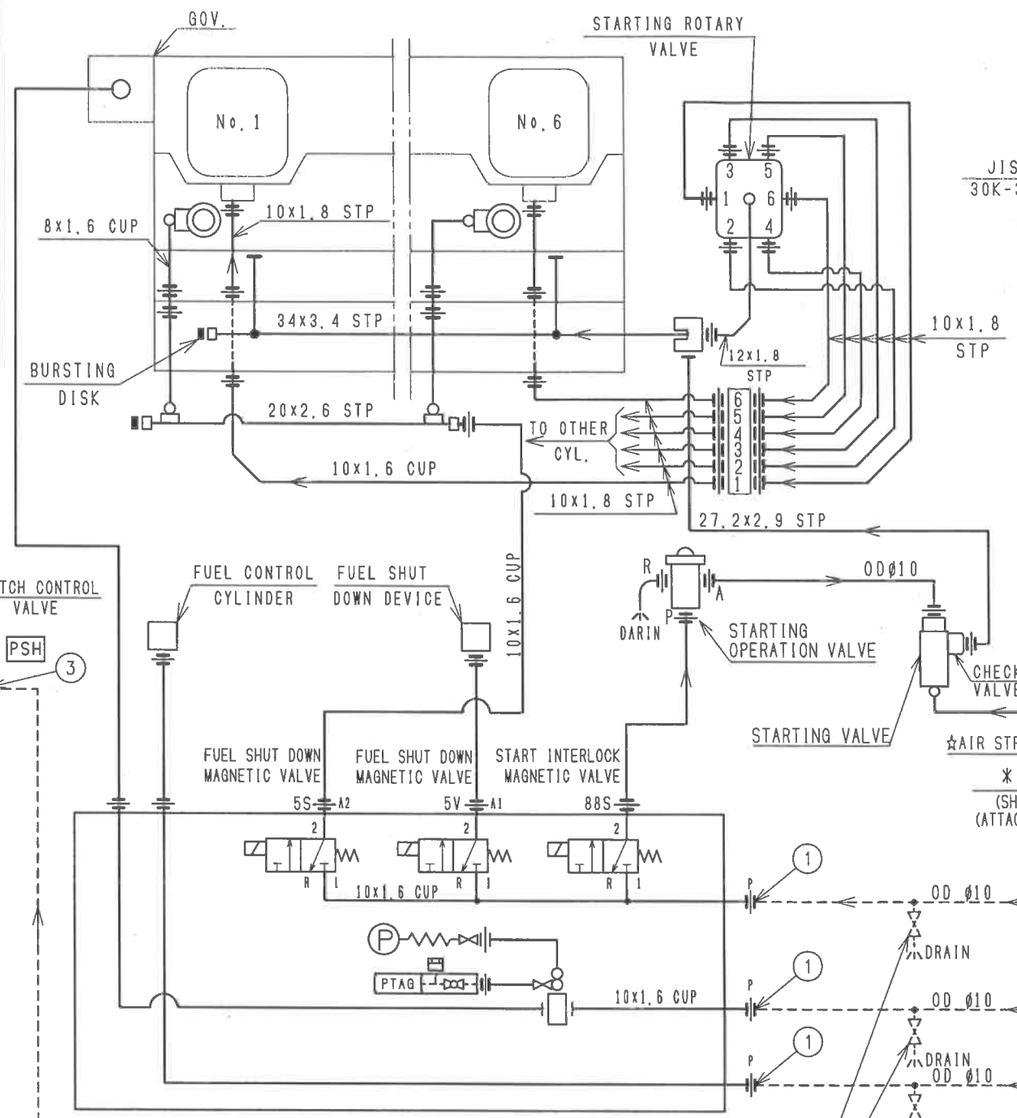
2017/05/09 TECHNICAL DEPARTMENT

REMARKS			
	FLANGE		VALVE
	UNION JOINT		CHECK VALVE
	EYE FLANGE		COCK
	JOINT		3 WAY COCK
	REDUCER	CUP	COPPER TUBE
	FLEX. TUBE	STP	STEEL TUBE
	DAIHATSU SUPPLY	SGP	GAS PIPE
	SHIPYARD	Ⓟ	PRESS. GAUGE
	LAGGING	Ⓣ	THERMOMETER
			PRESS. GAUGE









REMARKS
1. DRAIN VALVE (✱MARKED) SHOULD BE INSTALLATION THE LOWEST POSITION OF PIPING LINE.
2. ONLY MAIN PIPE :PARKERIZING
3. ☆MARKED STRAINER SHOULD BE INSTALLATION AS NEARLY AS POSSIBLE TO THE ENGINE.
4. AFTER FINISHING THE PIPING WORK, PLEASE CARRY OUT CLEANING THE PIPE SUFFICIENTLY.
5. PLEASE OPEN AND CHECK THE AIR STRAINER CONDITION AT BELOW POINT OF TIME.
   1) BEFORE INITIAL RUNNING 2) BEFORE ON BOARD TRIAL
   3) BEFORE SEA TRIAL 4) BEFORE DELIVERY

⚠	FIRST DRAWING	2016/05/27	M. T

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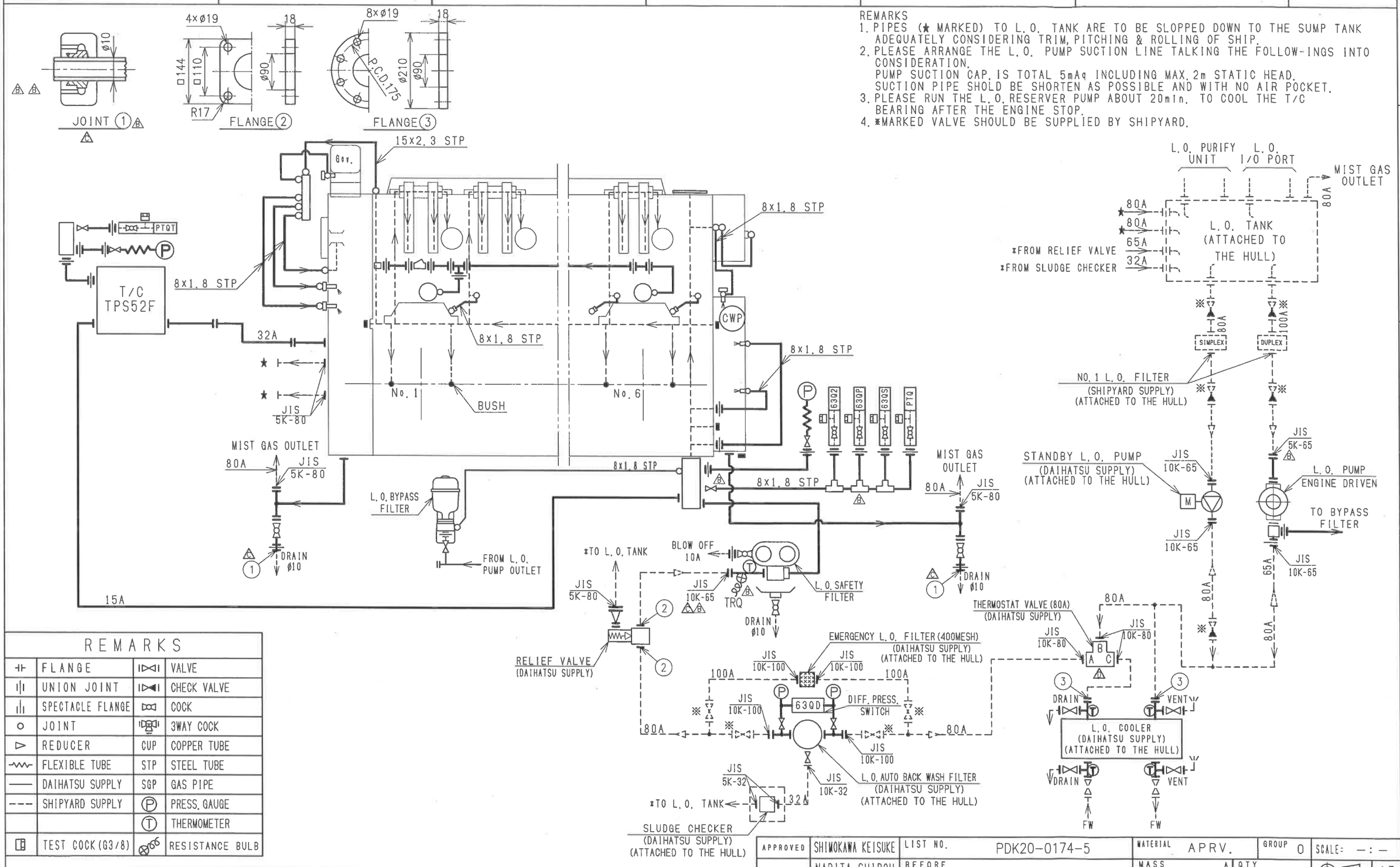
APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5	MATERIAL APRV.	GROUP 0	SCALE: —:—
CHECKED	NARITA SHIROU	BEFORE		MASS	QTY.	A3
	FUJIMOTO SYUHEI	REF.	BQP10000034A	TYPE DK-20e		LEVEL
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN		NAME SAケイトウズ::6DKM-20eF(L)		
DRAWN	(H)MORICHI YASUKO			S.A. PIPING SYSTEM::6DKM-20eF(L)		
DATE	2016/05/06			PARTS NO. AQP10008494	A	✱

2017/05/09 TECHNICAL DEPARTMENT



REMARKS			
⊣⊢	FLANGE	⊳⊲	VALVE
⊣⊢	UNION JOINT	⊳⊲	CHECK VALVE
⊣⊢	SPECTACLE FLANGE	⊠	COCK
○	JOINT	⊠	3WAY COCK
▷	REDUCER	CUP	COPPER TUBE
～～	FLEXIBLE TUBE	STP	STEEL TUBE
——	DAIHATSU SUPPLY	SGP	GAS PIPE
- - -	SHIPYARD SUPPLY	Ⓟ	PRESS. GAUGE
		Ⓣ	THERMOMETER
▯	TEST COCK (G3/8)	⊗ø6	RESISTANCE BULB

⚠x1	(6-B) REV: THERMOSTAT VALVE	2017/04/19	M. T
⚠x4	(1-D, 2-B, 4-B, 5-B) REV: FLANGE , FLANGE① →JOINT① , DRAIN PIPING	2016/09/05	M. T
⚠x8	(1-D/E, 4-C/B, 5-C, 6-C) REV: FLANGE①, PIPING, VALVE/TRQ/THERMOMETER POSITION	2016/07/05	M. T
⚠	FIRST DRAWING	2016/05/24	M. T

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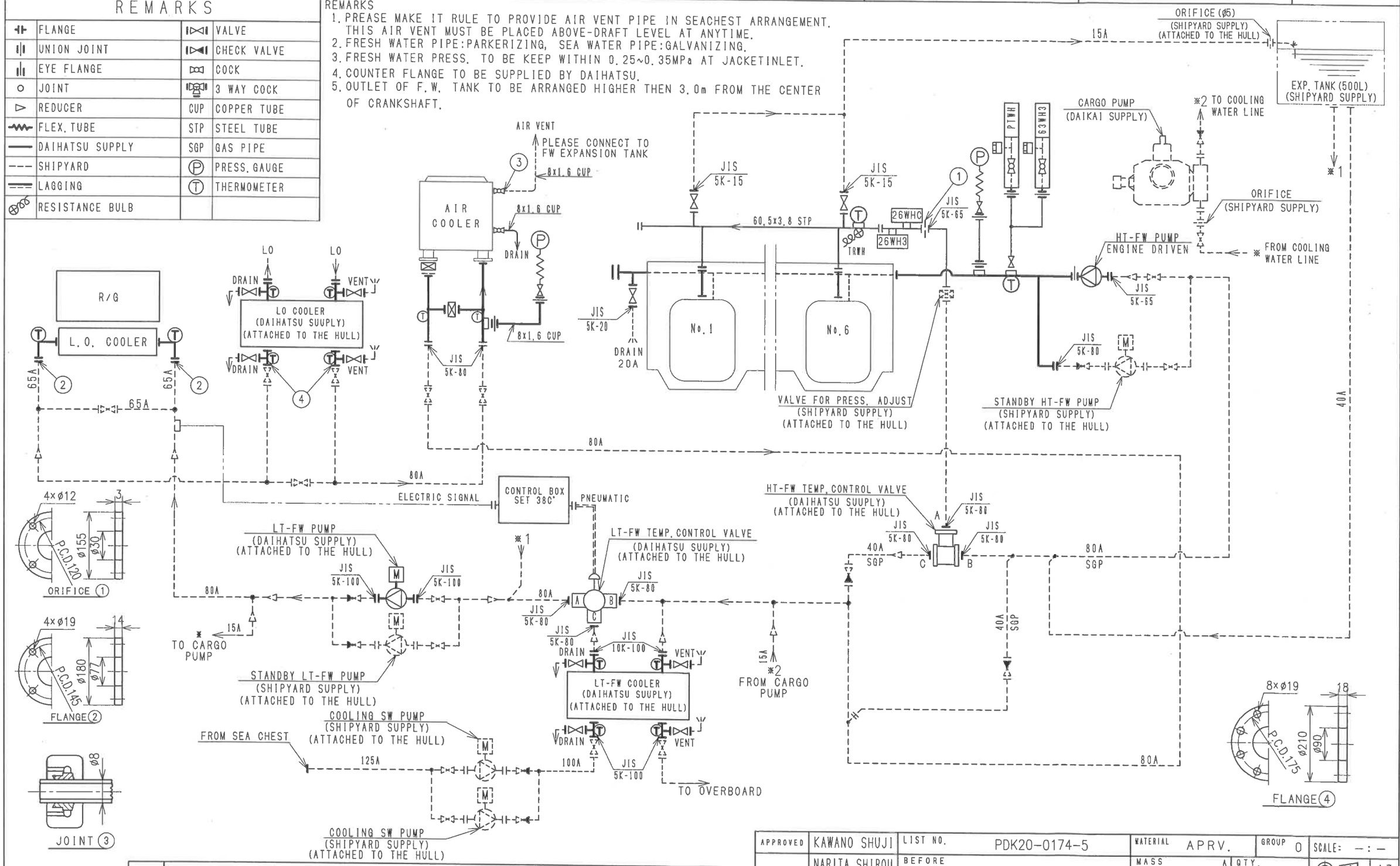
APPROVED	SHIMOKAWA KEISUKE	LIST NO.	PDK20-0174-5	MATERIAL APRV. / GROUP 0	SCALE: —:—
CHECKED	NARITA SHIROU	BEFORE		MASS / QTY.	A3
	FUJIMOTO SYUHEI	REF.	BQP10000035	TYPE DK-20e	LEVEL
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG. CO., LTD. OSAKA JAPAN		NAME LOケイトウズ::6DKM-20eF (L)	
DRAWN	(H) MORICHI YASUKO			L.O. PIPING SYSTEM::6DKM-20eF (L)	
DATE	2017/04/19			PARTS NO. AQP10008495	D *

2017/05/09 TECHNICAL DEPARTMENT

REMARKS			
	FLANGE		VALVE
	UNION JOINT		CHECK VALVE
	EYE FLANGE		COCK
	JOINT		3 WAY COCK
	REDUCER	CUP	COPPER TUBE
	FLEX. TUBE	STP	STEEL TUBE
	DAIHATSU SUPPLY	SGP	GAS PIPE
	SHIPYARD	Ⓟ	PRESS. GAUGE
	LAGGING	Ⓣ	THERMOMETER
	RESISTANCE BULB		

REMARKS
1. PREASE MAKE IT RULE TO PROVIDE AIR VENT PIPE IN SEACHEST ARRANGEMENT. THIS AIR VENT MUST BE PLACED ABOVE-DRAFT LEVEL AT ANYTIME.
2. FRESH WATER PIPE:PARKERIZING, SEA WATER PIPE:GALVANIZING.
3. FRESH WATER PRESS. TO BE KEEP WITHIN 0.25~0.35MPa AT JACKETINLET.
4. COUNTER FLANGE TO BE SUPPLIED BY DAIHATSU.
5. OUTLET OF F.W. TANK TO BE ARRANGED HIGHER THEN 3.0m FROM THE CENTER OF CRANKSHAFT.



⚠	FIRST DRAWING	2016/05/27	M. T

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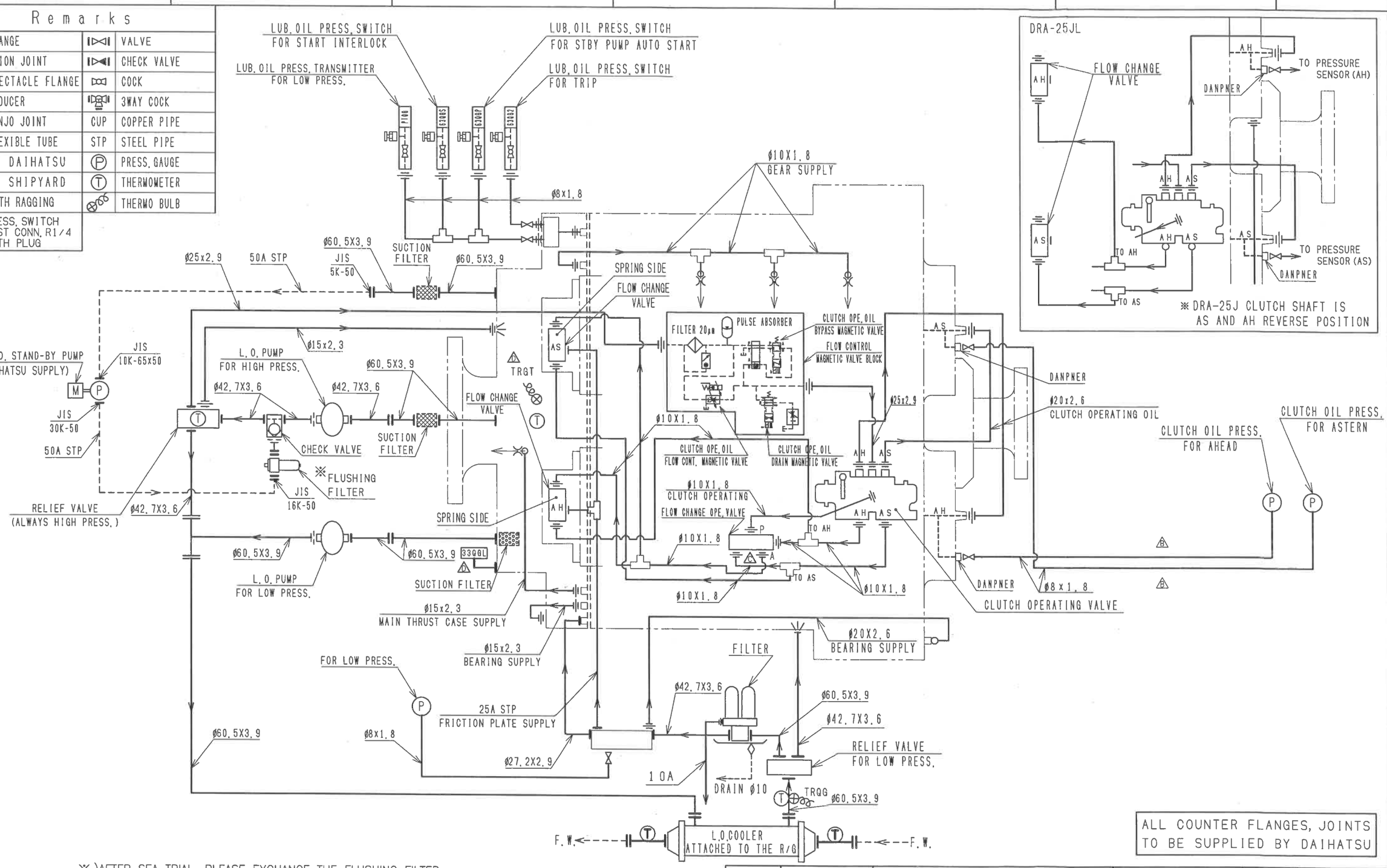
APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5	MATERIAL APRV.	GROUP 0
CHECKED	NARITA SHIROU	BEFORE		MASS	QTY.
	FUJIMOTO SYUHEI	REF.	BQP10000036B	TYPE DK-20e	SCALE: -:- A3
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN		NAME CWケイトウズ::6DKM-20eF(L)	LEVEL
DRAWN	(H)MORICHI YASUKO			C.W. PIPING SYSTEM::6DKM-20eF(L)	
DATE	2016/05/06			PARTS NO. AQP10008496	A *

2017/05/09 TECHNICAL DEPARTMENT

Remarks			
	FLANGE		VALVE
	UNION JOINT		CHECK VALVE
	SPECTACLE FLANGE		COCK
	REDUCER		3WAY COCK
	BANJO JOINT	CUP	COPPER PIPE
	FLEXIBLE TUBE	STP	STEEL PIPE
	BY DAIHATSU	Ⓟ	PRESS. GAUGE
	BY SHIPYARD	Ⓣ	THERMOMETER
	WITH RAGGING		THERMO BULB
	PRESS. SWITCH TEST CONN. R1/4 WITH PLUG		



※)AFTER SEA TRIAL, PLEASE EXCHANGE THE FLUSHING FILTER KIT FOR THE SHORT PIPE.

⚠2 x2	(3-C) ADD: TRGT, 33QGL	2016/08/30	M. T
⚠1 x1	(4-B) REV: PIPING	2016/08/02	M. T
⚠2 x2	(6-B/C) DEL: PTQGC AHD. / AST.	2016/07/05	M. T
⚠	FIRST DRAWING	2016/05/27	M. T

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ALL COUNTER FLANGES, JOINTS TO BE SUPPLIED BY DAIHATSU

APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5	MATERIAL APRV.	GROUP 0	SCALE: —:—
CHECKED	NARITA SHIROU	BEFORE		MASS	A Ⓒ QTY.	A3
	FUJIMOTO SYUHEI	REF.	BQP10000048A	TYPE DRA-25J		LEVEL
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG. CO., LTD. OSAKA JAPAN		NAME L.Oケイトウズ::DRA-25J(L)		
DRAWN	(H) SUMIDA YUKARI			L.O. PIPING SYSTEM::DRA-25J(L)		
DATE	2016/08/30			PARTS NO. AQP10008497	D	*

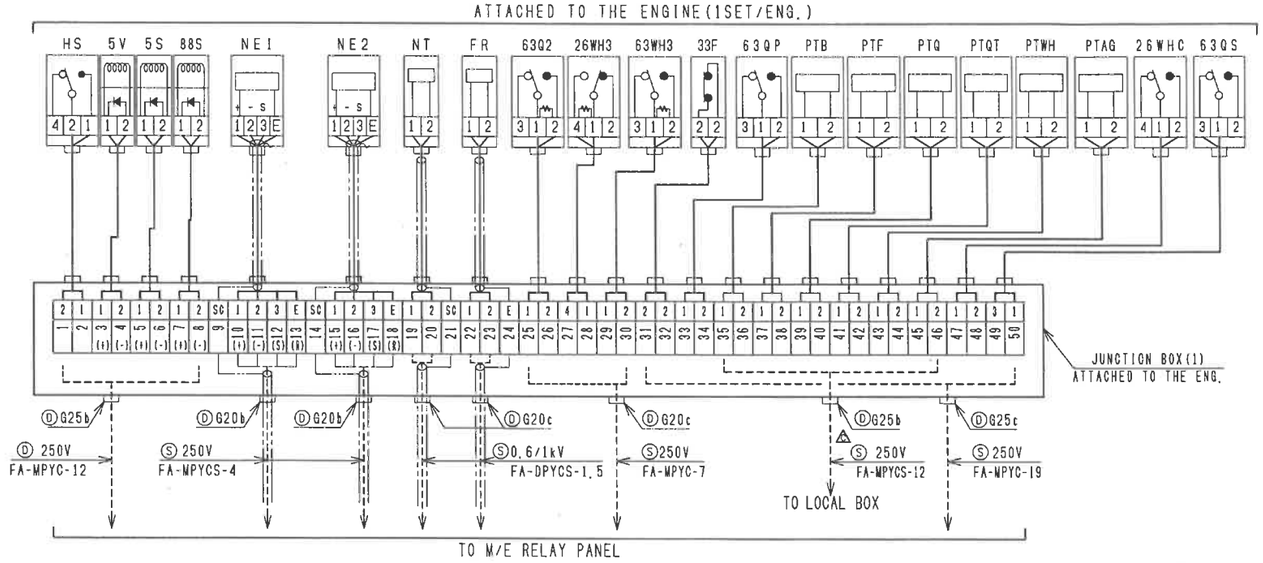
2017/05/09 TECHNICAL DEPARTMENT

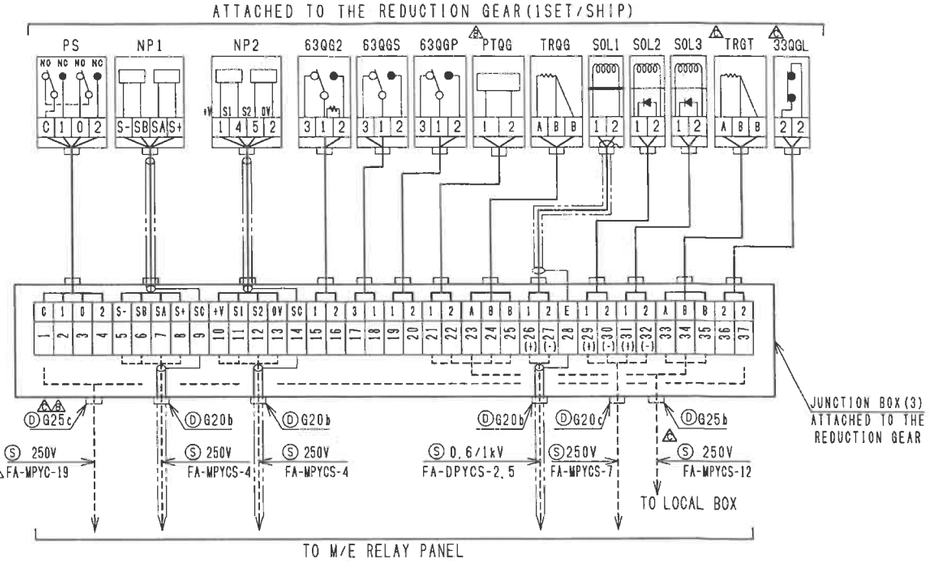
MARK	NAME	USE	NOTE
HS	HANDLE SWITCH	STAND-BY	AT RUNNING POSITION CONTACT (1-2) OFF
5V	FO SHUT DOWN MAGNETIC VALVE	CONTROL & SAFETY	DC24V±0.7A, FO CUT WHEN MAGNETIZED
5S	FO SHUT DOWN MAGNETIC VALVE	SAFETY	DC24V±0.7A, FO CUT WHEN MAGNETIZED
88S	START INTERLOCK MAGNETIC VALVE	INTERLOCK	DC24V±0.7A, READY TO START WHEN MAGNETIZED
NE1	PICK UP	CONTROL	FOR ENGINE REVOLUTION
NE2	PICK UP	SAFETY	FOR ENGINE REVOLUTION
NT	PICK UP	DETEC.	FOR TURBO-CHARGER REVOLUTION
FR	POTENTIOMETER	INDI.	

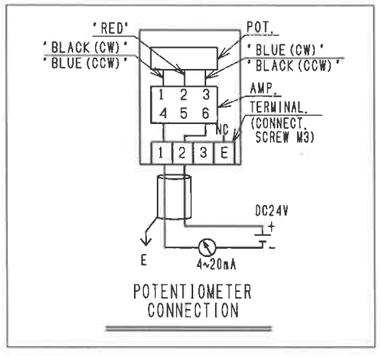
MARK	NAME	USE	NOTE
63Q2	LO PRESS. SWITCH	TRIP	BELOW 0.2 MPa (1-2) ON
26WH3	HT-FW TEMP. SWITCH	SLOWDOWN	OVER 90 ℃ (4-1) ON
63WH3	HT-FW PRESS. SWITCH	SLOWDOWN	BELOW 0.07 MPa (1-2) ON
33F	LEVEL SWITCH (LEAKED OIL TANK)	ALARM	WHEN F.O. HIGH PRESS. PIPE BROKEN CONTACT (2-2) OFF
63QP	L.O. PRESS. SWITCH	PUMP AUTO START	BELOW 0.27 MPa (1-2) ON
PTB	M/E BOOST AIR PRESS. TRANSMITTER	INDI.	RANGE:4-20mA/0-0.40MPa
PTF	M/E F.O. PRESS. TRANSMITTER	INDI. & ALARM	RANGE:4-20mA/0-0.60MPa ALARM:BELOW 0.1 MPa
PTQ	M/E L.O. PRESS. TRANSMITTER	INDI. & ALARM	RANGE:4-20mA/0-1.00MPa ALARM:BELOW 0.25MPa
PTQT	M/E T/C L.O. PRESS. TRANSMITTER	INDI. & ALARM	RANGE:4-20mA/0-1.00MPa ALARM:BELOW 0.20MPa
PTWH	M/E JACKET F.W. PRESS. TRANSMITTER	INDI. & ALARM	RANGE:4-20mA/0-0.60MPa ALARM:BELOW 0.1 MPa
PTAG	CONTROL AIR PRESS. TRANSMITTER	INDI.	RANGE:4-20mA/0-0.60MPa
63QS	L.O. PRESS. SWITCH	START INTERLOCK	OVER 0.27 MPa (3-1) ON
26WHC	F.W. HIGH TEMP. SWITCH	CONTROL	OVER 60 ℃ (2-1) OFF

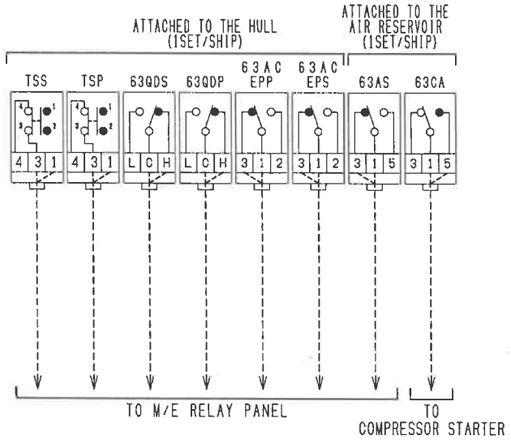
MARK	NAME	USE	NOTE
TSS	TURNING SAFETY LIMIT SWITCH FOR STBD. SIDE	STAND-BY	WHEN TURNING BAR ENGAGED CONTACT (3-1) ON
TSP	TURNING SAFETY LIMIT SWITCH FOR PORT SIDE	STAND-BY	WHEN TURNING BAR ENGAGED CONTACT (3-1) ON
63QDS	L.O. AUTO BACKWASH FILTER DIFF. PRESS. SWITCH FOR STBD. SIDE	ALARM	OVER 0.09 MPa (1-3) OFF
63QDP	L.O. AUTO BACKWASH FILTER DIFF. PRESS. SWITCH FOR PORT SIDE	ALARM	OVER 0.09 MPa (1-3) OFF
63AC EPP	CONTROL AIR PRESS. SWITCH	ALARM	BELOW 0.50 MPa (1-3) OFF
63AC EPS	CONTROL AIR PRESS. SWITCH	ALARM	BELOW 0.50 MPa (1-3) OFF
63AS	AIR RESERVOIR PRESS. SWITCH	ALARM	BELOW 1.50 MPa (4-1) OFF
63CA	AIR RESERVOIR PRESS. SWITCH	COMPRESSOR AUTO START & STOP	BELOW 1.8 MPa(1-5)ON, OVER 2.94 MPa(1-5)OFF (1.8~2.94 MPa AIR COMPRESSOR RUNNING)
TRB	M/E BOOST AIR TEMP.	INDI.	
TRC1~6	M/E EACH CYL. OUTLET EXH. GAS TEMP.	INDI. & ALARM	ALARM:OVER 480℃
TRT11, 2	M/E T/C INLET EXH. GAS TEMP.	INDI. & ALARM	ALARM:OVER 620℃
TRTO	M/E T/C OUTLET EXH. GAS TEMP.	INDI.	
TRQ	M/E L.O. TEMP.	INDI. & ALARM	ALARM:OVER 65℃
TRF	M/E F.O. INLET TEMP.	INDI. & ALARM	ALARM:OVER 60℃
TRWH	M/E H.T.-F.W. OUTLET TEMP.	INDI. & ALARM	ALARM:OVER 85℃

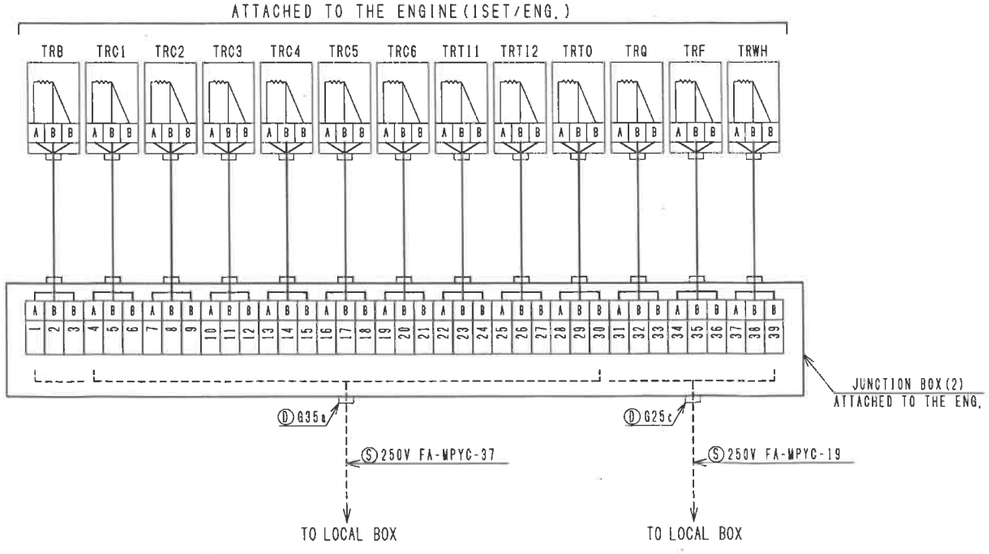
MARK	NAME	USE	NOTE
PS	CLUTCH OPERATING VALVE CHANGE OVER SWITCH	CONTROL	AHD:1-C, NET:3-C, AST:2-C
NP1	PICK UP	DETEC.	FOR PROPELLER REVOLUTION
NP2	PICK UP	CONTROL	FOR PROPELLER REVOLUTION
63QG2	L.O. PRESS. SWITCH	TRIP	BELOW 0.07 MPa (1-2) ON
63QGS	L.O. PRESS. SWITCH	START INTERLOCK	OVER 0.15 MPa (3-1) ON
63QGP	L.O. PRESS. SWITCH	PUMP AUTO START	BELOW 0.15 MPa (1-2) ON
PTQG	R/G L.O. PRESS. TRANSMITTER	INDI. & ALARM	RANGE:4-20mA/0-0.60MPa ALARM:BELOW 0.12MPa
⚠			
⚠			
TRQG	R/G L.O. TEMP.	INDI. & ALARM	ALARM:OVER 85℃
SOL1	MAGNETIC PROPORTIONAL FLOW CONTROL VALVE	CONTROL	DC24V±0.8A
SOL2	CLUTCH OPERATING OIL DRAIN MAGNETIC VALVE	CONTROL	DC24V±1.22A
SOL3	CLUTCH OPERATING OIL BYPASS MAGNETIC VALVE	CONTROL	DC24V±1.22A
⚠TRGT	R/G THRUST BEARING TEMP. THERMODULE	ALARM	ALARM:OVER 95 ℃
⚠33QGL	R/G L.O. SUMP TANK LEVEL SWITCH	ALARM	WHEN R/G L.O. TANK LOW CONTACT (2-2) OFF



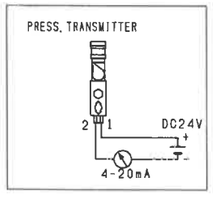








NOTE:
1. THIS DRAWING SHOWS THE STATE NORMAL OPERATION.
2. WIRING BETWEEN EACH DEVICE AND JUNCTION BOX SHOULD BE CARRIED OUT BY THE ENGINE MAKER.
3. Ⓢ MARKED CABLE AND THE WIRING WORKS SHOULD BE CARRIED OUT BY THE SHIPYARD.
4. AS FOR WIRING BETWEEN JUNCTION BOX AND PANEL, REFER TO THE DRAWING OF PANEL MAKER.
5. Diode for magnetic valve:correspond 10DDA40/NIHON INTER CO., LTD.
6. NO MARKED WIRE·····HABIA CABLE
7. ═══ WITH SEALED WIRE
8. ELECTRICAL RESSTANCE FOR TRIP SWITCH SHOULD HAVE 5.1kΩ/0.5W.



⚠x8	(4/7-D, 4/7/9-D, 7/9-F) ADD:TRQT/33QGL/GLAND REV:CABLE/GLAND SIZE , NOTE	2016/08/31	M. T
⚠x3	(7/8-F) REV:GLAND/CABLE SIZE DEL:PTQGC AHD/AST	2016/06/27	M. T
⚠	FIRST DRAWING	2016/05/27	M. T

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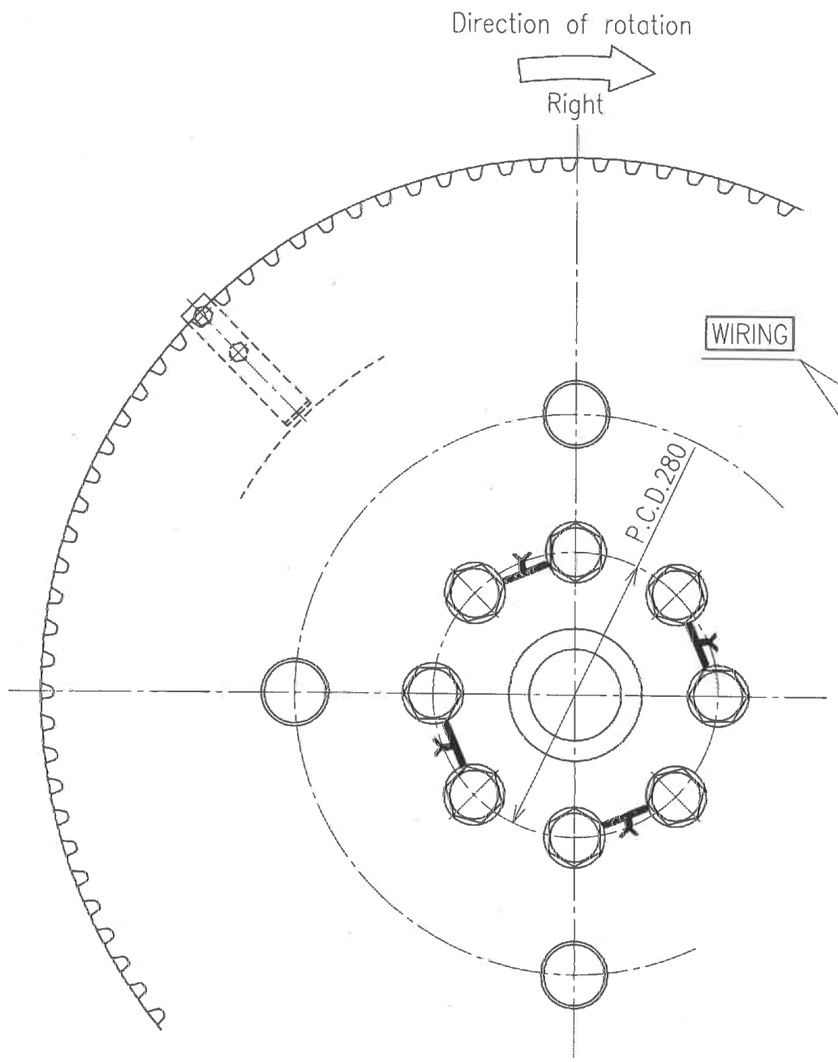
APPROVED	KAWANO SHUJI	LIST NO.	PDK20-0174-5
CHECKED	NARITA SHIROU	BEFORE	
	FUJIMOTO SYUHEI	REF.	BQP10000037A
	TOKUMOTO MANABU		
DRAWN	(H)SUMIDA YUKARI		
DATE	2016/08/30		

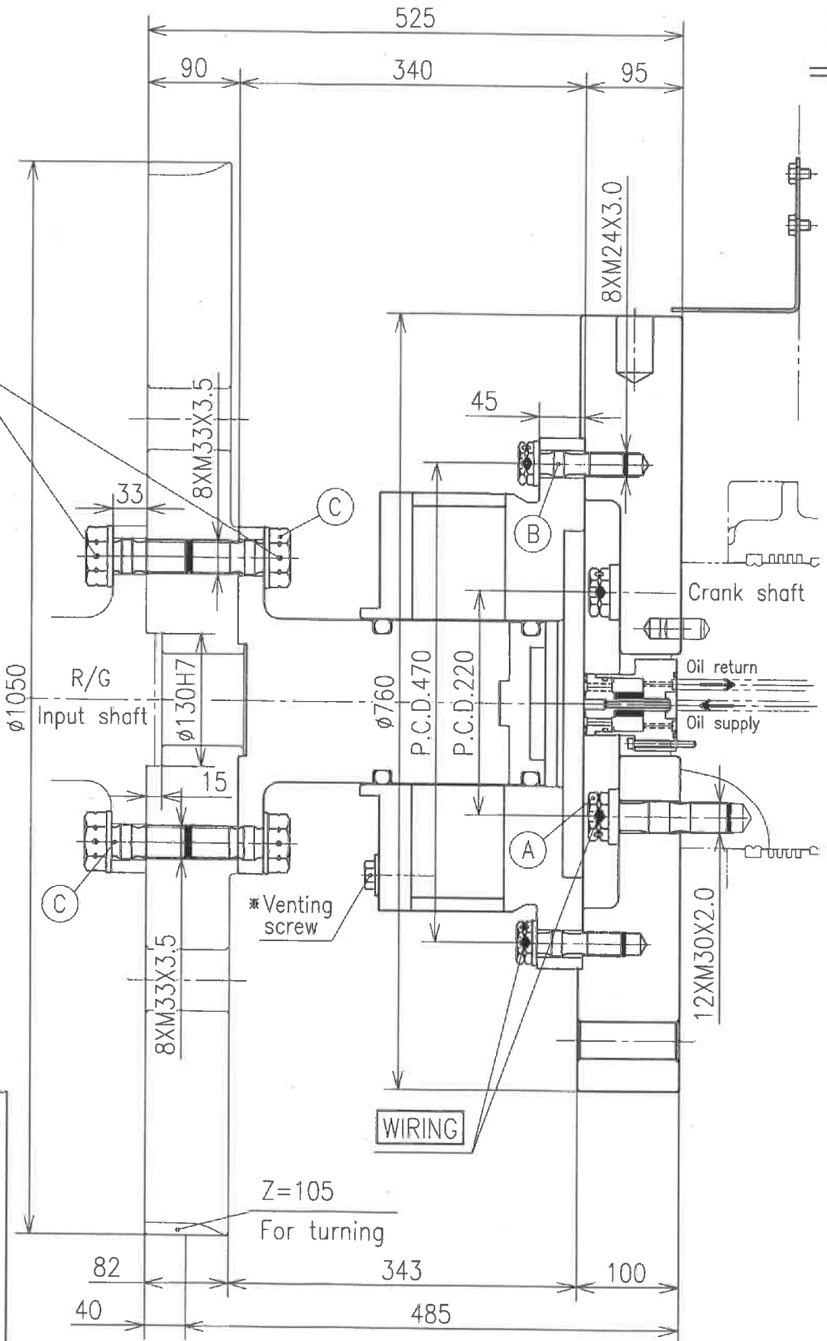
DAIHATSU DIESEL MFG.CO., LTD.
OSAKA JAPAN

MATERIAL	APRV.	GROUP	0
SCALE:	—:—		
MASS		Ⓐ QTY.	
TYPE	DK-20e		A2
LEVEL			
NAME	ケッセンズ::6DKM-20eF(L) x DRA-25J(L)		
	WIRING DIAGRAM::6DKM-20eF(L) x DRA-25J(L)		
PARTS NO.	AQP10008498	C	x

2017/05/09 TECHNICAL DEPARTMENT

FIXED ENG. TYPE : 6DKM-20eF

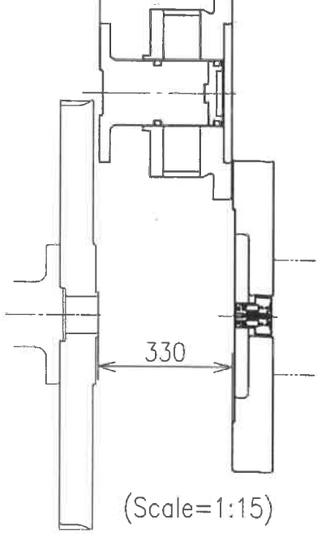


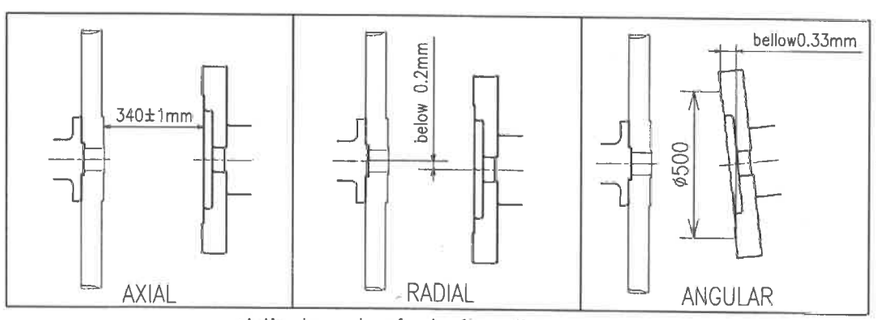


	WEIGHT
Eng.side	495kg
R/G side	598kg

SYMBOL	SIZE	TIGHTENING TORQUE	WIDTH ACROSS FLATS
A	M30X2.0	1275 N·m	41 mm
B	M24X3.0	640 N·m	41 mm
C	M33X3.5	1470 N·m	50 mm

- APPLY MOLYKOTE1000 SPRAY TO THE THREADED PORTIONS AND THE CONTACT SURFACES.
※Venting screw perform air extraction before initial running.





<Adjustment of shafts alignment>

⚠	FIRST DRAWING	(E020883) 2016/05/12	S.SAKATA

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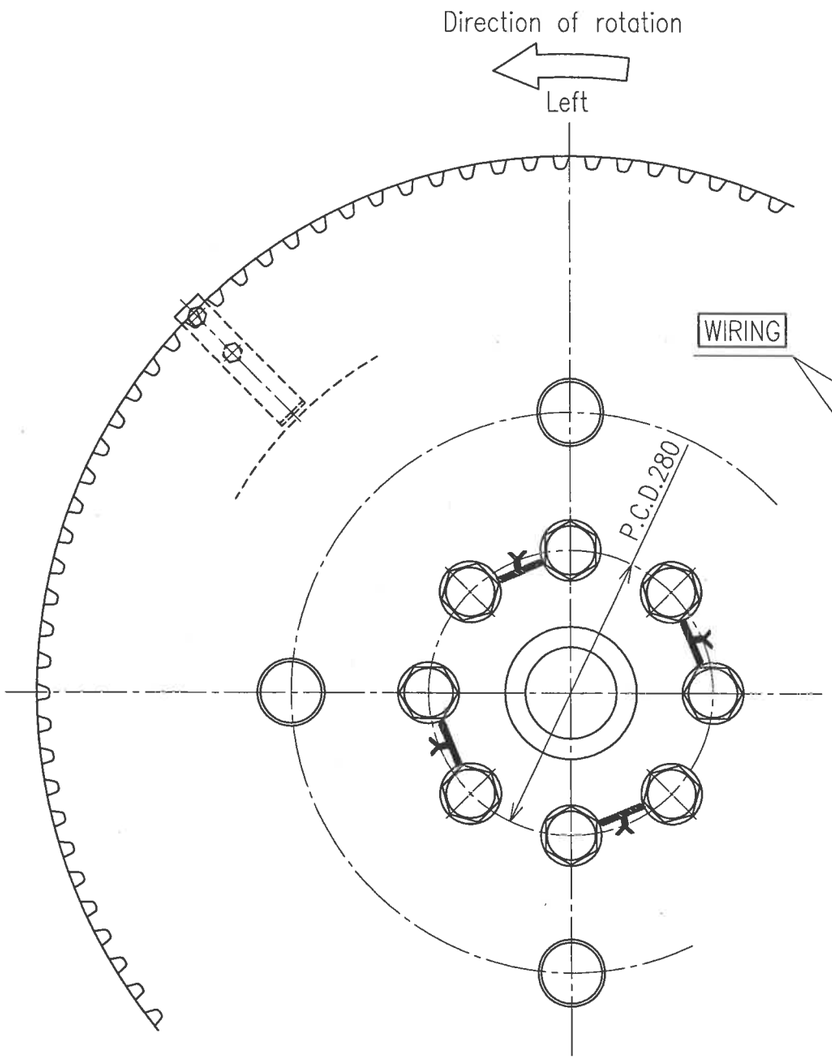
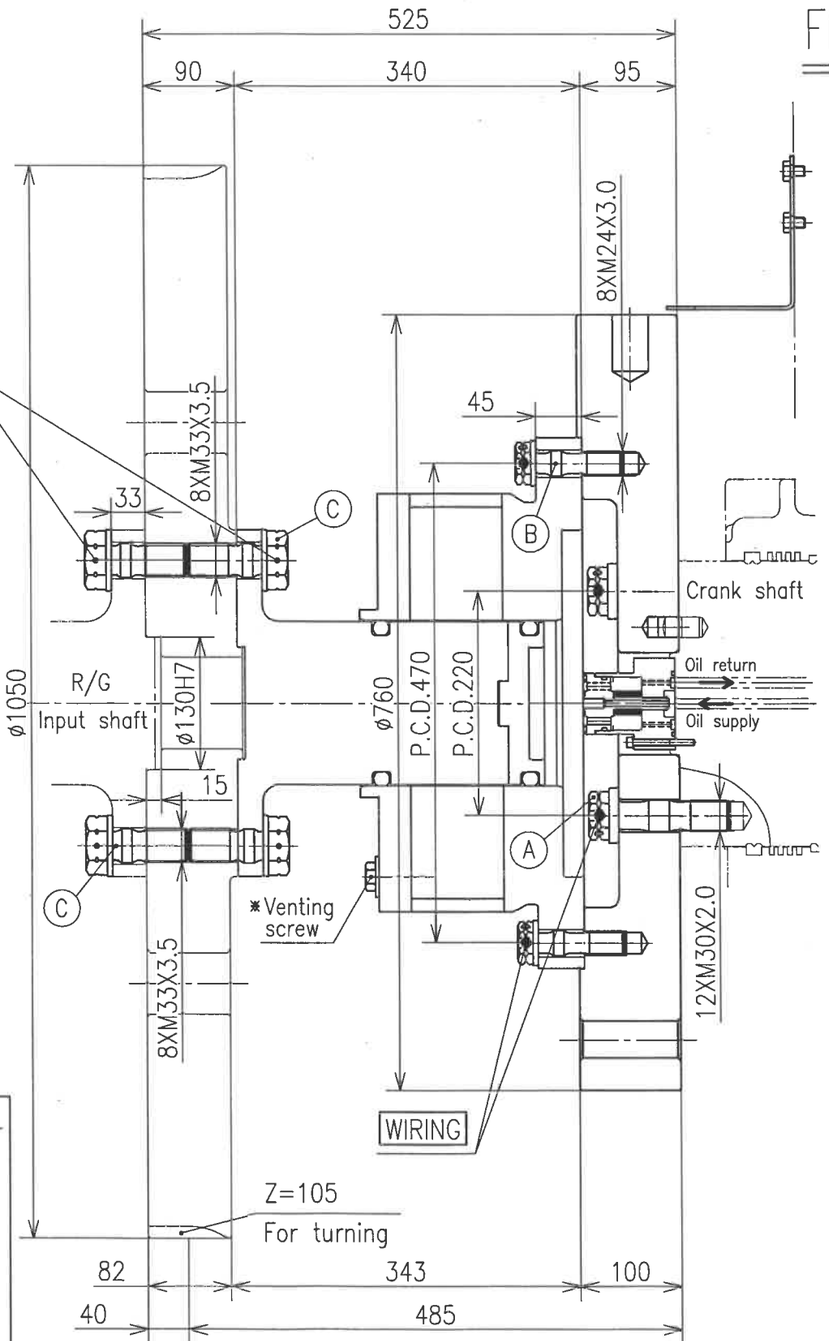
APPROVED	CHUJYOU JYUNYA	LIST NO. AG24005	MATERIAL KUMI	GROUP 0	SCALE:1:6(1:15)
CHECKED	SUSAMI NOBUYUKI	BEFORE	MASS 1093kg Ⓐ Ⓒ	QTY.	A3
	TAKASE YOSHIKUNI	REF.	TYPE DRA-19F		LEVEL
	YAMAGUCHI TAKANOBU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME ショウニンズ：カトウツギテS9：6DKM-20eF		
DRAWN	SAKATA SOHEI		APPROVAL:FLEXIBLE COUPLING S9:6DKM-20eF		
DATE	2016/05/12		PARTS NO. AG24005K013	A	*

2017/05/09 TECHNICAL DEPARTMENT

FIXED ENG. TYPE : 6DKM-20eFL



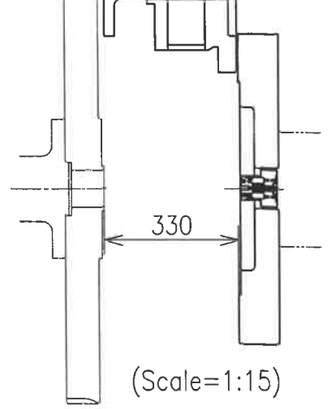


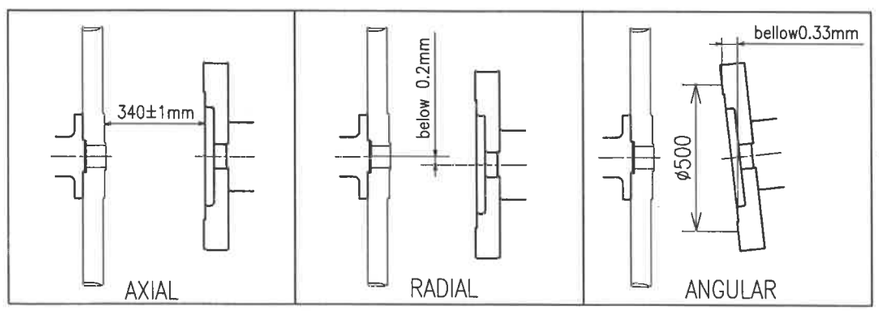
	WEIGHT
Eng.side	495kg
R/G side	598kg

SYMBOL	SIZE	TIGHTENING TORQUE	WIDTH ACROSS FLATS
A	M30X2.0	1275 N·m	41 mm
B	M24X3.0	640 N·m	41 mm
C	M33X3.5	1470 N·m	50 mm

・APPLY MOLYKOTE1000 SPRAY TO THE THREADED PORTIONS AND THE CONTACT SURFACES.

*Venting screw perform air extraction before initial running.





<Adjustment of shafts alignment>

⚠	FIRST DRAWING	(E020883)	2016/05/12 S.SAKATA

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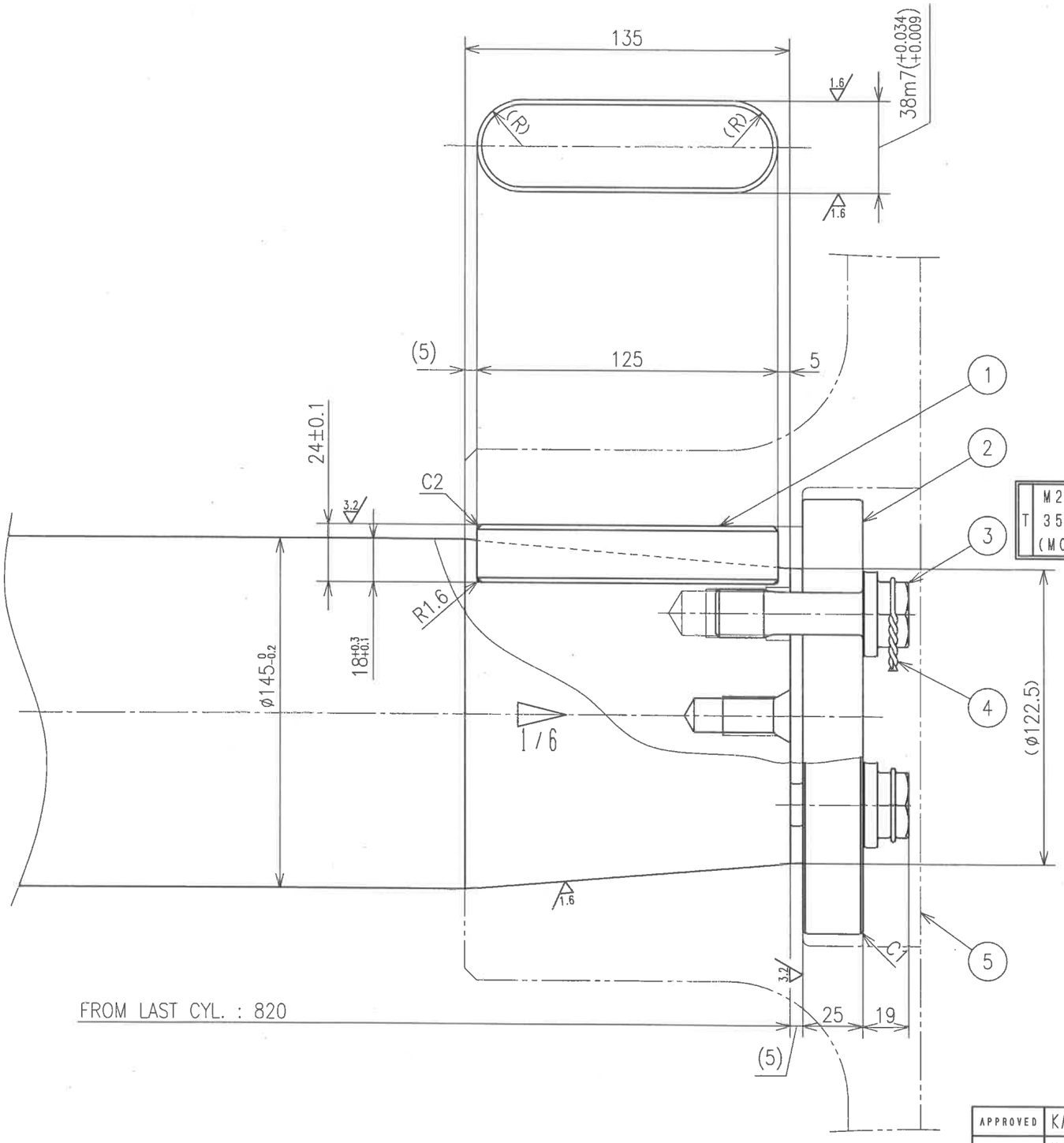
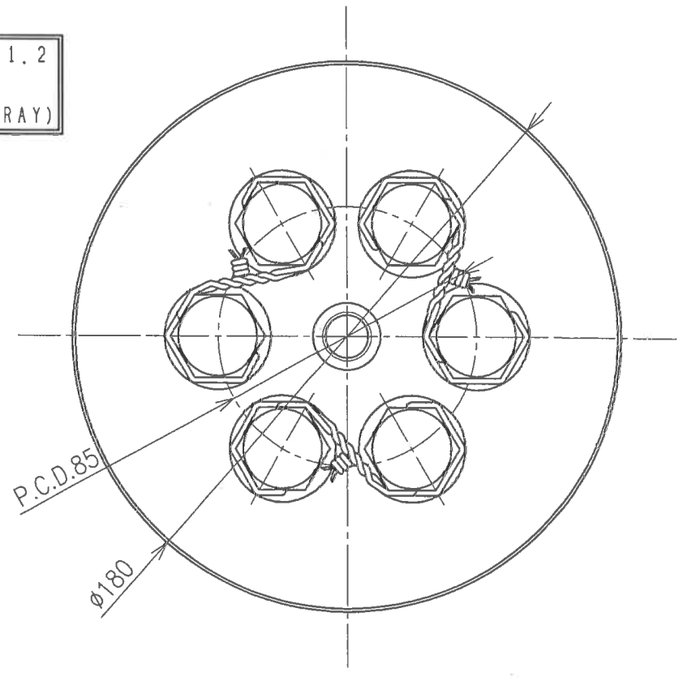
APPROVED	CHUJYOU JYUNYA	LIST NO. AG24005	MATERIAL KUMI	GROUP 0	SCALE=1:6(1:15)
CHECKED	SUSAMI NOBUYUKI	BEFORE	MASS 1093kg A C	QTY.	A3
	TAKASE YOSHIKUNI	REF.	TYPE DRA-19F		LEVEL
	YAMAGUCHI TAKANOBU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME ショウニンズ：カトウツギテS10:6DKM-20eFL		
DRAWN	SAKATA SOHEI		APPROVAL:FLEXIBLE COUPLING S10:6DKM-20eFL		
DATE	2016/05/12		PARTS NO. AG24005K014	A	*

2017/05/09 TECHNICAL DEPARTMENT

No.	NAME	MATERIAL	Q'ty/ENG.	SUPPLY
1	KEY	SCM435	1	DAIHATSU
2	SEAT PLATE	S45C	1	DAIHATSU
3	BOLT	SCM435	6	DAIHATSU
4	WIRE	SUS304-W1	3	DAIHATSU
5	COUPLING	–	1	SHIPYARD





T M20x1.5 HEX.27/31.2
350N·m
(MORYKOTE 1000 SPRAY)

⚠	FIRST DRAWING	2016/02/01	S.F

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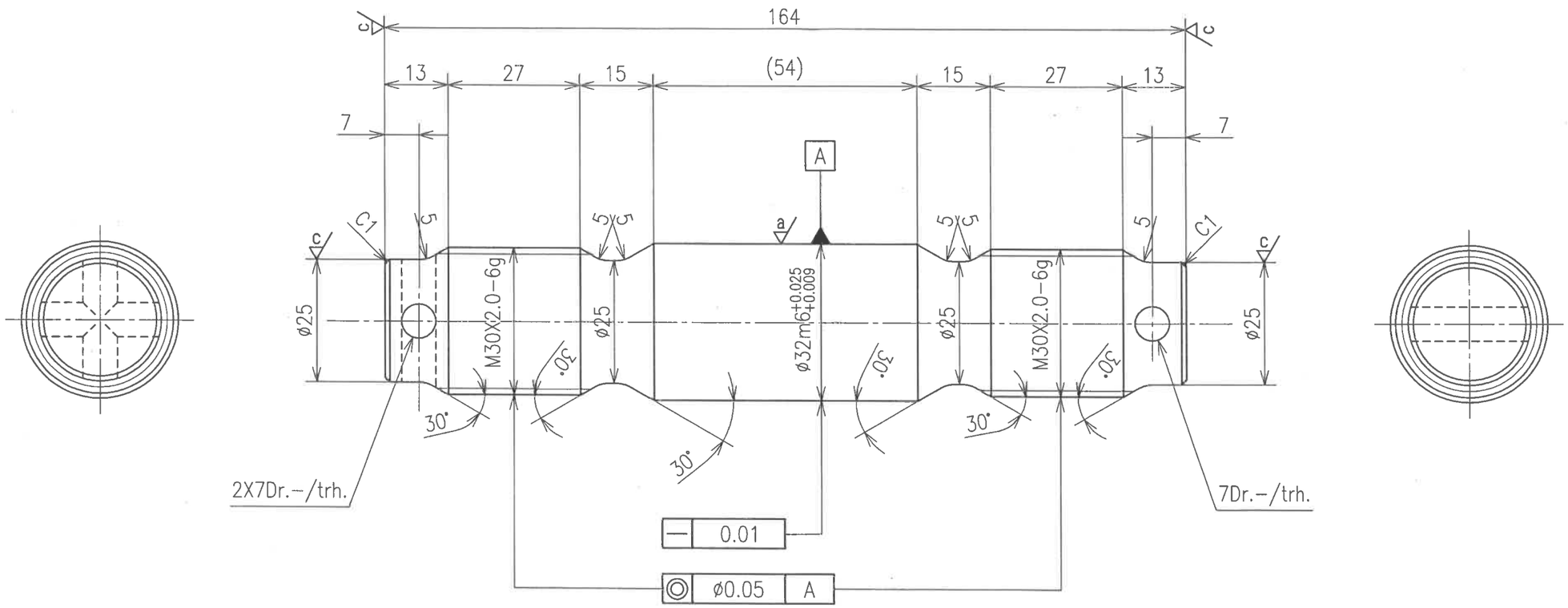
APPROVED	KAWANO SHUJI	LIST NO.	MATERIAL APRV.	GROUP 0	SCALE: 1:2
CHECKED	NARITA SHIROU	BEFORE	MASS	QTY.	A3
	FUJIMOTO SYUHEI	REF.	TYPE DK-20e		LEVEL
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME クミタテズ：ゼンタン：6DKM-20e		
DRAWN	(H)MORICHI YASUKO		ASSY. : P.T.O. : 6DKM-20e		
DATE	2016/02/01		PARTS NO. AQP10008255	A	*

2017/05/09 TECHNICAL DEPARTMENT



Yield point, 降伏点 ≧ 735 N/mm2
Tensile strngth, 引っ張り強さ ≧ 883 N/mm2

HQ,HT	HS 40~47

Alteration from last DWG.

・部品名称変更
(旧)『アタマナシリーマボルト 32x164』
↓
(新)『リーマボルト：ドウリョクデンタツ：32x164-M30x2.0』

・品質仕様書番号『E8-0221』追加

・降伏点、引っ張り強さ追加

⚠	FIRST DRAWING (E012242)	2011/09/29	藤原

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SPECIFIED BY (E8-0100), (E6-2010) (E8-0221)

DIMENSION	1 < 4 INCL	4 < 16	16 < 63	63 < 250	250 < 1000	1000 <	TOLERANCE OF ANGLE
▽~▽	±0.1	±0.2	±0.3	±0.5	±0.8	±1.5	±30'
▽~▽	±0.3	±0.5	±0.7	±1.2	±2.0		

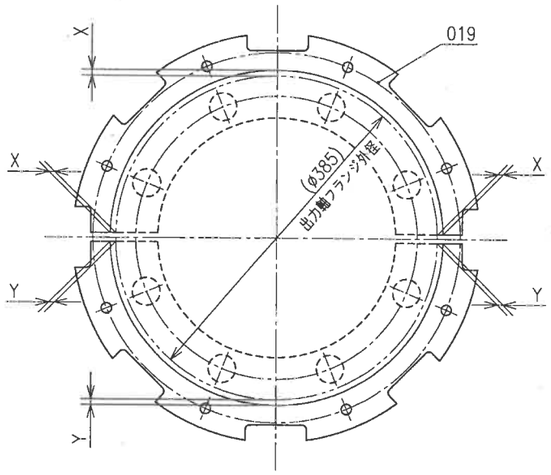
ROUGHNESS Ra: a ▽ 1.6, b ▽ 3.2, c ▽ 12.5 — CO.2 FOR NONINDICATED PART

APPROVED	塩田 雄二	LIST NO. G25322	MATERIAL D-SCM435 GROUP 4
CHECKED	山口 隆信	BEFORE	MASS. 0.9kg (A)(C) QTY.
	高瀬 好邦	REF.	TYPE DR-19F
DRAWN	(D) 藤原 邦夫	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME リーマボルト：ドウリョクデンタツ：32X164-M30X2.0
DATE	2011/09/29		REAMER BOLT:POWER TRANSMISSION:32X164-M30X2.0
			PARTS NO. G253220100A Z *

SCALE: 1:1 — A3

2017/05/09 TECHNICAL DEPARTMENT

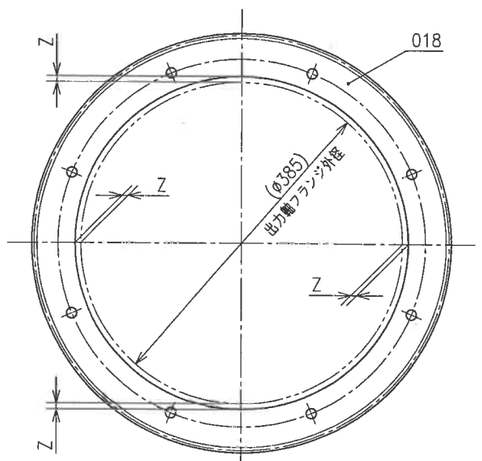
①



スキマXとYがそれぞれ同じ値になるよう、出力軸とプレート(019)を取付のこと

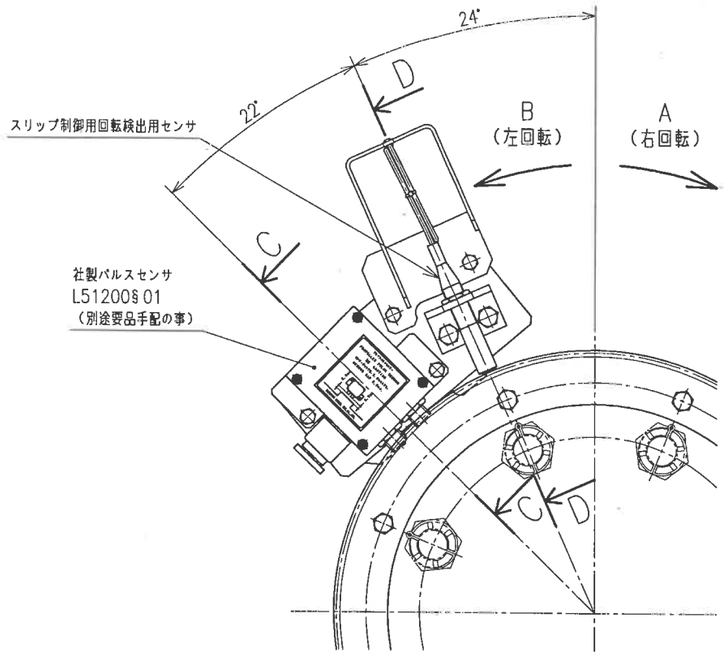
(標準半径スキマ 0.050~0.145)

②

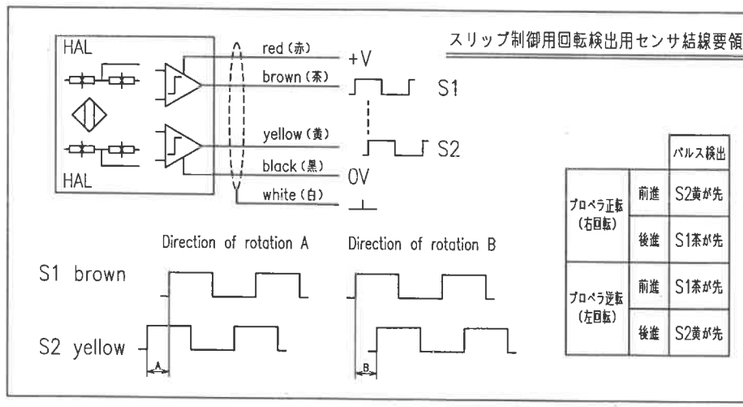


スキマZが同じ値になるよう、出力軸とギヤ(018)を取付のこと

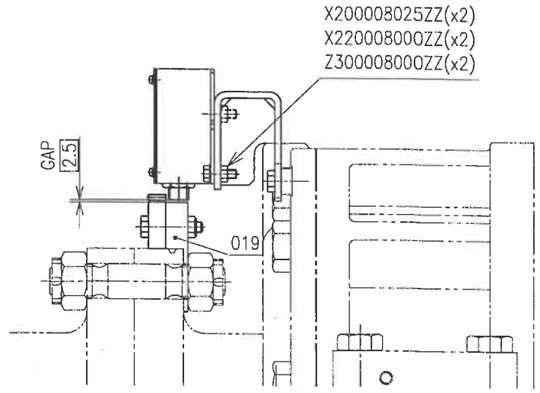
(標準半径スキマ 0.050~0.145)



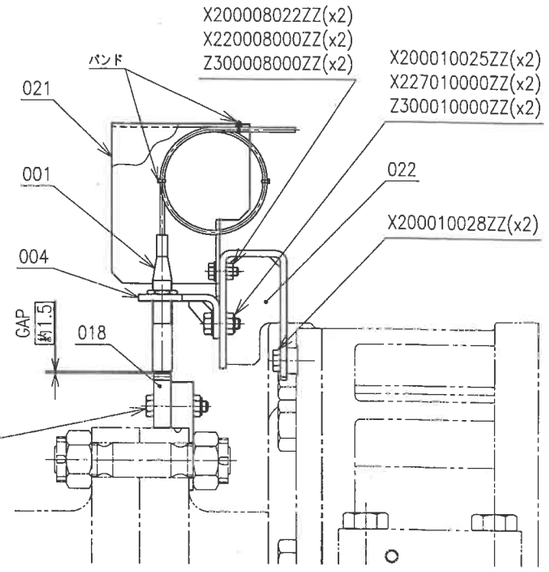
スリップ制御用回転検出用センサ結線要領



		パルス検出
プロペラ正転(右回転)	前進	S2黄が先
	後進	S1茶が先
プロペラ逆転(左回転)	前進	S1茶が先
	後進	S2黄が先

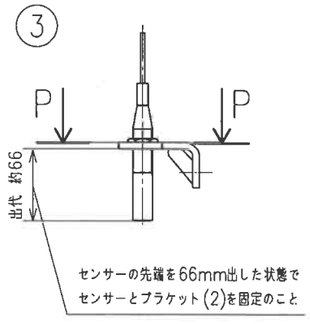


Section "C-C"



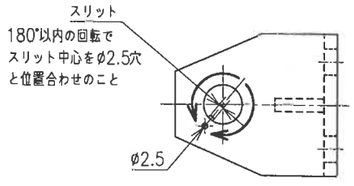
Section "D-D"

③



センサーの先端を66mm出した状態で
センサーとブラケット(2)を固定のこと

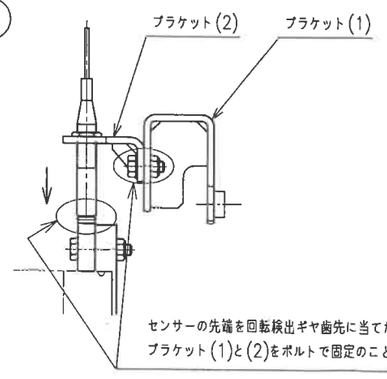
④



Section "P-P"

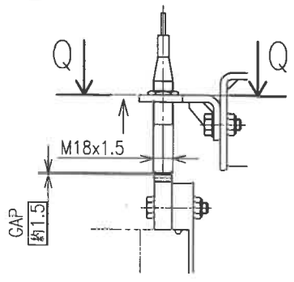
S=1:2

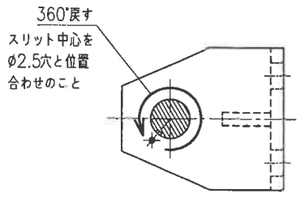
⑤



センサーの先端を回転検出ギヤ歯先に当てた状態で、
ブラケット(1)と(2)をボルトで固定のこと

⑥





Section "Q-Q"

S=1:2

	モジュール	歯数	
スリップ制御用	2.5	192	
プロペラ回転用	-	8	歯切欠き 70x20

EACH NO. REPRESENTS PART NO.
IN CONTROL LIST.
(LIST NO. NN00736 §13)

⚠ FIRST DRAWING	(E020862)	2016/4/28	S.SAKATA

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APPROVED	CHUJYOU JYUNYA	LIST NO.	NN00736
CHECKED	SUSAMI NOBUYUKI	BEFORE	
	TAKASE YOSHIKUNI	REF.	
	YAMAGUCHI TAKANOBU		
DRAWN	SAKATA SOHEI		
DATE	2016/04/28		

DAIHATSU DIESEL MFG.CO.,LTD.
OSAKA JAPAN

MATERIAL	KUMI	GROUP	0	SCALE:	1:4
MASS	–	QTY.			A2
TYPE	DR			LEVEL	

NAME クミタテズ：パルスセンサ：DRA-25JS
ASSY.：PULSE SENSOR：DRA-25JS

PARTS NO. NN00736K006 A

2017/05/09 TECHNICAL DEPARTMENT

5. How to install and adjust propeller pulse sensors

(1) Outline of how to install propeller pulse sensors

There are 2 types of the propeller pulse sensor: one for control, and the other for backup. Propeller rotation speed is detected by a rotation detection gear (for control) and a rotation detection plate (for backup) attached to the output shaft coupling.

The propeller pulse sensor is a proximity switch, and it can also distinguish ahead and astern rotational directions. This means that **if a propeller pulse sensor is not installed properly, the sensor may not be able to detect propeller rotation signal, and this may pose problems for clutch engagement and slip operations. Therefore, please make sure to install a propeller pulse sensor in accordance with the following instructions.**

* Note) A rotation detection gear is a single unit, and therefore it must be inserted to an output shaft before it is directly coupled to a middle shaft.

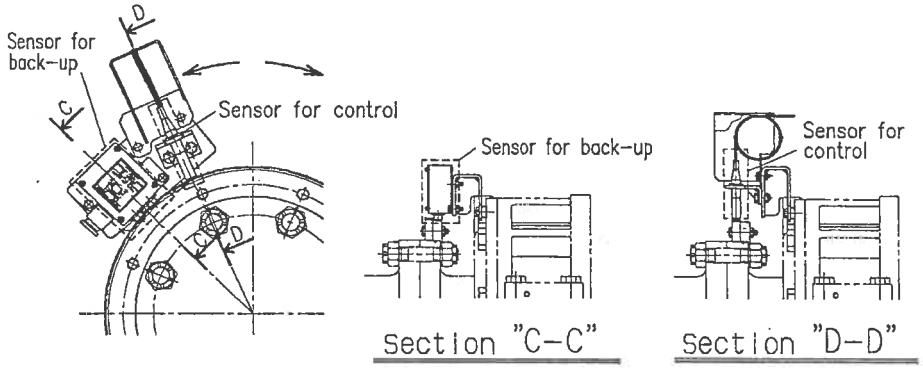


Fig. 7 Exterior view of propeller pulse sensor

(2) How to install propeller pulse sensors

1) How to install rotation detection gear (for control) and rotation detection plate (for back-up)

i) Set the rotation detection plate and the rotation detection gear to the output shaft coupling.

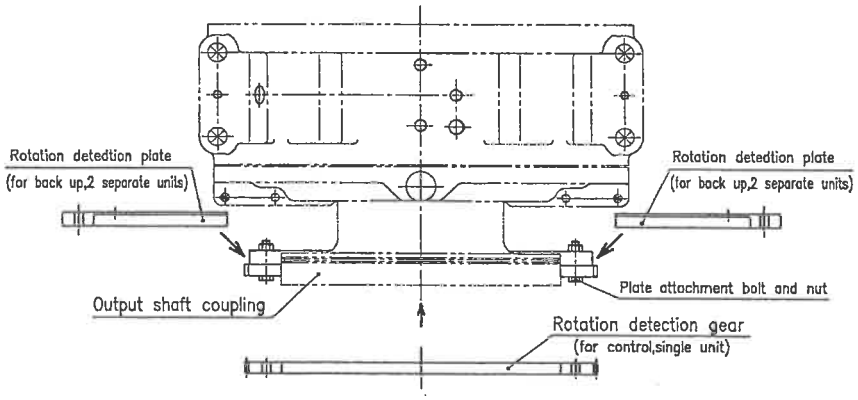


Fig. 8 Installing rotation detection gear/plate

ii) Adjust so that their clearances with the output shaft couplings in the places marked with "★" (10 places in total) will be uniform.

<Note> Standard radial clearance X,Y,Z:0.050~0.145

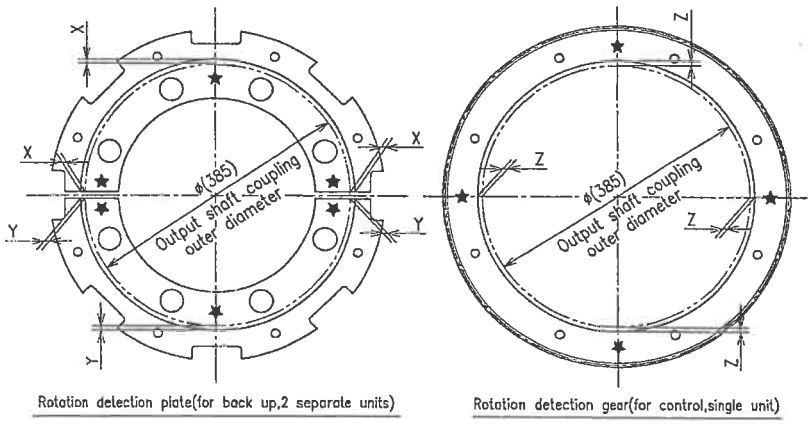


Fig. 9 Rotation detection gear/plate clearance adjustments

2) How to install propeller pulse sensor (for control)

i) Attach the sensor to a bracket so that its protrusion allowance is about 67 mm, and adjust the sensor's slit with the bracket's mark (a hole with the diameter of 2.5 mm). While keeping the sensor touching the rotation detection gear's tooth tip, fix a bracket (1) with a bolt.

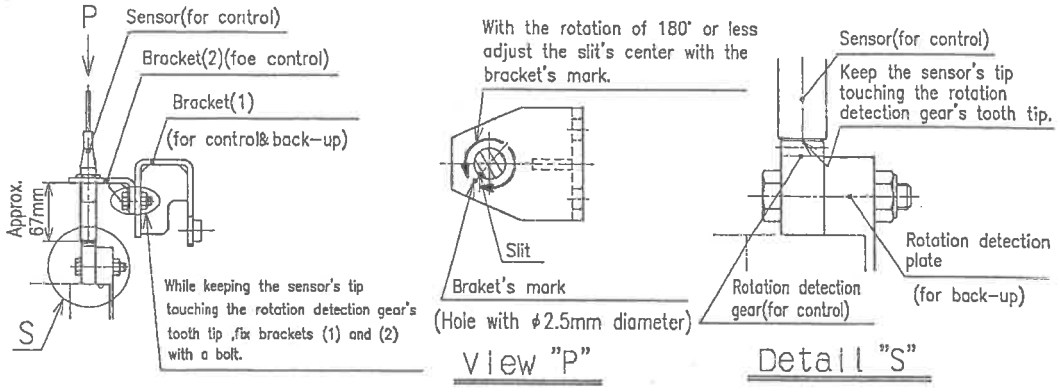


Fig. 10 Sensor position adjustments

ii) Rotate back the propeller pulse sensor (for control) one rotation, and adjust its clearance with its rotation detection gear to the value specified below.

A clearance between the propeller pulse sensor (for control) and its rotation detection gear must be 1.5

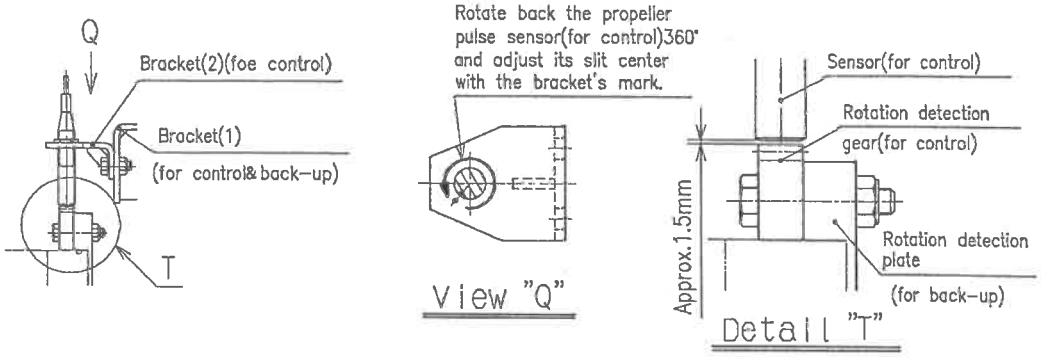
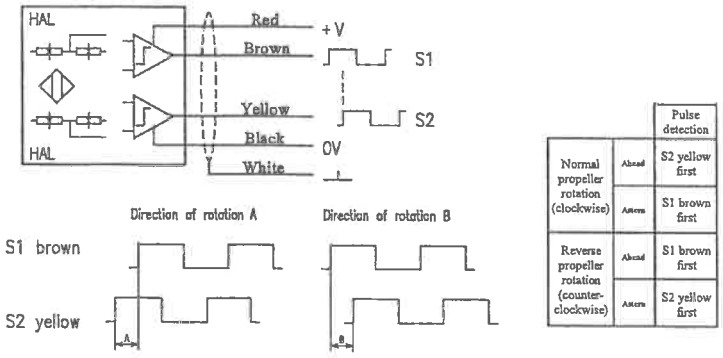


Fig. 11 Sensor clearance adjustment

iii) Connect wires in accordance with the following wire connection instructions.



		Pulse detection
Normal propeller rotation (clockwise)	Ahead	S2 yellow first
	Astern	S1 brown first
Reverse propeller rotation (counter-clockwise)	Ahead	S1 brown first
	Astern	S2 yellow first

Fig. 12 Wire connection instructions

iv) Points to be checked after installation

- Turn on an electronic controller.
- It can be confirmed whether the propeller pulse sensor (for control) is operating normally by measuring the voltage values of the control panel's terminal blocks No. 13 and No. 14.

3) How to install propeller pulse sensor (for backup)

The output signals of the propeller pulse sensor (for backup) are also square waves. It is used as a backup when the propeller pulse sensor (for control) fails.

2 sensors are installed in a sensor box to detect ahead and astern motions.

- It is a box construction. Therefore, use bracket attachment hole's looseness to adjust its clearance.

A clearance between the rotation detection plate and the propeller pulse sensor (for backup) must be 2.5 mm.

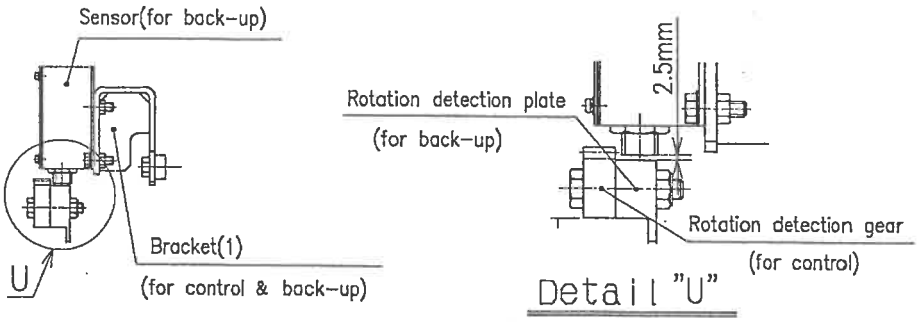


Fig. 13 Installing sensor for display

6. Emergency operations

(1) If the control panel for the proportional solenoid valve loses power or control signals to the proportional solenoid valve are lost, a "anomalous proportional solenoid valve controller (PLC)" alarm on a monitor panel is turned on. In that case,as back-up a prescribed current is separately sent from a main engine's panel to the proportional solenoid valve so that the proportional solenoid valve is opened by a prescribed open amount and clutch engagement operations can be performed by control handle operations.The prescribed current value is already Although a dial provided on the main engine's control panel can be used to change the prescribed current's value, **In case that the engine is running under the situation of "anomalous proportional solenoid valve controller (PLC)", make sure the engaging clutch oil must be within green mark range.**

Basically customers do not have to adjust the prescribed current, the engaging clutch oil pressure by any possibility is below blue mark range,turn clockwise (increase the proportional solenoid valve open angle) the dial located near the control panel for the proportional solenoid valve in main engine control panel in order to increase the pressure within blue mark range.

[Alarm reset]

Turn off the circuit breaker in the the control panel for the proportional solenoid valve, and then turn on it again. If the alram is not reset by above precedure, PLC might be failure. Please contact our service engineer.

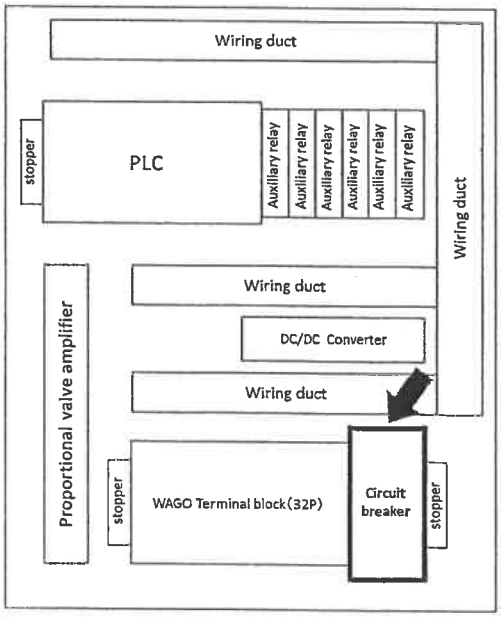
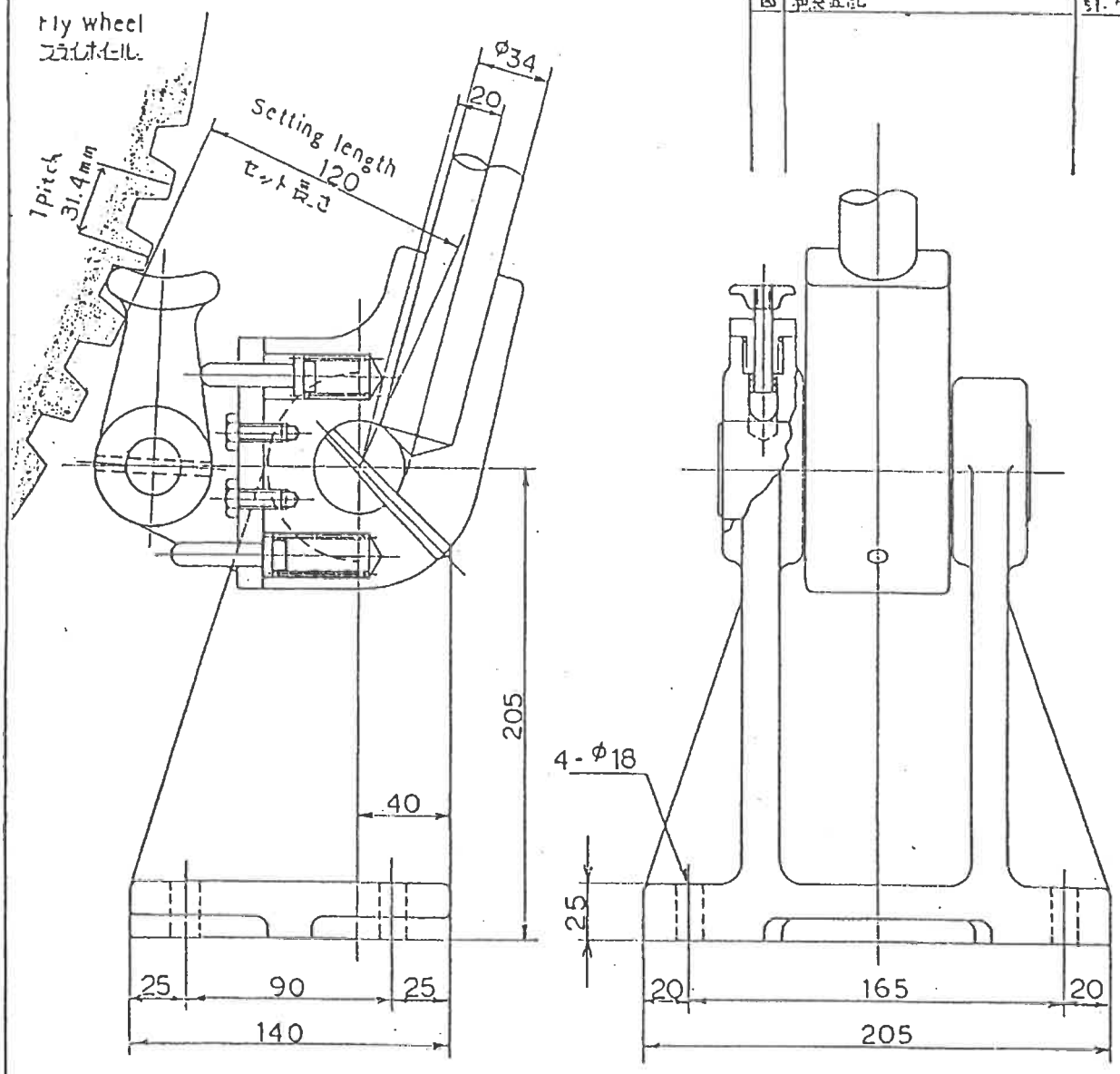


Fig. 14 The position of the circuit breaker in the the control

HISTORY OF DRG.	DATE
Ⓐ 初図作成	51.7.25



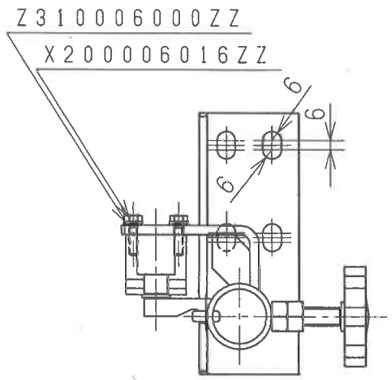
If the turning lever is turned by 21°.(To left or right)
ターニングレバーを左右どちらかに21°動かせば、フライホイールは1ピッチ
Fly wheel is driven for 1 pitch.(31.4 mm)
(31.4mm)回転します。
This device is capable of driving fly wheel for 2 pitches
この装置は、1回のターニングによって、最大2ピッチフライホイールを回転
as maximum for 1 stroke.
させられます。

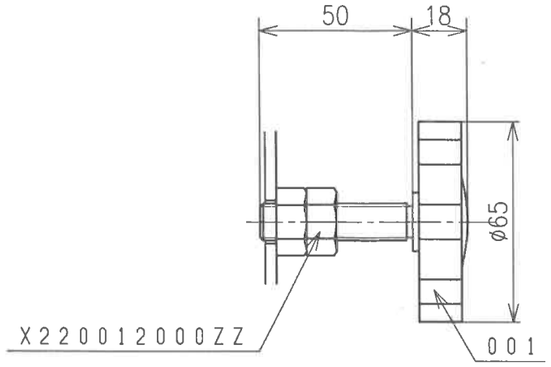
船体取付

TO BE FITTED BY SHIPYARD

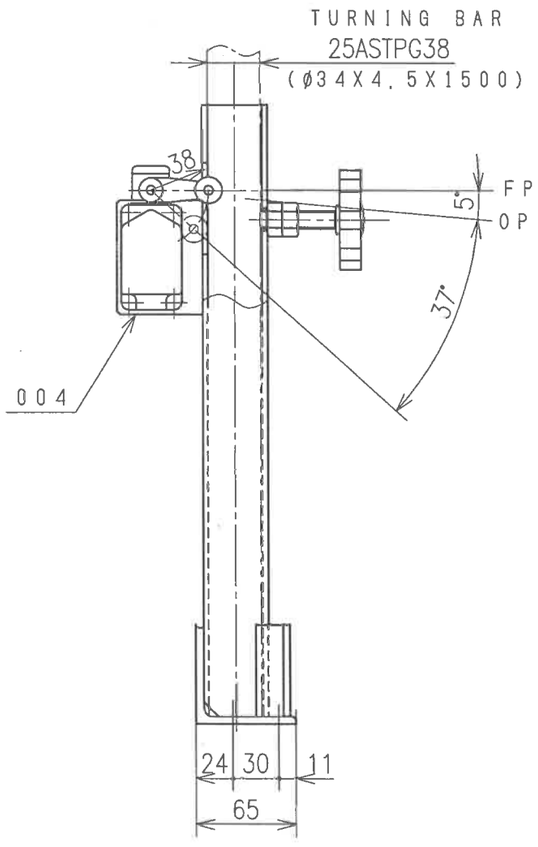
NO.	NAME OF PARTS	PARTS NO.	MATERIAL	WEIGHT kg	QTY.
APPROV-ED BY		LIST NO.	GROUP		
		BEFORE	SCALE —	TYPE 全形式	
CHECK-ED BY		AFTER		NAME フライホイールターニングソウテ	
		REF.		FLY WHEEL TURNING DEV.	
DRAWN BY		DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN		PARTS NO. A-5902011170	Z
DATE	SEP.-1.1981				

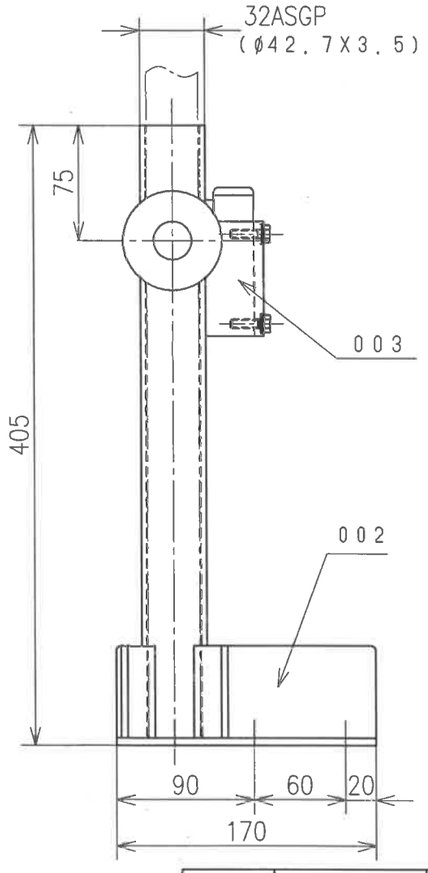
DRG. NO. 675685 A





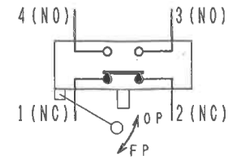
(S=1/2)





NN00022_01, 02, 21, 22

CONNECTION



NN00022_01, 02, 11, 12

CONTACT CAPACITY

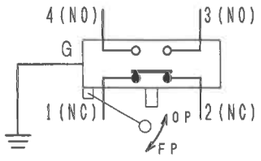
VOLTAGE		CURRENT
AC	250V	10A
	125V	10A
DC	125V	0.8A
	30V	6A

MINIMUM APPLICABLE LOAD

VOLTAGE		CURRENT
DC	24V	33mA

NN00022_11, 12, 31, 32

CONNECTION



NN00022_21, 22, 31, 32

RECOMMENDED LOAD RANGE

VOLTAGE		CURRENT
DC	24V	1~33mA

C	ADD MICROLOAD (E010151)	2010,09,22	T.NAKAMITSU
B	№002 Change (E003009)	2005,1,7	T.KAWANO
⚠	FIRST DRAWING		

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APPROVED	MOCHIDUKI MASARU	LIST NO.	NN00022
CHECKED	CHUJYOU JYUNYA	BEFORE	
CHECKED	CHUJYOU JYUNYA	REF.	
CHECKED	TORII YOSHIO	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	
DRAWN	NAKAMITSU TOSHIO		
DATE	2010/09/22		

MATERIAL	KUMI	GROUP	0
SCALE:	1:4		
MASS.		QTY.	
			A3
TYPE	ISA		
NAME	クミタテズ：ターニングセフティースイッチ：		
	ASSY.:TURNING SAFETY SWITCH:		
PARTS NO.	NN00022K001	C	*

2017/05/09 TECHNICAL DEPARTMENT

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チョックファスト計算書

DESIGN CALCULATION FOR "CHOCKFAST"

		DRAWN BY	M. Tokumoto
TYPE	6DKM-20eF	CHECKED BY	A. Tajimoto
LIST No.	AQP10008848A		
DATE	Jan.11,2017	APPROVED BY	S. Tsuru
REVISION			

チョックファスト面圧計算書
DESIGN CALCULATION FOR "CHOCKFAST"

・機関型式 / Engine type ····· 6DKM-20eF

・チョックファスト要目 / Principal items ····· ITW Philadelphia Resins Corporation
CHOCKFAST ORANGE

・機関の総質量 / Engine weight (W) W = 10,500 kg

・チョックファストの据付面積
/ Effective CHOCKFAST area (A) A = 281,032 mm^2

・ボルトの径及び数 / Bolt quantity & diameter
据え付けボルト / Hold down bolt (n1) n1 = 16 pcs. M20 (リーマー2本含 / with reamer bolt)

・ボルトの軸力 / Bolt tension (Torque)
据え付けボルト / Holding down bolt (T1) T1 = 30,793 N

・チョックのサイズ / CHOCK size 別紙参照 / Please see the attached sheet

・チョックファストの厚み / CHOCK thickness (mm) 35 mm

・機関質量による面圧 W / A = 0.3736 N/mm^2 (0.3736MPa)
/ Deadweight pressure on CHOCKFAST = 3.8110 kg/cm^2
船級規定値 / Regulation
NK,GL:0.7N/mm^2 LR,NV:0.9N/mm^2 KR:9kg/cm^2 BV:0.7MPa JG:7kg/cm^2

・機関質量及び据付ボルトによる面圧 / Total static pressure on CHOCKFAST

$$P = (W \times 9.81 + n1 \times T1) / A$$
$$= (10500 \times 9.81 + 16 \times 30{,}793) / 281{,}032$$
$$= 2.120 \text{ N/mm2}$$
$$= 21.62 \text{ kg/cm}^2$$

船級規定値 / Regulation
P.3,4を参照 / Reference to P.3,4

・検討結果 / Result

基準値以下 / Not over the rule

2017/05/09 TECHNICAL DEPARTMENT

4. 船級協会

- American Bureau of Shipping
- Det Norske Veritas
- Korean Resister of Shipping
- United States Coast Guard
- Dertsche Schifffs – Revision U. Klassifikation
- Polish Register
- Nippon Kaiji Kyokai
- Japan Goverment
- Bureau Veritas
- Germanischer Lloyd
- Lloyds Resister of Shipping
- Registrar Italiano Navals
- Registrar of U.S.S.R
- Biro Klasifikasisi Indonesia

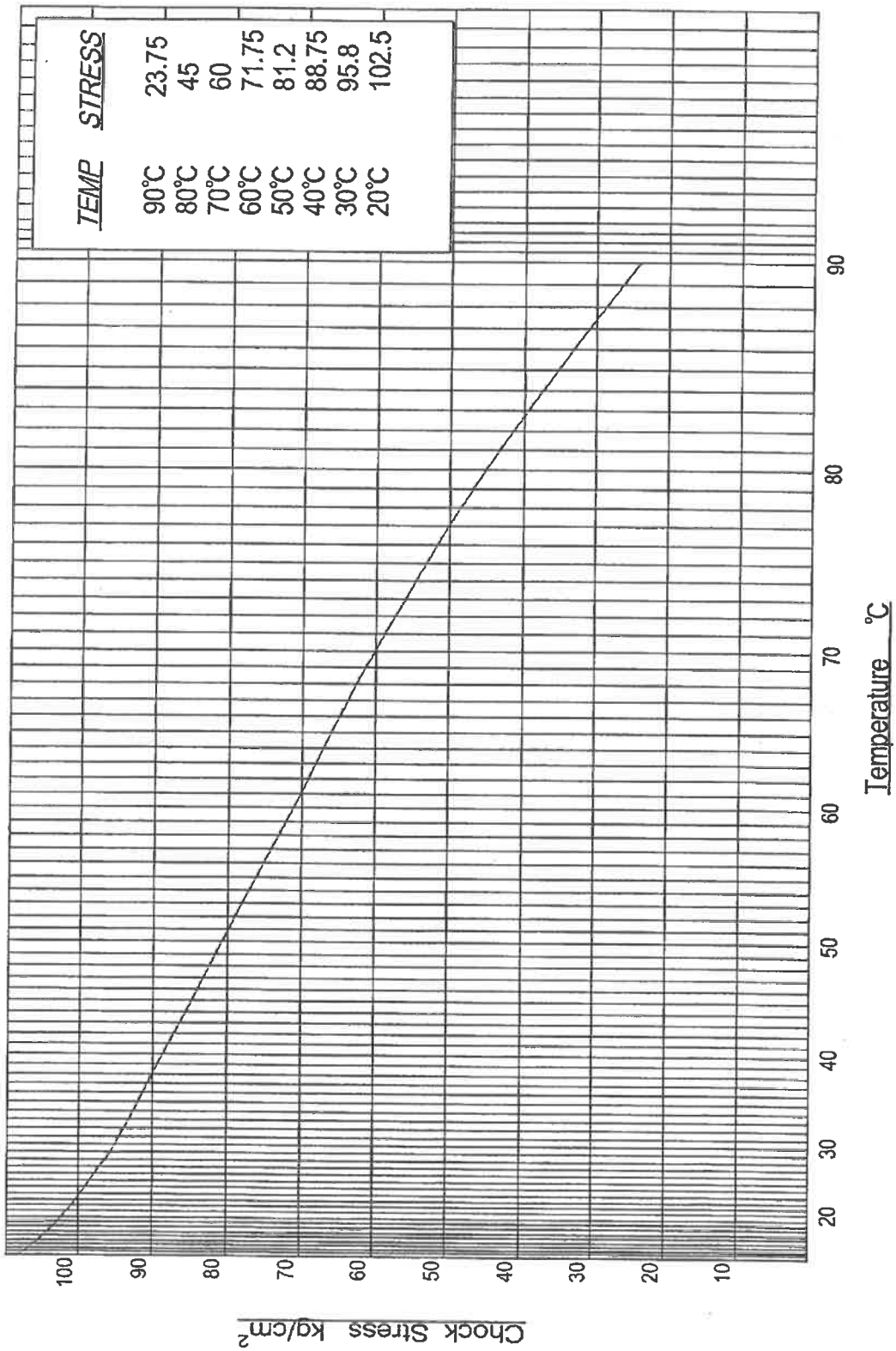
5. 主要船級協会 承認内容 （チョックファストオレンジ）

	機器運転時の最大温度	合計最大面圧	機器重量のみの最大面圧
NK	80℃	45kg/cm²	0.9N/mm²
LR	90℃	35kg/cm²	0.9N/mm²
ABS	80℃	45kg/cm²	—
NV	90℃	*	0.9N/mm²
CR	70℃	35kg/cm²	—
KR	80℃	45kg/cm²	9kg/cm²
USCG	80℃	45kg/cm²	—
BV	90℃	*	0.7Mpa
GL	80℃	4.5N/mm²	0.7N/mm²
ZG	80℃	35kg/cm²	—
JG	60℃	35kg/cm²	7kg/cm²

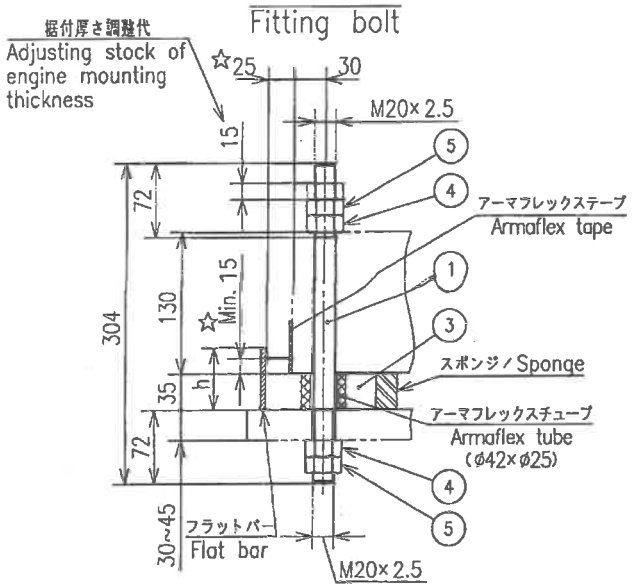
（*別紙チャートを参照）

注1）NK の主機の場合、チョック部の温度は 80℃以下、45 kg/cm² 以下とする。
注2）LR の場合、チョック部の運転時の温度が 80℃を超える場合又はチョックに掛かる動的荷重が 35kg/cm² を越える時は、流し込み後の樹脂の発熱温度又は、チョック部の温度を 40℃まで上げる。

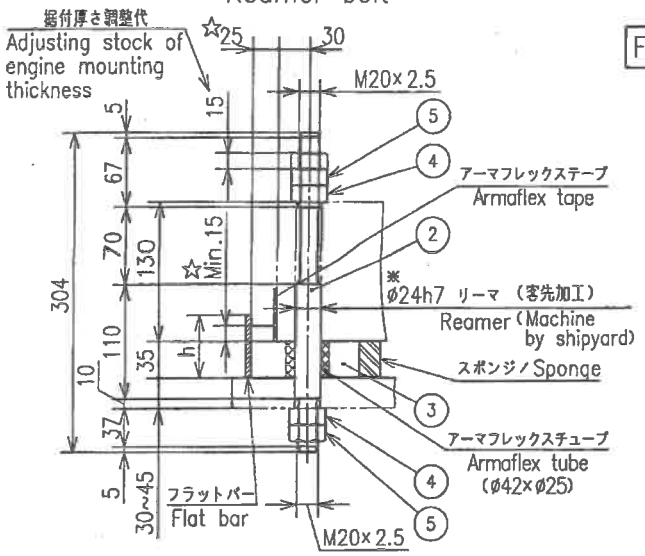
Permitted CHOCKFAST ORANGE chock stress



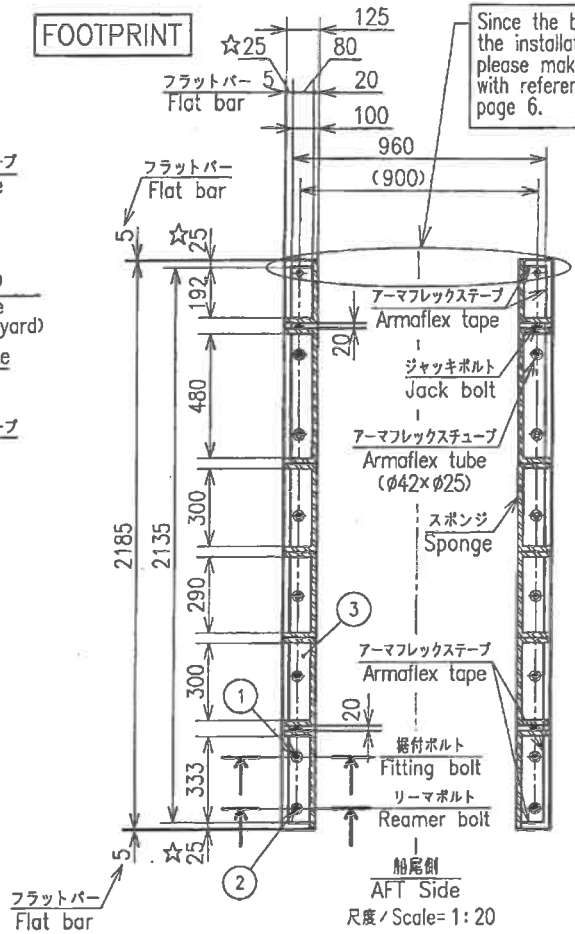
据付ボルト Fitting bolt



リーマボルト Reamer bolt



FOOTPRINT



Since the bolt hole and the installation end are close, please make the chock fast with reference AQP10008848 page 6.

尺度/Scale= 1:20

部品表 Parts list			
No.	名称 Name	材質 Material	数量/機関 Q'ty/Eng.
1	据付ボルト Fitting bolt	S45C	14
2	リーマボルト Reamer bolt	S45C	2
3	チョックファースト Chockfast	Chockfast orange	–
4	六角ナット(内側) Hexagon nut (Inner)	SS400	32
5	六角ナット(外側) Hexagon nut (Outer)	SS400	32

ボルト締め付けトルク Bolt tightening torque		
No.	ネジの呼び Nominal designation of thread	N·m
4	M20×2.5	83~98
5	M20×2.5	104~122

注：1) チョックファースト用ダムの寸法"h"は、船体の据付レーキ角度を考慮して下さい。
なお最小押し湯高さは15mmを確保して下さい。
2) ※部「リーマボルト」の外径は、リーマ寸法に+3mmの"仕上代付"にて納入します。
(船内にて現物合わせの上リーマ加工願います。)
3) ☆部寸法は、チョックファーストメーカー推薦値です。
4) 本図の機関台は、最小寸法で作図しています。

Note:1) The dam height "h" should be decided considering the rake of hull. The gate riser height should be more than 15mm.
2) The diameter of the reamer bolt (※ marked) is delivered to the reamer with the finishing stock of +3mm.
3) The demensions marked "☆" are chockfast maker recomendation.
4) The engine bed is the Min.size in this drawing.

補機ギヤケース：カコウと分割
AUX. gear case: Separate from frame

チョックファースト面積 The area of the chockfast	281,032mm²

⚠	FIRST DRAWING	2016/12/21	M. T

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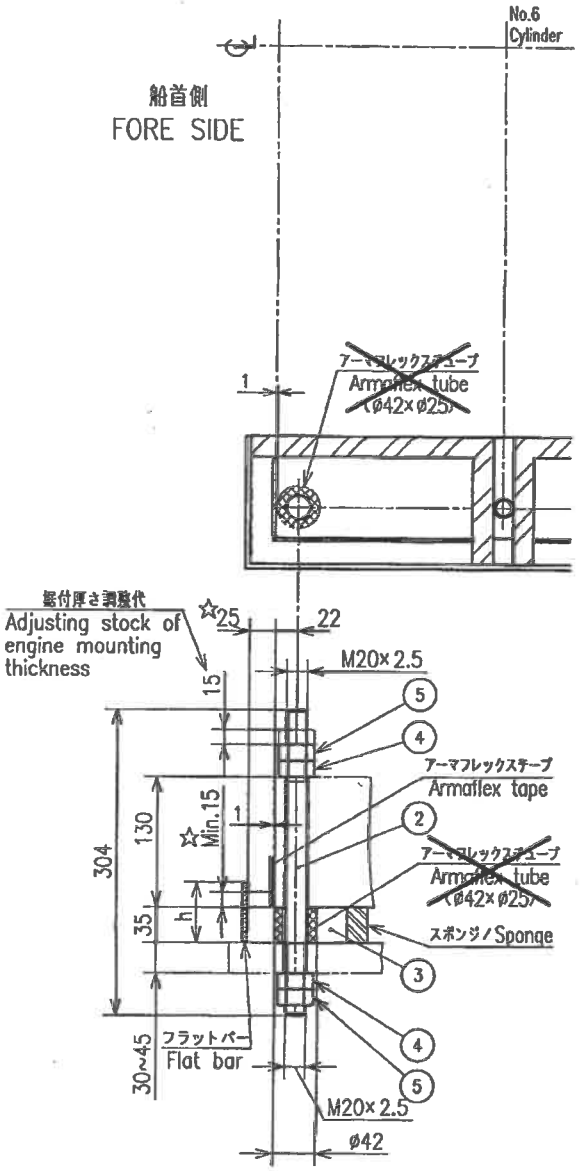
APPROVED		LIST NO.	
CHECKED		BEFORE	
		REF.	
DRAWN	(H) MORICHI YASUKO	DAIHATSU DIESEL MFG. CO., LTD. OSAKA JAPAN	
DATE	2016/12/21		

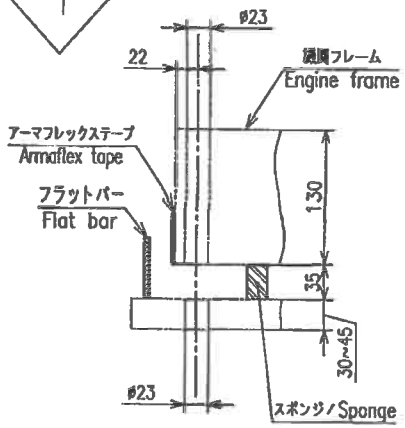
MATERIAL	APFV	GROUP 0	SCALE: 1:5
MASS		QTY.	A3
TYPE	DK-20e		LEVEL
NAME	クミタテズ：チョックファースト：6DKM-20eF		
	ASSY. : CHOCKFAST : 6DKM-20eF		
PARTS NO.	AQP10008831	A	*

AQP10008848A (5/6)
2017/05/09 TECHNICAL DEPARTMENT



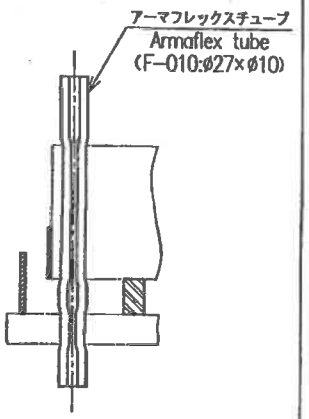
The most fore side 2 holes are tubeless.

1



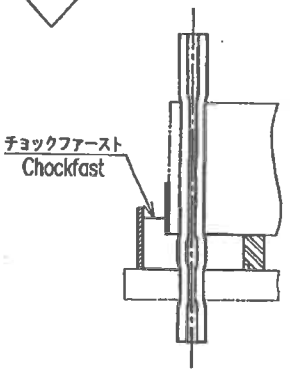
Extract the tube.

2



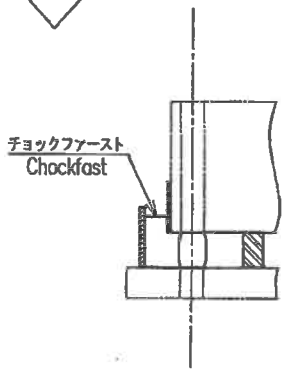
Please insert the tube into the bolt hole after grease the tube.

3



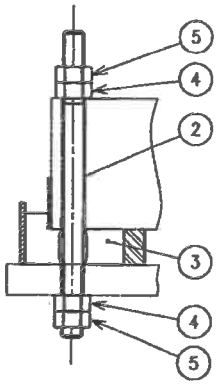
Construct Chock Fast in the usual way.

4



Tighten with bolts.

5



(AQP10008831_A-1)

〔部 品 表〕
〔PARTS LIST〕

No.	名 称 NAME	材 質 MATERIAL
1	スエツケボルト SETTING BOLT	S45C
2	リーマボルト REAMER BOLT	S45C
3	スエツケボルト用ライナー CHOCK LINER	FC200
4	リーマボルト用ライナー CHOCK LINER	FC200
5	六角ナット HEX. NUT	SS400
6	ロックナット LOCK NUT	SS400

〔ボルト締め付トルク〕
〔TIGHTNING TOLQUE〕

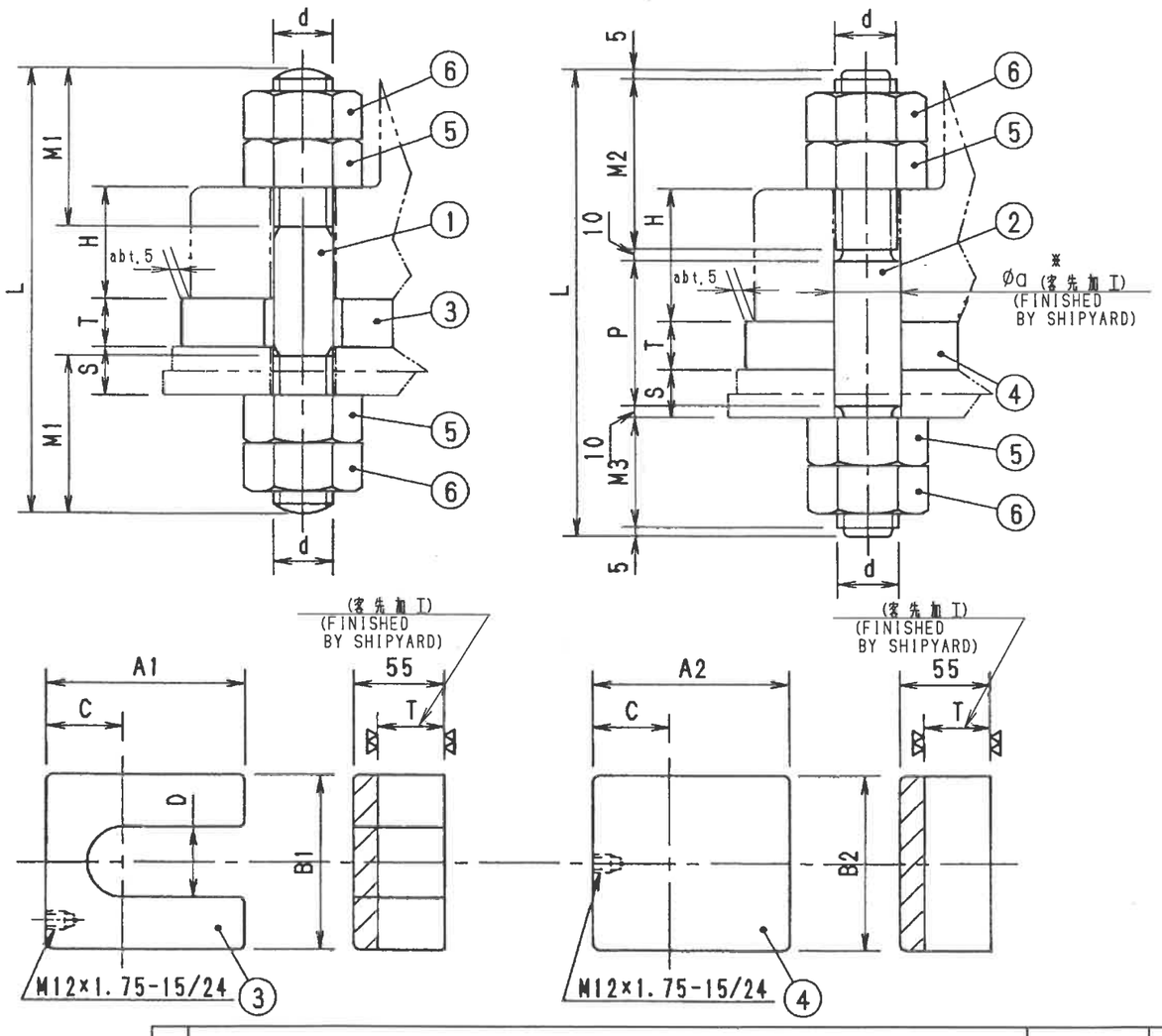
ネジの呼び SIZE OF SCREW	N・m	kg・m
M27x3.0	294~343	30~35
M33x3.5	540~638	55~65
M42x4.5	1128~1324	115~135
M48x5.0	1668~1962	170~200

〔据付及びリーマボルト寸法表〕
〔SETTING & REAMER BOLT SIZE TABLE〕

d	M27x3.0		M33x3.5			M42x4.5			M48x5.0
a	※32		※36			※48			※54
L	270		288	308		324		354	362
P	110		120			120			140
M1	80		86			104			113
M2	80		79	99		97		127	106
M3	50		59			77			86
H	60	70	70	80	90	60	70	100	90
呼び径x長さ SIZE x LENGTH	M27 x270		M33 x288	M33 x308		M42 x324		M42 x354	M48 x362
適用形式 APPLICATION FORM DRA		-11E	-16F	-19F -25H	-30F		-80F	-40F -50F	-120F
適用形式 APPLICATION FORM DRB	-35H	-16F -25H		-19F		-50F	-30F, 40F -80F		

〔チョックライナ寸法表〕

A1	115	135	185
A2	115	135	185
B1	100	110	130
B2	100	110	130
C	45	55	60
D	38	47	52
T	MAX. 45 ~ 30		
S	MAX. 45 ~ 35		
型番 SERIAL NUMBER	100x115	100x135	130x185
適用形式 APPLICATION FORM DRA	-11E, -16F~30F	-40F~80F	-120F
適用形式 APPLICATION FORM DRB	-16F, 19F, -25H, 35H		



注；1）※部「リーマボルト」の外径は"a"寸法に+3mmの"仕上代付"にて納入します。
（船内にて現物合わせの上リーマ加工願います）
REMARKS) ※THE DIA. OF THE REAMER BOLT (※ MARKED) TO BE SUPPLIED WITH +3mm MARGIN FOR "a" SIZE.

⚠B	CHG NO.1,2 MATELIAL S43C→S45C	2008.10.29	M.T
⚠A	FIRST DRAWING	2008.5.6	F.S

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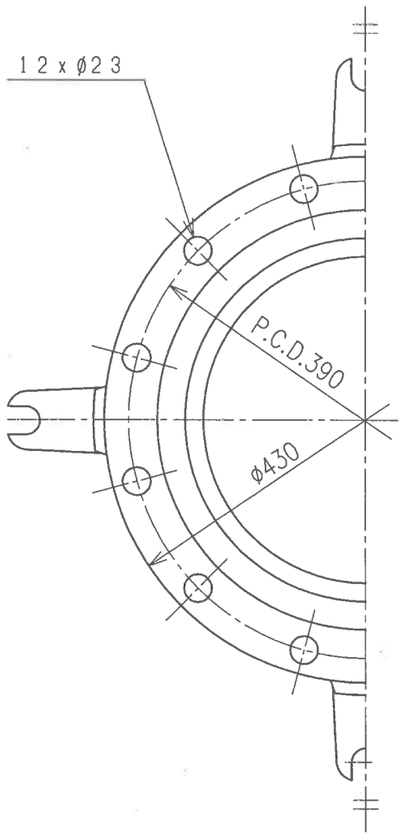
APPROVED	KAWANO SHUJI	LIST NO.	
CHECKED		BEFORE	
	NARITA SHIROU	REF. -	
	NISHIKAWA HIROSHI	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	
DRAWN	TOKUMOTO MANABU		
DATE	2008/10/29		

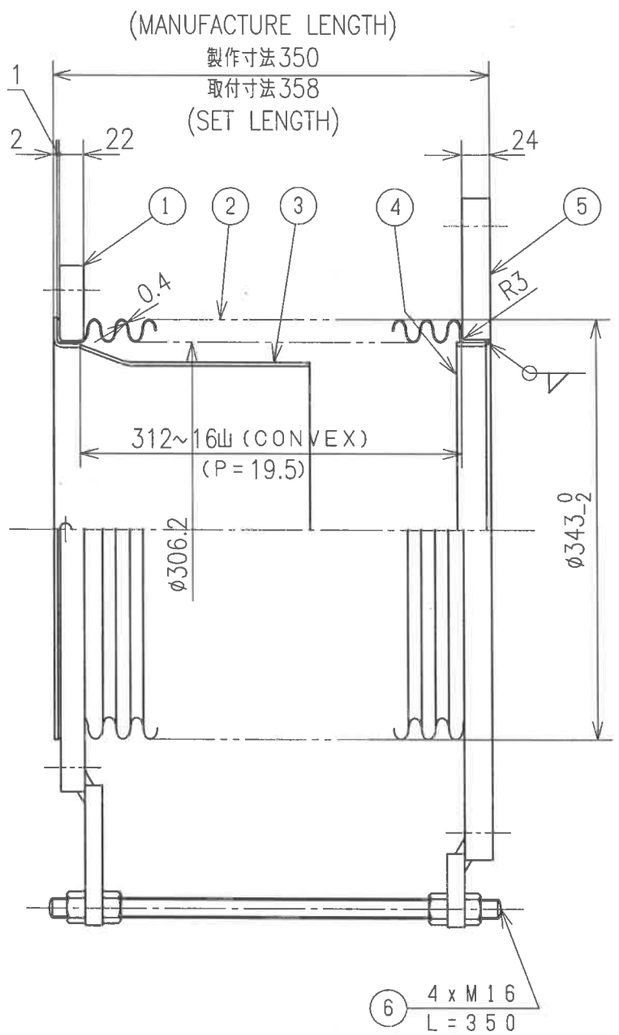
MATERIAL	APRV	GROUP 0	SCALE: -
MASS.	A Ⓒ	QTY.	3 A3
TYPE	ALL		
NAME	クミタテズ：チョックライナ：ゲンソクキ		
	ASSY.:CHOCK LINER:REDUCTION GEAR		
PARTS NO.	AQP10002495	B	※

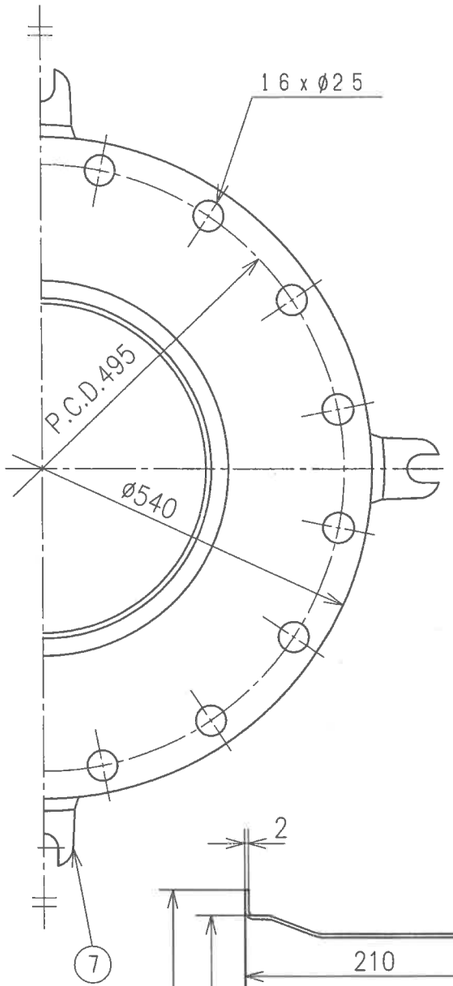
2017/05/09 TECHNICAL DEPARTMENT

BVルール受験品



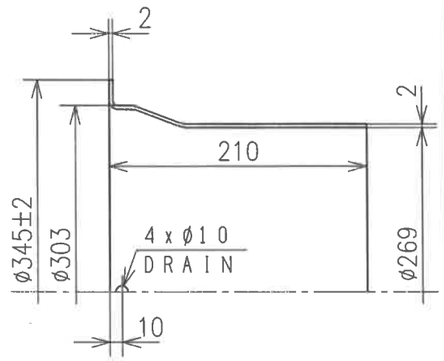
FLOW



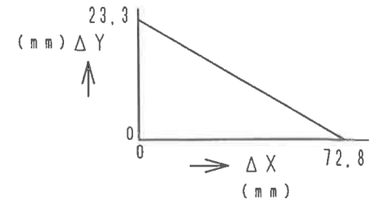


⑥ 4xM16 L=350

出荷時，取付寸法で固定，
取付完了後除去のこと
(TO BE REMOVED AFTER INSTALLTION)



Detail ③
(LINER：内筒)



許容変位線図
(ALLOWABLE MOVE CURVE)

仕様 SPECIFICATIONS		
AXIAL MOVE ΔX (SET LENGTH BASE) 軸方向変位量 ΔX（取付寸法基準）	EXTENSION 伸び	CONTRACTION 縮み
	20 mm	60 mm
LATERAL MOVE ΔY 横変位量 ΔY	偏心 ECCENTRIC	
	10 mm	
WORKING P. 使用圧力	0.003 MPaG	
WORKINFG T. 使用温度	MAX. 550 ℃	
SPRING RATON バネ定数	39.4 N/mm	

用途 APPLICATIONS	機関・排気ガス ENGINE EXHAUST GAS

ITEM 部品番号	PART NAME 部品名	MATERIAL 材質	REQD 数量	WEIGHT 単重	REMARK 備考
10					
9					
8					
7	HOLDER	SS400	各4		
6	SHIPPING BOLTS	SS400	4		
5	FLANGE	SS400	1		JIS5K 相当
4	WELD BAND	SUS304	1		t0.8x20
3	LINER	SUS304	1		
2	BELLOWS	SUS321	1		
1	FLANGE	SS400	1		JIS5K

⚠

注意/Caution

・排気伸縮接手取付の際は、芯ずれ防止、適正な面圏とするため
必ずセットボルトが取り付いた状態で行ってください。
・The installation work of turbocharger outlet expansion joint should be carried out with set bolts in order to avoid mis-aligment and keep the proper set length.

注）本図はNN00273011B シンシュクセッシュ：B-300Ax400Aに対し、「BVルール受験品」のみ異なる。

⚠	(1-B) ADD.CAUTION (E018623)	2015/03/27	Y.Tanaka
⚠	FIRST DRAWING (E009345)	2010/05/12	H.Yamane

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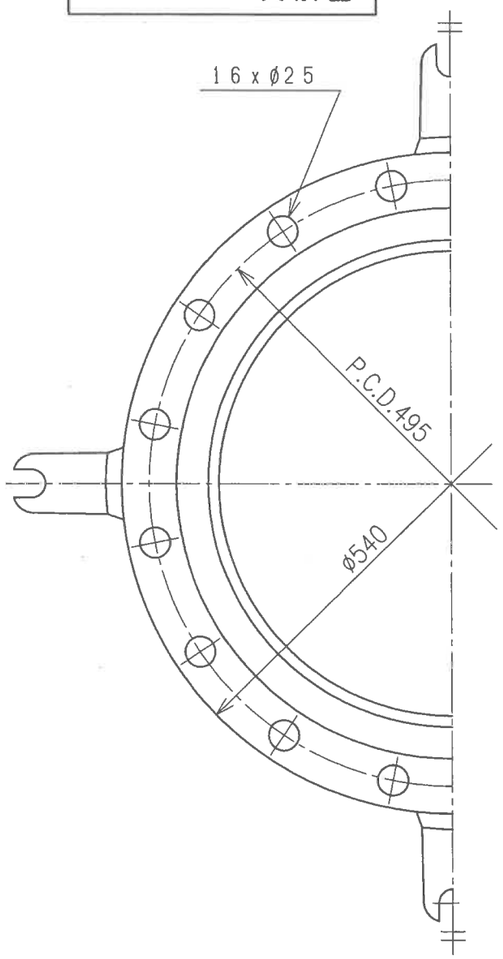
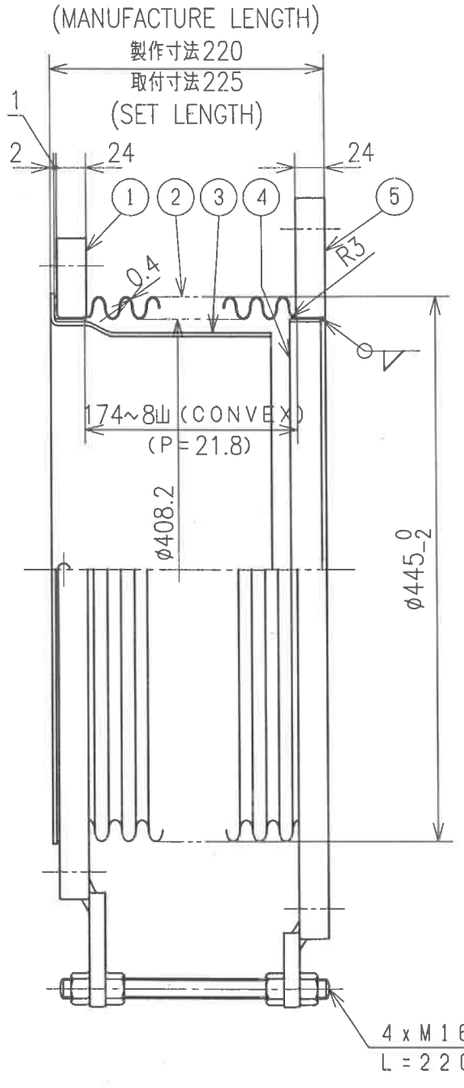
APPROVED	MIYAMOTO ICHIRO	LIST NO. B2599	MATERIAL PCHS	GROUP 7	SCALE:1:5
CHECKED		BEFORE	MASS. 46.8kg (A)	QTY. -	A3
	YAMANE HITOSHI	REF. 日本メタルホース M-16691(B2599)	TYPE ALL		C
	YAMADA KENGO	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME ハイキシンシュクセッシュ::B-300Ax400A-BV		
DRAWN	(H)TANAKA YUKIHO		EXPANSION JOINT::B-300Ax400A-BV		
DATE	2015/03/27		PARTS NO. B259905080Z	A	*

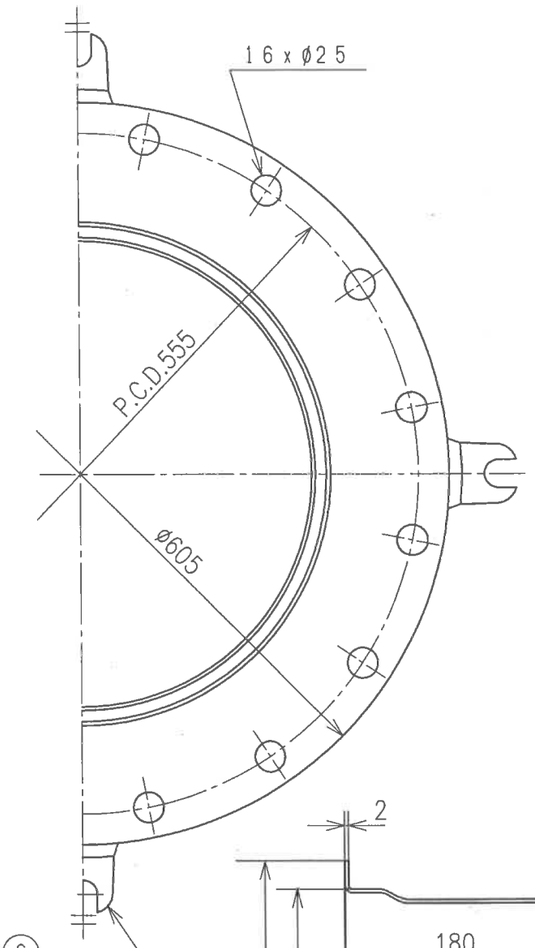
2017/05/09 TECHNICAL DEPARTMENT

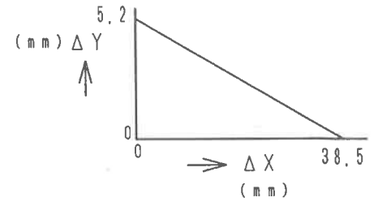
BVルール受験品



FLOW







許容変位線図
(ALLOWABLE MOVE CURVE)

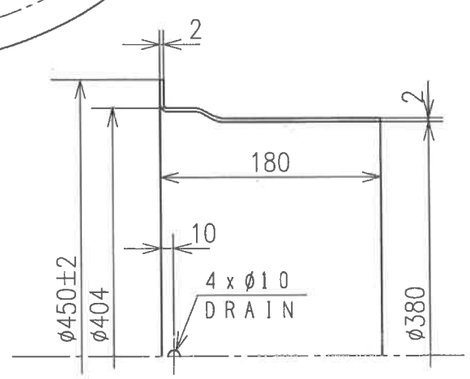
仕様 SPECIFICATIONS		
AXIAL MOVE ΔX (SET LENGTH BASE) 軸方向変位量 ΔX (取付寸法基準)	EXTENSION 伸び	CONTRACTION 縮み
	5 mm	30 mm
LATERAL MOVE ΔY 横変位量 ΔY	偏心 ECCENTRIC	
	5 mm	
WORKING P. 使用圧力	0.003	MPaG
WORKINFG T. 使用温度	MAX.550	℃
SPRING RATON バネ定数	97.9	N/mm

用途 APPLICATIONS	機関・排気ガス ENGINE EXHAUST GAS

ITEM 部品番号	PART NAME 部品名	MATERIAL 材質	REQD 数量	WEIGHT 単重	REMARK 備考
10					
9					
8					
7	HOLDER	SS400	各4		
6	SHIPPING BOLTS	SS400	4		
5	FLANGE	SS400	1		JIS5K 相当
4	WELD BAND	SUS304	1		t0.8x20
3	LINER	SUS304	1		
2	BELLOWS	SUS321	1		
1	FLANGE	SS400	1		JIS5K

4xM16
L=220 ⑥

出荷時，取付寸法で固定，
取付完了後除去のこと
(TO BE REMOVED
AFTER INSTALLTION)



③ DETAIL
(LINER・内筒)

⚠

注意/Caution

・排気伸縮接手取付の際は、芯ずれ防止、適正な面間とするため
必ずセットボルトが取り付いた状態で行ってください。
・The installation work of turbocharger outlet expansion joint
should be carried out with set bolts in order to avoid mis-aligment
and keep the proper set length.

注）本図はB259700210Z シンシュクセッシュ：400Ax450Aに対し、
「BVルール受験品」のみ異なる。

⚠	(1-B) ADD.CAUTION (E018547)	2015/03/11	Y.Tanaka
⚠	FIRST DRAWING (E009345)	2010/05/12	H.Yanase

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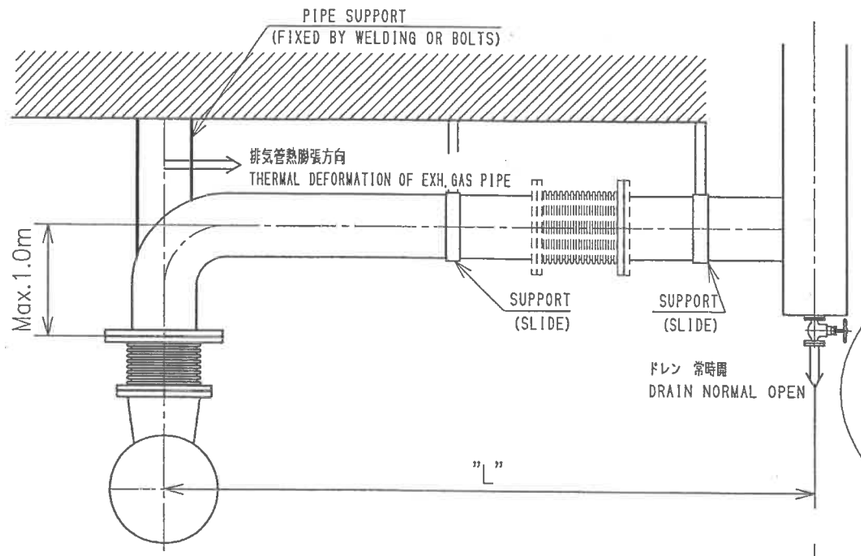
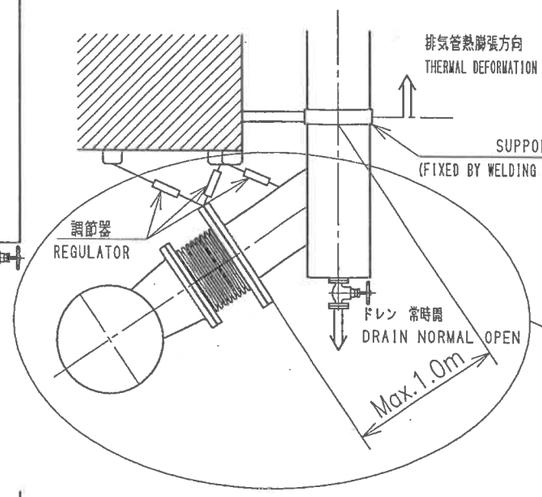
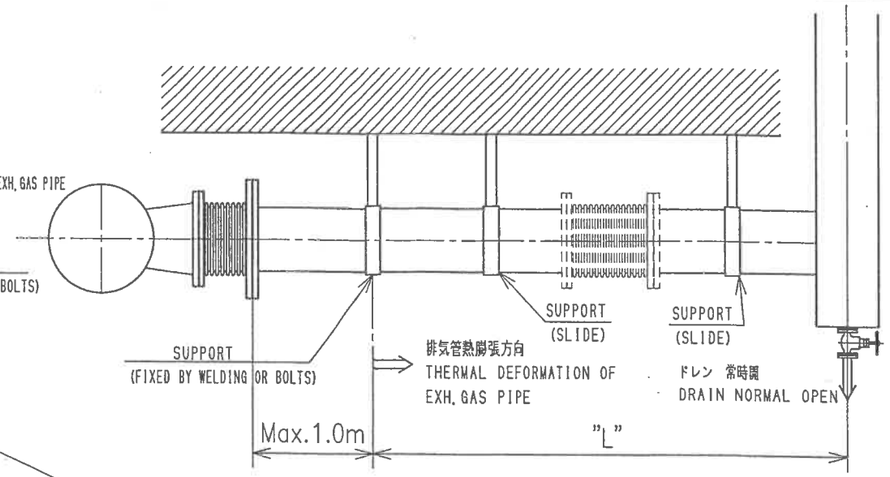
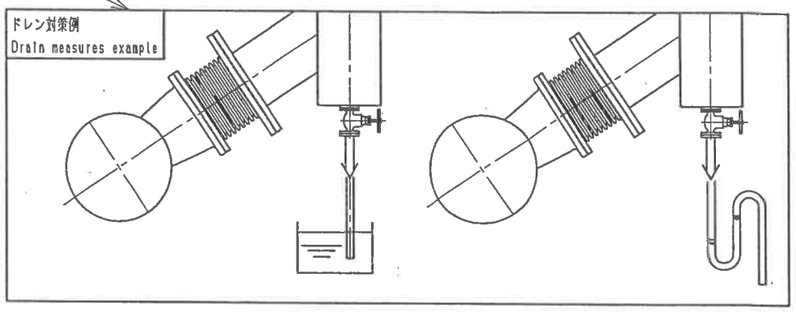
APPROVED	MIYAMOTO ICHIRO	LIST NO.	B2597	MATERIAL PCHS	GROUP 7	SCALE:1:5
CHECKED		BEFORE		MASS. 52.4kg	QTY. -	A3
	YAMANE HITOSHI	REF.	日本メタルネース M-16688	TYPE ALL		C
	YAMADA KENGO	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN		NAME ハイキシンシュクセッシュ::400x450-BV		
DRAWN	(H)TANAKA YUKIHO			EXPANSION JOINT::400x450-BV		
DATE	2015/03/11			PARTS NO. B259705130Z	A	*

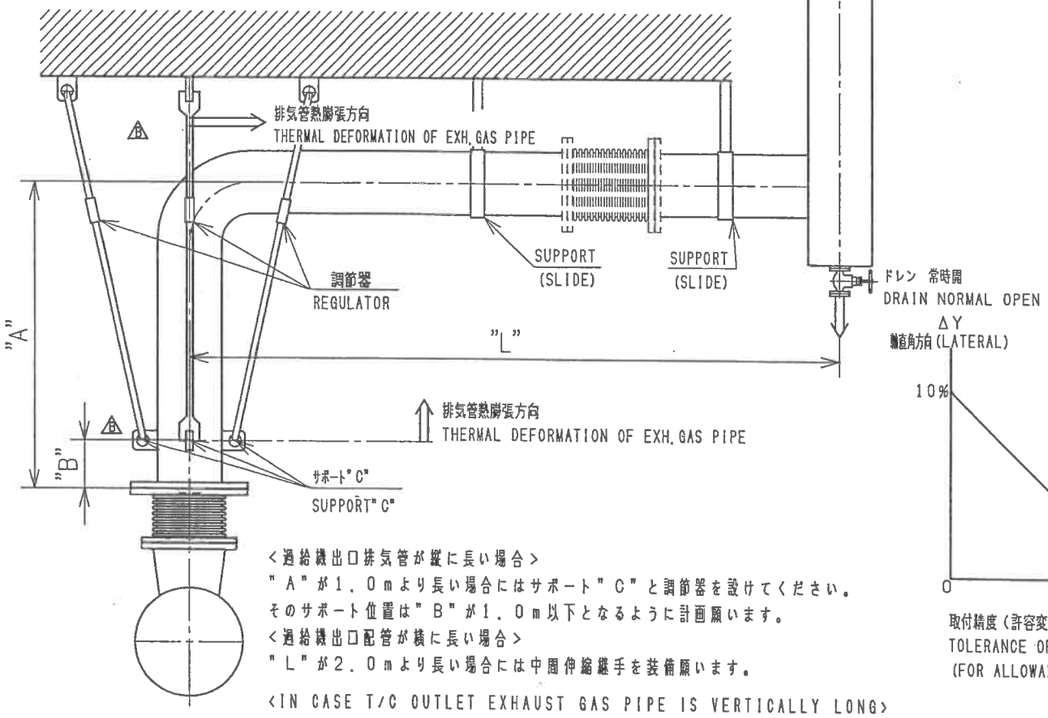
2017/05/09 TECHNICAL DEPARTMENT

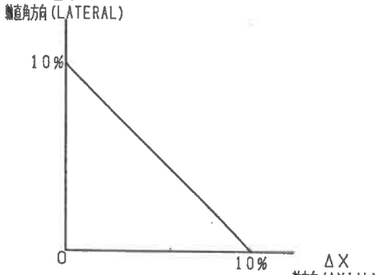












取付精度（許容変位線図に対する）
TOLERANCE OF EXPANSION JOINT FITTING
(FOR ALLOWABLE MOVE CURVE)

<過給機出口排気管が縦に長い場合>
"A"が1．0mより長い場合にはサポート"C"と調節器を設けてください。
そのサポート位置は"B"が1．0m以下となるように計画願います。
<過給機出口配管が横に長い場合>
"L"が2．0mより長い場合には中間伸縮継手を装備願います。

<IN CASE T/C OUTLET EXHAUST GAS PIPE IS VERTICALLY LONG>
IF THE LENGTH 'A' OF EXHAUST PIPE IS LONGER THAN 1.0M, INSTALL THE SUPPORT 'C' WITH REGULATOR.
THE SUPPORT 'C' SHALL BE DESIGNED SO THAT THE LENGTH 'B' IS NOT MORE THAN 1.0M.
<IN CASE T/C OUTLET EXHAUST GAS PIPE IS HORIZONTALLY LONG>
IF THE LENGTH 'L' OF EXHAUST PIPE IS LONGER THAN 2.0M, INSTALL ANOTHER EXPANSION JOINT.

1．過給機出口排気伸縮継手は機関側の熱膨張と振動から過給機を保護するためのものであり、
船体側排気ガス管の歪み，重量及び熱膨張による歪みを吸収するものではありません。
2．船体側排気ガス管の歪み，重量及び熱膨張による歪みが大きいと、過給機や排気伸縮継手が損傷する恐れがあります。
3．船体側排気ガス管は熱膨張があっても機関側に荷重が掛からないよう、排気伸縮継手の周辺で動かないように固定してください。
尚、排気伸縮継手の軸直角方向へ排気ガス管の熱膨張による歪みが生じないように中間伸縮継手を装備する等、考慮願います。
4．過給機排気伸縮継手の取付精度は、伸縮継手許容変位量の１０％以内としてください。詳細は取付許容線図をご参照ください。
伸縮継手許容変位量は承認図面内の伸縮継手図面を参照願います。
5．排気伸縮継手取付の際は、芯ずれ防止，適正な面間とする為、必ずセットボルトを付けた状態で行ってください。

1. T/C OUTLET EXPANSION JOINT IS FOR ABSORPTION OF VIBRATION AND THERMAL DEFORMATION OF D/G, AND NOT FOR THE MIS-ALIGNMENT, WEIGHT AND THERMAL DEFORMATION OF THE EXHAUST PIPE IN THE HULL.
2. T/C AND EXPANSION JOINT MIGHT BE DAMAGED IF THE MIS-ALIGNMENT, WEIGHT AND THERMAL DEFORMATION OF THE EXHAUST PIPE IN THE HULL ARE BIG.
3. EXHAUST GAS PIPE SHOULD BE FIXED IN THE HULL NEAR THE ENGINE SO AS NOT AFFECTED BY THE LOAD CAUSED BY THERMAL DEFORMATION. EXHAUST GAS PIPE SHALL BE DESIGNED SO THAT T/C OUTLET EXPANSION JOINT IS NOT AFFECTED BY LATERAL MOVEMENT CAUSED BY THERMAL DEFORMATION. INSTALLATION OF ANOTHER EXPANSION JOINT IN THE HULL SHALL BE CONSIDERED.
4. THE TOLERANCE OF EXPANSION JOINT FITTING IS 10% OF ΔY AND ΔX OF ALLOWABLE MOVE. THE ALLOWABLE MOVE OF EXPANSION JOINT IS REFERRED TO THE EXPANSION JOINT APPROVAL DRAWING.
5. THE INSTALLATION WORK OF T/C OUTLET EXPANSION JOINT SHOULD BE CARRIED OUT WITH SET BOLTS IN ORDER TO AVOID MIS-ALIGNMENT AND KEEP THE PROPER SET LENGTH.

B	DEL:PIPE SUPPORT REV:SKETCH	2011/02/01	H.S
A	FIRST DRAWING	2010/05/31	H.S

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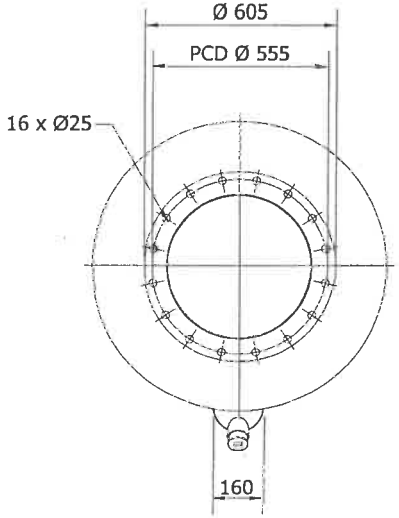
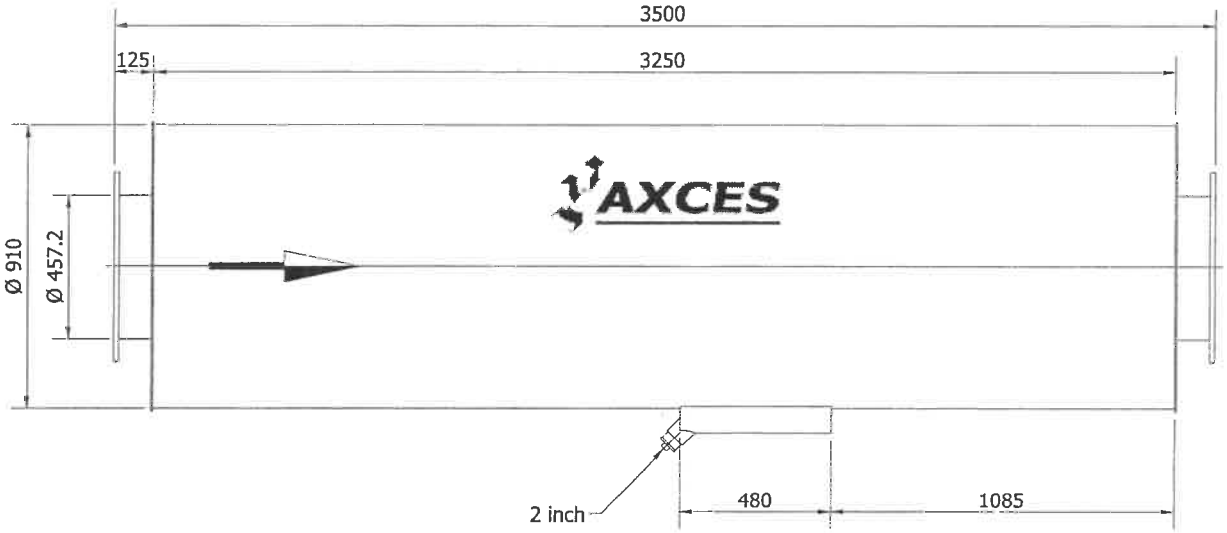
APPROVED	TERAUCHI YOICHI	LIST NO.	MATERIAL APRV	GROUP 0	SCALE:
CHECKED		BEFORE	MASS. A C	QTY.	A3
	KAWANO SHUJI	REF.	TYPE ALL		
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME ヨウリョウズ：ハイキシンシュクツギテトリツケ：		
DRAWN	SUGAI HIRONARI		POINT FIGURE:INSTALLATION OF EXPANSION JOINT:		
DATE	2011/02/01		PARTS NO. AQP10003709	B	*

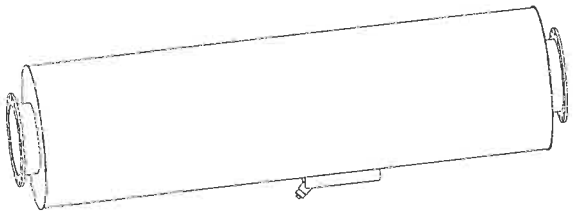
2016/11/14 TECHNICAL DEPARTMENT





SOUND PROGNSIS					
1/1 Octaves	Source {dB at 1m}	Source {dB at 1m}	T.L Silencer dB	Distance correction	Damped specturn (1m) dB(A)
31.5 Hz	109 dB	70 dB (A)	dB	66 dB (A)	66 dB (A)
63 Hz	114 dB	87 dB (A)	dB	72 dB (A)	72 dB (A)
125 Hz	120 dB	104 dB (A)	dB	78 dB (A)	78 dB (A)
250 Hz	114 dB	106 dB (A)	dB	69 dB (A)	69 dB (A)
500 Hz	106 dB	103 dB (A)	dB	42 dB (A)	42 dB (A)
1000 Hz	111 dB	111dB (A)	dB	66 dB (A)	66 dB (A)
2000 Hz	111 dB	112 dB (A)	dB	68 dB (A)	68 dB (A)
4000 Hz	103 dB	104 dB (A)	dB	61 dB (A)	61 dB (A)
Total	123 dB	116 dB (A)		80 dB (A)	80 dB (A)
Toatal damping (± 3dB):				36 dB (A)	36 dB (A)

EXHAUST SILENCER SPECIFICATION	
PROJECT	6000DWT TANKER (SINANJU TANKERS PTE LTD)
ENGINE MODEL	DAIHATSU MAIN ENGINE 6DKM-20e(L)
EXHAUST GAS FLOW RATE	15115 m³/h
EXHAUST GAS TEMP:	324 ° C
QUANTITY	2 PC/ SHIP



MATERIAL: MILD STEEL S235JR
FLANGES: JIS 5K
COATING: SILICONE PAINT - THERMORESISTANT UP TO 600°C, SILVER
WELDING QUALITY ACC. TO EN-ISO 5817 CLASS C
TOLERANCES ACC. TO EN-ISO 13920 CLASS B,F

DIMENSIONS	LENGTH	ANGLE	STRAIGHTNESS FLATNESS PARALLELISM
400 mm	+/-2 mm	+/-45'	+/-1,5 mm
1000 mm	+/-3 mm	+/-30'	+/-3 mm
2000 mm	+/-4 mm	+/-20'	+/-4,5 mm
4000 mm	+/-6 mm	+/-20'	+/-6 mm
>4000 mm	+/-8 mm	+/-20'	+/-8 mm

MOD	DESCRIPTION	DRAWN	DATE
B	---		
A	---		

AXCES

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The Netherlands
www.axces.com
+31 (0) 166 603 285

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TITLE	EXHAUST SILENCER
TYPE	REAB35-IS-18" (NB450)
PROJECTION	
DRAWING NUMBER	14091518A
SPARK ARRESTOR	CP35-18"(NB450)
WEIGHT	441 Kg
NUMBER OF SHEETS	1
DATE	2016-02-5
ARTICLE NUMBER	141200000180
DRAWN	ES
DAMPING	35 dB(A)
SCALE	1 / 16
FORMAT	A3

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FILTER:MERCURY:

MCY-2000MS

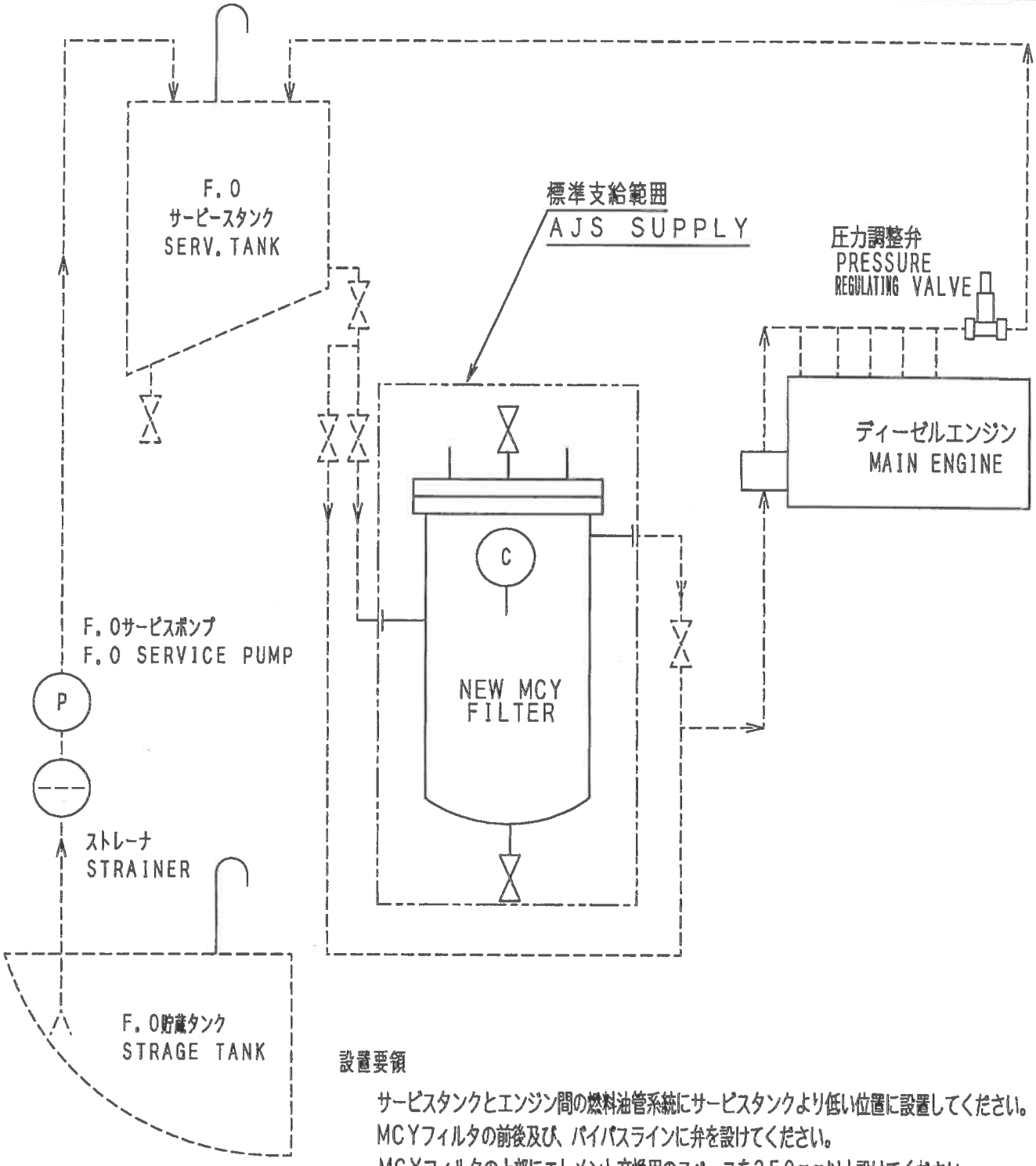
DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

JAPAN

SHIP NO.	—	DRAWN BY	K. Aoki
TYPE	ALL	CHECKED BY	K. Tajima, H. Kawauchi
LIST NO.	AQP20007600A		s. K M. Tokuoka
DATE	Apr.,15,2015	APPROVED BY	Y. Tachi
REVISION			

PDK26-0373-4/0375-6



設置要領

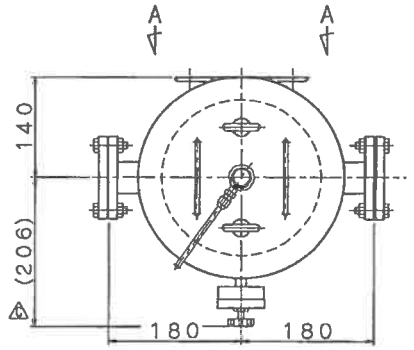
サービスタンクとエンジン間の燃料油管系統にサービスタンクより低い位置に設置してください。
MCYフィルタの前後及び、バイパスラインに弁を設けてください。
MCYフィルタの上部にエレメント交換用のスペースを３５０ｍｍ以上設けてください。

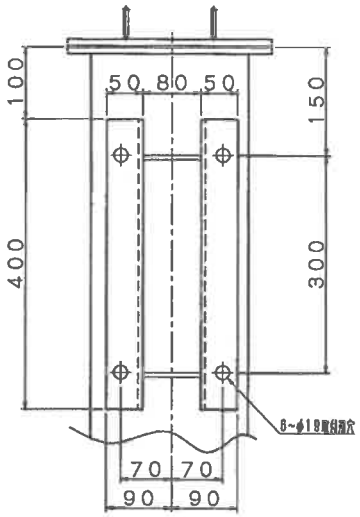
CAUTION ON INSTALLATION

TO INSTALL THE MCY FILTER BETWEEN A SERVICE TANK AND ENGINE.
IT IS RECOMMENDED THAT THE POSITION OF THE MCY FILTER IS BELOW THAN THE SERVICE TANK.
TO INSTALL A VALVE IN BEFORE THE MCY FILTER AND AFTER.
TO INSTALL A VALVE IN BYPASSLINE OF MCY FILTER.
TO PREPARE A SPACE OF 350mm AND MORE FOR ELEMENT REPLACEMENT.

MCY2000MS

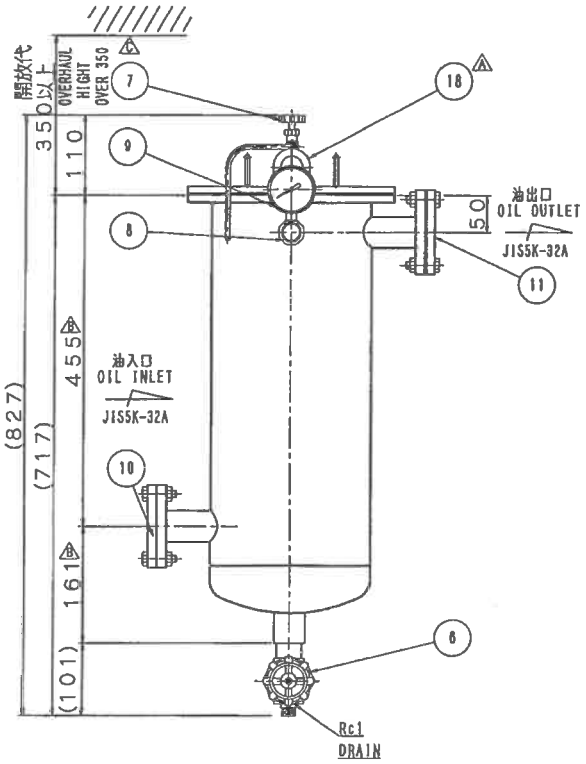
検図	設計	製図	MCYフィルタ配管要領 NEW MCY FILTER PIPING DIAGRAM	R度
		SEKI		
AMEROID		図面番号 5BK0002A		

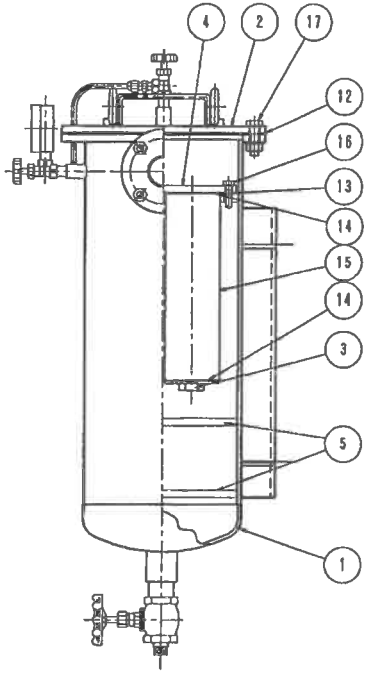




VEAW A-A

本体重量 WEIGHT OF FILTER	55 Kg
内油量 QUANTITY OIL CONTAIND	25 L
使用圧力(最大) WORKING OIL PRESS.	0.5MPa
テスト圧力 TEST PRESS.	0.75MPa
使用温度(最大) WORKING OIL TEMP.	60 ℃
通油量(最大) FLOW RATE (MAX)	800L/h
初期圧損(最大) PRESS. DROP	0.01MPa
ろ過精度 FILTERABILTY	10 μ





相フランジ付
WITH FLANGE

⚠	No.	PARTS NAME	MATERIAL	Q'TY	REMARKS
⚠	18	つり上げナット RIFTING EYE NUT	SS400	2	M12
	17	ボルト, ナット BOLT, NUT	SS400	12	M12
	16	ボルト BOLT	SS400	8	M8
	15	エレメント ELEMENT	SUS304	3	MS 250xφ75
	14	エレメント パッキン ELEMENT PACKING	NBR	6	φ30xφ70 t=2
	13	シールパッキン SEAL PACKING	NBR	1	φ185xφ200 t=2
	12	カバーパッキン COVER PACKING	NBR	1	φ218xφ275 t=2
	11	油出口フランジ OIL OUTLET FLANGE	SS400	1	JIS5K-32A
	10	油入口フランジ OIL INLET FLANGE	SS400	1	JIS5K-32A
	9	連成計 COMPOUND GAUGE	BC6	1	1/4x60x -0.1MPa~0.1MPa
	8	連成計元弁 COMPOUND GAUGE VALVE	C3604B	1	8A
	7	エア抜キ弁 AIR VENT VALVE	C3604B	1	8A
	6	ドレン弁 DRAIN VALVE	BC6	1	25A 青銅製仕様グローブバルブ
	5	じゃま板 BAFFLE PLATE	SS400	4	
	4	エレメント押さえ ELEMENT COVER	SS400	1	
	3	エレメントシャフト ELEMENT SHAFT	SGP SS400	3	20A
	2	本体カバー COVER	SS400	1	図番 2B000520
	1	本体 BODY	SGP SS400	1	200A/504mm 2BK00340

⚠			
⚠	表記統一	Apr-15-'15	梅田
⚠	入口配管位置変更	Feb-18-'15	竹内
⚠	(18)つり上げナット追加	Feb-18-'15	吉本
変更	内 容	変更日	変更者

変更履歴

検図	設計	製図	マーキュリフィルタ	尺度
Feb.04.'15	Feb.04.'15	Feb.04.'15	NEW-MCY2000MS	1/5
竹内	竹内	梅田	組立図	
AMEROID			図面番号 1BK0623C-R	

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FILTER:MERCURY:

MCY-2000MS-L

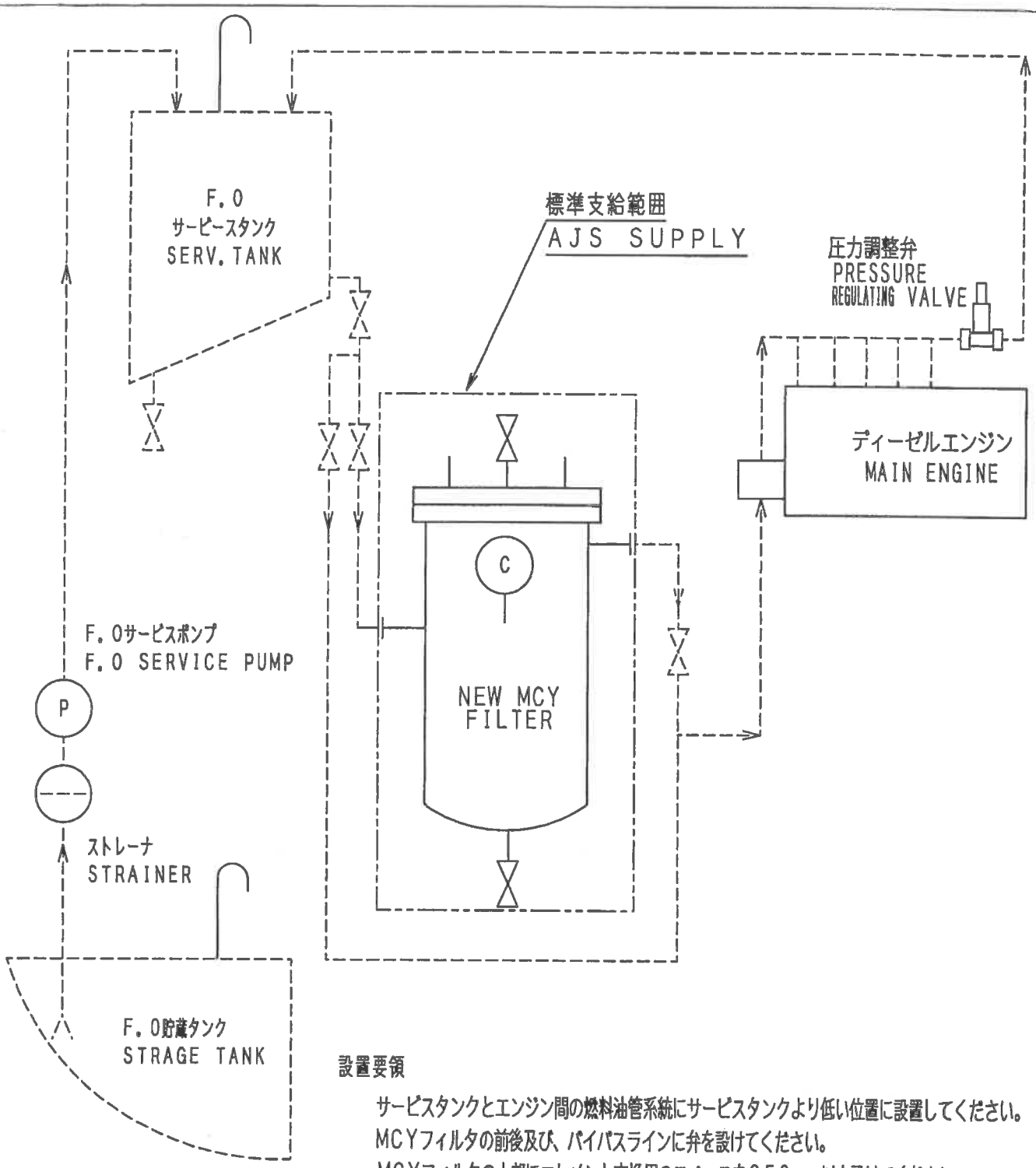
DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

JAPAN

SHIP NO.	—	DRAWN BY	K. Aoki
TYPE	ALL	CHECKED BY	S. Tsuzuki, H. Kawauchi
LIST NO.	AQP20007601A		S. Z M. Johmoto
DATE	Apr.,15,2015	APPROVED BY	Y. Tsuchi
REVISION			

PDK26-0373-4/0375-6



設置要領

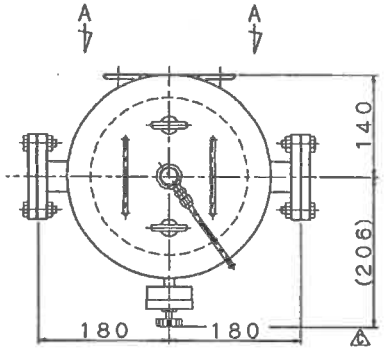
サービスタンクとエンジン間の燃料油管系統にサービスタンクより低い位置に設置してください。
MCYフィルタの前後及び、バイパスラインに弁を設けてください。
MCYフィルタの上部にエレメント交換用のスペースを３５０ｍｍ以上設けてください。

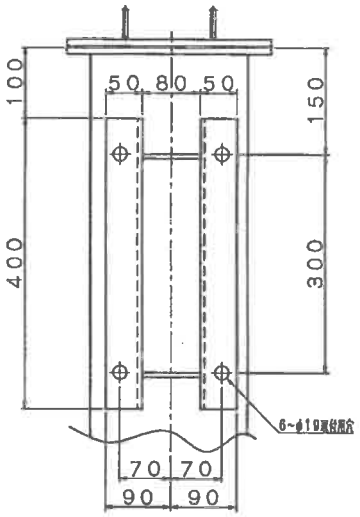
CAUTION ON INSTALLATION

TO INSTALL THE MCY FILTER BETWEEN A SERVICE TANK AND ENGINE.
IT IS RECOMMENDED THAT THE POSITION OF THE MCY FILTER IS BELOW THAN THE SERVICE TANK.
TO INSTALL A VALVE IN BEFORE THE MCY FILTER AND AFTER.
TO INSTALL A VALVE IN BYPASSLINE OF MCY FILTER.
TO PREPARE A SPACE OF 350mm AND MORE FOR ELEMENT REPLACEMENT.

MCY2000MS

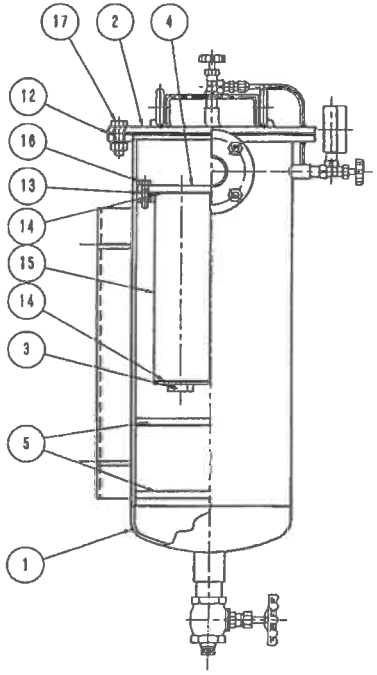
検図	設計	製図	MCYフィルタ配管要領 NEW MCY FILTER PIPING DIAGRAM	R度
		SEKI		
AMEROID		図面番号	5BK0002A	

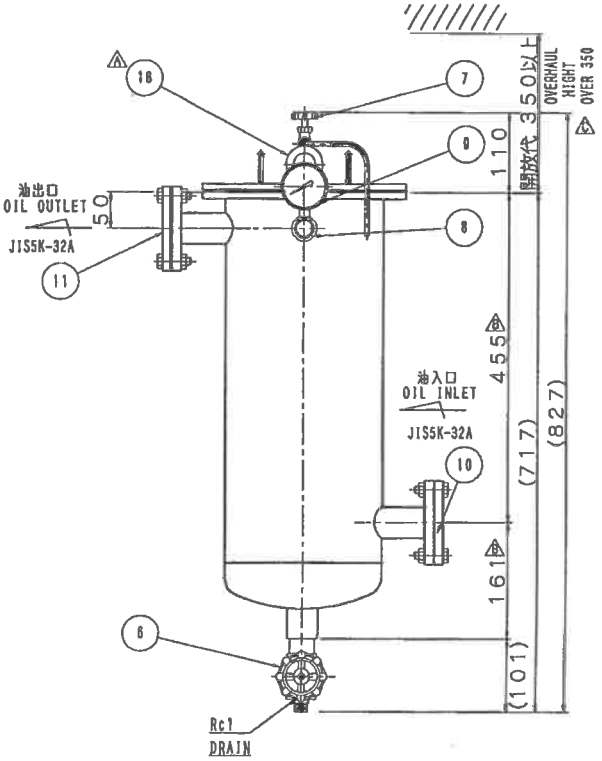




VEAW A-A

本体重量 WEIGHT OF FILTER	55 Kg
油内油量 QUANTITY OIL CONTAIND	25 L
使用圧力(最大) WORKING OIL PRESS.	0.5MPa
テスト圧力 TEST PRESS.	0.75MPa
使用温度(最大) WORKING OIL TEMP.	60 ℃
通油量(最大) FLOW RATE (MAX)	800L/h
初期圧損(最大) PRESS. DROP	0.01MPa
ろ過精度 FILTERABILTY	10 μ





No.	PARTS NAME	MATERIAL	Q'TY	REMARKS
⚠ 18	つり上げナット RIFTING EYE NUT	SS400	2	M12
17	ボルト, ナット BOLT, NUT	SS400	12	M12
16	ボルト BOLT	SS400	8	M8
15	エレメント ELEMENT	SUS304	3	MS 250xφ75
14	エレメント パッキン ELEMENT PACKING	NBR	6	φ30xφ70 t=2
13	シールパッキン SEAL PACKING	NBR	1	φ185xφ200 t=2
12	カバーパッキン COVER PACKING	NBR	1	φ218xφ275 t=2
11	油出口フランジ OIL OUTLET FLANGE	SS400	1	JIS5K-32A
10	油入口フランジ OIL INLET FLANGE	SS400	1	JIS5K-32A
9	連成計 COMPOUND GAUGE	BC6	1	1/4x80x -0.1MPa-0.1MPa
8	連成計元弁 COMPOUND GAUGE VALVE	C3604B	1	8A
7	エア抜キ弁 AIR VENT VALVE	C3604B	1	8A
6	ドレン弁 DRAIN VALVE	BC6	1	25A 1/4型青銅製グローブバルブ
5	じゃま板 BAFFLE PLATE	SS400	4	
4	エレメント押さえ ELEMENT COVER	SS400	1	
3	エレメントシャフト ELEMENT SHAFT	SGP SS400	3	20A
2	本体カバー COVER	SS400	1	図番 2B000520
1	本体 BODY	SGP SS400	1	200A/504mm 2BK00340

⚠			
⚠	表記統一	Apr-15-'15	梅田
⚠	入口配管位置変更	Feb-18-'15	竹内
⚠	⑱つり上げナット追加	Feb-18-'15	吉本
変更	内容	変更日	変更者

変更履歴

相フランジ付
WITH FLANGE

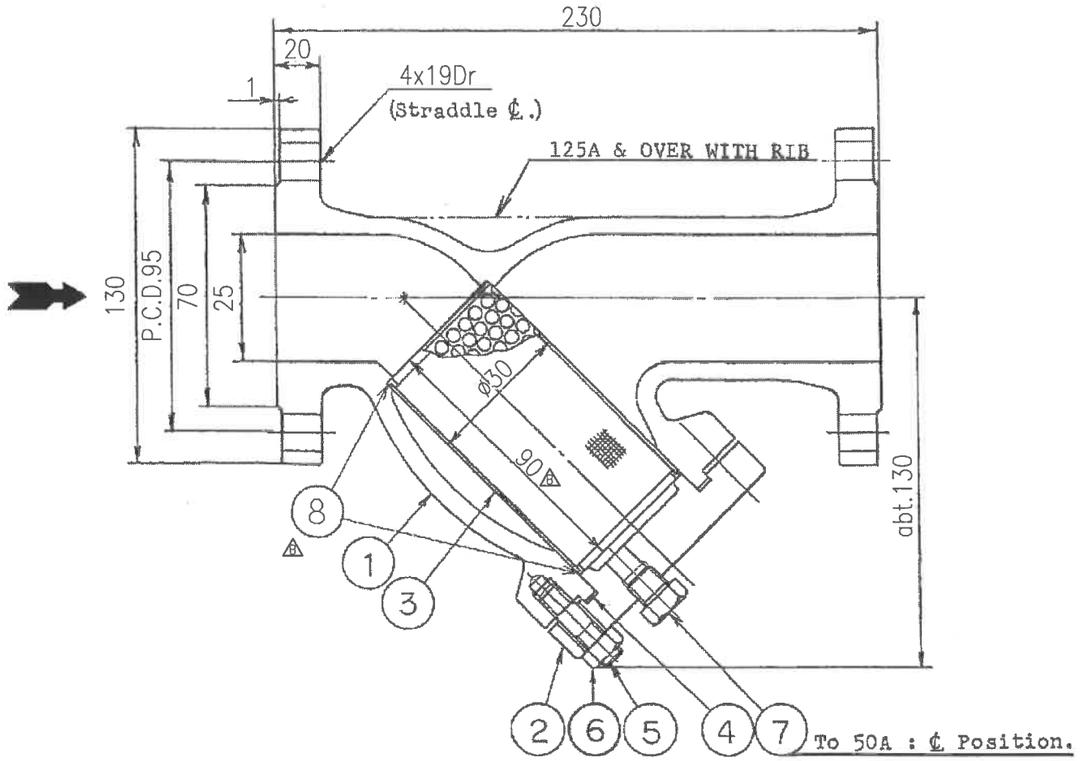
検図	設計	製図
Feb. 04. '15	Feb. 04. '15	Feb. 04. '15
竹内	竹内	梅田

マーキュリフィルタ
NEW-MCY2000MS-L
組立図

尺度 1/5

AMEROID

図面番号 1BK0623C-L



⑦ PLUG	⑤ BOLTS	④ GASKET
R3/8	M16x4	ø43xø31xt3.2

フランジ規格	FLANGE RATING	JIS 30K-RF
設計圧力	DESIGN PRESSURE	3.0 MPaG
設計温度	DESIGN TEMP.	200 °C
水圧試験	TEST PRESS. (WATER)	4.5 MPaG
気密試験	TEST PRESS. (AIR)	—
スクリーン	SCREEN	MESH WIRECLOTH & REINFORCEMENT PUNCHING PLATE

スクリーン補強筒（パンチングプレート）の
耐差圧強度：0.1MPa
Allowable differential pressure of
punching plate reinforcing screen: 0.1MPa

PART NO	PART NAME	MATERIAL	REQD NO	REMARKS
8	リング Rings	TEFLON	2	
7	ドレンプラグ Drain Plug	SUS304	1	15-50A(CENTER)
6	ナット Nuts	SUS304		
5	ボルト Bolts	SUS304		
4	ガスケット Gasket	T# 1804-NA	1	
3	スクリーン Screen	SUS316/SUS304	1	Mesh 300
2	カバー Cover	SUS304	1	
1	ボデー Body	SCS13A	1	

⚠x3	REV:POSITION OF A DIMENSION(3-C) ADD:RINGS(2-C), REMARK OF RING(5-B)	2016/03/24	W. N
⚠	FIRST DRAWING	2015/07/31	E. M

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APPROVED	KAWANO SHUJI	LIST NO.	MATERIAL SPEC	GROUP 7	SCALE: —
CHECKED		BEFORE	MASS	QTY.	A3
	TOKUMOTO MANABU	REF.	TYPE DK-20e		LEVEL
	SHIMOMURA YUYA	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME ストレーナー:SA:30K 25A		
DRAWN	NINOMIYA WATARU		STRAINER:SA:30K 25A		
DATE	2016/03/23		PARTS NO. AQA20012139	B	*

2017/05/09 TECHNICAL DEPARTMENT

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AIR RESERVOIR

200L X 2.94MPa

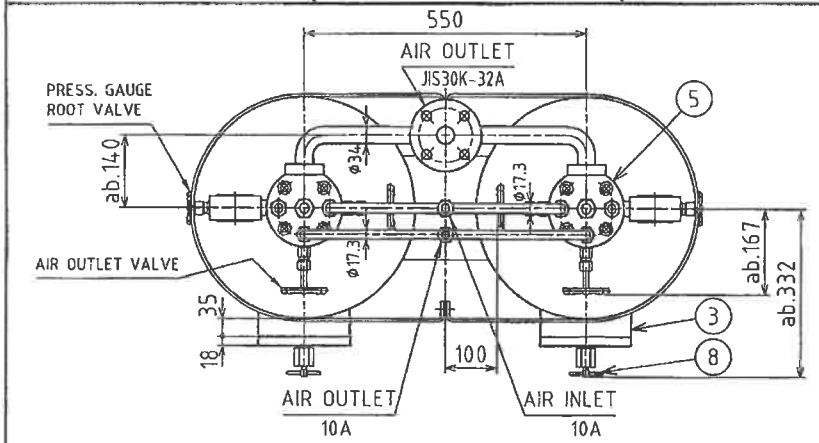
DAIHATSU DIESEL MFG.CO.,LTD.

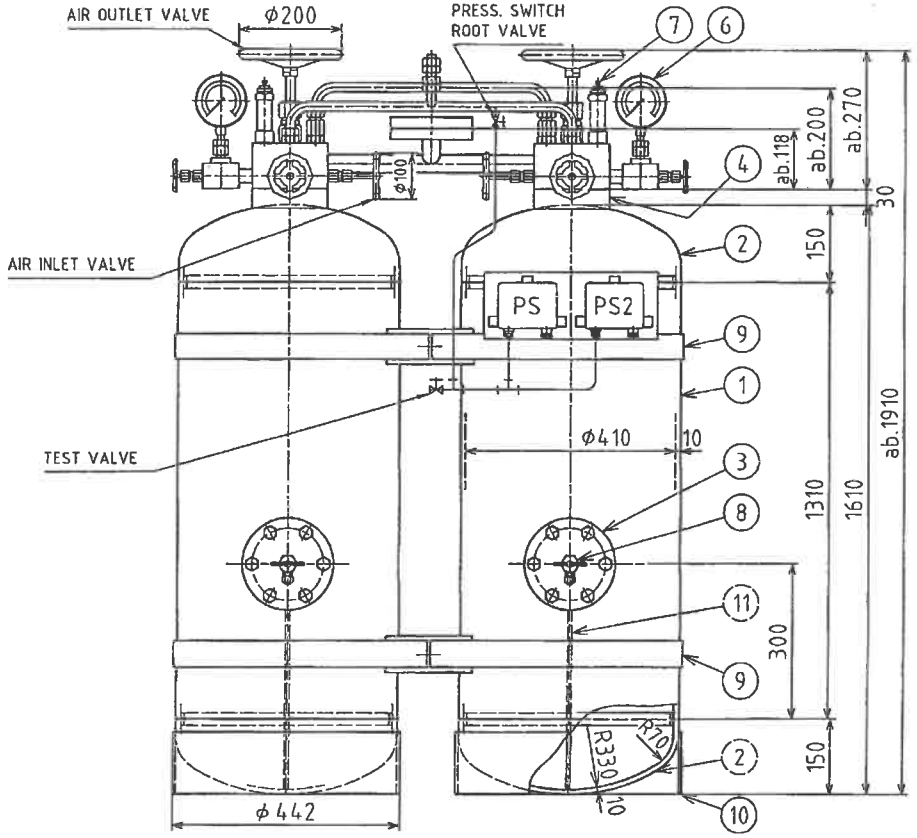
TECHNICAL DEPARTMENT

JAPAN

-	-	DRAWN BY	M. Johmoto
TYPE	6DKM-20eF(L)	CHECKED BY	S. Tajimoto
LIST NO.	AQP20008429A		
DATE	May.,31,2016	APPROVED BY	S. Tsuru
REVISION			

PDK20-0174-5





PARTICULAR	
RULE	BV
CAPACITY	200 LITERS
WORKING PRESS.	2.94 MPa
DESIGN PRESS.	3.14 MPa
HYD. TEST PRESS.	4.71 MPa

PRESS. SWITCH SET PRESS :
PS : 63AS
FOR ALARM
1.5MPa below (① - ③) OFF

PS2 : 63CA
FOR COMPRESSOR START & STOP
1.8MPa below (① - ⑤) ON
2.94MPa above (① - ⑤) OFF

QUANTITY PER TWO SETS GIVEN IN THIS TABLE.

NO.	NAME OF PARTS	REMARKS	MATERIAL	MASS	QTY.
11	DRAIN INNER PIPE	φ8×t1	CUPPER TUBE		2
10	STAND		SM400		2
9	BAND		SM400		2
8	DRAIN VALVE		S25C		2
7	SAFETY VALVE	Set Press.3.14MPa			2
6	PRESSURE GAUGE	AVS1/2×φ100×6MPa			2
5	HEADER VALVE		S25C		2
4	HEADER VALVE SEAT		SF440		2
3	INSPECTION HOLE		SF440		2
2	END PLATE		SB410		4
1	SHELL PLATE		SB410		2

APPROVED		LIST NO.	MATERIAL	GROUP	SCALE= 1:10
		BEFORE	MASS. 510 kg	QTY.	A3
CHECKED		REF. ヘンミ H-37485	TYPE		
		DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME	AIR RESERVOIR 200 L × 2.94 MPa	
DRAWN			PARTS NO.		
DATE	2016/06/03		DWG. NO.		

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AQP20008429A (2/2)

2017/05/09 TECHNICAL DEPARTMENT

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L.O. STBY PUMP

TYPE:ALGT-65

DAIHATSU DIESEL MFG.CO.,LTD.

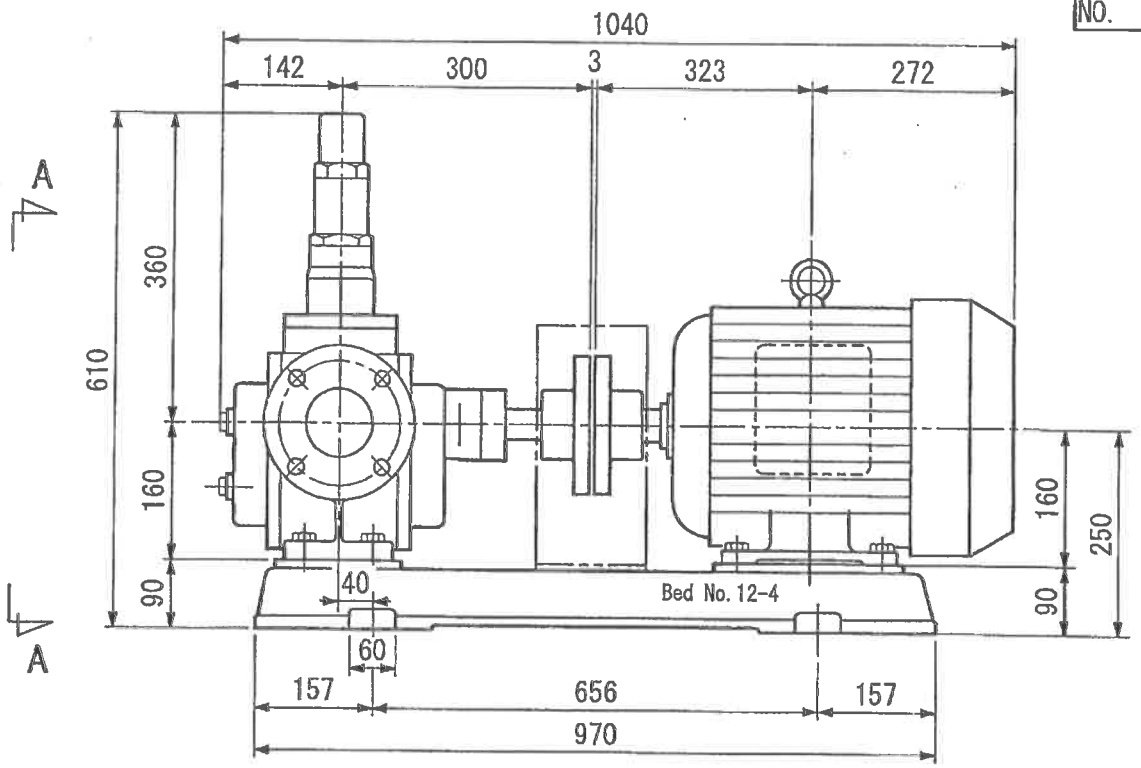
TECHNICAL DEPARTMENT

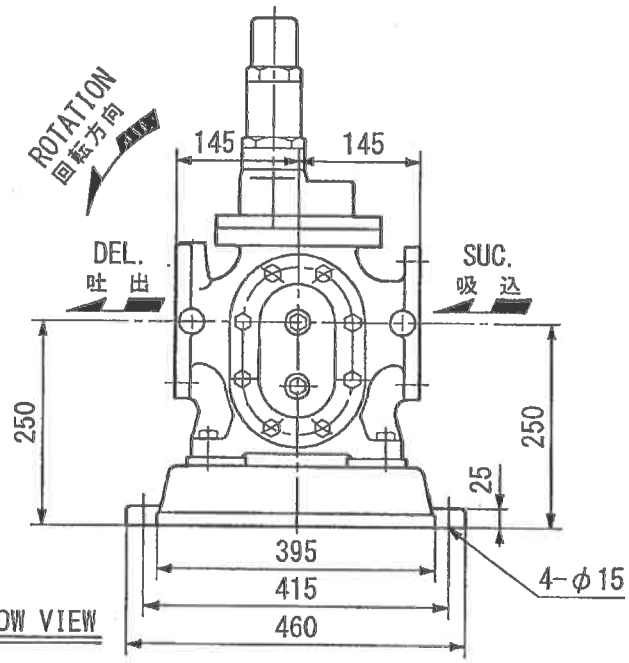
JAPAN

SHIP NO.		DRAWN BY	M. Tokumoto
TYPE	6DKM-20eF(L)	CHECKED BY	A. Tajimoto
LIST NO.	AQP10008526B		
DATE	2016/6/3	APPROVED BY	S. Tsuru
REVISION	B:REV. PUMP OUTLINE		

PDK20-0174-5

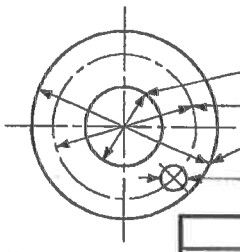
NO. 1





A-A ARROW VIEW

SUC. & DEL. FLANGE
吸込及吐出フランジ



JIS 10K SUC. 吸 込	JIS 10K DEL. 吐 出
φ 65	φ 65
φ140	φ140
φ175	φ175
4-φ19	4-φ19

MASS 質 量	
PUMP ポンプ	109 kg
MOTOR 電動機	75 kg
TOTAL 全質量	184 kg

ACCESSORIES FOR EACH 1 PUMP 付属品（ポンプ一台ニ付）	
DESCRIPTION 記	Q'TY 数 量
COUPLING JISφ160 カップリング	1set
COUPLING COVER カップリングカバー	1
COMMON BED 共通台	1
SAFETY VALVE (SET PRESSURE 0.7 MPa) 安全弁	1set
AIR COCK エアーコック	1
PRESSURE GAUGE 圧力計	1
COMPOUND GAUGE 連成計	1
GAUGE ROOT COCK 計器元コック	2
COUNTER FLANGE SUC. 65A DEL. 65A 相フランジ	1set

SPECIFICATION 仕 様	
SUCTION BORE 吸込口径	65 mm
DELIVERY BORE 吐出口径	65 mm
CAPACITY 容量	20.5 m^3/h
TOTAL PRESSURE 全圧力	0.6 MPa
SUCTION PRESSURE 吸込圧力	−0.05 MPa
REVOLVING SPEED 回転速度	950 min^{-1}
HYD. TEST PRESS. 耐圧試験	1.2 MPa
MOTOR OUTPUT 電動機出力	7.5kW , 6P
POWER SOURCE 電源	415V, 50Hz, 3ph
GUARANT. VIS. FOR POWER 500 mm^2/s GUARANT. VIS. FOR CAPA. 26 mm^2/s OIL TEMP = ℃	

PUMP NAME 名 称	L. O. AUX. PUMP
PUMP MODEL 型 式	ALGT-65
QUANTITY 台 数	1
RULE 検 査	BV
ANGLE PROJ. 角 法	3 rd
SCALE 尺 度	FREE
DRAWING NO. 図面番号	B6DG0340

NANIWA PUMP MFG.CO.,LTD.
TECH. DEPARTMENT DESIGN SECTION
株式会社浪速ポンプ製作所

naniwa pump

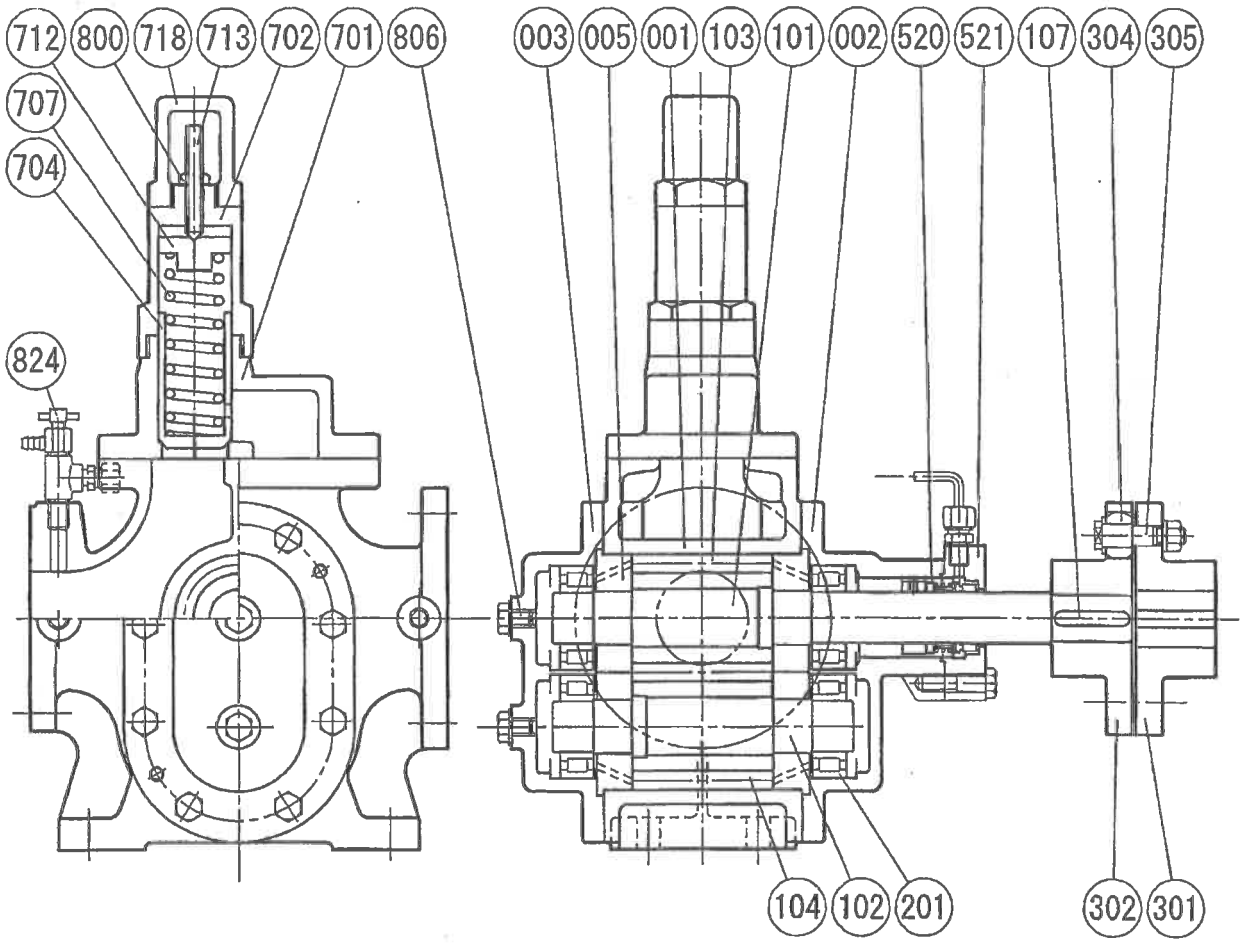
SECTIONAL DRAWING 構造断面図

HORIZONTAL GEAR ROTARY PUMP 横型電動歯車式

MODEL:型 式 ALGT-65

DRAWING NO. A6DS0200

DATE 10 AUG. 2006

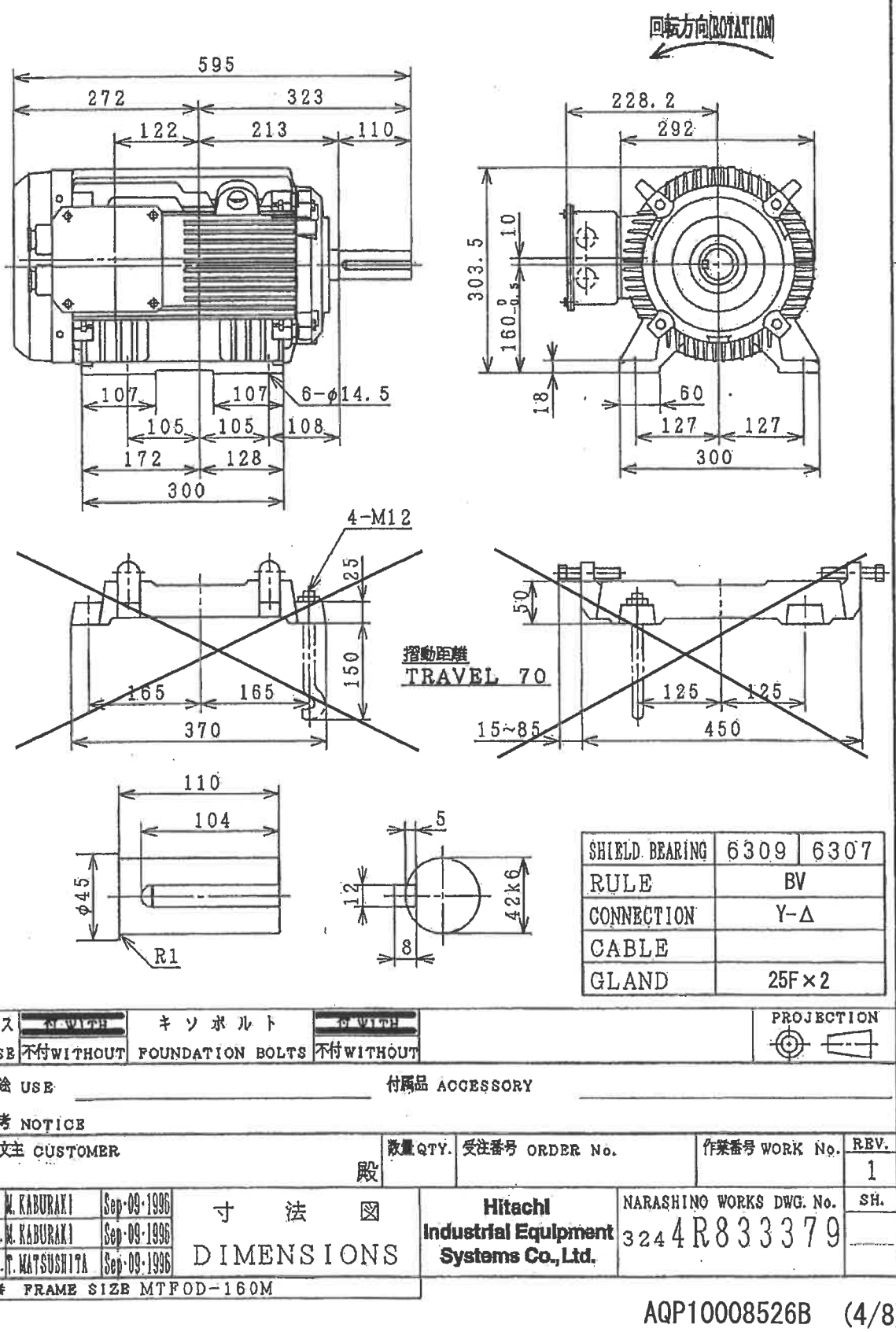


PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質		Q'TY 数量	PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質		Q'TY 数量
304	COUPLING RING 軸継手リング	RUBBER 合成ゴム		1 SET	824	AIR COCK 空気抜キコック	COPPER ALLOY ROD 銅合金棒	C3604	1
302	COUPLING 軸継手	CAST IRON 鋳鉄	FC200	1	806	BOLT PLUG ボルトプラグ	LOW CARBON STEEL WIRE 軟鋼線材	SWRM	2
301	COUPLING 軸継手	CAST IRON 鋳鉄	FC200	1	800	ADJUST NUT 調整ナット	MILD STEEL 一般構造用圧延鋼	SS400	1
201	ROLLER BEARING コロ軸受	SUJ 高炭素クロム軸受鋼		4	718	CAP キャップ	CAST IRON 鋳鉄	FC200	1
107	COUPLING KEY 軸継手キー	CARBON STEEL 機械構造用炭素鋼	S45C	1	713	ADJUST BOLT 調整ボルト	MILD STEEL 一般構造用圧延鋼	SS400	1
104	DRIVEN GEAR 遊歯車	CARBON STEEL 機械構造用炭素鋼	S45C	1	712	SPRING SEAT バネ座	MILD STEEL 一般構造用圧延鋼	SS400	1
103	DRIVE GEAR 主歯車	CARBON STEEL 機械構造用炭素鋼	S45C	1	707	VALVE SPRING 安全弁バネ	PIANO WIRE ピアノ線	SWP	1
102	DRIVEN SHAFT 遊軸	CARBON STEEL 機械構造用炭素鋼	S45C	1	704	SAFETY VALVE 安全弁	CARBON STEEL 機械構造用炭素鋼	S45C	1
101	DRIVE SHAFT 主軸	CARBON STEEL 機械構造用炭素鋼	S45C	1	702	VALVE COVER 安全弁カバー	CAST IRON 鋳鉄	FC200	1
005	SIDE PLATE サイドプレート	CAST IRON 鋳鉄	FC200	4	701	SAFETY VALVE BOX 安全弁箱	CAST IRON 鋳鉄	FC200	1
003	SIDE COVER サイドカバー	CAST IRON 鋳鉄	FC200	1	521	SEAL COVER シールカバー	CAST IRON 鋳鉄	FC200	1
002	SIDE COVER サイドカバー	CAST IRON 鋳鉄	FC200	1	520	MECHANICAL SEAL メカニカルシール			1 SET
001	CASING ケーシング	CAST IRON 鋳鉄	FC200	1	305	COUPRING BOLT & NUT 軸継手ボルト及ナット	MILD STEEL 一般構造用圧延鋼	SS400	1 SET

CAD

三相誘導電動機
THREE-PHASE INDUCTION MOTOR

出力 OUTPUT (kW)	形式 TYPE FORM	定格 RATING	電圧 VOLTAGE (V)	周波数 FREQ. (Hz)	同期回転速度 SYNC. SPEED (min⁻¹)	極数 POLES	耐熱クラス TH. CLASS	質量 MASS (kg)
7.5	TFO	KK	S1	415	50	1000	6	F	83



SHIELD BEARING	6309	6307
RULE	BV	
CONNECTION	Y-Δ	
CABLE		
GLAND	25F×2	

ベース BASE	有 WITH / 付WITHOUT	キソボルト FOUNDATION BOLTS	有 WITH / 不付WITHOUT	PROJECTION

用途 USE ＿＿＿＿＿＿＿＿ 付属品 ACCESSORY ＿＿＿＿＿＿＿＿

備考 NOTICE

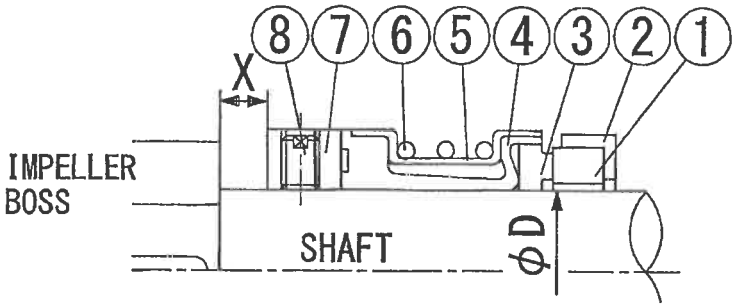
御注文主 CUSTOMER 殿	数量 QTY.	受注番号 ORDER No.	作業番号 WORK No.	REV.
				1

REGD 2002.04.01	DWN. M. KABURAKI	Sep-09-1996	寸法図 DIMENSIONS	Hitachi Industrial Equipment Systems Co., Ltd.	NARASHINO WORKS DWG. No. 3244R833379 SH.
	CHKD. M. KABURAKI	Sep-09-1996			
	APPD. T. MATSUSHITA	Sep-09-1996			

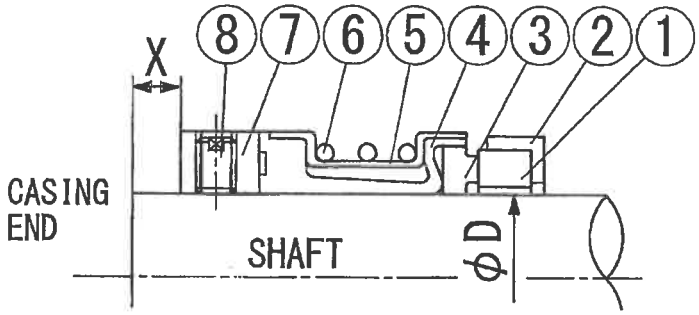
枠番 FRAME SIZE MTFOD-160M

AQP10008526B (4/8)

nariwa pump	SECTIONAL DRAWING MECHANICAL SEAL	DRAWING NO. 98DG0226 DATE 18 MAY 1998



FIG, 1



FIG, 2

NO.	PUMP NAME	MODEL	SHAFT DIA φ	SET DIMENSION "X"	FIG, NO.
1	L.O. STBY PUMP	ALGT-65	35	69	2
2	R/G L.O. STBY PUMP	ALGT-50H	30	58	2
3	COOL. F.W. PUMP	CRD-100-2	45	36	1

PART NO.	NAME OF PART	Q'TY
1	FLOATING SEAT	1
2	SEAT CAP	1
3	SEAL RING	1
4	BELLOWS	1
5	RETAINER	1
6	COIL SPRING	1
7	COLLAR	1
8	SET SCREW	1SET

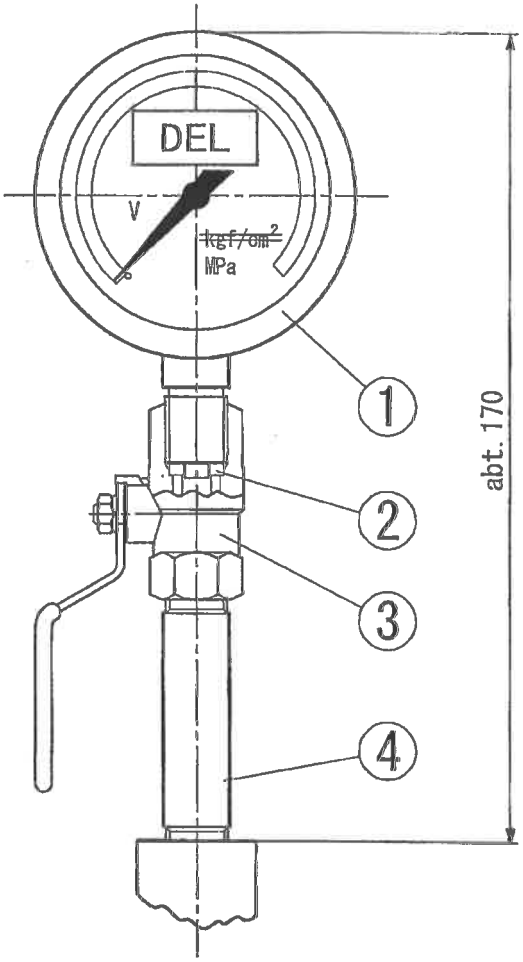
naniwa pump

GAUGE & GAUGE ROOT COCK
計器 及 計器元コック
HORIZONTAL PUMP TYPE
FITTING ON PUMP BODY 横型本体付タイプ

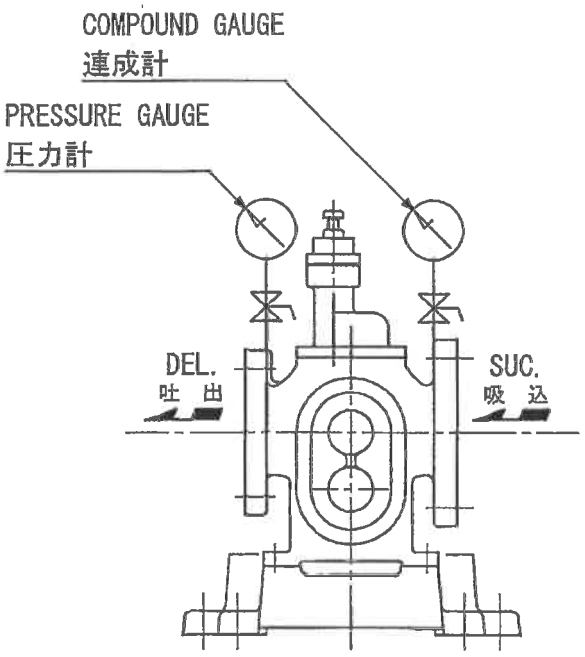
DRAWING NO. A4DH0010

DATE 4 FEB. 2004

JIS B7505



MIDDLE PRESSURE COCK
1.6 MPa and less.
中圧用コック 1.6 MPa 以下



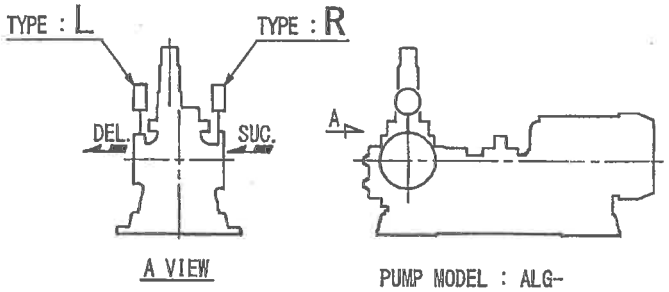
GEAR ROTARY PUMP
歯車ポンプ

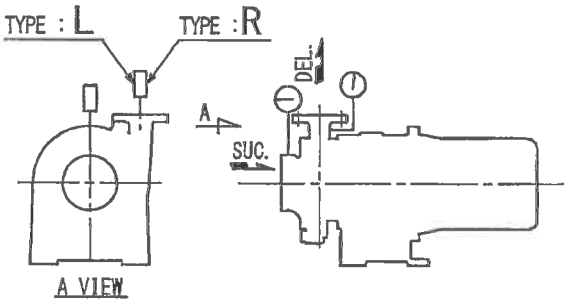
TYPE 型 式	AVT 1/4″ × φ60
UNIT 単 位	~~kgf/cm²~~, MPa
MASS 質 量	0.2 kg

PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質	Q'TY 数量	PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質	Q'TY 数量
2	GASKET ガスケット	RUBBER 合成ゴム	2	4	GAUGE PIPE 計器パイプ	SGP 配管用炭素鋼鋼管	2
1	GAUGE 計器		2	3	GAUGE ROOT COCK 計器元コック	BRASS 黄銅鋳物	2

naniwa pump	INDICATION LIST OF GAUGE SET POSITION HORIZONTAL PUMP TYPE FITTING ON PUMP	計器取付位置一覧 横型ポンプ ポンプ本体付き	DRAWING No. B2DG0421 DATE 30 OCT. 2012

CUSTOMER 御注文主：
PROJECT 向 け 先：
S.NO. 船 番：





Please indicate your required position in the blank
Remarks : 1. Please indicate "H" for hull fitting Gauge Boards. In this case, setting and piping for Gauge Boards are worked by shipyard. Therefore the Gauge Boards will be sent separately from pump.
2. Gauge Board size and the setting method are in accordance with maker's standard.

納入仕様書返却の際、空欄にご要求の計器方向を記入下さい。
備考：1．計器が船体付けの場合は空欄に"H"を記入下さい。その場合、計器の設置及び配管は造船所殿にて施工となりますので、計器はポンプ本体とは別送となります。
2．計器取付け寸法及び取付け要領はメーカー標準通りとなります。

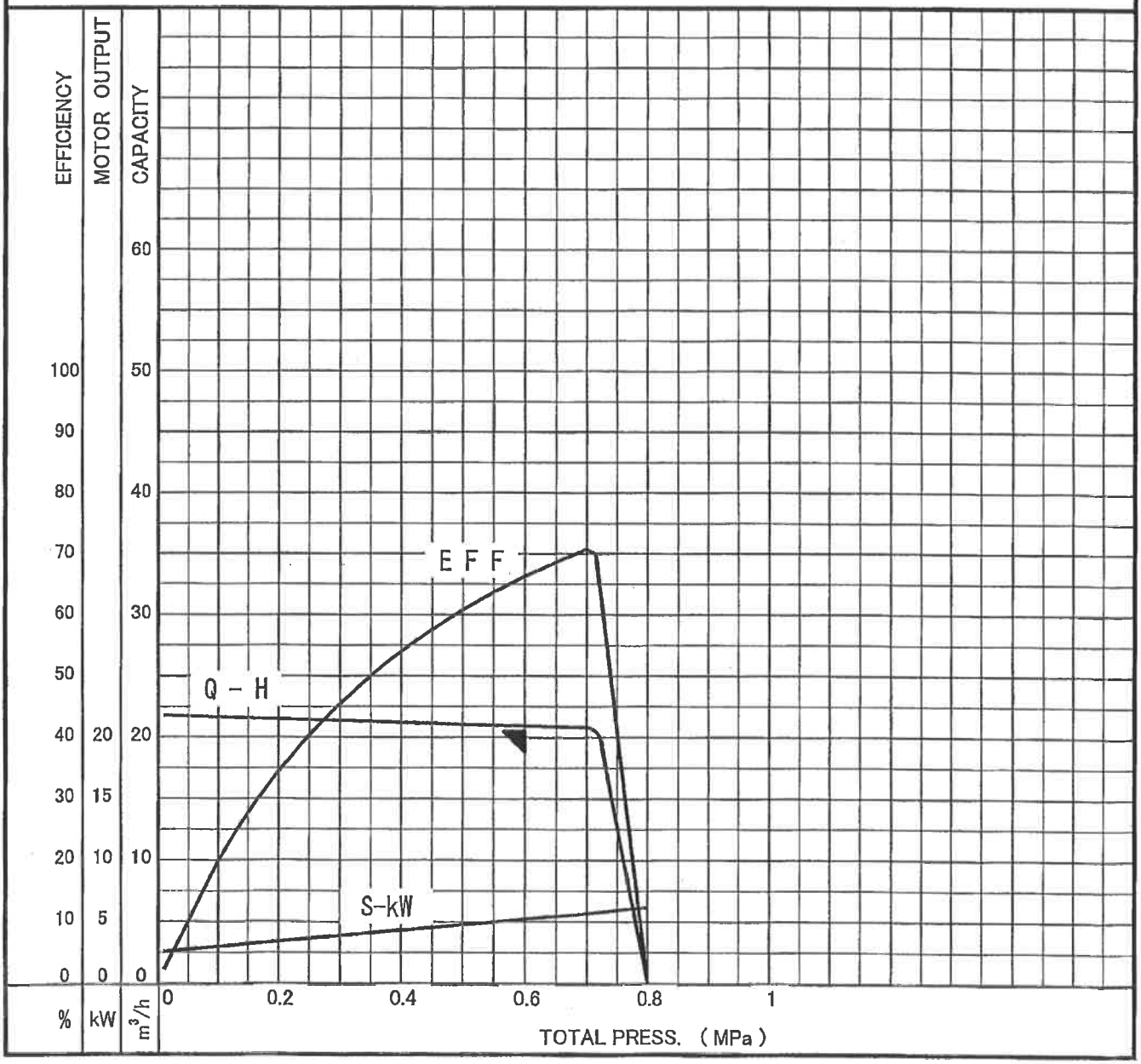
PUMP NAME 仕 様 名	PUMP MODEL 型 番	GAUGE SET POSITION 計器位置	NOTE 備 考
L.O. STBY PUMP	ALGT-65	L	
R/G L.O. STBY PUMP	ALGT-50H	L	
COOL.F.W. PUMP	CRD-100-2	L	

ESTIMATED PERFORMANCE CURVE

Page :

Date : 23 May. 2016

Customer	:	DAIHATSU DIESEL MFG.CO.,LTD				
Project	:					
Service As	:	L. O. STBY PUMP				
Pump Model	:	ALGT-65				
Bore	:	65×65				
Capacity	:	20.5	m^3/h	Total Press. :	0.6	MPa
Output	:	7.5	kW	950	min-1	
		50	Hz	415	V	
Test Liquid	:	L.O.				
H. Test Press.	:	1.2	MPa			
Remarks	:					



2017/05/09 TECHNICAL DEPARTMENT

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STANDBY L.O. PUMP

STARTER PANEL

DAIHATSU DIESEL MFG.CO.,LTD.

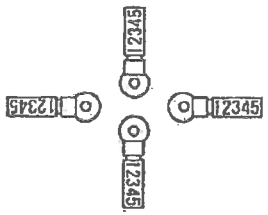
TECHNICAL DEPARTMENT

JAPAN

SHIP NO.		DRAWN BY	M. Tokumoto
TYPE	6DKM-20eF(L)	CHECKED BY	T. Tajima
LIST NO.	AQP20008443A		
DATE	2016/6/8	APPROVED BY	S. Tsuru
REVISION			

GENERAL SPECIFICATION

RULE & REGULATION	☐JG ☐NK ☐ABS ☐LR ☑BV ☐JIS・JEM
DRAWING & DOCUMENT	☑ENGLISH
NAME PLATE	LETTER/ ☐JAPANESE ☑ENGLISH ☐CHINESE GROUND-LETTER/ ☑WHITE-BLACK ☐WHITE-RED ☐RED-WHITE MATERIAL/ ACRYL RESIN
MATERIAL	☑STEEL ☐SUS ☐ALUMINUM
PROTECTION ANBIENT TEMP INSTALLATION	I P/ IP22 TEMP/ 45℃ ☐FLOOR MOUNTING TYPE ☑WALL MOUNTING TYPE ☐WALL FLUSH TYPE
PAINTING COLOR (MUNSELL NO.)	OUTSIDE/ ☑2.5G7/2 ☐7.5BG7/2 ☐N-3.0 ☐N-2.5 ☐(5R4/13) INSIDE/ ☑2.5G7/2 ☐7.5BG7/2 ☐N-3.0 ☐N-2.5 ☐(5R4/13) GLOSS/ ☐POLISHED ☑SEMI - POLISHED ☐NOT POLISHED ☐( )
BUS BAR	MATERIAL/ ~~COPPER (NICKEL PLATING)~~
INTERNAL WIRING COLOR OF WIRING CLOTHING	POWER CIRCUIT / BLACK CONTROL CIRCUIT/ GRAY EARTH CIRCUIT / YELLOW IS SPIRAL IN GREEN OR GREEN
COLOR OF INSULATION CAP	RED / R - U WHITE / S - V BLUE / T - W ~~GREEN/ R - U YELLOW/ S - V BROWN/ T - W~~ RED / P BLUE / N
CABLE SIZE	POWER CIRCUIT / MORE THAN 2.5mm² (SCP) CONTROL CIRCUIT/ AWG18, AWG16 (Style 1015) EARTH CIRCUIT / MORE THAN 2mm² (KIV)
TERMINAL PROCESSING	POWER CIRCUIT / INSULATION CAP(COLOR) CONTROL CIRCUIT/ WHITE MARK TUBE(CAP NOTHING)
SOLDERLESS TERMINAL	POWER CIRCUIT / R TYPE CONTROL CIRCUIT/ R TYPE
PHASE ARRAGEBENT	VIEWING FROM R-U S-V T-W (LEFT→RIGHT、UPPER→LOWER、NEAR→FAR) PANEL FRONT / P N (LEFT→RIGHT、UPPER→LOWER、NEAR→FAR)
SOURCE	POWER/ ☑AC415V 3Φ 50Hz ☐AC220V 3Φ 50Hz ☐AC220V 1Φ 50Hz ☐DC24V ☐AC24V ☐AC100V 3Φ 50Hz CONTROL/ ☐AC440V ☐AC220V ☑AC100V ☐DC24V ☐AC24V
OUTSIDE TERMINAL	☑PROVIDED ☐NOT PROVIDED
CONNECTION DIAGRAM	☐CARD CASE ☑VINYL BAG

FUSE	FUJI ☐FCF2 50kA UTSUNOMIYA(CELLOLITE) ☑UC-1 50kA ☐NC-1 20kA
ILLUMINATION LAMP	☑BA9S 6.3V 1W ☐E-12 18V 2W ☐E-10 30V 1W ☐BA9S 30V 1W ☐E-12 30V 2W ☐E-10 100V 1W ☐LED DC24V
SPARE PARTS	☑SUPPLIED ☑VINYL BAG ☐VINYL BOX ☐STEEL BOX (WITH KEY) ☐2.5G7/2 ☐7.5BG7/2 ☐NOT SUPPLIED ☐( )
WIRE INTRODUCTION	☐UPPER ☑LOWER ☐SIDE ☐BACK ☐GLAND ☑COMING
BOARD LIGHTING FLUORESCENT LIGHT SPARE LIGHT	☐PROVIDED ☑NOT PROVIDED (AC100V 20W) ☐PROVIDED ☑NOT PROVIDED (DC24V 20W)
METER	MAKER:DAIICHI ELECTRONICS(COVER BLACK FRAME) ☐110x110 ☐120x100 ☑60x60 ☐80x80 ☐96x96 ☐100x100
INSULATION OHMMETER	DAIICHI ELECTRONICS(COVER BLACK FRAME) ☐PROVIDED ☑NOT PROVIDED
UNIT	SI (METRIC SYSTEM)
CONNECTION FOR LOAD	☐DIRECT FOR MCB ☑WITH TB
CHANNEL BASE (ANGLE BASE)	☐PROVIDED ☑NOT PROVIDED
DIRECTION OF THE MARK TUBE	

CYUGOKU ELECTRIC SERVICE CO., LTD. SHIMONOSEKI JAPAN	STARTER PANEL	GENERAL SPECIFICATION	DWG. NO. 280603-01

部品表 PARTS LIST

P.

番号 No.	記号 SYMBOL	型式 TYPE	仕様 SPPECIFICATION	数量 QT.	メーカー MAKERS	備考 REMARKS
			STAND-BY L.O.PUMP STARTER PANEL			
1	MCB	S100-NF	100AF/40A 3P PM	1x2	TERASAKI	MOULDED CIRCUIT BREAKER
2			+T2PM12E3	1x2	TERASAKI	
3	A	PSK-60C	0-25-75A (R:17A)	1x2	DAIICHI	AMMETER
4	TR	NTP-10	AC415/100V 1Φ 50Hz 100VA	1x2	HISANO	TRANSFORMER
5	88M	SC-5-1+SZ-A20	COIL:AC100V 50Hz 1a1b+2a	1x2	FUJI	ELECTROMAGNETIC CONTACTOR
6	88D	SC-5-1+SZ-A20	COIL:AC100V 50Hz 1a1b+2a	1x2	FUJI	ELECTROMAGNETIC CONTACTOR
7	88S	SC-0	COIL:AC100V 50Hz 1a	1x2	FUJI	ELECTROMAGNETIC CONTACTOR
8	51	TR-5-1NH	12～18A 1a1b	1x2	FUJI	THERMAL RELAY
9	WL	DR30DOL-H4W	AC100/6.3V 1W BA9S	1x2	FUJI	INDICATION LAMP
10	GL	DR30DOL-H4G	AC100/6.3V 1W BA9S	1x2	FUJI	INDICATION LAMP
11	RL	DR30DOL-H4R	AC100/6.3V 1W BA9S	1x2	FUJI	INDICATION LAMP
12	3C	AR30FOR-10G	600V 10A 1a	1x2	FUJI	PUSHBUTTON SWITCH
13	3T	AR30FOR-01R	600V 10A 1b	1x2	FUJI	PUSHBUTTON SWITCH
14	43	AR30PR-222B	600V 10A 2a2b	1x2	FUJI	SELECTOR SWITCH
15	19	PM4S-A2C10M-AC120V	AC100V 0～10min	1x2	PANASONIC	TIMER
16	51X	RU2S-C-A100	AC100V 10A 2c	1x2	IDEC	RELAY
17	F	FCF2-3	500V 3A	5x2	FUJI	FUSE
18	FH	FH-30	600V 30A	5x2	YOSHIDA	FUSE HOLDER

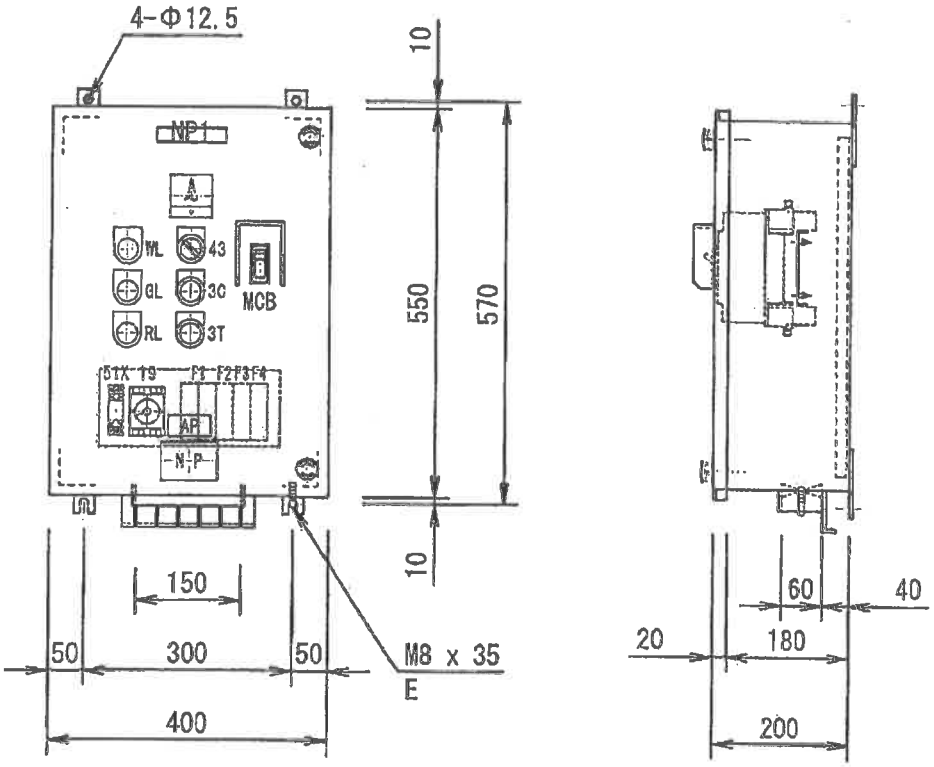
銘　板　表
NAME PLATE TABLE

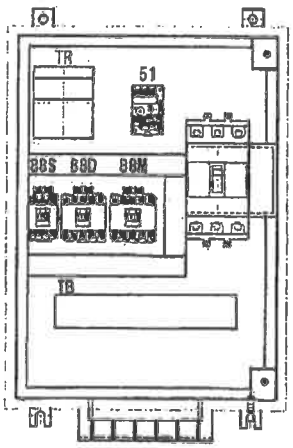
記号 No.	数量 Q'TY	寸法 TYPE	銘板名称 LETTERS ON NAME PLATE / 回路名称 CIRCUIT ON NAME	フレーム/セット FRAME/SET	ケーブルサイズ CABLE SIZE	色 COLOR 文字色 LETTER	色 COLOR 地色 GROUND	備考 REMARKS
NP1	2	100x20	STAND-BY L.O.PUMP			B	W	
WL	2	Φ30	SOURCE			B	W	
GL	2	Φ30	RUNNING			B	W	
RL	2	Φ30	OVER LOAD			B	W	
43	2	Φ30	MANU - AUTO			B	W	
3C	2	Φ30	START			B	W	
3T	2	Φ30	STOP			B	W	
AP	2		APPROVED MARK					
NP	2		MAKER NAME PLATE					

B:BLACK
W:WHITE
R:RED
MW:MILK WHITE

銘板表 NAME PLATE LIST	CYUGOKU ELECTRIC SERVICE CO.,LTD. SHIMONOSEKI JAPAN	DWG. No. 280603-04

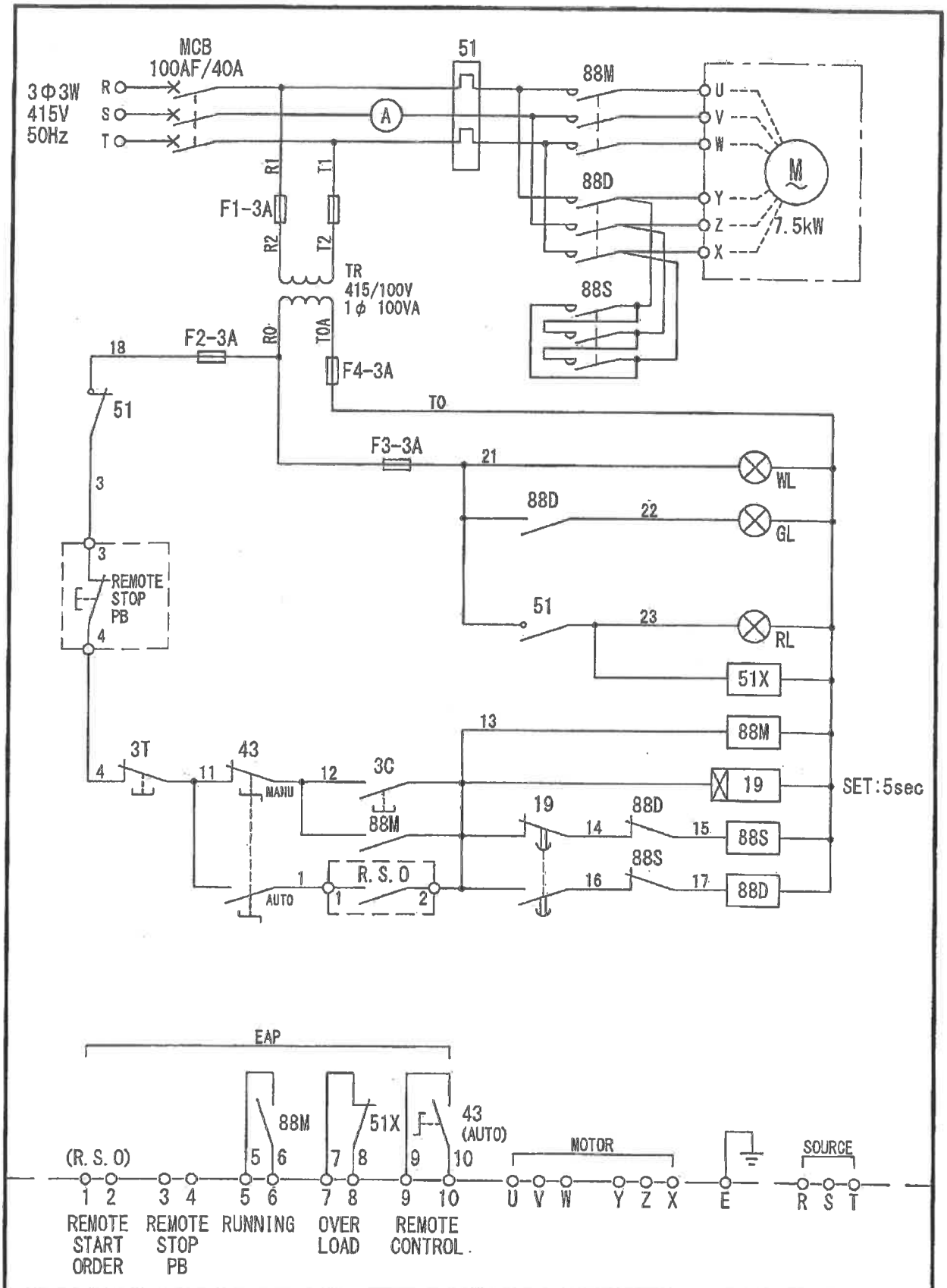
NP1	STAND-BY L.O. PUMP





PANEL PAINTED COLOR
OUT & INSIDE : 2.5G7/2
(MUNSEL No.) SEMI-POLISHED
WAIGHT PLANNED : 30 kg
FINISHED : kg
MATERIAL : STEEL ··· 2.3t
PROTECTION : IP22
Q'ty : 2

外形寸法図 OUTLINE VIEW	CYUGOKU ELECTRIC SERVICE CO.,LTD. SHIMONOSEKI JAPAN	DWG. No. 280603-05



NO.	CIRCUIT NAME	NO.	CIRCUIT NAME
ST	STAND-BY L.O. PUMP		

STARTER CIRCUIT	CYUGOKU ELECTRIC SERVICE CO.,LTD. SHIMONOSEKI JAPAN	DWG. No. 280603-06

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L.O.COOLER

DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

JAPAN

SHIP NO.		DRAWN BY	M. Tokumaba
TYPE	6DKM-20eF(L)	CHECKED BY	A. Iizimata
LIST NO.	AQP10008527A		
DATE	2016/6/3	APPROVED BY	S. Fum
REVISION			

PDK20-0174-5

①

PRINCIPAL ITEMS OF PLATE TYPE HEAT EXCHANGER		DOC. No.	MAN9571
MFG. No.	MODEL RX-185B-NHJM-43	QUANTITY	2 UNIT(S) /SHIP
AN-9571	EQPT. NAME M/E L.O.COOLER A/B		

1.

APPLICABLE CODE		BV
SHIP No.		
HEAT TRANSFER AREA[m^2]		10.46
HEAT EXCHANGED [kW]		148
MASS[kg]	DRY	400
	WET	430

2. PARTICULAR SHEET OF BASIC PLAN

FLUID NAME	L.O.	F.W.
INLET TEMP[°C]	64.82	38.50
OUTLET TEMP[°C]	55.00	41.68
FLOW RATE[m3/h]	32.00	40.00
PRESSURE DROP[MPa]	0.078	0.022
DESIGN PRESS[MPaG]	0.77	0.77
DESIGN TEMP. [°C]	90.00	90.00
TEST PRESS(HYDR.) [MPaG]	1.16	1.16
HOLDING VOLUME[lit]	15.5	15.5

3. TOOL (PER UNIT)

NAME	No. REQ'D
SPANNER	1
BRUSH FOR CLEANING	——
GAUGE PLATE	——
CHAIN	——
GAS CHARGING TOOLS	——

N2 GAS CHARGING PRESS[MPa]	——
WORKING PRESS[MPa]	——

4. ATTACHMENT (PER UNIT)

NAME	NO. REQ'D	
AIR VENT	2	
DRAIN BLOW	2	
NAME PLATE	1	
THERMOMETER	L type	4
	I type	——
STRAINER	——	
DUPLEX GAUGE	——	
ACCUMULATOR	——	

5. PAINTING

MUNSELL No. 2.5G7/2

APPROVED BY	S. HATANAKA	MAY, 31, 2016
REVIEWED BY	K. SHIMODAN	MAY, 31, 2016
PREPARED BY	N. MURATA	MAY, 31, 2016

SHEET No. 4

HISAKA WORKS, LTD.

MAY, 31, 2016

2017/05/09 TECHNICAL DEPARTMENT

ESTIMATE/AN95/AN9571/BAN9571.dwg
2017/05/09 TECHNICAL DEPARTMENT (3/5) AQP10008527A
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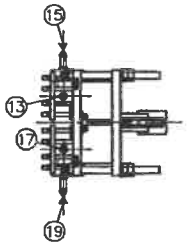
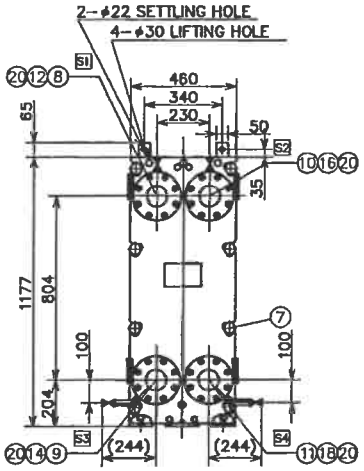
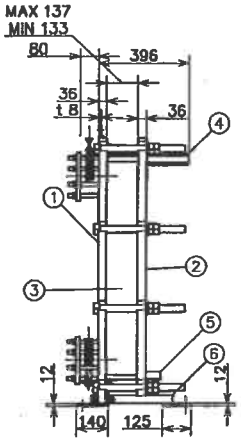
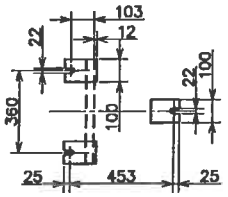
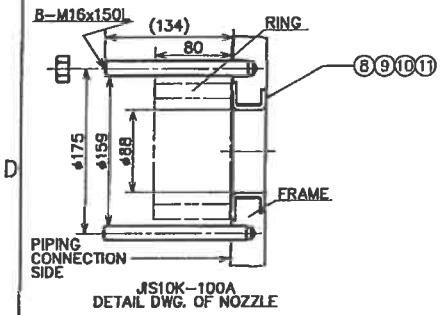










NOZZLE ORIENTATION			CONNECTION
S1	L.O.	INLET	JIS10K-100A×80A INCLUDE STUD B&N
S3	L.O.	OUTLET	JIS10K-100A×80A INCLUDE STUD B&N
S4	F.W.	INLET	JIS10K-100A×80A INCLUDE STUD B&N
S2	F.W.	OUTLET	JIS10K-100A×80A INCLUDE STUD B&N

MASS(EMPTY)	400	kg	MASS(FILLED)	430	kg

ATTACHMENT				
RATCHET SPANNER	C.S.	M30	1 PC
SET BOLT & NUT	SS400	M20×80L	3 SETS

SHIP NO.	SZ15045
EQUIP. NAME	M/E L.O.COOLER A/B
ITEM NO.	6DKM-20eF(L)
APPLICABLE CODE	BV

PART 7 FINISH: COATING OF ZINC & TIGHTENING BOLT WITH COVER

NO.	PART NAME	REQ'D	MATERIALS	REMARKS
20	COUNTER FLANGE	4	SS400	JIS10K-100A×80A
19	DRAIN BLOW	1	SGP	15A(PT 1/2) WITH G.V.(BC-6)
18	RING	1	STKM13A	
17	AIR VENT	1	SGP	15A(PT 1/2) WITH G.V.(BC-6)
16	RING	1	STKM13A	
15	DRAIN BLOW	1	SGP	15A(PT 1/2) WITH G.V.(BC-6)
14	RING	1	STKM13A	
13	AIR VENT	1	SGP	15A(PT 1/2) WITH G.V.(BC-6)
12	RING	1	STKM13A	
11	S.NOZZLE	1	NBR	
10	S.NOZZLE	1	NBR	
9	S.NOZZLE	1	NBR	
8	S.NOZZLE	1	NBR	
7	TIGHTENING BOLT & NUT	8 SETS	SNB7/S45C	M30×375L
6	FRAME SUPPORT	1	SS400	
5	LOWER GUIDE BAR	1	SUS304	
4	UPPER GUIDE BAR	1	SUS304	
3	PLATE	43	SUS316	
2	E.FRAME	1	SS400	
1	S.FRAME	1	SS400	

REV.	DATE	REVIEWED BY

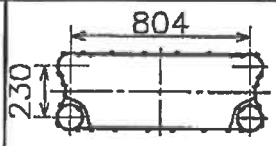
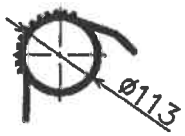
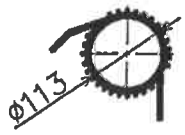
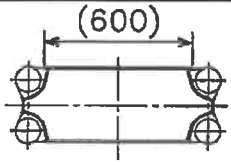
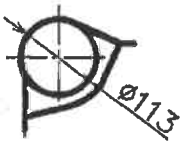
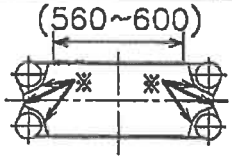
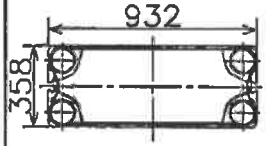
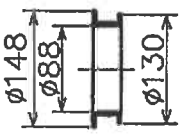
PURCHASER					
CUSTOMER					
MFG. NO.			QUANTITY	2	SHEET NO. 2
DESIGNED	N.MURATA	JUN.13.2016	SCALE	NONE	THIRD ANGLE PROJ.
DRAWN	N.MURATA	JUN.13.2016	NAME	RX-185B-NHJM-43 (H18)	
CHECKED	K.SHIMODAN	JUN.13.2016		ASSEMBLY DWG. OF PLATE TYPE HEAT EXCHANGER	
APPROVED	S.HATANAKA	JUN.13.2016	DWG. NO.	BAN9571	0

株式会社 日阪製作所 HISAKA WORKS, LTD.

AN9571

VER.200604

MFG.No.	–	PARTS LIST	Doc No.	PARAN9571
HISAKA PLATE HEAT EXCHANGER			SHIP No.	
MODEL RX-185B-NHJM-43		2UNIT(S)/SHIP	ITEM No.	6DKM-20eF(L)

No.	PARTS NAME	SKETCH (ØDIMENSION SHOWS CENTER OF DIAMETER.)	MATERIAL	QUANTITY WORKING PER 1 UNIT	QUANTITY SPARE PER 2UNITS	REMARKS
1	PLATE GASKET		G-NBR	42	–	
2	D.PLATE GASKET DA		NBR	2	–	SET ON D.PLATE
			–	–	–	SET ON E.PLATE BACK SIDE
3	D.PLATE GASKET DB		NBR	2	–	SET ON D.PLATE
			–	–	–	SET ON E.PLATE BACK SIDE
4	D.DISTANCE PIECE		NBR	1 SET	–	SET ON D.PLATE
5	E.PLATE GASKET		NBR	4	–	SET ON E.PLATE BACK SIDE
6	E.DISTANCE PIECE		NBR	1 SET	–	SET ON E.PLATE BACK SIDE ※If E.Nozzle installed
7	PLATE		SUS316	6	–	(RX-135A) WITH GASKET
				37	–	(RX-145A) WITH GASKET
				–	–	(RX-135A) WITHOUT GASKET
				–	–	(RX-145A) WITHOUT GASKET
8	RUBBER BOOTS		NBR	2	–	SET ON NOZZLE FOR L.O. SIDE
			NBR	2	–	SET ON NOZZLE FOR F.W. SIDE

APPROVED BY	S.HATANAKA	MAY,31,2016	HISAKA WORKS,LTD. OSAKA,JAPAN
PREPARED BY	T.OHNOKI	MAY,31,2016	

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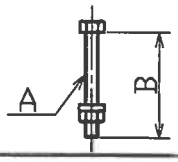
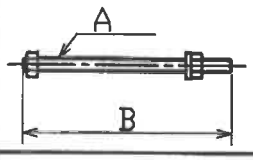
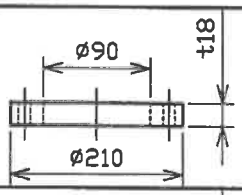
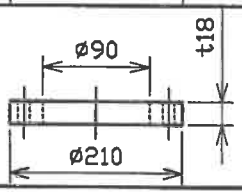
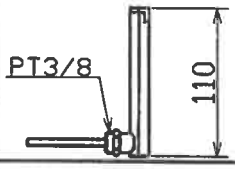
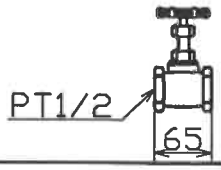
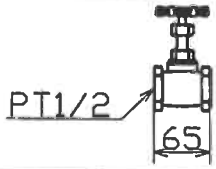
MFG.No.	-	ACCESSORIES LIST	Doc No.	ACTAN9571
HISAKA PLATE HEAT EXCHANGER			SHIP No.	SZ15045
MODEL RX-185B-NHJM-43		2UNIT(S)/SHIP	ITEM No.	6DKM-20eF(L)

No.	PARTS NAME	SKETCH (ØDIMENSION SHOWS CENTER OF DIAMETER.)	MATERIAL	QUANTITY WORKING PER 1 UNIT	QUANTITY SPARE PER 2UNITS	REMARKS
1	SET BOLT,NUT & WASHER		SS400	3 SET	–	A=M20 B=80
2	TIGHTENING BOLT & NUT		SNB7/S45C	8	–	A= M30 B= 400
3	COUNTER FLANGE		SS400	2	–	FOR F.W. SIDE JIS10K-100A×80A
4	COUNTER FLANGE		SS400	2	–	FOR L.O. SIDE JIS10K-100A×80A
5	L-TYPE THERMOMETER (WITH CASE)		SUS316	2	–	FOR L.O. SIDE
			SUS316	2	–	FOR F.W. SIDE
6	GLOBE VALVE		BC-6	1	–	FOR L.O. SIDE AIR VENT
				1	–	FOR F.W. SIDE AIR VENT
7	GLOBE VALVE		BC-6	1	–	FOR L.O. SIDE DRAIN BLOW
				1	–	FOR F.W. SIDE DRAIN BLOW
8						

APPROVED BY	S.HATANAKA	JUN,13,2016	HISAKA WORKS,LTD. OSAKA,JAPAN
PREPARED BY	T.OHNOKI	JUN,13,2016	

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L.O. FILTER FOR ENGINE INLET

DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

JAPAN

SHIP NO.		DRAWN BY	M. Tokumoto
TYPE	6DKM-20eF(L)	CHECKED BY	S. Tojimoto
LIST NO.	AQP10008539B		S. Naruto
DATE	2016/6/9	APPROVED BY	S. Kuno
REVISION	B: P5. CHANGE BODY MATERIAL FC250 → FCD400 CHANGE COVER MATERIAL FC250 → FCD400 2016/06/28 M.T		

PDK20-0174-5

CUSTOMER	Daihatsu Diesel Mfg Co., Ltd				
SHIPBUILDER	Zhejiang Shenzhou Shipbuilding Co., Ltd.				
SHIP No.	SZ15045	SZ15046		RULE	BV
WORK No.	PDK20-0174-5/PDK20-0176-7		ORDER No.		

K8E STRAINER

DWG. No.	K8E3V0-S30-X

USE	LUBE OIL STRAINER		LIQUID	Lub. Oil (SAE#30 or SAE#40)	
ENGINE TYPE	6DKM-20e		WORKING TEMP.	5 ~ 80 °C	
CAPACITY	MAX. 32	m³/h	CONNECTIONS: INLET & OUTLET	FLANGE	JIS-10K-100A
FILTRATING AREA	6300	cm^2		PIPE	100A
FILTRATING LIMIT	ABS. 30	μm	BACKWASHING LIQUID OUTLET	FLANGE	JIS-10K-32A
PRESSURE DROP	< 0.03	MPa		PIPE	32A
WORKING PRESSURE	NOR. 0.7 , DES. 0.77	MPa			
OIL TEST PRESSURE	1.16	MPa	PAINTING	STRAINER BODY MUNSELL	2.5G7/2
BACKWASHING CAPACITY	3.2	m³/h			
CLEANING METHOD	CONTINUOUS AUTOMATIC BACKWASHING		MASS	APPROX. 190	kg
DIFF. PRESS. SWITCH	ALARM POINT 0.09	MPa	QUANTITY	2	SETS/SHIP

REMARKS : COUNTER FLANGES , GASKETS , BOLTS & NUTS PROVIDED.

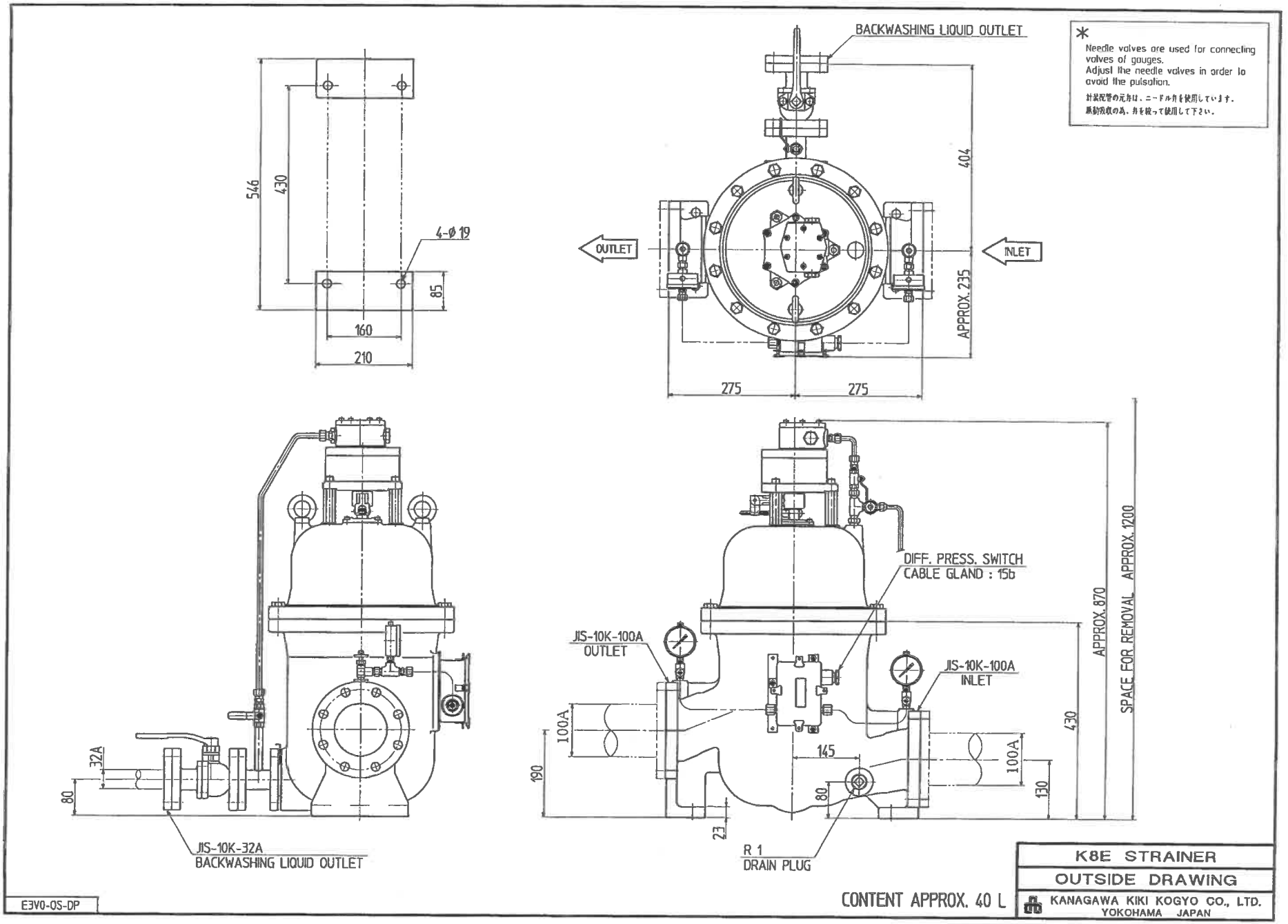
ORIFICE PLATE DIAMETER : φ6.1 (IN BACKWASHING PIPE)

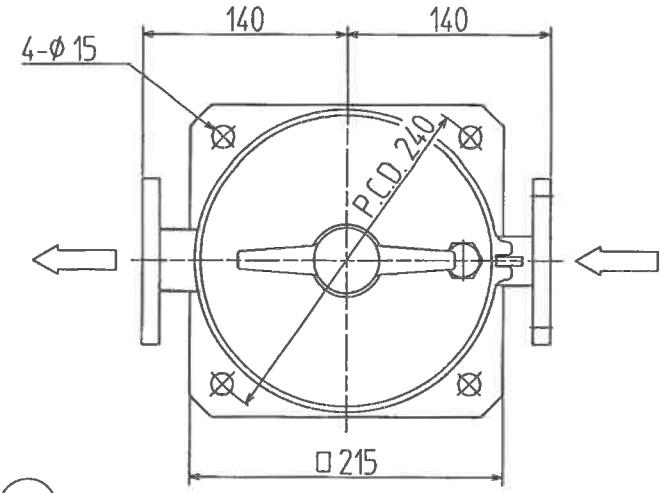
NAMEPLATE: ENGLISH & JAPANESE

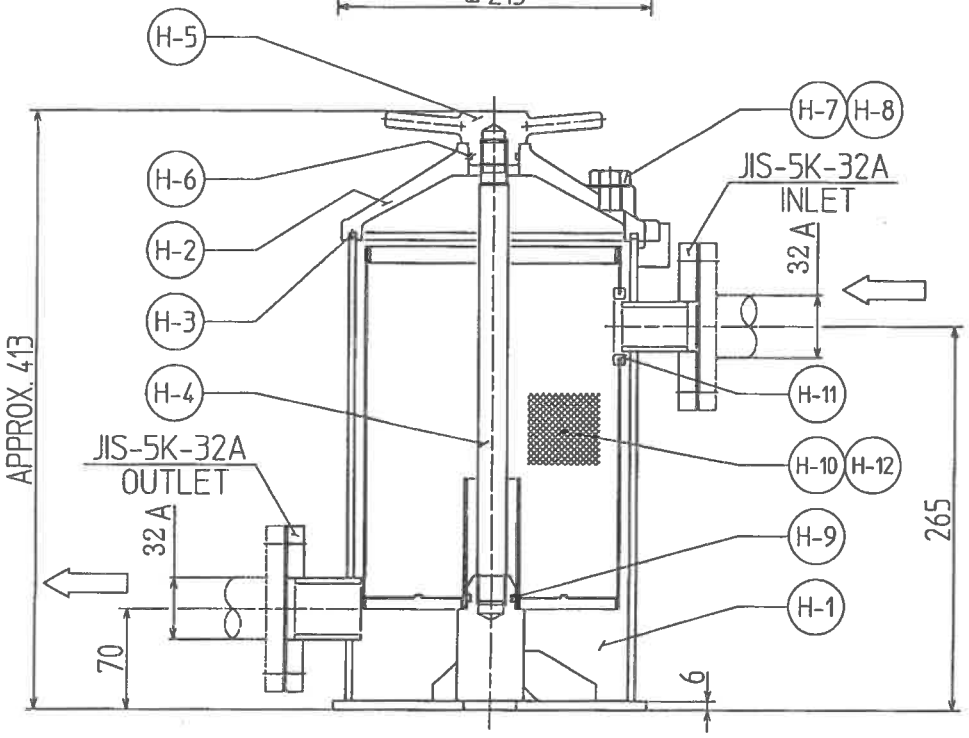
(ABS. = ABSOLUTE)

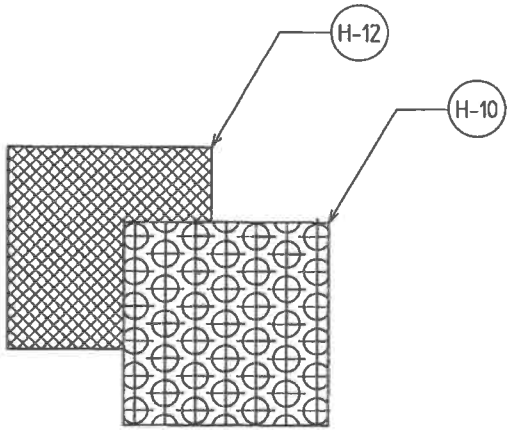
MFG. No.	M802041	APPROVED	CHECKED	DRAWN
		K.NONAKA 2016/02/16	K.NONAKA	M.TSUDA M.TSUDA
DATE	2016/2/16			

KANAGAWA KIKI KOGYO CO., LTD.
Main Office & Factory : 19-1 Okamura, 8-chome, Isogo-ku, Yokohama, Japan.
Tel. 045-761-4561(Rep.) FAX. 045-755-0089


BACKWASHING LIQUID OUTLET
OUTLET
INLET
404
APPROX. 235
275
275
546
430
4-ø19
85
160
210
DIFF. PRESS. SWITCH
CABLE GLAND : 15b
JIS-10K-100A
OUTLET
JIS-10K-100A
INLET
100A
100A
145
190
80
130
23
430
APPROX. 870
SPACE FOR REMOVAL APPROX. 1200
32A
80
JIS-10K-32A
BACKWASHING LIQUID OUTLET
R 1
DRAIN PLUG
* Needle valves are used for connecting valves of gauges.
Adjust the needle valves in order to avoid the pulsation.
計装配管の元方は、ニードル弁を使用しています。
脈動吸収の為、弁を絞って使用して下さい。
E3V0-0S-DP
CONTENT APPROX. 40 L
K8E STRAINER
OUTSIDE DRAWING
KANAGAWA KIKI KOGYO CO., LTD.
YOKOHAMA JAPAN







NO.	PARTS NAME		Q,TY	MATERIAL	REMARKS
H - 1	本体	BODY	1	SS400	
H - 2	蓋	COVER	1	AC4A	
H - 3	パッキン	PACKING	1	NBR	
H - 4	軸	SHAFT	1	SS400	
H - 5	締付ナット	LOCK NUT	1	FCD40	
H - 6	O-リング	O-RING	1	NBR	
H - 7	パッキン	PACKING	1	FIBER	
H - 8	プラグ	PLUG	1	SS400	
H - 9	O-リング	O-RING	1	NBR	
H - 10	濾枠	CAGE	1	SPC	
H - 11	パッキン	PACKING	1	NBR	
H - 12	金網	WIRE GAUZE	1	SUS304	100 MESH

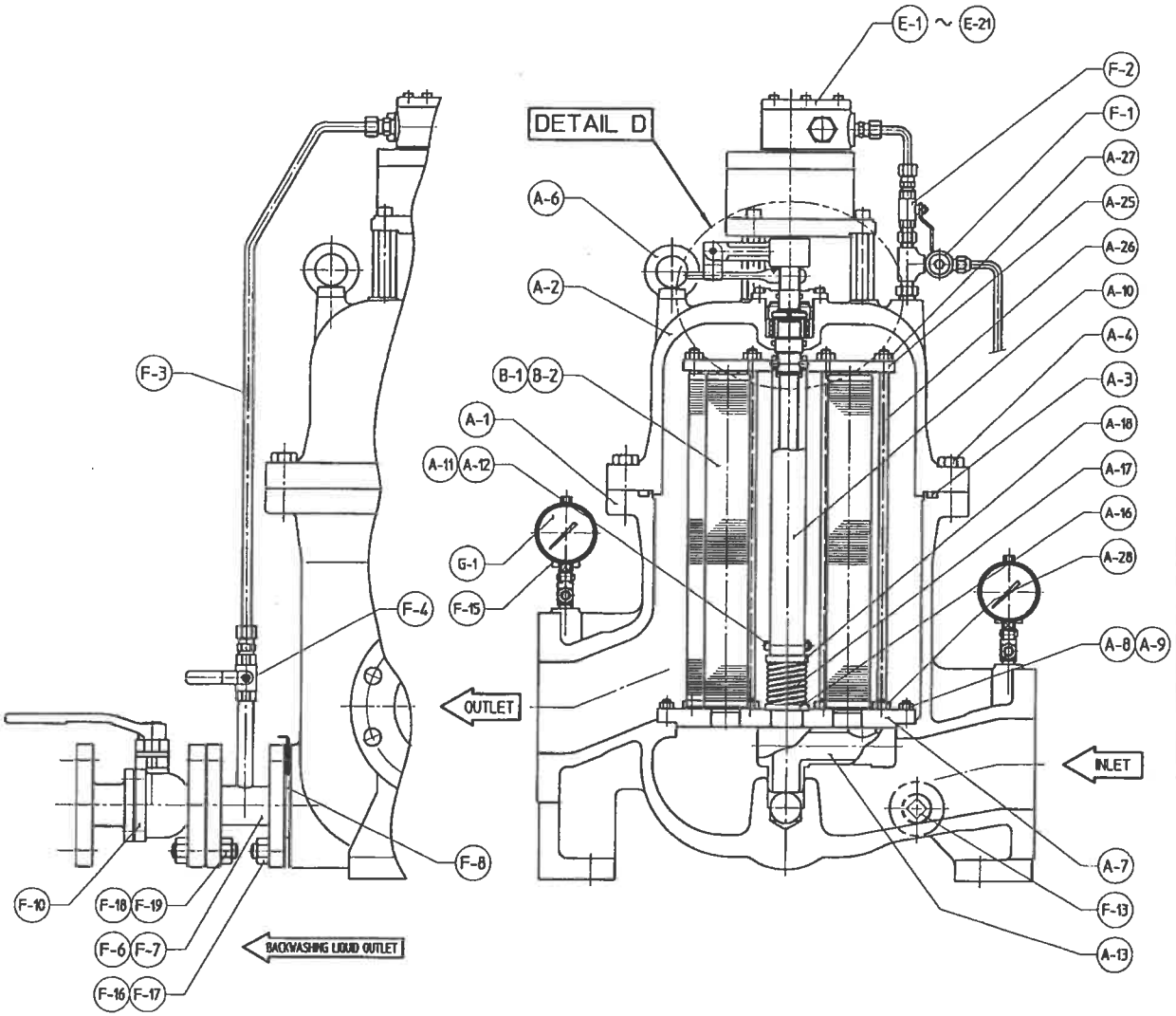
MASS : APPROX. 16 kg

SLUDGE CHECKER

ASSEMBLY DRAWING

KANAGAWA KIKI KOGYO CO., LTD.
YOKOHAMA JAPAN

X-2959D



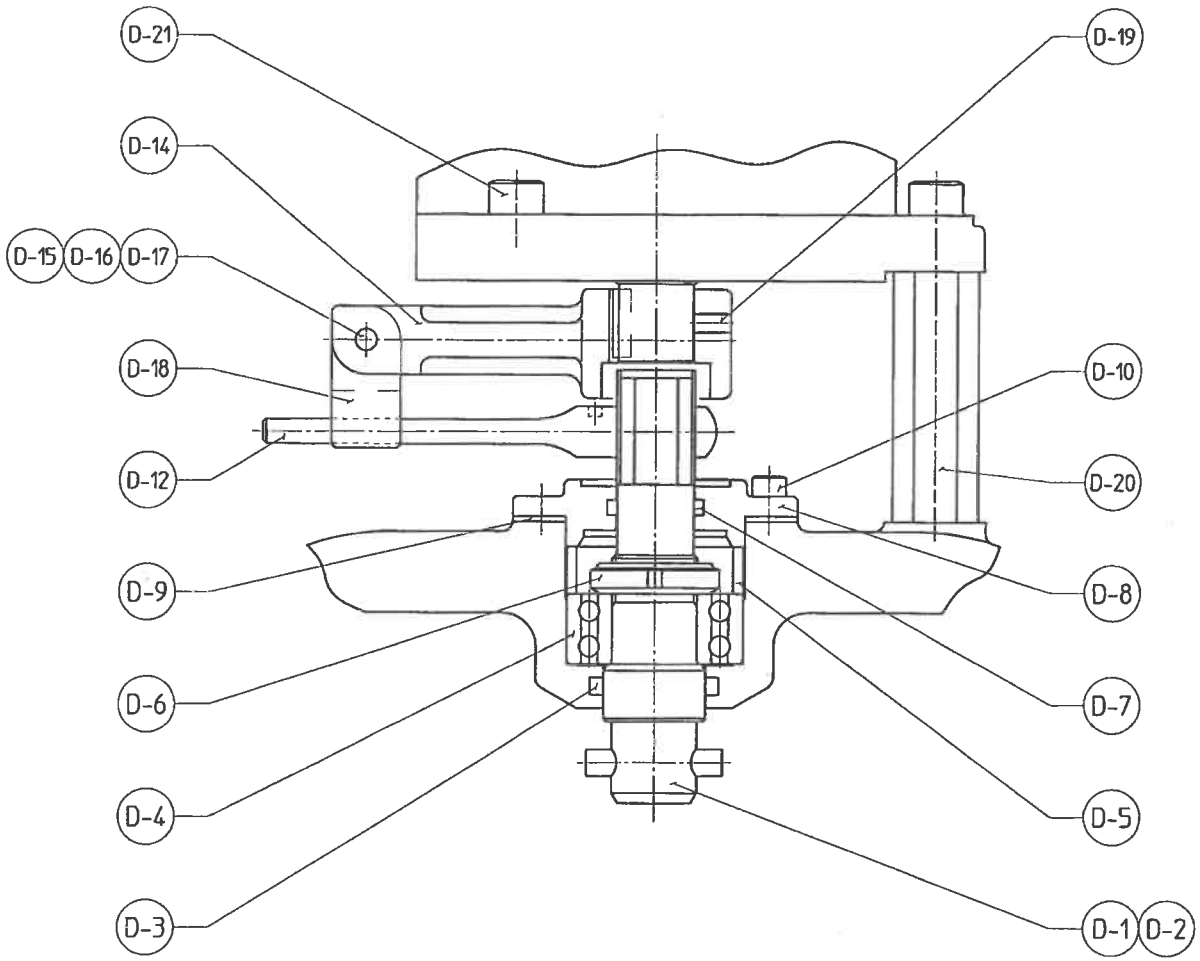
NO.	PARTS NAME		QTY	MATERIAL	REMARKS
A-1	本体	BODY	1	FCD400	
A-2	蓋	COVER	1	FCD400	
A-3	O-リング	O-RING	1	NBR	
A-4	六角ボルト	HEXAGON HEAD BOLT	12	SS400	
A-6	アイボルト	EYE BOLT	2	S25C	
A-7	集合体	ELEMENT STAND	1	SS400	
A-8	スタッドボルト	STUD BOLT	6	SS400	
A-9	U-ナット	U-NUT	6	SS400	
A-10	軸	SHAFT	1	STKM13	
A-11	六角穴付ボルト	CAP SCREW	1	SCM435	
A-12	U-ナット	U-NUT	1	SS400	
A-13	逆洗アーム	BACKWASHING ARM	1	FCD500	
A-16	バネ受け	WASHER	1	SS400	
A-17	バネ	SPRING	1	SWP-A	
A-18	バネ押え	WASHER	1	SS400	
A-25	エレメント押え板	ELEMENT HOLDING PLATE	1	SS400	
A-26	エレメント固定軸	ELEMENT HOLDING SHAFT	9	SS400	
A-27	U-ナット	U-NUT	9	SS400	
A-28	六角ナット	HEXAGON NUT	9	SS400	
B-1	エレメント	ELEMENT	12	SUS304 ALUMINUM	
B-2	パッキン	PACKING	12	NBR	
D-1 ~ D-21	DETAIL D				
E-1 ~ E-19	ACTUATOR ASSY				
F-1	ニードルバルブ	NEEDLE VALVE	1	C3771	
F-2	ボール弁	BALL VALVE	1	C3602	
F-3	アクチュエータ出口管	ACTUATOR OUTLET PIPE	1	OST	
F-4	ボールバルブ	BALL VALVE	1	BC	
F-6	集合管	PIPE	1	SGP	
F-7	パッキン	PACKING	3	NON ASBESTOS	
F-8	オリフィス板	ORIFICE PLATE	1	SS400	
F-10	ボール弁	BALL VALVE	1	FC250	JIS 10K 32A
F-13	プラグ	PLUG	1	FCMB35	
F-15	ニードルバルブ	NEEDLE VALVE	2	C3771	
F-16	スタッドボルト	STUD BOLT	4	SS400	
F-17	六角ナット	HEXAGON NUT	4	SS400	
F-18	スタッドボルト	STUD BOLT	4	SS400	
F-19	六角ナット	HEXAGON NUT	8	SS400	
G-2	圧力計	PRESSUER GAUGE	2		

E3V0-1-ASSY-7

K8E STRAINER
ASSEMBLY DRAWING
KANAGAWA KIKI KOGYO CO., LTD.
YOKOHAMA JAPAN

AQP10008539B (5/6)

2017/05/09 TECHNICAL DEPARTMENT



NO.	PARTS NAME		Q,TY	MATERIAL	REMARKS
D-1	駆動軸	DRIVE SHAFT	1	SCM435	
D-2	平行ピン	PARALLEL PIN	1	S45C	
D-3	X-リング	X-RING	1	NBR	
D-4	ベアリング	BEARING	1	SUJ2	
D-5	内輪カラー	COLLAR	1	SS400	
D-6	ファインUナット	FINE U-NUT	1	SS400	
D-7	O-リング	O-RING	1	NBR	
D-8	軸受箱蓋	BEARING BOX COVER	1	FCD400	
D-9	パッキン	PACKING	1	NON ASBESTOS	
D-10	六角穴付ボルト	CAP SCREW	4	SCM435	
D-12	ハンドル	HANDLE	1		
D-14	自動レバー (A)	AUTO LEVER (A)	1	FCD450	
D-15	ニードルローラー	NEEDLE ROLLER	1	SUJ2	
D-16	スプリングピン	SPRING PIN	1		
D-17	バネ	SPRING	1	SUS304	
D-18	自動レバー (B)	AUTO LEVER (B)	1	FCD400	
D-19	六角穴付止ネジ	SET SCREW	1	SCM435	
D-20	取付脚	STAND	3	SS400	
D-21	六角穴付ボルト	CAP SCREW	3	SCM435	

E-DETAIL-2-D

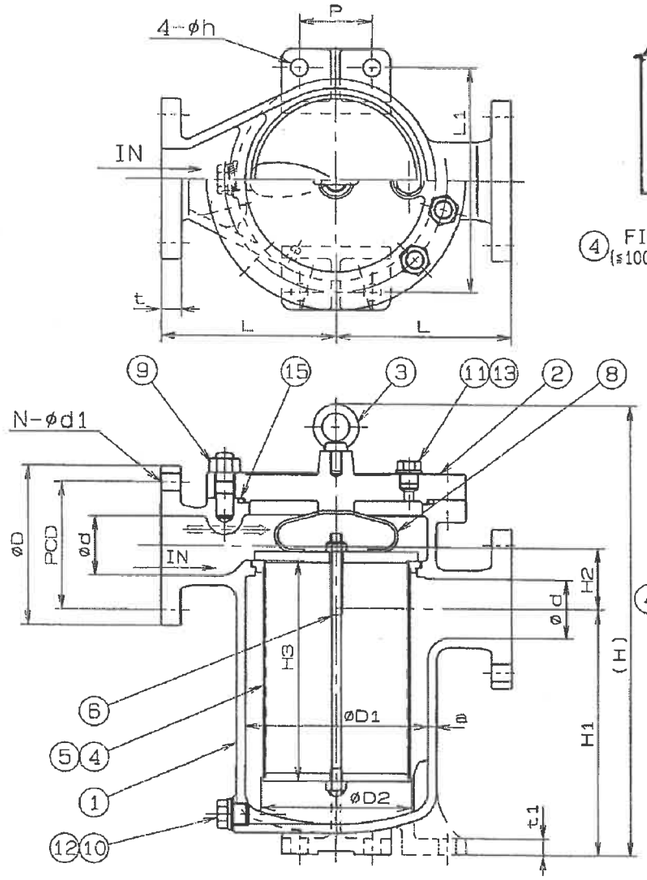
K8E STRAINER

DETAIL D

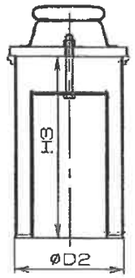
KANAGAWA KIKI KOGYO CO., LTD.
YOKOHAMA JAPAN

AQP10008539B (6/6)

2017/05/09 TECHNICAL DEPARTMENT



④ FILTER CYL.
(≤100A & ≤100Mesh)



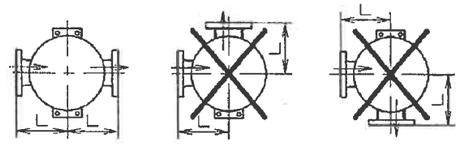
④ FILTER CYL.
(For ≧ 125A)

TYPE (型式)

S-TYPE

~~LA-TYPE~~ (For ≧ 65A)

~~LB-TYPE~~ (For ≧ 65A)



SPECIFICATION LIST (仕様表)

ITEM No. (符 号)	NOMI. DIA. (呼び径)	TYPE (形式)	FLUID NAME (流体)	FLOW RATE (m^3/h) (流量)	PRESS. (MPa) (設計圧)	TEMP. (℃) (温度)	FILT. DEGREE (メッシュ)	QUANT. (製作数)	SPARES (予備品)	FITTING (付属品)	REMARKS (備 考)
	10K-100A	S	L.O. (SAE#30)	32	(常用) 0.6 (最低) 1.0	(Max.) 80	400	⚠	–	–	–

⚠

15	O-RING	(O-リング)	FPM	1	
14	VOID				
13	GASKET	(ガスケット)	V#6500	1	
12	GASKET	(ガスケット)	V#6500	1	
11	AIR VENT PLUG	(空気抜きプラグ)	SS400	1	M20×1.5
10	DRAIN PLUG	(ドレンプラグ)	SS400	1	
9	BOLT & NUTS	(ボルト&ナット)	SS400	1set	
8	HANDLE OF CYLINDER	(こし両取手)	SS400	1	
7	VOID				
6	MAGNET HOLDING BAR	(磁石取付棒)	SS400	1	For≧50A
5	WIRE GAUZE	(こし網)	SUS304	1	
4	FILTER CYLINDER	(こし筒)	SS400	1	
3	EYEBOLT	(アイボルト)	SF440A	1	For≧40A
2	COVER	(ふ た)	SS400	1	or FCD400
1	BODY	(本 体)	FCD400	1	
ITEM	N A M E		MATERIAL	QUANT.	REMARK

DWG. NAME

JIS 10K-100A ⚠

SIMPLEX STRAINER

(丸フタ型単式こし器)

HYD. TEST PRESS	APPROVED BY	CHECK'D BY	DRAWN'D BY	SCALE
1.5 MPa	牙.ケ		牙.ケ	–

DATE	16. 5.30	DWG. NO.	MAD16520-01
DATE	97. 5	TYPE	MS11-2220D

MIZUNO STRAINER INDUSTRIES CO., LTD.

DIMENSION LIST (寸法表)

(Unit: mm)

NOMI. DIA. (公称)	d	L	FLANGE D	FLANGE BOLT PCD	FLANGE BOLT N	FLANGE BOLT d1	FLANGE t (*2)	(H)	BODY H1	BODY H2	BODY D1	BODY a	LEGES L1	LEGES P	LEGES t1	LEGES h	DR ⑩	O-RING ⑮	FILTER D2	FILTER H3	FILTER AR (*1)	CALC. MASS (Kg)
10K- 40A	40	130	140	105	4	19	20	300	145	43	110	8	–	–	–	–	M20*1.5	G115	87	148	16	25
10K- 50A	50	165	155	120	4	19	20	390	208	50	160	9	200	60	16	15	M24*2	G160	129	189	14	30
10K- 65A	65	195	175	140	4	19	22	480	270	67	200	10	245	80	18	19	M24*2	G195	164	242	15	50
10K- 80A	80	227	185	150	8	19	22	510	283	73	230	10	280	80	18	19	M24*2	G220	185	268	13	65
10K-100A	100	230	210	175	8	19	24	590	340	90	265	11	315	90	18	19	M24*2	G250	215	331	12	100
10K-125A	125	225	250	210	8	23	24	675	373	109	285	12	345	90	22	23	M24*2	G260	224	369	13	135
10K-150A	150	235	280	240	8	23	26	735	410	130	300	13	370	90	22	23	M24*2	G280	235	423	11	190
10K-200A	200	285	330	290	12	23	26	855	475	155	360	15	435	120	24	25	M24*2	1516/82	285	498	10	300
10K-250A	250	350	400	355	12	25	30	1025	529	200	440	17	520	150	26	27	M24*2	V430	340	563	8	440
10K-300A	300	400	445	400	16	25	32	1095	542	230	470	18	545	160	26	27	M24*2	G440	360	598	7	550
10K-350A	350	430	490	445	16	25	34	1280	590	265	520	18	595	160	26	27	M24*2	G490	410	654	7	700

*1 AR=(total area of holes)/(cross sectional area of inlet pipe)
*2 A flange thickness applies the size of FC.

⚠ x3	REV:QUANT DEL:DWG. NAME	2016/06/21	E. M
⚠	FIRST DRAWING	2016/06/03	E. M

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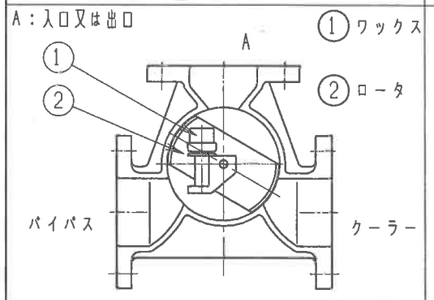
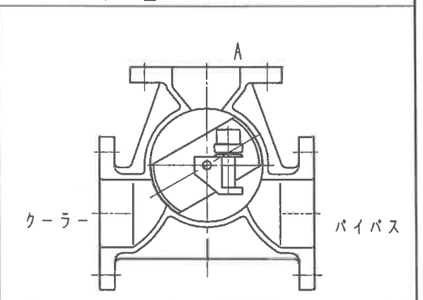
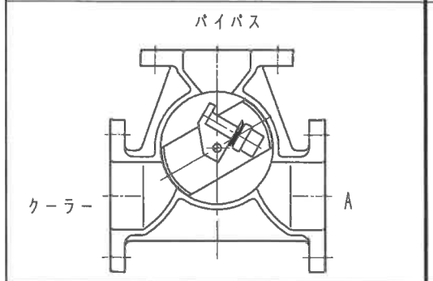
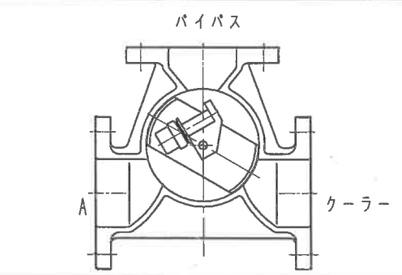
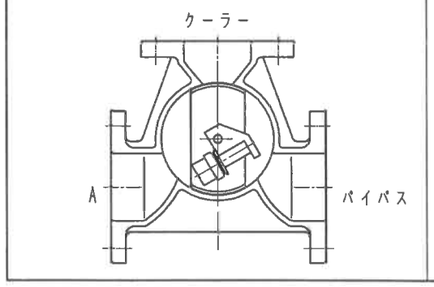
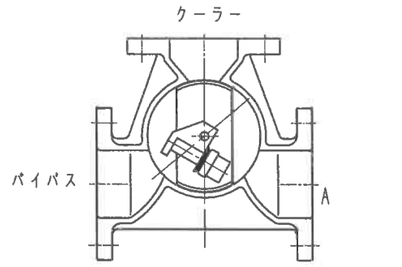
APPROVED	KAWANO SHUJI	LIST NO.	MATERIAL PCHS	GROUP 7	SCALE:
CHECKED		BEFORE	MASS	A QTY.	A3
		REF.	TYPE ALL		LEVEL
	SHIMOKAWA KEISUKE	DAIHATSU DIESEL MFG. CO., LTD. OSAKA JAPAN	NAME ロキ:タンシキジュンカツユ:10K-100A		
DRAWN	MATSUDA ETSUJI		FILTER:SIMPLEX:10K-100A		
DATE	2016/06/21		PARTS NO. AQA20013140	B	*

2017/05/09 TECHNICAL DEPARTMENT

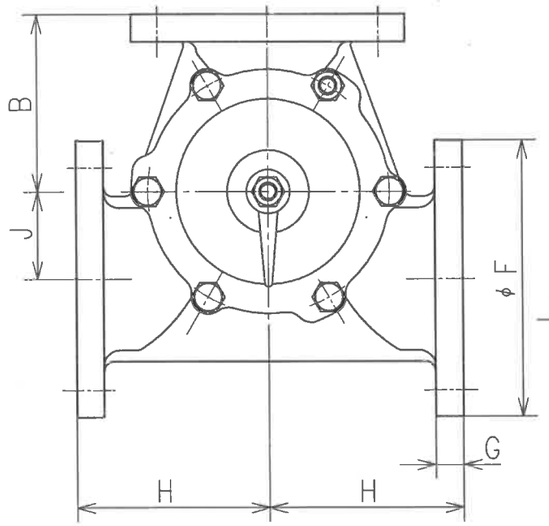
History Of Drawing			List No.	DATE	BY
A	First Drawing		-	2016.06.30	飯 野

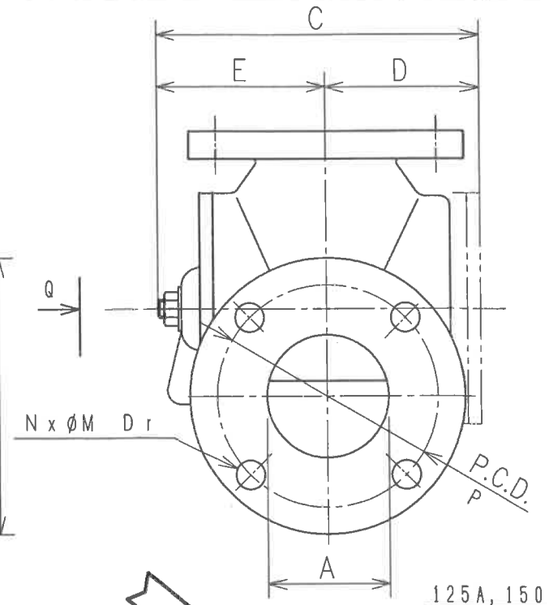
View Q

TP-ワックス式自動温度調整弁　（型式）

L. I型 TYPE	R. I型 TYPE
	
L. B型 TYPE	**R. B型 TYPE**
	
L. C型 TYPE	**R. C型 TYPE**
	

注記）相フランジ、ボルト、ナット、パッキン付き
アスベスト及びアスベストを含有する材料は絶対に使用しないこと





125A, 150A, 200Aは
下部カバー付き

A(SIZE)	40	50	65	80	100	125	150	200
B	115	115	115	115	148	165	178	242
C	160	160	198	195	246	282	335	464
D	67	67	84	81	104	124	150	222
E	93	93	114	114	142	158	185	242
F JIS10K	140	155	175	185	210	250	280	330
G	20	20	22	20	24	24	26	26
N～φM	4-19	4-19	4-19	8-19	8-19	8-23	8-23	12-23
P	105	120	140	150	175	210	240	290
H	127	127	127	127	148	165	178	242
J	57	57	57	57	57	64	76	102
質量 WEIGHT （概算 APPROX.） kg								
特殊鋳鉄 SPECIAL CAST.	14	16	20	24	40	51	82	150

制御方式 CONTROL	
分流型 DIVIDING	混合型 MIXING

NOTE

温度範囲 °C TEMP. RANGE			
25/35	37/47	57/66	79/88
27/40	41/49	63/71	85/93
31/42	43/54	68/78	
35/45	51/60	74/82	

用途 USE	潤滑油 LUB. OIL	清水 FRESH WATER
本体材質 BODY MATERIAL	特殊鋳鉄 SPECIAL CASTING	特殊鋳鉄 SPECIAL CASTING

低 リーク型

1. 図面に指示のない要目については、品質仕様書 E8-0100, E6-2010 によること。
UNLESS OTHERWISE SPECIFIED, BE IN ACCORDANCE WITH E8-0100, E6-2010

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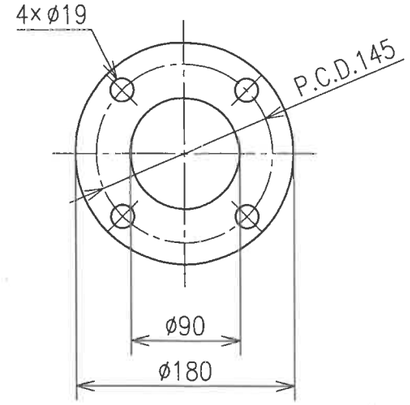
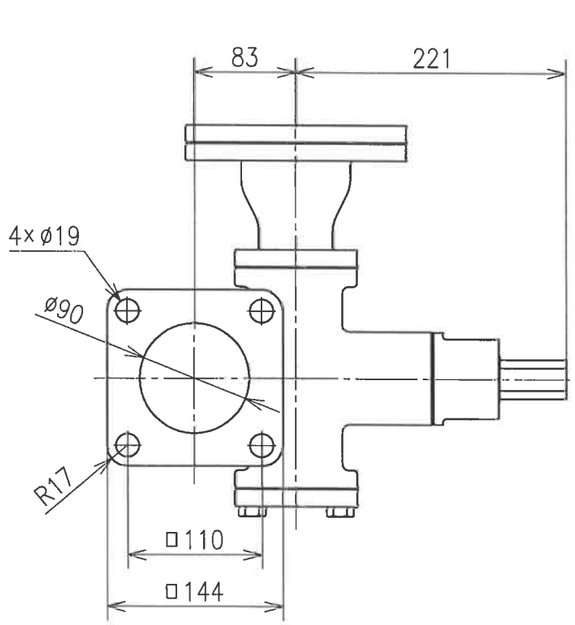
DIMENSION	1 < 4 INCL	4 < 16	16 < 63	63 < 250	250 < 1000	1000 <	TOLERANCE OF ANGLE
∀～∀	±0.1	±0.2	±0.3	±0.5	±0.8	±1.5	±30'
∀～∀	±0.3	±0.5	±0.7	±1.2	±2.0		

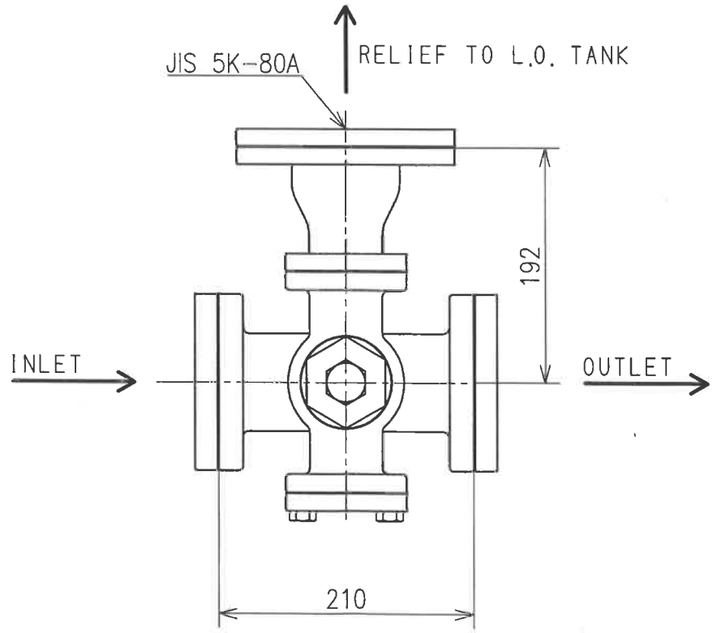
ROUGHNESS Ra	∀	∀	∀	CO.2 FOR NONINDICATED PART
	1.6	3.2	12.5	

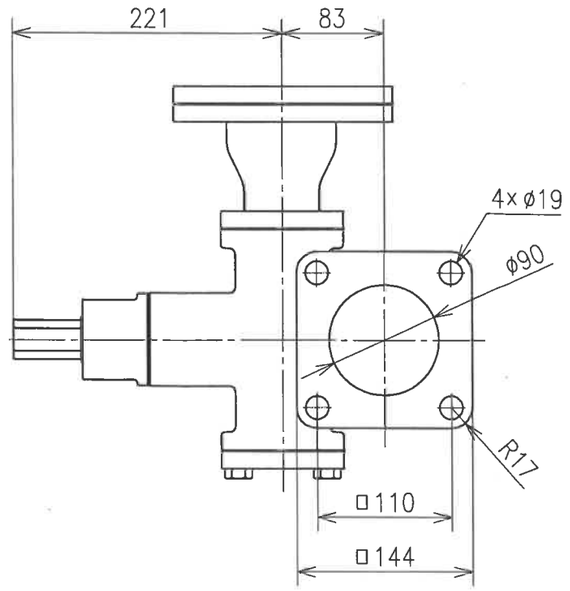
APPROVED	山根 一志	LIST NO.	MATERIAL PCHS / GROUP 7 / SCALE: -
CHECKED		BEFORE	MASS 24kg / QTY. 1/Eng / A3
	東内 伸嘉	REF.	TYPE ALL / LEVEL B
	吉井 知輝	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME オンチョウベン：TP-ウォルトンワックスシキ：LO-80A-RB
DRAWN	飯野 伸吾		THERMOSTAT VALVE:TP-WALTON WAX TYPE:LO-80A-RB
DATE	2016/06/30		PARTS NO. AQP20008498 / A / *

2017/05/09 TECHNICAL DEPARTMENT









⚠	FIRST DRAWING	2016/06/08	M. T

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APPROVED	KAWANO SHUJI	LIST NO.	MATERIAL APRV.	GROUP 0	SCALE: 1:5
CHECKED	NARITA SHIROU	BEFORE	MASS	A QTY.	A3
	FUJIMOTO SYUHEI	REF.	TYPE ALL		LEVEL
	TOKUMOTO MANABU	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME チョウアツベン：L.O.：80A		
DRAWN	(H)MORICHI YASUKO		RELIEF VALVE：L.O.：80A		
DATE	2016/06/08		PARTS NO. AQP10008538	A	*

2017/05/09 TECHNICAL DEPARTMENT

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L.O. STBY PUMP
FOR
REDUCTION GEAR

DAIHATSU DIESEL MFG.CO.,LTD.

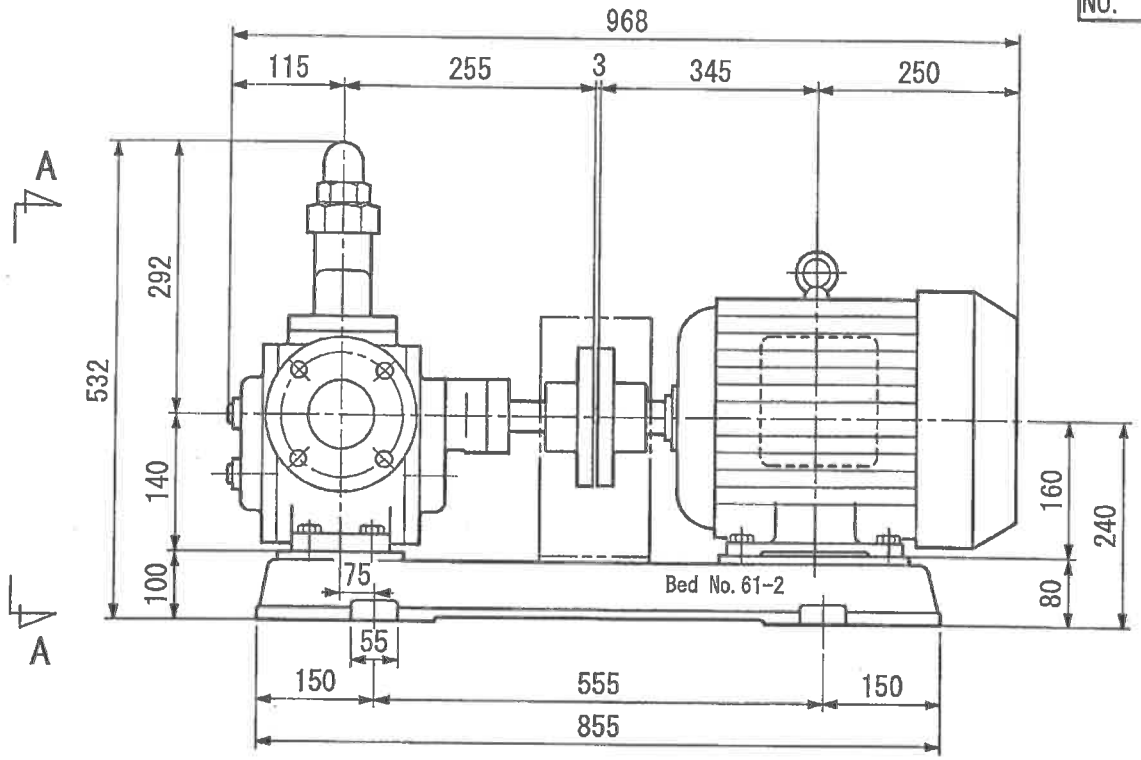
TECHNICAL DEPARTMENT

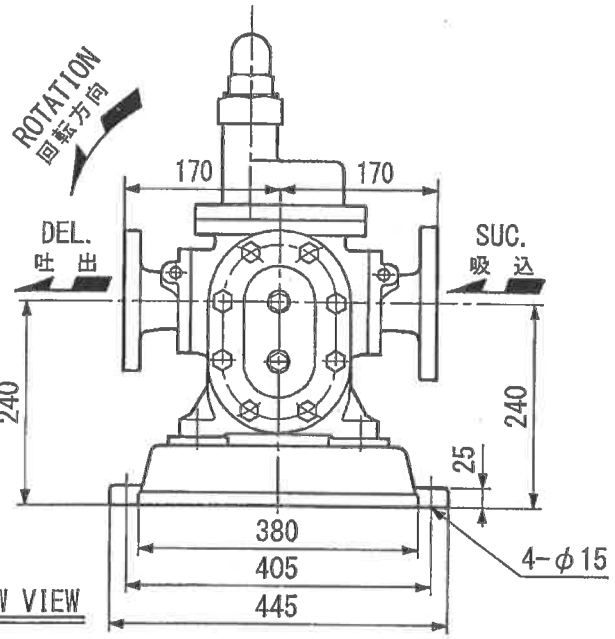
JAPAN

SHIP NO.		DRAWN BY	M. Tokuto
TYPE	6DKM-20eF(L)	CHECKED BY	T. Tajima
LIST NO.	AQP10008528A		
DATE	2016/6/3	APPROVED BY	S. Tsuru
REVISION			

PDK20-0174-5

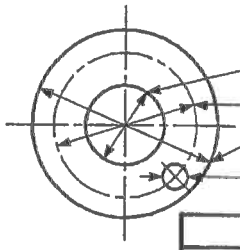
NO. 2





A-A ARROW VIEW

SUC. & DEL. FLANGE
吸込及吐出フランジ



JIS 10K SUC. 吸込	JIS 30K DEL. 吐出
φ 65	φ 50
φ140	φ130
φ175	φ165
4-φ19	8-φ19

MASS 質量	
PUMP ポンプ	90 kg
MOTOR 電動機	98 kg
TOTAL 全質量	188 kg

ACCESSORIES FOR EACH 1 PUMP 付属品（ポンプ一台ニ付）	
DESCRIPTION 記	Q'TY 数量
COUPLING JISφ160 カップリング	1set
COUPLING COVER カップリングカバー	1
COMMON BED 共通台	1
SAFETY VALVE (SET PRESSURE 2.5 MPa) 安全弁	1set
AIR COCK エアーコック	1
PRESSURE GAUGE 圧力計	1
COMPOUND GAUGE 連成計	1
GAUGE ROOT COCK 計器元コック	2
COUNTER FLANGE SUC. 50A 相フランジ DEL. 50A	1set

SPECIFICATION 仕様	
SUCTION BORE 吸込口径	65 mm
DELIVERY BORE 吐出口径	50 mm
CAPACITY 容量	6 m^3/h
TOTAL PRESSURE 全圧力	2.4 MPa
SUCTION PRESSURE 吸込圧力	−0.05 MPa
REVOLVING SPEED 回転速度	950 min^{-1}
HYD. TEST PRESS. 耐圧試験	4.0 MPa
MOTOR OUTPUT 電動機出力	11kW , 6P
POWER SOURCE 電源	415V, 50Hz, 3ph
GUARANT. VIS. FOR POWER 1000 mm^2/s	
GUARANT. VIS. FOR CAPA. 26 mm^2/s	
OIL TEMP = ℃	

PUMP NAME 名称

R/G L.O. STBY PUMP

PUMP MODEL 型式	ALGT-50H
QUANTITY 台数	1
RULE 検査	BV
ANGLE PROJ. 角法	3 rd
SCALE 尺度	FREE
DRAWING NO. 図面番号	A1DG0380

NANIWA PUMP MFG.CO.,LTD.
TECH. DEPARTMENT DESIGN SECTION
株式会社浪速ポンプ製作所

2017/05/09 TECHNICAL DEPARTMENT

naniwa pump

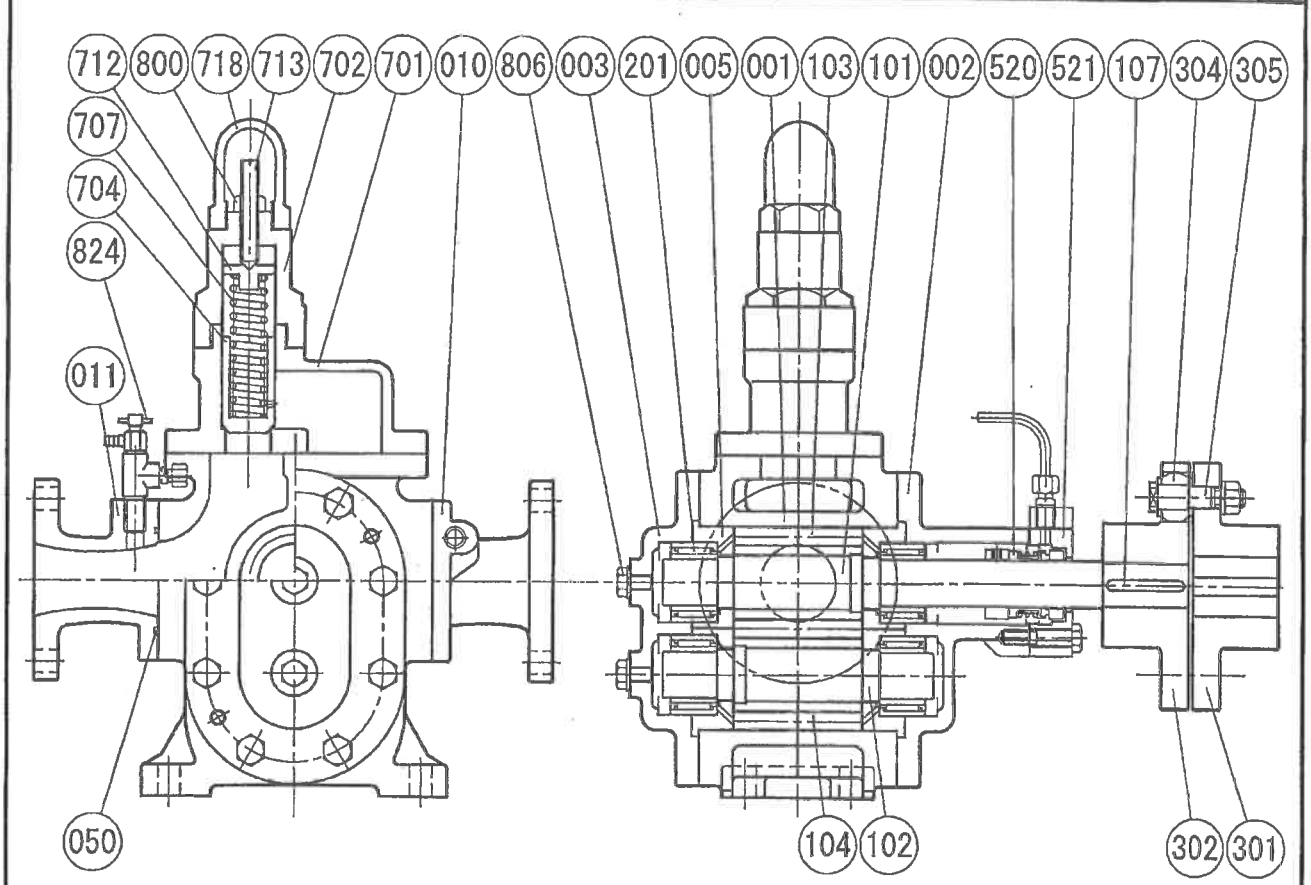
SECTIONAL DRAWING 構造断面図

HORIZONTAL GEAR ROTARY PUMP 横型電動歯車式

MODEL:型 式 ALGT-50H

DRAWING NO. A1DS0472

DATE 29 AUG. 2001

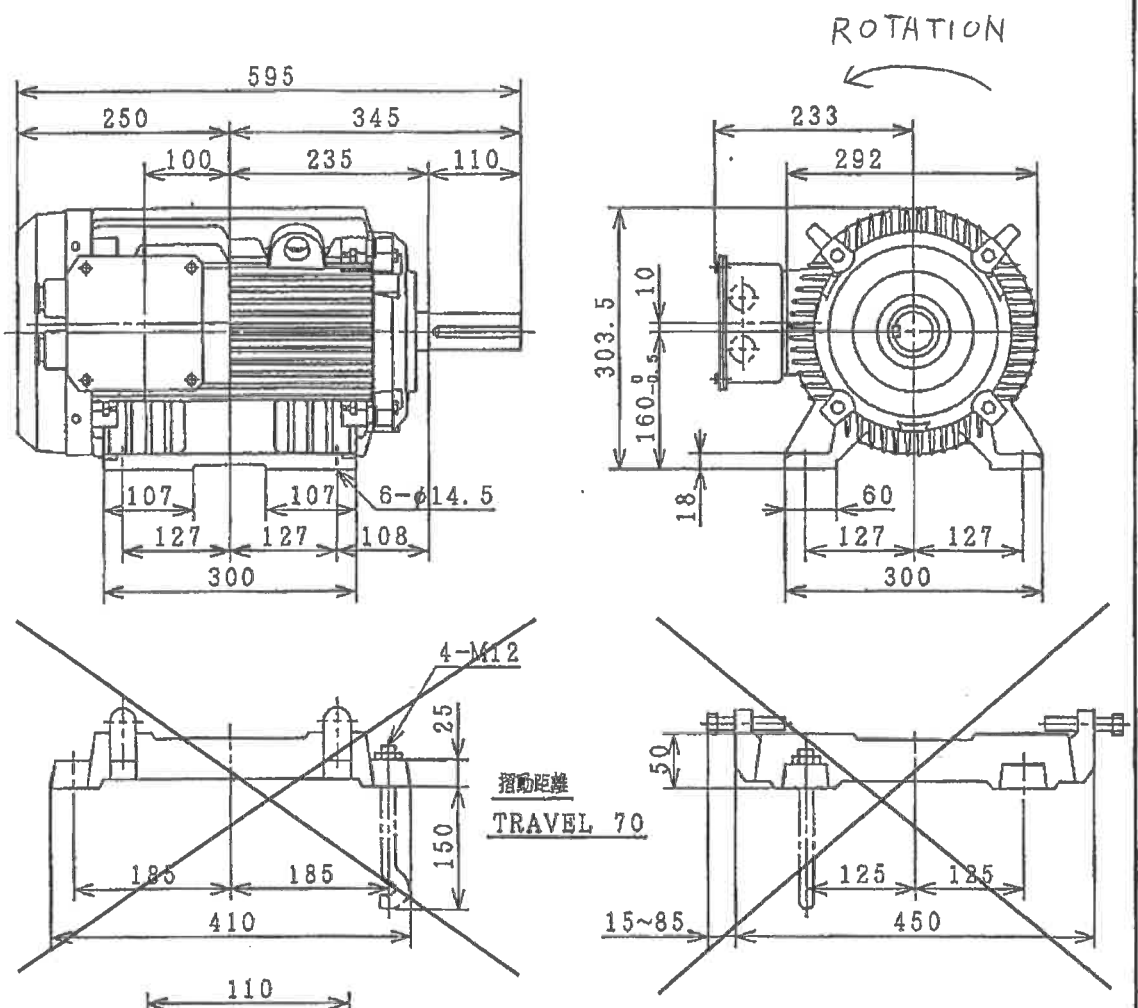


PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質		Q'TY 数量	PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質		Q'TY 数量
302	COUPLING 軸継手	CAST IRON 鋳鉄	FC200	1					
301	COUPLING 軸継手	CAST IRON 鋳鉄	FC200	1	824	AIR COCK 空気抜キコック	COPPER ALLOY ROD 銅合金棒	C3604	1
201	NEEDLE BEARING 針状コロ軸受	SUJ 高炭素クロム軸受鋼		4	806	BOLT PLUG ボルトプラグ	LOW CARBON STEEL WIRE 軟鋼線材	SWRM	2
107	COUPLING KEY 軸継手キー	CARBON STEEL 機械構造用炭素鋼	S45C	1	800	ADJUST NUT 調整ナット	MILD STEEL 一般構造用圧延鋼	SS400	1
104	DRIVEN GEAR 遊歯車	CARBON STEEL 機械構造用炭素鋼	S45C	1	718	CAP キャップ	CAST IRON 鋳鉄	FC200	1
103	DRIVE GEAR 主歯車	CARBON STEEL 機械構造用炭素鋼	S45C	1	713	ADJUST BOLT 調整ボルト	MILD STEEL 一般構造用圧延鋼	SS400	1
102	DRIVEN SHAFT 遊軸	CHROMI. MOLB-DEN クロムモリブデン鋼	SCM435	1	712	SPRING SEAT バネ座	MILD STEEL 一般構造用圧延鋼	SS400	1
101	DRIVE SHAFT 主軸	CHROMI. MOLB-DEN クロムモリブデン鋼	SCM435	1	707	VALVE SPRING 安全弁バネ	SPRING STEEL バネ鋼	SUP	1
050	O-RING Oリング	RUBBER 合成ゴム		2	704	SAFETY VALVE 安全弁	CARBON STEEL 機械構造用炭素鋼	S45C	1
011	DEL. FLANGE 吐出側フランジ	DUCTILE CAST IRON ダクタイル鋳鉄	FCD400	1	702	VALVE COVER 安全弁カバー	DUCTILE CAST IRON ダクタイル鋳鉄	FCD400	1
010	SUC. FLANGE 吸込側フランジ	DUCTILE CAST IRON ダクタイル鋳鉄	FCD400	1	701	SAFETY VALVE BOX 安全弁箱	DUCTILE CAST IRON ダクタイル鋳鉄	FCD400	1
005	SIDE PLATE サイドプレート	CAST IRON 鋳鉄	FC200	4	521	SEAL COVER シールカバー	CAST IRON 鋳鉄	FC200	1
003	SIDE COVER サイドカバー	DUCTILE CAST IRON ダクタイル鋳鉄	FCD400	1	520	MECHANICAL SEAL メカニカルシール			1 SET
002	SIDE COVER サイドカバー	DUCTILE CAST IRON ダクタイル鋳鉄	FCD400	1	305	COUPRING BOLT & NUT 軸継手ボルト及ナット	MILD STEEL 一般構造用圧延鋼	SS400	1 SET
001	CASING ケーシング	DUCTILE CAST IRON ダクタイル鋳鉄	FCD400	1	304	COUPLING RING 軸継手リング	RUBBER 合成ゴム		1 SET

CAD

三相誘導電動機
THREE-PHASE INDUCTION MOTOR

出力 OUTPUT (kW)	形 TYPE	式 FORM	定格 RATING	電圧 VOLTAGE (V)	周波数 FREQ. (Hz)	同期回転速度 SYNC. SPEED (min⁻¹)	極数 POLES	耐熱クラス TH. CLASS	質量 MASS (kg)
11	TFO	KK	S1	415	50	1000	6	F	98



SHIELD BEARING	6309 / 6307
RULE	BV
CONNECTION	Y-Δ
CABLE	
GLAND	A25F×2

ベース BASE		キソボルト FOUNDATION BOLTS		PROJECTION
	有 WITH / 不付 WITHOUT		有 WITH / 不付 WITHOUT	

用途 USE

付属品 ACCESSORY

備考 NOTICE

御注文主 CUSTOMER 殿 | 数量 QUANTITY | 受注番号 ORDER No. | 作業番号 WORK No.

1

DWN	H. ICHIKAWA	Nov-29-1995	寸法図
CHKD	M. KAMIZONO	Nov-29-1995	DIMENSIONS
APPD	T. MATSUSHITA	Nov-29-1995	

REGD 2002.04.01

Hitachi Industrial Equipment Systems Co., Ltd.

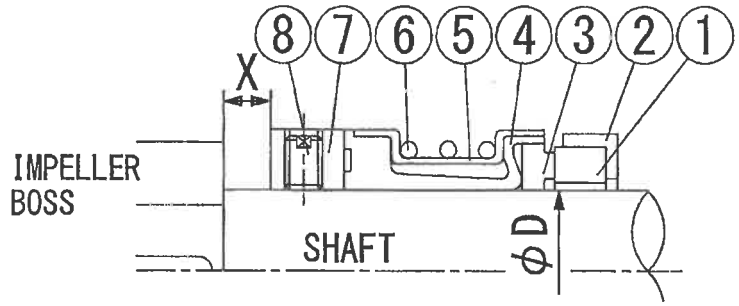
NARASHINO WORKS DWG. No. 324 4R831653

訂正番号 REV

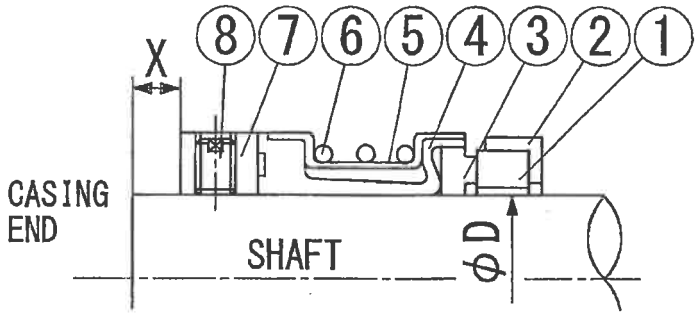
枠番 FRAME SIZE MTFOD-160L

AQP10008528A (4/8)

nariwa pump	SECTIONAL DRAWING MECHANICAL SEAL	DRAWING NO. 98DG0226 DATE 18 MAY 1998



FIG, 1



FIG, 2

NO.	PUMP NAME	MODEL	SHAFT DIA φ	SET DIMENSION "X"	FIG, NO.
1	L.O. STBY PUMP	ALGT-65	35	69	2
2	R/G L.O. STBY PUMP	ALGT-50H	30	58	2
3	COOL. F.W. PUMP	CRD-100-2	45	36	1

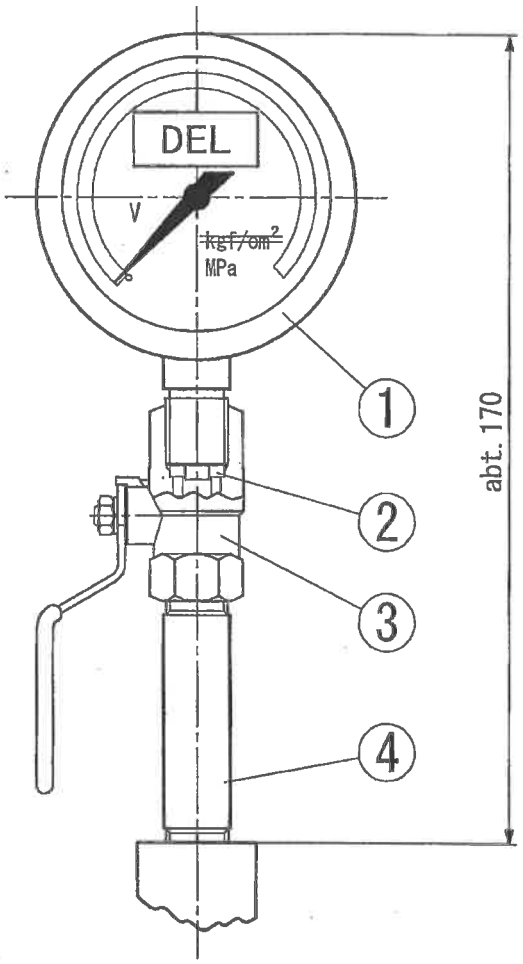
PART NO.	NAME OF PART	Q'TY
1	FLOATING SEAT	1
2	SEAT CAP	1
3	SEAL RING	1
4	BELLOWS	1
5	RETAINER	1
6	COIL SPRING	1
7	COLLAR	1
8	SET SCREW	1SET

GAUGE & GAUGE ROOT COCK
計器 及 計器元コック
HORIZONTAL PUMP TYPE
FITTING ON PUMP BODY 横型本体付タイプ

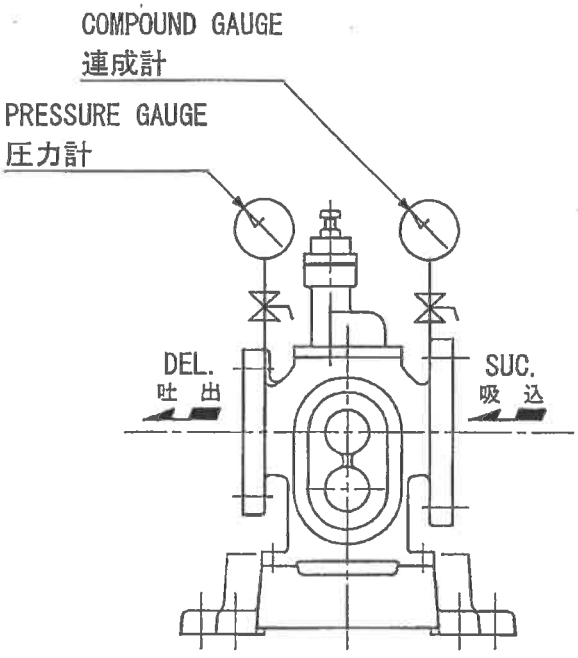
naniwa pump

DRAWING NO. A4DH0010
DATE 4 FEB. 2004

JIS B7505



MIDDLE PRESSURE COCK
1.6 MPa and less.
中圧用コック 1.6 MPa 以下



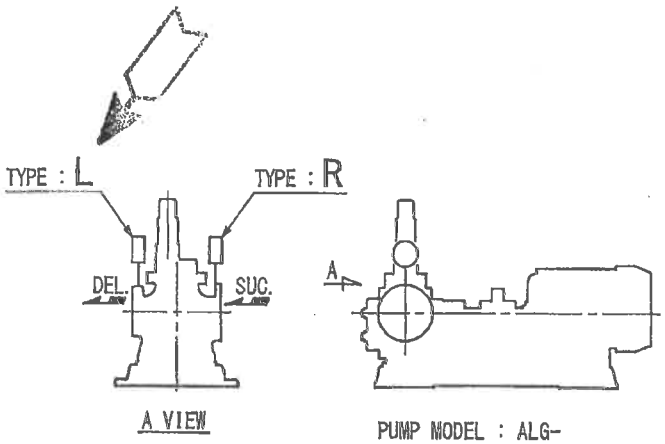
GEAR ROTARY PUMP
歯車ポンプ

TYPE 型 式	AVT 1/4″ × φ60
UNIT 単 位	~~kgf/cm²~~, MPa
MASS 質 量	0.2 kg

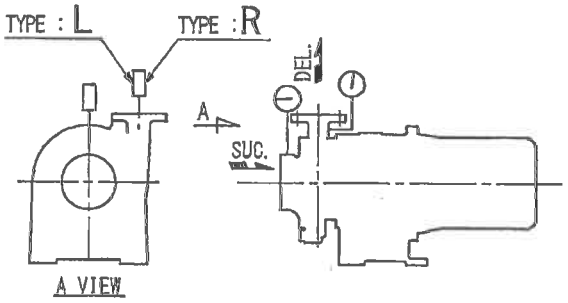
PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質	Q'TY 数量	PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質	Q'TY 数量
2	GASKET ガスケット	RUBBER 合成ゴム	2	4	GAUGE PIPE 計器パイプ	SGP 配管用炭素鋼鋼管	2
1	GAUGE 計器		2	3	GAUGE ROOT COCK 計器元コック	BRASS 黄銅鋳物	2

naniwa pump	INDICATION LIST OF GAUGE SET POSITION HORIZONTAL PUMP TYPE FITTING ON PUMP	計器取付位置一覧 横型ポンプ ポンプ本体付き	DRAWING No. B2DG0421 DATE 30 OCT. 2012

CUSTOMER 御注文主 :
PROJECT 向 け 先 :
S.NO. 船 番 :



PUMP MODEL : ALG-



PUMP MODEL : BHR, CRD

Please indicate your required position in the blank
Remarks : 1. Please indicate "H" for hull fitting Gauge Boards. In this case, setting and piping for Gauge Boards are worked by shipyard. Therefore the Gauge Boards will be sent separately from pump.
2. Gauge Board size and the setting method are in accordance with maker's standard.

納入仕様書返却の際、空欄にご要求の計器方向を記入下さい。
備考 : 1. 計器が船体付けの場合は空欄に"H"を記入下さい。その場合、計器の設置及び配管は造船所殿にて施工となりますので、計器はポンプ本体とは別送となります。
2. 計器取付け寸法及び取付け要領はメーカー標準通りとなります。

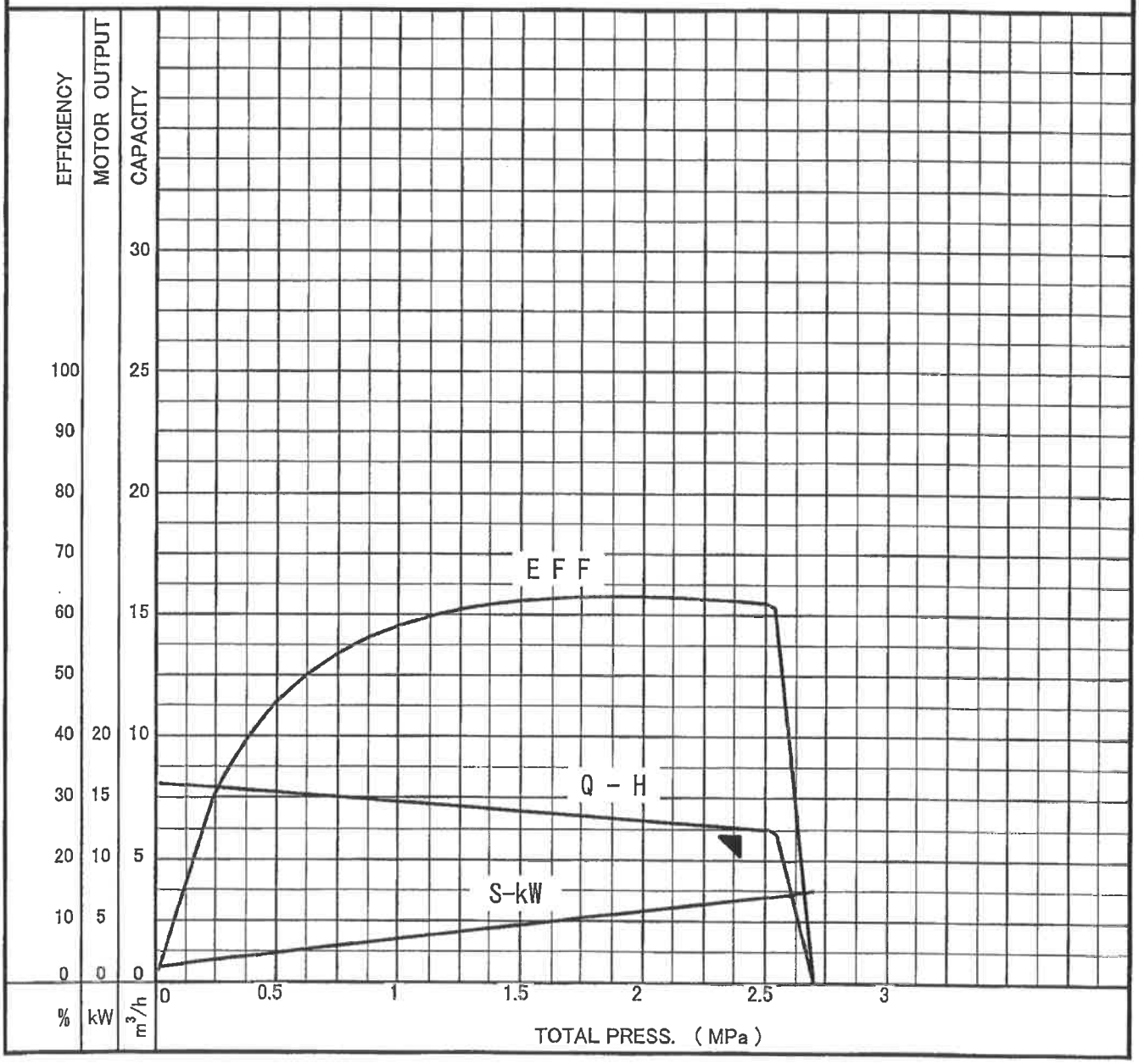
PUMP NAME 仕 様 名	PUMP MODEL 型 番	GAUGE SET POSITION 計器位置	NOTE 備 考
L.O. STBY PUMP	ALGT-65	L	
R/G L.O. STBY PUMP	ALGT-50H	L	
COOL.F.W. PUMP	CRD-100-2	L	

ESTIMATED PERFORMANCE CURVE

Page :

Date : 15 Feb. 2016

Customer	:	DAIHATSU DIESEL MFG.CO.,LTD				
Project	:					
Service As	:	R/G L.O. STBY PUMP				
Pump Model	:	ALGT-50H				
Bore	:	65×50				
Capacity	:	6	m^3/h	Total Press. :	2.4	MPa
Output	:	11	kW	950	min-1	
		50	Hz	415	V	
Test Liquid	:	L.O.				
H. Test Press.	:	4.0	MPa			
Remarks	:					



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R/G STANDBY L.O. PUMP

STARTER PANEL

DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

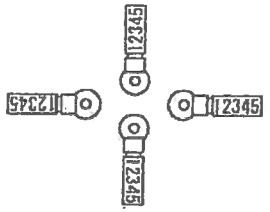
JAPAN

SHIP NO.		DRAWN BY	M. Tokumoto
TYPE	6DKM-20eF(L)	CHECKED BY	S. Tajima
LIST NO.	AQP20008445A		
DATE	2016/6/8	APPROVED BY	S. Furuse
REVISION			

PDK20-0174-5

GENERAL SPECIFICATION

RULE & REGULATION	☐JG ☐NK ☐ABS ☐LR ☑BV ☐JIS・JEM
DRAWING & DOCUMENT	☑ENGLISH
NAME PLATE	LETTER/ ☐JAPANESE ☑ENGLISH ☐CHINESE GROUND-LETTER/ ☑WHITE-BLACK ☐WHITE-RED ☐RED-WHITE MATERIAL/ ACRYL RESIN
MATERIAL	☑STEEL ☐SUS ☐ALUMINUM
PROTECTION ANBIENT TEMP INSTALLATION	I P/ IP22 TEMP/ 45℃ ☐FLOOR MOUNTING TYPE ☑WALL MOUNTING TYPE ☐WALL FLUSH TYPE
PAINTING COLOR (MUNSELL NO.)	OUTSIDE/ ☑2.5G7/2 ☐7.5BG7/2 ☐N-3.0 ☐N-2.5 ☐(5R4/13) INSIDE/ ☑2.5G7/2 ☐7.5BG7/2 ☐N-3.0 ☐N-2.5 ☐(5R4/13) GLOSS/ ☐POLISHED ☑SEMI - POLISHED ☐NOT POLISHED ☐( )
BUS BAR	MATERIAL/ ~~COPPER (NICKEL PLATING)~~
INTERNAL WIRING COLOR OF WIRING CLOTHING	POWER CIRCUIT / BLACK CONTROL CIRCUIT/ GRAY EARTH CIRCUIT / YELLOW IS SPIRAL IN GREEN OR GREEN
COLOR OF INSULATION CAP	RED / R - U WHITE / S - V BLUE / T - W ~~GREEN/ R - U YELLOW/ S - V BROWN/ T - W~~ RED / P BLUE / N
CABLE SIZE	POWER CIRCUIT / MORE THAN 2.5mm² (SCP) CONTROL CIRCUIT/ AWG18, AWG16 (Style 1015) EARTH CIRCUIT / MORE THAN 2mm² (KIV)
TERMINAL PROCESSING SOLDERLESS TERMINAL	POWER CIRCUIT / INSULATION CAP(COLOR) CONTROL CIRCUIT/ WHITE MARK TUBE(CAP NOTHING) POWER CIRCUIT / R TYPE CONTROL CIRCUIT/ R TYPE
PHASE ARRAGEBENT	VIEWING FROM R-U S-V T-W (LEFT→RIGHT、UPPER→LOWER、NEAR→FAR) PANEL FRONT / P N (LEFT→RIGHT、UPPER→LOWER、NEAR→FAR)
SOURCE	POWER/ ☑AC415V 3Φ 50Hz ☐AC220V 3Φ 50Hz ☐AC220V 1Φ 50Hz ☐DC24V ☐AC24V ☐AC100V 3Φ 50Hz CONTROL/ ☐AC440V ☐AC220V ☑AC100V ☐DC24V ☐AC24V
OUTSIDE TERMINAL	☑PROVIDED ☐NOT PROVIDED
CONNECTION DIAGRAM	☐CARD CASE ☑VINYL BAG

FUSE	FUJI ☐FCF2 50kA UTSUNOMIYA(CELLOLITE) ☑UC-1 50kA ☐NC-1 20kA
ILLUMINATION LAMP	☑BA9S 6.3V 1W ☐E-12 18V 2W ☐E-10 30V 1W ☐BA9S 30V 1W ☐E-12 30V 2W ☐E-10 100V 1W ☐LED DC24V
SPARE PARTS	☑SUPPLIED ☑VINYL BAG ☐VINYL BOX ☐STEEL BOX (WITH KEY) ☐2.5G7/2 ☐7.5BG7/2 ☐NOT SUPPLIED ☐( )
WIRE INTRODUCTION	☐UPPER ☑LOWER ☐SIDE ☐BACK ☐GLAND ☑COMING
BOARD LIGHTING FLUORESCENT LIGHT SPARE LIGHT	☐PROVIDED ☑NOT PROVIDED (AC100V 20W) ☐PROVIDED ☑NOT PROVIDED (DC24V 20W)
METER	MAKER:DAIICHI ELECTRONICS(COVER BLACK FRAME) ☐110x110 ☐120x100 ☑60x60 ☐80x80 ☐96x96 ☐100x100
INSULATION OHMMETER	DAIICHI ELECTRONICS(COVER BLACK FRAME) ☐PROVIDED ☑NOT PROVIDED
UNIT	SI (METRIC SYSTEM)
CONNECTION FOR LOAD	☐DIRECT FOR MCB ☑WITH TB
CHANNEL BASE (ANGLE BASE)	☐PROVIDED ☑NOT PROVIDED
DIRECTION OF THE MARK TUBE	

CYUGOKU ELECTRIC SERVICE CO., LTD. SHIMONOSEKI JAPAN	STARTER PANEL	GENERAL SPECIFICATION	DWG. NO. 280603-01

部品表 PARTS LIST

P.

番号 No.	記号 SYMBOL	型式 TYPE	仕様 SPPECIFICATION	数量 QT.	メーカー MAKERS	備考 REMARKS
			STAND-BY R/G L.O.PUMP STARTER PANEL			
1	MCB	S100-NF	100AF/50A 3P PM	1x2	TERASAKI	MOULDED CIRCUIT BREAKER
2			+T2PM12E3	1x2	TERASAKI	
3	A	PSK-60C	0-30-90A (R:23A)	1x2	DAIICHI	AMMETER
4	TR	NTP-10	AC415/100V 1Φ 50Hz 100VA	1x2	HISANO	TRANSFORMER
5	88M	SC-5-1+SZ-A20	COIL:AC100V 50Hz 1a1b+2a	1x2	FUJI	ELECTROMAGNETIC CONTACTOR
6	88D	SC-5-1+SZ-A20	COIL:AC100V 50Hz 1a1b+2a	1x2	FUJI	ELECTROMAGNETIC CONTACTOR
7	88S	SC-0	COIL:AC100V 50Hz 1a	1x2	FUJI	ELECTROMAGNETIC CONTACTOR
8	51	TR-N2H	18～26A 1a1b	1x2	FUJI	THERMAL RELAY
9	WL	DR30DOL-H4W	AC100/6.3V 1W BA9S	1x2	FUJI	INDICATION LAMP
10	GL	DR30DOL-H4G	AC100/6.3V 1W BA9S	1x2	FUJI	INDICATION LAMP
11	RL	DR30DOL-H4R	AC100/6.3V 1W BA9S	1x2	FUJI	INDICATION LAMP
12	3C	AR30FOR-10G	600V 10A 1a	1x2	FUJI	PUSHBUTTON SWITCH
13	3T	AR30FOR-01R	600V 10A 1b	1x2	FUJI	PUSHBUTTON SWITCH
14	43	AR30PR-222B	600V 10A 2a2b	1x2	FUJI	SELECTOR SWITCH
15	19	PM4S-A2C10M-AC120V	AC100V 0～10min	1x2	PANASONIC	TIMER
16	51X	RU2S-C-A100	AC100V 10A 2c	1x2	IDEC	RELAY
17	F	FCF2-3	500V 3A	5x2	FUJI	FUSE
18	FH	FH-30	600V 30A	5x2	YOSHIDA	FUSE HOLDER

部品表 PARTS LIST | CYUGOKU ELECTRIC SERVICE CO.,LTD. SHIMONOSEKI JAPAN | DWG No. 280603-03

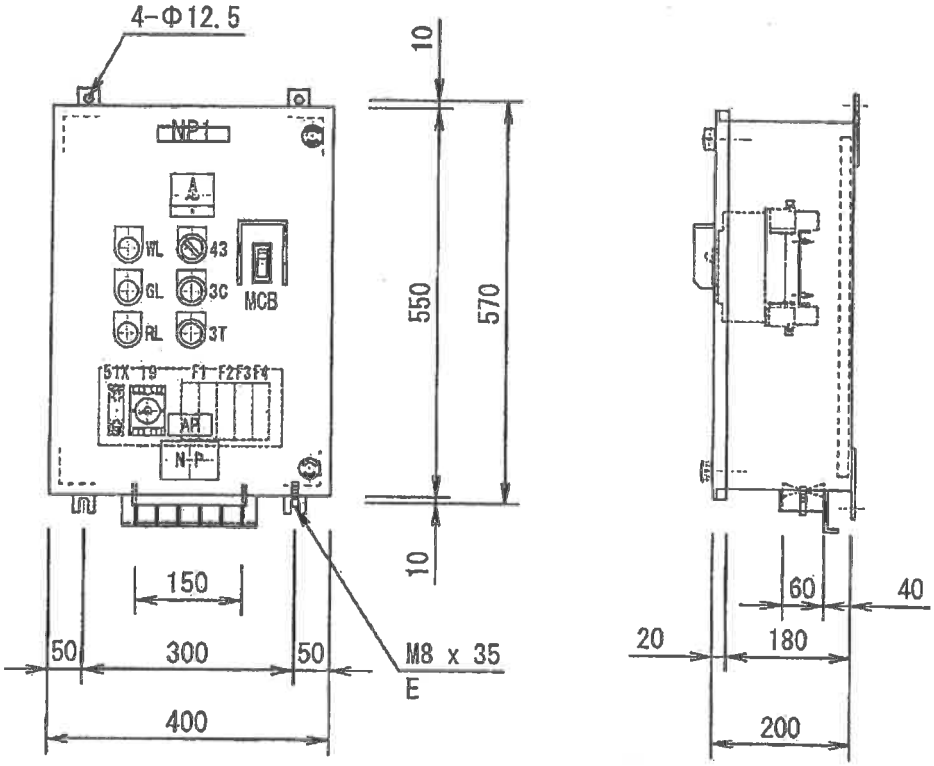
銘板表
NAME PLATE TABLE

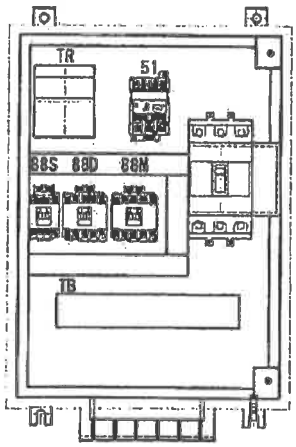
記号 No.	数量 Q'TY	寸法 TYPE	銘板名称 LETTERS ON NAME PLATE / 回路名称 CIRCUIT ON NAME	フレーム/セット FRAME/SET	ケーブルサイズ CABLE SIZE	色 COLOR 文字色 LETTER	色 COLOR 地色 GROUND	備考 REMARKS
NP1	2	100x20	STAND-BY R/G L.O.PUMP			B	W	
WL	2	Φ30	SOURCE			B	W	
GL	2	Φ30	RUNNING			B	W	
RL	2	Φ30	OVER LOAD			B	W	
43	2	Φ30	MANU - AUTO			B	W	
3C	2	Φ30	START			B	W	
3T	2	Φ30	STOP			B	W	
AP	2		APPROVED MARK					
NP	2		MAKER NAME PLATE					

B:BLACK
W:WHITE
R:RED
MW:MILK WHITE

銘板表 NAME PLATE LIST	CYUGOKU ELECTRIC SERVICE CO.,LTD. SHIMONOSEKI JAPAN	DWG No. 280603-07

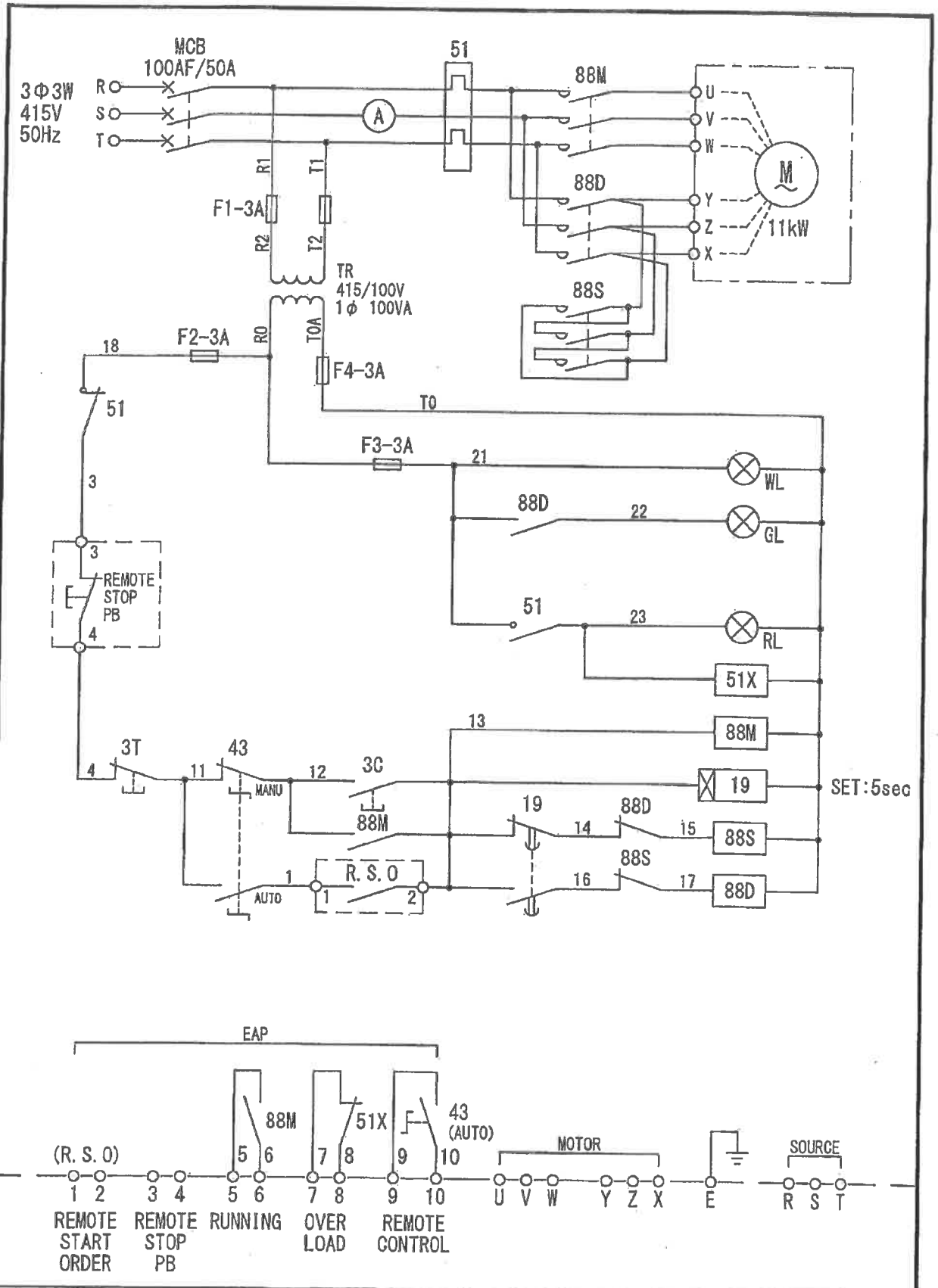
NP1	STAND-BY R/G L.O. PUMP





PANEL PAINTED COLOR
OUT & INSIDE : 2.5G7/2
(MUNSEL No.) SEMI-POLISHED
WAIGHT PLANNED : 30 kg
FINISHED : kg
MATERIAL : STEEL ··· 2.3t
PROTECTION : IP22
Q'ty : 2

外形寸法図 OUTLINE VIEW	CYUGOKU ELECTRIC SERVICE CO.,LTD. SHIMONOSEKI JAPAN	DWG No. 280603-08



NO.	CIRCUIT NAME	NO.	CIRCUIT NAME
ST	STAND-BY R/G L.O.PUMP		

STARTER CIRCUIT	CYUGOKU ELECTRIC SERVICE CO.,LTD. SHIMONOSEKI JAPAN	DWG. No. 280603-09

AQP20008445A (6/6)

2017/05/09 TECHNICAL DEPARTMENT

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LT F.W. PUMP

TYPE:CRD-100-2

DAIHATSU DIESEL MFG.CO.,LTD.

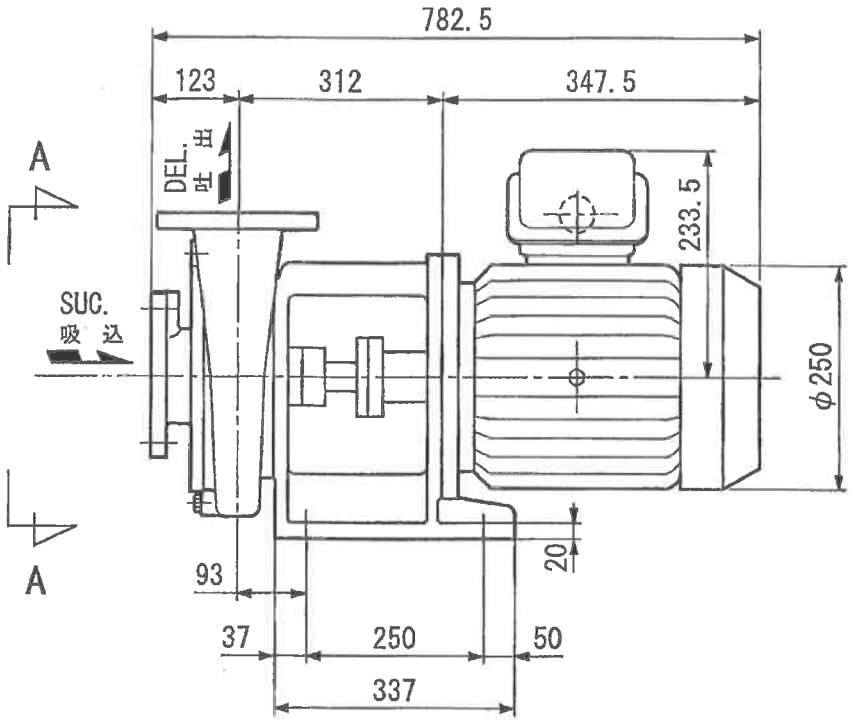
TECHNICAL DEPARTMENT

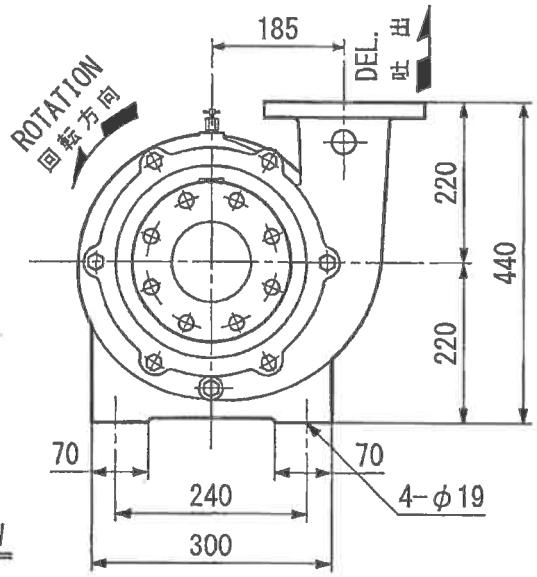
JAPAN

SHIP NO.		DRAWN BY	M. Fukumoto
TYPE	6DKM-20eF(L)	CHECKED BY	H. Fujimoto
LIST NO.	AQP10008529A		
DATE	2016/6/3	APPROVED BY	S. Kuwa
REVISION			

PDK20-0174-5

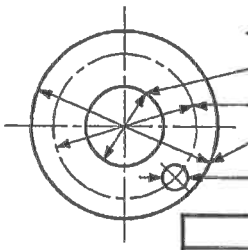
NO. 3





A-A ARROW VIEW

SUC. & DEL. FLANGE
吸込及吐出フランジ



JIS 5K SUC. 吸込	JIS 5K DEL. 吐出
φ100	φ100
φ165	φ165
φ200	φ200
8-φ19	8-φ19

MASS 質量		
PUMP ポンプ	75	kg
MOTOR 電動機	48	kg
TOTAL 全質量	123	kg

ACCESSORIES FOR EACH 1 PUMP 付属品（ポンプ一台ニ付）		SPECIFICATION 仕様	
DESCRIPTION 記	Q'TY 数量	SUCTION BORE 吸込口径	100 mm
COUPLING カップリング	1	DELIVERY BORE 吐出口径	100 mm
AIR COCK エアーコック	1	CAPACITY 容量	40 m^3/h
DRAIN JOINT WITH HOSE ドレン継手及びホース	1set	TOTAL HEAD 全揚程	20 m
PRESSURE GAUGE 圧力計	1	SUCTION HEAD 吸込揚程	m
COMPOUND GAUGE 連成計	1	REVOLVING SPEED 回転速度	1450 min^{-1}
GAUGE ROOT COCK 計器元コック	2	HYD. TEST PRESS. 耐圧試験	0.4 MPa
COUNTER FLANGE SUC. 100A 相フランジ DEL. 100A	1set	MOTOR OUTPUT 電動機出力	5.5kW , 4P
		POWER SOURCE 電源	415V, 50Hz, 3ph
		REMARKS 備考 WORKING TEMP.= ℃	

PUMP NAME 名称	COOL. F. W. PUMP
PUMP MODEL 型式	CRD-100-2
QUANTITY 台数	1
RULE 検査	BV
ANGLE PROJ. 角法	3 rd
SCALE 尺度	FREE
DRAWING NO. 図面番号	98DG0716

NANIWA PUMP MFG.CO.,LTD.
TECH. DEPARTMENT DESIGN SECTION
株式会社 浪速ポンプ製作所

naniwa pump

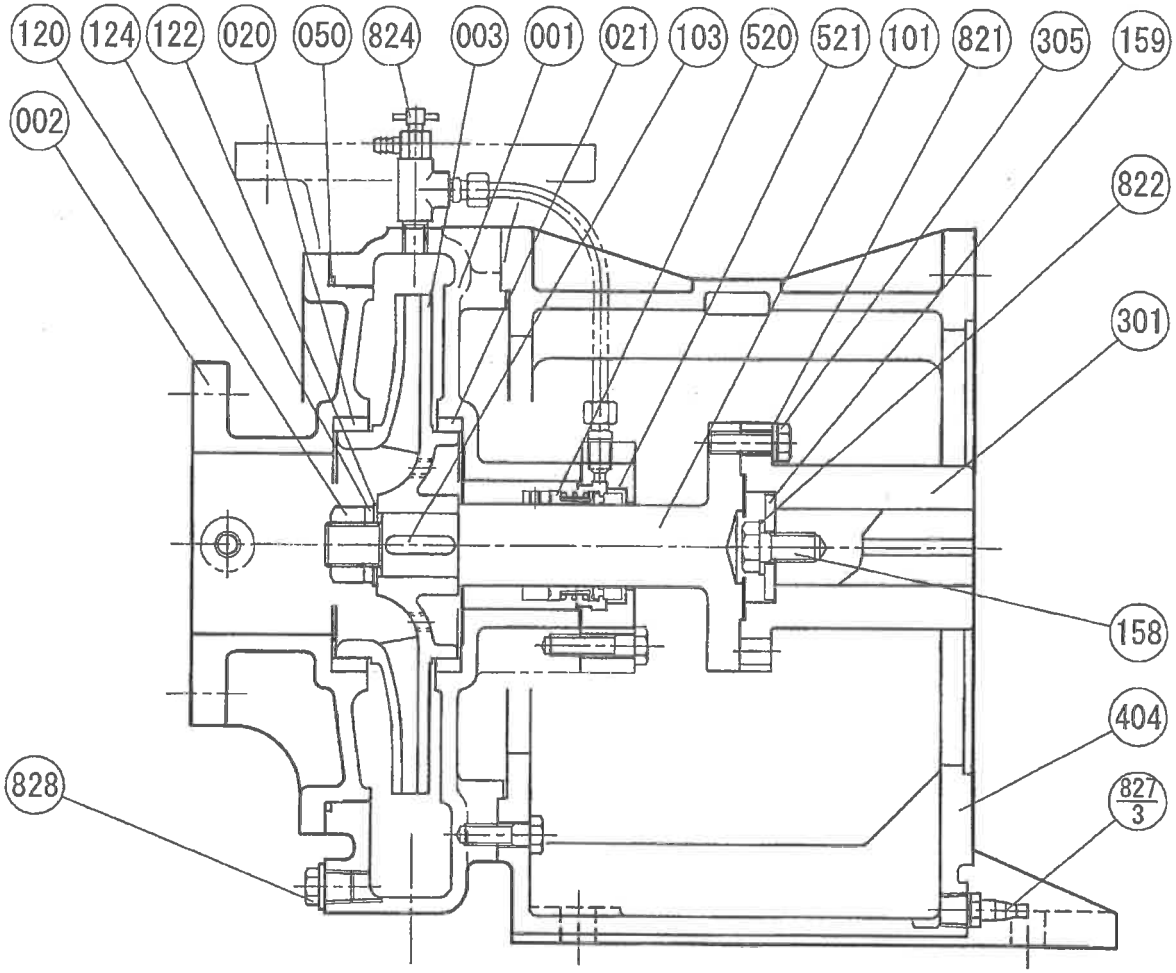
SECTIONAL DRAWING 構造断面図

HORIZONTAL CENTRIFUGAL PUMP 横型電動渦巻式

MODEL:型 式 CRD-100-2

DRAWING NO. A1DS0248

DATE 19 MAR. 2001



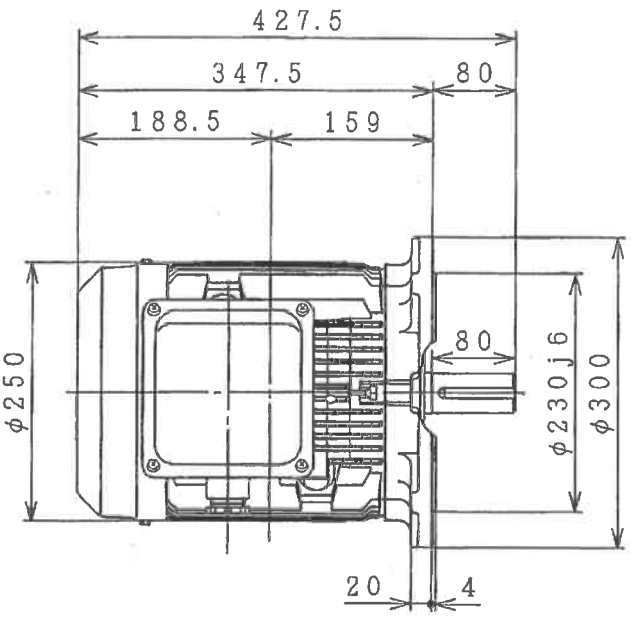
PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質		Q'TY 数量	PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質		Q'TY 数量
158	COUPLING SET BOLT 軸継手セットボルト	LOW CARBON STEEL WIRE 軟鋼線材	SWRM	1					
124	IMPELLER WASHER 羽根車ワッシャー	STAINLESS STEEL ステンレス鋼	SUS304	1	828	DRAIN PLUG ドレンプラグ	COPPER ALLOY ROD 銅合金棒	C3604	1
122	PLATE WASHER プレートワッシャー	STAINLESS STEEL ステンレス鋼	SUS304	1	827/3	DRAIN HOSE JOINT ドレンホース継手	COPPER ALLOY ROD 銅合金棒	C3604	1
120	IMPELLER NUT 羽根車ナット	COPPER ALLOY ROD 銅合金棒	C3604	1	824	AIR COCK 空気抜きコック	COPPER ALLOY ROD 銅合金棒	C3604	1
103	IMPELLER KEY 羽根車キー	STAINLESS STEEL ステンレス鋼	SUS304	1	822	SPRING LOCK WASHER バネ回り止めワッシャー	HIGH CARBON STEEL WIRE 硬鋼線材	SWRH	1
101	SHAFT 主軸	STAINLESS STEEL ステンレス鋼	SUS304	1	821	COUPLING WASHER 軸継手ワッシャー	HIGH CARBON STEEL WIRE 硬鋼線材	SWRH	1 SET
050	O-RING Oーリング	RUBBER 合成ゴム		1	521	SEAL COVER シールカバー	CAST BRONZE 青銅鋳物	CAC406	1
021	CASING RING ケーシングリング	CAST BRONZE 青銅鋳物	CAC406	1	520	MECHANICAL SEAL メカニカルシール			1 SET
020	CASING RING ケーシングリング	CAST BRONZE 青銅鋳物	CAC406	1	404	PUMP FRAME ポンプフレーム	CAST IRON 鋳鉄	FC200	1
003	IMPELLER 羽根車	CAST BRONZE 青銅鋳物	CAC406	1	305	COUPLING BOLT 軸継手ボルト	LOW CARBON STEEL WIRE 軟鋼線材	SWRM	1 SET
002	CASING COVER ケーシングカバー	CAST IRON 鋳鉄	FC200	1	301	COUPLING 軸継手	CAST IRON 鋳鉄	FC200	1
001	CASING ケーシング	CAST IRON 鋳鉄	FC200	1	159	COUPLING SET WASHER 軸継手セットワッシャー	MILD STEEL 一般構造用圧延鋼	SS400	1

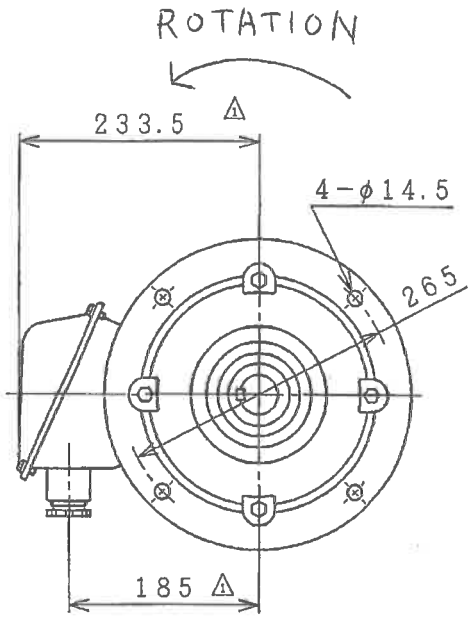
2017/05/09 TECHNICAL DEPARTMENT

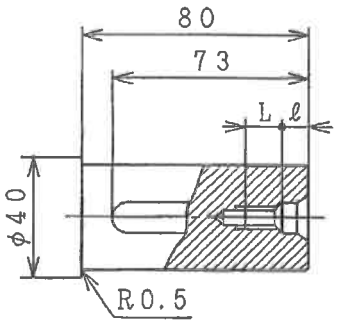
CAD

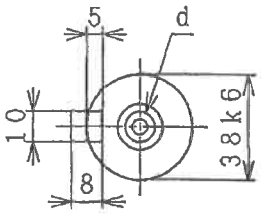
三相誘導電動機
THREE-PHASE INDUCTION MOTOR

出力 OUTPUT (kW)	形 TYPE	式 FORM	定格 RATING	電圧 VOLTAGE (V)	周波数 FREQ. (Hz)	同期回転速度 SYNC. SPEED (min^{-1})	極数 POLES	耐熱クラス TH. CLASS	質量 MASS (kg)
5.5	VTFO	KK	S1	415	50	1500	4	F	48









		d	L	ℓ
	A	M8	15	9
	B	M10	20	9
	C	M12	25	9
	D	M16	32	10
○	E	M12	22	9

SHIELD BEARING	6306	6308
RULE	BV	
CONNECTION	DOL	
CABLE		
GLAND	A25f	

PROJECTION

用途 USE

付属品 ACCESSORY

備考 NOTICE

御注文主 CUSTOMER (067170)
株式会社浪速ポンプ製作所 殿

数量 QTY.	受注番号 ORDER No.	作業番号 WORK No.	REV.
			2

REGD 2005.11.28

DWN.	S. Honma	Jun-21-2004
CHKD.	T. Abe	Jun-21-2004
APPD.	T. Oku	Jun-21-2004

寸法図
DIMENSIONS

Hitachi Industrial Equipment Systems Co., Ltd.

NARASHINO WORKS DWG. No. 3244R845070

SH.

枠番 FRAME SIZE MVTFOD-FF265 (132S)

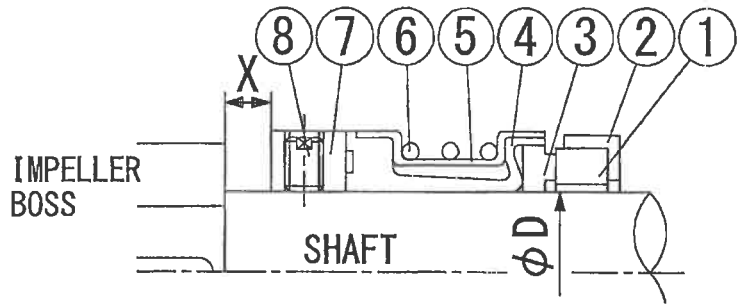
naniwa pump

SECTIONAL DRAWING

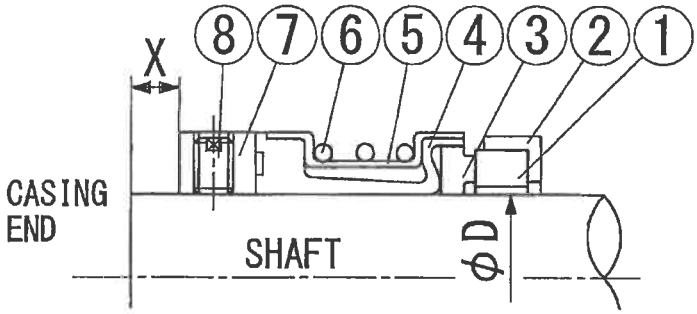
MECHANICAL SEAL

DRAWING NO. 98DG0226

DATE 18 MAY 1998



FIG, 1



FIG, 2

AQP10008526A (2/8)

NO.	PUMP NAME	MODEL	SHAFT DIA φ	SET DIMENSION "X"	FIG, NO.
1	L. O. AUX. PUMP	ALGT-65	35	69	2
2	R/G L. O. AUX. PUMP	ALGT-50H	30	58	2
3	COOL. F. W. PUMP	CRD-100-2	45	36	1

PART NO.	NAME OF PART	Q'TY
1	FLOATING SEAT	1
2	SEAT CAP	1
3	SEAL RING	1
4	BELLOWS	1
5	RETAINER	1
6	COIL SPRING	1
7	COLLAR	1
8	SET SCREW	1SET

AQP10008529A (5/8)

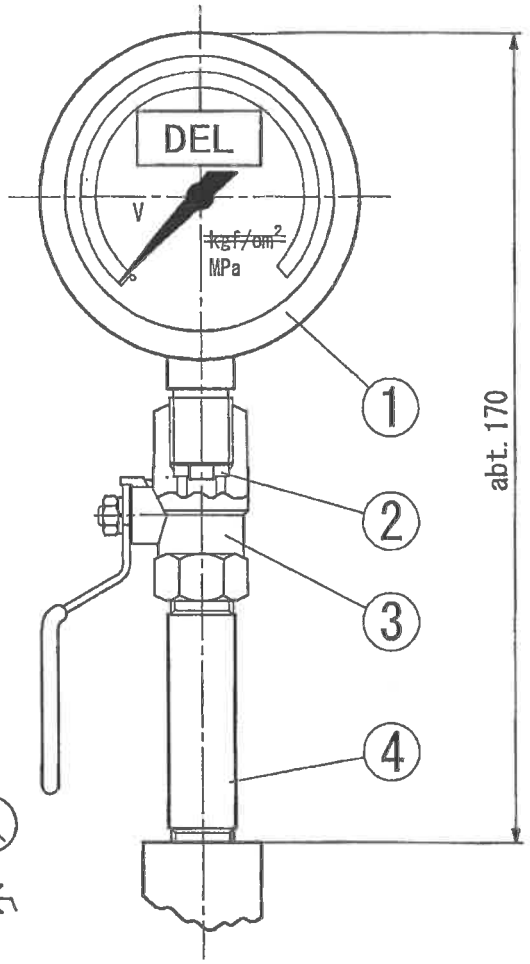
2017/05/09 TECHNICAL DEPARTMENT

naniwa pump

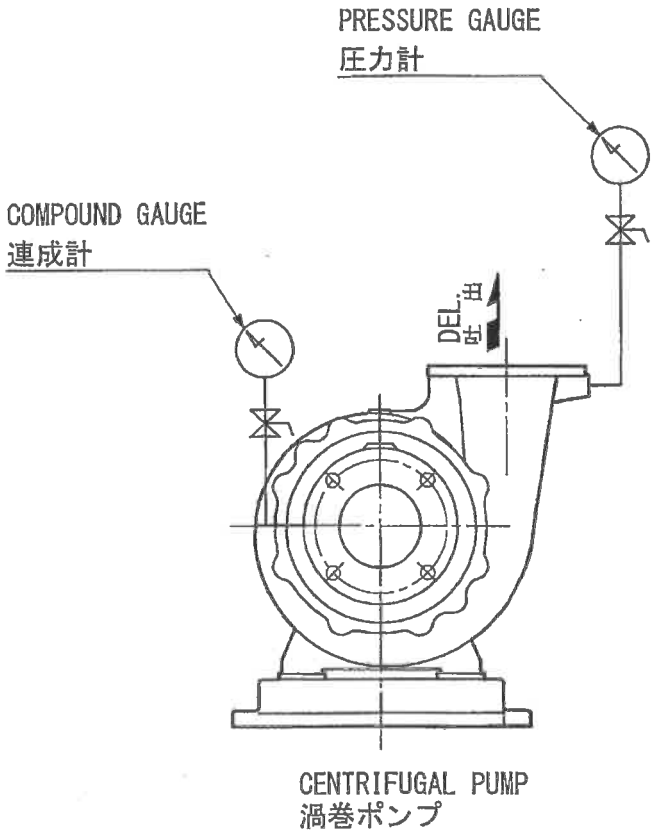
GAUGE & GAUGE ROOT COCK
計器 及 計器元コック
HORIZONTAL PUMP TYPE
FITTING ON PUMP BODY 横型本体付タイプ

DRAWING NO. A4DH0009
DATE 4 FEB. 2004

JIS B7505



MIDDLE PRESSURE COCK
1.6 MPa and less.
中圧用コック 1.6 MPa 以下

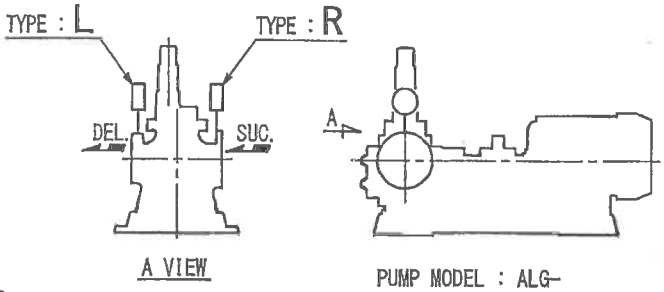


TYPE 型式	AVT 1/4″ × φ60
UNIT 単位	~~kgf/cm²~~, MPa
MASS 質量	0.2 kg

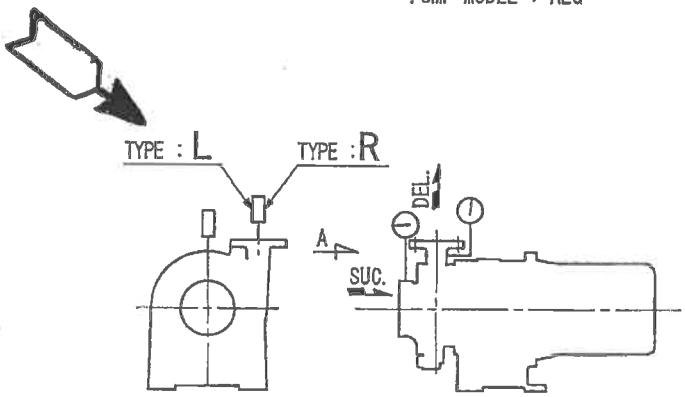
PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質	Q'TY 数量	PART NO. 品番	NAME OF PART 部品名称	MATERIAL 材質	Q'TY 数量
2	GASKET ガスケット	RUBBER 合成ゴム	2	4	GAUGE PIPE 計器パイプ	STAINLESS STEEL ステンレス鋼	2
1	GAUGE 計器		2	3	GAUGE ROOT COCK 計器元コック	BRASS 黄銅鋳物	2

naniwa pump	INDICATION LIST OF GAUGE SET POSITION HORIZONTAL PUMP TYPE FITTING ON PUMP	計器取付位置一覧 横型ポンプ ポンプ本体付き	DRAWING No. B2DG0421 DATE 30 OCT. 2012

CUSTOMER 御注文主 :
PROJECT 向 け 先 :
S.NO. 船 番 :



PUMP MODEL : ALG-



PUMP MODEL : BHR, CRD

Please indicate your required position in the blank
Remarks : 1. Please indicate "H" for hull fitting Gauge Boards. In this case, setting and piping for Gauge Boards are worked by shipyard. Therefore the Gauge Boards will be sent separately from pump.
2. Gauge Board size and the setting method are in accordance with maker's standard.

納入仕様書返却の際、空欄にご要求の計器方向を記入下さい。
備考 : 1. 計器が船体付けの場合は空欄に"H"を記入下さい。その場合、計器の設置及び配管は造船所殿にて施工となりますので、計器はポンプ本体とは別送となります。
2. 計器取付け寸法及び取付け要領はメーカー標準通りとなります。

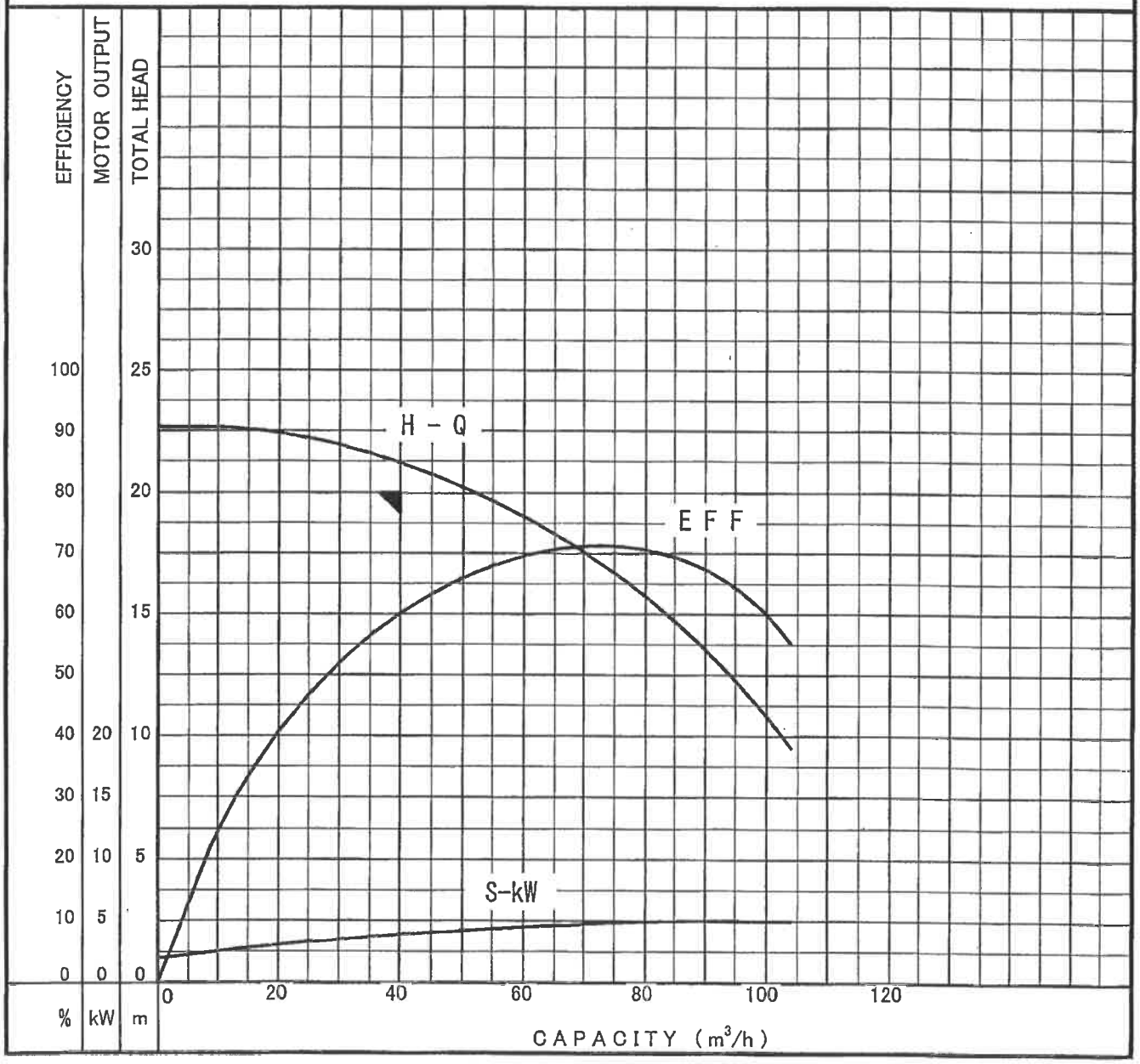
PUMP NAME 仕 様 名	PUMP MODEL 型 番	GAUGE SET POSITION 計器位置	NOTE 備 考
L.O. AUX. PUMP	ALGT-65	L	
R/G L.O. AUX. PUMP	ALGT-50H	L	
COOL. F.W. PUMP	CRD-100-2	L	

ESTIMATED PERFORMANCE CURVE

Page :

Date : 15 Feb. 2016

Customer	:	DAIHATSU DIESEL MFG.CO.,LTD				
Project	:					
Service As	:	COOL.F.W.PUMP				
Pump Model	:	CRD-100-2				
Bore	:	100×100				
Capacity	:	40	m^3/h	Total Head :	20	m
Output	:	5.5	kW	1450	min-1	
		50	Hz	415	V	
Test Liquid	:	FRESH WATER				
H. Test Press.	:	0.4	MPa			
Remarks	:					



2017/05/09 TECHNICAL DEPARTMENT

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LT-F.W. COOLER

DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

JAPAN

SHIP NO.		DRAWN BY	M. Tokuda
TYPE	6DKM-20eF(L)	CHECKED BY	A. Tajima
LIST NO.	AQP10008530A		
DATE	2016/6/3	APPROVED BY	S. Tsuru
REVISION			

PDK20-0174-5

PRINCIPAL ITEMS OF PLATE TYPE HEAT EXCHANGER		DOC. No.	MAP2976
MFG. No.	MODEL RX-135B-NPM-81	QUANTITY	2 UNIT(S) /SHIP
AP-2976	EQPT. NAME LT-FW COOLER		

1.

APPLICABLE CODE		BV
SHIP No.		
HEAT TRANSFER AREA[m^2]		20.15
HEAT EXCHANGED [kW]		744
MASS[kg]	DRY	340
	WET	400

2. PARTICULAR SHEET OF BASIC PLAN

FLUID NAME	F.W.	S.W.
INLET TEMP[°C]	54.00	32.00
OUTLET TEMP[°C]	38.00	45.28
FLOW RATE[m3/h]	40.00	50.00
PRESSURE DROP[MPa]	0.026	0.041
DESIGN PRESS[MPaG]	0.50	0.50
DESIGN TEMP. [°C]	90.00	60.00
TEST PRESS(HYDR.) [MPaG]	0.75	0.75
HOLDING VOLUME[lit]	28.9	29.2

3. TOOL (PER UNIT)

NAME	No. REQ'D
SPANNER	1
BRUSH FOR CLEANING	——
GAUGE PLATE	——
CHAIN	——
GAS CHARGING TOOLS	——

N2 GAS CHARGING PRESS[MPa]	——
WORKING PRESS[MPa]	——

4. ATTACHMENT (PER UNIT)

NAME	NO. REQ'D	
AIR VENT	2	
DRAIN BLOW	2	
NAME PLATE	1	
THERMOMETER	L type	4
	I type	——
STRAINER	1	
DUPLEX GAUGE	——	
ACCUMULATOR	——	

5. PAINTING

MUNSELL No. 2.5GBG7/2

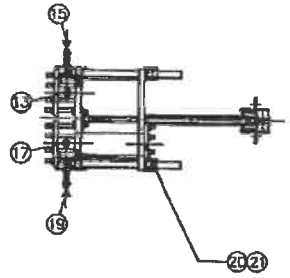
APPROVED BY	S. HATANAKA	MAY, 31, 2016
REVIEWED BY	K. SHIMODAN	MAY, 31, 2016
PREPARED BY	N. MURATA	MAY, 31, 2016

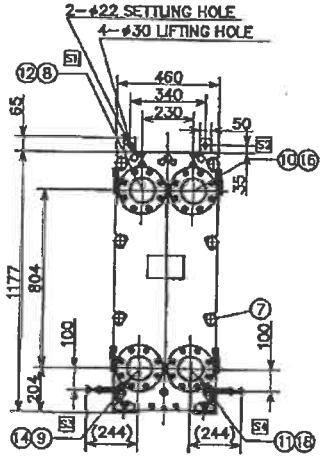
HISAKA WORKS, LTD.

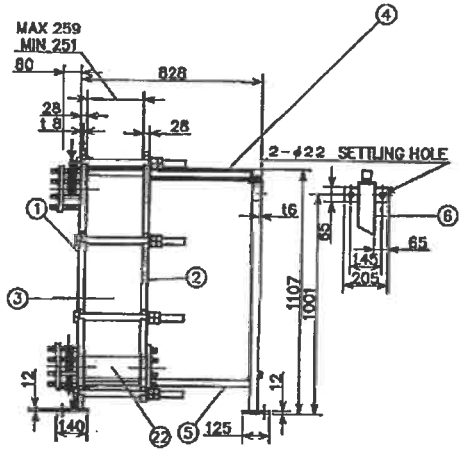
MAY, 31, 2016

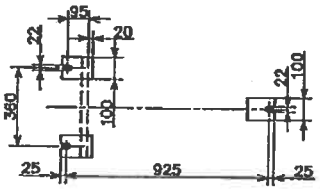
2017/05/09 TECHNICAL DEPARTMENT

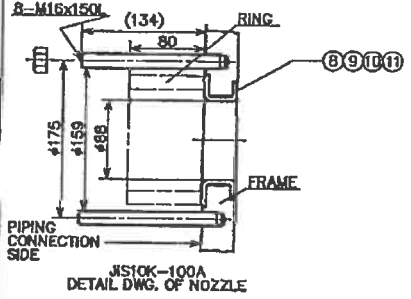
ESTIMATE / AP29 / AP2976 / BAP2976.dwg











NOZZLE ORIENTATION			CONNECTION
S1	F.W.	INLET	JIS10K-100A INCLUDE STUD B&N
S3	F.W.	OUTLET	JIS10K-100A INCLUDE STUD B&N
S4	S.W.	INLET	JIS10K-100A INCLUDE STUD B&N
S2	S.W.	OUTLET	JIS10K-100A INCLUDE STUD B&N

MASS(EMPTY)	340	kg	MASS(FILLED)	400	kg
ATTACHMENT	RATCHET SPANNER	C.S.	M30		1 PC
	SET BOLT & NUT	SS400	M20×80L		3 SETS

SHIP NO.	
EQUIP. NAME	LT-FW COOLER
ITEM NO.	6DKM-20eF(L)
APPLICABLE CODE	BV

PART 7 FINISH: COATING OF ZINC & TIGHTENING BOLT WITH COVER

NO.	PART NAME	REQ'D	MATERIALS	REMARKS
22	STRAINER	1	SUS316	PUNCHING METAL
21	BLIND FLANGE	1	SS400	JIS10K-100A
20	E.NOZZLE	1	NBR	JIS10K-100A INCLUDE STUD BANAPACKING
19	DRAIN BLOW	1	C6872	15A(PT 1/2) WITH G.V.(SC-6)
18	RING	1	CAC703	WITH COUNTER FLANGE(SS400)
17	AIR VENT	1	C6872	15A(PT 1/2) WITH G.V.(SC-6)
16	RING	1	CAC703	WITH COUNTER FLANGE(SS400)
15	DRAIN BLOW	1	SGP	15A(PT 1/2) WITH G.V.(SC-6)
14	RING	1	STKM13A	WITH COUNTER FLANGE(SS400)
13	AIR VENT	1	SGP	15A(PT 1/2) WITH G.V.(SC-6)
12	RING	1	STKM13A	WITH COUNTER FLANGE(SS400)
11	S.NOZZLE	1	NBR	
10	S.NOZZLE	1	NBR	
9	S.NOZZLE	1	NBR	
8	S.NOZZLE	1	NBR	
7	TIGHTENING BOLT & NUT	8 SETS	SS400/SS400	M30×475L
6	GUIDE BAR SUPPORT	1	SS400	
5	LOWER GUIDE BAR	1	SUS304	
4	UPPER GUIDE BAR	1	SUS304	
3	PLATE	81	TP270	
2	E.FRAME	1	SS400	
1	S.FRAME	1	SS400	

REV.	DATE	REVIEWED BY

PURCHASER						
CUSTOMER						
MFG. NO.			QUANTITY	2	SHEET NO.	2
DESIGNED	N.MURATA	MAY,31,2016	SCALE	NONE	THIRD ANGLE PROJ.	
DRAWN	N.MURATA	MAY,31,2016	NAME	RX-135B-NPM-81 (H18)		
CHECKED	K.SHIMODAN	MAY,31,2016	ASSEMBLY DWG. OF PLATE TYPE HEAT EXCHANGER			
APPROVED	S.HATANAKA	MAY,31,2016	DWG. NO.	BAP2976	0	

株式会社 日阪製作所 HISAKA WORKS, LTD.

AP2976 VER.200604

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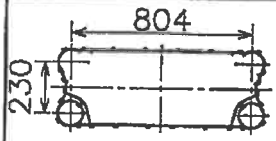
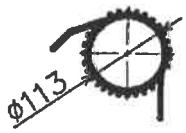
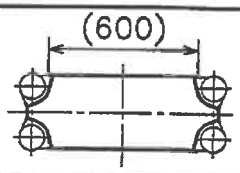
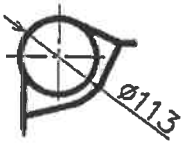
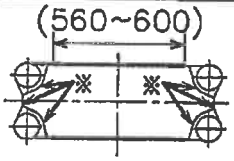
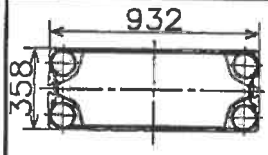
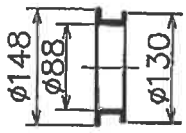
AQP10008530A (3/5)

2017/05/09 TECHNICAL DEPARTMENT

(3)

MFG.No.	–	PARTS LIST	Doc No.	PARAP2976
HISAKA PLATE HEAT EXCHANGER			SHIP No.	
MODEL RX-135B-NPM-81		2UNIT(S)/SHIP	ITEM No.	6DKM-20eF(L)

No.	PARTS NAME	SKETCH (ØDIMENSION SHOWS CENTER OF DIAMETER.)	MATERIAL	QUANTITY WORKING PER 1 UNIT	QUANTITY SPARE PER 2UNIT(S)	REMARKS
1	PLATE GASKET		G-NBR	80	–	
2	D.PLATE GASKET DA		NBR	2	–	SET ON D.PLATE
			NBR	1	–	SET ON E.PLATE BACK SIDE
3	D.PLATE GASKET DB		NBR	2	–	SET ON D.PLATE
			–	–	–	SET ON E.PLATE BACK SIDE
4	D.DISTANCE PIECE		NBR	1 SET	–	SET ON D.PLATE
5	E.PLATE GASKET		NBR	3	–	SET ON E.PLATE BACK SIDE
6	E.DISTANCE PIECE		NBR	1 SET	–	SET ON E.PLATE BACK SIDE ※If E.Nozzle installed
7	PLATE		TP270	81	–	(RX-135A) WITH GASKET
				–	–	(–) WITH GASKET
				–	–	(RX-135A) WITHOUT GASKET
				–	–	(–) WITHOUT GASKET
8	RUBBER BOOTS		NBR	2	–	SET ON NOZZLE FOR F.W. SIDE
			NBR	3	–	SET ON NOZZLE FOR S.W. SIDE

APPROVED BY	S.HATANAKA	MAY,31,2016	HISAKA WORKS,LTD. OSAKA,JAPAN
PREPARED BY	T.OHNOKI	MAY,31,2016	

HEADS VER.200604

RX-13

AQP10008530A (4/5)

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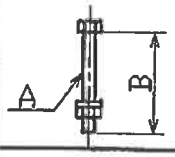
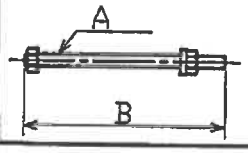
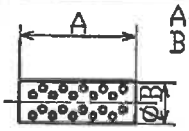
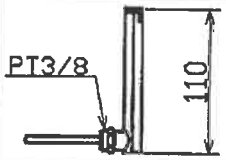
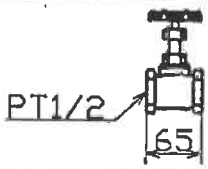
2017/05/09 TECHNICAL DEPARTMENT

(4)

MFG.No.	-	ACCESSORIES LIST	Doc No.	ACTAP2976
HISAKA PLATE HEAT EXCHANGER			SHIP No.	
MODEL RX-135B-NPM-81		2UNIT(S)/SHIP	ITEM No.	6DKM-20eF(L)

No.	PARTS NAME	SKETCH (ØDIMENSION SHOWS CENTER OF DIAMETER)	MATERIAL	QUANTITY WORKING PER 1 UNIT	QUANTITY SPARE PER 2UNITS	REMARKS
1	SET BOLT,NUT & WASHER		SS400	3 SET	–	A=M20 B=80
2	TIGHTENING BOLT & NUT		SS400/SS400	8	–	A= M30 B= 500
3	INNER STRAINER		-	–	–	PUNCHING METAL FOR F.W. SIDE
			SUS316	1	–	PUNCHING METAL FOR S.W. SIDE
4	COUNTER FLANGE		SS400	2	–	FOR S.W. SIDE JIS10K-100A
5	COUNTER FLANGE		SS400	2	–	FOR F.W. SIDE JIS10K-100A
6	BLIND FLANGE		SS400	1	–	FOR S.W. SIDE JIS10K-100A
7	L-TYPE THERMOMETER (WITH CASE)		SUS316	2	–	FOR F.W. SIDE
			TITANIUM CLASS2	2	–	FOR S.W. SIDE
8	GLOBE VALVE		BC-6	1	–	FOR F.W. SIDE AIR VENT
				1	–	FOR S.W. SIDE AIR VENT

APPROVED BY	S.HATANAKA	MAY,31,2016	HISAKA WORKS,LTD. OSAKA,JAPAN	
PREPARED BY	T.OHNOKI	MAY,31,2016		

HEADS VER.200604

HD_AccessoryList

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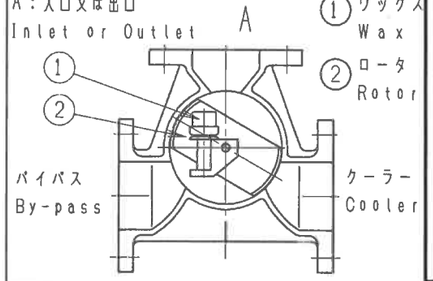
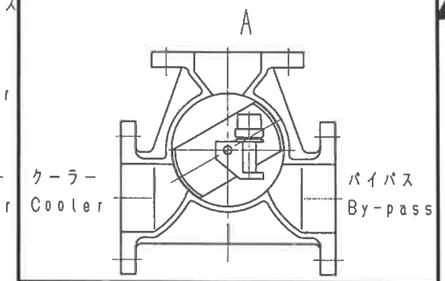
View 'Q'

TP-ワックス式自動温度調整弁 (型式)
TP-Wax type Thermostat valve (Type)

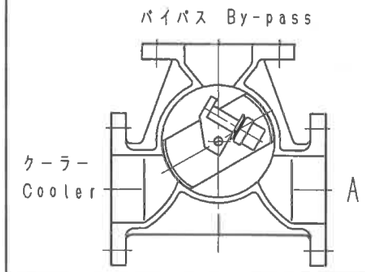
L.I型 TYPE



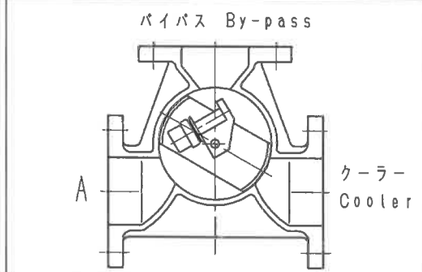
R.I型 TYPE



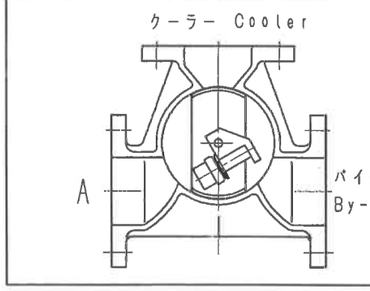
L.B型 TYPE



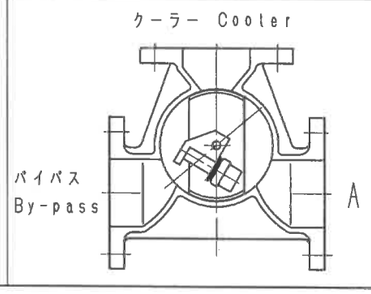
R.B型 TYPE

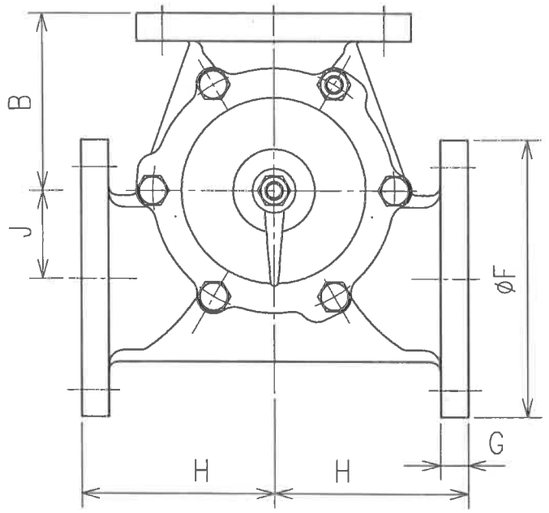


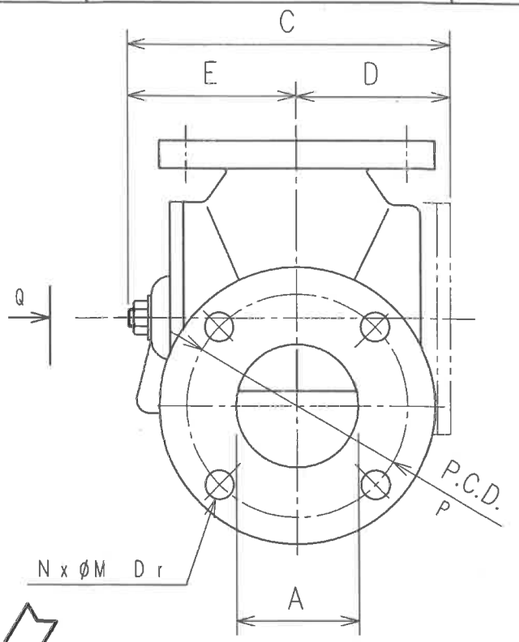
L.C型 TYPE



R.C型 TYPE







125A, 150A, 200Aは下部カバー付き
For 125~200A, with bottom cover

A(SIZE)		40	50	65	80	100	125	150	200
B		115	115	115	115	148	165	178	242
C		160	160	198	195	246	276	333	464
D		67	67	84	81	104	118	148	222
E		93	93	114	114	142	158	185	242
F	JIS5K	120	130	155	180	200	235	265	320
G		16	16	18	18	20	20	22	24
NxφM		4-15	4-15	4-15	4-19	8-19	8-19	8-19	8-23
P		95	105	130	145	165	200	230	280
H		127	127	127	127	148	165	178	242
J		57	57	57	57	57	64	76	102
質量 WEIGHT (概算 APPROX.) kg									
特殊鋳鉄 SPECIAL CAST		13	13	18	20	38	50	80	140

制御方式 CONTROL	
分流型 DIVIDING	混合型 MIXING

温度範囲 °C TEMP. RANGE			
25/35	37/47	57/66	79/88
27/40	41/49	63/71	85/93
31/42	43/54	68/78	
35/45	51/60	74/82	

用途 USE	潤滑油 LUB. OIL	清水 FRESH WATER
本体材質 BODY MATERIAL	特殊鋳鉄 SPECIAL CASTING	特殊鋳鉄 SPECIAL CASTING

相フランジ、ボルト、ナット、パッキン不要
Without counter flange, bolt, nut & Packing

⚠	FIRST DRAWING	2010/08/12	H.S		

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APPROVED	TERAUCHI YOICHI	LIST NO.	MATERIAL APRV	GROUP 7	SCALE:
CHECKED	KAWANO SHUJI	BEFORE	MASS.	QTY.	A3
	MATSUDA NAOKI	REF.	TYPE DK-26e		
DRAWN	KOKADO JYUNJI	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	NAME オンチョウベン:TP-ワックスシキ(5K):FW-80A-RI		
DATE	2010/08/12		THERMOSTAT VALVE:TP-WAX(5K):FW-80A-RI		
			PARTS NO. AQP20003993	A	*

2017/05/09 TECHNICAL DEPARTMENT

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LT-FW THERMO.CONTROL VALVE
ELECTRO PNEUMATIC TYPE

DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

JAPAN

SHIP NO.		DRAWN BY	M. Tokumoto
TYPE	6DKM-20eF(L)	CHECKED BY	S. Tsujimoto
LIST NO.	AQP10008537B		
DATE	2016/6/8	APPROVED BY	(Proxy) Y. Hamada
REVISION	B : Add. valve type RB M.T		

PDK20-0174-5

TPR熱学株式会社 '16.7.-5 技術

Rev.2

納入仕様書
DELIVERY SPECIFICATION

CUSTOMER 受注先: DAIHATSU DIESEL MFG. CO.,LTD ダイハツディーゼル 株式会社 殿

SPEC No. 仕様書番号: AQP20008442A

TYPE 機関型式: 6DKM-20e

SHIP No. 船番:

改定No.	項目 DESCRIPTION	型式番号 MODEL No.	P
2	SPECIFICATION OF ELECTRO-PNEUMATIC TYPE AUTOMATIC TEMPERATURE CONTROL VALVE 電ー空式自動温度調節弁 仕様書		1
	ELECTRO AIR PIPING 電気ーエア配管図	K85-900020-00	2
2	ELECTRO-PNEUMATIC TYPE AUTOMATIC TEMPERATURE CONTROL VALVE 電ー空式自動温度調節弁 本体駆動部	K85-080105-89	3
	CONTROL BOX コントロールボックス	K88-705065-00	4
	TEMPERATURE CONTROLLER 電子温度調節器(TTM-204-Q-IN-Z41)	K88-701022-00	5
	ELECTRO-PNEUMATIC REGULATOR 電ー空レギュレータ(ITV2010-012BL5-X87)	K88-702004-00	6
	FILTER REGULATOR フィルター付レギュレータ(AW20)	K88-327019-02	7
	MIST SEPARATOR ミストセパレータ(AM150)	K88-327015-02	8
	UNIT PANEL ユニットパネル(EXR2M)	K88-705033-02	9
	PLATINUM RESISTANCE THERMOMETER 白金測温抵抗体(R314 二重保護管)	K88-704027-00	10
	CONNECTION DIAGRAM 電気結線図	K88-723100-00	11

電-空式自動温度調節弁 ELECTRO-PNEUMATIC TYPE AUTOMATIC TEMPERATURE CONTROL VALVE 仕様書 SPECIFICATION		仕様書番号 DWG No.	AQP20008442A
		工事番号 JOB No.	
用途 USE	低温冷却清水温調弁	工事名称 PROJECT NAME	
リスト番号又は弁番号 LIST No. OR VALVE No.		数量 Q.T.Y 1SET	製造番号 SERIES No.

電-空式自動温度調節弁 ELECTRO-PNEUMATIC TYPE AUTOMATIC TEMPERATURE CONTROL VALVE				調節計 CONTROLLER			
使用条件 USE CONDITIONS	流体 FLUID		清水 FW	検出部流体 DETECTION PART FLUID	流体 FLUID		
	温度 TEMP.	℃	MAX NOR 38～54		温度 TEMP.	℃	MAX NOR
	圧力 PRESSURE	MPa	MAX NOR 0.2～0.35		圧力 PRESSURE	MPa	MAX NOR
	流量 FLOW RATE	m^3/h	MAX NOR 40	電子温度調節器 TEMPERATURE CONTROLLER	型式 TYPE		TTM-204-Q-IN-Z41
水圧試験圧力 HYDR'C TEST PRESS.		MPa	W.T.P 0.7		温度調節範囲 RANGE OF TEMP. ADJUSTMENT		0～100℃
構造 STRUCTURE	型式・流体流れ方向 HANDLING・FLOW DIRECTION		RB 混合型 MIXING ②2		設定温度 TEMP. SETTING		38℃
	呼び径(A) NOMINAL SIZE		80A		出力信号電流 OUTPUT SIGNAL		4～20mA
	接続フランジ CONNECTED FLANGE		JIS5K FF		供給電源 POWER SUPPLY		AC220V
	材質 MATERIAL		FC250	電-空レギュレータ ELECTRO-PNEUMATIC REGULATOR	型式 TYPE		ITV2010-012BL5-X87
	供給圧力 SUPPLY PRESSURE		0.3～0.5MPa		入力信号電流 INPUT SIGNAL		4～20mA
	信号圧力 SIGNAL PRESSURE		0.02～0.1MPa		供給圧力 SUPPLY PRESSURE		0.15～0.2MPa
塗装色(本体駆動部) PAINT COLOR (CONTROL VALVE)			マンセルNo. 2.5G7/2 MUNSELL		出力圧力 OUTPUT PRESSURE		0.02～0.1MPa
塗装色(ユニットパネル) PAINT COLOR (UNIT PANEL)			マンセルNo. 2.5G7/2 MUNSELL		供給電源 POWER SUPPLY		DC24V
銘板標示 BOARD OF DISPLAY			英文および和英 ENGLISH(THE CAUTION PLATE IS JAPANESE-ENGLISH)				

付属品・予備品

No.	名称	個数	型式等
1	ユニットパネル UNIT PANEL	1SET	EXR2M
2	白金測温抵抗体 PLATINUM RESISTANCE THERMOMETER	1	R314
3	φ10銅管継手 φ10 COPPER PIPE FITTING	5	組込 BUILDING IN
4	相フランジSET COMPANION FLANGE,BOLT,NUT,PACKING SET	1SET	80A/5K ①1
5			
6			

注記

・エアレス、エレクトロレス時はC側開とする。

"C"Port shall be fully opened when airless,electroless.

DA226

REV	REMARKS	DATE	PRED	APPD	CHKD	PRED
②2	型式RBとする	'16.07.04	大西			
①1	付属品に相フランジセット追加	'16.06.08	大西			
⓪0	新規作成	'16.06.07	大西	16.7.-4	16.7.-4	16.7.-4

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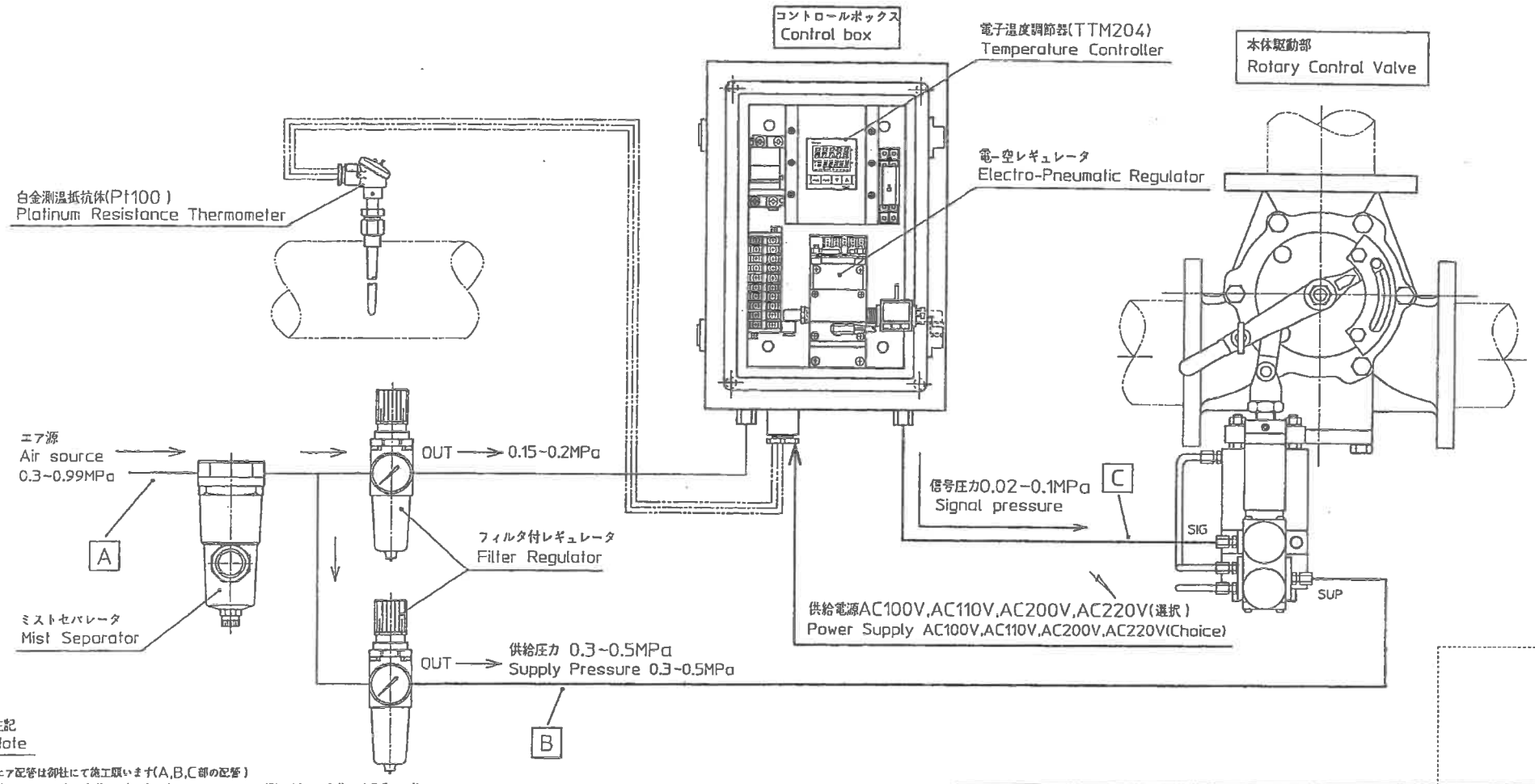
TPR THERMAL ENGINEERING CO., LTD.

DWG No.	D80-110002-01



注記
Note

1. エア配管は御社にて施工願います(A,B,C部の配管)
   Please construct the air pipe in your company.(Plumbing of the A,B,C part)
2. 清浄な空気を使用して下さい。
   Please use clean air.
3. ドレンを含んだ空気は、バルブや他の圧縮機器の作動不良の原因となります。
   エアドライヤやアフタークーラーなどを設置し、対策して下さい。
   Compressed air that contains moisture can cause malfunction of pneumatic equipment such as valves.
   Take measure to ensure air quality,installing an aftercooler,air dryer or water separator.
4. コントロールボックスへの電気配線は御社にて施工願います。
   The electric wiring to Control Box please undertake construction in your company.
5. 配線はシールドケーブル線を使用願います。
   The wiring please use a shield cable line.
6. シールド線被覆部は、片側のみをコントロールボックス内のアース部に配線ください。
   The shield line coating department please wire only one side in the ground department in the Control Box.

記号 NO	来歴 REVISION	年月日 DATE	製図 DWN
2			
1	注記追記	'15.08.31	大西
0	新規作成	'11.11.08	中尾

備考 REMARK		尺度 SCALE	Free
		制定日 DATE	2011-11-8
承認 APPD	検図 CHKD	製図 DWN	
15.8.31	15.8.31	15.8.31 大西	

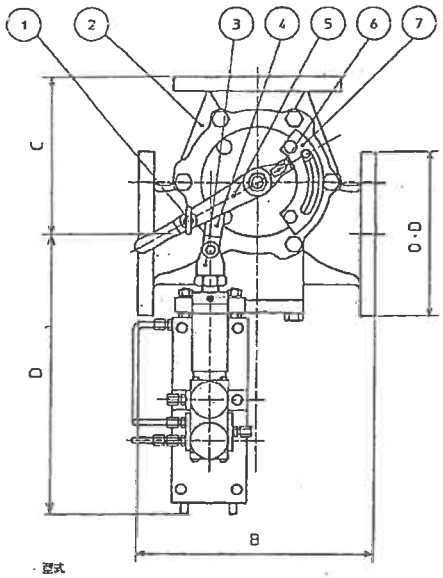
TPR 熱学株式会社
TPR THERMAL ENGINEERING CO.,LTD

ユニットパネル有り
With a unit panel

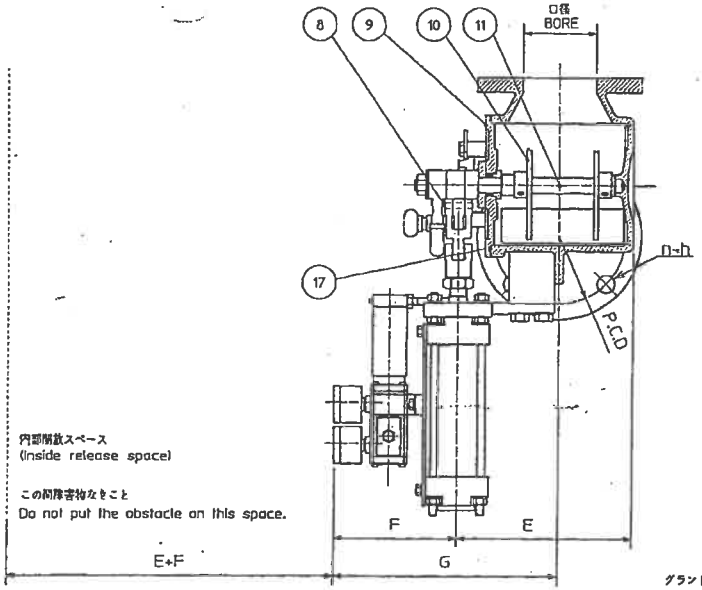
品名 TITLE
電-空式自動温度調節弁(PID)
電気-エア配管図
Electro-pneumatic type automatic temperature control valve
Electro-air piping

型式番号 MODEL No.
K85-900020-00

DWG No. D80-100054-00

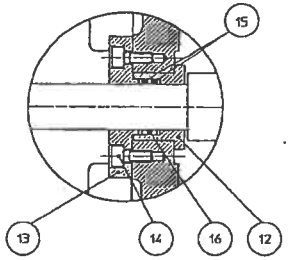


取扱
HANDLING

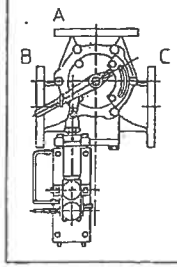
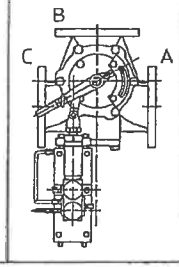
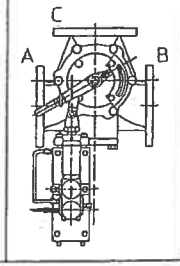
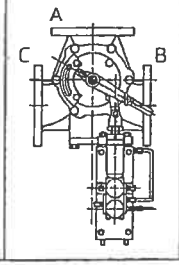
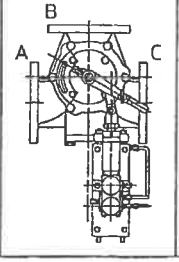
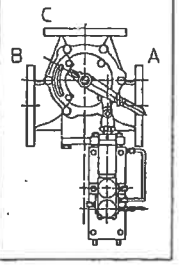


番号 No.	名称 NAME	材質 MATERIAL	個数 Q.T.Y	備考 REMARKS
1	非常ボルト EMER.BOLT	SS400	1	
2	本体 BODY	FC250	1	
3	シリンダリンク CYL.LINK	FC250	1	
4	リンク LINK	FC250	1	
5	ハンドル HANDLE	FC250	1	
6	レバー LEVER	FC250	1	
7	プレート PLATE	SS400	1	
8	スピンドル SPINDLE	SUS304	2	
9	上蓋カバー FRONT COVER	FC250	1	
10	ロータ ROTOR	CAC406	1	
11	シャフト SHAFT	SUS304	1	
12	上部カバーブッシュ FRONT COVER BUSH	C3604BD	1	
13	グランドフランジ GLAND FLANGE	SS400	1	
14	グランドボルト GLAND BOLT	SCM	2	
15	シャフトシール SHAFT SEAL	HNBR	2	
16	スペーサ SPACER	PTFE	1	
17	カバーシール COVER SEAL	NBR	1	

グランドフランジ部詳細
GLAND FLANGE DETAIL



エアレス・エレクトロレス時はC側全開とする。
"C" Port Shall be fully opened when airless,electroless.

	LI TYPE	LB TYPE	LC TYPE	RI TYPE	RB TYPE	RC TYPE
						

寸法一覧表(40A~100A JIS 5K FF)
DIMENSION LIST

D径 BORE	B	C	D ±10	E ±10	F ±10	G ±10	フランジ部詳細 FLANGE DETAIL O·D	P.C.D	n-h	BOLT	質量 MASS (kg)	エアシリンダ形式 ACTUATOR TYPE
40	254	172	332	157	135	225	120	95	4-φ15	M12	15	CPA2B50-50
50	254	172	332	157	135	225	130	105	4-φ15	M12	16	CPA2B50-50
65	254	172	332	195	135	246	155	130	4-φ15	M12	22	CPA2B50-50
80	254	172	332	192	135	246	180	145	4-φ19	M16	24	CPA2B50-50
100	296	205	326	235	135	267	200	165	8-φ19	M16	32	CPA2B50-50

	分流型 DIVIDING	混合型 MIXING
Aポート	入口 INLET	出口 OUTLET
Bポート	バイパス BYPASS	バイパス BYPASS
Cポート	クーラ COOLER	クーラ COOLER

記号 NO	改訂 REVISION	年月日 DATE	製図 DWN

備考 REMARKS		
	尺度 SCALE	Free
	制定日 DATE	'11-07-21
承認 APPD	検図 CHKD	製図 DWN
堅 '11.7.21 本	'11.7.21 口	大 '11.7.21 西

テーピ熱学技研株式会社
TEIPI THERMAL ENGINEERING CO.,LTD

品名 TITLE: Tp 電一空式自動温度調節弁 本体駆動部
Tp ELECTRO-PNEUMATIC TYPE AUTOMATIC TEMPERATURE CONTROL VALVE

型式番号 MODEL No.: K85-000100-89

AQP1008537B (5/13)

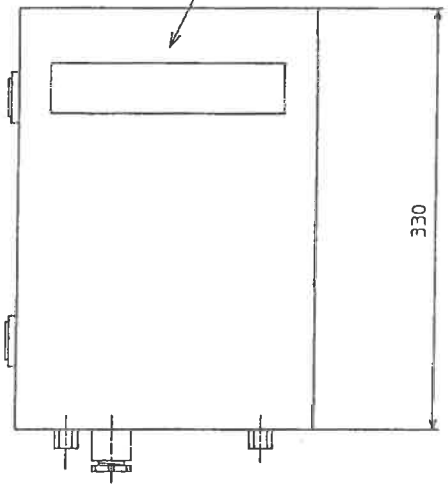
2017/05/09 TECHNICAL DEPARTMENT

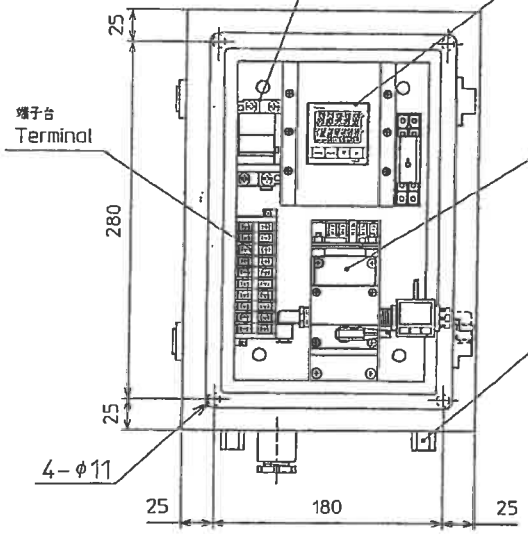
DWG No.	D80-110019-00

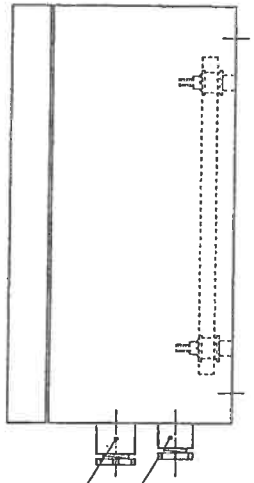
PID 自動温度調節弁 コントロールボックス

PID AUTOMATIC CONTROL VALVE CONTROL BOX

和文もしくは英文（選択）
A Japanese sentence or English sentence (choice)

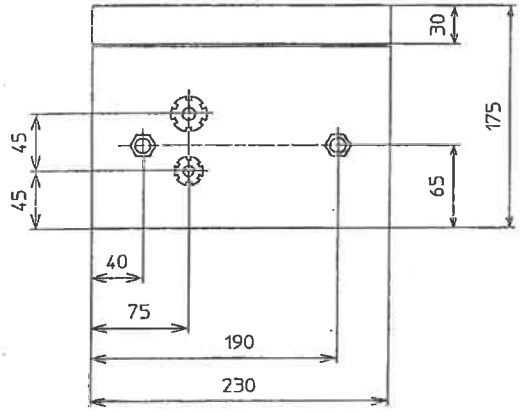






電線貫通金物20a
Cable gland 20a

電線貫通金物15b
Cable gland 15b



注記
Note

1.供給電源はAC100V、110V、200V、220Vより選択してください。
Please choose the supply power supply from AC100V, 110V, 200V, 220V.

2.コントロールボックス内への配線材料はシールドケーブル線を使用願います。
The wiring materials in the Control Box please use a shield cable line.

記号 NO	承認 REVISION	年月日 DATE	製図 DWN
△			
△			
△			

備考 REMARK		
	尺度 SCALE	Free
	制定日 DATE	2011-11-11

承認 APPD	検図 CHKD	製図 DWN
堅 11.11.16 本	人 11.11.11 口	中 11.11.11 尾

TPR 熱学株式会社
TPR THERMAL ENGINEERING CO.,LTD

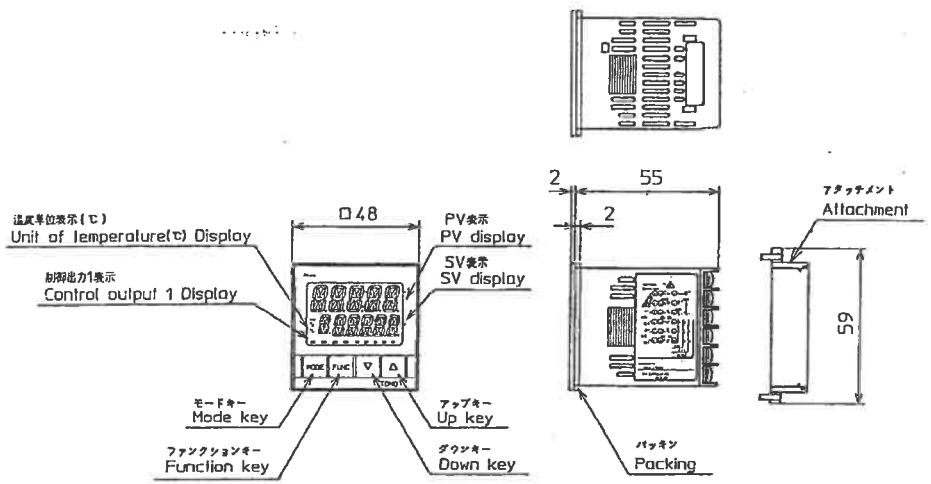
品名 TITLE

電-空式自動温度調節弁（PID）
コントロールボックス
Electro-pneumatic type automatic temperature control valve
Control box

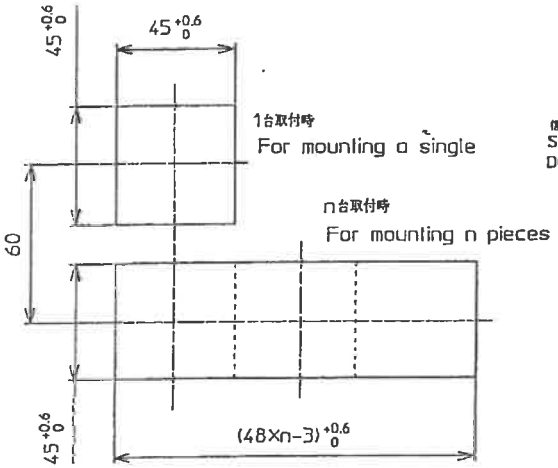
型式番号 MODEL No.

K88-705065-00

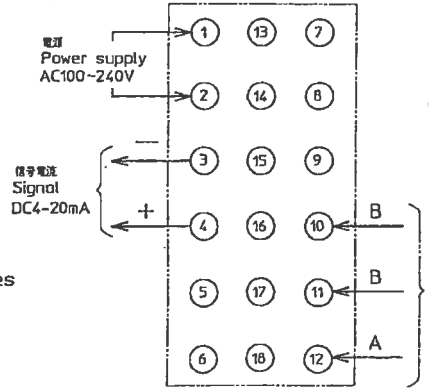
外形寸法・各部の名称
Outside dimensions・Names of parts and components



パネルカット寸法
Panel cutting dimensions



端子配列・配線
Terminal arrangement・wiring



仕様
Specification

入力種別 Input type	測温抵抗体 Resistance thermometer	Pt100(外部抵抗10Ω以下(1線あたり、3線とも同抵抗である事) Pt100,3-Wire system Allowable input lead wire resistance 10Ω or less per wire	
制御 Control	PID . (With auto tuning)	比例帯(P1) Proportional band (P1)	設定リミッタスパンの0.1~200.0% 0.1 to 200.0% of set limiter span
		積分時間(I) Integration time (I)	0~3600秒(「0」で積分動作OFF) 0 to 3600seconds (with "0" integral action OFF)
		微分時間(D) Differentiation time (D)	0~3600秒(「0」で微分動作OFF) 0 to 3600seconds (with "0" derivative control action OFF)
制御出力 Control output	電流 Current / voltage	DC 4~20mA(負荷抵抗600Ω以下) 4 to 20mADC (Load resistance 600Ω or less)	
サンプリング周期 Sampling cycle		0.2秒 0.2 seconds	
設定及び指示精度(周囲温度23℃±10℃にて) Setting and instruction accuracy (In ambient temperature 23℃±10 degree)	測温抵抗体 Resistance thermometer	指示値の±(0.3%+1 digit)又は±0.9℃の大きい方(23℃±10℃) Larger one±(0.3%+1digit) of the instruction value or ±0.9℃ (23℃±10℃)	
入力電源 Input power supply		AC 100~240V (-15%, +10%) 50/60Hz 100 to 240VAC (-15%, +10%), 50/60Hz,	
消費電力 Power consumption		AC 100~240V 10VA以下 less than 10VA (100 to 240VAC)	
付属品 Accessories		簡易取扱説明書とアタッチメント Simple instruction manual and Attachment (A handling description is sold separately.)	
標準周囲温湿度範囲(精度など特別範囲) Standard Range of surrounding temperature humidity (Compensating range such as precision)		23℃±10℃、45~75% RH 23℃±10℃ , 45 to 75% RH	
使用周囲温湿度範囲 Range of use surroundings temperature humidity		0~50℃、20~90% RH(結露なき事) 0 to 50℃ , 20 to 90% RH (Avoid dewiness)	
保存周囲温湿度範囲 Range of preservation surroundings temperature humidity		-20℃~70℃(氷結、結露なき事)、5~95% RH(結露なき事) -20 to 70℃ (Avoid dewiness and freezing), 5 to 95% RH (Avoid dewiness)	

仕様は取扱説明書にて、ご確認下さい。
Please confirm the specification with the manual.

電気配線図を確認のうえ、正しく配線して下さい。
Please wire correctly after confirming an electric wiring diagram.

備考 REMARK				質量 MASS	120g以下 120g or less
			尺度 SCALE	FREE	型式 TYPE	TTM-204-Q-IN-Z41 △1
			制定日 DATE	2011-11-11	仕様 SPECIFICATION	AC100~240V / 4~20mA
	承認 APPD	検図 CHKD	製図 DWN			
	15.7.22	15.7.22	15.7.22		品名 TITLE	電子温度調節器 TEMPERATURE CONTROLLER
記号 NO	変更 REVISION	年月日 DATE	製図 DWN	TPR 熱学株式会社 TPR THERMAL ENGINEERING CO.,LTD		型式番号 MODEL No.	K88-701022-00
△1	型式訳記訂正	'15.07.22	大西				
△0	新規作成	'11.11.11	中尾				

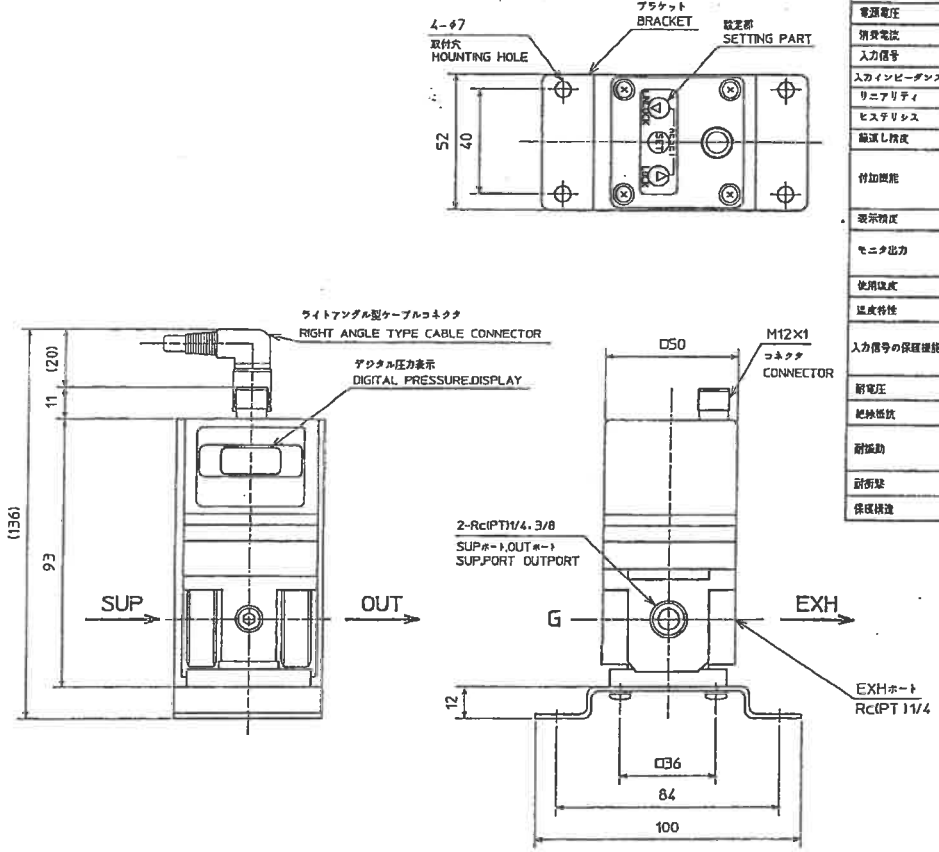
D80-702004-50

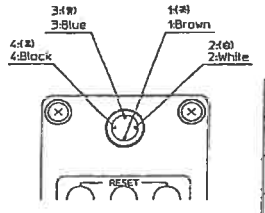
仕様

供給圧力	設定圧力+0.1MPaから最大1MPaまで
出力圧力	0.005~0.1MPa
最大流量	1500l/min(ANR)
電源電圧	24VDC±10%
消費電流	電源電圧 DC24Vタイプ:0.10A以下
入力信号	4~20mADC (ITV2000-01)
入力インピーダンス	250Ω以下 (ITV2000-01.ITV2000-11)
リニアリティ	±1.0% F.S.以下
ヒステリシス	0.5% F.S.以下
繰返し精度	±0.5% F.S.以下
付加機能	最少圧力調整:調整範囲 定格の0~50% 最大圧力調整:調整範囲 定格の10~100% キーロック機能:アップキーを2秒以上押し、セットキーを押すことでキー操作をロック状態にすることが可能
表示精度	±3% F.S.
モニタ出力	アナログ出力 1~5VDC(入力インピーダンス:1KΩ以上) 出力精度 ±5% F.S.
使用温度	0 ~50℃(結露しないこと)
温度特性	±0.12% F.S./℃以下
入力信号の保護機能	定格の150%をこえる信号が入力されると表示用LEDにエラー(Er1)が表示され機能停止 更に、電流入力タイプの場合は、40mADC以上で保護回路が作動
耐電圧	外部端子一括とケース間 AC1500V 50/60Hz 一分間
絶縁抵抗	外部端子一括とケース間 100MΩ(DC500Vメガ計にて)以上
耐振動	10~500Hz振幅1.5mmまたは加速度98m/s² いずれか小さいほうにてX,Y,Z方向各2時間
耐衝撃	294m/s² ・X・Y・Z方向各3回
保護構造	本体:IP65相当 ケーブルコネクタ:IP67

Specifications

Supply pressure	From set pressure +0.1MPa to Max.1MPa
Output pressure	0.005~0.1MPa
Max.flow rate	1500l/min(ANR) (SUP:at 0.7MPa)
Power voltage	24VDC±10%
Current consumption	Supply voltage 24VDC type:0.12A or less
Input signal	4~20mADC (ITV2000-01)
Input impedance	250Ω or less (ITV2000-01.ITV2000-11)
Linearity	±1.0% F.S.ro less
Hysteresis	0.5% F.S.or less
Repeatability	±0.5% F.S.or less
Additional functions	Min.pressure adjustment:adjusting range 0~50%of rated value Max.pressure adjustment:adjusting range 10~100%of rated value Key locking function:is possible to lock the keys by pushing UP key for 2 sec.or more and push SET key.
Display Accuracy	±3% F.S.
Monitor output	Analog output 1~5VDC(input impedance:1kΩ or more) Output Accuracy:±5% F.S or less
Operating temperature	0 ~50℃(Without condensation)
Temperature characteristics	±0.12% F.S.or less
Protective function for input signal	When a signal exceeding 150% of rated value is input ,an error(Er1) is indicated on LED and functions are stopped. In the case of current input type,protective circuit is operated at 40mADC of more
Withstand voltage	Between a batch of external terminals and case:1500VAC 50/60Hz For 1 minute
Insulation resistance	Between a batch of external terminals and case:100KΩ of morer 1 minute
Vibration proof	100~500Hz at either of smaller value of amplitude of 1.5mm or acceleration of 98m/s2 in X,Y, and Z directions for 2 hours each.
Impact resistance	294m/s2 in X,Y, and Z directions for 3 times each
Enclosure	Main unit:Equivalent to IP65 Cable connector:IP67
MASS	Approx.350g (No accessory)





1	茶 Brown	供給電源 Supply power
2	白 White	入力信号 Input signal
3	青 Blue	GND(COMMON)
4	黒 Black	モニタ出力 Monitor output

NO	REVISION	DATE	DWN

TEIPI THERMAL ENGINEERING CO.,LTD

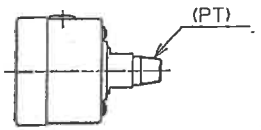
REMARK		MASS: 0.35kg
	SCALE: FREE	TYPE: ITV2010-012BL5-X87
	DATE: '10.7.27 JUL.27.'10	SPECIFICATION:
APPD: '10.7.27	CHKD: '10.7.27	DWN: '10.7.27
TITLE: 電-空レギュレータ Electro-pneumatic regulator		
MODEL No.: K88-702004-00		

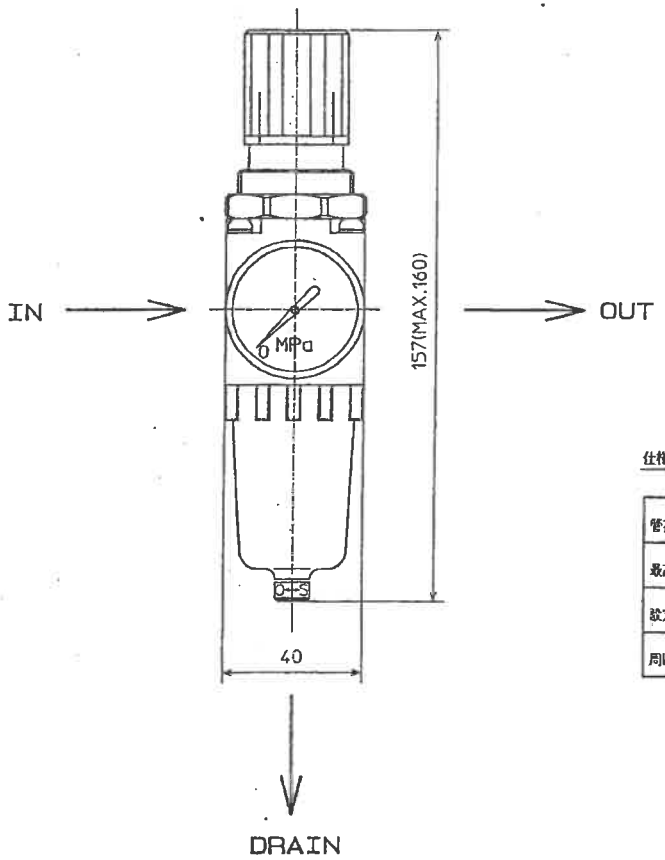
AQP1008537B (8/13)

2017/05/09 TECHNICAL DEPARTMENT

D80-727019-02

圧力計ネジサイズ詳細（各スケール共通）
The screw size of pressure gauge





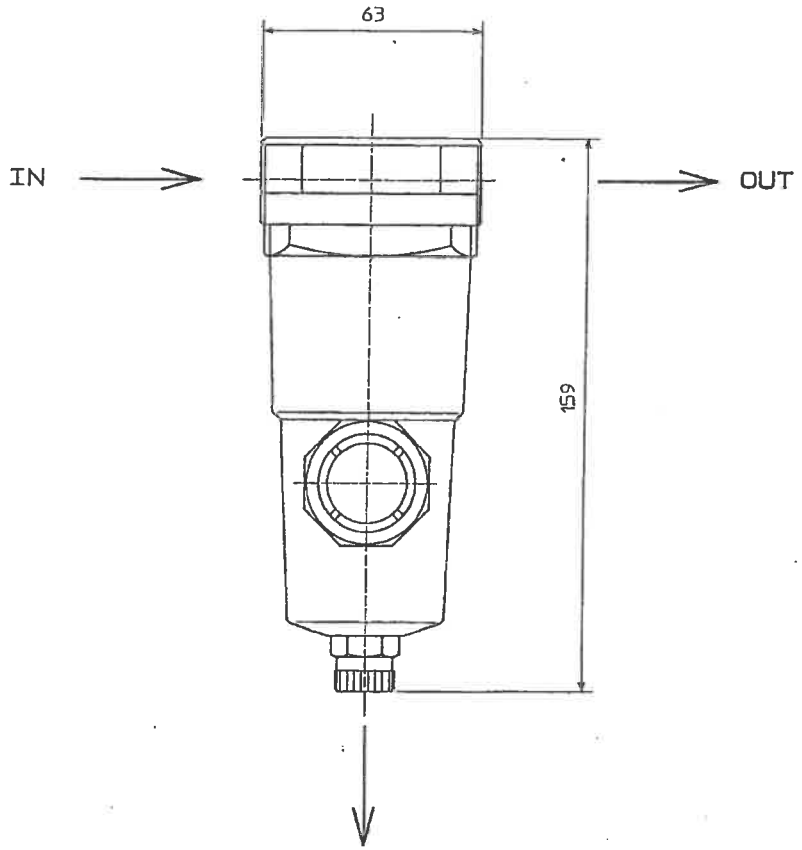
圧力計グリーンマーク表示
INDICATION OF A
PRESSURE GAUGE GREEN MARK

1MPa計 1MPa GAUGE	0.3~0.5MPa
0.25MPa計 0.25MPa GAUGE	0~0.1MPa
0.4MPa計 0.4MPa GAUGE	0.15~0.2MPa

仕様 SPECIFICATIONS

管接続口径	CONNECTION	Rc 1/4
最高使用圧力	MAXIMUM WORKING PRESSURE	1.0MPa (MAX)
設定圧力範囲	RANGE OF SETTING PRESSURE	0.05~0.85MPa
周囲温度	AMBIENT TEMPERATURE	-5~60℃

						質量 MASS	0.36kg
					尺度 SCALE	FREE	型式 TYPE	AW20-02
					制定日 DATE	'09.11.18 NOV.18.'09	仕様 SPECIFICATION	図中表示用
					承認 APPD	検図 CHD	製図 DWN	品名 TITLE	フィルター付レギュレータ Filter Regulator
NO	改定 REVISION		年月日 DATE	製図 DWN	テイピ熱学株式会社 TEIPI THERMAL ENGINEERING CO.,LTD			型式番号 MODEL No.	K88-327019-02



仕様 SPECIFICATIONS : AM150

管接続口径 CONNECTION	Rc 1/4
最高使用圧力 MAXIMUM WORKING PRESSURE	1.02MPa (MAX)
保証耐圧力 GUARANTEE ENDURANCE PRESSURE	1.53MPa
周囲温度 AMBIENT TEMPERATURE	5-60℃

						質量 MASS	0.36kg
					尺度 SCALE	FREE	型式 TYPE	AM150
					制定日 DATE	'09.11.18 NOV.18.'09	仕様 SPECIFICATION	図中表参照
				承認 APPD	製図 CHD	製図 DWN		
				樫 09.11.30 木	09.11.30 口	中 09.11.19 尾	品名 TITLE	ミストセパレーター Mist Separator
記号 NO	改版 REVISION	年月日 DATE	製図 DWN	アーピ熱学株式会社 TEIPI THERMAL ENGINEERING CO.,LTD			型式番号 MODEL No.	K88-327015-02

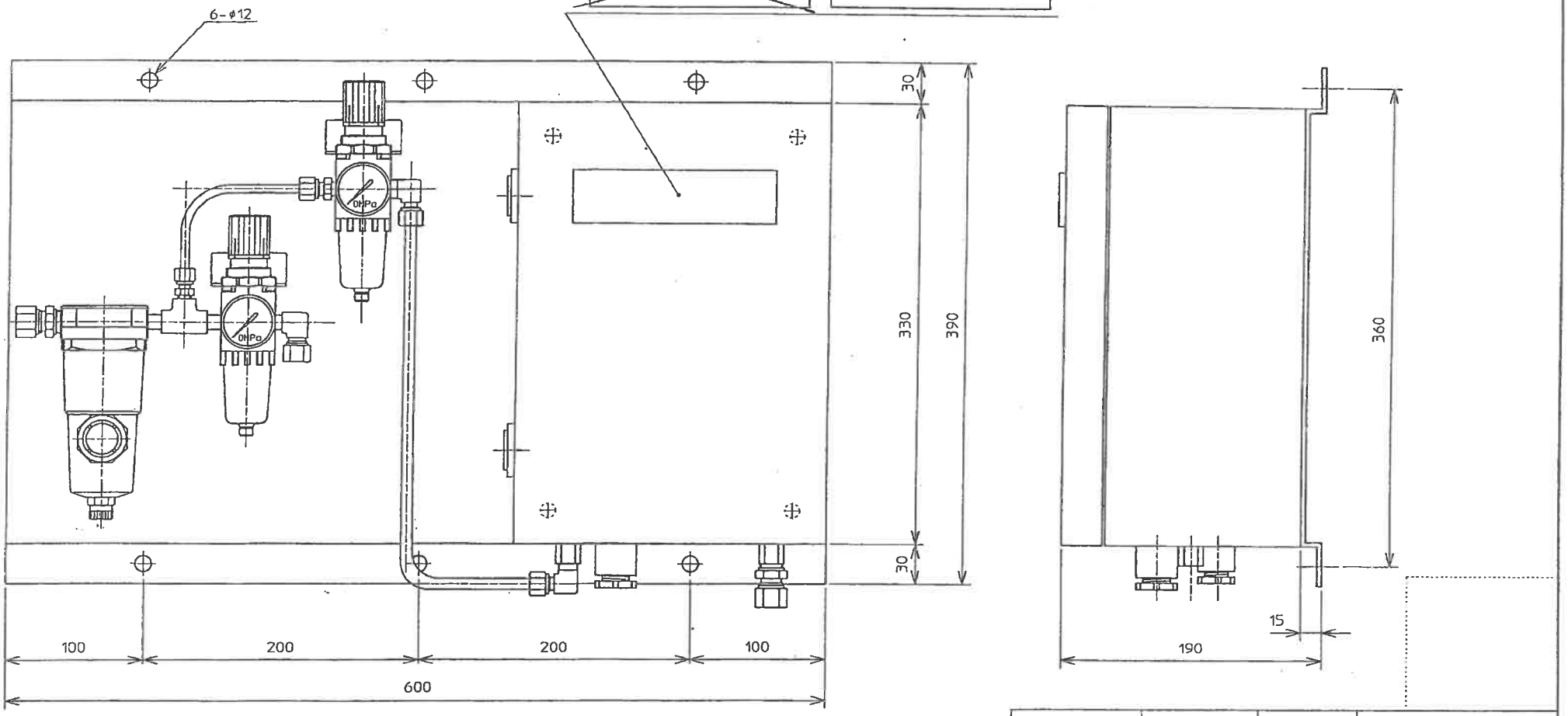
2017/05/09 TECHNICAL DEPARTMENT

D80-705033-03

和文、英文のどちらかを貼付

TP PID 自動温度調節弁 コントロールボックス

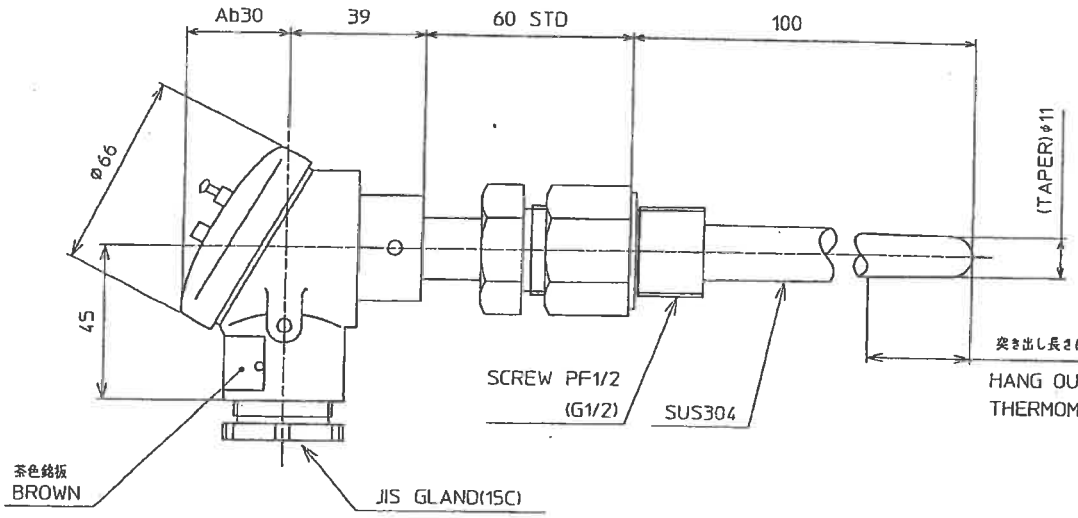
TP PID AUTOMATIC CONTROL VALVE CONTROL BOX



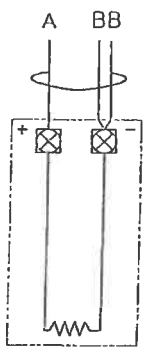
					REMARK	MASS	16.5kg
					SCALE	TYPE	EXR2M
					DATE '10.2.8 FEB.8.'10	SPECIFICATION	
				APPD	CHD	DWN	TITLE: ユニットパネル Unit Panel
NO	REVISION	DATE	DWN	TEIPI THERMAL ENGINEERING CO.,LTD			MODEL No. K88-705033-02

AQP1008537B (11/13)

2017/05/09 TECHNICAL DEPARTMENT



結線要領
CONNECTION



注記
NOTE

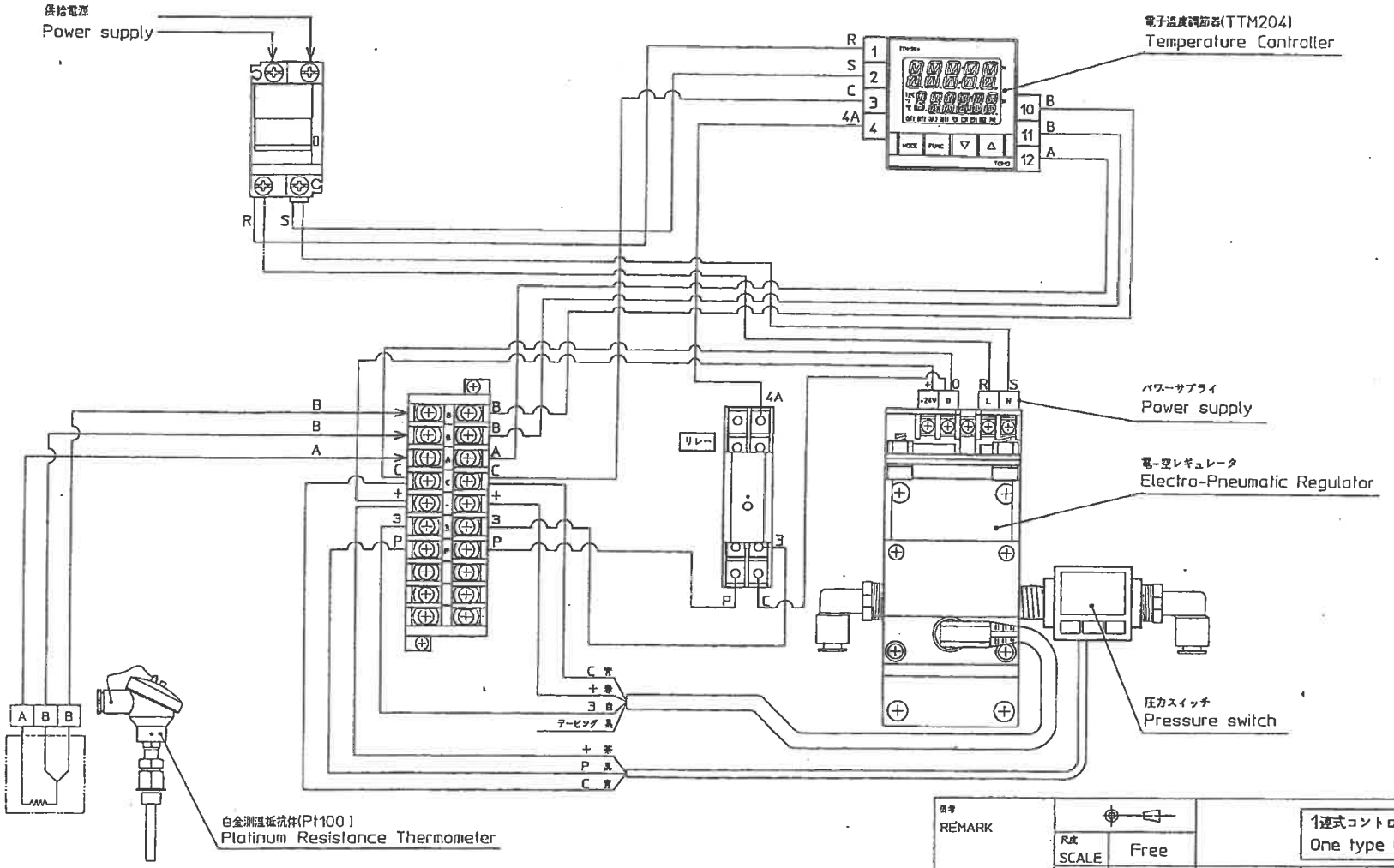
信号線にはシールド線をご使用願います。
Use shielded cables for signal line.

NO	REVISION	DATE	DWN

備考 REMARK		質量 MASS	
尺度 SCALE	FREE	型式 TYPE	R314
制定日 DATE	'10.4.2 APR.2.'10	仕様 SPECIFICATION	PT100 / 二重
承認 APPD	製図 CHD	製図 DWN	品名 TITLE
櫻 '10.4.-1 本	入 '10.4.-2 口	小 '10.4.-2 松	白金測温抵抗体 PLATINUM RESISTANCE THERMOMETER
テイピ熱学株式会社 TEIPI THERMAL ENGINEERING CO.,LTD			型式番号 MODEL No. K88-704027-00

2017/05/09 TECHNICAL DEPARTMENT

DWG No. D80-110032-00



注記
Note

1. 電源と白金測定温度抵抗体は御社にて結線願います。
Please connect a power supply and the platinum resistance thermometer in your company.

2. 結線はシールドケーブル線を使用願います。
The wiring please use a shield cable line.

備考 REMARK		尺度 SCALE	Free	1連式コントロールボックス One type Control box
		制定日 DATE	2011-11-14	
承認 APPD	検図 CHKD	製図 DWN	品名 TITLE	電一空式自動温度調節弁(PID) 電気結線図 Electro-pneumatic type automatic temperature control valve Electric connection diagram
櫻 11.11.16	入口 11.11.15	中尾 11.11.14		

記号 NO	改訂 REVISION	年月日 DATE	製図 DWN	TPR熱学株式会社 TPR THERMAL ENGINEERING CO.,LTD	型式番号 MODEL No.	K88-723100-00

Messrs. DAIHATSU DIESEL MFG. CO.,LTD.

OWNER	SINANJU TANKERS PTE LTD
SHIP YARD.	ZHENJIANG SHENZHOU SHIPBUILDING CO.,LTD.
HULL No.	SZ15045 / SZ15046 / SZ15047 / SZ15048

MAIN ENGINE REMOTE CONTROL SYSTEM

ENG. NO	:	PDK20-0174-5 / PDK20-0176-7 PDK20-0178-9 / PDK20-0181-2
ENG. TYPE	:	STBD 6DKM-20eF / PORT 6DKM-20eFL
VESSEL	:	1060kW x 900min-1
CONT.SYSTEM	:	ELECTRIC / PNEUMATIC CONTROL 4-20mA / 0.093-0.373MPa / 450-900min-1
JNO.		91-5228 / 91-5229 / 91-5283 / 91-5284
RULE		BV

NUNOTANI KEIKI SEISAKUSHO CO.,LTD.

2 - 6 - 10, NISHINAGASUCHO, AMAGASAKI. JAPAN

TEL. OSAKA (06) 6482-1132

FAX. OSAKA (06) 4868-3633

DRAWN BY	:	OKUDA
CHECKED BY	:	
CHIEF OF SEC.	:	REV. '16.11.07 K.T
MANAGER	:	生本部長 '16.11.7 原田
SECTION.	:	TECHNICAL

図面来歴表
＜REVISION HISTORY＞

変更記号 △MARK	DATE	変 更 概 要 / RIVISION NOTES
1	20,SEP,2016	誤記修正 (P11, 92, 128, 151, 163)
2	29,SEP,2016	1. N2010 BACKUP REPOSE信号をKEIに入力 (P66, 102, 145, 148) 2. 比例弁パネルへのFO RAKC信号入力位置を8番から9番へ変更 (P83, 119) 3. アラームセッター設定値追記 (P86, 92, 122, 128) 4. 誤記訂正 (P8, 124)
3	11,OCT,2016 ～ 19,OCT,2016	1. 比例弁電磁弁回路にサージ対策用バリスタを追加 (P83, 119) 2. 誤記訂正 (P87, 88, 146, 148, 149)
4	27,OCT,2016 ～ 18,NOV,2016	1. 誤記訂正(P20, 63, 64, 68, 69, 74, 75, 81, 89, 92, 94, 104, 105, 110, 111, 112, 117, 125, 127, 130, 143, 144, 148)
5	6,DEC,2016	誤記訂正 (P54, 81)
6	12,DEC,2016	2隻目 陸上運転時タイマー設定変更 2. 5秒→2秒 (P84, 120)

CAUTION

1. Don't touch charge parts of equipment to prevent an electric shock.

2. When the main engine start, do preparation of emergency stop to prevent over speed or another accident etc. by cause of control system defective.

3. Before departure, confirm equipment function is normal.
   Confirm normal function of system at regular intervals.
   (Example: every three-month or six month)

4. When abnormal occur, control state is held but, after confirmation of abnormal cause, shift control station to E/S and carry out repair/replace.
   If it is necessary to stop the main engine, it should be stopped.

5. Tighten up screw of terminal board/electric parts at regular intervals. (Example: every half-year.)

6. Governor control circuit and its printed circuit board are sensitive to electricity-static shock.
   Therefore, please care following matter.
   - Before handling, discharge electricity-static of human body.
     In the midst of handling, touch it on earthed working desk.
   - Keep the PCB from charged plastic, vinyl and chemical goods etc.
   - Don't touch the PCB in the control equipment by hand or conductible device.
   - Removed PCB from control equipment keep in electricity-static prevents cover.

7. Equipment's are consisting of electric parts / Electro-pneumatic converter, therefore avoid using at moisturized place.
   Packing is influenced by environment (Especially chemical moisture).
   We recommend overhaul or replacement of parts every three (3) years.

8. In case of inspection or checking inside of equipment turn off the electric power source switch (AC 220V/DC24V). The signal cables are recommended to be separated from AC power cables as far as possible in parallel wiring, or be crossed them.

9. If ship is equipped with handle, please operate with caution.
   - Avoid sudden and careless operation of handle.
   - Please reversing control advance to next step after confirm changing of clutch.(except clash astern and emergency operation.)

10. Any modification of the panel without authority of DAIHATSU is strictly forbidden.

ZHENJIANG SHINZHOU S/B
SZ15045/15046
SZ15047/15048

SYSTEM OUTLINE

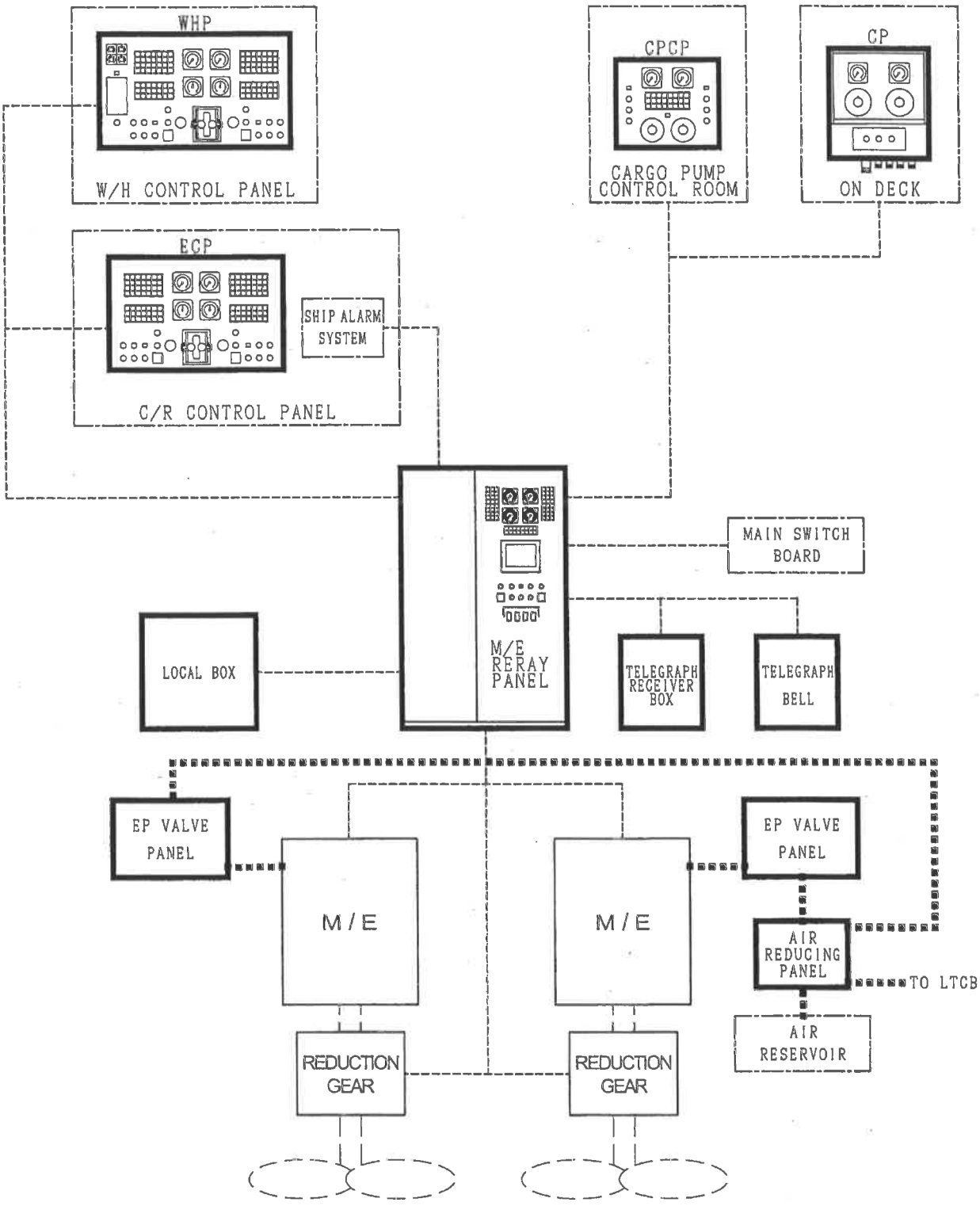
EQUIPMENT PROVIDED BY DAIHATSU(NUNOTANI)

EQUIPMENT PROVIDED BY DAIHATSU

EQUIPMENT PROVIDED BY SHIPYARD

ELECTRICAL CABLE PROVIDED BY SHIPYARD

AIR PIPING PROVIDED BY SHIPYARD



ZHENJIANG SHENZHOU SHIP BUILDING CO.,LTD. SNo.SZ15045 / SZ15046 / SZ15047 / SZ15048
Main engine remote control system overview

1. Electric power supply
   Electric power
   AC220V power supplied from main switchboard for control is converted to DC24V, which is always backed up by DC24V from MSB.

2. Pneumatic power
   2.94MPa compressed air supplied from air reservoir is reduced to 0.69MPa in the air reducing panel by one of two reducing valves.
   Control air is used for safety stop.

3. Control Position
   1. M/E RERAY PANEL (ERP)
   2. C/R CONTROL PANEL (ECP)
   3. W/H CONTROL PANEL (WHP)
   4. CARGO PUMP CONTROL PANEL (CPCP)
   5. CARGO PANEL (CP)

4. Control station change over
   1. ERP → ECP
      Interlock
      Propeller Clutch : disengage
      Clutch order : off
      ECP control handle positon : Neutral
      Position selector switch on ERP : REMOTE select
      Position selector switch on ECP : C/R select
      Confirm push button on ECP

      Upon satisfaction of the above conditions, control postion is transferred to W/H and indication lamp 「 C/R CONTROL 」 is lit and 「 E/R CONTROL 」 goes off.
      During unsatisfied, control postion indication lamp 「 C/R CONTROL 」 keeps flickering and buzzer on ECP sounds.

   2. ECP → WHP
      Interlock
      Propeller Clutch : disengage
      Clutch order : off
      ECP control handle positon : Neutral
      WHP control handle positon : Neutral
      Position selector switch on WHP : NAVI select
      Position selector switch on ECP : W/H select
      Confirm push button on WHP

      Upon satisfaction of the above conditions, control postion is transferred to W/H and indication lamp 「 W/H CONTROL 」 is lit and 「 C/R CONTROL 」 goes off.
      During unsatisfied, control postion indication lamp 「 W/H CONTROL 」 keeps flickering and buzzer on W/H sounds.

3. WHP → CPCP
   Interlock
   Propeller Clutch : disengage
   Clutch order : off
   WHP control handle positon : Neutral
   Cargo Clutch : disengage
   CPCP control dial "0" position
   Mode selector switch on W/H : CARGO select
   Position selector switch on CPCP : CPCP select
   Confirm push button on CPCP

   Upon satisfaction of the above conditions, control postion is transferred to CPCP and indication lamp 「 C/P CONTROL 」 is lit and 「 W/H CONTROL 」 goes off.
   During unsatisfied, control postion indication lamp 「 C/P CONTROL 」 keeps flickering and buzzer on CPCP sounds.

4. CPCP → CP
   Position selector switch on CPCP : CP select

   ※interlock is not applied.
   In order to change over shall be done at cargo clutch disengage, idling speed, CPCP dial "0"and CP dial"0".

5. CP → CPCP
   Position selector switch on CPCP : CPCP select

   ※interlock is not applied.
   In order to change over shall be done at cargo clutch disengage, idling speed, CPCP dial "0"and CP dial"0".

6. CPCP → WHP
   Mode selector switch on W/H : NAVI select

   ※interlock is not applied.
   In order to avoid altering of propelling thrust, change over shall be done at cargo clutch disengage, idling speed, CPCP dial "0"and WHP handle neutral.

7. WHP → ECP
   Position selector switch on ECP : C/R select

   ※interlock is not applied.
   In order to avoid altering of propelling thrust, change over shall be done at clutch disengage, idling speed, WHP handle neutral and ECP handle neutral.

8. ECP → ERP
   Position selector switch on ERP : LOCAL select

   ※interlock is not applied.
   In order to avoid altering of propelling thrust, change over shall be done at clutch disengage, idling speed and ECP handle neutral.

9. WHP(CP,CPCP,) → ERP
   Position selector switch on ERP : LOCAL select

   ※interlock is not applied.
   In order to avoid altering of propelling thrust, change over shall be done at clutch disengage, idling speed and WHP handle neutral.

5. Control Mode
   1. Main Mode and Back Up mode
      Micro computer N2010 controls M/E revolution speed and clutch.

      Control mode is changed automatically to back up N2010 at the following main N2010 abnormal conditions.
      a. N2010 communication abnormal
      b. N2010 AIO unit abnormal
      c. N2010 DIO unit abnormal
      d. control potentio power fail.

6. M/E start・stop・air run
   1. AIR RUN START at E/S
      Upon satisfaction of the following interlocks start interlock magnetic valve 88S is is energized and indication lamp「 AIR RUN START READY 」is lit.

      interlock
      a. E/S (ERP) control
      b. Actual Clutch : disengage
      c. Clutch engage order : off
      d. Turning : disengage
      e. Speed Switch : 14 off (M/E STOP)
      f. Stop magnetic valve 5S/5V : not active
      g. M/E LO press (63QS) : active
      h. R/G LO press (63QGS) : active

   2. M/E stop at E/S
      By selecting governor switch to stop postion, M/E stop order is output to governor. After that FO handle at M/E shall be shifted to stop postion.

7. FO rack limit control
   FO rack is limitted by air pressure from E-P Valve by 4～20mA DC current signal from micor computer during WHP control.

   For rack limit adjust, limit adjust dial is equipped on WHP

8. Main engine speed control

Micro computer equipped in M/E Relay Panel supplys speed order signal (4～20mADC) to E-P Valve which converted 4～20mA DC current signal to 0～0.6 MPa air pressure after increase/decrease speed adjustment. Hydraulic governor regurates M/E revolution speed by control air pressure.

Control air pressure
450～900 min-1 / 0.093～0.373 MPa
M/E revolution speed order at clutch disengaged.
ERP :450min-1
ECP :450min-1 (IDLE BOOST ON 480min-1)
WHP :450min-1 (IDLE BOOST ON 480min-1)
CPCP :450min-1
CP :450min-1

M/E revolution speed order shall be 450min-1 during clutch engaged.

1. WHP control
M/E revolution speed is controlled by control handl at WHP

2. ECP control
M/E revolution speed is controlled by control handl at ECP

3. CPCP control
M/E revolution speed is controlled by control dial at CPCP

4. CP control
M/E revolution speed is controlled by control dial at CP

5. Engine Room control
M/E revolution is controlled by governor knob.

6. Revolution speed increase order interlock
WHP and ECP control
Revolution speed increase order is available after Propeller clutch engaged.
(drain and bypass M.V.: close).

CPCP and CP control
Revolution speed increase order is available after CARGO clutch engaged.

9. Idle Boost Function

M/E speed holds idling M/E speed 480 min-1 at clutch disengage order.
M/E speed shall be 480 min-1 in 8 seconds after clutch engage order.
Idel Boost on/off switch shall be equipped on WHP and ECP.

10. Heating Mode

Heating (approximately 700min-1) operates for 10 minutes (initial set) on the following condition.
Heating on/off switches shall be equipped on ERP.

a. > 14
b. Clutch disengage
c. < 60℃ : M/E FW high temp SW (26WHC)

11. Clutch Control

1. WHP, ECP control
   By control air to Clutch from magnetic valve
   Since magnetic valve is double-solenoid type, engage and disengage order is output individually.
   Clutch is controlled by control handle on WHP or ECP
   Following M/E revoution interlock is applied.

   a. Clutch ahead engage interlock
      WHP or ECP control
      Propeller clutch disengaged
      Cargo clutch disengaged
      M/E RUN
      M/E revolution speed under 500 min-1(initial setting)

      Upon satisfaction of the above interlocks Ahead clutch engage order is output to EP Valve Box.

   b. Clutch ahead disengage interlock
      M/E revolution speed under 500 min-1(initial setting)

      Upon satisfaction of the above interlocks Ahead clutch engage order to clutch is cancelled by control handle operation to neutral position.

   c. Clutch astern engage interlock
      WHP or ECP control
      Propeller clutch disengaged
      Cargo clutch disengaged
      M/E RUN
      M/E revolution speed under 500 min-1(initial setting)
      Propeller revolution speed under ahed 70 min-1 (initial setting)

      Upon satisfaction of the above interlocks Astern clutch engage order is output to EP Valve Box.

   d. Clutch astern disengage interlock
      M/E revolution speed under 500 min-1(initial setting)

      Upon satisfaction of the above interlocks Astern clutch engage order to clutch is cancelled by control handle operation to neutral position.

2. Engine Room control
   By manual operation of the lever on clutch.

3. Clutch auto disengage
   When M/E revolution speed drops under 350 min-1, clutch is disengaged automatically until after 2sec and M/E recovers 440min-1.
   M/E revolution speed order is kept 480 min-1 during clutch disengaged when Idle Boost switch is ON

4. Clutch order is cancelled automatically when M/E shut down (manual or auto) is occurred.

12. Safety system

1. Automatic shutdown

Safety stop magnetic valve 5S & stop magnetic valve 5V are excited at the following events.

a. Over speed of main engine (1020min-1)
b. Main engine L.O. pressure too low
c. Reduction gear L.O. pressure too low

2. Manual Shutdown

Every push button "M/E EMERGENCY STOP" on ERP, ECP, WHP, CPCP or CP works independently of the control position.

3. Reset of Shutdown

Safety shutdown circuit is released after 20 seconds from M/E stop and by shifting FO handle at M/E to stop position.

4. Slowdown request (manual slowdown)

Slowdown request signal is output. at the following events.

a. M/E FW high temp (26WH3)
b. M/E FW low press (63WH3)
c. R/G shaft thrust bearing high temp (TRGT)

In this case, manually slowdown operated.

13. Safety Circuit System

Upon satisfaction of following conditions, safety circuit for M/E LO low press and and R/G LO low press shutdown is established and alarm repose signal for low press to monitor is released.

Stop magnetic valve 5S/5V: not active
20 seconds after M/E revolution speed is more than 200min-1.

14. Pump control
    1. M/E L.O. S/B PUMP
       Pump start interlock
       a. M/E RUN
       b. M/E L.O. low press switch (63QP) : on

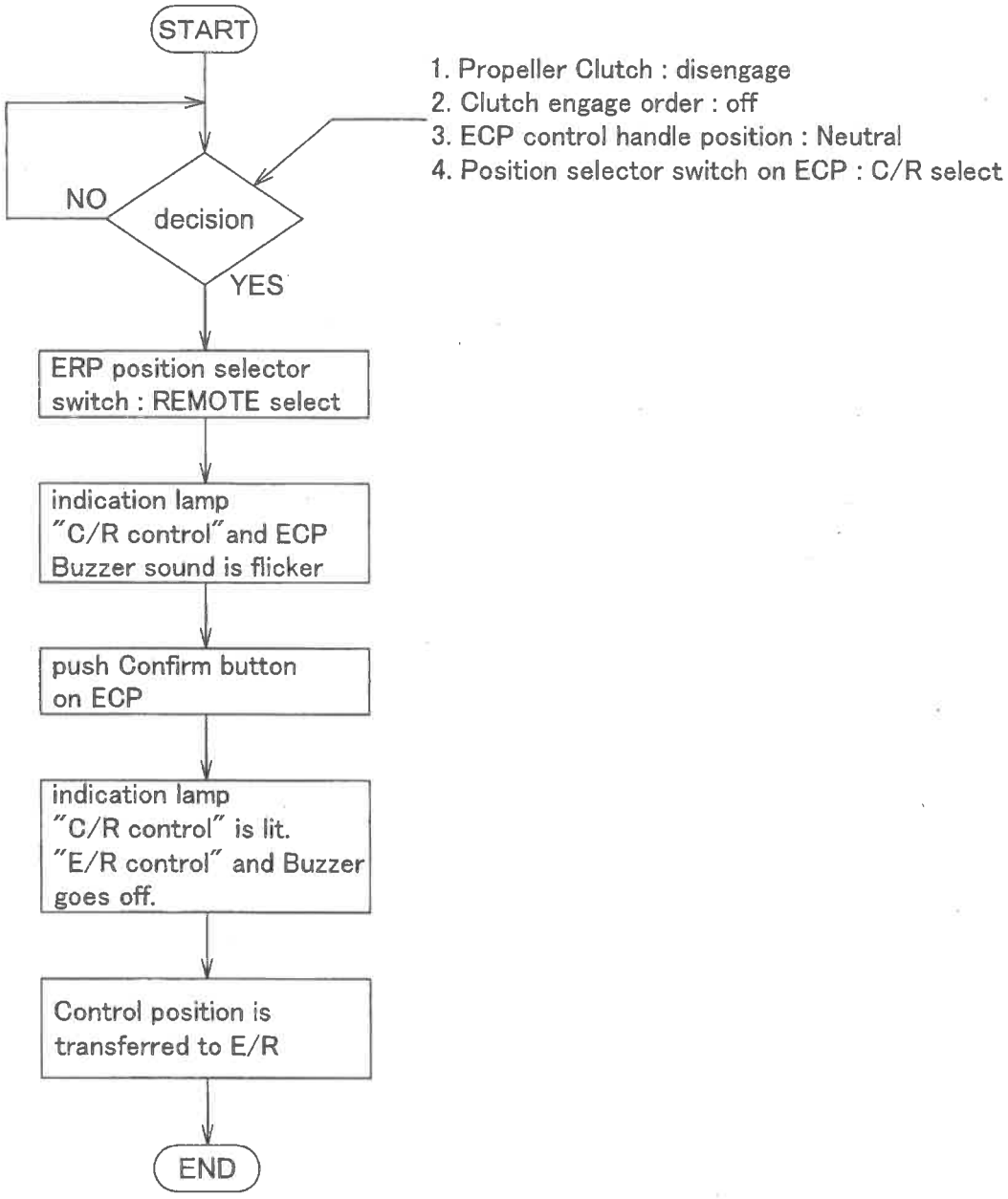
       Pump stop
       M/E L.O. S/B PUMP STOP push button on WHP

    2. R/G L.O. S/B PUMP
       Pump start interlock
       a. M/E RUN
       b. R/G L.O. low press switch (63QGP) : on

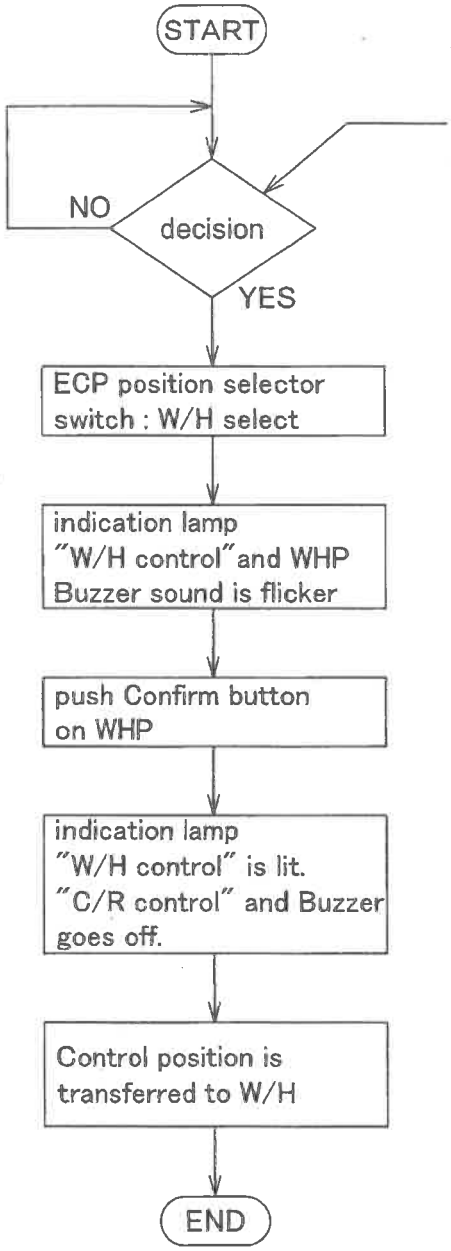
       Pump stop
       R/G L.O. S/B PUMP STOP push button on WHP

    2. After cooling (M/E L.O. S/B and R/G L.O. S/B PUMP)
       Out put after cooling pump run signal
       M/E 200min-1 > ON (10minute)

[change control position]
ERP → ECP



[change control position]
ECP → WHP



1. Propeller Clutch : disengage
2. Clutch engage order : off
3. ECP control handle position : Neutral
4. WHP control handle position : Neutral
5. Position selector switch on WHP : NAVI select

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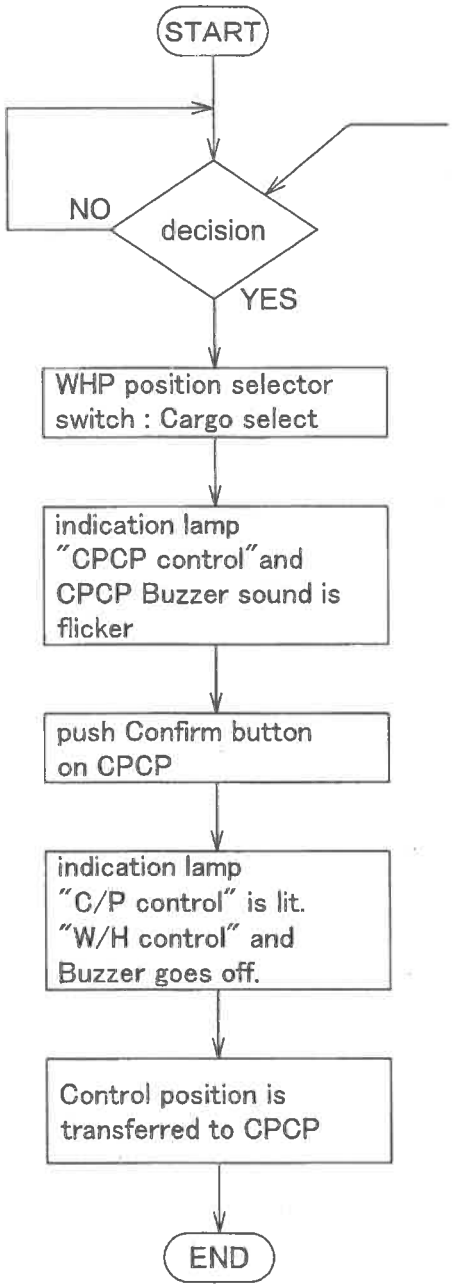
[change control position]
WHP → CPCP



1. Propeller Clutch : disengage
2. Clutch engage order : off
3. WHP control handle position : Neutral
4. Cargo Clutch : disengage
5. CPCP control dial "0" position
6. Position selector switch on CPCP : CPCP select

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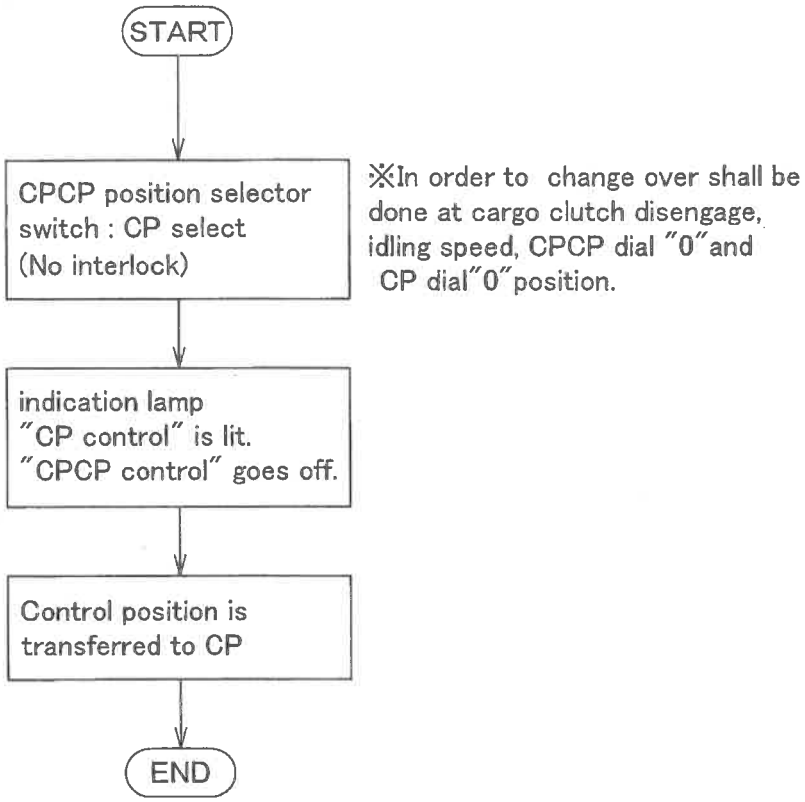
[change control position]

CPCP → CP



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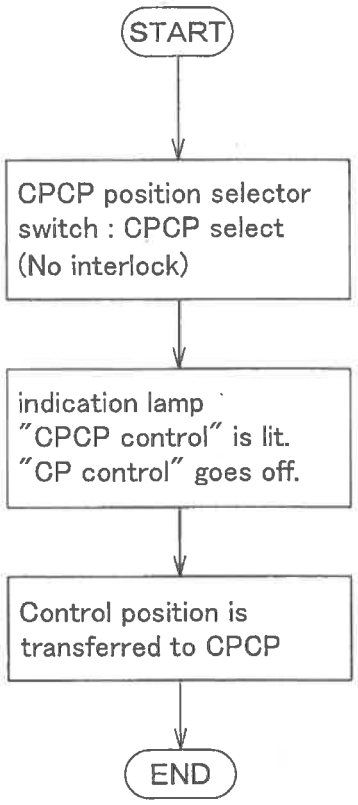
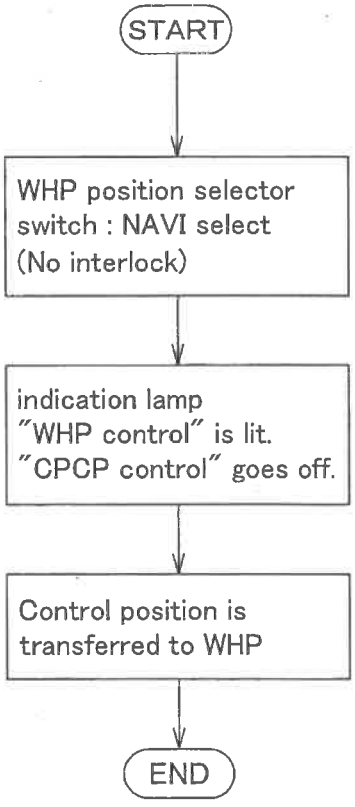
[change control position]
CP → CPCP



※In order to change over shall be done at cargo clutch disengage, idling speed, CPCP dial "0"and CP dial"0"position.

[change control position]
CPCP → WHP



※In order to avoid altering of propelling thrust, change over shall be done at cargo clutch disengage, idling speed, CPCP dial "0"and WHP handle neutral.

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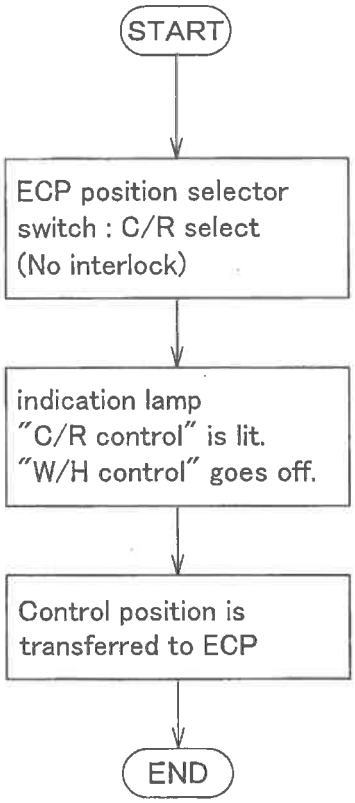
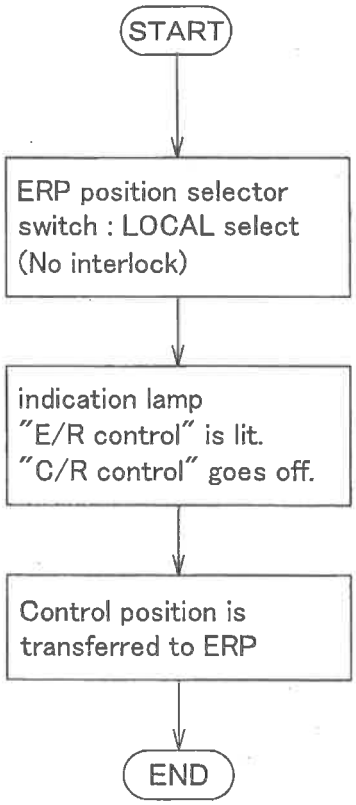
[change control position]
WHP → ECP



※In order to avoid altering of propelling thrust, change over shall be done at clutch disengage, idling speed, WHP handle neutral and ECP handle neutral.

[change control position]
ECP → ERP



※In order to avoid altering of propelling thrust, change over shall be done at clutch disengage, idling speed, WHP handle neutral and ECP handle neutral.

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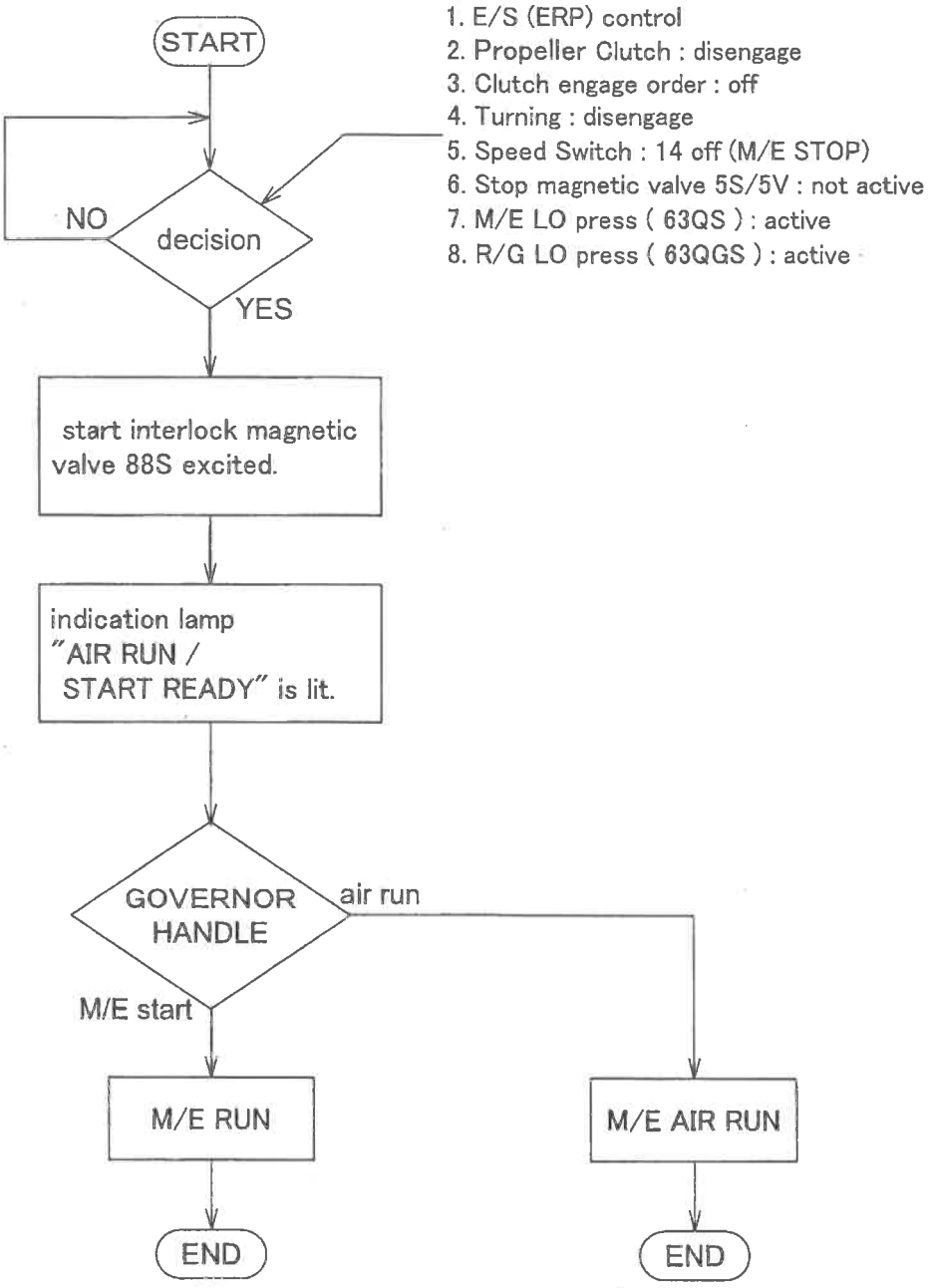
[M/E start・air run]



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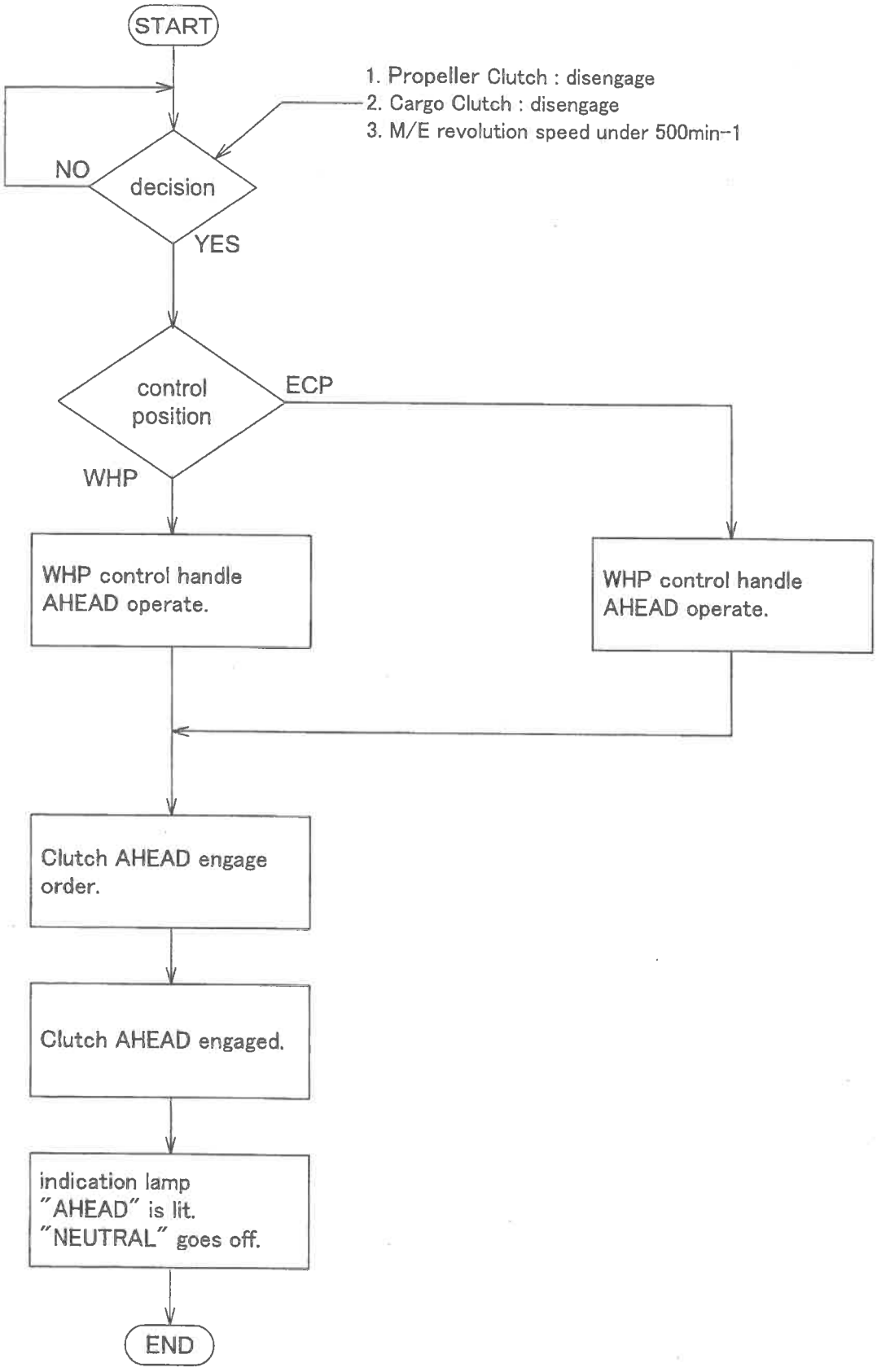
[Clutch control]
AHEAD engage operation



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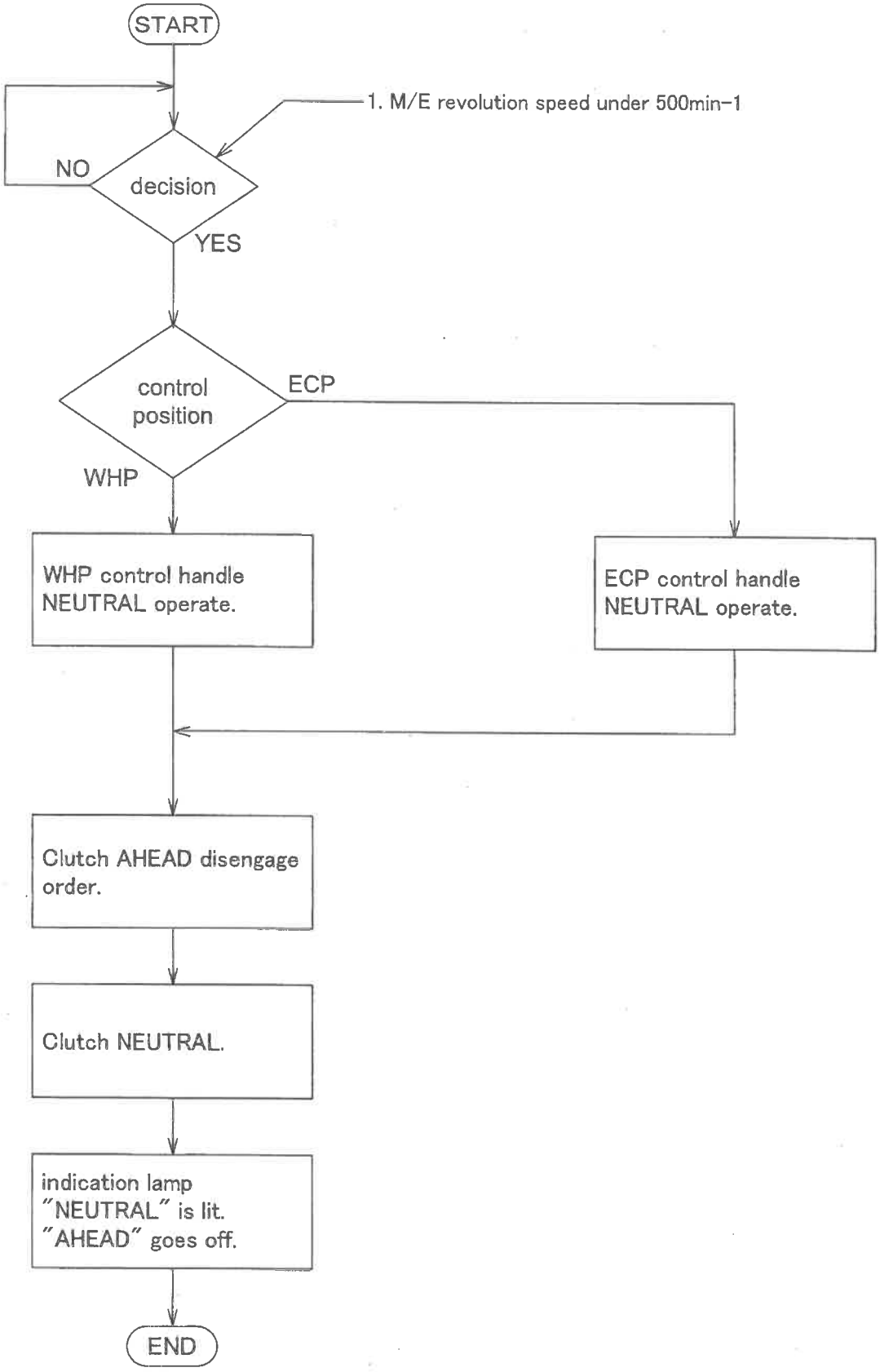
[Clutch control]
AHEAD disengage operation



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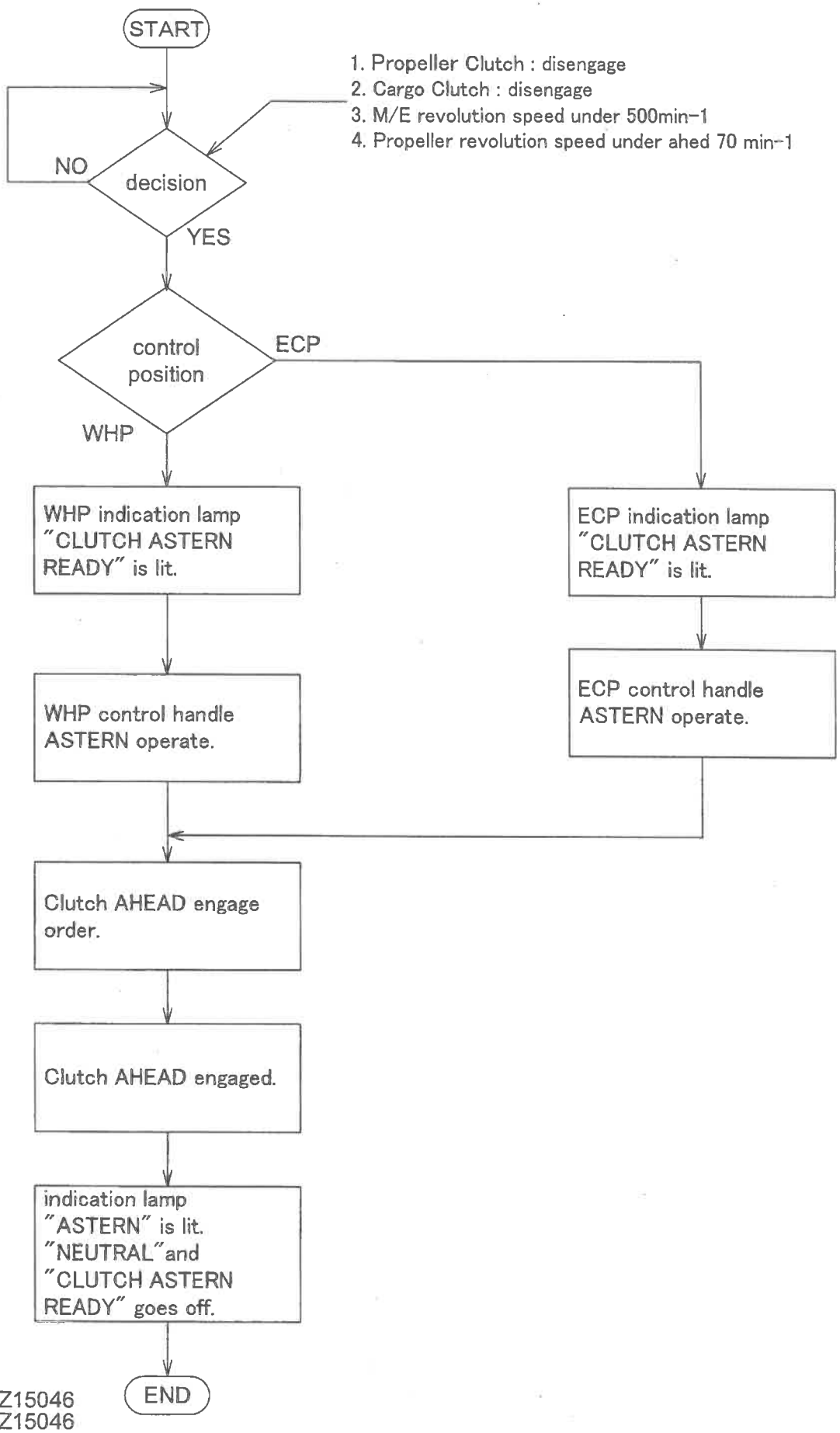
[Clutch control]
ASTERN engage operation



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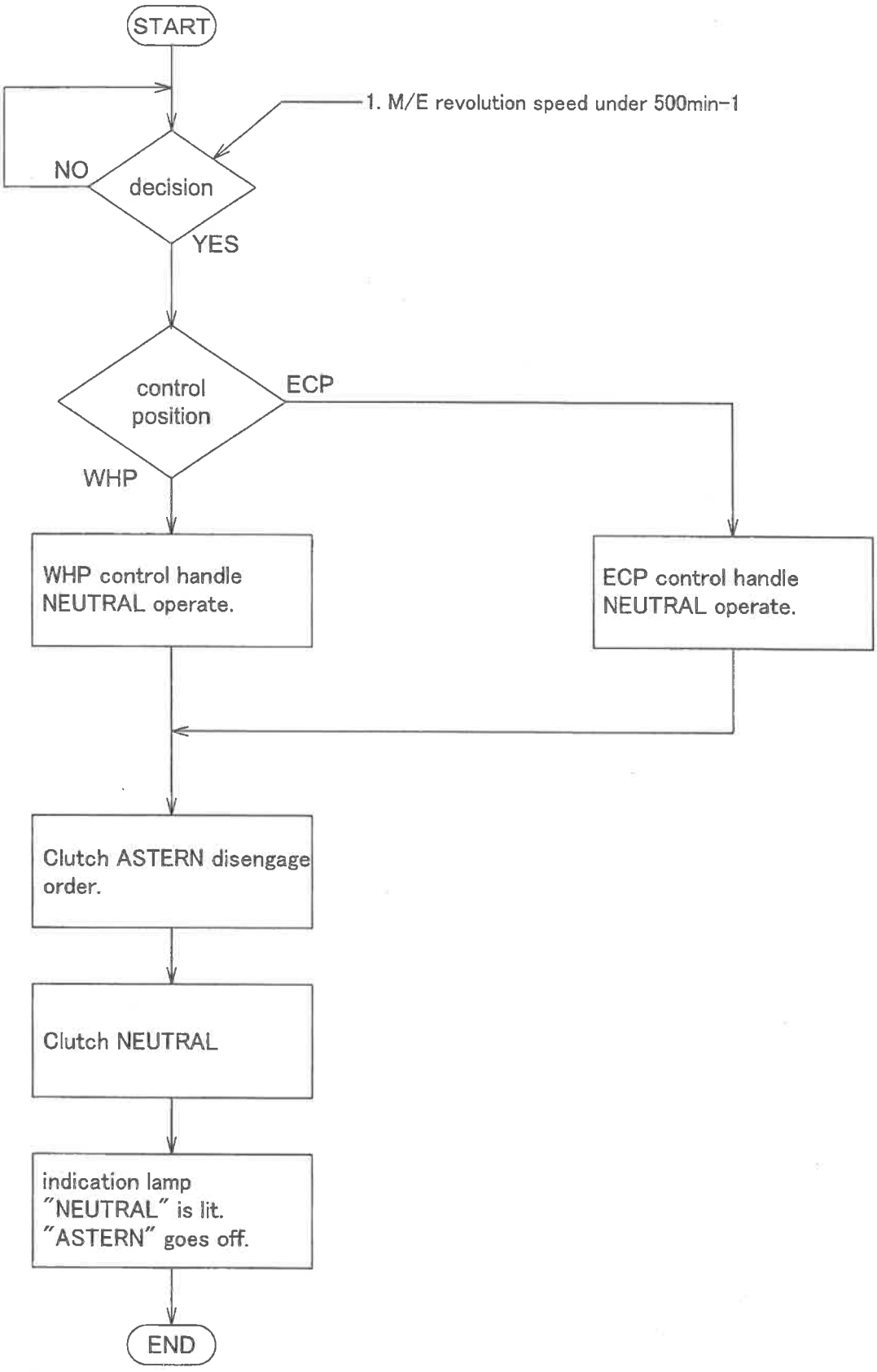
[Clutch control]
ASTERN disengage operation



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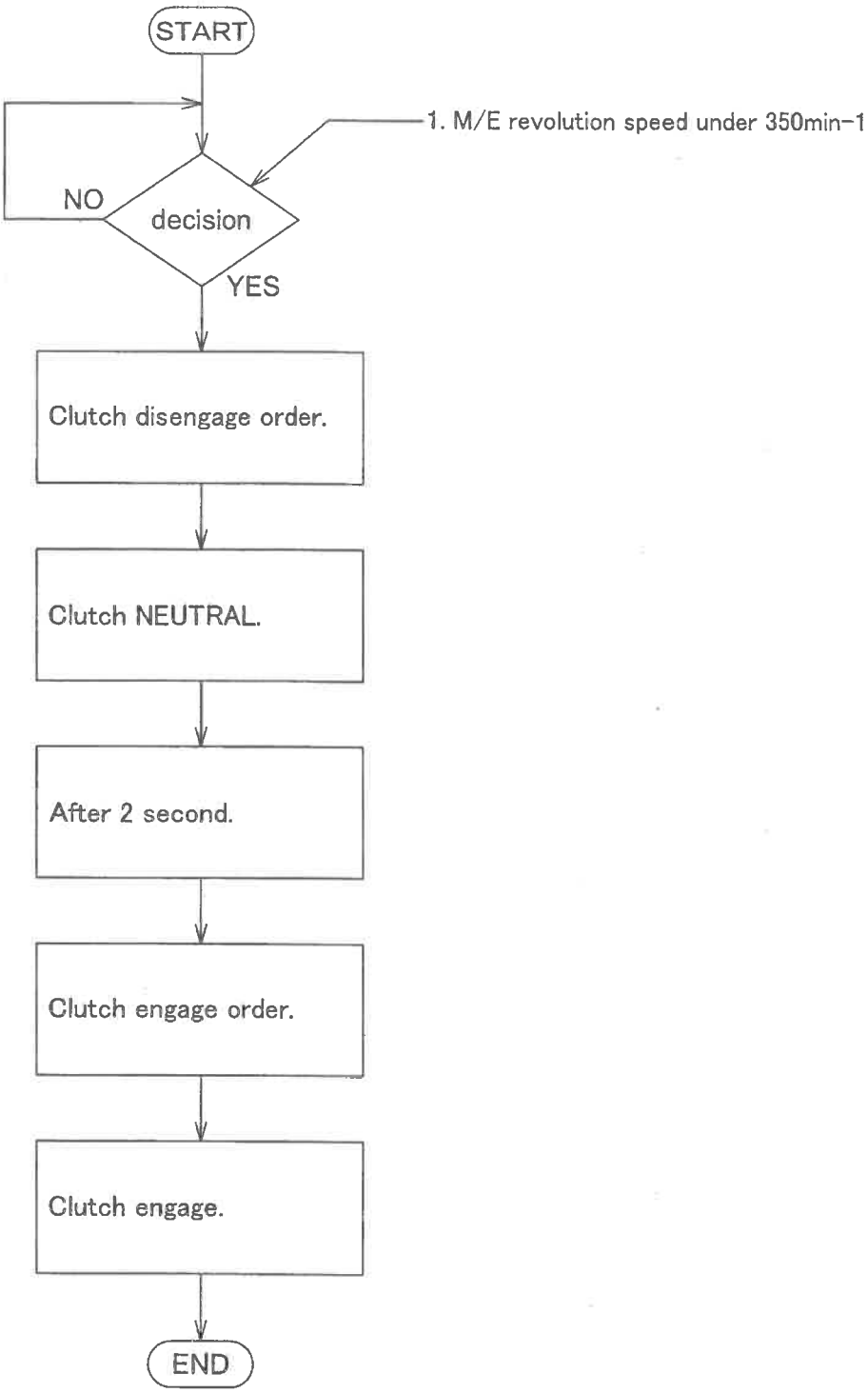
[Clutch control]
Clutch auto disengage



SZ15045 / SZ15046

MAIN TELEGRAPH OPERATION PATTERN(MAIN)

○:LIGHT ◎:FLICKER #:ALARM SOUND ●:ALARM SOUND STOP

CONTROL POSITION	TRANSMITTER RECEIVER	W/H TRANSMITTER	E/R RECEIVER	FUNCTION OF ALARM
W/H E/R	W/H ORDER REPLY E/R (MUTUAL COMMUNICATION) W/H REPLY ORDER E/R	STOP ○	STOP ○	———
		STOP → D/S ○ ORDER ⟹ ◎	STOP → D/S ○ ⟹ ◎	#
		D/S ◎ ⟶ ○	REPLY D/S ◎ ⟶ ○	STOP
		D/S → SOLW ○ ORDER ⟹ ◎	D/S → SOLW ○ ⟹ ◎	#
		SOLW ◎ ⟶ ○	REPLY SOLW ◎ ⟶ ○	STOP
		SOLW → HALF ○ ORDER ⟹ ◎	SOLW → HALF ○ ⟹ ◎	#
		HALF ◎ ⟶ ○	HALF REPLY ◎ ⟶ ○	STOP

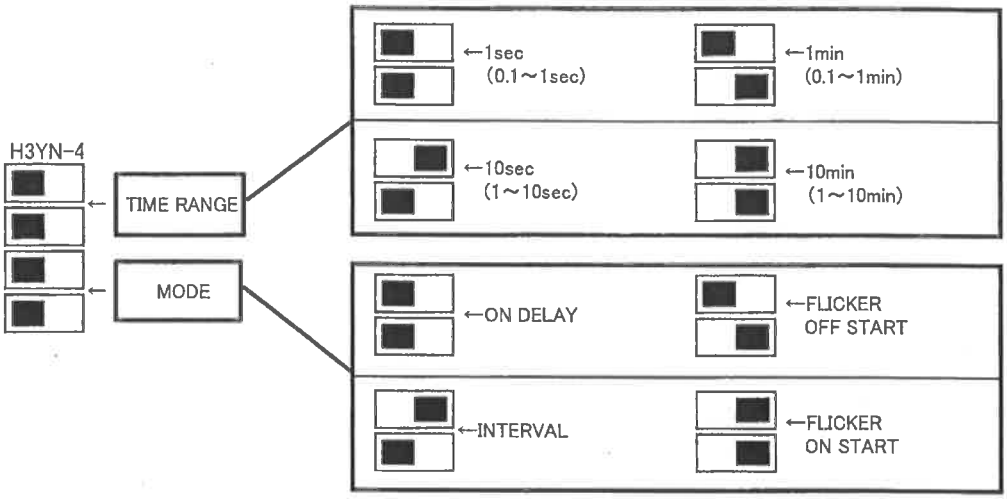
SUB TELEGRAPH OPERATION PATTERN(MAIN)

○:LIGHT ◎:FLICKER #:ALARM SOUND ●:ALARM SOUND STOP

* About S/BY instructions Only when the handle position of a receiver is F/E
An alarm stop can be carried out by S/BY alarm stop button
The alarm stop by stop button cannot be performed from an Engine room at the time of an order

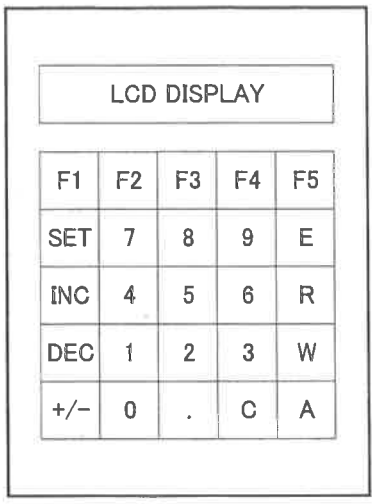
CONTROL POSITION	TRANSMITTER RECEIVER	W/H TRANSMITTER	E/R RECEIVER	FUNCTION OF ALARM
W/H E/R	W/H ORDER REPLY E/R (MUTUAL COMMUNICATION) W/H REPLY ORDER E/R	F/E ○	F/E ○	———
		S/B ORDER ◎	S/B ◎	#
		S/B ◎	S/B BELL STOP ◎ BUTTON	STOP
		S/B ○	S/B REPLY ○	↓ 2SEC STOP
		R/U ORDER ◎	R/U ◎	#
		R/U ○	R/U REPLY ○	STOP
		S/B ORDER ◎	S/B ◎	#
		S/B ○	S/B REPLY ○	STOP
		F/E ORDER ◎	F/E ◎	#
		F/E ○	F/E REPLY ○	STOP

TIMER LIST



NO.	TYPE	NAME	MODE	SET TIME	REMARKS
1	H3YN	TX M53S	INTERVAL	2 SEC	N2010 POWER
2	H3YN	FL 210S	FLICKER ON START	0.5 SEC	ECP FLICKER
3	H3YN	FL 231S	FLICKER ON START	0.5 SEC	WHP FLICKER
4	H3YN	FL 258S	FLICKER ON START	0.5 SEC	CPCP FLICKER
5	H3YN	FL 259S	FLICKER ON START	0.5 SEC	CPCP FLICKER
6	H3YN	TX 327S	INTERVAL	2 SEC	AHEAD CL ENG ORDER
7	H3YN	TX 330S	INTERVAL	2 SEC	AHEAD CL DIS-ENG ORDER
8	H3YN	TX 332S	INTERVAL	2 SEC	ASTERN CL ENG ORDER
9	H3YN	TX 335S	INTERVAL	2 SEC	ASTERN CL DIS-ENG ORDER
10	H3CR	TX 355S	E : (INTERVAL)	10 MIN	HEATING
11	H3CR	TX 366S	A : (ON DELAY)	10 MIN	M/E LO S/B PUMP AFTER COOLING
12	H3CR	TX 373S	A : (ON DELAY)	10 MIN	R/G LO S/B PUMP AFTER COOLING
13	H3CR	TX 377S	A : (ON DELAY)	15 SEC	63QS START INT
14	H3CR	TX 387S	A : (ON DELAY)	15 SEC	63QGS START INT
15	H3YN	TX 427S	ON DELAY	1 SEC	M.V.:OFF + NEUTRAL
16	H3CR	TX 430S	E : (INTERVAL)	15 SEC	CL AHEAD or ASTERN
17	H3CR	TX 431S	E : (INTERVAL)	2 SEC	CL AHEAD or ASTERN
18	H3YN	TX 520S	ON DELAY	0 SEC	M/E LO L-P SHUT/D REPOSE
19	H3YN	TX 521S	ON DELAY	0 SEC	R/G LO L-P SHUT/D REPOSE
20	H3YN	TX 523S	ON DELAY	20 SEC	HANDLE SW ABNL
21	H3YN	TX 552S	ON DELAY	0 SEC	FW HI-TMP SLOW/D REQUEST
22	H3YN	TX 553S	ON DELAY	0 SEC	FW LOW-TMP SLOW/D REQUEST
23	H3YN	TX 579S	ON DELAY	10 SEC	14L STOP
24	H3CR	TX 581S	A : (ON DELAY)	20 SEC	14 RUN
25	H3CR	TX 582S	A : (ON DELAY)	1 MIN	14 RUN

NO.	TYPE	NAME	MODE	SET TIME	REMARKS
26	H3CR	TX 638S	A : (ON DELAY)	2 MIN	M/E LO S/B PUMP OVERLAP RUN
27	H3CR	TX 640S	A : (ON DELAY)	2 MIN	R/G LO S/B PUMP OVERLAP RUN
28	H3YN	TX M53P	INTERVAL	2 SEC	N2010 POWER
29	H3YN	FL 210P	FLICKER ON START	0.5 SEC	ECP FLICKER
30	H3YN	FL 231P	FLICKER ON START	0.5 SEC	WHP FLICKER
31	H3YN	FL 258P	FLICKER ON START	0.5 SEC	CPCP FLICKER
32	H3YN	FL 259P	FLICKER ON START	0.5 SEC	CPCP FLICKER
33	H3YN	TX 327P	INTERVAL	2 SEC	AHEAD CL ENG ORDER
34	H3YN	TX 330P	INTERVAL	2 SEC	AHEAD CL DIS-ENG ORDER
35	H3YN	TX 332P	INTERVAL	2 SEC	ASTERN CL ENG ORDER
36	H3YN	TX 335P	INTERVAL	2 SEC	ASTERN CL DIS-ENG ORDER
37	H3CR	TX 355P	E : (INTERVAL)	10 MIN	HEATING
38	H3CR	TX 366P	A : (ON DELAY)	10 MIN	M/E LO S/B PUMP AFTER COOLING
39	H3CR	TX 373P	A : (ON DELAY)	10 MIN	R/G LO S/B PUMP AFTER COOLING
40	H3CR	TX 377P	A : (ON DELAY)	15 SEC	63QS START INT
41	H3CR	TX 387P	A : (ON DELAY)	15 SEC	63QGS START INT
42	H3YN	TX 427P	ON DELAY	1 SEC	M.V.:OFF + NEUTRAL
43	H3CR	TX 430P	E : (INTERVAL)	15 SEC	CL AHEAD or ASTERN
44	H3CR	TX 431P	E : (INTERVAL)	2 SEC	CL AHEAD or ASTERN
45	H3YN	TX 520P	ON DELAY	0 SEC	M/E LO L-P SHUT/D REPOSE
46	H3YN	TX 521P	ON DELAY	0 SEC	R/G LO L-P SHUT/D REPOSE
47	H3YN	TX 523P	ON DELAY	20 SEC	HANDLE SW ABNL
48	H3YN	TX 552P	ON DELAY	0 SEC	FW HI-TMP SLOW/D REQUEST
49	H3YN	TX 553P	ON DELAY	0 SEC	FW LOW-TMP SLOW/D REQUEST
50	H3YN	TX 579P	ON DELAY	10 SEC	14L STOP
51	H3CR	TX 581P	A : (ON DELAY)	20 SEC	14 RUN
52	H3CR	TX 582P	A : (ON DELAY)	1 MIN	14 RUN
53	H3CR	TX 638P	A : (ON DELAY)	2 MIN	M/E LO S/B PUMP OVERLAP RUN
54	H3CR	TX 640P	A : (ON DELAY)	2 MIN	R/G LO S/B PUMP OVERLAP RUN
55	H3YN	TX 901	ON DELAY	7 SEC	ALARM REPOSE
56	H3YN	FL 904	FLICKER ON START	0.5 SEC	ALARM UNIT ABNL
57	H3CR	TX 965	A : (ON DELAY)	5 SEC	ALARM AC/DC CONV ABNL
58	H3CR	TX 966	A : (ON DELAY)	5 SEC	ALARM DC/DC CONV ABNL
59	H3CR	TX 972	A : (ON DELAY)	5 SEC	STBD AC/DC CONV ABNL
60	H3CR	TX 973	A : (ON DELAY)	5 SEC	STBD DC/DC CONV ABNL
61	H3CR	TX 974	A : (ON DELAY)	5 SEC	PORT AC/DC CONV ABNL
62	H3CR	TX 975	A : (ON DELAY)	5 SEC	PORT DC/DC CONV ABNL
63					
64					
65					



FUNCTION KEY
F1 SELECT MONITOR DISPLAY

F2 SELECT PARAMETER DISPLAY

F3 TRANSMIT ABNORMAL HISTORY TO RS-422
(WITH PUSH [DP] KEY)

F4 SELECT COMMUNICATION DATA
(WITH PUSH [C] KEY)

F5 ABNORMAL RESET

KEY NAME

SET AUTO PARAMETER SET
* MARK ADDRESS ONLY

INC OUT OF WRIDE MODE : INCREASE ADDRESS +1
WRITE MODE : INCREASE DATA +1

DEC OUT OF WRIDE MODE : DECREASE ADDRESS -1
WRITE MODE : DECREASE DATA -1

+/- SETTING OF ± TO DATA WITHIN PARAMETER RANGE
RESET OF COMMUNICATION UNIT WITH PUSH [F5] KEY OPERATION

C ABNORMAL DISPLAY RESET
WRITE MODE : ZERO CLEAR OF DATA

A OUT OF WRIDE MODE : ADDRESS INPUT POSITION TO HOME
WRITE MODE : TO READ MODE
COMMUNICATION DATA MODE : DISPLAYAED CHANGE OF DATA

E ENTER KEY
WRITE MODE : PARAMETER IS MEMORIZED AND RETURNED TO READ MODE

R READ KEY
TO READ MODE

W WRITE KEY
TO WRITE MODE (PARAMETER SET IS AVAILABLE)

0~9 OUT OF WRIDE MODE : ADDRESS INPUT
WRITE MODE : PARAMETER SETTING
COMMUNICATION DATA MODE : DISPLAYAED CHANGE OF DATA

DECIMAL TRANSMIT ABNORMAL HISTORY TO RS-422
(WITH PUSH [F3] KEY)

WORK NO. PDK20-0178-9 RURE BV JNo.91-5283
ENGINE TYPE 6DKM-20eF 2M/E 2SHAF Y/M/D SWN/Ver
AIO
DISPLAY MONITOR DIO 2015/10/28 S022/01.02
ANALOGUE INPUT COMM

ADDRESS	CONTENT				
000	M/E CONTROL INFORMATION	1st LINE	CLUTCH INFO	BOOST INFO	CONTROL POS.
		2nd LINE	HANDLE ORD.	PROGRAM ORD.	GOV [mA]
		2rd LINE	M/E REV.	PELLER REV.	SLIP [mA]
		4th LINE			RACK LIMIT[mA]
001	INPUT INFORMATION	1st LINE			
		2nd LINE	W/H HANDLE REV. /SLIP.[mA]/ POTENTIO[V]		
		2rd LINE	CP DIAL REV./POTENTIO[V]		
		4th LINE	CPCP DIAL REV./POTENTIO[V]		
002	INPUT INFORMATION	1st LINE			
		2nd LINE	CR HANDLE REV./SLIP[mA]/POTENTIO[V]		
		2rd LINE	RACK LIMIT ADJ./ RACKLIMIT DIAL		
		4th LINE	POTENTIO[V]		
003	INPUT INFORMATION	1st LINE			
		2nd LINE	M/E REV. / INPUT[mA]		
		2rd LINE	PELLER REV. / INPUT[mA]		
		4th LINE			
004	SPARE	1st LINE			
		2nd LINE			
		2rd LINE			
		4th LINE			

AIO UNIT SW1=ALL OFF
DIO UNIT SW1=ALL OFF
COMM RSW1=0 RSW2=0

DISPLAY DIGITAL IN/OUT

ADDRESS	Bit	CONTENT	FROM	TERMINAL
006 AIO1 IN	7	M/E RUN		TB2-2-3
	6	CLUTCH AST. ENGAGE		TB2-2-2
	5	CLUTCH AHD. ENGAGE		TB2-2-1
	4	CP CONTROL		TB2-1-6
	3	CPCP CONTROL		TB2-1-5
	2	WH CONTROL		TB2-1-4
	1	CR CONTROL		TB2-1-3
	0	ERP CONTROL		TB2-1-2

ADDRESS	Bit	CONTENT	TO	TERMINAL	
006 AIO1 OUT	7	AIO UNIT ABNORMAL		TB2-3-5	Nor close
	6	COMMUNICATION ABNORMAL		TB2-3-4	Nor close
	5	POTENTIO POWER ABNL.		TB2-3-3	Nor close(Add.189,190)
	4	CLUTCH AST. ENGAGE READY		TB2-3-2	
	3	CLUTCH AST. DISENGAGE ORD.		TB2-3-1	
	2	CLUTCH AST. ENGAGE ORD.		TB2-2-6	
	1	CLUTCH AHD. DISENGAGE ORD.		TB2-2-5	
	0	CLUTCH AHD. ENGAGE ORD.		TB2-2-4	

ADDRESS	Bit	CONTENT	FROM	TERMINAL
006 AIO2 IN	7			TB2-2-3
	6			TB2-2-2
	5			TB2-2-1
	4			TB2-1-6
	3			TB2-1-5
	2			TB2-1-4
	1			TB2-1-3
	0			TB2-1-2

ADDRESS	Bit	CONTENT	TO	TERMINAL
006 AIO2 OUT	7			TB2-3-5
	6			TB2-3-4
	5			TB2-3-3
	4			TB2-3-2
	3			TB2-3-1
	2			TB2-2-6
	1			TB2-2-5
	0			TB2-2-4

ADDRESS	Bit	CONTENT	FROM	TERMINAL	COMMON:TB1-3-1,2
007 DIO1 IN Gr1	7	FO HANDLE SW.		TB1-1-6	
	6	STOP M.V.		TB1-1-5	
	5	CR CLUTCH ENGAGE		TB1-1-4	
	4	WH HANDLE AHD. LIMIT SW.		TB1-1-3	
	3	WH HANDLE AST. LIMIT SW.		TB1-1-2	
	2	CR HANDLE AHD. LIMIT SW.		TB1-1-1	
	1	CR HANDLE AST. LIMIT SW.		TB1-2-6	
	0	M/E REV. INCRAESE INTERLOCK		TB1-2-5	

ADDRESS	Bit	CONTENT	FROM	TERMINAL	COMMON:TB1-3-1,2
007 DIO1 IN Gr2	7			TB1-2-4	
	6	WH IDLE BOOST		TB1-2-3	
	5	CR IDLE BOOST		TB1-2-2	
	4			TB1-2-1	
	3			TB1-3-6	
	2			TB1-3-5	
	1			TB1-3-4	
	0	ABNORMAL RESET		TB1-3-3	

ADDRESS	Bit	CONTENT	TO	TERMINAL	COMMON:TB2-2-4
007 DIO1 OUT Gr1	7			TB2-2-5	
	6			TB2-2-6	
	5			TB2-1-1	
	4			TB2-1-2	
	3			TB2-1-3	
	2	IDLE BOOST OPERATE		TB2-1-4	
	1	M/E REV. ABNORMAL		TB2-1-5	
	0	PELLER REV. ABNORMAL		TB2-1-6	

ADDRESS	Bit	CONTENT	TO	TERMINAL	COMMON:TB2-3-1
007 DIO1 OUT Gr2	7	WH HANDLE VOLTAGE ABNL.		TB2-3-2	
	6	WH HANDLE RANGE ABNL.		TB2-3-3	
	5	CR HANDLE VOLTAGE ABNL.		TB2-3-4	
	4	CR HANDLE RANGE ABNL.		TB2-3-5	
	3	CPCP DIAL ABNORMAL		TB2-3-6	
	2	CP DIAL ABNORMAL		TB2-2-1	
	1			TB2-2-2	
	0	DIO UNIT ABNORMAL		TB2-2-3	

DISPLAY DATA

```
   N-2010 SYSTEM
  Version 02.00
    Nunotani Keiki
     Seisakusho Co.
```

INITIAL DISPLAY (POWER ON)
AFTER 2 SECONDS, SHIFT TO
MONITOR MODE

CONTENT

[H]:CLUTCH AHEAD ORDER / [S]:ASTERN ORDER / [N]:NEUTRAL
[H]:ACTUAL CLUTCH AHEAD / [S]:ASTERN / [N]:NEUTRAL
[B]:GOVERNOR BOOST ON

```
000 H H B 120 W/H
  H 420 O 420 G 04.00
  E 420 P 200 S 04.00
            L 04.00
```

ADDRESS OF FOLLOW UP SPEED CONTROL POS
HANDLE ORD, / PROGRAM ORD, / GOV [mA]
M/E REV. / PELLER REV. / SLIP[mA]
RACK LIMIT [mA]

```
          Rev Slip Pot
001 INPUT VALUE
DW/H 380 12.00 05.00
  C/P 380      05.00
  CCP 380      05.00
```

W/H M/E REV ORD. / SLIP[mA] / POTENTIO[V]
CP M/E REV ORD. / POTENTIO[V]
CPCP M/E REV ORD. / POTENTIO[V]
D:DEVIATION ABNL FROM Add.197、L:UPPER AND LOWER LIMIT ABNL、A:LIMIT AREA ABNL

```
002 INPUT VALUE
  C/P 380 12.00 05.00
  Src           10.00
```

C/R M/E REV ORD. / SLIP[mA] / POTENTIO[V]
RACK LIMIT ADJ. / POTENTIO[V]
POTENTIO[V]

```
003 M/E CONDITION
  M/E   500 12.00 mA
  P/P   120 13.00 mA
```

M/E REV. / INPUT[mA]
PELLER REV. / INPUT[mA]

```
004 SPARE
```

```
005 SPARE
```

MSU-133 UNIT No1, No2 IN/OUT DATA MONITOR

006		MSU-133	DI/O		
Bit7-No.1-0		7-No.2-0			
In	00000000	00000000			
Out	00000000	00000000			

[0]=OFF, [1]=ON

Bit7: 0 = AIO CPU ABNORMAL

0 = AIO CPU ABNORMAL

MSU-108 UNIT No1 INPUT DATA MONITOR

007	MQU-108	No.1		I/O
Bit7-Grp1-0		7-Grp2-0		
In	00000000	00000000		
Out	00000000	00000000		

[0]=OFF, [1]=ON

Gr2Bit0: 0 = DIO CPU ABNORMAL

MSU-108 UNIT No2 INPUT DATA MONITOR

008	MQU-108	No.2		I/O
Bit7-Grp1-0		7-Grp2-0		
In	00000000	00000000		
Out	00000000	00000000		

[0]=OFF, [1]=ON

Gr2Bit0: 0 = DIO CPU ABNORMAL

MSU-108 UNIT No3 INPUT DATA MONITOR

009	MQU-108	No.3		I/O
Bit7-Grp1-0		7-Grp2-0		
In	00000000	00000000		
Out	00000000	00000000		

[0]=OFF, [1]=ON

Gr2Bit0: 0 = DIO CPU ABNORMAL

MSU-133 UNIT & Key PROGRAM VERSION

016	MSU-133	1330	Prg	Ver.
Au1	09000	01000	14030	1
Au2	09000	01000	14030	1
Key	09000	01000	14030	1

SOFTWARE NUMBER, VERSION, CREATE DATE

MQU-108 UNIT PROGRAM VERSION

017	MQU-108	1080	Prg	Ver.
Du1	S0220	01000	15012	2
Du2	S0220	01000	15012	2
Du3	S0220	01000	15012	2

SOFTWARE NUMBER, VERSION, CREATE DATE

S:PROGRAM STANDERD

M/E REMOTE CONTROL
PARAMETER SETTING LIST
(STBD MAIN)
SZ15046

ADDRESS	CONTENT		UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D	
100	CONTROL HANDLE REV. SETTING	AHEAD MAX ORDER REV.	min-1	0	999	900			MAX
101		AHEAD NAV2 ORDER REV. (EQUAL 100%)	min-1	0	999	900			10
102		AHEAD NAV1 ORDER REV. (EQUAL 85%)	min-1	0	999	810			8
103		AHAED FULL ORDER REV. (EQUAL 75%)	min-1	0	999	720			6
104		AHEAD HALF ORDER REV. (EQUAL 50%)	min-1	0	999	630			4
105		AHEAD SLOW ORDER REV.	min-1	0	999	540			2
106		AHEAD D.SLOW ORDER REV.	min-1	0	999	450			0
107		STOP ORDER REV. (NEUTRAL)	min-1	0	999	450			
108		ASTERN D.SLOW ORDER REV.	min-1	0	999	450			0
109		ASTERN SLOW ORDER REV.	min-1	0	999	540			2
110		ASTERN HALF ORDER REV. (EQUAL 50%)	min-1	0	500	630			4
111		ASTERN FULL ORDER REV. (EQUAL 75%)	min-1	0	500	720			6
112		ASTERN EMC1 ORDER REV.	min-1	0	999	810			8
113		ASTERN EMC2 ORDER REV. (EQUAL 85%)	min-1	0	999	853			10
114		ASTERN MAX ORDER REV.	min-1	0	999	853			MAX
115	LIMIT M/E AHEAD REV.		min-1	0	999	900			
116	LIMIT M/E ASTERN REV.		min-1	0	999	900			
117	AUTO SLOW DOWN		min-1	0	999	500			
118	AUTO SLOW DOWN RESET REV.		min-1	0	999	500			
119	REVOLUTION INCREASE ORDER INTERLOCK 0: NO INTER LOCK 1: CLUTCH ENGAGE 2: CLUTCH ENGAGE AFTER SLIP		----	0	2	2			
120	AHD. FOLLOW UP SPEED	INCREASE TIME : Add.107～Add.126	sec	0.0	99.9	1.0			
121		INCREASE TIME : Add.126～Add.127	sec	0.0	99.9	1.8			
122		INCREASE TIME : Add.127～Add.128	sec	0.0	99.9	4.7			
123		INCREASE TIME : Add.128～Add.100	sec	0.0	99.9	0.0			
124		DECREASE TIME : Add.100～Add.129	sec	0.0	99.9	1.0			
125		DECREASE TIME : Add.129～Add.107	sec	0.0	99.9	0.0			
126		INC. SPEED CHANGE POINT (LOW SPEED ZONE)	min-1	0	999	818			
127		INC. SPEED CHANGE POINT (MIDDLE SPEED ZONE)	min-1	0	999	853			
128		INC. SPEED CHANGE POINT (HIGH SPEED ZONE)	min-1	0	999	900			
129		DECREASE SPEED CHANGE POINT	min-1	0	999	450			
130	AST. FOLLOW UP SPEED	INCREASE TIME : Add.107～Add.136	sec	0.0	99.9	1.0			
131		INCREASE TIME : Add.136～Add.137	sec	0.0	99.9	1.4			
132		INCREASE TIME : Add.137～Add.138	sec	0.0	99.9	3.8			
133		INCREASE TIME : Add.138～Add.114	sec	0.0	99.9	0.0			
134		DECREASE TIME : Add.114～Add.139	sec	0.0	99.9	1.0			
135		DECREASE TIME : Add.139～Add.107	sec	0.0	99.9	0.0			
136		INC. SPEED CHANGE POINT (LOW SPEED ZONE)	min-1	0	999	818			
137		INC. SPEED CHANGE POINT (MIDDLE SPEED ZONE)	min-1	0	999	853			
138		INC. SPEED CHANGE POINT (HIGH SPEED ZONE)	min-1	0	999	900			
139		DECREASE SPEED CHANGE POINT	min-1	0	999	450			
140	E/P VALVE CURRENT SETTING	IDLE REV. SETTING	min-1	0	999	450			
141		LOAD 25% REV. SETTING	min-1	0	999	567			
142		LOAD 50% REV. SETTING	min-1	0	999	714			
143		LOAD 75% REV. SETTING	min-1	0	999	818			
144		LOAD 85% REV. SETTING	min-1	0	999	853			
145		LOAD 100% REV. SETTING	min-1	0	999	900			
146		LOAD 110% REV. SETTING	min-1	0	999	929			
147		IDLE POSISION. CURRNT SETTING	mA	4.00	20.00	6.48			
148		LOAD 25% CURRENT SETTING	mA	4.00	20.00	8.42			
149		LOAD 50% CURRENT SETTING	mA	4.00	20.00	10.87			
150		LOAD 75% CURRENT SETTING	mA	4.00	20.00	12.58			
151		LOAD 85% CURRENT SETTING	mA	4.00	20.00	13.16			
152		LOAD 100% CURRENT SETTING	mA	4.00	20.00	13.95			
153		LOAD 110% CURRENT SETTING	mA	4.00	20.00	14.43			
154		E/S CONTROL CURRENT SETTING	mA	4.00	5.00	4.10			
155	GOV. / EP CURRENT OUTPUT (4mA BIT SCALING)		bit	0	4095	649			
156	GOV. / EP CURRENT OUTPUT (20mA BIT SCALING)		bit	0	4095	3221			

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
157	AHEAD NEUTRAL RECOVERY DETECTION REV.	min-1	0	999	350		
158	AHEAD NEUTRAL KEEP TIME (0 : NOT RECOVERY)	sec	0.0	99.9	2.0		
159	ASTERN NEUTRAL RECOVERY DETECTION REV.	min-1	0	999	350		
160	ASTERN NEUTRAL KEEP TIME (0 : NOT RECOVERY)	sec	0.0	99.9	2.0		
161	NEUTRAL RECOVERY CANCEL TIME (999 : NOT CANCEL)	sec	0	999	60		
162	AHEAD GOVERNOR BOOST REV. SET	min-1	0	200	0		
163	AHEAD GOVERNOR BOOST TIME SET (0 : NOT BOOST)	sec	0.0	99.9	0.0		
164	ASTERN GOVERNOR BOOST REV. SET	min-1	0	200	0		
165	ASTERN GOVERNOR BOOST TIME SET (0 : NOT BOOST)	sec	0.0	99.9	0.0		
166	GOVERNOR BOOST REV. SET AT AUTO DISENGAGE	min-1	0	200	0		
167	GOV BOOST TIME SET AT AUTO DISENGAGE (0 : DISABLED)	sec	0.0	99.9	0.0		
168	CHANGE GOV BOOST SET (0:CLUTCH ODR/ 1:DIRECT/ 2:BOTH)	---	0	2	1		
169	GOVERNOR BOOST APPLY PELLER REV. (ASTERN ONLY)	min-1	0	999	80		
170	M/E REV UPPER LIMIT AT AST CL ENG READY	min-1	0	999	500		
171	M/E REV LOWER LIMIT AT AST CL ENG READY	min-1	0	999	440		
172	M/E REV UPPER LIMIT AT AST CL DIS-ENG READY	min-1	0	999	500		
173	PELLER REV UPPER LIMIT AT AST CL ENG READY	min-1	0	999	70		
174	CLUTCH ENGAGE/DISENGAGE ORDER TIME (0 set : continuous)	sec	0.0	99.9	0.0		
175	M/E MAX REV. (20mA)	min-1	0	1600	1500		
176	M/E REV. 20mA SCALING (*)	bit	0	8192	7143		
177	M/E REV. 4mA SCALING (*)	bit	0	8192	1408		
178	PELLER MAX REVOLUTION (20mA)	min-1	0	999	400		
179	PELLER REV. 20mA SCALING (*)	bit	0	8192	7122		
180	PELLER REV. 4mA SCALING (*)	bit	0	8192	1417		
181	CH1 ABNL DETECTION CURRENT UPPER LIMIT(M/E REV.)	mA	0.00	21.00	20.30		
182	CH1 ABNL DETECTION CURRENT LOWER LIMIT(M/E REV.)	mA	0.00	21.00	3.70		
183	CH2 ABNL DETECTION CURRENT UPPER LIMIT(PELLER REV.)	mA	0.00	21.00	20.30		
184	CH2 ABNL DETECTION CURRENT LOWER LIMIT(PELLER REV.)	mA	0.00	21.00	3.70		
185	CH3 ABNL DETECTION VOLTAGE UPPER LIMIT(POTENTIO POWER)	volt	0.00	10.00	9.70		
186	CH3 ABNL DETECTION VOLTAGE LOWER LIMIT(POTENTIO POWER)	volt	0.00	10.00	0.30		
187	CH4 ABNL DETECTION VOLTAGE UPPER LIMIT(WH HANDLE)	volt	0.00	10.00	9.70		
188	CH4 ABNL DETECTION VOLTAGE LOWER LIMIT(WH HANDLE)	volt	0.00	10.00	0.30		
189	CH5 ABNL DETECTION VOLTAGE UPPER LIMIT(CPCP DIAL)	volt	0.00	10.00	9.70		
190	CH5 ABNL DETECTION VOLTAGE LOWER LIMIT(CPCP DIAL)	volt	0.00	10.00	0.30		
191	CH6 ABNL DETECTION VOLTAGE UPPER LIMIT(CP DIAL)	volt	0.00	10.00	9.70		
192	CH6 ABNL DETECTION VOLTAGE LOWER LIMIT(CP DIAL)	volt	0.00	10.00	0.30		
193	CH7 ABNL DETECTION VOLTAGE UPPER LIMIT(CR HANDLE)	volt	0.00	10.00	9.70		
194	CH7 ABNL DETECTION VOLTAGE LOWER LIMIT(CR HANDLE)	volt	0.00	10.00	0.30		
195	CH8 ABNL DETECTION VOLTAGE UPPER LIMIT(RACK LIMIT)	volt	0.00	10.00	9.70		
196	CH8 ABNL DETECTION VOLTAGE LOWER LIMIT(RACK LIMIT)	volt	0.00	10.00	0.30		
197	VOLTAGE DEVIATION AT LINE BREAK DETECTION	volt	0.00	10.00	1.00		
198	DEVIATION FOR LINE BREAK COUNT DETECTION(Add.197)	count	1	250	200		
199	HANDLE RANGE ABNL DETECTION TIME(0.0:DISABLED)	sec	0.0	99.9	2.0		

upper limit check is disabled when set max for abnormal detection upper limit.
lower limit check is disabled when set max for abnormal detection lower limit.

M/E REMOTE CONTROL
PARAMETER SETTING LIST
(STBD MAIN)
SZ15046

P.5-12
Ver.1.00
2016/12/26

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D	
200	W/H CONTROL HANDLE VOLTAGE SET	AHEAD MAX POS. VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	2.10			MAX
201		AHEAD NAV2 POS. VOLTAGE(＊)(related Add.101)	volt	0.00	10.00	2.10			10
202		AHEAD NAV1 POS. VOLTAGE(＊)(related Add.102)	volt	0.00	10.00	2.54			8
203		AHEAD FULL POS. VOLTAGE(＊)(related Add.103)	volt	0.00	10.00	3.00			6
204		AHEAD HALF POS. VOLTAGE(＊)(related Add.104)	volt	0.00	10.00	3.46			4
205		AHEAD SLOW POS. VOLTAGE(＊)(related Add.105)	volt	0.00	10.00	3.91			2
206		AHEAD D.SLOW POS. VOLTAGE(＊)(related Add.106)	volt	0.00	10.00	4.39			0
207		AHEAD CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	4.39			
208		ASTERN CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	5.39			
209		ASTERN D.SLOW POS. VOLTAGE(＊)(related Add.108)	volt	0.00	10.00	5.39			0
210		ASTERN SLOW POS. VOLTAGE(＊)(related Add.109)	volt	0.00	10.00	5.84			2
211		ASTERN HALF POS. VOLTAGE(＊)(related Add.110)	volt	0.00	10.00	6.28			4
212		ASTERN FULL POS. VOLTAGE(＊)(related Add.111)	volt	0.00	10.00	6.74			6
213		ASTERN EMC1 POS. VOLTAGE(＊)(related Add.112)	volt	0.00	10.00	7.20			8
214		ASTERN EMC2 POS. VOLTAGE(＊)(related Add.113)	volt	0.00	10.00	7.65			10
215		ASTERN MAX POS. VOLTAGE(＊)(related Add.114)	volt	0.00	10.00	7.65			MAX
216	W/H CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS		volt	0.00	10.00	0.20			
217	W/H CONT HANDLE RANGE, ABNL DETECTION DISABLED VOL RANGE		volt	0.00	10.00	1.00			
218	W/H GOVERNOR > CHANGE SLIP HYSTERISIS		volt	0.00	10.00	0.20			
219	W/H HANDLE SIGNAL CHANGE(0=INC 1=DEC)		----	0	1	1			
220	C/R CONTROL HANDLE VOLTAGE SET	AHEAD MAX POS. VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	2.19			MAX
221		AHEAD NAV2 POS. VOLTAGE(＊)(related Add.101)	volt	0.00	10.00	2.19			10
222		AHEAD NAV1 POS. VOLTAGE(＊)(related Add.102)	volt	0.00	10.00	2.66			8
223		AHEAD FULL POS. VOLTAGE(＊)(related Add.103)	volt	0.00	10.00	3.13			6
224		AHEAD HALF POS. VOLTAGE(＊)(related Add.104)	volt	0.00	10.00	3.57			4
225		AHEAD SLOW POS. VOLTAGE(＊)(related Add.105)	volt	0.00	10.00	4.03			2
226		AHEAD D.SLOW POS. VOLTAGE(＊)(related Add.106)	volt	0.00	10.00	4.50			0
227		AHEAD CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	4.50			
228		ASTERN CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	5.51			
229		ASTERN D.SLOW POS. VOLTAGE(＊)(related Add.108)	volt	0.00	10.00	5.51			0
230		ASTERN SLOW POS. VOLTAGE(＊)(related Add.109)	volt	0.00	10.00	5.96			2
231		ASTERN HALF POS. VOLTAGE(＊)(related Add.110)	volt	0.00	10.00	6.42			4
232		ASTERN FULL POS. VOLTAGE(＊)(related Add.111)	volt	0.00	10.00	6.87			6
233		ASTERN EMC1 POS. VOLTAGE(＊)(related Add.112)	volt	0.00	10.00	7.32			8
234		ASTERN EMC2 POS. VOLTAGE(＊)(related Add.113)	volt	0.00	10.00	7.77			10
235		ASTERN MAX POS. VOLTAGE(＊)(related Add.114)	volt	0.00	10.00	7.77			MAX
236	C/R CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS		volt	0.00	10.00	0.20			
237	C/R CONT HANDLE AREA, ABNL DETECTION DISABLE VOL RANGE		volt	0.00	10.00	1.00			
238	C/R GOVERNOR > SLIP CHANGE HYSTERISIS		volt	0.00	10.00	0.20			
239	C/R HANDLE SIGNAL CHANGE(0=INC 1=DEC)		----	0	1	1			

SHALL BE POWER RESET AT CHANGE Add.219, Add239 PARAMETER

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
240	CPCP DIAL VOLTAGE SET	MAX VOLTAGE(*)(related Add.280)	volt	0.00	10.00	8.07		
241								
242								
243								
244								
245								
246								
247		MINIMUM VOLTAGE(*)	volt	0.00	10.00	2.49		
248								
249								
250								
251								
252								
253								
254								
255								
256								
257								
258								
259								
260	CP DIAL VOLTAGE SET	MAX VOLTAGE(*)(related Add.100)	volt	0.00	10.00	7.40		
261								
262								
263								
264								
265								
266								
267		MINIMUM VOLTAGE(*)	volt	0.00	10.00	1.84		
268								
269								
270								
271								
272								
273								
274								
275								
276								
277								
278								
279								
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283								
284								
285								
286								
287								
288								
289								
290								
291								
292								
293								
294								
295		CARGO DIAL MAX REV.	min-1	0	999	900		
296		CARGO DIAL MINIMUM REV.	min-1	0	999	450		
297		CARGO DIAL FOLLU UP SPEED	min-1/sec	0	999	20		
298								
299								

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
300	AHEAD RACK LIMIT	LIMIT POSITION 1 REV. SET(related Add.305)	min-1	0	999	450		
301		LIMIT POSITION 2 REV. SET(related Add.306)	min-1	0	999	567		
302		LIMIT POSITION 3 REV. SET(related Add.307)	min-1	0	999	714		
303		LIMIT POSITION 4 REV. SET(related Add.308)	min-1	0	999	818		
304		LIMIT POSITION 5 REV. SET(related Add.309)	min-1	0	999	900		
305		LIMIT POS 1 CURRENT SET(related Add.300)	mA	4.00	20.00	6.50		
306		LIMIT POS 2 CURRENT SET(related Add.301)	mA	4.00	20.00	8.40		
307		LIMIT POS 3 CURRENT SET(related Add.302)	mA	4.00	20.00	10.90		
308		LIMIT POS 4 CURRENT SET(related Add.303)	mA	4.00	20.00	12.60		
309		LIMIT POS 5 CURRENT SET(related Add.304)	mA	4.00	20.00	14.22		
310	ASTERN RACK LIMIT	LIMIT POS 1 REV. SET(related Add.315)	min-1	0	999	450		
311		LIMIT POS 2 REV. SET(related Add.316)	min-1	0	999	567		
312		LIMIT POS 3 REV. SET(related Add.317)	min-1	0	999	714		
313		LIMIT POS 4 REV. SET(related Add.318)	min-1	0	999	818		
314		LIMIT POS 5 REV. SET(related Add.319)	min-1	0	999	853		
315		LIMIT POS 1 CURRENT SET(related Add.310)	mA	4.00	20.00	6.48		
316		LIMIT POS 2 CURRENT SET(related Add.311)	mA	4.00	20.00	8.42		
317		LIMIT POS 3 CURRENT SET(related Add.312)	mA	4.00	20.00	10.87		
318		LIMIT POS 4 CURRENT SET(related Add.313)	mA	4.00	20.00	12.58		
319		LIMIT POS 5 CURRENT SET(related Add.314)	mA	4.00	20.00	13.16		
320	LIMIT CURRENT at M/E STOP		mA	4.00	20.00	4.10		
321	LIMIT CURRENT at M/E RUN		mA	4.00	20.00	20.00		
322	RACK LIMIT ADJUSTMENT DIAL UPPER LIMIT		mA	0.00	5.00	3.00		
323	RACK LIMIT ADJUSTMENT DIAL LOWER LIMIT		mA	-0.00	-5.00	-3.00		
324	RACK LIMIT ADJ. DIAL INC. MAX POS VOLTAGE(*)		volt	0.00	10.00	8.52		
325	RACK LIMIT ADJ. DIAL "0" POS VOLTAGE(*)		volt	0.00	10.00	4.66		
326	RACK LIMIT ADJ. DIAL DEC. MAX POS VOLTAGE(*)		volt	0.00	10.00	0.82		
327	RACK LIMIT OUTPUT CURRENT:bit SCALING OF 4mA		bit	0	4095	645		
328	RACK LIMIT OUTPUT CURRENT:bit SCALING OF 20mA		bit	0	4095	3223		
329	SLIP ENABELED REV. ORDER UPPER LIMIT		min-1	0	999	450		
330	SLIP	AHEAD GOVERNOR RANGE CURRENT SET.	mA	4.00	20.00	20.00		
331		AHEAD SLIP END POS CURRENT SET.	mA	4.00	20.00	15.00		
332		AHEAD SLIP START POS CURRENT SET.	mA	4.00	20.00	13.00		
333		IDLE RANGE CURRENT SET.	mA	4.00	20.00	12.00		
334		ASTERN START POS CURRENT SET.	mA	4.00	20.00	11.00		
335		ASTERN END POS CURRENT SET.	mA	4.00	20.00	9.00		
336		ASTERN GOVERNOR RANGE CURRENT SET.	mA	4.00	20.00	4.00		
337		E/S:FO HANDLE SW. STOP CURRENT SET.	mA	4.00	20.00	12.00		
338		E/S:FO HANDLE SW. RUN CURRENT SET.	mA	4.00	20.00	12.00		
339		WAITING TIME OF SLIP ENABLED REV. UPPER LIMIT	sec	0.0	99.9	0.5		
340		SLIP ENABLED REV. UPPER LIMIT	min-1	0	999	400		
341		SLIP OUTPUT SELECT OF NEUTRAL(0:OFF, 1:ON)	---	0	1	0		
342		SLIP CURRENT OUTPUT:bit SCALING OF 4mA	bit	0	4095	649		
343		SLIP CURRENT OUTPUT:bit SCALING OF 20mA	bit	0	4095	3233		
344								
345								
346								
347								
348								
349								

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
350							
351							
352							
353							
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355							
356							
357							
358							
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366							
367							
368							
369							
370							
371							
372							
373							
374							
375							
376	CARGO REVOLUTION INCREASE ORDER INTERLOCK	---	0	1	1		
377							
378							
379							
380							
381							
382							
383							
384							
385							
386							
387							
388							
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398							
399							

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
400							
401	IDLE BOOST UP REV.	min-1	0	999	20		
402	BOOST UP HOLD TIME at AFTER CL ENGAGED	sec	0	999	10		
403							
404							
405							
406							
407							
408							
409	NOTUSE						
410	CH1 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
411	CH2 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
412	CH3 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
413	CH4 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
414	CH5 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
415	CH6 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
416	CH7 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
417	CH8 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
418	AHD. PELLER REV. OF CRASH AST. DETECTION	min-1	0	999	200		
419	RECOVER NEUTRAL DETECTION CONT AT CRASH AST	min-1	0	999	440		
420	AST. PELLER REV. OF CRASH AHD. DETECTION	min-1	0	999	150		
421	RECOVER NEUTRAL DETECTION CONT AT CRASH AHD	min-1	0	999	440		
422	AHD PLLER REV OF AHD CL ENG READY AT CRASH AST	min-1	0	999	70		
423	AST PLLER REV OF AST CL ENG READY AT CRASH AHD	min-1	0	999	999		
424	BOOST REV. AT CRASH AST. RECOVER NEUTRAL	min-1	0	999	0		
425	BOOST REV. AT CRASH AHD. RECOVER NEUTRAL	min-1	0	999	0		
426	M/E REV UPPER LIMIT AT AHD CL ENG READY	min-1	0	999	500		
427	M/E REV LOWER LIMIT AT AHD CL ENG READY	min-1	0	999	440		
428	M/E REV UPPER LIMIT AT AHD CL DIS-ENG READY	min-1	0	999	500		
429	PELLER REV UPPER LIMIT AT AHD CL ENG READY	min-1	0	999	999		

BOOST PRIORITY : HEATING (Add.400)>IDLE BOOST(Add.401,402)>GOV. BOOST(Add.162~167)

: RED ADDRESS DOES NOT BECOME "NOT USE",AND DISABLE THE FEATURE BY PARAMETER
: YELLOW ADDRESS DOES NOT BECOME "NOT USE",BUT NOT INCLUDED IN THE PROGRAM FEATURE
: GREEN ADDRESS IS "NOT USE",AND NOT INCLUDED IN THE PROGRAM FEATURE

M/E REMOTE CONTROL
PARAMETER SETTING LIST
(STBD BACK UP)
SZ15046

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D	
100	CONTROL HANDLE REV. SETTING	AHEAD MAX ORDER REV.	min-1	0	999	900			MAX
101		AHEAD NAV2 ORDER REV. (EQUAL 100%)	min-1	0	999	900			10
102		AHEAD NAV1 ORDER REV. (EQUAL 85%)	min-1	0	999	810			8
103		AHAED FULL ORDER REV. (EQUAL 75%)	min-1	0	999	720			6
104		AHEAD HALF ORDER REV. (EQUAL 50%)	min-1	0	999	630			4
105		AHEAD SLOW ORDER REV.	min-1	0	999	540			2
106		AHEAD D.SLOW ORDER REV.	min-1	0	999	450			0
107		STOP ORDER REV. (NEUTRAL)	min-1	0	999	450			
108		ASTERN D.SLOW ORDER REV.	min-1	0	999	450			0
109		ASTERN SLOW ORDER REV.	min-1	0	999	540			2
110		ASTERN HALF ORDER REV. (EQUAL 50%)	min-1	0	500	630			4
111		ASTERN FULL ORDER REV. (EQUAL 75%)	min-1	0	500	720			6
112		ASTERN EMC1 ORDER REV.	min-1	0	999	810			8
113		ASTERN EMC2 ORDER REV. (EQUAL 85%)	min-1	0	999	853			10
114		ASTERN MAX ORDER REV.	min-1	0	999	853			MAX
115	LIMIT M/E AHEAD REV.		min-1	0	999	900			
116	LIMIT M/E ASTERN REV.		min-1	0	999	900			
117	AUTO SLOW DOWN		min-1	0	999	500			
118	AUTO SLOW DOWN RESET REV.		min-1	0	999	500			
119	REVOLUTION INCREASE ORDER INTERLOCK 0: NO INTER LOCK 1: CLUTCH ENGAGE 2: CLUTCH ENGAGE AFTER SLIP		—	0	2	2			
120	AHD. FOLLOW UP SPEED	INCREASE TIME : Add.107～Add.126	sec	0.0	99.9	1.0			
121		INCREASE TIME : Add.126～Add.127	sec	0.0	99.9	1.8			
122		INCREASE TIME : Add.127～Add.128	sec	0.0	99.9	4.7			
123		INCREASE TIME : Add.128～Add.100	sec	0.0	99.9	0.0			
124		DECREASE TIME : Add.100～Add.129	sec	0.0	99.9	1.0			
125		DECREASE TIME : Add.129～Add.107	sec	0.0	99.9	0.0			
126		INC. SPEED CHANGE POINT (LOW SPEED ZONE)	min-1	0	999	818			
127		INC. SPEED CHANGE POINT (MIDDLE SPEED ZONE)	min-1	0	999	853			
128		INC. SPEED CHANGE POINT (HIGH SPEED ZONE)	min-1	0	999	900			
129		DECREASE SPEED CHANGE POINT	min-1	0	999	450			
130	AST. FOLLOW UP SPEED	INCREASE TIME : Add.107～Add.136	sec	0.0	99.9	1.0			
131		INCREASE TIME : Add.136～Add.137	sec	0.0	99.9	1.4			
132		INCREASE TIME : Add.137～Add.138	sec	0.0	99.9	3.8			
133		INCREASE TIME : Add.138～Add.114	sec	0.0	99.9	0.0			
134		DECREASE TIME : Add.114～Add.139	sec	0.0	99.9	1.0			
135		DECREASE TIME : Add.139～Add.107	sec	0.0	99.9	0.0			
136		INC. SPEED CHANGE POINT (LOW SPEED ZONE)	min-1	0	999	818			
137		INC. SPEED CHANGE POINT (MIDDLE SPEED ZONE)	min-1	0	999	853			
138		INC. SPEED CHANGE POINT (HIGH SPEED ZONE)	min-1	0	999	900			
139		DECREASE SPEED CHANGE POINT	min-1	0	999	450			
140	E/P VALVE CURRENT SETTING	IDLE REV. SETTING	min-1	0	999	450			
141		LOAD 25% REV. SETTING	min-1	0	999	567			
142		LOAD 50% REV. SETTING	min-1	0	999	714			
143		LOAD 75% REV. SETTING	min-1	0	999	818			
144		LOAD 85% REV. SETTING	min-1	0	999	853			
145		LOAD 100% REV. SETTING	min-1	0	999	900			
146		LOAD 110% REV. SETTING	min-1	0	999	929			
147		IDLE POSISION. CURRNT SETTING	mA	4.00	20.00	6.48			
148		LOAD 25% CURRENT SETTING	mA	4.00	20.00	8.42			
149		LOAD 50% CURRENT SETTING	mA	4.00	20.00	10.87			
150		LOAD 75% CURRENT SETTING	mA	4.00	20.00	12.58			
151		LOAD 85% CURRENT SETTING	mA	4.00	20.00	13.16			
152		LOAD 100% CURRENT SETTING	mA	4.00	20.00	13.95			
153		LOAD 110% CURRENT SETTING	mA	4.00	20.00	14.43			
154		E/S CONTROL CURRENT SETTING	mA	4.00	5.00	4.10			
155	GOV. / EP CURRENT OUTPUT (4mA BIT SCALING)		bit	0	4095	636			
156	GOV. / EP CURRENT OUTPUT (20mA BIT SCALING)		bit	0	4095	3202			

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
157	AHEAD NEUTRAL RECOVERY DETECTION REV.	min-1	0	999	350		
158	AHEAD NEUTRAL KEEP TIME (0 : NOT RECOVERY)	sec	0.0	99.9	2.0		
159	ASTERN NEUTRAL RECOVERY DETECTION REV.	min-1	0	999	350		
160	ASTERN NEUTRAL KEEP TIME (0 : NOT RECOVERY)	sec	0.0	99.9	2.0		
161	NEUTRAL RECOVERY CANCEL TIME (999 : NOT CANCEL)	sec	0	999	60		
162	AHEAD GOVERNOR BOOST REV. SET	min-1	0	200	0		
163	AHEAD GOVERNOR BOOST TIME SET (0 : NOT BOOST)	sec	0.0	99.9	0.0		
164	ASTERN GOVERNOR BOOST REV. SET	min-1	0	200	0		
165	ASTERN GOVERNOR BOOST TIME SET (0 : NOT BOOST)	sec	0.0	99.9	0.0		
166	GOVERNOR BOOST REV. SET AT AUTO DISENGAGE	min-1	0	200	0		
167	GOV BOOST TIME SET AT AUTO DISENGAGE (0 : DISABLED)	sec	0.0	99.9	0.0		
168	CHANGE GOV BOOST SET (0:CLUTCH ODR/ 1:DIRECT/ 2:BOTH)	---	0	2	1		
169	GOVERNOR BOOST APPLY PELLER REV. (ASTERN ONLY)	min-1	0	999	80		
170	M/E REV UPPER LIMIT AT AST CL ENG READY	min-1	0	999	500		
171	M/E REV LOWER LIMIT AT AST CL ENG READY	min-1	0	999	440		
172	M/E REV UPPER LIMIT AT AST CL DIS-ENG READY	min-1	0	999	500		
173	PELLER REV UPPER LIMIT AT AST CL ENG READY	min-1	0	999	70		
174	CLUTCH ENGAGE/DISENGAGE ORDER TIME (0 set : continuous)	sec	0.0	99.9	0.0		
175	M/E MAX REV. (20mA)	min-1	0	1600	1500		
176	M/E REV. 20mA SCALING (*)	bit	0	8192	7120		
177	M/E REV. 4mA SCALING (*)	bit	0	8192	1386		
178	PELLER MAX REVOLUTION (20mA)	min-1	0	999	400		
179	PELLER REV. 20mA SCALING (*)	bit	0	8192	7077		
180	PELLER REV. 4mA SCALING (*)	bit	0	8192	1361		
181	CH1 ABNL DETECTION CURRENT UPPER LIMIT (M/E REV.)	mA	0.00	21.00	20.30		
182	CH1 ABNL DETECTION CURRENT LOWER LIMIT (M/E REV.)	mA	0.00	21.00	3.70		
183	CH2 ABNL DETECTION CURRENT UPPER LIMIT (PELLER REV.)	mA	0.00	21.00	20.30		
184	CH2 ABNL DETECTION CURRENT LOWER LIMIT (PELLER REV.)	mA	0.00	21.00	3.70		
185	CH3 ABNL DETECTION VOLTAGE UPPER LIMIT (POTENTIO POWER)	volt	0.00	10.00	9.70		
186	CH3 ABNL DETECTION VOLTAGE LOWER LIMIT (POTENTIO POWER)	volt	0.00	10.00	0.30		
187	CH4 ABNL DETECTION VOLTAGE UPPER LIMIT (WH HANDLE)	volt	0.00	10.00	9.70		
188	CH4 ABNL DETECTION VOLTAGE LOWER LIMIT (WH HANDLE)	volt	0.00	10.00	0.30		
189	CH5 ABNL DETECTION VOLTAGE UPPER LIMIT (CPCP DIAL)	volt	0.00	10.00	9.70		
190	CH5 ABNL DETECTION VOLTAGE LOWER LIMIT (CPCP DIAL)	volt	0.00	10.00	0.30		
191	CH6 ABNL DETECTION VOLTAGE UPPER LIMIT (CP DIAL)	volt	0.00	10.00	9.70		
192	CH6 ABNL DETECTION VOLTAGE LOWER LIMIT (CP DIAL)	volt	0.00	10.00	0.30		
193	CH7 ABNL DETECTION VOLTAGE UPPER LIMIT (CR HANDLE)	volt	0.00	10.00	9.70		
194	CH7 ABNL DETECTION VOLTAGE LOWER LIMIT (CR HANDLE)	volt	0.00	10.00	0.30		
195	CH8 ABNL DETECTION VOLTAGE UPPER LIMIT (RACK LIMIT)	volt	0.00	10.00	9.70		
196	CH8 ABNL DETECTION VOLTAGE LOWER LIMIT (RACK LIMIT)	volt	0.00	10.00	0.30		
197	VOLTAGE DEVIATION AT LINE BREAK DETECTION	volt	0.00	10.00	1.00		
198	DEVIATION FOR LINE BREAK COUNT DETECTION (Add.197)	count	1	250	200		
199	HANDLE RANGE ABNL DETECTION TIME(0.0:DISABLED)	sec	0.0	99.9	2.0		

upper limit check is disabled when set max for abnormal detection upper limit.
lower limit check is disabled when set max for abnormal detection lower limit.

ADDRESS	CONTENT		UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D	
200	W/H CONTROL HANDLE VOLTAGE SET	AHEAD MAX POS. VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	2.02			MAX
201		AHEAD NAV2 POS. VOLTAGE(＊)(related Add.101)	volt	0.00	10.00	2.02			10
202		AHEAD NAV1 POS. VOLTAGE(＊)(related Add.102)	volt	0.00	10.00	2.48			8
203		AHEAD FULL POS. VOLTAGE(＊)(related Add.103)	volt	0.00	10.00	2.94			6
204		AHEAD HALF POS. VOLTAGE(＊)(related Add.104)	volt	0.00	10.00	3.39			4
205		AHEAD SLOW POS. VOLTAGE(＊)(related Add.105)	volt	0.00	10.00	3.84			2
206		AHEAD D.SLOW POS. VOLTAGE(＊)(related Add.106)	volt	0.00	10.00	4.31			0
207		AHEAD CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	4.31			
208		ASTERN CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	5.32			
209		ASTERN D.SLOW POS. VOLTAGE(＊)(related Add.108)	volt	0.00	10.00	5.32			0
210		ASTERN SLOW POS. VOLTAGE(＊)(related Add.109)	volt	0.00	10.00	5.76			2
211		ASTERN HALF POS. VOLTAGE(＊)(related Add.110)	volt	0.00	10.00	6.21			4
212		ASTERN FULL POS. VOLTAGE(＊)(related Add.111)	volt	0.00	10.00	6.67			6
213		ASTERN EMC1 POS. VOLTAGE(＊)(related Add.112)	volt	0.00	10.00	7.12			8
214		ASTERN EMC2 POS. VOLTAGE(＊)(related Add.113)	volt	0.00	10.00	7.57			10
215		ASTERN MAX POS. VOLTAGE(＊)(related Add.114)	volt	0.00	10.00	7.57			MAX
216	W/H CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS		volt	0.00	10.00	0.20			
217	W/H CONT HANDLE RANGE, ABNL DETECTION DISABLED VOL RANGE		volt	0.00	10.00	1.00			
218	W/H GOVERNOR > CHANGE SLIP HYSTERISIS		volt	0.00	10.00	0.20			
219	W/H HANDLE SIGNAL CHANGE(0=INC 1=DEC)		---	0	1	1			
220	C/R CONTROL HANDLE VOLTAGE SET	AHEAD MAX POS. VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	2.16			MAX
221		AHEAD NAV2 POS. VOLTAGE(＊)(related Add.101)	volt	0.00	10.00	2.16			10
222		AHEAD NAV1 POS. VOLTAGE(＊)(related Add.102)	volt	0.00	10.00	2.63			8
223		AHEAD FULL POS. VOLTAGE(＊)(related Add.103)	volt	0.00	10.00	3.08			6
224		AHEAD HALF POS. VOLTAGE(＊)(related Add.104)	volt	0.00	10.00	3.53			4
225		AHEAD SLOW POS. VOLTAGE(＊)(related Add.105)	volt	0.00	10.00	4.00			2
226		AHEAD D.SLOW POS. VOLTAGE(＊)(related Add.106)	volt	0.00	10.00	4.46			0
227		AHEAD CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	4.46			
228		ASTERN CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	5.47			
229		ASTERN D.SLOW POS. VOLTAGE(＊)(related Add.108)	volt	0.00	10.00	5.47			0
230		ASTERN SLOW POS. VOLTAGE(＊)(related Add.109)	volt	0.00	10.00	5.93			2
231		ASTERN HALF POS. VOLTAGE(＊)(related Add.110)	volt	0.00	10.00	6.38			4
232		ASTERN FULL POS. VOLTAGE(＊)(related Add.111)	volt	0.00	10.00	6.82			6
233		ASTERN EMC1 POS. VOLTAGE(＊)(related Add.112)	volt	0.00	10.00	7.27			8
234		ASTERN EMC2 POS. VOLTAGE(＊)(related Add.113)	volt	0.00	10.00	7.74			10
235		ASTERN MAX POS. VOLTAGE(＊)(related Add.114)	volt	0.00	10.00	7.74			MAX
236	C/R CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS		volt	0.00	10.00	0.20			
237	C/R CONT HANDLE AREA, ABNL DETECTION DISABLE VOL RANGE		volt	0.00	10.00	1.00			
238	C/R GOVERNOR > SLIP CHANGE HYSTERISIS		volt	0.00	10.00	0.20			
239	C/R HANDLE SIGNAL CHANGE(0=INC 1=DEC)		---	0	1	1			

SHALL BE POWER RESET AT CHANGE Add.219, Add239 PARAMETER

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
240	CPCP DIAL VOLTAGE SET	MAX VOLTAGE(＊)(related Add.280)	volt	0.00	10.00	8.04		
241								
242								
243								
244								
245								
246								
247		MINIMUM VOLTAGE(＊)	volt	0.00	10.00	2.46		
248								
249								
250								
251								
252								
253								
254								
255								
256								
257								
258								
259								
260	CP DIAL VOLTAGE SET	MAX VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	7.34		
261								
262								
263								
264								
265								
266								
267		MINIMUM VOLTAGE(＊)	volt	0.00	10.00	1.77		
268								
269								
270								
271								
272								
273								
274								
275								
276								
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283								
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286								
287								
288								
289								
290								
291								
292								
293								
294								
295		CARGO DIAL MAX REV.	min-1	0	999	900		
296		CARGO DIAL MINIMUM REV.	min-1	0	999	450		
297		CARGO DIAL FOLLU UP SPEED	min-1/sec	0	999	20		
298								
299								

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
300	AHEAD RACK LIMIT	LIMIT POSITION 1 REV. SET(related Add.305)	min-1	0	999	450		
301		LIMIT POSITION 2 REV. SET(related Add.306)	min-1	0	999	567		
302		LIMIT POSITION 3 REV. SET(related Add.307)	min-1	0	999	714		
303		LIMIT POSITION 4 REV. SET(related Add.308)	min-1	0	999	818		
304		LIMIT POSITION 5 REV. SET(related Add.309)	min-1	0	999	900		
305		LIMIT POS 1 CURRENT SET(related Add.300)	mA	4.00	20.00	6.50		
306		LIMIT POS 2 CURRENT SET(related Add.301)	mA	4.00	20.00	8.40		
307		LIMIT POS 3 CURRENT SET(related Add.302)	mA	4.00	20.00	10.90		
308		LIMIT POS 4 CURRENT SET(related Add.303)	mA	4.00	20.00	12.60		
309		LIMIT POS 5 CURRENT SET(related Add.304)	mA	4.00	20.00	14.22		
310	ASTERN RACK LIMIT	LIMIT POS 1 REV. SET(related Add.315)	min-1	0	999	450		
311		LIMIT POS 2 REV. SET(related Add.316)	min-1	0	999	567		
312		LIMIT POS 3 REV. SET(related Add.317)	min-1	0	999	714		
313		LIMIT POS 4 REV. SET(related Add.318)	min-1	0	999	818		
314		LIMIT POS 5 REV. SET(related Add.319)	min-1	0	999	853		
315		LIMIT POS 1 CURRENT SET(related Add.310)	mA	4.00	20.00	6.48		
316		LIMIT POS 2 CURRENT SET(related Add.311)	mA	4.00	20.00	8.42		
317		LIMIT POS 3 CURRENT SET(related Add.312)	mA	4.00	20.00	10.87		
318		LIMIT POS 4 CURRENT SET(related Add.313)	mA	4.00	20.00	12.58		
319		LIMIT POS 5 CURRENT SET(related Add.314)	mA	4.00	20.00	13.16		
320		LIMIT CURRENT at M/E STOP	mA	4.00	20.00	4.10		
321		LIMIT CURRENT at M/E RUN	mA	4.00	20.00	20.00		
322		RACK LIMIT ADJUSTMENT DIAL UPPER LIMIT	mA	0.00	5.00	3.00		
323		RACK LIMIT ADJUSTMENT DIAL LOWER LIMIT	mA	-0.00	-5.00	-3.00		
324		RACK LIMIT ADJ. DIAL ING. MAX POS VOLTAGE(*)	volt	0.00	10.00	8.44		
325		RACK LIMIT ADJ. DIAL "0" POS VOLTAGE(*)	volt	0.00	10.00	4.59		
326		RACK LIMIT ADJ. DIAL DEC. MAX POS VOLTAGE(*)	volt	0.00	10.00	0.78		
327		RACK LIMIT OUTPUT CURRENT:bit SCALING OF 4mA	bit	0	4095	641		
328		RACK LIMIT OUTPUT CURRENT:bit SCALING OF 20mA	bit	0	4095	3202		
329		SLIP ENABELED REV. ORDER UPPER LIMIT	min-1	0	999	450		
330	SLIP	AHEAD GOVERNOR RANGE CURRENT SET.	mA	4.00	20.00	20.00		
331		AHEAD SLIP END POS CURRENT SET.	mA	4.00	20.00	15.00		
332		AHEAD SLIP START POS CURRENT SET.	mA	4.00	20.00	13.00		
333		IDLE RANGE CURRENT SET.	mA	4.00	20.00	12.00		
334		ASTERN START POS CURRENT SET.	mA	4.00	20.00	11.00		
335		ASTERN END POS CURRENT SET.	mA	4.00	20.00	9.00		
336		ASTERN GOVERNOR RANGE CURRENT SET.	mA	4.00	20.00	4.00		
337		E/S:FO HANDLE SW. STOP CURRENT SET.	mA	4.00	20.00	12.00		
338		E/S:FO HANDLE SW. RUN CURRENT SET.	mA	4.00	20.00	12.00		
339		WAITING TIME OF SLIP ENABLED REV. UPPER LIMIT	sec	0.0	99.9	0.5		
340		SLIP ENABLED REV. UPPER LIMIT	min-1	0	999	400		
341		SLIP OUTPUT SELECT OF NEUTRAL(0:OFF, 1:ON)	---	0	1	0		
342		SLIP CURRENT OUTPUT:bit SCALING OF 4mA	bit	0	4095	642		
343		SLIP CURRENT OUTPUT:bit SCALING OF 20mA	bit	0	4095	3203		
344								
345								
346								
347								
348								
349								

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
350							
351							
352							
353							
354							
355							
356							
357							
358							
359							
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366							
367							
368							
369							
370							
371							
372							
373							
374							
375							
376	CARGO REVOLUTION INCREASE ORDER INTERLOCK	—	0	1	1		
377							
378							
379							
380							
381							
382							
383							
384							
385							
386							
387							
388							
389							
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397							
398							
399							

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y	SHOP TRIAL M	SHOP TRIAL D	SEA TRIAL Y	SEA TRIAL M	SEA TRIAL D
400											
401	IDLE BOOST UP REV.	min-1	0	999	20						
402	BOOST UP HOLD TIME at AFTER CL ENGAGED	sec	0	999	10						
403											
404											
405											
406											
407											
408											
409	NOTUSE										
410	CH1 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200						
411	CH2 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200						
412	CH3 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200						
413	CH4 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200						
414	CH5 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200						
415	CH6 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200						
416	CH7 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200						
417	CH8 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200						
418	AHD. PELLER REV. OF CRASH AST. DETECTION	min-1	0	999	200						
419	RECOVER NEUTRAL DETECTION CONT AT CRASH AST	min-1	0	999	440						
420	AST. PELLER REV. OF CRASH AHD. DETECTION	min-1	0	999	150						
421	RECOVER NEUTRAL DETECTION CONT AT CRASH AHD	min-1	0	999	440						
422	AHD PLLER REV OF AHD CL ENG READY AT CRASH AST	min-1	0	999	70						
423	AST PLLER REV OF AST CL ENG READY AT CRASH AHD	min-1	0	999	999						
424	BOOST REV. AT CRASH AST. RECOVER NEUTRAL	min-1	0	999	0						
425	BOOST REV. AT CRASH AHD. RECOVER NEUTRAL	min-1	0	999	0						
426	M/E REV UPPER LIMIT AT AHD CL ENG READY	min-1	0	999	500						
427	M/E REV LOWER LIMIT AT AHD CL ENG READY	min-1	0	999	440						
428	M/E REV UPPER LIMIT AT AHD CL DIS-ENG READY	min-1	0	999	500						
429	PELLER REV UPPER LIMIT AT AHD CL ENG READY	min-1	0	999	999						

BOOST PRIORITY : HEATING (Add.400)>IDLE BOOST(Add.401,402)>GOV. BOOST(Add.162～167)

: RED ADDRESS DOES NOT BECOME "NOT USE",AND DISABLE THE FEATURE BY PARAMETER
: YELLOW ADDRESS DOES NOT BECOME "NOT USE",BUT NOT INCLUDED IN THE PROGRAM FEATURE
: GREEN ADDRESS IS "NOT USE",AND NOT INCLUDED IN THE PROGRAM FEATURE

M/E REMOTE CONTROL
PARAMETER SETTING LIST
(PORT MAIN)
SZ15046

P.5-24
Ver.1.00
2016/12/26

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D	
100	CONTROL HANDLE REV. SETTING	AHEAD MAX ORDER REV.	min-1	0	999	900			MAX
101		AHEAD NAV2 ORDER REV. (EQUAL 100%)	min-1	0	999	900			10
102		AHEAD NAV1 ORDER REV. (EQUAL 85%)	min-1	0	999	810			8
103		AHAED FULL ORDER REV. (EQUAL 75%)	min-1	0	999	720			6
104		AHEAD HALF ORDER REV. (EQUAL 50%)	min-1	0	999	630			4
105		AHEAD SLOW ORDER REV.	min-1	0	999	540			2
106		AHEAD D.SLOW ORDER REV.	min-1	0	999	450			0
107		STOP ORDER REV. (NEUTRAL)	min-1	0	999	450			
108		ASTERN D.SLOW ORDER REV.	min-1	0	999	450			0
109		ASTERN SLOW ORDER REV.	min-1	0	999	540			2
110		ASTERN HALF ORDER REV. (EQUAL 50%)	min-1	0	500	630			4
111		ASTERN FULL ORDER REV. (EQUAL 75%)	min-1	0	500	720			6
112		ASTERN EMC1 ORDER REV.	min-1	0	999	810			8
113		ASTERN EMC2 ORDER REV. (EQUAL 85%)	min-1	0	999	853			10
114		ASTERN MAX ORDER REV.	min-1	0	999	853			MAX
115	LIMIT M/E AHEAD REV.		min-1	0	999	900			
116	LIMIT M/E ASTERN REV.		min-1	0	999	900			
117	AUTO SLOW DOWN		min-1	0	999	500			
118	AUTO SLOW DOWN RESET REV.		min-1	0	999	500			
119	REVOLUTION INCREASE ORDER INTERLOCK 0: NO INTER LOCK 1: CLUTCH ENGAGE 2: CLUTCH ENGAGE AFTER SLIP		——	0	2	2			
120	AHD. FOLLOW UP SPEED	INCREASE TIME : Add.107～Add.126	sec	0.0	99.9	1.0			
121		INCREASE TIME : Add.126～Add.127	sec	0.0	99.9	1.8			
122		INCREASE TIME : Add.127～Add.128	sec	0.0	99.9	4.7			
123		INCREASE TIME : Add.128～Add.100	sec	0.0	99.9	0.0			
124		DECREASE TIME : Add.100～Add.129	sec	0.0	99.9	1.0			
125		DECREASE TIME : Add.129～Add.107	sec	0.0	99.9	0.0			
126		INC. SPEED CHANGE POINT (LOW SPEED ZONE)	min-1	0	999	818			
127		INC. SPEED CHANGE POINT (MIDDLE SPEED ZONE)	min-1	0	999	853			
128		INC. SPEED CHANGE POINT (HIGH SPEED ZONE)	min-1	0	999	900			
129		DECREASE SPEED CHANGE POINT	min-1	0	999	450			
130	AST. FOLLOW UP SPEED	INCREASE TIME : Add.107～Add.136	sec	0.0	99.9	1.0			
131		INCREASE TIME : Add.136～Add.137	sec	0.0	99.9	1.4			
132		INCREASE TIME : Add.137～Add.138	sec	0.0	99.9	3.8			
133		INCREASE TIME : Add.138～Add.114	sec	0.0	99.9	0.0			
134		DECREASE TIME : Add.114～Add.139	sec	0.0	99.9	1.0			
135		DECREASE TIME : Add.139～Add.107	sec	0.0	99.9	0.0			
136		INC. SPEED CHANGE POINT (LOW SPEED ZONE)	min-1	0	999	818			
137		INC. SPEED CHANGE POINT (MIDDLE SPEED ZONE)	min-1	0	999	853			
138		INC. SPEED CHANGE POINT (HIGH SPEED ZONE)	min-1	0	999	900			
139		DECREASE SPEED CHANGE POINT	min-1	0	999	450			
140	E/P VALVE CURRENT SETTING	IDLE REV. SETTING	min-1	0	999	450			
141		LOAD 25% REV. SETTING	min-1	0	999	567			
142		LOAD 50% REV. SETTING	min-1	0	999	714			
143		LOAD 75% REV. SETTING	min-1	0	999	818			
144		LOAD 85% REV. SETTING	min-1	0	999	853			
145		LOAD 100% REV. SETTING	min-1	0	999	900			
146		LOAD 110% REV. SETTING	min-1	0	999	929			
147		IDLE POSISION. CURRNT SETTING	mA	4.00	20.00	6.48			
148		LOAD 25% CURRENT SETTING	mA	4.00	20.00	8.42			
149		LOAD 50% CURRENT SETTING	mA	4.00	20.00	10.87			
150		LOAD 75% CURRENT SETTING	mA	4.00	20.00	12.58			
151		LOAD 85% CURRENT SETTING	mA	4.00	20.00	13.16			
152		LOAD 100% CURRENT SETTING	mA	4.00	20.00	13.95			
153		LOAD 110% CURRENT SETTING	mA	4.00	20.00	14.43			
154		E/S CONTROL CURRENT SETTING	mA	4.00	5.00	4.10			
155	GOV. / EP CURRENT OUTPUT (4mA BIT SCALING)		bit	0	4095	634			
156	GOV. / EP CURRENT OUTPUT (20mA BIT SCALING)		bit	0	4095	3227			

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
157	AHEAD NEUTRAL RECOVERY DETECTION REV.	min-1	0	999	350		
158	AHEAD NEUTRAL KEEP TIME (0 : NOT RECOVERY)	sec	0.0	99.9	2.0		
159	ASTERN NEUTRAL RECOVERY DETECTION REV.	min-1	0	999	350		
160	ASTERN NEUTRAL KEEP TIME (0 : NOT RECOVERY)	sec	0.0	99.9	2.0		
161	NEUTRAL RECOVERY CANCEL TIME (999 : NOT CANCEL)	sec	0	999	60		
162	AHEAD GOVERNOR BOOST REV. SET	min-1	0	200	0		
163	AHEAD GOVERNOR BOOST TIME SET (0 : NOT BOOST)	sec	0.0	99.9	0.0		
164	ASTERN GOVERNOR BOOST REV. SET	min-1	0	200	0		
165	ASTERN GOVERNOR BOOST TIME SET (0 : NOT BOOST)	sec	0.0	99.9	0.0		
166	GOVERNOR BOOST REV. SET AT AUTO DISENGAGE	min-1	0	200	0		
167	GOV BOOST TIME SET AT AUTO DISENGAGE (0 : DISABLED)	sec	0.0	99.9	0.0		
168	CHANGE GOV BOOST SET (0:CLUTCH ODR/ 1:DIRECT/ 2:BOTH)	----	0	2	1		
169	GOVERNOR BOOST APPLY PELLER REV. (ASTERN ONLY)	min-1	0	999	80		
170	M/E REV UPPER LIMIT AT AST CL ENG READY	min-1	0	999	500		
171	M/E REV LOWER LIMIT AT AST CL ENG READY	min-1	0	999	440		
172	M/E REV UPPER LIMIT AT AST CL DIS-ENG READY	min-1	0	999	500		
173	PELLER REV UPPER LIMIT AT AST CL ENG READY	min-1	0	999	70		
174	CLUTCH ENGAGE/DISENGAGE ORDER TIME (0 set : continuous)	sec	0.0	99.9	0.0		
175	M/E MAX REV. (20mA)	min-1	0	1600	1500		
176	M/E REV. 20mA SCALING (*)	bit	0	8192	7186		
177	M/E REV. 4mA SCALING (*)	bit	0	8192	1435		
178	PELLER MAX REVOLUTION (20mA)	min-1	0	999	400		
179	PELLER REV. 20mA SCALING (*)	bit	0	8192	7174		
180	PELLER REV. 4mA SCALING (*)	bit	0	8192	1441		
181	CH1 ABNL DETECTION CURRENT UPPER LIMIT(M/E REV.)	mA	0.00	21.00	20.30		
182	CH1 ABNL DETECTION CURRENT LOWER LIMIT(M/E REV.)	mA	0.00	21.00	3.70		
183	CH2 ABNL DETECTION CURRENT UPPER LIMIT(PELLER REV.)	mA	0.00	21.00	20.30		
184	CH2 ABNL DETECTION CURRENT LOWER LIMIT(PELLER REV.)	mA	0.00	21.00	3.70		
185	CH3 ABNL DETECTION VOLTAGE UPPER LIMIT(POTENTIO POWER)	volt	0.00	10.00	9.70		
186	CH3 ABNL DETECTION VOLTAGE LOWER LIMIT(POTENTIO POWER)	volt	0.00	10.00	0.30		
187	CH4 ABNL DETECTION VOLTAGE UPPER LIMIT(WH HANDLE)	volt	0.00	10.00	9.70		
188	CH4 ABNL DETECTION VOLTAGE LOWER LIMIT(WH HANDLE)	volt	0.00	10.00	0.30		
189	CH5 ABNL DETECTION VOLTAGE UPPER LIMIT(CPCP DIAL)	volt	0.00	10.00	9.70		
190	CH5 ABNL DETECTION VOLTAGE LOWER LIMIT(CPCP DIAL)	volt	0.00	10.00	0.30		
191	CH6 ABNL DETECTION VOLTAGE UPPER LIMIT(CP DIAL)	volt	0.00	10.00	9.70		
192	CH6 ABNL DETECTION VOLTAGE LOWER LIMIT(CP DIAL)	volt	0.00	10.00	0.30		
193	CH7 ABNL DETECTION VOLTAGE UPPER LIMIT(CR HANDLE)	volt	0.00	10.00	9.70		
194	CH7 ABNL DETECTION VOLTAGE LOWER LIMIT(CR HANDLE)	volt	0.00	10.00	0.30		
195	CH8 ABNL DETECTION VOLTAGE UPPER LIMIT(RACK LIMIT)	volt	0.00	10.00	9.70		
196	CH8 ABNL DETECTION VOLTAGE LOWER LIMIT(RACK LIMIT)	volt	0.00	10.00	0.30		
197	VOLTAGE DEVIATION AT LINE BREAK DETECTION	volt	0.00	10.00	1.00		
198	DEVIATION FOR LINE BREAK COUNT DETECTION(Add.197)	count	1	250	200		
199	HANDLE RANGE ABNL DETECTION TIME(0.0:DISABLED)	sec	0.0	99.9	2.0		

upper limit check is disabled when set max for abnormal detection upper limit.
lower limit check is disabled when set max for abnormal detection lower limit.

ADDRESS	CONTENT		UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D	
200	W/H CONTROL HANDLE VOLTAGE SET	AHEAD MAX POS. VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	7.80			MAX
201		AHEAD NAV2 POS. VOLTAGE(＊)(related Add.101)	volt	0.00	10.00	7.80			10
202		AHEAD NAV1 POS. VOLTAGE(＊)(related Add.102)	volt	0.00	10.00	7.34			8
203		AHEAD FULL POS. VOLTAGE(＊)(related Add.103)	volt	0.00	10.00	6.88			6
204		AHEAD HALF POS. VOLTAGE(＊)(related Add.104)	volt	0.00	10.00	6.44			4
205		AHEAD SLOW POS. VOLTAGE(＊)(related Add.105)	volt	0.00	10.00	5.98			2
206		AHEAD D.SLOW POS. VOLTAGE(＊)(related Add.106)	volt	0.00	10.00	5.52			0
207		AHEAD CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	5.52			
208		ASTERN CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	4.50			
209		ASTERN D.SLOW POS. VOLTAGE(＊)(related Add.108)	volt	0.00	10.00	4.50			0
210		ASTERN SLOW POS. VOLTAGE(＊)(related Add.109)	volt	0.00	10.00	4.05			2
211		ASTERN HALF POS. VOLTAGE(＊)(related Add.110)	volt	0.00	10.00	3.60			4
212		ASTERN FULL POS. VOLTAGE(＊)(related Add.111)	volt	0.00	10.00	3.15			6
213		ASTERN EMC1 POS. VOLTAGE(＊)(related Add.112)	volt	0.00	10.00	2.68			8
214		ASTERN EMC2 POS. VOLTAGE(＊)(related Add.113)	volt	0.00	10.00	2.22			10
215		ASTERN MAX POS. VOLTAGE(＊)(related Add.114)	volt	0.00	10.00	2.22			MAX
216	W/H CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS		volt	0.00	10.00	0.20			
217	W/H CONT HANDLE RANGE, ABNL DETECTION DISABLED VOL RANGE		volt	0.00	10.00	1.00			
218	W/H GOVERNOR > CHANGE SLIP HYSTERISIS		volt	0.00	10.00	0.20			
219	W/H HANDLE SIGNAL CHANGE(0=INC 1=DEC)		---	0	1	0			
220	C/R CONTROL HANDLE VOLTAGE SET	AHEAD MAX POS. VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	7.79			MAX
221		AHEAD NAV2 POS. VOLTAGE(＊)(related Add.101)	volt	0.00	10.00	7.79			10
222		AHEAD NAV1 POS. VOLTAGE(＊)(related Add.102)	volt	0.00	10.00	7.34			8
223		AHEAD FULL POS. VOLTAGE(＊)(related Add.103)	volt	0.00	10.00	6.86			6
224		AHEAD HALF POS. VOLTAGE(＊)(related Add.104)	volt	0.00	10.00	6.42			4
225		AHEAD SLOW POS. VOLTAGE(＊)(related Add.105)	volt	0.00	10.00	5.96			2
226		AHEAD D.SLOW POS. VOLTAGE(＊)(related Add.106)	volt	0.00	10.00	5.49			0
227		AHEAD CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	5.49			
228		ASTERN CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	4.48			
229		ASTERN D.SLOW POS. VOLTAGE(＊)(related Add.108)	volt	0.00	10.00	4.48			0
230		ASTERN SLOW POS. VOLTAGE(＊)(related Add.109)	volt	0.00	10.00	4.01			2
231		ASTERN HALF POS. VOLTAGE(＊)(related Add.110)	volt	0.00	10.00	3.56			4
232		ASTERN FULL POS. VOLTAGE(＊)(related Add.111)	volt	0.00	10.00	3.10			6
233		ASTERN EMC1 POS. VOLTAGE(＊)(related Add.112)	volt	0.00	10.00	2.63			8
234		ASTERN EMC2 POS. VOLTAGE(＊)(related Add.113)	volt	0.00	10.00	2.19			10
235		ASTERN MAX POS. VOLTAGE(＊)(related Add.114)	volt	0.00	10.00	2.19			MAX
236	C/R CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS		volt	0.00	10.00	0.20			
237	C/R CONT HANDLE AREA, ABNL DETECTION DISABLE VOL RANGE		volt	0.00	10.00	1.00			
238	C/R GOVERNOR > SLIP CHANGE HYSTERISIS		volt	0.00	10.00	0.20			
239	C/R HANDLE SIGNAL CHANGE(0=INC 1=DEC)		---	0	1	0			

SHALL BE POWER RESET AT CHANGE Add.219, Add239 PARAMETER

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
240	CPCP DIAL VOLTAGE SET	MAX VOLTAGE(＊)(related Add.280)	volt	0.00	10.00	8.16		
241								
242								
243								
244								
245								
246								
247		MINIMUM VOLTAGE(＊)	volt	0.00	10.00	2.59		
248								
249								
250								
251								
252								
253								
254								
255								
256								
257								
258								
259								
260	CP DIAL VOLTAGE SET	MAX VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	8.23		
261								
262								
263								
264								
265								
266								
267		MINIMUM VOLTAGE(＊)	volt	0.00	10.00	2.64		
268								
269								
270								
271								
272								
273								
274								
275								
276								
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283								
284								
285								
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287								
288								
289								
290								
291								
292								
293								
294								
295		CARGO DIAL MAX REV.	min-1	0	999	900		
296		CARGO DIAL MINIMUM REV.	min-1	0	999	450		
297		CARGO DIAL FOLLU UP SPEED	min-1/sec	0	999	20		
298								
299								

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
300	AHEAD RACK LIMIT	LIMIT POSITION 1 REV. SET(related Add.305)	min-1	0	999	450		
301		LIMIT POSITION 2 REV. SET(related Add.306)	min-1	0	999	567		
302		LIMIT POSITION 3 REV. SET(related Add.307)	min-1	0	999	714		
303		LIMIT POSITION 4 REV. SET(related Add.308)	min-1	0	999	818		
304		LIMIT POSITION 5 REV. SET(related Add.309)	min-1	0	999	900		
305		LIMIT POS 1 CURRENT SET(related Add.300)	mA	4.00	20.00	6.60		
306		LIMIT POS 2 CURRENT SET(related Add.301)	mA	4.00	20.00	8.42		
307		LIMIT POS 3 CURRENT SET(related Add.302)	mA	4.00	20.00	9.50		
308		LIMIT POS 4 CURRENT SET(related Add.303)	mA	4.00	20.00	10.50		
309		LIMIT POS 5 CURRENT SET(related Add.304)	mA	4.00	20.00	12.00		
310	ASTERN RACK LIMIT	LIMIT POS 1 REV. SET(related Add.315)	min-1	0	999	450		
311		LIMIT POS 2 REV. SET(related Add.316)	min-1	0	999	567		
312		LIMIT POS 3 REV. SET(related Add.317)	min-1	0	999	714		
313		LIMIT POS 4 REV. SET(related Add.318)	min-1	0	999	818		
314		LIMIT POS 5 REV. SET(related Add.319)	min-1	0	999	853		
315		LIMIT POS 1 CURRENT SET(related Add.310)	mA	4.00	20.00	6.48		
316		LIMIT POS 2 CURRENT SET(related Add.311)	mA	4.00	20.00	8.42		
317		LIMIT POS 3 CURRENT SET(related Add.312)	mA	4.00	20.00	10.87		
318		LIMIT POS 4 CURRENT SET(related Add.313)	mA	4.00	20.00	12.58		
319		LIMIT POS 5 CURRENT SET(related Add.314)	mA	4.00	20.00	13.16		
320		LIMIT CURRENT at M/E STOP	mA	4.00	20.00	4.10		
321		LIMIT CURRENT at M/E RUN	mA	4.00	20.00	20.00		
322		RACK LIMIT ADJUSTMENT DIAL UPPER LIMIT	mA	0.00	5.00	3.00		
323		RACK LIMIT ADJUSTMENT DIAL LOWER LIMIT	mA	-0.00	-5.00	-3.00		
324		RACK LIMIT ADJ. DIAL INC. MAX POS VOLTAGE(*)	volt	0.00	10.00	8.50		
325		RACK LIMIT ADJ. DIAL "0" POS VOLTAGE(*)	volt	0.00	10.00	4.54		
326		RACK LIMIT ADJ. DIAL DEC. MAX POS VOLTAGE(*)	volt	0.00	10.00	0.66		
327		RACK LIMIT OUTPUT CURRENT:bit SCALING OF 4mA	bit	0	4095	646		
328		RACK LIMIT OUTPUT CURRENT:bit SCALING OF 20mA	bit	0	4095	3220		
329		SLIP ENABELED REV. ORDER UPPER LIMIT	min-1	0	999	450		
330	SLIP	AHEAD GOVERNOR RANGE CURRENT SET.	mA	4.00	20.00	20.00		
331		AHEAD SLIP END POS CURRENT SET.	mA	4.00	20.00	15.00		
332		AHEAD SLIP START POS CURRENT SET.	mA	4.00	20.00	13.00		
333		IDLE RANGE CURRENT SET.	mA	4.00	20.00	12.00		
334		ASTERN START POS CURRENT SET.	mA	4.00	20.00	11.00		
335		ASTERN END POS CURRENT SET.	mA	4.00	20.00	9.00		
336		ASTERN GOVERNOR RANGE CURRENT SET.	mA	4.00	20.00	4.00		
337		E/S:FO HANDLE SW. STOP CURRENT SET.	mA	4.00	20.00	12.00		
338		E/S:FO HANDLE SW. RUN CURRENT SET.	mA	4.00	20.00	12.00		
339		WAITING TIME OF SLIP ENABLED REV. UPPER LIMIT	sec	0.0	99.9	0.5		
340		SLIP ENABLED REV. UPPER LIMIT	min-1	0	999	400		
341		SLIP OUTPUT SELECT OF NEUTRAL(0:OFF, 1:ON)	---	0	1	0		
342		SLIP CURRENT OUTPUT:bit SCALING OF 4mA	bit	0	4095	637		
343		SLIP CURRENT OUTPUT:bit SCALING OF 20mA	bit	0	4095	3192		
344								
345								
346								
347								
348								
349								

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
350							
351							
352							
353							
354							
355							
356							
357							
358							
359							
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365							
366							
367							
368							
369							
370							
371							
372							
373							
374							
375							
376	CARGO REVOLUTION INCREASE ORDER INTERLOCK	----	0	1	1		
377							
378							
379							
380							
381							
382							
383							
384							
385							
386							
387							
388							
389							
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393							
394							
395							
396							
397							
398							
399							

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
400							
401	IDLE BOOST UP REV.	min-1	0	999	20		
402	BOOST UP HOLD TIME at AFTER CL ENGAGED	sec	0	999	10		
403							
404							
405							
406							
407							
408							
409	NOTUSE						
410	CH1 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
411	CH2 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
412	CH3 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
413	CH4 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
414	CH5 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
415	CH6 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
416	CH7 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
417	CH8 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
418	AHD. PELLER REV. OF CRASH AST. DETECTION	min-1	0	999	200		
419	RECOVER NEUTRAL DETECTION CONT AT CRASH AST	min-1	0	999	440		
420	AST. PELLER REV. OF CRASH AHD. DETECTION	min-1	0	999	150		
421	RECOVER NEUTRAL DETECTION CONT AT CRASH AHD	min-1	0	999	440		
422	AHD PLLER REV OF AHD CL ENG READY AT CRASH AST	min-1	0	999	70		
423	AST PLLER REV OF AST CL ENG READY AT CRASH AHD	min-1	0	999	999		
424	BOOST REV. AT CRASH AST. RECOVER NEUTRAL	min-1	0	999	0		
425	BOOST REV. AT CRASH AHD. RECOVER NEUTRAL	min-1	0	999	0		
426	M/E REV UPPER LIMIT AT AHD CL ENG READY	min-1	0	999	500		
427	M/E REV LOWER LIMIT AT AHD CL ENG READY	min-1	0	999	440		
428	M/E REV UPPER LIMIT AT AHD CL DIS-ENG READY	min-1	0	999	500		
429	PELLER REV UPPER LIMIT AT AHD CL ENG READY	min-1	0	999	999		

BOOST PRIORITY : HEATING (Add.400)>IDLE BOOST(Add.401,402)>GOV. BOOST(Add.162～167)

:RED ADDRESS DOES NOT BECOME "NOT USE",AND DISABLE THE FEATURE BY PARAMETER
:YELLOW ADDRESS DOES NOT BECOME "NOT USE",BUT NOT INCLUDED IN THE PROGRAM FEATURE
:GREEN ADDRESS IS "NOT USE",AND NOT INCLUDED IN THE PROGRAM FEATURE

M/E REMOTE CONTROL
PARAMETER SETTING LIST
(PORT BACK UP)
SZ15046

P.5-31
Ver.1.00
2016/12/26

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D	
100	CONTROL HANDLE REV. SETTING	AHEAD MAX ORDER REV.	min-1	0	999	900			MAX
101		AHEAD NAV2 ORDER REV. (EQUAL 100%)	min-1	0	999	900			10
102		AHEAD NAV1 ORDER REV. (EQUAL 85%)	min-1	0	999	810			8
103		AHAED FULL ORDER REV. (EQUAL 75%)	min-1	0	999	720			6
104		AHEAD HALF ORDER REV. (EQUAL 50%)	min-1	0	999	630			4
105		AHEAD SLOW ORDER REV.	min-1	0	999	540			2
106		AHEAD D.SLOW ORDER REV.	min-1	0	999	450			0
107		STOP ORDER REV. (NEUTRAL)	min-1	0	999	450			
108		ASTERN D.SLOW ORDER REV.	min-1	0	999	450			0
109		ASTERN SLOW ORDER REV.	min-1	0	999	540			2
110		ASTERN HALF ORDER REV. (EQUAL 50%)	min-1	0	500	630			4
111		ASTERN FULL ORDER REV. (EQUAL 75%)	min-1	0	500	720			6
112		ASTERN EMC1 ORDER REV.	min-1	0	999	810			8
113		ASTERN EMC2 ORDER REV. (EQUAL 85%)	min-1	0	999	853			10
114		ASTERN MAX ORDER REV.	min-1	0	999	853			MAX
115		LIMIT M/E AHEAD REV.	min-1	0	999	900			
116		LIMIT M/E ASTERN REV.	min-1	0	999	900			
117		AUTO SLOW DOWN	min-1	0	999	500			
118		AUTO SLOW DOWN RESET REV.	min-1	0	999	500			
119		REVOLUTION INCREASE ORDER INTERLOCK 0: NO INTER LOCK 1: CLUTCH ENGAGE 2: CLUTCH ENGAGE AFTER SLIP	——	0	2	2			
120	AHD. FOLLOW UP SPEED	INCREASE TIME : Add.107～Add.126	sec	0.0	99.9	1.0			
121		INCREASE TIME : Add.126～Add.127	sec	0.0	99.9	1.8			
122		INCREASE TIME : Add.127～Add.128	sec	0.0	99.9	4.7			
123		INCREASE TIME : Add.128～Add.100	sec	0.0	99.9	0.0			
124		DECREASE TIME : Add.100～Add.129	sec	0.0	99.9	1.0			
125		DECREASE TIME : Add.129～Add.107	sec	0.0	99.9	0.0			
126		INC. SPEED CHANGE POINT (LOW SPEED ZONE)	min-1	0	999	818			
127		INC. SPEED CHANGE POINT (MIDDLE SPEED ZONE)	min-1	0	999	853			
128		INC. SPEED CHANGE POINT (HIGH SPEED ZONE)	min-1	0	999	900			
129		DECREASE SPEED CHANGE POINT	min-1	0	999	450			
130	AST. FOLLOW UP SPEED	INCREASE TIME : Add.107～Add.136	sec	0.0	99.9	1.0			
131		INCREASE TIME : Add.136～Add.137	sec	0.0	99.9	1.4			
132		INCREASE TIME : Add.137～Add.138	sec	0.0	99.9	3.8			
133		INCREASE TIME : Add.138～Add.114	sec	0.0	99.9	0.0			
134		DECREASE TIME : Add.114～Add.139	sec	0.0	99.9	1.0			
135		DECREASE TIME : Add.139～Add.107	sec	0.0	99.9	0.0			
136		INC. SPEED CHANGE POINT (LOW SPEED ZONE)	min-1	0	999	818			
137		INC. SPEED CHANGE POINT (MIDDLE SPEED ZONE)	min-1	0	999	853			
138		INC. SPEED CHANGE POINT (HIGH SPEED ZONE)	min-1	0	999	900			
139		DECREASE SPEED CHANGE POINT	min-1	0	999	450			
140	E/P VALVE CURRENT SETTING	IDLE REV. SETTING	min-1	0	999	450			
141		LOAD 25% REV. SETTING	min-1	0	999	567			
142		LOAD 50% REV. SETTING	min-1	0	999	714			
143		LOAD 75% REV. SETTING	min-1	0	999	818			
144		LOAD 85% REV. SETTING	min-1	0	999	853			
145		LOAD 100% REV. SETTING	min-1	0	999	900			
146		LOAD 110% REV. SETTING	min-1	0	999	929			
147		IDLE POSISION. CURRNT SETTING	mA	4.00	20.00	6.48			
148		LOAD 25% CURRENT SETTING	mA	4.00	20.00	8.42			
149		LOAD 50% CURRENT SETTING	mA	4.00	20.00	10.87			
150		LOAD 75% CURRENT SETTING	mA	4.00	20.00	12.58			
151		LOAD 85% CURRENT SETTING	mA	4.00	20.00	13.16			
152		LOAD 100% CURRENT SETTING	mA	4.00	20.00	13.95			
153		LOAD 110% CURRENT SETTING	mA	4.00	20.00	14.43			
154		E/S CONTROL CURRENT SETTING	mA	4.00	5.00	4.10			
155		GOV. / EP CURRENT OUTPUT (4mA BIT SCALING)	bit	0	4095	645			
156		GOV. / EP CURRENT OUTPUT (20mA BIT SCALING)	bit	0	4095	3226			

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
157	AHEAD NEUTRAL RECOVERY DETECTION REV.	min-1	0	999	350		
158	AHEAD NEUTRAL KEEP TIME (0 : NOT RECOVERY)	sec	0.0	99.9	2.0		
159	ASTERN NEUTRAL RECOVERY DETECTION REV.	min-1	0	999	350		
160	ASTERN NEUTRAL KEEP TIME (0 : NOT RECOVERY)	sec	0.0	99.9	2.0		
161	NEUTRAL RECOVERY CANCEL TIME (999 : NOT CANCEL)	sec	0	999	60		
162	AHEAD GOVERNOR BOOST REV. SET	min-1	0	200	0		
163	AHEAD GOVERNOR BOOST TIME SET (0 : NOT BOOST)	sec	0.0	99.9	0.0		
164	ASTERN GOVERNOR BOOST REV. SET	min-1	0	200	0		
165	ASTERN GOVERNOR BOOST TIME SET (0 : NOT BOOST)	sec	0.0	99.9	0.0		
166	GOVERNOR BOOST REV. SET AT AUTO DISENGAGE	min-1	0	200	0		
167	GOV BOOST TIME SET AT AUTO DISENGAGE (0 : DISABLED)	sec	0.0	99.9	0.0		
168	CHANGE GOV BOOST SET (0:CLUTCH ODR/ 1:DIRECT/ 2:BOTH)	---	0	2	1		
169	GOVERNOR BOOST APPLY PELLER REV. (ASTERN ONLY)	min-1	0	999	80		
170	M/E REV UPPER LIMIT AT AST CL ENG READY	min-1	0	999	500		
171	M/E REV LOWER LIMIT AT AST CL ENG READY	min-1	0	999	440		
172	M/E REV UPPER LIMIT AT AST CL DIS-ENG READY	min-1	0	999	500		
173	PELLER REV UPPER LIMIT AT AST CL ENG READY	min-1	0	999	70		
174	CLUTCH ENGAGE/DISENGAGE ORDER TIME (0 set : continuous)	sec	0.0	99.9	0.0		
175	M/E MAX REV. (20mA)	min-1	0	1600	1500		
176	M/E REV. 20mA SCALING (*)	bit	0	8192	7165		
177	M/E REV. 4mA SCALING (*)	bit	0	8192	1410		
178	PELLER MAX REVOLUTION (20mA)	min-1	0	999	400		
179	PELLER REV. 20mA SCALING (*)	bit	0	8192	7105		
180	PELLER REV. 4mA SCALING (*)	bit	0	8192	1402		
181	CH1 ABNL DETECTION CURRENT UPPER LIMIT(M/E REV.)	mA	0.00	21.00	20.30		
182	CH1 ABNL DETECTION CURRENT LOWER LIMIT(M/E REV.)	mA	0.00	21.00	3.70		
183	CH2 ABNL DETECTION CURRENT UPPER LIMIT(PELLER REV.)	mA	0.00	21.00	20.30		
184	CH2 ABNL DETECTION CURRENT LOWER LIMIT(PELLER REV.)	mA	0.00	21.00	3.70		
185	CH3 ABNL DETECTION VOLTAGE UPPER LIMIT(POTENTIO POWER)	volt	0.00	10.00	9.70		
186	CH3 ABNL DETECTION VOLTAGE LOWER LIMIT(POTENTIO POWER)	volt	0.00	10.00	0.30		
187	CH4 ABNL DETECTION VOLTAGE UPPER LIMIT(WH HANDLE)	volt	0.00	10.00	9.70		
188	CH4 ABNL DETECTION VOLTAGE LOWER LIMIT(WH HANDLE)	volt	0.00	10.00	0.30		
189	CH5 ABNL DETECTION VOLTAGE UPPER LIMIT(CPCP DIAL)	volt	0.00	10.00	9.70		
190	CH5 ABNL DETECTION VOLTAGE LOWER LIMIT(CPCP DIAL)	volt	0.00	10.00	0.30		
191	CH6 ABNL DETECTION VOLTAGE UPPER LIMIT(CP DIAL)	volt	0.00	10.00	9.70		
192	CH6 ABNL DETECTION VOLTAGE LOWER LIMIT(CP DIAL)	volt	0.00	10.00	0.30		
193	CH7 ABNL DETECTION VOLTAGE UPPER LIMIT(CR HANDLE)	volt	0.00	10.00	9.70		
194	CH7 ABNL DETECTION VOLTAGE LOWER LIMIT(CR HANDLE)	volt	0.00	10.00	0.30		
195	CH8 ABNL DETECTION VOLTAGE UPPER LIMIT(RACK LIMIT)	volt	0.00	10.00	9.70		
196	CH8 ABNL DETECTION VOLTAGE LOWER LIMIT(RACK LIMIT)	volt	0.00	10.00	0.30		
197	VOLTAGE DEVIATION AT LINE BREAK DETECTION	volt	0.00	10.00	1.00		
198	DEVIATION FOR LINE BREAK COUNT DETECTION(Add.197)	count	1	250	200		
199	HANDLE RANGE ABNL DETECTION TIME(0.0:DISABLED)	sec	0.0	99.9	2.0		

upper limit check is disabled when set max for abnormal detection upper limit.
lower limit check is disabled when set max for abnormal detection lower limit.

M/E REMOTE CONTROL
PARAMETER SETTING LIST
(PORT BACK UP)
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P.5-33
Ver.1.00
2016/12/26

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D	
200	W/H CONTROL HANDLE VOLTAGE SET	AHEAD MAX POS. VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	7.79			MAX
201		AHEAD NAV2 POS. VOLTAGE(＊)(related Add.101)	volt	0.00	10.00	7.79			10
202		AHEAD NAV1 POS. VOLTAGE(＊)(related Add.102)	volt	0.00	10.00	7.32			8
203		AHEAD FULL POS. VOLTAGE(＊)(related Add.103)	volt	0.00	10.00	6.87			6
204		AHEAD HALF POS. VOLTAGE(＊)(related Add.104)	volt	0.00	10.00	6.41			4
205		AHEAD SLOW POS. VOLTAGE(＊)(related Add.105)	volt	0.00	10.00	5.96			2
206		AHEAD D.SLOW POS. VOLTAGE(＊)(related Add.106)	volt	0.00	10.00	5.51			0
207		AHEAD CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	5.51			
208		ASTERN CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	4.49			
209		ASTERN D.SLOW POS. VOLTAGE(＊)(related Add.108)	volt	0.00	10.00	4.49			0
210		ASTERN SLOW POS. VOLTAGE(＊)(related Add.109)	volt	0.00	10.00	4.05			2
211		ASTERN HALF POS. VOLTAGE(＊)(related Add.110)	volt	0.00	10.00	3.60			4
212		ASTERN FULL POS. VOLTAGE(＊)(related Add.111)	volt	0.00	10.00	3.13			6
213		ASTERN EMC1 POS. VOLTAGE(＊)(related Add.112)	volt	0.00	10.00	2.68			8
214		ASTERN EMC2 POS. VOLTAGE(＊)(related Add.113)	volt	0.00	10.00	2.21			10
215		ASTERN MAX POS. VOLTAGE(＊)(related Add.114)	volt	0.00	10.00	2.21			MAX
216	W/H CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS		volt	0.00	10.00	0.20			
217	W/H CONT HANDLE RANGE, ABNL DETECTION DISABLED VOL RANGE		volt	0.00	10.00	1.00			
218	W/H GOVERNOR > CHANGE SLIP HYSTERISIS		volt	0.00	10.00	0.20			
219	W/H HANDLE SIGNAL CHANGE(0=INC 1=DEC)		---	0	1	0			
220	C/R CONTROL HANDLE VOLTAGE SET	AHEAD MAX POS. VOLTAGE(＊)(related Add.100)	volt	0.00	10.00	7.79			MAX
221		AHEAD NAV2 POS. VOLTAGE(＊)(related Add.101)	volt	0.00	10.00	7.79			10
222		AHEAD NAV1 POS. VOLTAGE(＊)(related Add.102)	volt	0.00	10.00	7.33			8
223		AHEAD FULL POS. VOLTAGE(＊)(related Add.103)	volt	0.00	10.00	6.89			6
224		AHEAD HALF POS. VOLTAGE(＊)(related Add.104)	volt	0.00	10.00	6.43			4
225		AHEAD SLOW POS. VOLTAGE(＊)(related Add.105)	volt	0.00	10.00	5.97			2
226		AHEAD D.SLOW POS. VOLTAGE(＊)(related Add.106)	volt	0.00	10.00	5.51			0
227		AHEAD CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	5.51			
228		ASTERN CLUTCH ENGAGE POS. VOLTAGE(＊)	volt	0.00	10.00	4.49			
229		ASTERN D.SLOW POS. VOLTAGE(＊)(related Add.108)	volt	0.00	10.00	4.49			0
230		ASTERN SLOW POS. VOLTAGE(＊)(related Add.109)	volt	0.00	10.00	4.04			2
231		ASTERN HALF POS. VOLTAGE(＊)(related Add.110)	volt	0.00	10.00	3.59			4
232		ASTERN FULL POS. VOLTAGE(＊)(related Add.111)	volt	0.00	10.00	3.14			6
233		ASTERN EMC1 POS. VOLTAGE(＊)(related Add.112)	volt	0.00	10.00	2.68			8
234		ASTERN EMC2 POS. VOLTAGE(＊)(related Add.113)	volt	0.00	10.00	2.22			10
235		ASTERN MAX POS. VOLTAGE(＊)(related Add.114)	volt	0.00	10.00	2.22			MAX
236	C/R CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS		volt	0.00	10.00	0.20			
237	C/R CONT HANDLE AREA, ABNL DETECTION DISABLE VOL RANGE		volt	0.00	10.00	1.00			
238	C/R GOVERNOR > SLIP CHANGE HYSTERISIS		volt	0.00	10.00	0.20			
239	C/R HANDLE SIGNAL CHANGE(0=INC 1=DEC)		---	0	1	0			

SHALL BE POWER RESET AT CHANGE Add.219, Add239 PARAMETER

ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
240	CPCP DIAL VOLTAGE SET	MAX VOLTAGE(*)(related Add.280)	volt	0.00	10.00	8.14		
241								
242								
243								
244								
245								
246								
247		MINIMUM VOLTAGE(*)	volt	0.00	10.00	2.57		
248								
249								
250								
251								
252								
253								
254								
255								
256								
257								
258								
259								
260	CP DIAL VOLTAGE SET	MAX VOLTAGE(*)(related Add.100)	volt	0.00	10.00	8.16		
261								
262								
263								
264								
265								
266								
267		MINIMUM VOLTAGE(*)	volt	0.00	10.00	2.59		
268								
269								
270								
271								
272								
273								
274								
275								
276								
277								
278								
279								
280								
281								
282								
283								
284								
285								
286								
287								
288								
289								
290								
291								
292								
293								
294								
295		CARGO DIAL MAX REV.	min-1	0	999	900		
296		CARGO DIAL MINIMUM REV.	min-1	0	999	450		
297		CARGO DIAL FOLLU UP SPEED	min-1/sec	0	999	20		
298								
299								

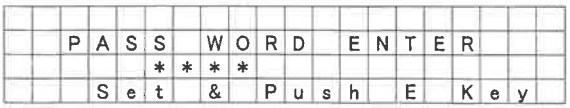
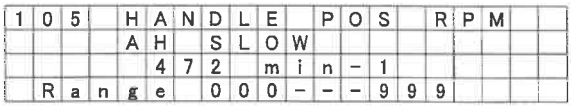
ADDRESS		CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
300	AHEAD RACK LIMIT	LIMIT POSITION 1 REV. SET(related Add.305)	min-1	0	999	450		
301		LIMIT POSITION 2 REV. SET(related Add.306)	min-1	0	999	567		
302		LIMIT POSITION 3 REV. SET(related Add.307)	min-1	0	999	714		
303		LIMIT POSITION 4 REV. SET(related Add.308)	min-1	0	999	818		
304		LIMIT POSITION 5 REV. SET(related Add.309)	min-1	0	999	900		
305		LIMIT POS 1 CURRENT SET(related Add.300)	mA	4.00	20.00	6.60		
306		LIMIT POS 2 CURRENT SET(related Add.301)	mA	4.00	20.00	8.42		
307		LIMIT POS 3 CURRENT SET(related Add.302)	mA	4.00	20.00	9.50		
308		LIMIT POS 4 CURRENT SET(related Add.303)	mA	4.00	20.00	10.50		
309		LIMIT POS 5 CURRENT SET(related Add.304)	mA	4.00	20.00	12.00		
310	ASTERN RACK LIMIT	LIMIT POS 1 REV. SET(related Add.315)	min-1	0	999	450		
311		LIMIT POS 2 REV. SET(related Add.316)	min-1	0	999	567		
312		LIMIT POS 3 REV. SET(related Add.317)	min-1	0	999	714		
313		LIMIT POS 4 REV. SET(related Add.318)	min-1	0	999	818		
314		LIMIT POS 5 REV. SET(related Add.319)	min-1	0	999	853		
315		LIMIT POS 1 CURRENT SET(related Add.310)	mA	4.00	20.00	6.48		
316		LIMIT POS 2 CURRENT SET(related Add.311)	mA	4.00	20.00	8.42		
317		LIMIT POS 3 CURRENT SET(related Add.312)	mA	4.00	20.00	10.87		
318		LIMIT POS 4 CURRENT SET(related Add.313)	mA	4.00	20.00	12.58		
319		LIMIT POS 5 CURRENT SET(related Add.314)	mA	4.00	20.00	13.16		
320		LIMIT CURRENT at M/E STOP	mA	4.00	20.00	4.10		
321		LIMIT CURRENT at M/E RUN	mA	4.00	20.00	20.00		
322		RACK LIMIT ADJUSTMENT DIAL UPPER LIMIT	mA	0.00	5.00	3.00		
323		RACK LIMIT ADJUSTMENT DIAL LOWER LIMIT	mA	-0.00	-5.00	-3.00		
324		RACK LIMIT ADJ. DIAL INC. MAX POS VOLTAGE(*)	volt	0.00	10.00	8.48		
325		RACK LIMIT ADJ. DIAL "0" POS VOLTAGE(*)	volt	0.00	10.00	4.51		
326		RACK LIMIT ADJ. DIAL DEC. MAX POS VOLTAGE(*)	volt	0.00	10.00	0.62		
327		RACK LIMIT OUTPUT CURRENT:bit SCALING OF 4mA	bit	0	4095	644		
328		RACK LIMIT OUTPUT CURRENT:bit SCALING OF 20mA	bit	0	4095	3223		
329		SLIP ENABELED REV. ORDER UPPER LIMIT	min-1	0	999	450		
330	SLIP	AHEAD GOVERNOR RANGE CURRENT SET.	mA	4.00	20.00	20.00		
331		AHEAD SLIP END POS CURRENT SET.	mA	4.00	20.00	15.00		
332		AHEAD SLIP START POS CURRENT SET.	mA	4.00	20.00	13.00		
333		IDLE RANGE CURRENT SET.	mA	4.00	20.00	12.00		
334		ASTERN START POS CURRENT SET.	mA	4.00	20.00	11.00		
335		ASTERN END POS CURRENT SET.	mA	4.00	20.00	9.00		
336		ASTERN GOVERNOR RANGE CURRENT SET.	mA	4.00	20.00	4.00		
337		E/S:FO HANDLE SW. STOP CURRENT SET.	mA	4.00	20.00	12.00		
338		E/S:FO HANDLE SW. RUN CURRENT SET.	mA	4.00	20.00	12.00		
339		WAITING TIME OF SLIP ENABLED REV. UPPER LIMIT	sec	0.0	99.9	0.5		
340		SLIP ENABLED REV. UPPER LIMIT	min-1	0	999	400		
341		SLIP OUTPUT SELECT OF NEUTRAL(0:OFF, 1:ON)	---	0	1	0		
342		SLIP CURRENT OUTPUT:bit SCALING OF 4mA	bit	0	4095	646		
343		SLIP CURRENT OUTPUT:bit SCALING OF 20mA	bit	0	4095	3223		
344								
345								
346								
347								
348								
349								

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
350							
351							
352							
353							
354							
355							
356							
357							
358							
359							
360							
361							
362							
363							
364							
365							
366							
367							
368							
369							
370							
371							
372							
373							
374							
375							
376	CARGO REVOLUTION INCREASE ORDER INTERLOCK	—	0	1	1		
377							
378							
379							
380							
381							
382							
383							
384							
385							
386							
387							
388							
389							
390							
391							
392							
393							
394							
395							
396							
397							
398							
399							

ADDRESS	CONTENT	UNIT	LOWER	UPPER	SET VALUE	SHOP TRIAL Y M D	SEA TRIAL Y M D
400							
401	IDLE BOOST UP REV.	min-1	0	999	20		
402	BOOST UP HOLD TIME at AFTER CL ENGAGED	sec	0	999	10		
403							
404							
405							
406							
407							
408							
409	NOTUSE						
410	CH1 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
411	CH2 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
412	CH3 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
413	CH4 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
414	CH5 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
415	CH6 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
416	CH7 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
417	CH8 UPPER AND LOWER LIMIT ABNL DETECTION COUNT	count	1	250	200		
418	AHD. PELLER REV. OF CRASH AST. DETECTION	min-1	0	999	200		
419	RECOVER NEUTRAL DETECTION CONT AT CRASH AST	min-1	0	999	440		
420	AST. PELLER REV. OF CRASH AHD. DETECTION	min-1	0	999	150		
421	RECOVER NEUTRAL DETECTION CONT AT CRASH AHD	min-1	0	999	440		
422	AHD PLLER REV OF AHD CL ENG READY AT CRASH AST	min-1	0	999	70		
423	AST PLLER REV OF AST CL ENG READY AT CRASH AHD	min-1	0	999	999		
424	BOOST REV. AT CRASH AST. RECOVER NEUTRAL	min-1	0	999	0		
425	BOOST REV. AT CRASH AHD. RECOVER NEUTRAL	min-1	0	999	0		
426	M/E REV UPPER LIMIT AT AHD CL ENG READY	min-1	0	999	500		
427	M/E REV LOWER LIMIT AT AHD CL ENG READY	min-1	0	999	440		
428	M/E REV UPPER LIMIT AT AHD CL DIS-ENG READY	min-1	0	999	500		
429	PELLER REV UPPER LIMIT AT AHD CL ENG READY	min-1	0	999	999		

BOOST PRIORITY : HEATING (Add.400)>IDLE BOOST(Add.401,402)>GOV. BOOST(Add.162～167)

: RED ADDRESS DOES NOT BECOME "NOT USE",AND DISABLE THE FEATURE BY PARAMETER
: YELLOW ADDRESS DOES NOT BECOME "NOT USE",BUT NOT INCLUDED IN THE PROGRAM FEATURE
: GREEN ADDRESS IS "NOT USE",AND NOT INCLUDED IN THE PROGRAM FEATURE

DISPLAY DATA	CONTENT
	PASS WARD REQUEST AT PARAMETER CHANGE PARAMETER CHANGE IS AVAILABLE AFTER PASSWORD INPUT PASS WARD [1131]
	Add.100 AHEAD MAX ORDER REV.
	Add.101 AHEAD NAV2 ORDER REV. (EQUAL 100%)
	Add.102 AHEAD NAV1 ORDER REV. (EQUAL 85%)
	Add.103 AHAED FULL ORDER REV. (EQUAL 75%)
	Add.104 AHEAD HALF ORDER REV. (EQUAL 50%)
	Add.105 AHEAD SLOW ORDER REV.
	Add.106 AHEAD D.SLOW ORDER REV.
	Add.107 STOP ORDER REV. (NEUTRAL)
	Add.108 ASTERN D.SLOW ORDER REV.
	Add.109 ASTERN SLOW ORDER REV.
	Add.110 ASTERN HALF ORDER REV. (EQUAL 50%)
	Add.111 ASTERN FULL ORDER REV. (EQUAL 75%)
	Add.112 ASTERN EMC1 ORDER REV.

113	HANDLE POS RPM
	AS EMC2
	666 min-1
Range	0000---999

Add.113
ASTERN EMC2 ORDER REV. (EQUAL 85%)

114	HANDLE POS RPM
	AS MAX.
	692 min-1
Range	0000---999

Add.114
ASTERN MAX ORDER REV.

115	HANDLE POS RPM
	AH LIMIT
	750 min-1
Range	0000---999

Add.115
LIMIT M/E AHEAD REV.

116	HANDLE POS RPM
	AS LIMIT
	692 min-1
Range	0000---999

Add.116
LIMIT M/E ASTERN REV.

117	SLOW DOWN
	SET RPM
	400 min-1
Range	0000---999

Add.117
AUTO SLOW DOWN

118	SLOW DOWN
	RESET RPM
	400 min-1/sec
Range	0000---999

Add.118
AUTO SLOW DOWN RESET REV.

119	REVO INCREASE
	INTERLOCK
	1 INTERLOCK
Range	0 or 1

Add.119
REVOLUTION INCREASE ORDER INTERLOCK
0: NO INTER LOCK 1: CLUTCH ENGAGE
2: CLUTCH ENGAGE AFTER SLIP

120	REVO INC TIME
	AH IDLE-P1
	10.0 sec
Range	00.0---99.9

Add.120
INCREASE TIME : Add.107～Add.126

121	REVO INC TIME
	AH P1-P2
	10.0 sec
Range	00.0---99.9

Add.121
INCREASE TIME : Add.126～Add.127

122	REVO INC TIME
	AH P2-P3
	10.0 sec
Range	00.0---99.9

Add.122
INCREASE TIME : Add.127～Add.128

123	REVO INC TIME
	AH P3-RATIO
	10.0 sec
Range	00.0---99.9

Add.123
INCREASE TIME : Add.128～Add.100

124	REVO DEC TIME
	AH RATIO-P4
	10.0 sec
Range	00.0---99.9

Add.124
DECREASE TIME : Add.100～Add.129

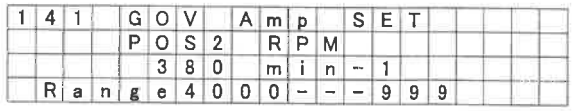
125	REVO DEC TIME
	AH P4-IDLE
	10.0 sec
Range	00.0---99.9

Add.125
DECREASE TIME : Add.129～Add.107

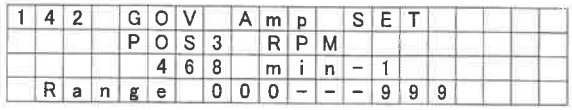
126	SPEED CHANGE
	INC POINT AH P1
	500 min-1
Range	0000---999

Add.126
INC. SPEED CHANGE POINT (LOW SPEED ZONE)

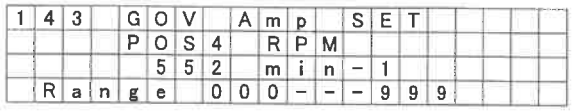
| | | |



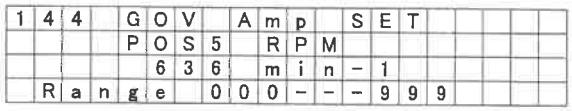
Add.141
LOAD 25% REV. SETTING



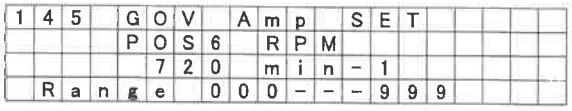
Add.142
LOAD 50% REV. SETTING



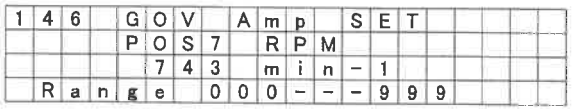
Add.143
LOAD 75% REV. SETTING



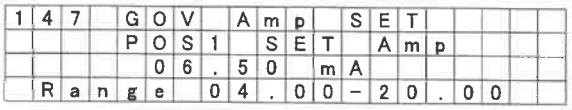
Add.144
LOAD 85% REV. SETTING



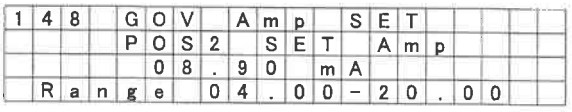
Add.145
LOAD 100% REV. SETTING



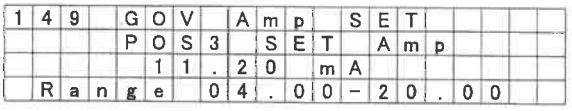
Add.146
LOAD 110% REV. SETTING



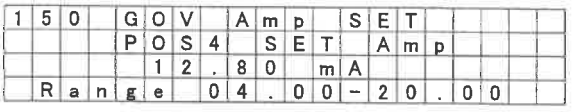
Add.147
IDLE POSISION. CURRNT SETTING



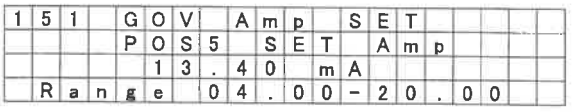
Add.148
LOAD 25% CURRENT SETTING



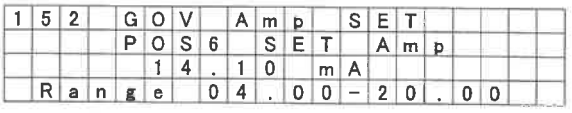
Add.149
LOAD 50% CURRENT SETTING



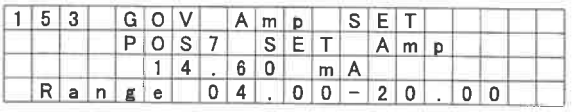
Add.150
LOAD 75% CURRENT SETTING



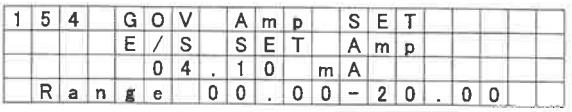
Add.151
LOAD 85% CURRENT SETTING



Add.152
LOAD 100% CURRENT SETTING



Add.153
LOAD 110% CURRENT SETTING



Add.154
E/S CONTROL CURRENT SETTING

| | | |

169	GOV BOOST SET	
	APPLY PELLA RPM	
	080 min-1	
Range	000--999	

Add.169
GOVERNOR BOOST APPLY PELLER REV. (ASTERN ONLY)

170	AST CLUTCH CONT	
	ENGU Hi REVO	
	468 min-1	
Range	000--999	

Add.170
CLUTCH ENGAGE POSSIBLE M/E REV. UPPER LIMIT

171	AST CLUTCH CONT	
	ENGU Lo REVO	
	552 min-1	
Range	000--999	

Add.171
CLUTCH ENGAGE POSSIBLE M/E REV. LOWER LIMIT

172	AST CLUTCH CONT	
	DISE Hi REVO	
	636 min-1	
Range	000--999	

Add.172
CLUTCH DISENGAGE POSSIBLE M/E REV. UPPER LIMIT

173	AST CLUTCH CONT	
	ENGU PELL REVO	
	552 min-1	
Range	000--999	

Add.173
CLUTCH ENGAGE POSSIBLE PELLER REV. UPPER LIMIT

174	CLUTCH CONT SET	
	CONT TIME	
	00.0 sec	
Range	00.0--99.9	

Add.174
CLUTCH ENGAGE/DISENGAGE ORDER TIME (0 set : continuous)

175	M/E STATUS	
	MAX REVOLUTION	
	1000 min-1	
Range	0000--1600	

Add.175
M/E MAX REV. (20mA)

176	M/E STATUS	
	20mA SCALING	
(*)	8100 bit (x2)	
Range	0000--8192	

Add.176
M/E REV. 20mA SCALING (*)

177	M/E STATUS	
	4mA SCALING	
(*)	1500 bit (x2)	
Range	0000--8192	

Add.177
M/E REV. 4mA SCALING (*)

178	PELLA STATUS	
	MAX REVOLUTION	
	2500 min-1	
Range	0000--2500	

Add.178
PELLER MAX REVOLUTION (20mA)

179	PARAMETER SET	
	20mA SCALING	
(*)	8100 bit (x2)	
Range	0000--8192	

Add.179
PELLER REV. 20mA SCALING (*)

180	PARAMETER SET	
	4mA SCALING	
(*)	1500 bit (x2)	
Range	0000--8192	

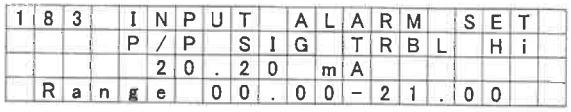
Add.180
PELLER REV. 4mA SCALING (*)

181	INPUT ALARM SET	
	M/E SIG TRBL Hi	
	20.20 mA	
Range	00.00--21.00	

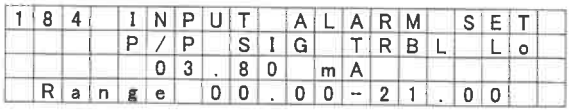
Add.181
M/E REV. SIGNAL
CURRENT UPPER LIMIT AT LINE BREAK DETECTION

182	INPUT ALARM SET	
	M/E SIG TRBL Lo	
	03.80 mA	
Range	00.00--21.00	

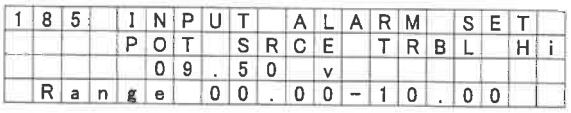
Add.182
M/E REV. SIGNAL
CURRENT LOWER LIMIT AT LINE BREAK DETECTION



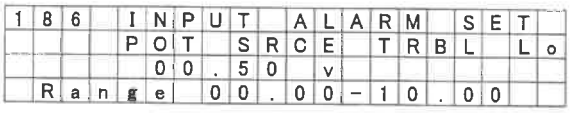
Add.183
PROPELLER REV. SIGNAL
CURRENT UPPER LIMIT AT LINE BREAK DETECTION



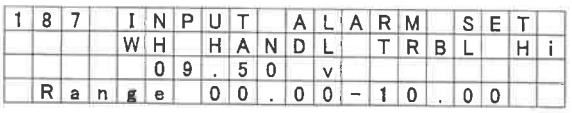
Add.184
PROPELLER REV. SIGNAL
CURRENT LOWER LIMIT AT LINE BREAK DETECTION



Add.187
POTENTIO POWER
VOLTAGE UPPER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE

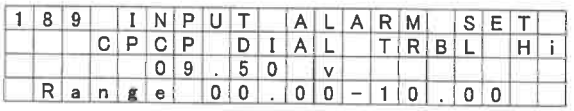


Add.188
POTENTIO POWER
VOLTAGE LOWER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE



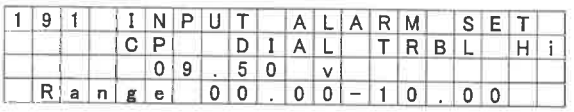
Add.187
W/H CONTROL HANDLE POTENTIO
VOLTAGE UPPER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE

Add.188
W/H CONTROL HANDLE POTENTIO
VOLTAGE LOWER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE



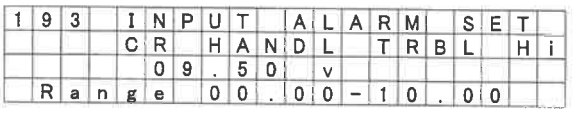
Add.189
CPCP DIAL POTENTIO (D/L POT.)
VOLTAGE UPPER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE

Add.190
CPCP DIAL POTENTIO (D/L POT.)
VOLTAGE LOWER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE

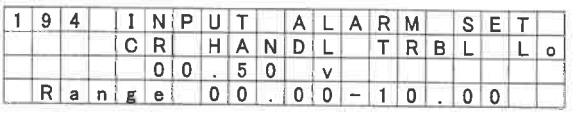


Add.191
CP DIAL POTENTIO
VOLTAGE UPPER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE

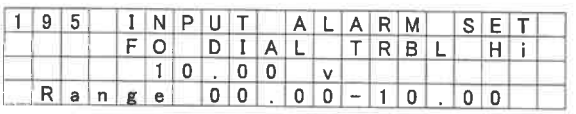
Add.192
CP DIAL POTENTIO
VOLTAGE LOWER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE



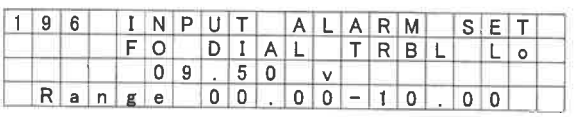
Add.193
C/R CONTROL HANDLE POTENTIO
VOLTAGE UPPER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE



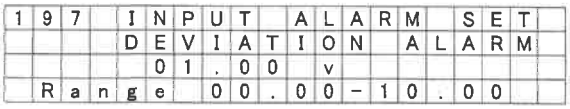
Add.194
C/R CONTROL HANDLE POTENTIO
VOLTAGE LOWER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE



Add.195
FO RACK DIAL POTENTIP
VOLTAGE UPPER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE

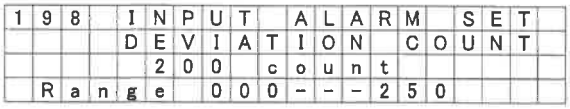


Add.196
FO RACK DIAL POTENTIP
VOLTAGE LOWER LIMIT AT LINE BREAK DETECTION
VOLTAGE RANGE



Add.197
VOLTAGE DEVIATION AT LINE BREAK DETECTION

VOLTAGE RANGE

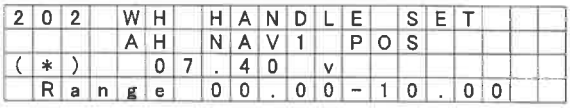


Add.198
DEVIATION FOR LINE BREAK COUNT DETECTION (Add.197)

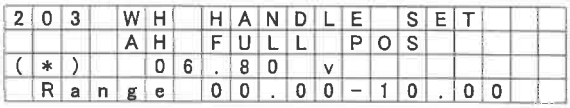
Add.199
HANDLE RANGE ABNL DETECTION TIME(0.0:DISABLED)

Add.200
CONTROL HANDLE VOLTAGE AT AHEAD MAX POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD MAX POS. SETTING BY [SET] KEY

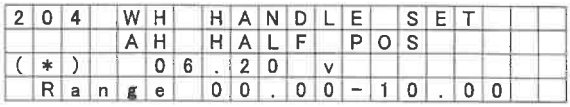
Add.201
CONTROL HANDLE VOLTAGE AT AHEAD NAV2 POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD NAV2 POS. SETTING BY [SET] KEY



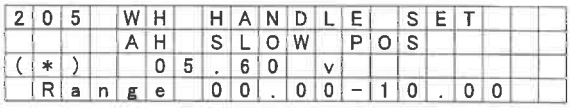
Add.202
CONTROL HANDLE VOLTAGE AT AHEAD NAV1 POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD NAV1 POS. SETTING BY [SET] KEY



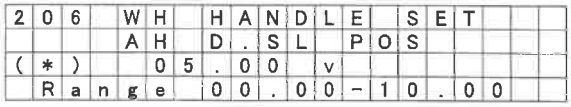
Add.203
CONTROL HANDLE VOLTAGE AT AHEAD FULL POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD FULL POS. SETTING BY [SET] KEY



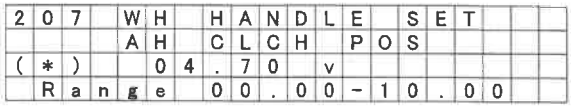
Add.204
CONTROL HANDLE VOLTAGE AT AHEAD HALF POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD HALF POS. SETTING BY [SET] KEY



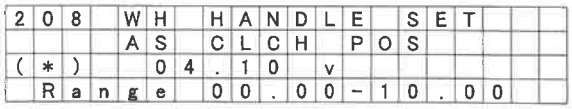
Add.205
CONTROL HANDLE VOLTAGE AT AHEAD SLOW POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD SLOW POS. SETTING BY [SET] KEY



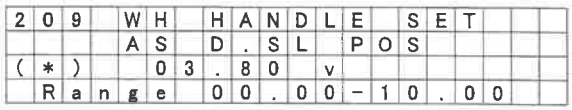
Add.206
CONTROL HANDLE VOLTAGE AT AHEAD D.SLOW POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD D.SLOW POS. SETTING BY [SET] KEY



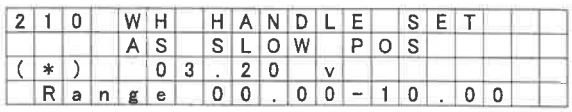
Add.207
CONTROL HANDLE VOLTAGE AT AHEAD CLUTCH ENGAGE POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD CLUCH ENGAGE POS. SETTING BY [SET] KEY



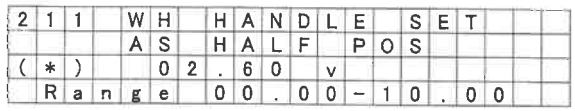
Add.208
CONTROL HANDLE VOLTAGE AT ASTERN CLUTCH ENGAGE POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN CLUCH ENGAGE POS. SETTING BY [SET] KEY



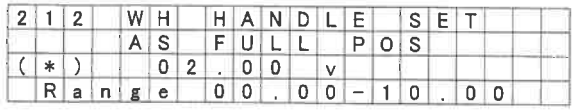
Add.209
CONTROL HANDLE VOLTAGE AT ASTERN D.SLOW POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN D.SLOW POS. SETTING BY [SET] KEY



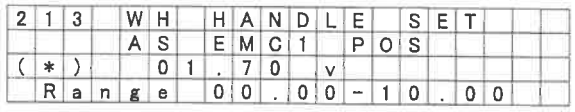
Add.210
CONTROL HANDLE VOLTAGE AT ASTERN SLOW POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN SLOW POS. SETTING BY [SET] KEY



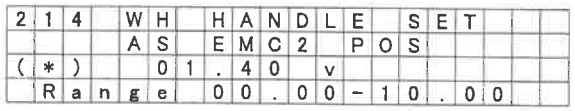
Add.211
CONTROL HANDLE VOLTAGE AT ASTERN HALF POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN HALF POS. SETTING BY [SET] KEY



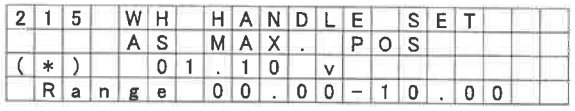
Add.212
CONTROL HANDLE VOLTAGE AT ASTERN FULL POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN FULL POS. SETTING BY [SET] KEY



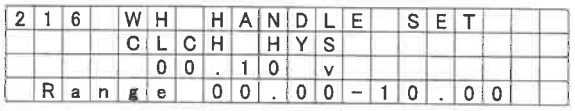
Add.213
CONTROL HANDLE VOLTAGE AT ASTERN EMC1 POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN EMC1 POS. SETTING BY [SET] KEY



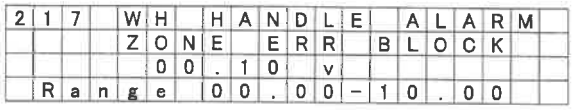
Add.214
CONTROL HANDLE VOLTAGE AT ASTERN EMC2 POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN EMC2 POS. SETTING BY [SET] KEY



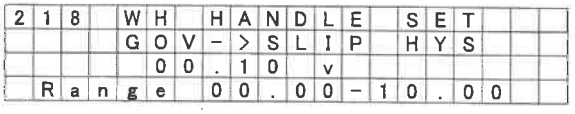
Add.215
CONTROL HANDLE VOLTAGE AT ASTERN MAX POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN MAX POS. SETTING BY [SET] KEY



Add.216
W/H CONT HANDLE CLUTCH ENGAGE/DISENGAGE HYSTERESIS

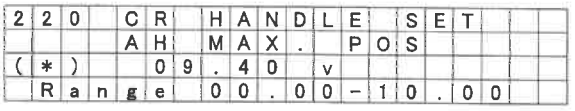


Add.217
W/H CONT HANDLE RANGE, ABNL DETECTION DISABLED VOL RANGE

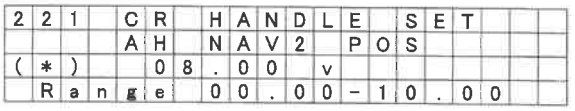


Add.218
W/H GOVERNOR > CHANGE SLIP HYSTERISIS

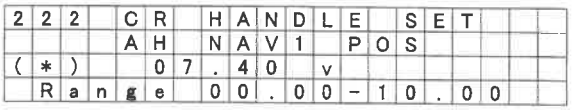
Add.219
W/H HANDLE SIGNAL CHANGE(0=INC 1=DEC)



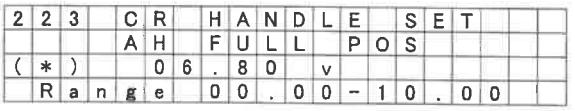
Add.220
CONTROL HANDLE VOLTAGE AT AHEAD MAX POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD MAX POS. SETTING BY [SET] KEY



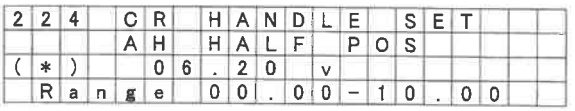
Add.221
CONTROL HANDLE VOLTAGE AT AHEAD NAV2 POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD NAV2 POS. SETTING BY [SET] KEY



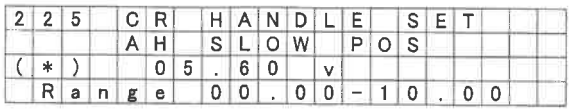
Add.222
CONTROL HANDLE VOLTAGE AT AHEAD NAV1 POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD NAV1 POS. SETTING BY [SET] KEY



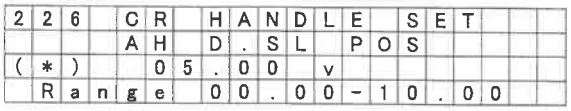
Add.223
CONTROL HANDLE VOLTAGE AT AHEAD FULL POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD FULL POS. SETTING BY [SET] KEY



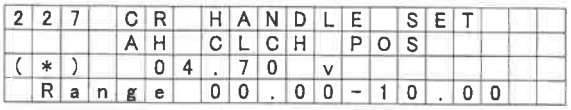
Add.224
CONTROL HANDLE VOLTAGE AT AHEAD HALF POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD HALF POS. SETTING BY [SET] KEY



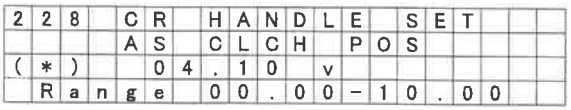
Add.225
CONTROL HANDLE VOLTAGE AT AHEAD SLOW POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD SLOW POS. SETTING BY [SET] KEY



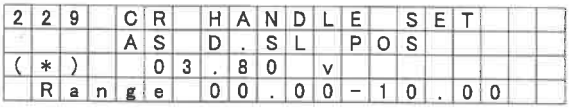
Add.226
CONTROL HANDLE VOLTAGE AT AHEAD D.SLOW POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD D.SLOW POS. SETTING BY [SET] KEY



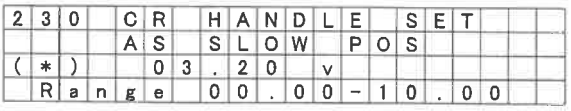
Add.227
CONTROL HANDLE VOLTAGE AT AHEAD CLUTCH ENGAGE POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO AHEAD CLUTCH ENGAGE POS. SETTING BY [SET] KEY



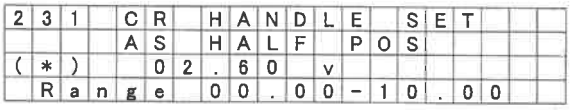
Add.228
CONTROL HANDLE VOLTAGE AT ASTERN CLUTCH ENGAGE POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN CLUTCH ENGAGE POS. SETTING BY [SET] KEY



Add.229
CONTROL HANDLE VOLTAGE AT ASTERN D.SLOW POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN D.SLOW POS. SETTING BY [SET] KEY

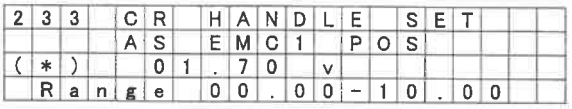


Add.230
CONTROL HANDLE VOLTAGE AT ASTERN SLOW POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN SLOW POS. SETTING BY [SET] KEY

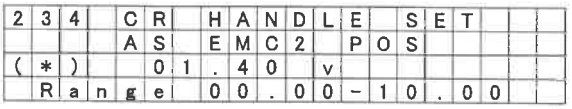


Add.231
CONTROL HANDLE VOLTAGE AT ASTERN HALF POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN HALF POS. SETTING BY [SET] KEY

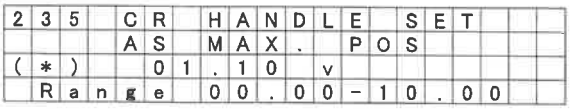
Add.232
CONTROL HANDLE VOLTAGE AT ASTERN FULL POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN FULL POS. SETTING BY [SET] KEY



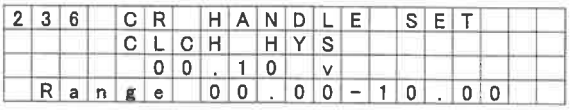
Add.233
CONTROL HANDLE VOLTAGE AT ASTERN EMC1 POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN EMC1 POS. SETTING BY [SET] KEY



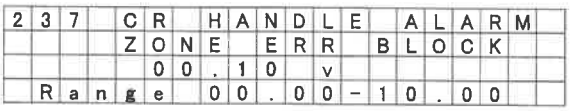
Add.234
CONTROL HANDLE VOLTAGE AT ASTERN EMC2 POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN EMC2 POS. SETTING BY [SET] KEY



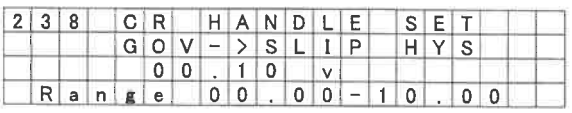
Add.235
CONTROL HANDLE VOLTAGE AT ASTERN MAX POS.
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
HANDLE SET TO ASTERN MAX POS. SETTING BY [SET] KEY



Add.236
CONTROL HANDLE CLUTCH ENGAGE・DISENGAGE HYSTERESIS

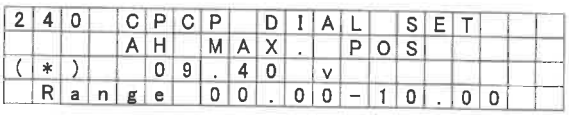


Add.237
C/R CONT HANDLE AREA, ABNL DETECTION DISABLE VOL RANGE

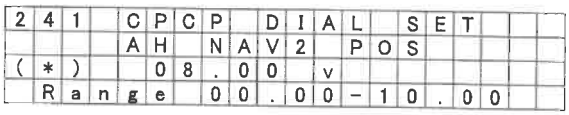


Add.238
C/R GOVERNOR > SLIP CHANGE HYSTERISIS

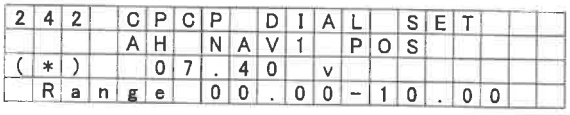
Add.239
C/R HANDLE SIGNAL CHANGE(0=INC 1=DEC)



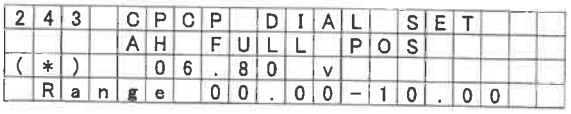
Add.240
DIAL MAX VOLTAGE
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
DIAL SET TO MAX POS. SETTING BY [SET] KEY



Add.241

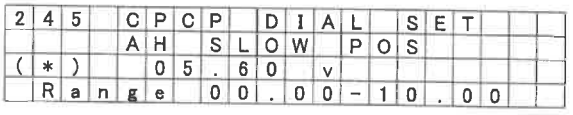


Add.242

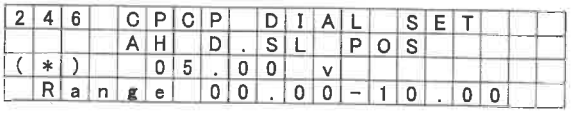


Add.243

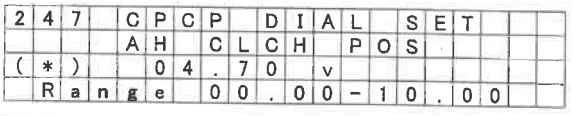
Add.244



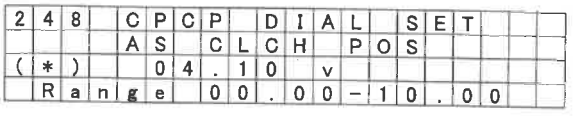
Add.245



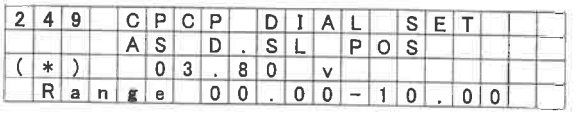
Add.246



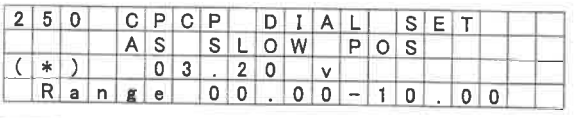
Add.247
DIAL MINIMUM VOLTAGE
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
DIAL SET TO "0" POS. SETTING BY [SET] KEY



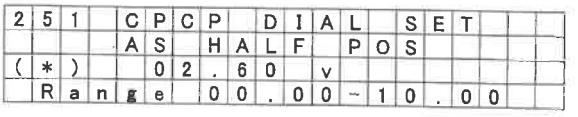
Add.248



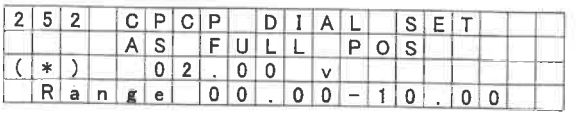
Add.249



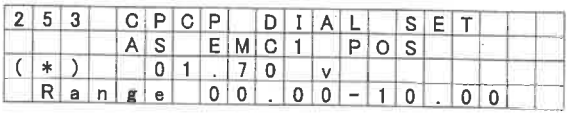
Add.250



Add.251



Add.252



Add.253

254	CPCP	DIAL	SET
	AS	EMC2	POS
(*)	01.40	v	
	Range	00.00-10.00	

Add.254

255	CPCP	DIAL	SET
	AS	MAX.	POS
(*)	01.10	v	
	Range	00.00-10.00	

Add.255

256	CPCP	DIAL	SET
	CLCH	HYS	
	00.10	v	
	Range	00.00-10.00	

Add.256

257	CPCP	DIAL	ALARM
	ZONE	ERR	BLOCK
	00.10	v	
	Range	00.00-10.00	

Add.257

258	CPCP	DIAL	SET
	GOV->SLIP		HYS
	00.10	v	
	Range	00.00-10.00	

Add.258

259	CPCP	DIAL	SET
	INPUT	CURV	
	0	INCREASE	
	Range	0 or 1	

Add.259

260	CP	DIAL	SET
	AH	MAX.	POS
(*)	09.40	v	
	Range	00.00-10.00	

Add.260
DIAL MAX VOLTAGE
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
DIAL SET TO MAX POS. SETTING BY [SET] KEY

261	CP	DIAL	SET
	AH	NAV2	POS
(*)	08.00	v	
	Range	00.00-10.00	

Add.261

262	CP	DIAL	SET
	AH	NAV1	POS
(*)	07.40	v	
	Range	00.00-10.00	

Add.262

263	CP	DIAL	SET
	AH	FULL	POS
(*)	06.80	v	
	Range	00.00-10.00	

Add.263

264	CP	DIAL	SET
	AH	HALF	POS
(*)	06.20	v	
	Range	00.00-10.00	

Add.264

265	CP	DIAL	SET
	AH	SLOW	POS
(*)	05.60	v	
	Range	00.00-10.00	

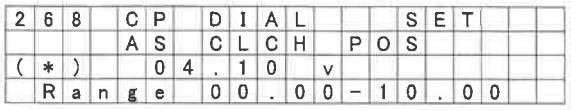
Add.265

266	CP	DIAL	SET
	AH	D.SL	POS
(*)	05.00	v	
	Range	00.00-10.00	

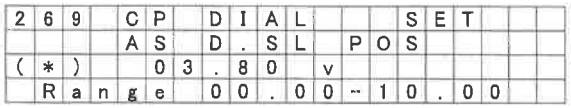
Add.266

267	CP	DIAL	SET
	AH	CLCH	POS
(*)	04.70	v	
	Range	00.00-10.00	

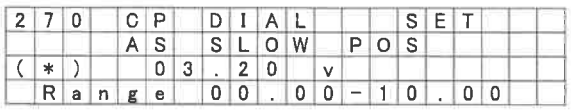
Add.267
DIAL MINIMUM VOLTAGE
「(*)」IS REPRESENT AUTO WRITING SUPPORT PARAMETER
DIAL SET TO "0" POS. SETTING BY [SET] KEY



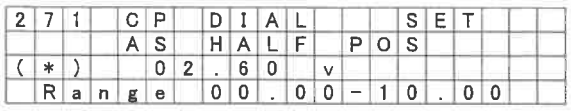
Add.268



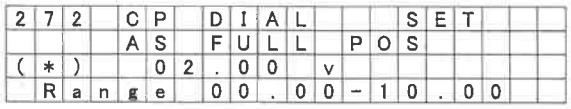
Add.269



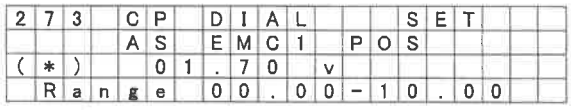
Add.270



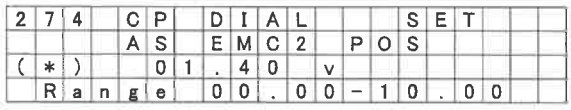
Add.271



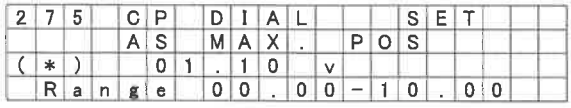
Add.272



Add.273



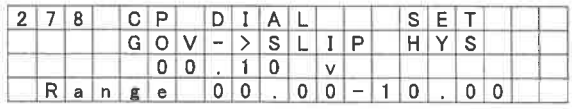
Add.274



Add.275

Add.276

Add.277



Add.278

Add.279

Add.280

Add.281

Add.282

Add.283

Add.284

Add.285

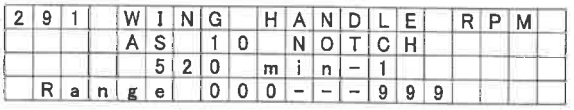
Add.286

Add.287

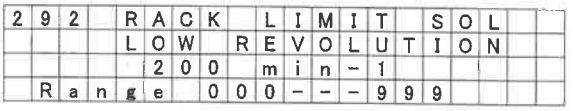
Add.288

Add.289

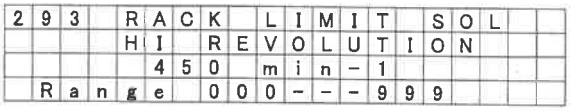
Add.290



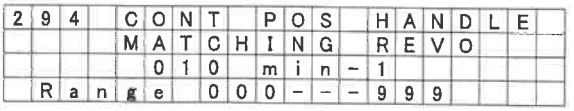
Add.291



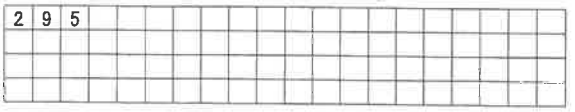
Add.292



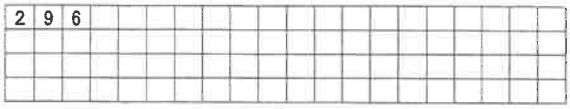
Add.293



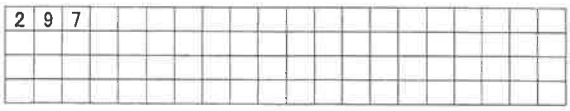
Add.294



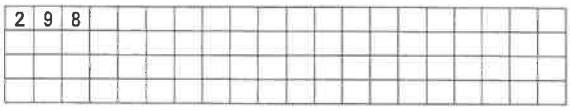
Add.295
CARGO DIAL MAX REV.



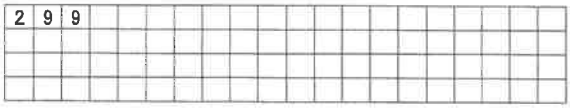
Add.296
CARGO DIAL MINIMUM REV.



Add.297
CARGO DIAL FOLLU UP SPEED

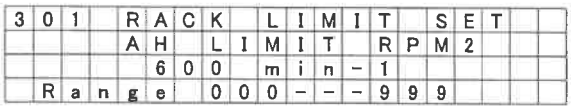


Add.298

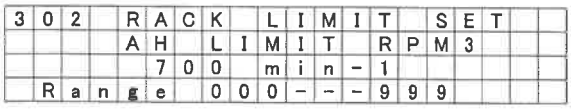


Add.299

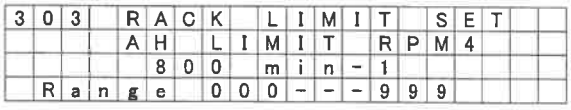
Add.300
LIMIT POSITION 1 REV. SET (related Add.305)



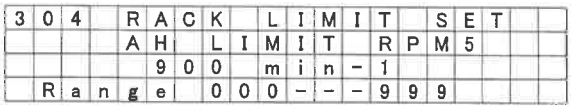
Add.301
LIMIT POSITION 2 REV. SET (related Add.306)



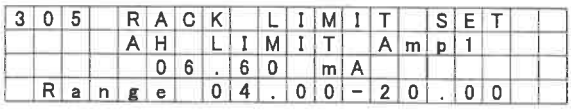
Add.302
LIMIT POSITION 3 REV. SET (related Add.307)



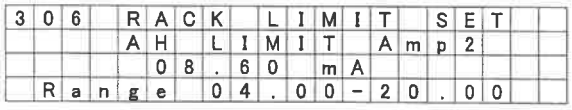
Add.303
LIMIT POSITION 4 REV. SET (related Add.308)



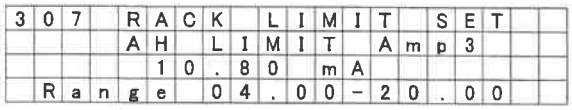
Add.304
LIMIT POSITION 5 REV. SET (related Add.309)



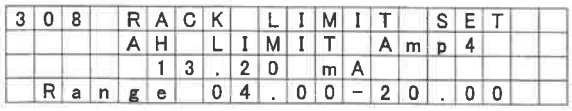
Add.305
LIMIT POS 1 CURRENT SET (related Add.300)



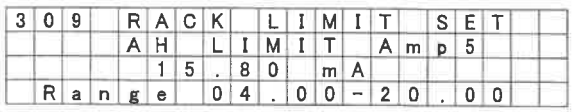
Add.306
LIMIT POS 2 CURRENT SET (related Add.301)



Add.307
LIMIT POS 3 CURRENT SET (related Add.302)



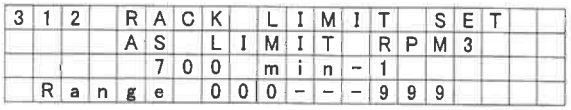
Add.308
LIMIT POS 4 CURRENT SET (related Add.303)



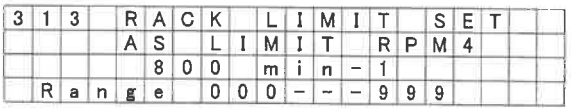
Add.309
LIMIT POS 5 CURRENT SET (related Add.304)

Add.310
LIMIT POS 1 REV. SET (related Add.315)

Add.311
LIMIT POS 2 REV. SET (related Add.316)

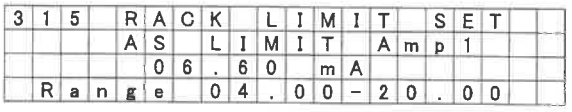


Add.312
LIMIT POS 3 REV. SET (related Add.317)

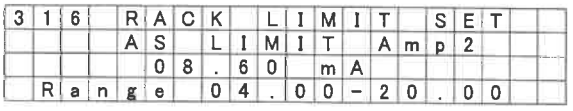


Add.313
LIMIT POS 4 REV. SET (related Add.318)

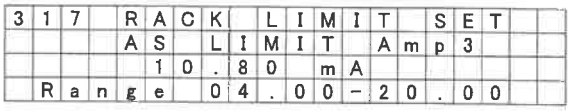
Add.314
LIMIT POS 5 REV. SET (related Add.319)



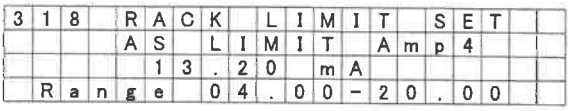
Add.315
LIMIT POS 1 CURRENT SET (related Add.310)



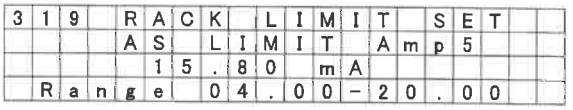
Add.316
LIMIT POS 2 CURRENT SET (related Add.311)



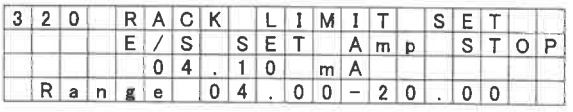
Add.317
LIMIT POS 3 CURRENT SET (related Add.312)



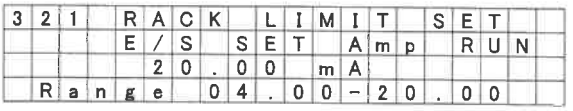
Add.318
LIMIT POS 4 CURRENT SET (related Add.313)



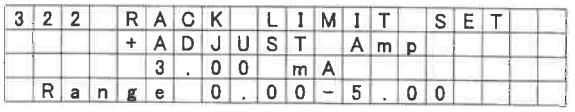
Add.319
LIMIT POS 5 CURRENT SET (related Add.314)



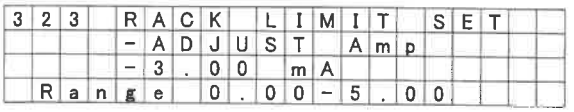
Add.320
LIMIT CURRENT at M/E STOP



Add.321
LIMIT CURRENT at M/E RUN



Add.322
RACK LIMIT ADJUSTMENT DIAL UPPER LIMIT



Add.323
RACK LIMIT ADJUSTMENT DIAL LOWER LIMIT

| | | |

338 SLIP CONT SET
E/S SET Amp RUN
20.00 mA
Range 04.00-20.00

Add.338
E/S:FO HANDLE SW. RUN CURRENT SET.

339 SLIP ABLE RPM
BLK WAITE TIME
02.0 sec
Range 00.0---99.9

Add.339
WAITING TIME OF SLIP ENABLED REV. UPPER LIMIT

340 SLIP CONT SET
SLIP ABLE RPM
400 min-1
Range 0000---999

Add.340
SLIP ENABLED REV. UPPER LIMIT

341 SLIP CONT SET
NEUT SLIP SELE
1 SLIP ON
Range 0 or 1

Add.341
SLIP OUTPUT SELECT OF NEUTRAL(0:OFF, 1:ON)

342 SLIP CONT SCALE
4mA bit SET
0642 bit
Range 0000---4095

Add.342
SLIP CURRENT OUTPUT:bit SCALING OF 4mA

343 SLIP CONT SCALE
20mA bit SET
3268 bit
Range 0000---4095

Add.343
SLIP CURRENT OUTPUT:bit SCALING OF 20mA

344 JET ASIST HANDLE
QUICK OPERATE
*** min-1/0.1s
Range 000---999

Add.344

345 JET ASIST TIME
AHD ORDER/GOV
005 sec
Range 000---999

Add.345

346 JET ASIST TIME
AST ORDER/GOV
005 sec
Range 000---999

Add.346

347 JET ASIST TIME
AFTER STALL BLK
006 sec
Range 000---999

Add.347

348 JET ASIST TIME
ASIST ABLE RPM
450 min-1
Range 000---999

Add.348

349 JET ASIST TIME
OPE MODE SELECT
0 CL ORDER
Range 0, 1 or 2

Add.349

350 INS CONTROL
SPEED SET1
01.0 Knt
Range 00.0---20.0

Add.350

351 INS CONTROL
SPEED SET2
02.0 Knt
Range 00.0---20.0

Add.351

| | | |

366	INS CONTROL
	NO RECEIVE TIME
	005 sec
	Range 000---999

Add.366

367	INS CONTROL
	ERR MODE SELE
	0 HDLE POS
	Range 0 or 1

Add.367

368	COMM. DATA
	CHECK SUM SELE
	0 NO CHECK
	Range 0 or 1

Add.368

369	CG DIAL SET
	INPUT CURV
	0 INCREASE
	Range 0 or 1

Add.369

370	CRITICAL ZONE
	Hi REVOLUTION
	300 min-1
	Range 000---999

Add.370

371	CRITICAL ZONE
	Lo REVOLUTION
	200 min-1
	Range 000---999

Add.371

372	CRITICAL JUMP
	JUMP MODE SELE
	0 BOOST RPM
	Range 0 or 1 or 2

Add.372

373	RACK STATUS
	MAX STROKE
	050 mm
	Range 000---999

Add.373

374	RACK STATUS
	20mA SCALING
	8115 bit
	Range 0000---8192

Add.374

375	RACK STATUS
	4mA SCALING
	1583 bit
	Range 0000---8192

Add.375

376	NOT USE

377	HANDLE SIGNAL
	4mA bit SET
	0642 bit
	Range 0000---4095

Add.377

378	HANDLE SIGNAL
	20mA bit SET
	0642 bit
	Range 0000---4095

Add.378

379	NOT USE

380	START/STOP	CONT
	START	FAIL TIME
	005	sec
Range	000	---999

Add.380

381	START/STOP	CONT
	STOP	TIME
	040	sec
Range	000	---999

Add.381

382	START/STOP	CONT
	NOT	START READY
	040	sec
Range	000	---999

Add.382

383	START/STOP	CONT
	FW	PUMP READY
	005	sec
Range	000	---999

Add.383

384	START/STOP	CONT
	AFTER	COOLING
	10.0	min
Range	00.0	---99.9

Add.384

385	START/STOP	CONT
	AFTER	COOLING
	15.0	min
Range	00.0	---99.9

Add.385

386	START/STOP	CONT
	ES	START INHIBIT
	010	sec
Range	000	---999

Add.386

390	ORDER	ADJUST
	+ADJUST	REVO
	050	min-1
Range	000	---100

Add.390

391	ORDER	ADJUST
	-ADJUST	REVO
	-050	min-1
Range	000	---100

Add.391

392	ORDER	ADJUST
	DIAL	+MAX POS
(*)	09.50	v
Range	00.00	-10.00

Add.392

393	ORDER	ADJUST
	DIAL	NEUT POS
(*)	05.00	v
Range	00.00	-10.00

Add.393

394	ORDER	ADJUST
	DIAL	-MAX POS
(*)	00.50	v
Range	00.00	-10.00

Add.394

395	Fi-Fi	MODE	SET
	10	NOCH	RPM
	010	min-1	
Range	000	---999	

Add.395

396	Fi-Fi	MODE	SET
	0	NOCH	RPM
	010	min-1	
Range	000	---999	

Add.396

397	Fi-Fi	MODE				
	INC/DEC	SPEED				
	010	min-1/sec				
Range	000	---	999			

Add.397

398	Fi-Fi	MODE	SET			
	10	NOCH	POS			
(*)	09.00	v				
Range	00.00	-10.00				

Add.398

399	Fi-Fi	MODE	SET			
	0	NOCH	POS			
(*)	01.00	v				
Range	00.00	-10.00				

Add.399

400	WARM	UP	SET			
	WARM	UP	REVO			
	100	min-1				
Range	000	---	999			

Add.400

401	IDLE	BOOST	MODE			
	WARM	UP	REVO			
	100	min-1				
Range	000	---	999			

Add.401

402	IDLE	BOOST	MODE			
	KEEP	TIME				
	005	sec				
Range	000	---	999			

Add.402

403	CLUTCH	INCHING				
	AH	ENGAGE	TIME			
	01.0	sec				
Range	00.0	---	99.9			

Add.403

404	CLUTCH	INCHING				
	AH	DISENG	TIME			
	01.0	sec				
Range	00.0	---	99.9			

Add.404

405	CLUTCH	INCHING				
	AH	END	PELL	REVO		
	100	min-1				
Range	000	---	999			

Add.405

406	CLUTCH	INCHING				
	AS	ENGAGE	TIME			
	01.0	sec				
Range	00.0	---	99.9			

Add.406

407	CLUTCH	INCHING				
	AS	DISENG	TIME			
	01.0	sec				
Range	00.0	---	99.9			

Add.407

408	CLUTCH	INCHING				
	AS	END	PELL	REVO		
	100	min-1				
Range	000	---	999			

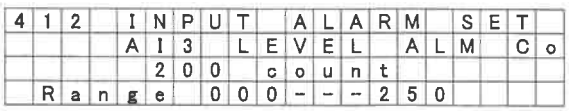
Add.408

410	INPUT	ALARM	SET			
	AI1	LEVEL	ALM	Co		
	200	count				
Range	000	---	250			

Add.410
CH1 UPPER AND LOWER LIMIT ABNL DETECTION COUNT
related Add.181,182

411	INPUT	ALARM	SET			
	AI2	LEVEL	ALM	Co		
	200	count				
Range	000	---	250			

Add.411
CH2 UPPER AND LOWER LIMIT ABNL DETECTION COUNT
related Add.183,184



Add.412
CH3 UPPER AND LOWER LIMIT ABNL DETECTION COUNT
related Add.185,186

Add.413
CH4 UPPER AND LOWER LIMIT ABNL DETECTION COUNT
related Add.187,188

Add.414
CH5 UPPER AND LOWER LIMIT ABNL DETECTION COUNT
related Add.189,190

Add.415
CH6 UPPER AND LOWER LIMIT ABNL DETECTION COUNT
related Add.191,192

Add.416
CH7 UPPER AND LOWER LIMIT ABNL DETECTION COUNT
related Add.193,194

Add.417
CH8 UPPER AND LOWER LIMIT ABNL DETECTION COUNT
related Add.195,196

Add.418
AHD. PELLER REV. OF CRASH AST. DETECTION

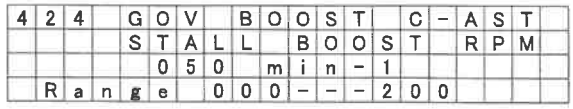
Add.419
RECOVER NEUTRAL DETECTION CONT AT CRASH AST

Add.420
AST. PELLER REV. OF CRASH AHD. DETECTION

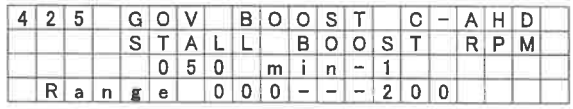
Add.421
RECOVER NEUTRAL DETECTION CONT AT CRASH AHD

Add.422
AHD PLLER REV OF AHD CL ENG READY AT CRASH AST

Add.423
AST PLLER REV OF AST CL ENG READY AT CRASH AHD



Add.424
BOOST REV. AT CRASH AST. RECOVER NEUTRAL



Add.425
BOOST REV. AT CRASH AHD. RECOVER NEUTRAL

426	AHD	CLUTCH	CONT
	ENGU	Hi	REVO
	420	min	-1
Range	000	---	999

Add.426
M/E REV UPPER LIMIT AT AHD CL ENG READY

427	AHD	CLUTCH	CONT
	ENGU	Lo	REVO
	330	min	-1
Range	000	---	999

Add.427
M/E REV LOWER LIMIT AT AHD CL ENG READY

428	AHD	CLUTCH	CONT
	DISE	Hi	REVO
	500	min	-1
Range	000	---	999

Add.428
M/E REV UPPER LIMIT AT AHD CL DIS-ENG READY

429	AHD	CLUTCH	CONT
	ENGU	PELL	REVO
	030	min	-1
Range	000	---	999

Add.429
PELLER REV UPPER LIMIT AT AHD CL ENG READY

430	AHD	Hi	CLUTCH
	ENGU	Hi	REVO
	370	min	-1
Range	000	---	999

Add.430

431	AHD	Hi	CLUTCH
	ENGU	Lo	REVO
	330	min	-1
Range	000	---	999

Add.431

432	AHD	Hi	CLUTCH
	DISE	Hi	REVO
	380	min	-1
Range	000	---	999

Add.432

433	AHD	Hi	CLUTCH
	ENGU	PELL	REVO
	030	min	-1
Range	000	---	999

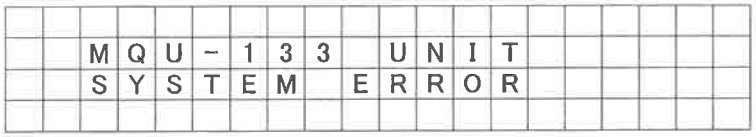
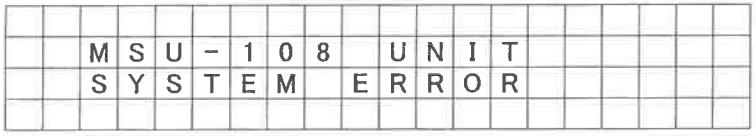
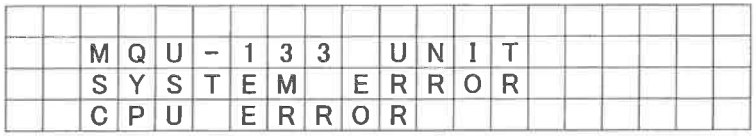
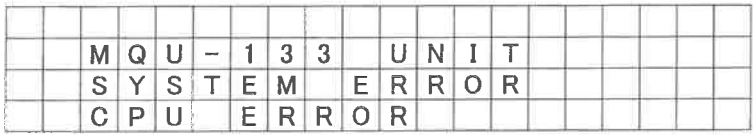
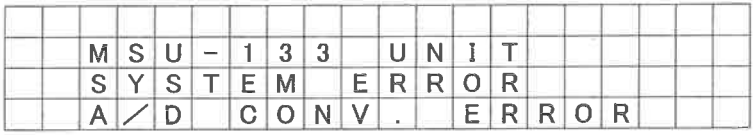
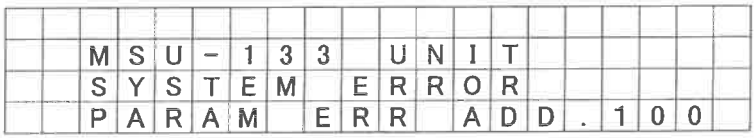
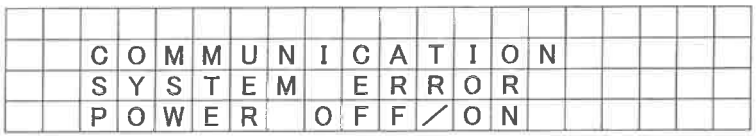
Add.433

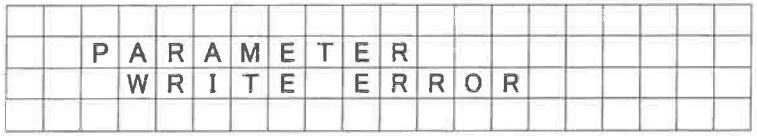
434	AHD	Hi	CLUTCH
	KEEP	ORDER	REVO
	480	min	-1
Range	000	---	999

Add.434

435	AHD	Hi	CLUTCH
	KEEP	ORDER	TIME
	010	sec	
Range	000	---	999

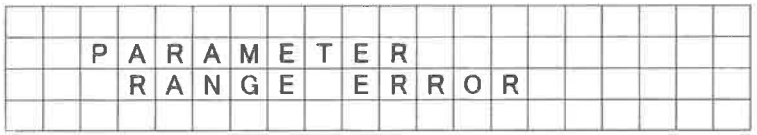
Add.435

ABNORMAL DISPLAY	CONTENT	COUNTER MEASURE
	MQU-133 AIO UNIT ABNORMAL	RESET BY KEY [C]、[F5] WHEN ABNORMAL IS REPEATED REPLACEMENT OF PCB MAY BE NECESSARY
	MSU-108 DIO UNIT ABNORMAL	RESET BY KEY [C]、[F5] WHEN ABNORMAL IS REPEATED REPLACEMENT OF PCB MAY BE NECESSARY
	MQU-133 AIO UNIT CPU ABNROMAL	POWER SUPPLY TO BE OFF AND ON WHEN ABNORMAL IS REPEATED REPLACEMENT OF PCB MAY BE NECESSARY
	MQU-133 AIO UNIT CPU ABNROMAL	POWER SUPPLY TO BE OFF AND ON WHEN ABNORMAL IS REPEATED REPLACEMENT OF PCB MAY BE NECESSARY
	MQU-133 AIO UNIT A/D CONVERTOR ABNROMAL	POWER SUPPLY TO BE OFF AND ON WHEN ABNORMAL IS REPEATED REPLACEMENT OF PCB MAY BE NECESSARY
	PARAMETER ABNORMAL DISPLAYED ADDRESS PRAMETER SET ABNORMAL	RESET BY KEY [C] CORRECT DATA SHALL BE SET AFTER CHECK PARAMETER OF DISPLAYED ADDRESS
	COMMUNICATION ABNROMAL	POWER SUPPLY TO BE OFF AND ON



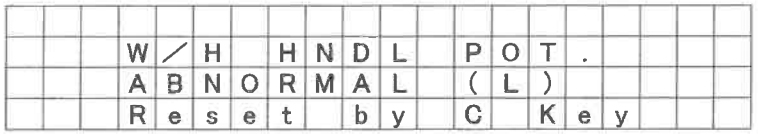
PARAMETER WRITE FAULT

REWRITE DATA AFTER RESET BY KEY [C]
WHEN ABNORMAL IS REPEATED
REPLACEMENT OF PCB MAY BE NECESSARY
AIO(MSU-133)



PARAMETER OUT OF RANGE

RESET BY KEY [C]
CORRECT DATA SHALL BE SET
AFTER CHECK PARAMETER



HANDLE POT. SIG. ABNL.

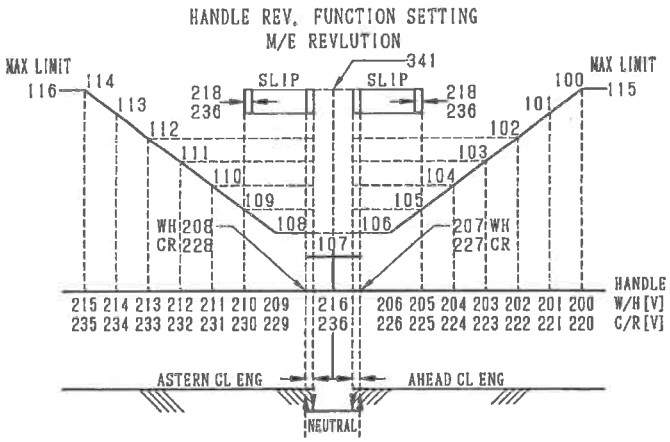
RESET BY KEY [C]
CHECK POT. VOLTAGE AND POT. POWER
REPLACE POTENTIO in case of ABNORMAL POT. VOLTAGE
REPLACE PCB in case of ABNORMAL POT. POWER
IF NORMAL,RESET BY KEY [F5]

(L): OVER UPPER AND LOWER LIMIT
(D): SCAN PER DEVIATION
(A): RANGE ABNORMAL

SZ15045/15406/15047/15048

BLOCK DIAGRAM(1/2)

M/E CONTROL



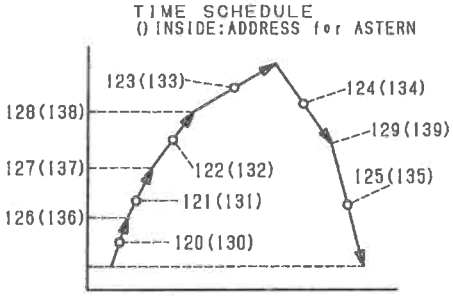
[HANDLE VOLTAGE SETTING]
Add. 219=0 WH HANDLE SET. (Add. 200~215)
Add. 200>=201>=・・・>=214>=215

Add. 219=1 WH HANDLE SET. (Add. 200~215)
Add. 200<=201<=・・・<=214<=215

Add. 239=0 CR HANDLE SET. (Add. 220~235)
Add. 220>=221>=・・・>=234>=235

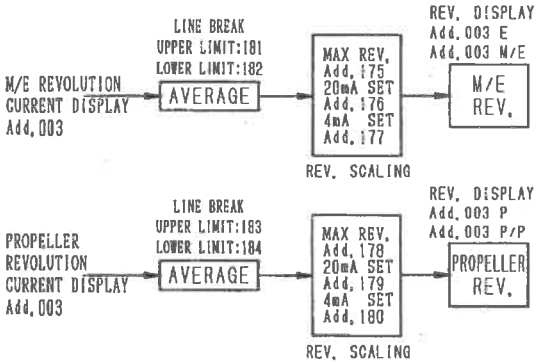
Add. 239=1 CR HANDLE SET. (Add. 220~235)
Add. 220<=221<=・・・<=234<=235

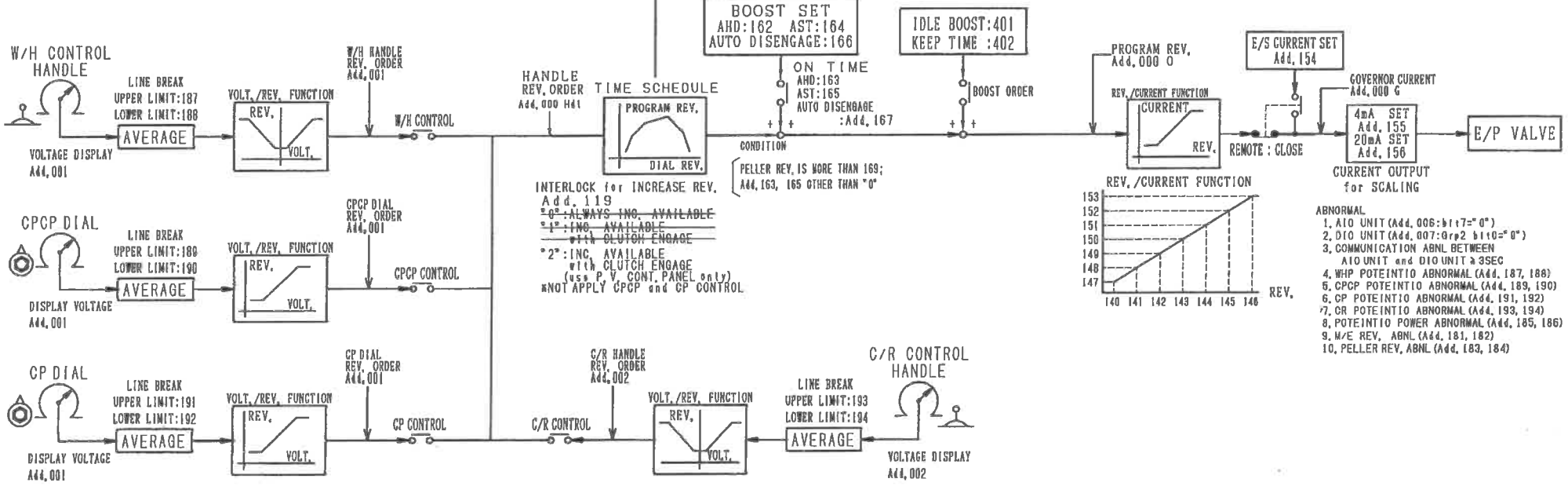
CL ATUO DISENGAGE (DURING M/E RUN)
・M/E REV. ≤ CL ATUO DISENGAGE DITECT REV. (Add. 157, 159)
・KEEP TIME for NEUTRAL (Add. 158, 160)
CL RE-ENGAGE AFTER THIS SET TIME
・OPERATE GOVERNOR BOOST AT THIS TIME (Add. 166, 167)
・CL AUTO DISENGAGE CANCEL TIME AFTER CL ENGAGE (Add. 161)
・Add. 161 = SET [999] NOT CANCEL



PROGRAM REV. ORDER FOLLOW UP SPEED
() INSIDE: ADDRESS for ASTERN
Add. 120 (130) INC. FOLLOW UP SPEED: Add. 107 (107) ~126 (136)
Add. 121 (131) INC. FOLLOW UP SPEED: Add. 126 (136) ~127 (137)
Add. 122 (132) INC. FOLLOW UP SPEED: Add. 127 (137) ~128 (138)
Add. 123 (133) INC. FOLLOW UP SPEED: Add. 128 (138) ~100 (114)
Add. 124 (134) DEC. FOLLOW UP SPEED: Add. 100 (114) ~129 (139)
Add. 125 (135) DEC. FOLLOW UP SPEED: Add. 129 (139) ~107 (107)

FOLLOW UP SPEED CHANGE POINT
Add. 126 (136) INC. CHANGE POINT (LOW)
Add. 127 (137) INC. CHANGE POINT (MIDDLE)
Add. 128 (138) INC. CHANGE POINT (HIGH)
Add. 129 (139) DEC. CHANGE POINT



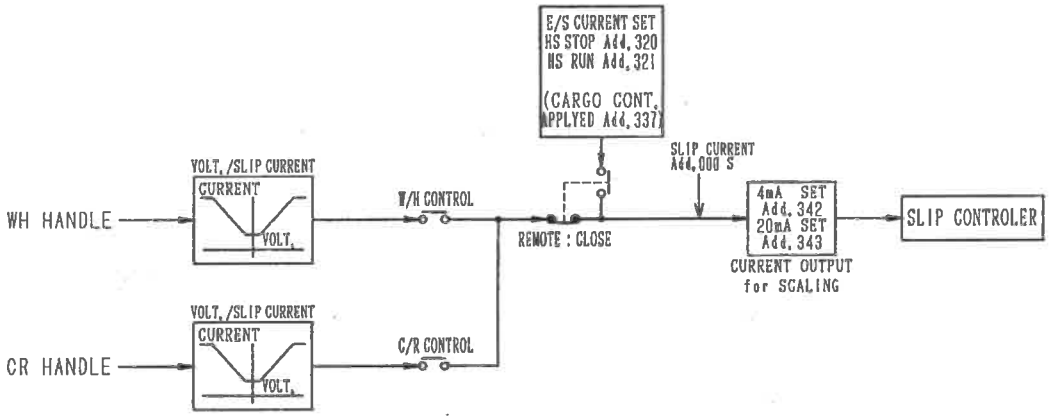


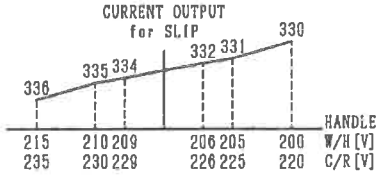
ABNORMAL
1. AIO UNIT (Add. 006: bit7="0")
2. DIO UNIT (Add. 007: Grp2 bit0="0")
3. COMMUNICATION ABNL BETWEEN AIO UNIT and DIO UNIT ≧3SEC
4. WHP POTEINTIO ABNORMAL (Add. 187, 188)
5. CPCP POTEINTIO ABNORMAL (Add. 189, 190)
6. CP POTEINTIO ABNORMAL (Add. 191, 192)
7. CR POTEINTIO ABNORMAL (Add. 193, 194)
8. POTEINTIO POWER ABNORMAL (Add. 185, 186)
9. M/E REV. ABNL (Add. 181, 182)
10. PELLER REV. ABNL (Add. 183, 184)

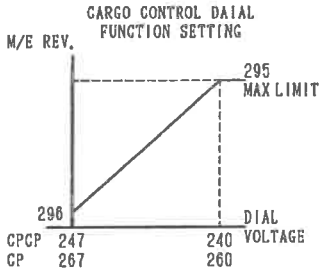
SZ15045/15406/15047/15048

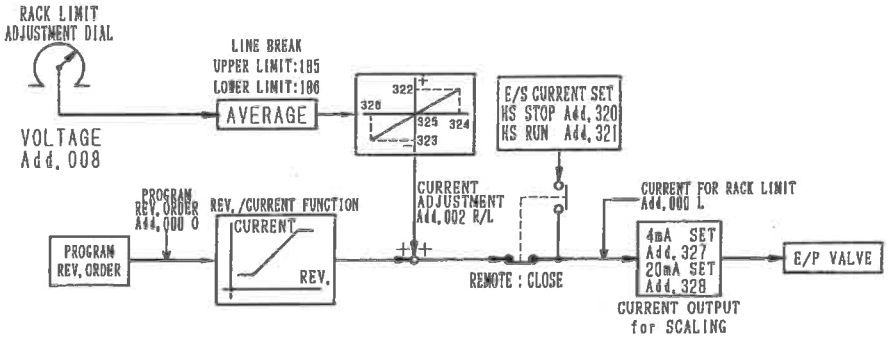
BLOCK DIAGRAM(2/2)

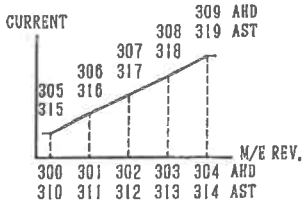
SLIP・RACK CONTROL









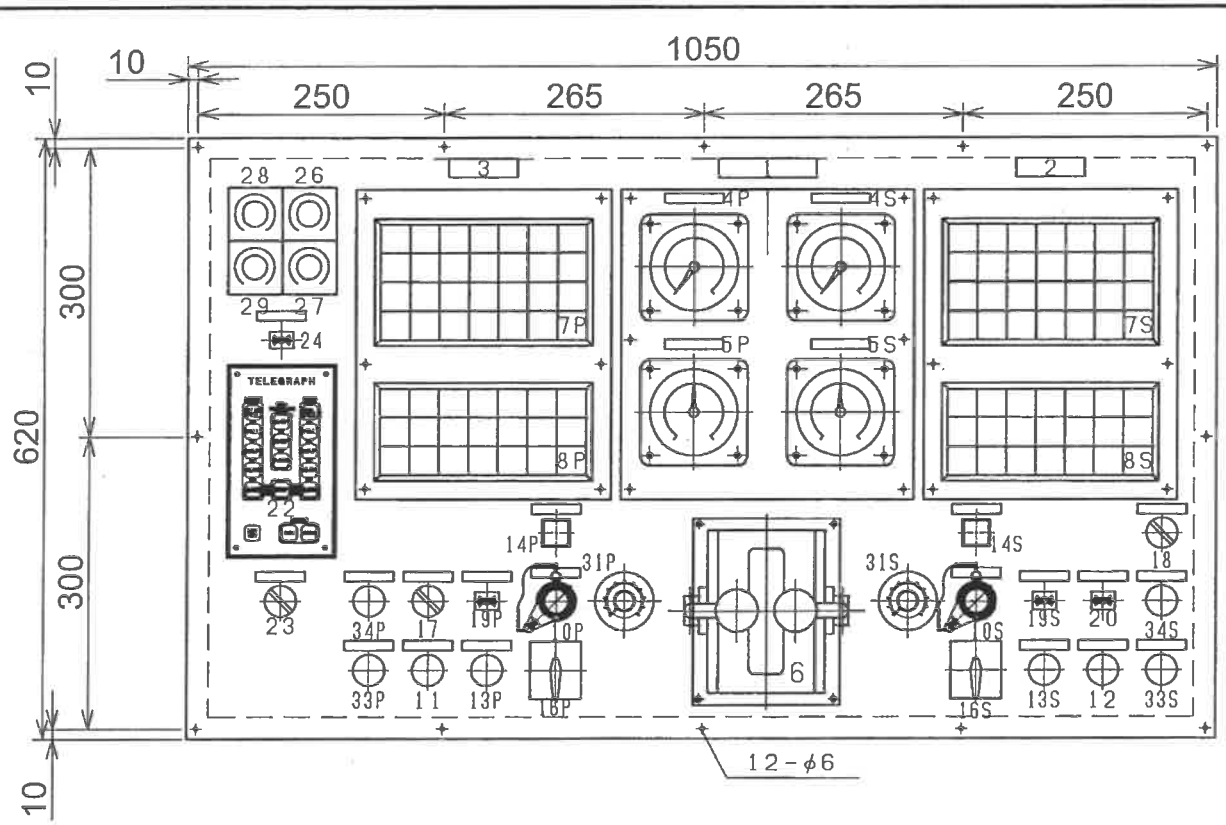


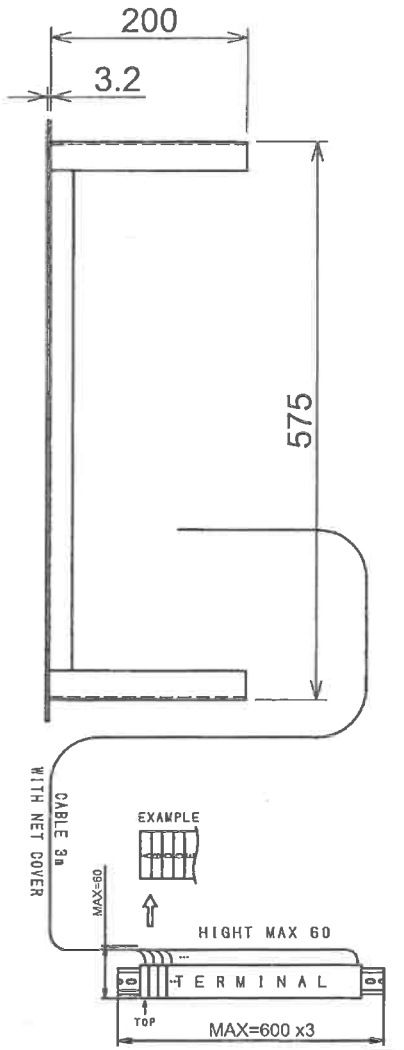
構 成 一 覧 表
COMPONENTS LIST

船 番 SHIP NO.	SZ15045/15046 SZ15047/15048	品 名 NAME OF DEVICE	M/E REMOTE CONTROL SYSTEM	数 量 NO. OF SET	1

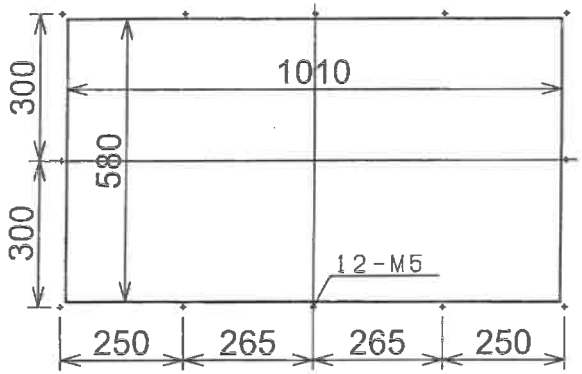
NO.	名 称 NAME OF COMPONENT	個 数 Q'TY	WEIGHT (重量) APPROX.(予定)	WEIGHT (重量) ACTUAL(完成)	図面番号 DWG. NO.	備 考 REMARKS
1	W/H CONTROL PANEL	1			3M1-141136	
2	C/R CONTROL PANEL	1			3M1-141139	
3	M/E RELAY PANEL	1			3M1-141144	
4	CARGO PUMP CONTROL PANEL	1			3M1-141142	
5	CARGO PANEL	1			3M1-141143	
6	REDUCING VALVE PANEL	1			3M1-134959	
7	EP VALVE PANEL	2			3M1-141146	
8	TELEGRAPH BOX	1			3M1-139768	
9	TELEGRAPH GONG	1			3M1-141410	
10	LOCAL BOX	1				
11	SPARE PARTS	1				
12						
13						
14						
15						
16						
17						

1, SEP., 2016





PANEL CUT (1/10)



IP:22

WHP

PAINTING COLOR	WORK NO.		WORK SHOP		W/H CONTROL PANEL
2.5G7/2	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER	CHIEF	CHECK	DRAWN	
1/5				Y. O	DATE 28, JUL., 2016 DWG. NO. 3M1-141136

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

△2 6, OCT., 2016

NO.	名 称 / NAME	記 名 / LETTERING	備 考 / REMARKS	Q'TY
1	NAME PLATE	W/H CONTROL PANEL	20x100	1
2	NAME PLATE	STBD	20x70	1
3	NAME PLATE	PORT	20x70	1
4	INDICATOR	M/E REVOLUTION SPEED	□110 DVF11VL 0~10V / 0~1500min-1	2
5	INDICATOR	PROPELLER REVOLUTION SPEED	□110 DVF11VL -10~0~+10V -10~0~+10V / -400~0~+400min-1	2
6	CONTROL HANDLE		YEGL9D	1
7	INDICATION LAMP		SLC30N-0407(P&S)	2
8	INDICATION LAMP	(ALARM)	SLC30N-0307(P&S)	2
9				
10	PUSH BUTTON	EMERGENCY STOP	ABN8P10	2
11	PUSH BUTTON	LAMP・BUZZER TEST	ABW120B	1
12	PUSH BUTTON	ALARM STOP	ABW111R	1
13	PUSH BUTTON	CONFIRM	ABW110Y	2
14	PUSH BUTTON	SYSTEM RESET	HW2B-M120R / HW9Z-K1(with COVER)	2
15				
16	CAM SWITCH	CONT。MODE NAVI - CARGO	BNSF0-C201 SKK	2
17	SELECT SWITCH	IDL BOOST OFF - ON	ASW220	1
18	SELECT SWITCH	LAMP POWER OFF-ON	ASW220	1
19	BUZZER	CONFIRM	M2BJ-BH24D	2
20	BUZZER	ALARM	M2BJ-BH24D	1
21				
22	TELEGRAPH	(TELEGRAPH TRANSMITTER)		1
23	SELECT SWITCH	TELEGRAPH POWER OFF - ON	ASW220	1
24	BUZZER	TELEGRAPH	M2BJ-BH24D	1
25				
26	DIMMER	INDICATOR	1k25W *2 △2	1
27	DIMMER	HANDLE	1k25W	1
28	DIMMER	INDICATION LAMP	500Ω25W *2	1
29	DIMMER	ALARM LAMP	500Ω25W	1
30				
31	LIMIT DIAL	RACK LIMIT	RA30Y20SB202	2
32				
33	PUSH BUTTON	M/E LO S/B PUMP STOP	ABW111R	2
34	PUSH BUTTON	R/G LO S/B PUMP STOP	ABW111R	2
35				
36				
37				
38				
39				
40				
41				
42				
43				
44				

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. '16.11.7 YH	REV. '16.11.07 K.T		Y. O

W/H CONTROL PANEL
部品表 / PARTS LIST

DATE	15, AUG., 2016	DWG. NO.	4M1-141137

4M1-141137

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

WHP INDICATION

7P

(SPARE) WP3 (W)	PORT AC220V (W)	CONTROL POWER (W)	M/E S/B LO PUMP RUN (G)	CLUTCH AHEAD (G)	NAVI MODE (W)	W/H CONTROL (W)
(SPARE) WP4 (W)	CONTROL MAIN (W)	SAFETY POWER (W)	R/G S/B LO PUMP RUN (G)	CLUTCH NEUTRAL (W)	CARGO MODE (W)	C/P CONTROL (W)
(SPARE) WP5 (W)	CONTROL BACK UP (W)	BATTERY DC24V (W)	IDLE BOOST CONTROL ON (W)	CLUTCH ASTERN (R)	M/E RUN (G)	C/R CONTROL (W)
(SPARE) WP6 (W)	(SPARE) WP2 (W)	(SPARE) WP7 (W)	IDLE BOOST CONTROL OPERATE (G)	CLUTCH ASTERN READY (W)	(SPARE) WP1 (W)	E/R CONTROL (W)

7S

W/H CONTROL (W)	NAVI MODE (W)	CLUTCH AHEAD (G)	M/E S/B LO PUMP RUN (G)	CONTROL POWER (W)	STBD AC220V (W)	(SPARE) WS3 (W)
C/P CONTROL (W)	CARGO MODE (W)	CLUTCH NEUTRAL (W)	R/G S/B LO PUMP RUN (G)	SAFETY POWER (W)	CONTROL MAIN (W)	(SPARE) WS4 (W)
C/R CONTROL (W)	M/E RUN (G)	CLUTCH ASTERN (R)	IDLE BOOST CONTROL ON (W)	ALARM POWER (W)	CONTROL BACK UP (W)	(SPARE) WS5 (W)
E/R CONTROL (W)	(SPARE) WS1 (W)	CLUTCH ASTERN READY (W)	IDLE BOOST CONTROL OPERATE (G)	ALARM AC220V (W)	(SPARE) WS2 (W)	(SPARE) WS6 (W)

8P

(SPARE) WAP6 (R)	OTHER ABNORMAL (R)	M/E S/B L.O. PUMP OVERLAP RUN (R)	M/E ABNORMAL (R)	M/E F.W. LOW PRESS SLOWDOWN REQUEST (R)	M/E L.O. LOW PRESS SHUTDOWN (R)	MANUAL EMERGENCY STOP (R)
(SPARE) WAP7 (R)	(SPARE) WAP9 (R)	R/G S/B L.O. PUMP OVERLAP RUN (R)	POWER SUPPLY ABNORMAL (R)	M/E F.W. HI-TEMP SLOWDOWN REQUEST (R)	R/G L.O. LOW PRESS SHUTDOWN (R)	M/E OVERSPEED SHUTDOWN (R)
(SPARE) WAP8 (R)	(SPARE) WAP5 (R)	M/E L.O. LOW PRESS (R)	CONTROL AIR LOW PRESS (R)	SHAFT BEARING HI-TEMP SLOWDOWN REQUEST (R)	R/G L.O. HI-TEMP SLOWDOWN REQUEST (R)	(SPARE) WAP1 (R)

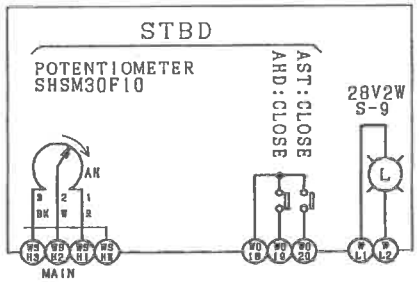
8S

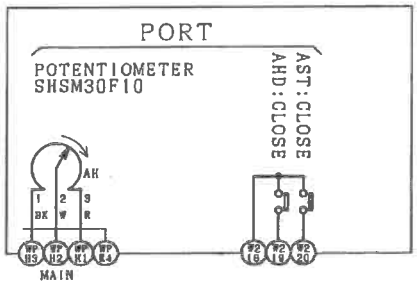
MANUAL EMERGENCY STOP (R)	M/E L.O. LOW PRESS SHUTDOWN (R)	M/E F.W. LOW PRESS SLOWDOWN REQUEST (R)	M/E ABNORMAL (R)	M/E S/B L.O. PUMP OVERLAP RUN (R)	ALARM POWER SUPPLY ABNORMAL (R)	(SPARE) WAS6 (R)
M/E OVERSPEED SHUTDOWN (R)	R/G L.O. LOW PRESS SHUTDOWN (R)	M/E F.W. HI-TEMP SLOWDOWN REQUEST (R)	POWER SUPPLY ABNORMAL (R)	R/G S/B L.O. PUMP OVERLAP RUN (R)	ALARM UNIT ABNORMAL (R)	(SPARE) WAS7 (R)
(SPARE) WAS1 (R)	R/G L.O. HI-TEMP SLOWDOWN REQUEST (R)	SHAFT BEARING HI-TEMP SLOWDOWN REQUEST (R)	CONTROL AIR LOW PRESS (R)	M/E L.O. LOW PRESS (R)	(SPARE) WAS5 (R)	(SPARE) WAS8 (R)

PAINTING COLOR	WORK NO.	WORK SHOP		W/H CONTROL PANEL INDICATION	
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER	CHIEF	CHECK	DRAWN	
	6.11.7			Y. O	DATE 1, SEP., 2016 / DWG. NO. 3M1-141138

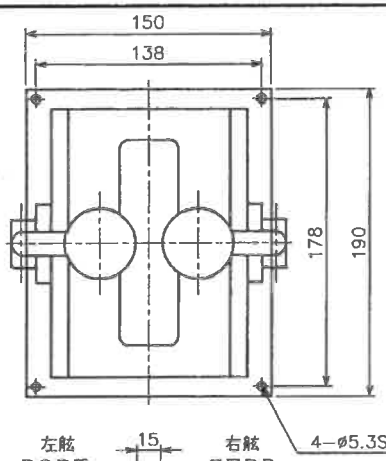
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

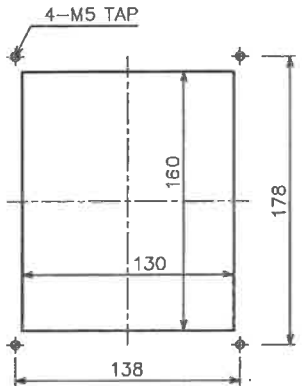
TS-052

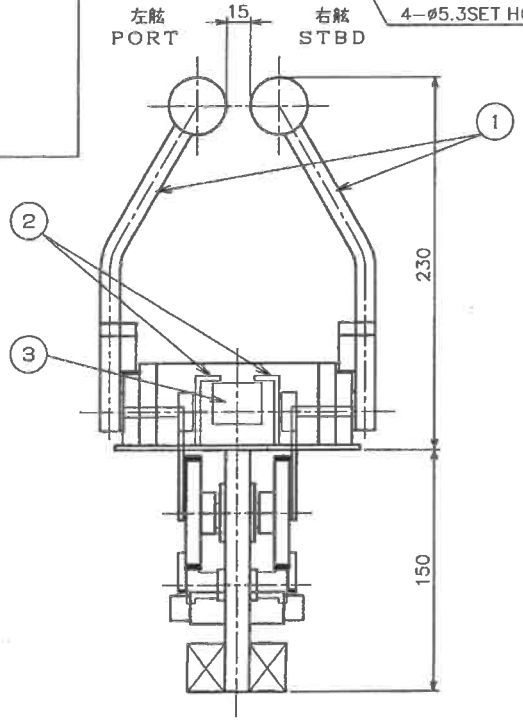


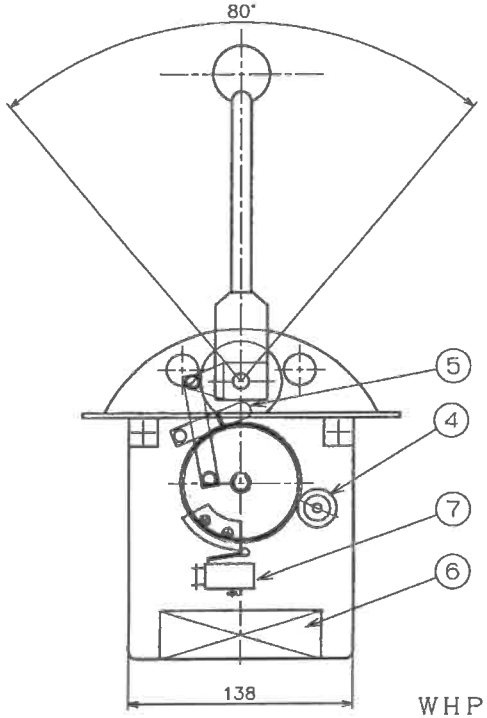


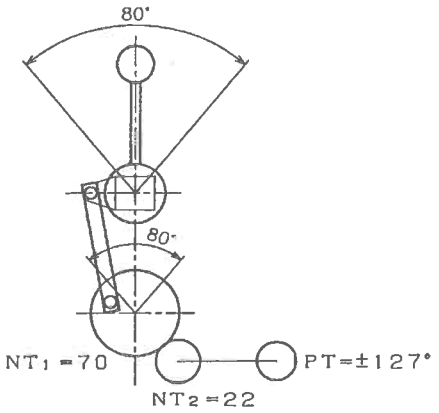
BK:BLACK
R :RED
G :GREEN
W :WHITE











$PT=\pm 40^\circ \times \frac{70}{22}=\pm 127^\circ$

No.	NAME OF PARTS	REQ.	REMARKS
1	レバー LEVER	2	
2	指針 POINTER	2	
3	文字板 DIAL PLATE	1	
4	ポテンショメータ POTENTIOMETER	2	SHSM30F10 -
5	電球 LAMP	1	28V 2W S-9
6	端子 TERMINAL	2	TS615 14P
7	リミットスイッチ LIMIT SWITCH	4	D2VW-5L2A-1M

P 10-1

WHP

TYPE	YEGL9D

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

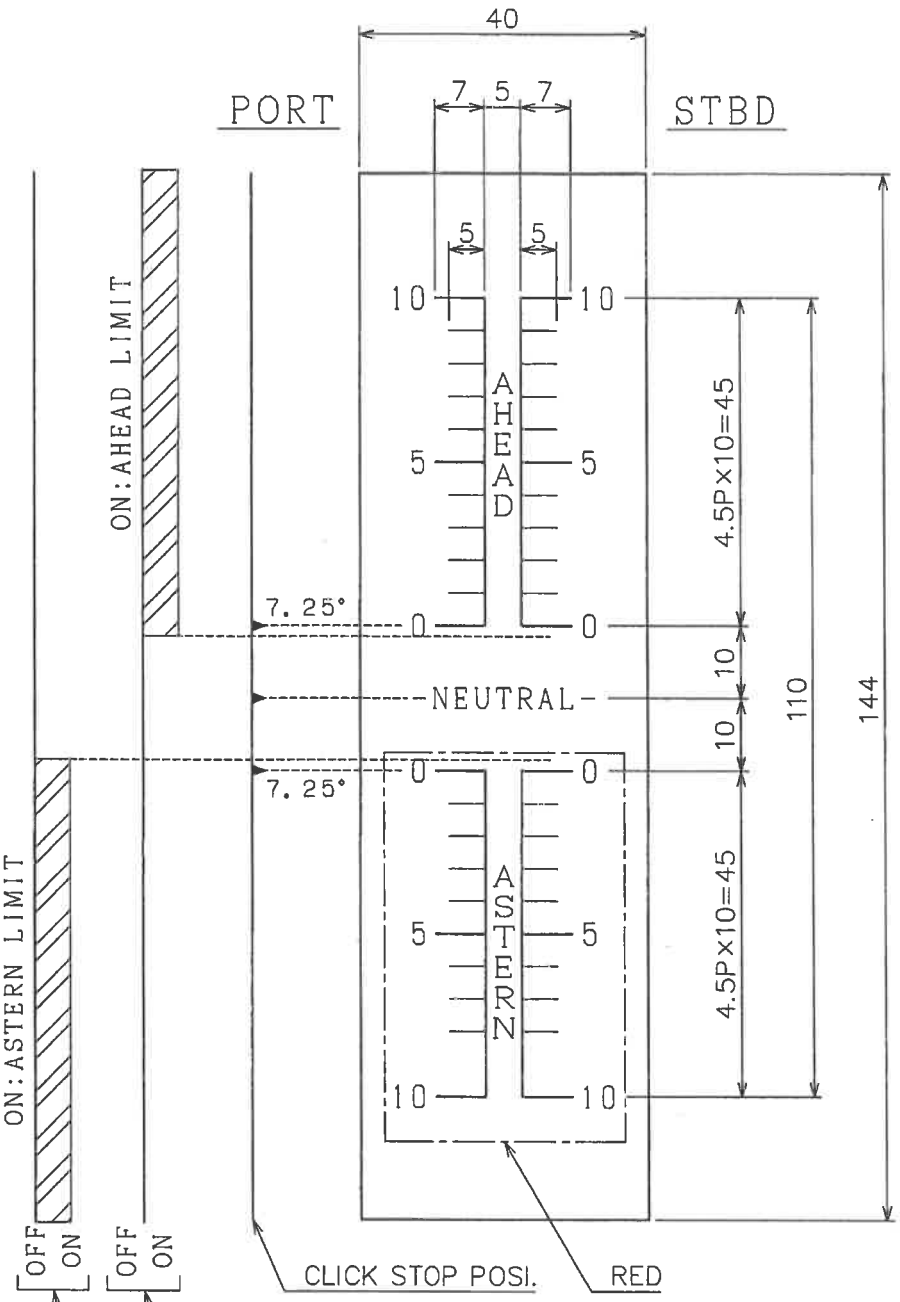
SCALE	MANAGER	CHIEF	CHECK	DRAWN
1/3	6.11.7	14-7		Y.O

主機関遠隔操縦装置 操縦ハンドル
M/E CONTROL HANDLE

DATE 18, JUL, 2014 | DWG. NO. 3M1-139304

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M1-139304



MATERIAL

SHEET RESIN:1t

SCULPTURE:UNDER SURFACE

GROUND:WHITE

LETTERING&LINE:AHEAD, NEUTRAL:BLACK

:ASTERN:RED

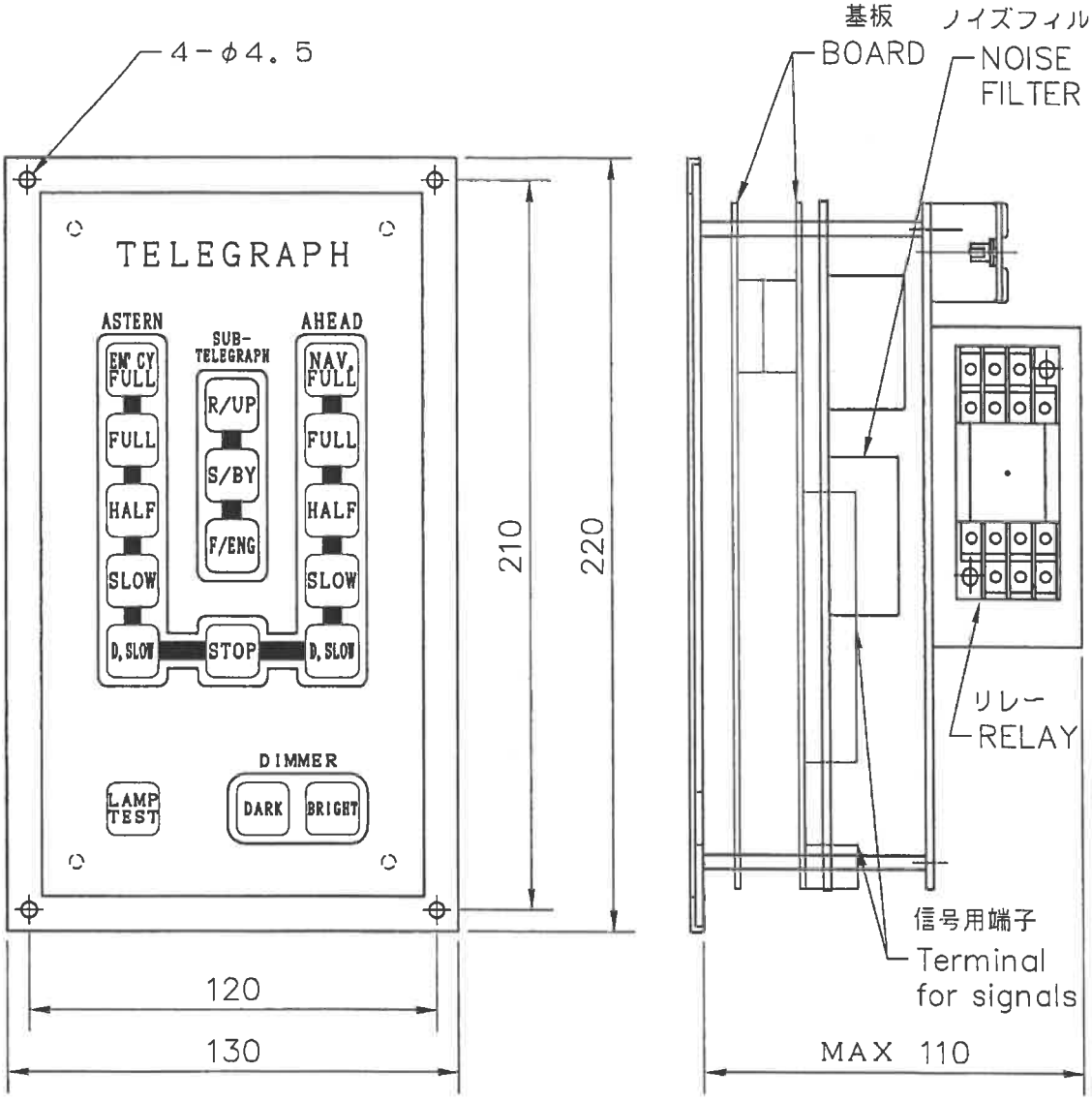
FOR WHP HANDLE

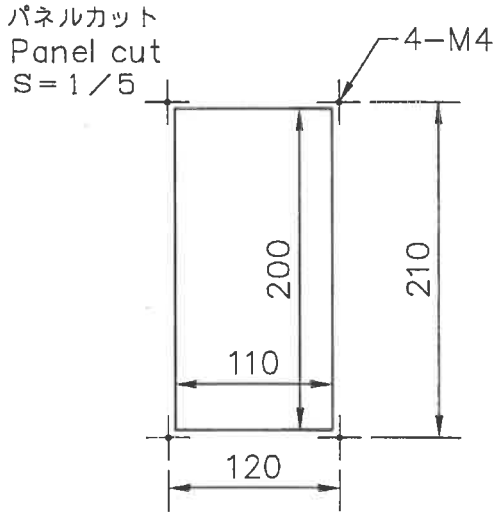
TYPE	YEGL-9

PAINTING COLOR	WORK NO.		WORK SHOP	
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER	CHIEF	CHECK	DRAWN
1 / 1	REV. '16.11.7 Y.H	REV. '16.11.07 K.T		K.O

DIAL FOR CONTROL HANDLE

DATE	6, OCT., 2010	DWG. NO.	4M1-113392A





LUMINOUS COLOR

"AHEAD" SIDE: GREEN

"ASTERN" SIDE: RED

"STOP": YELLOW

"SUB TELE": WHITE

発光色

AHEAD 側：緑色

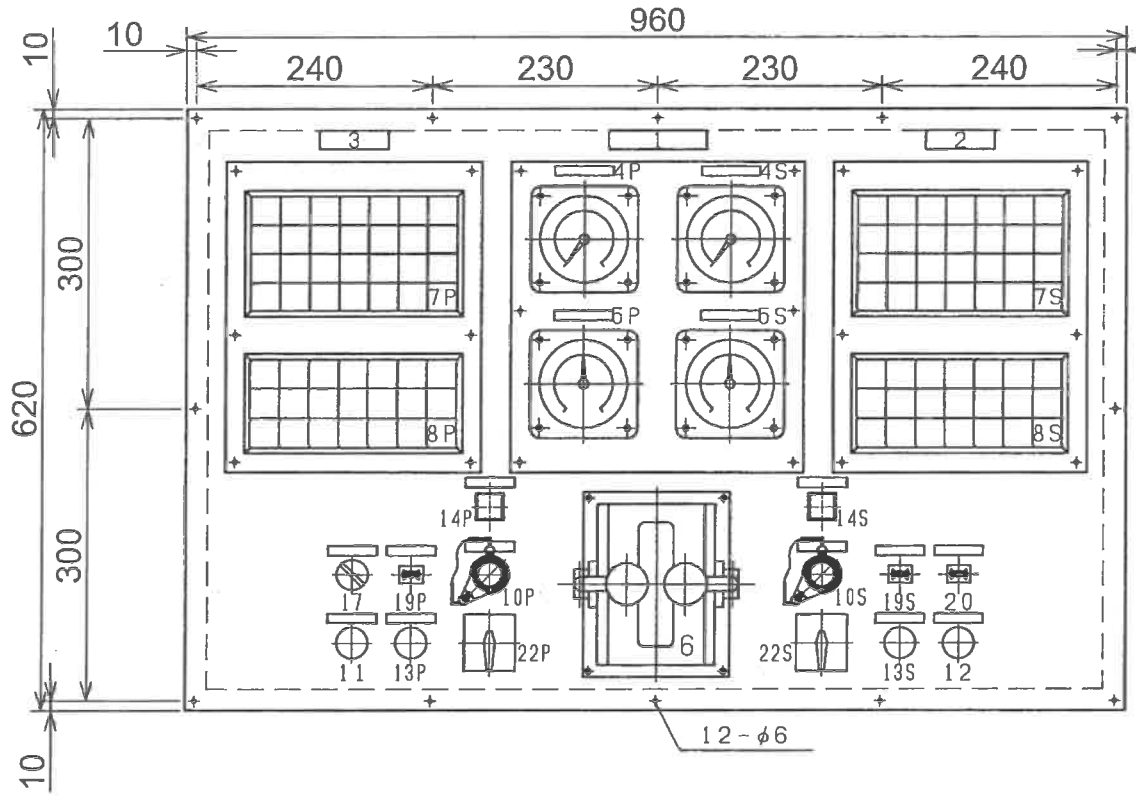
ASTERN 側：赤色

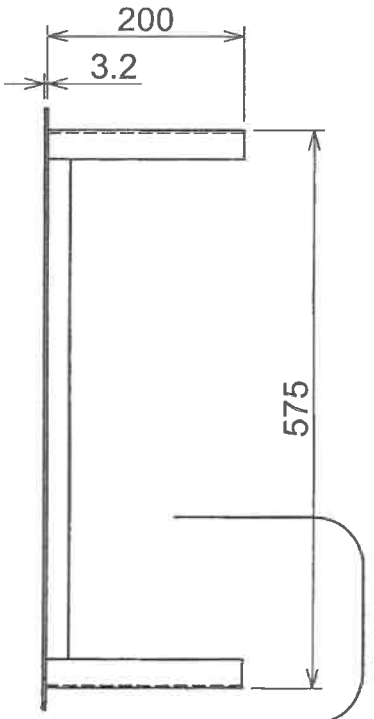
STOP ：黄色

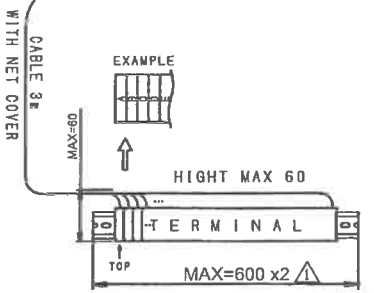
サブテレ ：白色

FOR W/H

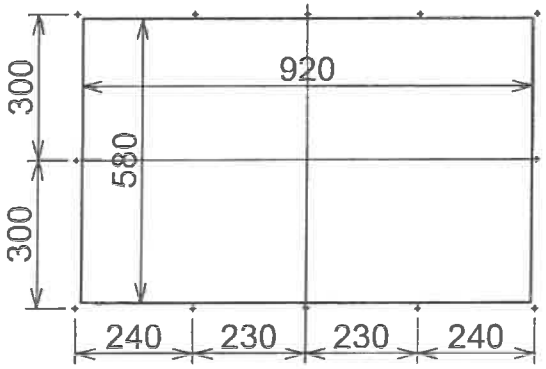
PAINTING COLOR	WORK NO.	WORK SHOP	エンジンテレグラフ発信器 ENGINE TELEGRAPH TRNSMITTER			
2.5G7/2						
SCALE	MANAGER	CHIEF	CHECK	DRAWN		
1/2	REV. '16.8.22 Y.H	REV. '16.8.22 K.T	REV. '16.8.22 T.M	Fukuoka	DATE 26-JULY-2012	DWG. NO. 4M1-137781







PANEL CUT (1/10)



IP:22

ECP

P 11

PAINTING COLOR	WORK NO.	WORK SHOP			C/R CONTROL PANEL		
2.5G7/2	SZ15045/15046 SZ15047/15048						
SCALE	MANAGER	CHIEF	CHECK	DRAWN			
1/5	16.11.7			Y. O	DATE 1, SEP., 2016	DWG. NO.	3M1-141139

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

⚠ 20, SEP., 2016

NO.	名 称 / NAME	記 名 / LETTERING	備 考 / REMARKS	Q'TY
1	NAME PLATE	C/R CONTROL PANEL	20x100	1
2	NAME PLATE	STBD	20x70	1
3	NAME PLATE	PORT	20x70	1
4	INDICATOR	M/E REVOLUTION SPEED	□110 DVF11VL 0~10V / 0~1500min-1	2
5	INDICATOR	PROPELLER REVOLUTION SPEED	□110 DVF11VL -10~0~+10V -10~0~+10V / -400~0~+400min-1	2
6	CONTROL HANDLE		YEGL9D	1
7	INDICATION LAMP		SLC30N-0407(P&S)	2
8	INDICATION LAMP	(ALARM)	SLC30N-0307(P&S)	2
9				
10	PUSH BUTTON	EMERGENCY STOP	ABN8P10	2
11	PUSH BUTTON	LAMP・BUZZER TEST	ABW120B	1
12	PUSH BUTTON	ALARM STOP	ABW111R	1
13	PUSH BUTTON	CONFIRM	ABW110Y	2
14	PUSH BUTTON	SYSTEM RESET	HW2B-M120R / HW9Z-K1(with COVER)	2
15				
16				
17	SELECT SWITCH	IDL BOOST OFF - ON	ASW220	1
18				
19	BUZZER	CONFIRM	M2BJ-BH24D	2
20	BUZZER	ALARM	M2BJ-BH24D	1
21				
22	CAM SWITCH	CONT. POSITION C/R - W/H	BNSF0-C201 SKK	2
23				
24				
25				
26				
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28				
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4M1-141140

PAINTING COLOR	WORK NO.		WORK SHOP	
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. '16.11.7 Y.H	REV. '16.11.07 K.T		Y. O

C/R CONTROL PANEL
部品表 / PARTS LIST

DATE	1, SEP., 2016	DWG. NO.	4M1-141140

ECP INDICATION

7P

(SPARE) EP6 (W)	(SPARE) EP2 (W)	PORT AC220V (W)	CONTROL POWER (W)	M/E S/B LO PUMP RUN (G)	CLUTCH AHEAD (G)	W/H CONTROL (W)
(SPARE) EP7 (W)	(SPARE) EP3 (W)	CONTROL MAIN (W)	SAFETY POWER (W)	R/G S/B LO PUMP RUN (G)	CLUTCH NEUTRAL (W)	C/P CONTROL (W)
(SPARE) EP8 (W)	(SPARE) EP4 (W)	CONTROL BACK UP (W)	BATTERY DC24V (W)	IDLE BOOST CONTROL ON (W)	CLUTCH ASTERN (R)	C/R CONTROL (W)
(SPARE) EP9 (W)	(SPARE) EP5 (W)	M/E RUN (G)	(SPARE) EP10 (W)	IDLE BOOST CONTROL OPERATE (G)	CLUTCH ASTERN READY (W)	E/R CONTROL (W)

7S

W/H CONTROL (W)	CLUTCH AHEAD (G)	M/E S/B LO PUMP RUN (G)	CONTROL POWER (W)	STBD AC220V (W)	(SPARE) ES2 (W)	(SPARE) ES6 (W)
C/P CONTROL (W)	CLUTCH NEUTRAL (W)	R/G S/B LO PUMP RUN (G)	SAFETY POWER (W)	CONTROL MAIN (W)	(SPARE) ES3 (W)	(SPARE) ES7 (W)
C/R CONTROL (W)	CLUTCH ASTERN (R)	IDLE BOOST CONTROL ON (W)	ALARM POWER (W)	CONTROL BACK UP (W)	(SPARE) ES4 (W)	(SPARE) ES8 (W)
E/R CONTROL (W)	CLUTCH ASTERN READY (W)	IDLE BOOST CONTROL OPERATE (G)	ALARM AC220V (W)	M/E RUN (G)	(SPARE) ES5 (W)	(SPARE) ES9 (W)

8P

(SPARE) EAP6 (R)	OTHER ABNORMAL (R)	M/E S/B L.O. PUMP OVERLAP RUN (R)	M/E ABNORMAL (R)	M/E F.W. LOW PRESS SLOWDOWN REQUEST (R)	M/E L.O. LOW PRESS SHUTDOWN (R)	MANUAL EMERGENCY STOP (R)
(SPARE) EAP7 (R)	(SPARE) EAP9 (R)	R/G S/B L.O. PUMP OVERLAP RUN (R)	POWER SUPPLY ABNORMAL (R)	M/E F.W. HI-TEMP SLOWDOWN REQUEST (R)	R/G L.O. LOW PRESS SHUTDOWN (R)	M/E OVERSPEED SHUTDOWN (R)
(SPARE) EAP8 (R)	(SPARE) EAP5 (R)	M/E L.O. LOW PRESS (R)	CONTROL AIR LOW PRESS (R)	SHAFT BEARING HI-TEMP SLOWDOWN REQUEST (R)	R/G L.O. HI-TEMP SLOWDOWN REQUEST (R)	(SPARE) EAP1 (R)

8S

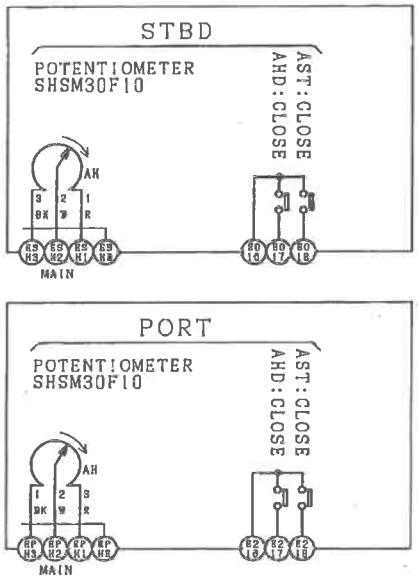
MANUAL EMERGENCY STOP (R)	M/E L.O. LOW PRESS SHUTDOWN (R)	M/E F.W. LOW PRESS SLOWDOWN REQUEST (R)	M/E ABNORMAL (R)	M/E S/B L.O. PUMP OVERLAP RUN (R)	ALARM POWER SUPPLY ABNORMAL (R)	(SPARE) EAS6 (R)
M/E OVERSPEED SHUTDOWN (R)	R/G L.O. LOW PRESS SHUTDOWN (R)	M/E F.W. HI-TEMP SLOWDOWN REQUEST (R)	POWER SUPPLY ABNORMAL (R)	R/G S/B L.O. PUMP OVERLAP RUN (R)	ALARM UNIT ABNORMAL (R)	(SPARE) EAS7 (R)
(SPARE) EAS1 (R)	R/G L.O. HI-TEMP SLOWDOWN REQUEST (R)	SHAFT BEARING HI-TEMP SLOWDOWN REQUEST (R)	CONTROL AIR LOW PRESS (R)	M/E L.O. LOW PRESS (R)	(SPARE) EAS5 (R)	(SPARE) EAS8 (R)

P 13

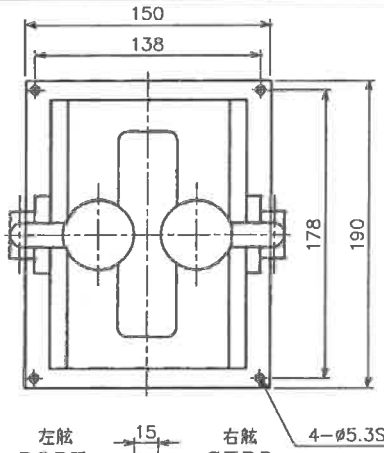
PAINTING COLOR	WORK NO.	WORK SHOP		C/R CONTROL PANEL INDICATION		
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	DWG. NO.
	REV. '16.11.7			Y. O	1, SEP., 2016	3M1-141141

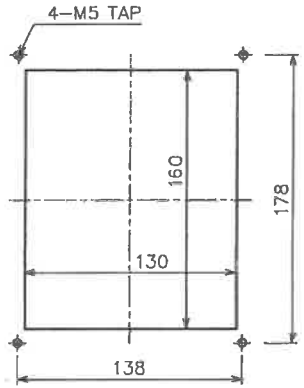
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

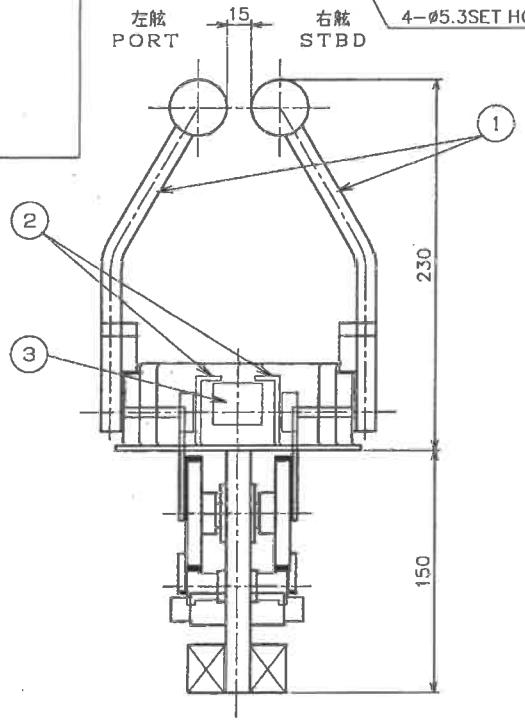
TS-052

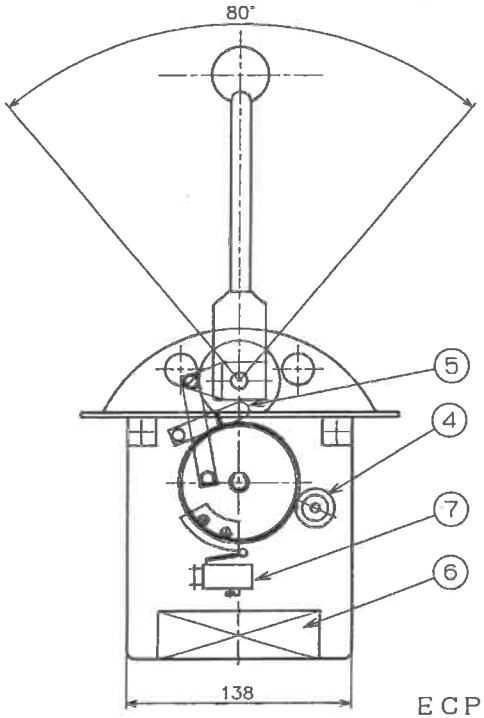


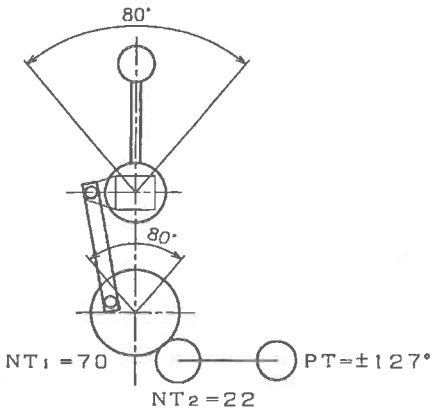
BK:BLACK
R :RED
G :GREEN
W :WHITE











$PT=\pm 40^{\circ}\times\frac{70}{22}=\pm 127^{\circ}$

No.	NAME OF PARTS	REQ.	REMARKS
1	レバー LEVER	2	
2	指針 POINTER	2	
3	文字板 DIAL PLATE	1	
4	ポテンショメータ POTENTIOMETER	2	SHSM30F10 -
5	電球 LAMP	1	28V 2W S-9
6	端子 TERMINAL	2	TS615 14P
7	リミットスイッチ LIMIT SWITCH	4	D2VW-5L2A-1M

P 14-1

ECP

TYPE	YEGL9D

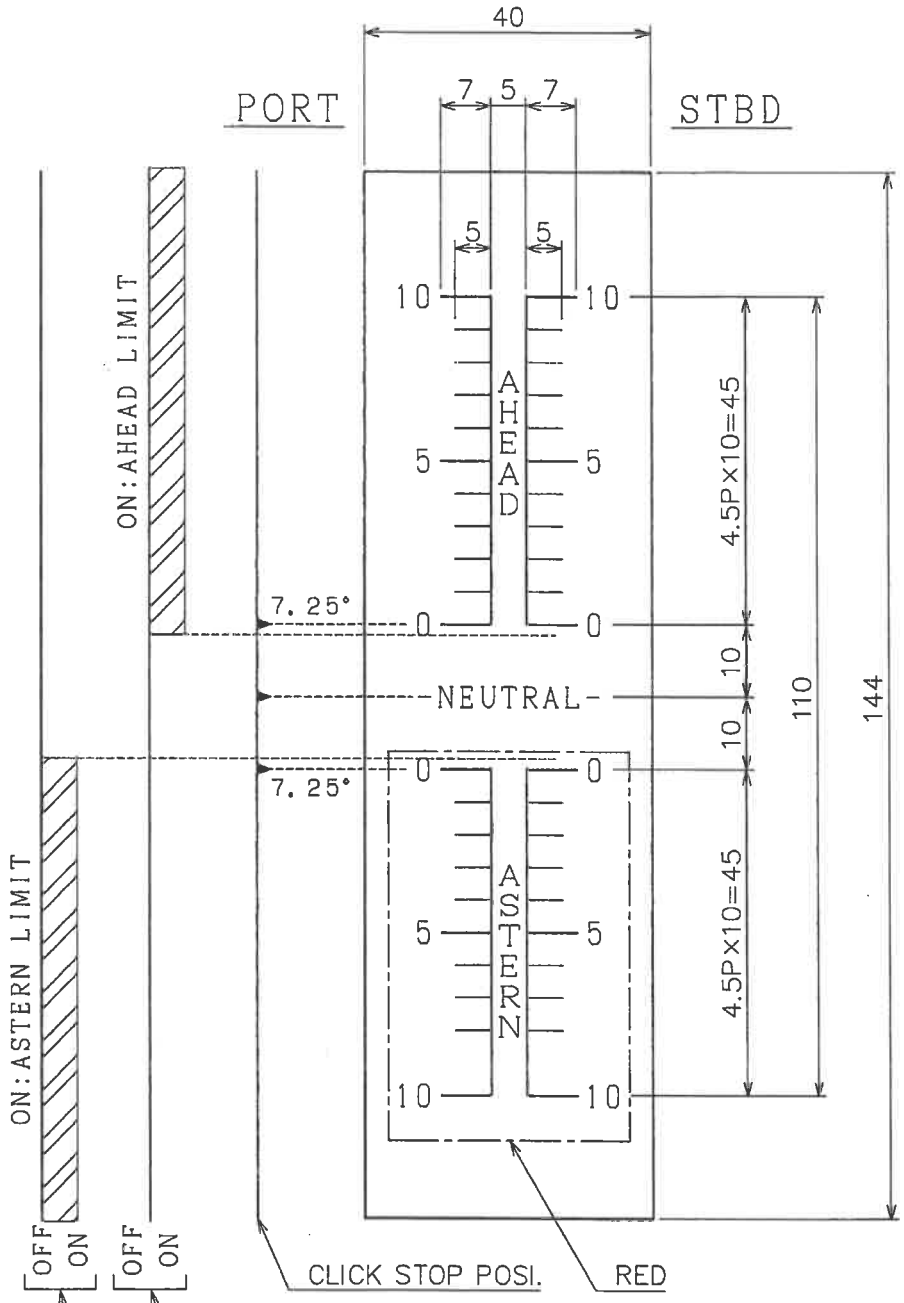
PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER	CHIEF	CHECK	DRAWN
1/3	14.11.7			Y.O

主機関遠隔操縦装置 操縦ハンドル
M/E CONTROL HANDLE

DATE 24, AUG, 2014 DWG. NO. 3M1-139304

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M1-139304



MATERIAL

SHEET RESIN:1t

SCULPTURE:UNDER SURFACE

GROUND:WHITE

LETTERING&LINE:AHEAD, NEUTRAL:BLACK

:ASTERN:RED

FOR ECP HANDLE

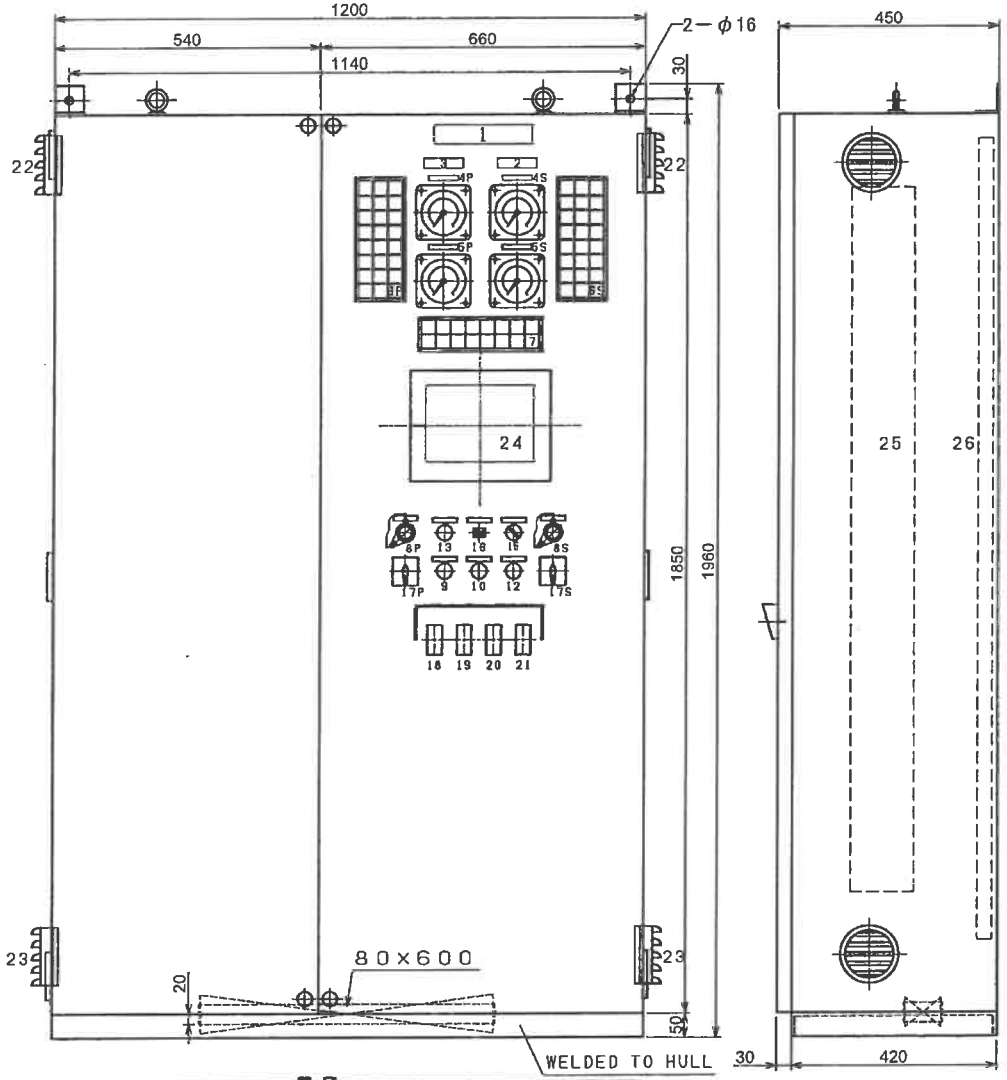
TYPE	YEGL-9

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
1 / 1		REV. '16.11.07 K.T		K.O

DIAL FOR CONTROL HANDLE

DATE	6, OCT., 2010	DWG. NO.	4M1-113392A



NO.	NAME	LETTERING	REMARKS	Q'TY
1	NAME PLATE	M/E RELAY PANEL		1
2	NAME PLATE	STBD	20x70	1
3	NAME PLATE	PORT	20x70	1
4	INDICATOR	M/E REVOLUTION SPEED	DVF-11V 0~10V/0~1500min-1	2
5	INDICATOR	T/C REVOLUTION SPEED	DVF-11V 4~20mA/0~60000min-1	2
6	INDICATION LAMP		SLC30N-0803	2
7	ALARM INDICATION LAMP		SLC30N-0208	1
8	PUSH BUTTON	EMERGENCY STOP	ABN8P10	2
9	PUSH BUTTON	LAMP·BUZZER TEST	ABW120B	1
10	PUSH BUTTON	ALARM STOP	ABW111R	1
11				
12	PUSH BUTTON	FLICKER STOP	ABW111Y	1
13	PUSH BUTTON	MONITOR TEST	ABW111B	1
14				
15	SELECT SWITCH	HEATING OFF - ON	ASW220	1
16	BUZZER	ALARM	M2BJ-BH24D	1
17	CAM SWITCH	CONT. POSITION LOCAL - REMOTE	BNSF0-C201 SKK	2
18	POWER SWITCH	DC24V	S125-SF 2P FC 75A	1
19	POWER SWITCH	PORT AC220V	E50-SF 2P FC 10A	1
20	POWER SWITCH	STBD AC220V	E50-SF 2P FC 10A	1
21	POWER SWITCH	ALARM AC220V	E50-SF 2P FC 10A	1
22	LOUVER with FAN		WLP-12K-2	2
23	LOUVER		WLP-12	2
24	LCD		10.4 inch	1
25	RELAY UNIT			1
26	TERMINAL BOARD			1

IP:22

ERP

Please be put to out of water-spray or water-mist nozzle area.

P 15

PAINTING COLOR	WORK NO.	WORK SHOP	M/E RELAY PANEL	
2.5G7/2	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE	DWG. NO.
1/10		Y. O	1, SEP., 2016	3M1-141144

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

ERP INDICATION

6P

CONTROL POWER (W)	CLUTCH AHEAD (G)	W/H CONTROL (W)
SAFETY POWER (W)	CLUTCH NEUTRAL (W)	C/P CONTROL (W)
PORT AC220V (W)	CLUTCH ASTERN (R)	C/R CONTROL (W)
BATTERY DC24V (W)	CONTROL MAIN (W)	E/R CONTROL (W)
(SPARE) ERP6 (W)	CONTROL BACK UP (W)	TURNING ENGAGE (R)
HEATING CONTROL ON (W)	AIR RUN /START READY (G)	TURNING DISENGAGE (W)
HEATING CONTROL OPERATE (G)	(SPARE) ERP3 (W)	(SPARE) ERP1 (W)
(SPARE) ERP5 (W)	(SPARE) ERP4 (W)	(SPARE) ERP2 (W)

6S

W/H CONTROL (W)	CLUTCH AHEAD (G)	CONTROL POWER (W)
C/P CONTROL (W)	CLUTCH NEUTRAL (W)	SAFETY POWER (W)
C/R CONTROL (W)	CLUTCH ASTERN (R)	STBD AC220V (W)
E/R CONTROL (W)	CONTROL MAIN (W)	ALARM POWER (W)
TURNING ENGAGE (R)	CONTROL BACK UP (W)	ALARM AC220V (W)
TURNING DISENGAGE (W)	AIR RUN /START READY (G)	HEATING CONTROL ON (W)
(SPARE) ERS1 (W)	(SPARE) ERS3 (W)	HEATING CONTROL OPERATE (G)
(SPARE) ERS2 (W)	(SPARE) ERS4 (W)	(SPARE) ERS5 (W)

P 16

PAINTING COLOR	WORK NO.	WORK SHOP	M/E RELAY PANEL INDICATION	
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER / CHIEF	CHECK / DRAWN		
		Y. O	DATE 1, SEP., 2016	DWG. NO. 3M1-141145

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

ERP ALARM INDICATION

7

(SPARE) ERAP2 Ⓡ	(SPARE) ERAP4 Ⓡ	OTHER ABNORMAL Ⓡ	MANUAL EMERGENCY STOP Ⓡ	MANUAL EMERGENCY STOP Ⓡ	ALARM POWER SUPPLY ABNORMAL Ⓡ	ALARM UNIT ABNORMAL Ⓡ	(SPARE) ERAS2 Ⓡ
(SPARE) ERAP3 Ⓡ	SLOWDOWN REQUEST Ⓡ	M/E ABNORMAL Ⓡ	M/E AUTO SHUTDOWN Ⓡ	M/E AUTO SHUTDOWN Ⓡ	M/E ABNORMAL Ⓡ	SLOWDOWN REQUEST Ⓡ	(SPARE) ERAS3 Ⓡ

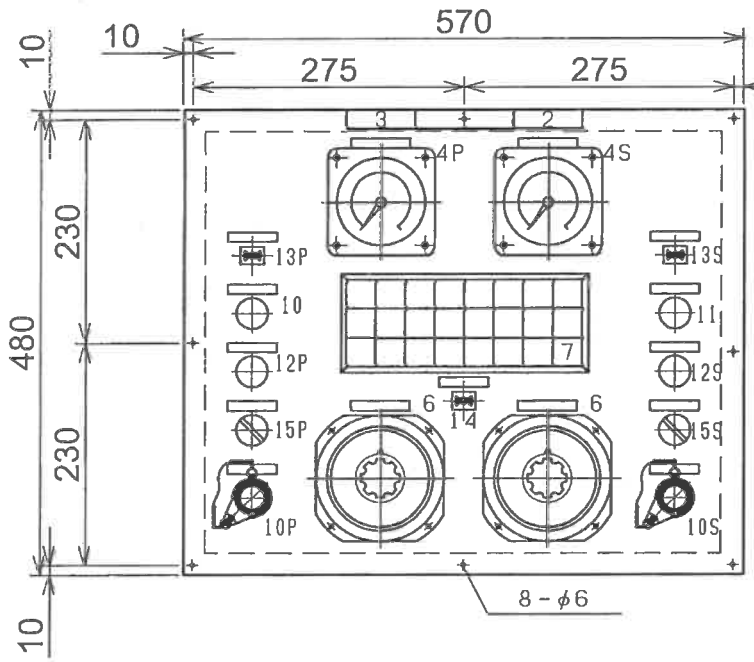
P 17

PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK	DRAWN
			Y. O

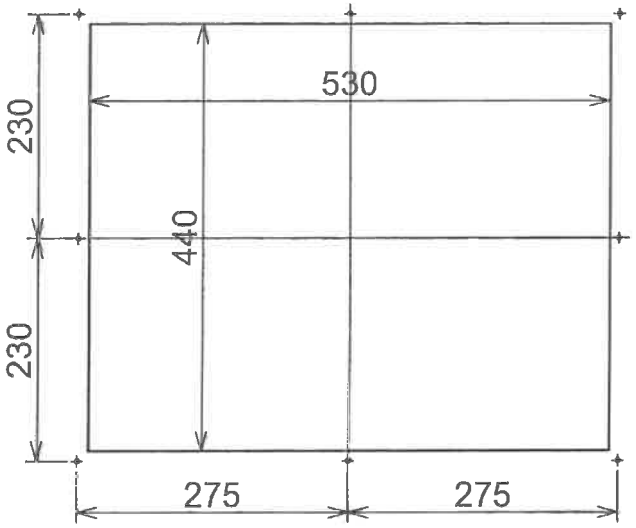
M/E RELAY PANEL
ALARM INDICATION

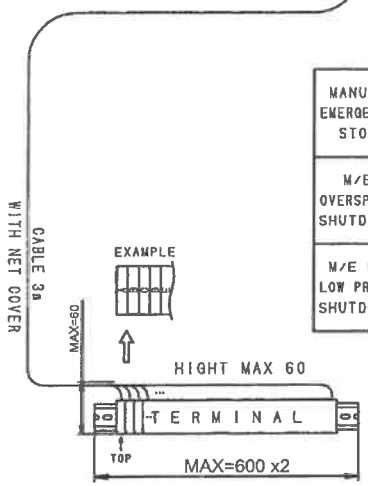
DATE 1, SEP., 2016 DWG. NO. 3MI-141415

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PANEL CUT





NO.	名 称 / NAME	記 名 / LETTERING	備 考 / REMARKS	Q'TY
1	NAME PLATE	CARGO PUMP CONTROL PANEL	20x100	1
2	NAME PLATE	STBD	20x70	1
3	NAME PLATE	PORT	20x70	1
4	INDICATOR	M/E REVOLUTION SPEED	DI10 DVF-11 0-10V/0~1500min-1	2
5				
6	DIAL	M/E CONTROL DIAL	DL-11 DI30	2
7	INDICATION LAMP		SLC30N-0308	1
8				
9	PUSH BUTTON	EMERGENCY STOP	ABNBP10	2
10	PUSH BUTTON	LAMP·BUZZER TEST	ABW120B	1
11	PUSH BUTTON	ALARM STOP	ABW111R	1
12	PUSH BUTTON	CONFIRM	ABW11DY	2
13	BUZZER	CONFIRM	M20J-BH24D	2
14	BUZZER	ALARM	M20J-BH24D	1
15	SELECT SWITCH	CONT.POSITION CPCP - CP	CG4-1 A721-600 FS1	2
16				

7

MANUAL EMERGENCY STOP ®	R/G L.O. LOW PRESS SHUTDOWN ®	NAVI MODE ®	CP CONTROL ®	CP CONTROL ®	NAVI MODE ®	R/G L.O. LOW PRESS SHUTDOWN ®	MANUAL EMERGENCY STOP ®
M/E OVERSPEED SHUTDOWN ®	(SPARE) CPP1 ®	CARGO MODE ®	CPCP CONTROL ®	CPCP CONTROL ®	CARGO MODE ®	(SPARE) CPS1 ®	M/E OVERSPEED SHUTDOWN ®
M/E LO LOW PRESS SHUTDOWN ®	(SPARE) CPPA1 ®	DIAL ZERO POSITION ®	M/E RUN ®	M/E RUN ®	DIAL ZERO POSITION ®	ALARM UNIT ABNORMAL ®	M/E LO LOW PRESS SHUTDOWN ®

IP:22

CPCP

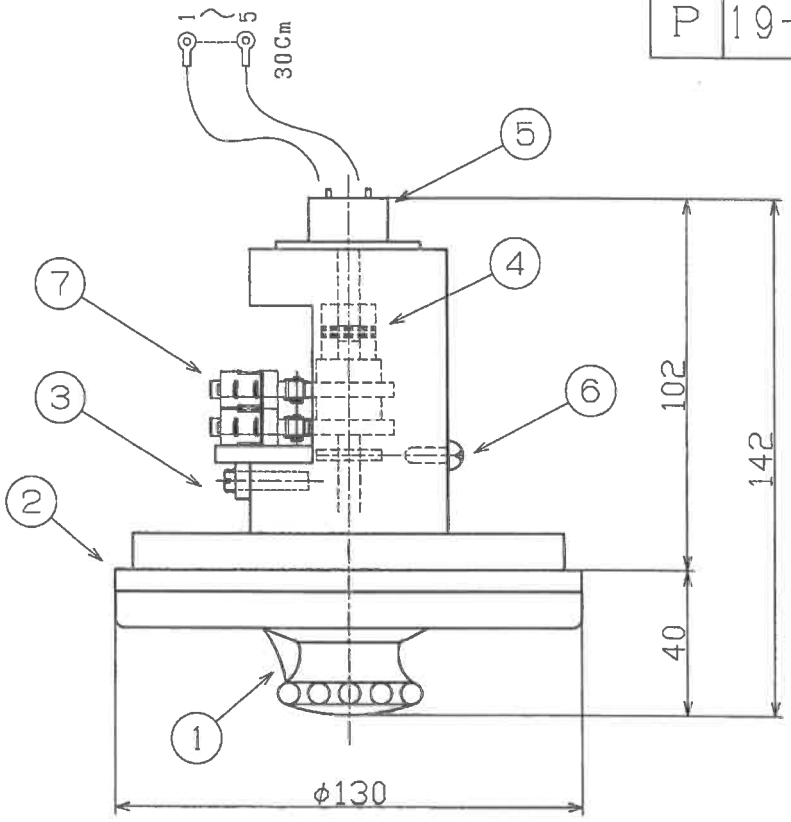
P 18

PAINTING COLOR	WORK NO.	WORK SHOP	CARGO PUMP CONTROL PANEL	
2.5G7/2	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE	DWG. NO.
1/5		Y. O	7, SEP., 2016	3M1-141142

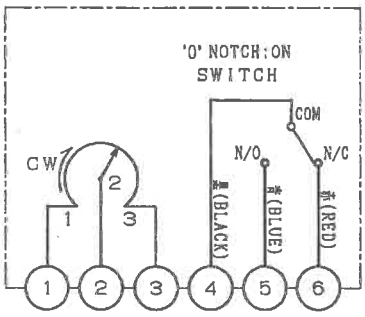
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 19-1

No.	NAME 名称	REQ.No. 個数	MATERIAL 材質	REMARKS 備考
1	KNOB ツマミ	1		JRM 50Nϕ8
2	BODY 本体	1	AC7A-F	
3	BRAKE ブレーキ	1		ADJUSTABLE 調整可能
4	COUPLING 継手	1	BSBM2	
5	POTENTIOMETER ポテンショメーター	1		SHSM30F10
6	STOPPER OF REV. ANGLE 回転角度設定器	2		DIAL ADJUST. RANGE 210°~350° ダイヤル設定範囲 210°~350°
7	MICRO SWITCH マイクロスイッチ	1		V-55-1A3
8				

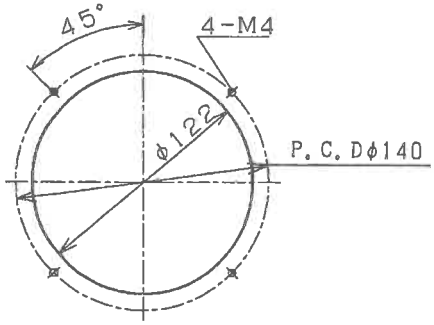


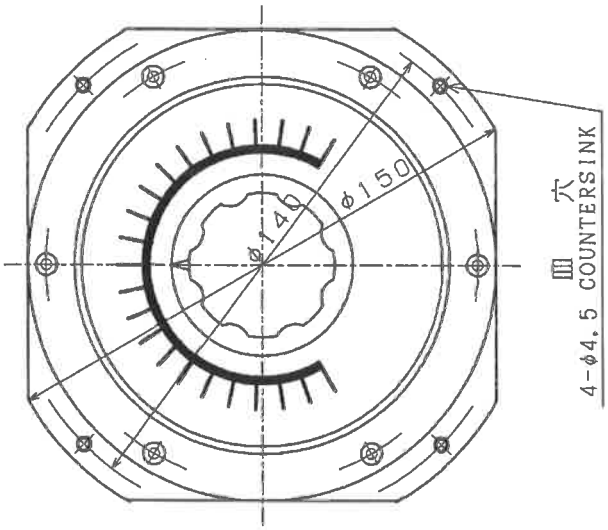
CONNECT. DIAGRAM 接続図



PANEL CUT (パネルカット)

SCALE 1/4





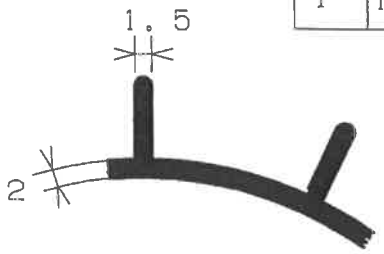
CPCP

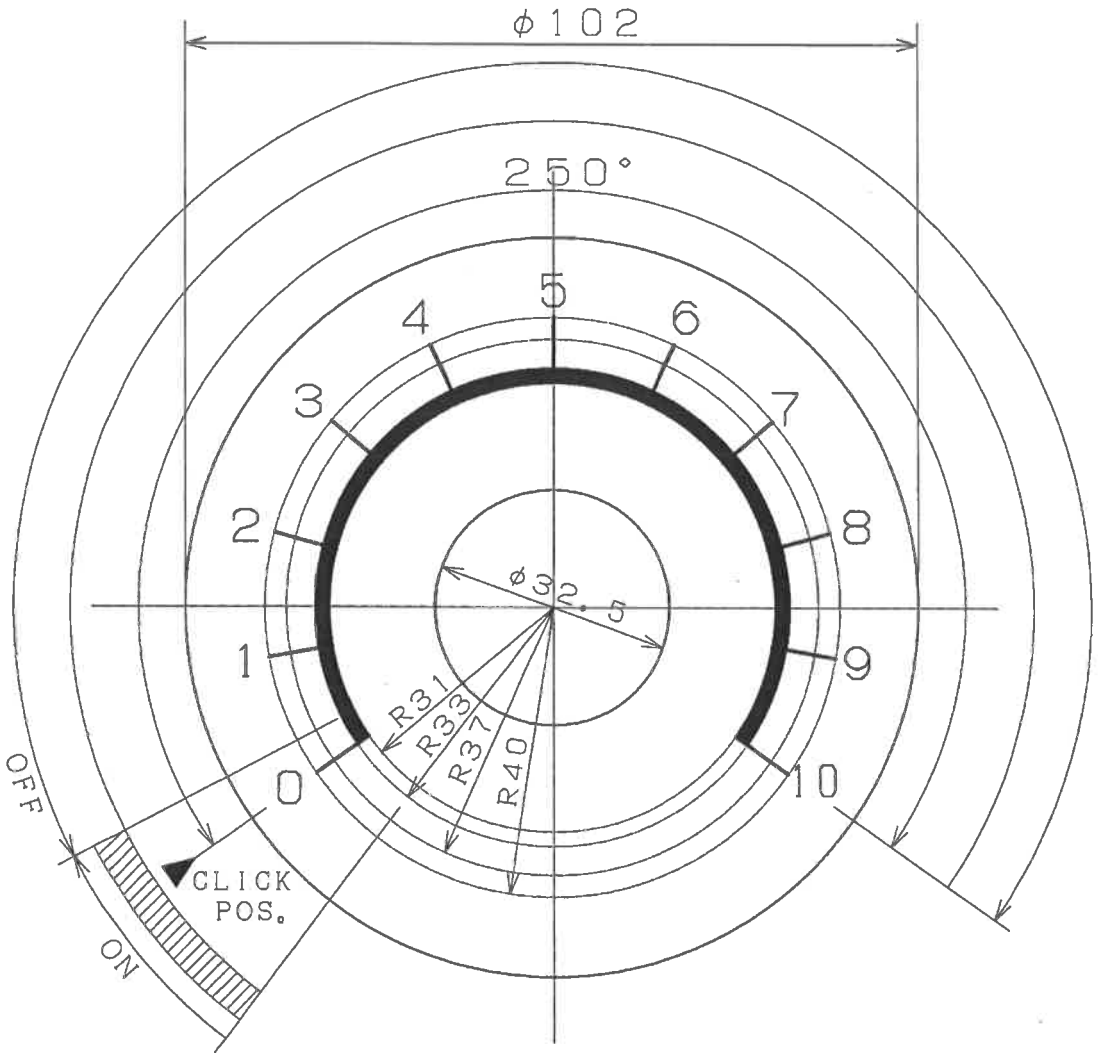
STBD/PORT EACH ONE

TYPE	DL-11 □130		
PAINTING COLOR	WORK No.	WORK SHOP	CONTROL DIAL 制御ダイヤル リミットスイッチ付
2.5G 7/2	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF / CHECK / DRAWN		
1/2	REV. '16.11.7 Y.T / REV. '16.11.07 K.T /		J.S
DATE	2, MAR, 2012	DWG. No.	4M1-137112

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

TS-054

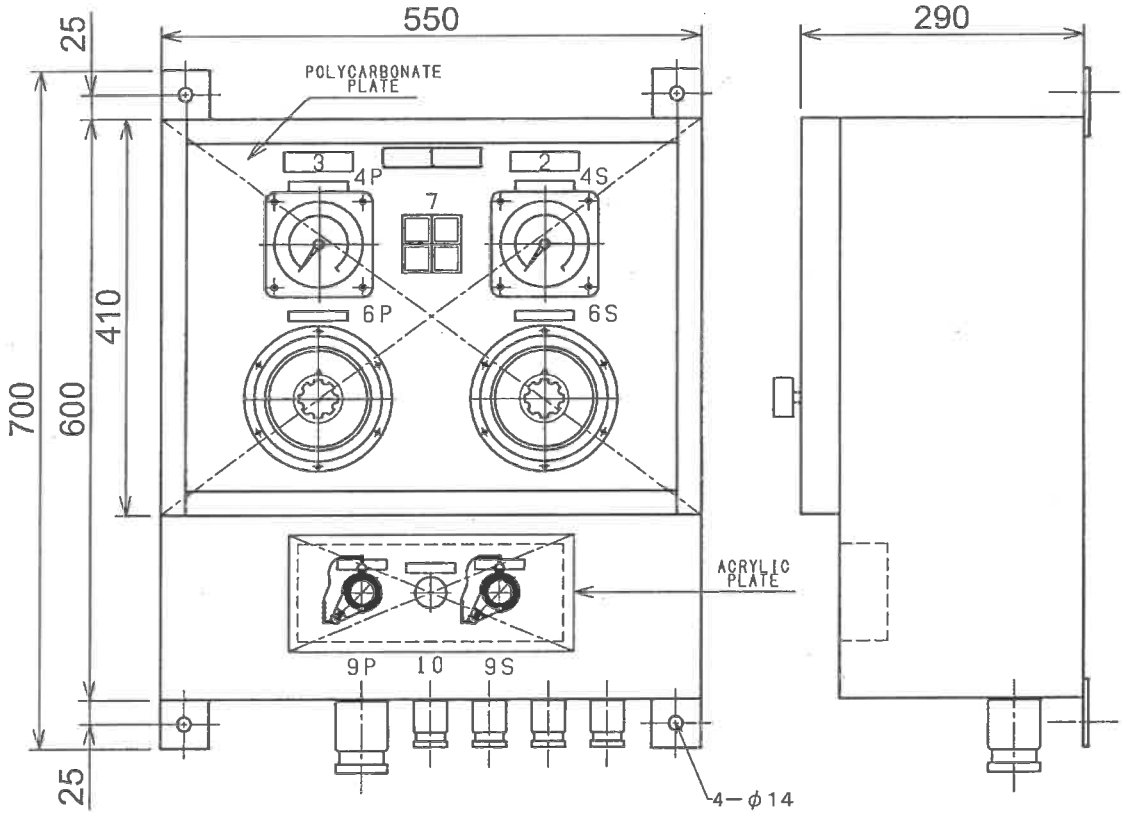




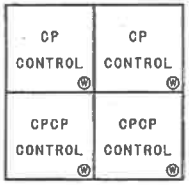
LIMIT SWITCH

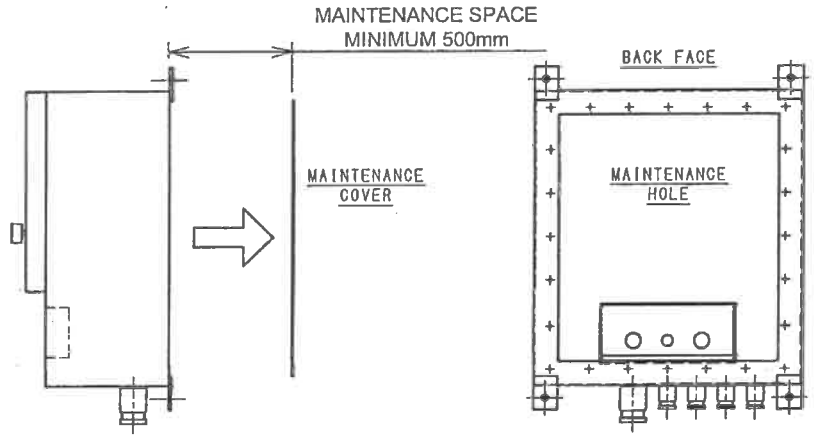
4M1-134545

PAINTING COLOR	WORK NO.	WORK SHOP	φ150 DIAL(DL-1) DIAL SCALE PLATE			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN		
1/1	REV. '16.11. 7 Y.H	REV. '16.11.07 K.T		T. S	DATE: 3, SEP, 2007	DWG. NO.: 4M1-134545



NO.	名 称 / NAME	記 名 / LETTERING	備 考 / REMARKS	Q'TY
1	NAME PLATE	CARGO PANEL	20x100	1
2	NAME PLATE	STBD	20x70	1
3	NAME PLATE	PORT	20x70	1
4	INDICATOR	M/E REVOLUTION SPEED	□110 0VF-11 0~10V/0~1500min-1	2
5				
6	DIAL	M/E CONTROL DIAL	DL-11 φ150	2
7	INDICATION LAMP		DR22F3M-E4	4
8				
9	PUSH BUTTON	EMERGENCY STOP	ABN8P10	2
10	PUSH BUTTON	LAMP TEST	ABW1208	1
11				
12				
13				
14				





※Make a maintenance space at least 500mm between the panel back face and wall.
If it's difficult, finish wiring before covering the panel.

IP:44 SUS304 CP

P 20

△4 18, NOV., 2016

PAINTING COLOR	WORK NO.	WORK SHOP
2.5G7/2	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
1/5	REV. 16.11.24		REV. 16.11.24	Y. O

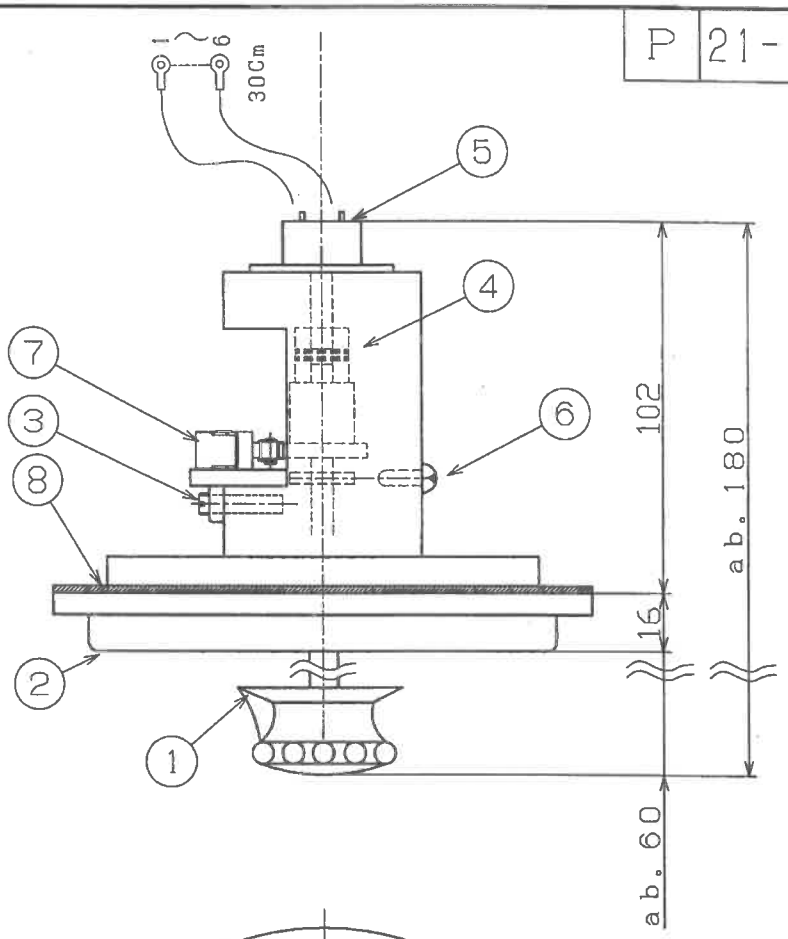
CARGO PANEL
(explosion proof area / On deck)

DATE 7, SEP., 2016 DWG. NO. 3M1-141143

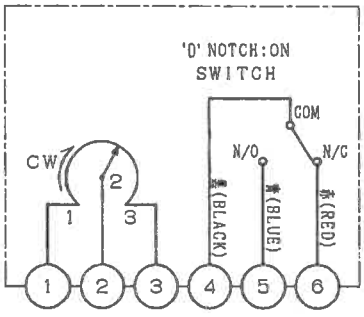
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 21-1

No.	NAME 名称	REQ.No. 個数	MATERIAL 材質	REMARKS 備考
1	KNOB ツマミ	1		JRM 50NΦ8
2	BODY 本体	1	AC7A-F	
3	BRAKE ブレーキ	1		ADJUSTABLE 調整可能
4	COUPLING 継手	1	BSBM2	
5	POTENTIOMETER ポテンショメーター	1		SHSM30F10
6	STOPPER OF REV. ANGLE 回転角度設定器	2		DIAL ADJUST. RANGE 210°~350° ダイヤル設定範囲 210°~350°
7	MICRO SWITCH マイクロスイッチ	1		D2VW-5L2A-1M
8	PACKING パッキン	1		NEOPRENE ネオプレン t=2

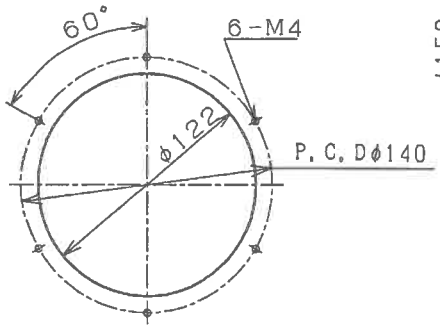


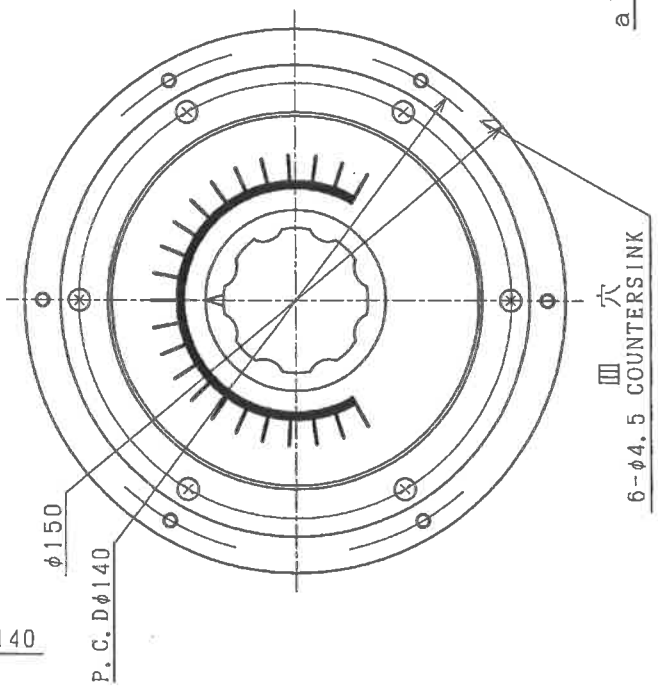
CONNECT. DIAGRAM 接続図



PANEL CUT (パネルカット)

SCALE 1/4





CP

PROTECTION : IP56
STBD/PORT EACH ONE

TYPE	DL-11 (Φ150)

PAINTING COLOR	WORK No.	WORK SHOP
2.5G 7/2	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
1/2	REV. '16.11.7 Y.H	REV. K.T		Y.O

CONTROL DIAL
制御ダイヤル リミット付

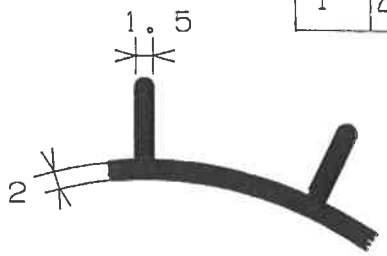
DATE	2-AUG.-2016	DWG. No.	4M1-141476

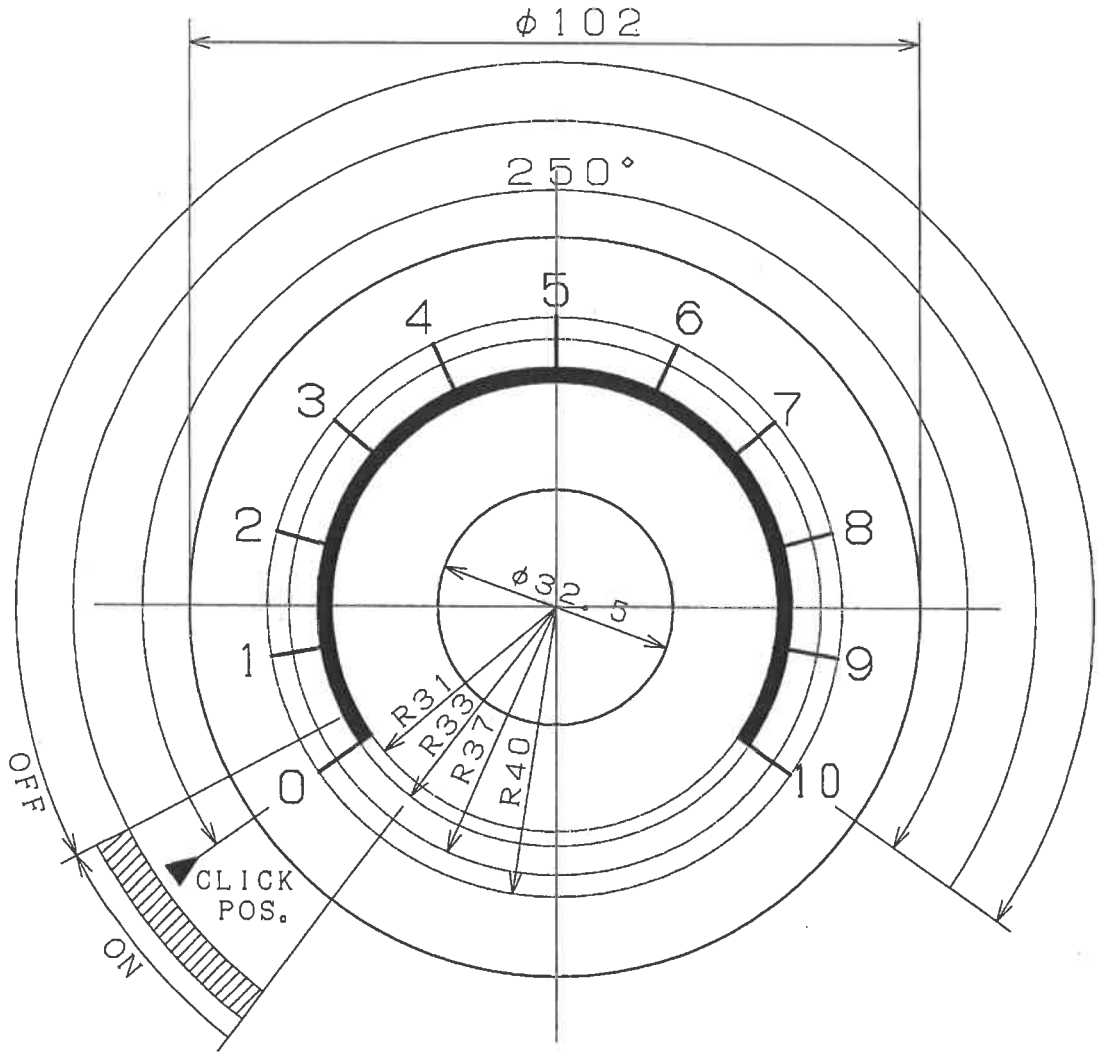
4M1-141476

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

TS-054

P 21-2



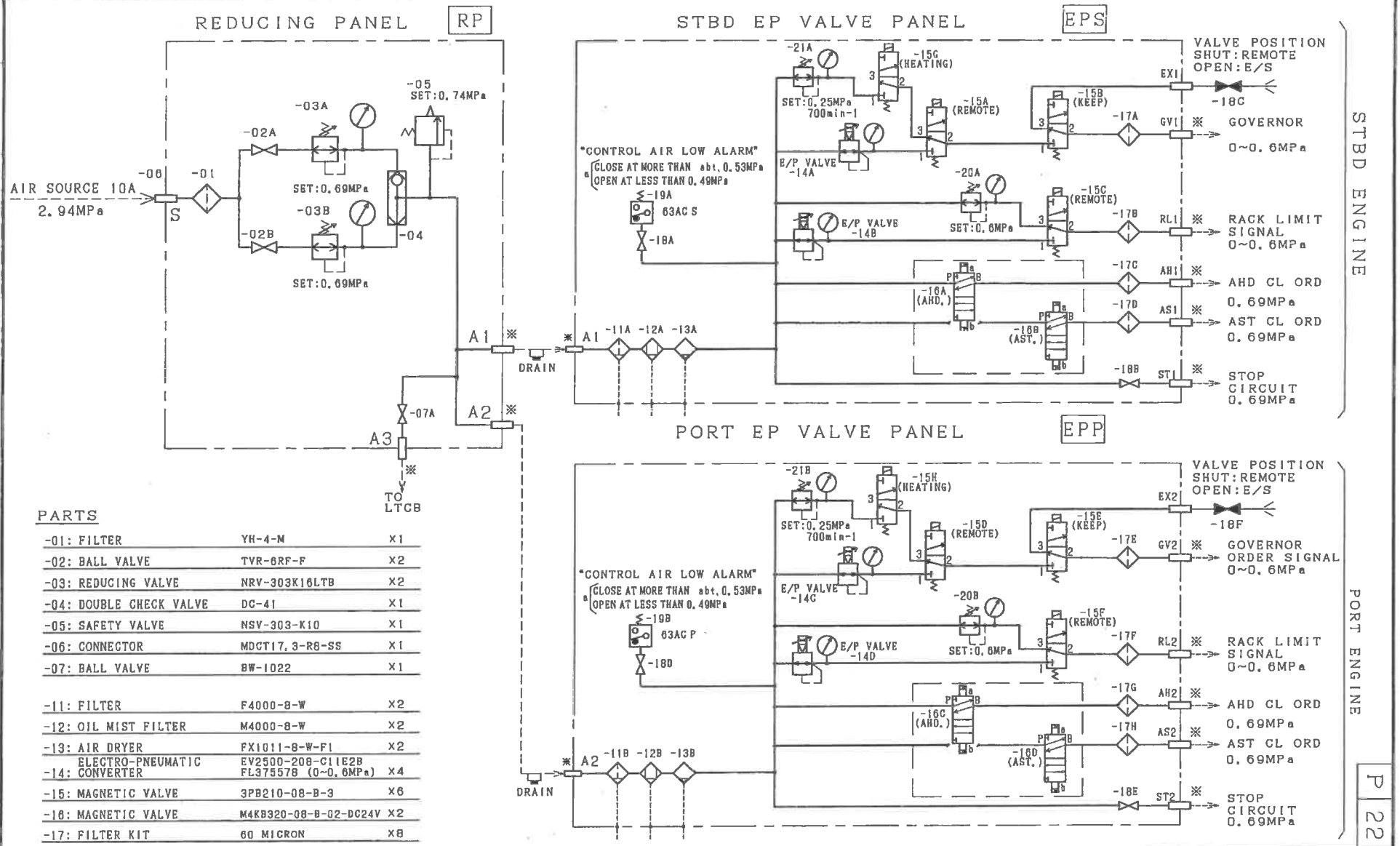


LIMIT SWITCH

4M1-134545

PAINTING COLOR	WORK NO.		WORK SHOP		φ150 DIAL (DL-1) DIAL SCALE PLATE			
	SZ15045/15046 SZ15047/15048							
SCALE	MANAGER	CHIEF	CHECK	DRAWN				
1/1	REV. '16.11. Y.H	REV. '16.11.07 K.T		T. S	DATE	3, SEP, 2007	DWG. NO.	4M1-134545

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PARTS

-01: FILTER	YH-4-M	X1
-02: BALL VALVE	TVR-6RF-F	X2
-03: REDUCING VALVE	NRV-303K16LTB	X2
-04: DOUBLE CHECK VALVE	DC-41	X1
-05: SAFETY VALVE	NSV-303-K10	X1
-06: CONNECTOR	MDCT17.3-R8-SS	X1
-07: BALL VALVE	BW-1022	X1
-11: FILTER	F4000-8-W	X2
-12: OIL MIST FILTER	M4000-8-W	X2
-13: AIR DRYER	FX1011-8-W-F1	X2
-14: ELECTRO-PNEUMATIC CONVERTER	EV2500-208-C11E2B FL375578 (0~0.6MPa)	X4
-15: MAGNETIC VALVE	3PB210-08-B-3	X6
-16: MAGNETIC VALVE	M4KB320-08-B-02-DC24V	X2
-17: FILTER KIT	60 MICRON	X8
-18: BALL VALVE	BW-1022	X7
-19: PRESSURE SWITCH WITH/TEST VALVE	MBC5100-061B0102 (MBV5000-1201)	X2
-20: REDUCING VALVE	R4000-8-W	X4
-21: REDUCING VALVE	R4000-8-W-L	X2

REMARKS PIPES MARKED ※ ARE NOT SUPPLIED BY MAKER. (OUT DIA φ10 Cu TUBE)

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. 6.11.7			Y.O

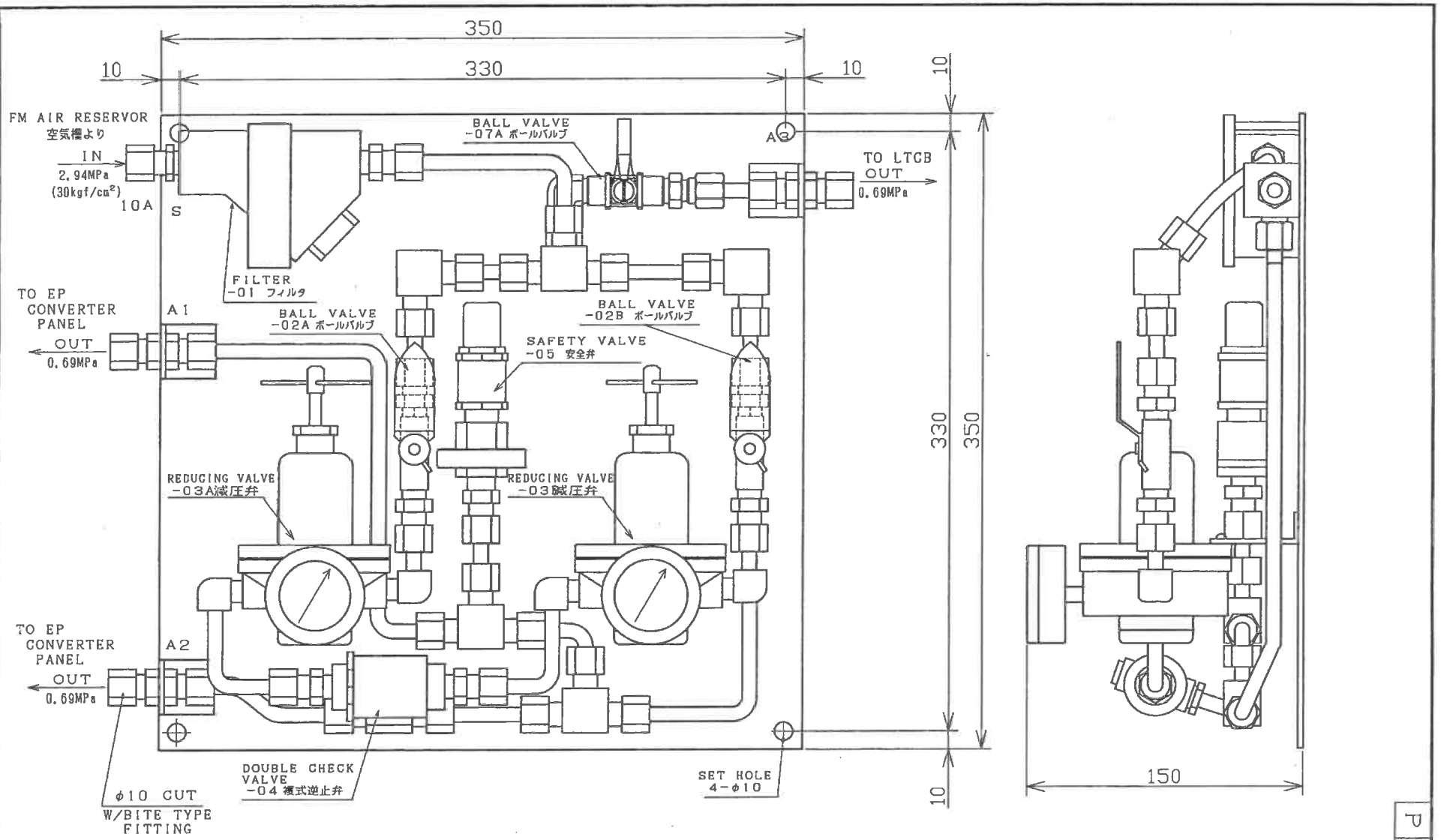
AIR SOURCE PANEL PIPING DIAGRAM

DATE 10, AUG., 2016 DWG. NO. 3M4-101620

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M4-101620

P 22

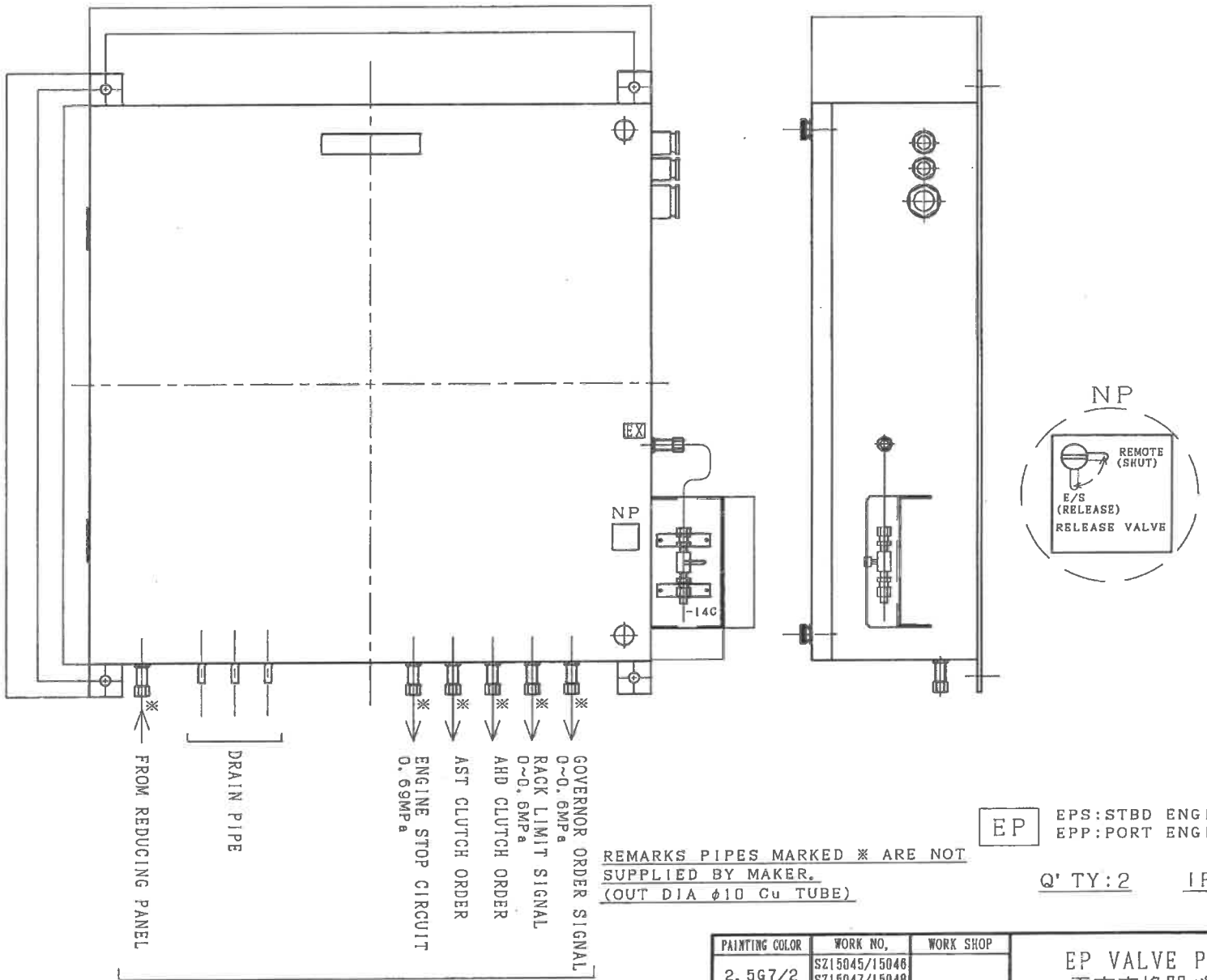


RP

PAINTING COLOR	WORK NO.	WORK SHOP	REDUCING PANEL 減圧弁パネル 3出力 / 3 OUT PUT			
2.5G7/2	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	DWG. NO.
1/2	REV. '16.11.7	K-T		Y.O	1. JUL. 2016	3M1-141437

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M1-141437



BITE TYPE JOINT
(※PIPE OUTSIDE DIAMETER Ø10)

REMARKS PIPES MARKED ※ ARE NOT SUPPLIED BY MAKER.
(OUT DIA ϕ10 Cu TUBE)

EP EPS:STBD ENGINE
EPP:PORT ENGINE

Q'TY:2 IP44

P 24

3M1-141146

PAINTING COLOR	WORK NO,	WORK SHOP	
2.5G7/2	SZ15045/15046 SZ15047/15048		

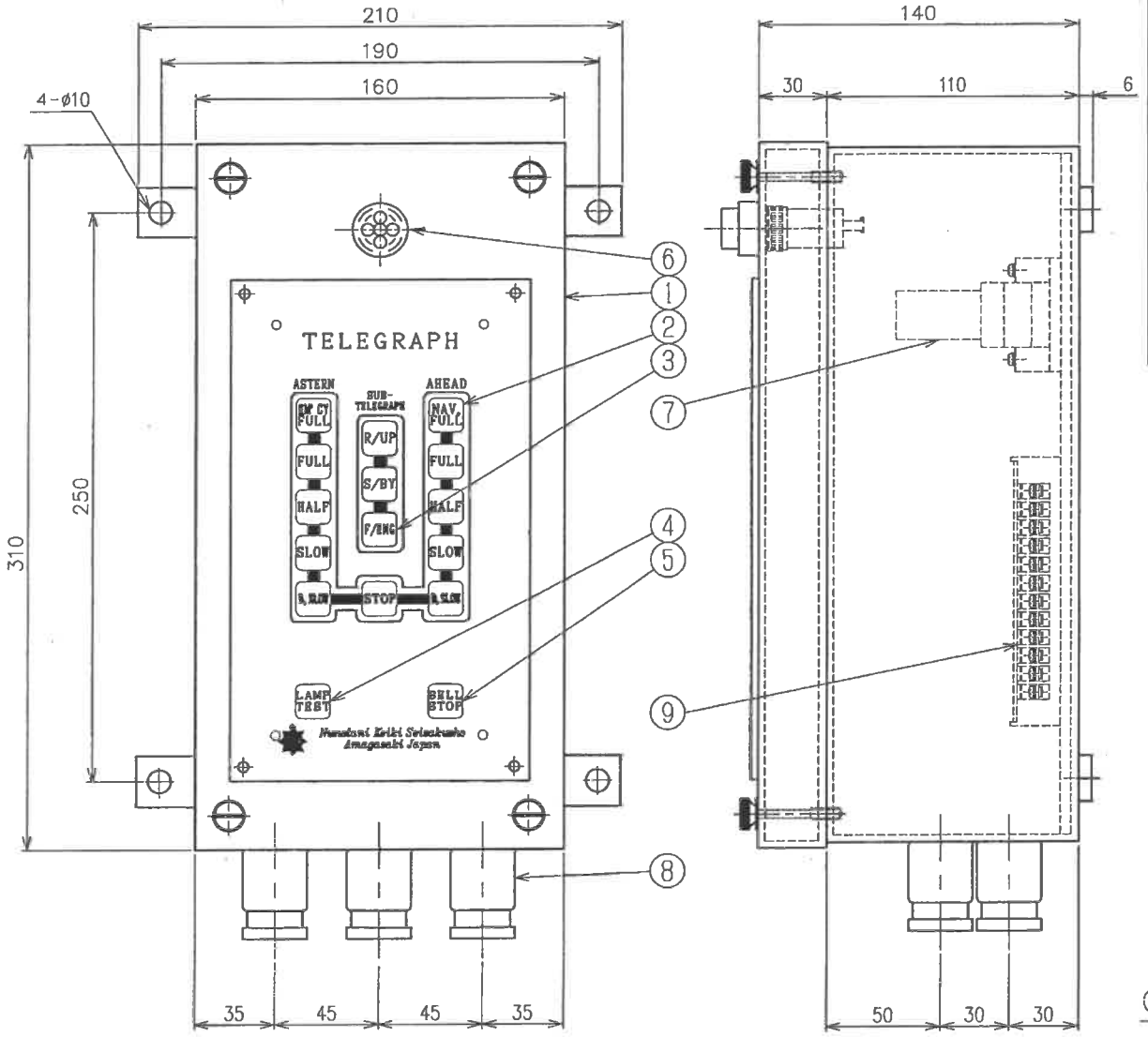
SCALE	MANAGER	CHIEF	CHECK	DRAWN
1/6	'16.11.7			Y.O

EP VALVE PANEL
電空変換器パネル

DATE 26, JUL., 2016 DWG. NO. 3M1-141146

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

TS-052



No.	名称 NAME	材質 MATERIAL	備考 REMARKS
1	テレグラフ箱 TELEGRAPH BOX	SS400P	
2	メインテレグラフ用ＬＥＤ LED for MAIN TELEGRAPH		AHEAD SIDE:GREEN ASTERN SIDE, STOP:RED
3	サブテレグラフ用ＬＥＤ LED for SUB TELEGRAPH		SUB TELE:WHITE
4	ランプテスト押釦 PUSH BUTTON for LAMP TEST		
5	ゴングストップ押釦 PUSH BUTTON for GONG STOP		
6	テレグラフブザー BUZZER for SUB TELEGRAPH		FUJI ELECTRIC AH165-XBE
7	リレー RELAY		OMRON LY-2
8	グランド GROUND		15
8	端子盤 TERMINAL BOARD		KIMUDEN TS-615, 12P

Q' TY 1

IP44

P 25

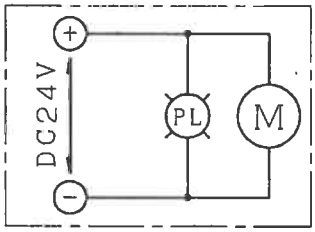
PAINTING COLOR	WORK No.	WORK SHOP	ENGINE ROOM TELEGRAPH BOX			
2.5G7/2	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN		
1/2	'14.11.7	K.T		Fukuoka	DATE H26-6-10	DWG. NO. 3MI-139768

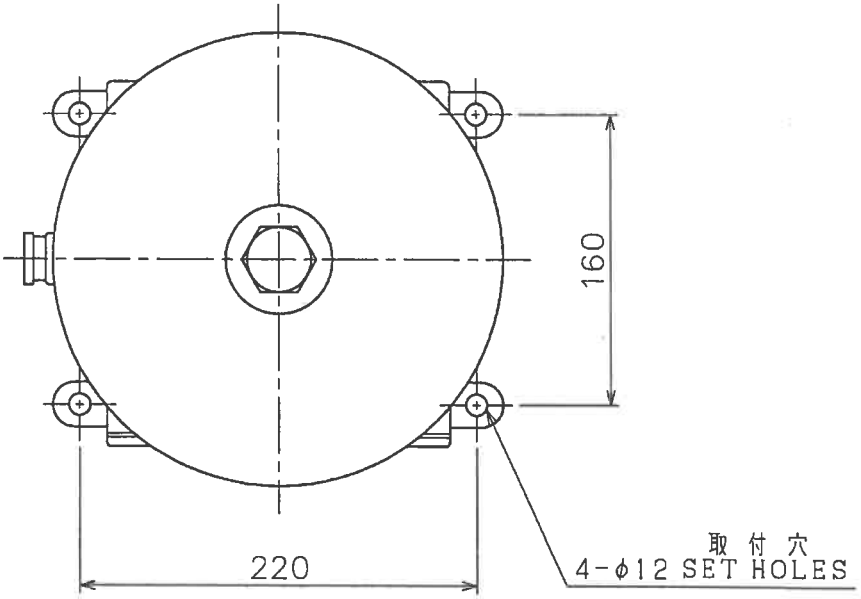
TML-243

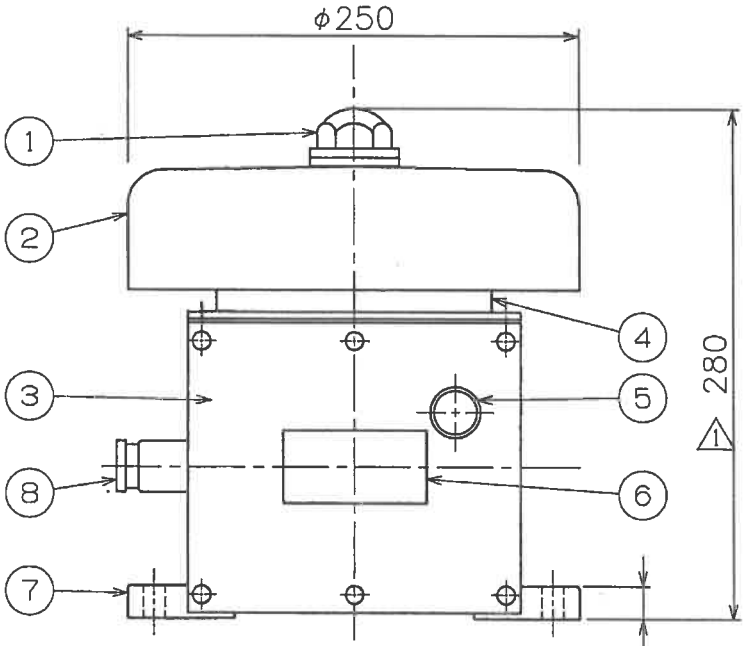
No.	NAME OF PARTS（名称）		No. REQ.（個数）	MATERIAL （材質）		REMARKS（備考）
1	NUT	（ナット）	1	BRASS CASTING	YBsC3	
2	GONG	（鐘）	1	BELL METAL	———	
3	COVER	（フタ）	1	ALUMINUM ALLOY PLATE	A5052P-1/2H	
4	SUPPORT	（鐘支持台）	1	CAST IRON	FC20	
5	PILOT LANP	（表示灯）	1		———	GREEN
6	NAME PLATE	（銘板）	1	BRASS PLATE	C2801P	
7	BODY	（本体）	1	ALUMINUM ALLOY CASTING	AC7A-F	
8	GLAND	（グランド）	1		PBT	15

P	26

結線図
CONNECTION DIAGRAM







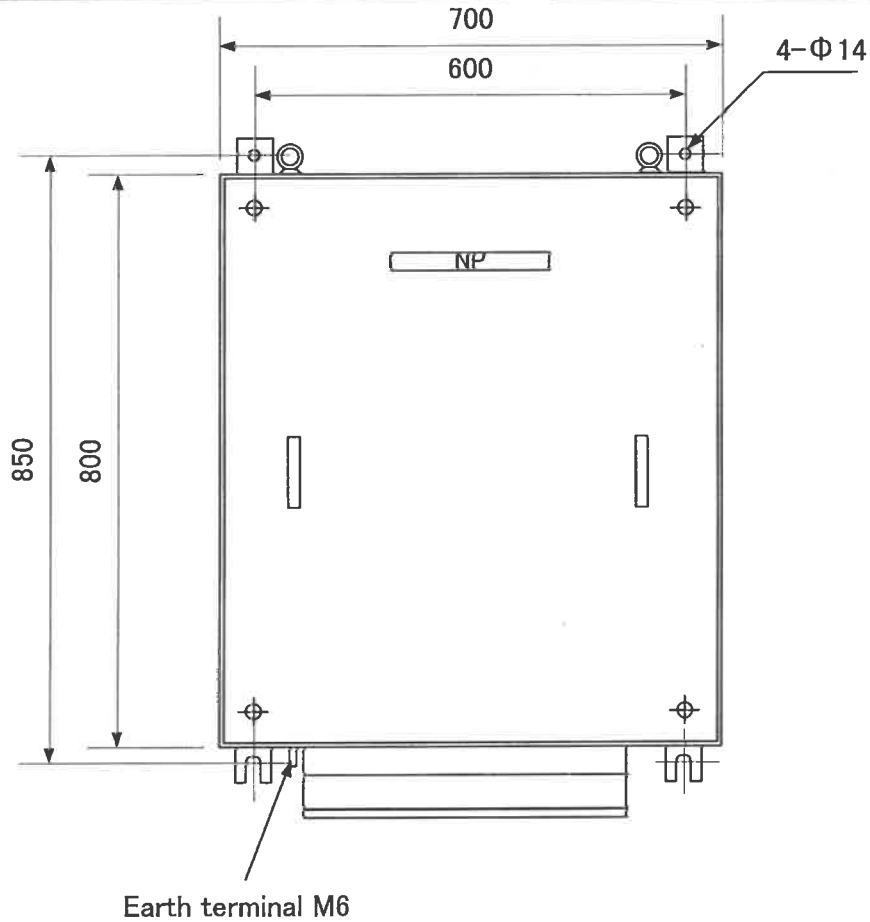
Q' TY 1

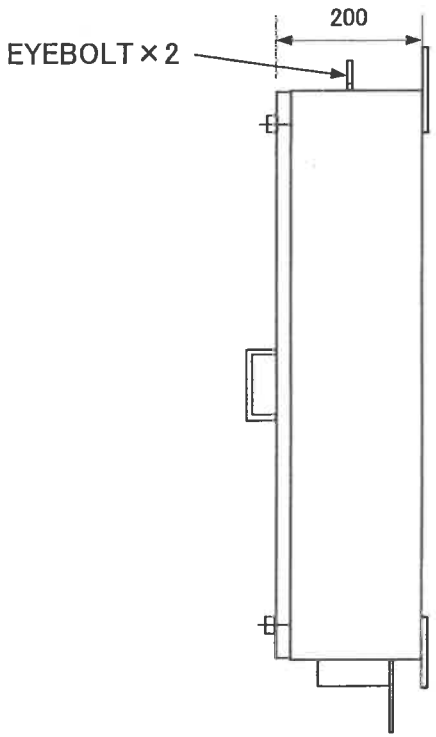
IP44

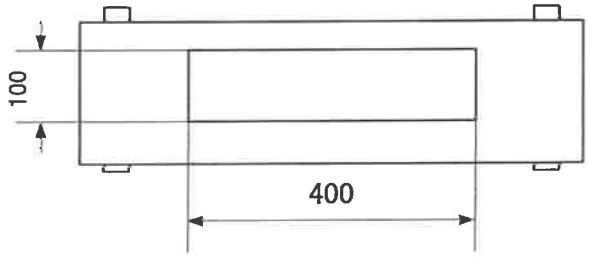
⚠1 2, JUN, 2016 ライナー変更につき寸法変更

PAINTING COLOR	WORK NO.		WORK SHOP		エンジンテレグラフ ゴング GONG OF ENGINE TELEGRAPH			
2.5G7/2	SZ15045/15046 SZ15047/15048							
SCALE	MANAGER	CHIEF	CHECK	DRAWN				
1／4	REV. '16.11.? Y.H	REV. '16.11.07 K.T		K.O	DATE	9, JUN, 2016	DWG. NO.	4M1-141410

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.





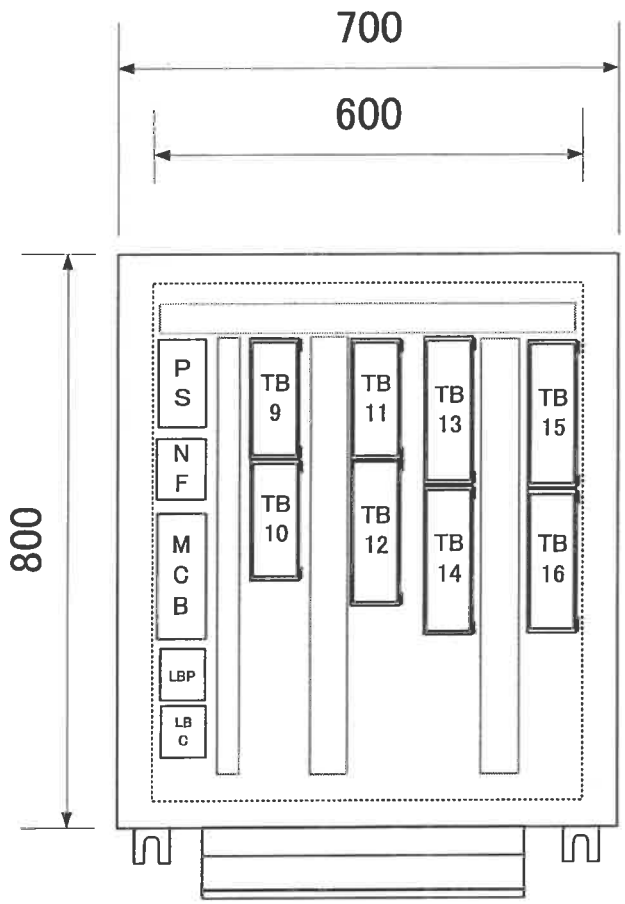


CONSTRUCTION : SPRAY PROOF
MATERIAL : BODY SPCC t2.3
DOOR SPCC t1.6
PAINTING COLOR : MUNSEL 2.5G7/2 SG
NAME PLATE : LOCAL BOX

LB

P27-1

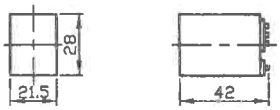
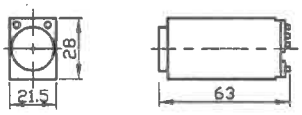
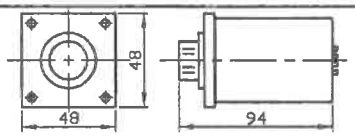
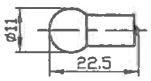
CUSTOMER 客先		TITLE 図面名称	LOCAL BOX (TYPE-RC) OUTLINE
DRAWN 作成 C.Miyazaki	BY	JOB No. 工事番号	DRAWING No. 図面番号 K-2917



P27-2

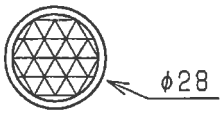
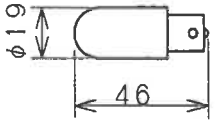
CUSTOMER 客先		TITLE 図面名称	LOCAL BOX (TYPE-RC) INNER ARRANGEMENT
DRAWN 作成 C.Miyazaki	BY	JOB No. 工事番号	DRAWING No. 図面番号 K-2917

SHIP NO.	SPARE PARTS LIST FOR	U S E	SETS PER VESSEL
SZ15045 SZ15046 SZ15047 SZ15048	M/E REMOTE CONTROL SYSTEM		1

ITEM NO	NAME OF PART	OUTLINE	QUANTITY WORKING PER SET	QUANTITY WORKING PER VESS.	QUANTITY SPARE	REMARKS
1	RELAY		290	290	29	オムロン MY4N-D2 DC24V
2	RELAY		6	6	1	オムロン MY4 AC220V
3	TIMER		36	36	4	オムロン H3YN-4 DC24V
4	TIMER		28	28	3	オムロン H3CR-A DC24V
5	LAMP		288	288	288	IDEC LS-3 DC30V1W
6	–					
7	–					
8	–					
9	–					
10	–					

MFR' S NAME	NUNOTANI KEIKI SEISAKUSHO CO.,LTD.	DWG.NO.	

SHIP No.	SPARE PARTS LIST FOR	U S E	SETS PER VESSEL
SZ15045 SZ15046 SZ15047 SZ15048	エンジンテレグラフ ENGINE TELEGRAPH		1

ITEM NO	NAME OF PART	OUTLINE	QUANTITY WORKING PER SET	QUANTITY WORKING PER VESS.	QUANTITY SPARE	REMARKS
1	グローブ GLOBE		1	1	1	緑 ゴング用 Green for GONG
2	電球 LAMP		1	1	1	DC28V 5W S-15 FOR GONG
3						
4						
5						
6						
7						
8						
9						
10						

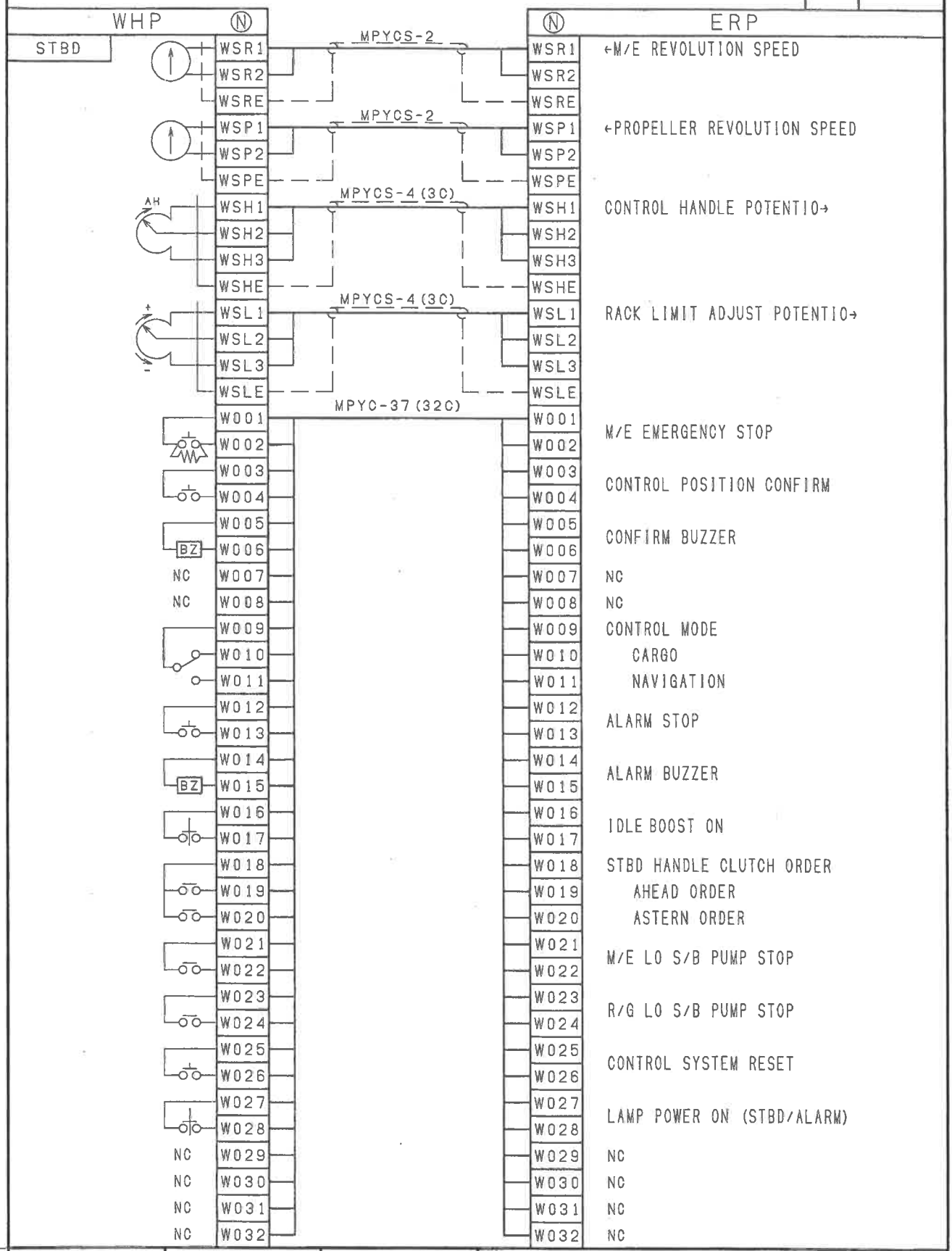
MFR'S NAME	NUNOTANI KEIKI SEISAKUSHO CO.,LTD	DRAW NO.	

TL-SP5

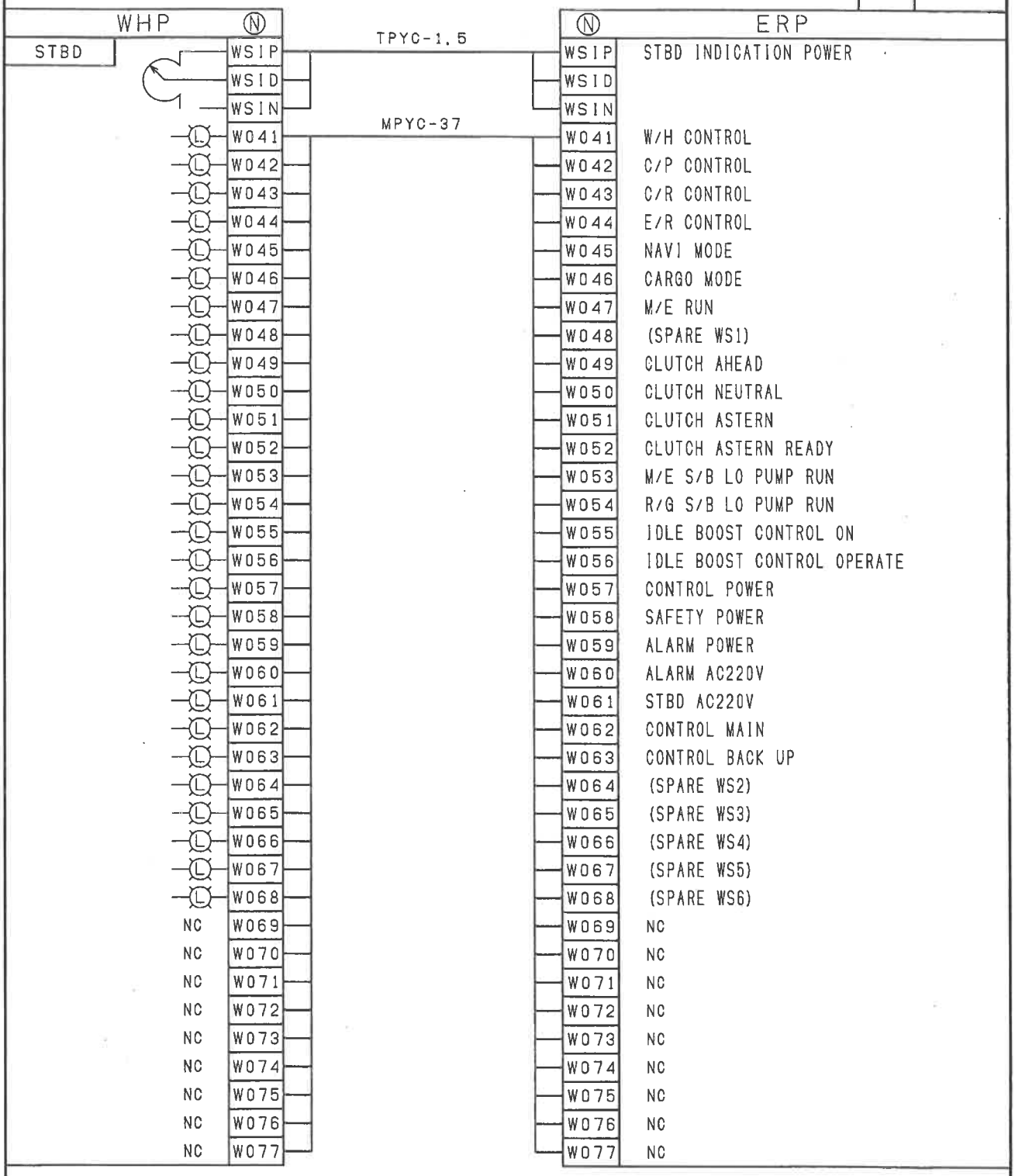
SYMBOL LIST

WHP	W/H CONTROL PANEL
ECP	C/R CONTROL PANEL
ERP	M/E RELAY PANEL
CPCP	CARGO PUMP CONTROL PANEL
CP	CARGO PANEL
EP*	S:STBD E-P VALVE PANEL P:PORT E-P VALVE PANEL
MJ*	S:STBD M/E JUNCTION BOX P:PORT M/E JUNCTION BOX
RJ*	S:STBD R/G JUNCTION BOX P:PORT R/G JUNCTION BOX
LB	LOCAL BOX
MSB	MAIN SWITCHBOARD

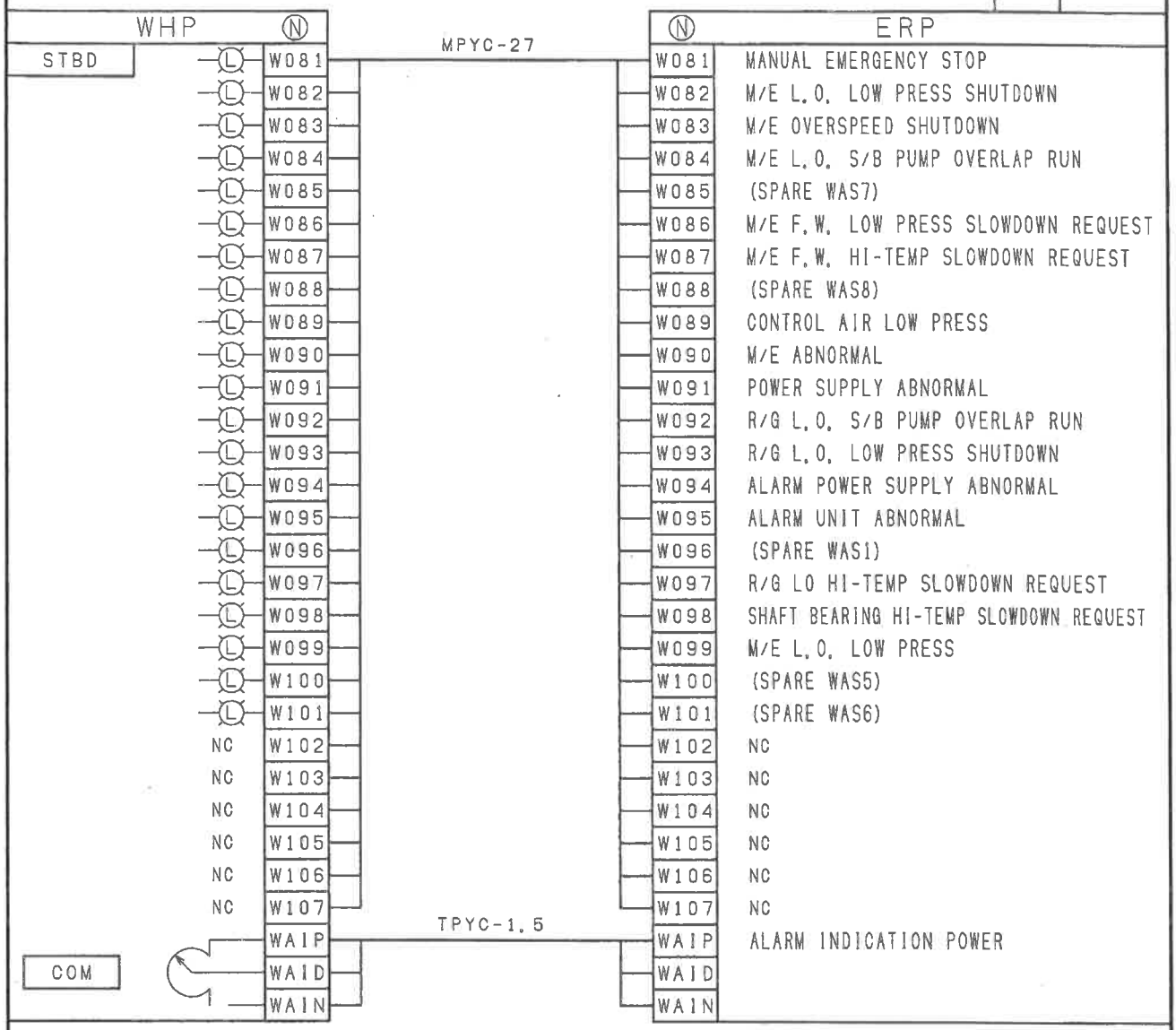
(S)	SHIPYARD SUPPLY
(D)	DAIHATSU SUPPLY
(N)	NUNOTANI SUPPLY



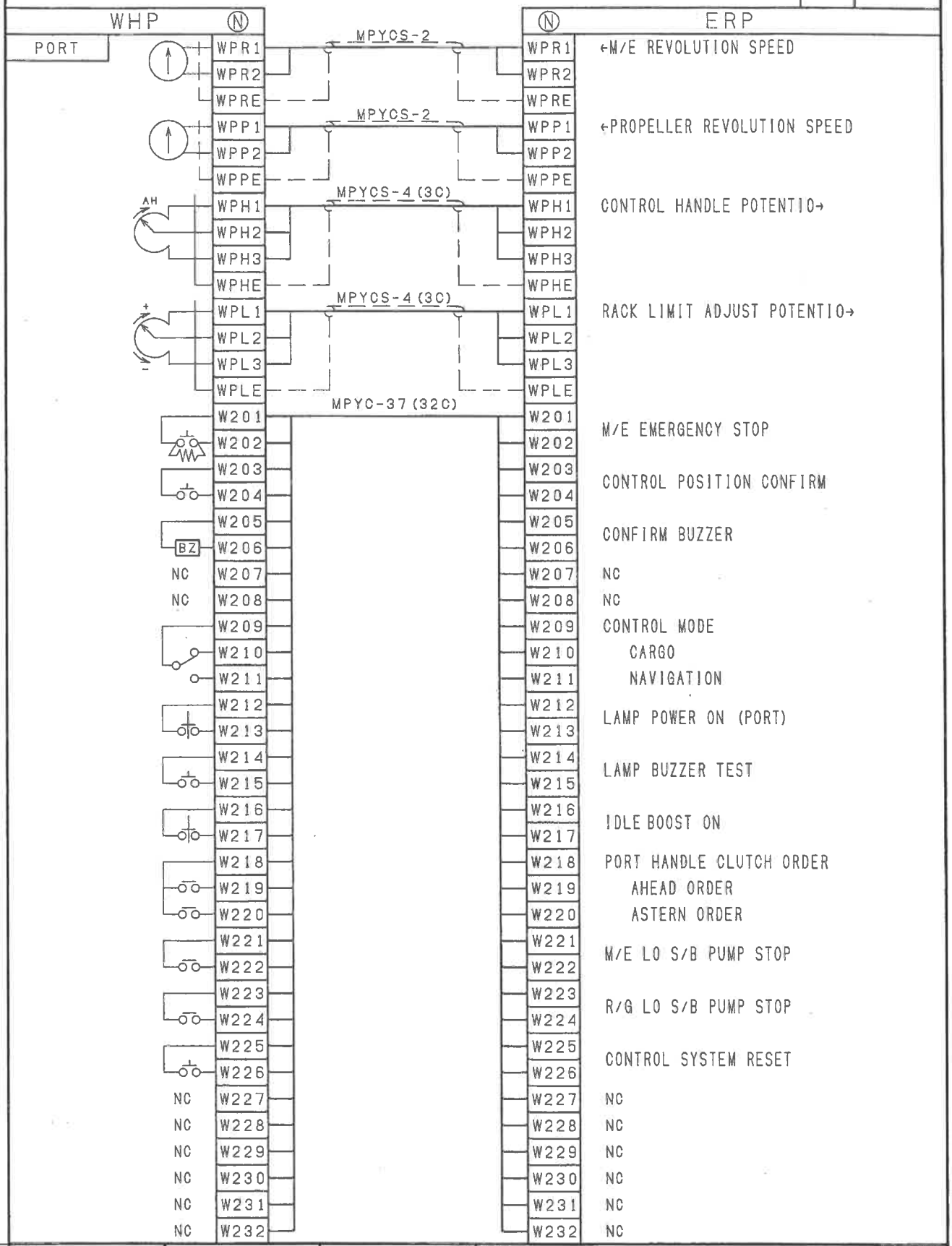
4M3-194565	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 01
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	
		REV. '16.11.7 Y.H	REV. '16.11.07 K.T	Y. O	DATE: 1, SEP., 2016 / DWG. NO.: 4M3-194565



4M3-194566	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 02
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	
		REV. 16.11. Y.H	REV. 16.11.07 K.T	Y. O	DATE: 1, SEP., 2016 / DWG. NO.: 4M3-194566

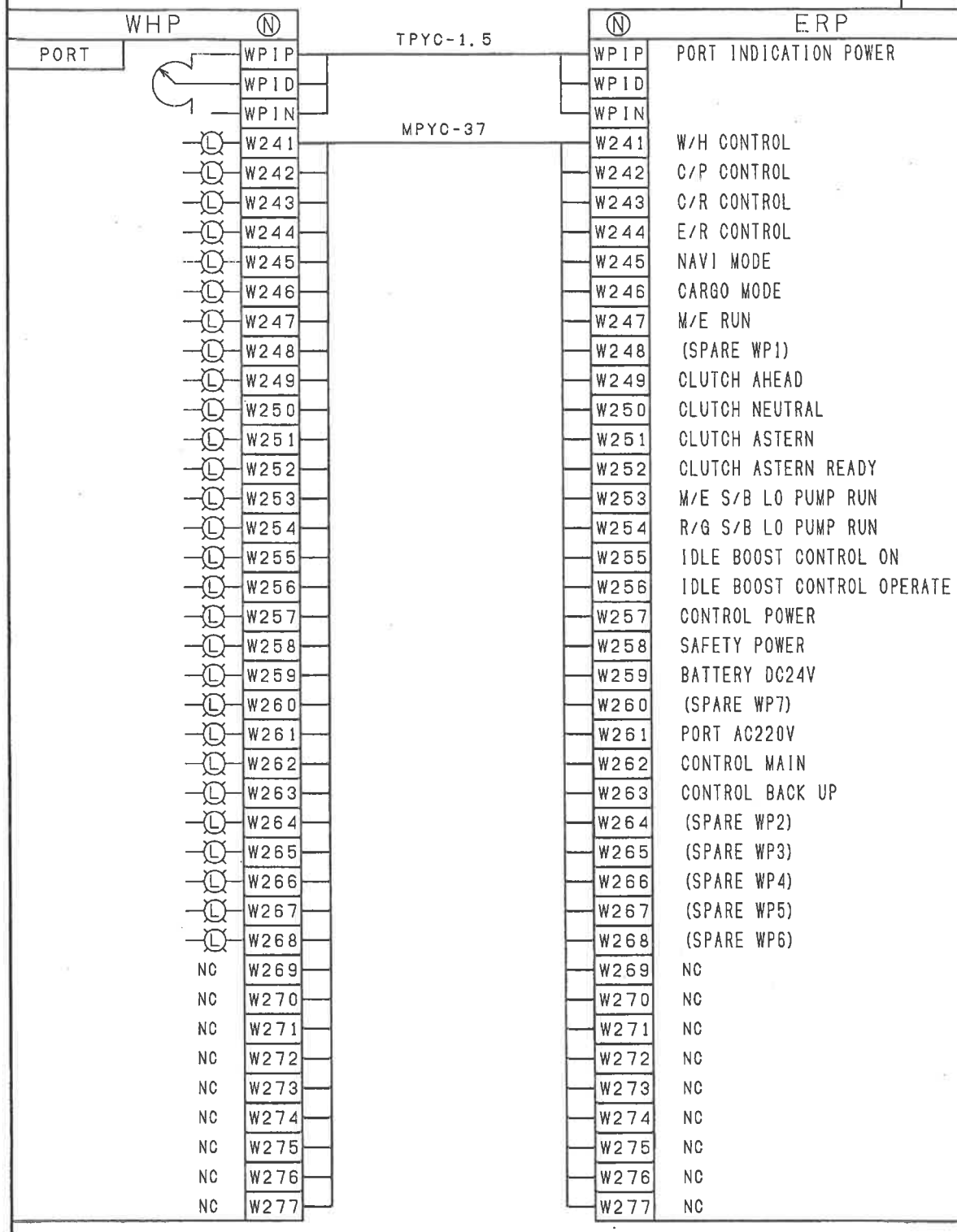


4M3-194567	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 03		
		SZ15045/15046 SZ15047/15048					
	SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	1, SEP., 2016
		REV. '16.11.7 Y.H	REV. '16.11.07 K.T		Y. O	DWG. NO.	4M3-194567



4M3-194568	PAINTING COLOR	WORK NO.	WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 04			
		SZ15045/15046 SZ15047/15048					
	SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	1, SEP., 2016
		REV. 16.11.7 YH	REV. 16.11.07 K.T		Y. O	DWG. NO.	4M3-194568

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



4M3-194569	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 05
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	
				Y. O	DATE 1, SEP., 2016 / DWG. NO. 4M3-194569

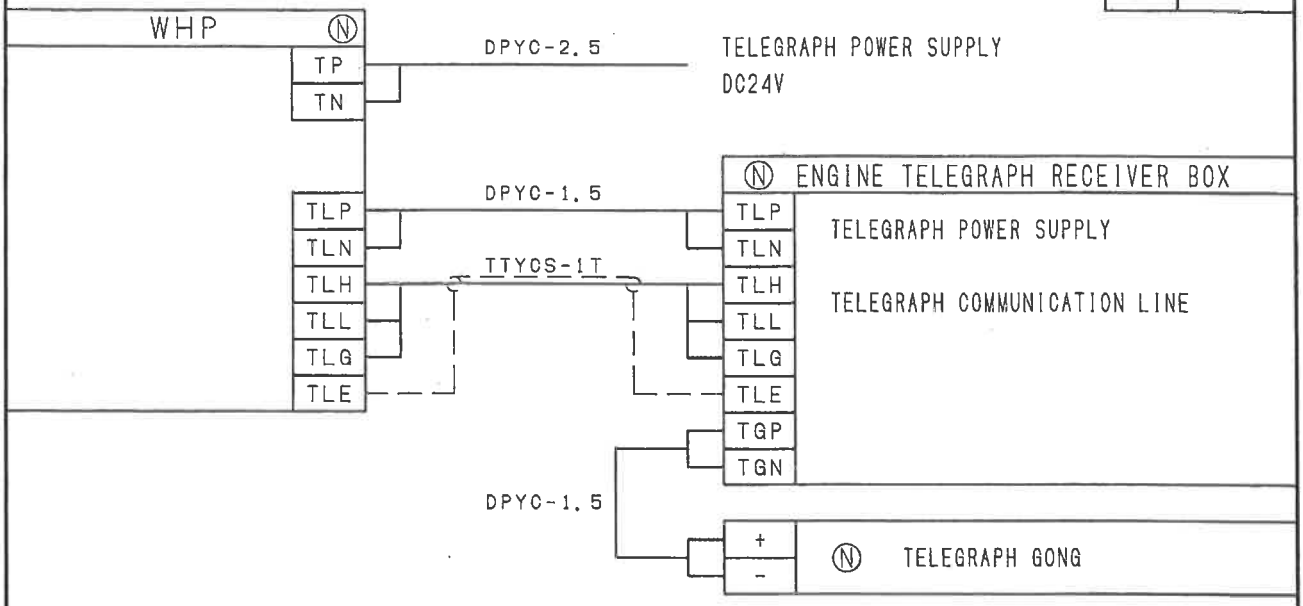
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

WHP	Ⓝ	MPYC-37	Ⓝ	ERP
Ⓛ	W281		W281	MANUAL EMERGENCY STOP
Ⓛ	W282		W282	M/E L.O. LOW PRESS SHUTDOWN
Ⓛ	W283		W283	M/E OVERSPEED SHUTDOWN
Ⓛ	W284		W284	M/E L.O. S/B PUMP OVERLAP RUN
Ⓛ	W285		W285	(SPARE WAP7)
Ⓛ	W286		W286	M/E F.W. LOW PRESS SLOWDOWN REQUEST
Ⓛ	W287		W287	M/E F.W. HI-TEMP SLOWDOWN REQUEST
Ⓛ	W288		W288	(SPARE WAP8)
Ⓛ	W289		W289	CONTROL AIR LOW PRESS
Ⓛ	W290		W290	M/E ABNORMAL
Ⓛ	W291		W291	POWER SUPPLY ABNORMAL
Ⓛ	W292		W292	R/G L.O. S/B PUMP OVERLAP RUN
Ⓛ	W293		W293	R/G L.O. LOW PRESS SHUTDOWN
Ⓛ	W294		W294	OTHER ABNORMAL
Ⓛ	W295		W295	(SPARE WAP9)
Ⓛ	W296		W296	(SPARE WAP1)
Ⓛ	W297		W297	R/G LO HI-TEMP SLOWDOWN REQUEST
Ⓛ	W298		W298	SHAFT BEARING HI-TEMP SLOWDOWN REQUEST
Ⓛ	W299		W299	M/E L.O. LOW PRESS
Ⓛ	W300		W300	(SPARE WAP5)
Ⓛ	W301		W301	(SPARE WAP6)
NC	W302		W302	NC
NC	W303		W303	NC
NC	W304		W304	NC
NC	W305		W305	NC
NC	W306		W306	NC
NC	W307		W307	NC

4M3-194570

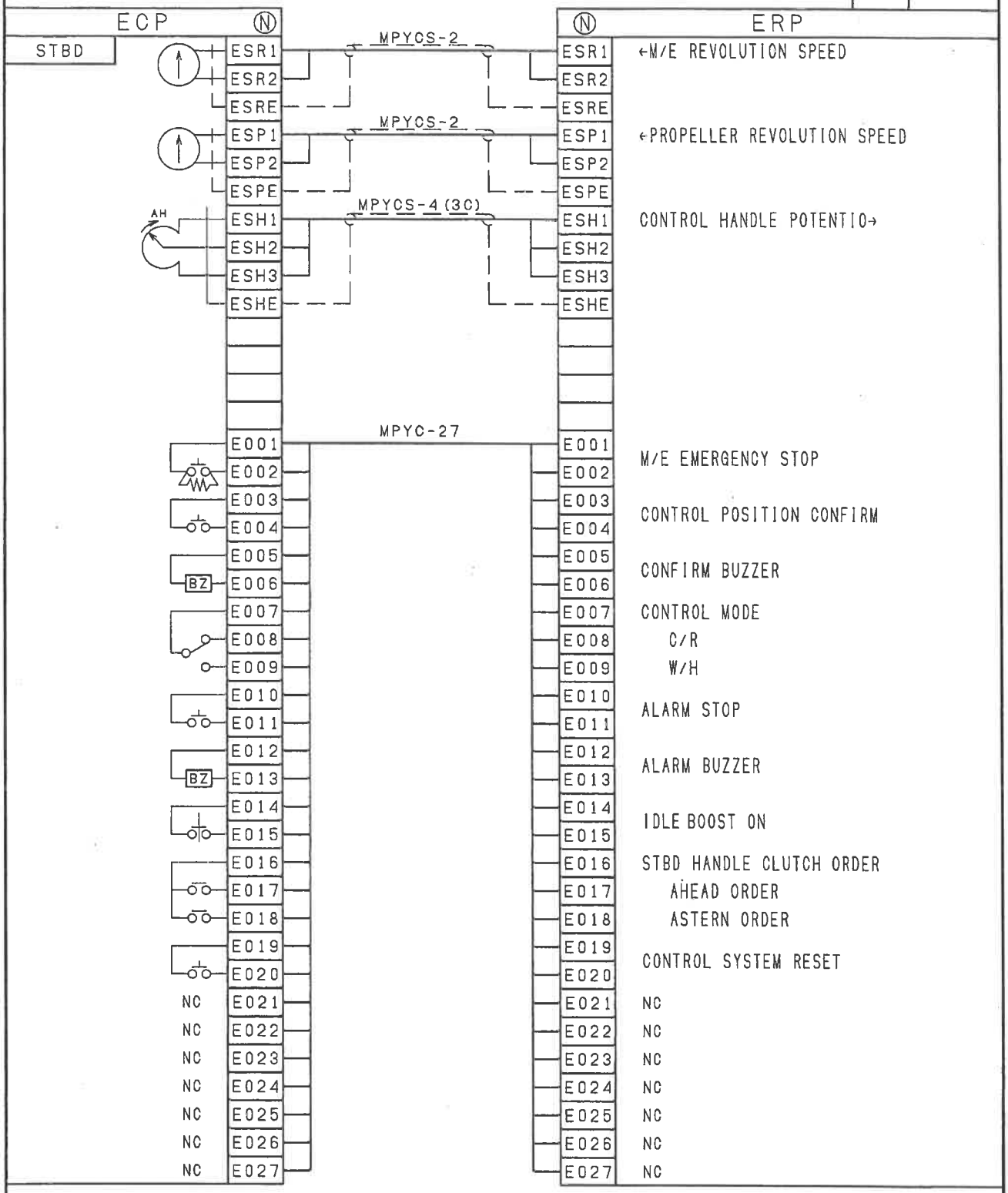
PAINTING COLOR	WORK NO.	WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 06			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN		
				Y. O	DATE: 1, SEP., 2016	DWG. NO.: 4M3-194570

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



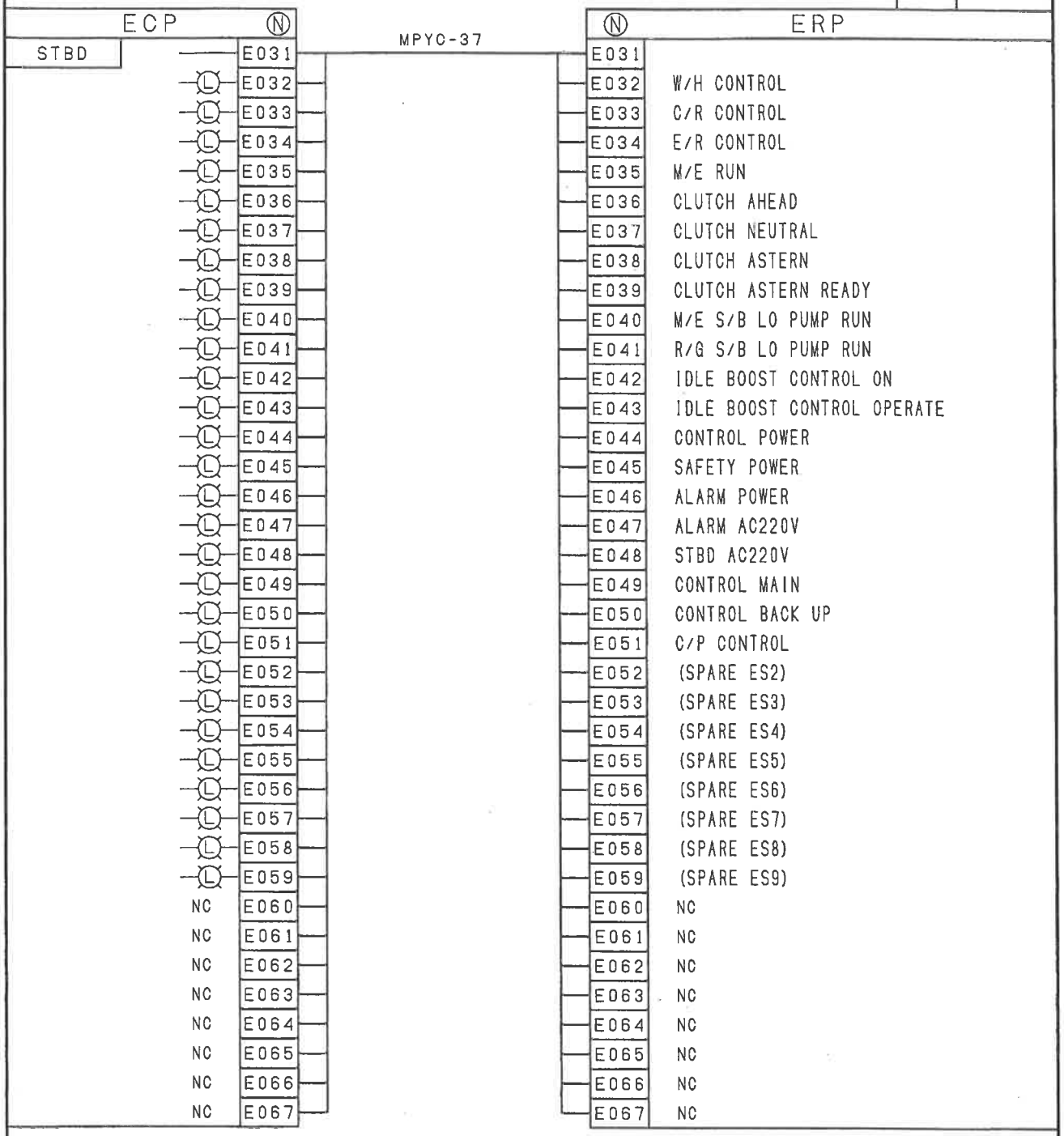
4M3-194571	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 07
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	
				Y. O	DATE: 1, SEP., 2016 / DWG. NO.: 4M3-194571

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



4M3-194572	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 08		
		SZ15045/15046 SZ15047/15048					
	SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	1, SEP., 2016
					Y. O	DWG. NO.	4M3-194572

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



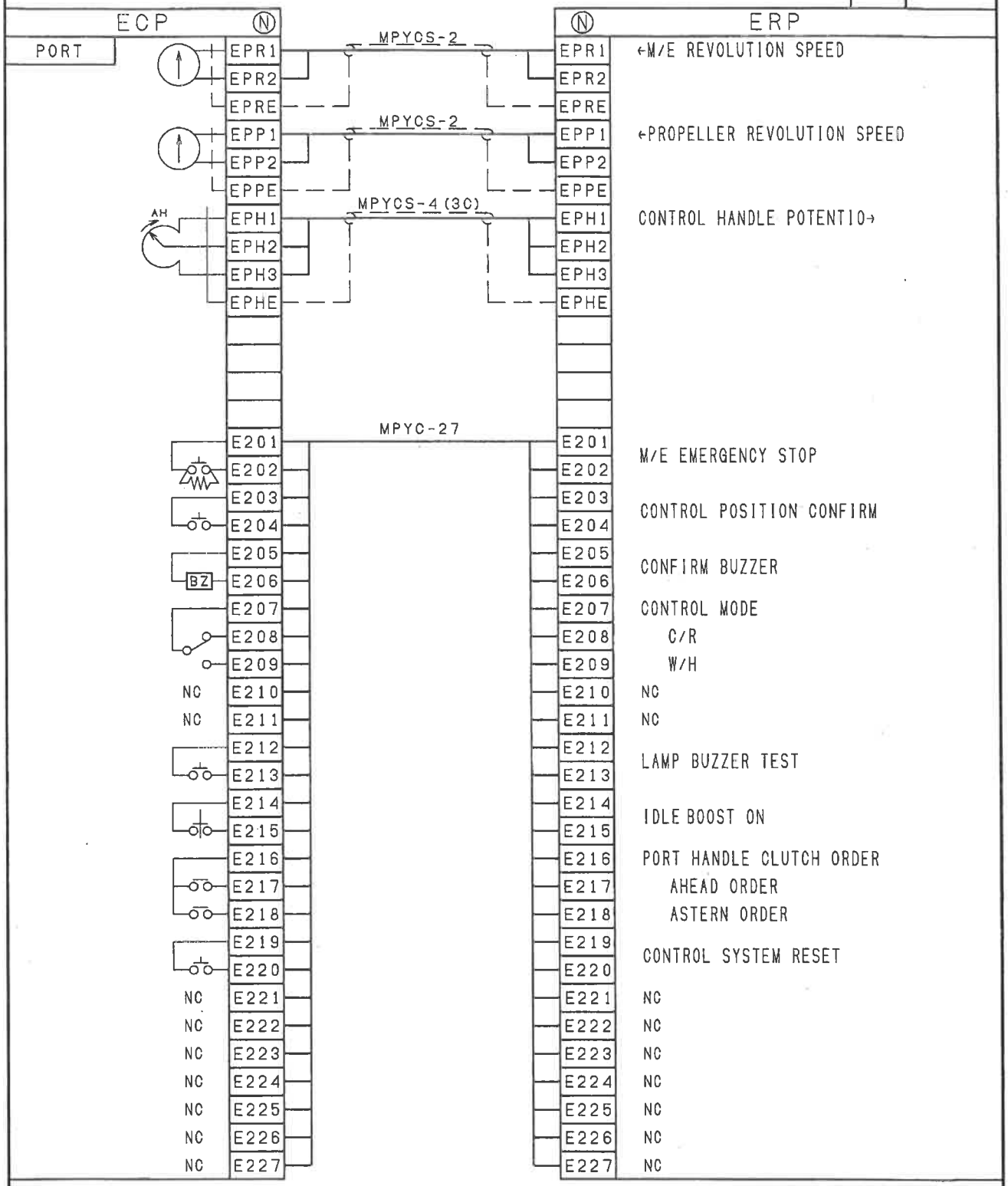
4M3-194573

PAINTING COLOR	WORK NO.	WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 09			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN		
				Y. O	DATE 1, SEP., 2016	DWG. NO. 4M3-194573

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

ECP			MPYC-27		ERP	
STBD		E071			E071	
	(L)	E072			E072	MANUAL EMERGENCY STOP
	(L)	E073			E073	M/E L.O. LOW PRESS SHUTDOWN
	(L)	E074			E074	M/E OVERSPEED SHUTDOWN
	(L)	E075			E075	M/E L.O. S/B PUMP OVERLAP RUN
	(L)	E076			E076	(SPARE EAS7)
	(L)	E077			E077	M/E F.W. LOW PRESS SLOWDOWN REQUEST
	(L)	E078			E078	M/E F.W. HI-TEMP SLOWDOWN REQUEST
	(L)	E079			E079	(SPARE EAS8)
	(L)	E080			E080	CONTROL AIR LOW PRESS
	(L)	E081			E081	M/E ABNORMAL
	(L)	E082			E082	POWER SUPPLY ABNORMAL
	(L)	E083			E083	R/G L.O. S/B PUMP OVERLAP RUN
	(L)	E084			E084	R/G L.O. LOW PRESS SHUTDOWN
	(L)	E085			E085	ALARM POWER SUPPLY ABNORMAL
	(L)	E086			E086	ALARM UNIT ABNORMAL
	(L)	E087			E087	(SPARE EAS1)
	(L)	E088			E088	R/G LO HI-TEMP SLOWDOWN REQUEST
	(L)	E089			E089	SHAFT BEARING HI-TEMP SLOWDOWN REQUEST
	(L)	E090			E090	M/E L.O. LOW PRESS
	(L)	E091			E091	(SPARE EAS5)
	(L)	E092			E092	(SPARE EAS6)
	NC	E093			E093	NC
	NC	E094			E094	NC
	NC	E095			E095	NC
	NC	E096			E096	NC
	NC	E097			E097	NC

4M3-194574	PAINTING COLOR	WORK NO.		WORK SHOP		WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 10			
		SZ15045/15046 SZ15047/15048							
	SCALE	MANAGER	CHIEF	CHECK	DRAWN				
		REV. '16.11.7 Y.H	REV.		Y. O	DATE	1, SEP., 2016	DWG. NO.	4M3-194574



4M3-194575	PAINTING COLOR	WORK NO.	WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 11	
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE: 1, SEP., 2016	DWG. NO.: 4M3-194575
		REV. '16.11.7 Y.H	Y. O		

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ECP PORT	Ⓝ	MPYC-37	Ⓝ	ERP
	E231		E231	
(L)	E232		E232	W/H CONTROL
(L)	E233		E233	C/R CONTROL
(L)	E234		E234	E/R CONTROL
(L)	E235		E235	M/E RUN
(L)	E236		E236	CLUTCH AHEAD
(L)	E237		E237	CLUTCH NEUTRAL
(L)	E238		E238	CLUTCH ASTERN
(L)	E239		E239	CLUTCH ASTERN READY
(L)	E240		E240	M/E S/B LO PUMP RUN
(L)	E241		E241	R/G S/B LO PUMP RUN
(L)	E242		E242	IDLE BOOST CONTROL ON
(L)	E243		E243	IDLE BOOST CONTROL OPERATE
(L)	E244		E244	CONTROL POWER
(L)	E245		E245	SAFETY POWER
(L)	E246		E246	BATTERY DC24V
(L)	E247		E247	(SPARE EP10)
(L)	E248		E248	PORT AC220V
(L)	E249		E249	CONTROL MAIN
(L)	E250		E250	CONTROL BACK UP
(L)	E251		E251	C/P CONTROL
(L)	E252		E252	(SPARE EP2)
(L)	E253		E253	(SPARE EP3)
(L)	E254		E254	(SPARE EP4)
(L)	E255		E255	(SPARE EP5)
(L)	E256		E256	(SPARE EP6)
(L)	E257		E257	(SPARE EP7)
(L)	E258		E258	(SPARE EP8)
(L)	E259		E259	(SPARE EP9)
NC	E260		E260	NC
NC	E261		E261	NC
NC	E262		E262	NC
NC	E263		E263	NC
NC	E264		E264	NC
NC	E265		E265	NC
NC	E266		E266	NC
NC	E267		E267	NC

4M3-194576

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

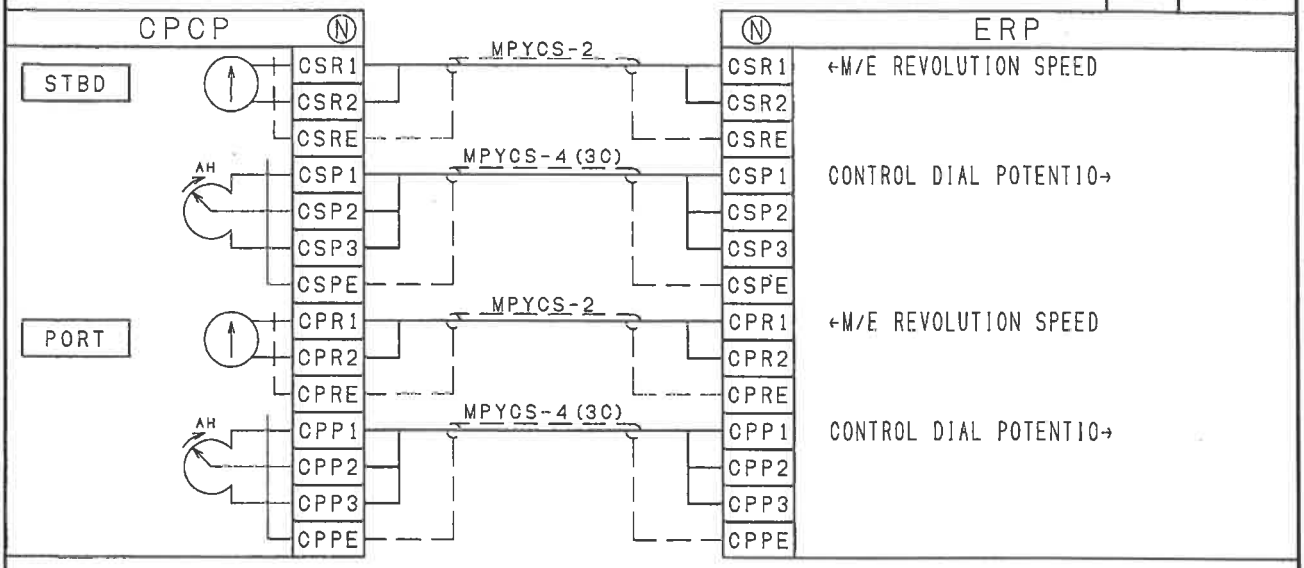
SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. 16.11.7 Y.H	REV.		Y. O

WIRING DIAGRAM FOR
REMOTE CONT. SYSTEM 12

DATE	1, SEP., 2016	DWG. NO.	4M3-194576

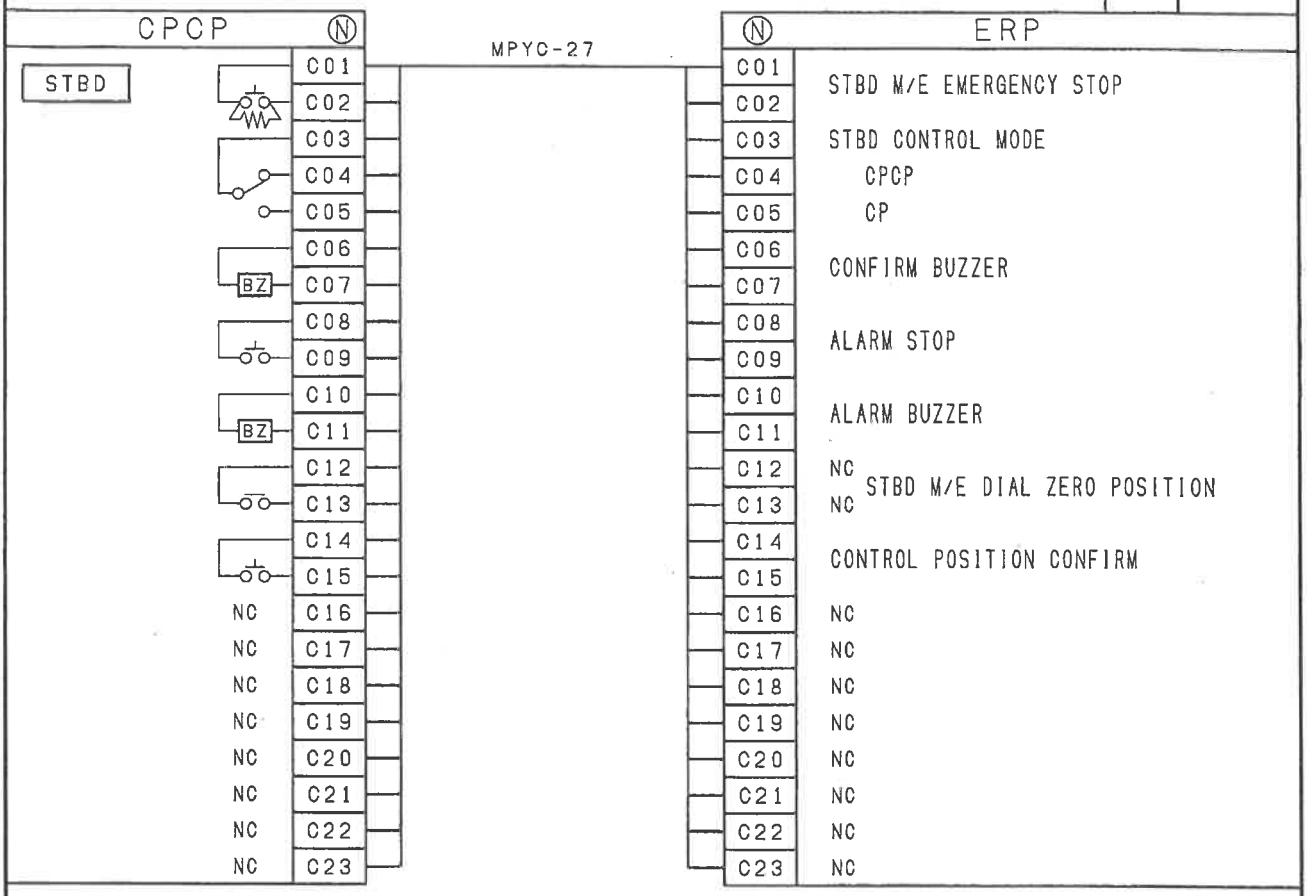
ECP	Ⓝ	MPYC-37	Ⓝ	ERP
PORT	NC	E271	E271	NC
Ⓛ	E272		E272	MANUAL EMERGENCY STOP
Ⓛ	E273		E273	M/E L.O. LOW PRESS SHUTDOWN
Ⓛ	E274		E274	M/E OVERSPEED SHUTDOWN
Ⓛ	E275		E275	M/E L.O. S/B PUMP OVERLAP RUN
Ⓛ	E276		E276	(SPARE EAP7)
Ⓛ	E277		E277	M/E F.W. LOW PRESS SLOWDOWN REQUEST
Ⓛ	E278		E278	M/E F.W. HI-TEMP SLOWDOWN REQUEST
Ⓛ	E279		E279	(SPARE EAP8)
Ⓛ	E280		E280	CONTROL AIR LOW PRESS
Ⓛ	E281		E281	M/E ABNORMAL
Ⓛ	E282		E282	POWER SUPPLY ABNORMAL
Ⓛ	E283		E283	R/G L.O. S/B PUMP OVERLAP RUN
Ⓛ	E284		E284	R/G L.O. LOW PRESS SHUTDOWN
Ⓛ	E285		E285	OTHER ABNORMAL
Ⓛ	E286		E286	(SPARE EAP9)
Ⓛ	E287		E287	(SPARE EAP1)
Ⓛ	E288		E288	R/G LO HI-TEMP SLOWDOWN REQUEST
Ⓛ	E289		E289	SHAFT BEARING HI-TEMP SLOWDOWN REQUEST
Ⓛ	E290		E290	M/E L.O. LOW PRESS
Ⓛ	E291		E291	(SPARE EAP5)
Ⓛ	E292		E292	(SPARE EAP6)
NC	E293		E293	NC
NC	E294		E294	NC
NC	E295		E295	NC
NC	E296		E296	NC
NC	E297		E297	NC

4M3-194577	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 13		
		SZ15045/15046 SZ15047/15048					
	SCALE	MANAGER	CHIEF	CHECK	DRAWN		
		REV. 16.11.9 Y.I	REV. 17.7		Y. O	DATE 1, SEP., 2016	DWG. NO. 4M3-194577



4M3-194578	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 14		
		SZ15045/15046 SZ15047/15048					
	SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	1, SEP., 2016
					Y. O	DWG. NO.	4M3-194578

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



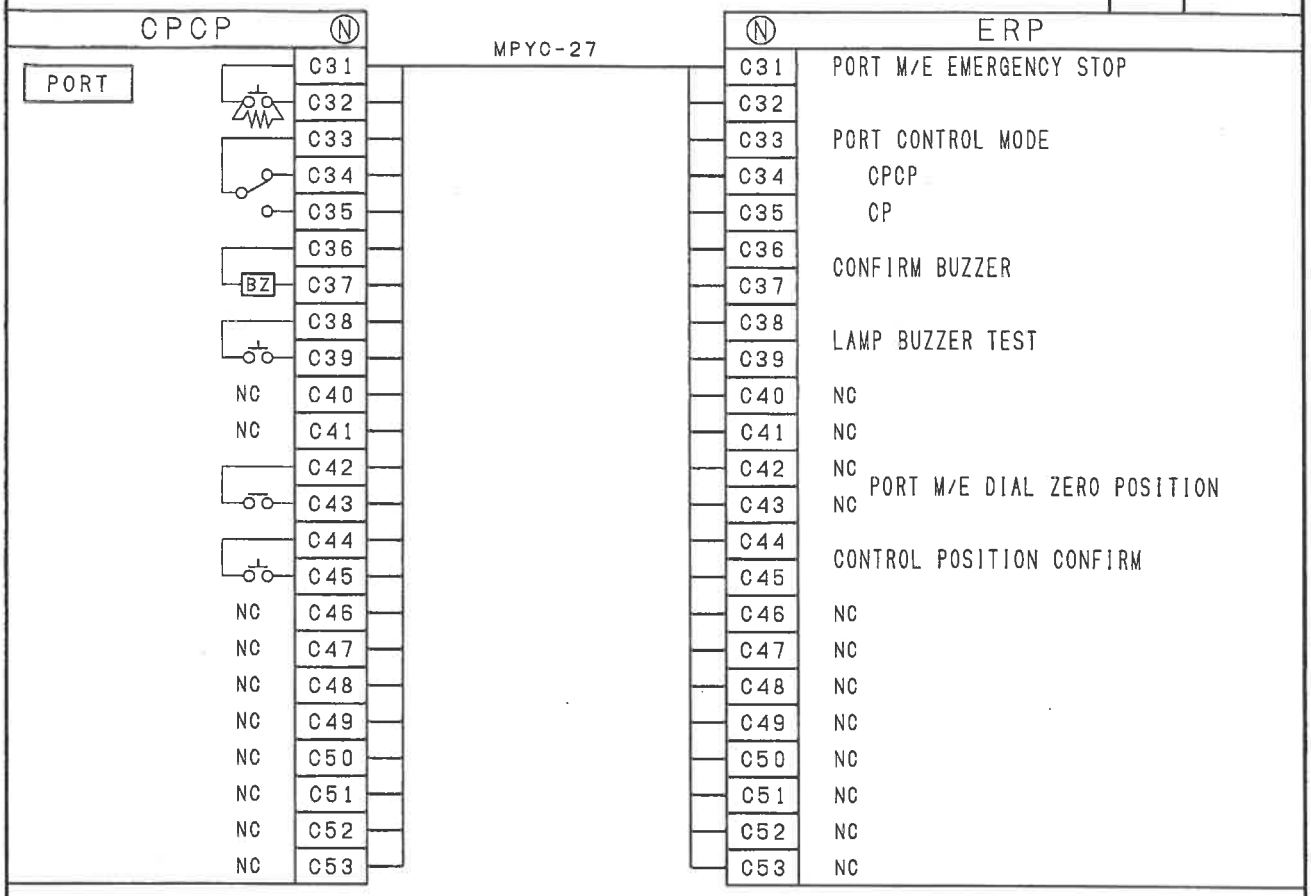
PAINTING COLOR	WORK NO.		WORK SHOP	
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER	CHIEF	CHECK	DRAWN
				Y. O

WIRING DIAGRAM FOR
REMOTE CONT. SYSTEM 15

DATE	1, SEP., 2016	DWG. NO.	4M3-194579

4M3-194579

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



4M3-194580	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 16
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	
				Y. O	DATE: 1, SEP., 2016 / DWG. NO.: 4M3-194580

CPCP	Ⓝ	MPYC-27	Ⓝ	ERP
STBD	C61		C61	
(L)	C62		C62	CP CONTROL
(L)	C63		C63	CPCP CONTROL
(L)	C64		C64	M/E RUN
(L)	C65		C65	NAVI MODE
(L)	C66		C66	CARGO MODE
(L)	C67		C67	DIAL ZERO POSITION
(L)	C68		C68	(SPARE CPS1)
NC	C69		C69	NC
NC	C70		C70	NC
NC	C71		C71	NC
NC	C72		C72	NC
NC	C73		C73	NC
NC	C74		C74	NC
NC	C75		C75	NC
NC	C76		C76	NC
NC	C77		C77	NC
NC	C78		C78	NC
NC	C79		C79	NC
NC	C80		C80	NC
NC	C81		C81	NC
(L)	C82		C82	R/G L.O. LOW PRESS SHUTDOWN
(L)	C83		C83	ALARM UNIT ABNORMAL
(L)	C84		C84	M/E OVERSPEED SHUTDOWN
(L)	C85		C85	M/E LO LOW PRESS SHUTDOWN
(L)	C86		C86	M/E EMERGENCY STOP
	C87		C87	

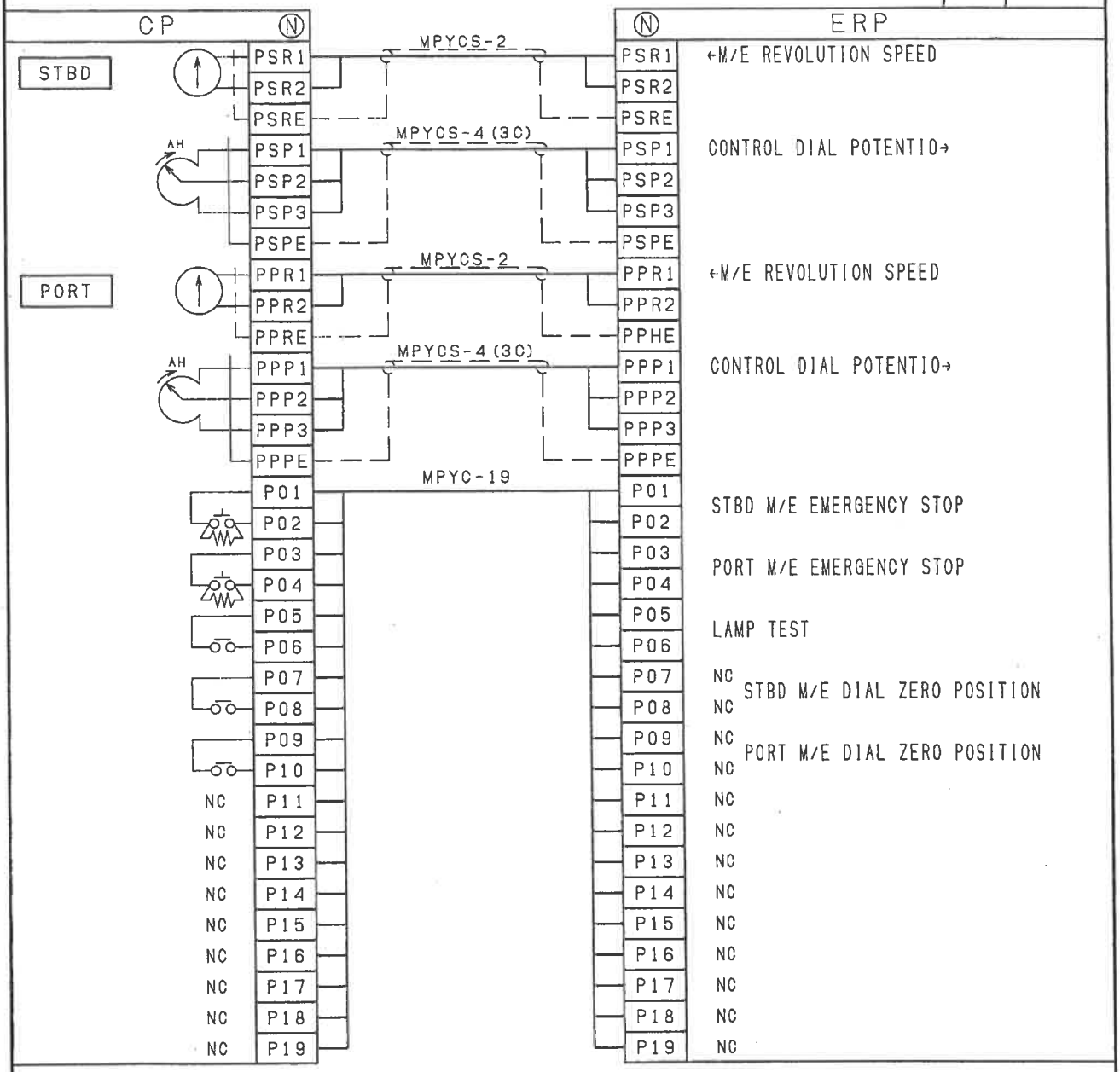
4M3-194581	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 17
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	
			REV. 16.11.07 K.T	Y. O	DATE: 1, SEP., 2016 / DWG. NO.: 4M3-194581

CPCP			MPYC-27		ERP	
PORT		C91			C91	
	(L)	C92			C92	CP CONTROL
	(L)	C93			C93	CPCP CONTROL
	(L)	C94			C94	M/E RUN
	(L)	C95			C95	NAVI MODE
	(L)	C96			C96	CARGO MODE
	(L)	C97			C97	DIAL ZERO POSITION
	(L)	C98			C98	(SPARE CPP1)
	NC	C99			C99	NC
	NC	C100			C100	NC
	NC	C101			C101	NC
	NC	C102			C102	NC
	NC	C103			C103	NC
	NC	C104			C104	NC
	NC	C105			C105	NC
	NC	C106			C106	NC
	NC	C107			C107	NC
	NC	C108			C108	NC
	NC	C109			C109	NC
	NC	C110			C110	NC
	NC	C111			C111	NC
	(L)	C112			C112	R/G L.O. LOW PRESS SHUTDOWN
	(L)	C113			C113	(SPARE CPPA1)
	(L)	C114			C114	M/E OVERSPEED SHUTDOWN
	(L)	C115			C115	M/E LO LOW PRESS SHUTDOWN
	(L)	C116			C116	M/E EMERGENCY STOP
	NC	C117			C117	NC

PAINTING COLOR	WORK NO.		WORK SHOP		WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 18			
	SZ15045/15046 SZ15047/15048							
SCALE	MANAGER	CHIEF	CHECK	DRAWN				
	SEV 16.11.7	SEV 16.11.07		Y. O	DATE	1, SEP., 2016	DWG. NO.	4M3-194582

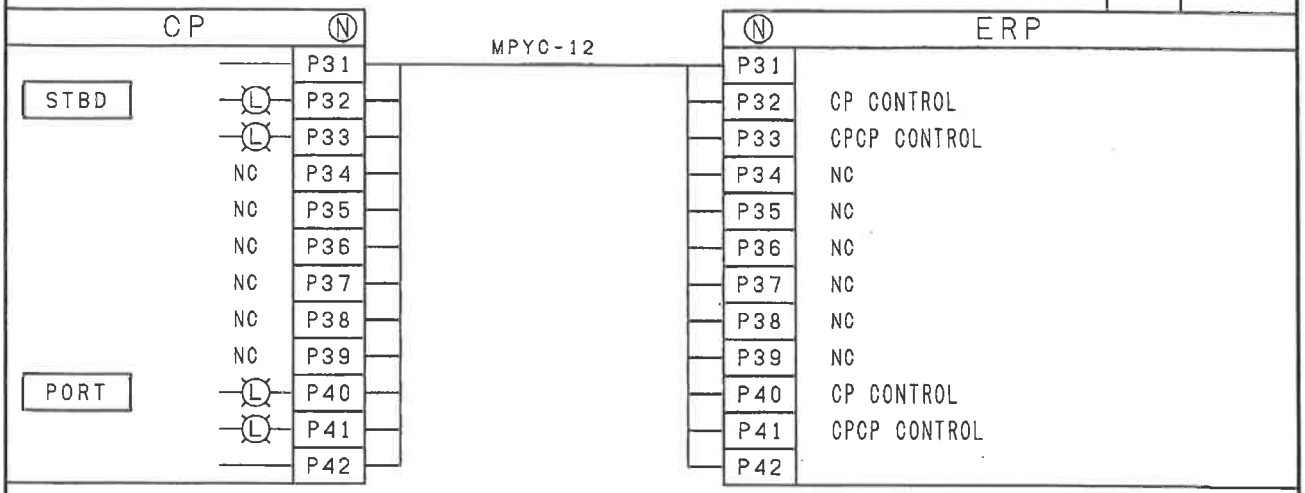
4M3-194582

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



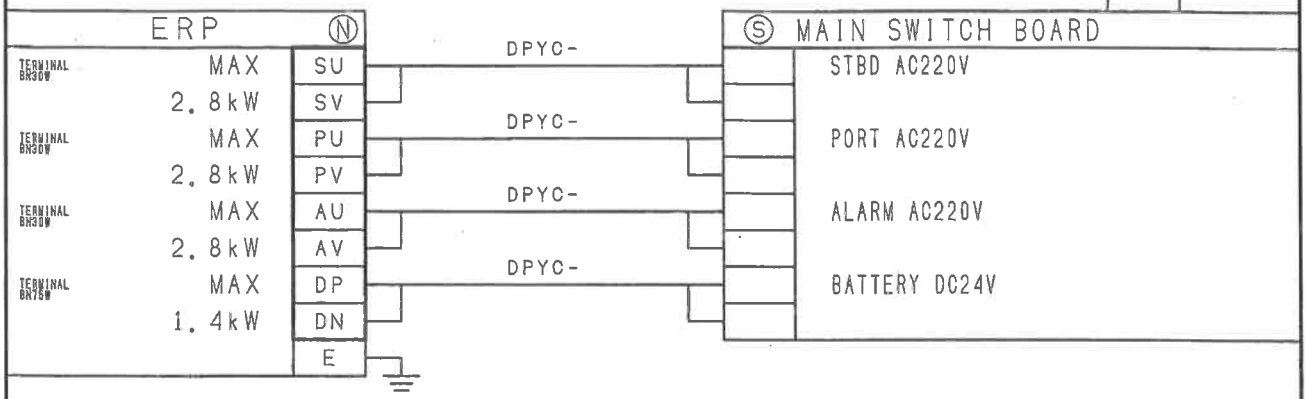
4M3-194583	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 19
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	DATE: 1, SEP., 2016 / DWG. NO.: 4M3-194583
				Y. O	

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



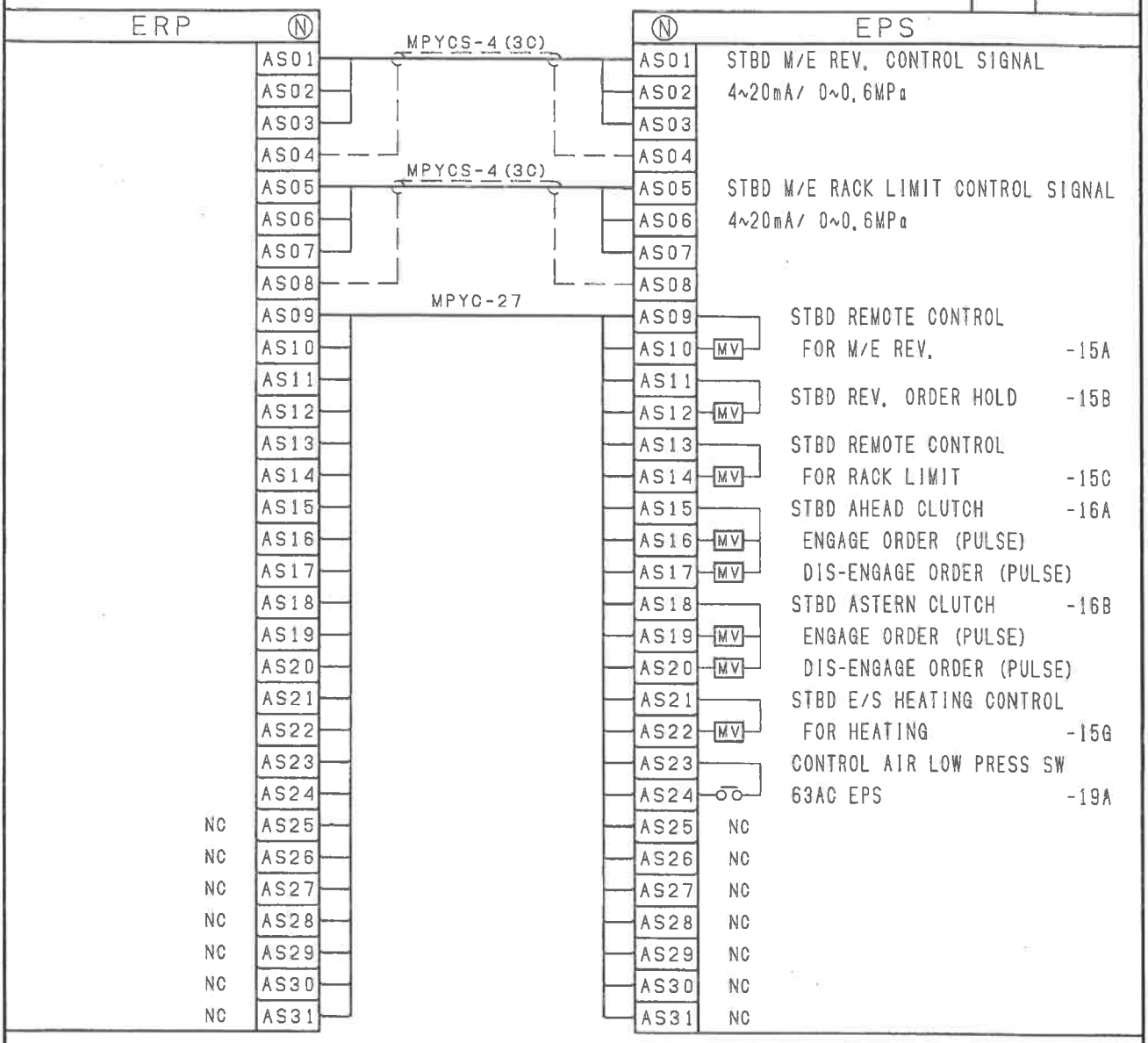
PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 20	
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER	CHIEF	CHECK / DRAWN		
			Y. O	DATE: 1, SEP., 2016	DWG. NO.: 4M3-194584

4M3-194584



4M3-194585	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 21
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	
		REV. '16.11.7 Y.H		Y. O	DATE: 5, SEP., 2016 / DWG. NO.: 4M3-194585

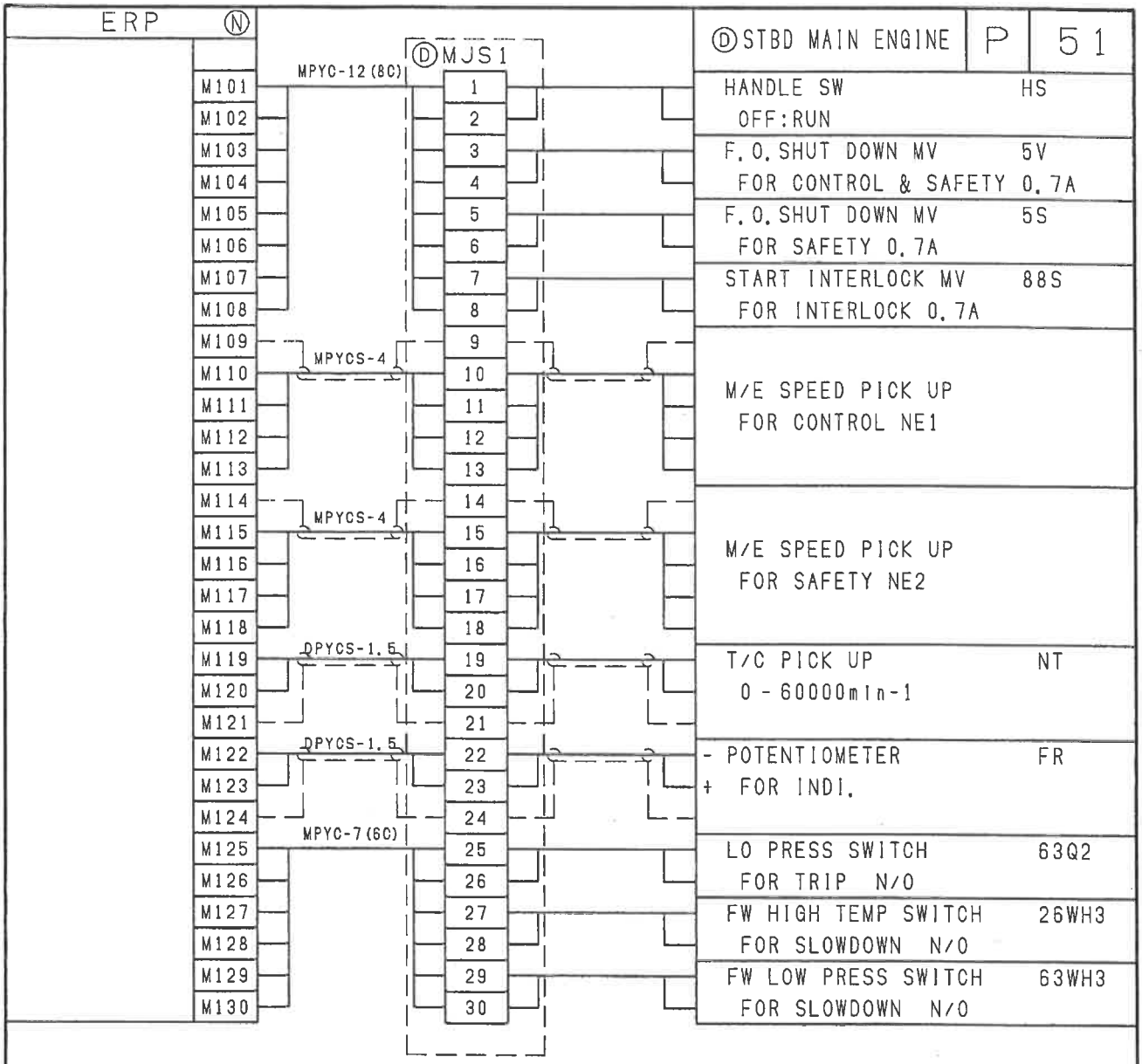
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.		WORK SHOP		WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 22			
	SZ15045/15046 SZ15047/15048							
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	1, SEP., 2016	DWG. NO.	4M3-194586
	REV. '16.11.7			Y. O				

4M3-194586

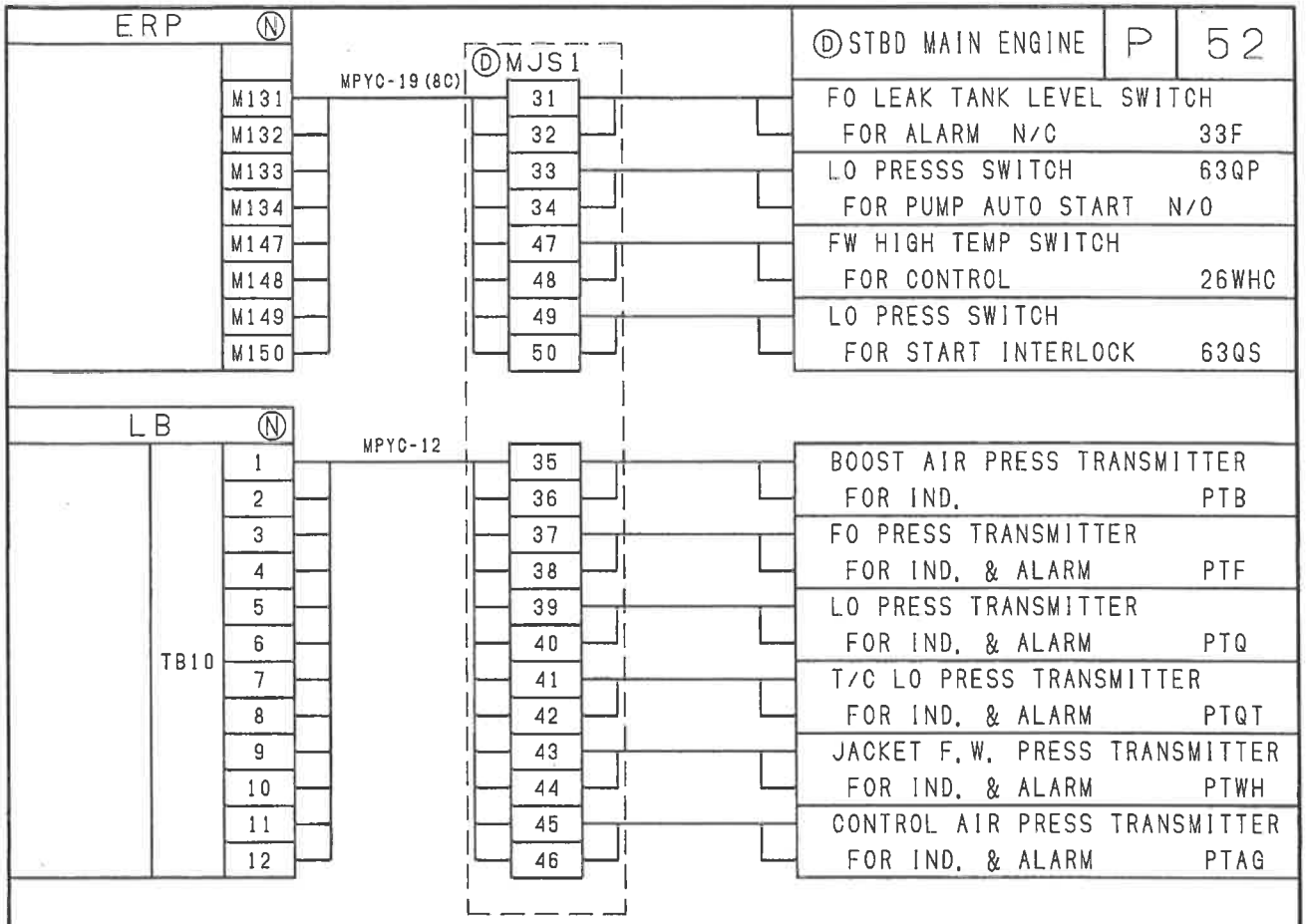
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 23			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	DWG. NO.
	REV. '16.11.7 Y.H			Y.O	1, SEP., 2016	4M3-194587

4M3-194587

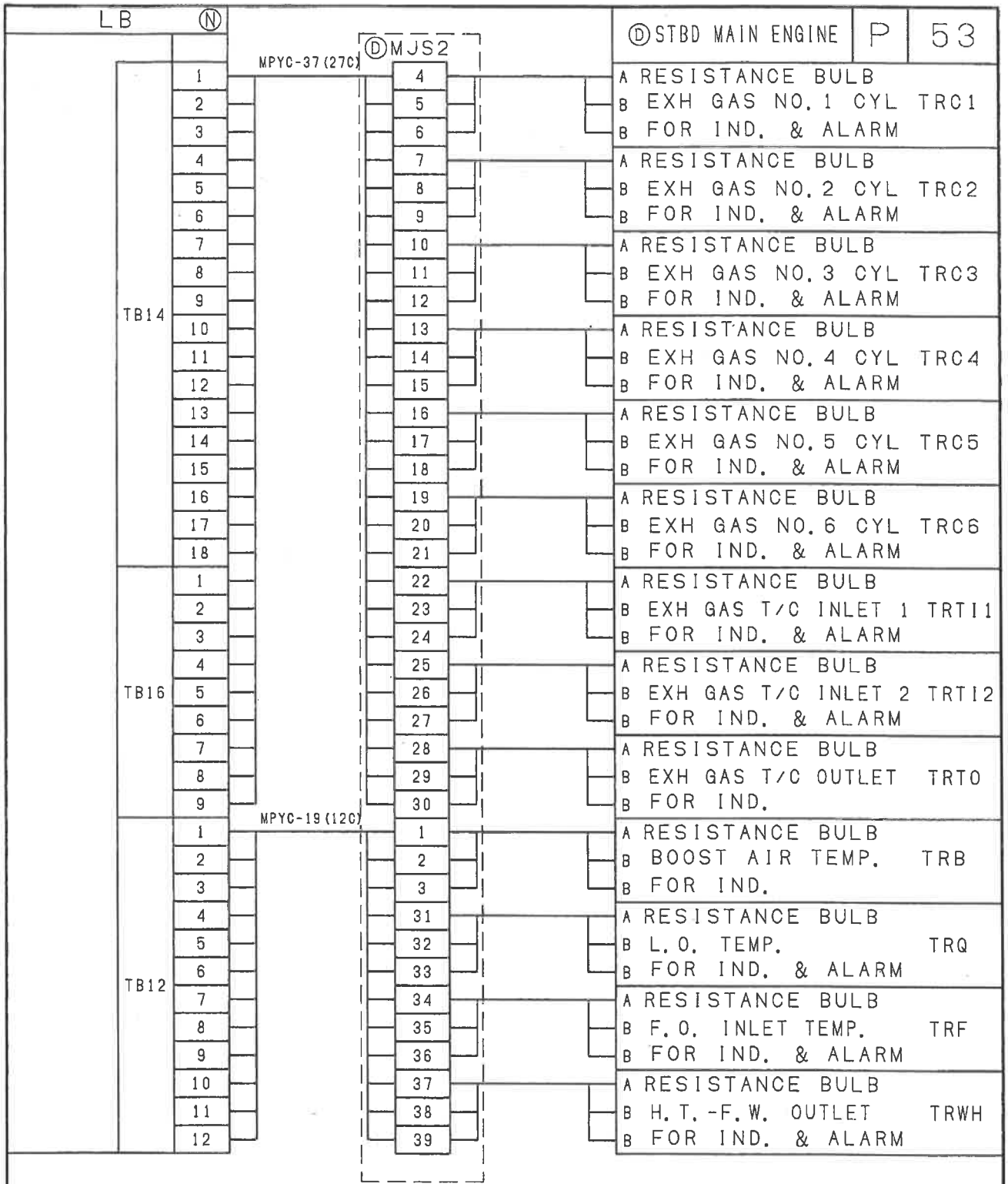
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



4M3-194588	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 24
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK	DRAWN
		REV. 19.1.9 Y.H			Y. O

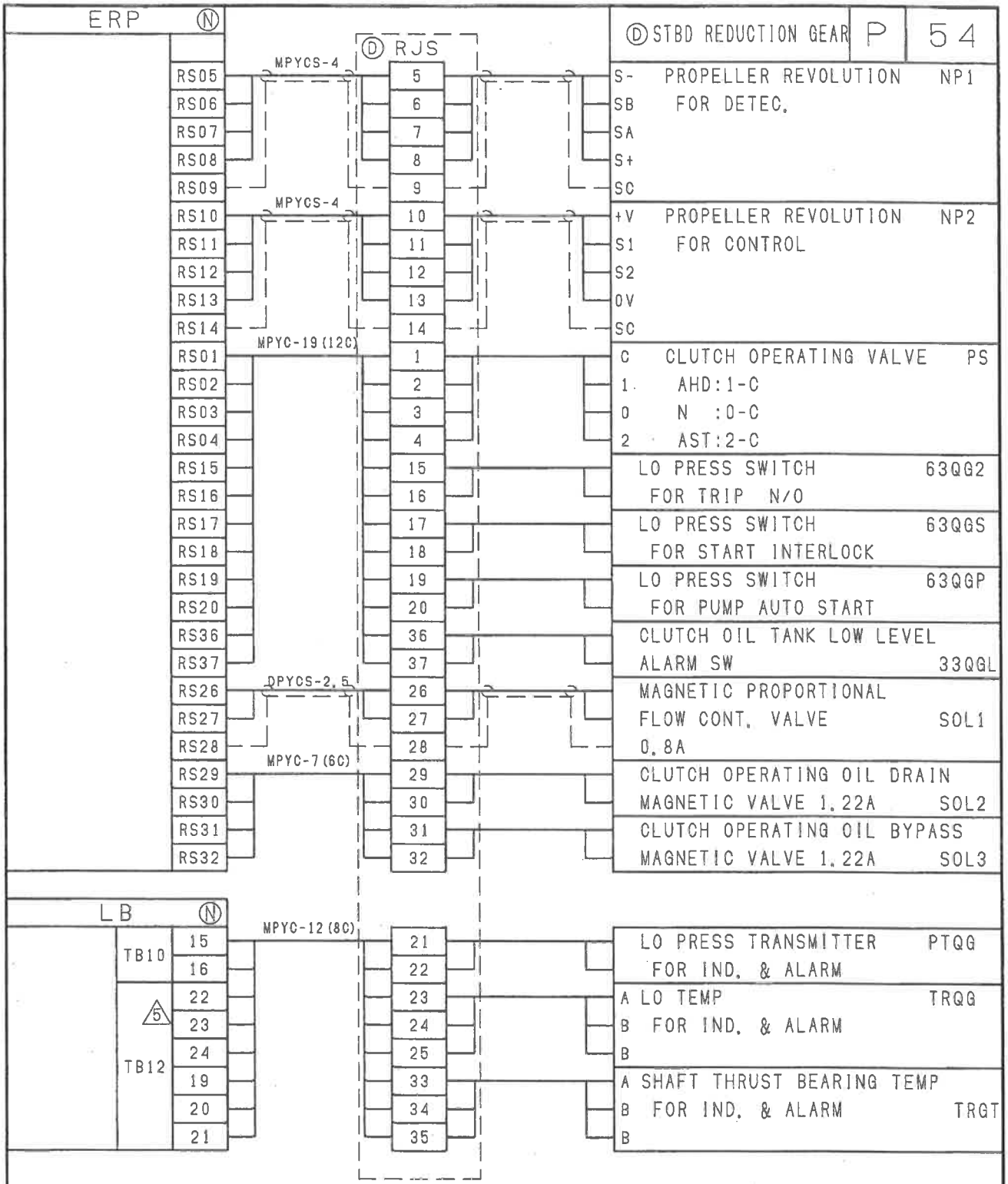
DATE	2, SEP., 2016	DWG. NO.	4M3-194588

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



4M3-194589	PAINTING COLOR	WORK NO.	WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 25			
		SZ15045/15046 SZ15047/15048					
	SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	1, SEP., 2016
		REV.			Y. O	DWG. NO.	4M3-194589

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

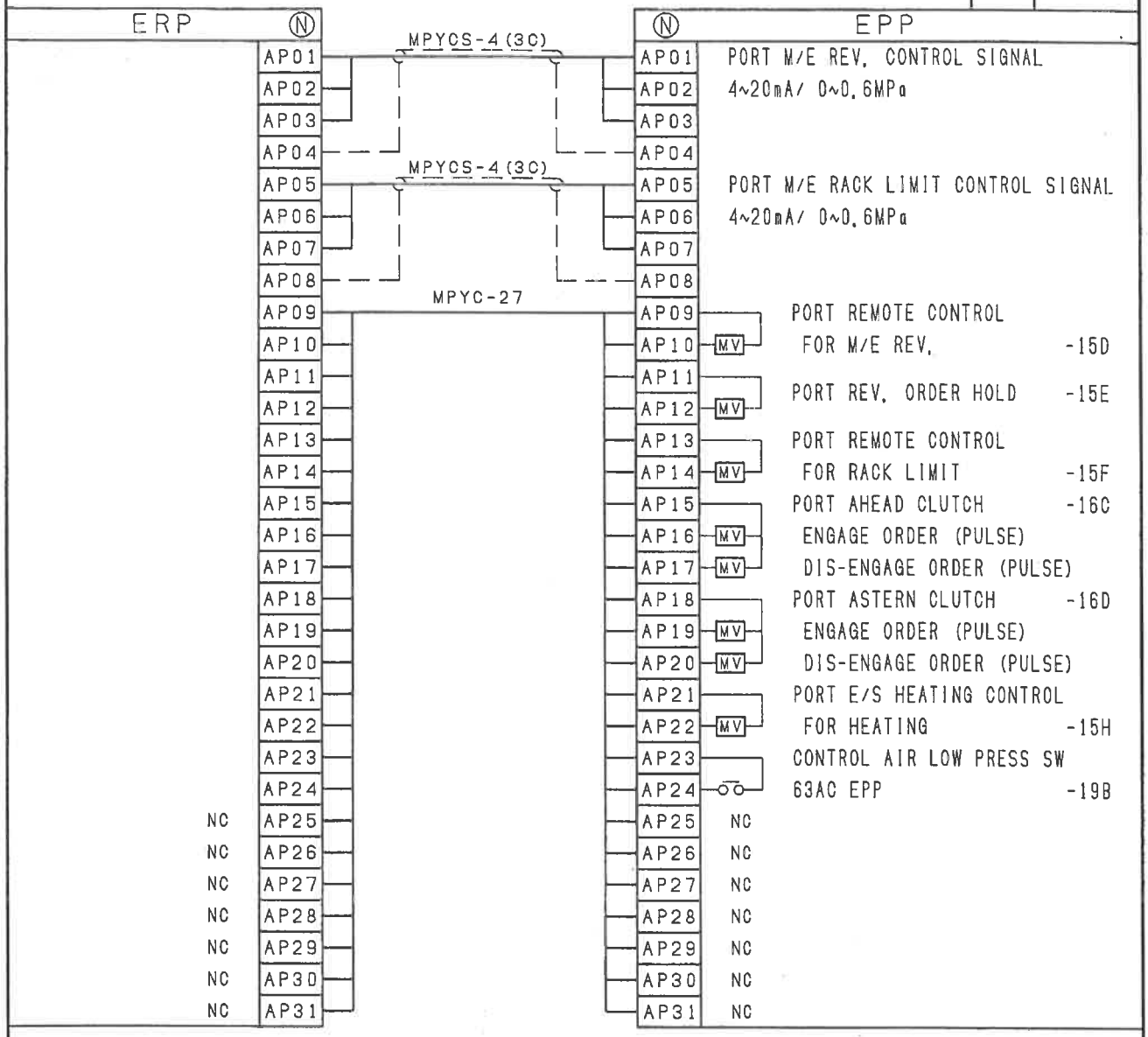


5 6, DEC., 2016

PAINTING COLOR	WORK NO.	WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 26	
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. '16.12 6 Y.H	REV. '16.12.06 K.T	REV. '16.12.05 K.T	Y. O
	DATE	2, SEP., 2016	DWG. NO.	4M3-194590

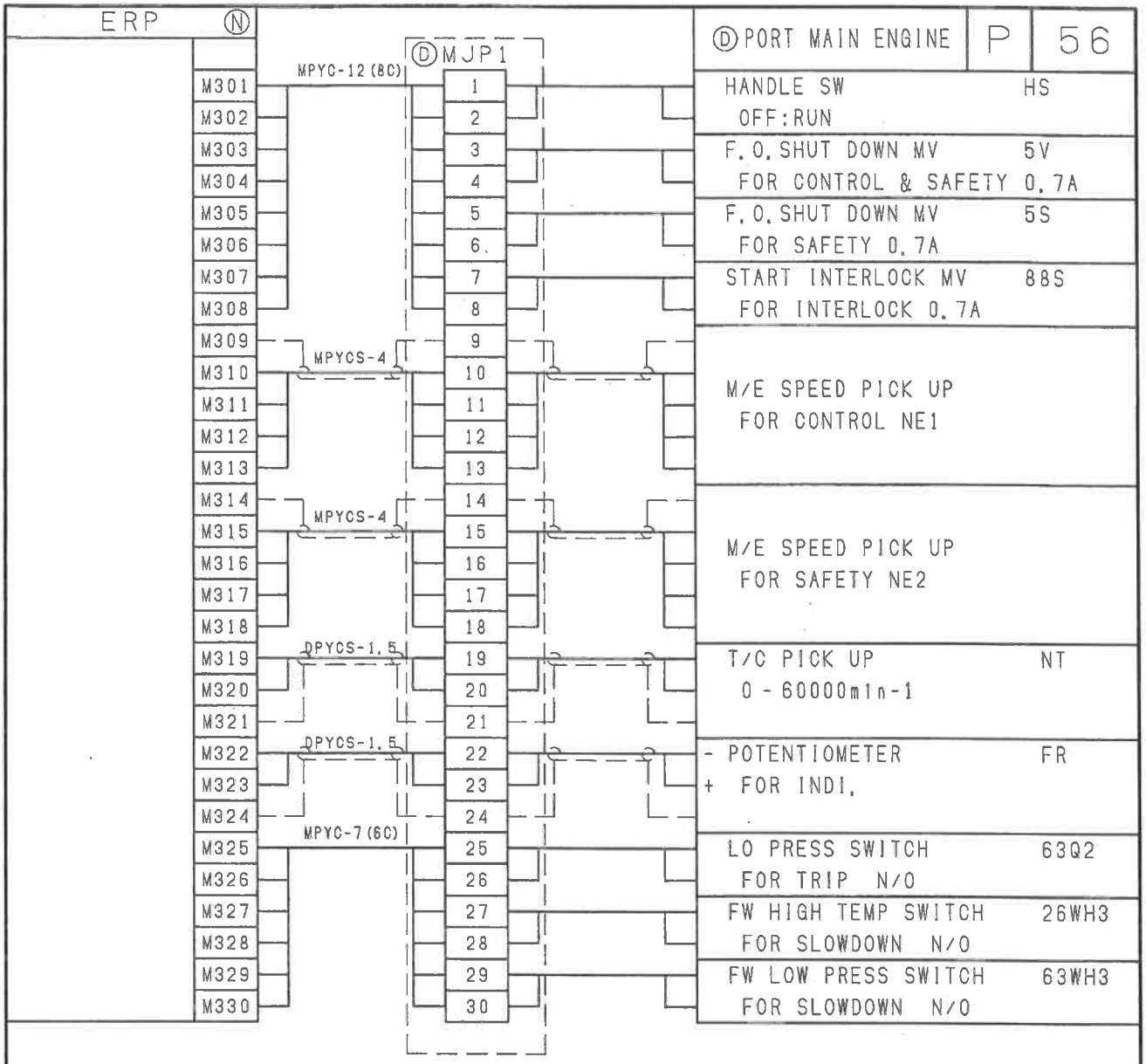
4M3-194590

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 27	
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER	CHIEF	CHECK	DRAWN	
		REV. 15.11.07		Y. O	DATE 1, SEP., 2016 / DWG. NO. 4M3-194591

4M3-194591



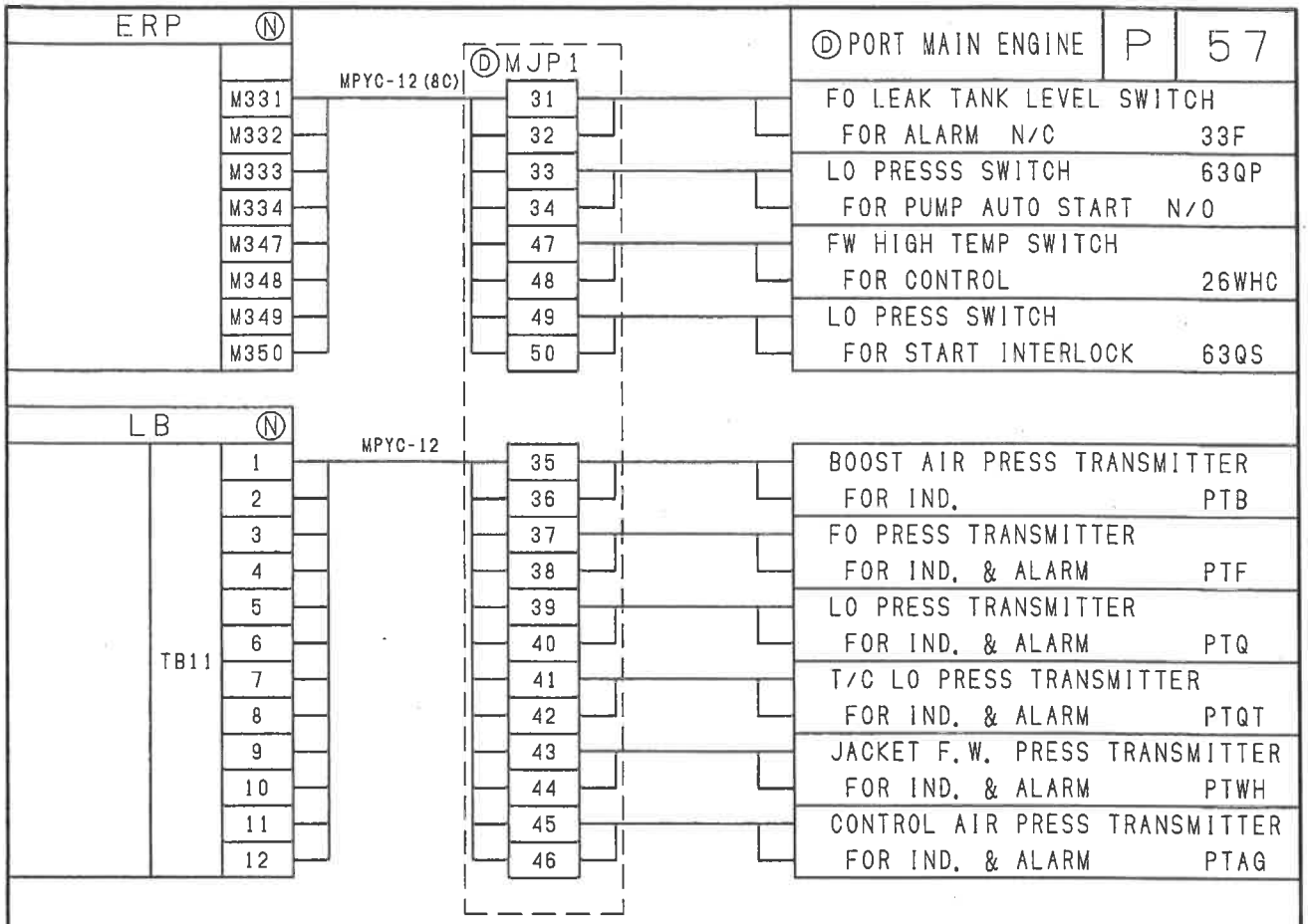
ⒹPORT MAIN ENGINE	P	56
HANDLE SW OFF:RUN		HS
F.O.SHUT DOWN MV FOR CONTROL & SAFETY 0.7A		5V
F.O.SHUT DOWN MV FOR SAFETY 0.7A		5S
START INTERLOCK MV FOR INTERLOCK 0.7A		88S
M/E SPEED PICK UP FOR CONTROL NE1		
M/E SPEED PICK UP FOR SAFETY NE2		
T/C PICK UP 0-60000min-1		NT
- POTENTIOMETER + FOR INDI.		FR
LO PRESS SWITCH FOR TRIP N/O		63Q2
FW HIGH TEMP SWITCH FOR SLOWDOWN N/O		26WH3
FW LOW PRESS SWITCH FOR SLOWDOWN N/O		63WH3

PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK	DRAWN
	REV. '16.11.7 Y.H / REV. '16.11.17 K.T		Y.O

WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 28

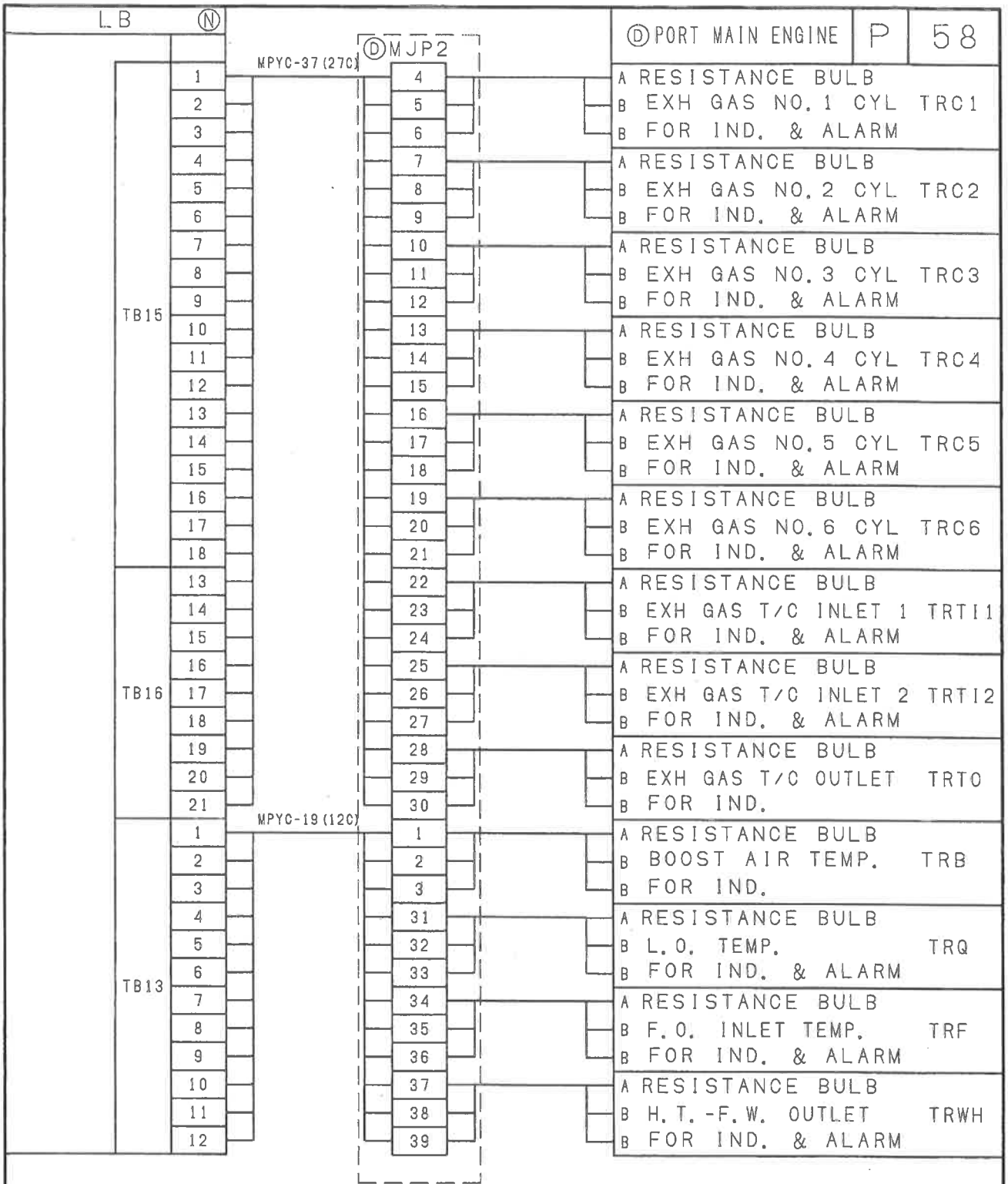
DATE	1, SEP., 2016	DWG. NO.	4M3-194592

4M3-194592



4M3-194593	PAINTING COLOR	WORK NO.		WORK SHOP		WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 29		
		SZ15045/15046 SZ15047/15048						
	SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	2, SEP., 2016	DWG. NO. 4M3-194593
		REV. 16.11.7			Y. O			

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

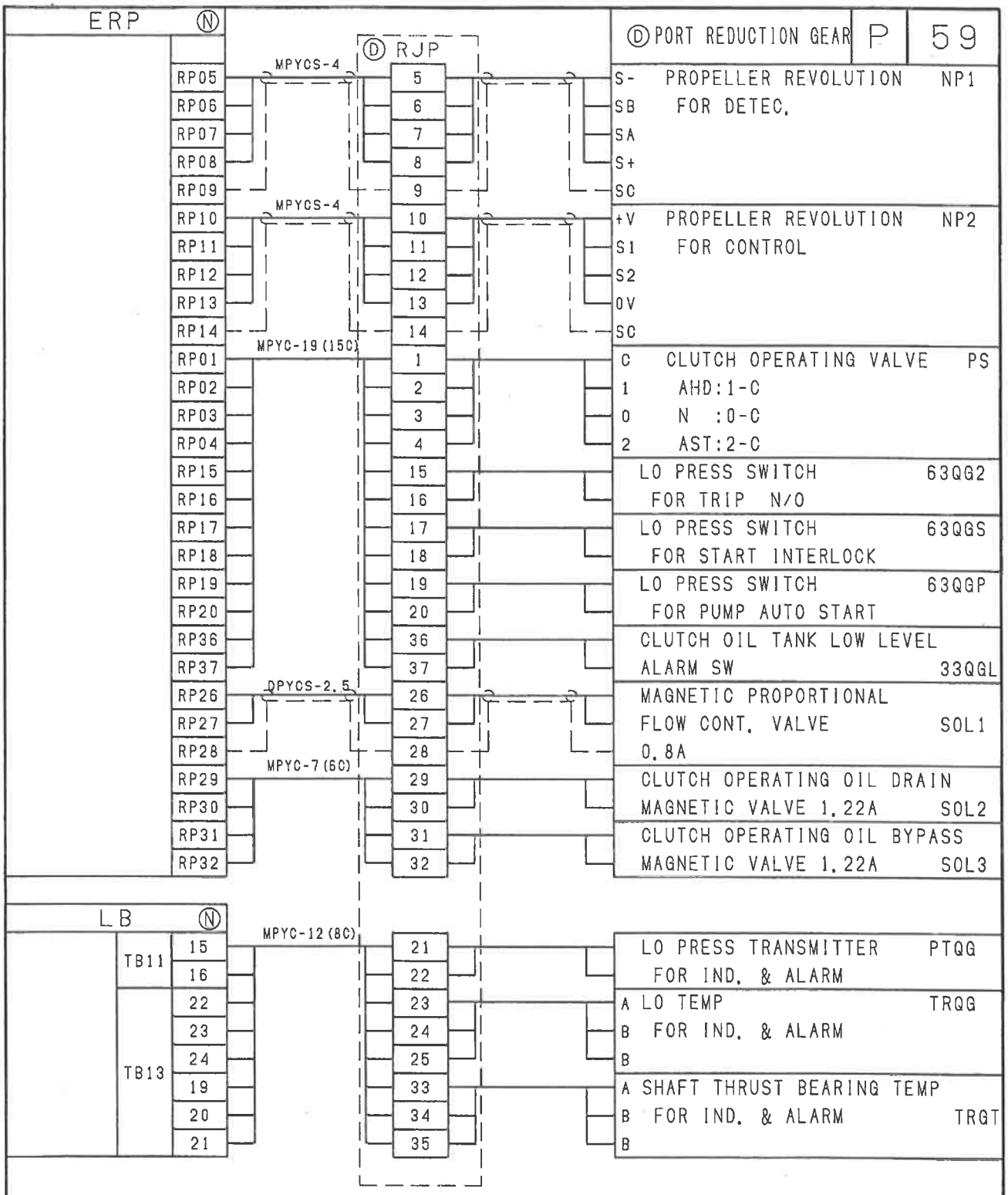
SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. '16.11.7 Y.H			Y. O

WIRING DIAGRAM FOR
REMOTE CONT. SYSTEM 30

DATE	1, SEP., 2016	DWG. NO.	4M3-194594

4M3-194594

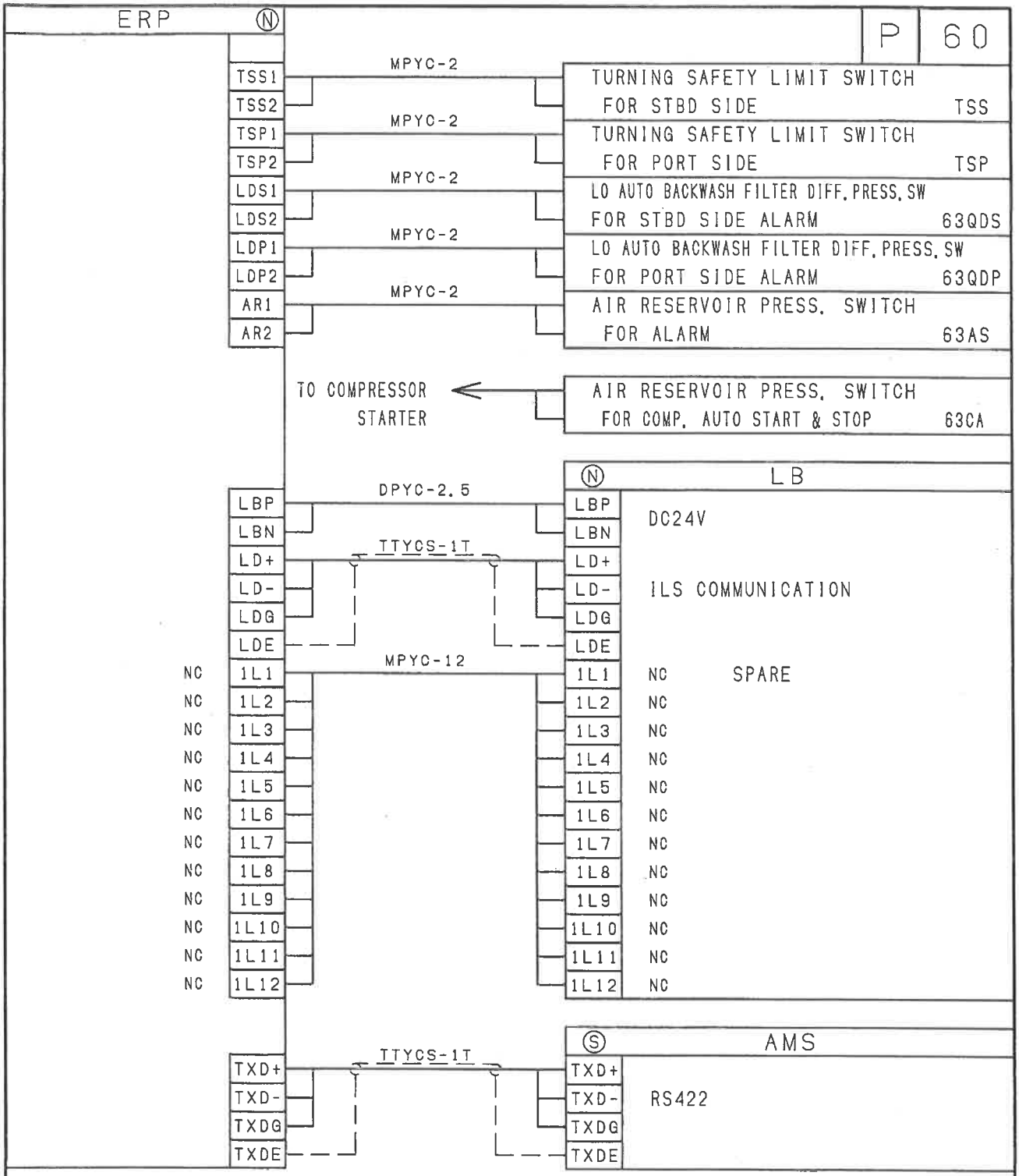
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 31
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER	CHIEF	CHECK / DRAWN	
	REV. 16.11.7 Y.H		Y. O	DATE 2, SEP., 2016 / DWG. NO. 4M3-194595

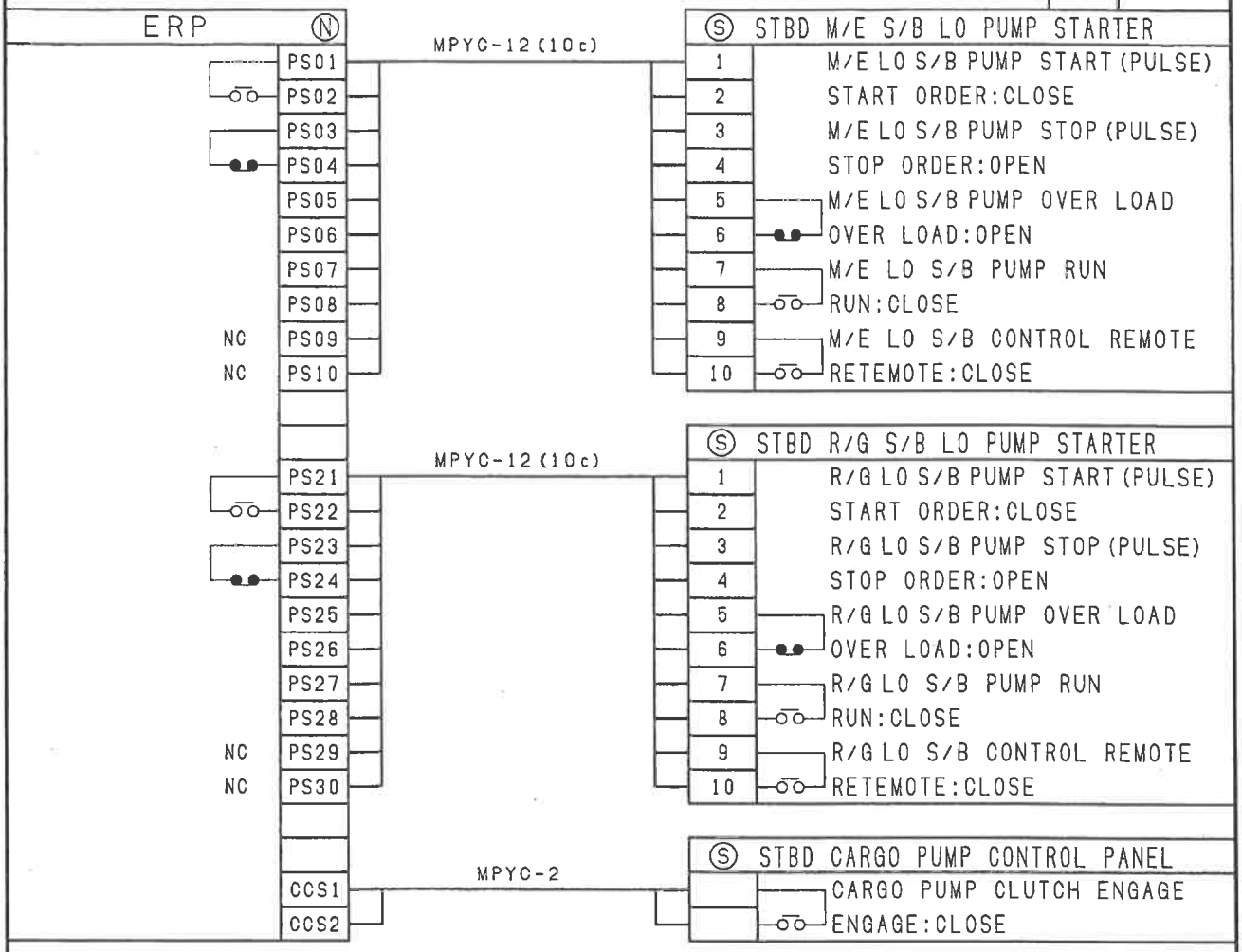
4M3-194595

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



4M3-194596	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 32
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK / DRAWN	DATE: 1, SEP., 2016 / DWG. NO.: 4M3-194596
				Y. O	

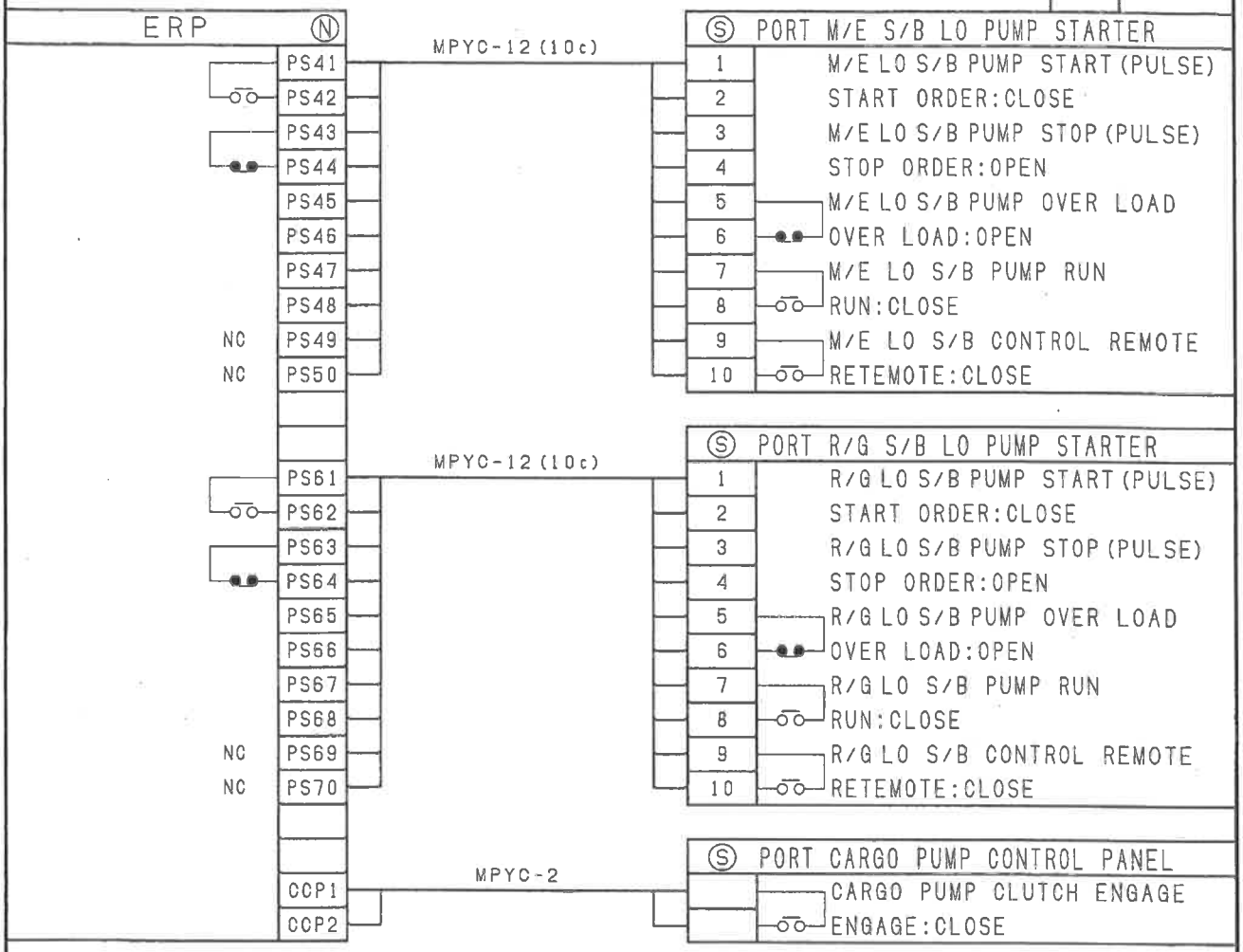
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



4M3-194597	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 33
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK	DRAWN
					Y. O

DATE	1, SEP., 2016	DWG. NO.	4M3-194597

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



4M3-194598	PAINTING COLOR	WORK NO.		WORK SHOP	WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 34
		SZ15045/15046 SZ15047/15048			
	SCALE	MANAGER	CHIEF	CHECK	DRAWN
					Y. O
		DATE	1, SEP., 2016	DWG. NO.	4M3-194598

STBD MEASURING TERMINAL

P 62A

ERP	Ⓝ	
STBD HANDLE POSITION 4~20mA/1~5V	MS01	NC
	MS02	NC
STBD M/E REV. ORDER 4~20mA/1~5V	MS03	NC
	MS04	NC
STBD M/E REVOLUTION 4~20mA/1~5V	MS05	NC
	MS06	NC
STBD PROPELLER REV. 4~20mA/1~5V	MS07	NC
	MS08	NC
STBD T/C REVOLUTION 4~20mA/1~5V	MS09	NC
	MS10	NC
STBD FO RACK 4~20mA/1~5V	MS11	NC
	MS12	NC
STBD DRAIN VALVE CLOSE:ON	MS13	NC
	MS14	NC
STBD BYPASS VALVE CLOSE:ON	MS15	NC
	MS16	NC
STBD AHD CL-ENGAGE ENGAGE:ON	MS17	NC
	MS18	NC
STBD AST CL-ENGAGE ENGAGE:ON	MS19	NC
	MS20	NC
NC	MS21	NC
NC	MS22	NC
NC	MS23	NC
NC	MS24	NC

PORT MEASURING TERMINAL

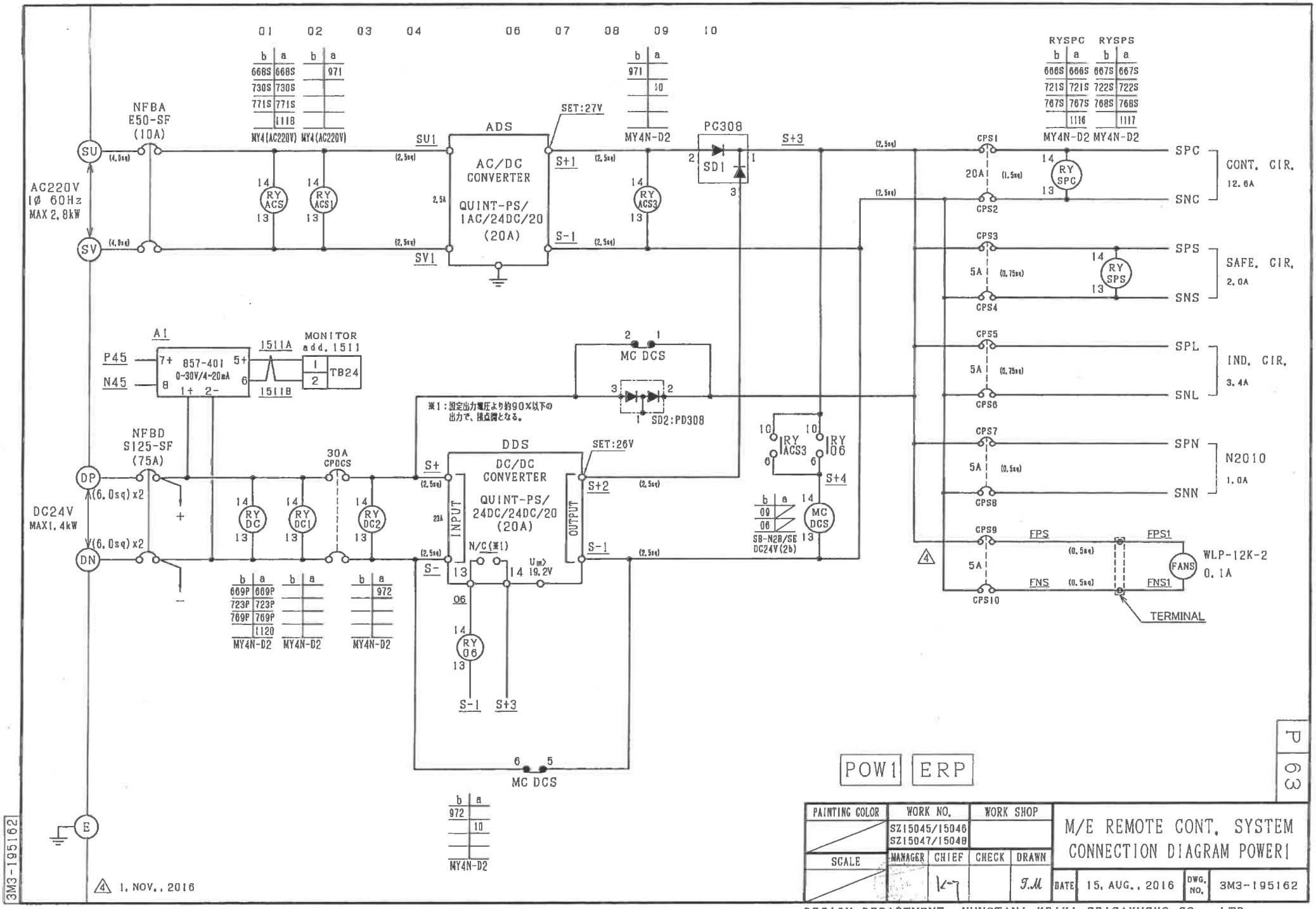
PORT HANDLE POSITION 4~20mA/1~5V	MP01	NC
	MP02	NC
PORT M/E REV. ORDER 4~20mA/1~5V	MP03	NC
	MP04	NC
PORT M/E REVOLUTION 4~20mA/1~5V	MP05	NC
	MP06	NC
PORT PROPELLER REV. 4~20mA/1~5V	MP07	NC
	MP08	NC
PORT T/C REVOLUTION 4~20mA/1~5V	MP09	NC
	MP10	NC
PORT FO RACK 4~20mA/1~5V	MP11	NC
	MP12	NC
PORT DRAIN VALVE CLOSE:ON	MP13	NC
	MP14	NC
PORT BYPASS VALVE CLOSE:ON	MP15	NC
	MP16	NC
PORT AHD CL-ENGAGE ENGAGE:ON	MP17	NC
	MP18	NC
PORT AST CL-ENGAGE ENGAGE:ON	MP19	NC
	MP20	NC
NC	MP21	NC
NC	MP22	NC
NC	MP23	NC
NC	MP24	NC

PAINTING COLOR	WORK NO.		WORK SHOP	
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER	CHIEF	CHECK	DRAWN
				Y. O

WIRING DIAGRAM FOR REMOTE CONT. SYSTEM 35

DATE 1, SEP., 2016 DWG. NO. 4M3-194599

4M3-194599



△4 1, NOV., 2016

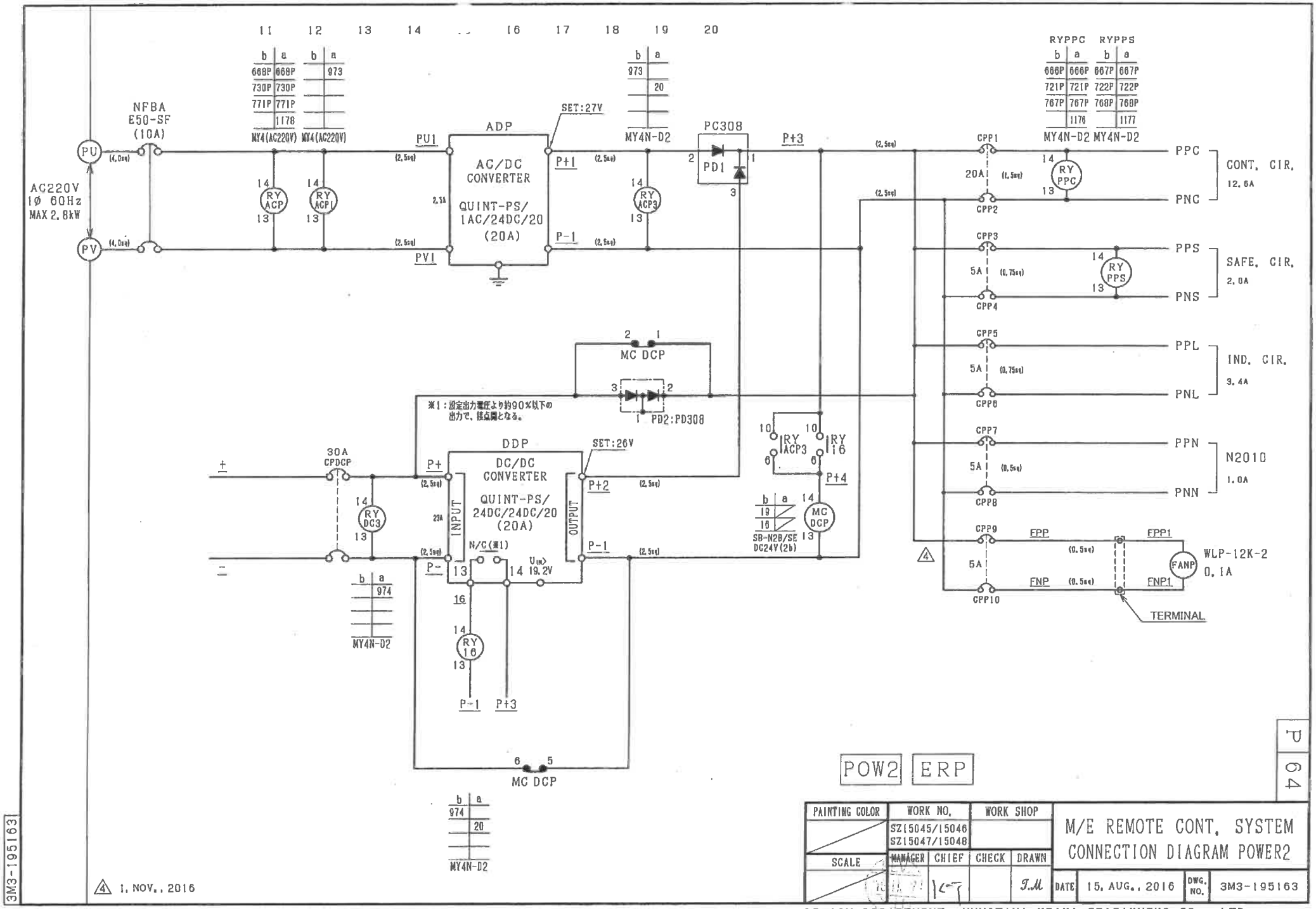
3M3-195162

POW1 ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM POWER1			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	DWG. NO.
				T.M	15, AUG., 2016	3M3-195162

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 63



POW2 ERP

P 64

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

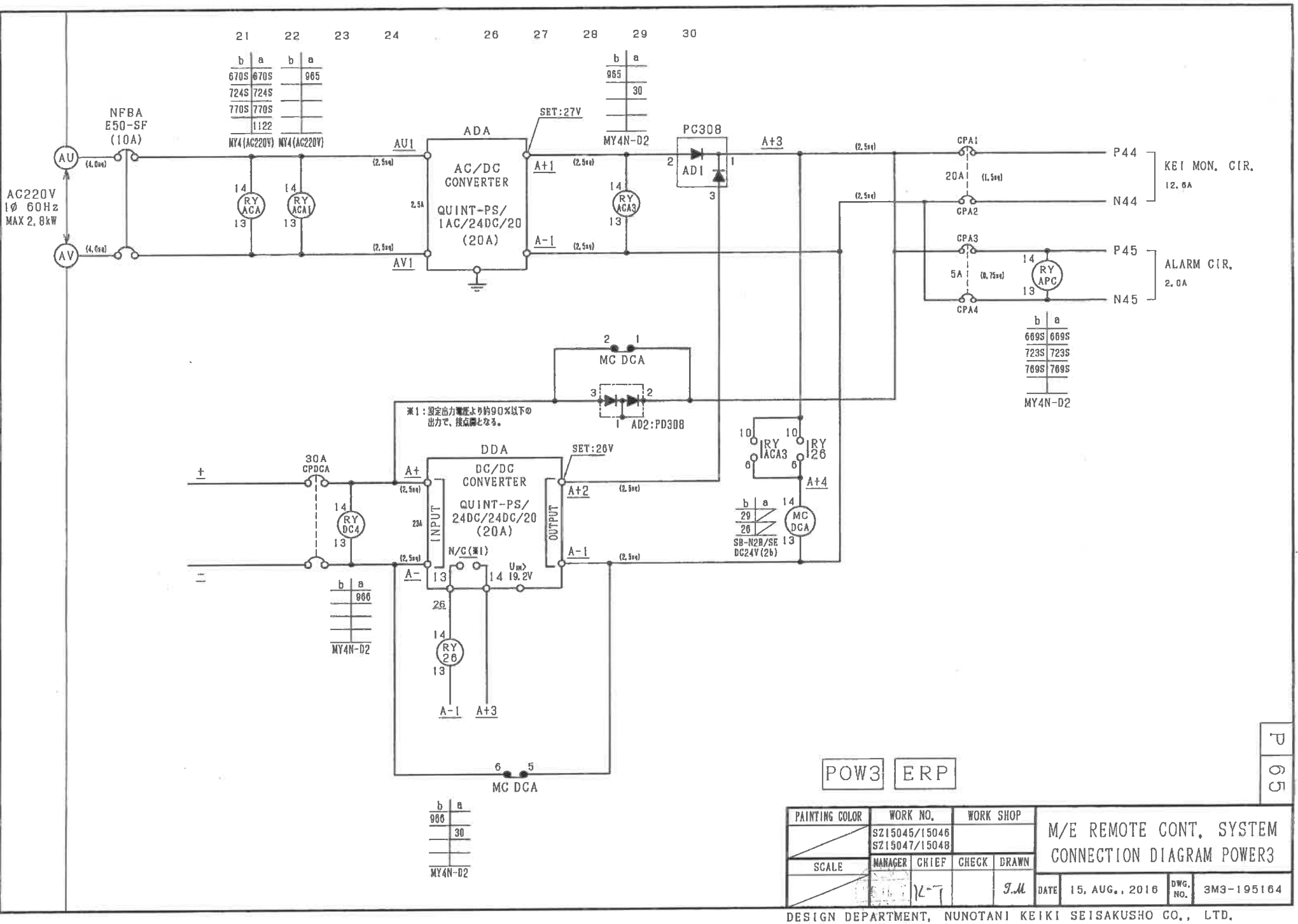
SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM POWER2

DATE 15. AUG., 2016 | DWG. NO. 3M3-195163

3M3-195163

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



POW3 ERP

P 65

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM POWER3

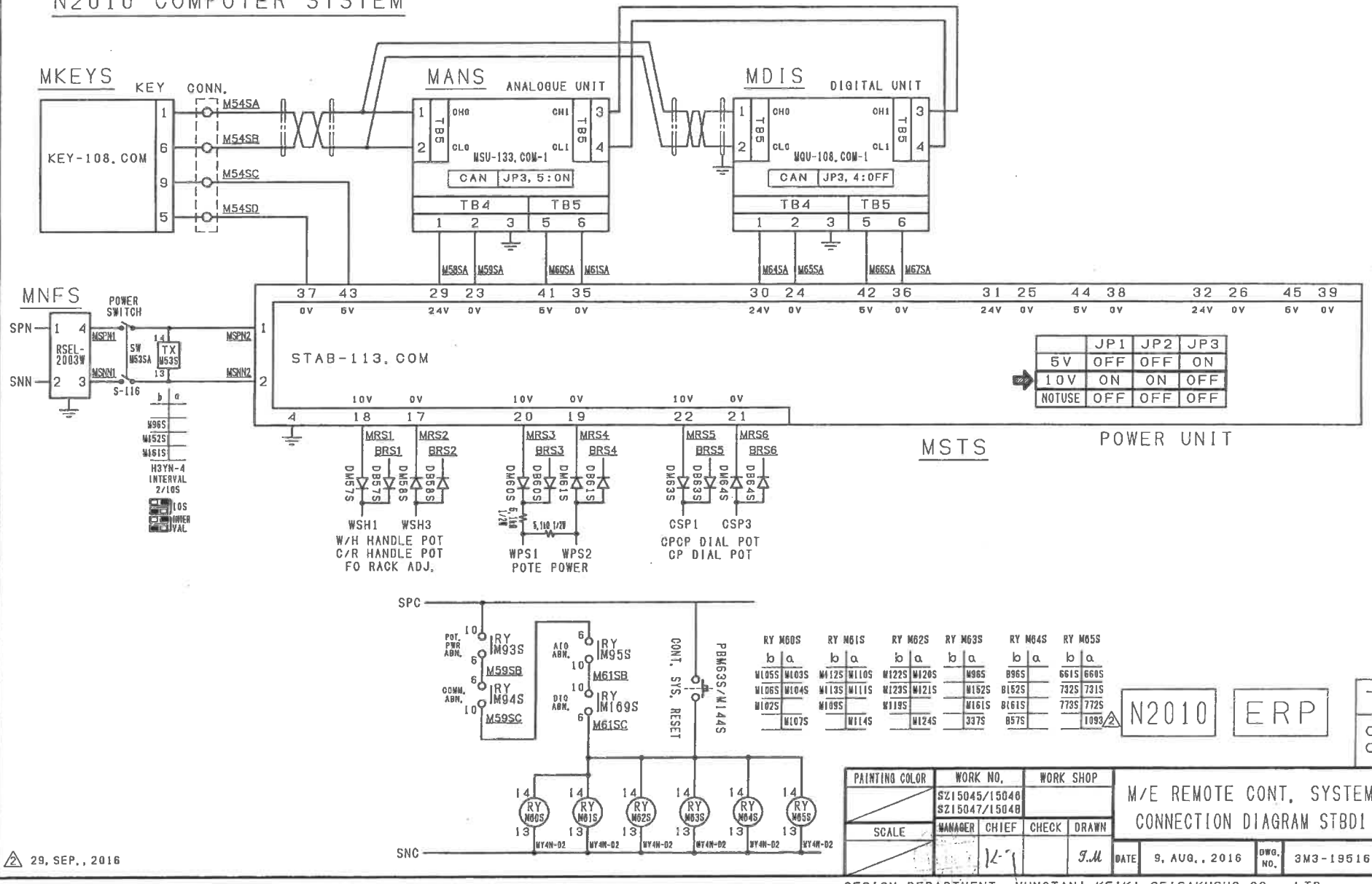
DATE 15. AUG., 2016 | DWG. NO. 3M3-195164

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195164

M51S M52S M53S M54S M55S M56S M57S M58S M59S M60S M61S M62S M63S M64S M65S M66S M67S M68S M69S M70S M71S M72S M73S M74S M75S

N2010 COMPUTER SYSTEM



RY M60S		RY M61S		RY M62S		RY M63S		RY M64S		RY M65S	
b	a	b	a	b	a	b	a	b	a	b	a
M105S	M103S	M112S	M110S	M122S	M120S		M96S	B96S		661S	660S
M106S	M104S	M113S	M111S	M123S	M121S		M152S	B152S		732S	731S
M102S		M109S		M119S			M161S	B161S		773S	772S
	M107S		M114S		M124S		337S	B57S			1093

N2010 ERP

P 99

3M3-195165

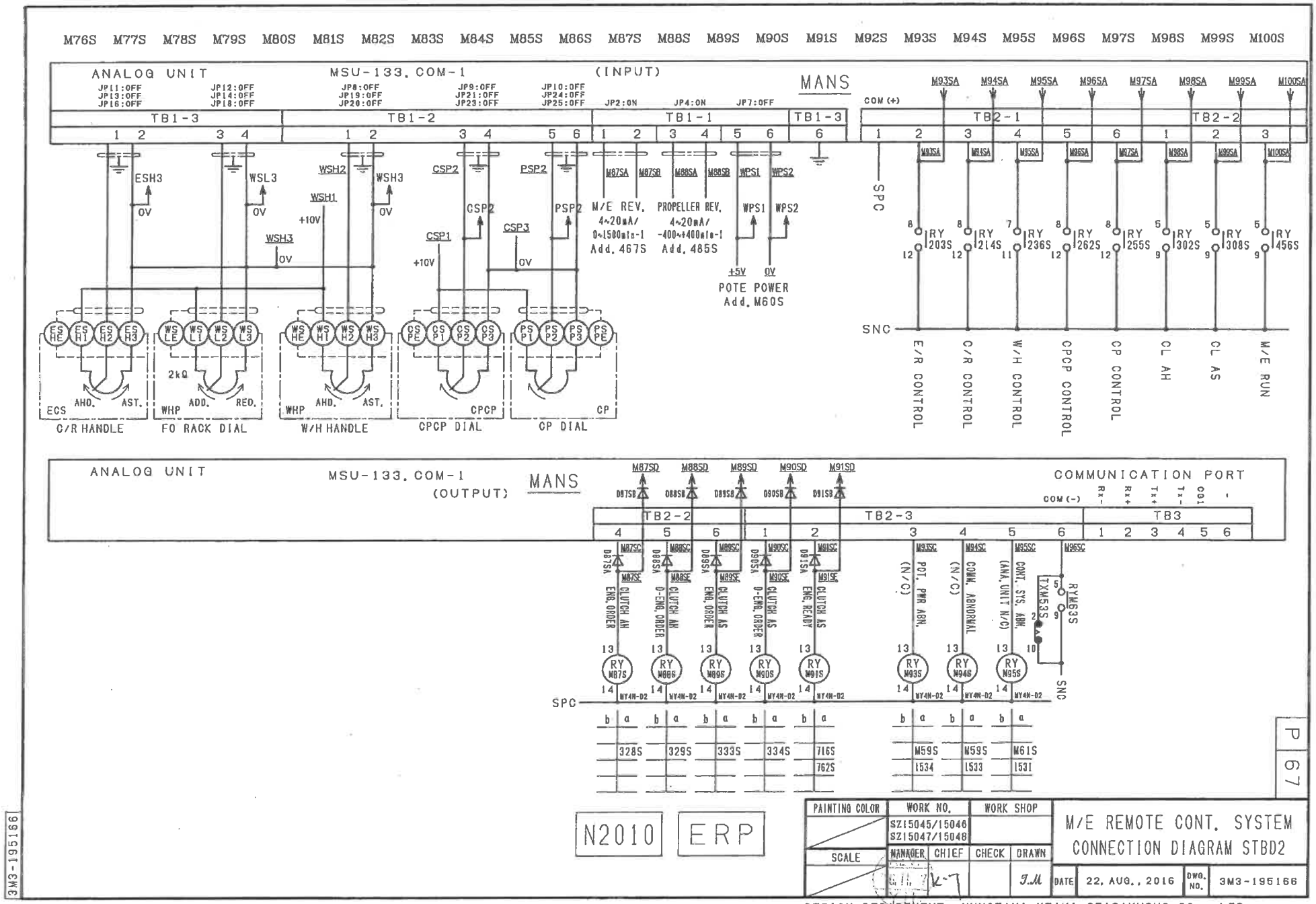
2 29, SEP., 2016

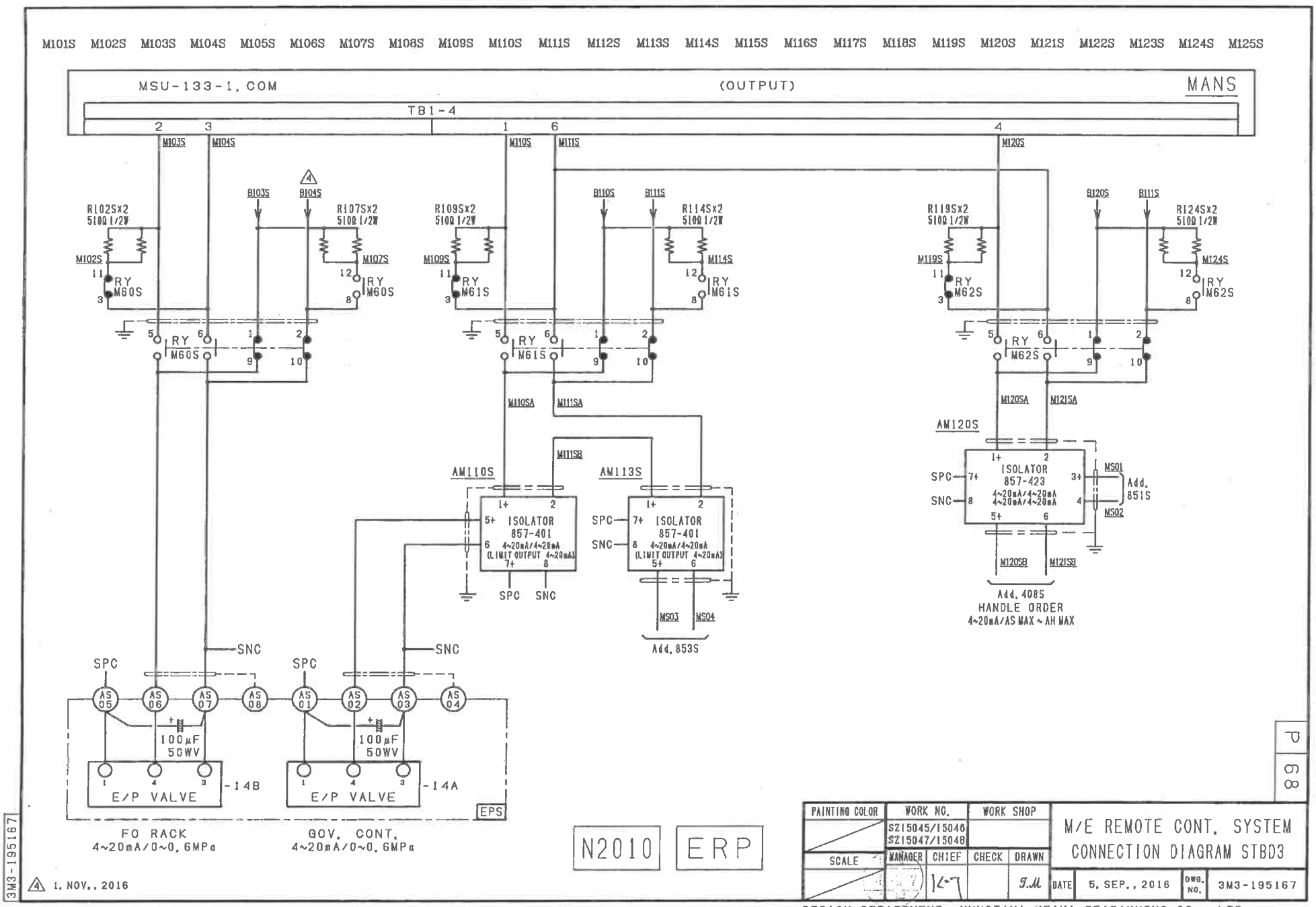
PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER	CHIEF	CHECK / DRAWN
			J.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD1

DATE 9, AUG., 2016 DWG. NO. 3M3-195165

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.


M76S M77S M78S M79S M80S M81S M82S M83S M84S M85S M86S M87S M88S M89S M90S M91S M92S M93S M94S M95S M96S M97S M98S M99S M100S
ANALOG UNIT
MSU-133. COM-1
(INPUT)
MANS
JP11:OFF JP12:OFF JP8:OFF JP9:OFF JP10:OFF
JP13:OFF JP14:OFF JP19:OFF JP21:OFF JP24:OFF
JP16:OFF JP18:OFF JP20:OFF JP23:OFF JP25:OFF
JP2:ON JP4:ON JP7:OFF
TB1-3 TB1-2 TB1-1 TB1-3
1 2 3 4 1 2 3 4 5 6 1 2 3 4 5 6 6
ESH3 WSL3 WSH2 WSH1 WSH3 CSP2 CSP1 CSP3 PSP2
0V +10V
M87SA M87SB M/E REV. 4~20mA/ 0~1500min-1 Add. 467S
M88SA M88SB PROPELLER REV. 4~20mA/ -400~+400min-1 Add. 485S
WPS1 WPS2 +5V 0V POTE POWER Add. M60S
ES HE ES H1 ES H2 ES H3 ECS AHD. AST. C/R HANDLE
WS LE WS L1 WS L2 WS L3 2kΩ WHP ADD. RED. FO RACK DIAL
WS HE WS H1 WS H2 WS H3 WHP AHD. AST. W/H HANDLE
CS PE CS P1 CS P2 CS P3 CPCP CPCP DIAL
PS P1 PS P2 PS P3 PS PE CP CP DIAL
COM (+)
TB2-1 TB2-2
1 2 3 4 5 6 1 2 3
M93SA M94SA M95SA M96SA M97SA M98SA M99SA M100SA
SPC
RY 203S RY 214S RY 236S RY 262S RY 255S RY 302S RY 308S RY 456S
SNC
E/R CONTROL C/R CONTROL W/H CONTROL CPCP CONTROL CP CONTROL CL AH CL AS M/E RUN
ANALOG UNIT
MSU-133. COM-1
(OUTPUT)
MANS
COMMUNICATION PORT
M87SD M88SD M89SD M90SD M91SD
D87SB D88SB D89SB D90SB D91SB
COM (-) Rx- Rx+ Tx+ Tx- CG1 -
TB2-2 TB2-3 TB3
4 5 6 1 2 3 4 5 6 1 2 3 4 5 6
M87SC M88SC M89SC M90SC M91SC M93SC M94SC M95SC M96SC
D87SA D88SA D89SA D90SA D91SA
M87SE M88SE M89SE M90SE M91SE
ENG. ORDER GLUTCH AH D-ENG. ORDER GLUTCH AH ENG. ORDER GLUTCH AS D-ENG. ORDER GLUTCH AS ENG. READY GLUTCH AS
POT. PWR ABN. (N/C) COMM. ABNORMAL (N/C) CONT. SYS. ABN. (ANA. UNIT N/C)
RY M87S RY M88S RY M89S RY M90S RY M91S RY M93S RY M94S RY M95S
MY4N-D2
TX453S RY M63S SNC
SPC
b a b a b a b a b a b a b a b a
328S 329S 333S 334S 716S 762S M59S 1534 M59S 1533 M61S 1531
P 67
N2010 ERP
PAINTING COLOR
WORK NO. SZ15045/15046 SZ15047/15048
WORK SHOP
SCALE
MANAGER CHIEF CHECK DRAWN
T.M
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD2
DATE 22. AUG., 2016
DWG. NO. 3M3-195166
3M3-195166
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD3			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	5, SEP., 2016
		12-7		T.M	DWG. NO.	3M3-195167

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

4 1, NOV., 2016

3M3-195167

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M126S M127S M128S M129S M130S M131S M132S M133S M134S M135S M136S M137S M138S M139S M140S M141S M142S M143S M144S M145S M146S M147S M148S M149S M150S

DIGITAL UNIT(1/3) MQU-108.COM-1 MQU

TB1-3	入力端子	M129SA	M130SA	M131SA	M132SA	M133SA	M134SA	M135SA	M136SA	TB1-2 M137SA	M138SA ⚠4	M139SA ⚠4	TB1-3	M144SA
1 2		6 5 4 3 2 1	6 5 4 3 2 1										6 5 4 3	
(004P IN Gr1/2)	アドレス			(004P IN Gr1)								(004P IN Gr2)		
COM COM	BIT	7 6 5 4 3 2 1 0								7 6 5	4 3 2 1 0			

TB3	1	2	3	4	5	6
	GPS +	GPS −	VDR +	VDR −	VDR 0V	

SPC — SPC

SNC — SNC

	RY		RY		RY		RY		RY		RY		RY		TX		RY		RY	
	3	11	6	10	5	9	5	9	5	9	5	9	5	9	6	10	10	6	10	6
	501S		558S		269S		240S		241S		218S		219S		427S		390S		391S	
	F0 HANDLE		5V ON		CARGO CL ENG		HANDLE LS AHEAD (M/H)		HANDLE LS ASTERN (M/H)		HANDLE LS AHEAD (C/R)		HANDLE LS ASTERN (C/R)		M.V.: OFF + NEUTRAL (For DRAIN/BYPASS)		IDLE BOOST ON (M/H)		IDLE BOOST ON (C/R)	

PBW144S/W63S — SYS. ABNORMAL RESET

PB144S: WO 25 / EO 19 (JHM) — PB145S: WO 26 / EO 20 (ECR)

DIGITAL UNIT(2/3) MQU-108.COM-1 MQU

TER	TB3 1	2	3	4	5	6	TB4 1	2	3	4	5	6
	GPS +	GPS −	VDR +	VDR −	VDR 0V		24V P	0V N	FG	24V P	0V N	FG

M64SA — M65SA

30 — 24

POWER UNIT
(STAB-113.COM)

SNC — SNC

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3M3-195168

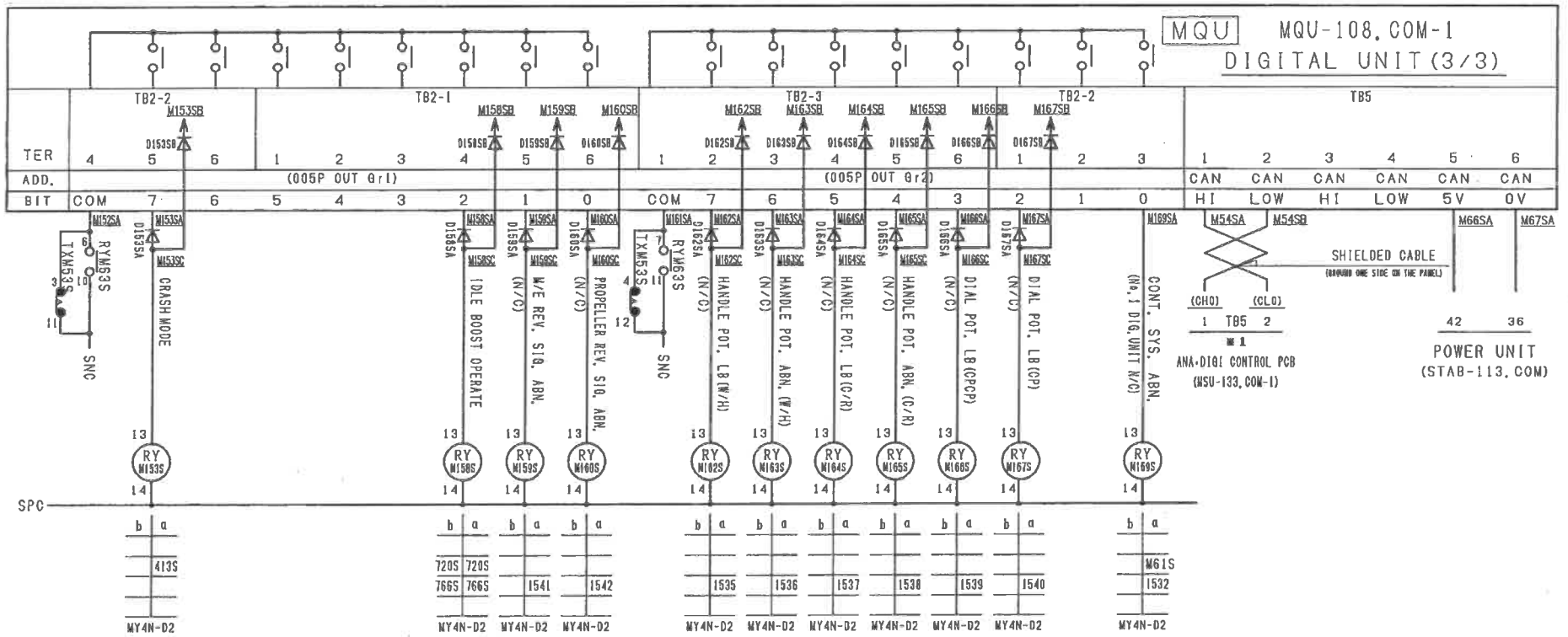
⚠4 17, NOV., 2016

N2010 ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD4		
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER REV. 16.11.24	CHIEF	CHECK REV. 16.11.24 T.M	DRAWN	DATE 24, AUG., 2016 / DWG. NO. 3M3-195168

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

M151S M152S M153S M154S M155S M156S M157S M158S M159S M160S M161S M162S M163S M164S M165S M166S M167S M168S M169S M170S M171S M172S M173S M174S M175S



P 70

PAINTING COLOR	WORK NO.	WORK SHOP		M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD5		
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	8. SEP., 2016
	15.11.7	K-7		T.M	DWG. NO.	3M3-195169

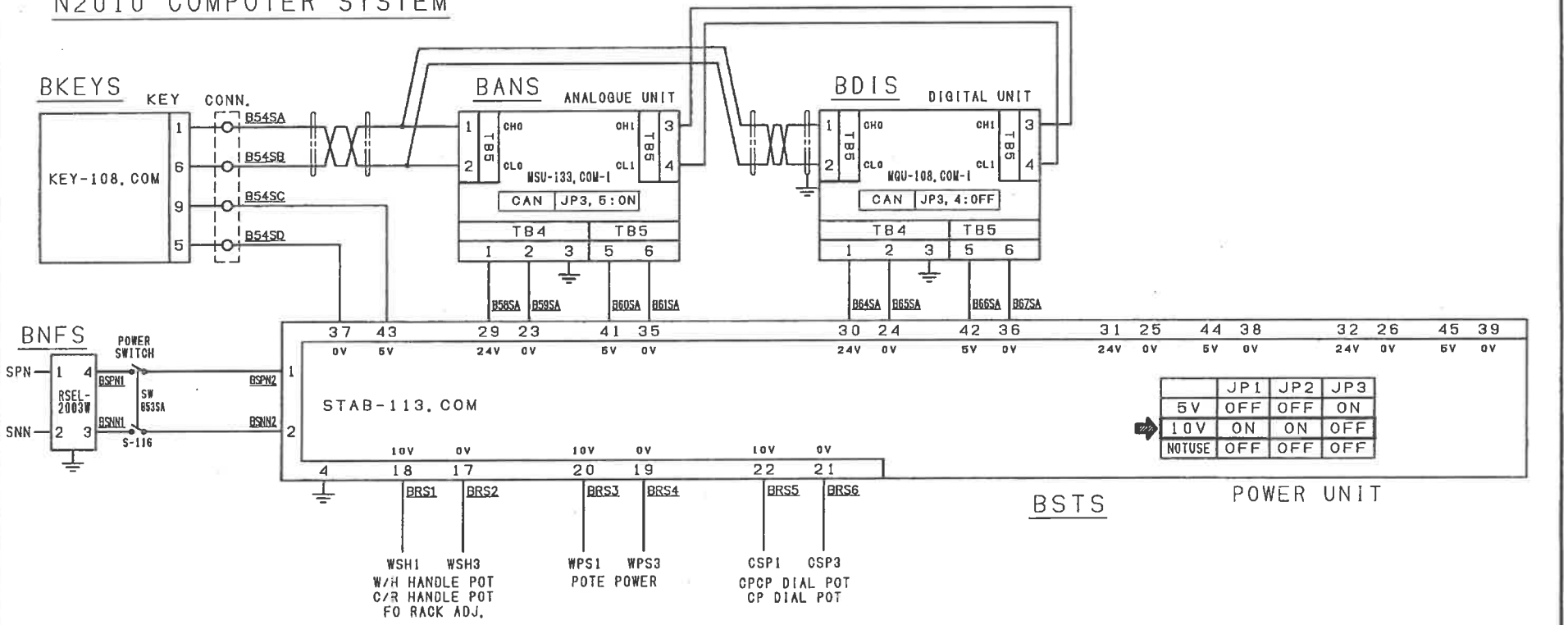
N2010 ERP

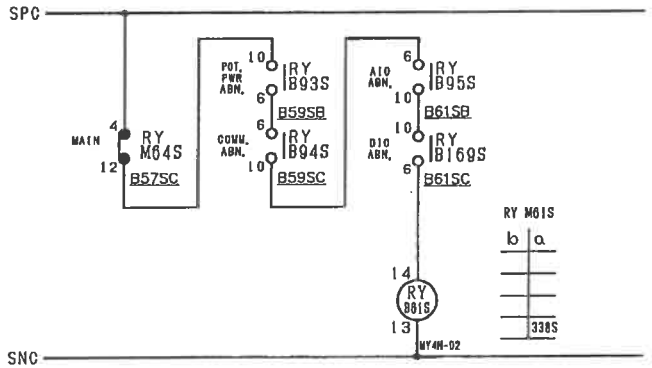
3M3-195169

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

B51S B52S B53S B54S B55S B56S B57S B58S B59S B60S B61S B62S B63S B64S B65S B66S B67S B68S B69S B70S B71S B72S B73S B74S B75S

N2010 COMPUTER SYSTEM





N2010 ERP

P 71

PAINTING COLOR	WORK NO.	WORK SHOP		
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD6

DATE 9. AUG., 2016 DWG. NO. 3M3-195170

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195170

B76S B77S B78S B79S B80S B81S B82S B83S B84S B85S B86S B87S B88S B89S B90S B91S B92S B93S B94S B95S B96S B97S B98S B99S B100S

ANALOG UNIT MSU-133. COM-1 (INPUT) MANS

JP11:OFF JP13:OFF JP16:OFF | JP12:OFF JP14:OFF JP18:OFF | JP8:OFF JP19:OFF JP20:OFF | JP9:OFF JP21:OFF JP23:OFF | JP10:OFF JP24:OFF JP25:OFF | JP2:ON | JP4:ON | JP7:OFF

TB1-3 (1 2 3 4) | TB1-2 (1 2 3 4 5 6) | TB1-1 (1 2 3 4 5 6) | TB1-3 (6)

ESH2 WSL2 WSH2 CSP2 PSP2 B87SA B87SB B88SA B88SB

WSH3 0V CSP3 0V

M/E REV. 4~20mA/ 0~1500min-1 Add. 467S

PROPELLER REV. 4~20mA/ -400~+400min-1 Add. 485S

ESH2 WSL2 WSH2 WSH3 CSP2 CSP3 PSP2 WPS1 WPS2

COM (+) | TB2-1 (1 2 3 4 5 6) | TB2-2 (1 2 3)

SPC

M93SA M94SA M95SA M96SA M97SA M98SA M99SA M100SA

SNC

E/R CONTROL
C/R CONTROL
W/H CONTROL
CPCP CONTROL
CP CONTROL
CL AH
CL AS
M/E RUN

ANALOG UNIT MSU-133. COM-1 (OUTPUT) MANS

COMMUNICATION PORT

COM (-) | TB2-2 (4 5 6) | TB2-3 (1 2 3 4 5 6) | TB3 (1 2 3 4 5 6) Rx- Rx+ Tx+ Tx- OSI -

CLUTCH AH ENG. ORDER — M87SD
CLUTCH AH D-ENG. ORDER — M88SD
CLUTCH AS ENG. ORDER — M89SD
CLUTCH AS D-ENG. ORDER — M90SD
CLUTCH AS ENG. READY — M91SD

B93SC POT. PWR ABN. (N/C) — RY B93S — MY4N-D2
B94SC COMN. ABNORMAL (N/C) — RY B94S — MY4N-D2
B95SC CONT. SYS. ABN. (ANA. UNIT N/C) — RY B95S — MY4N-D2
B96SC — RYM64S — SNC

SPC

b	a	b	a	b	a
	B59S		B59S		B61S
	1574		1573		1571

P 72

N2010 ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. 6.11.7			T.M

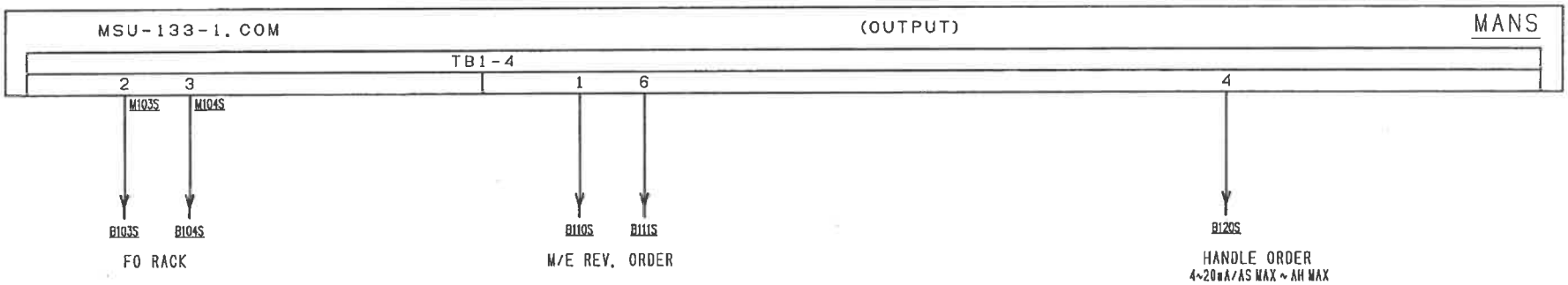
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD7

DATE 22. AUG., 2016 | DWG. NO. 3M3-195171

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195171

B101S B102S B103S B104S B105S B106S B107S B108S B109S B110S B111S B112S B113S B114S B115S B116S B117S B118S B119S B120S B121S B122S B123S B124S B125S



P 73

3M3-195172

N2010 ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD8
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE 22. AUG., 2016 / DWG. NO. 3M3-195172
		J.M	

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

B126S B127S B128S B129S B130S B131S B132S B133S B134S B135S B136S B137S B138S B139S B140S B141S B142S B143S B144S B145S B146S B147S B148S B149S B150S

DIGITAL UNIT(1/3) MQU-108, COM-1

MQU

TB1-3 1	TB1-3 2	入力端子	TB1-1 6	5	4	3	2	1	TB1-2 6	5	4	3	2	1	TB1-3 6	5	4	3	TB3 1	2	3	4	5	6
(004P IN Gr1/2)		アドレス	(004P IN Gr1)								(004P IN Gr2)								GPS	GPS	VDR	VDR	VDR	
COM	COM	BIT	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	+	-	+	-	0V	

SPC — SPC

M129SA M130SA M131SA M132SA M133SA M134SA M135SA M136SA M138SA ⚠4 M139SA ⚠4 M144SA

SNC — SNC

FO HANDLE
5V ON
CARGO CL ENG
HANDLE LS AHEAD (M/H)
HANDLE LS ASTERN (M/H)
HANDLE LS AHEAD (C/R)
HANDLE LS ASTERN (C/R)
M.V.:OFF+NEUTRAL (F'ti DRAIN/BYPASS)
IDLE BOOST ON (M/H)
IDLE BOOST ON (C/R)
SYS. ABNORMAL RESET

DIGITAL UNIT(2/3) MQU-108, COM-1

MQU

TER	TB3 1	2	3	4	5	6	TB4 1	2	3	4	5	6
	GPS +	GPS -	VDR +	VDR -	VDR 0V		24V P	0V N	FG	24V P	0V N	FG

B64SA B65SA

30 24

POWER UNIT
(STAB-113, COM)

SNC — SNC

P 74

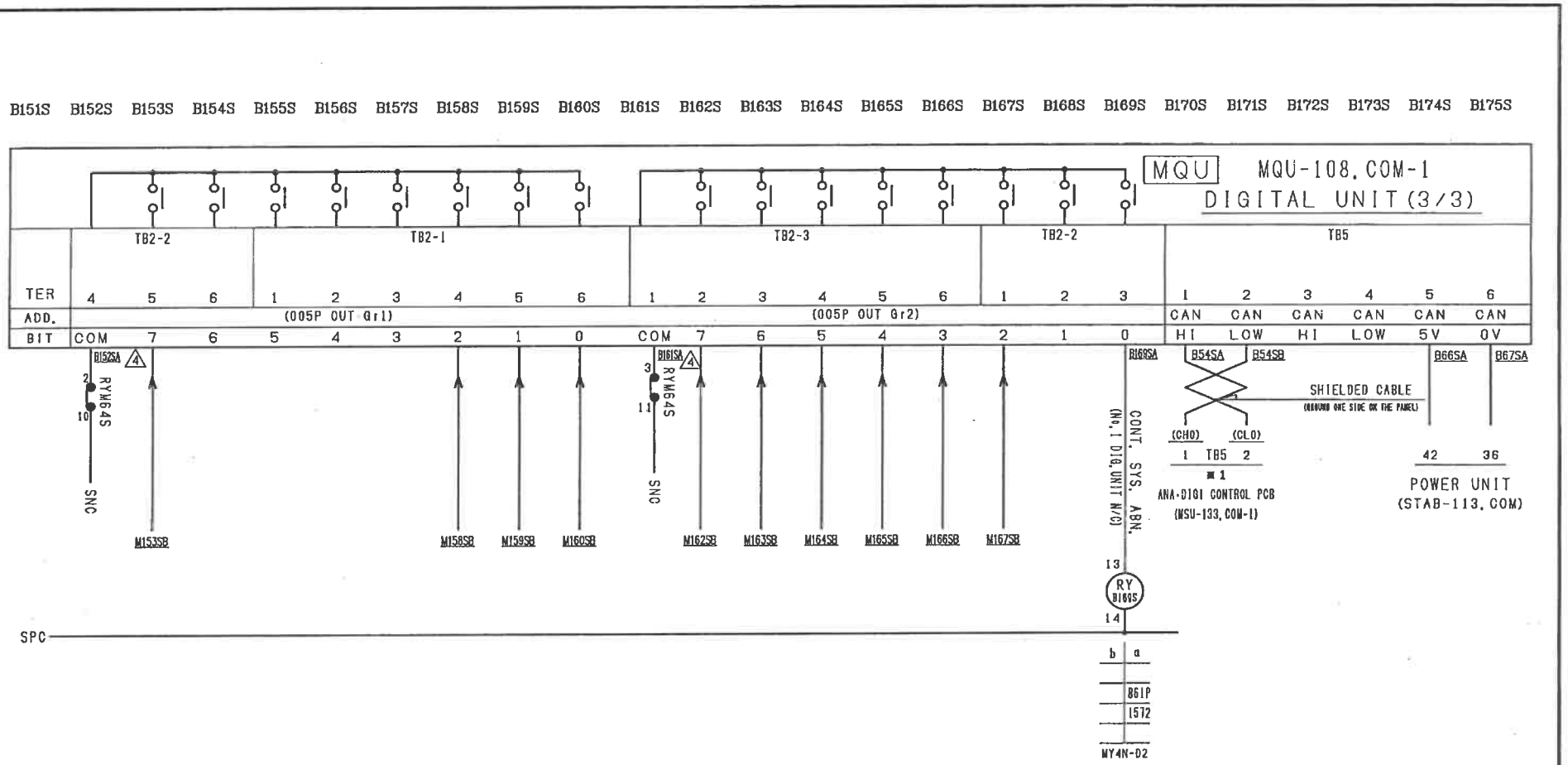
N2010 ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD9
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER CHIEF	CHECK DRAWN	DATE 24, AUG., 2016
	REV. '16.11.24	REV. '16.11.24 T.M	DWG. NO. 3M3-195173

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195173

⚠4 17, NOV., 2016



P 75

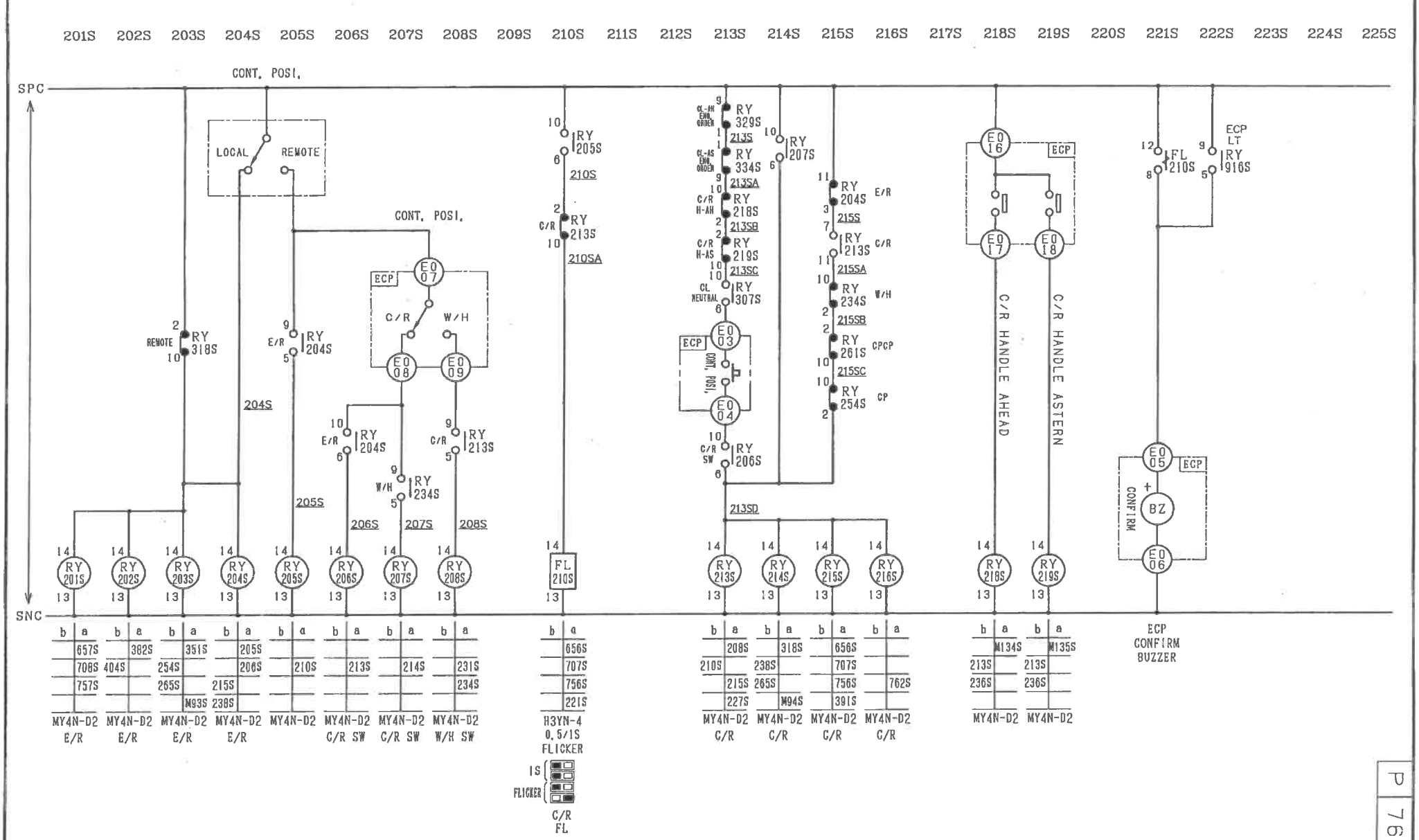
PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD10			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	8. SEP., 2016
		K-7		T.M	DWG. NO.	3M3-195174

N2010 ERP

3M3-195174

4 1. NOV., 2016

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
		K.T		T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD11

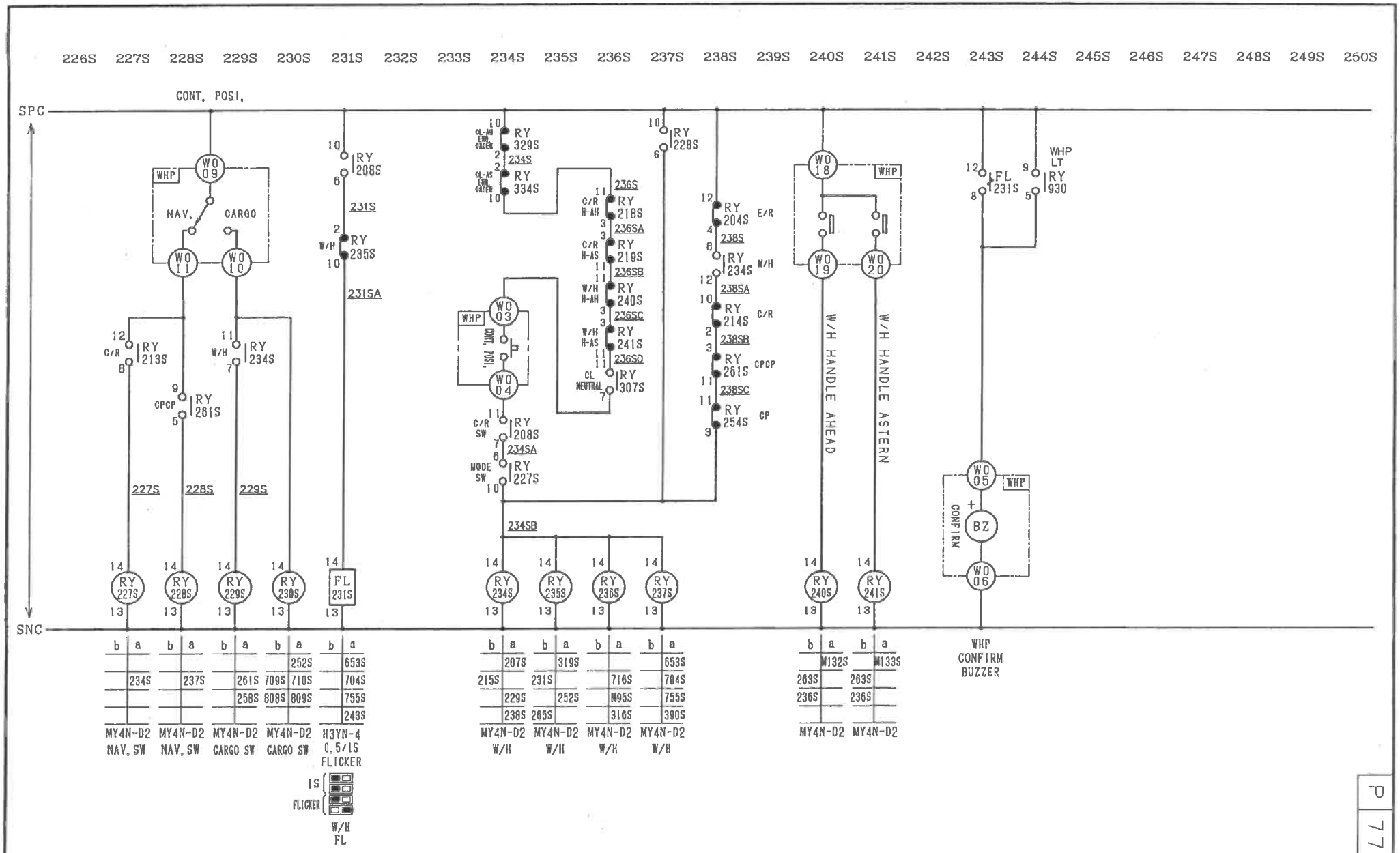
DATE 6. SEP., 2016 | DWG. NO. 3M3-195175

STBD ERP

P 76

3M3-195175

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD12

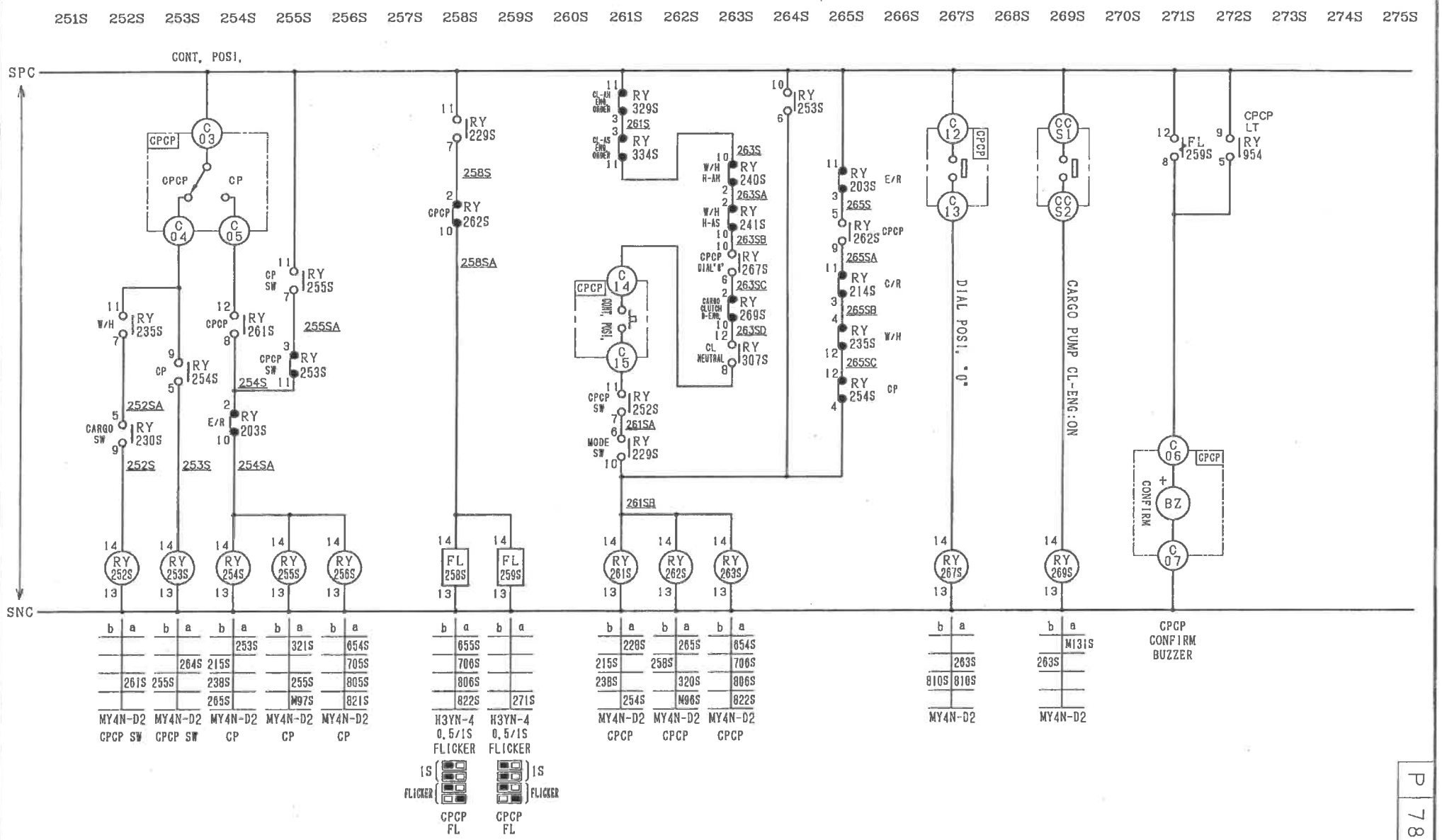
DATE 6. SEP., 2016 | DWG. NO. 3M3-195176

STBD ERP

P 77

3M3-195176

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

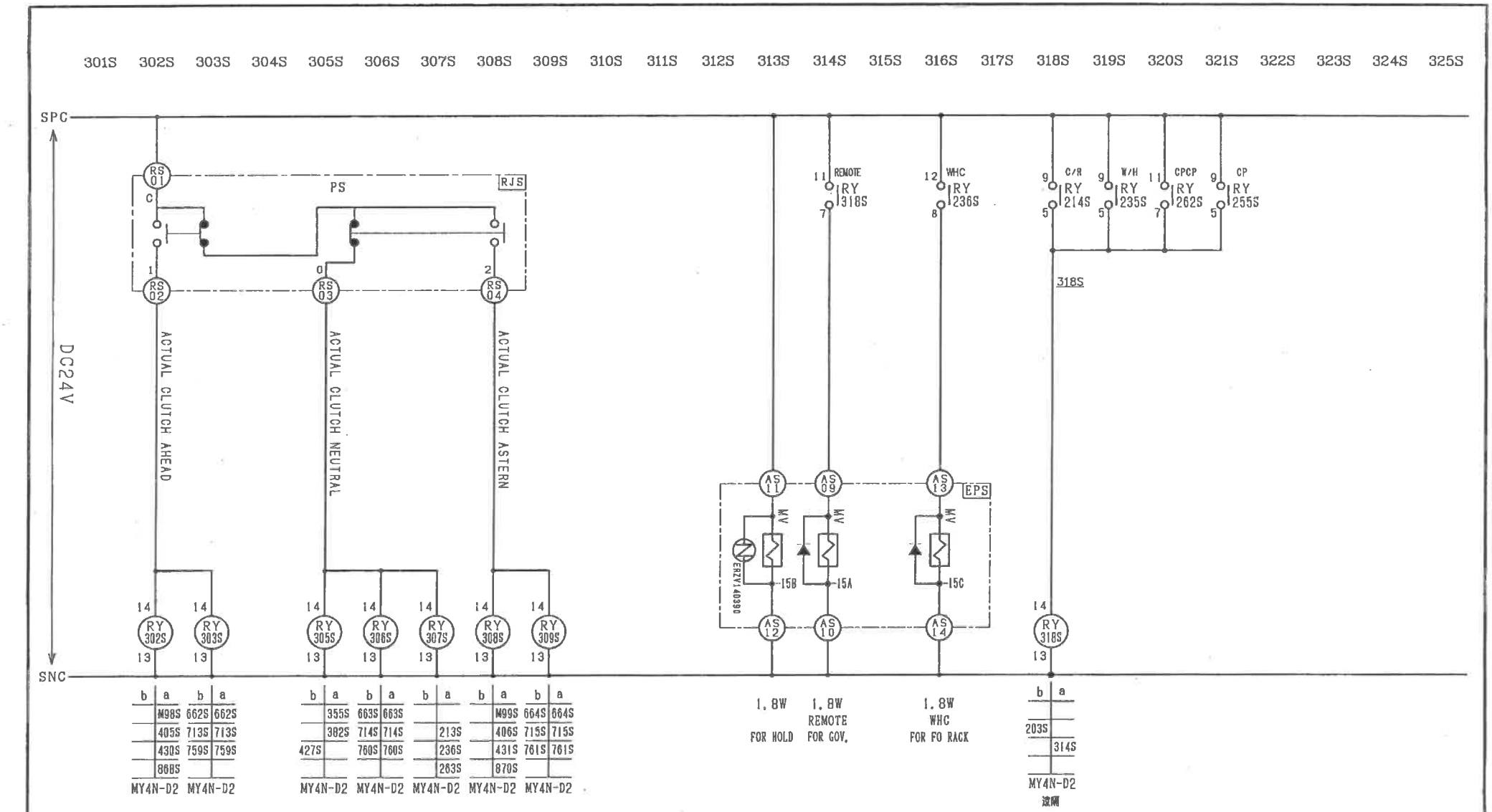
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD13

DATE 6, SEP., 2016 | DWG. NO. 3M3-195177

STBD ERP

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195177



P 79

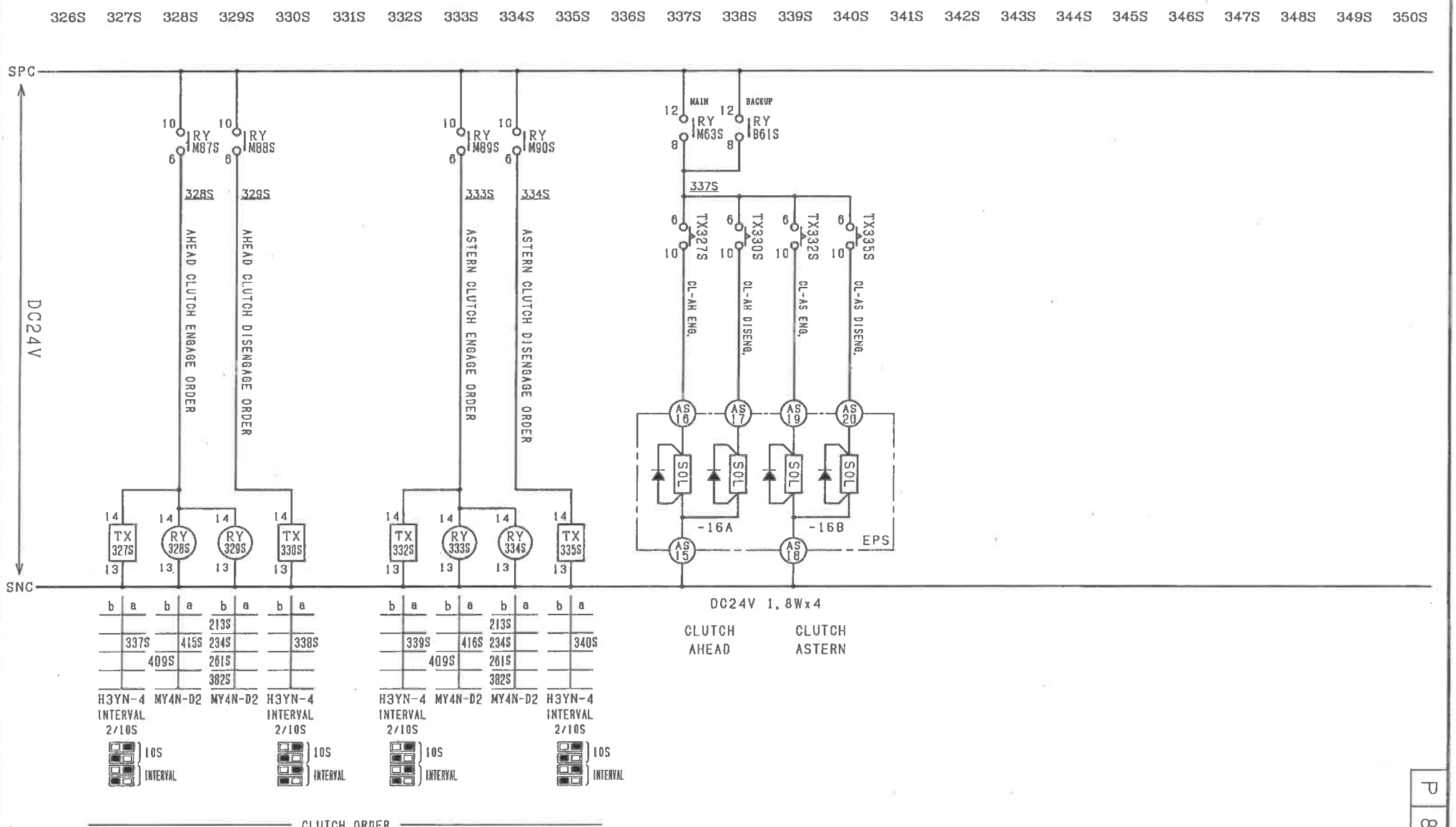
PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD14		
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER	CHIEF	CHECK	DRAWN	
				J.M	DATE 12. AUG., 2016

DWG. NO. 3M3-195178

STBD ERP

3M3-195178

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
		K-T		T.M

STBD ERP

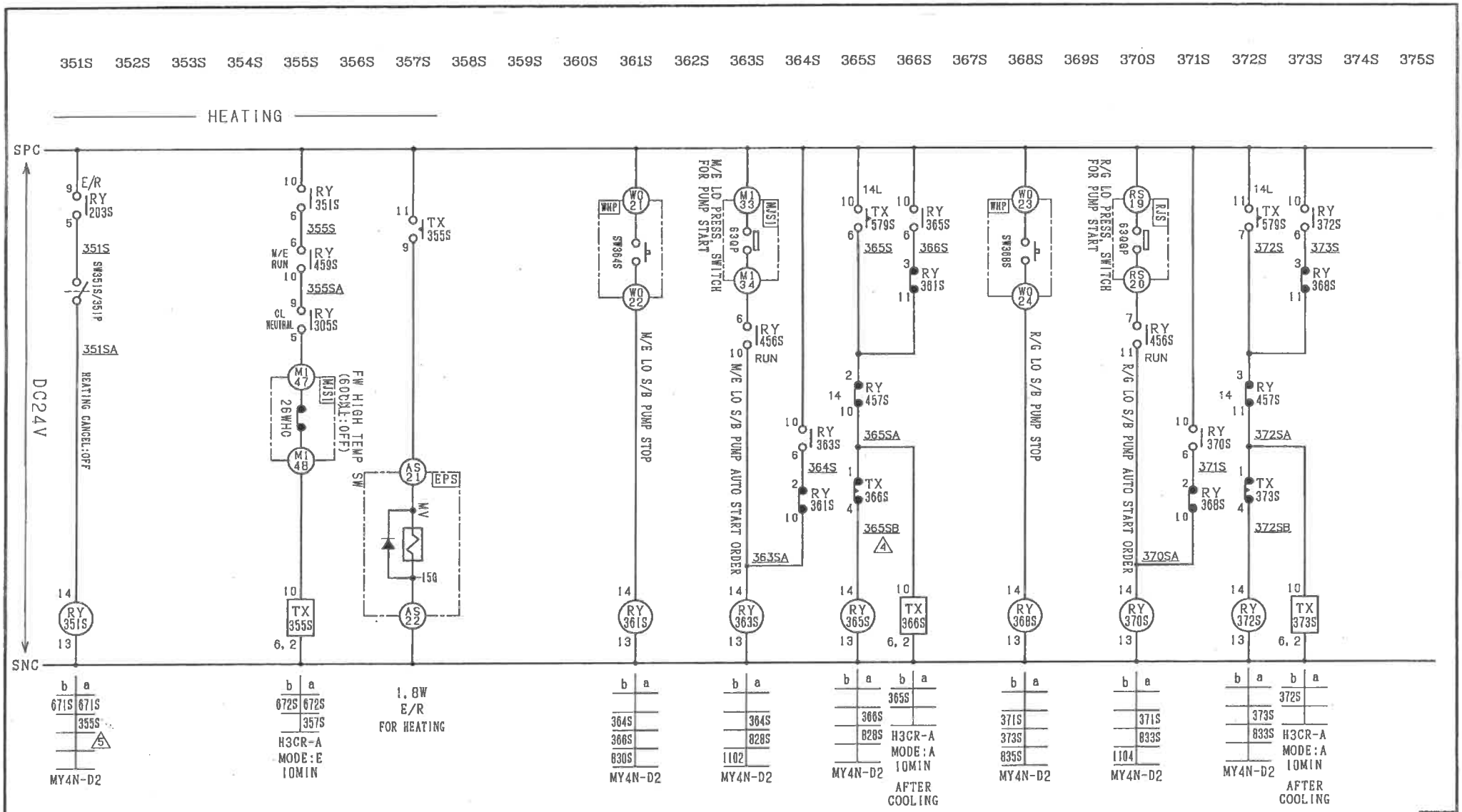
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD15

DATE 6. SEP., 2016 DWG. NO. 3M3-195179

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 80

3M3-195179



P 81

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. 16.12 b	K-T	REV. 16.12.06 K-T	T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD16

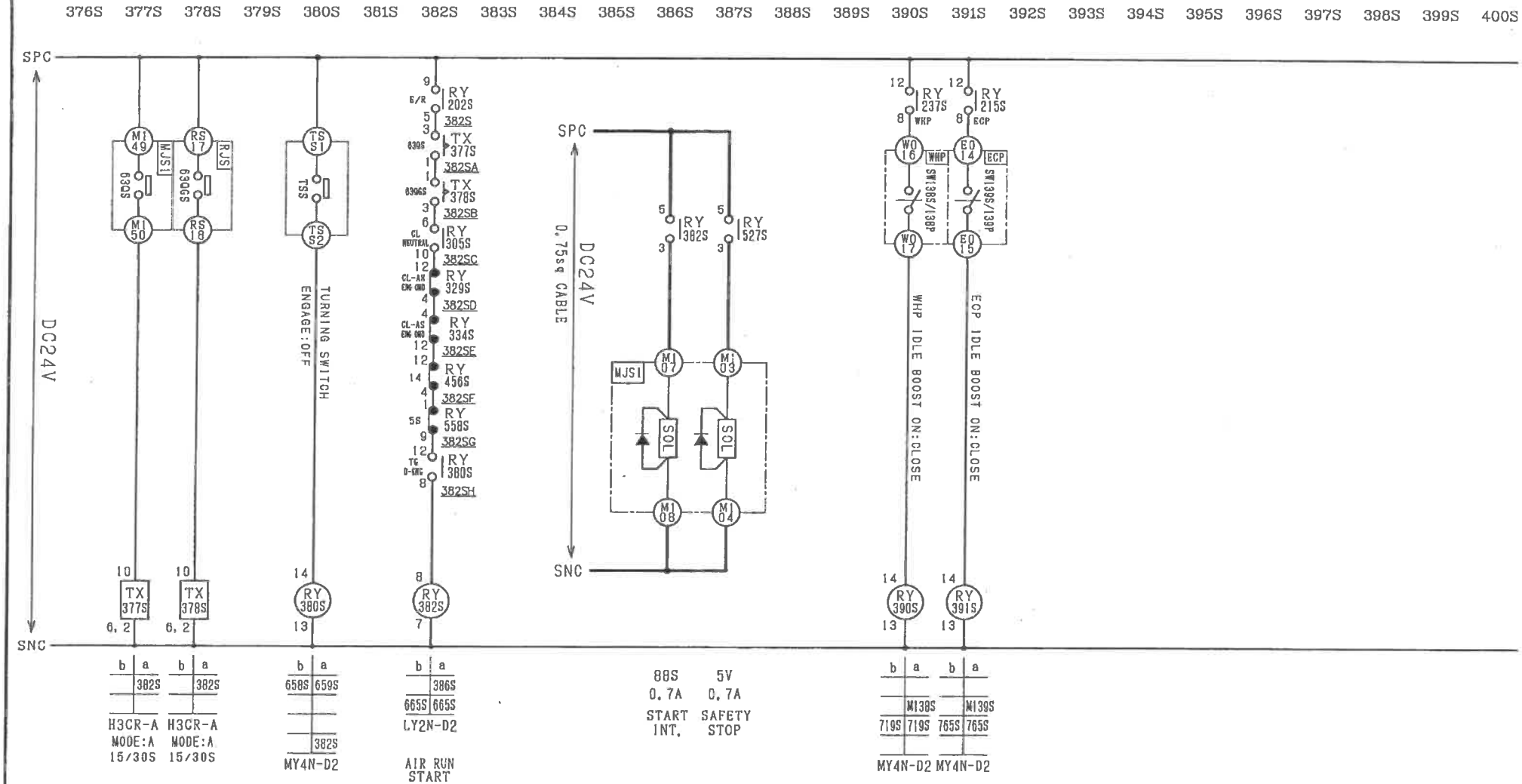
DATE	2, SEP., 2016	DWG. NO.	3M3-195180

STBD ERP

5 6, DEC., 2016
4 1, NOV., 2016

3M3-195180

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	
SCALE	MANAGER / CHIEF	CHECK / DRAWN
		T.M

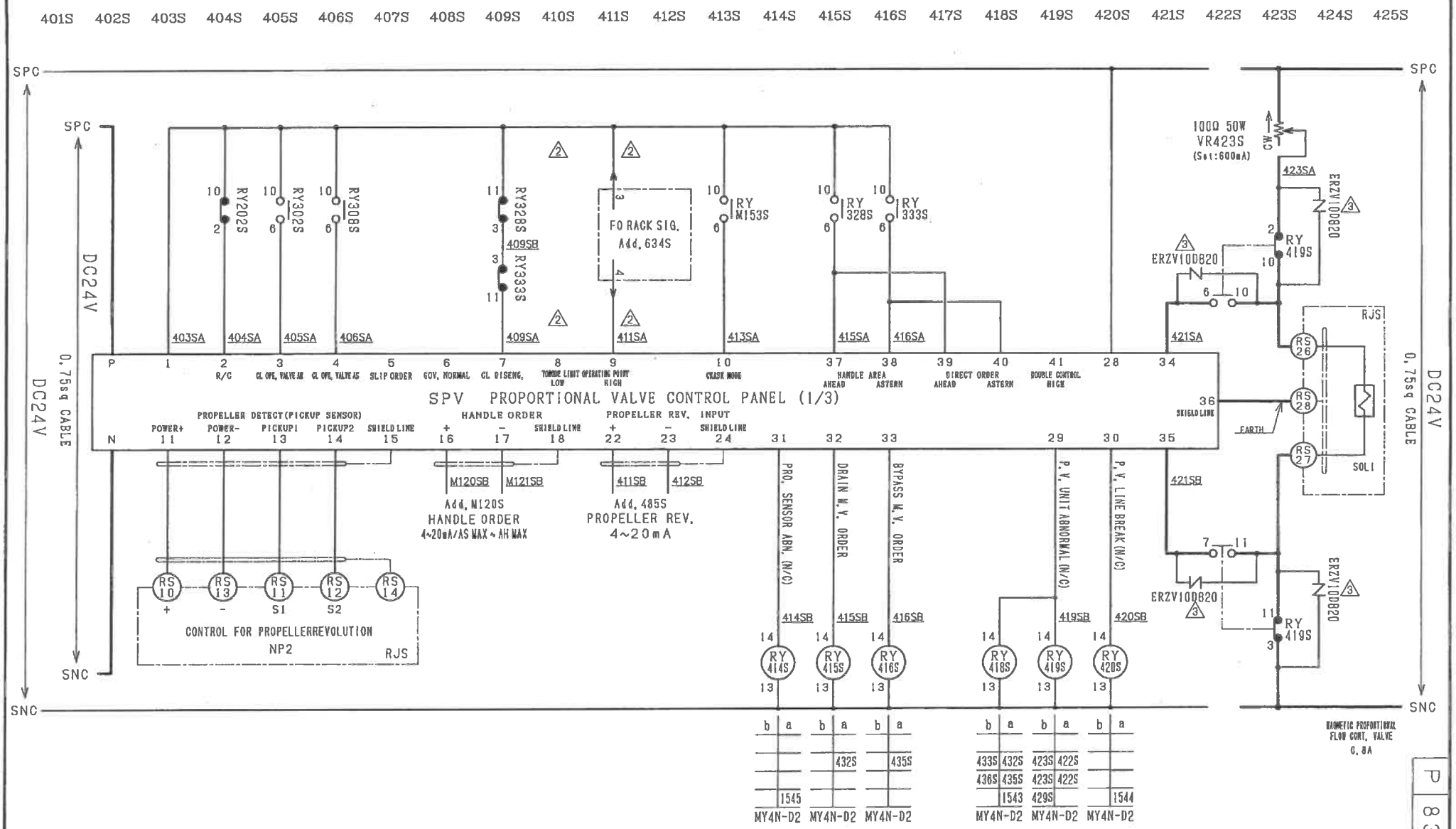
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD17

DATE 6. SEP., 2016 | DWG. NO. 3M3-195181

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 82

3M3-195181



3M3-195182

3 17, OCT., 2016
2 29, SEP., 2016

STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	S.H.			J.M

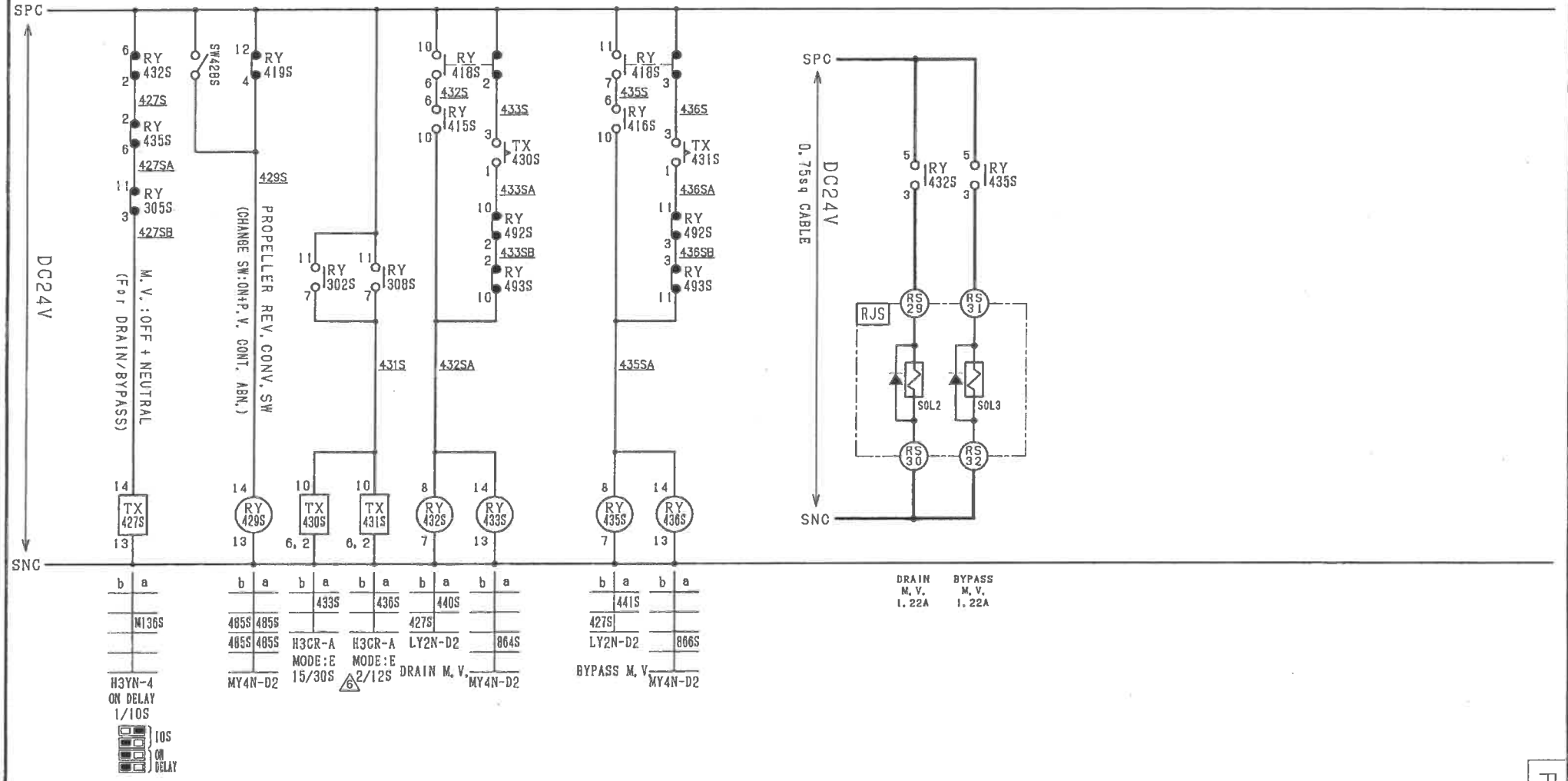
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD18

DATE 8. SEP., 2016 DWG. NO. 3M3-195182

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 83

426S 427S 428S 429S 430S 431S 432S 433S 434S 435S 436S 437S 438S 439S 440S 441S 442S 443S 444S 445S 446S 447S 448S 449S 450S



P 84

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	
SCALE	MANAGER / CHIEF / CHECK / DRAWN	

STBD ERP

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM STBD19

DATE 8. SEP., 2016 | DWG. NO. 3M3-195183

6 12. DEC., 2016

3M3-195183

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

451S 452S 453S 454S 455S 456S 457S 458S 459S 460S 461S 462S 463S 464S 465S 466S 467S 468S 469S 470S 471S 472S 473S 474S 475S

M/E REVOLUTION

SPC

SNC

PULSE SENSOR

NE1

MJS1

SENSOR ABNORMAL

SPEED SWITCH UNIT
for CONTROL

SS1C
MDP-REV-D
DC:24V 2.5W

(ON:150min-1)
(OFF:143min-1) (200min-1) (9999min-1) (9999min-1) (ABNORMAL)

FG

456S

ADD. 1112

461SA

461SB

MY4N-D2

A470SA SPC SNC
ISOLATOR 857-400 4~20mA/0~10V
0~10V/0~1500min-1
A472SA
A472SB

462S
A464SA SPC SNC
ISOLATOR 857-400 4~20mA/0~10V
WHP
462SA

A470SB SPC SNC
ISOLATOR 857-400 4~20mA/0~10V
ECP

462SB
A464SB SPC SNC
ISOLATOR 857-400 4~20mA/0~10V
CPCP
462SC

A470SC SPC SNC
ISOLATOR 857-400 4~20mA/0~10V
CP

462SD
A464SC SPC SNC
ISOLATOR 857-423 4~20mA/4~20mA 4~20mA/4~20mA
M87SA
M87SB
ADD. M87S 4~20mA/0~1500min-1
462SE
B87SA
B87SB
ADD. B87S 4~20mA/0~1500min-1

A470SD SPC SNC
ISOLATOR 857-423 4~20mA/4~20mA 4~20mA/4~20mA
473SA
473SB
MONITOR TB9 CH1+ CH1-
4~20mA/0~1500min-1
Add. 1211

462SF
A464SD SPC SNC
ISOLATOR 857-423 4~20mA/4~20mA 4~20mA/4~20mA
PVS
466SA
466SB
PROPORTIONAL VALVE CONTROL PANEL (2/3)
MS05
MS06
ADD. 855S 4~20mA/0~1500min-1

STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
1/1, 1/2				T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD20

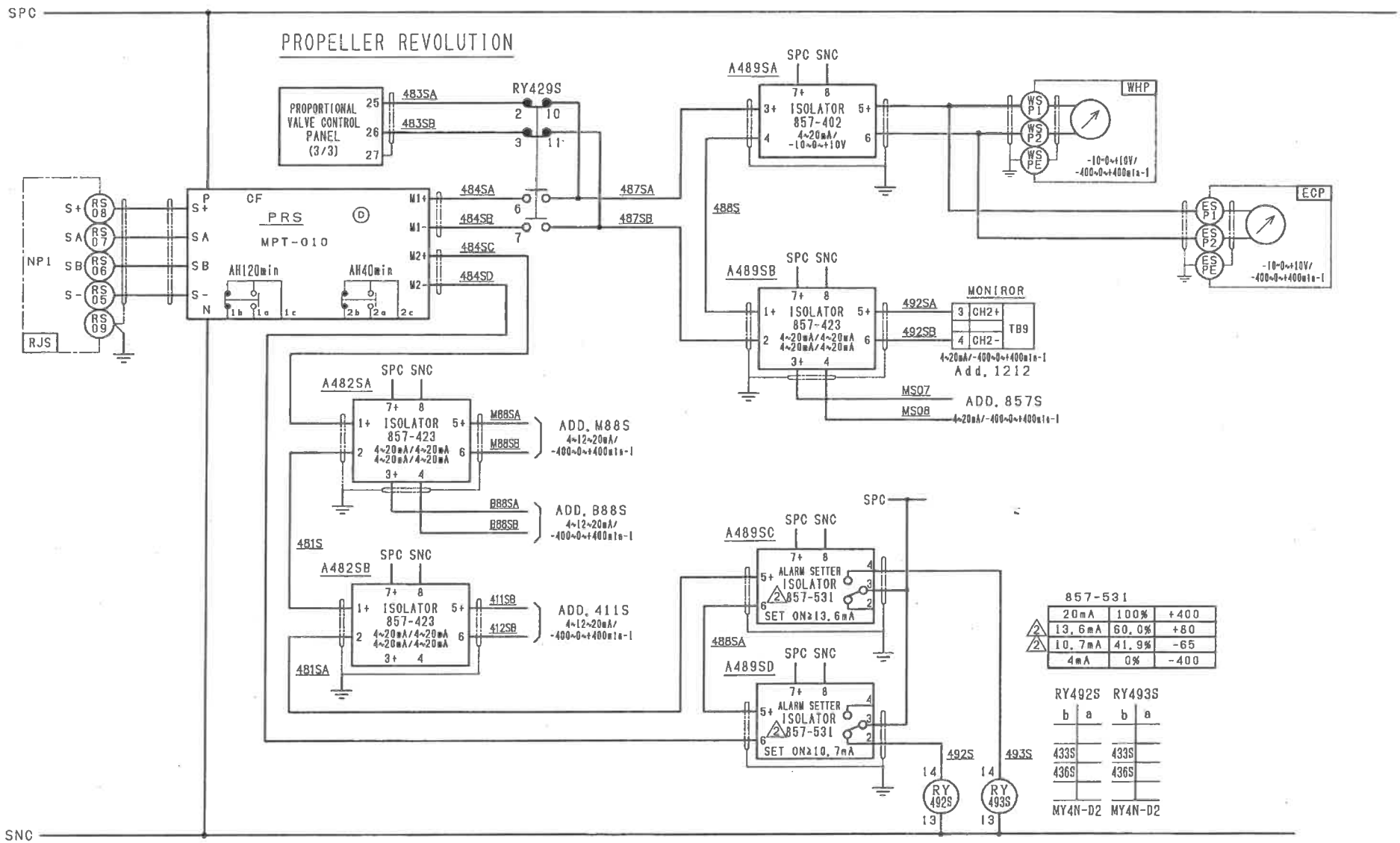
DATE 9. SEP., 2016 DWG. NO. 3M3-195184

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195184

P 85

476S 477S 478S 479S 480S 481S 482S 483S 484S 485S 486S 487S 488S 489S 490S 491S 492S 493S 494S 495S 496S 497S 498S 499S 500S



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	
SCALE	MANAGER / CHIEF	CHECK / DRAWN
		J.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD21

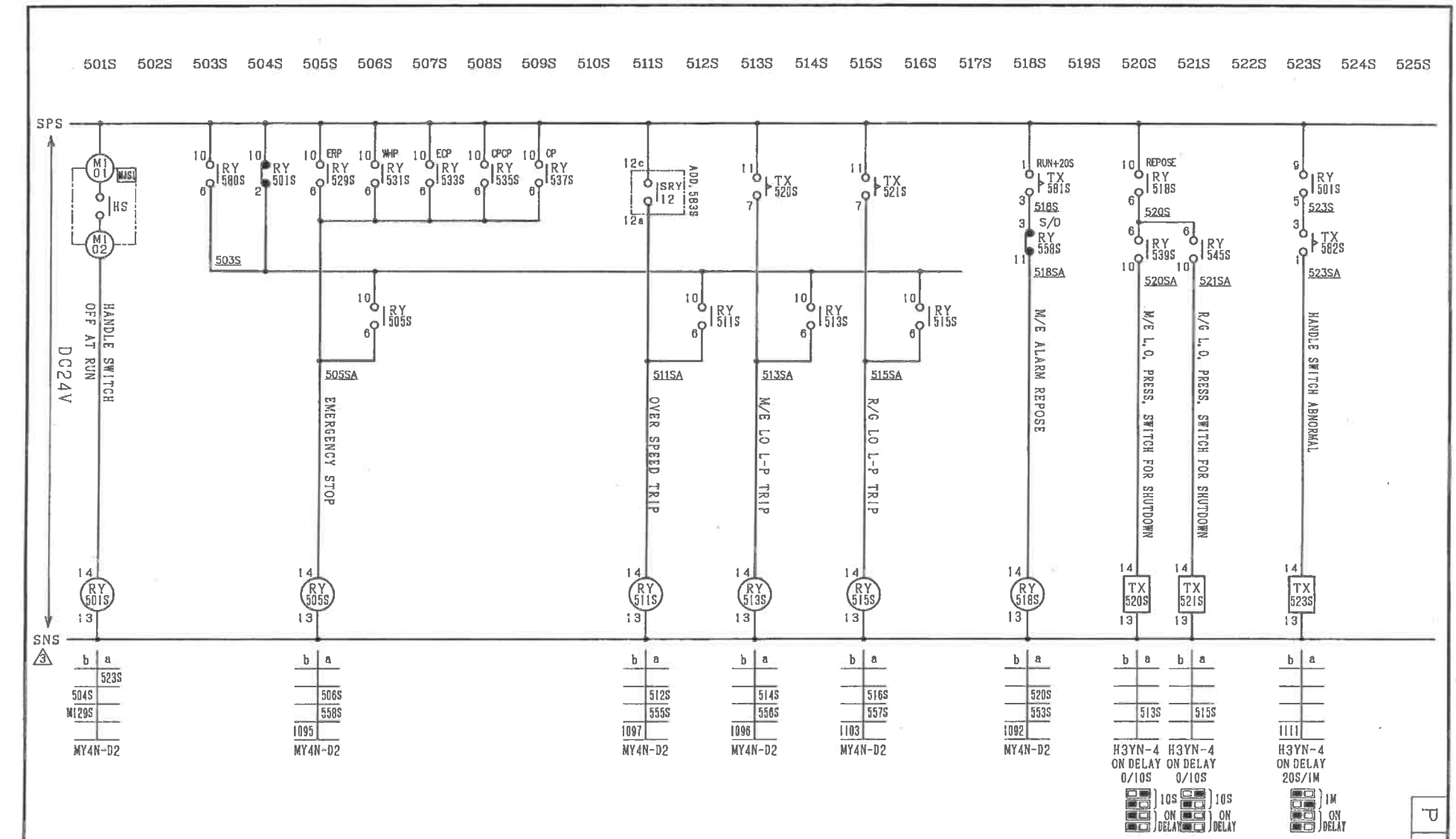
DATE 8. SEP., 2016 | DWG. NO. 3M3-195185

STBD ERP

2 29. SEP., 2016

3M3-195185

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	
SCALE	MANAGER / CHIEF	CHECK / DRAWN
		T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD22

DATE 12. AUG., 2016 | DWG. NO. 3M3-195186

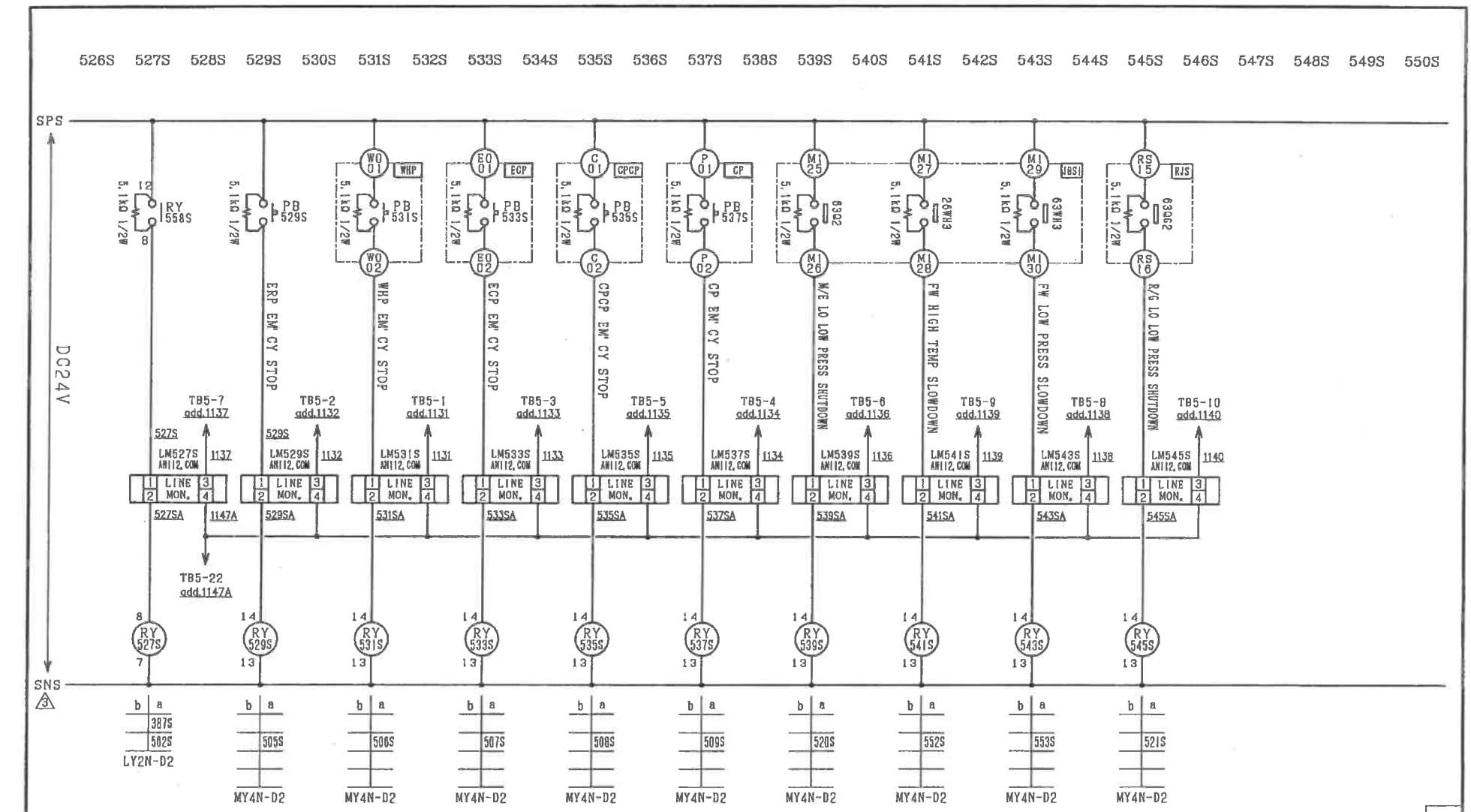
STBD ERP

3M3-195186

③ 18. OCT., 2016

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 87



3M3-195187

△3 18, OCT., 2016

P 88

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

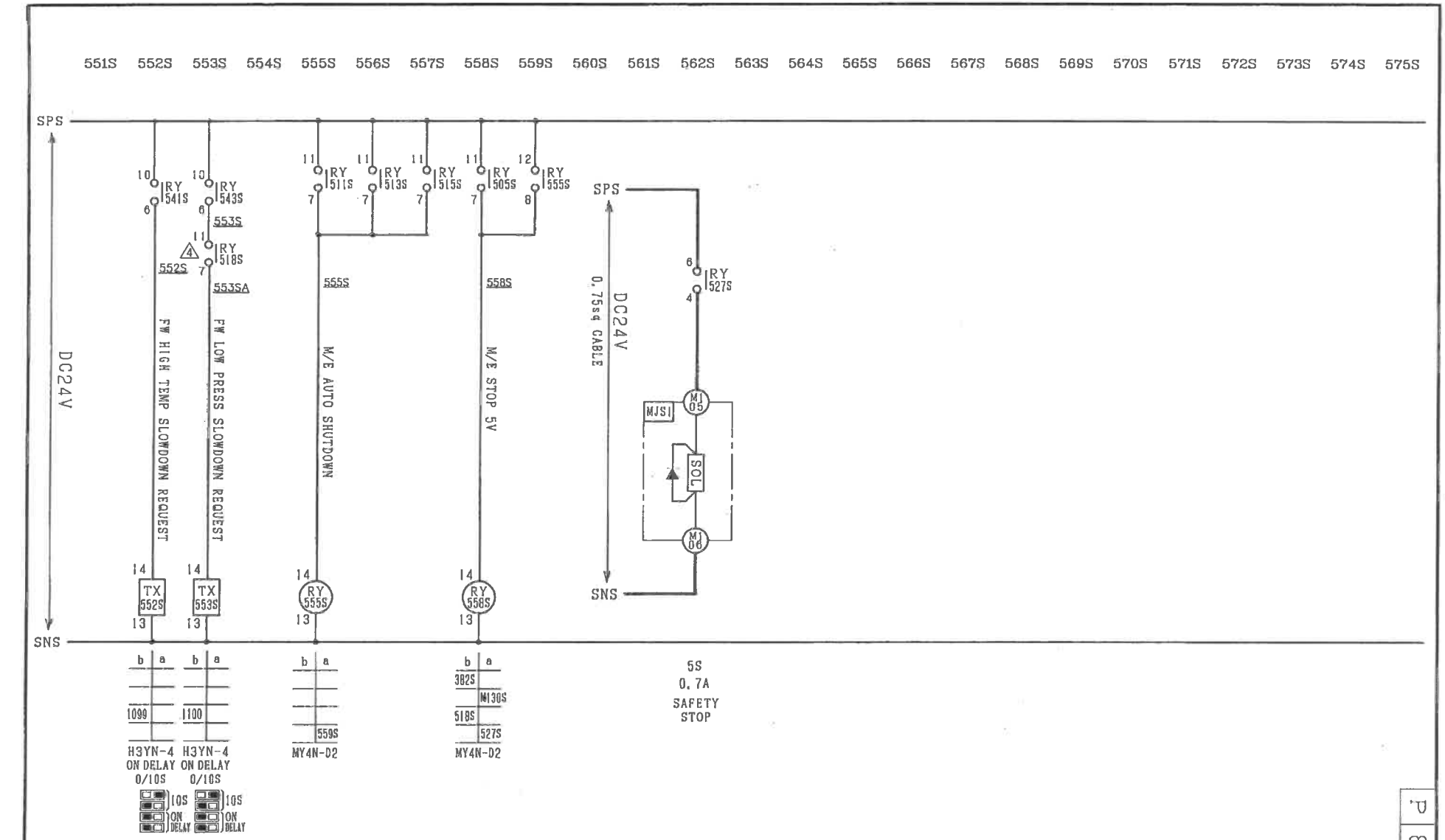
SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV.			T.M

STBD ERP

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD23

DATE 10, AUG., 2016 DWG. NO. 3M3-195187

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



3M3-195188

△4 1, NOV., 2016

STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

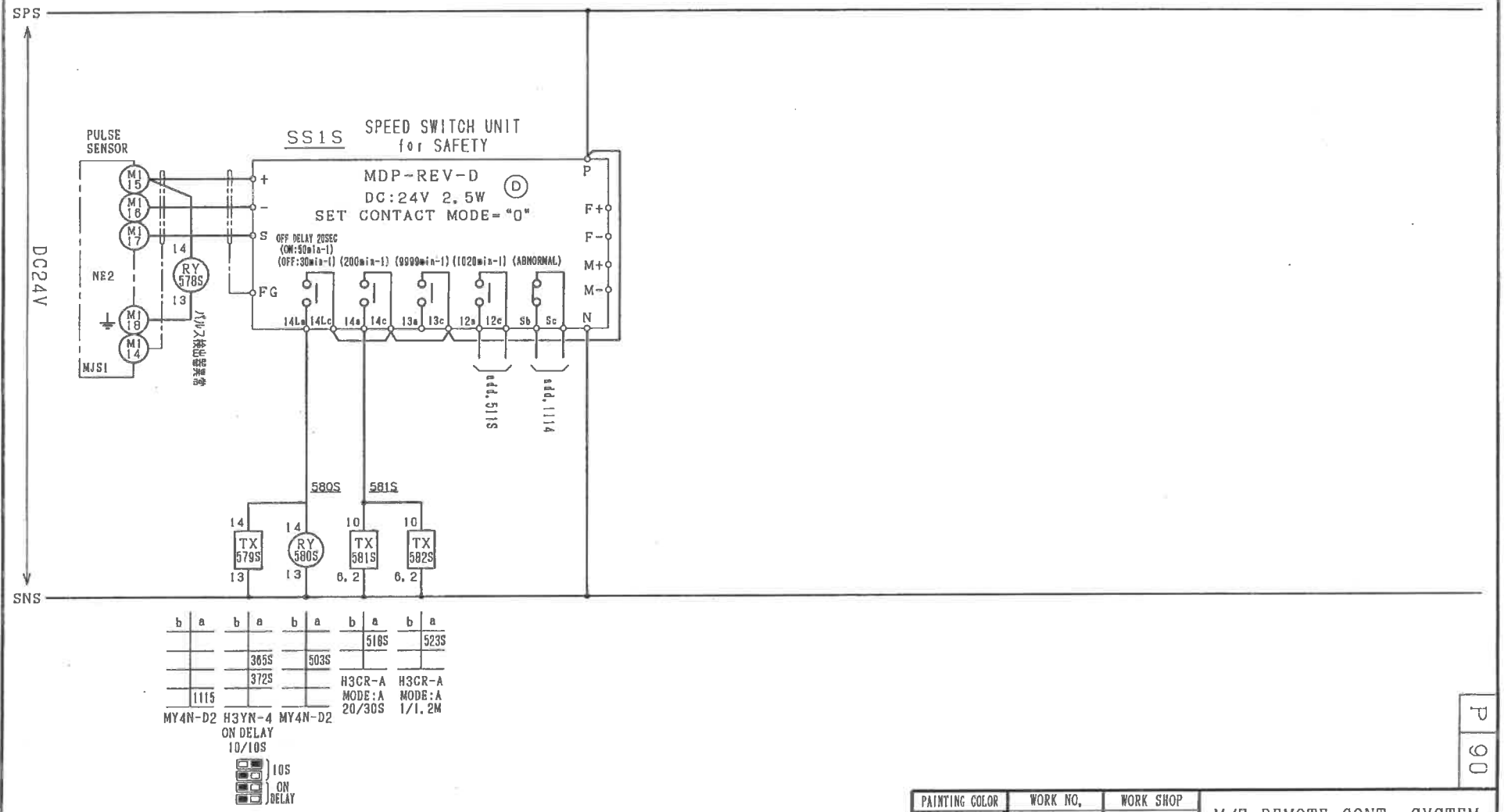
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD24

DATE 12. AUG., 2016 | DWG. NO. 3M3-195188

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 89

576S 577S 578S 579S 580S 581S 582S 583S 584S 585S 586S 587S 588S 589S 590S 591S 592S 593S 594S 595S 596S 597S 598S 599S 600S



CONTACT MODE "0" (14L:14:13:12) = (A:A:A:A)
CONTACT MODE "1" (14L:14:13:12) = (A:A:A:B)

STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER	CHIEF	CHECK / DRAWN
			T.M

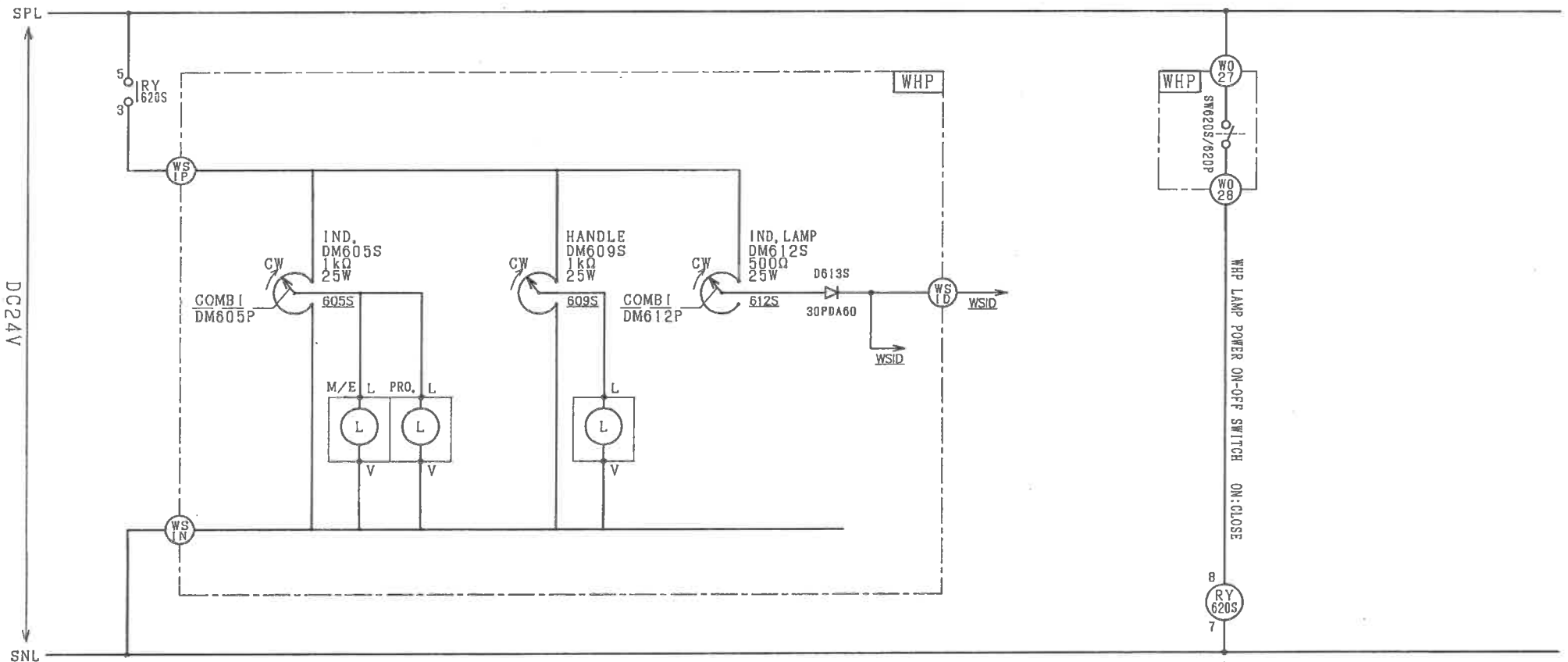
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD25

DATE 12. AUG., 2016 DWG. NO. 3M3-195189

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195189

601S 602S 603S 604S 605S 606S 607S 608S 609S 610S 611S 612S 613S 614S 615S 616S 617S 618S 619S 620S 621S 622S 623S 624S 625S



3M3-195190

STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

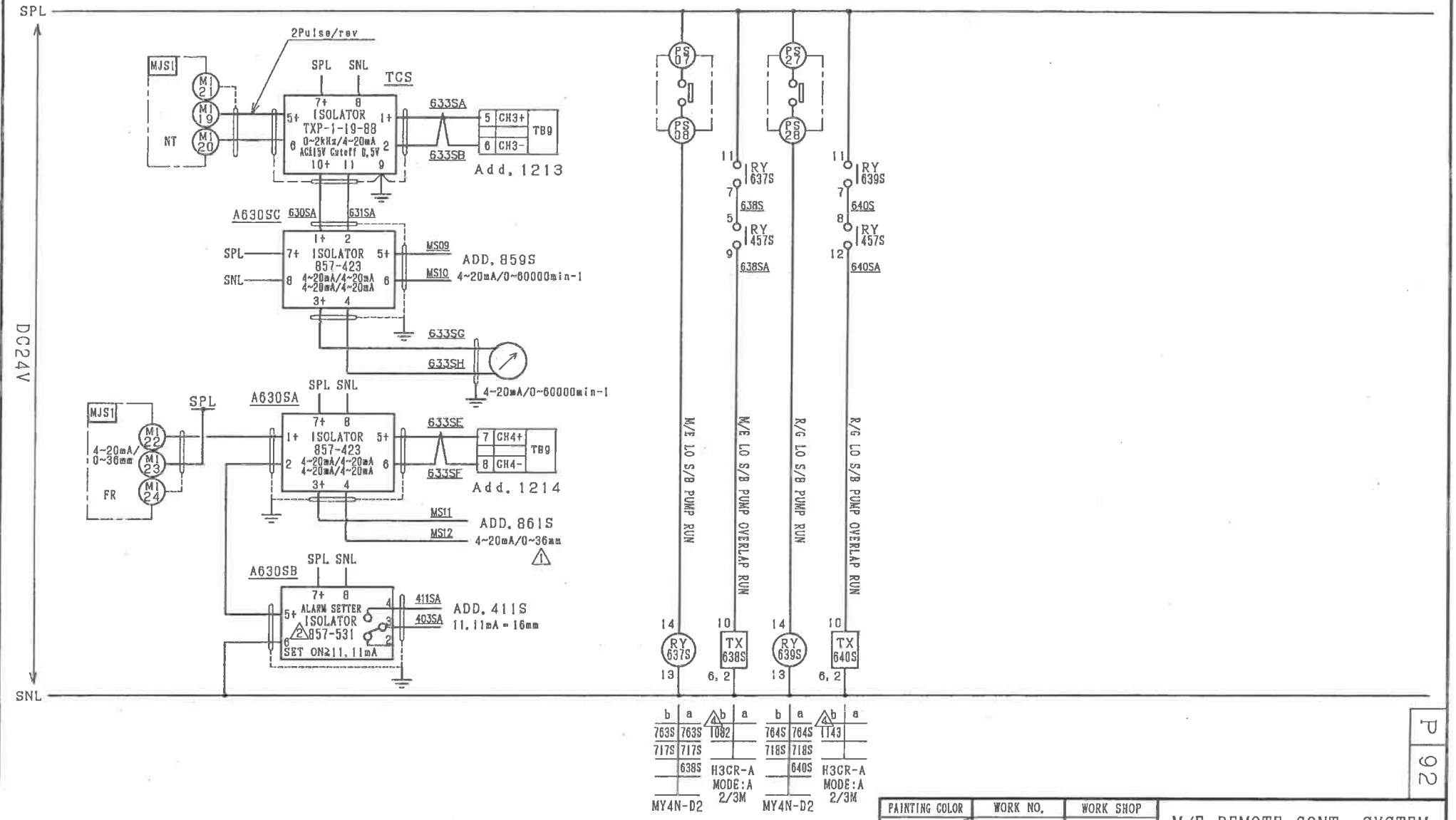
SCALE	MANAGER	CHIEF	CHECK	DRAWN
	6.11.7			T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD26

DATE 10. AUG., 2016 DWG. NO. 3M3-195190

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

626S 627S 628S 629S 630S 631S 632S 633S 634S 635S 636S 637S 638S 639S 640S 641S 642S 643S 644S 645S 646S 647S 648S 649S 650S



STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD27
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER REV	CHIEF	CHECK DRAWN
		REV. 16.11.24 T.M	T.M
			DATE 1, SEP., 2016 DWG. NO. 3M3-195191

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

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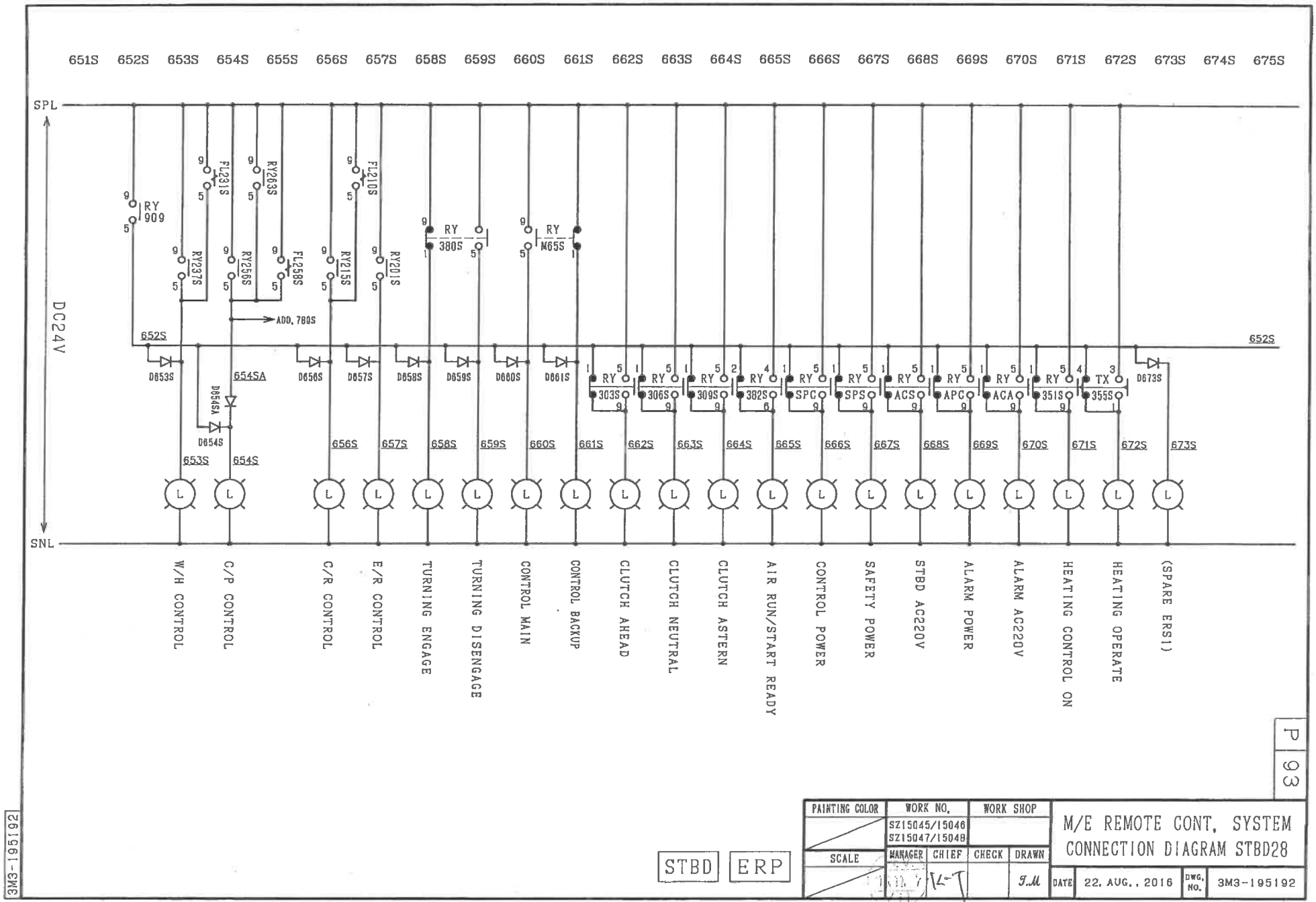
3M3-195191

△4 27, OCT., 2016
△2 29, SEP., 2016
△1 13, SEP., 2016



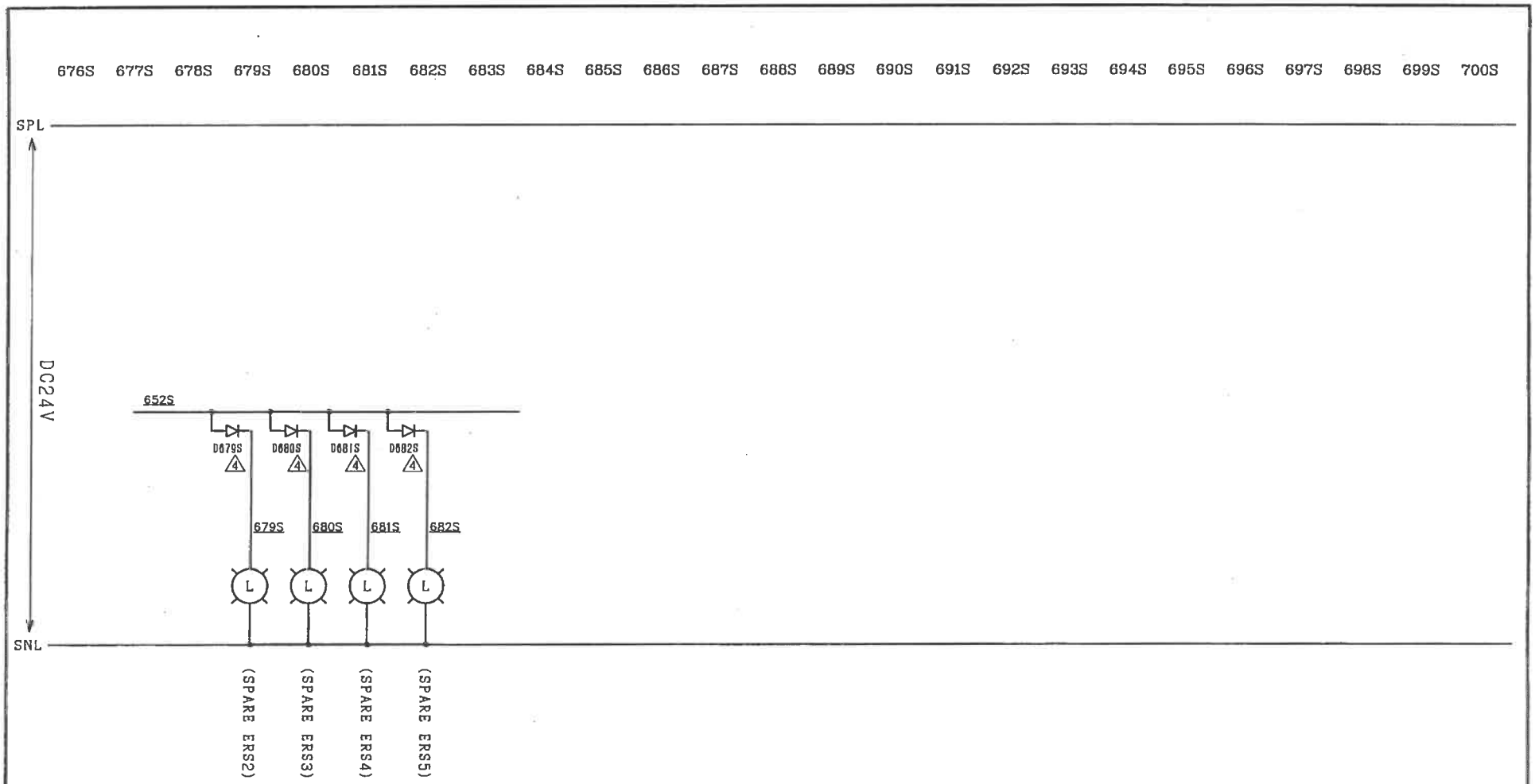
P 93

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD28		
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER	CHIEF	CHECK	DRAWN	
				T.M	DATE 22. AUG., 2016
					DWG. NO. 3M3-195192

STBD ERP

3M3-195192

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P 94

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

STBD ERP

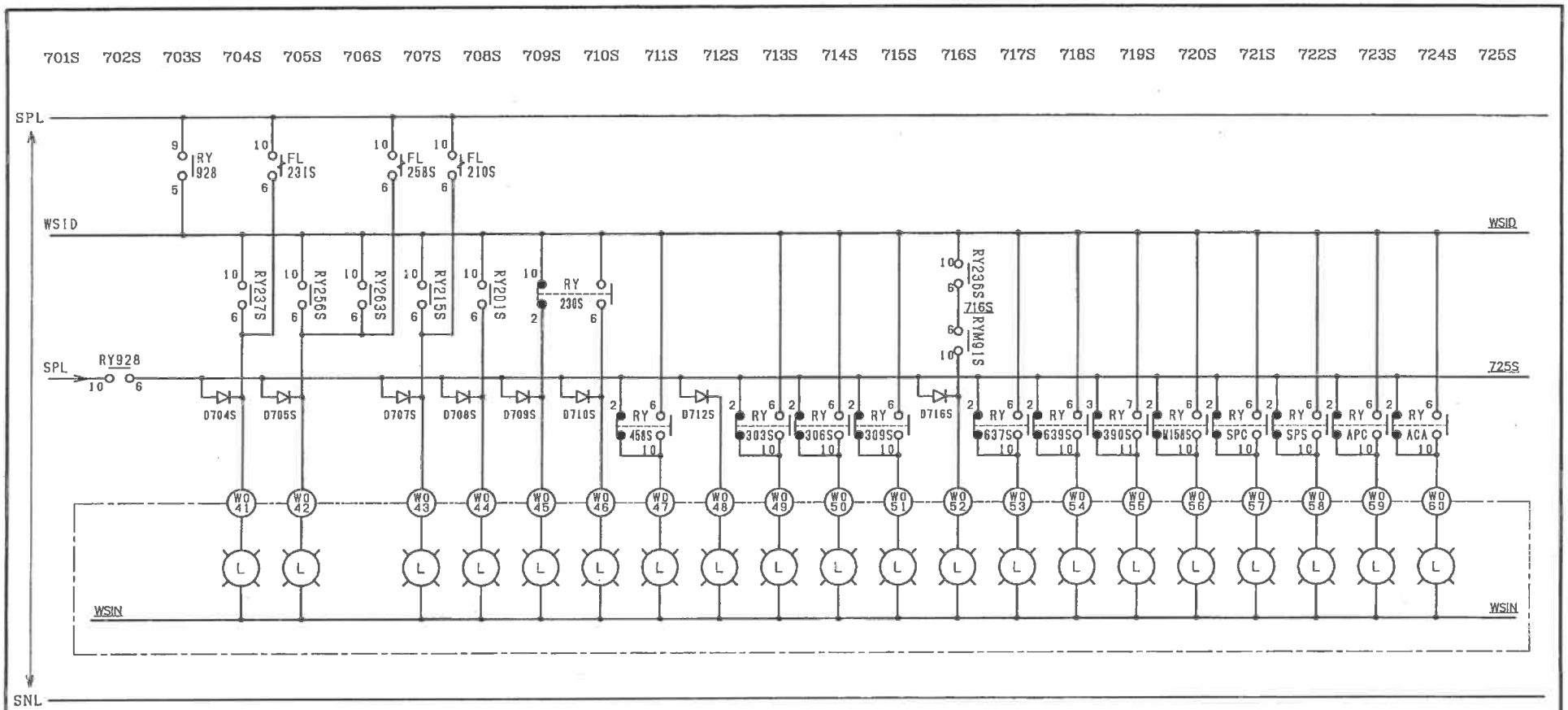
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM STBD29

DATE 3. AUG., 2016 | DWG. NO. 3M3-195193

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195193

④ 1. NOV., 2016



STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

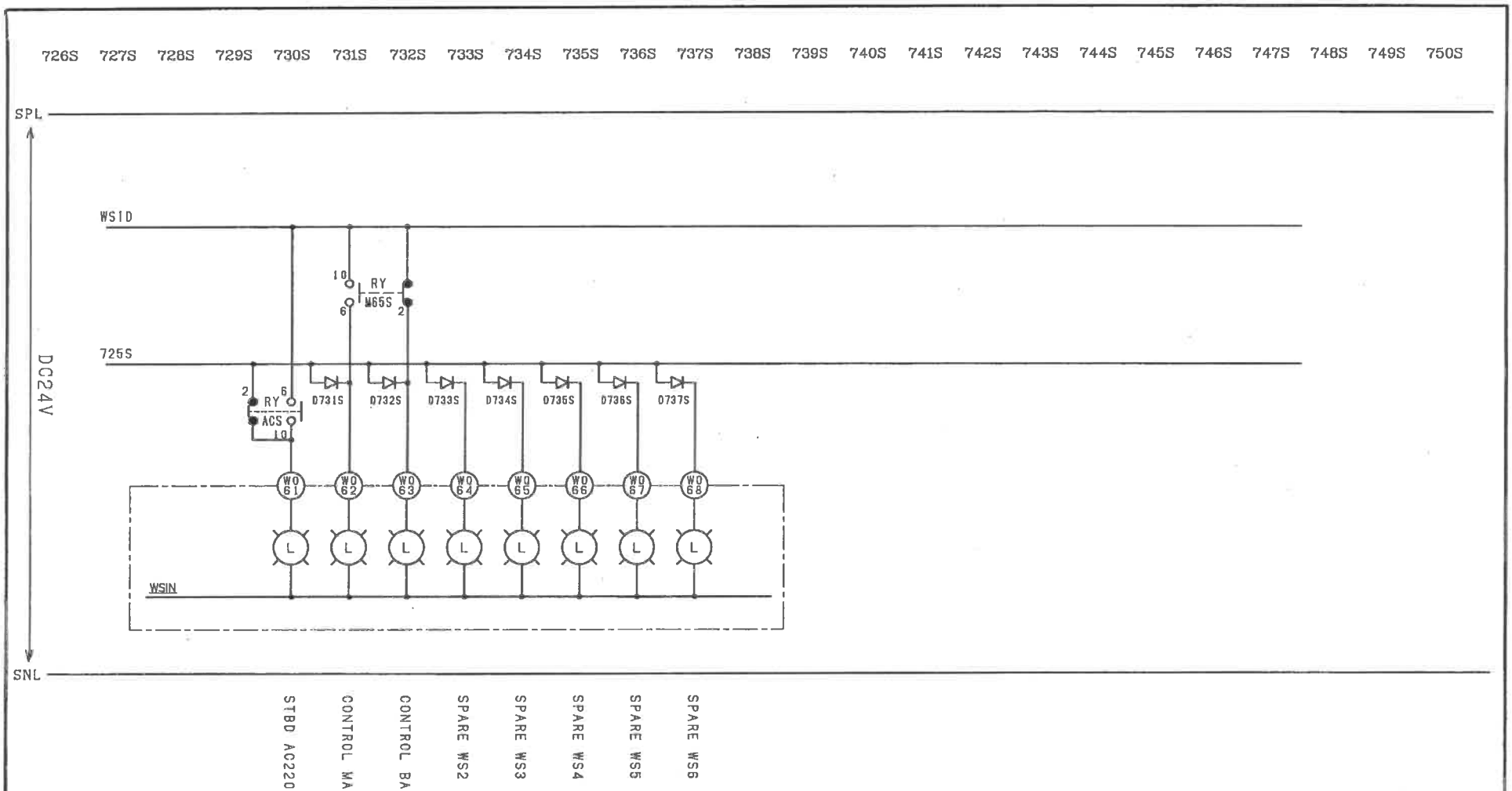
SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD30

DATE 18. AUG., 2016 DWG. NO. 3M3-195194

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195194



3M3-195195

STBD ERP

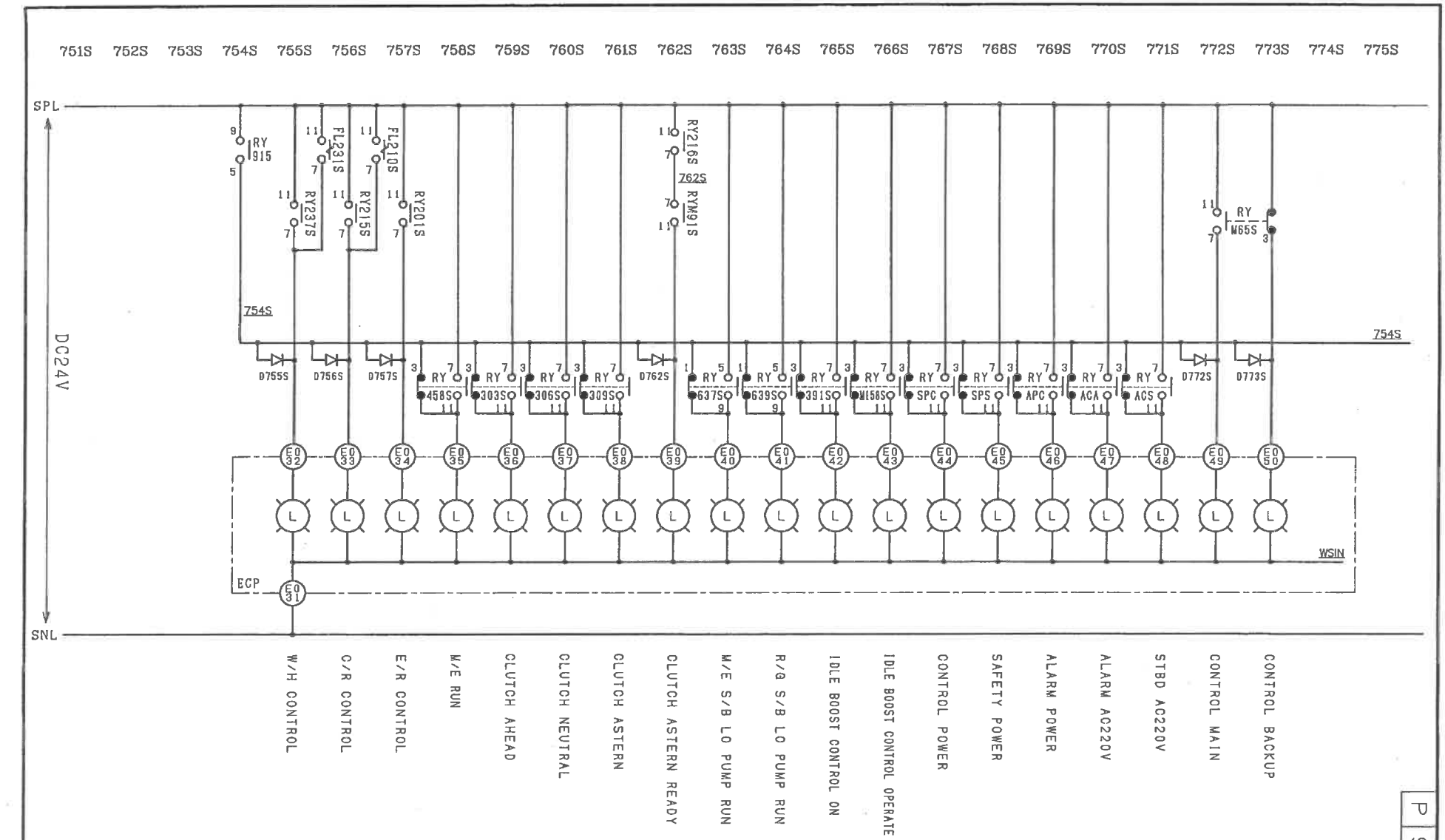
PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM STBD31

DATE 18. AUG., 2016 DWG. NO. 3M3-195195

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK	DRAWN
			T.M

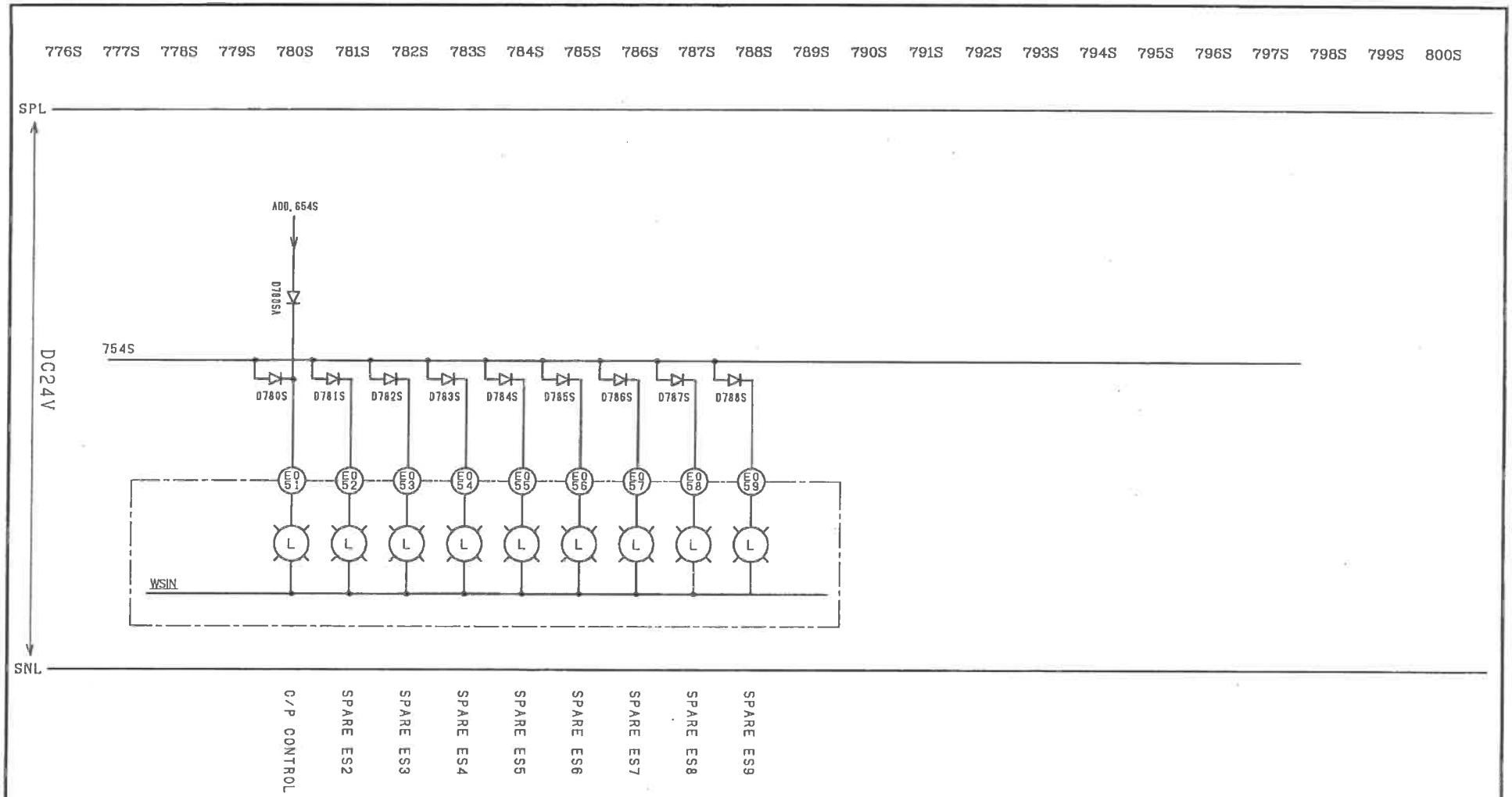
STBD ERP

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD32

DATE 15. AUG., 2016 DWG. NO. 3M3-195196

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195196



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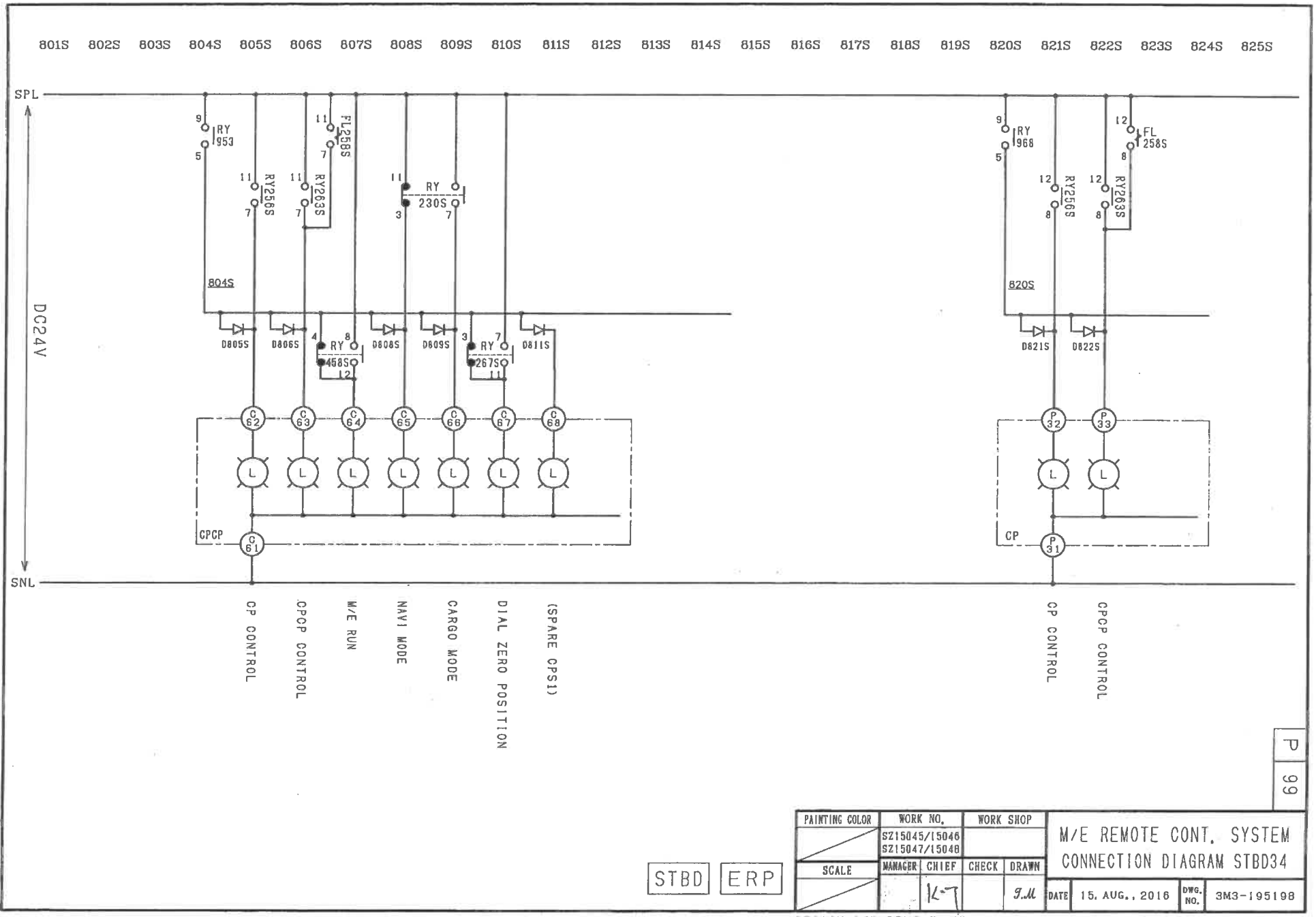
3M3-195197

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD33
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK / DRAWN	
	REV. 16.11.	J.M	DATE 19. AUG., 2016 / DWG. NO. 3M3-195197

STBD ERP

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



STBD ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

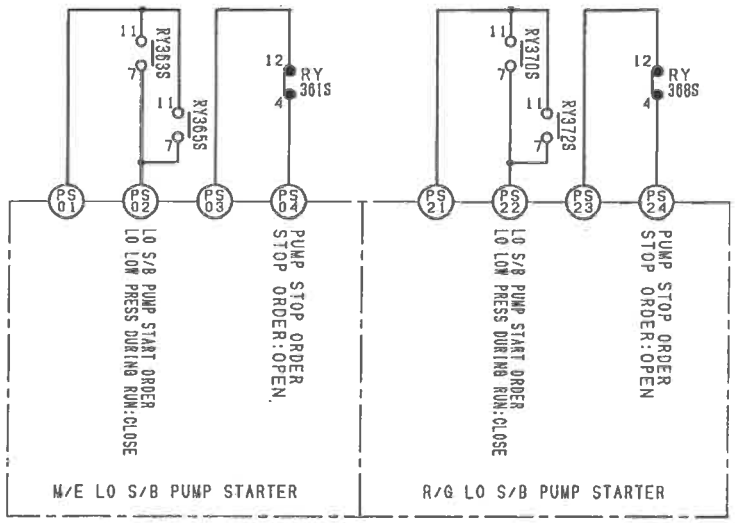
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD34

DATE 15. AUG., 2016 DWG. NO. 3M3-195198

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195198

826S 827S 828S 829S 830S 831S 832S 833S 834S 835S 836S 837S 838S 839S 840S 841S 842S 843S 844S 845S 846S 847S 848S 849S 850S



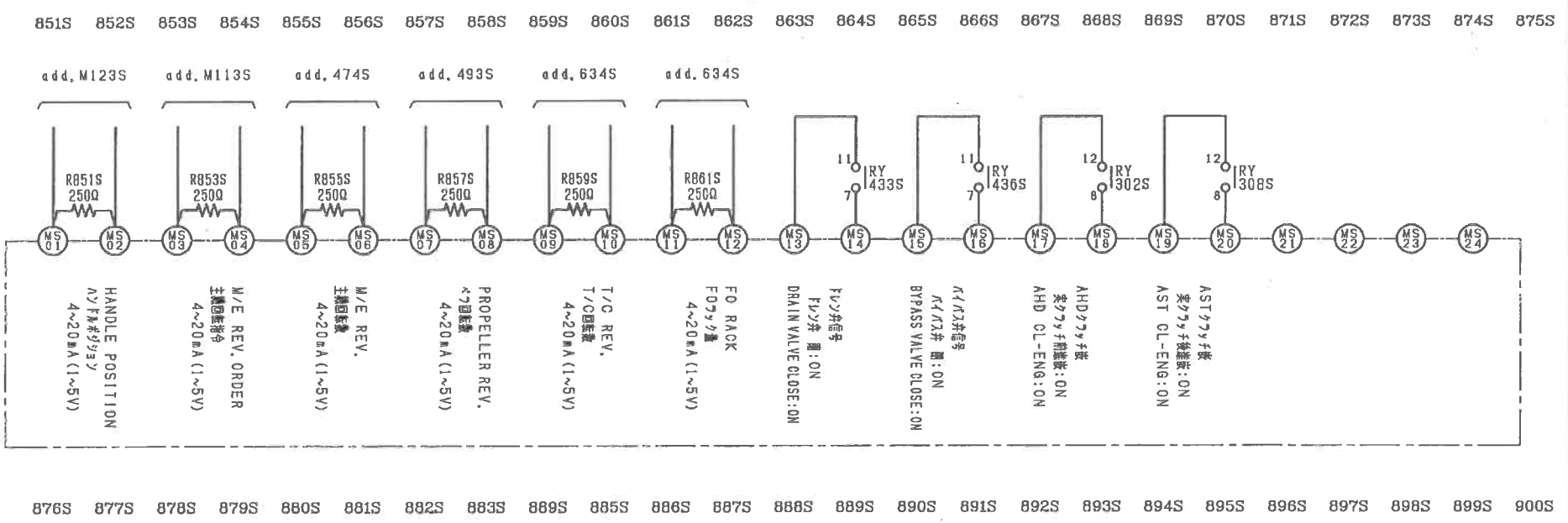
P 100

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD35		
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE	DWG. NO.	
	6.11.	Y.M	3. AUG.. 2016	3M3-195199	

STBD ERP

3M3-195199

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P 101

3M3-195200

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	6.11.7			T.M

STBD ERP

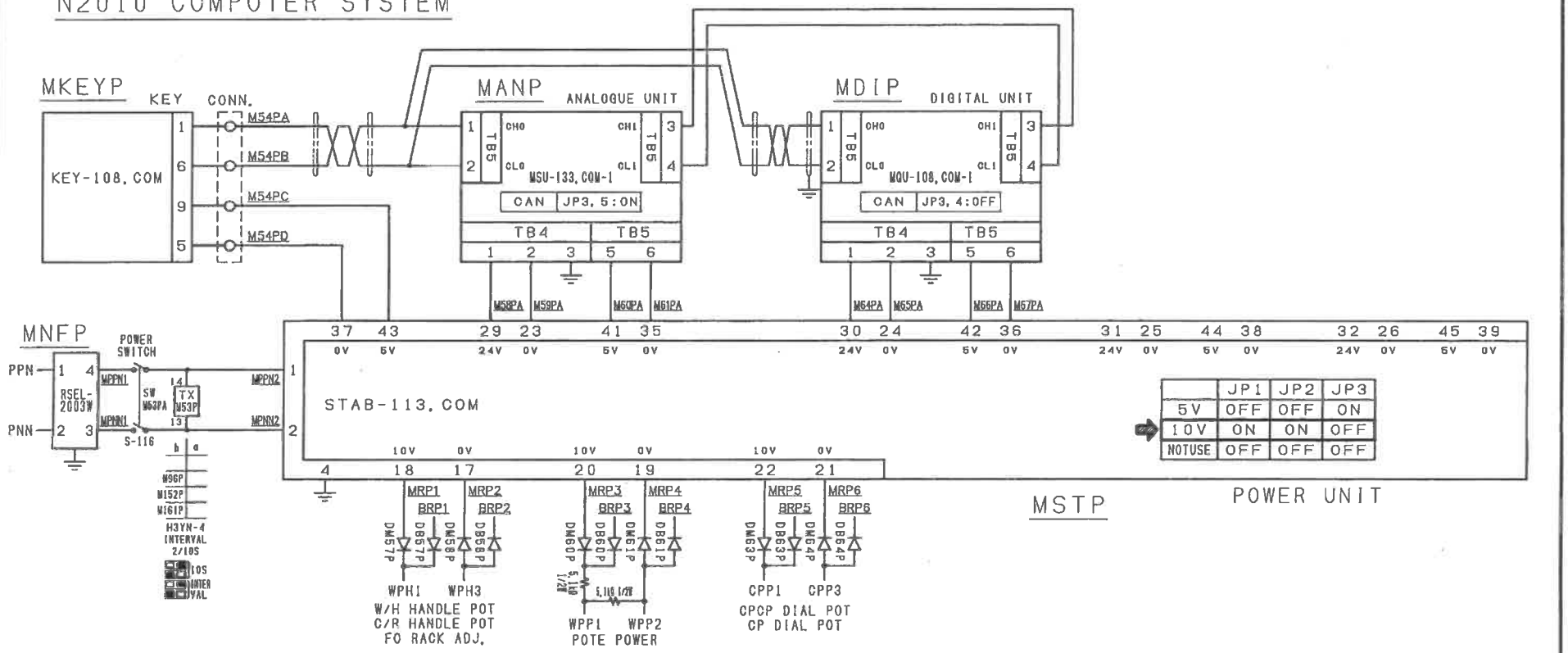
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM STBD36

DATE 15. AUG., 2016 DWG. NO. 3M3-195200

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

M51P M52P M53P M54P M55P M56P M57P M58P M59P M60P M61P M62P M63P M64P M65P M66P M67P M68P M69P M70P M71P M72P M73P M74P M75P

N2010 COMPUTER SYSTEM



PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT1
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF / CHECK / DRAWN		DATE 9, AUG., 2016 / DWG. NO. 3M3-195201
	6.11. / T.M		

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

N2010 ERP P 102

3M3-195201

2 29, SEP., 2016

M76P M77P M78P M79P M80P M81P M82P M83P M84P M85P M86P M87P M88P M89P M90P M91P M92P M93P M94P M95P M96P M97P M98P M99P M100P

ANALOG UNIT MSU-133. COM-1 (INPUT) MANS

JP11:OFF JP13:OFF JP18:OFF JP12:OFF JP14:OFF JP18:OFF JP8:OFF JP19:OFF JP20:OFF JP9:OFF JP21:OFF JP23:OFF JP10:OFF JP24:OFF JP25:OFF JP2:ON JP4:ON JP7:OFF

TB1-3 | TB1-2 | TB1-1 | TB1-3

1 2 | 3 4 | 1 2 | 3 4 | 5 6 | 1 2 | 3 4 | 5 6 | 6

EPH3 WPL3 WPH2 WPH1 WPH3 CPP2 PPP2 CPP1 CPP3 M87PA M87PB M88PA M88PB WPP1 WPP2

0V +10V +5V

M/E REV. 4~20mA/ 0~1500min-1 Add. 467P

PROPELLER REV. 4~20mA/ -400~+400min-1 Add. 485P

POTE POWER Add. M60P

EP HE EP H1 EP H2 EP H3 — C/R HANDLE (ECS, AHD., AST.)

WP LE WP L1 WP L2 WP L3 — F0 RACK DIAL (2kΩ, WHP, ADD., RED.)

WP HE WP H1 WP H2 WP H3 — W/H HANDLE (WHP, AHD., AST.)

CP PE CP P1 CP P2 CP P3 — CPCP DIAL (CPCP)

PP P1 PP P2 PP P3 PP PE — CP DIAL (CP)

COM (+) TB2-1 TB2-2

M93PA M94PA M95PA M96PA M97PA M98PA M99PA M100PA

PPC PNC

RY 203P RY 214P RY 236P RY 262P RY 255P RY 302P RY 308P RY 456P

E/R CONTROL C/R CONTROL W/H CONTROL CPCP CONTROL CP CONTROL CL AH CL AS M/E RUN

ANALOG UNIT MSU-133. COM-1 (OUTPUT) MANS

M87PD M88PD M89PD M90PD M91PD

D87PB D88PB D89PB D90PB D91PB

COMMUNICATION PORT

COM (-) Rx- Rx+ Tx+ Tx- CG1 -

TB2-2 | TB2-3 | TB3

4 5 6 | 1 2 | 3 4 5 6 | 1 2 3 4 5 6

M87PC M88PC M89PC M90PC M91PC M93PC M94PC M95PC M96PC

D87SA D88SA D89SA D90SA D91SA

M87PE M88PE M89PE M90PE M91PE

ENG. ORDER (M87P) | Ø-ENG. ORDER (M88P) | ENG. ORDER (M89P) | Ø-ENG. ORDER (M90P) | ENG. READY (M91P)

CLUTCH AH | CLUTCH AH | CLUTCH AS | CLUTCH AS | CLUTCH AS

POT. PWR ABN. (N/C) | COMM. ABNORMAL (N/C) | CONT. STS. ABN. (ANA. UNIT N/C)

TXM53P RYM63P

RY M87P RY M88P RY M89P RY M90P RY M91P RY M93P RY M94P RY M95P

MY4N-D2

PPC PNC

328P	329P	333P	334P	716P	762P

M59P	M59P	M61P
1554	1553	1551

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N2010 ERP

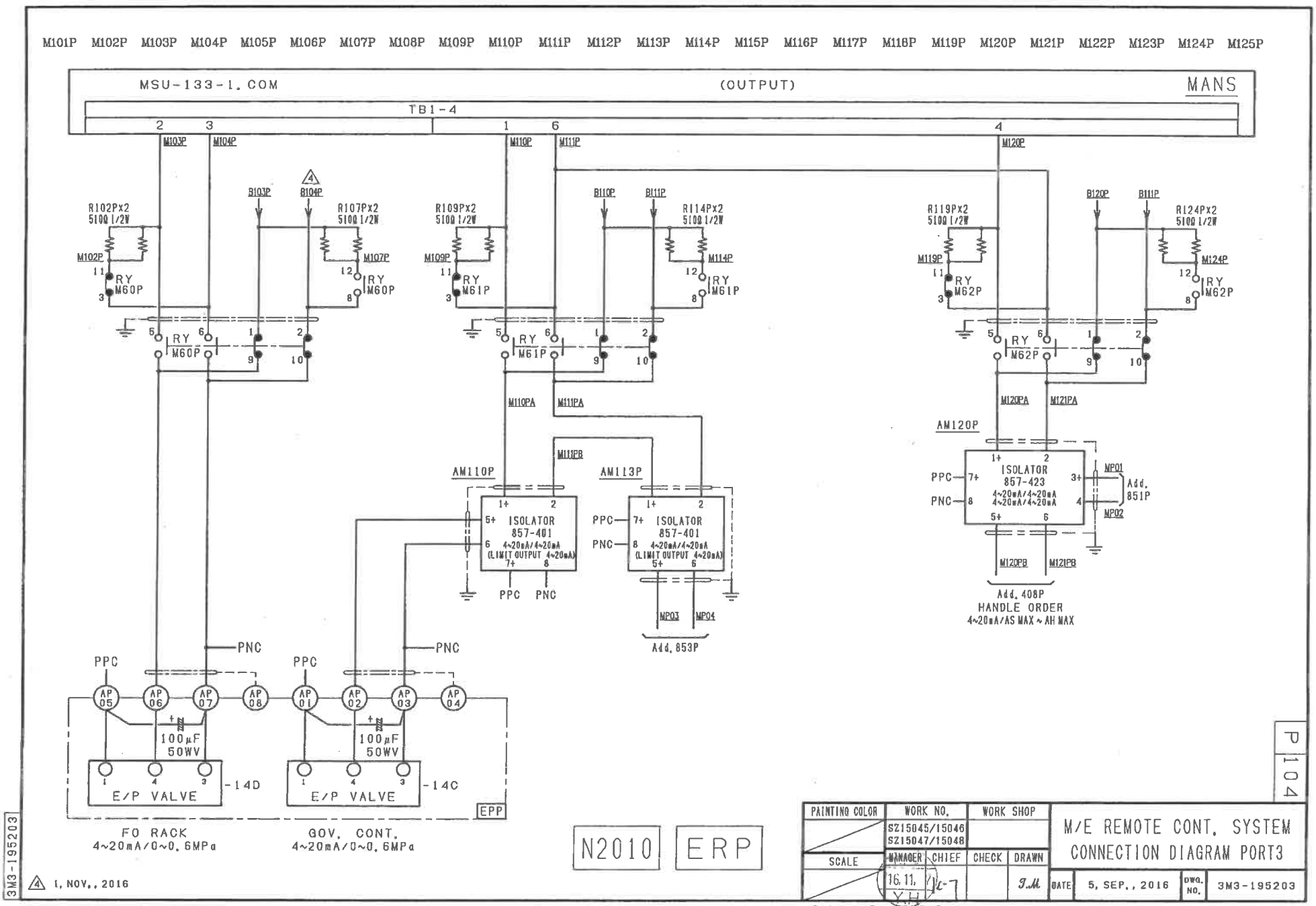
PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER	CHIEF	CHECK	DRAWN
	'16.11.7 Y.H			T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT2

DATE 22. AUG., 2016 DWG. NO. 3M3-195202

3M3-195202

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.


M101P M102P M103P M104P M105P M106P M107P M108P M109P M110P M111P M112P M113P M114P M115P M116P M117P M118P M119P M120P M121P M122P M123P M124P M125P
MSU-133-1.COM
(OUTPUT)
MANS
TB1-4
M103P
M104P
M110P
M111P
M120P
B103P
B104P
B110P
B111P
B120P
B111P
R102Px2 510Ω 1/2W
R107Px2 510Ω 1/2W
R109Px2 510Ω 1/2W
R114Px2 510Ω 1/2W
R119Px2 510Ω 1/2W
R124Px2 510Ω 1/2W
M102P
M107P
M109P
M114P
M119P
M124P
RY M60P
RY M61P
RY M62P
M110PA
M111PA
M120PA
M121PA
AM110P
AM113P
AM120P
M113PB
ISOLATOR 857-401
4~20mA/4~20mA
(LIMIT OUTPUT 4~20mA)
ISOLATOR 857-423
4~20mA/4~20mA
4~20mA/4~20mA
PPC
PNC
MP01
MP02
MP03
MP04
Add. 851P
Add. 853P
Add. 408P
HANDLE ORDER
4~20mA/AS MAX ~ AH MAX
M120PB
M121PB
AP 05
AP 06
AP 07
AP 08
AP 01
AP 02
AP 03
AP 04
100μF 50WV
E/P VALVE
-14D
-14C
EPP
FO RACK
4~20mA/0~0.6MPa
GOV. CONT.
4~20mA/0~0.6MPa
N2010
ERP
3M3-195203
4 1, NOV., 2016
P 104
PAINTING COLOR
WORK NO.
SZ15045/15046
SZ15047/15048
WORK SHOP
SCALE
MANAGER
CHIEF
CHECK
DRAWN
16.11.
Y.H
T.M
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM PORT3
DATE
5, SEP., 2016
DWG. NO.
3M3-195203
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

M126P M127P M128P M129P M130P M131P M132P M133P M134P M135P M136P M137P M138P M139P M140P M141P M142P M143P M144P M145P M146P M147P M148P M149P M150P

DIGITAL UNIT(1/3) MQU-108, COM-1

MQU

TB1-3 1	TB1-3 2	入力端子	M129PA 6	M130PA 5	TB1-1 M131PA 4	M132PA 3	M133PA 2	M134PA 1	M135PA 6	M136PA 5	TB1-2 M137PA 4	3	M138PA 4 2	M139PA 4 1	TB1-3 6	5	4	3	M144PA	TB3 1	2	3	4	5	6
(004P IN Gr1/2)		アドレス				(004P IN Gr1)								(004P IN Gr2)						GPS +	GPS -	VDR +	VDR -	VDR 0V	
COM	COM	BIT	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0							

PPC

PPC

PNC

PNC

RY 501P (3/11)
RY 558P (6/10)
RY 269P (5/9)
RY 240P (5/9)
RY 241P (5/9)
RY 218P (5/9)
RY 219P (5/9)
TX 427P (6/10)
RY 390P (10/6)
RY 391P (10/6)

PBM144P/M63P

W2 25 — PB1469P — W2 26
E2 19 — PB1468P — E2 20
FWD
EGD

F.O HANDLE
5V ON
CARGO CL ENG
HANDLE LS AHEAD (W/H)
HANDLE LS ASTERN (W/H)
HANDLE LS AHEAD (C/R)
HANDLE LS ASTERN (C/R)
W.V. OFF + NEUTRAL (For DRAIN/BYPASS)
IDLE BOOST ON (W/H)
IDLE BOOST ON (C/R)
SYS. ABNORMAL RESET

DIGITAL UNIT(2/3) MQU-108, COM-1

MQU

TER	TB3 1	2	3	4	5	6	TB4 1	2	3	4	5	6
	GPS +	GPS -	VDR +	VDR -	VDR 0V		24V P	0V N	FG	24V P	0V N	FG

M64PA M65PA

30 24

POWER UNIT
(STAB-113, COM)

PNC

PNC

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3M3-195204

4 17, NOV., 2016

N2010 ERP

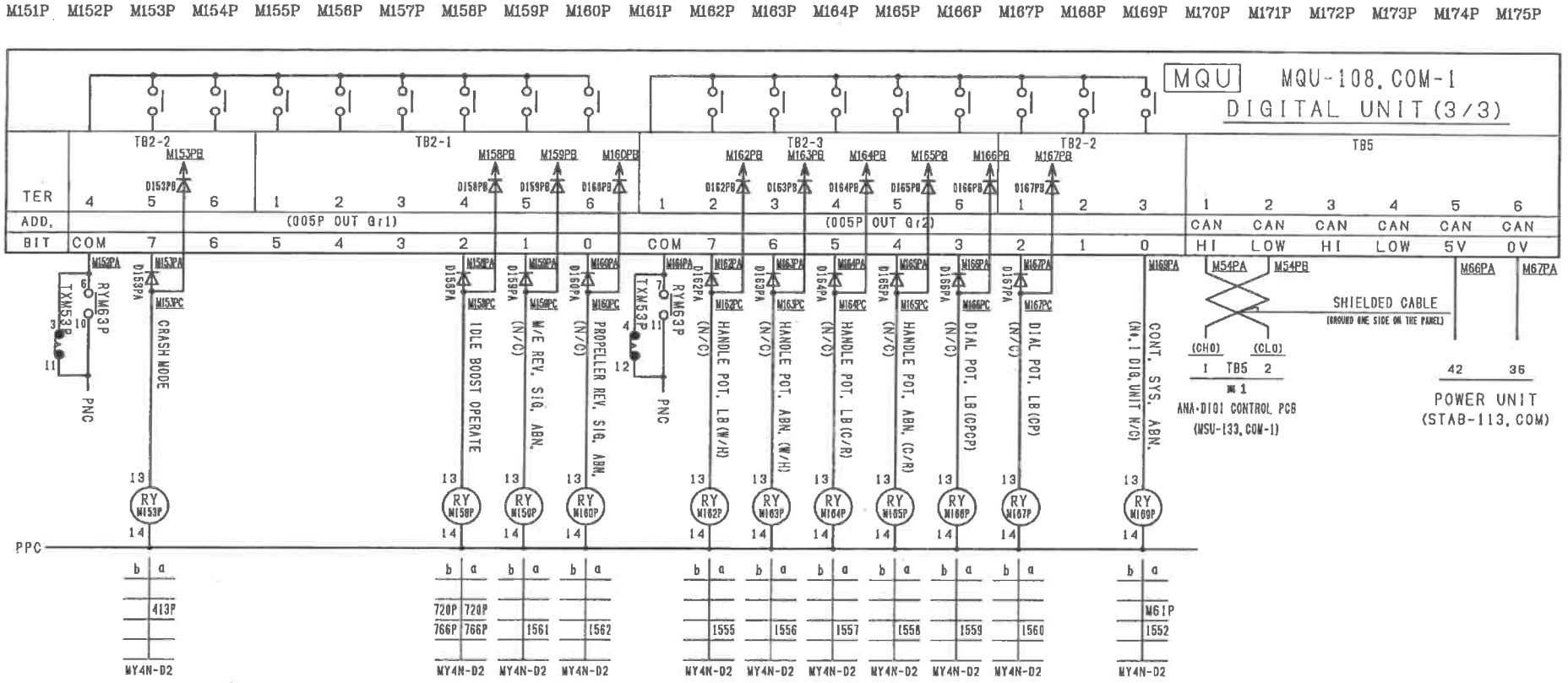
PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT4

DATE 24, AUG., 2016 DWG. NO. 3M3-195204

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



N2010 ERP

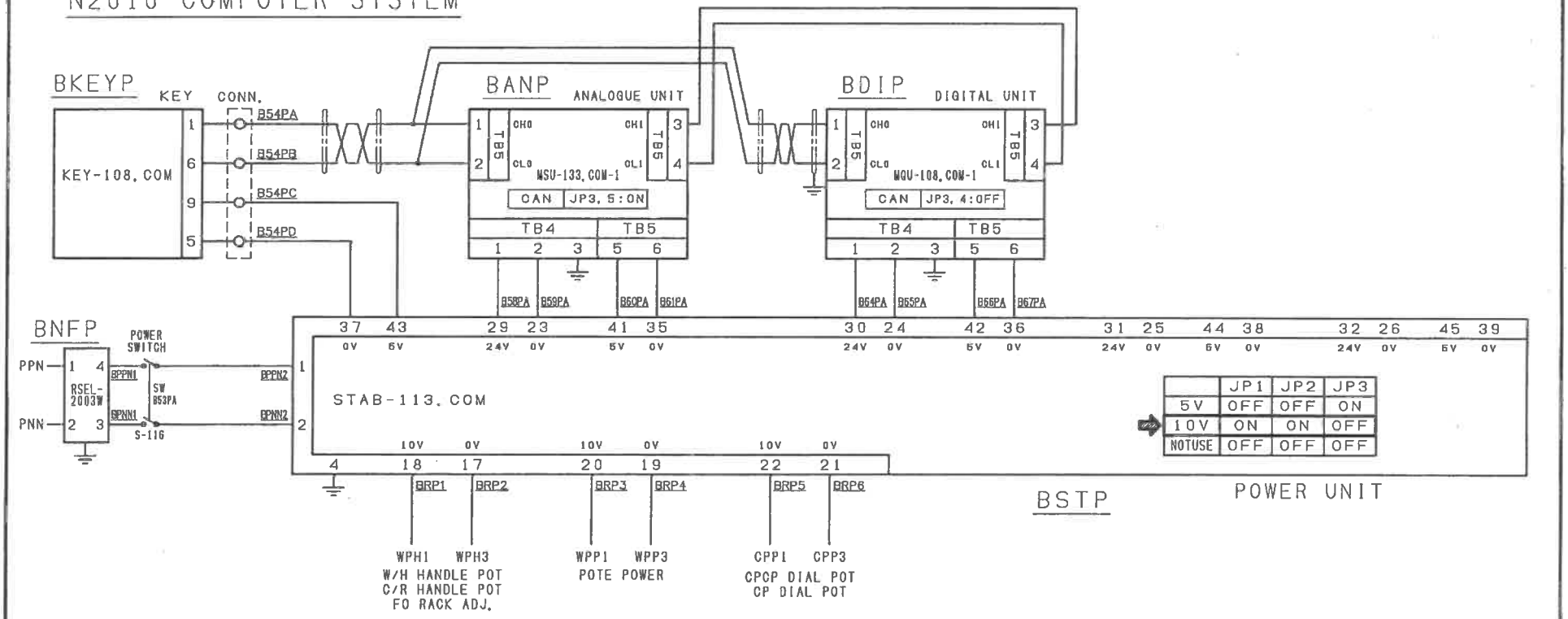
PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT5			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	8. SEP., 2016
	16. 11. 7			T.M	DWG. NO.	3M3-195205

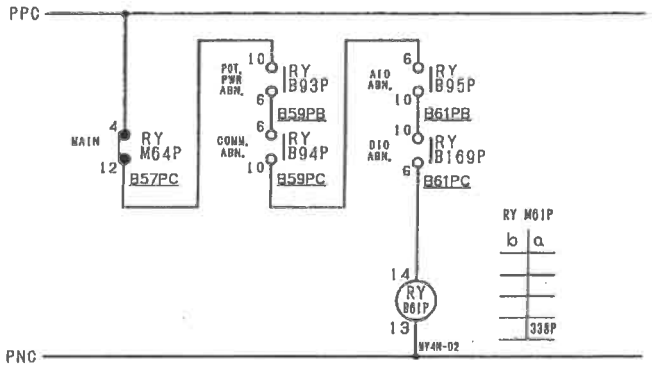
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195205

B51P B52P B53P B54P B55P B56P B57P B58P B59P B60P B61P B62P B63P B64P B65P B66P B67P B68P B69P B70P B71P B72P B73P B74P B75P

N2010 COMPUTER SYSTEM





N2010 ERP P107

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT6	
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE: 9, AUG., 2016	DWG. NO.: 3M3-195206
	'15.11.7	J.M		

3M3-195206

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

B76P B77P B78P B79P B80P B81P B82P B83P B84P B85P B86P B87P B88P B89P B90P B91P B92P B93P B94P B95P B96P B97P B98P B99P B100P

ANALOG UNIT MSU-133, COM-1 (INPUT) MANS

JP11:OFF JP13:OFF JP16:OFF JP12:OFF JP14:OFF JP18:OFF JP8:OFF JP19:OFF JP20:OFF JP9:OFF JP21:OFF JP23:OFF JP10:OFF JP24:OFF JP25:OFF JP2:ON JP4:ON JP7:OFF

TB1-3 | TB1-2 | TB1-1 | TB1-3

1 2 | 3 4 | 1 2 | 3 4 | 5 6 | 1 2 | 3 4 | 5 6 | 6

EPH2 WPL2 WPH2 CPP2 PPP2 B87PA B87PB B88PA B88PB

WPH3 0V CPP3 0V

M/E REV. 4~20mA/ 0~1500min-1 Add. 467P

PROPELLER REV. 4~20mA/ -400~+400min-1 Add. 485P

EPH2 WPL2 WPH2 WPH3 CPP2 CPP3 PPP2 WPP1 WPP2

COM (+)

TB2-1 | TB2-2

1 2 3 4 5 6 | 1 2 3

PPC

M93PA M94PA M95PA M96PA M97PA M98PA M99PA M100PA

PNC

E/R CONTROL
C/R CONTROL
W/H CONTROL
CPCP CONTROL
CP CONTROL
CL AH
CL AS
M/E RUN

ANALOG UNIT MSU-133, COM-1 (OUTPUT) MANS

COMMUNICATION PORT

COM (-) Rx- Rx+ Tx+ Tx- CGI -

TB2-2 | TB2-3 | TB3

4 5 6 | 1 2 | 3 4 5 6 | 1 2 3 4 5 6

CLUTCH AH ENG. ORDER
CLUTCH AH D-ENG. ORDER
CLUTCH AS ENG. ORDER
CLUTCH AS D-ENG. ORDER
CLUTCH AS ENG. READY

M87PD M88PD M89PD M90PD M91PD

B93PC POT. PWR ABN. (N/C)
B94PC COMM. ABNORMAL (N/C)
B95PC CONT. SYS. ABN. (ANA. UNIT N/C)
B96PC RYM64P

RY B93P RY B94P RY B95P

13 14 MY4N-02

PPC PNC

b a | b a | b a

B59P 1578 | B59P 1577 | B61P 1575

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N2010 ERP

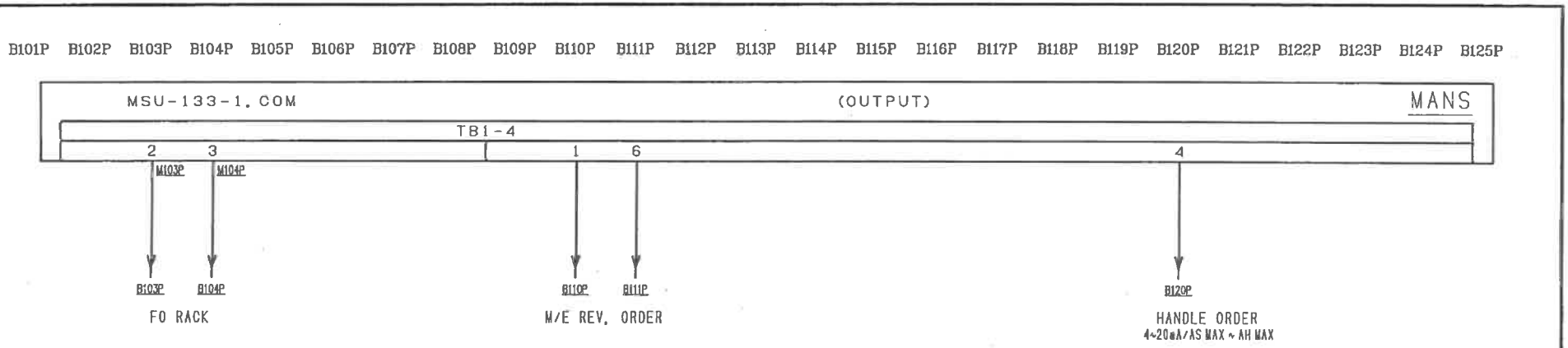
PAINTING COLOR	WORK NO.	WORK SHOP		
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER	CHIEF	CHECK	DRAWN
	16.11. 7 YH			T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT7

DATE 22, AUG., 2016 DWG. NO. 3M3-195207

3M3-195207

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P109

3M3-195208

N2010 ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT8	
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER CHIEF	CHECK DRAWN	DATE 22. AUG., 2016	DWG. NO. 3M3-195208
	6.11.7	T.M		

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

B126P B127P B128P B129P B130P B131P B132P B133P B134P B135P B136P B137P B138P B139P B140P B141P B142P B143P B144P B145P B146P B147P B148P B149P B150P

DIGITAL UNIT(1/3) MQU-108, COM-1

MQU

TB1-3 1	TB1-3 2	入力端子	TB1-1 6	5	4	3	2	1	TB1-2 6	5	4	3	2	1	TB1-3 6	5	4	3	TB3 1	2	3	4	5	6
(004P IN Gr1/2)		アドレス	(004P IN Gr1)								(004P IN Gr2)								GPS	GPS	VDR	VDR	VDR	
COM	COM	BIT	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	+	-	+	-	0V	

PPC — PPC

M129PA M130PA M131PA M132PA M133PA M134PA M135PA M136PA M138PA ④ M139PA ④ M144PA

PNC — PNC

FO HANDLE
5V ON
CARGO CL ENG
HANDLE LS AHEAD (W/H)
HANDLE LS ASTERN (W/H)
HANDLE LS AHEAD (C/R)
HANDLE LS ASTERN (C/R)
W.V. :OFF + NEUTRAL (F.O. DRAIN/BYPASS)
IDLE BOOST ON (W/H)
IDLE BOOST ON (C/R)
SYS. ABNORMAL RESET

DIGITAL UNIT(2/3) MQU-108, COM-1

MQU

TER	TB3 1	2	3	4	5	6	TB4 1	2	3	4	5	6
	GPS +	GPS -	VDR +	VDR -	VDR 0V		24V P	0V N	FG	24V P	0V N	FG

B64PA B65PA

30 24

POWER UNIT
(STAB-113, COM)

PNC — PNC

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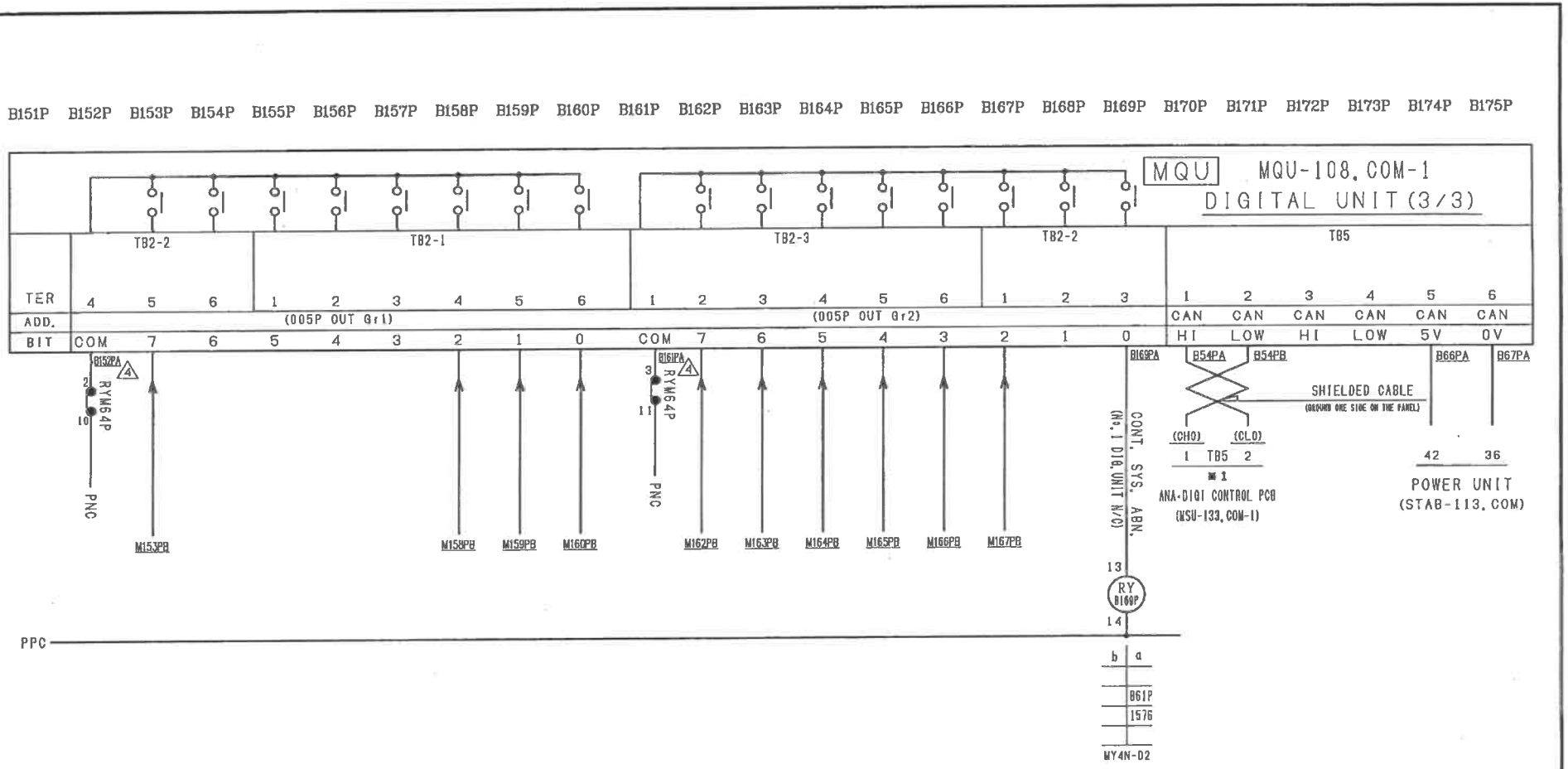
PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT9
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF / CHECK / DRAWN		DATE 24, AUG., 2016 DWG. NO. 3M3-195209
	Y.H / T.M		

N2010 ERP

3M3-195209

④ 17, NOV., 2016

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



N2010 ERP

PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER. CHIEF	CHECK	DRAWN
	6.11.7		T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT10

DATE 8. SEP.. 2016 DWG. NO. 3M3-195210

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

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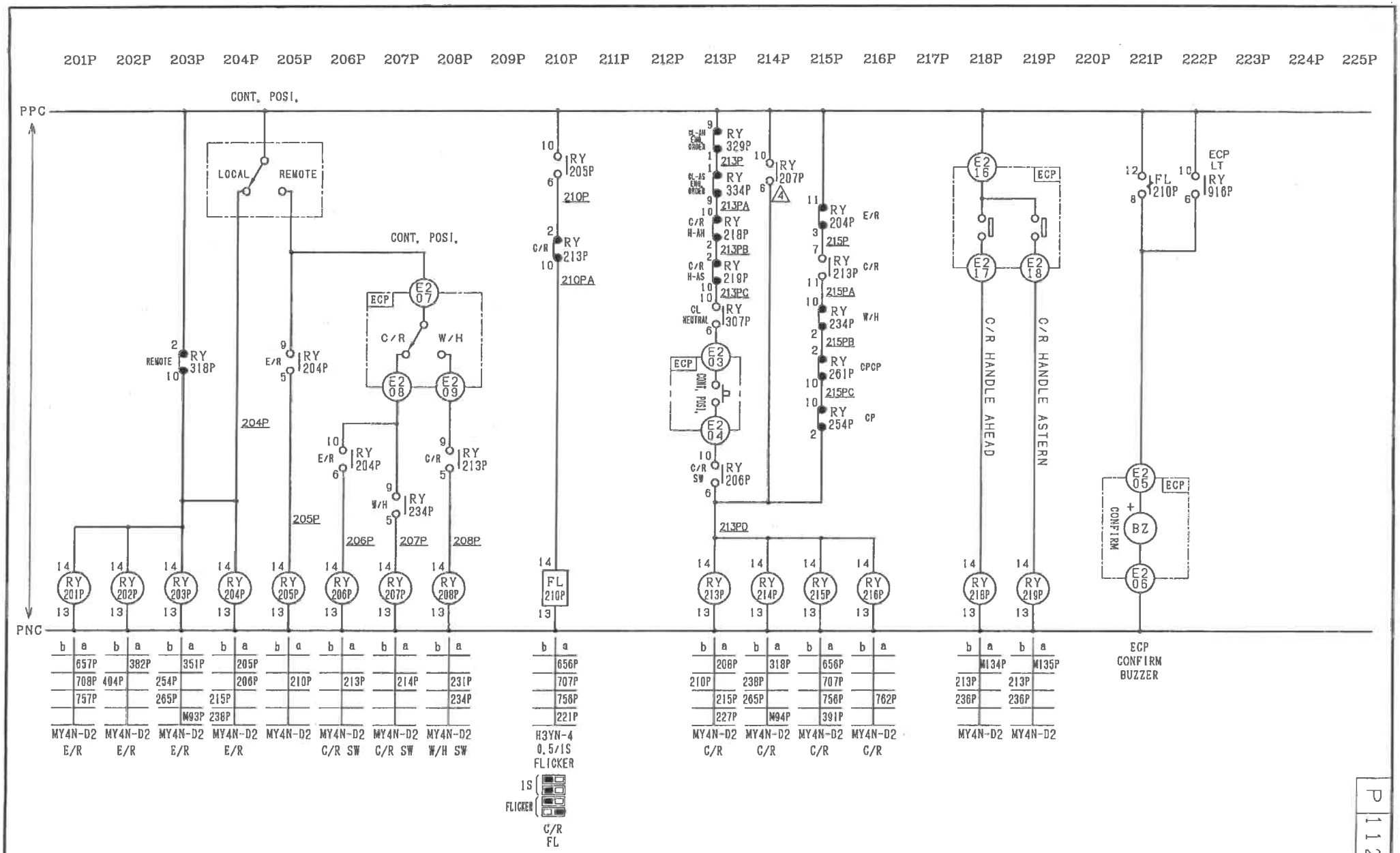
3M3-195210

△4 1. NOV.. 2016



P 112

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	6.11.7			T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT11

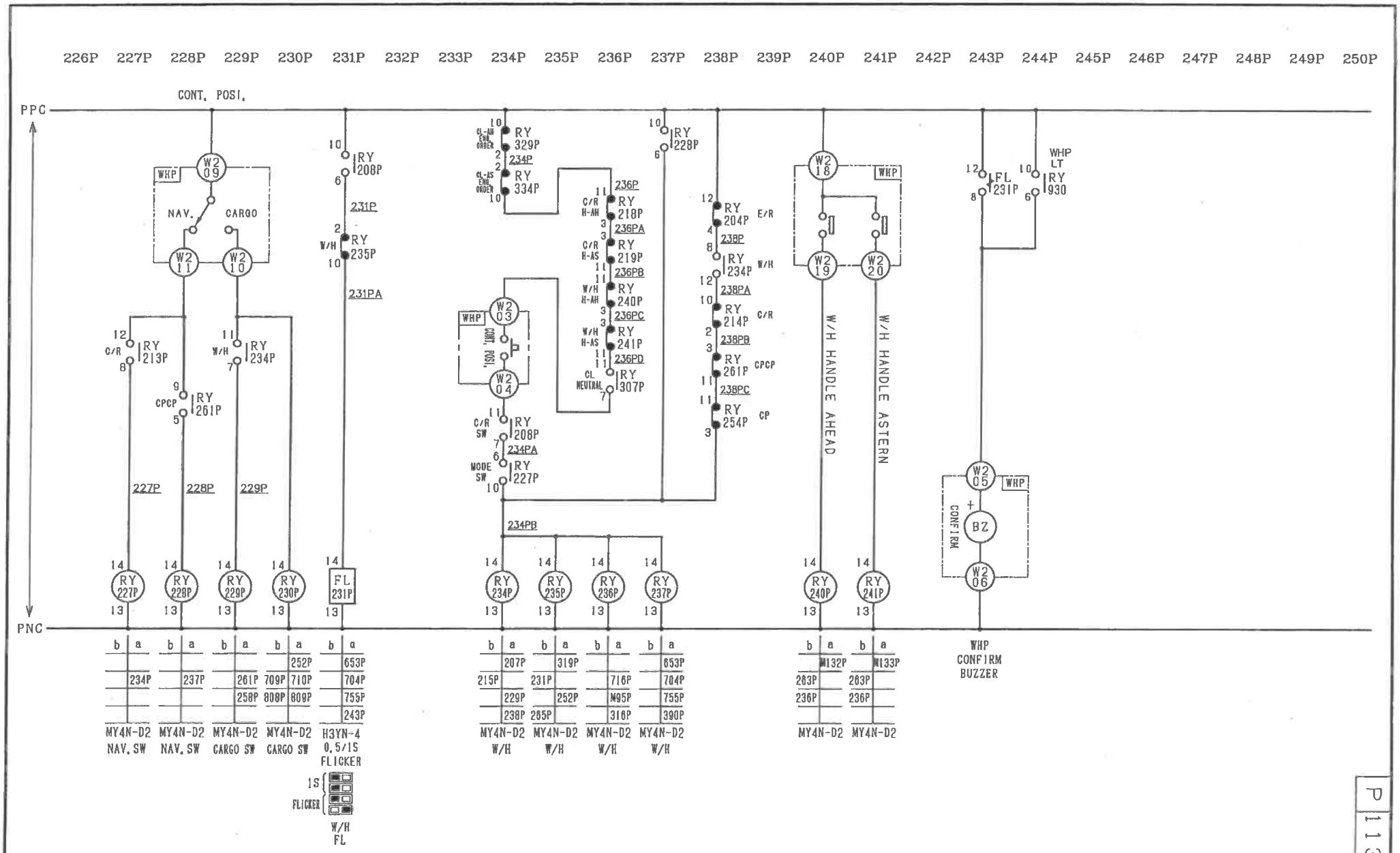
DATE 6, SEP., 2016 | DWG. NO. 3M3-195211

PORT ERP

3M3-195211

△4 1, NOV., 2016

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PORT ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	6.11.7			T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT12

DATE 6. SEP., 2016 DWG. NO. 3M3-195212

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

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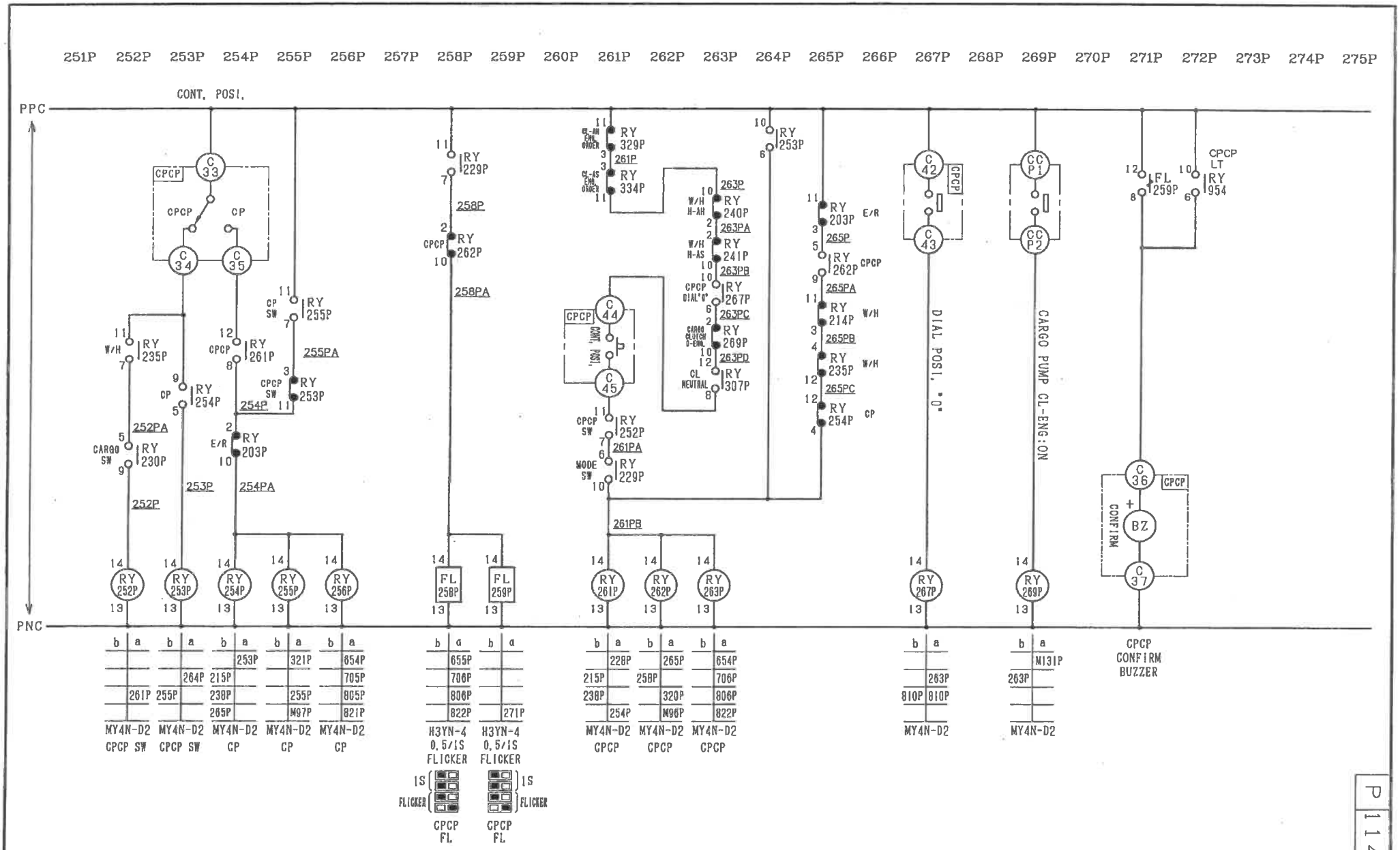
3M3-195212



PORT ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	'16.11.7 YH	K-T		T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT13

DATE	6, SEP., 2016	DWG. NO.	3M3-195213

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

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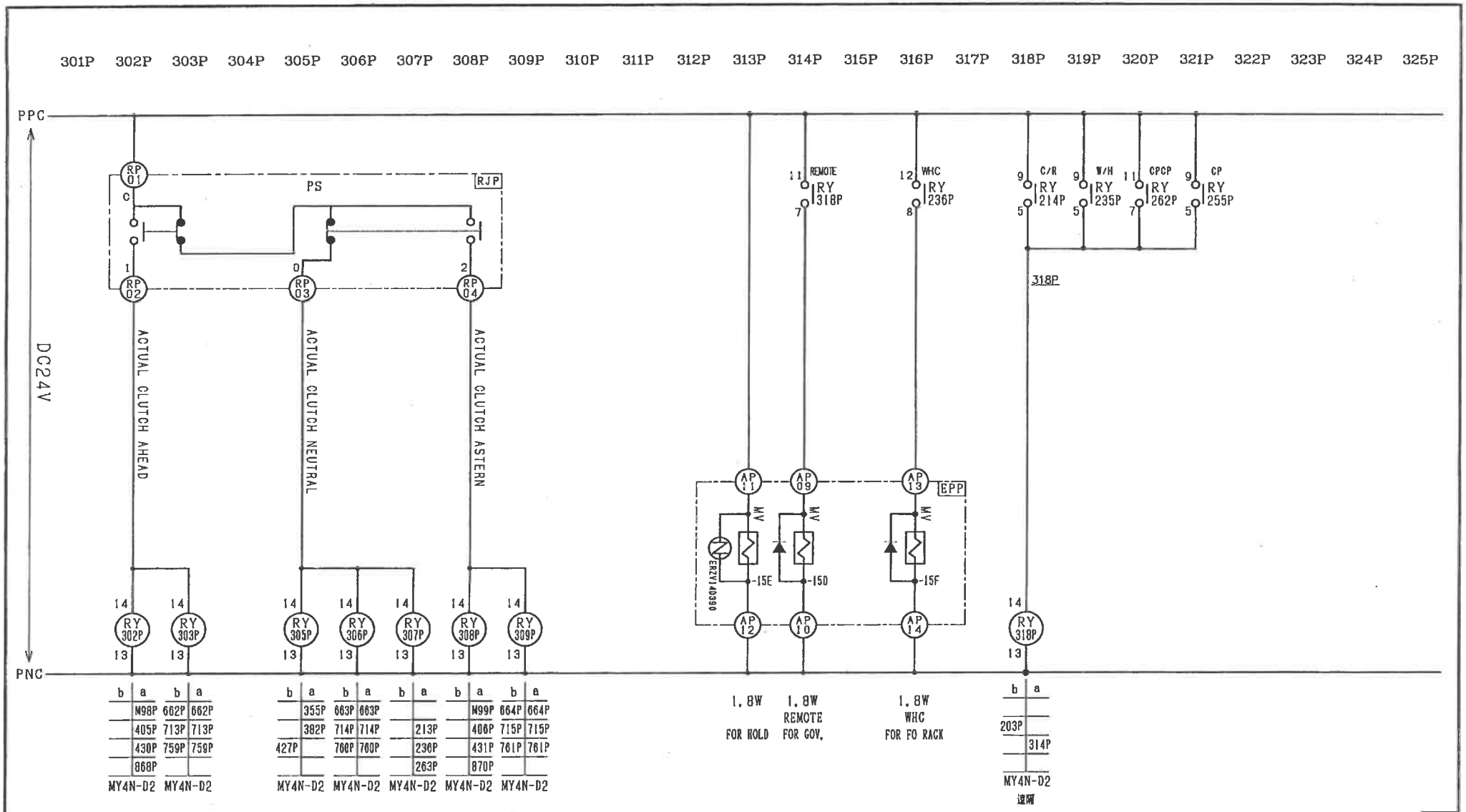
3M3-195213



PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT14
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE: 12. AUG., 2016 / DWG. NO.: 3M3-195214
	16.11.7 YH		J.M

PORT ERP

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

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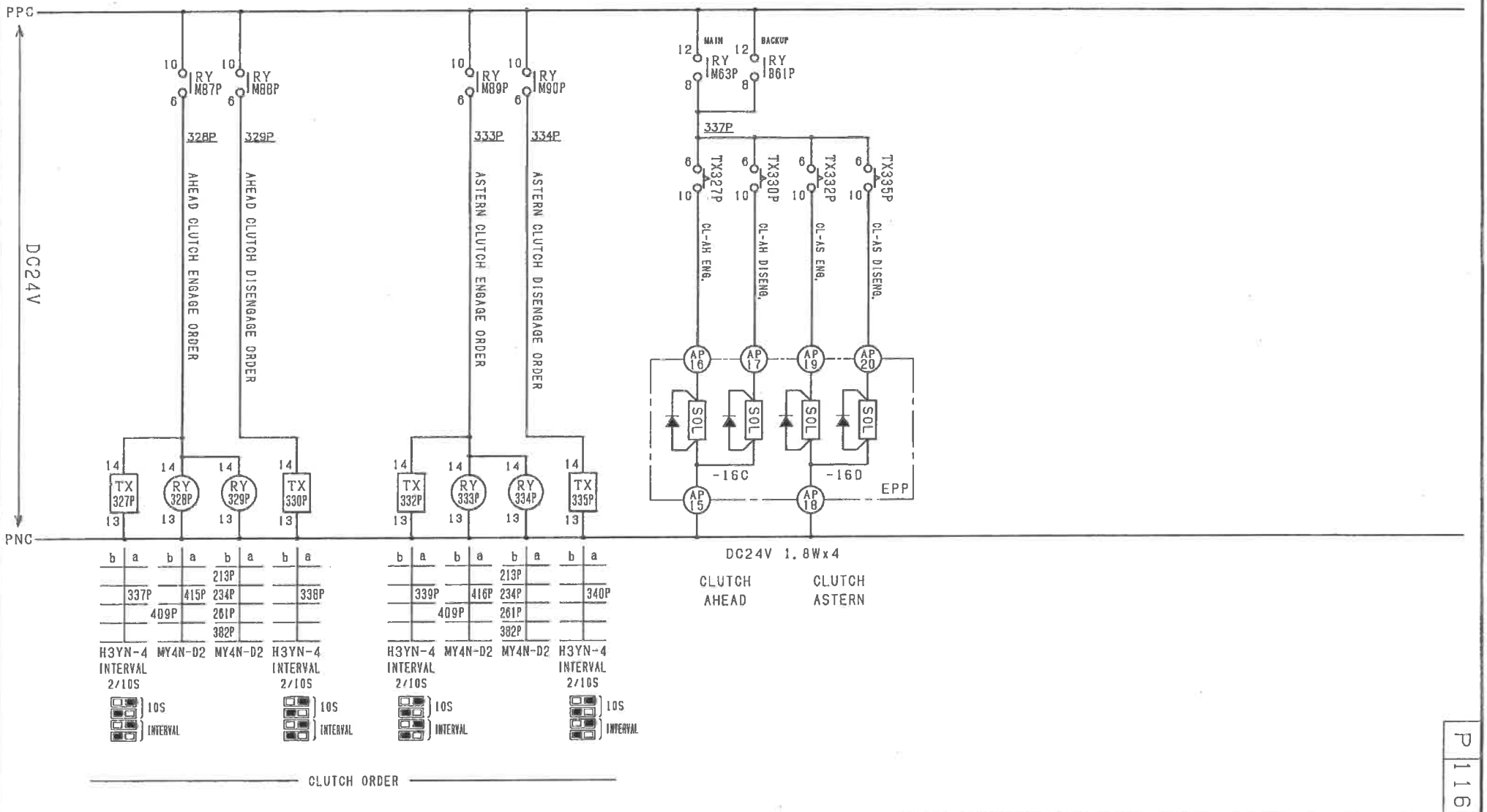
3M3-195214

326P 327P 328P 329P 330P 331P 332P 333P 334P 335P 336P 337P 338P 339P 340P 341P 342P 343P 344P 345P 346P 347P 348P 349P 350P



PAINTING COLOR	WORK NO. SZ15045/15046 SZ15047/15048	WORK SHOP			M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT15
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE 6. SEP., 2016 / DWG. NO. 3M3-195215
	16.11.			T.M	

PORT ERP

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

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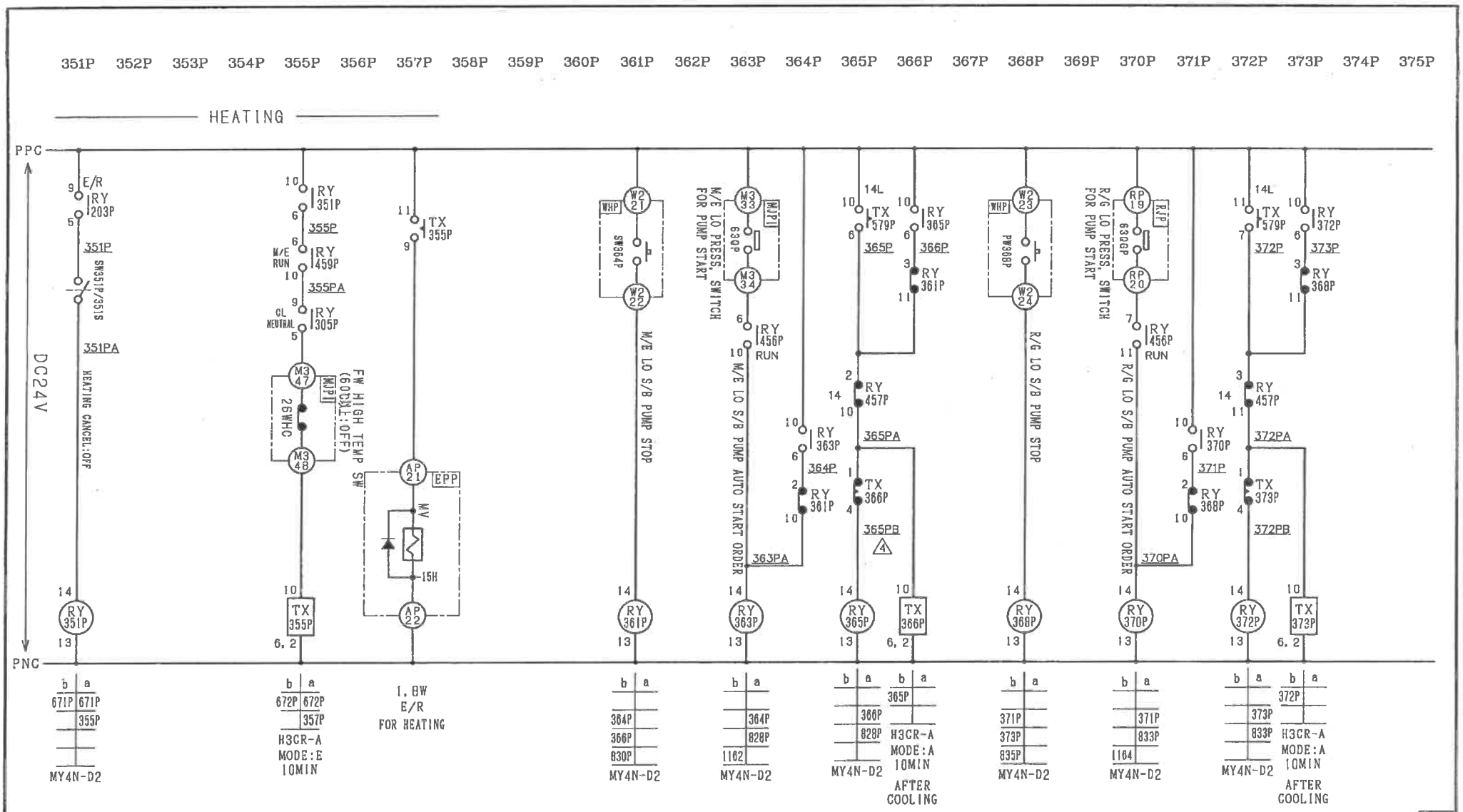
3M3-195215



4 1. NOV., 2016

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	
SCALE	MANAGER / CHIEF	CHECK / DRAWN
	16.11.7	J.M

PORT ERP

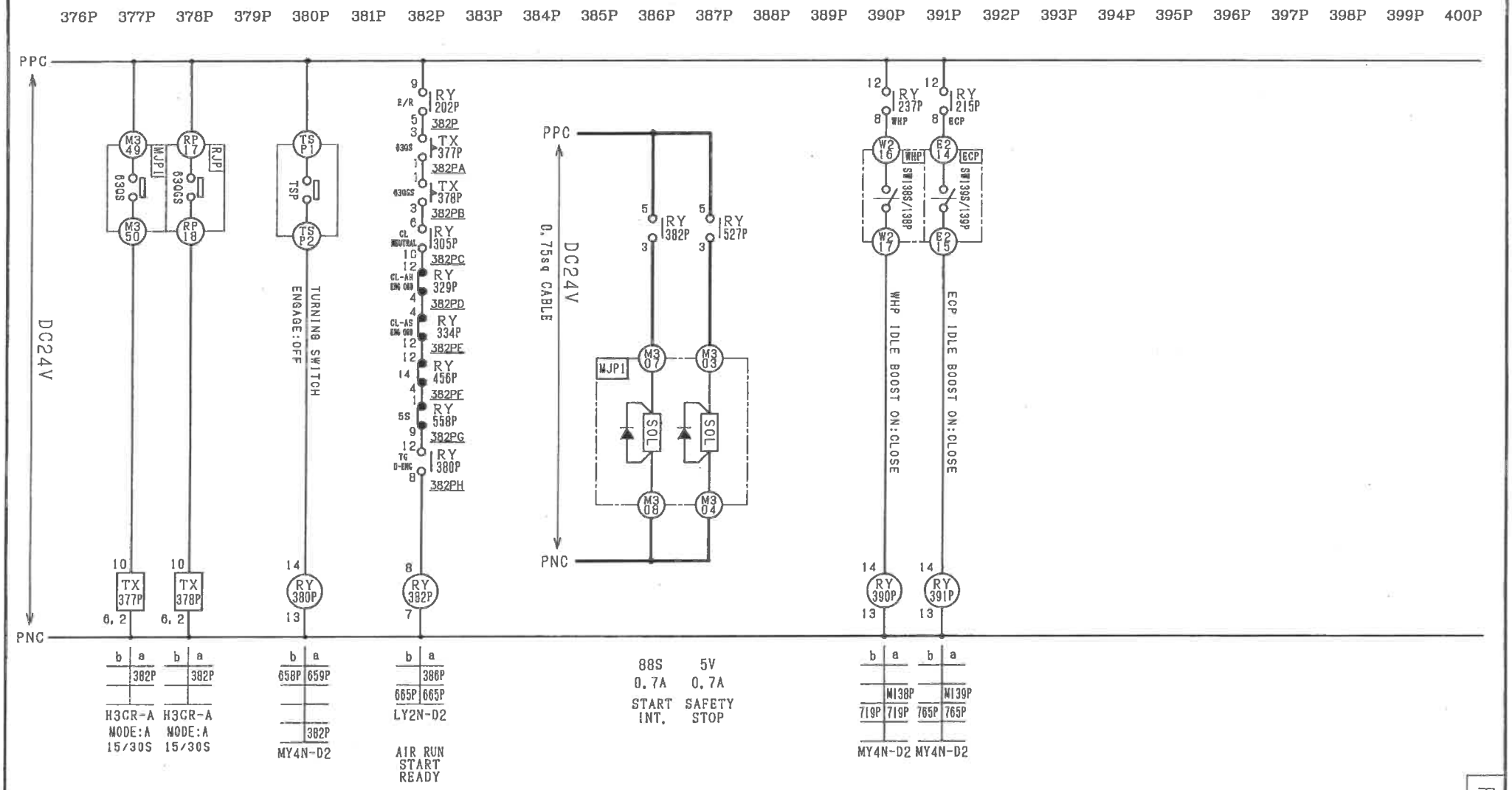
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT16

DATE 2. SEP., 2016 DWG. NO. 3M3-195216

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195216

P 117



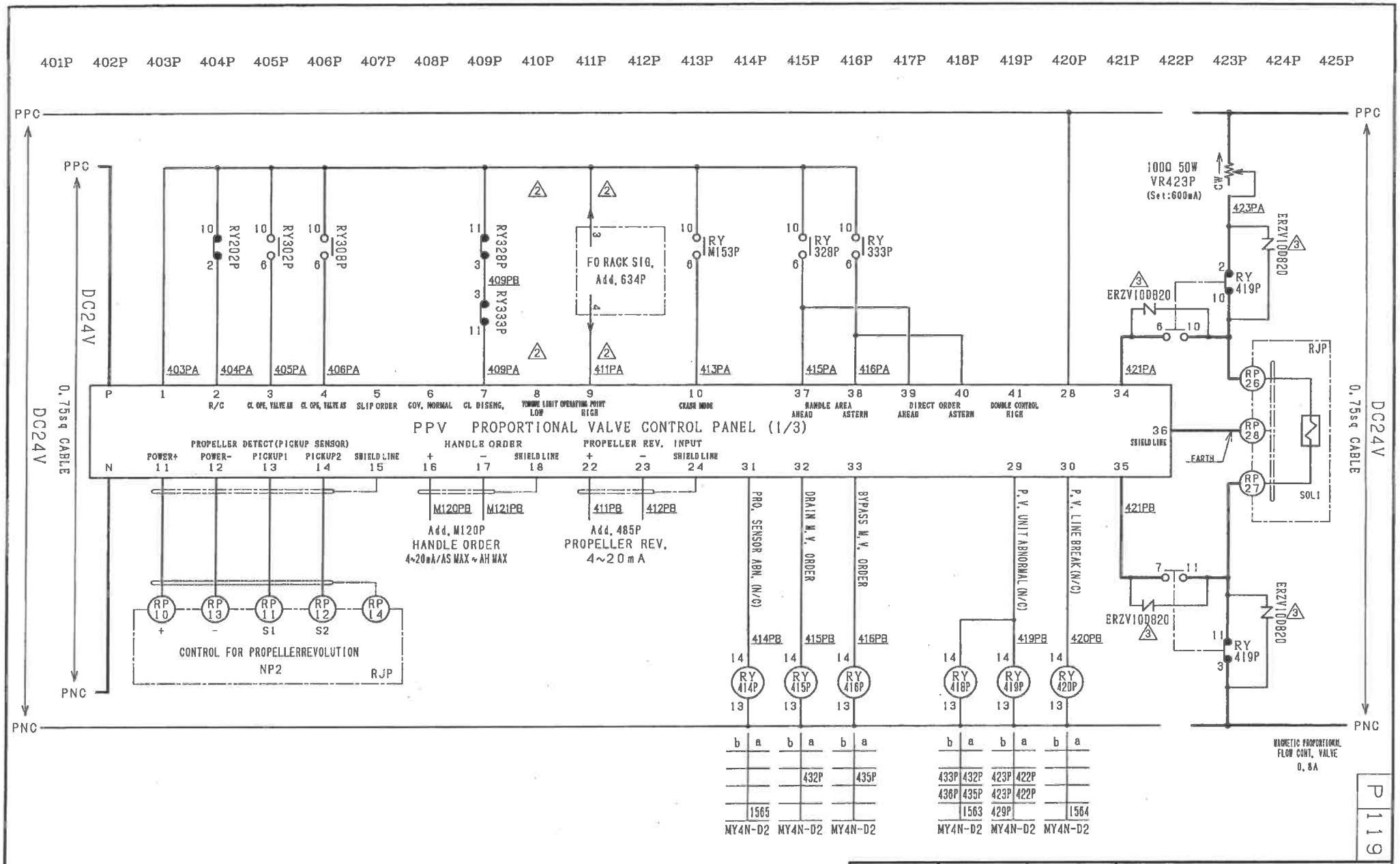
P 118

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT17		
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER / CHIEF	CHECK	DRAWN	DATE	DWG. NO.
	'16.11.7		T.M	6. SEP., 2016	3M3-195217

PORT ERP

3M3-195217

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



3M3-195218

△3 17, OCT., 2016
△2 29, SEP., 2016

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	REV. 6.11.7			T.M

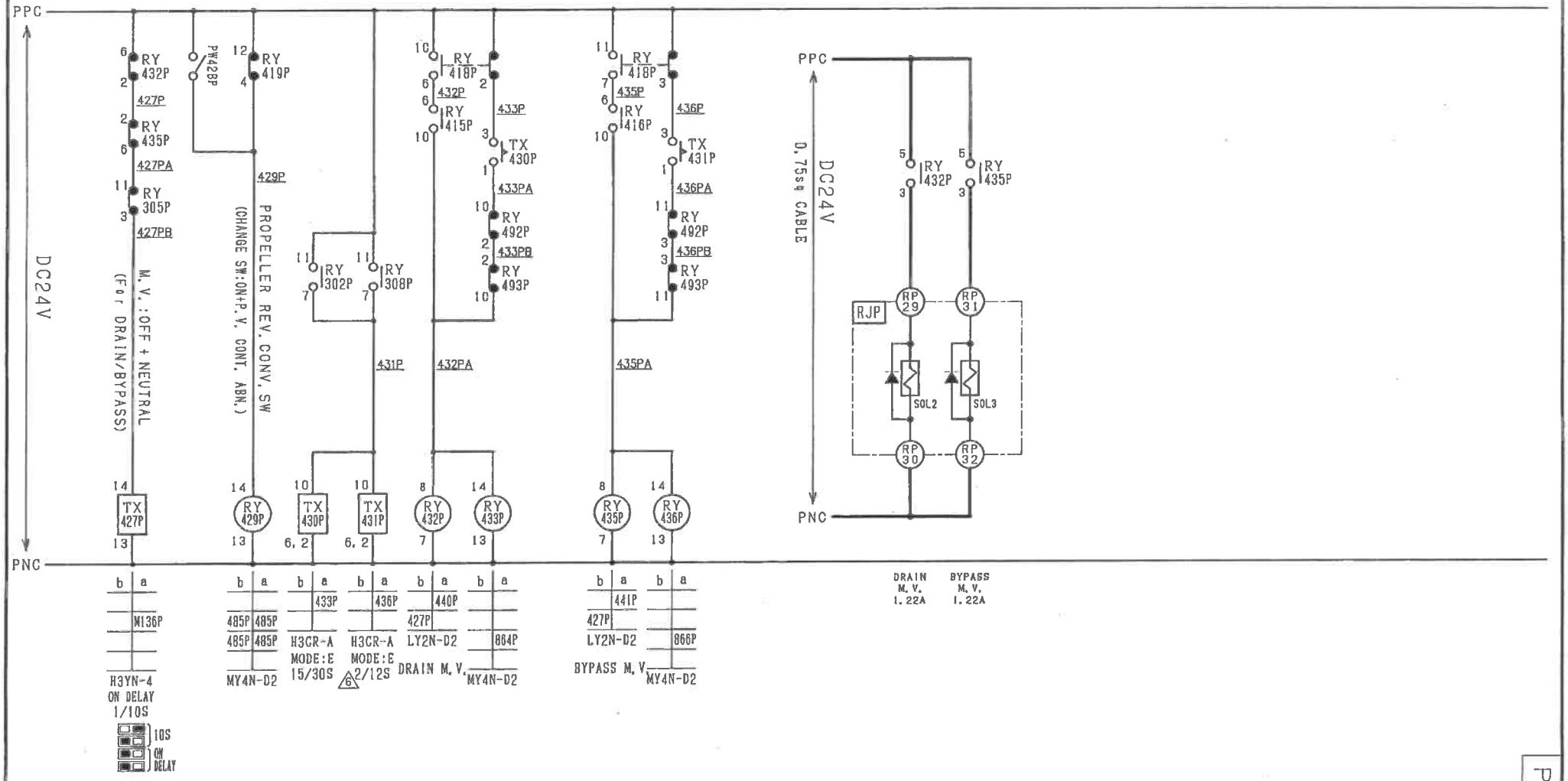
PORT ERP

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT18

DATE	8. SEP., 2016	DWG. NO.	3M3-195218

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

426P 427P 428P 429P 430P 431P 432P 433P 434P 435P 436P 437P 438P 439P 440P 441P 442P 443P 444P 445P 446P 447P 448P 449P 450P



DRAIN M.V.	BYPASS M.V.
1.22A	1.22A

△6 12, DEC., 2016

PORT ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	
SCALE	MANAGER / CHIEF	CHECK / DRAWN
	REV. 16.12.12	REV. 16.12.12 T.M

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM PORT19

DATE 8, SEP., 2016 | DWG. NO. 3M3-195219

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

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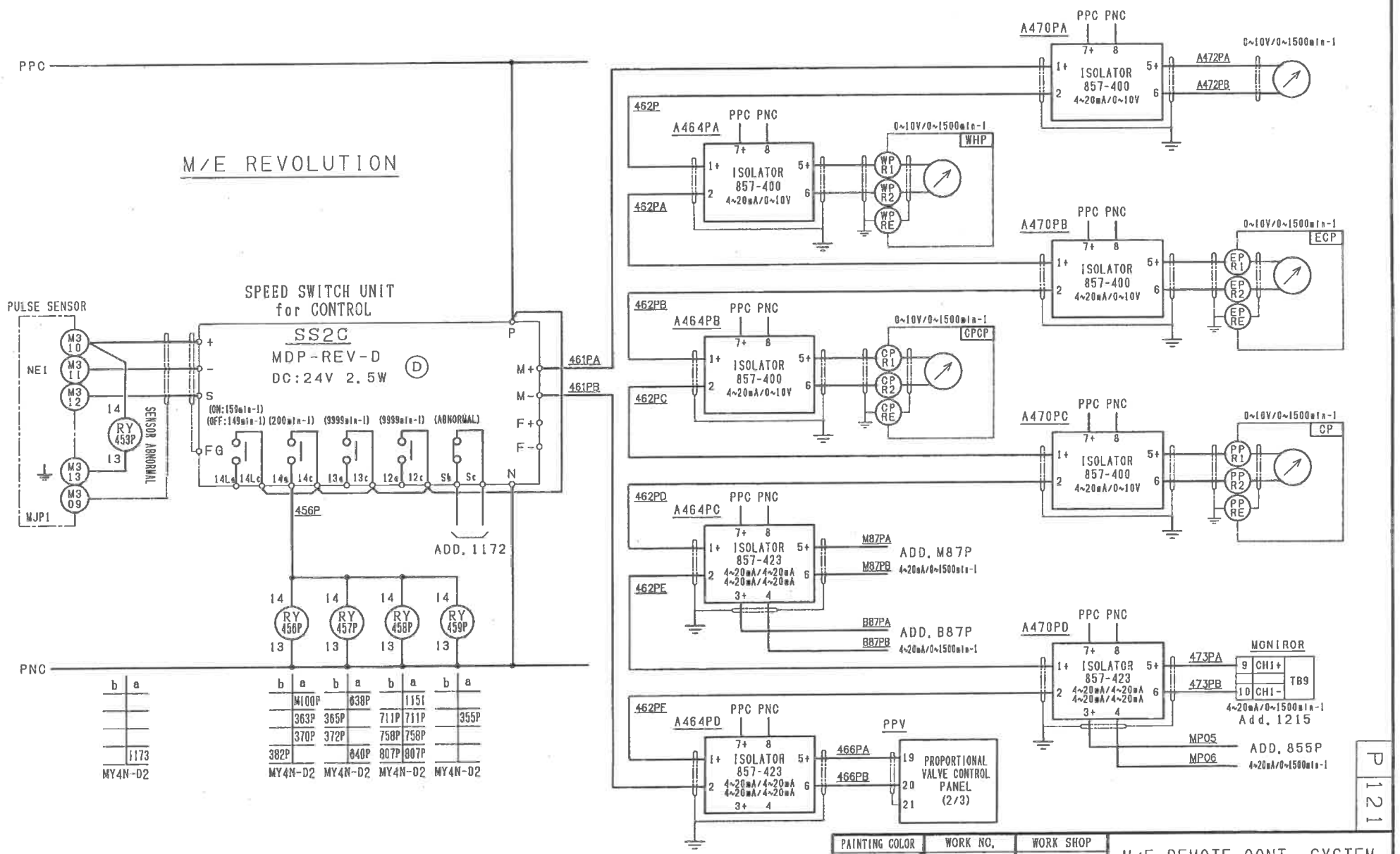
3M3-195219

451P 452P 453P 454P 455P 456P 457P 458P 459P 460P 461P 462P 463P 464P 465P 466P 467P 468P 469P 470P 471P 472P 473P 474P 475P



P 121

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

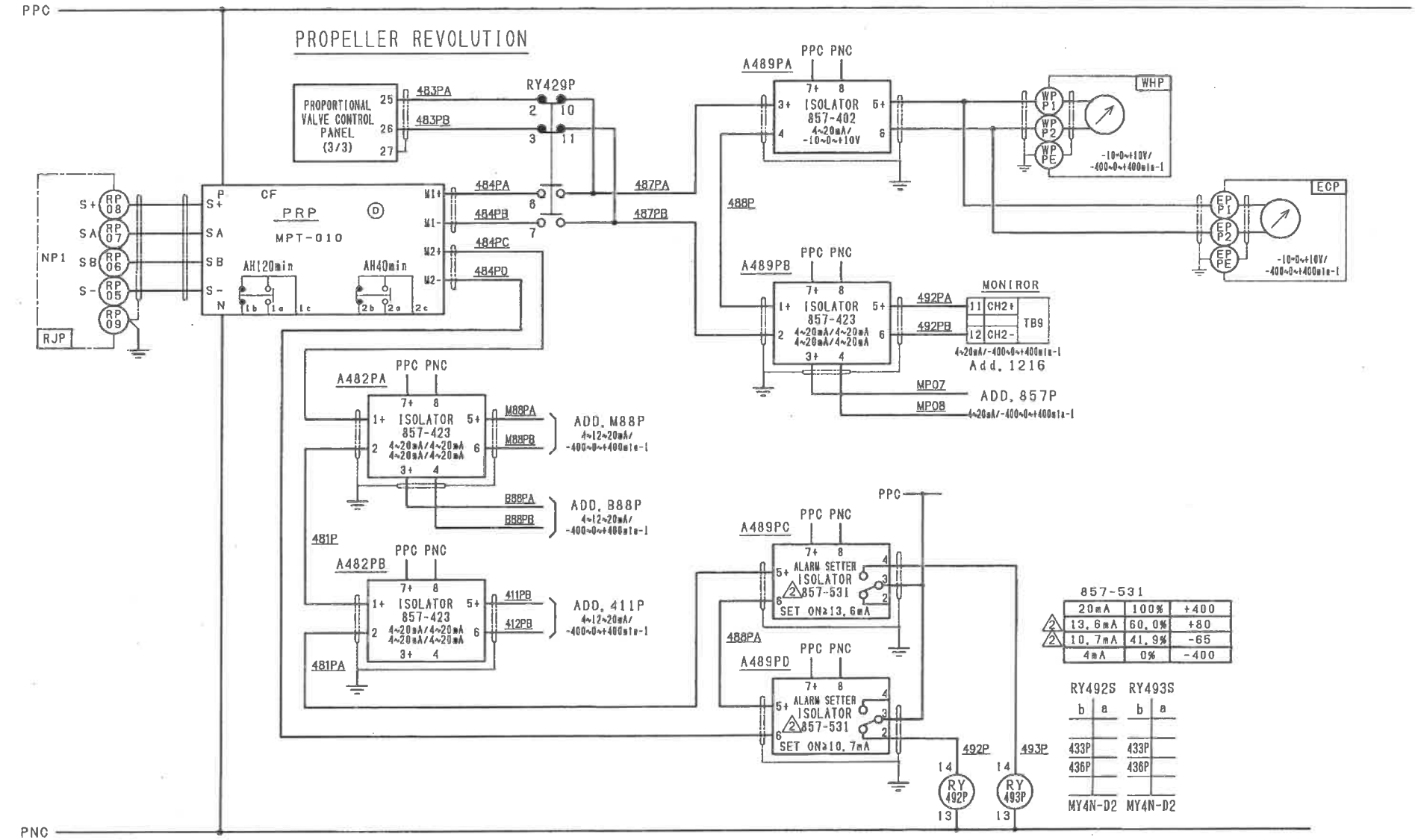
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT20

DATE 9. SEP., 2016 | DWG. NO. 3M3-195220

3M3-195220

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

476P 477P 478P 479P 480P 481P 482P 483P 484P 485P 486P 487P 488P 489P 490P 491P 492P 493P 494P 495P 496P 497P 498P 499P 500P



P 122

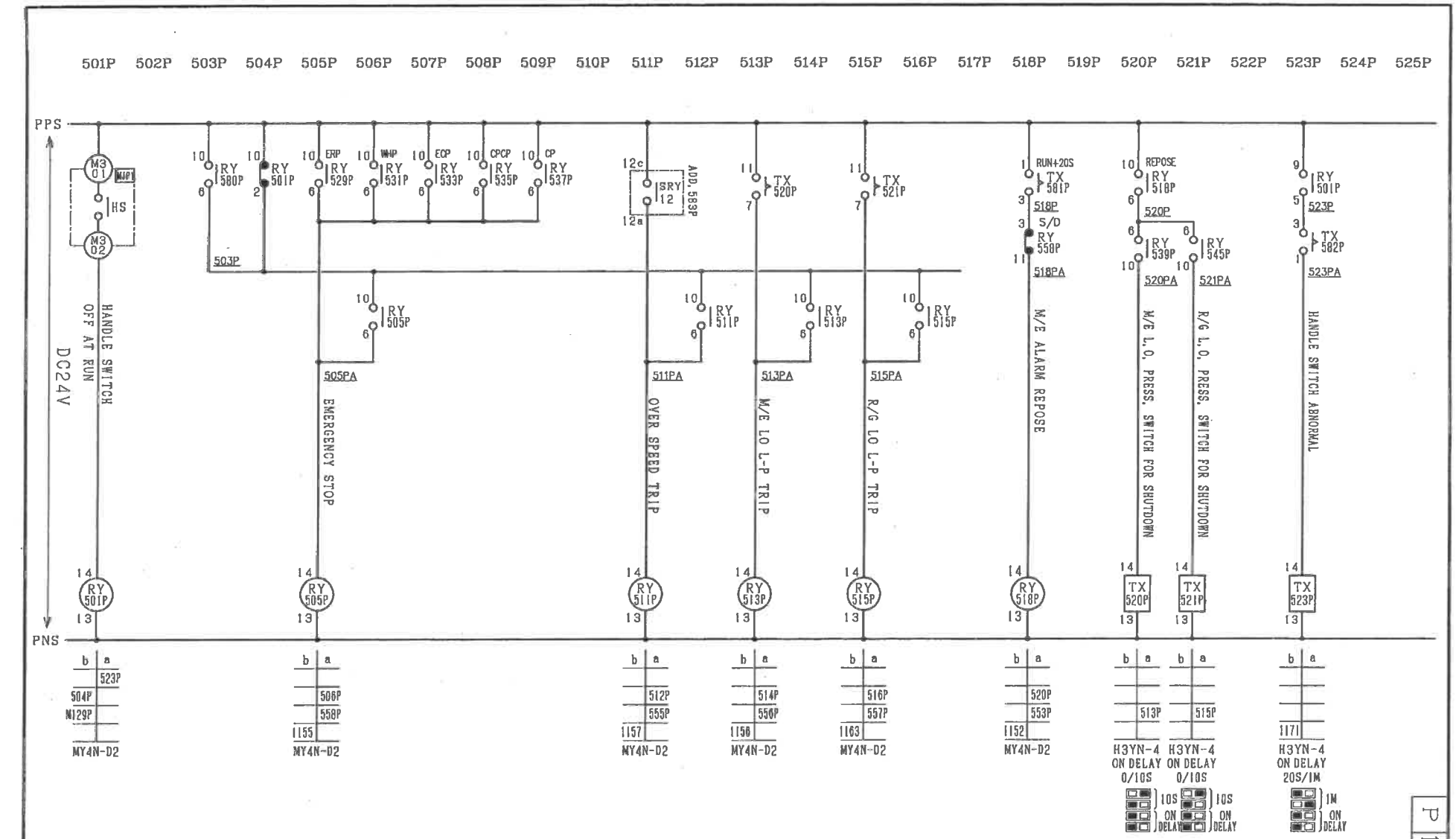
PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT21
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE: 8, SEP., 2016
		T.M	DWG. NO.: 3M3-195221

PORT ERP

3M3-195221

2 29, SEP., 2016

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

PORT ERP

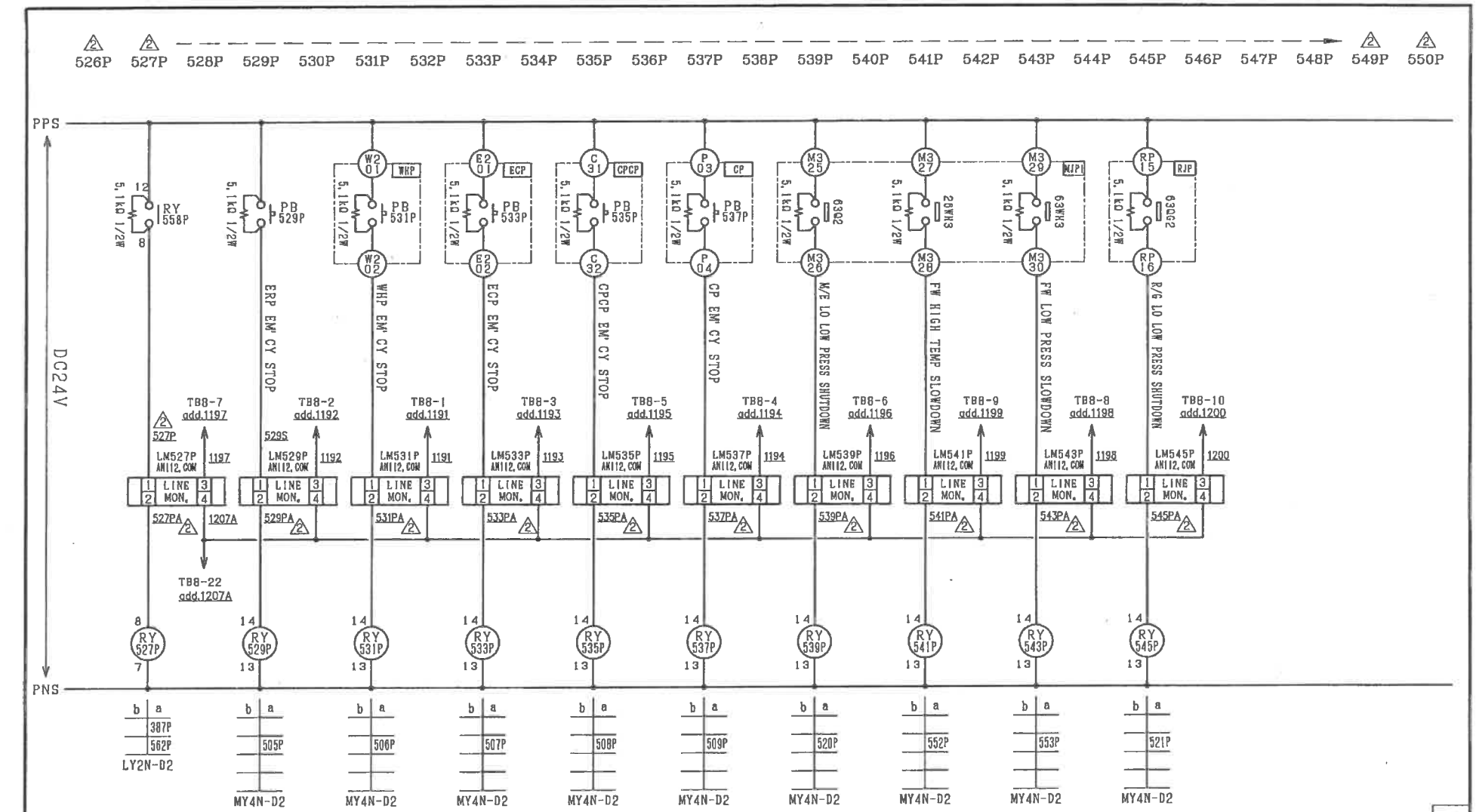
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT22

DATE 12. AUG., 2016 DWG. NO. 3M3-195222

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195222

P 123



PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

PORT ERP

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT23

DATE 10. AUG., 2016 DWG. NO. 3M3-195223

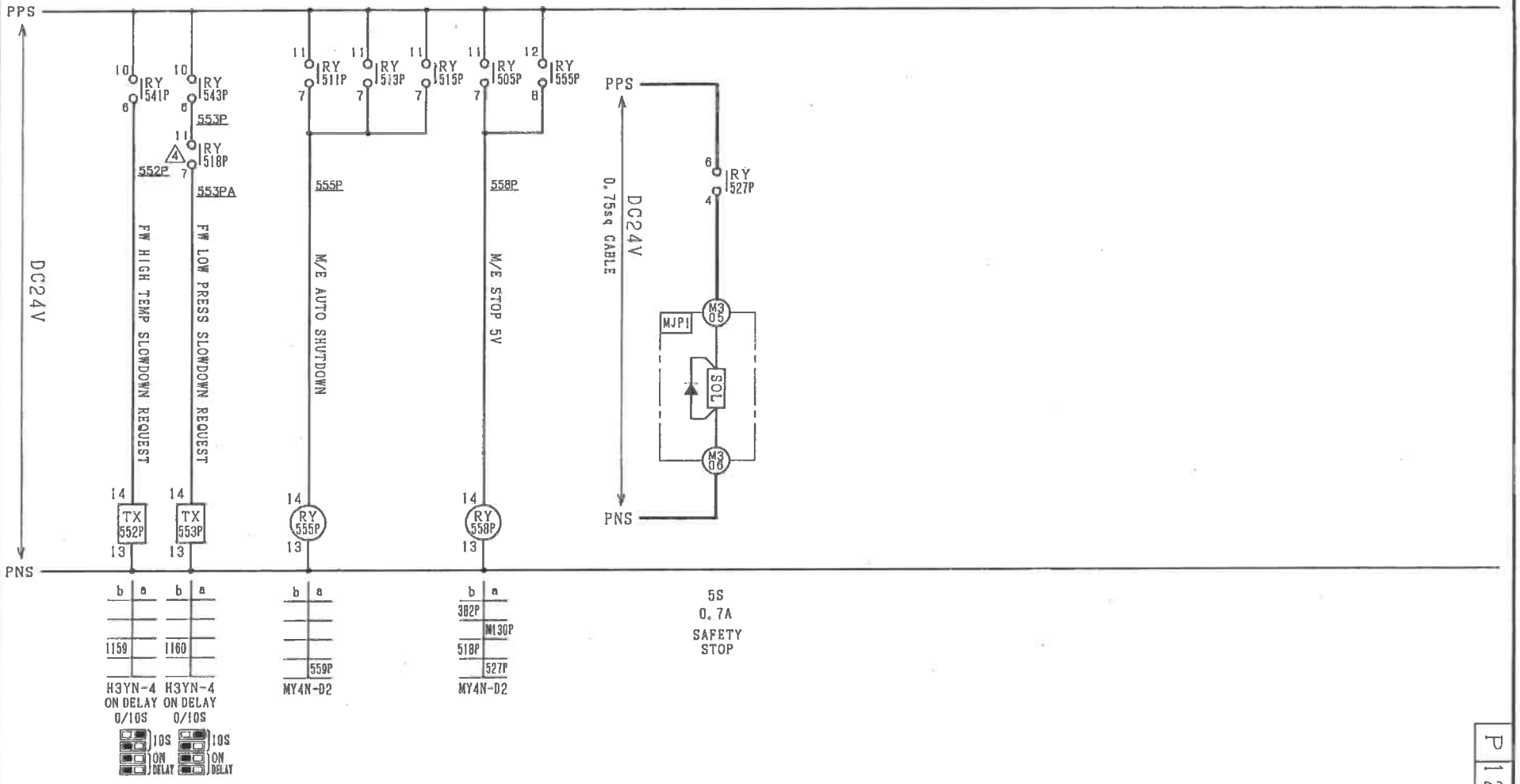
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 124

3M3-195223

2 29. SEP., 2016

551P 552P 553P 554P 555P 556P 557P 558P 559P 560P 561P 562P 563P 564P 565P 566P 567P 568P 569P 570P 571P 572P 573P 574P 575P



P 125

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT24			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	12. AUG., 2016
				T.M	DWG. NO.	3M3-195224

PORT ERP

3M3-195224

4 1. NOV., 2016

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

576P 577P 578P 579P 580P 581P 582P 583P 584P 585P 586P 587P 588P 589P 590P 591P 592P 593P 594P 595P 596P 597P 598P 599P 600P

PNS

DC24V

PULSE SENSOR

M3 15 (−)
M3 16
M3 17
NE2
M3 18
M3 14
MJP1

14
RY 578P
13

パルス検出器異常

SS2S SPEED SWITCH UNIT for SAFETY

MDP-REV-D
DC:24V 2.5W
SET CONTACT MODE="0"

+
−
S
FG

OFF DELAY 20SEC
(ON:50min-1)
(OFF:30min-1) (200min-1) (9999min-1) (1020min-1) (ABNORMAL)

14Lb 14Lc 14a 14c 13a 13c 12a 12c Sb Sc

P
F+
F−
M+
M−
N

add. 511P
add. 1174

580P 581P

14 TX 579P 13
14 RY 580P 13
10 TX 581P 6, 2
10 TX 582P 6, 2

b a b a b a b a b a

365P
372P
1175
503P
518P
523P

MY4N-D2
H3YN-4 ON DELAY 10/10S
MY4N-D2
H3CR-A MODE:A 20/30S
H3CR-A MODE:A 1/1.2M

10S ON DELAY

CONTACT MODE "0" (14L:14:13:12) = (A:A:A:A)
CONTACT MODE "1" (14L:14:13:12) = (A:A:A:B)

PORT ERP

PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK	DRAWN
			T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT25

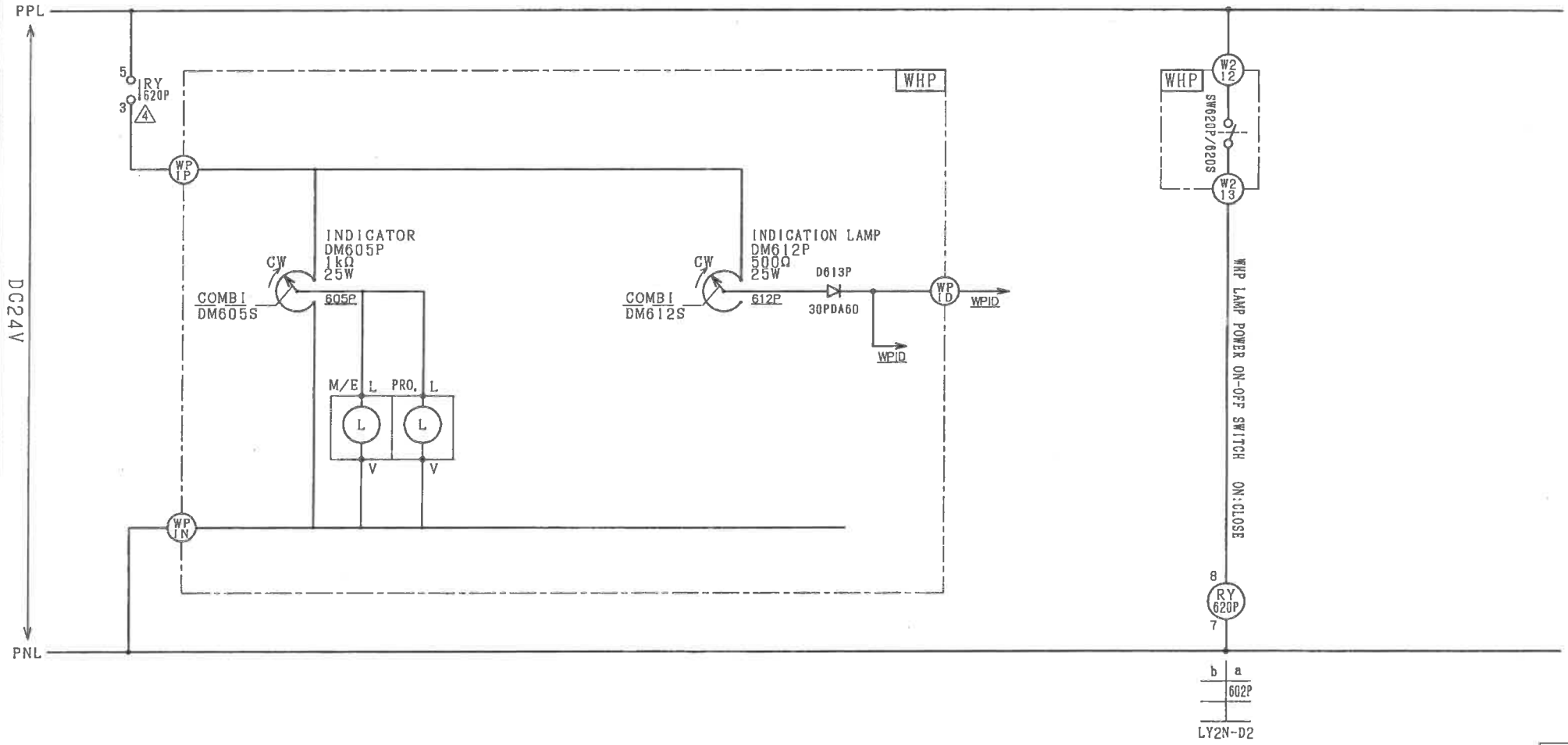
DATE 12. AUG., 2016 DWG. NO. 3M3-195225

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 126

3M3-195225

601P 602P 603P 604P 605P 606P 607P 608P 609P 610P 611P 612P 613P 614P 615P 616P 617P 618P 619P 620P 621P 622P 623P 624P 625P



P 127

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT26			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	10. AUG., 2016
	16.11.			T.M	DWG. NO.	3M3-195226

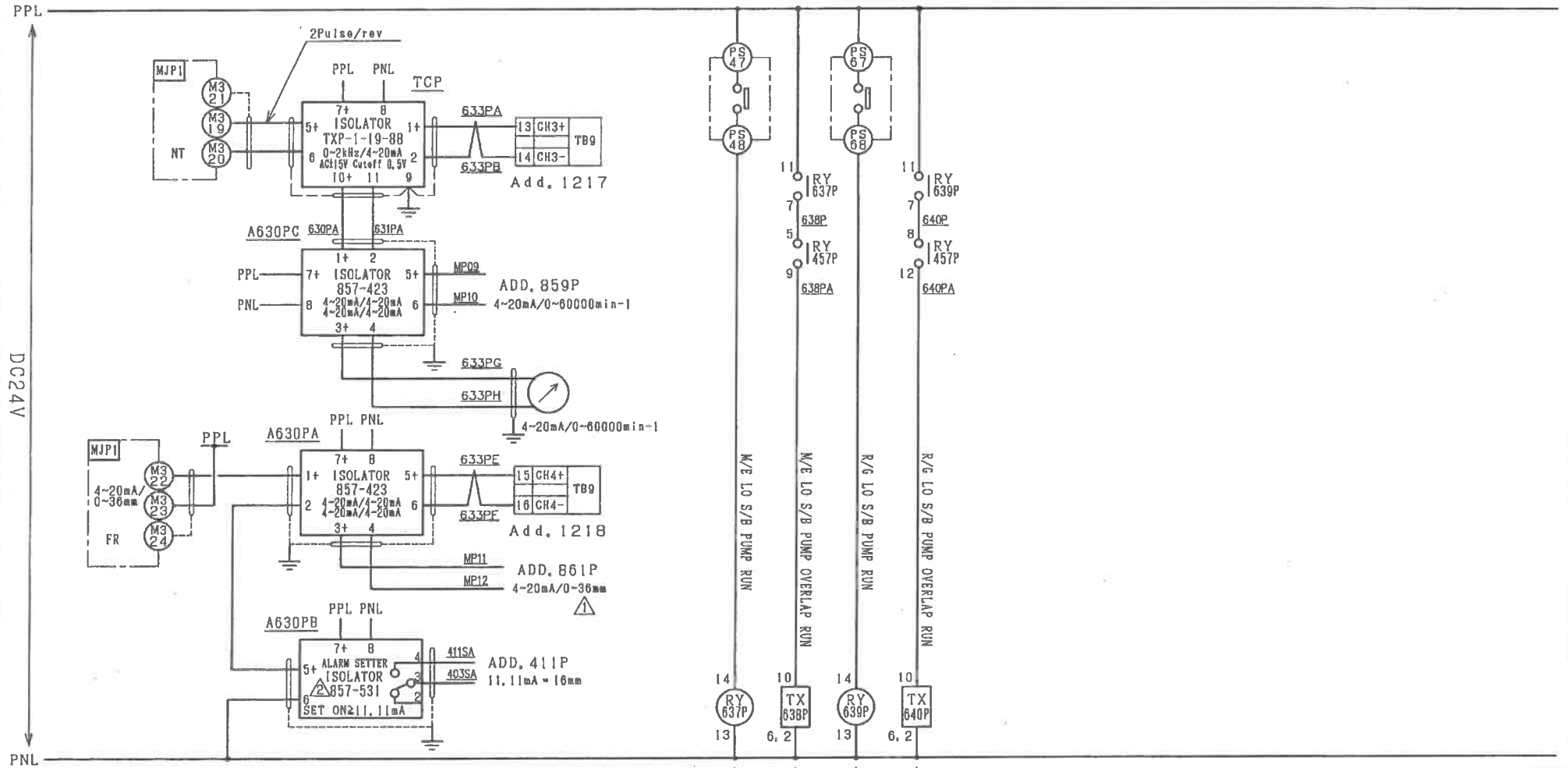
PORT ERP

3M3-195226

4 1, NOV., 2016

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

626P 627P 628P 629P 630P 631P 632P 633P 634P 635P 636P 637P 638P 639P 640P 641P 642P 643P 644P 645P 646P 647P 648P 649P 650P



PORT ERP

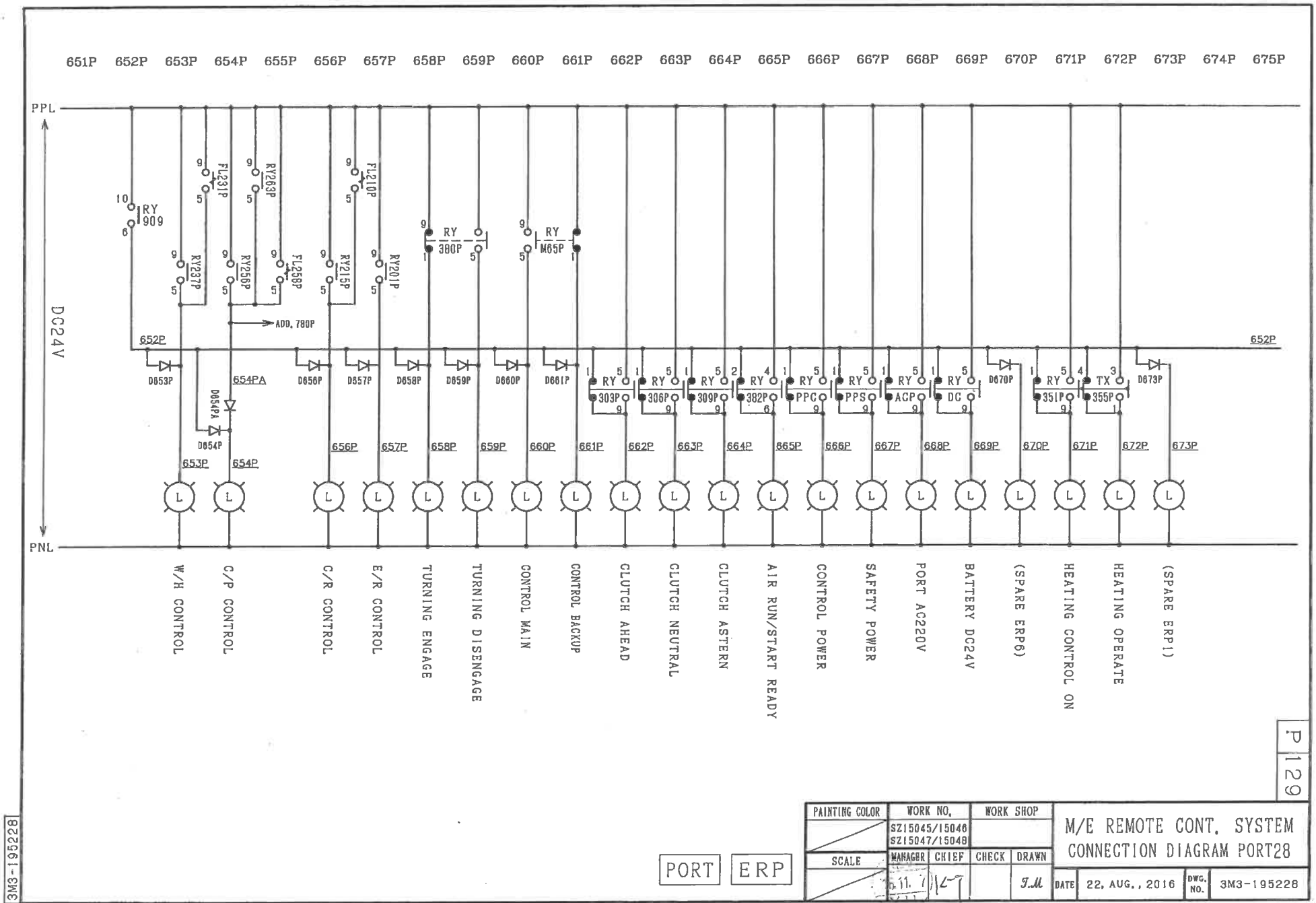
P 128

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT27			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE	DWG. NO.
		17-7		T.M	1, SEP., 2016	3M3-195227

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195227

△2 29, SEP., 2016
△1 13, SEP., 2016



P129

3M3-195228

PORT ERP

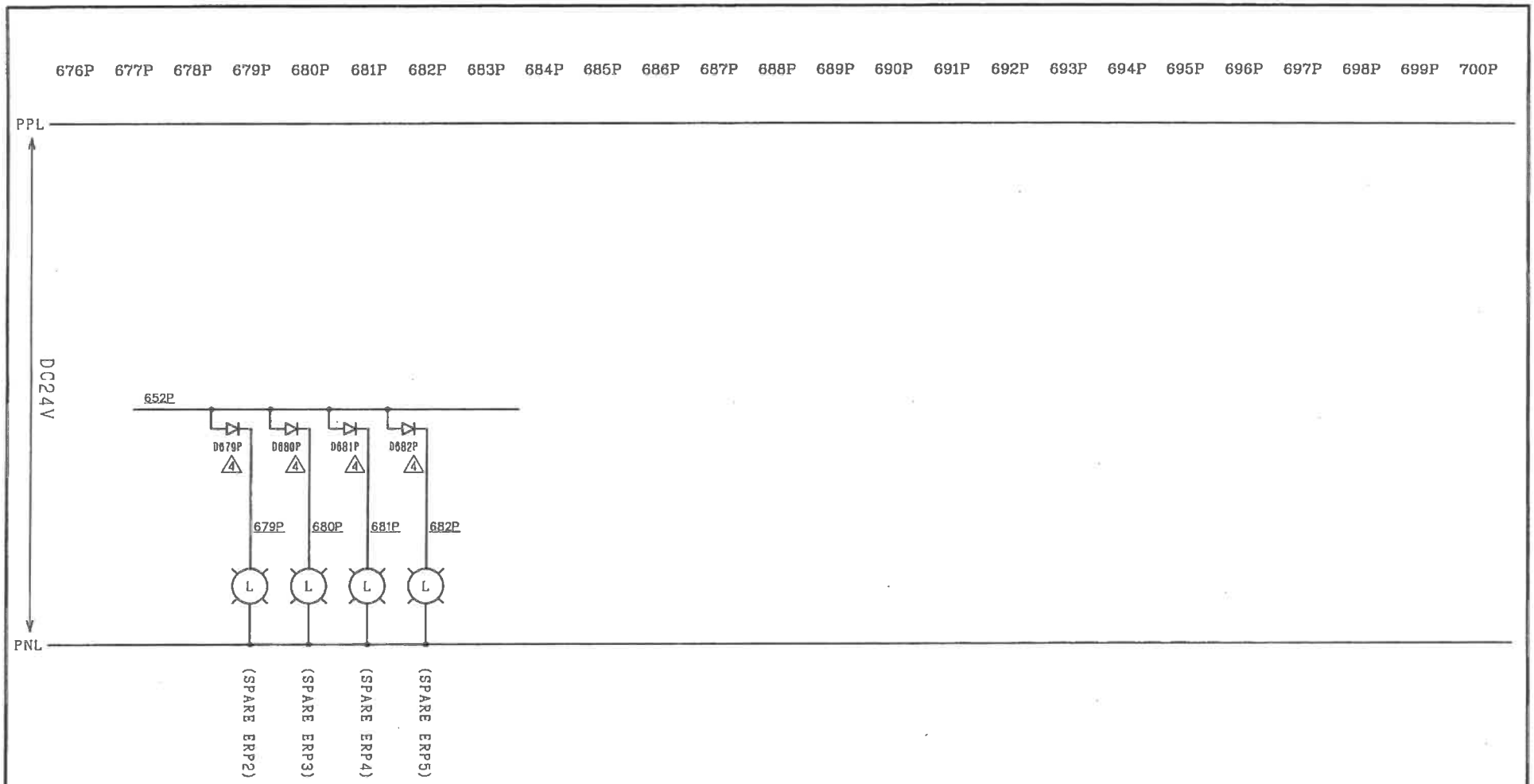
PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT28

DATE 22. AUG., 2016 DWG. NO. 3M3-195228

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P 130

3M3-195229

△4 1, NOV., 2016

PORT ERP

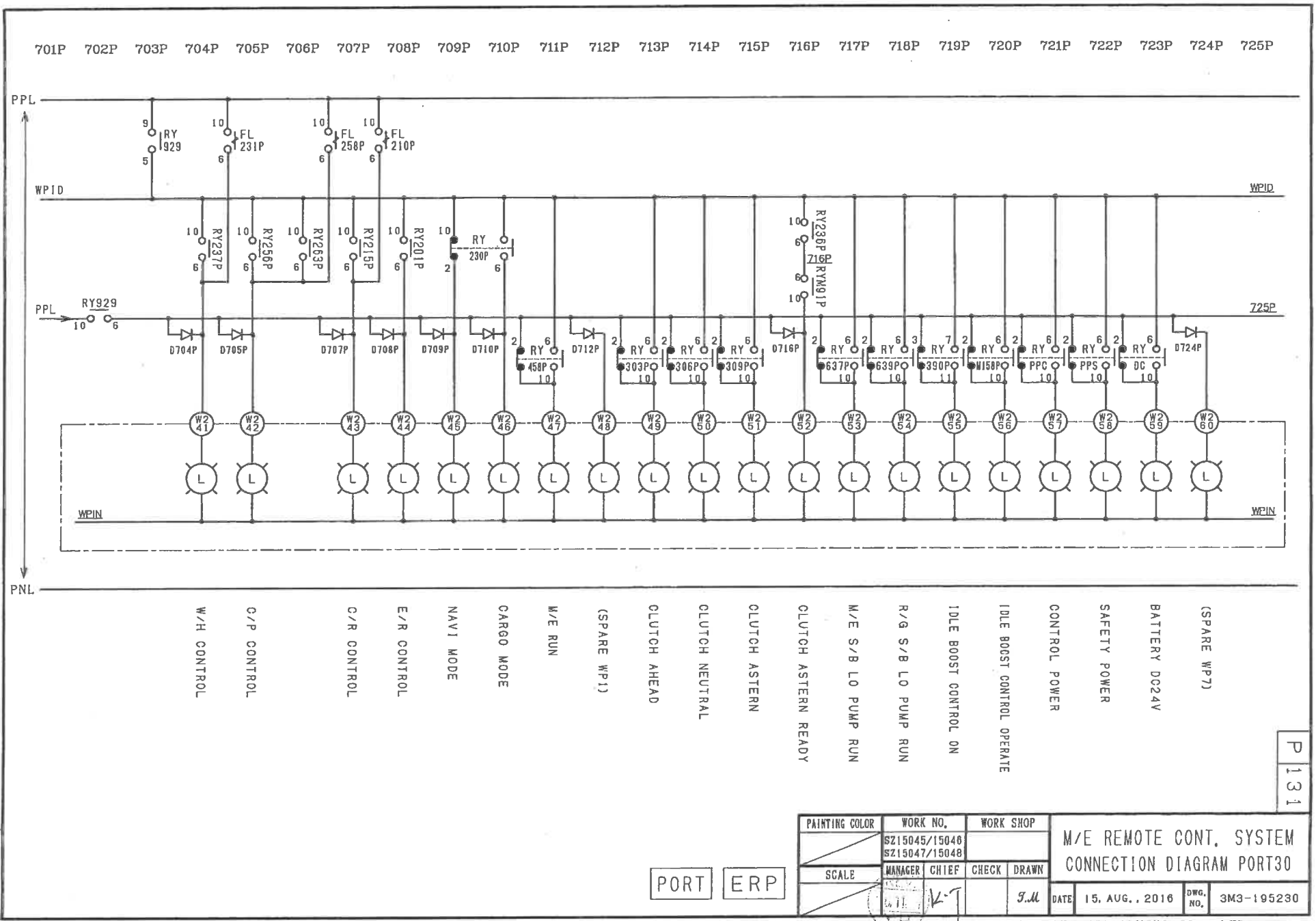
PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
	16.11.			T.M.

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT29

DATE 3, AUG., 2016 DWG. NO. 3M3-195229

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P 131

PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER	CHIEF	CHECK / DRAWN
			J.M

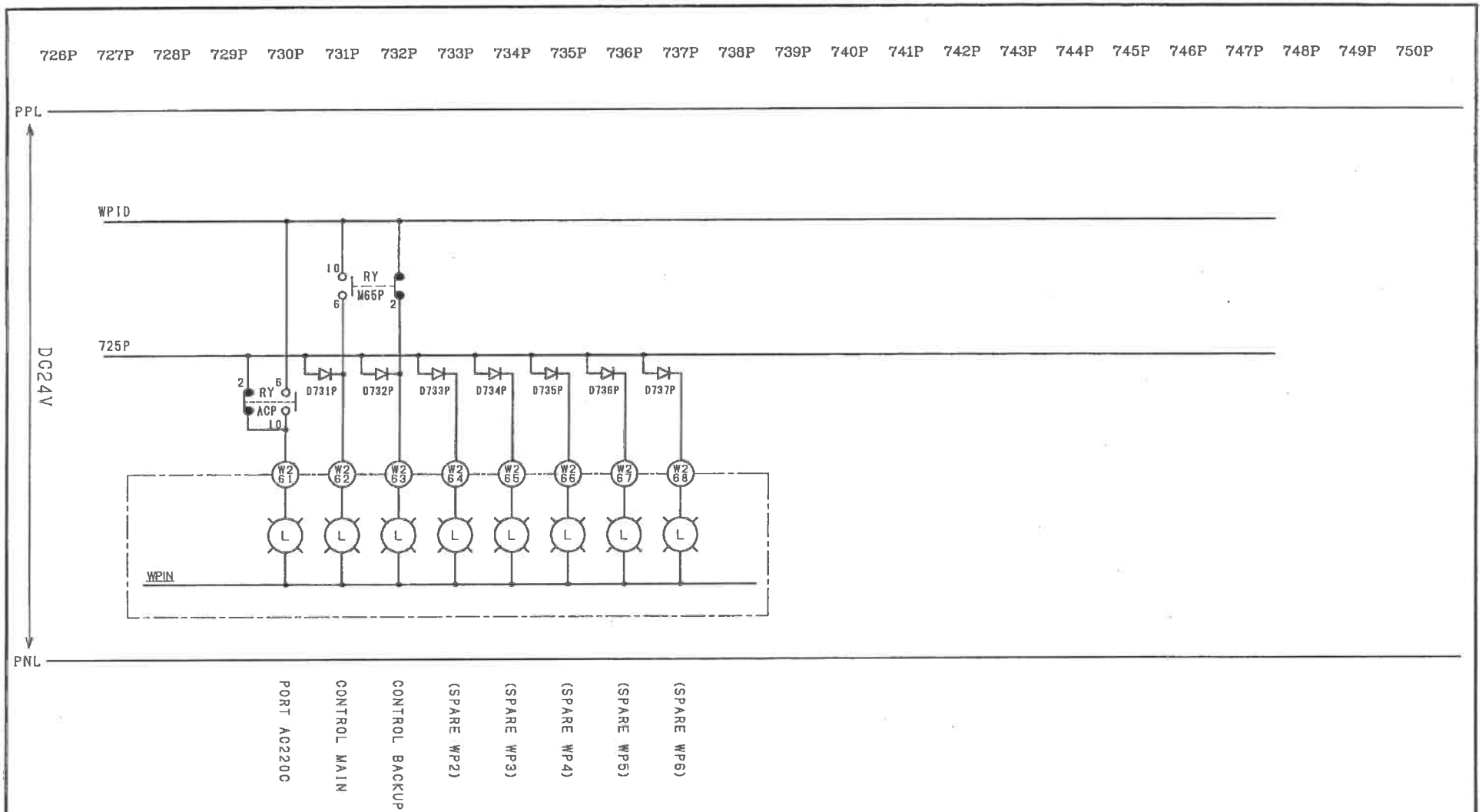
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT30

DATE 15. AUG., 2016 | DWG. NO. 3M3-195230

PORT ERP

3M3-195230

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P 132

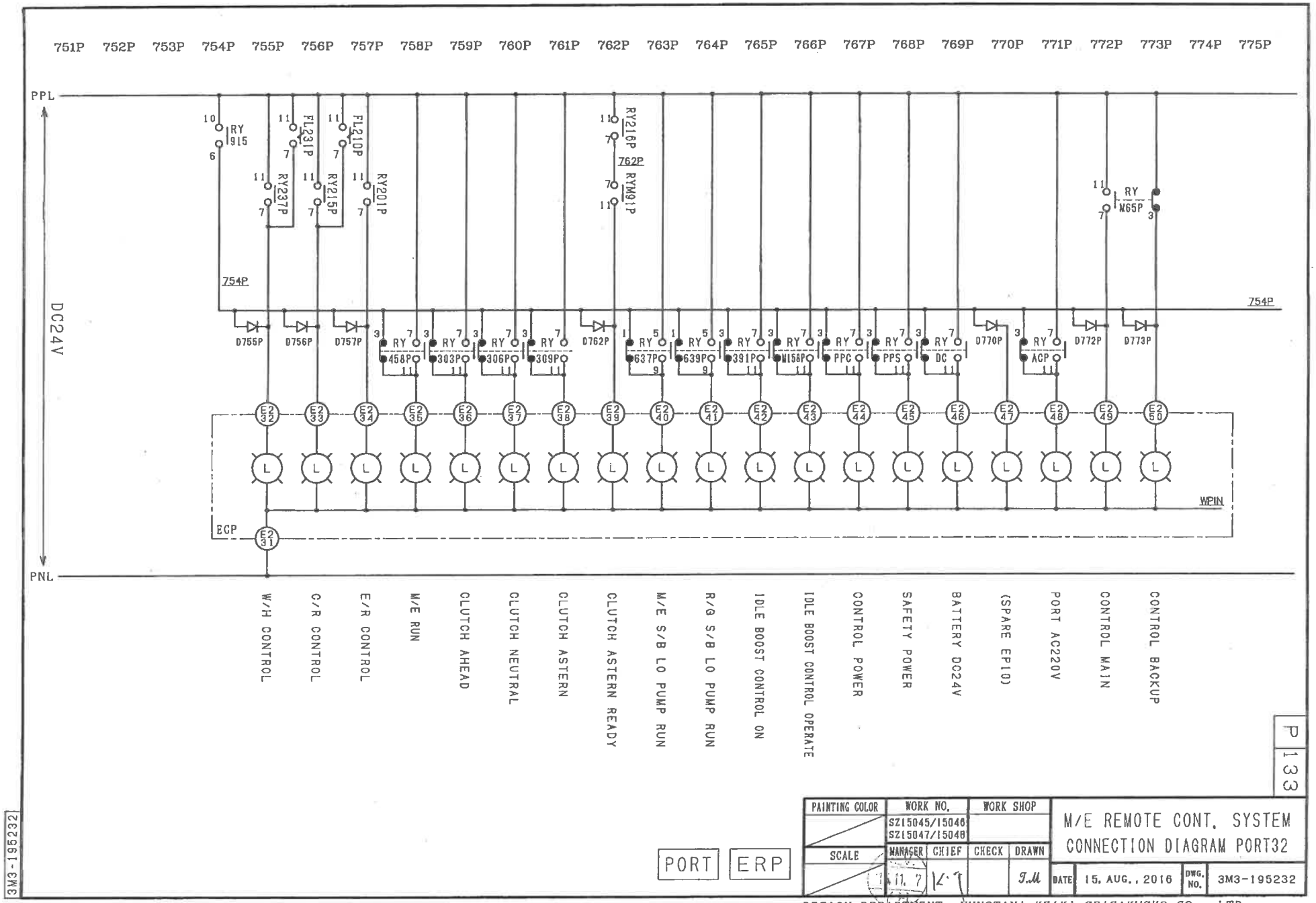
3M3-195231

PORT ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT31
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK / DRAWN	
	6.11. / 12-7		T.M

DATE 10. AUG., 2016 | DWG. NO. 3M3-195231

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



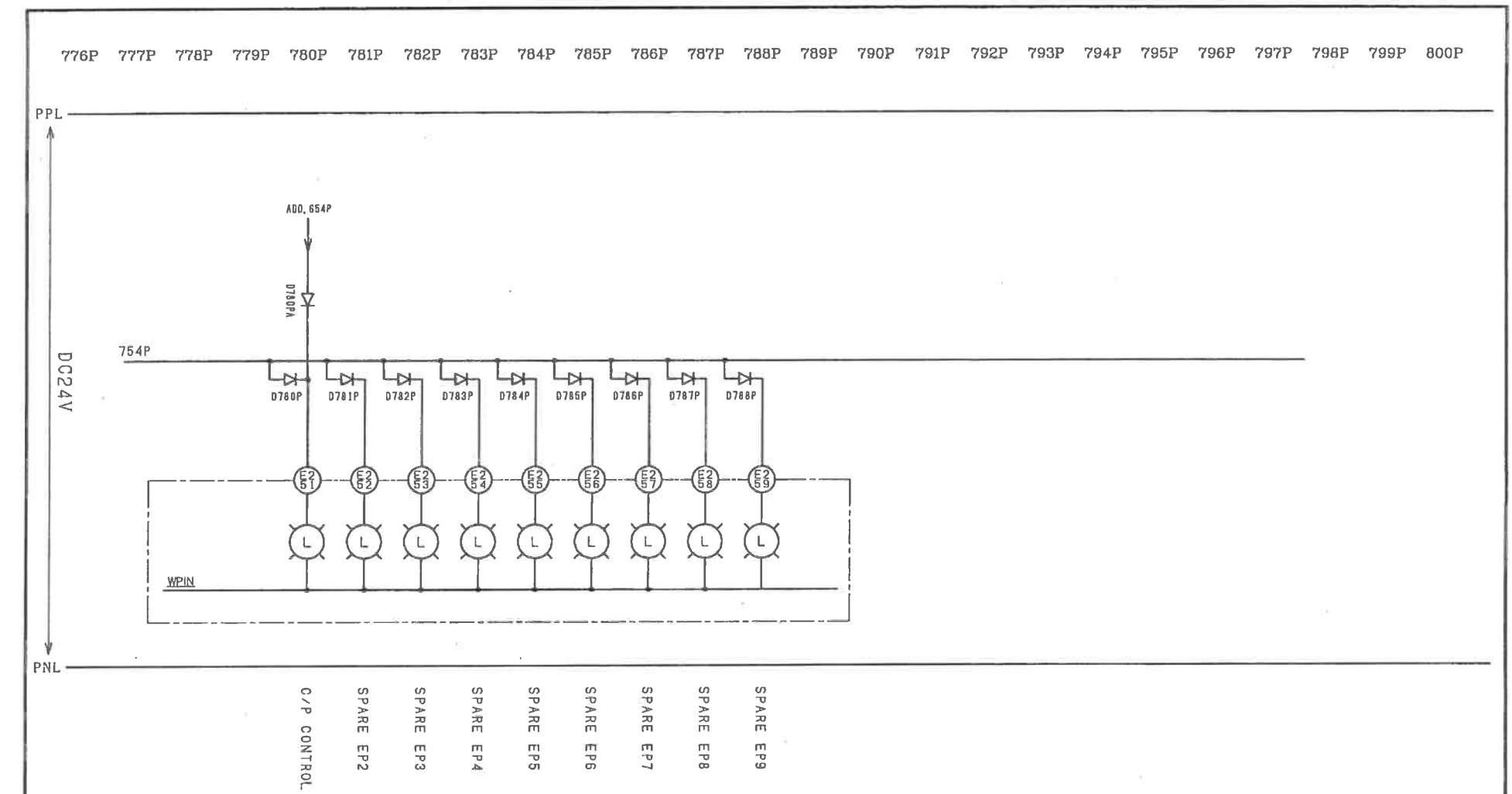
P 133

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT32
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE: 15. AUG., 2016
	11. 7 /		DWG. NO.: 3M3-195232

PORT ERP

3M3-195232

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P 134

PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK	DRAWN
	6.11.		T.M

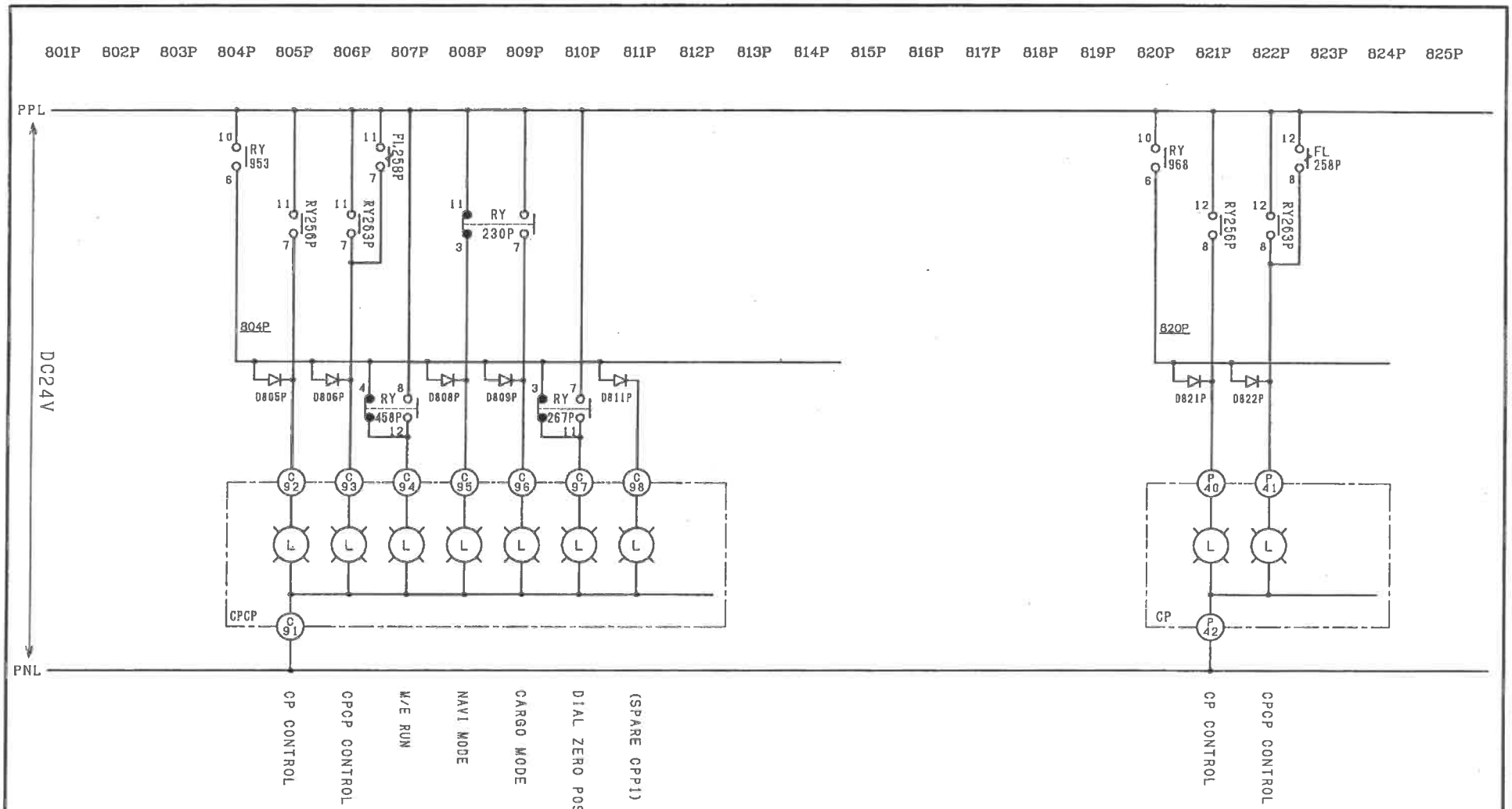
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT33

DATE 22. AUG.. 2016 | DWG. NO. 3M3-195233

PORT ERP

3M3-195233

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P 135

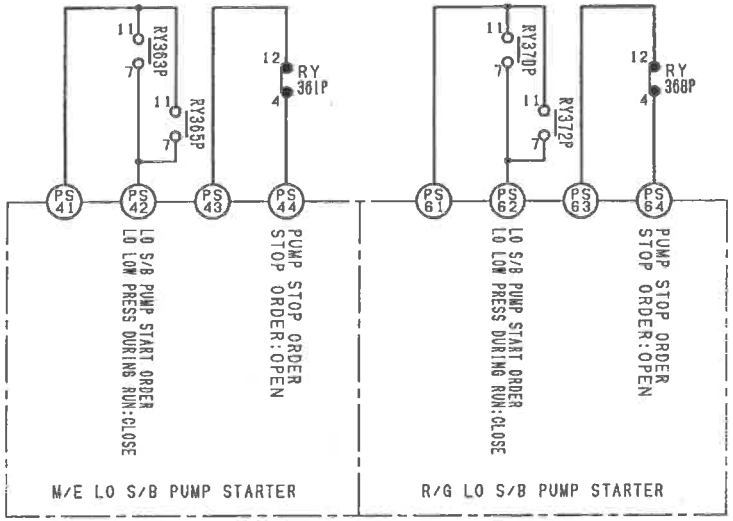
3M3-195234

PORT ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT34		
	SZ15045/15048 SZ15047/15048				
SCALE	MANAGER CHIEF	CHECK	DRAWN	DATE	DWG. NO.
	16.11.		T.M	15, AUG., 2016	3M3-195234

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

826P 827P 828P 829P 830P 831P 832P 833P 834P 835P 836P 837P 838P 839P 840P 841P 842P 843P 844P 845P 846P 847P 848P 849P 850P



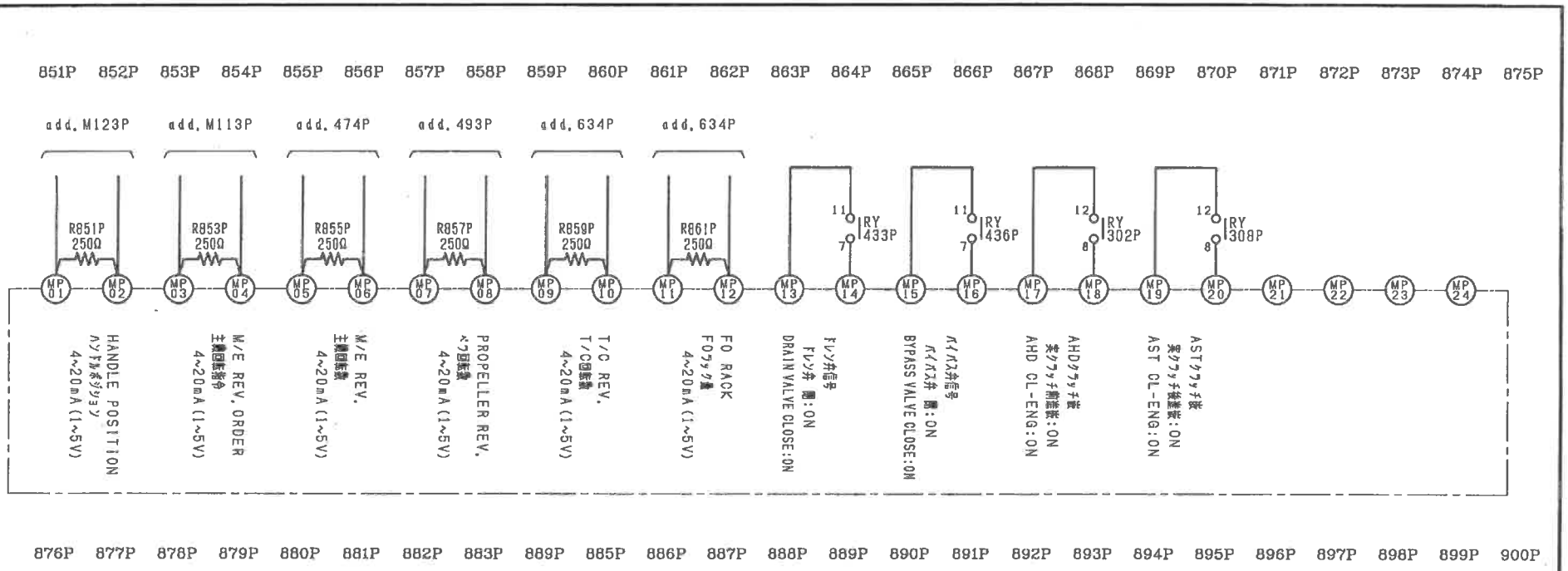
P 136

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT35			
	SZ15045/15046 SZ15047/15048					
SCALE	MANAGER / CHIEF	CHECK	DRAWN	DATE	3, AUG., 2016	DWG. NO.	3M3-195235
	REV. 6.11.7		T.M			

PORT ERP

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195235



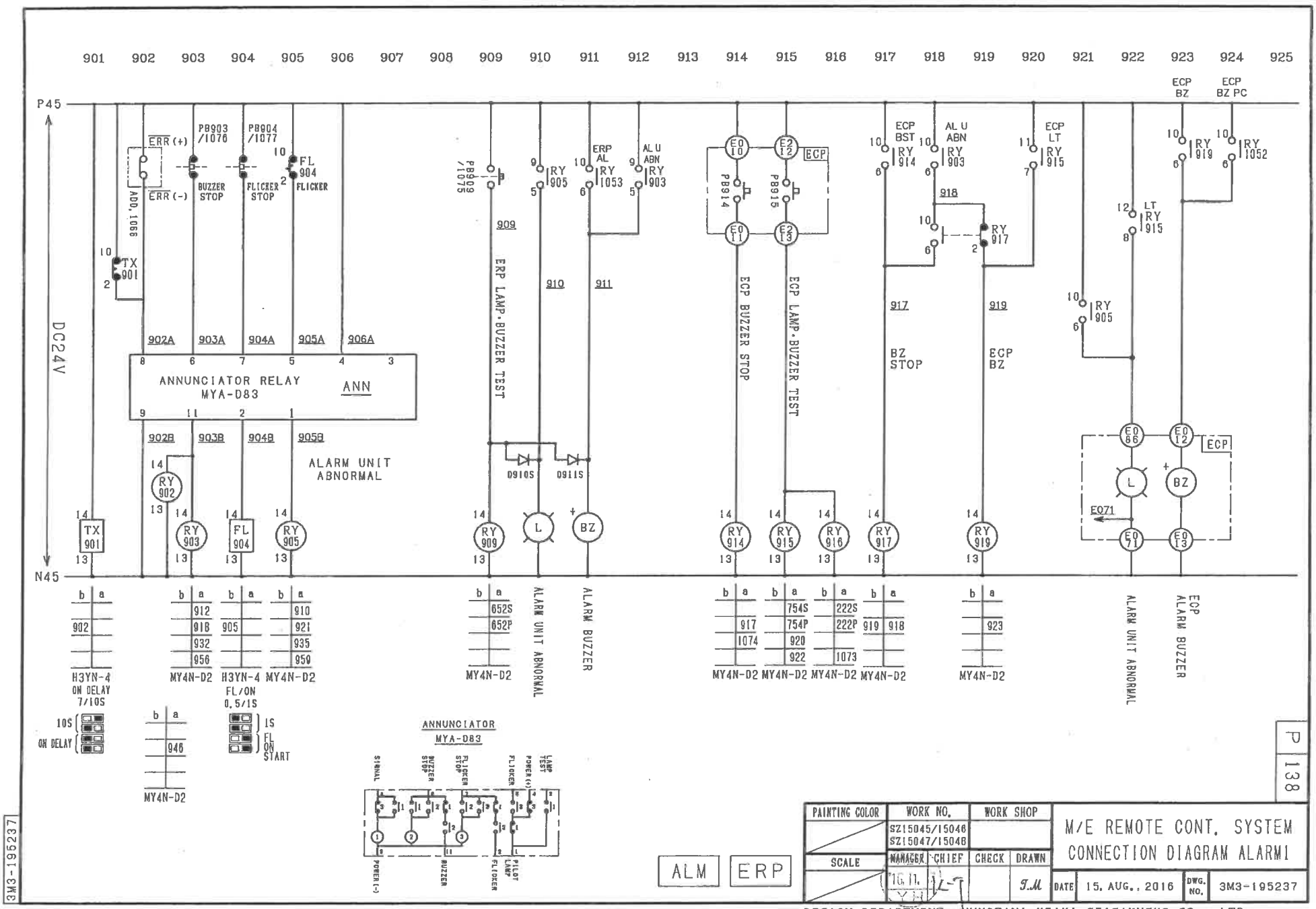
P 137

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM PORT36		
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER / CHIEF	CHECK	DRAWN	DATE	15. AUG., 2016
	16.11.		T.M	DWG. NO.	3M3-195236

PORT ERP

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195236



PAINTING COLOR	WORK NO. SZ15045/15046 SZ15047/15048	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM ALARM1		
SCALE	MANAGER	CHIEF	CHECK	DRAWN	DATE 15. AUG., 2016 / DWG. NO. 3M3-195237
	'16.11.			T.M	

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

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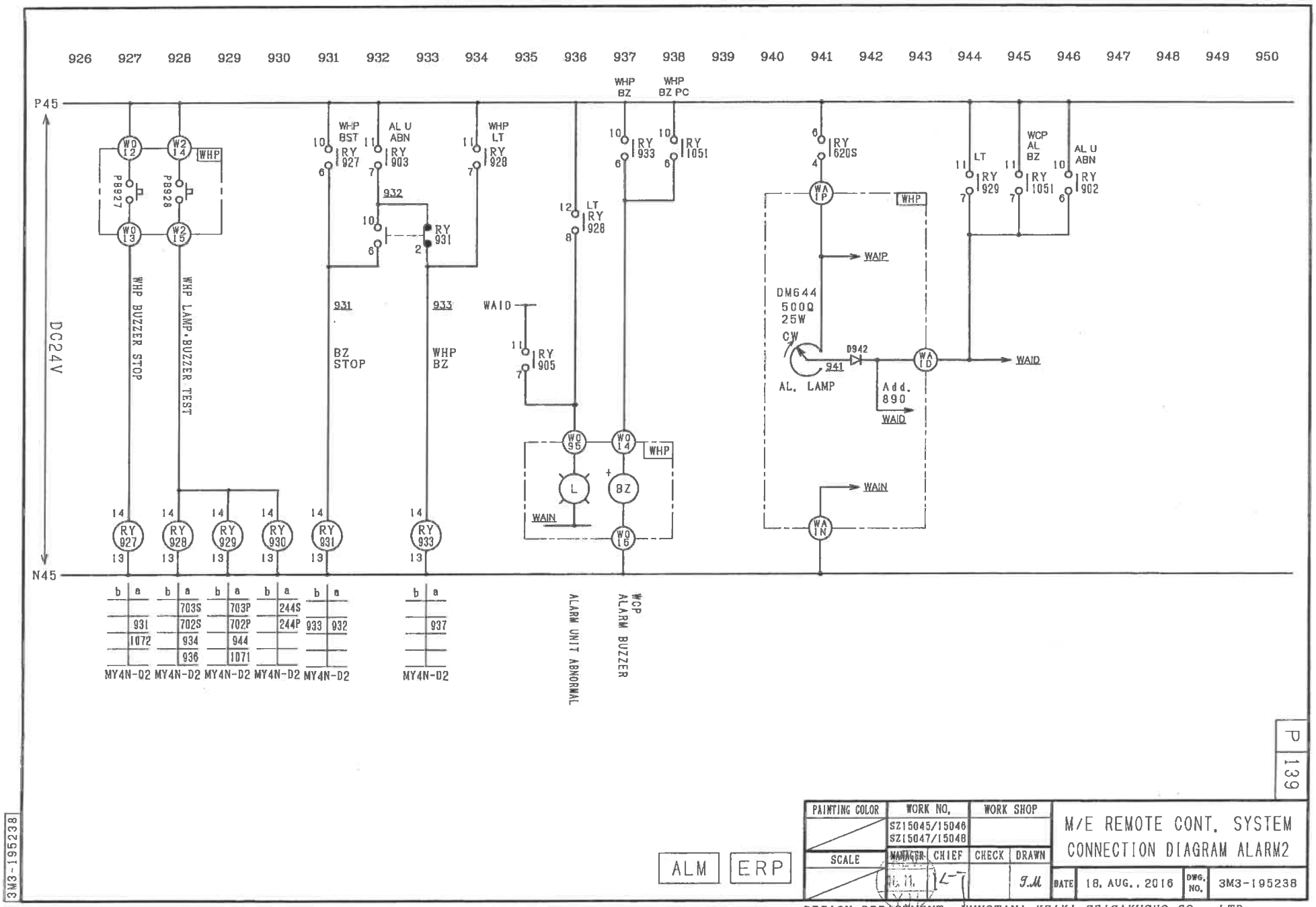
3M3-195237



ALM ERP

PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM ALARM2
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE: 18. AUG., 2016 / DWG. NO.: 3M3-195238

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195238

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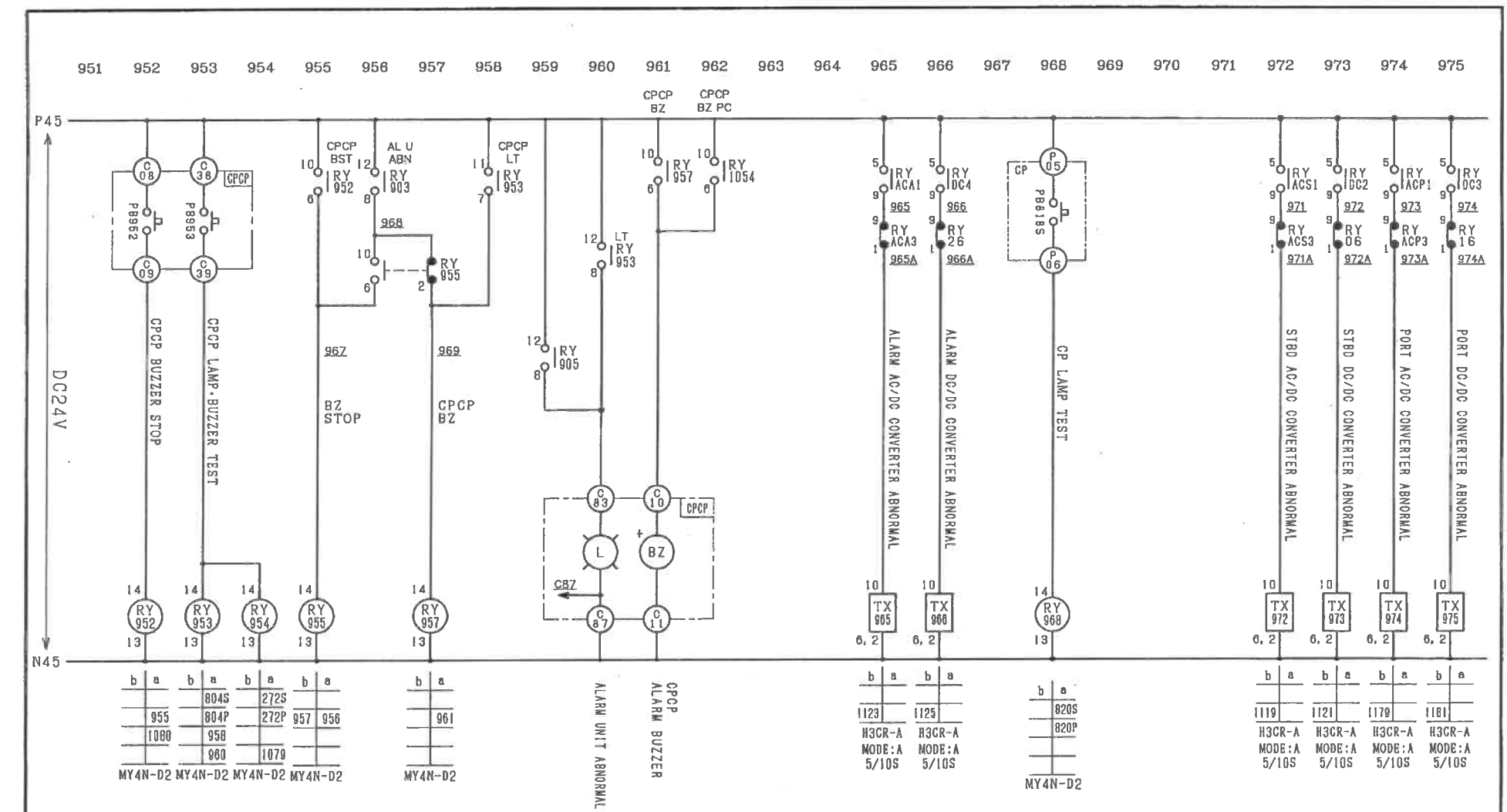

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM ALARM3

DATE 9. SEP., 2016 DWG. NO. 3M3-195239

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

3M3-195239

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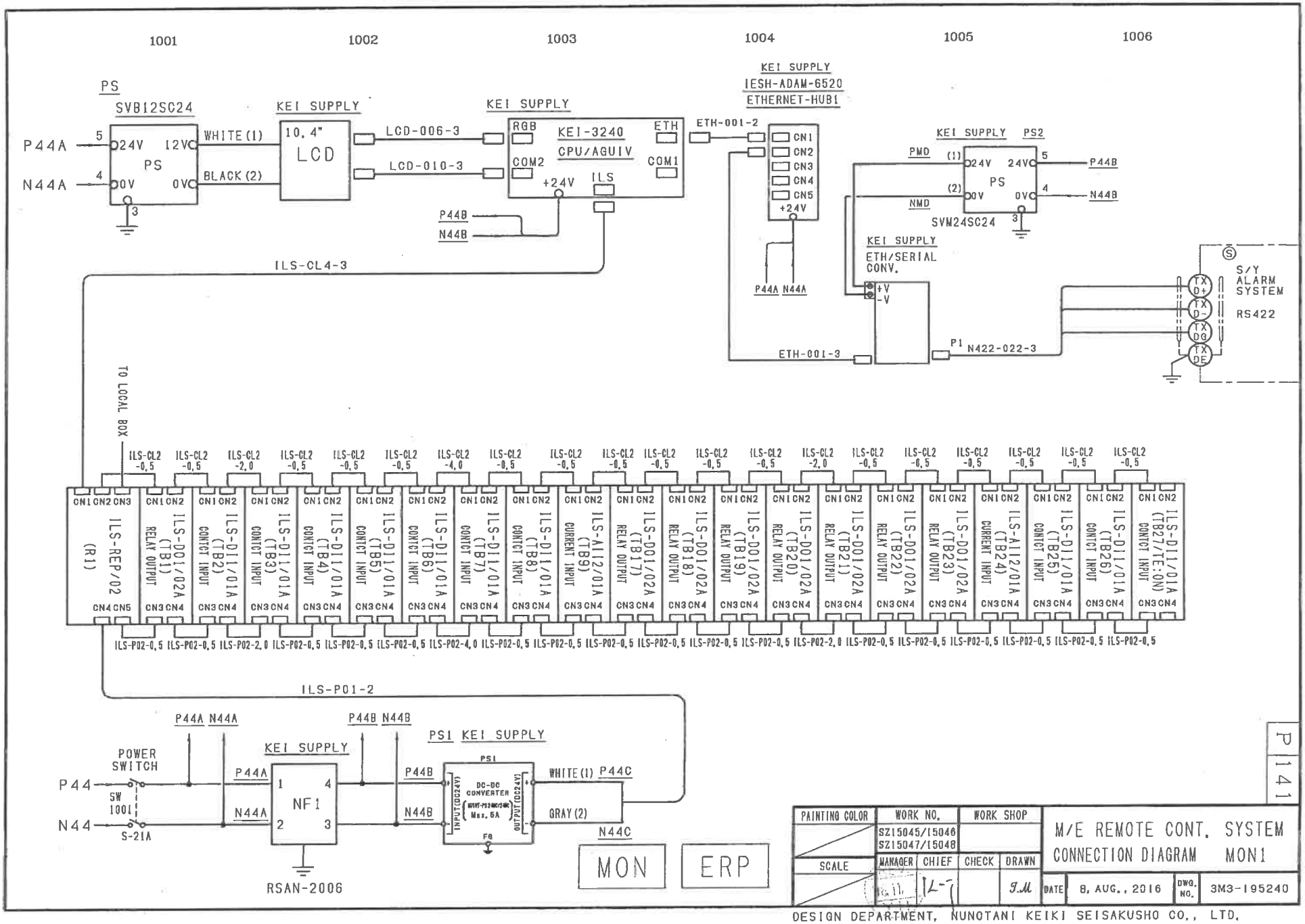

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	
SCALE	MANAGER / CHIEF / CHECK / DRAWN	
	T.M	

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON1

DATE 8, AUG., 2016 | DWG. NO. 3M3-195240

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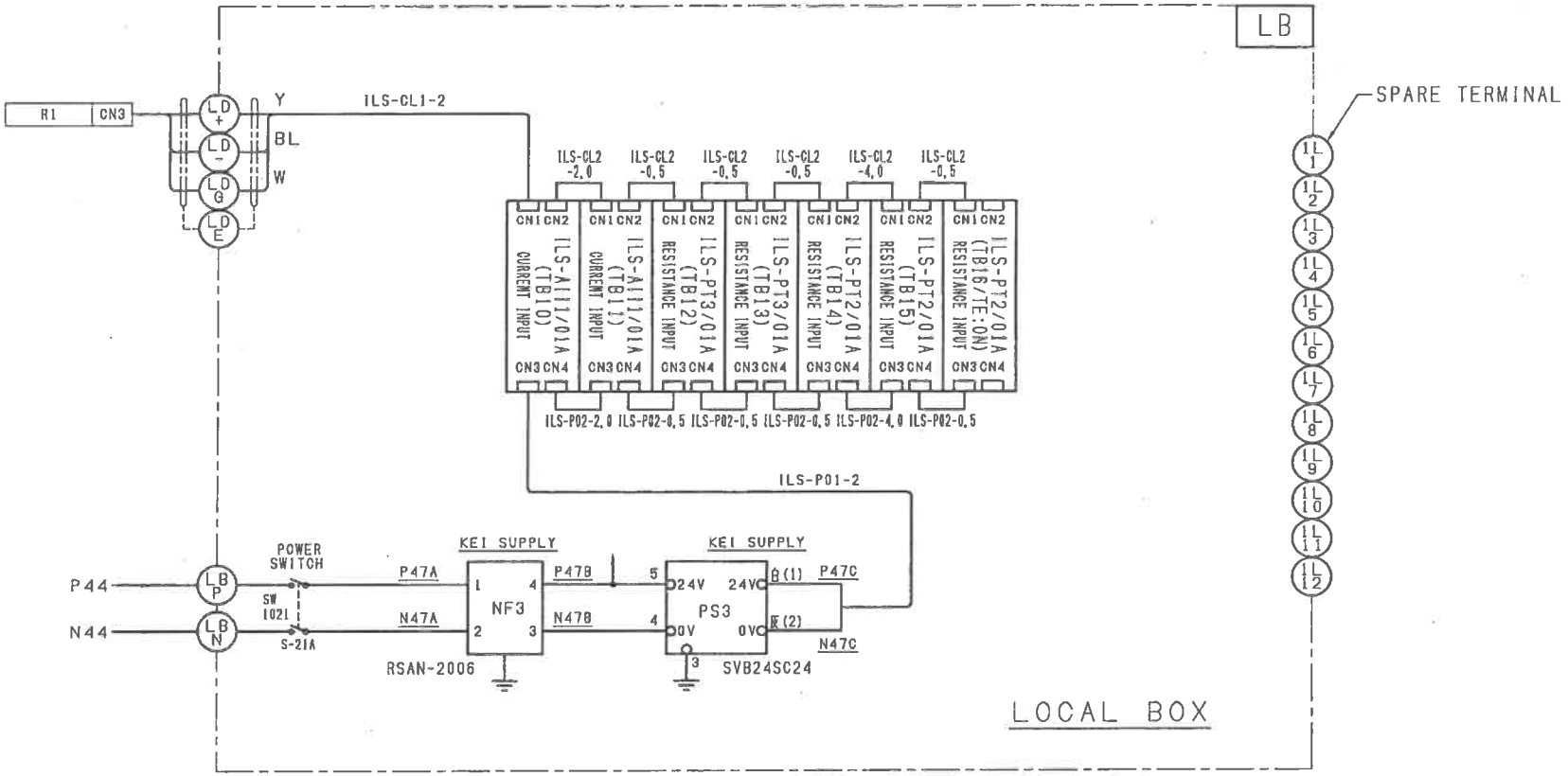
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

1001 1002 1003 1004 1005 1006



MON ERP

PAINTING COLOR	WORK NO.	WORK SHOP
	SZ15045/15046 SZ15047/15048	

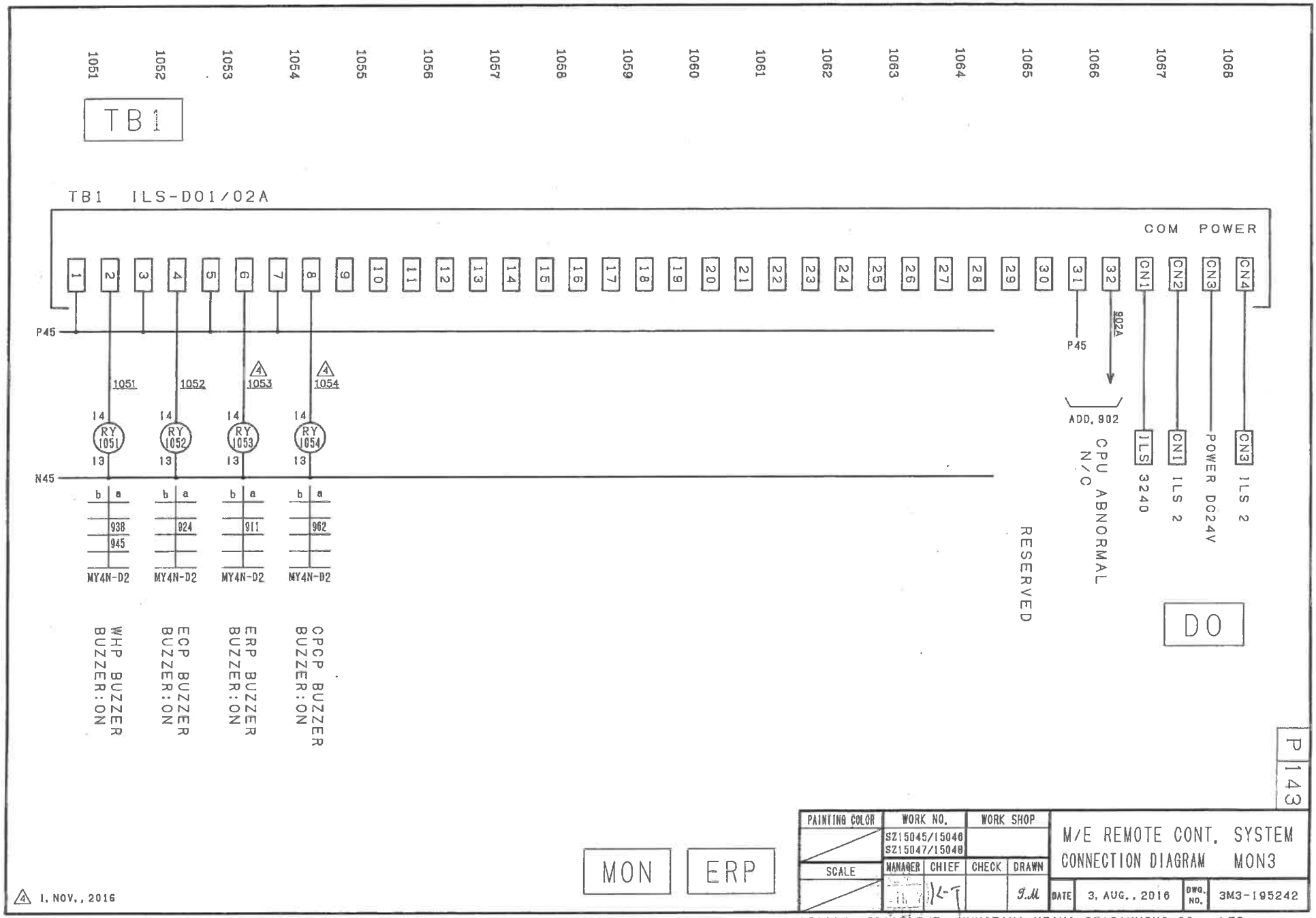
SCALE	MANAGER	CHIEF	CHECK	DRAWN
				T.M

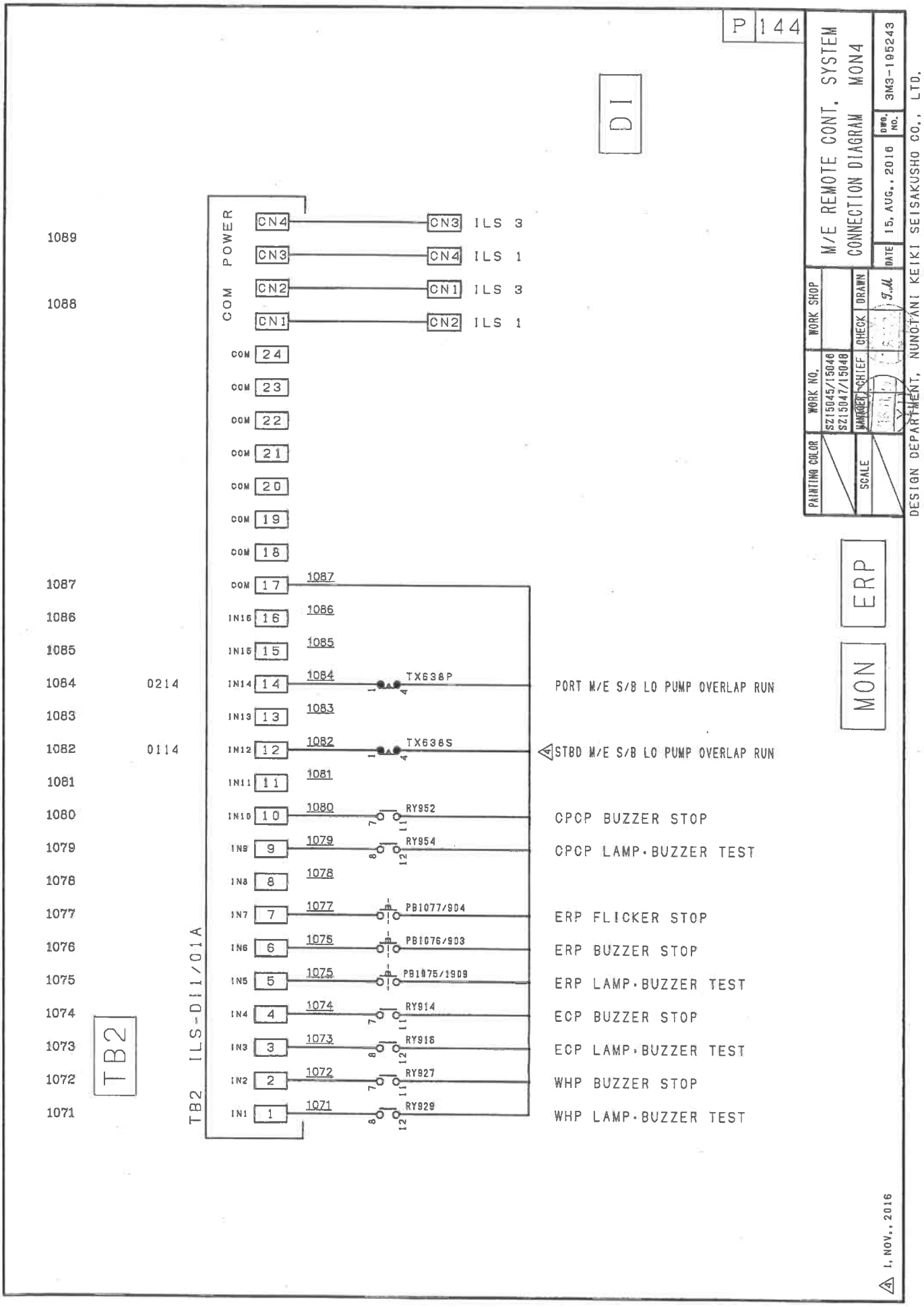
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON2

DATE 3, AUG., 2016 DWG. NO. 3M3-195241

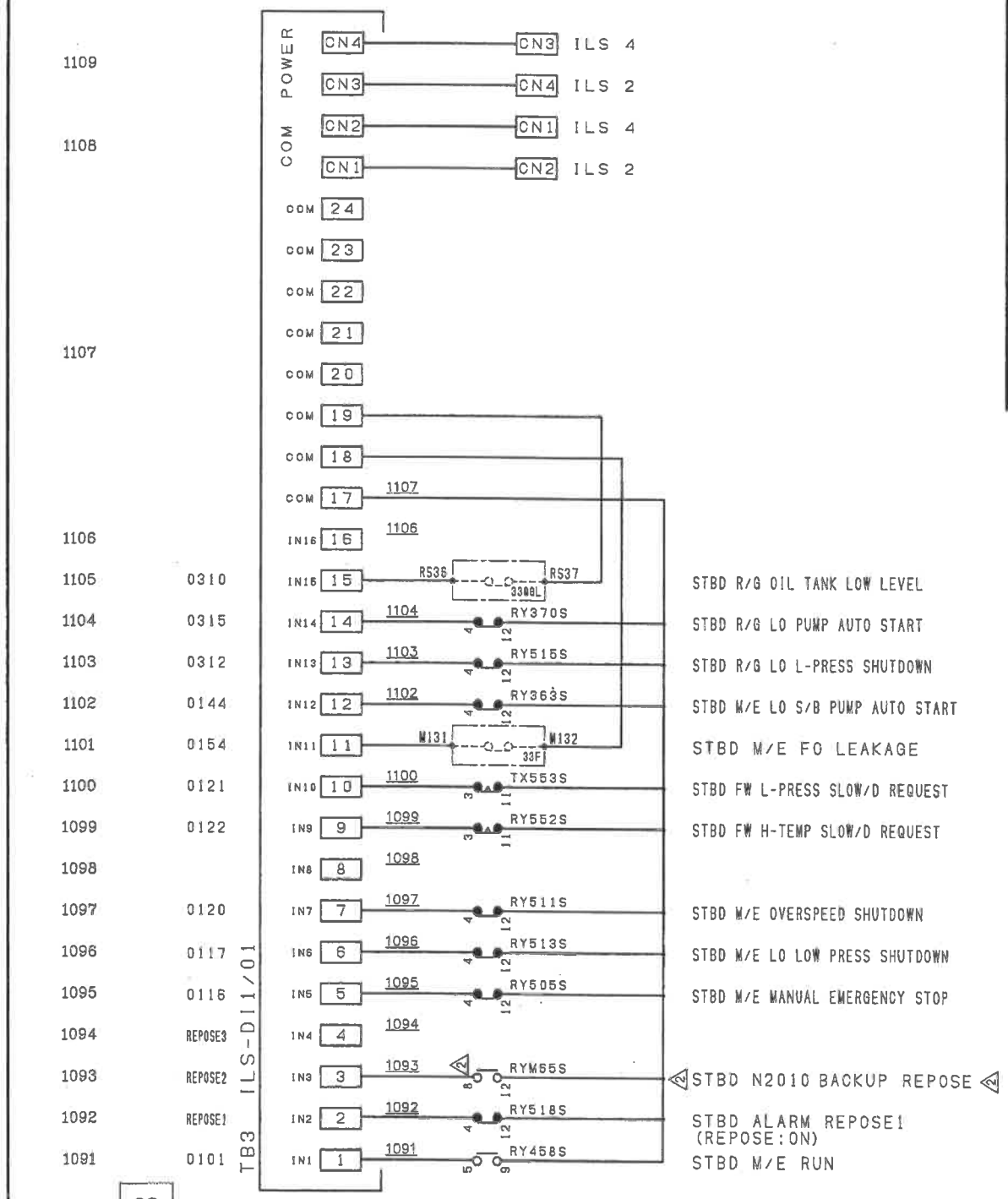
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

P 142


1051 1052 1053 1054 1055 1056 1057 1058 1059 1060 1061 1062 1063 1064 1065 1066 1067 1068
TB1
TB1 ILS-D01/02A
COM POWER
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 CN1 CN2 CN3 CN4
P45
902A
P45
ADD. 902
CPU ABNORMAL N/C
ILS 3240
CN1 ILS 2
POWER DC24V
CN3 ILS 2
RESERVED
1051 1052 4 1053 4 1054
14 RY 1051 13
14 RY 1052 13
14 RY 1053 13
14 RY 1054 13
N45
b a
938 945
924
911
962
MY4N-D2
WHP BUZZER BUZZER:ON
ECP BUZZER BUZZER:ON
ERP BUZZER BUZZER:ON
CPCP BUZZER BUZZER:ON
DO
P143
MON
ERP
4 1. NOV., 2016
PAINTING COLOR
WORK NO.
SZ15045/15046
SZ15047/15048
WORK SHOP
SCALE
MANAGER
CHIEF
CHECK
DRAWN
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON3
DATE 3. AUG., 2016
DWG. NO. 3M3-195242
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.


P 144
DI
ERP
MON
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON4
DWG. NO. 3M3-195243
DATE 15. AUG., 2016
WORK SHOP
DRAWN
CHECK
CHIEF
MANAGER
WORK NO.
SZ15045/15046
SZ15047/15048
PAINTING COLOR
SCALE
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
COM POWER
CN4
CN3
ILS 3
CN3
CN4
ILS 1
CN2
CN1
ILS 3
CN1
CN2
ILS 1
COM 24
COM 23
COM 22
COM 21
COM 20
COM 19
COM 18
COM 17
IN16 16
IN15 15
IN14 14
IN13 13
IN12 12
IN11 11
IN10 10
IN9 9
IN8 8
IN7 7
IN6 6
IN5 5
IN4 4
IN3 3
IN2 2
IN1 1
1089
1088
1087
1086
1085
1084
1083
1082
1081
1080
1079
1078
1077
1076
1075
1074
1073
1072
1071
0214
0114
TB2
TB2 ILS-DI1/01A
TX638P
TX638S
RY952
RY954
PB1077/904
PB1076/903
PB1075/1909
RY914
RY916
RY927
RY929
PORT M/E S/B LO PUMP OVERLAP RUN
4 STBD M/E S/B LO PUMP OVERLAP RUN
CPCP BUZZER STOP
CPCP LAMP・BUZZER TEST
ERP FLICKER STOP
ERP BUZZER STOP
ERP LAMP・BUZZER TEST
ECP BUZZER STOP
ECP LAMP・BUZZER TEST
WHP BUZZER STOP
WHP LAMP・BUZZER TEST
4 1, NOV., 2016

DI



TB3

ERP

MON

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON5

DWG. NO. 3M3-185244

DATE 9. SEP., 2016

DRAWN

CHECK

CHIEF

MANAGER

WORK NO. SZ15045/15046 SZ15047/15048

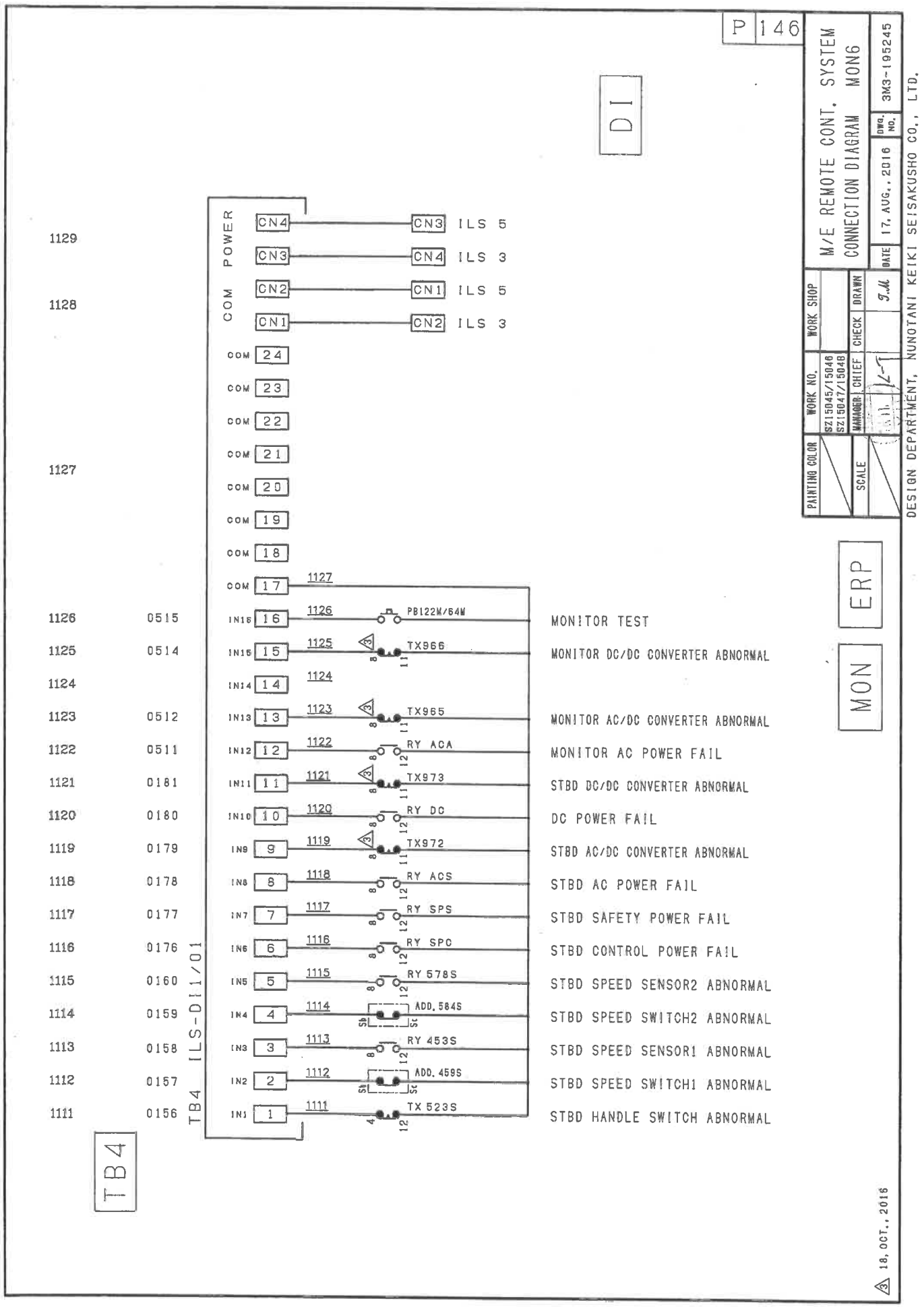
WORK SHOP

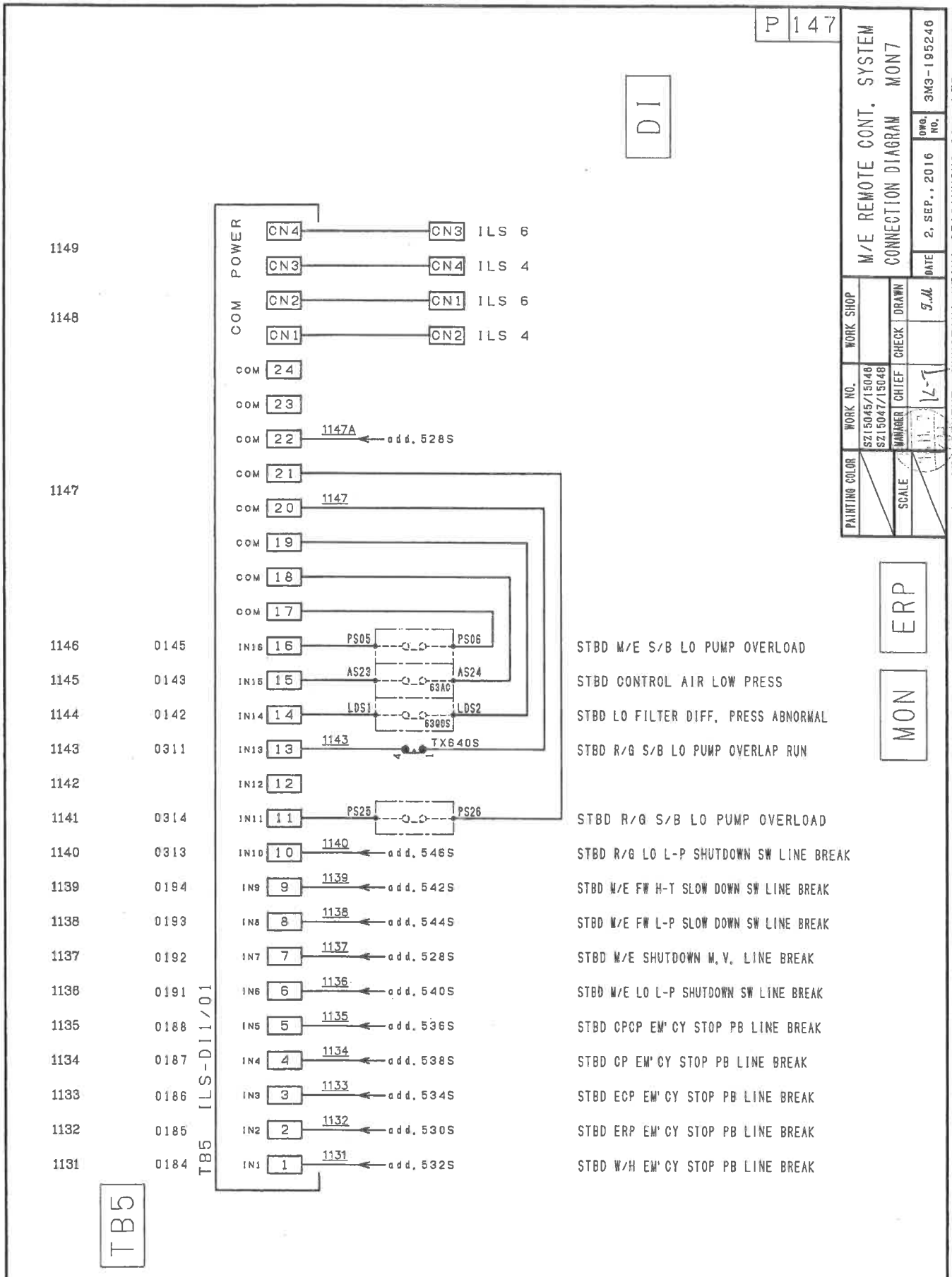
PAINTING COLOR

SCALE

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

2 29, SEP., 2016


P 146
DI
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON6
DWG. NO. 3M3-195245
DATE 17. AUG., 2016
WORK SHOP
DRAWN
CHECK
CHIEF
MANAGER
WORK NO. SZ15045/15046 SZ15047/15048
PAINTING COLOR
SCALE
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
ERP
MON
TB4
TB4 ILS-DI1/01
POWER
COM
CN4 CN3 ILS 5
CN3 CN4 ILS 3
CN2 CN1 ILS 5
CN1 CN2 ILS 3
1129
1128
1127
1126 0515 IN16 16 PB122M/64M MONITOR TEST
1125 0514 IN15 15 TX966 MONITOR DC/DC CONVERTER ABNORMAL
1124 IN14 14
1123 0512 IN13 13 TX965 MONITOR AC/DC CONVERTER ABNORMAL
1122 0511 IN12 12 RY ACA MONITOR AC POWER FAIL
1121 0181 IN11 11 TX973 STBD DC/DC CONVERTER ABNORMAL
1120 0180 IN10 10 RY DC DC POWER FAIL
1119 0179 IN9 9 TX972 STBD AC/DC CONVERTER ABNORMAL
1118 0178 IN8 8 RY ACS STBD AC POWER FAIL
1117 0177 IN7 7 RY SPS STBD SAFETY POWER FAIL
1116 0176 IN6 6 RY SPC STBD CONTROL POWER FAIL
1115 0160 IN5 5 RY 578S STBD SPEED SENSOR2 ABNORMAL
1114 0159 IN4 4 ADD. 584S STBD SPEED SWITCH2 ABNORMAL
1113 0158 IN3 3 RY 453S STBD SPEED SENSOR1 ABNORMAL
1112 0157 IN2 2 ADD. 459S STBD SPEED SWITCH1 ABNORMAL
1111 0156 IN1 1 TX 523S STBD HANDLE SWITCH ABNORMAL
COM 24
COM 23
COM 22
COM 21
COM 20
COM 19
COM 18
COM 17
18, OCT., 2016



M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON7

PAINTING COLOR	WORK NO.	WORK SHOP		DATE	DWG. NO.
SCALE	SZ15045/15046 SZ15047/15048	CHECK	DRAWN	2. SEP., 2016	3M3-195246
	MANAGER CHIEF				

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

DI

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON8

DWG. NO. 3M3-195247

DATE 9, SEP., 2016

WORK SHOP | CHECK | DRAWN

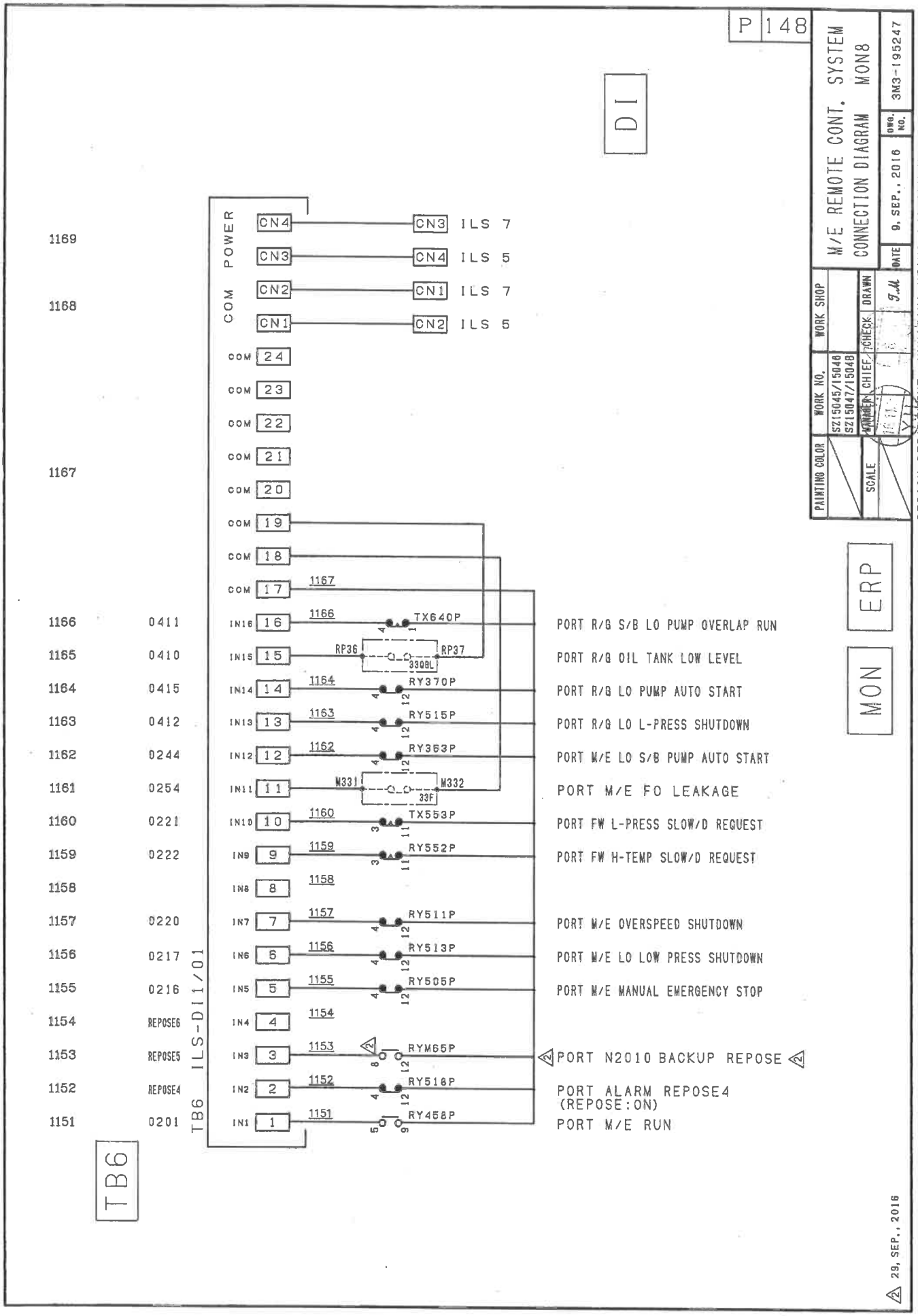
WORK NO. SZ15045/15048 SZ15047/15048

PAINTING COLOR | SCALE

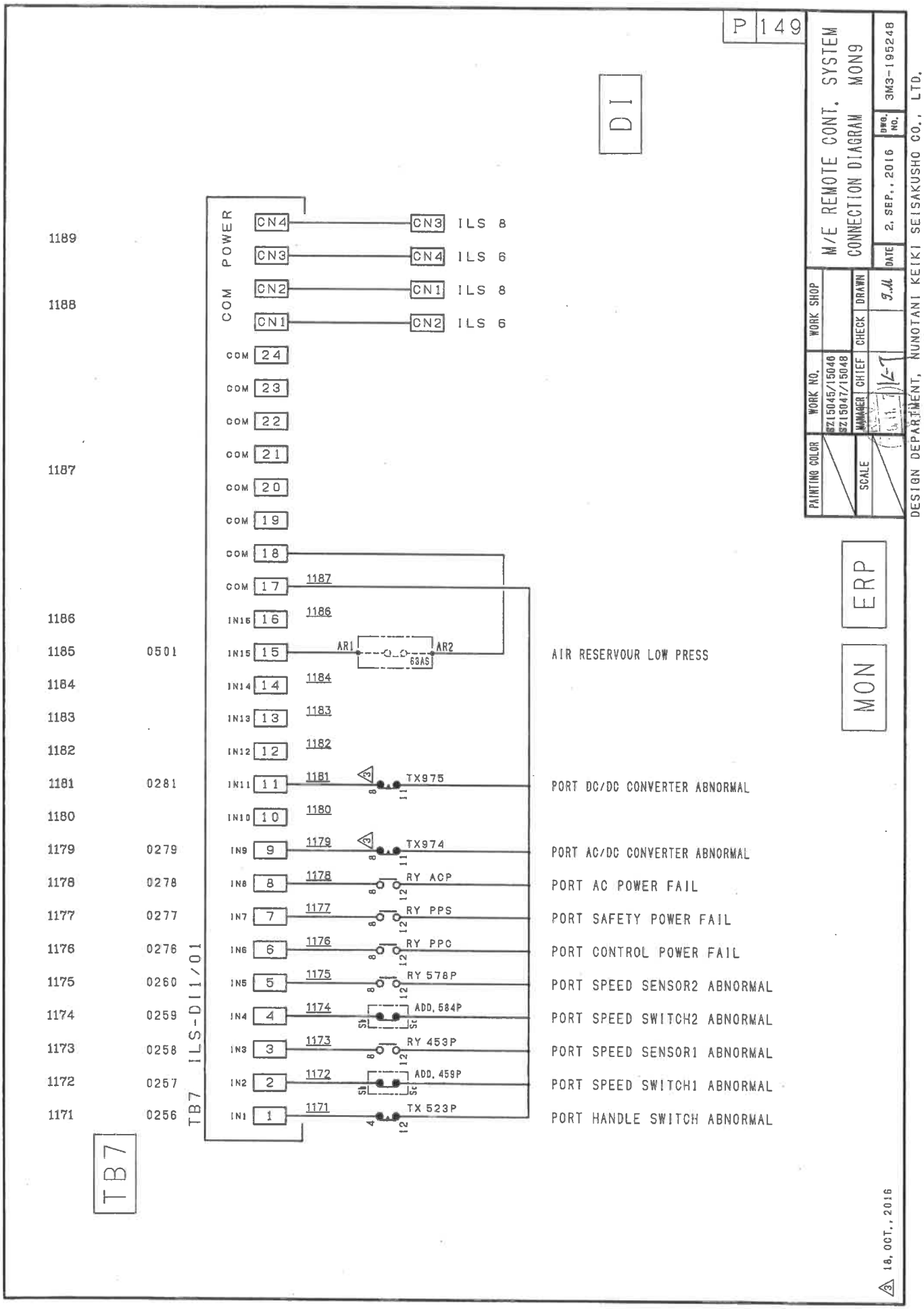
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

ERP

MON



⚠2 29, SEP., 2016


P 149
DI
ERP
MON
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON9
DWG. NO. 3M3-195248
DATE 2. SEP.. 2016
WORK SHOP
DRAWN
CHECK
CHIEF
MANAGER
WORK NO. SZ15045/15048 SZ15047/15048
SCALE
PAINTING COLOR
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
TB7
TB7 ILS-DI1/01
POWER
COM
CN4 CN3 ILS 8
CN3 CN4 ILS 6
CN2 CN1 ILS 8
CN1 CN2 ILS 6
COM 24
COM 23
COM 22
COM 21
COM 20
COM 19
COM 18
COM 17
IN16 16
IN15 15
IN14 14
IN13 13
IN12 12
IN11 11
IN10 10
IN9 9
IN8 8
IN7 7
IN6 6
IN5 5
IN4 4
IN3 3
IN2 2
IN1 1
1189
1188
1187
1186
1185 0501
1184
1183
1182
1181 0281
1180
1179 0279
1178 0278
1177 0277
1176 0276
1175 0260
1174 0259
1173 0258
1172 0257
1171 0256
AR1 AR2 63AS
AIR RESERVOUR LOW PRESS
TX975
PORT DC/DC CONVERTER ABNORMAL
TX974
PORT AC/DC CONVERTER ABNORMAL
RY ACP
PORT AC POWER FAIL
RY PPS
PORT SAFETY POWER FAIL
RY PPC
PORT CONTROL POWER FAIL
RY 578P
PORT SPEED SENSOR2 ABNORMAL
ADD. 584P
PORT SPEED SWITCH2 ABNORMAL
RY 453P
PORT SPEED SENSOR1 ABNORMAL
ADD. 459P
PORT SPEED SWITCH1 ABNORMAL
TX 523P
PORT HANDLE SWITCH ABNORMAL
3 18. OCT., 2016

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON10

DWG. NO.	DATE	DRAWN	CHECK	CHIEF MANAGER	WORK NO.	WORK SHOP	SCALE	PAINTING COLOR
3M3-195249	2. SEP., 2016	T.M			SZ15045/15046 SZ15047/15048			

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

DI

ERP

MON

TB8

TB8 ILS-D11/01

COM POWER

CN4 → CN3 ILS 9

CN3 → CN4 ILS 7

CN2 → CN1 ILS 9

CN1 → CN2 ILS 7

1209		COM	24		
		COM	23		
1208		COM	22	1207A	add. 528P
		COM	21		
1207		COM	20		
		COM	19		
		COM	18		
		COM	17		
1206	0245	IN16	16	PS45 — PS46	PORT S/B LO PUMP OVERLOAD
1205	0243	IN15	15	AP23 — AP24 (63AC)	PORT CONTROL AIR LOW PRESS
1204	0242	IN14	14	LDP1 — LDP2 (63LDP)	PORT LO FILTER DIFF. PRESS ABNORMAL
1203		IN13	13	1203	
1202		IN12	12	1202	
1201	0414	IN11	11	PS65 — PS66	PORT R/G S/B LO PUMP OVERLOAD
1200	0413	IN10	10	1200 add. 546P	PORT R/G LO L-P SHUTDOWN SW LINE BREAK
1199	0294	IN9	9	1199 add. 542P	PORT M/E FW H-T SLOWDOWN SW LINE BREAK
1198	0293	IN8	8	1198 add. 544P	PORT M/E FW L-P SLOWDOWN SW LINE BREAK
1197	0292	IN7	7	1197 add. 528P	PORT M/E SHUTDOWN M.V. LINE BREAK
1196	0291	IN6	6	1196 add. 540P	PORT M/E LO L-P SHUTDOWN SW LINE BREAK
1195	0288	IN5	5	1195 add. 536P	PORT CPCP EM'CY STOP PB LINE BREAK
1194	0287	IN4	4	1194 add. 538P	PORT CP EM'CY STOP PB LINE BREAK
1193	0286	IN3	3	1193 add. 534P	PORT ECP EM'CY STOP PB LINE BREAK
1192	0285	IN2	2	1192 add. 530P	PORT ERP EM'CY STOP PB LINE BREAK
1191	0284	IN1	1	1191 add. 532P	PORT W/H EM'CY STOP PB LINE BREAK

AI

ERP

MON

TB9

PAINTING COLOR		WORK NO.	SZ15045/15040 SZ15047/15048	WORK SHOP		
SCALE		MARKER	CHIEF	CHECK	DRAWN	DATE
					J.M	9, AUG., 2016

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON11

DWG. NO. 3M3-19525D

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

△ 13, SEP., 2016

COM POWER

CN4	→	CN3	ILS 10	1220
CN3	→	CN4	ILS 8	
CN2	→	CN1	ILS 10	1219
CN1	→	CN2	ILS 8	

TB9 ILS-AII2/01A

CH8-	16	633PF	Add. 634P	0209	1218	PORT FO RACK 4~20mA/0~36mm △
CH8+	15	633PE				
CH7-	14	633PB	Add. 634P	0208	1217	PORT T/C REVOLUTION SPEED 4~20mA/0~60000min-1
CH7+	13	633PA				
CH6-	12	492PB	Add. 493P	0207	1216	PORT PROPELLER REVOLUTION SPEED 4~20mA/0~±400min-1
CH6+	11	492PA				
CH5-	10	473PB	Add. 474P	0206	1215	PORT M/E REVOLUTION SPEED 4~20mA/0~1000min-1
CH5+	9	473PA				
CH4-	8	633SF	Add. 634S	0109	1214	STBD FO RACK 4~20mA/0~36mm △
CH4+	7	633SE				
CH3-	6	633SB	Add. 634S	0108	1213	STBD T/C REVOLUTION SPEED 4~20mA/0~60000min-1
CH3+	5	633SA				
CH2-	4	492SB	Add. 493S	0107	1212	STBD PROPELLER REVOLUTION SPEED 4~20mA/0~±400min-1
CH2+	3	492SA				
CH1-	2	473SB	Add. 474S	0106	1211	STBD M/E REVOLUTION SPEED 4~20mA/0~1000min-1
CH1+	1	473SA				

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON12
3M3-195251
8. SEP., 2016
DATE
DWG. NO.
WORK SHOP
DRAWN
CHECK
CHIEF
MANAGER
WORK NO.
SZI5045/15046
SZI5047/15048
SCALE
PAINTING COLOR
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
A1
LB
MON
TB10
TB10 ILS-AIII/01A
COM POWER
CN4
CN3
CN2
CN1
ILS 11
ILS 9
ILS 11
ILS 9
1240
1239
1238
1237
1236
1235
1234
1233
1232
1231
0301
0136
0134
0132
0131
0133
0137
CH8- 16
CH8+ 15
CH7- 14
CH7+ 13
CH6- 12
CH6+ 11
CH5- 10
CH5+ 9
CH4- 8
CH4+ 7
CH3- 6
CH3+ 5
CH2- 4
CH2+ 3
CH1- 2
CH1+ 1
SIG
+24V
PT
PTQG
PTAG
PTWH
PTQT
PTQ
PTF
PTB
JBS1
RJS
STBD R/G LO PRESS
4~20mA/0~0.6MPa
STBD M/E GOV-CONTROL AIR PRESS
4~20mA/0~0.6MPa
STBD M/E FW PRESS
4~20mA/0~0.6MPa
STBD M/E T/C LO PRESS
4~20mA/0~1.0MPa
STBD M/E LO PRESS
4~20mA/0~1.0MPa
STBD M/E FO PRESS
4~20mA/0~0.6MPa
STBD M/E BOOST AIR PRESS
4~20mA/0~0.4MPa

A I

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON13

PAINTING COLOR	WORK NO.	WORK SHOP			
	SZ15045/15046 SZ15047/15048	MANAGER	CHIEF	CHECK	DRAWN
SCALE					

DATE 8. SEP., 2016
DWG. NO. 3M3-195252

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

LB

MON

TB11

TB11 ILS-AIII/01A

COM POWER

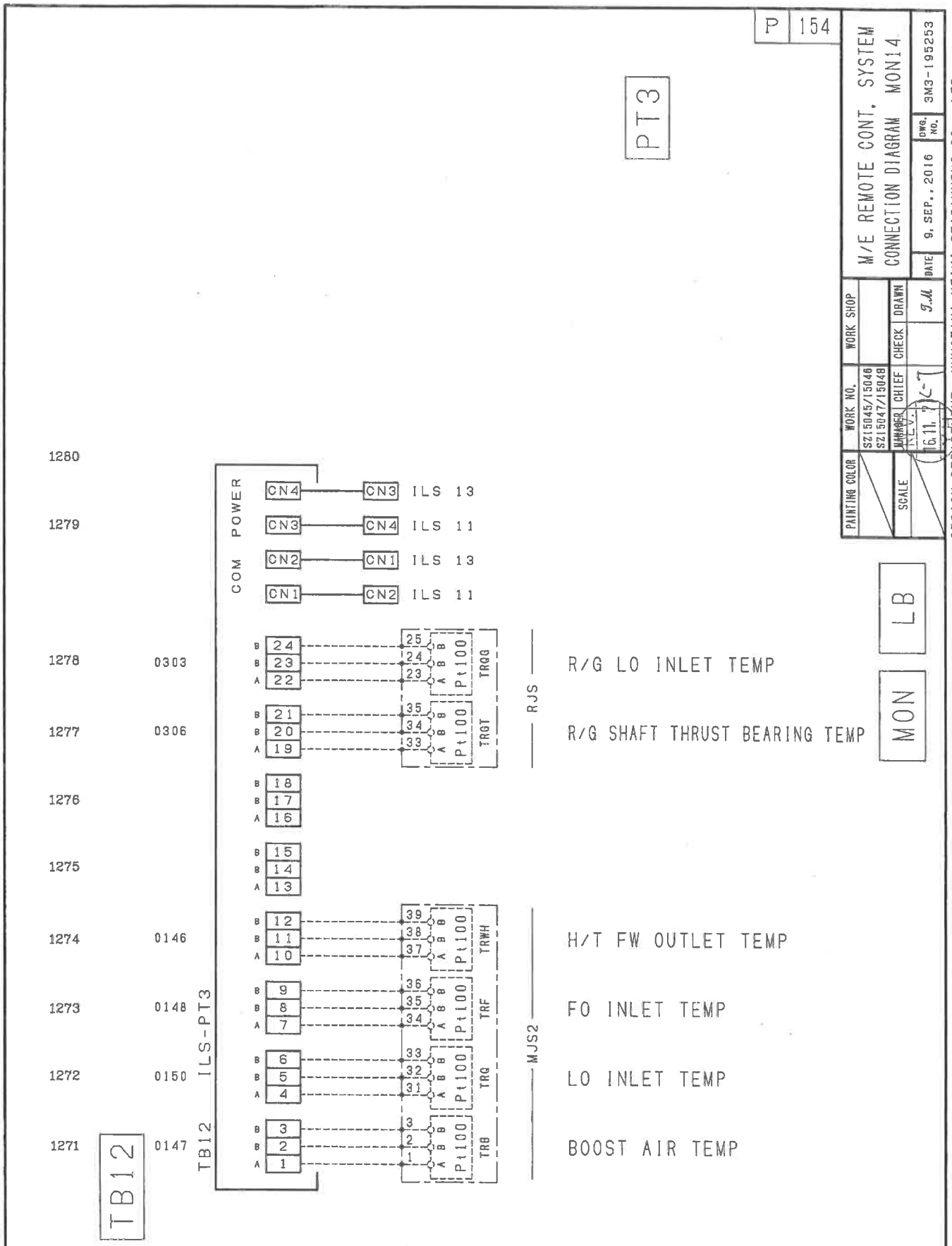
CN4 — CN3 ILS 12
CN3 — CN4 ILS 10
CN2 — CN1 ILS 12
CN1 — CN2 ILS 10

1260		CH8- 16	PTQG	PORT R/G LO PRESS 4~20mA/0~0.6MPa
1259	0401	CH8+ 15		
1258		CH7- 14		
1257		CH7+ 13		
1256	0236	CH6- 12	PTAG	PORT M/E GOV-CONTROL AIR PRESS 4~20mA/0~0.6MPa
1255	0234	CH6+ 11		
1254	0232	CH5- 10	PTWH	PORT M/E FW PRESS 4~20mA/0~0.6MPa
1253	0231	CH5+ 9		
1252	0233	CH4- 8	PTQT	PORT M/E T/C LO PRESS 4~20mA/0~1.0MPa
1251	0237	CH4+ 7		
		CH3- 6	PTQ	PORT M/E LO PRESS 4~20mA/0~1.0MPa
		CH3+ 5		
		CH2- 4	PTF	PORT M/E FO PRESS 4~20mA/0~0.6MPa
		CH2+ 3		
		CH1- 2	PTB	PORT M/E BOOST AIR PRESS 4~20mA/0~0.4MPa
		CH1+ 1		

RJP

JBP1

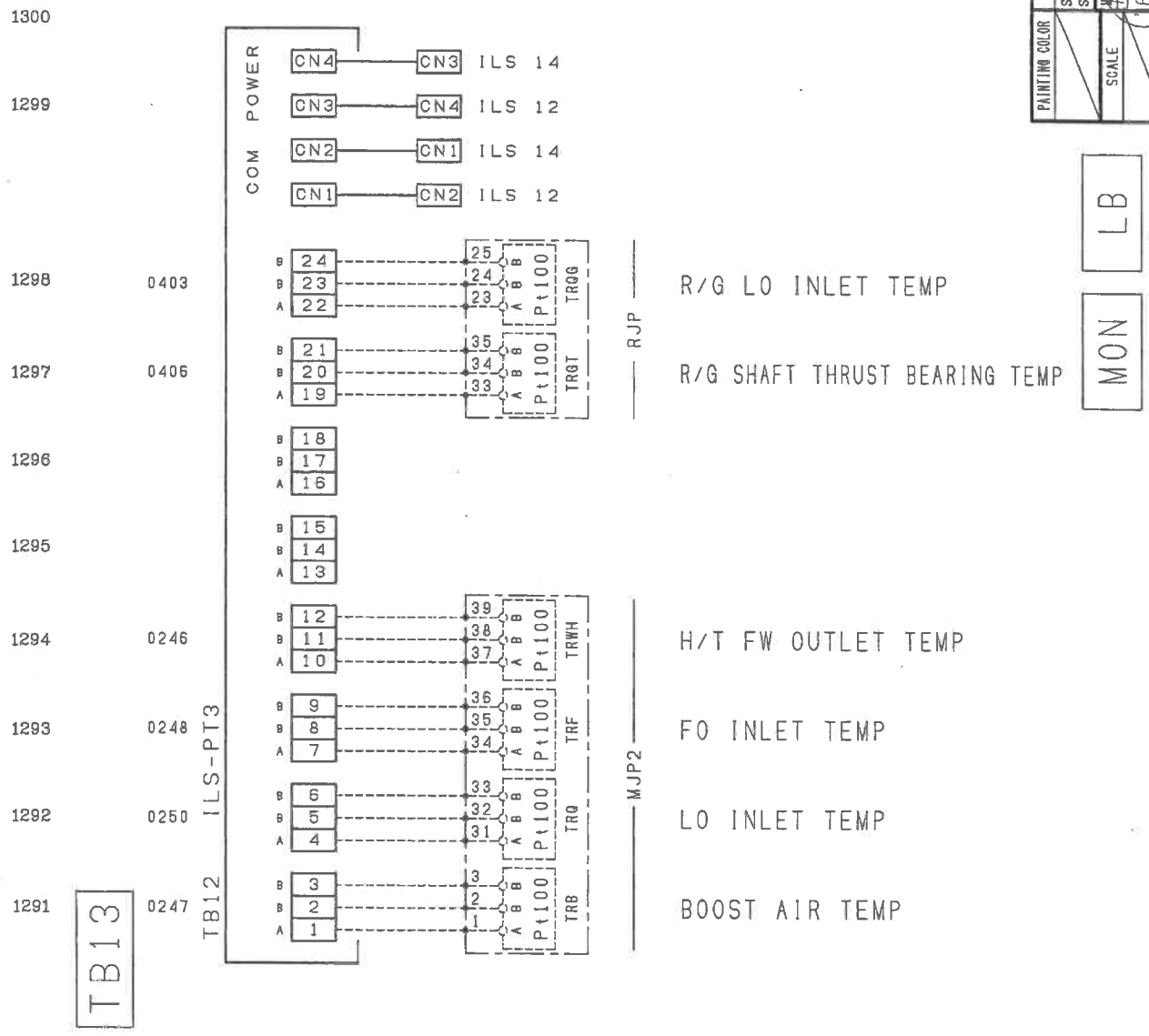
S16 +24V PT



PAINTING COLOR	WORK NO.	WORK SHOP		M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON14	
	SZ15045/15046 SZ15047/15048	CHIEF	CHECK	DRAWN	DATE 9, SEP., 2016
SCALE	NUMBER REV.	16.11.		J.M	DWG. NO. 3M3-195253

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

PT3



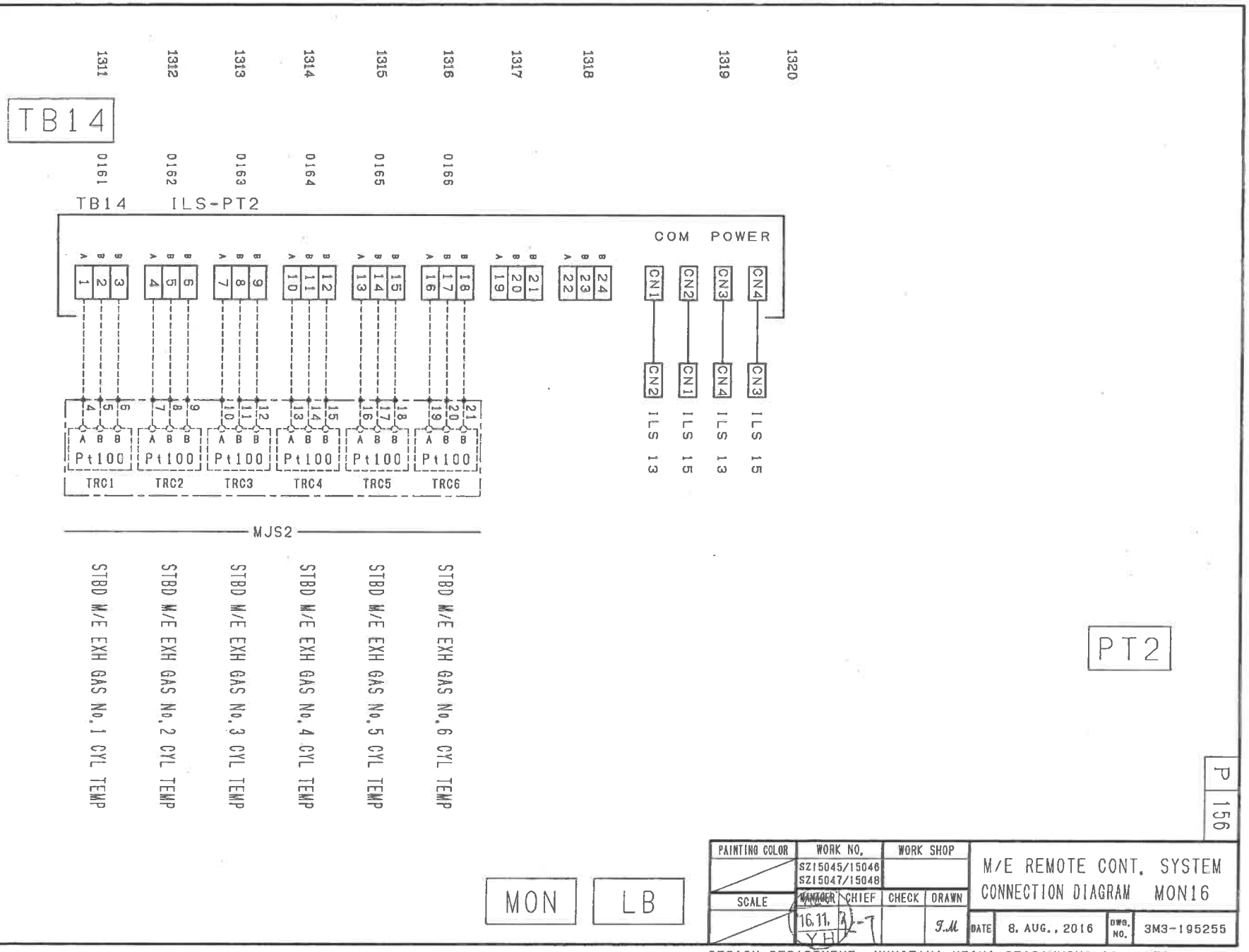
MON LB

PAINTING COLOR	WORK NO. SZ15045/15046 SZ15047/15048	WORK SHOP			
SCALE	MANAGER	CHIEF	CHECK	DRAWN	
	REV. 6.11.			J.M	

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON15

DATE 9. SEP.. 2016 DWG. NO. 3M3-195254

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



P 156

PAINTING COLOR	WORK NO.	WORK SHOP	
	SZ15045/15046 SZ15047/15048		
SCALE	MANAGER / CHIEF	CHECK	DRAWN
	16.11. YH		T.M

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON16

DATE 8. AUG., 2016 DWG. NO. 3M3-195255

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

PT2

PAINTING COLOR	WORK NO.	WORK SHOP		M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON17		
	SZ15045/15048 SZ15047/15048	NUMBER	CHIEF	CHECK	DRAWN	DATE
SCALE			16.11.7		J.M	8. AUG., 2016

DWG. NO. 3M3-195256

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

MON

LB

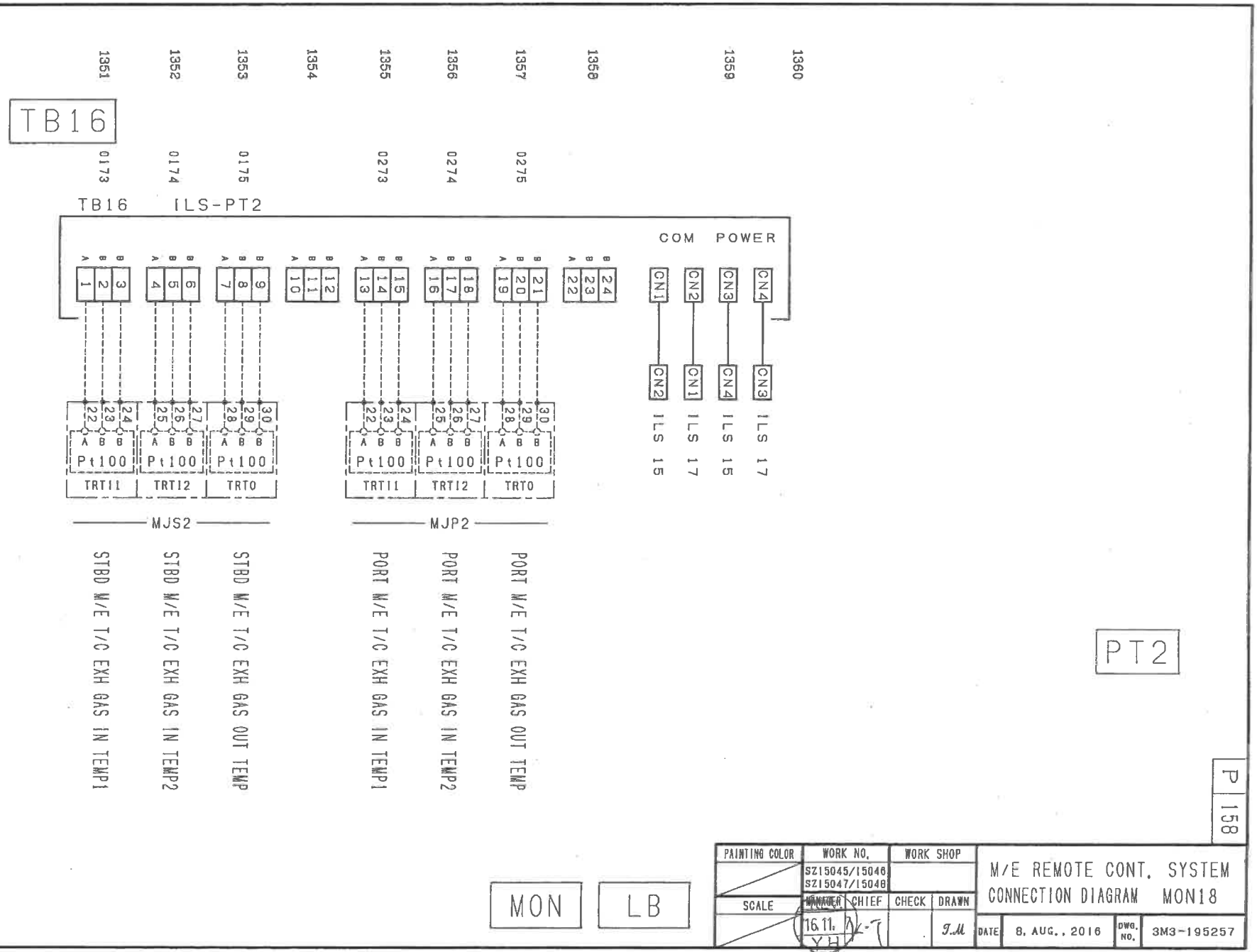
TB15

TB15 ILS-PT2

COM POWER

CN4	CN3	ILS 16	1340
CN3	CN4	ILS 14	1339
CN2	CN1	ILS 16	
CN1	CN2	ILS 14	

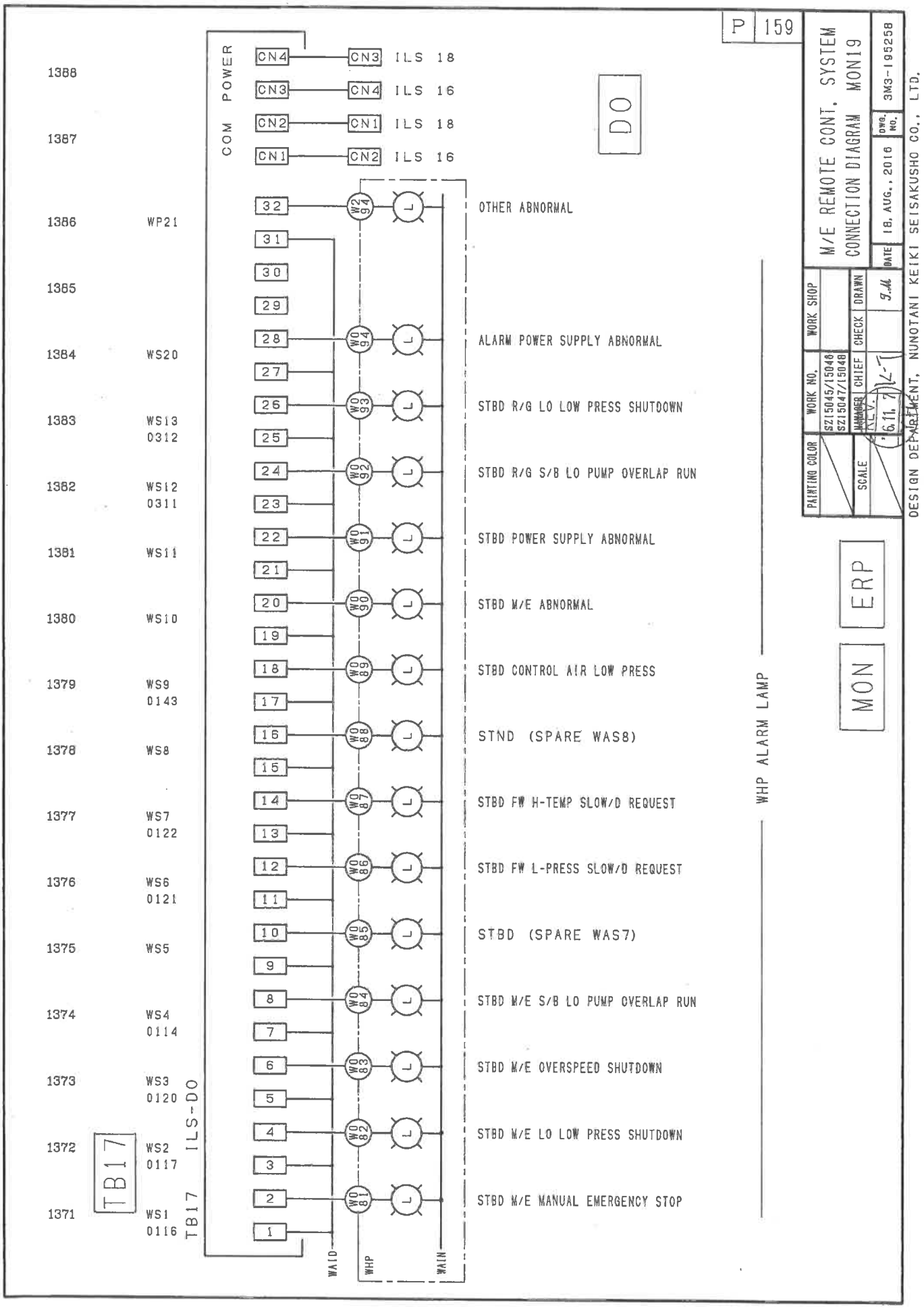
B 24 / B 23 / A 22							1338
B 21 / B 20 / A 19							1337
B 18 / B 17 / A 16	0266	21 / 20 / 19	Pt100	TRC6	MJP2	PORT M/E EXH GAS No. 6 CYL TEMP	1336
B 15 / B 14 / A 13	0265	18 / 17 / 16	Pt100	TRC5		PORT M/E EXH GAS No. 5 CYL TEMP	1335
B 12 / B 11 / A 10	0264	15 / 14 / 13	Pt100	TRC4		PORT M/E EXH GAS No. 4 CYL TEMP	1334
B 9 / B 8 / A 7	0263	12 / 11 / 10	Pt100	TRC3		PORT M/E EXH GAS No. 3 CYL TEMP	1333
B 6 / B 5 / A 4	0262	9 / 8 / 7	Pt100	TRC2		PORT M/E EXH GAS No. 2 CYL TEMP	1332
B 3 / B 2 / A 1	0261	6 / 5 / 4	Pt100	TRC1		PORT M/E EXH GAS No. 1 CYL TEMP	1331

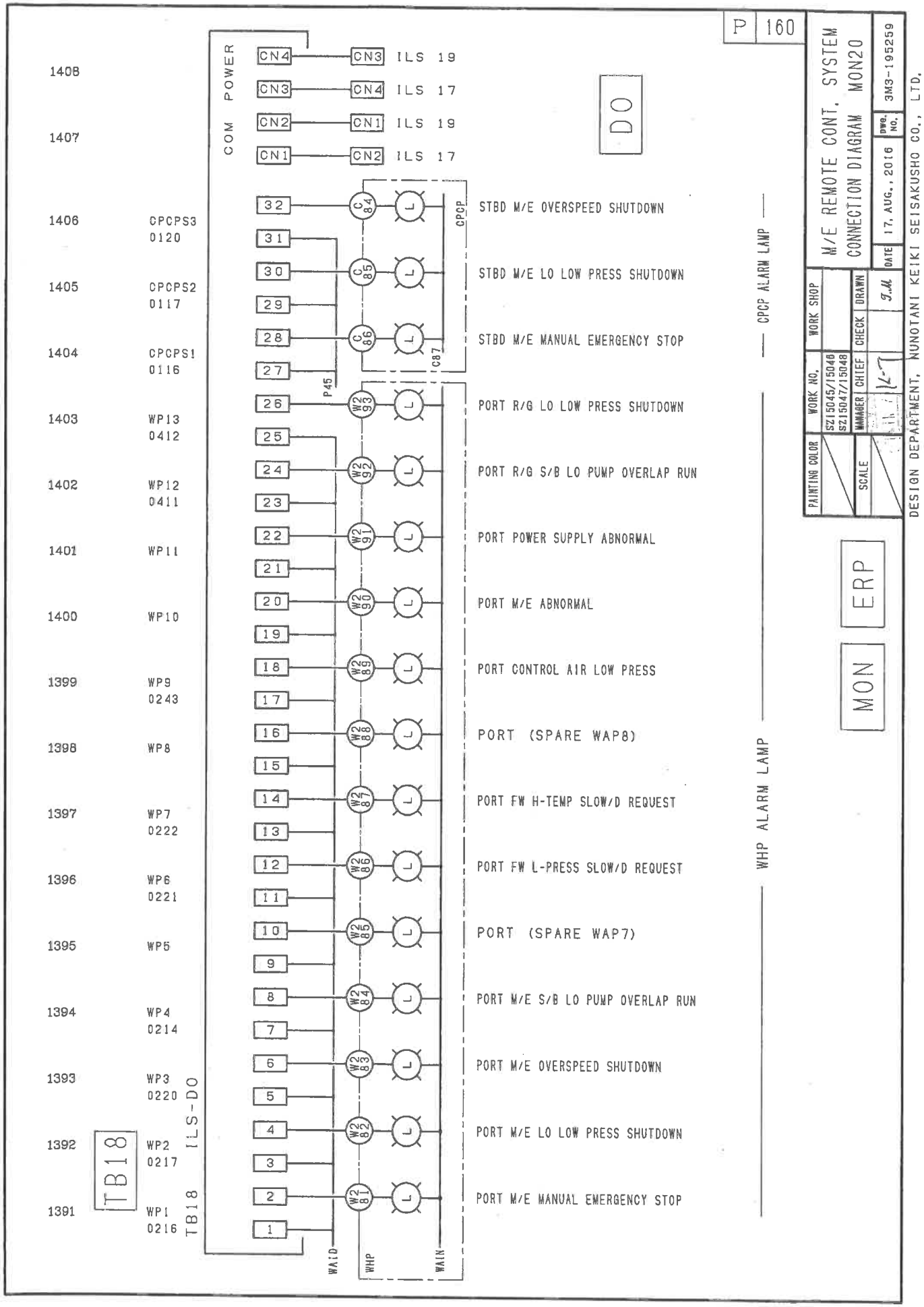


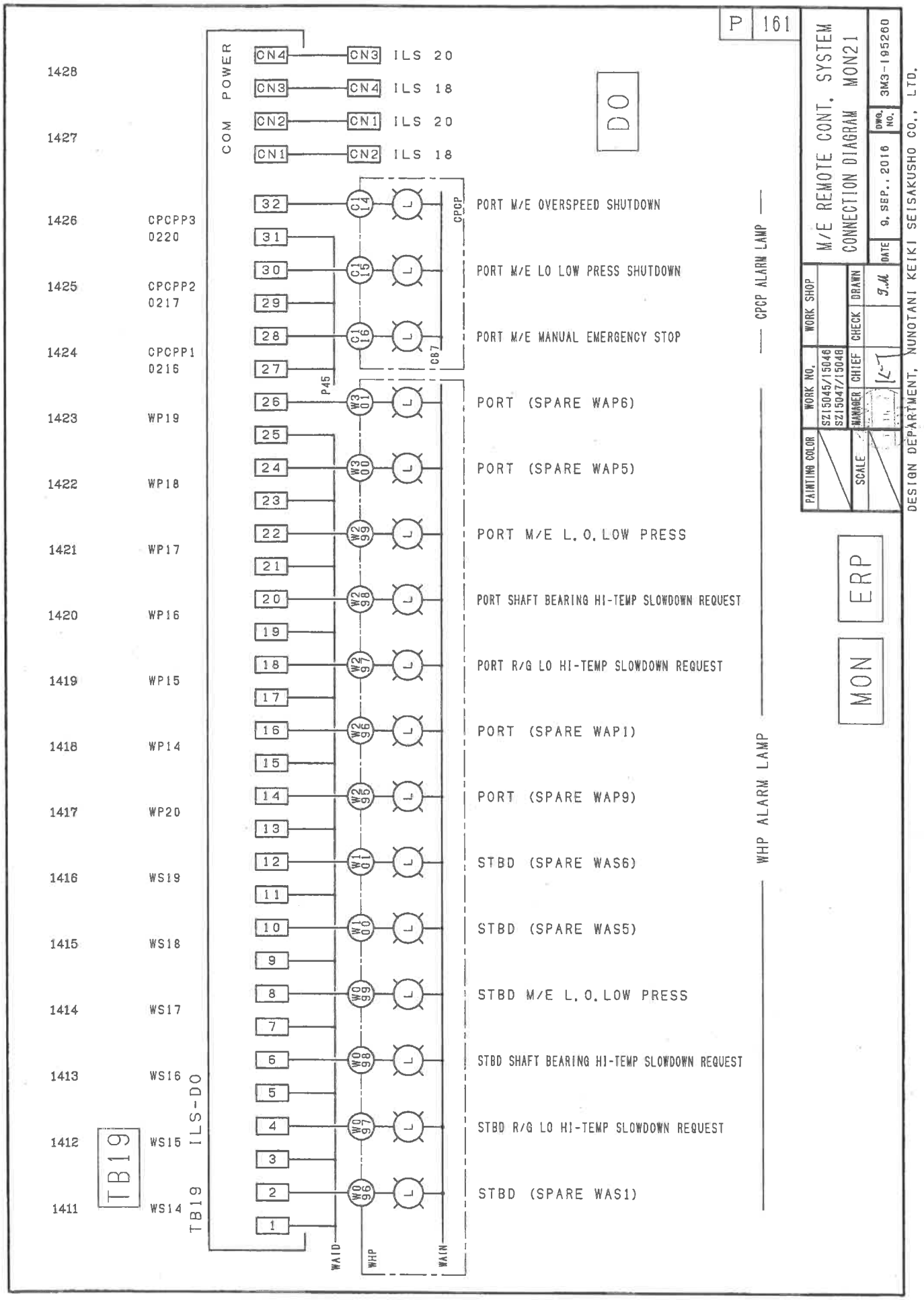
P 158

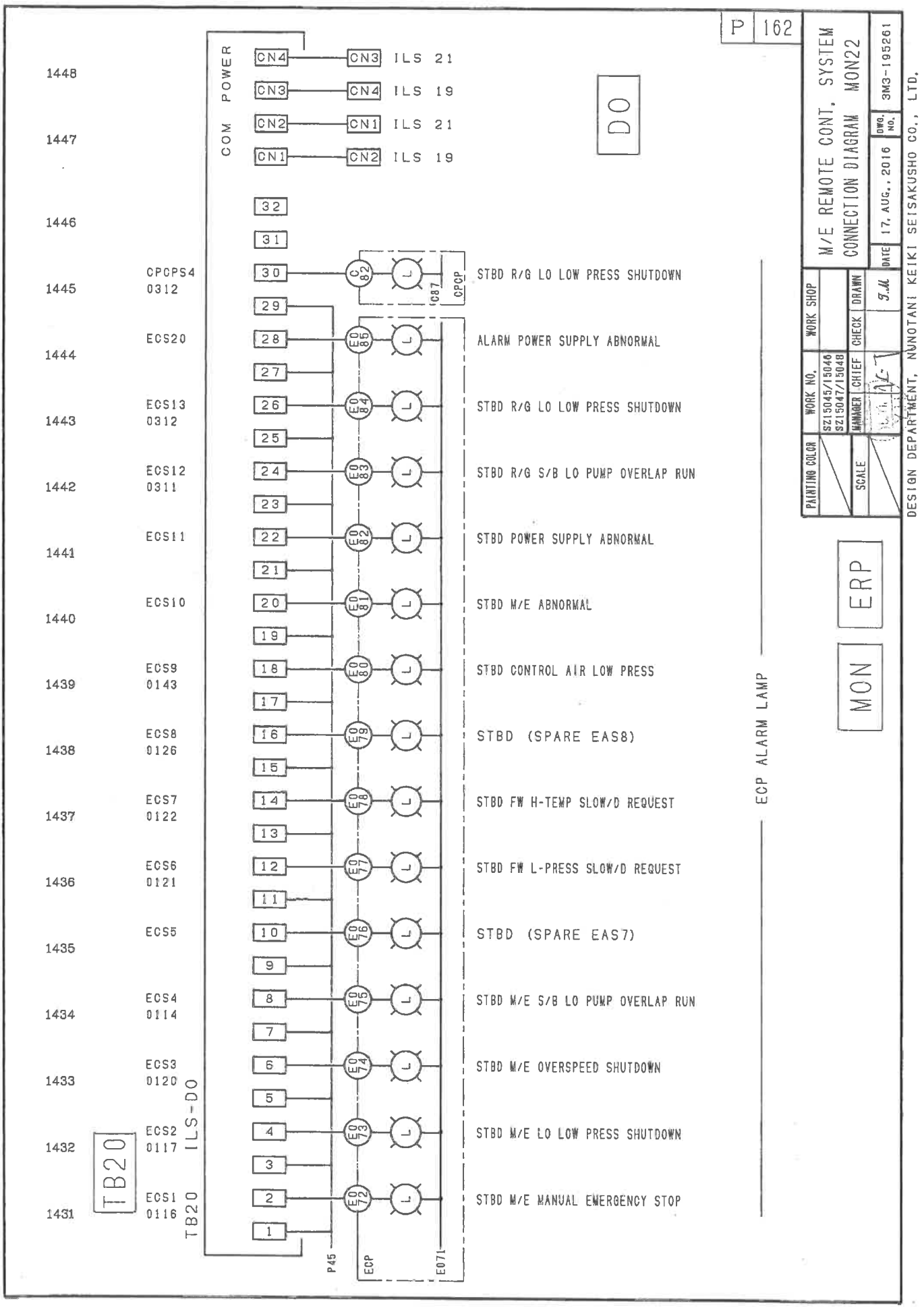
PAINTING COLOR	WORK NO.	WORK SHOP	M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON18		
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER	CHIEF	CHECK	DRAWN	
	16.11. YH			T.M	DATE 8. AUG., 2016
					DWG. NO. 3M3-195257

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.


P 159
M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON19
DWG. NO. 3M3-19525B
DATE 18. AUG., 2016
WORK SHOP
DRAWN
CHECK
CHIEF
MANAGER
WORK NO. SZ15045/15048 SZ15047/15048
PAINTING COLOR
SCALE
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
DO
ERP
MON
WHP ALARM LAMP
COM POWER
CN4 CN3 ILS 18
CN3 CN4 ILS 16
CN2 CN1 ILS 18
CN1 CN2 ILS 16
TB17 ILS-DO
TB17
1388
1387
1386 WP21
1385
1384 WS20
1383 WS13 0312
1382 WS12 0311
1381 WS11
1380 WS10
1379 WS9 0143
1378 WS8
1377 WS7 0122
1376 WS6 0121
1375 WS5
1374 WS4 0114
1373 WS3 0120
1372 WS2 0117
1371 WS1 0116
32 OTHER ABNORMAL
31
30
29
28 ALARM POWER SUPPLY ABNORMAL
27
26 STBD R/G LO LOW PRESS SHUTDOWN
25
24 STBD R/G S/B LO PUMP OVERLAP RUN
23
22 STBD POWER SUPPLY ABNORMAL
21
20 STBD M/E ABNORMAL
19
18 STBD CONTROL AIR LOW PRESS
17
16 STND (SPARE WAS8)
15
14 STBD FW H-TEMP SLOW/D REQUEST
13
12 STBD FW L-PRESS SLOW/D REQUEST
11
10 STBD (SPARE WAS7)
9
8 STBD M/E S/B LO PUMP OVERLAP RUN
7
6 STBD M/E OVERSPEED SHUTDOWN
5
4 STBD M/E LO LOW PRESS SHUTDOWN
3
2 STBD M/E MANUAL EMERGENCY STOP
1
WO 94
WO 93
WO 92
WO 91
WO 90
WO 89
WO 88
WO 87
WO 86
WO 85
WO 84
WO 83
WO 82
WO 81
L
WAID
WHP
WAIN


P 160
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON20
DWG. NO. 3M3-195259
DATE 17. AUG., 2016
WORK SHOP
DRAWN
CHECK
CHIEF
MANAGER
WORK NO. SZ15045/15046 SZ15047/15048
SCALE
PAINTING COLOR
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
MON
ERP
DO
COM POWER
CN4 CN3 ILS 19
CN3 CN4 ILS 17
CN2 CN1 ILS 19
CN1 CN2 ILS 17
TB18 ILS-DO
TB18
1408
1407
1406 CPCPS3 0120
1405 CPCPS2 0117
1404 CPCPS1 0116
1403 WP13 0412
1402 WP12 0411
1401 WP11
1400 WP10
1399 WP9 0243
1398 WP8
1397 WP7 0222
1396 WP6 0221
1395 WP5
1394 WP4 0214
1393 WP3 0220
1392 WP2 0217
1391 WP1 0216
32 C84 L STBD M/E OVERSPEED SHUTDOWN
31
30 C85 L STBD M/E LO LOW PRESS SHUTDOWN
29
28 C86 L STBD M/E MANUAL EMERGENCY STOP
27
26 W2 93 L PORT R/G LO LOW PRESS SHUTDOWN
25
24 W2 92 L PORT R/G S/B LO PUMP OVERLAP RUN
23
22 W2 91 L PORT POWER SUPPLY ABNORMAL
21
20 W2 90 L PORT M/E ABNORMAL
19
18 W2 89 L PORT CONTROL AIR LOW PRESS
17
16 W2 88 L PORT (SPARE WAP8)
15
14 W2 87 L PORT FW H-TEMP SLOW/D REQUEST
13
12 W2 86 L PORT FW L-PRESS SLOW/D REQUEST
11
10 W2 85 L PORT (SPARE WAP7)
9
8 W2 84 L PORT M/E S/B LO PUMP OVERLAP RUN
7
6 W2 83 L PORT M/E OVERSPEED SHUTDOWN
5
4 W2 82 L PORT M/E LO LOW PRESS SHUTDOWN
3
2 W2 81 L PORT M/E MANUAL EMERGENCY STOP
1
P45
C87
CPCP
CPCP ALARM LAMP
WHP ALARM LAMP
WAID
WHP
WAIN


P 161
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON21
DWG. NO. 3M3-195260
DATE 9. SEP.. 2016
WORK SHOP
DRAWN
CHECK
CHIEF
MANAGER
WORK NO. SZ15045/15046 SZ15047/15048
SCALE
PAINTING COLOR
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
DO
ERP
MON
COM POWER
CN4 CN3 ILS 20
CN3 CN4 ILS 18
CN2 CN1 ILS 20
CN1 CN2 ILS 18
TB19 ILS-DO
TB19
1428
1427
1426 CPCPP3 0220
1425 CPCPP2 0217
1424 CPCPP1 0216
1423 WP19
1422 WP18
1421 WP17
1420 WP16
1419 WP15
1418 WP14
1417 WP20
1416 WS19
1415 WS18
1414 WS17
1413 WS16
1412 WS15
1411 WS14
32 C1 I4
31
30 C1 I5
29
28 C1 I6
27
26 W3 01
25
24 W3 00
23
22 W2 99
21
20 W2 98
19
18 W2 97
17
16 W2 96
15
14 W2 95
13
12 W1 01
11
10 W1 00
9
8 W0 99
7
6 W0 98
5
4 W0 97
3
2 W0 96
1
CPCP
CB7
P45
WAID
WHP
WAIN
PORT M/E OVERSPEED SHUTDOWN
PORT M/E LO LOW PRESS SHUTDOWN
PORT M/E MANUAL EMERGENCY STOP
PORT (SPARE WAP6)
PORT (SPARE WAP5)
PORT M/E L.O. LOW PRESS
PORT SHAFT BEARING HI-TEMP SLOWDOWN REQUEST
PORT R/G LO HI-TEMP SLOWDOWN REQUEST
PORT (SPARE WAP1)
PORT (SPARE WAP9)
STBD (SPARE WAS6)
STBD (SPARE WAS5)
STBD M/E L.O. LOW PRESS
STBD SHAFT BEARING HI-TEMP SLOWDOWN REQUEST
STBD R/G LO HI-TEMP SLOWDOWN REQUEST
STBD (SPARE WAS1)
CPCP ALARM LAMP
WHP ALARM LAMP


P 162
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON22
DWG. NO. 3M3-195261
DATE 17. AUG., 2016
WORK SHOP
DRAWN
CHECK
CHIEF
MANAGER
SCALE
PAINTING COLOR
WORK NO. SZ15045/15046 SZ15047/15048
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
DO
ERP
MON
COM POWER
CN4 CN3 ILS 21
CN3 CN4 ILS 19
CN2 CN1 ILS 21
CN1 CN2 ILS 19
TB20 ILS-DO
TB20
CPCPS4 0312
ECS20
ECS13 0312
ECS12 0311
ECS11
ECS10
ECS9 0143
ECS8 0126
ECS7 0122
ECS6 0121
ECS5
ECS4 0114
ECS3 0120
ECS2 0117
ECS1 0116
1448
1447
1446
1445
1444
1443
1442
1441
1440
1439
1438
1437
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21
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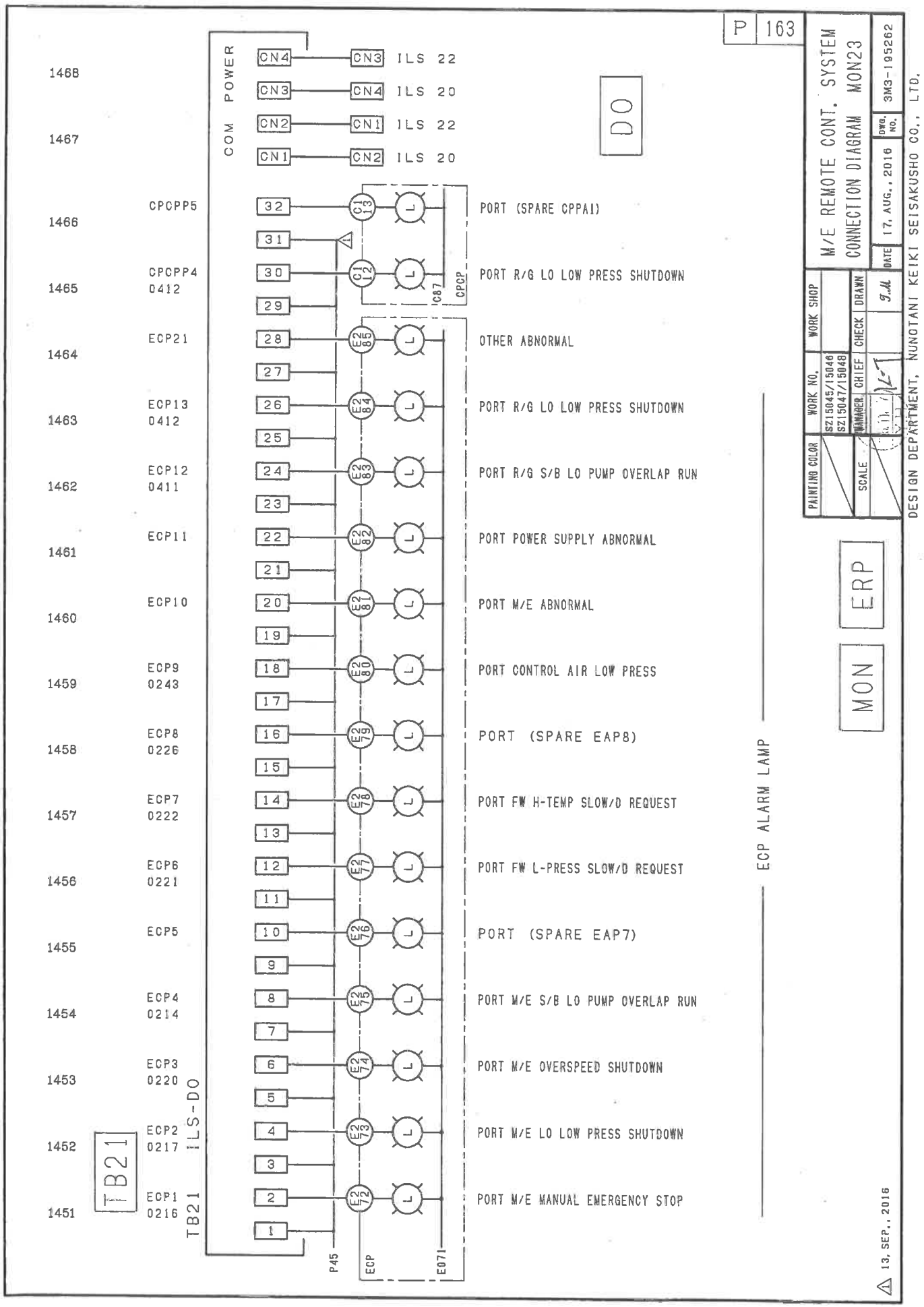
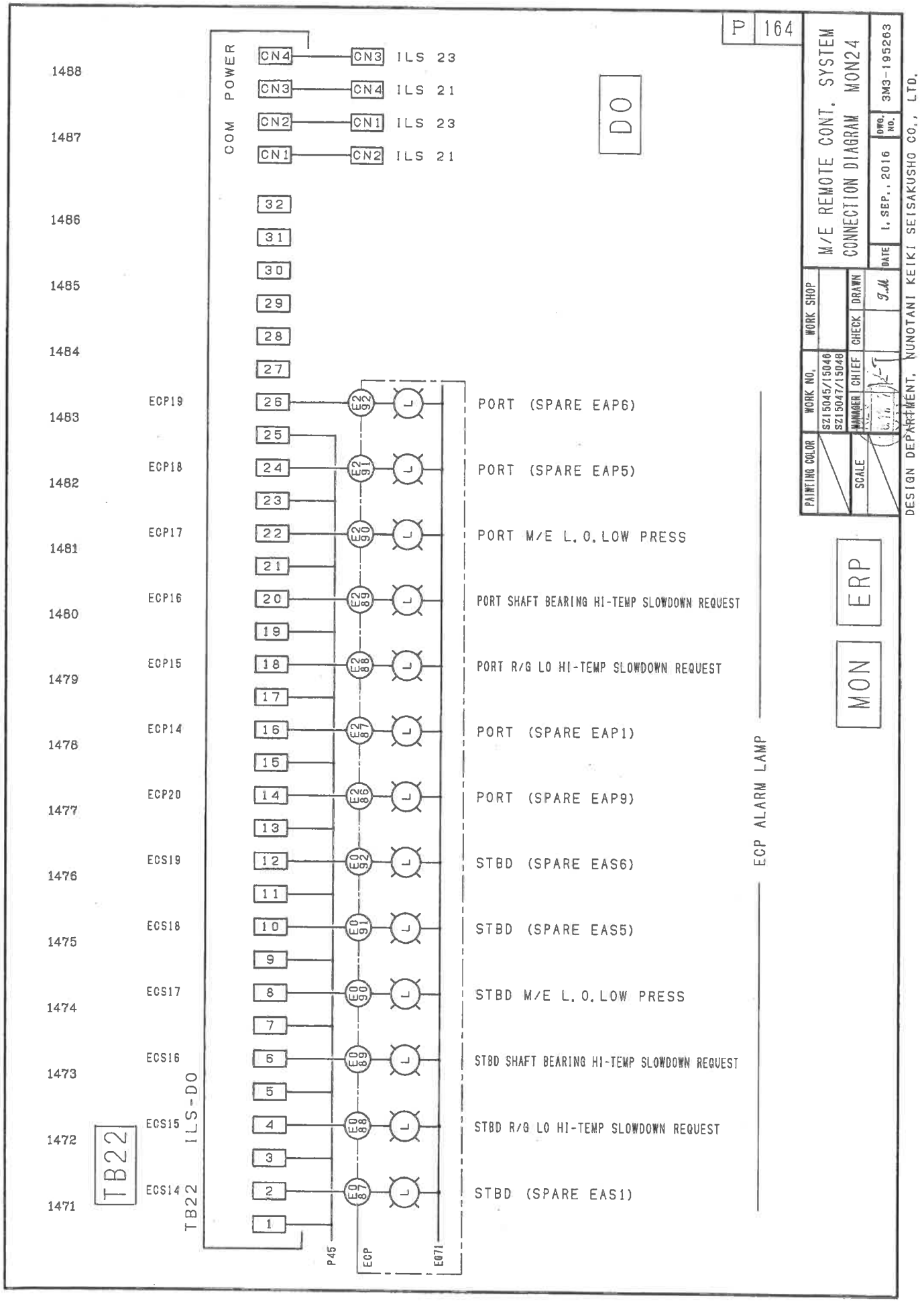
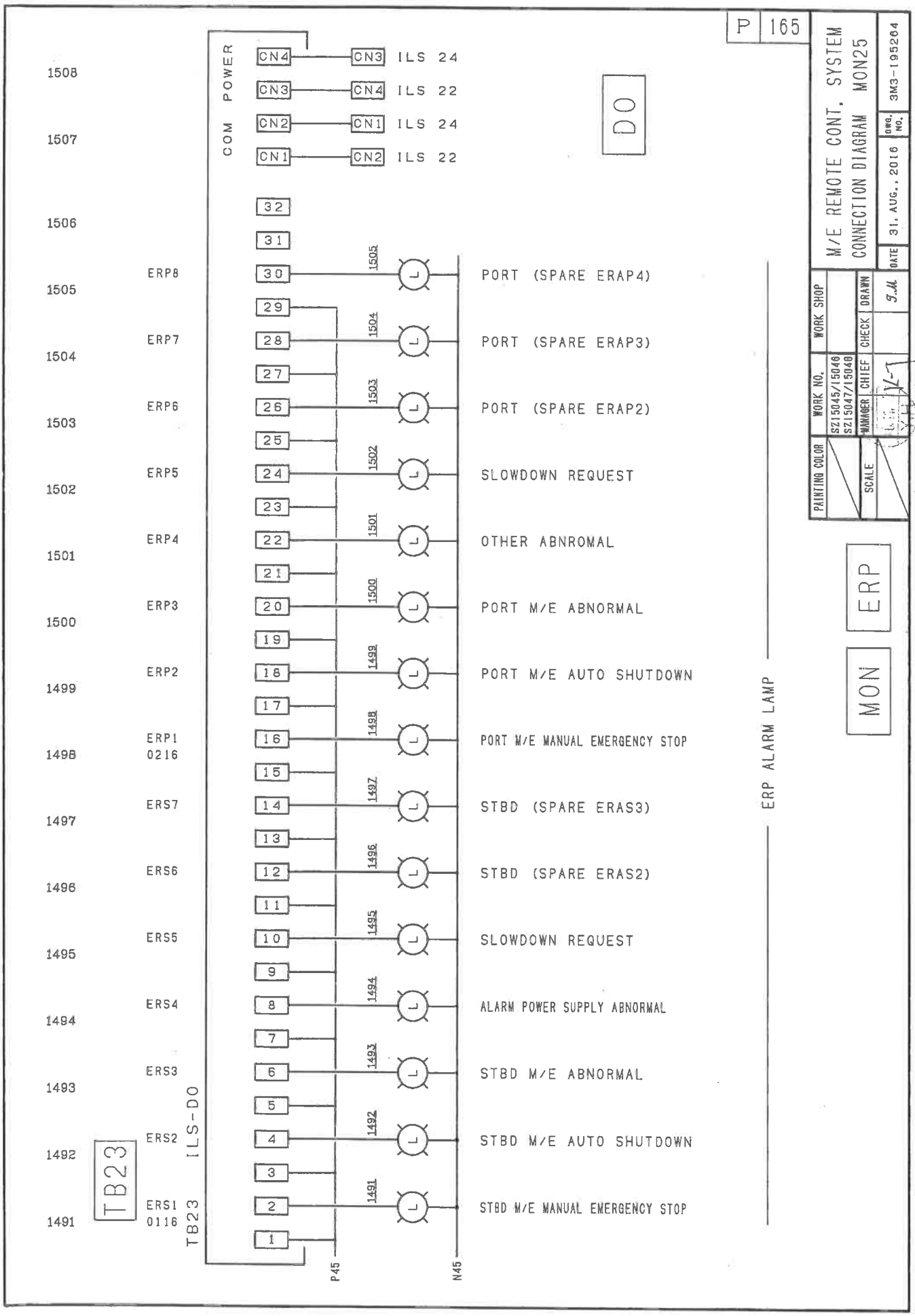
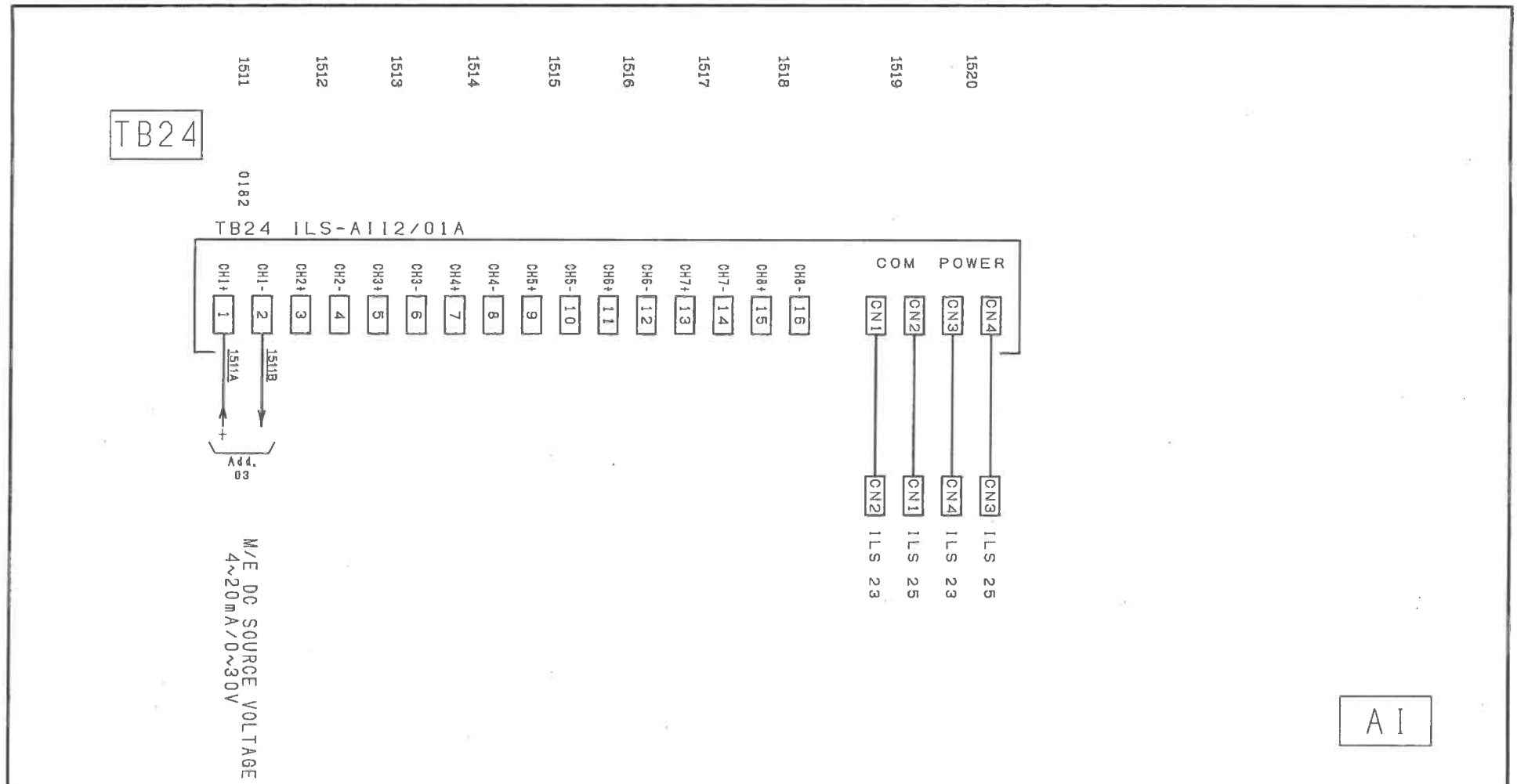
19
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
C82
C87
CPCP
E085
E084
E083
E082
E081
E080
E079
E078
E077
E076
E075
E074
E073
E072
L
P45
ECP
E071
STBD R/G LO LOW PRESS SHUTDOWN
ALARM POWER SUPPLY ABNORMAL
STBD R/G LO LOW PRESS SHUTDOWN
STBD R/G S/B LO PUMP OVERLAP RUN
STBD POWER SUPPLY ABNORMAL
STBD M/E ABNORMAL
STBD CONTROL AIR LOW PRESS
STBD (SPARE EAS8)
STBD FW H-TEMP SLOW/D REQUEST
STBD FW L-PRESS SLOW/D REQUEST
STBD (SPARE EAS7)
STBD M/E S/B LO PUMP OVERLAP RUN
STBD M/E OVERSPEED SHUTDOWN
STBD M/E LO LOW PRESS SHUTDOWN
STBD M/E MANUAL EMERGENCY STOP
ECP ALARM LAMP


P 163
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON23
DWG. NO. 3M3-195262
DATE 17. AUG., 2016
WORK SHOP
WORK NO. SZ15045/15046 SZ15047/15048
MANAGER CHIEF CHECK DRAWN
SCALE
PAINTING COLOR
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
13. SEP., 2016
MON
ERP
DO
COM POWER
CN4 CN3 ILS 22
CN3 CN4 ILS 20
CN2 CN1 ILS 22
CN1 CN2 ILS 20
TB21 ILS-DO
TB21
CPCPP5 0412
CPCPP4 0412
ECP21
ECP13 0412
ECP12 0411
ECP11
ECP10
ECP9 0243
ECP8 0226
ECP7 0222
ECP6 0221
ECP5
ECP4 0214
ECP3 0220
ECP2 0217
ECP1 0216
1468
1467
1466
1465
1464
1463
1462
1461
1460
1459
1458
1457
1456
1455
1454
1453
1452
1451
32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
C1 13
C1 12
E2 85
E2 84
E2 83
E2 82
E2 81
E2 80
E2 79
E2 78
E2 77
E2 76
E2 75
E2 74
E2 73
E2 72
L
C87
CPCP
P45
ECP
E071
PORT (SPARE CPPA1)
PORT R/G LO LOW PRESS SHUTDOWN
OTHER ABNORMAL
PORT R/G LO LOW PRESS SHUTDOWN
PORT R/G S/B LO PUMP OVERLAP RUN
PORT POWER SUPPLY ABNORMAL
PORT M/E ABNORMAL
PORT CONTROL AIR LOW PRESS
PORT (SPARE EAP8)
PORT FW H-TEMP SLOW/D REQUEST
PORT FW L-PRESS SLOW/D REQUEST
PORT (SPARE EAP7)
PORT M/E S/B LO PUMP OVERLAP RUN
PORT M/E OVERSPEED SHUTDOWN
PORT M/E LO LOW PRESS SHUTDOWN
PORT M/E MANUAL EMERGENCY STOP
ECP ALARM LAMP


P 164
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON24
DWG. NO. 3M3-195263
DATE 1. SEP. . 2016
WORK SHOP
CHECK
DRAWN
WORK NO. SZ15045/15046 SZ15047/15048
CHIEF
MANAGER
SCALE
PAINTING COLOR
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
DO
ERP
MON
COM POWER
CN4 CN3 ILS 23
CN3 CN4 ILS 21
CN2 CN1 ILS 23
CN1 CN2 ILS 21
1488
1487
1486
1485
1484
1483
1482
1481
1480
1479
1478
1477
1476
1475
1474
1473
1472
1471
32
31
30
29
28
27
26
25
24
23
22
21
20
19
18
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1
ECP19
ECP18
ECP17
ECP16
ECP15
ECP14
ECP20
ECS19
ECS18
ECS17
ECS16
ECS15
ECS14
E2 92
E2 91
E2 90
E2 89
E2 88
E2 87
E2 86
E0 92
E0 91
E0 90
E0 89
E0 88
E0 87
L
PORT (SPARE EAP6)
PORT (SPARE EAP5)
PORT M/E L.O. LOW PRESS
PORT SHAFT BEARING HI-TEMP SLOWDOWN REQUEST
PORT R/G LO HI-TEMP SLOWDOWN REQUEST
PORT (SPARE EAP1)
PORT (SPARE EAP9)
STBD (SPARE EAS6)
STBD (SPARE EAS5)
STBD M/E L.O. LOW PRESS
STBD SHAFT BEARING HI-TEMP SLOWDOWN REQUEST
STBD R/G LO HI-TEMP SLOWDOWN REQUEST
STBD (SPARE EAS1)
ECP ALARM LAMP
TB22
TB22 ILS-DO
P45
ECP
E071


P 165
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON25
DWG. NO. 3M3-195264
DATE 31. AUG.. 2016
WORK SHOP
WORK NO. SZ15045/15046 SZ15047/15048
PAINTING COLOR
SCALE
MANAGER CHIEF CHECK DRAWN
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
DO
ERP
MON
COM POWER
CN4 CN3 ILS 24
CN3 CN4 ILS 22
CN2 CN1 ILS 24
CN1 CN2 ILS 22
TB23 ILS-DO
TB23
P45
N45
ERP ALARM LAMP
ERP8 PORT (SPARE ERAP4)
ERP7 PORT (SPARE ERAP3)
ERP6 PORT (SPARE ERAP2)
ERP5 SLOWDOWN REQUEST
ERP4 OTHER ABNROMAL
ERP3 PORT M/E ABNORMAL
ERP2 PORT M/E AUTO SHUTDOWN
ERP1 0216 PORT M/E MANUAL EMERGENCY STOP
ERS7 STBD (SPARE ERAS3)
ERS6 STBD (SPARE ERAS2)
ERS5 SLOWDOWN REQUEST
ERS4 ALARM POWER SUPPLY ABNORMAL
ERS3 STBD M/E ABNORMAL
ERS2 STBD M/E AUTO SHUTDOWN
ERS1 0116 STBD M/E MANUAL EMERGENCY STOP
1508
1507
1506
1505
1504
1503
1502
1501
1500
1499
1498
1497
1496
1495
1494
1493
1492
1491



P 166

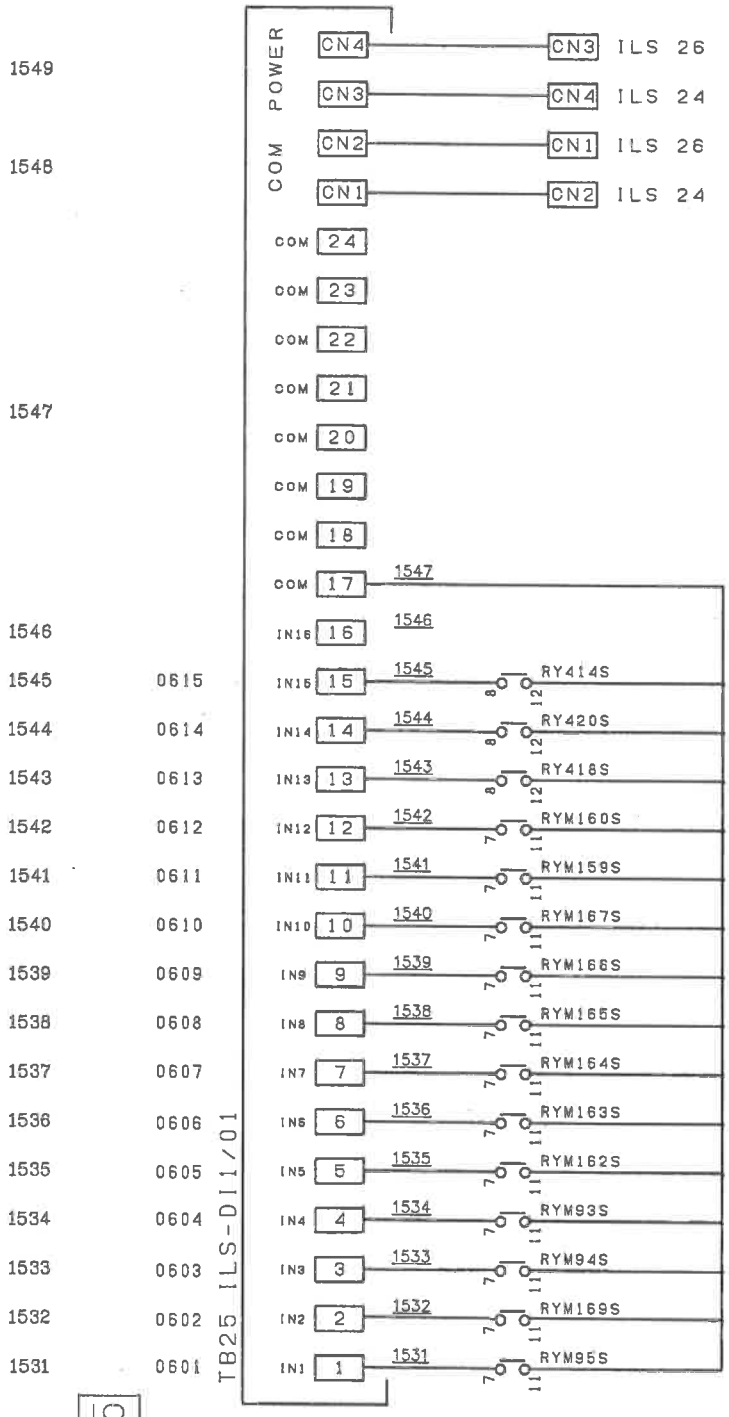
PAINTING COLOR	WORK NO.	WORK SHOP		
	SZ15045/15046 SZ15047/15048			
SCALE	MANAGER / CHIEF	CHECK	DRAWN	
	16.11.		T.M	

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON26

DATE 3. AUG., 2016 | DWG. NO. 3M3-195265

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

DI



1549		
1548		
1547		
1546		
1545	0615	STBD CLUTCH CONTROL PROPELLER PULSE SENSOR ABNORMAL
1544	0614	STBD P.V. LINE BREAK
1543	0613	STBD P.V. UNIT ABNORMAL
1542	0612	STBD PROPELLER REVOLUTION SIGNAL FAIL
1541	0611	STBD M/E REVOLUTION SIGNAL FAIL
1540	0610	STBD CP DIAL POTENSIO LINE BREAK
1539	0609	STBD CPCP DIAL POTENSIO LINE BREAK
1538	0608	STBD ECP HANDLE POTENSIO ABNORMAL
1537	0607	STBD ECP HANDLE POTENSIO LINE BREAK
1536	0606	STBD WHP HANDLE POTENSIO ABNORMAL
1535	0605	STBD WHP HANDLE POTENSIO LINE BREAK
1534	0604	STBD POTENSIO POWER SUPPLY FAIL
1533	0603	STBD COMMUNICATION ABNORMAL
1532	0602	STBD DIO UNIT ABNORMAL
1531	0601	STBD AIO UNIT ABNORMAL

ERP

MON

M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON27

DWG. NO. 3M3-195266

DATE 17. AUG., 2016

WORK SHOP

DRAWN

CHECK

CHIEF

MANAGER

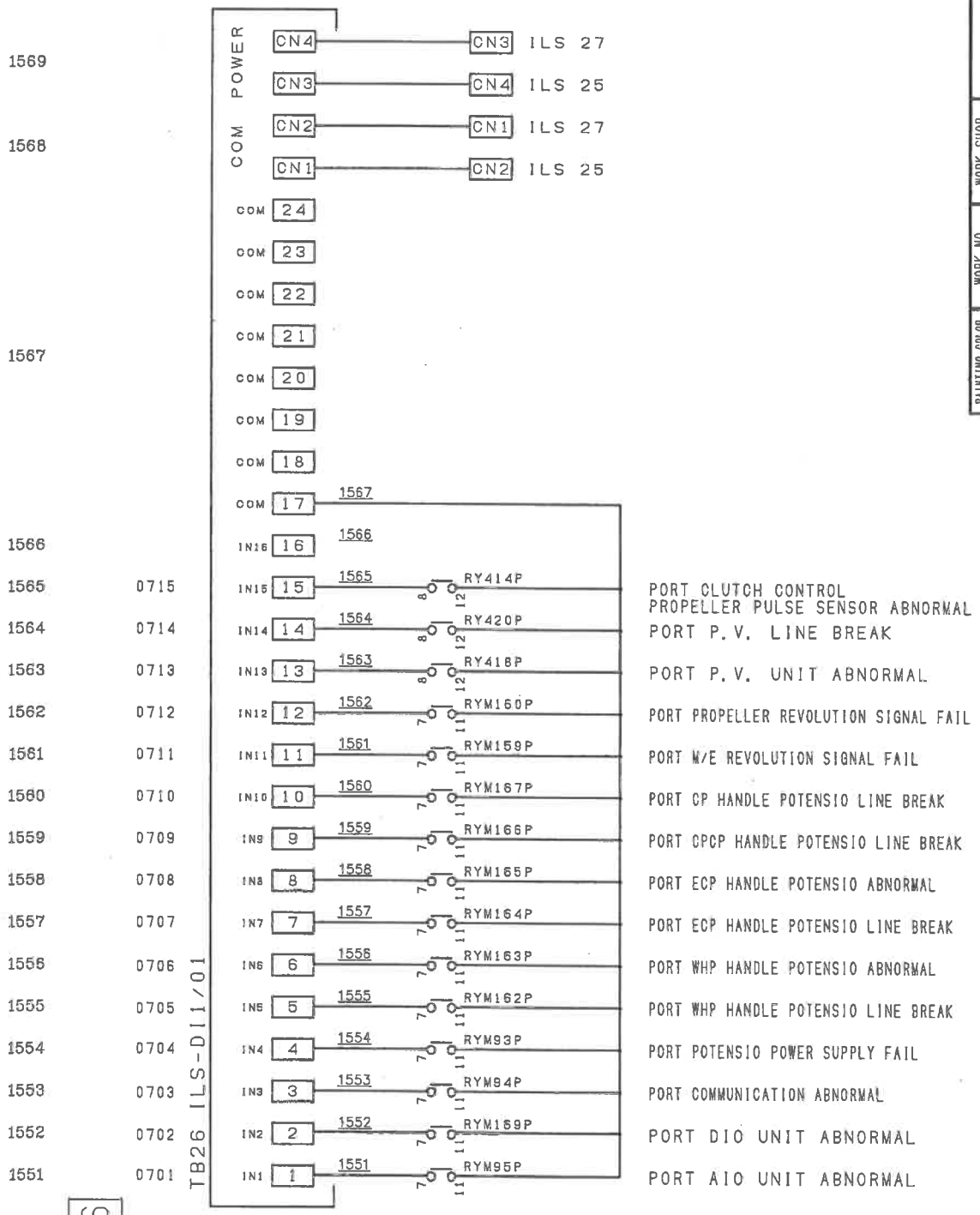
WORK NO. SZ15045/15046 SZ15047/15048

PAINTING COLOR

SCALE

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.

DI



TB26

ERP

MON

M/E REMOTE CONT. SYSTEM CONNECTION DIAGRAM MON28

DWG. NO. 3M3-195267

DATE 17. AUG.. 2016

WORK SHOP

DRAWN J.M

CHECK

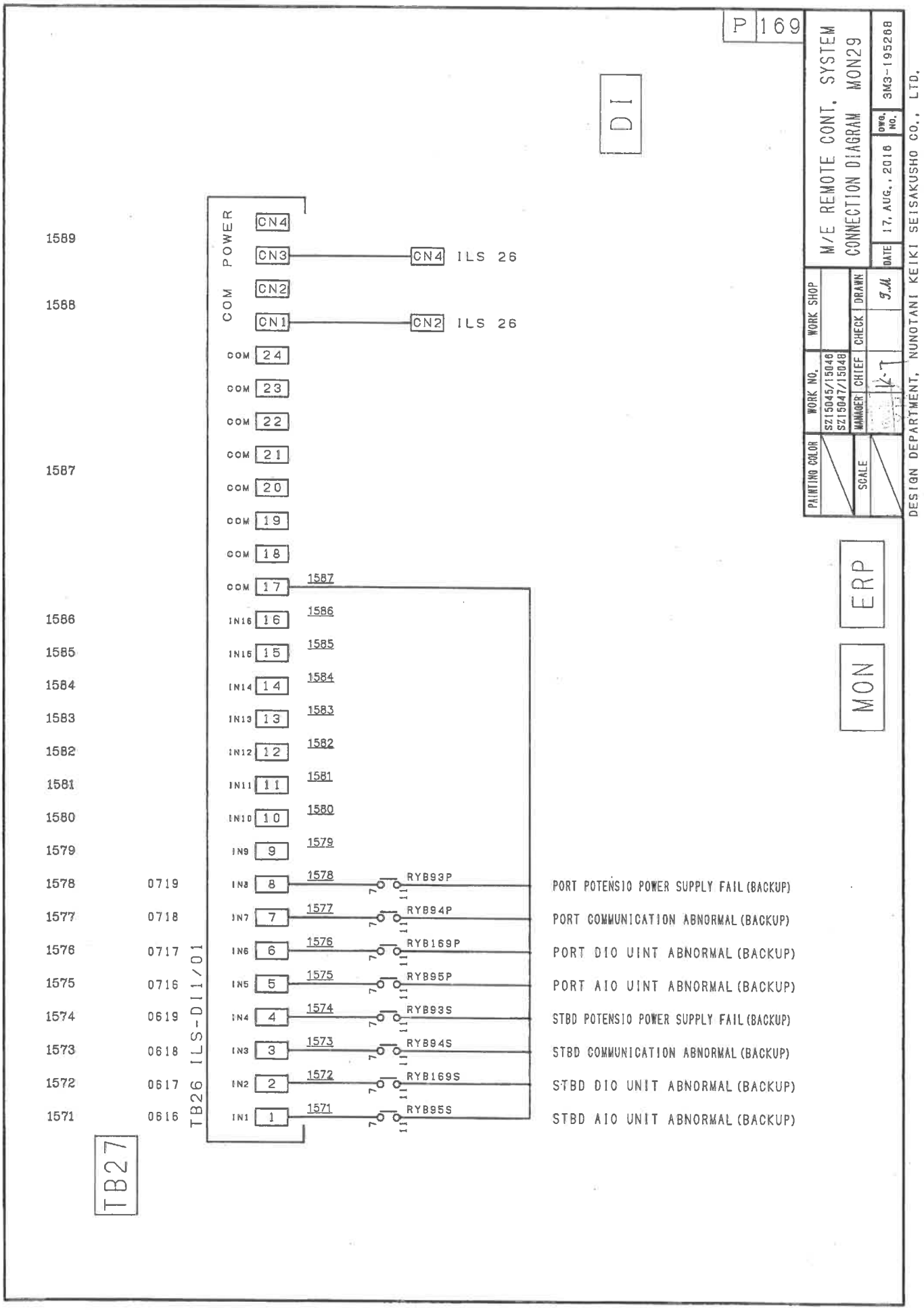
CHIEF MANAGER

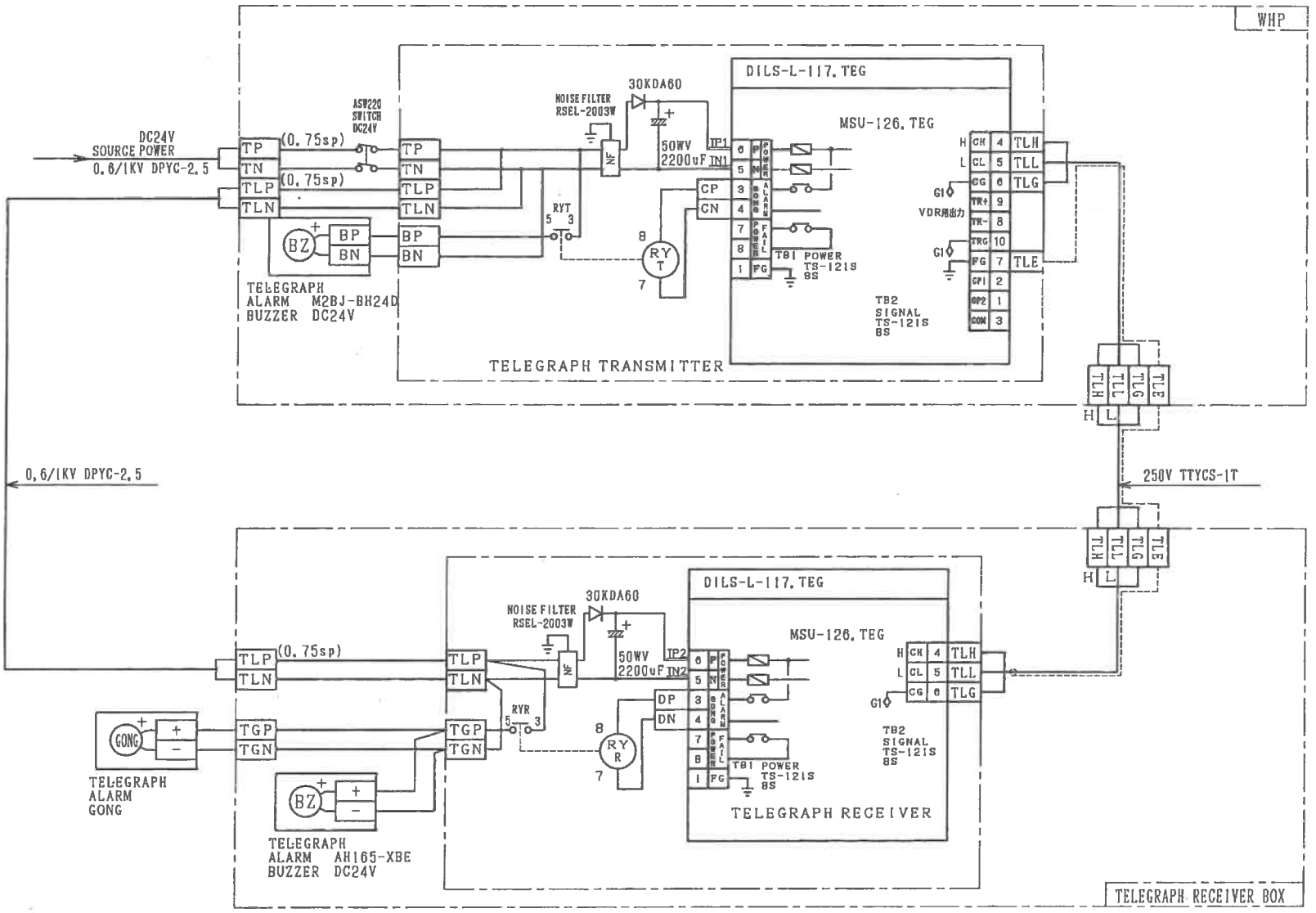
WORK NO. SZ15045/15046 SZ15047/15048

SCALE

PAINTING COLOR

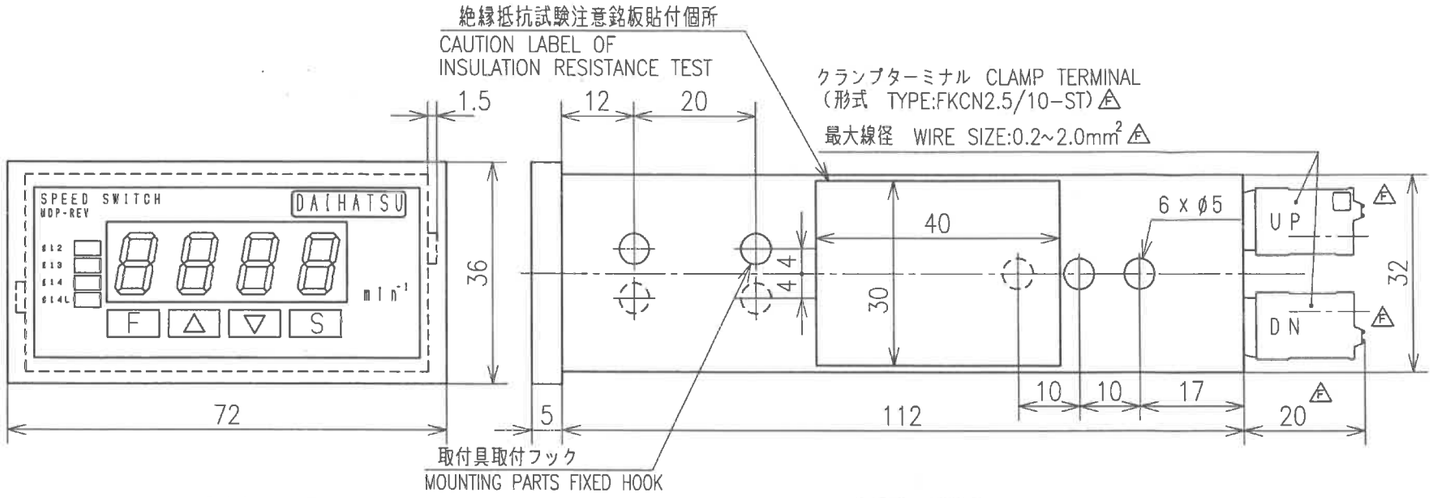
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.


P 169
M/E REMOTE CONT. SYSTEM
CONNECTION DIAGRAM MON29
DATE 17, AUG., 2016
DWG. NO. 3M3-195268
WORK SHOP
WORK NO. SZ15045/15046 SZ15047/15048
CHIEF MANAGER
CHECK
DRAWN
PAINTING COLOR
SCALE
DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.
DI
ERP
MON
TB27
TB26 ILS-D11/01
POWER
COM
CN4
CN3
CN2
CN1
CN4 ILS 26
CN2 ILS 26
1589
1588
1587
1586
1585
1584
1583
1582
1581
1580
1579
1578
1577
1576
1575
1574
1573
1572
1571
0719
0718
0717
0716
0619
0618
0617
0616
RYB93P
RYB94P
RYB169P
RYB95P
RYB93S
RYB94S
RYB169S
RYB95S
PORT POTENSIO POWER SUPPLY FAIL (BACKUP)
PORT COMMUNICATION ABNORMAL (BACKUP)
PORT DIO UINT ABNORMAL (BACKUP)
PORT AIO UINT ABNORMAL (BACKUP)
STBD POTENSIO POWER SUPPLY FAIL (BACKUP)
STBD COMMUNICATION ABNORMAL (BACKUP)
STBD DIO UNIT ABNORMAL (BACKUP)
STBD AIO UNIT ABNORMAL (BACKUP)



PAINTING COLOR	WORK NO.	WORK SHOP	エンジンテレグラフ結線図 CONNECTION DIAGRAM OF ENGINE TELEGRAPH		
	SZ15045/15046 SZ15047/15048				
SCALE	MANAGER / CHIEF	CHECK / DRAWN	DATE	DWG. NO.	
		Y.O	30-AUG.-2016		3M3-195116

DESIGN DEPARTMENT, NUNOTANI KEIKI SEISAKUSHO CO., LTD.



入出力コネクタ
I/O CONNECTOR

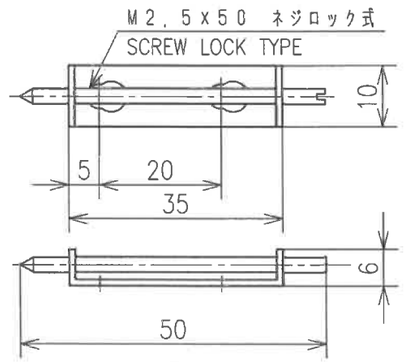
	24Vdc		SENSOR			FREQUENCY OUTPUT		ANALOG OUTPUT (4~20mA)	
FG	P	N	+	S	−	F+	F−	M+	M−
Sb	Sc	12a	12c	13a	13c	14a	14c	14L a	14L c
CPU TRB		OVER		RATED		LOW		LOWLOW	

OUTPUT CONTACT

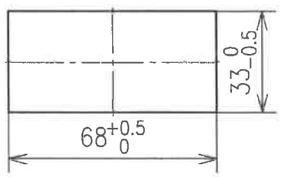
上側コネクタ UPPER CONNECTOR
下側コネクタ LOWER CONNECTOR

端子番号 TERMINAL NUMBER	用途 OBJECT	
FG	:フレームグランド	FRAME GRAND
P	:DC24V	
N	:0V	
+	:DC24V	PULSE INPUT (センサ SENSOR)
S	:パルス入力	
−	:COM	
F+	:周波数出力	FREQUENCY OUTPUT
F−	:COM	
M+	:電流出力	CURRENT OUTPUT (オプション OPTION)
M−	:COM	
Sb	:CPU TRB	CPU TROUBLE DETECTION CONTACT OUTPUT
Sc	:	
12a	:#12検知接点	#12 DETECTION CONTACT OUTPUT
12c	:	
13a	:#13検知接点	#13 DETECTION CONTACT OUTPUT
13c	:	
14a	:#14検知接点	#14 DETECTION CONTACT OUTPUT
14c	:	
14La	:#14Lタイマー接点	#14L TIMMER CONTACT OUTPUT
14Lc	:(※使用にあたっては「マニュアル」参照のこと REFER TO THE MANUAL.)	

取付具(付属品)
MOUNTING PART(ACCESSORIES)



パネルカット
PANEL CUT(1:2)



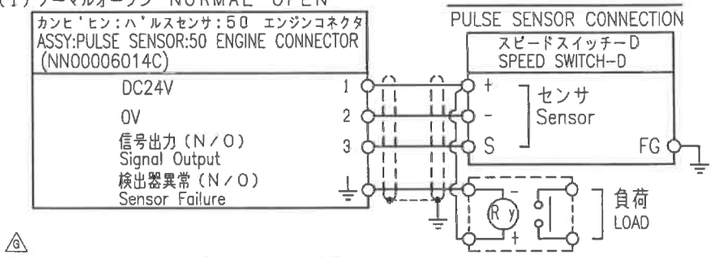
ALTERATION FROM LAST DRAWING
出力信号追加
・CPU TROUBLE追加(端子台:Sb、Sc)
(E008316)

電源	POWER VOLTAGE	DC24V(±25%)リップル10%以下	24VDC(±25%)RIPPLE 10% OR LESS	
消費電力	POWER CONSUMPTION	最大3.2W	3.2W MAX.	
計測精度	MEASURING ACCURACY	0.05%±1digit atF.S. ±0.5% atF.S.(アナログ出力)	0.05% 1digit atF.S. ±0.5% atF.S.(analog output)	
計測範囲	MEASURING RANGE	2Hz~7kHz/0~9990min⁻¹	2Hzto7kHZ/0to9990min-1/	
入力 INPUT		パルスセンサ出力信号 (ロー:8V以下 ハイ:15V以上)	PULSE SENSOR OUTPUT SIGNAL (LOW:under 8V HI.:over 15V	
出力 OUTPUT	アナログ出力 ANALOG	4~20mA (負荷抵抗最大500Ω)	4to20mA (load registance500Ω MAX.)	
	接点出力 CONTACT	AC・DC100V 0.1A (無電圧半導体接点)	100VAC/VDC 0.1A (voltage free contact)	
	周波数出力 FREQUENCY	1Hz/1回転(60Hz以下は除く) 0~15V(負荷抵抗5kΩ以上)	1Hz/1ROTATION(without under 60Hz) 0to15V(load resistance5kΩover)	
表示 DISPLAY	デジタル表示 DIGITAL DISPLAY	7セグメントLED(赤)x4桁(輝度調整有り)	7SEGMENTS LEDx4DIGITS(WITH DIMMER)	
	LED表示 LED LAMP	#12(赤), #13(緑) #14(緑), #14L(緑)	#12(RED),#13(GREEN) #14(GREEN),#14L(GREEN)	
絶縁抵抗	INSULATION RESISTANCE	導体部-箱体間 20MΩ以上 DC50V	CONDUCTOR-BODY OVER20MΩ 50VDC	
耐電圧	DIELECTRIC STRENGTH	接点出力-箱体間 AC1500V 1分間	CONTACT OUTPUT-BODY 1500VAC 1min.	
使用温度	OPERATING TEMP.	+5~+55℃	+5to+55℃	
保存温度	STORAGE TEMP.	-20~+65℃	-20to+65℃	
相対湿度	RELATIVE HUMIDITY	95%以内(但し、結露がないこと)	WITHIN 95%(NON CONDENSATION)	
雰囲気	ATMOSPHERE	腐食性ガスがないこと	WITHOUT CORROSIVE GAS	
振動	VIBRATION RESISTANCE	5~13.2Hz ±1mm, 13.2~100Hz 0.7G		
衝撃	SHOCK RESISTANCE	10G以内	WITHIN 10G	
材質	MATERIALS	グラスファイバー強化樹脂	PLASTIC WITH GLASS FIBER	
塗装色	BODY COLOR	樹脂自然色(黒)	BLACK(PLASTIC NATURAL COLOR)	
重量	WEIGHT	0.2kg		

(1) ノーマルオープン NORMAL OPEN

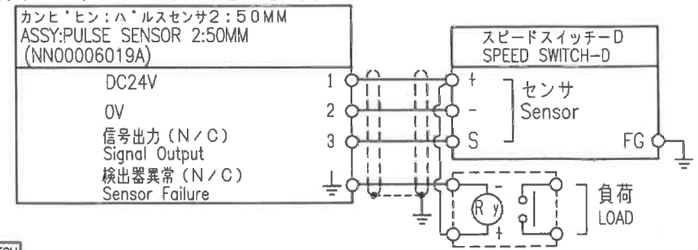
カンヒ゛ヒン:ハ゛ルスセンサ:50 エンジンコネクタ
ASSY:PULSE SENSOR:50 ENGINE CONNECTOR
(NN00006014C)

パルスセンサ接続図
PULSE SENSOR CONNECTION



A (2) ノーマルクローズ NORMAL CLOSE

カンヒ゛ヒン:ハ゛ルスセンサ2:50MM
ASSY:PULSE SENSOR 2:50MM
(NN00006019A)



※パルスセンサの検出異常を利用する時は、リレーを使用して下さい。
リレーはオムロンMY4-D相当品(コイル定格電流30MA以下)のものを選定下さい。

*PLEASE USE A RELAY,
when you use the sensor failure output on the pulse sensor.
A relay should select the " OMRON MY4-D " or equivalent it (30mA or less of relay coil rated current).

G	誤記訂正(ノーマルクローズ追記)他	(E019920)	2017/02/23	NAKAMITSU
F	コネクタ変更Ver._7_	(E018787)	2015/04/24	NAKAMITSU
E	過速度検出動作ソフト改善Ver.7	(E017287)	2015/04/16	資松
D	誤記訂正	(E017065)	2014/04/21	NAKAMITSU
C	内部構成部品変更によるV6仕様出現	(E016145)	2013/10/24	NAKAMITSU
B	内部構成部品変更によるV5仕様出現	(E014744)	2013/02/19	NAKAMITSU
A	FIRST DRAWING	(E008316)	2009/08/25	資松

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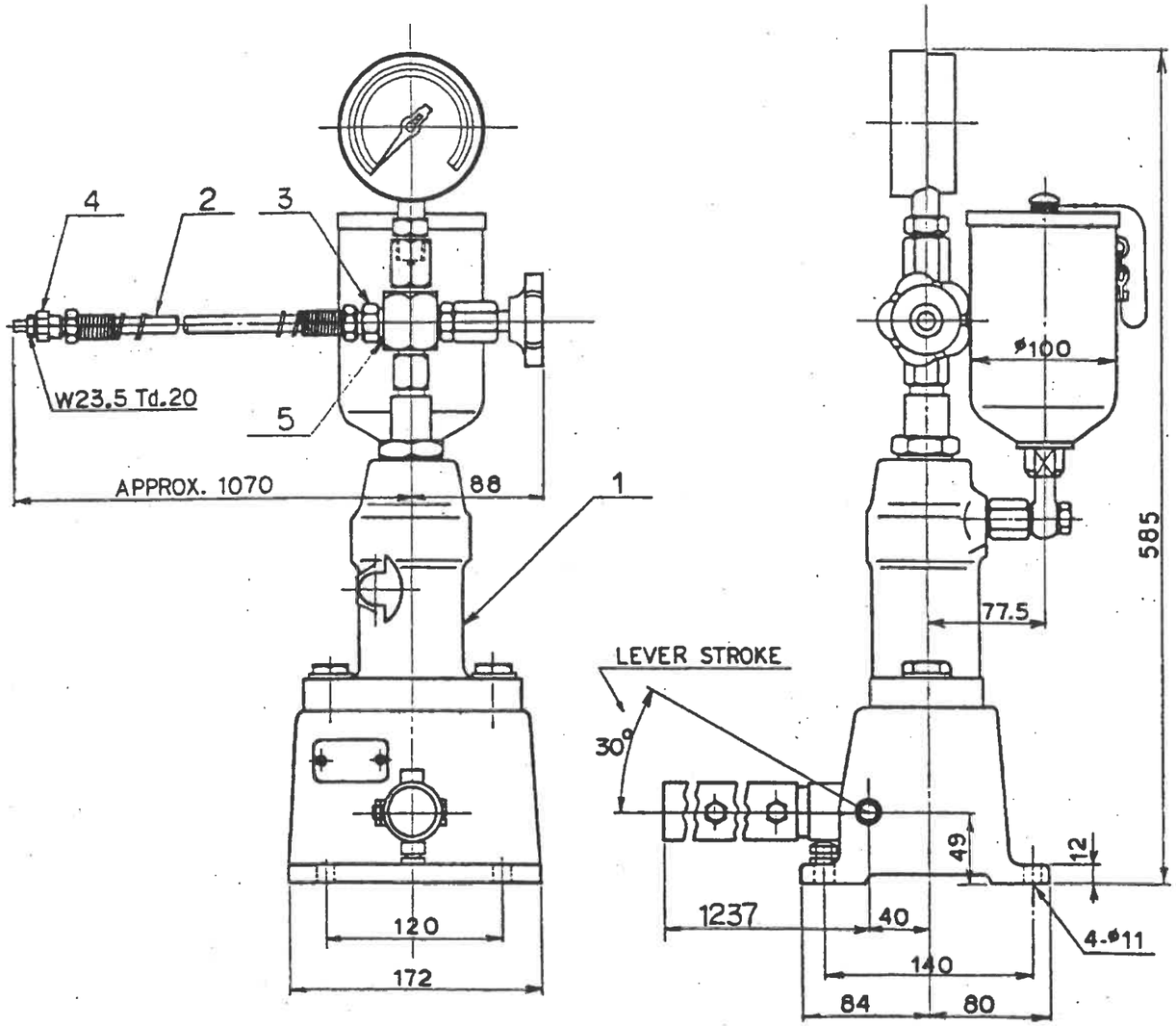
APPROVED	望月 勝	LIST NO.	NN00262	MATERIAL ASSY.	GROUP 9
CHECKED		BEFORE		MASS	QTY.
	望月 勝	REF.		TYPE MES	
DRAWN	中光 俊夫	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN		NAME カンビヒン:ソクドスイッチユニット:-D	
DATE	2017/02/23			ASSY.:SPEED SWITCH UNIT:-D	
				PARTS NO. NN00262001B	G *

SCALE:1:1 A3

2017/05/09 TECHNICAL DEPARTMENT

DRG. NO. 476616 ΣABC

	HISTORY OF DRG.	DATE
A	NO.5 追加	63.2.19 伊藤
B	#001, B改 フランジ φ16 → φ20 レバー長 937 → 1237	MAR.10
C	C091300060 B → C C091390100 A → B (ISC00-09(7))	'97.10.13 H.NISHIKA
D	C091390100 B → C C091300060 C → D (SI変更) (ISC10-0999)	Oct. 1. 99 A. KATO



SPECIFICATION	
PLUNGER DIA.	φ20
CAPACITY	4000 mm³/st
TANK CAPACITY	1000 mℓ
PRESS. GAUGE	A(D)1/2x100x50MPa

NO.	NAME OF PARTS	PARTS NO.	MATERIAL	WEIGHT A C	QTY.
5	GASKET 18	C091300340Z	C1100P-O		1
4	PLUG COUPLING	E339600100Z	PCHS		1
3	JOINT PT3/8 x M18	C091300310Z	S43C		1
2	HIGH PRESS. HOSE 1000	C091300320Z	SPEC		1
1	NOZZLE TEST PUMP (K)	C091300060D	SPEC		1
	NOZZLE TEST PUMP ASSY (K)	C091390100C			O
					§10

APPROVED BY		LIST NO. C0913		GROUP	
CHECKED BY		BEFORE	SCALE 1:4	TYPE DL, DK	
DRAWN BY		AFTER		NAME NOZZLE TEST PUMP ASSY.	
DATE	MAR.10,1988	REF.	DAIHATSU DIESEL MFG.CO.,LTD. OSAKA JAPAN	PARTS NO. C091131-09816Z	

DRG. NO. 476616 ΣABC

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TOOLS LIST
FOR
DIESEL ENGINE

DAIHATSU DIESEL MFG. CO., LTD.

TECHNICAL DEPARTMENT

OSAKA JAPAN

SHIP No.	SZ15045/15046	DRAWN BY	M. Tokumaba
TYPE	6DKM-20eF (L)	CHECKED BY	S. Tajimoto
LIST No.	AQP10008522A		
DATE	2016.06.10	APPROVED BY	S. Tanaka

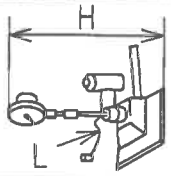
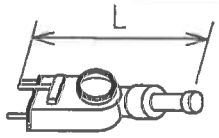
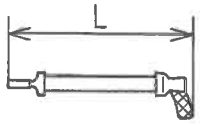
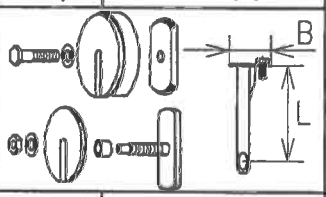
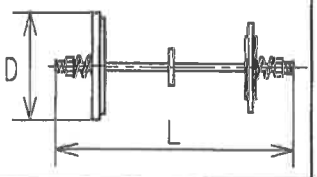
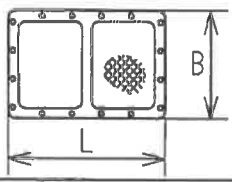
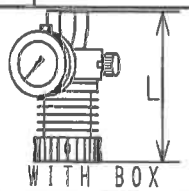
2017/05/09 TECHNICAL DEPARTMENT

REVISION

REVISION	DATE	HISTORY OF REVISION

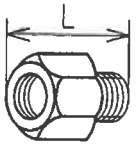
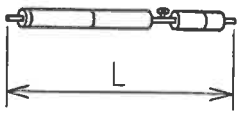
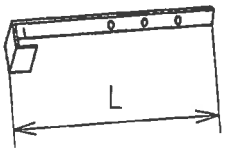
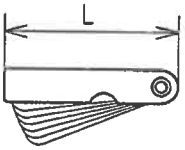
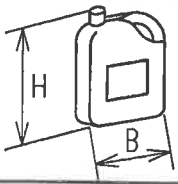
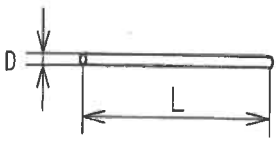
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TOOLS		PAGE	
TYPE 6DKM-20eF(L)		BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
1	NOZZLE TEST PUM P-G	No. 70309 	ASSY	1		AE01199052		H= 580 L= 620 WEIGHT 20.000 Kg
2	DEFLECTION GAUG E ASSY 85-106	No. 70500 	ASSY	1		C091590030		L= 106 WEIGHT 0.240 Kg
3	BLOWER SYRINGE	No. 71500 	ASSY	1		C044090010		L= 526 WEIGHT 1.740 Kg
4	CAMSHAFT METAL REMOVER DK-20	No. 71803 	ASSY	1		C095690110		B= 150 L= 207 WEIGHT 0.500 Kg
5	TURBO-CHARGER B LANK PLATE ASSY	No. 70101 	ASSY. カンピ	1		C012390010		L= 490 D= 255 WEIGHT 13.000 Kg
6	WIRE GAUZE DK20 INTAKE PORT	No. 70115 	SPCC SWM -G1	1		C015600150		B= 322 L= 470 WEIGHT Kg
7	MAX. PRESS. GAUGE MT31-20M	No. 70206 	PCHS	1		C091170052		L= 133 WEIGHT 2.390 Kg

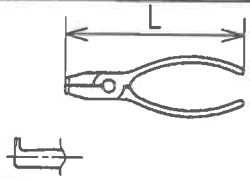
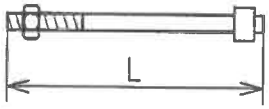
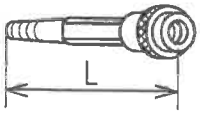
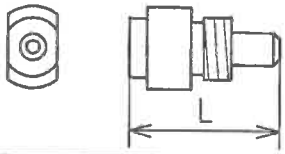
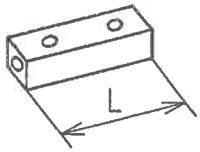
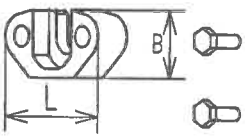
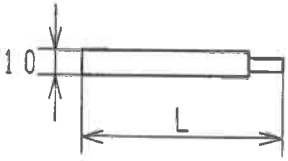
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TOOLS						PAGE	
TYPE 6DKM-20eF(L)						BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
8	JOINT M18XM22	No. 70405 	SS400B-D	1		C091440040		L= 41 WEIGHT 0.100 Kg
9	INSIDE MICROMETER 200-225	No. 70700 	PCHS	1		C091870040		L= 225 WEIGHT 0.190 Kg
10	POSITION GAUGE DK20	No. 70801 	SS400P	1		C092870180		L= 420 WEIGHT 0.180 Kg
11	THICKNESS GAUGE 70M	No. 70900 	PCHS	1		AE01199059		L= 80 WEIGHT 0.025 Kg
12	TORQUE WRENCH N 5600QLK	No. 71000	PCHS	1		C092070020		L= 1195 WEIGHT 8.500 Kg
13	BLOWER WASH	No. 71700 	PCHS	1		NN00295001		H= 280 B= 220 WEIGHT 4.500 Kg
14	TURNING BAR 34X 1000	No. 83300 	STPG370(SCH80)	2		NN00053002		D= 34 L= 1000 WEIGHT 3.300 Kg

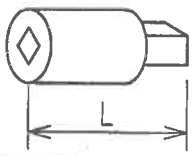
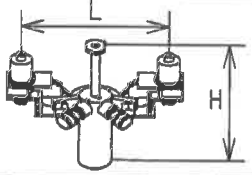
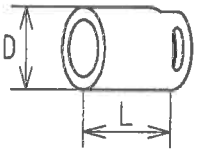
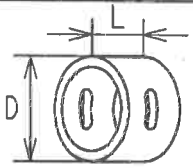
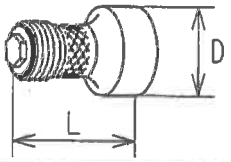
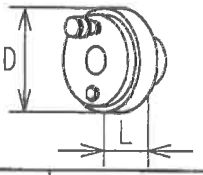
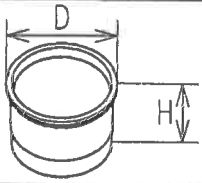
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TOOLS						PAGE	
TYPE 6DKM-20eF(L)						BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
15	SNAP RING PLIER	No. 97110 	PCHS	1		AE01199003		L= 212 WEIGHT 0.250 Kg
16	HIGH PRESS. HOSE 2000	No. 83001 	PCHS	6		AE01199013		L= 2000 WEIGHT Kg
17	SOCKET COUPLING	No. 83002 	PCHS	2		AE01199026		L= 58 WEIGHT 0.250 Kg
18	PLUG COUPLING	No. 83003 	PCHS	7		AE01199027		L= 48 WEIGHT 0.200 Kg
19	TERMINAL 4T	No. 83008 	SS400	1		AE01199028		L= 150 WEIGHT 0.320 Kg
20	PULL PIECE OIL PRESS. RAM	No. 90708 	ASSY	1		AE01199106		L= 83 B= 42 WEIGHT 0.330 Kg
21	HANDLE HYDRAURIC JACK	No. 83105 	SCM435-D	1		AE02195010		L= 135 WEIGHT 0.080 Kg

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TOOLS						PAGE	
TYPE 6DKM-20eF(L)						BOX No.	

No	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
22	ADAPTOR 19X12.7	No. 87600 	PCHS	1		AE02199103		L= 43 WEIGHT 0.420 Kg
23	OIL PRESS. RAM ASSY	No. 90712 	SPEC	1		E219459010		L= 265 H= 417 WEIGHT 11.000 Kg
24	JACK STAND-HEAD	No. 83113 	S45C	4		E219550050		D= 90 L= 190 WEIGHT 4.570 Kg
25	JACK STAND-CAP	No. 83118 	STKM13A	2		E219550060		D= 90 L= 65 WEIGHT 1.420 Kg
26	TENSION BOLT	No. 83114 	S43C	4		E219550090		D= 60 L= 164 WEIGHT 1.680 Kg
27	HYDRAULIC JACK(HEAD & CAP)	No. 83111 	ASSY	4		E219550330		D= 135 L= 87 WEIGHT 6.000 Kg
28	PISTON INSERT	No. 80113 	FC200	1		E219950020		H= 180 D= 245 WEIGHT 13.300 Kg

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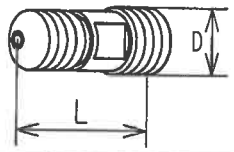
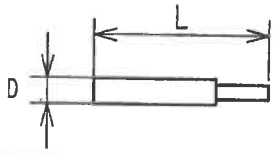
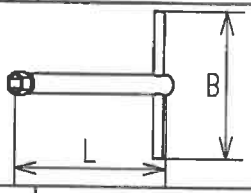
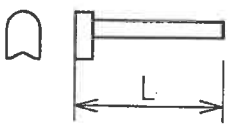
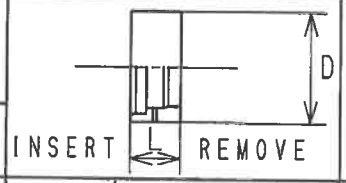
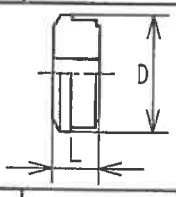
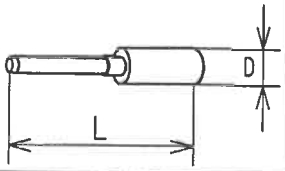
No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
29	NOZZLE HOLDER CONNECTOR	No. 84302 	S43C	1		E219950220		L= 50 D= 18 WEIGHT 0.130 Kg
30	REMOVER VALVE GUIDE	No. 87200 	SS400B	1		E219950250		L= 360 D= 23 WEIGHT 1.100 Kg
31	PIPE SPANNER 19 X20A	No. 80506 	SGP	1		E219950260		L= 457 B= 330 WEIGHT 2.000 Kg
32	REMOVER NOZZLE HOLDER	No. 92801 	SS400P, B	1		E219950270		L= 550 WEIGHT 1.800 Kg
33	COOLING VALVE SEAT JIG	No. 93409 	SS400	1		E219950400		D= 90 L= 40 WEIGHT Kg
34	VALVE SEAT INSERT(1)	No. 93407 	SS400P	1		E219950410		L= 26 D= 65 WEIGHT 0.540 Kg
35	VALVE SEAT INSERT(2)	No. 93408 	SS400B	1		E219950420		D= 32 L= 330 WEIGHT 1.100 Kg

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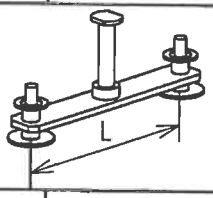
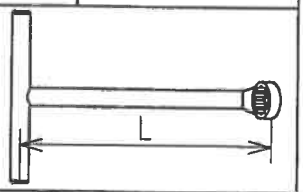
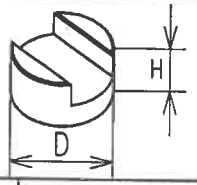
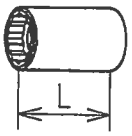
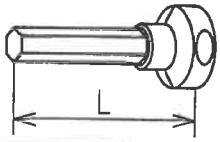
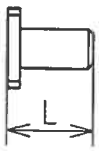
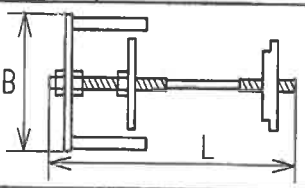
No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
36	SETTING TOOL HYDRAULIC JACK	No. 88702 	ASSY	1		E219950450		L= 265 WEIGHT 2.900 Kg
37	F.O. PUMP SPANNER 24	No. 80505 	SGP	1		E219950530		L= 580 WEIGHT 1.200 Kg
38	F.O. PUMP TAPPET REMOVER	No. 92001 	SS400	1		E219950540		H= 30 D= 70 WEIGHT 0.600 Kg
39	SOCKET 24X19L	No. 83208 	PCHS	1		E219950620		L= 100 WEIGHT 0.400 Kg
40	HEX. WRENCH KEY 8X12.7L	No. 83206 	PCHS	1		E219950630		L= 100 WEIGHT 0.090 Kg
41	MAIN BEARING PIN	No. 81000 	SS400B	1		E219951080		L= 23 WEIGHT 0.030 Kg
42	REMOVER CYLINDER LINER	No. 80715 	SPEC	1		E219959010		B= 400 L= 930 WEIGHT 18.000 Kg

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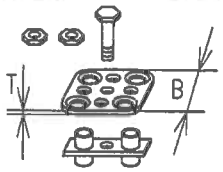
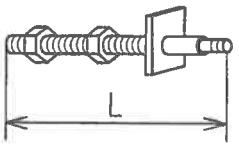
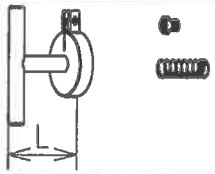
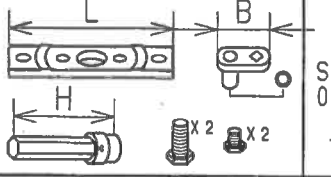
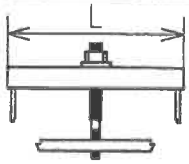
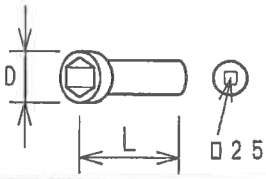
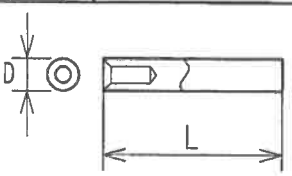
No	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
43	REMOVER VALVE SPRING	No. 86404 	SPEC/SS400/STPG	1		E219959020		B= 154 T= 19 WEIGHT 2.200 Kg
44	SUPPORTING TOOL PISTON	No. 80601 	SPEC/SS400	1		E219959030		L= 415 WEIGHT 0.600 Kg
45	GRINDING TOOL IN. EX. VALVE	No. 81402 	SPEC/SS400/SGP	1		E219959100		L= 144 WEIGHT 0.700 Kg
46	REMOVER IDOLE GEAR SHAFT	No. 88902 	SPEC/SS400	1		E219959110		L= 174 B= 146 H= 145 WEIGHT 2.550 Kg
47	REMOVER PROTECT RING	No. 80119 	SPEC	1		E219959150		L= 285 WEIGHT 10.00 Kg
48	BOX SPANNER 65X 130	No. 96101 	S43C	1		E239900390		L= 130 D= 89 WEIGHT 2.730 Kg
49	NOZZLE REMOVER	No. 86701 	C3604BE	1		E239950280		D= 18 L= 100 WEIGHT 0.220 Kg

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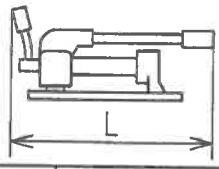
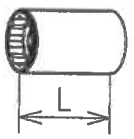
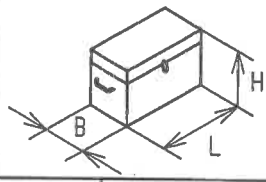
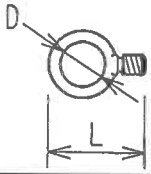
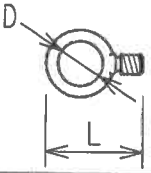
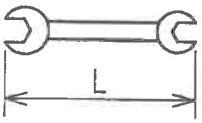
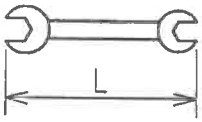
No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No. / PART No.	REMARKS
50	OIL PRESS. PUMP	No. 83000 	ASSY	1		E339600050	L= 1030 WEIGHT 14.000 Kg
51	SOCKET 27X19XL100	No. 83208 	PCHS	1		AE01199056	L= 100 WEIGHT 0.500 Kg
52	BOX-12TYPE WITH PAD LOCK	No. 99800 	SS400	1		Y910148300ZA	B= 400 L= 800 H= 300 WEIGHT 16.750 Kg
53	EYE BOLT M10	No. 95000 	PCHS	1		Z270010000ZZ	D= 25 L= 60 WEIGHT 0.300 Kg
54	EYE BOLT M16	No. 95000 	PCHS	1		Z270016000ZZ	D= 35 L= 87 WEIGHT 0.500 Kg
55	SPANNER(2) 8X10	No. 95100 	PCHS	1		Z900208100ZZ	L= 120 WEIGHT 0.030 Kg
56	SPANNER(2) 11X13	No. 95100 	PCHS	1		Z900211130ZZ	L= 135 WEIGHT 0.050 Kg
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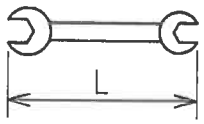
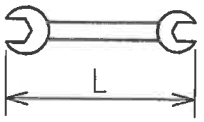
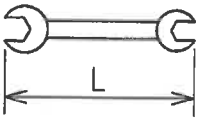
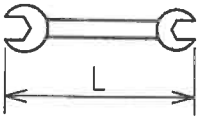
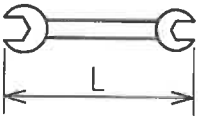
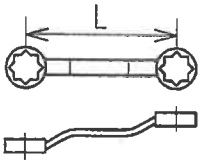
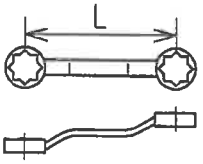
57	SPANNER (2) 17X19	No. 95100 	PCHS	1		Z900217190ZZ		L= 185 WEIGHT 0.140 Kg
58	SPANNER (2) 19X22	No. 95100 	PCHS	1		Z900219220ZZ		L= 205 WEIGHT 0.180 Kg
59	SPANNER (2) 22X24	No. 95100 	PCHS	1		Z900222240ZZ		L= 226 WEIGHT 0.220 Kg
60	SPANNER (2) 24X27	No. 95100 	PCHS	1		Z900224270ZZ		L= 245 WEIGHT 0.320 Kg
61	SPANNER (2) 27X30	No. 95100 	PCHS	1		Z900227300ZZ		L= 270 WEIGHT 0.425 Kg
62	OFFSET WRENCH 17X19	No. 95302 	PCHS	1		Z904117190LZ		L= 245 WEIGHT 0.210 Kg
63	OFFSET WRENCH	No. 95302 	PCHS	1		Z904122240LZ		L= 280 WEIGHT Kg

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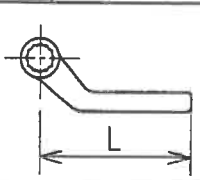
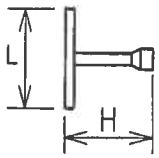
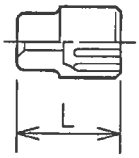
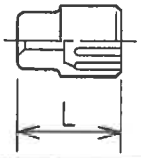
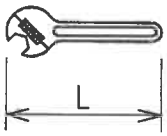
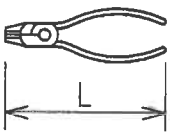
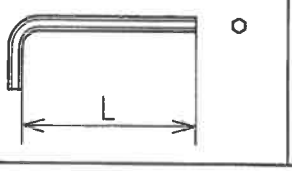
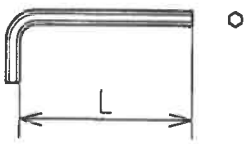
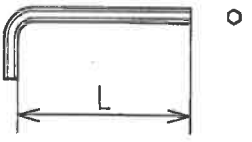
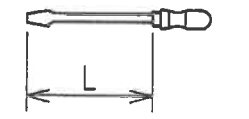
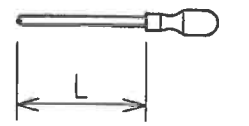
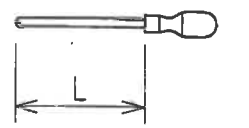
64	OFFSET WRENCH 36	No. 95400 	PCHS	1		Z90603600OZZ		L= 290 WEIGHT 0.500 Kg
65	BOX SPANNER(1) 24X180	No. 95600 	PCHS	1		Z91012418OZZ		L= 190 H= 190 WEIGHT 0.400 Kg
66	SOCKET(2) 24X19	No. 96300 	PCHS	1		Z91522400OZZ		L= 55 WEIGHT 0.200 Kg
67	SOCKET(2) 27X19	No. 96300 	PCHS	1		Z91522700OZZ		L= 48 WEIGHT 0.245 Kg
68	ADJUST. ANGLE WRENCH 250	No. 97000 	PCHS	1		Z92002500OZZ		L= 260 WEIGHT 0.330 Kg
69	CUTTING PLIER 150	No. 97200 	PCHS	1		Z92401500OZZ		L= 155 WEIGHT 0.180 Kg
70	HEX. WRENCH KEY 4	No. 97300 	PCHS	1		Z92500400OZZ		L= 72 WEIGHT 0.010 Kg

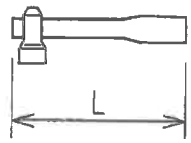
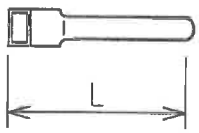
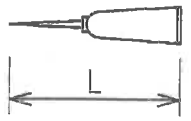
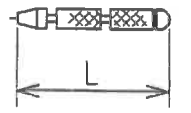
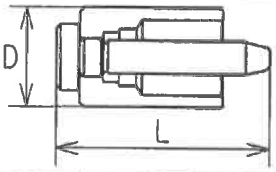
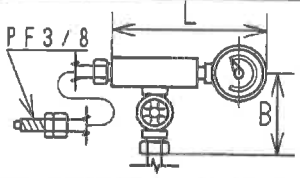
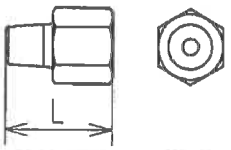
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TOOLS						PAGE	
TYPE 6DKM-20eF(L)						BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
71	HEX. WRENCH KEY 5	No. 97300 	PCHS	1		Z92500500OZZ		L= 80 WEIGHT 0.050 Kg
72	HEX. WRENCH KEY 8	No. 97300	PCHS	1		Z92500800OZZ		L= 100 WEIGHT 0.060 Kg
73	HEX. WRENCH KEY 10	No. 97300	PCHS	1		Z92501000OZZ		L= 112 WEIGHT 0.105 Kg
74	HEX. WRENCH KEY 14	No. 97300 	PCHS	1		Z92501400OZZ		L= 140 WEIGHT 0.270 Kg
75	SCREW DRIVER 8X 150	No. 97500 	PCHS	1		Z92800815OZZ		L= 150 WEIGHT 0.300 Kg
76	PLUS SCREW DRIVER(1) 2	No. 97600 	PCHS	1		Z93010020OZZ		L= 100 WEIGHT 0.053 Kg
77	PLUS SCREW DRIVER(1) 4	No. 97600 	PCHS	1		Z93010040OZZ		L= 200 WEIGHT 0.190 Kg

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TOOLS						PAGE	
TYPE 6DKM-20eF(L)						BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
78	HAMMER 1 1/2 POUND	No. 97700 	PCHS	1		Z93201200OZZ		L= 352 WEIGHT 0.820 Kg
79	PIPE HANDLE12X32X600	No. 97800 	STPG370	1		Z93401232OZZ		L= 600 WEIGHT 2.000 Kg
80	OIL FEEDER 150ML	No. 98200 	PCHS	1		Z93850150OZZ		L= 200 WEIGHT 0.025 Kg
81	NOZZLE CLEANING TOOL	No. 98501 	PCHS	1		Z940019313ZA		L= 60 WEIGHT 0.010 Kg
82	INSERT STEM SEAL TOOL ASSY	No. 82207 	S45C	1		E219955060		D= 42 L= 90 WEIGHT 0.510 Kg
83	PRESSURE SWITCH TESTING DEVICE	No. 72201 	ASSY	1		NN00277004		L= 215 B= 78 WEIGHT 1.050 Kg
84	JOINT SEAT G3/8 XR1/4	No. 10510 	PCHS	1		AG01557015		L= 36 WEIGHT 0.060 Kg

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

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TOOLS LIST
FOR
REDUCTION GEAR

DAIHATSU DIESEL MFG. CO., LTD.

TECHNICAL DEPARTMENT

OSAKA JAPAN

SHIP No.	SZ15045/15046	DRAWN BY	M. Tokumoto
TYPE	DRA-25J(L)	CHECKED BY	
LIST No.	AQP10008523A		
DATE	2016.06.02	APPROVED BY	(Prod) K. Ito

PDK20-0174-5

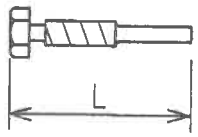
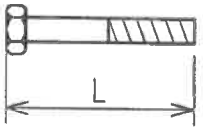
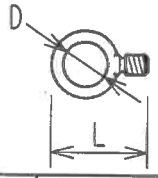
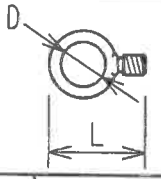
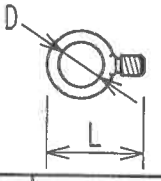
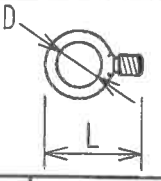
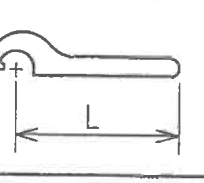
2017/05/09 TECHNICAL DEPARTMENT

REVISION

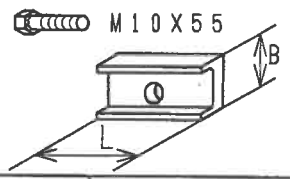
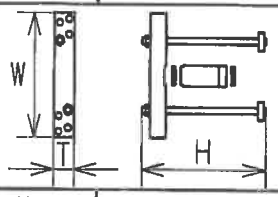
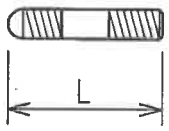
REVISION	DATE	HISTORY OF REVISION

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TOOLS						PAGE	
TYPE DRA-25J(L)						BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
1	JACK BOLT	No. 60100 	SS400B-D	2		G800600230		L= 140 WEIGHT 0.200 Kg
2	SET BOLT M10X1.5X138	No. 60400 	S43C	6		G006200220		L= 145 WEIGHT Kg
3	EYE BOLT M12	No. 95000 	PCHS	2		Z270012000ZZ		D= 30 L= 73 WEIGHT 0.190 Kg
4	EYE BOLT M16	No. 95000 	PCHS	2		Z270016000ZZ		D= 35 L= 87 WEIGHT 0.500 Kg
5	EYE BOLT M10	No. 95000 	PCHS	2		Z270010000ZZ		D= 25 L= 60 WEIGHT 0.300 Kg
6	EYE BOLT M20	No. 95000 	PCHS	1		Z270020000ZZ		D= 40 L= 101 WEIGHT Kg
7	HOOK SPANNER WRENCH (2) 175	No. 96900 	PCHS	1		Z916217500ZZ		L= 380 WEIGHT Kg

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TOOLS						PAGE	
TYPE DRA-25J(L)						BOX No.	
No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER SHIP	SUPPLY SPARE PER SHIP	DRAWING No. / PART No.	REMARKS
8	METAL CAP SUPPORT PLATE 80	No. 60301 	ASSY	2		NN00391001	B= 35 L= 80 WEIGHT Kg
9	PULLING OUT BEARING TOOL(FRONT)	No. 60706 	ASSY	1		NN00391002	H= 405 T= 53 W= 410 WEIGHT Kg
10	STAD M20X2.5X660	No. 10400 	SS400B-D	2		G002900080	L= 660 WEIGHT 1.600 Kg
NO		No.					WEIGHT Kg
NO		No.					WEIGHT Kg
NO		No.					WEIGHT Kg
NO		No.					WEIGHT Kg
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN						

TURBO-CHARGER(TYPE:TPS52-F)

LIST OF TOOLS

TPS52-F形 過給機

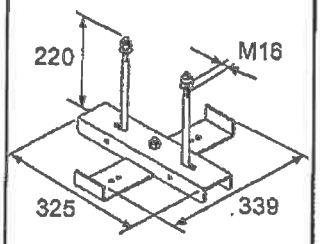
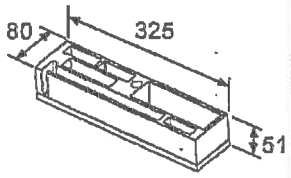
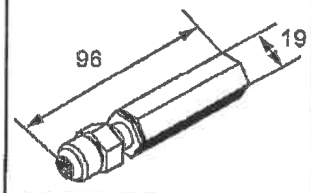
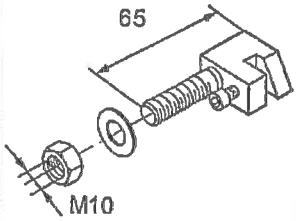
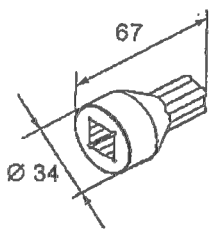
工 具 リ ス ト

DAIHATSU DIESEL MFG. CO., LTD.
TECHNICAL DEPARTMENT
JAPAN

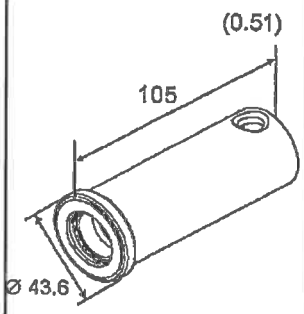
SHIP No.	—	DRAWN BY	A. Fujimoto
TYPE	ALL	CHECKED BY	T. Kawaji
LIST No.	B684300140FC		K. Ishida
DATE	Jul.27,2005	APPROVED BY	Y. Tenchi
REVISION	A : 2010/7/6 B : REV.UP ONLY 2012/4/3 T.K C : 2012/5/9 S.F		

B6843		D41527	(1／3)

Tools 分解要具						Page	(1/2)
TPS 52-F.. 形排気ガスタービン過給機 Type TPS 52-F.. EXAUST GAS TURBOCHARGER						Ship NO.	
						Box NO.	

No.	Name	Sketch (Mass) [kg]	Material	Supply		Drawing		Remarks
				Work-ing	Spare	No.	Part no.	
1	Service support 支え板 (NV51N26N21)	(6.30) 		1			90012	
2	Insert 分解要具入れ (NV51N26N26)	(0.05) 		1			90013	
3	Press-off tool 押上要具 (NV51N26N04)	(0.20) 		1			90042	
4	Extractor device 引抜要具 (NV51N26N06)	(0.10) 		2			90070	
5	Compressor wheel key コンプレッサホイール キー (NV51N26N12)	(0.25) 		1			90245	

MFR's name address	IHI Corporation 株式会社 IHI	IHI

Tools 分解要具								
						Page	(2/2)	
TPS 52-F.. 形排気ガスタービン過給機						Ship NO.		
Type TPS 52-F.. EXAUST GAS TURBOCHARGER						Box NO.		
No.	Name	Sketch (Mass) [kg]	Material	Supply		Drawing		Remarks
				Work-ing	Spare	No.	Part no.	
6	Press-on sleeve 押込ブッシュ (NV51N26N11)	(0.51)  		1			90380	
MFR's name address		IHI Corporation 株式会社 IHI						IHI

HYDRAULIC GOVERNOR
LIST OF TOOLS
RHD型ガバナー工具リスト

MODEL RHD6/10-MC/PC

DAIHATSU DIESEL MFG. CO., LTD.
TECHNICAL DEPARTMENT
OSAKA, JAPAN

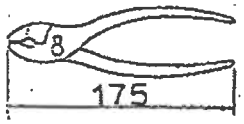
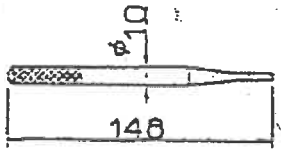
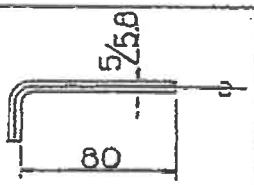
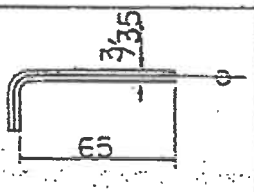
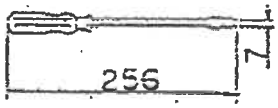
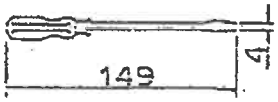
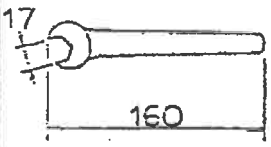
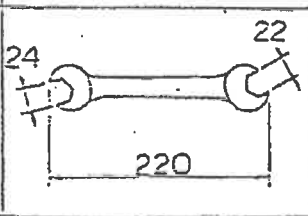
ZEXEL LIST NO. 307610-0010

SHIP No.		DRAWN BY	L. Hiyamoto
TYPE	RHD6/10-MC/PC	CHECKED BY	
LIST No.	B682500250FA		M. Yamauchi
DATE	MAR. 20, 1987	APPRPVED BY	E. Takemura
REVISION	A : Revised List NO(RH-490-14-4→307610-0010) Sep. 16, 1997		

B6825 #25

2017/05/09 TECHNICAL DEPARTMENT

TOOLS	1~3	4~17	18	PAGE	1
				SHIP NO.	
TYPE RHD6/10				BOX NO.	

NO. (19~22)	NAME	SKETCH	MATERIAL	SUPPLY PER SHIP WORKING (23~25)	SUPPLY PER SHIP SPARE (26~28)	DRAWING NO. (29~40)	DRAWING PART NO.	REMARKS
1	Pliers プライヤ		SCr4	1		376100-2400		
2	Extractor エキストラクタ		SK2	1		376100-2800		
3	Hexagon Wrench ヘクザゴンレンチ		SCM3 or SNCM6	1		376100-2700		
4	Hexagon Wrench ヘクザゴンレンチ		SCM3 or SNCM6	1		376100-2600		
5	Driver ドライバー			1		376100-2200		
6	Driver ドライバー			1		376100-2300		
7	Spanner スパナ		SCM3	1		376100-2500		
8	Spanner スパナ		SCM3	1		376100-2100		

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG.CO.,LTD. 1-87, 1-CHOME, OYODO-NAKA, OYODO-KU, OSAKA, JAPAN

B6825002500 (2/2)

2017/05/09 TECHNICAL DEPARTMENT

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TOOLS LIST
FOR
F.W. COOLER

DAIHATSU DIESEL MFG.CO.,LTD.

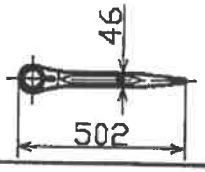
TECHNICAL DEPARTMENT

JAPAN

SHIP NO.		DRAWN BY	M. Tokuto
TYPE		CHECKED BY	T. Tajima
LIST NO.	AQP20008437B		
DATE	2016/6/3	APPROVED BY	S. Kuwa
REVISION	B:DEL. ITEM NO. MODEL 2016/10/18 M.T		

(5)

MFG.No.	–	TOOL LIST	Doc No.	TOLAP2976
HISAKA PLATE HEAT EXCHANGER			SHIP No.	
			ITEM No.	

No.	PARTS NAME	SKETCH (ØDIMENSION SHOWS CENTER OF DIAMETER.)	MATERIAL	QUANTITY WORKING PER 1 UNIT	QUANTITY SPARE PER 2UNITS	REMARKS
1	RATCHET SPANNER		C.S.	1	—	FOR M30 MASS 2.1kg
2						
3						
4						
5						
6						
7						
8						

APPROVED BY	S.HATANAKA	MAY,31,2016	HISAKA WORKS,LTD. OSAKA,JAPAN	
PREPARED BY	T.OHNOKI	MAY,31,2016		

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HEADS VER.200604

HD_ToolList

2017/05/09 TECHNICAL DEPARTMENT

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TOOLS LIST
FOR
L.O. COOLER

DAIHATSU DIESEL MFG.CO.,LTD.

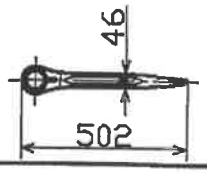
TECHNICAL DEPARTMENT

JAPAN

SHIP NO.		DRAWN BY	M. Tokumoto
TYPE		CHECKED BY	T. Tojimoto
LIST NO.	AQP20008438B		
DATE	2016/6/3	APPROVED BY	S. Kawa.
REVISION	B:DEL. ITEM NO. MODEL 2016/10/18 M.T		

(5)

MFG.No.	-	TOOL LIST	Doc No.	TOLAN9571
HISAKA PLATE HEAT EXCHANGER			SHIP No.	
			ITEM No.	

No.	PARTS NAME	SKETCH (ØDIMENSION SHOWS CENTER OF DIAMETER.)	MATERIAL	QUANTITY WORKING PER 1 UNIT	QUANTITY SPARE PER 2UNITS	REMARKS
1	RATCHET SPANNER		C.S.	1	—	FOR M30 MASS 2.1kg
2						
3						
4						
5						
6						
7						
8						

APPROVED BY	S.HATANAKA	MAY,31,2016	HISAKA WORKS,LTD. OSAKA,JAPAN
PREPARED BY	T.OHNOKI	MAY,31,2016	

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SPARE PARTS LIST
FOR
DIESEL ENGINE

DAIHATSU DIESEL MFG. CO., LTD.

TECHNICAL DEPARTMENT

OSAKA JAPAN

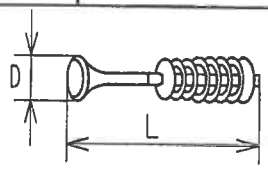
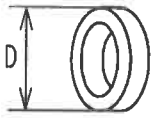
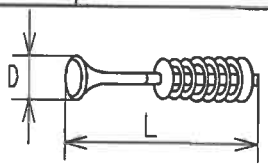
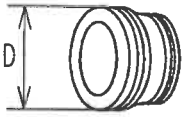
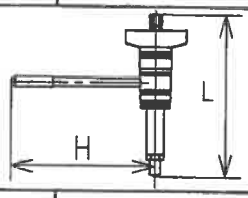
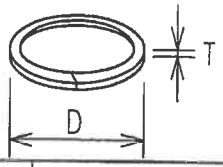
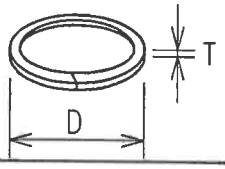
SHIP No.	SZ15045/15046/15047/15048	DRAWN BY	M. Johnouchi
TYPE	6DKM-20eF (L)	CHECKED BY	K. Fujimoto
LIST No.	AQP10008524B		
DATE	2016. 06. 13	APPROVED BY	E. Kuwa

2017/05/09 TECHNICAL DEPARTMENT

REVISION

REVISION	DATE	HISTORY OF REVISION
△B	2016/07/27	REV:No. 25 M. T

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

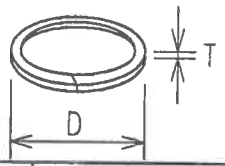
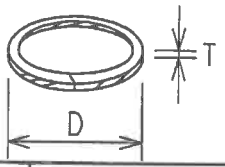
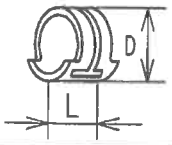
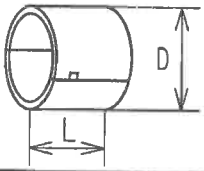
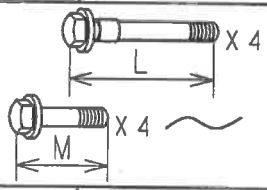
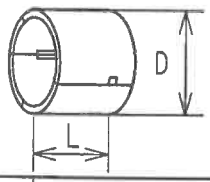
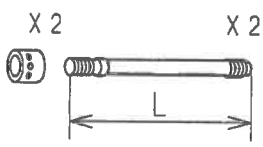
SPARE PARTS							PAGE	
TYPE 6DKM-20eF(L)							BOX No.	
No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
1	INTAKE VALVE ASSY	No. 00100 	ASSY	12	2	DK-20-001		D= 64 L= 292 WEIGHT 1.500 Kg
2	INTAKE VALVE SEAT / CYLINDER HEAD	No. 00300 	SPE. CAST (E83)	12	2	E205650060		D= 76 WEIGHT 0.300 Kg
3	EXHAUST VALVEASSY	No. 00100 	ASSY	12	4	DK-20-006		D= 64 L= 292 WEIGHT 1.500 Kg
4	EXHAUST VALVE SEAT / CYLINDER HEAD	No. 00307 	SUH3 STELLITE	12	2	E205650080		D= 76 WEIGHT 0.350 Kg
5	NOZZLEHOLDER DLL140TE3010N5 / F.O. INJECTION SYSTEM	No. 01107 	F 01G 00 9 02E	6	6	E206255210		L= 299 H= 267 WEIGHT 4.100 Kg
6	PISTON RING(3) / PISTON	No. 01500 	RIK42A	6	1	E205150170		D= 200 T= 5 WEIGHT 0.163 Kg
7	PISTON RING(2) / PISTON	No. 01500 	RIK42A	6	1	E205150200		D= 200 T= 5 WEIGHT 0.163 Kg
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN							

SPARE PARTS

PAGE

TYPE 6DKM-20eF(L)

BOX No.

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
8	PISTON RING(1) PISTON	No. 01500 	RIK21A	6	1	E205150240		D= 200 T= 5 WEIGHT 0.163 Kg
9	OIL RING(COIL-8.0) PISTON	No. 01600 	RIK21A SWPA	6	1	E205150230		D= 200 T= 7 WEIGHT 0.091 Kg
10	PISTON PIN BUSH CONNECTING ROD	No. 01804 	DS10 L10	6	1	E205255030		D= 100 L= 97 WEIGHT 0.740 Kg
11	CRANK PIN BEARING SHELL CONNECTING ROD	No. 02300 	SD10 A40	6	1	E205250010		D= 181 L= 78 WEIGHT 1.600 Kg
12	CRANK PIN BOLT ASSY	No. 02419 	ASSY	6	1	DK-20-024		L= 195 M= 99 WEIGHT 3.200 Kg
13	MAIN BEARING SHELL MAIN BEARING	No. 02700 	A40 S10C	7	1	E200750010		D= 210 L= 77 WEIGHT 1.770 Kg
14	METAL CAP BOLT ASSY	No. 02920 	ASSY	7	1	DK-20-029		L= 485 WEIGHT 8.600 Kg

MFR'S NAME & ADDRESS

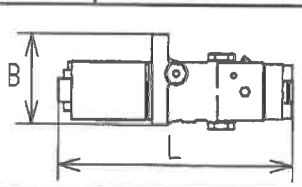
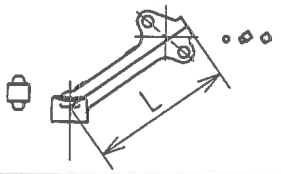
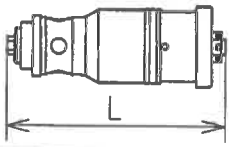
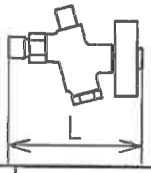
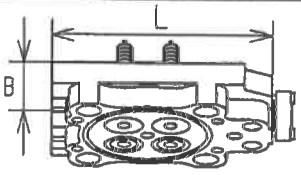
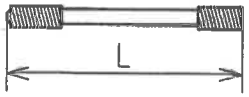
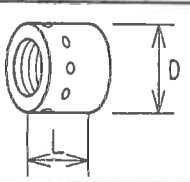
DAIHATSU DIESEL MFG. CO., LTD.
1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

SPARE PARTS

PAGE

TYPE 6DKM-20eF(L)

BOX No.

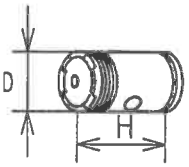
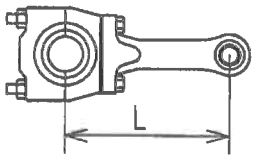
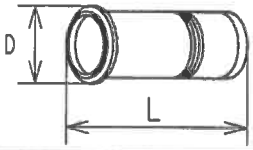
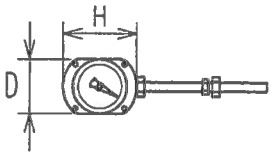
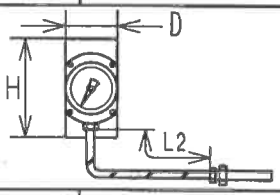
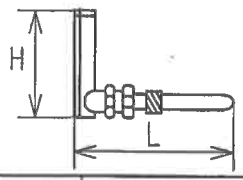
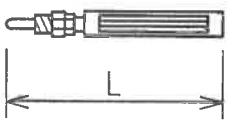
No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
15	F.O. PUMP W/C F.O. INJECTION PUMP	No. 03203 	ASSY	6	1	E206455030		L= 378 B= 141 WEIGHT 16.800 Kg
16	F.O. INJECTION PIPE WITH JOINT	No. 03406 	ASSY	6	1	DK-20-034		L= 129 WEIGHT 3.300 Kg
17	STARTING VALVE ASSY	No. 03716 	ASSY	6	1	DK-20-037		L= 137 WEIGHT 1.100 Kg
18	CYLINDER PRESSURE VALVE ASSY INDICATOR & SAFETY VALVE	No. 04119 	ASSY	6	1	NN00002039		L= 167 WEIGHT 1.500 Kg
19	CYL. HEAD ASSY (WITH VALVE)	No. 05204 	ASSY	6	1	DK-20-052		B= 235 L= 445 WEIGHT 130.000 Kg
20	CYLINDER HEAD BOLT FRAME	No. 05306 	D-SCM435	24	2	E200250130		L= 578 WEIGHT 4.230 Kg
21	CIRCULAR NUT M39X3.0 CYLINDER HEAD	No. 10302 	S43C	24	2	AE02008003		D= 64 L= 40 WEIGHT 0.600 Kg
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN							

SPARE PARTS

PAGE

TYPE 6DKM-20eF(L)

BOX No.

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
22	PISTON ASSY	No. 05401 	ASSY	6	1	DK-20-054		D= 200 H= 275 WEIGHT 27.000 Kg
23	CONNECTING ROD ASSY	No. 05501 	ASSY	6	1	DK-20-055		L= 550 WEIGHT 46.000 Kg
24	CYLINDER LINER	No. 05600 	ASSY	6	1	DK-20-056		D= 275 L= 535 WEIGHT 69.000 Kg
25	THERMOMETER DIAL TYPE G1/2-X100 -L95-650C EX. THERMOMETER CYL. OUTLET	No. 06804 	PCHS	6	1	NN00612001		D= 50 H= 66 WEIGHT Kg ⚠
26	THERMOMETER DIAL TYPE L95	No. 	PCHS.	3	1	AQA20011343/A		D= 50 H= 100 L2= 3000 WEIGHT Kg
27	THERMOMETER 100 CX1/2L THERMOMETER (100C)	No. 06901 	PCHS	1	1	Y24100400ZB		L= 157 H= 110 WEIGHT 0.300 Kg
28	THERMOMETER THERMOMETER (100C)	No. 06900 	PCHS	2	1	Y29100500ZZ		L= 203 WEIGHT 0.200 Kg

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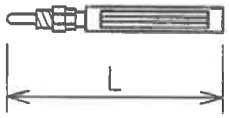
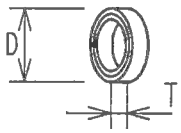
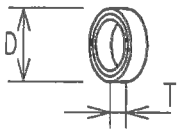
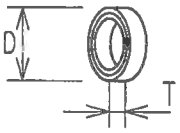
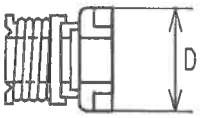
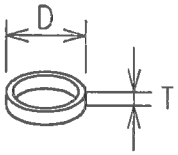
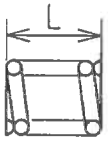
29	THERMOMETER 100 CX1/2P THERMOMETER (100C)	No. 06900 	PCHS	1	1	Y23100400OZC		L= 169 WEIGHT 0.190 Kg
30	OIL SEAL SC406212 F.O.FEED PUMP FITTING	No. 07000 	PCHS	1	1	Z4614062125C		D= 62 T= 12 WEIGHT Kg
31	OIL SEAL SB406088 L.O. PUMP	No. 07000 	PCHS	1	1	Z461406008SB		D= 60 T= 8 WEIGHT Kg
32	OIL SEAL TC355008 C.W. PUMP-70A	No. 07000 	PCHS	1	1	Z461355008TC		D= 50 T= 8 WEIGHT Kg
33	MECHANICAL SEAL (560-30)NT31 C.W. PUMP-70A	No. 07200 	PCHS	1	1	AE01102010		D= 50 WEIGHT 0.100 Kg
34	MOUTH RING C.W. PUMP-70A	No. 07400 	BC3	2	1	C038420180		D= 115 T= 12 WEIGHT 0.170 Kg
35	VALVE SPRING(INNER) INTAKE VALVE	No. 07600 	SWOSC-V	12	4	E205750140		L= 77 WEIGHT Kg

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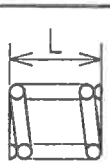
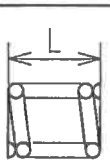
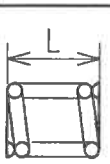
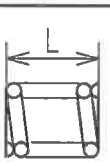
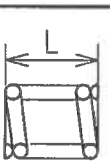
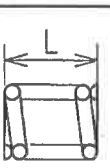
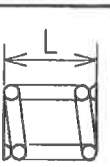
36	VALVE SPRING(OUTER) INTAKE VALVE	No. 07600 	SWOSC-V	12	4		E205750150	L= 84 WEIGHT Kg
37	VALVE SPRING(INNER) EXHAUST VALVE	No. 07600 	SWOSC-V	12	4		E205750140	L= 77 WEIGHT Kg
38	VALVE SPRING(OUTER) EXHAUST VALVE	No. 07600 	SWOSC-V	12	4		E205750150	L= 84 WEIGHT Kg
39	NOZZLE SPRING F.O. INJECTION SYSTEM	No. 07600 	SPEC	6	2		AE01062023	L= 30 WEIGHT Kg
40	SPRING 4.0X25.5X165 OIL RELIEF VALVE	No. 07600 	SWPA	1	1		C037570201	L= 165 WEIGHT 0.140 Kg
41	SPRING 5.5X36.5X183 OIL RELIEF VALVE	No. 07600 	SWPA	1	1		C037570400	L= 183 WEIGHT 0.200 Kg
42	SPRING 2.0X13.0X75.8 OIL RELIEF VALVE	No. 07600 	SWPA	1	1		NN00017010	L= 76 WEIGHT 0.030 Kg

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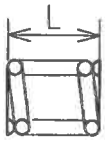
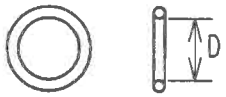
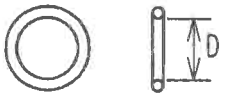
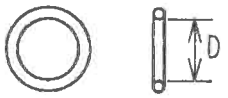
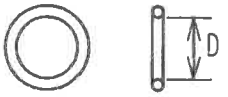
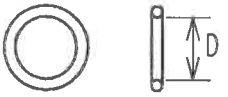
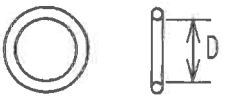
43	SPRING 4.0X30.5 X69.2 STARTING VALVE	No. 07600 	SWPA	6	2	E206050110		L= 69 WEIGHT Kg
44	O-RING G115 IDLE GEAR	No. 08200 	NBR	1	1	Z56021153IZZ		D= 115 WEIGHT Kg
45	O-RING P30 IDLE GEAR	No. 08200 	NBR	1	1	Z56010303SZZ		D= 30 WEIGHT Kg
46	O-RING G90 IDLE GEAR	No. 08200 	NBR	1	1	Z56020903IZZ		D= 90 WEIGHT Kg
47	O-RING P31 F.O.PIPING	No. 08200 	FPM	2	1	Z56010313SDZ		D= 31 WEIGHT Kg
48	O-RING G35 FRAME	No. 08200 	NBR	24	8	Z56020353IZZ		D= 35 WEIGHT Kg
49	O-RING 258X6.0 CYLINDER LINER	No. 08200 	SPEC	6	2	E205050090		D= 258 WEIGHT Kg

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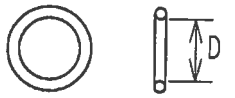
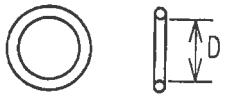
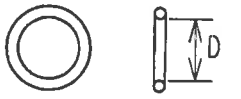
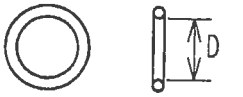
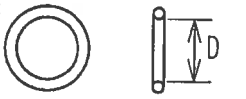
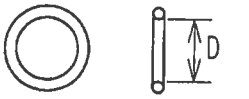
50	O RING G230 CYLINDER LINER	No. 08200		18	6		E205050100	D= 229 WEIGHT Kg
51	O-RING 62.5X2.0 CYLINDER HEAD	No. 08200 	FP33	12	4		E205650270	D= 63 WEIGHT Kg
52	O-RING(SI)S71 CYLINDER HEAD	No. 08200 	SI	12	4		E205650260	D= 71 WEIGHT Kg
53	O-RING G40(4D) CYLINDER HEAD	No. 08200 	FPM	6	2		E205650300	D= 39 WEIGHT Kg
54	O-RING P22A CYLINDER HEAD	No. 08200 	NBR	6	2		Z560102235ZZ	D= 22 WEIGHT Kg
55	O-RING P14 ROCKER ARM	No. 08200 	NBR	6	2		Z5601014247Z	D= 14 WEIGHT Kg
56	O-RING P42 STARTING VALVE	No. 08200 	NBR	6	2		Z560104235ZZ	D= 42 WEIGHT Kg

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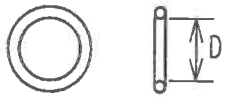
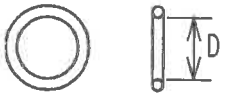
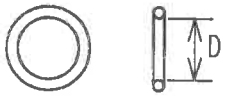
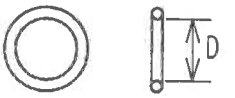
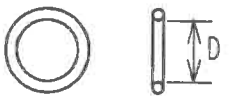
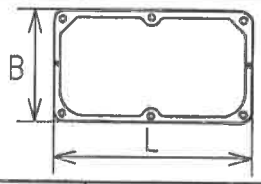
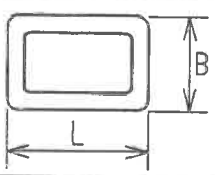
57	O-RING 34X5.4 F.O. INJECTION SYSTEM	No. 08200 	FPM	6	2	E206250120	D= 34 WEIGHT Kg
58	O-RING P50 F.O. INJECTION SYSTEM	No. 08200 	FPM	18	6	Z560105035DZ	D= 50 WEIGHT Kg
59	O-RING P32 F.O. INJECTION SYSTEM	No. 08200 	FPM	12	4	Z560103235DZ	D= 32 WEIGHT Kg
60	O-RING G30 F.O. INJECTION SYSTEM	No. 08200 	FPM	6	2	Z560203031DZ	D= 30 WEIGHT Kg
61	O-RING P18 F.O. PIPING	No. 08200 	FPM	12	4	Z560101824DZ	D= 18 WEIGHT Kg
62	GASKET SIDE COVER(4) SIDE COVER	No. 08422 	CORTEX-NO.25	3	1	E200350150	B= 262 L= 610 WEIGHT Kg
63	GASKET SIDE COVER(5) SIDE COVER	No. 08403 	CORTEX-NO.25	1	1	E200350160	B= 180 L= 270 WEIGHT Kg
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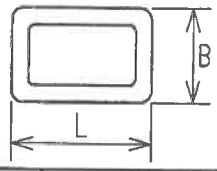
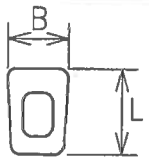
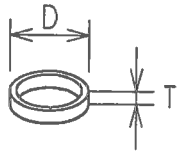
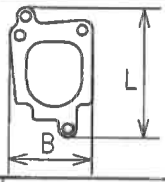
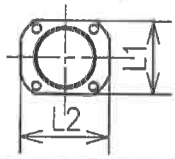
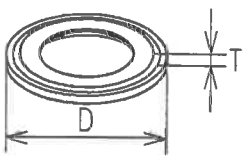
No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
64	GASKET CAM GEAR COVER / SIDE COVER	No. 08403 	SPEC	1	1		E200350170	B= 180 L= 455 WEIGHT Kg
65	GASKET INTAKE MANIFOLD BEND / INTAKE MANIFOLD	No. 08315 	SPEC	6	2		E202050070	B= 128 L= 145 WEIGHT Kg
66	RUBBER SEAL 105 X126X21.5 / INTAKE MANIFOLD	No. 07400 	NBR	6	2		AE02026011	D= 126 T= 20 WEIGHT Kg
67	EXHAUST PIPE GASKET / EX. MANIFOLD M/E	No. 08528 	SUS301H	6	2		E202150270	B= 113 L= 176 WEIGHT Kg
68	EX. MANIFOL GASKET 90A / EX. MANIFOLD M/E	No. 08805 	SUS304 CERAMIC B	8	4		A592000290	L1= 146 L2= 164 WEIGHT Kg
69	GASKET CYLINDER HEAD / CYLINDER HEAD	No. 08600	S10C	6	2		E205650180	D= 256 WEIGHT 0.130 Kg
70	GASKET FOR C.W. CONNECTOR / CYLINDER HEAD	No. 08800 	NBR SUS 304	24	8		E205650560	D= 36 T= 5 WEIGHT Kg

MFR'S NAME & ADDRESS

DAIHATSU DIESEL MFG. CO., LTD.
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SPARE PARTS

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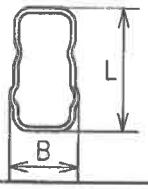
TYPE 6DKM-20eF(L)

BOX No.

No	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
71	RUBBER SEAL HEAD COVER CYLINDER HEAD COVER	No. 08610 	SPEC	6	2	E206350100		B= 196 L= 331 WEIGHT Kg
72	GASKET 30 STARTING VALVE	No. 08600 	SPCC-A MFNI-10microM	6	2	Z665030000CD		D= 38 WEIGHT 0.006 Kg
73	GASKET 38 STARTING VALVE	No. 08600 	C1100P-0	6	2	Z665038000ZZ		D= 47 WEIGHT 0.010 Kg
74	GASKET 19 F.O. INJECTION SYSTEM	No. 08600 	C1100P-0	6	2	Z565001920ZZ		D= 26 WEIGHT 0.004 Kg
75	GASKET F.O. INJECTION SYSTEM	No. 08600 	PCHS	12	4	AE01062016		D= 18 WEIGHT 0.003 Kg
76	GASKET 34 S.A. PIPING SYSTEM	No. 08600 	C1100P-0	2	10	Z665034000ZZ		D= 42 WEIGHT 0.008 Kg
77	GASKET 28 S.A. PIPING SYSTEM	No. 08600 	C1100P-0	1	10	Z665028000ZZ		D= 36 WEIGHT 0.006 Kg

MFR'S NAME & ADDRESS

DAIHATSU DIESEL MFG. CO., LTD.
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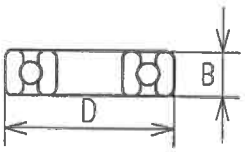
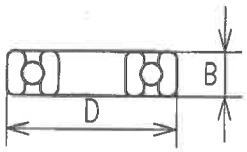
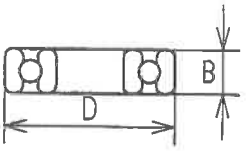
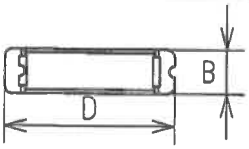
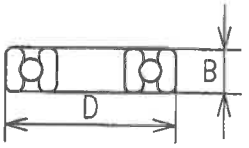
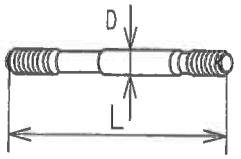
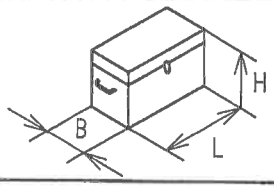
SPARE PARTS						PAGE	
TYPE 6DKM-20eF(L)						BOX No.	
No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No. / PART No.	REMARKS
78	GASKET 13 S.A. PIPING SYSTEM	No. 08600 	SPCC NON AS COMPOUND	12	10	Z565001300EE	D= 20 WEIGHT 0.001 Kg
79	GASKET 17 S.A. PIPING SYSTEM	No. 08600 	SPCC NON AS COMPOUND	6	10	Z565001700EE	D= 24 WEIGHT 0.001 Kg
80	GASKET 21 F.O. PIPING	No. 08600 	SPCC NON AS COMPOUND	6	2	Z565002100EE	D= 29 WEIGHT 0.002 Kg
81	GASKET 17 F.O. PIPING	No. 08600 	SPCC NON AS COMPOUND	1	2	Z565001700EE	D= 24 WEIGHT 0.001 Kg
82	GASKET 10 F.O. PIPING	No. 08600 	SPCC NON AS COMPOUND	12	2	Z565001000EE	D= 15 WEIGHT 0.001 Kg
83	GASKET 13 F.O. PIPING	No. 08600 	SPCC NON AS COMPOUND	20	2	Z565001300EE	D= 20 WEIGHT 0.001 Kg
84	BEARING (1) 6007 GOVERNOR MOUNTING PARTS	No. 09000 	PCHS	1	1	Z45116007ZZ	D= 62 B= 14 WEIGHT 0.150 Kg
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN						

SPARE PARTS							PAGE	
TYPE 6DKM-20eF(L)							BOX No.	

No	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
85	BEARING(1)6306 F.O.FEED PUMP FITTING	No. 09000 	PCHS	1	1	Z45116306OZZ		D= 72 B= 19 WEIGHT 0.100 Kg
86	BEARING(1)6206 F.O.FEED PUMP FITTING	No. 09000 	PCHS	1	1	Z45116206OZZ		D= 62 B= 16 WEIGHT 0.200 Kg
87	NEEDLE BEARING 40X55X30 L.O. PUMP	No. 09001 	SPEC	5	1	H616503040		D= 55 B= 30 WEIGHT 0.190 Kg
88	BEARING 35 C.W. PUMP-70A	No. 09000 	PCHS	2	1	CD38500570		D= 72 B= 17 WEIGHT 0.400 Kg
89	BURSTING DISK(SAFETY PLUG) STARTING AIR PIPE SAFETY PLUG	No. 10000	C1220T-1/4H	1	3	NN00160003		D= 24 WEIGHT 0.001 Kg
90	REAMER BOLT OVER 27X304	No. 04312 	S45C(H)	2	1	NN00744028		L= 304 D= 27 WEIGHT 0.900 Kg
91	BOX-12TYPE WITH PAD LOCK	No. 99800 	SS400		1	Y910148300ZA		B= 400 L= 800 H= 300 WEIGHT 16.750 Kg

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SPARE PARTS LIST
FOR
REDUCTION GEAR

DAIHATSU DIESEL MFG. CO., LTD.

TECHNICAL DEPARTMENT

OSAKA JAPAN

SHIP No.	SZ15045/46/47/48	DRAWN BY	M. Tokumoto
TYPE	DRA-25J(L)	CHECKED BY	A. Tajima
LIST No.	AQP10008525C		
DATE	2016.06.10	APPROVED BY	

2017/05/09 TECHNICAL DEPARTMENT

REVISION

REVISION	DATE	HISTORY OF REVISION
B	2016/09/30	REV:No. 22, DEL:No. 23 M. T
C	2016/12/05	ADD:No. 51, 52 M. T

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

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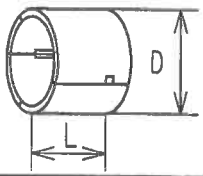
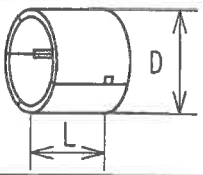
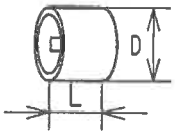
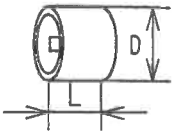
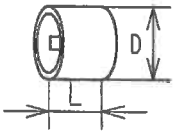
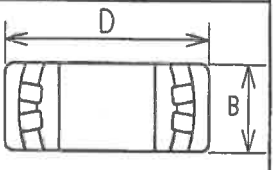
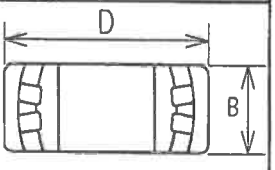
TYPE DRA-25J(L)						BOX No.	

No	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
1	METAL 250X260X130 BEARINGS	No. 02700 	S10C WJ2	1	1	G253320010		D= 260 L= 130 WEIGHT 3.900 Kg
2	METAL 155X163X115 BEARINGS	No. 02700 	STEEL WJ 2	1	1	G253320020		D= 163 L= 115 WEIGHT 1.800 Kg
3	BUSH 190X211X90 BEARINGS	No. 01802 	STEEL WJ 2	1	1	G253320030		D= 211 L= 90 WEIGHT 4.500 Kg
4	BUSH 190X211X135 BEARINGS	No. 01802 	S10C/STPG38 WJ2	1	1	G253320040		D= 211 L= 135 WEIGHT 8.500 Kg
5	GEAR BUSH 155X169X100 BEARINGS	No. 01802 	S10C WJ2	4	1	G193320050		D= 169 L= 100 WEIGHT 2.500 Kg
6	BEARING 22228CAME4S11 BEARINGS	No. 30100 	SPEC	2	1	G253325010		D= 250 B= 68 WEIGHT Kg
7	BEARING 23228CAME4S11 BEARINGS	No. 30100 	SPEC	2	1	G253325020		D= 250 B= 88 WEIGHT Kg

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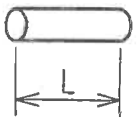
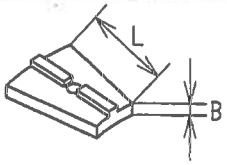
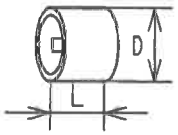
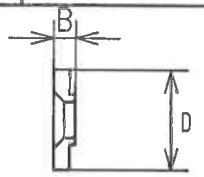
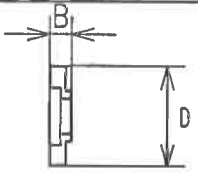
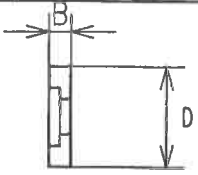
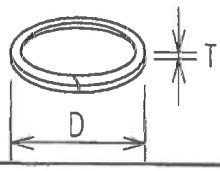
No	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
8	PAD PIN GEAR BOX	No. 10103 	S43C	10	5	G254120340		L= 103 WEIGHT 0.400 Kg
9	PAD 240X442X35.9 PAD	No. 30300 	SS41/ SF45, WJ2	20	10	G005000020		L= 101 B= 36 WEIGHT Kg
10	PAD PIN BUSH PAD	No. 01802 	S43C	20	10	G005000150		D= 35 L= 20 WEIGHT Kg
11	THRUST METAL 156X250X15 BEARINGS	No. 30201 	STEEL WJ 2	2	1	G042200120		D= 250 B= 19 WEIGHT Kg
12	THRUST METAL 140X225X20 BEARINGS	No. 30202 	S10C LBC4/LBC3	2	1	G193320080		D= 225 B= 20 WEIGHT 3.000 Kg
13	THRUST METAL 158X207X10 BEARINGS	No. 30203 	STEEL LBC4/LBC3	2	1	G193320090		D= 207 B= 10 WEIGHT Kg
14	OIL PRESS. RING 30X2.5X1.5 GEAR BOX	No. 01500 	SPEC CAST	4	1	G001500100		D= 30 T= 3 WEIGHT Kg

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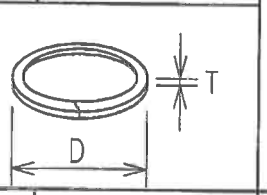
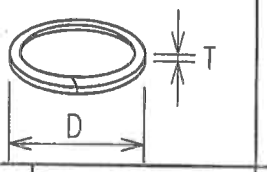
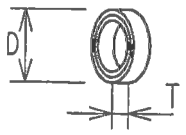
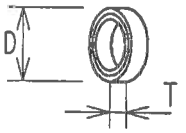
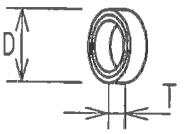
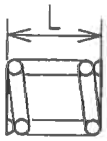
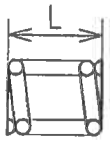
SPARE PARTS							PAGE	
TYPE DRA-25J(L)							BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
15	OIL PRESS. RING 220X7 CLUTCH	No. 01500 	SPE CAST	2	1	G011400300		D= 220 T= 7 WEIGHT Kg
16	OIL PRESS. RING3 60X8 CLUTCH	No. 01500 	SPE CAST	2	1	G014500300		D= 360 T= 8 WEIGHT Kg
17	OIL SEAL SB223507 CHUTCH OPERATING VALVE	No. 07000 	PCHS	1	1	Z46122350 7SD		D= 35 T= 7 WEIGHT 0.009 Kg
18	OIL SEAL SB406212 L.O. PUMP	No. 07000 	PCHS	1	1	Z46140621 2SB		D= 62 T= 12 WEIGHT Kg
19	OIL SEAL SB507212 L.O. PUMP	No. 07000 	PCHS	1	1	Z46150721 2SB		D= 72 T= 12 WEIGHT 0.035 Kg
20	SPRING (PISTON) CLUTCH	No. 07600 	SWPB	12	1	G253120160		L= 85 WEIGHT Kg
21	SPRING 3.5X32.5 X81 L.O. RELIEF VALVE (HIGH PRESS)	No. 07600 	SWPB	1	1	G005300191		L= 81 WEIGHT Kg

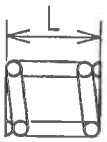
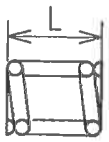
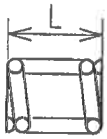
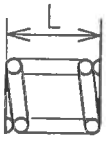
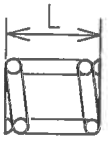
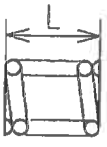
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

SPARE PARTS

PAGE

TYPE DRA-25J(L)

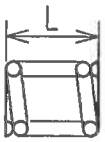
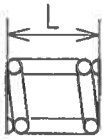
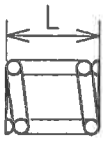
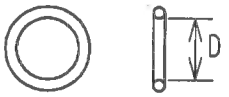
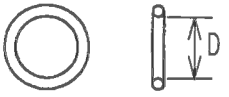
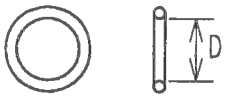
BOX No.

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
22	SPRING 2.6x24.2x82 L.O. RELIEF VALVE (HIGH PRESS)	No. 07600 	SWPB	1	1	G005300192		L= 82 WEIGHT Kg
23								
24	SPRING 5.5X36.5 X209 OIL RELIEF VALVE	No. 07600 	SWPA	1	1	C037570151		L= 209 WEIGHT 0.380 Kg
25	ROD SPRING CHUTCH OPERATING VALVE	No. 07600 	SWPA	2	1	A362079801		L= 24 WEIGHT Kg
26	LEVER SPRING CHUTCH OPERATING VALVE	No. 07600 	SWPA	1	1	A362079802		L= 21 WEIGHT Kg
27	PISTON SPRING CHUTCH OPERATING VALVE	No. 07600 	SWPA	2	1	A362079807		L= 73 WEIGHT Kg
28	SAFETY VALVE SPRING L.O. PUMP	No. 07600 	SWPB	1	1	G811400361		L= 81 WEIGHT Kg

MFR'S NAME & ADDRESS

DAIHATSU DIESEL MFG. CO., LTD.
1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

SPARE PARTS						PAGE	
TYPE DRA-25J(L)						BOX No.	

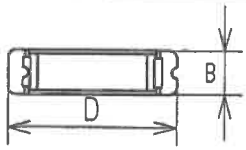
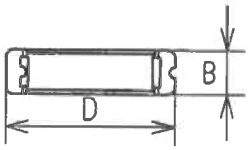
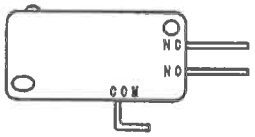
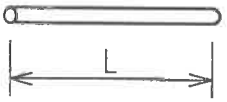
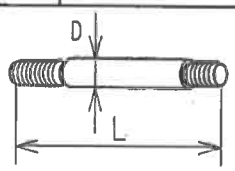
No	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
29	SPRING 5.5X23X103.5 L.O. PUMP	No. 07600 	SWPB	1	1	G811400861		L= 104 WEIGHT Kg
30	SPRING 2.3X20X86 CHECK VALVE	No. 07600 	SWPA	1	1	G005700071		L= 86 WEIGHT Kg
31	SPRING 1.2x11.2x78 FLOW CHANGE CONTROL VALVE	No. 07600 	SWPB	1	1	NN00738003		L= 78 WEIGHT Kg
32	O-RING P48A CHUTCH OPERATING VALVE	No. 08200 	NBR	2	1	Z560104857ZZ		D= 48 WEIGHT Kg
33	O-RING P10A L.O. PUMP	No. 08200 	NBR	2	1	Z560101024ZZ		D= 10 WEIGHT Kg
34	O-RING P32 L.O. PUMP	No. 08200 	NBR(1B)	2	1	Z560103235AZ		D= 32 WEIGHT Kg
35	GASKET 17 L.O. RELIEF VALVE(HIGH PRESS)	No. 08600 	SPCC NON AS COMPOUND	2	1	Z565001700EE		D= 24 WEIGHT 0.001 Kg
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN							

SPARE PARTS						PAGE	
TYPE DRA-25J(L)						BOX No.	

No	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
36	GASKET 21 L.O.RELIEF VALVE(HIGH PRESS)	No. 08600 	SPCC NON AS COMPOUND	2	1	Z565002100EE		D= 29 WEIGHT 0.002 Kg
37	GASKET 17 CHUTCH OPERATING VALVE	No. 08600 	SPCC NON AS COMPOUND	2	1	Z565001700EE		D= 24 WEIGHT 0.001 Kg
38	GASKET 21 CHUTCH OPERATING VALVE	No. 08600 	SPCC NON AS COMPOUND	3	1	Z565002100EE		D= 29 WEIGHT 0.002 Kg
39	GASKET 8 CHUTCH OPERATING VALVE	No. 08600 	C1100P-0	2	1	Z665008000ZZ		D= 12 WEIGHT 0.001 Kg
40	GASKET 10 CHUTCH OPERATING VALVE	No. 08600 	C1100P-0	4	1	Z665010000ZZ		D= 14 WEIGHT 0.001 Kg
41	GASKET 14 CHUTCH OPERATING VALVE	No. 08600 	C1100P-0	1	1	Z665014000ZZ		D= 19 WEIGHT 0.002 Kg
42	GASKET 34 FLOW CHANGE CONTROL VALVE	No. 08600 	SPCC NON AS COMPOUND	1	1	Z565003400EE		D= 42 WEIGHT 0.004 Kg

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG.CO.,LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

SPARE PARTS						PAGE	
TYPE DRA-25J(L)						BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
43	GASKET 25 FLOW CHANGE CONTROL VALVE	No. 08600 	C1100P-0 NON-AS.	1	1	Z565002500HZ		D= 32 WEIGHT Kg
44	NEEDLE BEARING NA6908 L.O. PUMP	No. 09004 	SPEC	4	1	G811400950		D= 62 B= 40 WEIGHT 0.435 Kg
45	BEARING NA6910X 265 L.O. PUMP	No. 09001 	PCHS	4	1	G003100440		D= 72 B= 40 WEIGHT Kg
46	MICRO SWITCH V-15-1E5 CHUTCH OPERATING VALVE	No. 32500 	PCHS	2	1	A001070161		WEIGHT 0.006 Kg
47	O BELT 7X555 O VELT	No. 11000 	PCHS	1	1	G001701895		L= 555 WEIGHT Kg
48	O BELT 7X810 O VELT	No. 11000 	PCHS	1	1	G001701973		L= 810 WEIGHT Kg
49	REAMER BOLT OVER 39X308	No. 	S45C	4	1	AQP20002146/A		L= 308 D= 39 WEIGHT 2.500 Kg

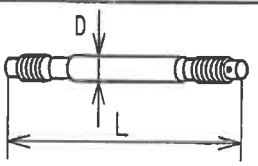
MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

SPARE PARTS

PAGE

TYPE DRA-25J(L)

BOX No.

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
50	REAMER BOLT	No. 04301 	D-SCM435	8	1		G253220100	L= 164 D= 32 WEIGHT 0.900 Kg
51	GASKET 13 L.O. RELIEF VALVE (HIGH PRESS)	No. 08600	SPCC NON AS COMPOUND	2	1		Z565001300EE	D= 20 WEIGHT 0.001 Kg ⚠
52	GASKET 10 L.O. RELIEF VALVE (HIGH PRESS)	No. 08600	SPCC NON AS COMPOUND	2	1		Z565001000EE	D= 15 WEIGHT 0.001 Kg ⚠
NO		No.						WEIGHT Kg
NO		No.						WEIGHT Kg
NO		No.						WEIGHT Kg
NO		No.						WEIGHT Kg

MFR'S NAME & ADDRESS

DAIHATSU DIESEL MFG. CO., LTD.
1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

TURBO-CHARGER(TYPE:TPS52-D/E/F)

LIST OF SPARE PARTS

TPS52-D/E/F形 過給機

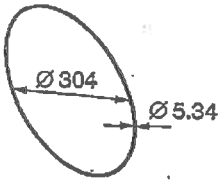
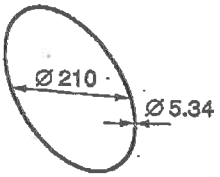
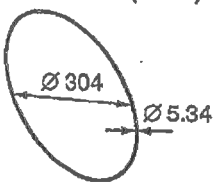
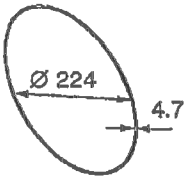
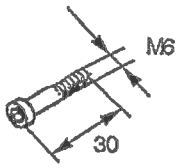
予 備 品 リ ス ト

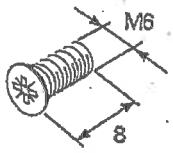
(STANDARD)

DAIHATSU DIESEL MFG. CO., LTD.
TECHNICAL DEPARTMENT
JAPAN

SHIP No.	–	DRAWN BY	T. Kawaji
TYPE	ALL	CHECKED BY	M. Kawauchi
LIST No.	B684305100ZZ		K. Shikano
DATE	Jan.07,2011	APPROVED BY	Y. Tsuchi
REVISION			

Spare 予備品							Page	(1/2)
TPS 52 形排気ガスタービン過給機 Type TPS 52 EXAUST GAS TURBOCHARGER							Ship NO.	
							Box NO.	

	No.	Name	Sketch (Mass) [kg]	Material	Supply		Drawing		Remarks
					Work'/Unit	Spare	No.	Part no.	
(1) 1	1	O-ring Oリング (NV51N10N99)	(0.045) 		1	1		81010 82010	Part of service kit 1 97070 サービスキット1 (Black)
( )1	2	O-ring Oリング (NV51N12N99)	(0.030) 		1	1		77005	Part of service kit 1 97070 サービスキット1
(1) 1	3	O-ring Oリング (NV51N10N93)	(0.040) 		1	1		42012	Part of service kit 1 97070 サービスキット1 (Green)
1	4	Gasket パッキン (NV51N05N99)	(0.050) 		1	1		52406	Part of service kit 1 97070 サービスキット1
(1) 1	5	Socket screw 六角穴付ボルト (NV51N10N96)	(0.009) 		1	1		42008	Part of service kit 1 97070 サービスキット1 (F31/F32では不使用)
MFR's name adress			IHI Corporation 株式会社 IHI						IHI

Spare 予備品							Page	(2/2)
TPS 52 形排気ガスタービン過給機							Ship NO.	
Type TPS 52 EXAUST GAS TURBOCHARGER							Box NO.	
No.	Name	Sketch (Mass) [kg]	Material	Supply		Drawing		Remarks
				Work' /Unit	Spare	No.	Part no.	
(2) 2 6	Screw 十字穴付き皿小ネジ (NV51N27N99)	(0.003)  		2	2		72041	Part of service kit 1 97070 サービスキット1 (D/E/F33では不使用)
MFR's name adress		IHI Corporation 株式会社 IHI						IHI

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SPARE PARTS LIST
FOR
INTER COOLER

DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

OSAKA JAPAN

SHIP No.		DRAWN BY	K.Aoki
TYPE	DK-20e	CHECKED BY	N.Matsuda
LIST No.	AQP10007605A		
DATE	2014.12.26	APPROVED BY	T.Tachi

2017/05/09 TECHNICAL DEPARTMENT

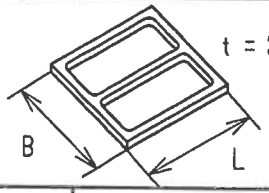
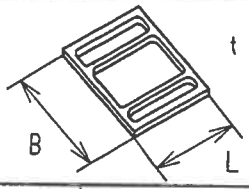
REVISION

REVISION	DATE	HISTORY OF REVISION
A	2014/12/26	FIRST DRAWING. K. A

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

2017/05/09 TECHNICAL DEPARTMENT

SPARE PARTS						PAGE	
TYPE DK-20e						BOX No.	

No.	NAME / USE	SKETCH	MATERIAL	SUPPLY WORKING PER ENG	SUPPLY SPARE PER SHIP	DRAWING No.	DRAWING PART No.	REMARKS
1	GASKET (2)	No. 72508 	SPEC	1	1	C048470900		L= 384 B= 503 WEIGHT 0.000 Kg
2	GASKET (3)	No. 72507 	SPEC	1	1	C048470970		L= 384 B= 503 WEIGHT 0.000 Kg
NO		No.						WEIGHT Kg
NO		No.						WEIGHT Kg
NO		No.						WEIGHT Kg
NO		No.						WEIGHT Kg
NO		No.						WEIGHT Kg

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG. CO., LTD. 1-1-30, OYODONAKA, KITA-KU, OSAKA, JAPAN

HYDRAULIC GOVERNOR

LIST OF SPARE PARTS

RHD型ガバナー予備品リスト

MODEL RHD6-PC & RHD10-PC

DAIHATSU DIESEL MFG. CO., LTD.

ENGINEERING DEPARTMENT

OSAKA, JAPAN

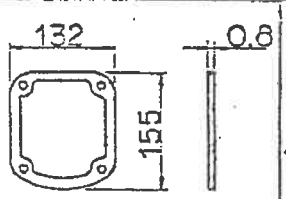
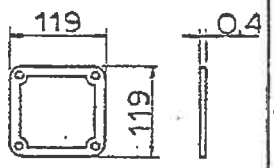
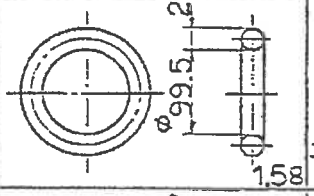
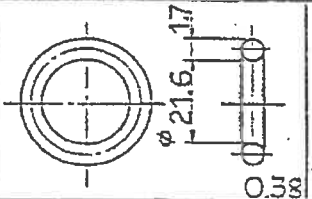
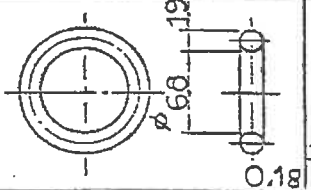
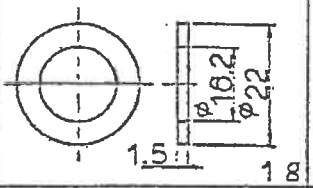
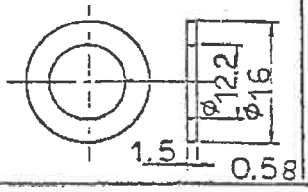
DIESEL KIKI LIST NO.RH-490-14-2

SHIP No.		DRAWN BY	H. Autsuki
TYPE	RHC6-PC/RHD10-PC	CHECKED BY	
LIST No.	B6825001 40FZ		N. Nomura
DATE	MAR 20 1987	APPROVED BY	T. Nishimura
REVISION			

B6825 | #14

2017/05/09 TECHNICAL DEPARTMENT

SPARE PARTS	1~3	4~17	18	PAGE	1
				SHIP NO.	
TYPE RHD6-PC/RHD10-PC				BOX NO.	

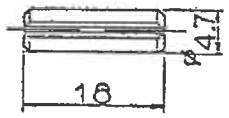
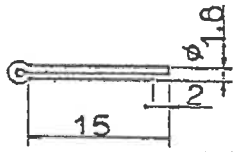
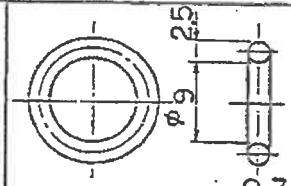
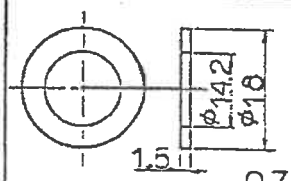
NO. 19~22	NAME	SKETCH WEIGHT 8	MATERIAL	SUPPLY PER SHIP WORKING 23~25	SUPPLY PER SHIP SPARE 26~28	DRAWING NO. 29~40	DRAWING PART NO.	REMARKS
1	Gasket ガスケット	 4 g	ASTMD -1147-59T		1	158017 -0400		
2	Gasket ガスケット	 3 g	ASTMD -1147-59T		1	158017 -0600		
3	O Ring O リング		Synthtic Rubber ニトリルゴム		1	158028 -0000		
4	O Ring O リング		Synthtic Rubber ニトリルゴム		1	029632 -2030		
5	O Ring O リング		Synthtic Rubber ニトリルゴム		1	029630 -7040		
6	Gasket ガスケット	 1 g	Copper Cup		1	029331 -6040		
7	Gasket ガスケット	 8	Copper Cup		2	026512 -1640		

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG.CO.,LTD. 1-87, 1-CHOME, OYODO-NAKA, OYODO-KU, OSAKA, JAPAN

B6825-00140FZ (2/3)

2017/05/09 TECHNICAL DEPARTMENT

SPARE PARTS	1~3	4~17	18	PAGE	2
				SHIP NO.	
TYPE RHD6-PC/RHD10-PC				BOX NO.	

NO. 19~22	NAME	SKETCH WEIGHT g	MATERIAL	SUPPLY PER SHIP WORKING 23~25	SUPPLY PER SHIP SPARE 26~28	DRAWING NO. 29~40	DRAWING PART NO.	REMARKS
8	Spring Pin スプリングピン	 1.5 g	SK 5		1	029404 -5010		
9	Split Pin スプリットピン	 0.5 g	SWRM3		3	025520 -1510		
10	O Ring O リング	 0.3 g	Synthtic Rubber ニトリルゴム		1	029630 -9050		
11	Gasket ガスケット	 0.7 g	Copper Cup		1	029331 -4120		

MFR'S NAME & ADDRESS	DAIHATSU DIESEL MFG.CO.,LTD. 1-87, 1-CHOME, OYODO-NAKA, OYODO-KU, OSAKA, JAPAN

2017/05/09 TECHNICAL DEPARTMENT

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SPARE PARTS LIST

FOR

MOTOR DRIVEN PUMP

DAIHATSU DIESEL MFG.CO.,LTD.

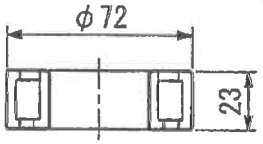
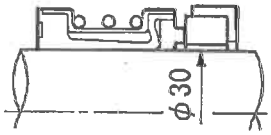
TECHNICAL DEPARTMENT

JAPAN

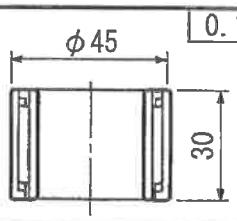
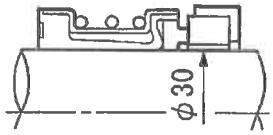
SHIP NO.		DRAWN BY	M. Tokuda
TYPE	6DKM-20eF(L)	CHECKED BY	S. Tajima
LIST NO.	AQP20008439A		
DATE	2016/6/3	APPROVED BY	S. Kawa
REVISION			

PDK20-0174-5

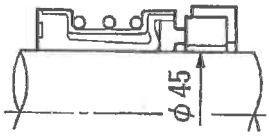
SPARE PARTS LIST 予備品目録			PAGE ページ	1
L. O. AUX. PUMP	HORIZONTAL GEAR ROTARY PUMP MODEL: ALGT-65 1 SET		SHIP NO. 番 船	
			BOX NO. 予備品箱番号	

Item NO. 項	NAME 部品名称	MASS : Kg SKETCH 写図	MATERIAL 材質	SUPPLY WORK-ING PER PUMP	SUPPLY SPARE PER SHIP	DRAWING NO. B6DS0007 NO.	DRAWING NO. B6DS0007 PART NO.	REMARKS
1	ROLLER BEARING コロ軸受	0.4 	SUJ	4	1		201	
1	MECHANICAL SEAL メカニカルシール	0.1 		1 SET	1 SET		520	

MFR'S NAME & ADDRESS	NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO. A1DP0541

SPARE PARTS LIST 予備品目録							PAGE ページ		2
R/G L.O.AUX. PUMP		HORIZONTAL GEAR ROTARY PUMP MODEL: ALGT-50H 1 SET					SHIP NO. 番船		
							BOX NO. 予備品箱番号		
Item NO. 項	NAME 部品名称	MASS : Kg SKETCH 写図	MATERIAL 材質	SUPPLY WORK-ING PER PUMP	SUPPLY SPARE PER SHIP	DRAWING NO. A1DS0472 NO.	DRAWING NO. A1DS0472 PART NO.	REMARKS	
1	NEEDLE BEARING 針状コロ軸受	0.14 	SUJ 高炭素クロム軸受鋼	4	1		201		
1	MECHANICAL SEAL メカニカルシール	0.1 		1 SET	1 SET		520		

MFR'S NAME & ADDRESS	NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO. A1DP0541

SPARE PARTS LIST 予備品目録							PAGE ページ		3
COOL. F. W. PUMP HORIZONTAL CENTRIFUGAL PUMP MODEL: CRD-100-2 1 SET							SHIP NO. 番船		
							BOX NO. 予備品箱番号		
Item NO. 項	NAME 部品名称	MASS : Kg SKETCH 写図	MATERIAL 材質	SUPPLY WORK-ING PER PUMP	SUPPLY SPARE PER SHIP	DRAWING NO. A1DS0248 NO.	DRAWING NO. A1DS0248 PART NO.	REMARKS	
1	MECHANICAL SEAL メカニカルシール	0.14 		1 SET	1 SET		520		

MFR' S NAME & ADDRESS	NANIWA PUMP MFG.CO.,LTD. 11-5, SHINMACHI 3-CHOME, NISHI-KU OSAKA JAPAN	DRAWING NO. A1DP0370

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SPARE PARTS LIST FOR STANDBY L.O. PUMP & R/G STANDBY L.O. PUMP STARTER

DAIHATSU DIESEL MFG.CO.,LTD.

TECHNICAL DEPARTMENT

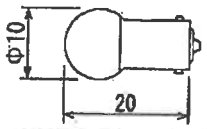
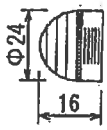
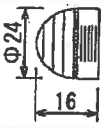
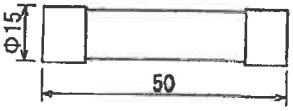
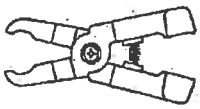
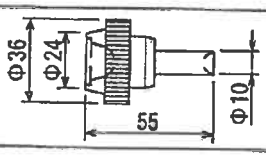
JAPAN

SHIP NO.		DRAWN BY	M. Johmoto
TYPE	6DKM-20eF(L)	CHECKED BY	M. Fujimoto
LIST NO.	AQP20008444A		
DATE	2016/6/8	APPROVED BY	S. Fuma
REVISION			

PDK20-0174-5

BOX No. P

船番 SHIP No.	予備品表 SPARE PARTS LIST FOR	用途 USE	台数 SETS PER VESS.
	STAND-BY L.O. PUMP STARTER PANEL STAND-BY R/G L.O. PUMP STARTER PANEL		

項目 ITEM No.	名称 NAME OF PARTS	外形図 OUT LINE	数量 QUANTITY 1セット PER SET	数量 QUANTITY 1船 PER VESS	数量 QUANTITY SPARE	備考 REMARKS
1	LAMP		3	12	12	6.3V 1W BA9S
2	GLOBE		1	4	1	FUJI DR9C001-W WL
3	GLOBE		1	4	1	FUJI DR9C001-G GL
4	GLOBE		1	4	1	FUJI DR9C001-R RL
5	FUSE		5	20	20	UTSUNOMIYA UC-1 500V 3A
6	FUSE NIPPER				1	YOSHIDA FH-P
7	LAMP TOOL				1	FUJI AHX702
8						
9						
	SPARE PARTS BAG		1			VINYL BAG

製造社名 MFR'S NAME	CYUGOKU ELECTRIC SERVICE CO.,LTD. SHIMONOSEKI JAPAN	DWG. No. 280603-10

AQP20008444A (2/2)

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SPARE PARTS LIST FOR L.O. FILTER

DAIHATSU DIESEL MFG.CO.,LTD.

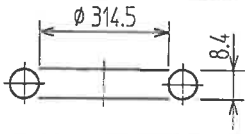
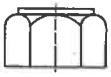
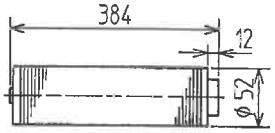
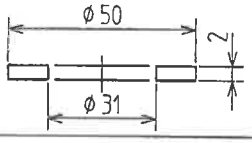
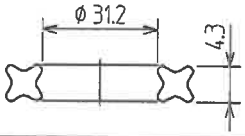
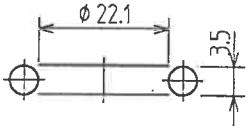
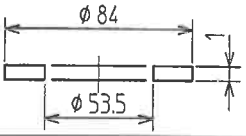
TECHNICAL DEPARTMENT

JAPAN

SHIP NO.		DRAWN BY	M. Tokunaka
TYPE	6DKM-20eF(L)	CHECKED BY	A. Tajima
LIST NO.	AQP20008440A		
DATE	2016/6/3	APPROVED BY	S. Tsuru
REVISION			

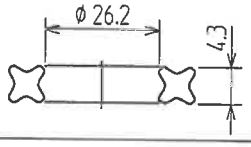
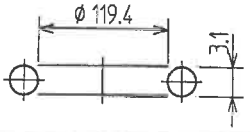
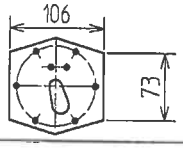
PDK20-0174-5

		PAGE	
SPARE PARTS LIST / 予備品リスト		SHIP NO.	
LUBE OIL STRAINER		BOX NO.	

NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SETS		DRAWING		REMARKS
				WORK-ING	SPARE	NO.	PART NO.	
1	O-リング O-RING		NBR	1X2	2		A-3	P315-BR
2	U-ナット U-NUT		SS400	9X2	4		A-27	M10-U-S41
3	エレメント ELEMENT		SUS304 ALUMINUM NBR	12X2	2		B-1 B-2	E4X-EL-03
4	パッキン PACKING		NBR	12X2	24		B-2	P050/31/2-BRC
5	X-リング X-RING		NBR	1X2	2		D-3	R30-BR
6	O-リング O-RING		NBR	1X2	2		D-7	P022.4-BR
7	パッキン PACKING		NON ASBESTOS	1X2	2		D-9	P-300-100004A
8								
9								
10								

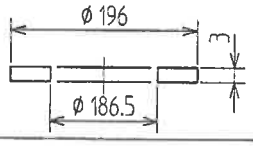
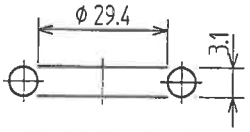
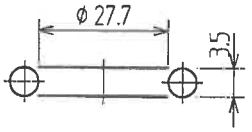
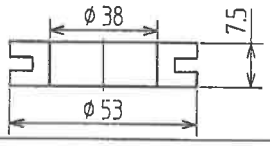
MFR'S NAME & ADDRESS	KANAGAWA KIKI KOGYO CO,. LTD. Yokohama,Japan	LIST NO.	E3X-2SP
		M.F.G. NO.	

		PAGE	
ACTUATOR SPARE PARTS LIST / アクチュエータ予備品リスト		SHIP NO.	
LUBE OIL STRAINER		BOX NO.	

NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SETS		DRAWING		REMARKS
				WORK-ING	SPARE	NO.	PART NO.	
1	X-リング X-RING		NBR	1X2	2		E-4	R25-BR
2	O-リング O-RING		NBR	2X2	2		E-7	G120-BR
3	パッキン PACKING		NON ASBESTOS	2X2	2		E-13	P-300-100040A
4								
5								
6								
7								
8								
9								
10								

MFR'S NAME & ADDRESS	KANAGAWA KIKI KOGYO CO,. LTD. Yokohama,Japan	LIST NO.	X-3050-2SP
		M.F.G. NO.	

		PAGE	
SLUDGE CHECKER SPARE PARTS LIST / スラッジ・チェッカー予備品リスト		SHIP NO.	
LUBE OIL STRAINER		BOX NO.	

NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SETS		DRAWING		REMARKS
				WORK-ING	SPARE	NO.	PART NO.	
1	パッキン / PACKING		NBR	1X2	2		H-3	AP05A-R
2	O-リング / O-RING		NBR	1X2	2		H-6	G030-BR
3	O-リング / O-RING		NBR	1X2	2		H-9	P028-BR
4	パッキン / PACKING		NBR	1X2	2		H-11	P-200-007275
5								
6								
7								
8								
9								
10								

MFR'S NAME & ADDRESS	KANAGAWA KIKI KOGYO CO,. LTD. Yokohama,Japan	LIST NO.	X-2959-2SP
		M.F.G. NO.	

2017/05/09 TECHNICAL DEPARTMENT

SHOP TRIAL PROCEDURE

: MAIN ENGINE :

BV RULE

DAIHATSU DIESEL MFG.CO.,LTD.
TECHNICAL DEPARTMENT
JAPAN

		DRAWN BY	H. Nishikawa
TYPE	ALL	CHECKED BY	S. Narita
LIST No.	AQP10001649A		
DATE	Apr., 24, 2007	APPROVED BY	R. Mori
REVISION			

1. GENERAL

1.1 Applied rule

BUREAU VERITAS (BV)

2. ITEMS OF INSPECTION

The diesel engine are designed, manufactured and shall be tested in accordance with the specified applied rule regulation. Tems are as follows ;

(1) Material inspection
(2) Parts inspection
(3) Running test
(4) Overhaul inspection
(5) Spare parts inspection

2.1 Material inspection

Material inspection shall be carried out in accordance with the maker regulation.

2.2 Parts inspection

Parts inspection, include hydraulic pressure test, shall be carried out in accordance with the maker regulation.

2.3 Running test

Running test shall be carried out under the attendance of inspector of the applied rule, ship owner and shipyard.
The engine shall be coupled with water dynamometer.
Marked * shall be carried out as manufacturer's test and data shall be submitted.

* 2.3.1 Measurement deflection. (cold condition)

Measurement shall be carried out in advance of running test.

* 2.3.2 Starting test

Starting performance test shall be carried out by actual starting air tank or the same capacity or less one installed in the factory for test.

2.3.3 Load test

Load test shall be carried out in accordance with attached "Running test date sheet". Output shows engine only. Test mode is as following list.

MARINE PROPELLER CHARACTERISTIC TEST

	TEST MODE	LOAD (%)	DURATION	MEASUREMENT
* 1	Dead slow	Min.	15 min.	1 time
* 2	Partial load	25	20 min.	1 time
* 3	Partial load	50	20 min.	1 time
* 4	Partial load	75	20 min.	1 time
* 5	Partial load	85	30 min.	1 time
* 6	Partial load	90	20 min	1 time
7	Full load	100	1 hr.	2 times
* 8	Over load	110	30 min.	1 time

* 2.3.4 Measurement of fuel consumption at 100% load

* 2.3.5 Governor test

Permanent speed variation test shall be carried out by changing the load from full load to no load by means of water dynamometer.

2.3.6 Working test for engine protecting device and automatic control device.

The working of the above devices specified ' Specifications for Daihatsu main engine " shall be tested.

* **2.3.8 Temperature measurement after running test**

Temperature measurement shall be carried out in accordance with the items specified in attached "Running test data sheet".

2.4 Overhaul inspection

Overhaul inspection shall be carried out in accordance with attached ' Table for overhaul inspection " under the condition of engine flame fitted on the sole plate. (1 eng. only)
Items not specified in the Table shall not be overhauled.
This inspection can omit on condition that the rule examiner judge needless from the past achievement.

2.5 Spare parts inspection

Quantity and dimension of spare parts shall be inspected as manufacturer.
Requesting check anything, please get in touch with us in advance.

Table for overhaul inspection

No.	Name of part	Overhaul scope	Remarks
1	Cylinder cover	1 cyl.	
2	Intake & exhaust valves	1 cyl.	
3	Piston	1 cyl.	Piston & oil ring are not overhauled
4	Piston pin metal	1 cyl.	
5	Connecting rod	1 cyl.	Piston pin metal shall be overhauled at No.4.Crank pin metal shall be overhauled at No.4.
6	Crank pin metal	1 cyl.	
7	Crank pin bolt	1 cyl.	
8	Crank shaft	1 Eng.	Crank shaft shall be fitted to the sole plate and inspected by turning.
9	Main bearing & the fitting bolts.	1 cyl.	The lower metal shall not be overhauled as long as the upper metal is not in the abnormal condition.

LOAD RUNNING TEST

Date of test :　　ENG.TYPE :　　ENG.NO. :

Kind of load				%	Standard at 100%	MIN	D.S.	25	50	75	85	100	100	110	110
Time of recording				o'clock-mh	-										
Running test time				min	-										
Speed	Engine		n	min^{-1}											
Engine output				kW											
Dynamometer read			W	kN											
Operating air pressure				MPa	For reference										
Fuel consumption				** kg/h											
				** g/kWh	≦ +5%										
Fuel pump rack reading															
Cooling water pressure			Jacket	MPa	For reference										
			Cooler	MPa	For reference										
Lub. oil pressure	Bearing			MPa											
	Turbo-charger			MPa											
Boost air pressure				MPa											
Fuel oil pressure	Eng. inlet			MPa	For reference										
Lub. oil temperature	Oil cooler		Inlet	℃											
			Outlet	℃											
	Engine inlet			℃											
Cooling water temperature	Engine		Inlet	℃											
			Outlet	℃											
	Air cooler		Inlet	℃											
			Outlet	℃											
	Oil cooler		Inlet	℃											
			Outlet	℃											
Exhaust gas temperature	Cylinderhead outlet		1	℃	Difference ≦10%										
			2	℃											
			3	℃											
			4	℃											
			5	℃											
			6	℃											
	Turbo-charger inlet		Upper	℃											
			Lower	℃											
	Turbo-charger outlet			℃											
Boost air temperature	Air cooler outlet			℃											
Maximum combustion pressure			1	MPa	Difference ≦±0.3										
			2	MPa											
			3	MPa											
			4	MPa											
			5	MPa											
			6	MPa											
W.W.Governor load indicator															
Turbo-charger speed (×100)				min^{-1}											
Other Data															
Air temp. at Turbo-charger inlet				℃											
Room temperature				℃											
F.O. temp. of F.O.flowmeter inlet				℃											
C.W. temp. of Air cooler inlet				℃											
Humidity				%											
Atmospheric pressure				hPa											

* Inspected by DAIHATSU
**Corrected value with Low heating value 42.7 MJ/kg , Accordance with ISO

(/)

FOR REFERENCE

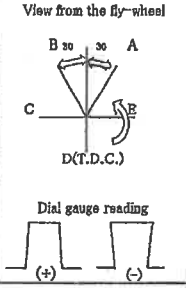
ENG.TYPE :　　ENG.NO. :

Crank-shaft deflection　　Date of test ;

Cold condition (Standard +4.8～—14.4) Unit: 1/100mm

	1	2	3	4	5	6		
A								
B	±0	±0	±0	±0	±0	±0		
C								
D								
E								

View from the fly-wheel



Governor test　　Date of test ;

	Load (%)	Description Rated	Instant	Permanent	Regulation (%) Inst.	Perma.
Engine speed (min^{-1})	100→0					
Standard					≦	≦

Engine starting test　　Date of test ;

Air receiver capacity	time	press.	time	press.	time	press.
L	1	2.45	7		13	
Initial pressuer	2		8		14	
2.45 MPa	3		9		15	
Room temp.	4		10		16	
℃	5		11		Standard	
	6		12		≧6	≦1.18

Temperature after continuous operation (℃)　　Date of test ;

Diesel engine

Cylinder NO.	1	2	3	4	5	6	7		
Main bearing									
Crank pin bearing							–		
Lub. Oil									

Protecting Device test

Date of test ;

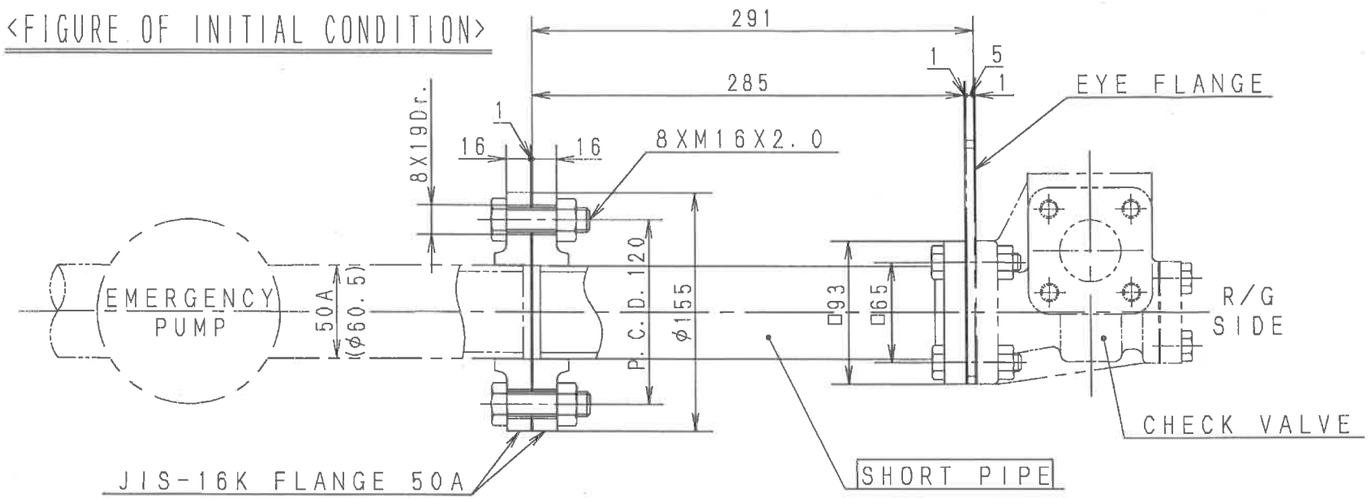
Over speed (stop)	(672～ 690) min^{-1}
L.O. pressure down (stop)	(0.20±0.01) MPa
L.O. pressure down (alarm)	(0.25±0.01) MPa
L.O. pressure down for T/C (alarm)	(0.06±0.005) MPa
F.O. pressure down (alarm)	(0.20±0.01) MPa
F.W. pressure down (alarm)	(0.15±0.01) MPa
Turning safty press. Switch (interlock)	(0.50±0.01) MPa
C.W. temp. rise (stop)	(90±3/2) ℃
C.W. temp. rise (alarm)	(85±3/2) ℃
L.O. temp. rise (alarm)	(70±3/2) ℃

Fuel oil ; Exxon mobil oil "A"	
Density ; g/cm^3 at 15℃	
Low heating value ; MJ/kg	Drawing ;
Lub. oil ; Exxon mobil motor oil #30	Checked ;
Dynamometer constant ; KW=Wn×0.7355×10^{-3}/9.80665	

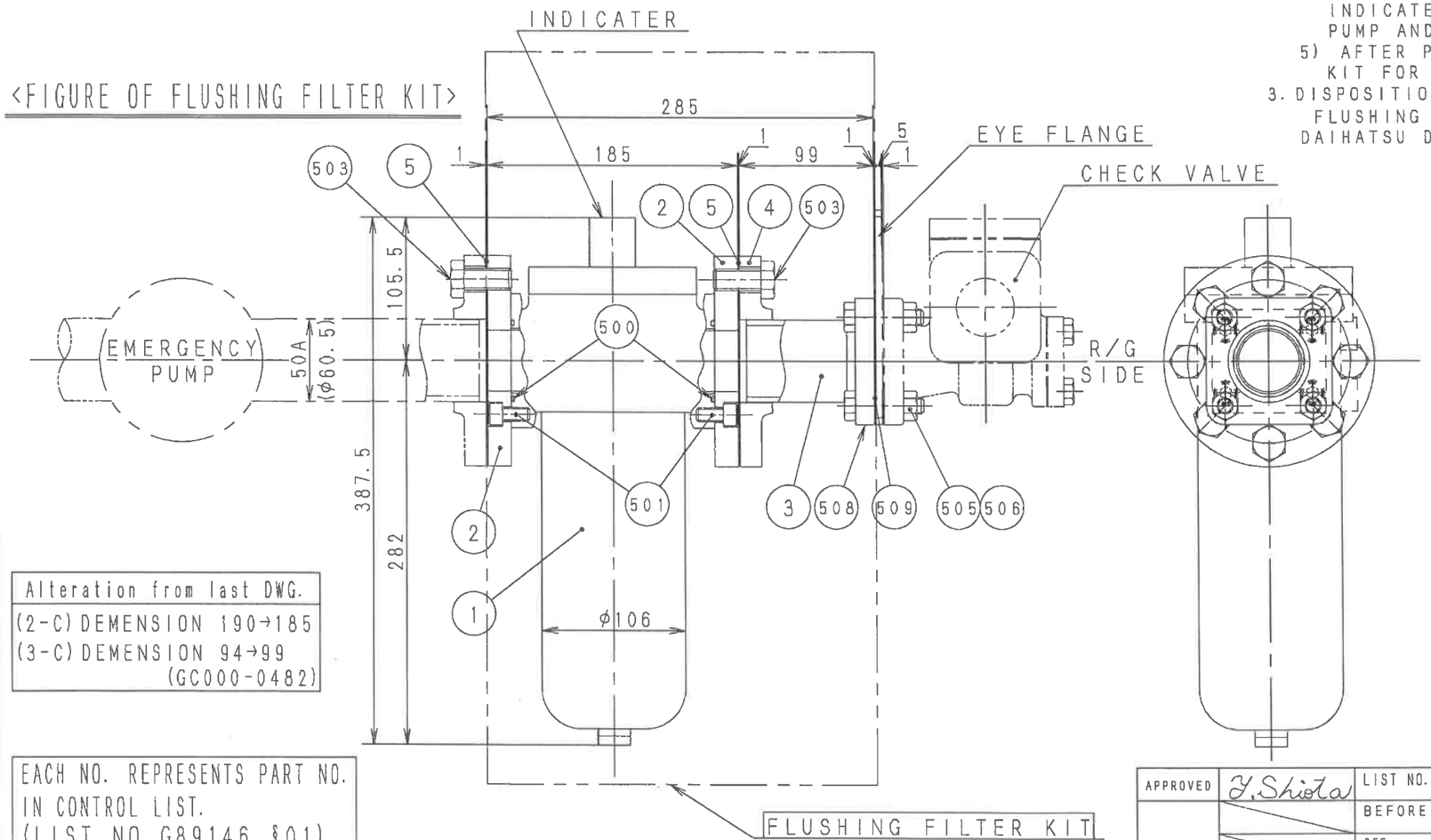
(/)

2017/05/09 TECHNICAL DEPARTMENT

<FIGURE OF INITIAL CONDITION>



<FIGURE OF FLUSHING FILTER KIT>



1. PURPOSE
   WELDDING SCALES, RUST AND DUST MAY CAUSE TROUBLE, FOR EXAMPLE STICK, AT CONTROL DEVICES OF THE REDUCTION GEAR (FOR SHORT "R/G").
   PLEASE MAKE SURE TO CONDUCT PIPE FLUSHING TO AVOID THE ABOVE FACT.
2. PROCEDURE OF PIPE FLUSHING
   1) R/G AND PIPES HAVE ALREADY BEEN CLEANED AT OUR FACTORY, BUT PLEASE CONFIRM AGAIN THAT THE DUST DOES NOT ADHERE TO THE FITTING SURFACE OF THE PIPE FLANGE.
      PLEASE CONDUCT ACID CLEANING TO ALL PIPES INSTALLED BY THE SHIPYARD (COMPLETELY REMOVE RUST AND WELDDING SCALES FROM PIPES).
   2) ASSEMBLY OF THE FLUSHING FILTER KIT (150 MESH)
      PLEASE EXCHANGE THE SHORT PIPE (L=285) FOR THE FLUSHING FILTER KIT (SEE FIG.).
   3) IT IS ALLOWABLE TO USE LUBRICATING OIL OF R/G FOR PIPE FLUSHING.
   4) PLEASE OPERATE EMERGENCY PUMP MORE THAN 60 MINUTES AND HAMMER PIPES PROPERLY TO REMOVE INNER DUST DURING PUMP OPERATING.
      FLUSHING FILTER HAS INDICATER TO CONFIRM CLOGGING. IF INDICATER SHOWS RED (USUALLY GREEN), PLEASE STOP EMERGENCY PUMP AND CLEAN ELEMENT OF THE FLUSHING FILTER.
   5) AFTER PIPE FLUSHING, PLEASE EXCHANGE THE FLUSHING FILTER KIT FOR THE SHORT PIPE AGAIN.
3. DISPOSITION OF THE FLUSHING FILTER
   FLUSHING FILTER KIT IS LENDED. PLEASE RETURN IT TO DAIHATSU DIESEL AFTER FLUSHING WORK.

NO.	PARTS NAME	QTY.
1	FLUSHING FILTER	1
2	FLANGE, FLUSHING FILTER	2
3	PIPE, FLUSHING FILTER	1
4	JIS-16K FLANGE, 50A	1
5	JIS-16K GASKET, 50A	2
500	O-RING G55	2
501	HEX. SOCKET BOLT M10X20	8
503	BOLT M16X2.0X35	16
505	BOLT M12X1.75X45	4
506	NUT M12X1.75	4
508	FLANGE (2) 60X65	1
509	GASKET (2) 60X65	1

Alteration from last DWG.
(2-C) DEMENSION 190→185
(3-C) DEMENSION 94→99
(GC000-0482)

EACH NO. REPRESENTS PART NO. IN CONTROL LIST. (LIST NO. G89146 $01)

APPROVED	T. Shiota	LIST NO. G89146	MATERIAL KUMI	GROUP 0	SCALE=1:5
CHECKED	T. Yamaguchi	BEFORE 818169 Z	MASS. 15kg	QTY. 1/ENG.	A3
		REF. –	TYPE ALL		
DRAWN	Y. TAKASE	DAIHATSU DIESEL MFG. CO., LTD. OSAKA JAPAN	NAME フラッシング ヨウリョウズ (エイゴ)		
DATE	2006.2.1		POINT FIGURE, FLUSHING		
			PARTS NO. G89146-981	A	Z
			DWG. NO. 818238		

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2017/05/09 TECHNICAL DEPARTMENT

BV-RULE

CALCULATION SHEET OF TORSIONAL VIBRATION

IN CASE OF ONE CYLINDER MISFIRING,
THE BARRED SPEED RAGES ARE "0 - 450min^{-1}" AND "ABOVE 650min^{-1}"

DAIHATSU DIESEL MFG. CO.,LTD.

TECHNICAL DEPARTMENT

MORIYAMA,JAPAN

TYPE	6DKM-20eF(L)	DRAWN BY	J. Tomabe
		CHECKED BY	T. Suzuki
LIST No.	CAL001084/A		K. Kuroki
DATE	AUG.-5,2016	APROVED BY	R. Sawa
REVISION			

PDK20-0174-5	#

§ 1. Specification for the Diesel Engine

(1) Diesel Engine

Type	: Vertical 4 cycle, Single-Acting, Solid and Direct injection with turbo cherger & inter cooler
No.of cylinder	: 6
Bore and Stroke	: 200 mm × 300 mm
Rated speed	: 900 min^{-1}
Normal output	: 1060 kWm
Firing order	: 1 - 2 - 4 - 6 - 5 - 3 (STBD SIDE)
	: 1 - 3 - 5 - 6 - 4 - 2 (PORT SIDE)
Manufacture	: DAIHATSU DIESEL MFG CO.,LTD.

(2) Flexible coupling

Bow side

Type	: DFA850
Dynamic torsional stiffness	: 0.166 MNm/rad
Transient allowable torque	: 26.2 kNm
Continuous allowable torque	: 2.45 kNm
Manufacture	: DAIHATSU DIESEL MFG CO.,LTD.

Stern side

Type	: BC41
Static torsional stiffness	: 0.116 MNm/rad
Manufacture	: GEISLINGER

(3) Reduction Gear

Type	: DRA-25J(L)
Gear ratio	: 3.752/4.612
Manufacture	: DAIHATSU DIESEL MFG CO.,LTD.

(4) Propeller shaft

Tensile strength	: 600 N/mm2

§ 2. Equivalent systems

See page 4

§ 3. The natural frequencies

- CLUTCH OFF

The 1 node vibration	N1 =	10.1	Hz	(Table 1)
The 2 node vibration	N2 =	34.6	Hz	(Table 2)
The 3 node vibration	N3 =	104.6	Hz	(Table 3)

- BOW SIDE CLUTCH ON
  STERN SIDE CLUTCH OFF

The 1 node vibration	N1 =	8.5	Hz	(Table 4)
The 2 node vibration	N2 =	14.5	Hz	(Table 5)
The 3 node vibration	N3 =	37.5	Hz	(Table 6)
The 4 node vibration	N4 =	104.2	Hz	(Table 7)

- BOW SIDE CLUTCH OFF
  STERN SIDE CLUTCH ON

The 1 node vibration	N1 =	8.7	Hz	(Table 8)
The 2 node vibration	N2 =	12.6	Hz	(Table 9)
The 3 node vibration	N3 =	34.6	Hz	(Table 10)
The 4 node vibration	N4 =	88.5	Hz	(Table 11)
The 5 node vibration	N5 =	104.6	Hz	(Table 12)

ELASTIC MASS SYSTEM

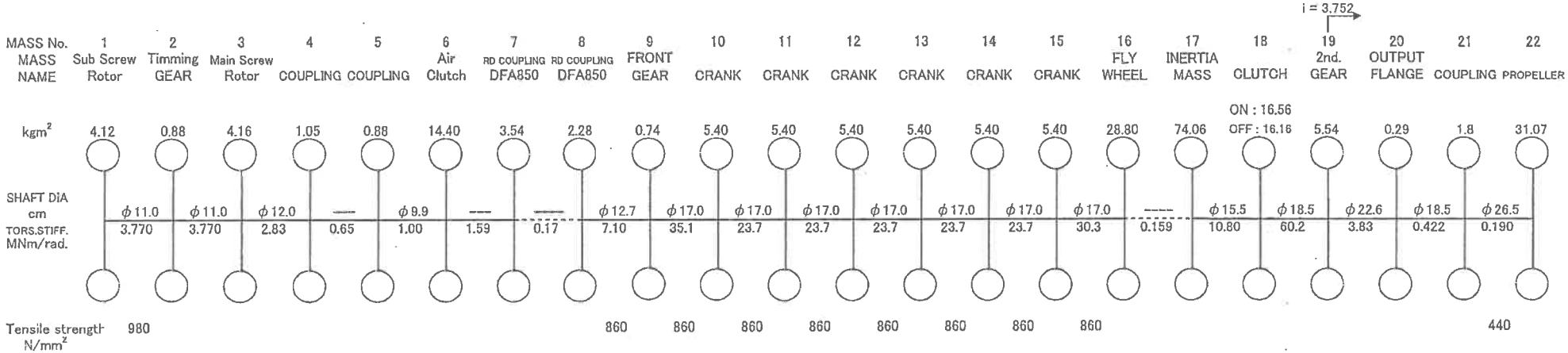


TABLE 1 HOLZER TABLE AND STRESS, TORQUE

THE 1 NODE VIBRATION

N= 10.1 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
RD COUPLING	1	3.54	0.166	---	1.000	0.1	---
RD COUPLING	2	2.28	7.10	12.7	0.913	0.2	56.9
FRONT GEAR	3	0.74	35.1	17.0	0.910	0.3	26.6
CRANK	4	5.40	23.7	17.0	0.909	0.5	47.3
CRANK	5	5.40	23.7	17.0	0.907	0.7	67.9
CRANK	6	5.40	23.7	17.0	0.905	0.9	88.5
CRANK	7	5.40	23.7	17.0	0.901	1.1	109.0
CRANK	8	5.40	23.7	17.0	0.896	1.2	129.5
CRANK	9	5.40	30.3	17.0	0.891	1.4	149.8
FLYWHEEL	10	28.80	0.159	---	0.886	2.5	---
INERTIAMASS	11	74.06	10.8	15.5	-0.676	0.4	61.1
CLUTCH OFF	12	16.16			-0.680	0.0000	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
3	203.0	5.410

TABLE 2 HOLZER TABLE AND STRESS, TORQUE

THE 2 NODE VIBRATION

N= 34.6 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
RD COUPLING	1	3.54	0.166	---	1.000	1.7	---
RD COUPLING	2	2.28	7.10	12.7	-0.011	1.7	413.6
FRONT GEAR	3	0.74	35.1	17.0	-0.034	1.7	171.2
CRANK	4	5.40	23.7	17.0	-0.039	1.6	160.8
CRANK	5	5.40	23.7	17.0	-0.046	1.4	148.7
CRANK	6	5.40	23.7	17.0	-0.052	1.3	135.0
CRANK	7	5.40	23.7	17.0	-0.057	1.2	119.8
CRANK	8	5.40	23.7	17.0	-0.062	1.0	103.4
CRANK	9	5.40	30.3	17.0	-0.066	0.8	85.8
FLYWHEEL	10	28.80	0.159	---	-0.069	-0.1	---
INERTIAMASS	11	74.06	10.8	15.5	0.003	0.0	-3.0
CLUTCH OFF	12	16.16			0.003	-0.0001	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
3	692.5	0.322

TABLE 3 HOLZER TABLE AND STRESS, TORQUE

THE 3 NODE VIBRATION

N= 104.6 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
RD COUPLING	1	3.54	0.166	---	1.000	15.3	---
RD COUPLING	2	2.28	7.10	12.7	-8.220	-65.8	-16355.7
FRONT GEAR	3	0.74	35.1	17.0	-7.293	-89.1	-9238.2
CRANK	4	5.40	23.7	17.0	-7.039	-253.3	-26255.9
CRANK	5	5.40	23.7	17.0	-5.970	-392.5	-40688.5
CRANK	6	5.40	23.7	17.0	-4.313	-493.1	-51115.1
CRANK	7	5.40	23.7	17.0	-2.231	-545.1	-56509.1
CRANK	8	5.40	23.7	17.0	0.070	-543.5	-56339.4
CRANK	9	5.40	30.3	17.0	2.365	-488.3	-50622.8
FLYWHEEL	10	28.80	0.159	---	3.978	6.3	---
INERTIAMASS	11	74.06	10.8	15.5	-0.012	2.4	331.0
CLUTCH OFF	12	16.16			-0.035	0.0000	

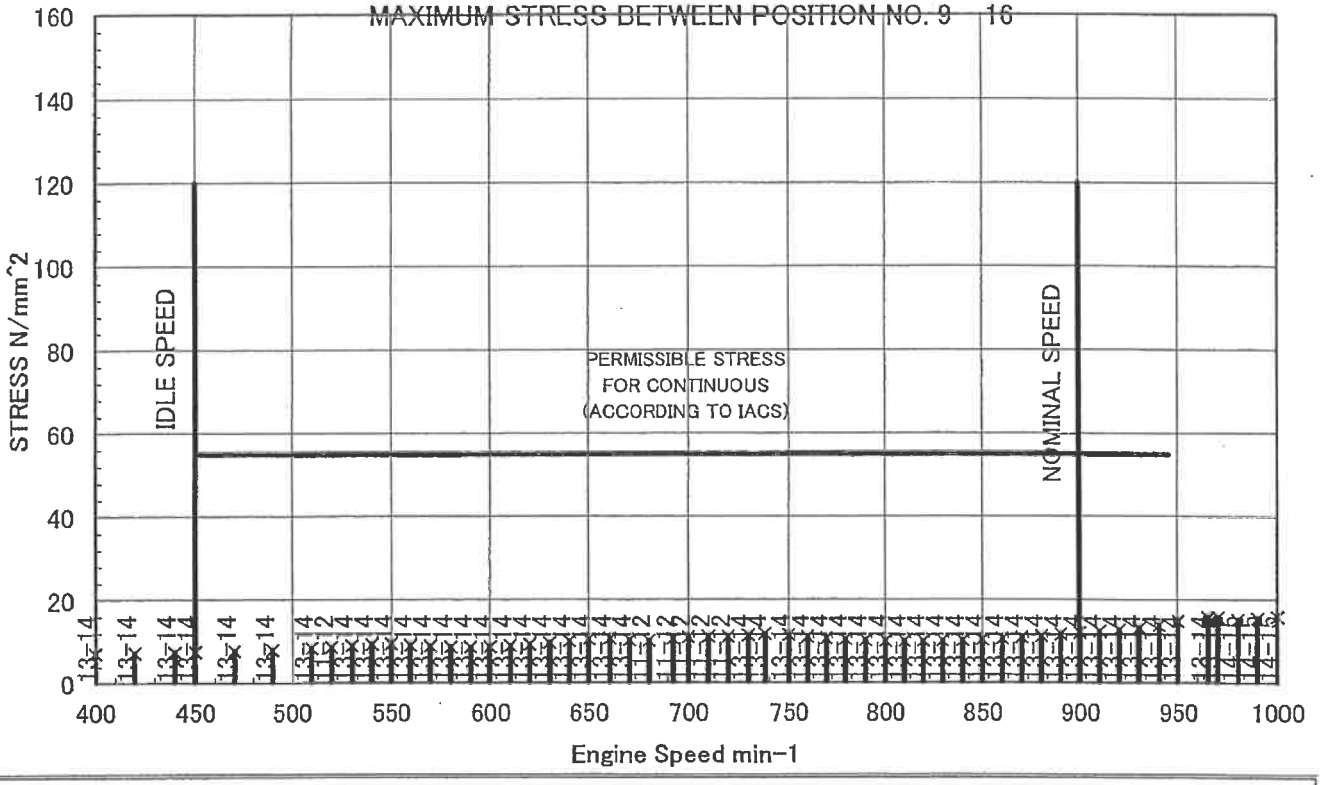
ORDRE NUMBER	ENGINE min-1	VECTOR SUM
6.5	965.2	13.894
8.5	738.1	13.894
9	697.1	17.118
9.5	660.4	13.894

CLUTCH OFF

STRESS-SPEED-DIAGRAM

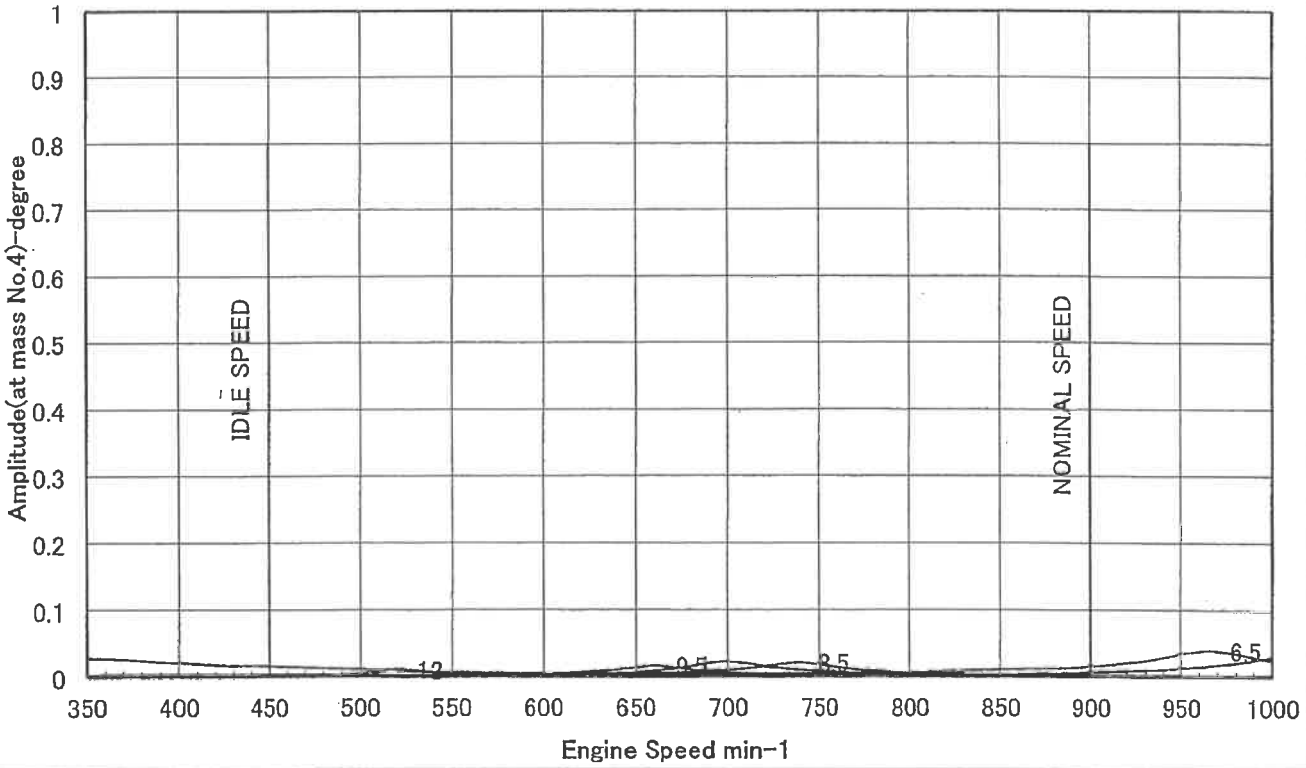
HARMONIC SYNTHESIS

MAXIMUM STRESS BETWEEN POSITION NO. 9 16



CLUTCH OFF

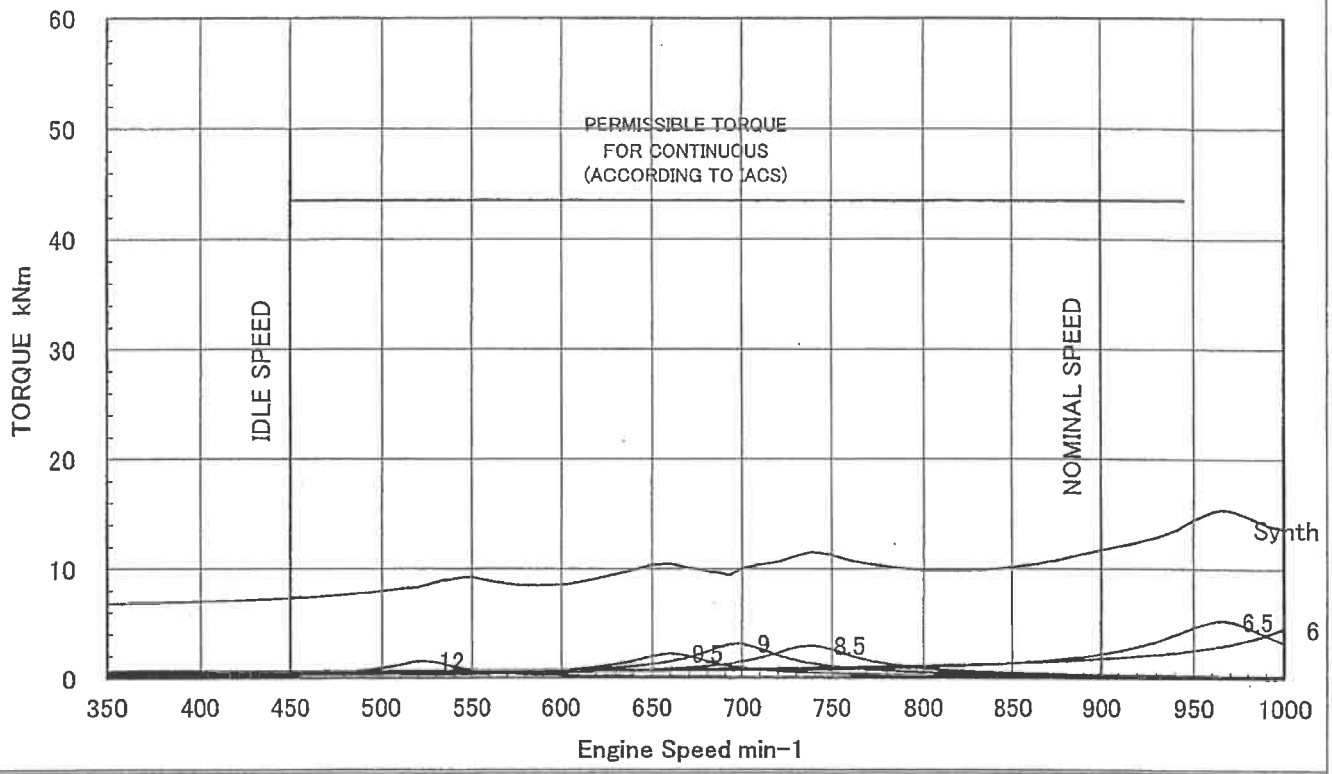
ANGULAR AMPLITUDE AT MASS No.9



CLUTCH OFF

VIBRATORY TORQUE DIAGRAM

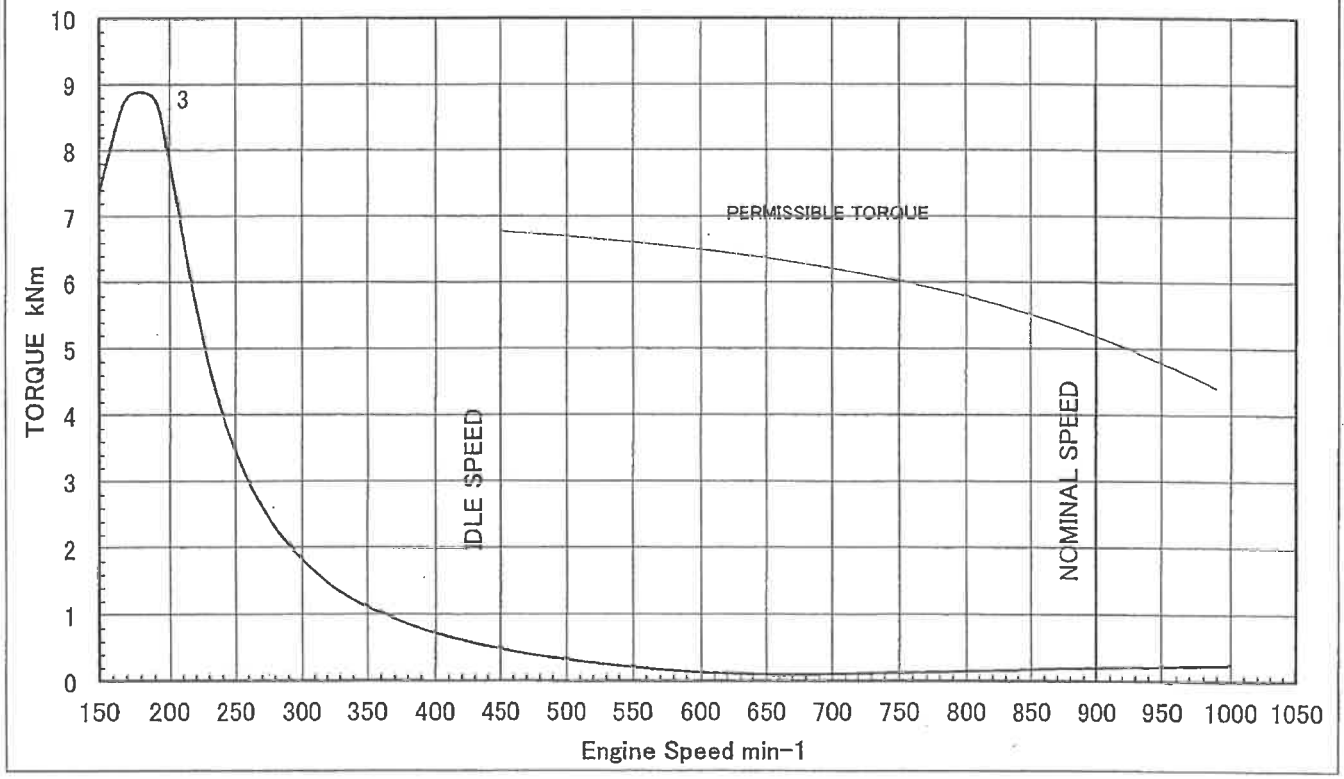
Mass No. 13 – 14 (CRANK SHAFT)



CLUTCH OFF

VIBRATORY TORQUE DIAGRAM

Mass No. 16 – 17 (GEISLINGER COUPLING : STERN SIDE)



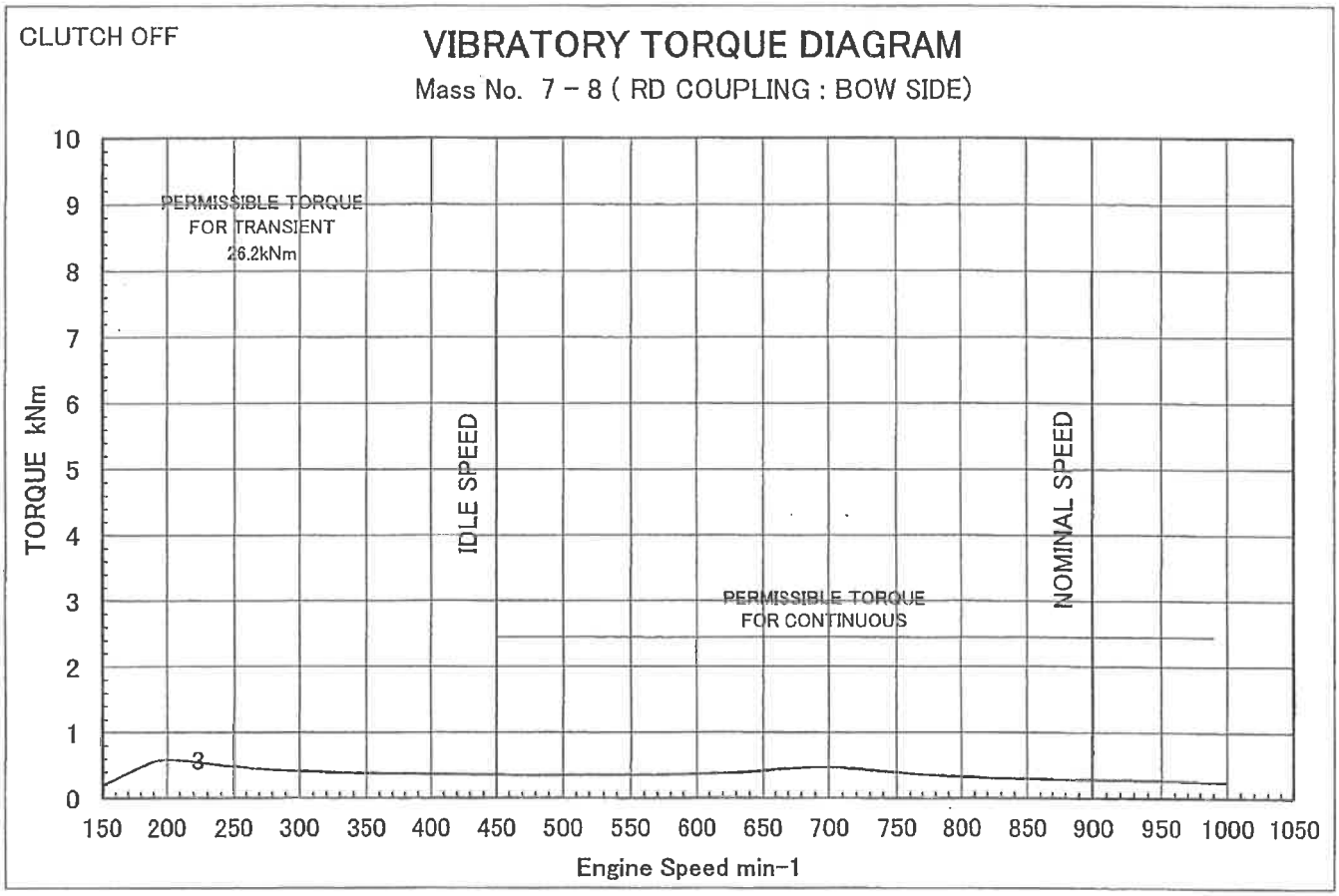

CLUTCH OFF
VIBRATORY TORQUE DIAGRAM
Mass No. 7 – 8 (RD COUPLING : BOW SIDE)
PERMISSIBLE TORQUE
FOR TRANSIENT
26.2kNm
IDLE SPEED
NOMINAL SPEED
PERMISSIBLE TORQUE
FOR CONTINUOUS
3
TORQUE kNm
0
1
2
3
4
5
6
7
8
9
10
150 200 250 300 350 400 450 500 550 600 650 700 750 800 850 900 950 1000 1050
Engine Speed min-1

TABLE 4 HOLZER TABLE AND STRESS, TORQUE

THE 1 NODE VIBRATION

N= 8.5 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
SUB SCREW ROTOR	1	4.12	3.77	11.0	1.000	0.1	45.3
TIMMING GEAR	2	0.88	3.77	11.0	0.997	0.1	55.0
MAIN SCREW ROTOR	3	4.16	2.83	12.0	0.993	0.3	77.4
COUPLING	4	1.05	0.645	---	0.984	0.3	---
COUPLING	5	0.88	1.000	9.9	0.938	0.3	165.8
AIR CLUTCH	6	14.40	1.59	---	0.907	0.7	---
RD COUPLING	7	3.54	0.166	---	0.863	0.8	---
RD COUPLING	8	2.28	7.10	12.7	0.393	0.8	200.2
FRONT GEAR	9	0.74	35.1	17.0	0.382	0.8	84.3
CRANK	10	5.40	23.7	17.0	0.379	0.9	90.4
CRANK	11	5.40	23.7	17.0	0.376	0.9	96.5
CRANK	12	5.40	23.7	17.0	0.372	1.0	102.5
CRANK	13	5.40	23.7	17.0	0.368	1.0	108.4
CRANK	14	5.40	23.7	17.0	0.363	1.1	114.3
CRANK	15	5.40	30.3	17.0	0.359	1.2	120.0
FLYWHEEL	16	28.80	0.159	---	0.355	1.5	---
INERTIAMASS	17	74.06	10.8	15.5	-0.559	0.3	35.8
CLUTCH OFF	18	16.16			-0.561	0.0007	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
3	170.7	2.216

TABLE 5 HOLZER TABLE AND STRESS, TORQUE

THE 2 NODE VIBRATION

N= 14.5 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
SUB SCREW ROTOR	1	4.12	3.77	11.0	1.000	0.3	130.1
TIMMING GEAR	2	0.88	3.77	11.0	0.991	0.4	157.7
MAIN SCREW ROTOR	3	4.16	2.83	12.0	0.980	0.7	220.6
COUPLING	4	1.05	0.645	---	0.954	0.8	---
COUPLING	5	0.88	1.000	9.9	0.825	0.9	467.8
AIR CLUTCH	6	14.40	1.59	---	0.736	1.8	---
RD COUPLING	7	3.54	0.166	---	0.625	1.9	---
RD COUPLING	8	2.28	7.10	12.7	-0.551	1.8	458.6
FRONT GEAR	9	0.74	35.1	17.0	-0.577	1.8	187.6
CRANK	10	5.40	23.7	17.0	-0.582	1.5	160.6
CRANK	11	5.40	23.7	17.0	-0.589	1.3	133.4
CRANK	12	5.40	23.7	17.0	-0.594	1.0	105.9
CRANK	13	5.40	23.7	17.0	-0.599	0.8	78.3
CRANK	14	5.40	23.7	17.0	-0.602	0.5	50.4
CRANK	15	5.40	30.3	17.0	-0.604	0.2	22.5
FLYWHEEL	16	28.80	0.159	---	-0.604	-1.2	---
INERTIAMASS	17	74.06	10.8	15.5	0.163	-0.2	-30.2
CLUTCH OFF	18	16.16			0.165	-0.0002	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
3	289.2	3.569

TABLE 6 HOLZER TABLE AND STRESS, TORQUE

THE 3 NODE VIBRATION

N= 37.5 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
SUB SCREW ROTOR	1	4.12	3.77	11.0	1.000	2.3	876.8
TIMMING GEAR	2	0.88	3.77	11.0	0.939	2.8	1053.2
MAIN SCREW ROTOR	3	4.16	2.83	12.0	0.866	4.8	1401.8
COUPLING	4	1.05	0.645	---	0.698	5.2	---
COUPLING	5	0.88	1.000	9.9	-0.102	5.1	2684.0
AIR CLUTCH	6	14.40	1.59	---	-0.614	0.2	---
RD COUPLING	7	3.54	0.166	---	-0.626	-1.0	---
RD COUPLING	8	2.28	7.10	12.7	0.001	-1.0	-258.0
FRONT GEAR	9	0.74	35.1	17.0	0.015	-1.0	-106.9
CRANK	10	5.40	23.7	17.0	0.018	-1.0	-101.3
CRANK	11	5.40	23.7	17.0	0.022	-0.9	-94.3
CRANK	12	5.40	23.7	17.0	0.026	-0.8	-86.1
CRANK	13	5.40	23.7	17.0	0.030	-0.7	-76.9
CRANK	14	5.40	23.7	17.0	0.033	-0.6	-66.7
CRANK	15	5.40	30.3	17.0	0.036	-0.5	-55.6
FLYWHEEL	16	28.80	0.159	---	0.037	0.1	---
INERTIAMASS	17	74.06	10.8	15.5	-0.001	0.0	1.6
CLUTCH OFF	18	16.16			-0.001	0.0002	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
3	750.8	0.165
3.5	643.5	0.025
6	375.4	0.165

TABLE 7 HOLZER TABLE AND STRESS, TORQUE

THE 4 NODE VIBRATION

N= 104.2 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
SUB SCREW ROTOR	1	4.12	3.77	11.0	1.000	17.7	6756.4
TIMMING GEAR	2	0.88	3.77	11.0	0.532	19.7	7526.5
MAIN SCREW ROTOR	3	4.16	2.83	12.0	0.010	19.9	5851.7
COUPLING	4	1.05	0.645	---	-0.692	16.7	---
COUPLING	5	0.88	1.000	9.9	-3.287	4.3	2259.1
AIR CLUTCH	6	14.40	1.59	---	-3.717	-225.3	---
RD COUPLING	7	3.54	0.166	---	10.429	-67.0	---
RD COUPLING	8	2.28	7.10	12.7	50.857	431.2	107203.3
FRONT GEAR	9	0.74	35.1	17.0	44.780	573.5	59450.4
CRANK	10	5.40	23.7	17.0	43.149	1573.0	163064.6
CRANK	11	5.40	23.7	17.0	36.508	2418.7	250732.2
CRANK	12	5.40	23.7	17.0	26.297	3027.9	313880.0
CRANK	13	5.40	23.7	17.0	13.514	3340.9	346332.5
CRANK	14	5.40	23.7	17.0	-0.590	3327.3	344916.2
CRANK	15	5.40	30.3	17.0	-14.636	2988.2	309769.4
FLYWHEEL	16	28.80	0.159	---	-24.509	-39.1	---
INERTIAMASS	17	74.06	10.8	15.5	0.076	-14.8	-2024.2
CLUTCH OFF	18	16.16			0.214	-0.0005	

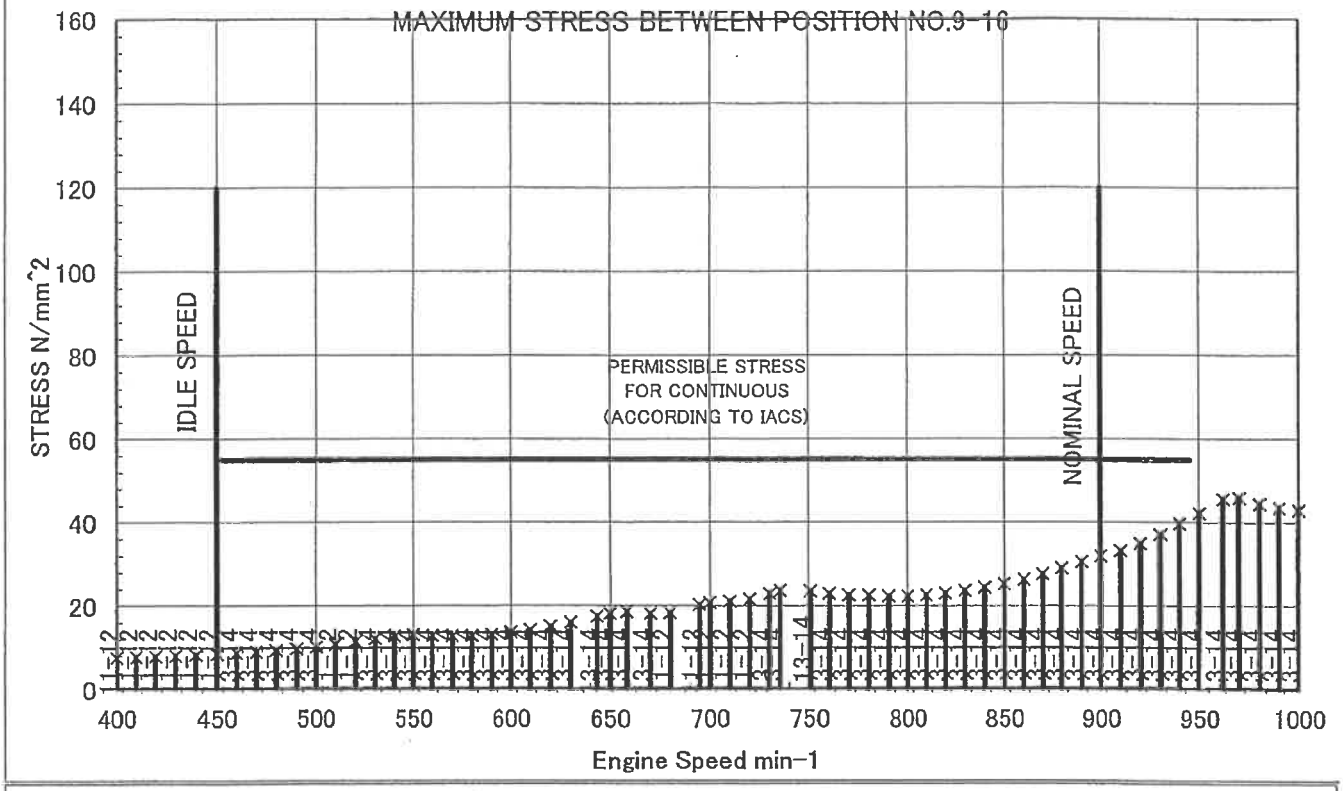
ORDRE NUMBER	ENGINE min-1	VECTOR SUM
6.5	961.9	85.363
8.5	735.6	85.363
9	694.7	104.242
9.5	658.2	85.363

BOW SIDE CLUTCH : ON
STERN SIDE CLUTCH : OFF

STRESS-SPEED-DIAGRAM

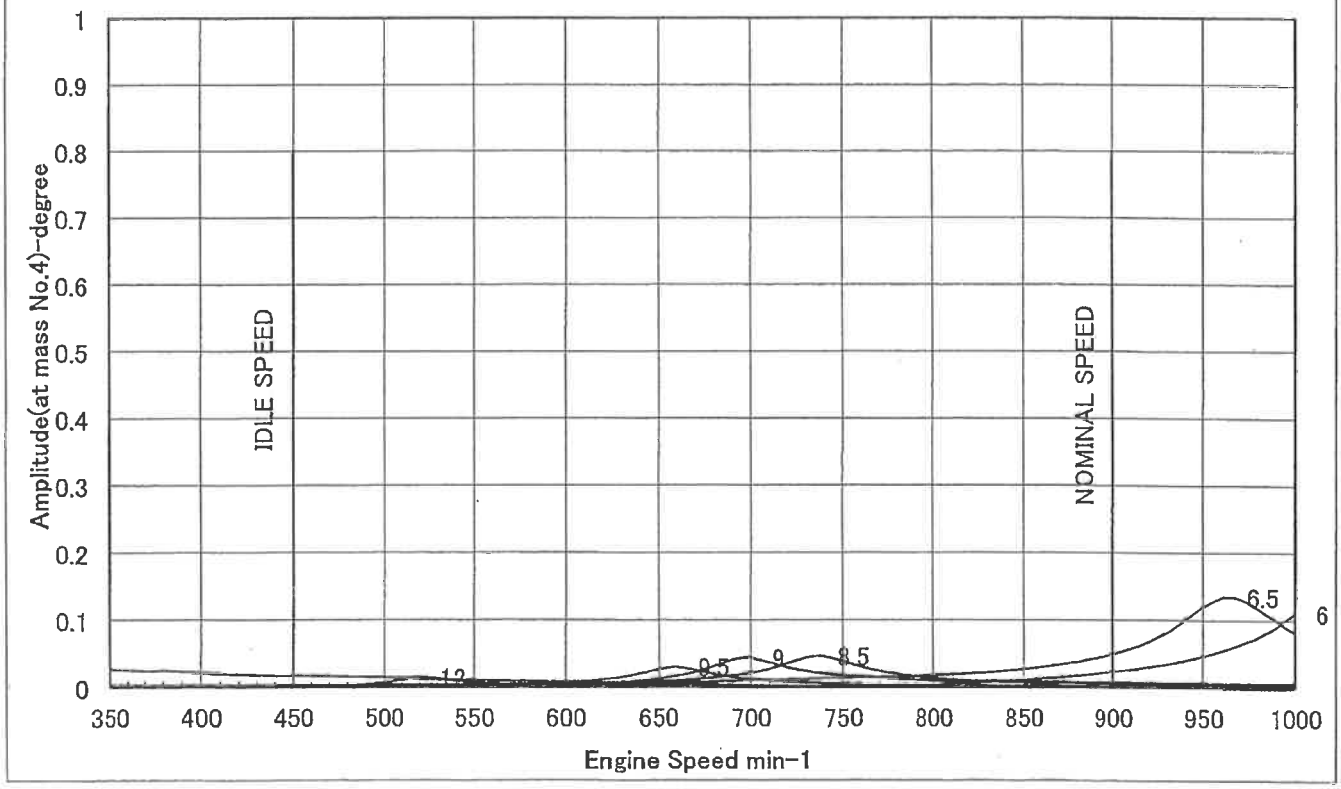
HARMONIC SYNTHESIS

MAXIMUM STRESS BETWEEN POSITION NO.9-16



BOW SIDE CLUTCH : ON
STERN SIDE CLUTCH : OFF

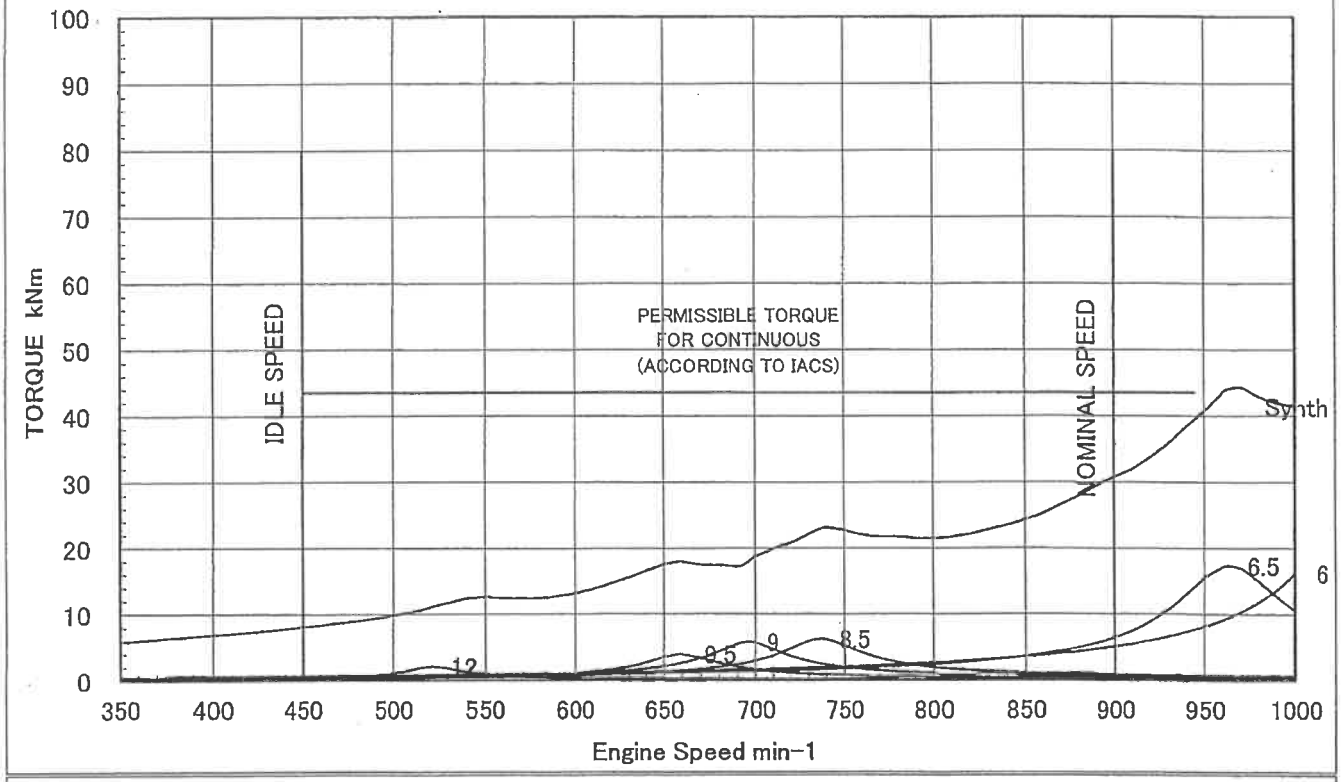
ANGULAR AMPLITUDE AT MASS No.9



BOW SIDE CLUTCH : ON
STERN SIDE CLUTCH : OFF

VIBRATORY TORQUE DIAGRAM

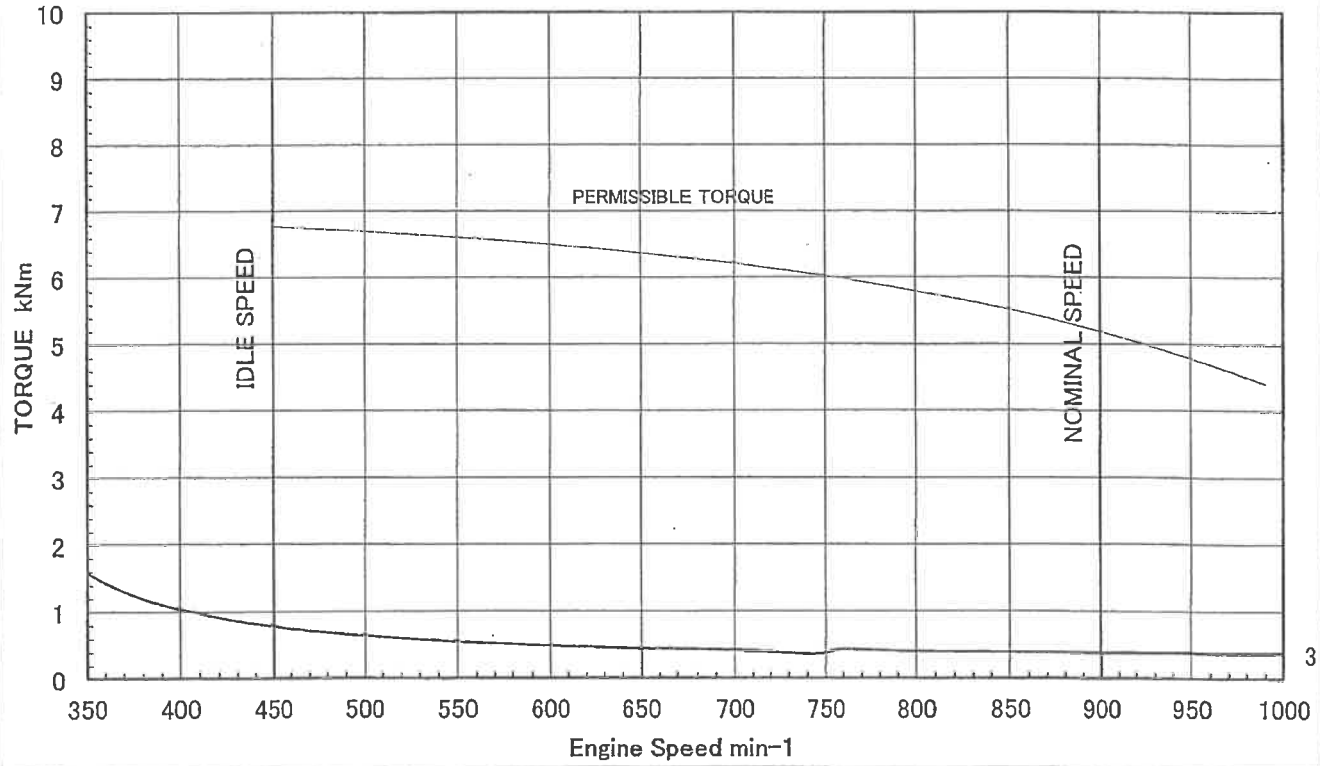
Mass No. 13 – 14 (CRANK SHAFT)



BOW SIDE CLUTCH : ON
STERN SIDE CLUTCH : OFF

VIBRATORY TORQUE DIAGRAM

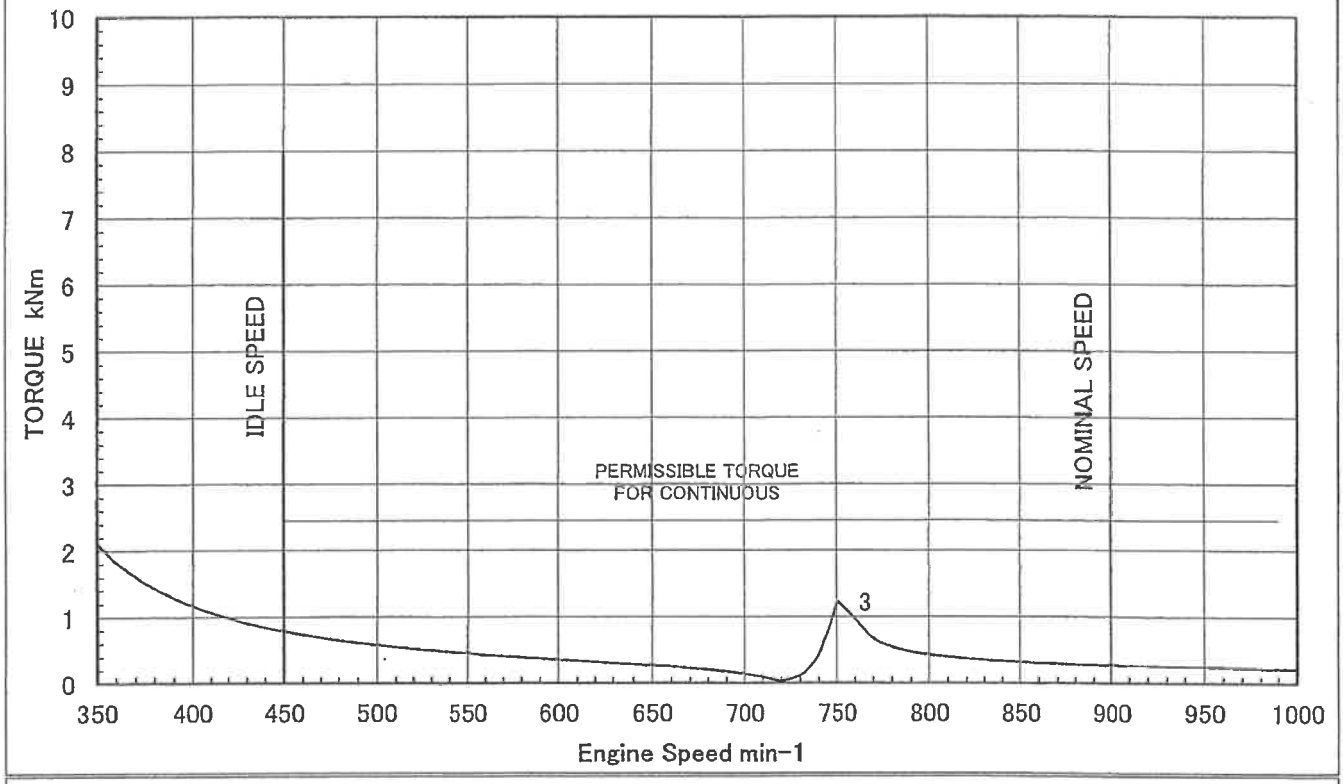
Mass No. 16 – 17 (GEISLINGER COUPLING : STERN SIDE)



BOW SIDE CLUTCH : ON
STERN SIDE CLUTCH : OFF

VIBRATORY TORQUE DIAGRAM

Mass No. 7 – 8 (RD COUPLING : BOW SIDE)



BOW SIDE CLUTCH : ON
STERN SIDE CLUTCH : OFF

VIBRATORY TORQUE DIAGRAM

Mass No. 2 – 3 (PUMP SHAFT)

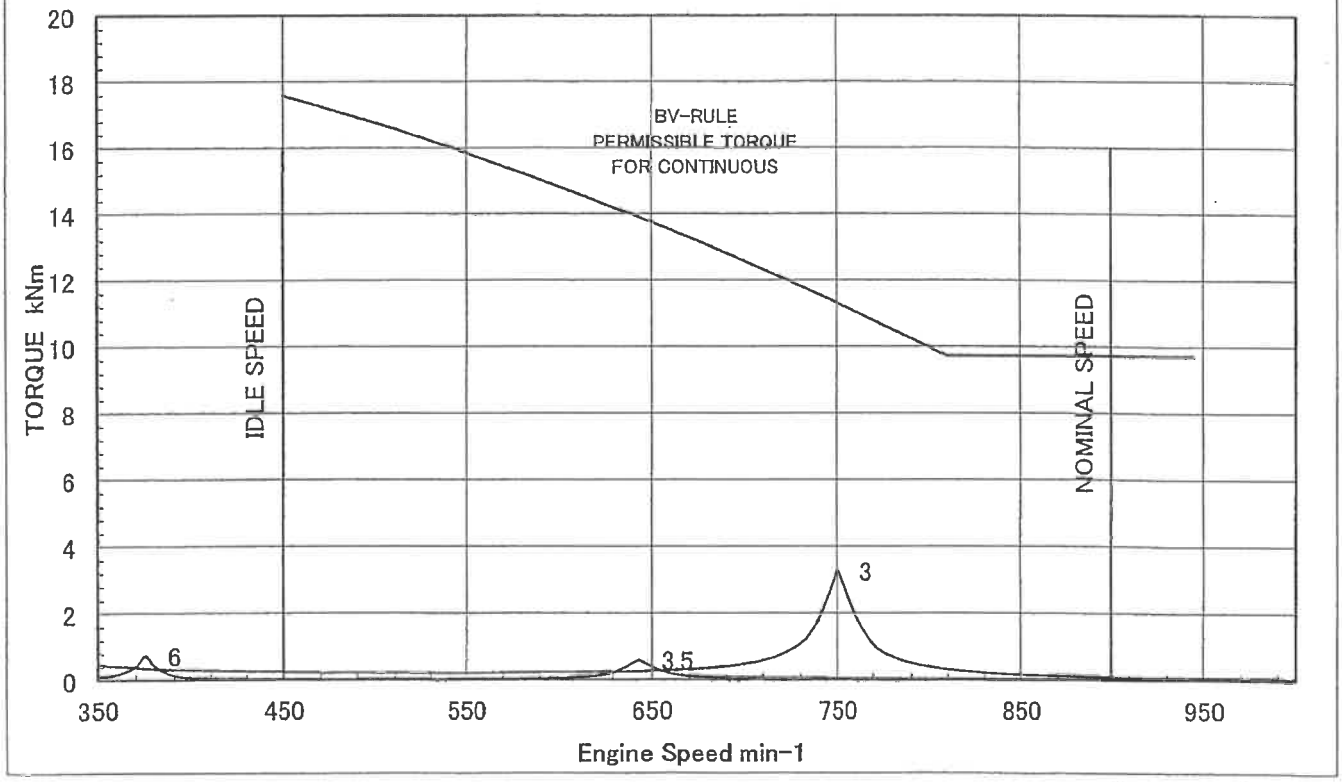


TABLE 8 HOLZER TABLE AND STRESS, TORQUE

THE 1 NODE VIBRATION

N= 8.7 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
RD COUPLING	1	3.54	0.166	---	1.000	0.1	---
RD COUPLING	2	2.28	7.10	12.7	0.936	0.2	42.5
FRONT GEAR	3	0.74	35.1	17.0	0.933	0.2	19.9
CRANK	4	5.40	23.7	17.0	0.933	0.3	35.6
CRANK	5	5.40	23.7	17.0	0.931	0.5	51.3
CRANK	6	5.40	23.7	17.0	0.929	0.6	66.9
CRANK	7	5.40	23.7	17.0	0.927	0.8	82.5
CRANK	8	5.40	23.7	17.0	0.923	0.9	98.1
CRANK	9	5.40	30.3	17.0	0.919	1.1	113.6
FLYWHEEL	10	28.80	0.159	---	0.916	1.9	---
INERTIAMASS	11	74.06	10.8	15.5	-0.273	1.3	175.2
CLUTCH ON	12	16.56	60.2	18.5	-0.285	1.1	92.1
2nd.GEAR	13	5.54	3.83	22.6	-0.287	1.1	48.1
OUTPUT FLANGE	14	0.29	0.422	18.5	-0.315	1.1	87.5
COUPLING	15	1.80	0.190	26.5	-0.573	1.1	28.9
PROPELLER	16	31.07			-1.131	0.0001	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
3	174.6	5.562

TABLE 9 HOLZER TABLE AND STRESS, TORQUE

THE 2 NODE VIBRATION

N= 12.6 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
RD COUPLING	1	3.54	0.166	---	1.000	0.2	---
RD COUPLING	2	2.28	7.10	12.7	0.866	0.3	86.0
FRONT GEAR	3	0.74	35.1	17.0	0.861	0.4	40.0
CRANK	4	5.40	23.7	17.0	0.860	0.7	70.2
CRANK	5	5.40	23.7	17.0	0.857	1.0	100.3
CRANK	6	5.40	23.7	17.0	0.853	1.3	130.3
CRANK	7	5.40	23.7	17.0	0.848	1.5	160.0
CRANK	8	5.40	23.7	17.0	0.841	1.8	189.6
CRANK	9	5.40	30.3	17.0	0.834	2.1	218.8
FLYWHEEL	10	28.80	0.159	---	0.827	3.6	---
INERTIAMASS	11	74.06	10.8	15.5	-1.441	-3.1	-422.1
CLUTCH ON	12	16.56	60.2	18.5	-1.412	-4.6	-368.0
2nd.GEAR	13	5.54	3.83	22.6	-1.405	-5.0	-222.4
OUTPUT FLANGE	14	0.29	0.422	18.5	-1.273	-5.1	-407.3
COUPLING	15	1.80	0.190	26.5	-0.073	-5.1	-138.8
PROPELLER	16	31.07			2.604	-0.0002	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
3	252.0	5.094

TABLE 10 HOLZER TABLE AND STRESS, TORQUE

THE 3 NODE VIBRATION

N= 34.6 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
RD COUPLING	1	3.54	0.166	---	1.000	1.7	---
RD COUPLING	2	2.28	7.10	12.7	-0.011	1.7	413.6
FRONT GEAR	3	0.74	35.1	17.0	-0.034	1.7	171.2
CRANK	4	5.40	23.7	17.0	-0.039	1.6	160.8
CRANK	5	5.40	23.7	17.0	-0.046	1.4	148.7
CRANK	6	5.40	23.7	17.0	-0.052	1.3	135.0
CRANK	7	5.40	23.7	17.0	-0.057	1.2	119.9
CRANK	8	5.40	23.7	17.0	-0.062	1.0	103.4
CRANK	9	5.40	30.3	17.0	-0.066	0.8	85.8
FLYWHEEL	10	28.80	0.159	---	-0.069	-0.1	---
INERTIAMASS	11	74.06	10.8	15.5	0.002	0.0	-3.6
CLUTCH ON	12	16.56	60.2	18.5	0.003	0.0	-0.4
2nd.GEAR	13	5.54	3.83	22.6	0.003	0.0	0.1
OUTPUT FLANGE	14	0.29	0.422	18.5	0.003	0.0	0.2
COUPLING	15	1.80	0.190	26.5	0.002	0.0	0.1
PROPELLER	16	31.07			0.000	0.0000	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
3	692.5	0.322

TABLE 11 HOLZER TABLE AND STRESS, TORQUE

THE 4 NODE VIBRATION

N= 88.5 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
RD COUPLING	1	3.54	0.166	---	1.000	11.0	---
RD COUPLING	2	2.28	7.10	12.7	-5.610	-28.7	-7139.7
FRONT GEAR	3	0.74	35.1	17.0	-5.206	-40.7	-4214.8
CRANK	4	5.40	23.7	17.0	-5.090	-125.8	-13037.9
CRANK	5	5.40	23.7	17.0	-4.559	-202.0	-20940.7
CRANK	6	5.40	23.7	17.0	-3.706	-264.0	-27365.2
CRANK	7	5.40	23.7	17.0	-2.592	-307.3	-31857.9
CRANK	8	5.40	23.7	17.0	-1.294	-329.0	-34101.7
CRANK	9	5.40	30.3	17.0	0.094	-327.4	-33938.2
FLYWHEEL	10	28.80	0.159	---	1.176	-222.5	---
INERTIAMASS	11	74.06	10.8	15.5	141.207	32148.6	4396810.0
CLUTCH ON	12	16.56	60.2	18.5	-156.786	24112.3	1948979.3
2nd.GEAR	13	5.54	3.83	22.6	-196.864	20736.7	914924.9
OUTPUT FLANGE	14	0.29	0.422	18.5	-738.825	20064.1	1613892.5
COUPLING	15	1.80	0.190	26.5	-5495.703	-10623.7	-290742.8
PROPELLER	16	31.07			110.466	0.0000	

ORDRE NUMBER	ENGINE min-1	VECTOR SUM
6.5	817.3	7.605
8.5	625.0	7.605
9	590.2	17.147
9.5	559.2	7.605

TABLE 12 HOLZER TABLE AND STRESS, TORQUE

THE 5 NODE VIBRATION

N= 104.6 Hz

MASS NAME	MASS NO.	EQ.MASS kgm2	TOR.STIFF. MNm/rad.	SHAFT DIA cm	AMP. radian	TORQUE x1E5 Nm	STRESS N/mm2
RD COUPLING	1	3.54	0.166	---	1.000	15.3	---
RD COUPLING	2	2.28	7.10	12.7	-8.220	-65.8	-16355.5
FRONT GEAR	3	0.74	35.1	17.0	-7.293	-89.1	-9238.1
CRANK	4	5.40	23.7	17.0	-7.039	-253.3	-26255.6
CRANK	5	5.40	23.7	17.0	-5.970	-392.5	-40688.2
CRANK	6	5.40	23.7	17.0	-4.313	-493.1	-51114.7
CRANK	7	5.40	23.7	17.0	-2.231	-545.1	-56508.7
CRANK	8	5.40	23.7	17.0	0.070	-543.5	-56339.1
CRANK	9	5.40	30.3	17.0	2.365	-488.3	-50622.6
FLYWHEEL	10	28.80	0.159	---	3.978	6.3	---
INERTIAMASS	11	74.06	10.8	15.5	-0.009	3.6	489.3
CLUTCH ON	12	16.56	60.2	18.5	-0.042	0.6	47.7
2nd.GEAR	13	5.54	3.83	22.6	-0.043	-0.4	-19.1
OUTPUT FLANGE	14	0.29	0.422	18.5	-0.031	-0.5	-38.0
COUPLING	15	1.80	0.190	26.5	0.081	0.2	4.2
PROPELLER	16	31.07			-0.001	0.0000	

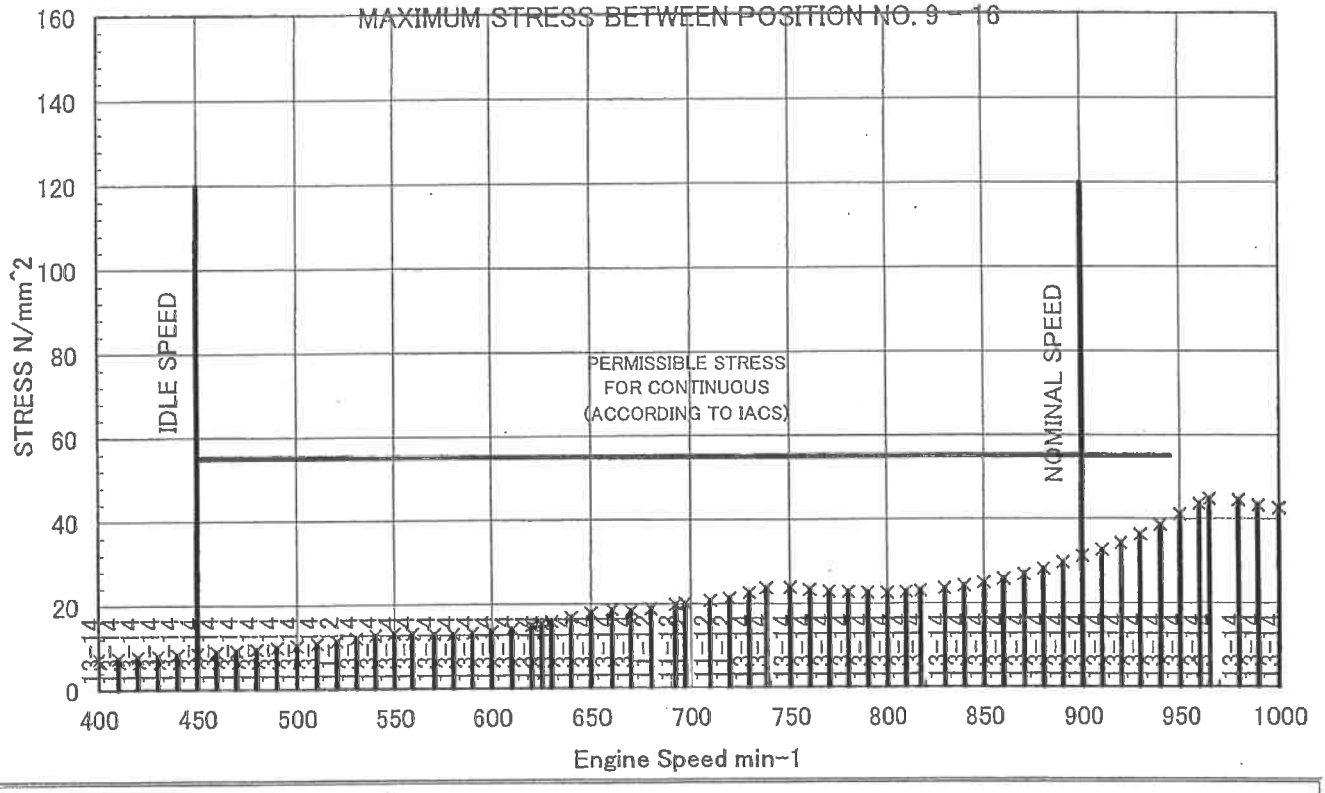
ORDRE NUMBER	ENGINE min-1	VECTOR SUM
6.5	965.2	13.894
8.5	738.1	13.894
9	697.1	17.118
9.5	660.4	13.894

BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

STRESS-SPEED-DIAGRAM

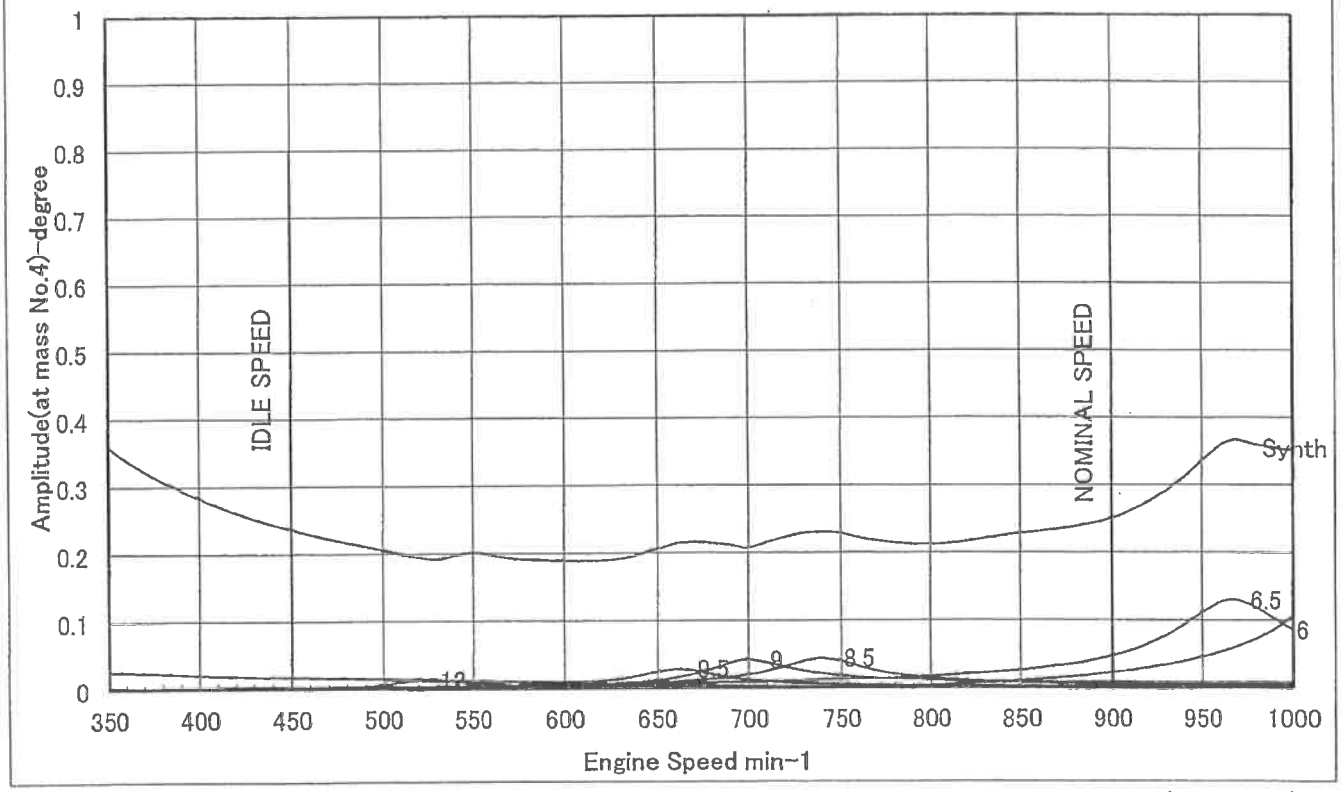
HARMONIC SYNTHESIS

MAXIMUM STRESS BETWEEN POSITION NO. 9 - 16



BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

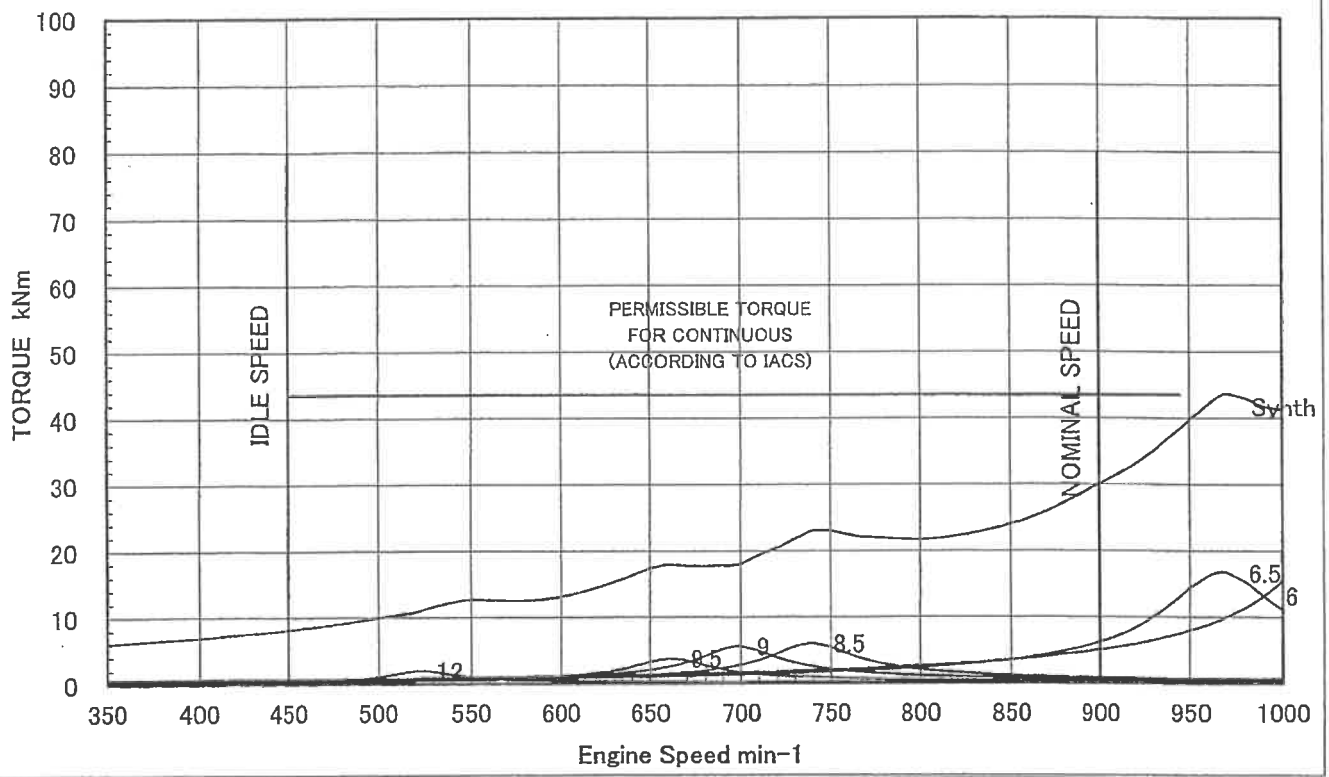
ANGULAR AMPLITUDE AT MASS No.9



BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

VIBRATORY TORQUE DIAGRAM

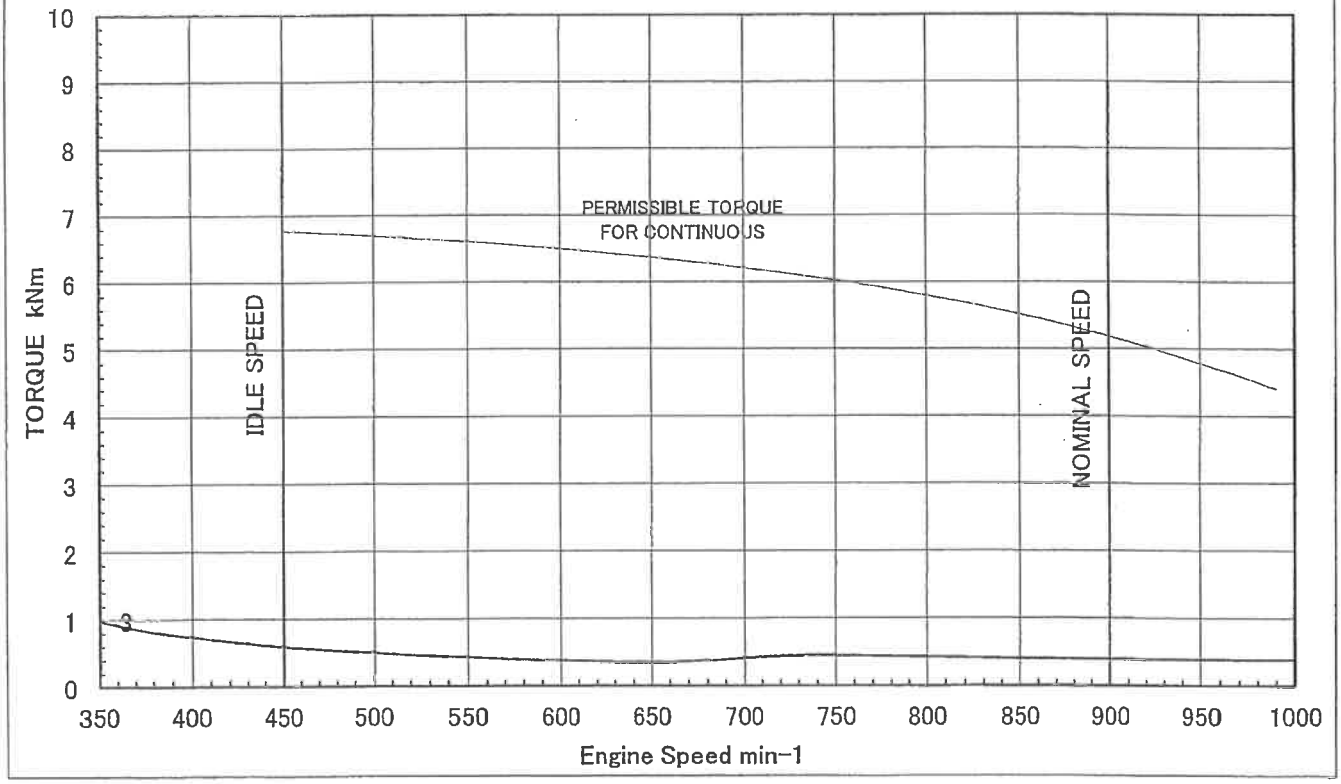
Mass No. 13 - 14 (CRANK SHAFT)



BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

VIBRATORY TORQUE DIAGRAM

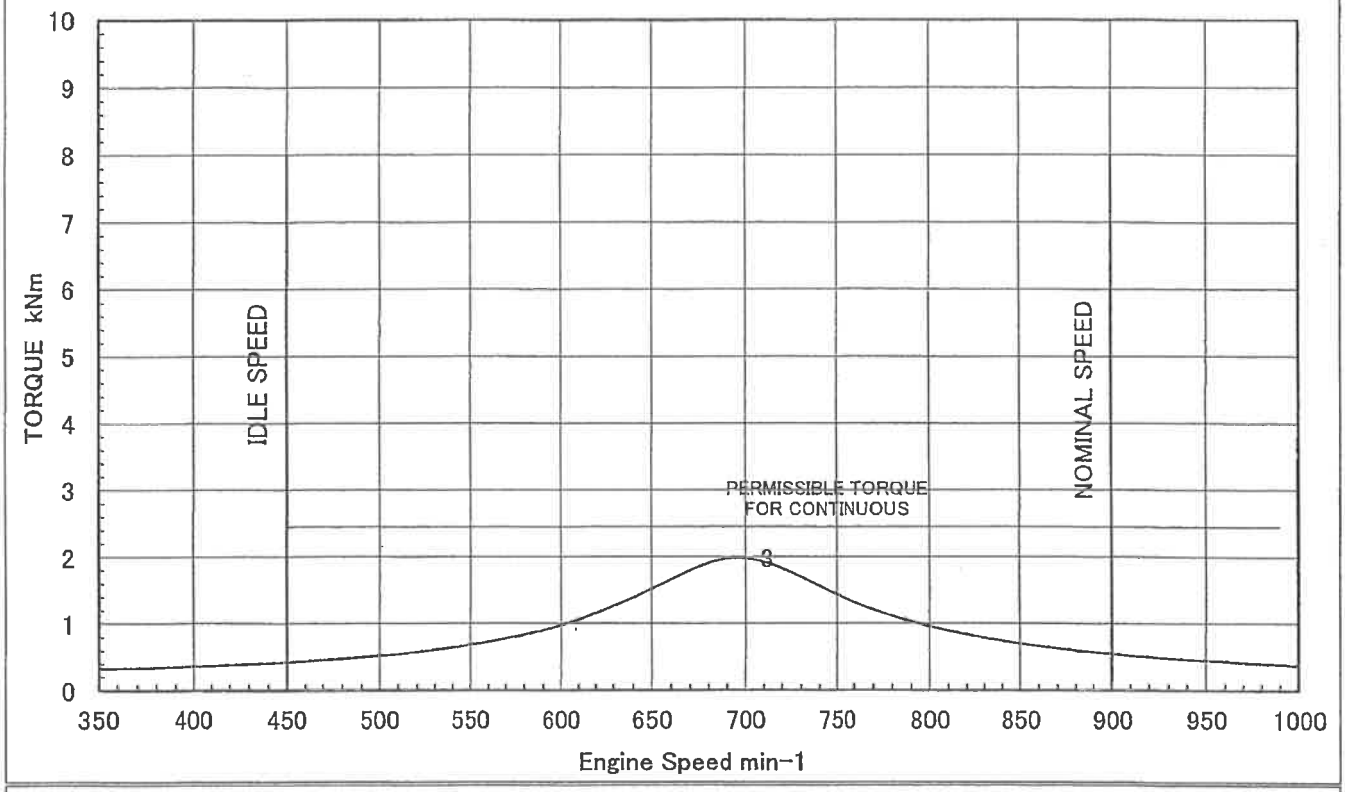
Mass No. 11 - 12 (GEISLINGER COUPLING : STERN SIDE)



BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

VIBRATORY TORQUE DIAGRAM

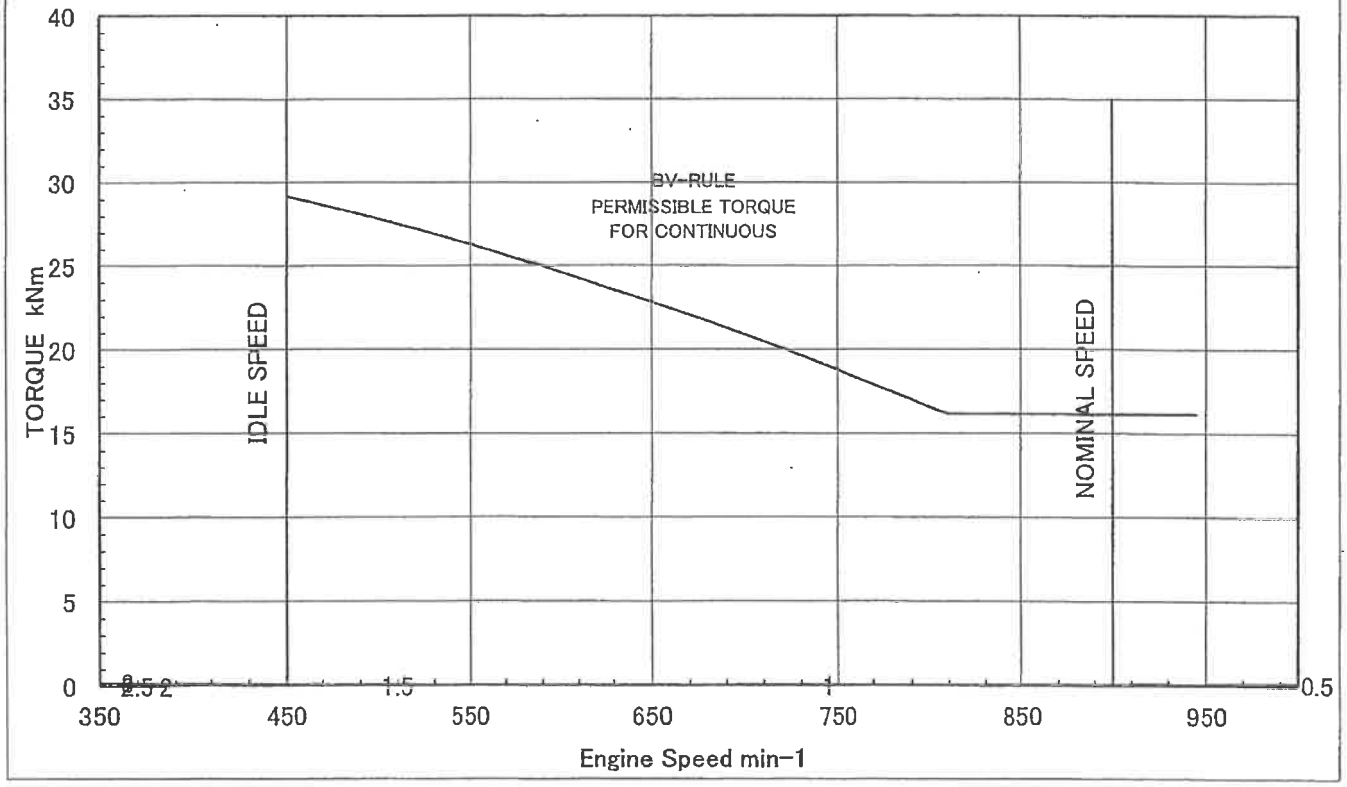
Mass No. 7 – 8 (RD COUPLING : BOW SIDE)



BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

VIBRATORY TORQUE DIAGRAM

Mass No. 21 – 22 (PROPELLER SHAFT)

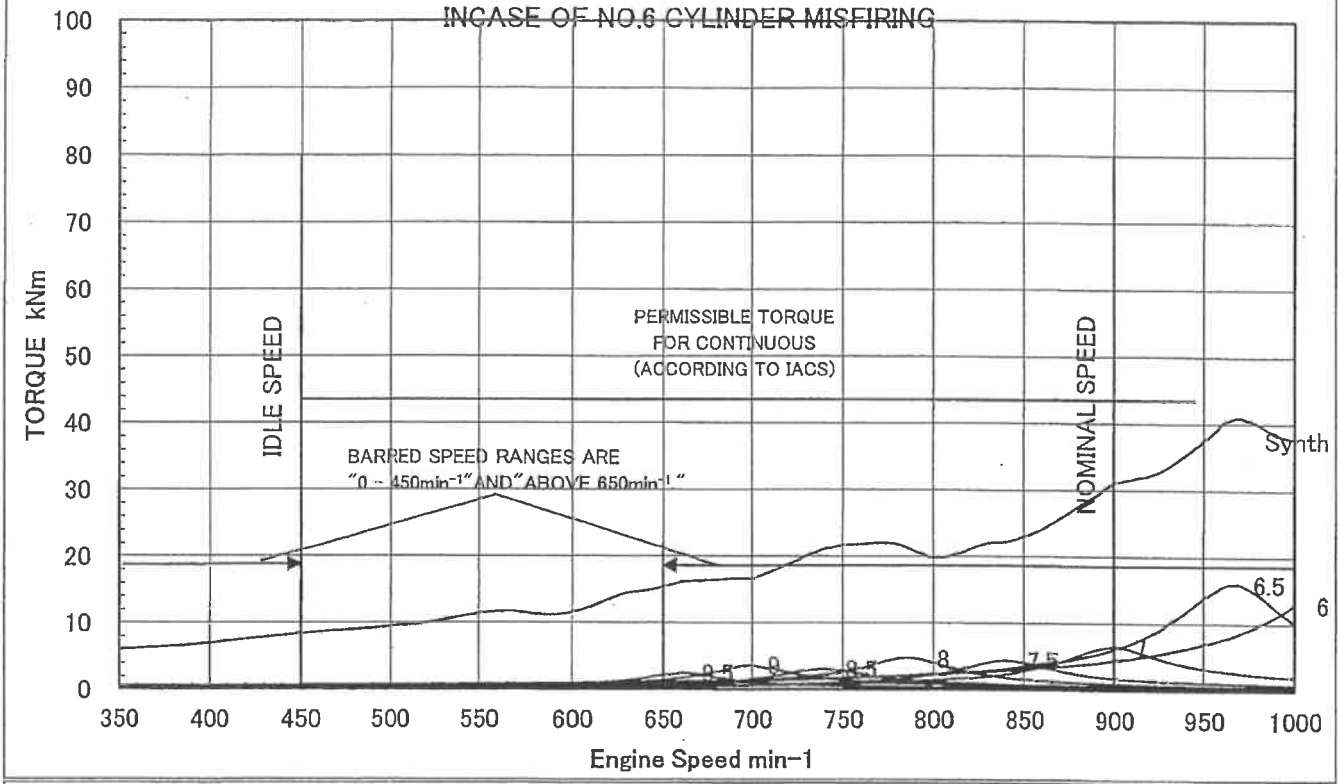


BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

VIBRATORY TORQUE DIAGRAM

Mass No. 13 – 14 (CRANK SHAFT)

INCASE OF NO.6 CYLINDER MISFIRING

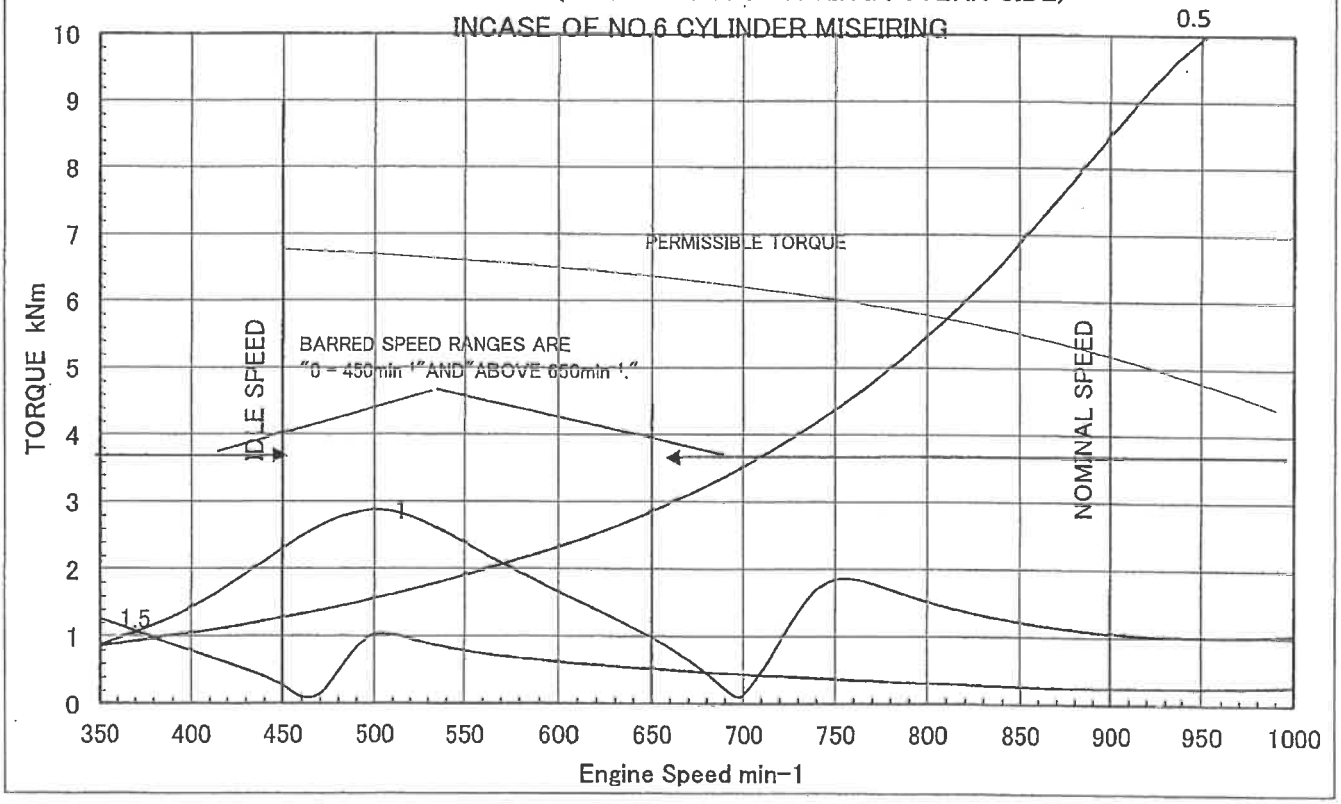


BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

VIBRATORY TORQUE DIAGRAM

Mass No. 16 – 17 (GEISLINGER COUPLING : STERN SIDE)

INCASE OF NO.6 CYLINDER MISFIRING

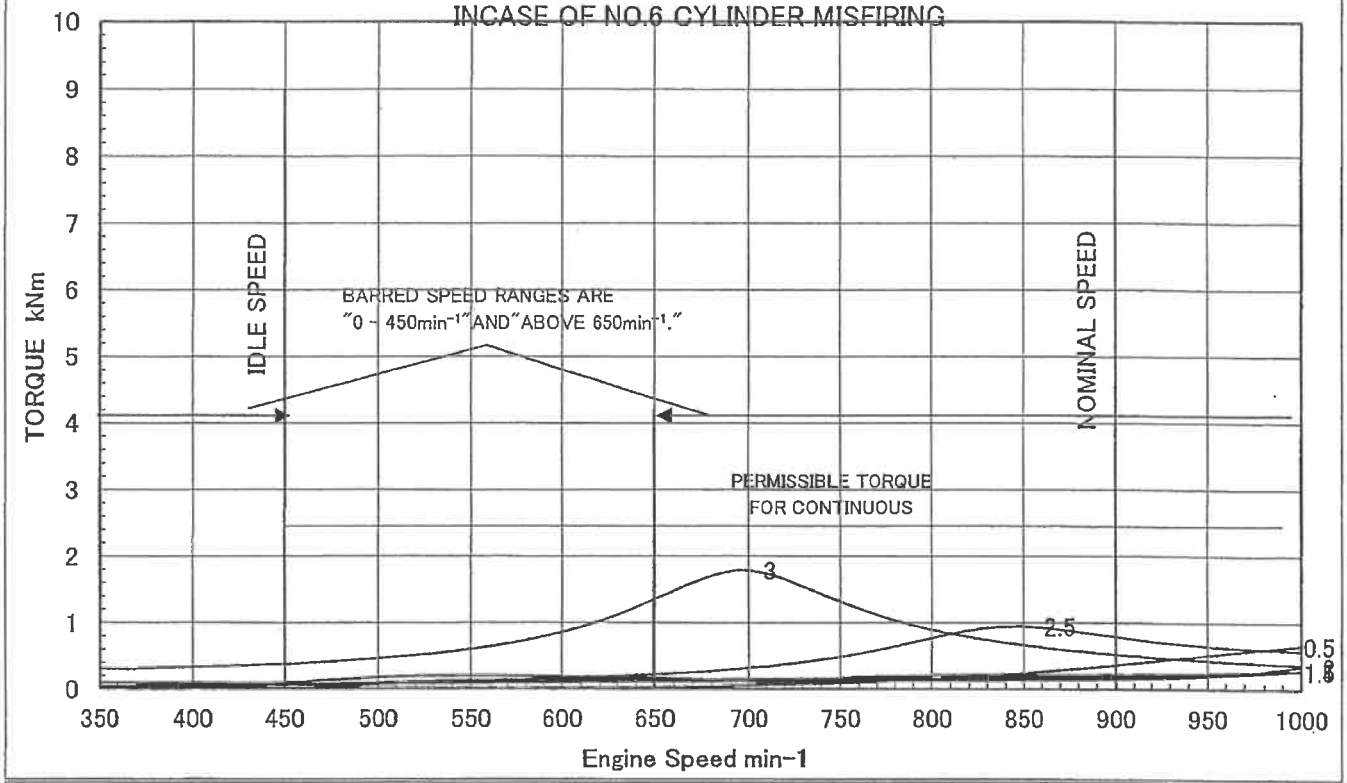


BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

VIBRATORY TORQUE DIAGRAM

Mass No. 7 – 8 (RD COUPLING : BOW SIDE)

INCASE OF NO.6 CYLINDER MISFIRING

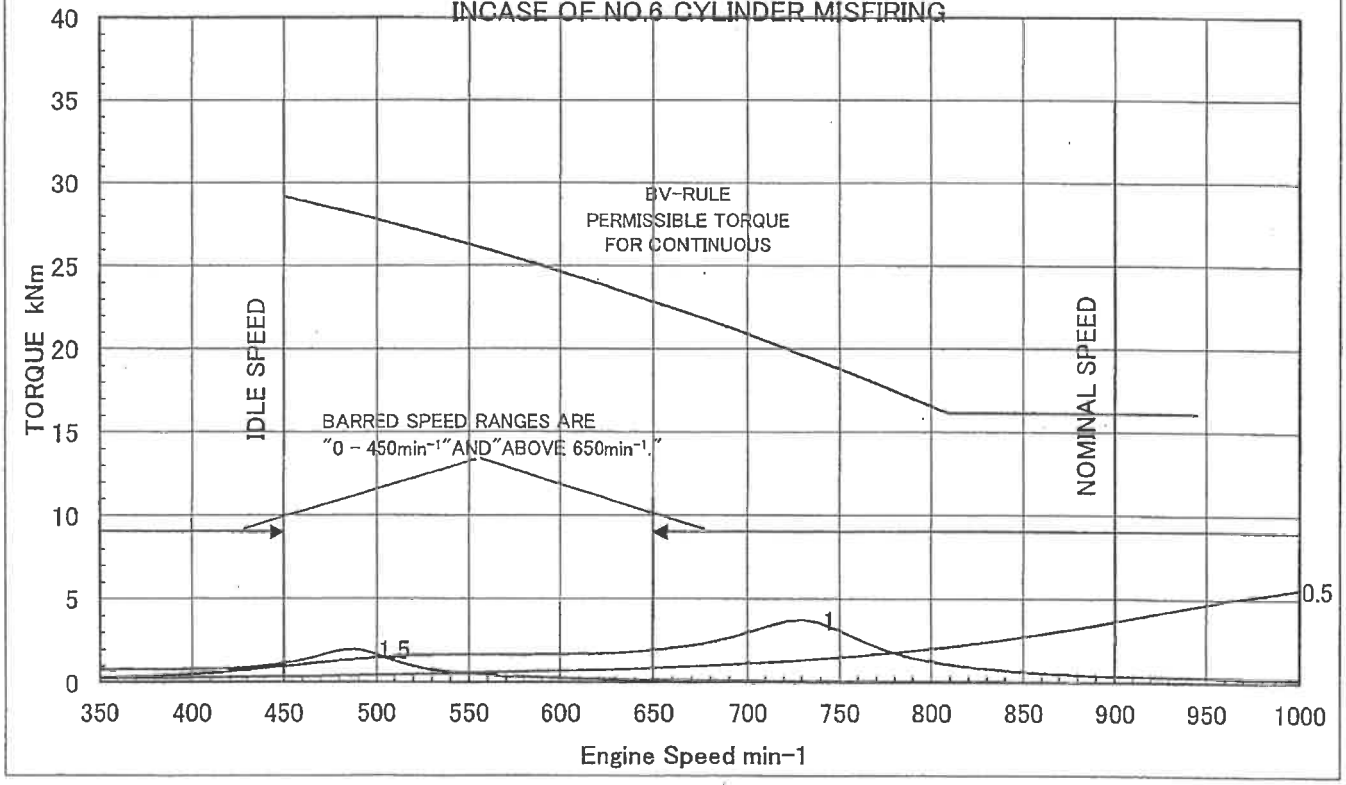


BOW SIDE CLUTCH : OFF
STERN SIDE CLUTCH : ON

VIBRATORY TORQUE DIAGRAM

Mass No. 21 – 22 (PROPELLER SHAFT)

INCASE OF NO.6 CYLINDER MISFIRING



Standard Procedure for Flushing Diesel Engines Piping

- This document is a standard procedure that applies to marine main engines.
- The flushing procedure, flushing time and other particulars vary greatly depending on conditions such as the treatment of pipes, pipe length, length of welds, quantity, piping condition and conditions ambient to the engine.
- It is necessary to choose an optimal method commensurate with your vessel.
- Use this procedure as a guideline to devise an optimal procedure and flush the piping accordingly.
- Attached drawing (on page 17/17) NAS Standard - For reference purposes

DAIHATSU DIESEL MFG. CO., LTD.
TECHNICAL DEPARTMENT
OSAKA, JAPAN

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DATE	Feb.16,2015	APPROVED BY	Y. Tsuchi
REVISION			

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1. General

Purpose

This procedure describes the processes for removing rust, welding spatters, burnt scale caused by welding and other foreign matter deposited on the inner surfaces of the pipes for steel oil piping such as fuel oil piping and lubricant piping manufactured in a shipyard as well as for removing dust, scale, sand and water that have contaminated the pipes during the piping installation so that the diesel engine operates normally as intended. Since this procedure describes the general flushing process, it needs to be edited and modified according to the specific purpose of flushing to make it suit each individual piece of equipment to achieve optimal results.

Scope

This procedure applies to the lubricant pipes and fuel oil pipes produced in a shipyard after they have been cleaned by an acid pickling. Note that it does not apply to the engine main body that constitutes the diesel engine equipment. As for devices and components having been completely flushed clean at the respective device manufacturers, unless otherwise specified, be sure to provide them with bypass piping so that the flushing oil does not run through them. Note that this does not apply to drain piping and air flow piping.

Fundamentals

To get the most from flushing, it is advisable to clean the inner surfaces of a pipe as thoroughly as possible before carrying out the flushing and be sure to follow the instructions below for piping installation.

- Bevel the portion to be welded in advance.
- Finish the beveled surfaces thoroughly with a grinder.
- After the welding, clean the pipe with an acid pickling.
- Spray the inner surfaces of the pipe with oil mists to create a corrosion preventive film.
- Cap both ends of the pipe with pieces of vinyl tape until its installation.
- Be sure to remove the vinyl tape before piping installation.

Although the above instructions are very basic, they are often ignored or forgotten only to result in insufficient flushing. Be sure to have technicians follow them thoroughly.

Criteria to judge completion of flushing

- Quantitatively describing the successful completion of flushing may be difficult. The most common practice may be that vessel owners, shipyards, lubricant manufacturers and engine manufacturers compare the dirtiness of the lubricant suction pump and sludge generation

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coming out of cleaners with the past records and experiences to mutually determine the criteria.

- The following are some criteria according to NAS standard classes.
  Lubricant piping system NAS standard Class 7 or better
  Fuel oil piping system NAS standard Class 5 or better

Preparation for flushing

Preparation

- Inspect the inner surfaces of the tank to make sure that they are free from water and other foreign matter. If rust is present, remove it completely.
- Carefully clean the inner parts of the filter.
- Do not use a piece of cloth for cleaning. Instead, use a vinyl sponge or a vacuum cleaner.
- Pay special attention to the oil pan, filters and tank to clean them thoroughly.

Preparation of necessary items

No.	Necessary item	Purpose
1	Temporary filter	LO line: 300 mesh (46 μm) FO line: 350 mesh (42 μm) and special filter, Example: AION PVA sponge (90 μm)"
2	Vinyl sponge	Cleaning tank
3	Vinyl filter	To eliminate fine foreign matter in the filter during flushing.
4	Washing oil	To wash clean filter elements.
5	Magnet	To be placed in the filter.
6	Temporary piping	To bypass the oil flow. (Rubber hoses reinforced with steel wire or vinyl hoses are easy to use.)
7	Blind flange	To stop the oil flow and cap the pipe openings.
8	Sample can container	To sample the oil for analysis.
9	Wooden hammer	Hammering
10	NAS (National Aerospace Standard) cleanliness class test instrument	Cleanliness test instrument used when completing the flushing. Example: Contamination Kit (YC-100-22) made by Yuken Kogyo Co., Ltd.
11	Plastic cap or vinyl sheet	To closely cap the ends of each pipe to prevent foreign matter contamination after flushing.
12	Temporary heater	For heating where a heater is not installed.

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2. Procedure for flushing lubricant piping unit

This section of the document describes the particulars regarding the flushing of a lubricant piping unit when mounting the engine.

General cautionary notes

- Flush the unit two times – primary flushing and secondary flushing.
- As the primary flushing, flush the piping unit manufactured by the shipyard. At this stage, do not let the flushing oil flow through the equipment (such as the engine, reduction gear unit, lubricant cooler, filtration unit, temperature regulating valve, pressure regulating valve and heater) that have been tested at their respective manufacturers. In other words, let the flushing oil bypass these units and equipment.
- In addition to providing these units and equipment with bypass pipes, you may have to remove some of the internal parts of these units and equipment as appropriate depending on the location of unit installation, structure and conditions of the flushing.
- Before the secondary flushing, detach the bypass pipes and put the removed parts back on to restore the original piping and assembly. Then, flush the entire lubricant piping unit.
- Use either the "flushing oil" or the "system oil", which is used on the engine, for flushing.
- Even if using the "flushing" oil for the primary flushing, be sure to use the "system oil" for the secondary flushing. Before using the flushing oil, thoroughly consult the oil manufacturer to make sure that the intended flushing oil is suitable for and compatible with the engine system oil.
- During the flushing, use the cleaner installed on your vessel to remove impurities.
- If using the flushing oil, be sure to completely remove the flushing oil after the flushing process.
- If it is necessary to remove a pipe or a device of the lubricant piping unit in the middle of the flushing process, be sure not to let the dust and other foreign matter contaminate the lubricant piping unit.
- If you use a temporary filter or a gauze wire, consider the use of a reinforcement to prevent a problem caused by damaged or broken element.

	Flushing oil	System oil	Items to be flushed	Guideline flushing duration
Primary flushing	○	○	Piping manufactured and installed by shipyard (Be sure to let flushing oil bypass the units and devices	20 hours or longer

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			supplied by manufacturers.)	
Secondary flushing	×	○	Entire lubricant piping unit with original pipes	10 hours or longer

Preparation

Preparing auxiliary devices and piping

- Refer to [Figure – 1] as attached for example.
- Prepare and install temporary pipes or temporary hoses that bypass the flushing oil so that the flushing oil does not run through the units and equipment (such as the engine, reduction gear unit, lubricant oil cooler, filtration units, temperature regulating valve, pressure regulating valve and heaters) that have been tested at the respective manufacturers.
- Use the temporary pipes and temporary hoses that are the same or similar sizes as the corresponding original pipes and hoses.
- Install the temporary filtration units immediately before the flushing pump inlet, flushing pump outlet and inlet of the engine.
- The mesh grades of the filter are as follows.

Temporary filtration unit position	Filter
At pump inlet	Mesh 60 to 90
At pump outlet	Mesh 90 to 120
At engine inlet	Mesh 300

- Prepare and install the flushing pump and heaters.
- If appropriate and practical depending on the situation, consider the use of the actual filtration units, motor driven pump, heaters and other devices originally installed on your ship. If so decided, be sure to thoroughly clean the inner portions of the equipment and restore the original condition after finishing the flushing.
- Check that the auxiliary units (such as the lubricant pump, lubricant feed pump, lubricant cleaner and other) driven by electric motors are in line with the correct respective centerlines and rotate in the intended directions. Also check that the attachments are installed appropriately on the respective units.
- Check that the thermometer, valve cocks, plugs and other piping attachments are all installed with the bolts and nuts securely tightened. Pay particular attention to the check valves (backflow prevention valves) to make sure that they are at the right locations and operate normally.
- Get the lubricant cleaner ready for operation.
- Check that the valves are correctly opened and closed. Lock them.

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- Install a baffle plate at the end of the return pipe to make sure that the flushing process takes place with the pipes completely filled up with the flushing oil. Determine, however, the dimensions and the installation location of the baffle place in a manner that it does not excessively hinder the flow of the flushing oil.

Primary flushing

Filling piping with oil

Flushing with system oil

- Use the feeder pump to fill the lubricant storage tank with the lubricant.
- While feeding the oil, check the level gauge of the tank to see the amount of oil.
- Fill the sump tank with the oil. Temporarily mark the level gauge of the sump tank to make it easy to check the oil level. Record the trim of the vessel.
- Unless otherwise specified, fill up the sump tank with the same amount of oil as that for actually operating the engine.
- The system oil in general is heated to about 30°C to 40°C for ease of feeding. Be sure to start the flushing before the oil temperature drops.

Flushing with flushing oil

- Follow the same steps as those for the flushing with the system oil except that, in this case, you directly fill up the sump tank with the flushing oil.

Preparatory steps before flushing

- As described above, since the flushing oil has to bypass the units and equipment (such as the engine, reduction gear unit, lubricant cooler, filtration units, temperature regulating valve, pressure regulating valve and heaters) that have been tested at their respective manufacturers, install the temporary pipes or temporary hoses so that the flushing oil does not run through these units and equipment.
- Install blind flanges at the inlet and outlet of lubricant on each unit to prevent foreign matter contamination.
- Remove the short pipe before the lubricant inlet of the engine and connect a temporary piping up to the engine side cover. Install a temporary filtration unit (with mesh 300 filter) at the free end of temporary piping A.
- In a similar manner, install a blind flange at the end of the piping to the engine.
- Since you have used the flushing oil, completely remove the flushing oil directly from the sump tank after the flashing process. Then, fill it up with the system oil.

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Heating temperature and heating method during flushing

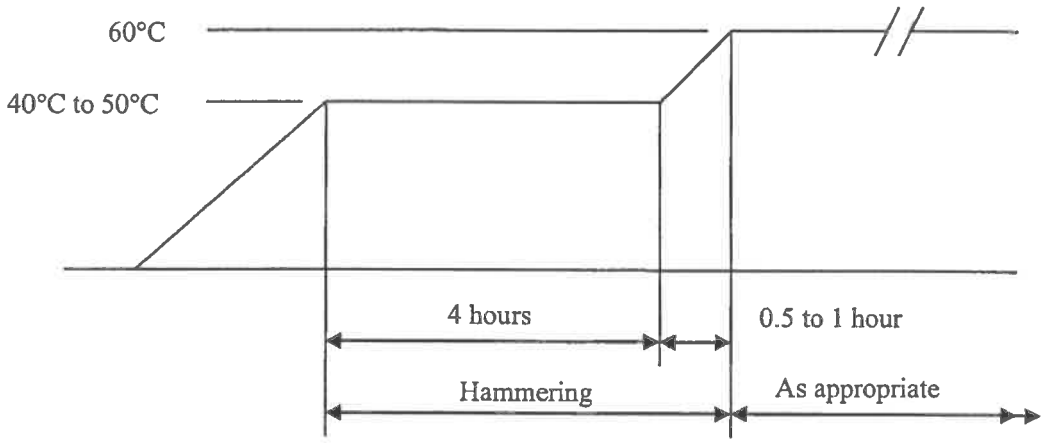
- Heating temperature

  40°C to 50°C

- Heating method

  Unless otherwise specified, continuously heat up the oil to a constant temperature. Note that changing the heating temperatures may be effective to get the better results out of the hammering process that remove scale from the piping.

  Hold at a certain temperature for 4 hours. Then, heat it up further for 0.5 to 1.0 hour. Be sure to keep the oil temperature at 60°C or below during the hot hammering.



- Heating method

  Use an electric heater or steam.

Cautionary notes on flushing

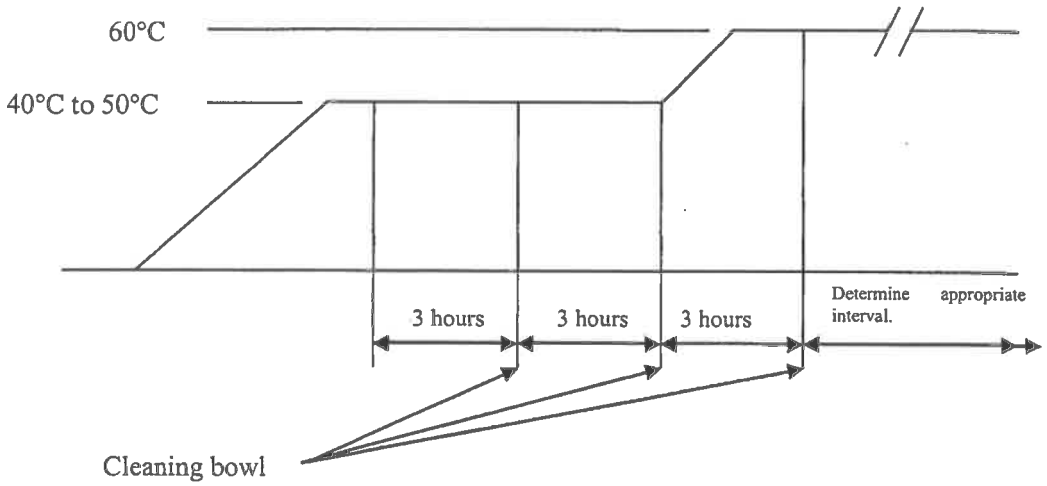
Lubricant pump

- Start up the pump while the ambient is still quiet. This is to check the lubricant leaks as well as to make it easy to stop the pump immediately in an emergency after first starting up the pump.
- At an early stage after starting up the pump, a higher electric current tends to flow through the pump drive motor because of the low temperature of the lubricant. To cope with this situation, adjust the bypass valve at the discharge line of the pump.
- Maintain the pump discharge pressure at about 0.4 MPa.

Cleaner

- Start operating the lubricant cleaner at the same time as starting the heating.

- In an early stage, clean the bowl once about every 3 hours.
- Having clean the bowl 2 to 3 times, determine the interval between further cleaning depending on the amount of sludge coming out of the cleaner.



Piping

- Immediately after starting up the lubricant pump, check the entire piping for any leak and observe the pump for normal operation. If any abnormality is present, stop the pump immediately.
- Unless otherwise specified, install the bypass pipes on the products that have been completely flushed clean at their respective manufacturers so that the flushing oil does not run through them.

Heating

- Heat up the lubricant gradually. Be sure to vent air.
- Refer to page 6 for the heating temperatures and heating method.

Cleaning and hammering

- While heating up the lubricant, hit the bypass pipes with a hammer so that the hammering impact causes the scale to come off the inner surface of the pipes.
- Hit the welded portions and bent portions of the piping carefully with the hammer.

Records

- Before starting the lubricant pump, use an appropriate method to measure the amount of oil in

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the sump tank. Start the lubricant pump and stop it immediately after starting it up. Use an appropriate method to measure the change in the amount of oil in the sump tank. At the same time, measure the trim of the vessel. Record and retain all these measurements before and after starting the lubricant pump as they will provide the base data for estimating the amount of oil remaining in the piping after the flushing.

- Use the temporary mesh 300 filter to collect foreign matter every 2 hours. Retaining the foreign matter or photographing it is recommended.

Process after finishing the primary flushing

- Remove the oil from the lubricant system. Put the removed oil in a settling tank or a storage tank.
- Be sure to remove the oil before the oil temperature drops so as to avoid water condensation. If necessary, use compressed air to remove the residual oil.
- After finishing removal of the oil, use a vinyl sponge to clean the inner surfaces of the sump tank.
- Remove the blind flanges and temporary pipes.
- Install the short pipe and other parts that have been removed to restore the original piping. Be sure to clean the inner surfaces of the pipes carefully before installation.
- After the flushing, analyze the oil to see that the flushing process has not change the characters of the oil.

Secondary flushing

- Remove the bypass pipes from the lubricant piping unit and restore the piping to the original condition. Flush the entire lubricant piping unit.

Cautionary notes on flushing

Lubricant pump

- Use a spare motor drive pump to carry out flushing.
- Refer to the cautionary notes set out for the primary flushing.

Cleaner

- Refer to the cautionary notes set out for the primary flushing.

Filtration units

- Where a backwashing duplex filter is provided, first carry out the backwashing for about 30 minutes after starting the flushing. After that, while observing how the sludge deposits,

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carry out backwash approximately every 2 hours.

- Where an automatic backwash filter is provided, remove the sludge in a similar manner as above.

Process after finishing the secondary flushing

- Remove the oil from the lubricant system. Put the removed oil in a settling tank or a storage tank.
- Be sure to remove the oil before the oil temperature drops so as to avoid water condensation. If necessary, use compressed air to remove the residual oil.
- After finishing removal of the oil, use a vinyl sponge to clean the inner surfaces of the sump tank.
- Overhaul all the filtration units to clean them. While it is a normal practice to remove the temporary vinyl filters, you can leave them there until finishing the dock trial operation. Determine which one to take depending on the filter condition.
- Do the same with the magnet.
- After the flushing, analyze the oil to see that the flushing process has not change the characters of the oil.

Records

- Refer to the cautionary notes set out for the primary flushing.

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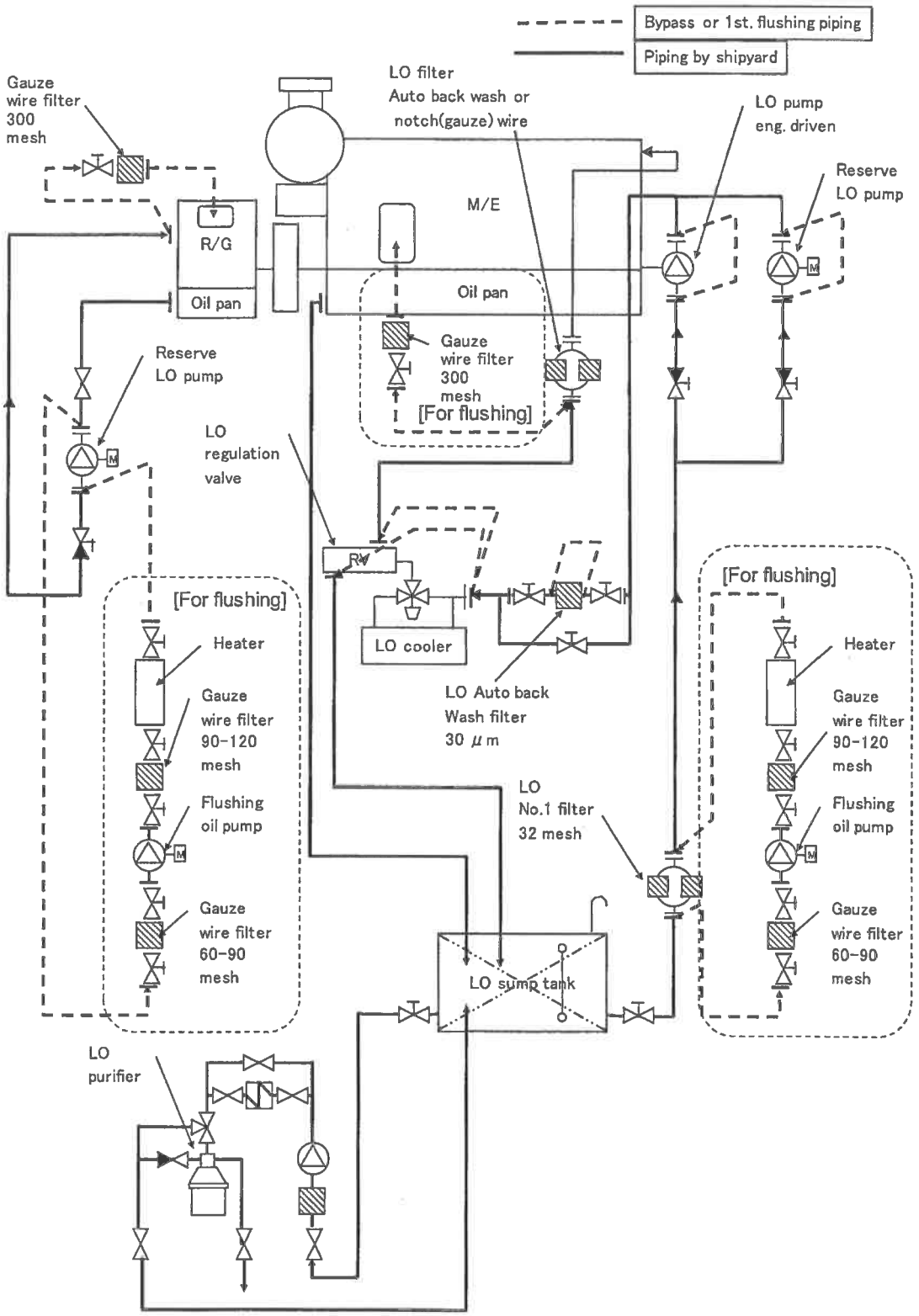

[Figure - 1] An example of lubricant system flushing

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3. Procedure for flushing fuel oil piping unit

This section of the document explains the procedure for flushing the fuel oil piping unit before installing the engine on the ship.

Introduction

This section describes the general cautionary notes on the flushing of the fuel oil piping system before installing the engine on the newly built vessel.

Before starting the processes, be sure to read through the procedure for flushing the lubricant piping unit that describes various cautionary notes and preparatory steps regarding the flushing

Preparation

- Prepare and install temporary pipes or temporary hoses that bypass the flushing oil so that the flushing oil does not run through the units and equipment (such as the engine, flow meters, filtration units and viscosity controller) that have been tested at the respective manufacturers.
- Install a blind flange at the fuel oil inlet and outlet of each device to prevent foreign matter contamination.
- Install a bypass pipe between the inlet and outlet of the engine.
- Install temporary pipes in a manner that the fuel oil for flushing does not return from the air separator to the fuel oil pump but returns to the fuel oil service tank for circulation.
- Temporarily provide the first fuel oil filter with gauze wire of 60 to 90 mesh for flushing.
- Where the fuel oil piping unit has an automatic backwash filter, install a temporary bypass pipe.
- Temporarily provide the bypass filter for the automatic backwash filter with gauze wire of 120 mesh for flushing.
- Get the fuel oil cleaner ready for operate.
- Check that all the attachments such as the thermometer, valve and plug are completely installed.
- Get the fuel oil heater ready for startup.

Refer to attached [Figure - 2] as an example.

Flushing

- Fill the DO service tank with fuel oil.
- Start up the fuel oil feed pump.
- Maintain the pump discharge pressure at about 0.45 MPa.
- Use the heater bypass line for about 3 hours after starting the flushing.

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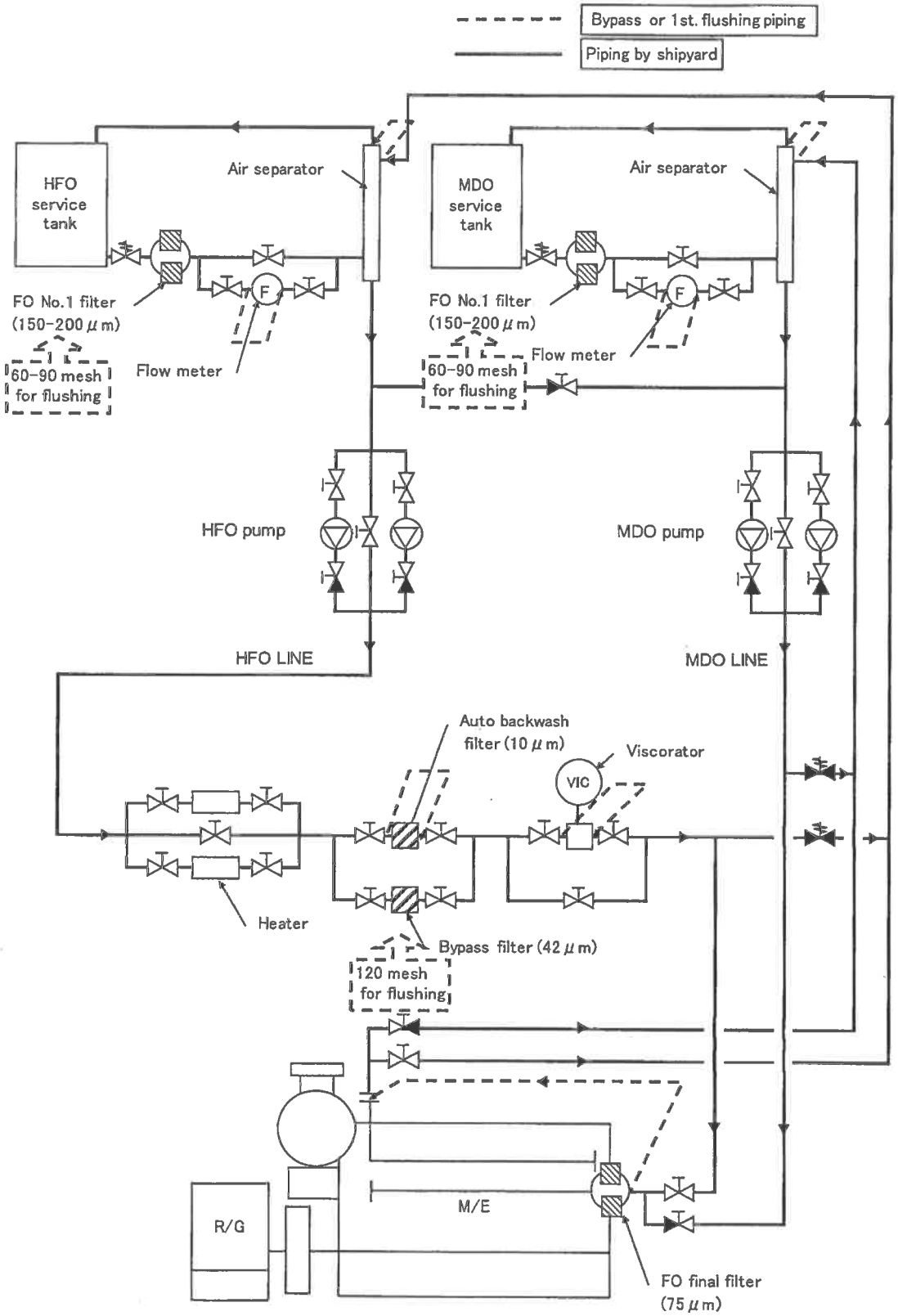
- In 3 hours, switch over to the heater line and maintain the fuel oil at a temperature between 40°C and 60°C.
- Start up the fuel oil cleaner.
- For the duration of about 15 hours after starting the flushing, clean the primary and secondary filters every 2 hours.
- After that, clean the primary and secondary filters every 4 hours.
- Hit the piping, bent portions and welded portions in particular, except for the thermally insulated portions, with a hammer so that the impact helps the scale come off the inner surfaces of the pipes.
- Where the devices such as the flow meter, heater and viscosity controller has a bypass line, flush both the main pipes and bypass pipes alternately.
- Flush the fuel oil piping unit for about 30 hours as a guideline. Adjust the flushing duration depending on the condition of the depositing on the filters and the sludge coming out of the cleaner.

Procedure after flushing

- Remove the temporary pipes to restore the original piping of each unit.
- Install and adjust the flow meter and viscosity controller according to the respective operation manuals.
- Remove the temporary filters.
- Disassemble the cleaner and clean the inner parts.
- After completing the flushing, circulate fuel oil through the fuel oil system to make sure that there is no leak.
- As much as possible, circulate fuel oil every day until starting the dock trial operation.
- The fuel oil filter installed on the engine is equipped with a special filter installed before shipping from the manufacturer. Remove the special filter after about 200 hours of operation or before HFO operation.



[Figure - 2] An example of fuel oil system flushing

4. Check list for pipe flushing

Date of issue (Month, date and year)

Issued by

Customer

Engine model × number of units

□ Engine application □ Engine for service generator □ Main engine □ Other ( )

Lubricant piping

1) Has pipes been washed with acid agent? □ Yes □ Not yet

2) Is engine provided with filter? □ Yes ( ) mesh □ No

3) Is temporary filter provided? □ Yes ( ) mesh □ No
Type ( )

4) Special filter □ Yes ( ) mesh □ No
Type ( )

5) Capacity of temporary pump ( ) m3/h

6) Type of flushing oil ( )

7) Is temporary heater provided? □ Yes ( ) kW □ No

8) Heating temperature (oil temperature) ( ) °C

9) Hammering frequency and interval Every ( ) hours
( ) Total times of hammering

10) Duration of flushing Primary flushing ( ) hours
Secondary flushing ( ) hours

11) Number of inspection and cleaning operations of filter
Primary flushing ( ) hours
Secondary flushing ( ) hours

12) Photographing foreign matter deposit □ Yes □ No

13) Results of cleanliness measurement per to NAS International Particle Count Standard
First: Class Second: Class
Third: Class Fourth: Class
Fifth: Class Sixth: Class

14) Final judgment (Final NAS class) □ Pass (Class) □ Fail
(Class)

15) Process after flushing

16) Special remarks

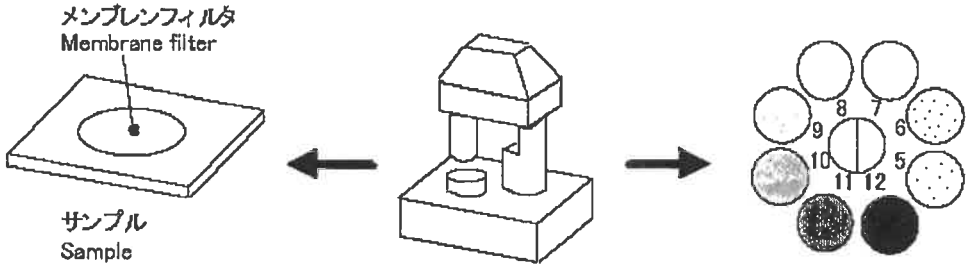
17) Time schedule of work (guideline)

- Primary flushing 20 hours
- Secondary flushing 10 hours
- Cleaning cleaner bowl 3 hours + as necessary
- Backwashing filter and discharging sludge 30 minutes after starting + every 2 hours

Fuel oil piping

1) Has pipes been washed with acid agent? □ Yes □ Not yet
2) Is engine provided with filter? □ Yes ( ) mesh □ No
3) Is temporary filter provided? □ Yes ( ) mesh □ No
Type ( )
4) Special filter □ Yes ( ) mesh □ No
Type ( )
5) Capacity of temporary pump ( ) m3/h
6) Type of flushing oil ( )
7) Is temporary heater provided? □ Yes ( ) kW □ No
8) Heating temperature (oil temperature) ( ) °C
9) Hammering frequency and interval Every ( ) hours
( ) Total times of hammering
10) Duration of flushing ( ) hours
11) Number of inspection and cleaning operations of filter
( ) hours
12) Photographing foreign matter deposit □ Yes □ No
13) Results of cleanliness measurement per to NAS International Particle Count Standard
First: Class Second: Class
Third: Class Fourth: Class
Fifth: Class Sixth: Class
14) Final judgment (Final NAS class) □ Pass (Class) □ Fail
(Class)
15) Process after flushing
16) Special remarks
17) Time schedule of work (guideline)

- Flushing 30 hours
- Backwashing filter and discharging sludge 30 minutes after starting + every 2 hours



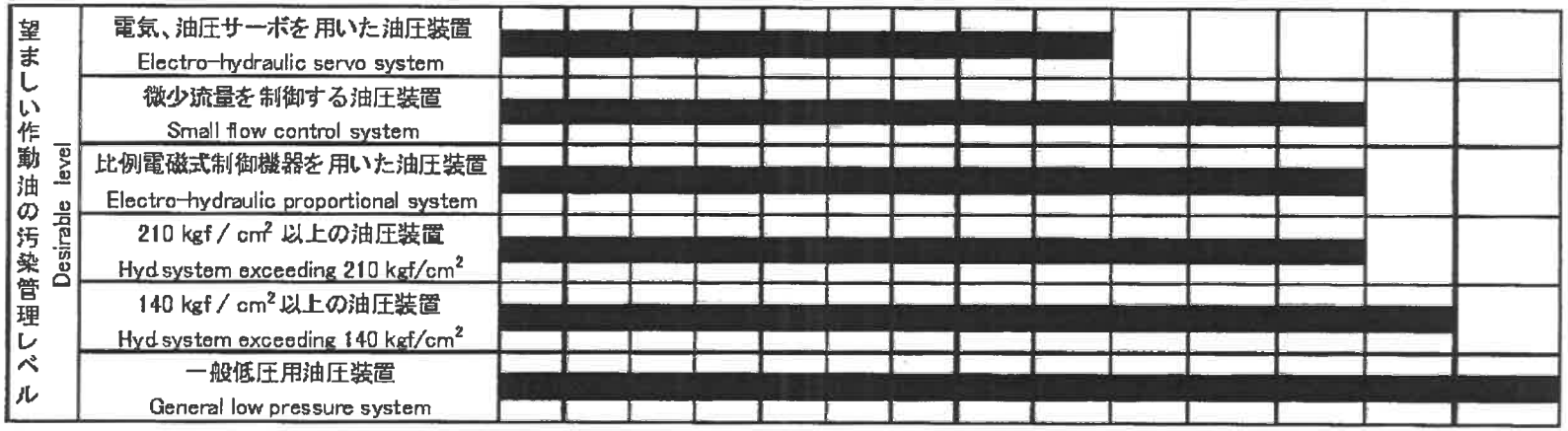
NAS 規格準拠標準汚染板
Standard contamination plate in accordance with NAS grade

NAS(National Aerospace Standard)1638

作動油中の許容粒子数 (NAS 1638) Permissible numbers of particle in Hydraulic Oil (NAS 1638)

(100ml 中の個数 Number / 100 ml)

サイズ分類 Size classification(μm)	級 Class													
	00	0	1	2	3	4	5	6	7	8	9	10	11	12
5—15	125	250	500	1,000	2,000	4,000	8,000	16,000	32,000	64,000	128,000	256,000	512,000	1,024,000
15—25	22	44	89	178	356	712	1,425	2,850	5,700	11,400	22,800	45,600	91,200	182,400
25—50	4	8	16	32	63	126	253	506	1,012	2,025	4,050	8,100	16,200	32,400
50—100	1	2	3	6	11	22	45	90	180	360	720	1,440	2,880	5,760
Over 100	0	0	1	1	2	4	8	16	32	64	128	256	512	1,024



DAIHATSU

SINGLE-ENGINE, SINGLE-SHAFT TYPE REDUCTION GEAR

INSTRUCTION BOOK

DRA《J-type》
DRB《J-type》

DAIHATSU DIESEL MFG. CO., LTD

INB000112A (1/24)

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Before operation

Before Operating Daihatsu Diesel Engine, Please be sure to read the following items for safety and correct operation.

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§1 Outline

Daihatsu's DRA and DRB series of hydraulic reverse reduction gear units are those for marine main engines with designs mainly aiming to be structurally simple, easy to use, durable, and compact and light-weight, and have built-in thrust bearings.

The input shaft of the series' reverse reduction gear unit is directly coupled with a diesel engine through a flexible coupling and its output shaft is coupled with a propeller shaft system.

The gear box of the series' reverse reduction gear unit contains reduction gears and a hydraulic multiple-disc clutch, can reduce a main engine's rotation and transfer the reduced rotation to a shaft system, and can change to an ahead, neutral or astern sailing mode.

For your information, if the series' reduction gear unit is directly coupled with Daihatsu's diesel engine, a remote controller can be used to remotely control the engine's speed and change the clutch operation to an ahead or astern mode.

This document only describes how to handle the series' reduction gear unit. Therefore, refer to the separately provided manuals shown below with regard to their associated instruments.

(1) Proportional Solenoid Valve Controller Manual
(2) Flexible Coupling Manual

§2 Structure

1. Outline of Structure (Refer to Figures 1, 2, 3 and 4.)

The outline, gear system, hydraulic system and cross-section drawings of the series' unit are provided in an associated annex.

The gear box consists of a lower gearbox, an upper gearbox and a cover, and contains reduction gears and a hydraulic clutch inside. Its lower part forms an oil pan.

In the forward part of the gearbox (the bow side), there is an input shaft coupling, which is coupled to the main engine through the flexible coupling.

In the rear part of the gearbox (the stern side), there is an output shaft, which forms Mitchell's thrust bearing system, and transfers propeller thrust to a vessel's body through the gearbox.

The input and output shafts are coaxial for a DRA type unit and off axial for a DRB type unit.

On the gearbox's side surface, accessories related to hydraulic pressure such as an LO pump, an LO filter, an oil cooler, a relief control valve, a pressure gauge and a clutch control valve are located.

In addition, a pressure alarm switch, a wing pump and other equipment may be arranged depending on respective specifications.

The hydraulic and lubricating systems of the series' unit have independent pipe lines which have no relation to those of the engine.

2. Power transmission (Refer to Figure 2.)

Switching between the ahead and astern modes can be done by engaging/disengaging ahead/ astern clutches .

- For the ahead mode:

  The force will be transferred through the route shown by the solid line.

  The ahead clutch is engaged, and the astern clutch is disengaged. The input and output shafts rotate in the same direction for a DRA type, and they rotate in the opposite directions for a DRB type.

- For the astern mode:

  The force will be transferred through the route shown by the dotted line.

  The astern clutch is engaged, and the ahead clutch is disengaged. The input and output shafts rotate in the opposite directions for a DRA type, and they rotate in the same direction for a DRB type.

3. Reduction Gears (Refer to Figure 4.)

The tooth surfaces of the reduction gears are quenched and polished, and they are finely finished.

Figure 2 shows how gear trains are arranged, and the gear trains are divided into first, reverse and second gear trains.

The first gear train is a stage to use speed reduction ratios to increase or decrease speed. The reverse gear train is combined with a clutch to switch between the ahead and astern modes. The second gear train is a stage to reduce speed.

Each gear is fixed to a shaft with shrinkage fits or bolts.

4. Reduction Gear Shaft (Refer to Figure 4.)

A gear shaft consists of input, output and clutch shafts.

The clutch shaft is integrated with the reduction gears and the shrinkage fits, which also use a clutch piston case. On the other hand, a 2-stage pinion slides along the clutch shaft, and a hydraulic clutch is installed between the reverse gear and the 2-stage pinion.

Attached to the end of the clutch shaft is a gear coupling, which drives oil supply equipment and an LO pump which transfer clutch-operated high hydraulic pressure and lubricating oil pressure.

5. Bearings (Refer to Figure 4.)

The structure of each bearing is such that a lower bearing part is integrated with a casing and a metal cap is used to fasten the bearing.

A bearing is either thin wall or round metal, and is completed with white metal casted into a forge steel back metal. Appropriate tightening

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and stretch margins are provided so that a standard bearing clearance is realized when assembled.

If a bearing is damaged due to careless handling or it is not fastened properly during assembly, it cannot perform fully. Therefore, it must be handled with care, and this document's fastening standard must be strictly followed.

6. Thrust Bearing (Refer to Figure 4.)

A pad to receive ahead thrust and that to receive astern thrust are respectively placed facing a collar, which is machined from an output shaft.

Propeller thrust is transferred from the output shaft collar to sections in the following order: the pad, the main thrust case, the gear box, and a vessel's body.

White metal is casted into the pad's thrust surface, whose back has a protrusion. When the shaft rotates, this protrusion functions as a fulcrum, and the pad inclines to the sliding surface appropriately, and forms oil films well.

The sliding surface is soft, and special care must be taken when it is handled.

The pad is held by a pin, but the pin merely holds the pad and does not function as a fulcrum.

Lubricant oil can be supplied through a main thrust case. Lubricant oil lubricates the pad's surface from the inner circle, and returns to the oil pan from the collar's upper part through a bypass port.

The main thrust case is divided into 2 sections, that is, upper and lower parts.

When the main thrust case is to be disassembled, unscrew a mating surface's bolts and pull the case upward, then it can be disassembled easily.

7. Hydraulic Clutch (Refer to Figure 4.)

Ahead and astern clutches are wet hydraulic multiple disc ones, and are located in parallel on the clutch shaft above the gear box, and either or both of them can be easily hoisted upward if the gear box's cover is removed.

When oil pressure is applied to the oil piston's back, friction plates are firmly attached to each other, and thrust is transferred in the following order: the first stage gear (the input shaft), the reverse gear, the clutch drum, the friction plate and the second stage pinion gear.

If oil pressure is applied to either one of the ahead and astern clutches, relevant clutch is engaged; then, oil pressure will not be applied to the other clutch, its friction plates run idle, and that clutch is disengaged.

When a clutch is disengaged, oil pressure decreases, and the action of a piston spring moves its piston backward. Hence, clearance between the friction plates is secured, and the clutch is well disengaged.

8. Flexible Coupling (Refer to Figure 5.)

The flexible coupling is installed between an engine and the reduction gear unit's input shaft, and absorbs harmful shocks induced by such forces as torque fluctuations and torsional vibrations which are generated in the engine to protect the reduction gears and the clutch.

Its outer tube is fastened by bolts to a flywheel, and it is a high elasticity coupling with a flat spring being used in its inside to achieve high elasticity as well as with part of the main engine's lubricant oil being filled to its inside to realize better damping effects. Refer to its manufacture's manual for details.

9. Oil pressure system

1) Oil circuit (Refer to Figure 3.)

The oil circuit is divided into two circuits, that is, high pressure and low pressure circuits.

The high pressure oil circuit drives clutches.

The low pressure oil circuit lubricates individual bearings, gears and the clutches' friction plates.

Lubricant oil and clutch operating oil are absorbed by the LO pump from the oil pan through a suction strainer.

A circuit from the LO pump's discharge opening to a high pressure relief valve and

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the clutch operating valve forms the high pressure circuit.

The high pressure circuit's pressure becomes higher when the ahead or astern clutch is engaged by the high pressure control valve.

During idling, the high pressure control valve's piston is returned and the valve is opened, and hydraulic pressure becomes lower.

A circuit from the high pressure control valve to the oil cooler, a low pressure control valve, a filter and a LO main pipe forms the low pressure circuit, which lubricates relevant sections.

Lubricant oil released from the low pressure control valve is piped to the gear box's upper part and drained in the gear case, and is returned to the oil pan.

2) Oil pump (Refer to Figure 6.)

The hydraulic pump is gear type equipment, which is attached from the clutch shaft's rear part through a drive coupling. The pump's discharge rate is determined to ensure that the clutch operates securely.

When operating the pump, pay attention to the pressure gauge and check the change of hydraulic pressure.

3) Oil cooler (Refer to Figure 7.)

The oil cooler is a multitubular type, whose structure is such that lubricant oil flows through cooling tubes' outer circumferences, and cooling water flows through the tubes' inside.

If seawater is used for cooling, zinc protector is applied to the inside of the oil cooler case to protect each section of the oil cooler from electric corrosion.

4) Filter (Refer to Figure 8.)

The filter is attached to the side of the gear box, and is 200 mesh metal plate multiple layer type or 100 mesh notch wire type.

The notch wire type can be changed to use-both-sides, use-one-side and blow-off (back blowing) operations by handle operations even if it is being operated by a switch cock. However, the use-one-side operation should only be allowed during the open blowing operation.

5) Pressure relief valve
(Refer to Figures 10 and 11.)

The high pressure control valve is installed on the gear box's upper part, and controls the high pressure circuit. The low pressure control valve is installed on the oil cooler's upper part, and controls the low pressure circuit.

6) Gauge (Refer to Figure 13.)

A gauge board is installed on the gear box's upper part, which has high, low and other pressure gauges. A LO temperature gauge is installed between the high pressure control valve and the oil cooler's entry.

7) Clutch operating valve (Refer to Figure 14.)

The clutch control valve is installed on the gear box's upper part along with a hydraulic pressure detection valve (refer to 8-2). A oil pipe from the high pressure valve is connected to the entry of the clutch control valve, and is piped from the ahead and astern exits to the oil pressure entries of the ahead and astern clutches, respectively.

(7-1) Standard air-assisted control
(Refer to Figure 14.)

Two ends function as ahead and astern air cylinders, and are piped to respective pilot valves in a bridge. A spool type valve stem (19) inserted inside is moved by control air pressure which acts on a piston (15).

That is, when a control handle is in a neutral position, because control air pressure does not act on an ahead and astern piston (15), a piston spring (48) locates the valve stem (19) at the center as indicated in Figure 14, and both conduit pipes running from the exit to the ahead and astern clutches are in

direct communication with the gear box's inside from an oil discharge hole.

Next, if the control handle is moved to the ahead position, an ahead pilot valve is opened, and when air pressure acting on the ahead piston becomes higher than or equal to about 0.08 MPa (0.8 kg/cm^2), the piston spring (48) is compressed and the valve stem (19) moves to the left. Due to this movement, a channel of an ahead oil pressure exit to the oil discharge hole is closed by the valve stem, and at the same time, to communicate with a oil pressure entry, oil is transferred to the ahead clutch so that the ahead clutch is engaged.

Conversely, if the control handle is moved to the astern position, the valve stem moves to the right to engage the astern clutch, and the ahead clutch is disengaged because the ahead hydraulic pressure exit is in communication with the oil discharge hole.

The position to which the valve stem moved is informed to a lever (4) through a rack and a pinion mechanism, and the position can be confirmed from outside. Furthermore, 2 micro switches (39) are embedded on an instruction plate (3)'s lower part, which electrically detect the ahead/astern position of the lever or the valve stem to display the detected position on a bridge's control panel.

If operated from an engine room, first, the lever (4) must be moved to the ahead or astern position, and a stopper handle (18) must be pushed in and fixed to the instruction plate (3) with a stopper. **However, when operations from an engine room are finished, please make sure that the stopper handle will be pulled up and further rotated 90 degrees. If not, it may become impossible to remotely operate the unit.**

8) Pressure switch

When control air pressure, reduction gear hydraulic pressure and other pressure abnormally decrease, a pressure switch is activated and sends electronic signal to an alarm panel separately installed so that an alarm will be displayed.

Please do not change contact-making pressure (preset value) abruptly because it is pre-adjusted according to relevant specifications.

10. Special Specifications (Refer to Figure 3.)

(1) LO cooler bypass

A bypass circuit with a stopper valve can be provided to the oil cooler to prevent lubricant oil temperature from becoming very low in a cold region.

(2) Backup pump circuit

If emergency, separately placed priming, wing and other pumps are equipped, their suction and discharge openings will be made in the reduction gear unit.

Furthermore, check valves (Figure 9) will be attached to the discharge openings to the reduction gear unit.

11. Valves and Cocks

How to operate valves and cock are as described below.

(1) Pressure gauges and pressure switches (Outlet master valves)

Normally: Open

When changing a pressure gauge and a pressure switch: Close

(2) LO cooler bypass (Stop valve)

Normally: Close

When used normally: Open

(3) Backup Pump Circuit

(A galvanized blind flange attached to a check valve)

- Galvanized blind flange at an entry to an LO pump
  Normally: Open
  When using a backup pump and changing the pump: Close. But when priming with the backup pump: Open
- Galvanized blind flange at an entry to a backup pump
  Normally: Close
  When using a backup pump: Open

§3 Operations, Remote Control

1. Preparation for Operations

1) Inspection of individual sections

Inspect whether bolts and nuts are not loosened and plumbing is normal.

Especially, carefully inspect sections which were overhauled after previous operations.

Check whether valves and cocks are in their normal positions (refer to §2-11). Especially, be careful when the emergency line of a backup pump is used.

2) Oil level validation

An oil-level gauge is placed at the reduction gear unit's stern side.

Level marks indicated on the oil-level gauge are those used for operations. Therefore, make sure that a current oil level during operations is between the marks of U.L and L.L. Remind that **levels during a vessel's stop and those during the vessel's operations are different from each other (the difference is about 20 to 50 mm).**

3) Priming

A priming pump attached to the reduction gear or a backup pump attached to the reduction gear lubricates individual bearings and gears.

4) Clutch operating valve

The clutch operating valve will not work if its handle lever is fixed. Therefore, a stopper needs to be pulled up from a stopper holder and turned by a half rotation so that the stopper is locked (for an air type), or a press button needs to be unlocked (for a solenoid valve type).

5) Control air pressure validation (for the air-assisted control)

Confirm whether the pressure indication of the control panel is normal.

6) Confirm whether power indicators on the control panel and the alarm panel are turned on.

2. Startup

1) Position confirmation of control air change valve and control handle

A control air change valve must be switched to an engine position and the control handle must be in the neutral position.

2) Confirmation of governor manual handle

A governor manual handle must be returned and locked completely after engine control is performed.

3) Confirmation of front end load

If force is extracted from an engine's front end, a front end clutch must be disengaged. If there is no clutch, the front end must be in unloaded condition.

4) Warming-up

If a clutch is engaged immediately after an engine is started-up, the engine may not be sufficiently warmed-up, and may be unreasonably forced to work, and the engine's life may be negatively impacted. Therefore, about 10 minute warming-up must be performed.

5) Hand-over to remote control

During the warming-up, respective sections must be inspected. After it is confirmed that there is no anomaly, control authority will be handed over to a vessel's bridge.

3. Caution during operation

1) Oil pressure confirmation

After an engine is started, confirm whether oil pressure is within the range of prescribed values.

High hydraulic pressure occurs only when the clutch is engaged. Therefore, confirm whether high oil pressure occurs after the clutch is engaged.

Be careful if oil pressure is significant below a prescribed value. It may mean that clutch operations might not have been performed satisfactorily.

Because hydraulic pressure was adjusted during pre-shipment factory operations, it need not be adjusted again. **If oil pressure adjustment is necessary after the pressure operating valve or other section is overhauled, it must be performed during dead slow sailing.**

If it is conducted during full-speed sailing, oil pressure decreases when an engine's rotation speed is decreased, engaging or disengaging the clutch may become impossible, and the clutch may be seized. After the adjustment is performed, confirm whether oil pressure at full-speed sailing is within the range of prescribed values.

2) Inspection of oil temperature

Oil temperature depends on cooling water temperature, load and other condition. Normally, 30 minutes to 1 hour after start-up, its value becomes stable and enters in the range of 40 to 60 °C at the LO cooler's entry. However, if the clutch is engaged and disengaged frequently, it increases by 5 to 10 °C.

Lubricant oil temperature must be less than or equal to 70 °C during operations.

3) Inspection of oil and water leakage

4) Inspection of individual sections' temperatures, abnormal sounds and mist density changes

At the initial stage of failure, in a corresponding section, its temperature may rapidly rise, abnormal sounds may be generated there and the amount of mist there from the breather may increase. Attention must be paid to individual sections' temperatures, abnormal sounds and mist.

Individual sections' temperatures are nearly equal to oil temperature at the cooler's entry, but the temperatures of bearings and the cover of the clutch's upper part are slightly higher than the oil temperature.

5) Lubricant oil filter

Metal plate multiple layer type

It must be tuned once a day, as well as it must be drained once a week, and opened and cleaned once 1 to 2 months.

Notch wire type

For normal operations, filters must be placed in a position for common use so that both right and left filters can be used. Switch to a right or left filter during cleaning and switch operations.

When changing filters, pay attention to handle stopper and marking positions. Change them during an idling or low load mode, and pay attention to LO pressure.

6) Filter for pressure relief valve

Open and clean the filter to prevent it from being clogged, as necessary.

7) Others

Record pre-shipment factory operation data and parameter values during normal sailing (such as the values of hydraulic pressure, oil temperature, and clutch engagement and disengagement time), and compare currently measured values with those of the parameters during normal sailing so that any malfunction can be detected as early as possible.

For malfunction, refer to §5 "Reduction Gear Unit Failures and Appropriate Measures."

4. Maintenance during Vessel Stop

1) Lubricant oil exchange

When exchanging lubricant oil, use the priming or backup pump to pump out old lubricant oil, remove the oil pan's cover to wash away residue on the oil pan's bottom, and fill new lubricant oil.

Refer to the section "Standard Overhaul and Maintenance Interval."

2) Lubricant oil filter

The lubricant oil filter's elements must be cleaned always when lubricant oil is exchanged.

3) Inspection of individual sections

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To inspect gears, bearings and other sections, the section "Standard Overhaul and Maintenance Interval" must be referred to.

5. Remote Control

1) Confirmation of ahead/astern indicator and propeller revolution counter

The control handle must be operated carefully, and every time it is operated, an ahead/astern indicator and a propeller revolution counter must be checked to confirm its actuation response.

2) Relation between handle operations and vessel speed

When increasing a vessel's speed from slow ahead, gradually increase the speed while watching it. Be cautious especially when a vessel's speed is slow because the vessel is fully loaded or weather is stormy, or when the vessel is just warmed up.

3) Vessel operation when entering into port

When a vessel is entering into port after it has continuously sailed for a long time, decrease the vessel's speed sufficiently, and do not perform astern operations hastily.

If a clutch cannot be engaged, return the clutch to the neutral position, and try the same operations again.

4) Control handle operations between neutral and dead slow positions

If the control handle is between the neutral and dead slow (0 notch) positions, **it is unclear whether the pilot valve is open or close. Therefore, the control handle must be operated swiftly, and a vessel should not be operated with the control handle positioned between the neutral and dead slow positions.**

The clutch will be engaged swiftly and securely if the control handle is returned to the dead slow position as required after it has been moved to a position which is not very close to the dead slow position but with some margin (2 to 3 notches).

5) Crash astern

When astern sailing is to be performed with a vessel's speed is faster than half ahead sailing, move the control handle to the neutral position, wait for a while so that propeller rotation speed decreases, and activate astern sailing.

The propeller rotation speed when astern sailing is activated must be less than or equal to two thirds of dead slow sailing.

However, if instructed otherwise, follow the instruction. When crash astern sailing is performed, there may be a possibility that a vessel's engine is overloaded momentarily and stalls because the vessel is moving with inertia, and its propeller is rotated by water flow.

Because crash astern sailing may negatively affect the clutch's life, do not perform it except in case of emergency.

6) Early detection of malfunction

It may be a sign of trouble if response to a command by the control handle is slow, or if difference between a commanded notch and propeller rotation speed is large.

If malfunction is found, check a failed section and take appropriate measures as soon as possible.

In normal condition, control air is consumed only when the control handle is returned to the slowdown and neutral positions, and air is exhausted from the control valve.

7) Communication between bridge and engine room

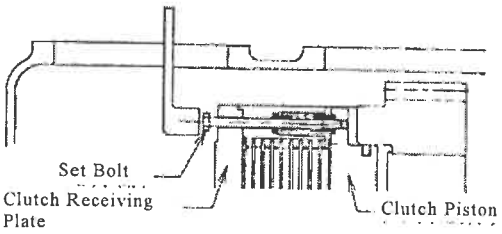
Always be reminded that deck and engine departments make preparatory discussions sufficiently so that when something wrong happens to operations from a bridge, a vessel's control can be handed over to operations from an engine room immediately, or so that when contingency or emergency happens, right measures can be taken appropriately.

6. Clutch Emergency Bolt

During a vessel's sailing, an emergency bolt can be used to continue its sailing when the hydraulic system failed and the clutch cannot be engaged, when clutch slip is sever and its sailing might become impossible, or when other emergency cases happen.

How to use clutch emergency bolt

1) After an engine is stopped, remove a blind box on the gear box's upper part.
2) Pick a set bolt from fixtures, insert the set bolt to a clutch receiving plate's set bolt hole, and screw it into a clutch piston.
3) There are 6 set bolts, and they must be fastened equally.
   This will connect the clutch.
   Be careful not to fasten a set bolt with force stronger than prescribed torque.
   (For fastening torque, refer to drawings in the manual separately provided.)
4) **If emergency bolts are used, refrain from unreasonable vessel operations, and use the clutch with load to the clutch less than or equal to 50%.**



Clutch Emergency Bolt

§4 Maintenance and Service

1. General Instructions

In general, time interval between the reduction gear unit's maintenance services depends on how it is operated. Therefore, it is desirable if time interval between its overhaul and parts exchange services is determined how it is operated.
"Standard Overhaul Service Interval" in the following table should serve as a baseline.

[General instructions for open overhaul]

1) Parts should be handled with care with safety first in mind.
2) Be reminded of a part's assembling sequences not to assemble the part erroneously.
3) Bolts and nuts must be fastened equally without applying insufficient or excessive torque.
   Bolt fastening torque applied to major parts must be based on the fastening torque standard in the annex (drawings in the manual).
4) Inner bolts, nuts and others which are not opened must be checked regularly.
5) The clearance gaps of individual bearings and thrust bearings must be compared with those in a wear limit table of the annex (drawings in the manual) for their inspection.
6) New split pins, washers and seals must be used for assembly.
7) Be careful not to drop foreign objects such as bolts and nuts into the case during open overhaul.
8) Be careful for rust protection.
9) If backup parts are used, supply new backup parts immediately.
10) Please use Daihatsu's genuine parts for parts compatibility and accuracy.
11) If you have any questions about handling and maintenance, please contact Daihatsu's service engineers.

2. Adjustment Standard Values (At an Engine's Rated Rotation)

LO oil pressure

High pressure	1.9 to 2.1 MPa (19 to 21 kg/cm^2)
Low pressure	0.25 to 0.4 MPa (2.5 to 4 kg/cm^2)

Other standard values for bearing clearances and bolt fastening forces are described in the annex (drawings in the manual).

3. Standard intervals for overhaul and maintenance

Inspected parts	Work contents		Operation Time (Hr.)	Note
Exterior of reduction gear unit	Pipe, fastening, loosening, and oil leak inspection		Everyday	
L.O. management	Oil amount check		Everyday	
	Filter cleaning		500 - 1000	
	Lubricant oil exchange	First time	200 - 500	
		Second time	4000 - 6000	(1 year)
Gears	Inspection of gear tooth surface		10000 - 15000	(3 years)
Main thrust bearing	Inspection of pad surface		20000 - 30000	(5 years)
Plain bearing/thrust bearing	Inspection of interior and rear stopper		20000 - 30000	(5 years)
Roller bearing	Replacement		20000	(5 years)
Needle bearing (Lubricant oil pump)	Opening/inspection		4000 - 6000	(1 year)
	Replacement		10000 - 15000	(3 years)
Main bolts	Clutch Drum bolt	Loosening inspection, retightening	10000 - 15000	(3 years)
		Replacement	20000 - 30000	(5 years)
	Other bolts	Loosening inspection, retightening	10000 - 15000	(3 years)
Hydraulic equipment Clutch control valve L.O. pressure relief valve	Opening, inspection and cleaning		4000 - 6000	(1 year)
L.O. high pressure relief valve spring	Replacement		20000 - 30000	(5 years)
Clutch	Inspection of friction plate, and 2-stage pinion spline section		20000 - 30000	(5 years)
L.O. cooler	Opening, inspection and cleaning	L.O. side	10000 - 15000	(3 years)
		C.W side	4000 - 6000	(1 year)
L.O. cooler zinc protector	Inspection, scale removal, and replacing old protector with new one if more than 30% is damaged		1000 - 1500	(3 months)
Geislinger coupling Note: For instructions and wear limit, refer to separately provided Geislinger manual.	• Remove the coupling, and disassemble and inspect inner members (do not disassemble its outer members). • Exchange rubber seals, disc springs and lock washers.		18000 - 20000	(5 years)
	• Disassemble the coupling's outer member, and inspect and repair its spring pack and inner star. • Exchange rubber seals, disc springs and lock washers.		40000 - 60000	(10 years)

4. Disassembly and Assembly

The reduction gear unit shall be disassembled in the following order.

Pipes → accessories related to hydraulic pressure such as the LO pump and the oil cooler → a side cover on the gear box's end face → the gear box's cover → the clutch shaft → the main thrust case → a gear box (2) → the input and output shafts' bearing crown → the input shaft's thrust metal → the input and output shafts.

・Assembly

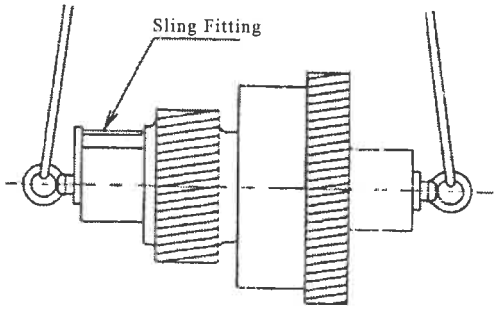
The reduction gear unit can be assembled in the reverse order of the disassembly sequence mentioned above. However, individual parts must be cleaned well so that either contaminants or chips will not enter into the reduction gear unit. In addition, make sure that the following instructions are observed carefully.

1) Check match marks.
2) Always grease O-rings and other rubber parts when they are assembled.
3) Fasten bolts and nuts with appropriate torque values. Refer to the annex "Major Bolts Fastening Standard."
   Check if there are any rotation locks.
4) Clean mating surfaces and apply a thin layer of liquid packing to each of them when assembling the gear box.
   (The liquid packing must be Threebond 1104 or its equivalents except when appropriate packing is already prepared.)
5) O-rings, oil seals, packing, rotation locks and other consumable parts must be replaced with new ones every time when they are disassembled.

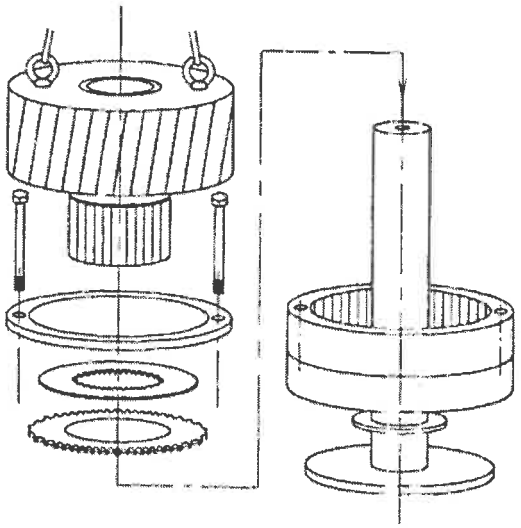
(4-1) How to disassemble and assemble clutch shaft assembly

・Disassembly Procedure

1) Attach hanging hooks and eye bolts, which are accessories, to the clutch shaft's both ends, and pull out the clutch shaft assembly from the gear case.
2) Remove bushes, the thrust metal and a secondary pinion gear.
3) Remove drum bolts, and then the clutch receiving plate.
4) Remove the clutch drum. (There is a knock-pin, and use a tap bolt.)
5) Remove the piston spring, the friction plate, the thrust metal, the piston and a oil pressure ring.



Lifting Schematic



Assembly Schematic

・Assembly Procedure

1) Attach the oil pressure ring to the piston and the reverse gear. Then, install the piston, the thrust metal, the clutch spring, the clutch drum, the friction

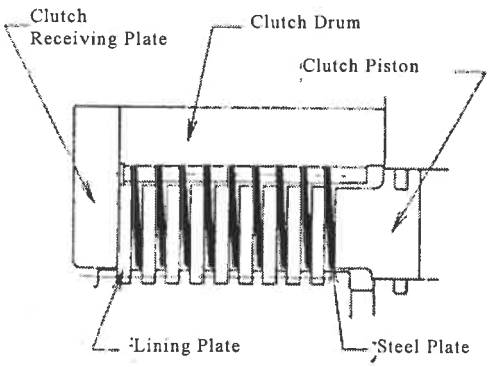
plate, the clutch receiving plate, and the drum bolts in this order. The drum bolt must always be fastened with a prescribed fastening torque value.

2) Install the secondary pinion gear, the thrust metal and a radial metal.

3) Attach the hanging hooks and the eye bolts, which are accessories, to the clutch shaft's both ends, and install the clutch shaft with the 2-stage pinion gear attached to it, while paying attention to prevent the clutch shaft from touching the gear case. (**The hanging hooks must always be used.**)

(Cautions)

(a) The friction plate must be assembled in the following manner: steel and lining plates must be inserted one after the other such that the first steel plate is inserted to the clutch piston's side and the last lining plate is inserted to the clutch receiving plate's side. **If the friction plate is not assembled in this manner, it may burn out. Therefore, sufficient care must be taken when the friction plate is assembled.**

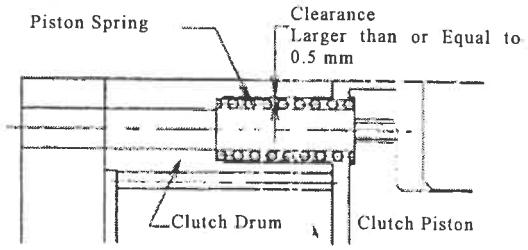


Friction Plate Assembly Schematic

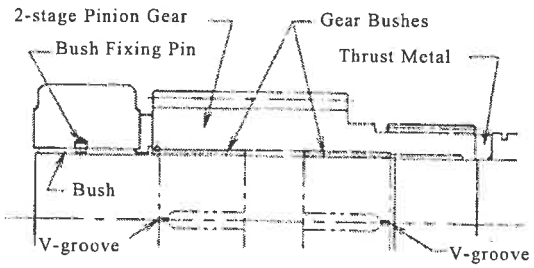
(b) The steel plates are cone-shaped to enable the clutch to be disengaged smoothly. When the steel plates are installed, the directions of their cone shapes must be aligned. Inspect the friction plate, while paying attention to its deformations and discolorations. Replace parts with new ones if they have any anomaly. When assembling the friction plate, make sure that no contaminations will enter into the friction plate.

(c) Note that, to assemble the reverse gear with the clutch drum, a stepped dowel pin is used for positioning (which must be driven to the clutch piston case's side) and that a match marks is provided to the stepped dowel pin (while another corresponding match mark is provided to the clutch piston). It is arranged in such a way that the friction plate will not come off an associated drum spline even if the reverse gear is assembled with the clutch drum in such a state that the clutch drum's end slightly pushing the stepped dowel pin's head. However, operability will improve if the drum bolt is used to inert the clutch drum to the stepped dowel pin. Make sure that the piston spring will not be installed if it is deformed when the clutch drum is installed. If any anomaly is found, replace the piston spring with a new one.

(d) Move the piston spring and the clutch piston while being careful not to make any scars, to **verify that a clearance between the piston spring's outer circumference and the clutch drum's spring hole is larger than or equal to 0.5 mm at worst.**



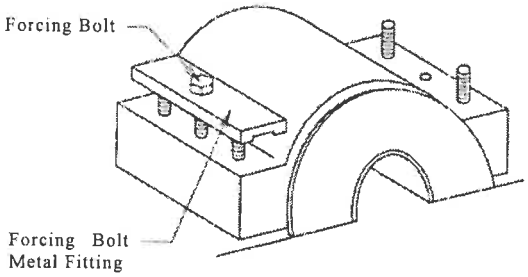
(e) Gear Bush Replacement (Refer to the Figure Below)
Usually, it is not necessary to replace a pair of gear bushes fixed to the secondary pinion gear by shrinkage fitting with a pair of new ones. If it becomes necessary to replace the gear bushes with new ones by any chance, after having removed the current ones, heat the secondary pinion gear to 170 °C in an oil tank, and install new ones while verifying V-grooves' directions and installation locations. **The 2-stage pinion gear shall not be heated to more than 170 °C by any chance.**



(f) Locking Rotations of Input and Clutch Shafts with Stern-side Bush
The bush fixing pin shown in the above picture is used to lock the rotation of the input and clutch shafts. Therefore, be careful to verify its location and not to omit it when disassembling or assembling the bearing crown.

(4-2) Bearing Disassembly and Assembly

1) Remove U-nuts for clamp bolts.
2) Attach a forcing bolt for disassembly, which is an accessary, and screw in the forcing bolt to remove the bearing crown.
3) When the thrust metal is to be removed from the bearing crown and the gear box, attach a wood piece to a claw part of the thrust metal's back metal and strike the wood piece softly to make a clearance between the thrust metal and its housing so that it can be removed easily (if it consists of thin-wall completely split metals).



Metal Cap Disassembly Schematic

(Cautions)

(a) **Always use the forcing bolt metal fitting and the forcing bolt for disassembly when disassembling the metal cap.** If a hammer or other tools are used to strike and move the bearing crown to the left or to the right, or forward or backward, bearing alignment surface positions may be distorted, possibly resulting in poor fit and bearing seizure. Therefore, the forcing bolt metal fitting and the forcing bolt must always be used.

(b) Appropriate tightening and stretch margins are provided so that a standard bearing clearance is realized when the bearing is assembled. Therefore, never try to conduct filing or adjustment using shims when the bearing is being assembled.

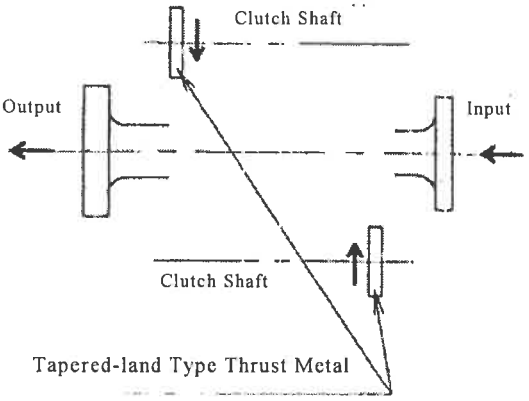
(c) When assembling the bearing, wipe off contaminants on the housing and the thrust metal completely.

(d) Handle the bearing with care so that the bearing will not be damaged. Fastening must be done in accordance with the fastening torque standard in the annex.

(e) When assembling or disassembling the stern side of the input shaft, the bow side of the output shaft or the bearing

crown, confirm the positions of bushes or fixing pins for rotation locks and be careful not to omit any of them.

(f) A tapered-land type thrust metal must be installed in a position with respect to the clutch shaft as shown in the figure below, irrespective of an engine's rotation speed and direction.
Furthermore, the outer circumference of a tapered-land type thrust metal's flange



is marked with an arrow* to indicate gear rotation direction. Therefore, confirm the arrow when installing it.
*: It indicates gear rotation direction when a clutch with whose clutch shaft a tapered-land type thrust metal is integrated is turned on.

(4-3) Main Thrust Bearing
Disassembly Procedure

1) Remove mating surface reamer bolts and gear box attachment bolts installed to the upper and lower cases of the main thrust bearing.
2) Hoist the upper case upward and move the lower case downward. (For 80F, only hoist the upper box because the lower box is integrated with the gear box.)
3) Hoist the main thrust bearing's retainer. (Required only for 80F)
4) Remove the main thrust bearing's pad from the upper and lower cases or the half-split retainer.

(Handling Instructions)
The main thrust bearing's pad is supported in a free state by pins fixed to the upper and lower boxes or the retainer. Therefore, the pad may come off the pins and be damaged when the main thrust bearing is tilted during its hoisting. Be careful not to tilt the main thrust bearing before removing the pad. If it comes off the pins by mistake and its white metal surface is scratched, remove its burrs carefully.

(4-4) Flexible Coupling
Removal of Geislinger Coupling Assembly

1) Remove bolts which attach an inner flange to an inertial disc, slide the inner flange along a shaft to remove its inside low part.
2) Remove bolts which attach an outer flange plate to a flywheel, slide the outer flange plate along a shaft to remove its inside low part.
3) Attach a sling fitting to the outer flange plate, and slowly hoist the Geislinger coupling assembly to remove the assembly as a whole.

*: For the assembly and disassembly procedures of the Geislinger coupling's main unit, refer to the separately provided Geislinger Coupling Manual.

(4-5) Oil Filter
(a) Metal Plate Multiple Layer Type
Disassembly Procedure

1) Remove retainer attachment bolts.
2) Remove a case.
3) Remove element attachment bolts.

(Cautions)

a) If it is removed together with a cover assembly, attention must be paid to an arrow marked on the cover's surface when it is being installed. (The arrow indicates flow direction.)

b) If its elements are disassembled, it may become difficult to assemble them again. Therefore, it is better not to disassemble the elements.

c) After its installation is completed, enough air must be bled from a plug located on the cover's upper part.

(b) Notch Wire Type

Opening and Cleaning Procedure

1) Rotate its handle 90 degrees counterclockwise from its normal position to move it to its halt position.

2) Loosen an air vent plug on its case's upper part to drain lubricant oil.

3) Unscrew attachment nuts and remove the case, and then pull out elements.

4) Clean the elements and install them again.

5) Return the handle to its normal position.

6) Use the same procedure to open and clean its right side elements.

(Handling Instructions)

a) The handle's notch must be correctly adjusted to a hole.

b) After having pulled out the elements, check their contamination condition carefully and verify whether there is any metal powder.

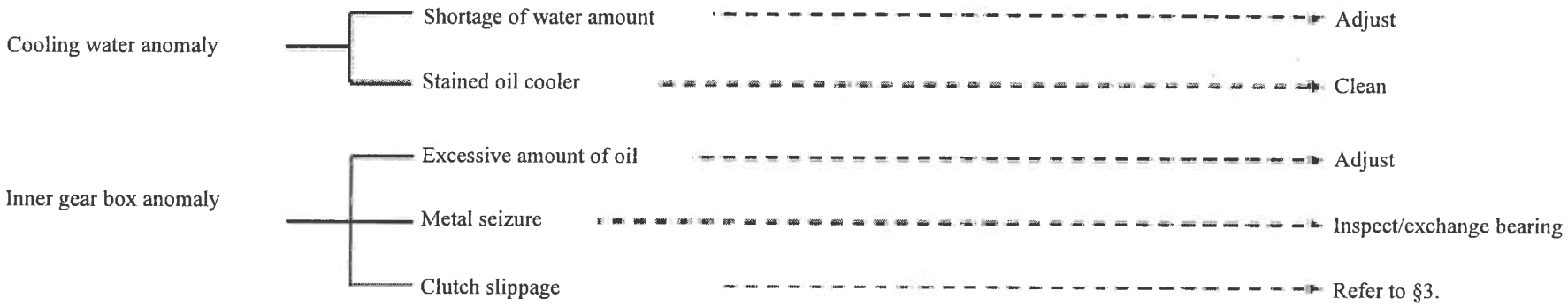
c) How to open and clean the filter is also described on a name plate attached to it.

(4-6) Oil Cooler

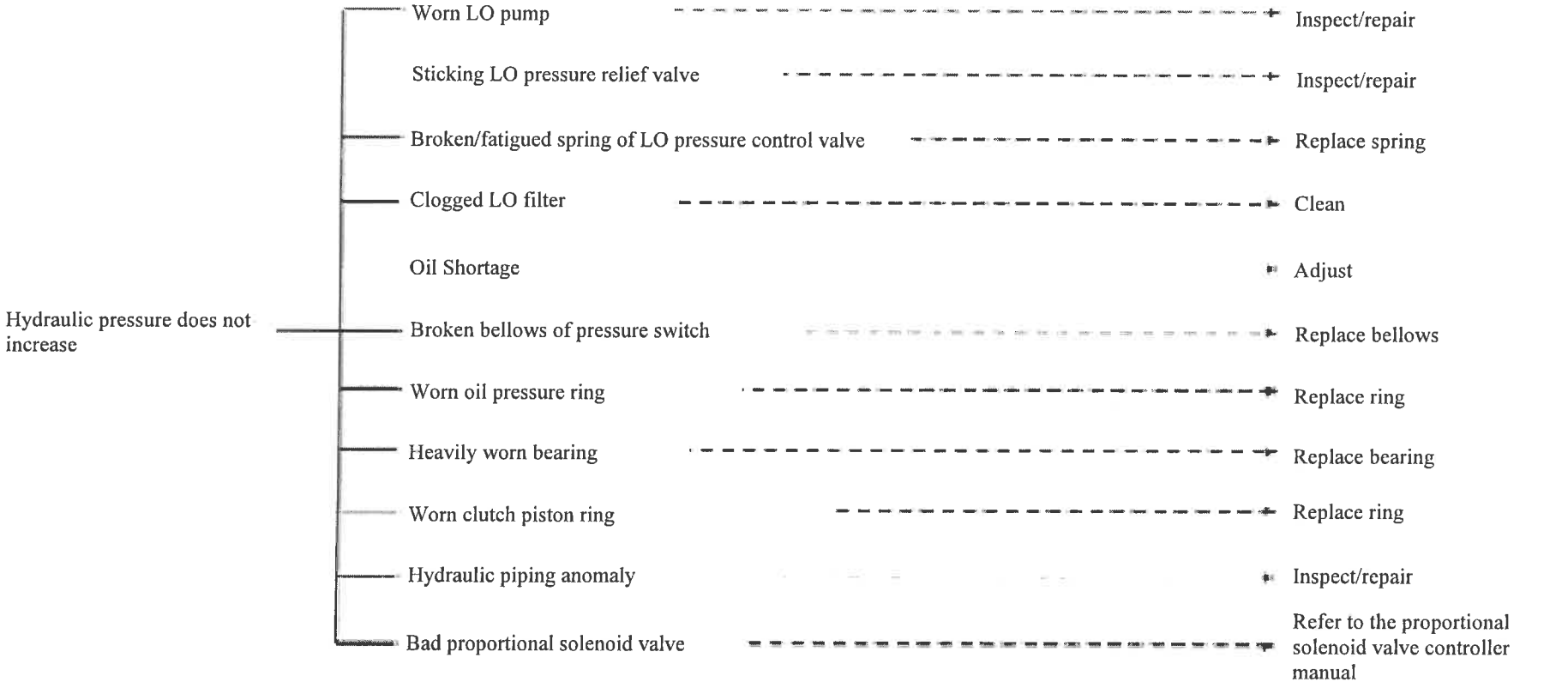
To open and inspect the oil cooler, refer to the Engine Manual.

§5 Troubleshooting for Reduction Gear

1. Oil temperature increases abnormally.



2. Hydraulic pressure does not become a prescribed value.



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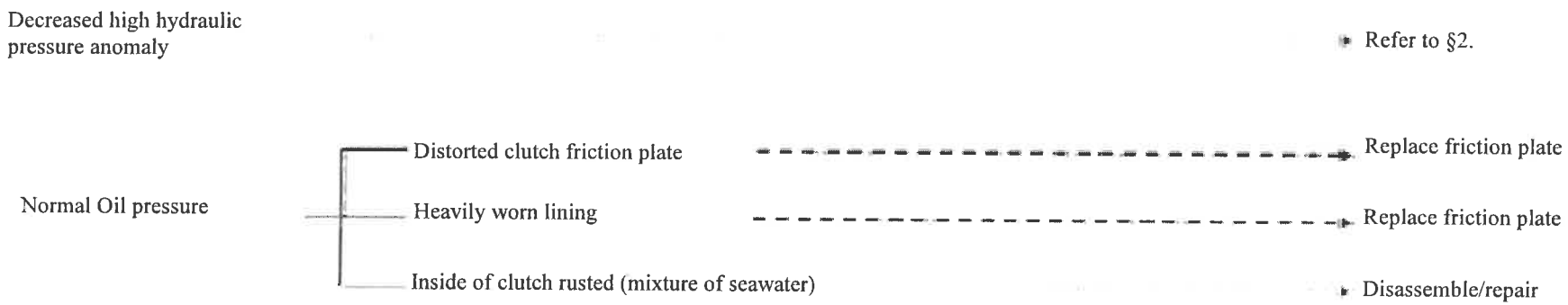
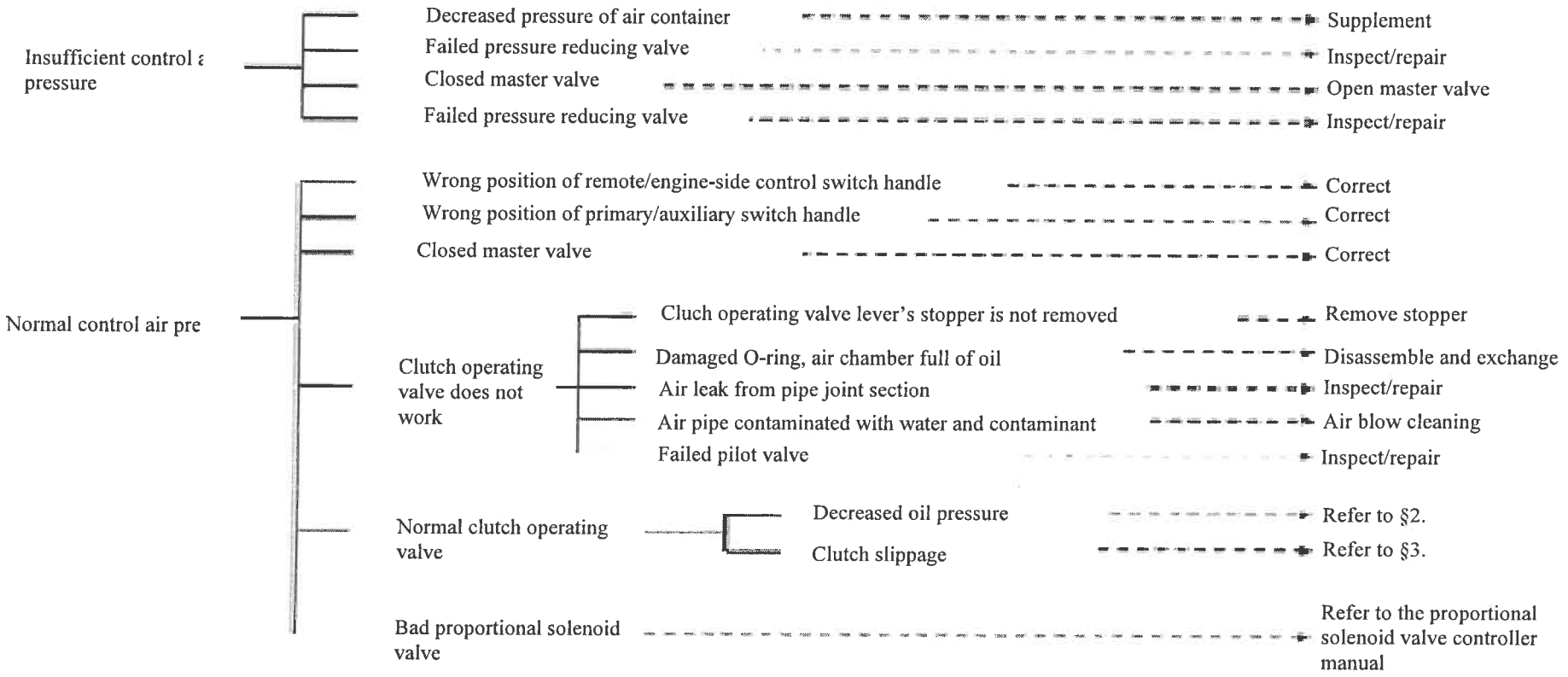
3. Slipping clutch



4. The clutch cannot be engaged even though the control handle is operated.



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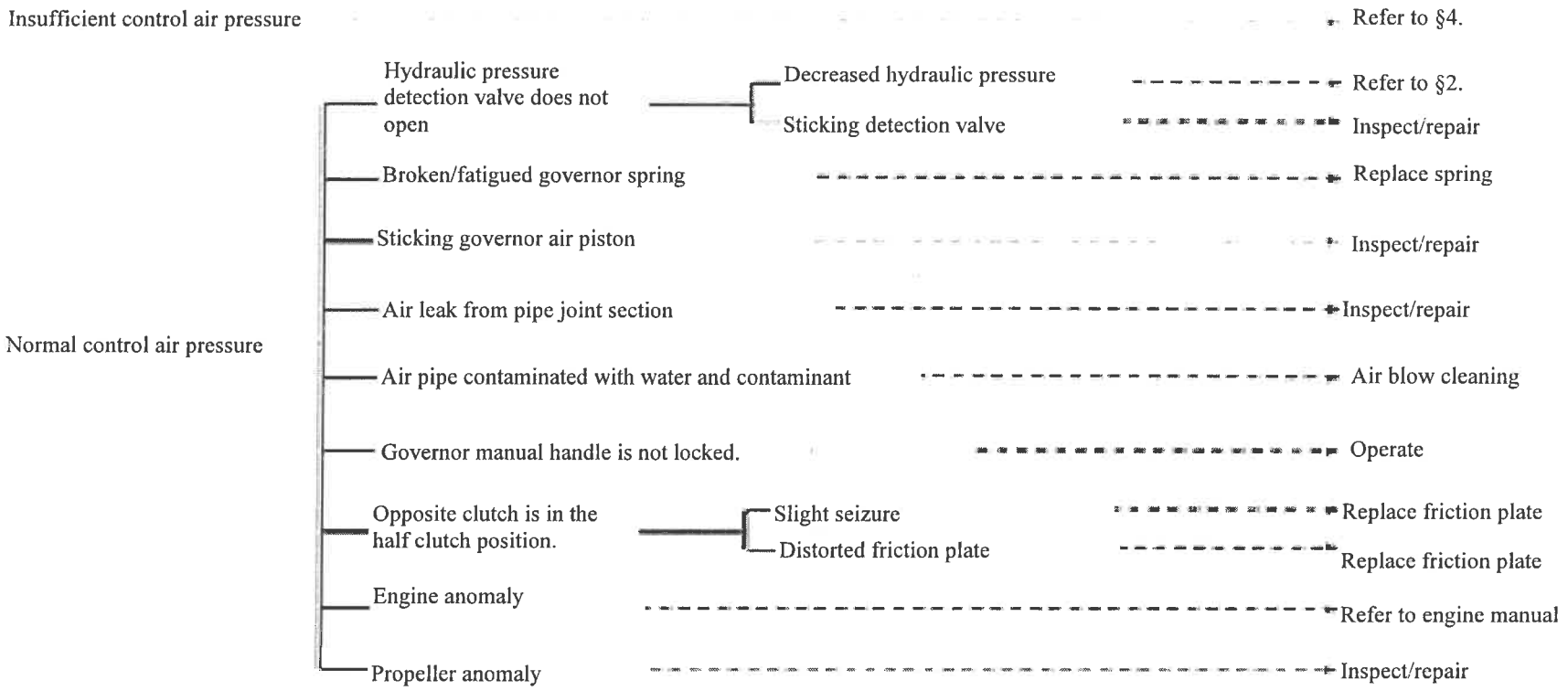
5. A clutch can be engaged but rotation speed does not increase.



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DAIHATSU

減速機取扱説明書図面

INSTRUCTION BOOK [DRAWINGS]

DRA-25J(L)

(6DKM-20eF(L))

ダイハツディーゼル株式会社

DAIHATSU DIESEL MFG. CO., LTD

PAL000071A(1/49)

2017/05/10 TECHNICAL DEPARTMENT

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DRA-25J(L)型減速機
【主要目表】

入力軸最大回転力	
減速比(前進/後進)	
全　　長	1600mm
全　　高	1771mm
全　　幅	1645mm
重　　量 (乾燥)	4100kg
潤滑油ポンプ	歯車式
油冷却器(多管式)	伝熱面積 8.28m2
潤滑油濾器(ノッチワイヤ切替式)	200 メッシュ
使用潤滑油	#250 ディーゼルエンジン油 (SAE#30番相当)
初期注入量	150L

【主要部締付標準】
(ボルト, ナット座面及びネジ面にモリコート1000を塗布して締め付けのこと)

ネジ部位置		ネジ径 (六角二面幅)	締付トルク	ネジ部位置	ネジ径 (六角二面幅)	締付トルク
メタルキャップ	入力軸 前後	M24X3.0 (36mm)	440Nm (45kgfm)	入力軸カップリング	M33X3.5 (50mm)	1470Nm (150kgfm)
	クラッチ軸 前	M24X3.0 (36mm)	440Nm (45kgfm)	出力軸カップリング	M30X2.0 (46mm)	880Nm (90kgfm)
	クラッチ軸 後	M30X3.5 (46mm)	735Nm (75kgfm)	メインスラストケース取付	M27X3.0 (41mm)	590Nm (60kgfm)
	出力軸前後	M24X3.0 (36mm)	440Nm (45kgfm)	メインスラストケース (リーマボルト)	M24X3.0 (36mm)	315Nm (32kgfm)
1段ギヤ(2)		M24X2.0 (36mm)	735Nm (75kgfm)	ギヤツキテ (潤滑油ポンプ駆動)	M16X2.0 (24mm)	130Nm (13kgm)
クラッチドラムボルト		M20X2.0 (30mm)	345Nm (35kgfm)	非常用クラッチ 締結ボルト	M10X1.5 (17mm)	25Nm (2.5kgfm)

DRA-25J(L)
【SPECIFICATIONS】

Input shaft maximum torque	
Reduction ratio	
Length	1600mm
Height	1771mm
Width	1645mm
Weight	4100kg
Oil pump	Gear type
Oil cooler(Multi−tubular)	8.28m2
Filter	200Mesh
Lubricating oil	Diesel engine oil (SAE. #30)
L. O. initial quantity	150L

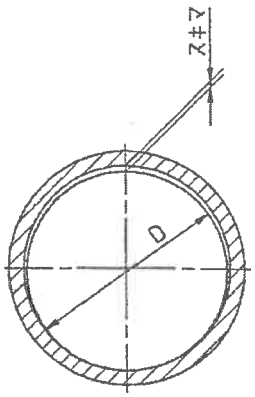
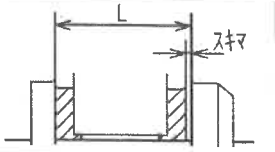
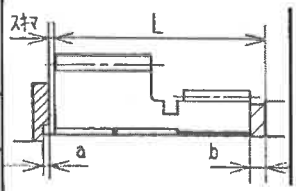
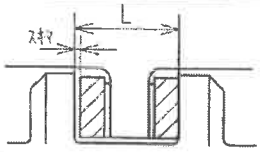
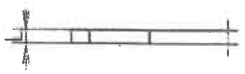
【Standard tightening torque】

Anti-seizure agent(MOLYKOTE 1000 Spray type) should be applied the seat and threads of bolts and nuts.

Thread Situation		Thread size (Width across flat)	Tightening torque	Thread Situation	Thread size (Width across flat)	Tightening torque
Metal cap	Input shaft (front,rear)	M24X3.0 (36mm)	440Nm (45kgfm)	Input shaft coupling	M33X3.5 (50mm)	1470Nm (150kgfm)
	Clutch shaft (front)	M24X3.0 (36mm)	440Nm (45kgfm)	Output shaft coupling	M30X2.0 (46mm)	880Nm (90kgfm)
	Clutch shaft (rear)	M30X3.5 (46mm)	735Nm (75kgfm)	Main thrust case (Fitting)	M27X3.0 (41mm)	590Nm (60kgfm)
	Output shaft (front,rear)	M24X3.0 (36mm)	440Nm (45kgfm)	Main thrust case (Reamer bolt)	M24X3.0 (36mm)	315Nm (32kgfm)
Primary gear (2)		M24X2.0 (36mm)	735Nm (75kgfm)	Gear coupling (L.O.pump drive)	M16X2.0 (24mm)	130Nm (13kgfm)
Clutch drum bolt		M20X2.0 (30mm)	345Nm (35kgfm)	Emergency clutch set bolt	M10X1.5 (17mm)	25Nm (2.5kgfm)

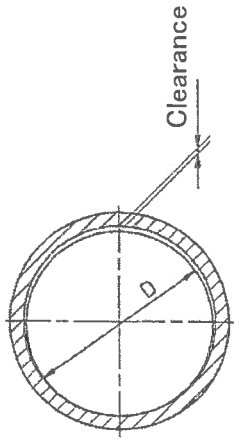
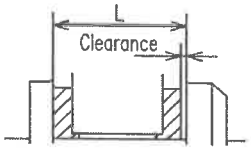
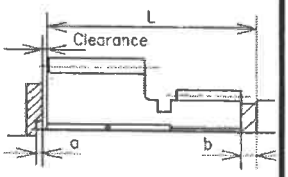
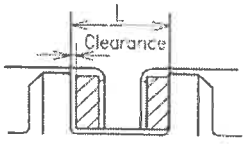
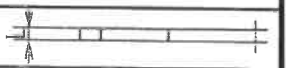
嵌合および摩耗限度表

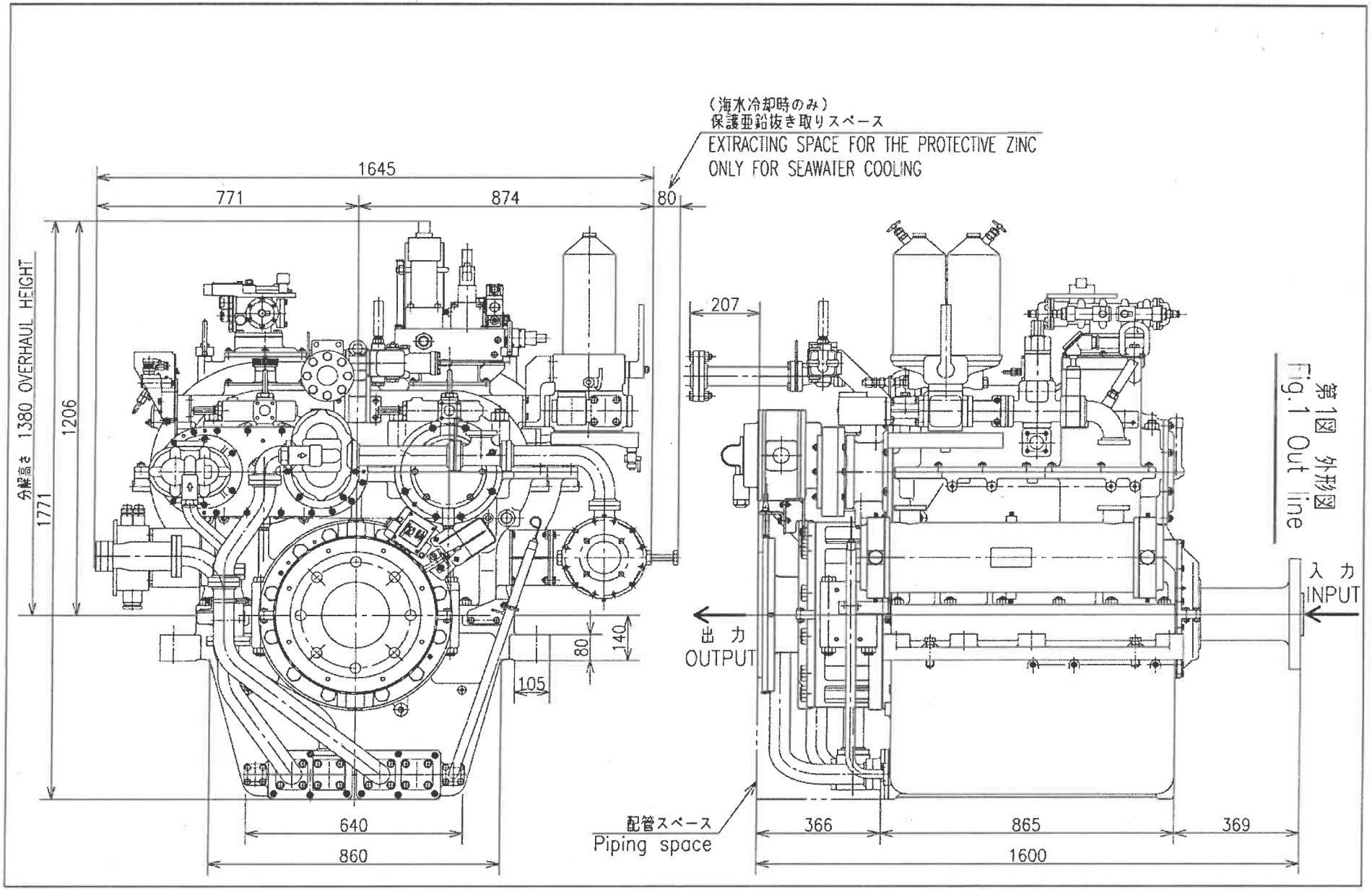
DRA-25J(L)

部品			D、L	標準すきま	交換限度	測定箇所
歯車軸受	入力軸 前	軸外径	155	0.11～0.19	厚さ3.85で交換	
		組立時内径				
		裏金厚さ	4			
	入力軸 後	軸外径	190	0.18～0.29	すきま0.40で交換	
		組立時内径				
		ブッシュ厚さ	10.5			
	出力軸 前	軸外径	190	0.18～0.29	すきま0.40で交換	
		組立時内径				
		ブッシュ厚さ	10.5			
	出力軸 後	軸外径	250	0.21～0.30	厚さ4.85で交換	
		組立時内径				
		裏金厚さ	5			
	ｷﾞﾔﾌﾞｯｼｭ	軸外径	155	0.16～0.26	すきま0.40で交換	
		組立時内径				
		ブッシュ厚さ	7			
推力軸受	入力軸	軸側	155	0.25～0.55		
		軸受側				
		ｽﾗｽﾄﾒﾀﾙ厚さ(前後)	15		厚さ14.5で交換	
	2段小歯車	軸側	403	0.50～0.95		
		軸受側				
		ｽﾗｽﾄﾒﾀﾙ厚さ a=	1.5		厚さa=1.1で交換	
		b=	10		厚さb=9.7で交換	
	出力軸	軸側	132	0.40～0.73		
		軸受側				
		ﾊﾟｯﾄﾞ厚さ	35.9		厚さ35.0で交換	
クラッチ摩擦板		ｽﾁｰﾙﾌﾟﾚｰﾄ厚さ	3.0±0.16		厚さ2.5で交換	
		ﾗｲﾆﾝｸﾞﾌﾟﾚｰﾄ厚さ	7.5±0.10		厚さ7.0または油溝深さ0.12で交換	

Clearance and wear limits

DRA-25J(L)

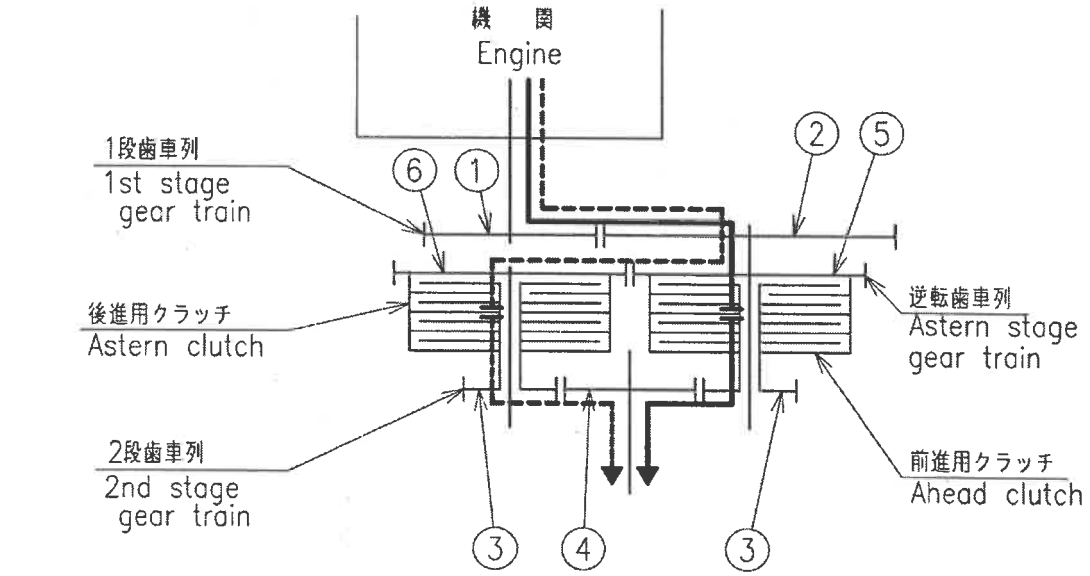
Part			D、L	Standard clearance	Wear limits	Measurement spot
Radial bearings	Input shaft (front)	Shaft diameter	155	0.11 ～0.19	Exchange at 3.85 thickness	
		Inner diameter as assembled				
		Thickness of metal	4			
	Input shaft (rear)	Shaft diameter	190	0.18 ～0.29	Exchange at 0.40 clearance	
		Inner diameter as assembled				
		Thickness of bush	10.5			
	Output shaft (front)	Shaft diameter	190	0.18 ～0.29	Exchange at 0.40 clearance	
		Inner diameter as assembled				
		Thickness of bush	10.5			
	Output shaft (rear)	Shaft diameter	250	0.21 ～0.30	Exchange at 4.85 thickness	
		Inner diameter as assembled				
		Thickness of metal	5			
	Gear bush	Shaft diameter	155	0.16 ～0.26	Exchange at 0.40 clearance	
		Inner diameter as assembled				
		Thickness of bush	7			
Thrust bearings	Input shaft	Shaft side	155	0.25 ～0.55		
		Bearing side				
		Thickness of thrust metal (front)(rear)	15		Exchange at 14.5 thickness	
	Secondary pinion gear	Shaft side	403	0.50 ～0.95	Exchange at a=1.1, b=9.7 thickness	
		Bearing side				
		Thickness of thrust metal	a=1.5			
			b=10			
	Output shaft	Shaft side	132	0.40 ～0.73		
		Bearing side				
		Thickness of thrust pads	35.9		Exchange at 35.0 thickness	
Clutch friction plate		Thickness of steel plate	3.0±0.16		Exchange at 2.5 thickness	
		Thickness of lining plate	7.5±0.10		Exchange at 7.0(thickness) or 0.12(groove)	

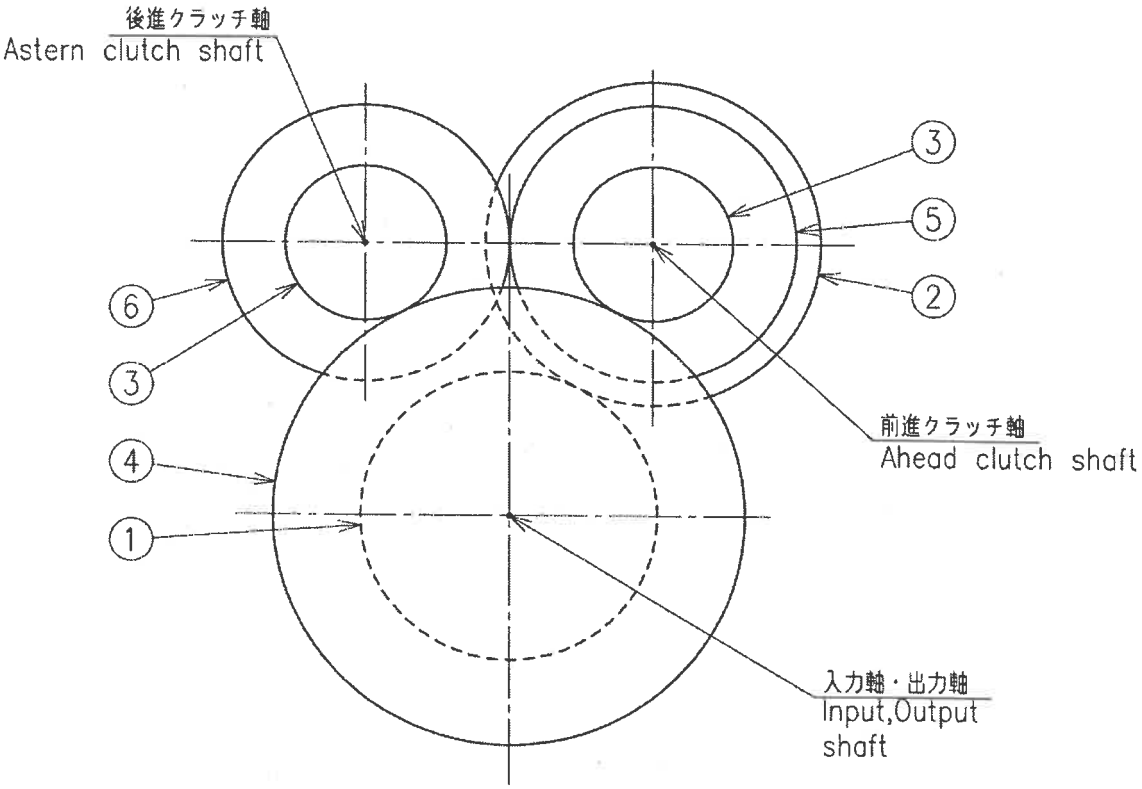


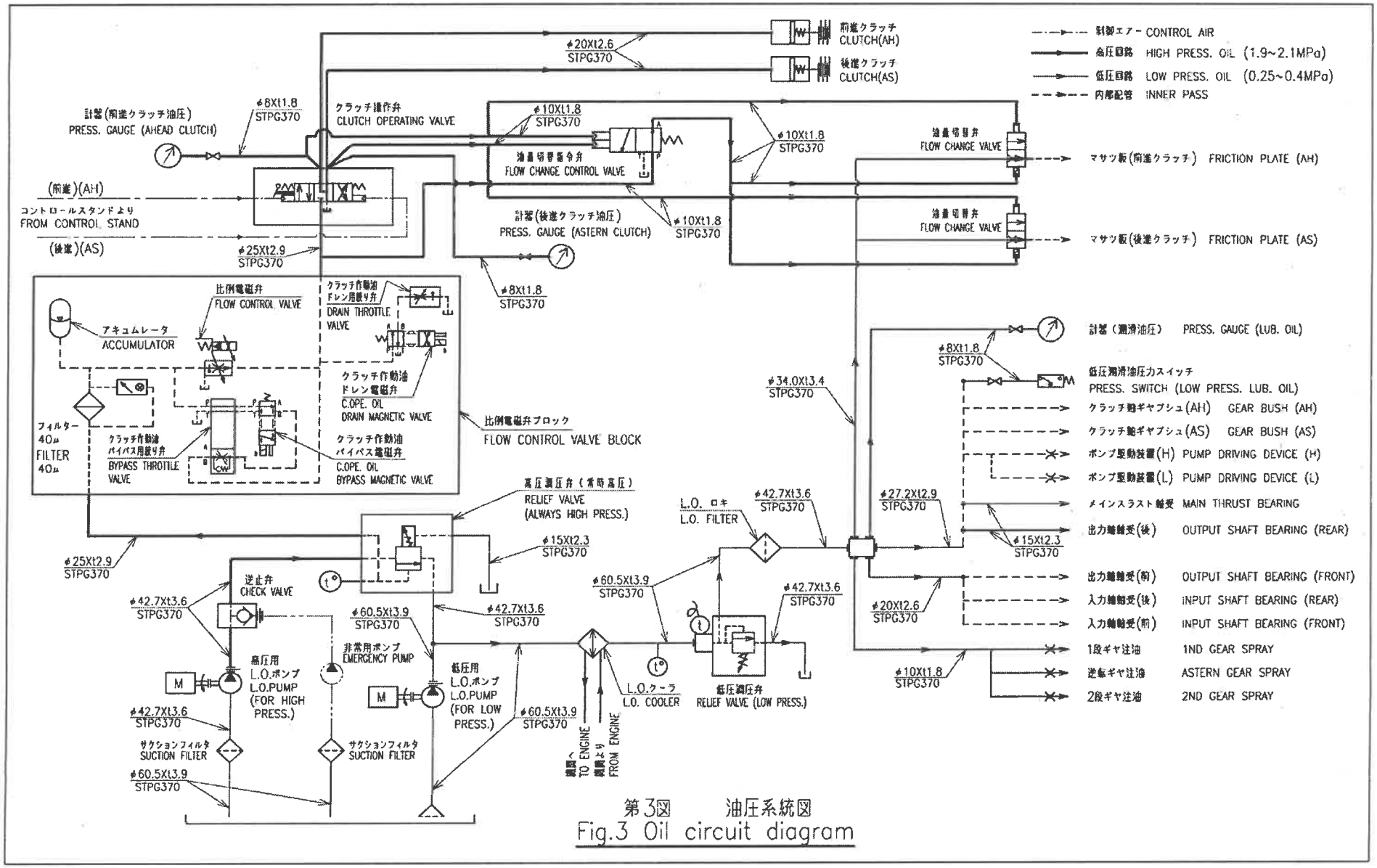
第1図 外形図
Fig.1 Out line

第２図　歯車系統図
Fig.2 Gear train

動力伝達経路
(Power transmission)
前進時 (Ahead) ①→②→③→④
後進時 (Astern) ①→②→⑤→⑥→③→④





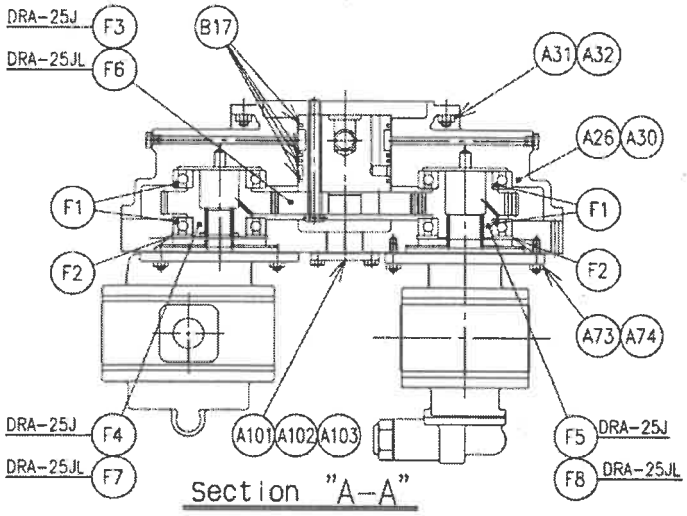


第3図　油圧系統図

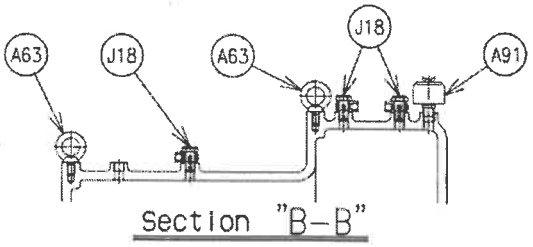
Fig.3 Oil circuit diagram

第4－1図 断面図

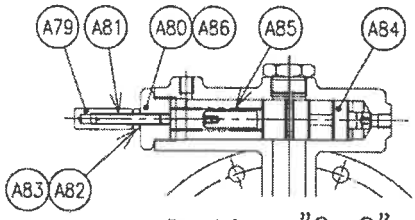
Fig.4－1 Sectional view



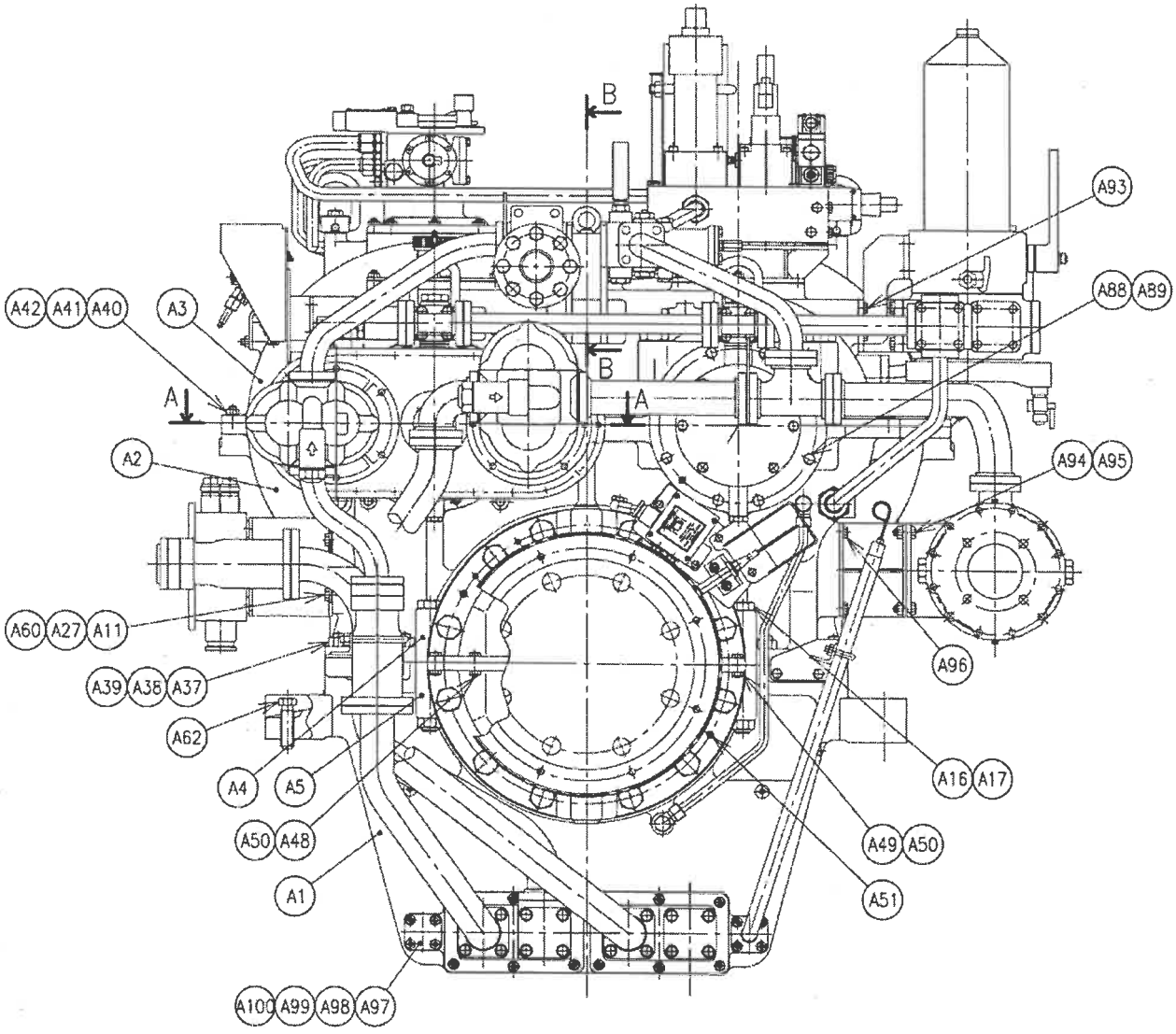
Section "A－A"



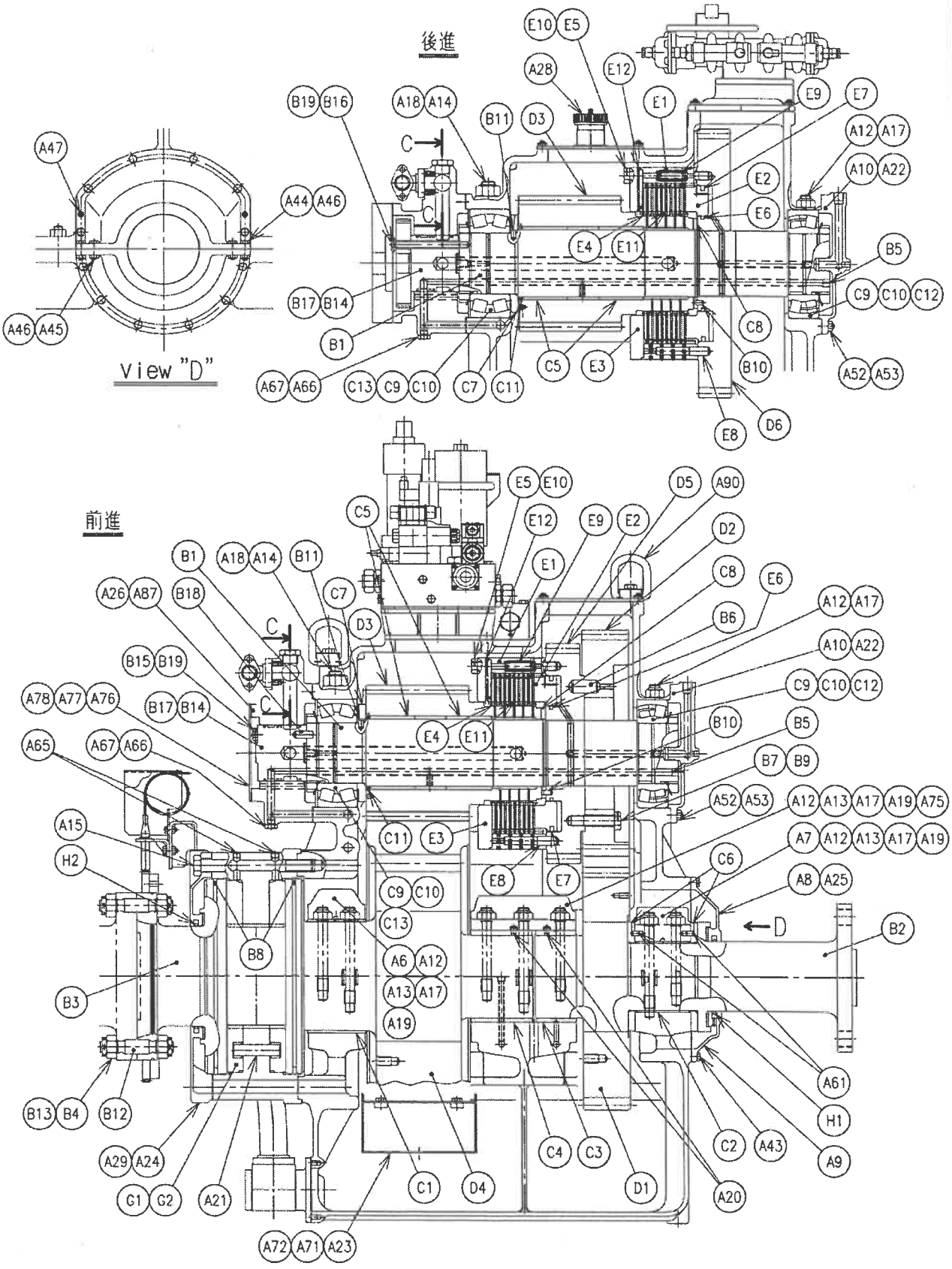
Section "B－B"



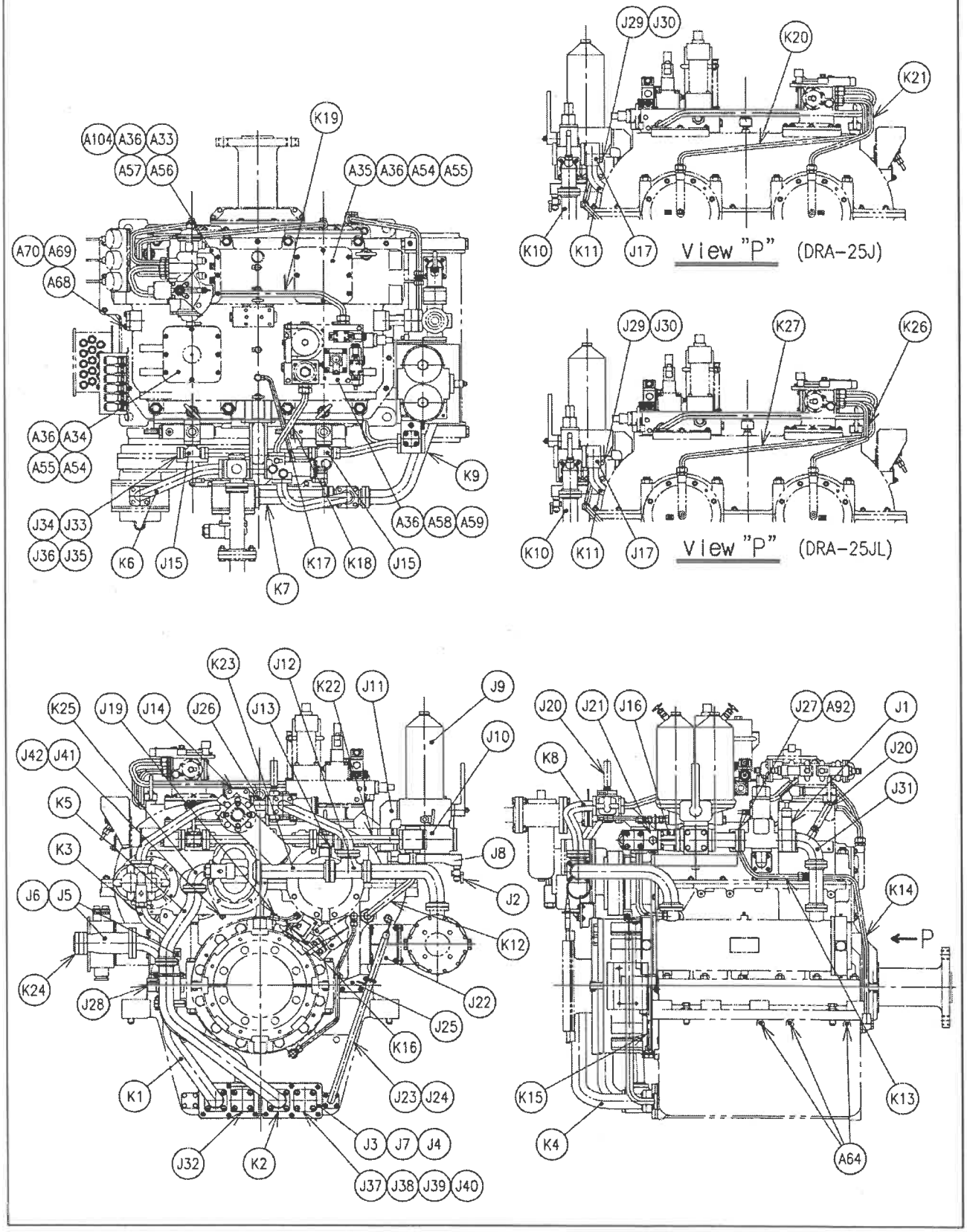
Section "C－C"



第4−2図 断面図
Fig.4−2 Sectional view



第4－3図　断面図
Fig.4－3 Sectional view



第4図　断面図
Fig.4　Sectional view

番号 No.	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	数量 Qty. DRA-25JL
A 1	AG89001001A	ギヤボックス:(1):DRA-25J	GEAR BOX:(1):DRA-25J	1	1
A 2	AG89001002B	ギヤボックス:(2):DRA-25J	GEAR BOX:(2):DRA-25J	1	1
A 3	AG89001003A	ギヤボックスフタ::DRA-25J	GEAR BOX COVER::DRA-25J	1	1
A 4	G254120051Z	-:メインスラストケース(ウエ):	-:MAIN THRUST CASE(UPPER):	1	1
A 5	G254120052Z	-:メインスラストケース(シタ):	-:MAIN THRUST CASE(UNDER):	1	1
A 6	G254120080Z	メタルキャップ::260X126	METAL CAP::260X126	1	1
A 7	G254120100Z	-:メタルキヤツプ 163X125:	-:METAL CAP 163X125:	1	1
A 8	G254120141Z	-:ニユウリヨクジクマエフタ(ウエ):	-:INPUT SHAFT FRONT COVER(UPPER):	1	1
A 9	G254120142Z	-:ニユウリヨクジクマエフタ(シタ):	-:INPUT SHAFT FRONT COVER(UNDER):	1	1
A 10	G254120180A	フタ:クラッチジク(マエ):DRA-25J	COVER:CLUTCH SHAFT (FRONT):DRA-25J	2	2
A 11	G254120220Z	-:テンケンマドフタ(3):	INSPECTION COVER(3)	2	2
A 12	G254120240A	ボルト:キャップボルト:M24X3.0X214	BOLT:CAP BOLT:M24X3.0X214	12	12
A 13	G254120250A	ボルト:キャップボルト:M24X3.0X242	BOLT:CAP BOLT:M24X3.0X242	6	6
A 14	G194120250Z	ボルト:キャップボルト:M30X3.5X252	BOLT:CAP BOLT:M30X3.5X252	4	4
A 15	G254120270A	-:ボルト M27X3.0X250:	-:BOLT M27X3.0X250:	12	12
A 16	G112200450Z	-:メインスラストケースリーマボルト:	-:REAMER BOLT MAIN THRUST CASE:	4	4
A 17	G121700310Z	-:Uナツト M24X3.0:	-:U-NUT M24X3.0:	22	22
A 18	G113500710Z	-:Uナツト M30X3.5:	-:U-NUT M30X3.5:	4	4
A 19	G121700590Z	-:リングノツク34X35:	-:RING KNOCK 34X35:	6	6
A 20	G194120330Z	-:ブシユコテイピン:	-:BUSH PIN:	2	2
A 21	G254120340Z	-:パツドピン 25X103:	-:PAD PIN 25X103:	10	10
A 22	G001500100Z	-:ユアツリング 30X2.5:	-:OIL PRESS.RING 30X2.5X1.5:	4	4
A 23	G254120360Z	-:2ダンギヤカバー:	-:COVER SECONDARY WHEEL:	1	1
A 24	G254120370Z	-:シユツリヨクジクフタパツキン:	GASKET OUTPUT SHAFT COVER:	1	1
A 25	G254120380Z	-:ニユウリヨクジクフタパツキン:	-:GASKET INPUT SHAFT FRONT COVER:	1	1
A 26	G194120390Z	-:ポンプトリツケフタパツキン:	-:GASKET PUMP FITTING COVER:	2	2
A 27	G254120420Z	-:テンケンマドフタ(3)パツキン:	-:GASKET INSPECTION COVER(3):	2	2
A 28	B078170010Z	-:キヤツプ (PT2):	CAP(PT2):	1	1
A 29	G254125010A	カンビヒン:シュツリョクジク フタ:DR-25F	ASSY.:OUTPUT SHAFT COVER:DR-25F	1	1
A 30	AG89001004B	カンビヒン:ポンプトリツケフタ:DRA-25J	ASSY.:PUMP FITTING COVER:DRA-25J	1	1
A 31	X210012046ZZ	スタッド::M12X1.75X46	STUD::M12X1.75X46	8	8
A 32	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	8	8
A 33	AG25001005A	フタ:ソウサベントリツケ:240X240	COVER:OPERATING VALVE FIT:240X240	1	1
A 34	AG25001006A	フタ:L.Oホキュウカン:240X240	COVER:L.O REPLENISH PIPE:240X240	1	1
A 35	AG25001007A	フタ:テンケンマド:240X240	COVER:INSPECTION COVER:240X240	1	1
A 36	AG25001008A	パッキン:テンケンマド:240X240	PACKING:INSPECTION COVER:240X240	4	4
A 37	X210020080ZZ	スタッド::M20X2.5X80	STUD::M20X2.5X80	18	18
A 38	X220020000ZZ	ナット::M20X2.5	NUT::M20X2.5	18	18
A 39	Z336016040ZZ	ダンツキピン::16X40	STEPPED PIN::16X40	2	2
A 40	X210012048ZZ	スタッド::M12X1.75X48	STUD::M12X1.75X48	15	15
A 41	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	15	15
A 42	Z336010030ZZ	ダンツキピン::10X30	STEPPED PIN::10X30	2	2
A 43	X200010022ZZ	ボルト::M10X1.5X22	BOLT::M10X1.5X22	12	12

番号 No.	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	数量 Qty. DRA-25JL
A 44	X200010038ZZ	ボルト::M10X1.5X38	BOLT::M10X1.5X38	2	2
A 45	X345012022ZZ	リーマボルト::12X22	REAMER BOLT::12X22	2	2
A 46	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	4	4
A 47	X341008025ZZ	ノックボルトASSY.::8X25	KNOCK BOLT ASSY.::8X25	2	2
A 48	X200010038ZZ	ボルト::M10X1.5X38	BOLT::M10X1.5X38	2	2
A 49	X345012022ZZ	リーマボルト::12X22	REAMER BOLT::12X22	2	2
A 50	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	4	4
A 51	X341008030ZZ	ノックボルトASSY.::8X30	KNOCK BOLT ASSY.::8X30	2	2
A 52	X210012046ZZ	スタッド::M12X1.75X46	STUD::M12X1.75X46	16	16
A 53	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	16	16
A 54	X210010035ZZ	スタッド::M10X1.5X35	STUD::M10X1.5X35	16	16
A 55	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	16	16
A 56	X210010043ZZ	スタッド::M10X1.5X43	STUD::M10X1.5X43	8	8
A 57	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	8	8
A 58	X210010038ZZ	スタッド::M10X1.5X38	STUD::M10X1.5X38	8	8
A 59	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	8	8
A 60	X200012020ZZ	ボルト::M12X1.75X20	BOLT::M12X1.75X20	12	12
A 61	Z335610016ZZ	ヘイコウピン:(6):10X16	STRAIGHT PIN:(6):10X16	2	2
A 62	X213024130ZZ	オシボルト::M24X3.0X130	JACK BOLT::M24X3.0X130	4	4
A 63	Z270012000ZZ	アイボルト::M12	EYE BOLT::M12	2	2
A 64	Z571503000ZZ	テーパプラグ:(5):3/8	TAPER PLUG:(5):3/8	3	3
A 65	Z571503000ZZ	テーパプラグ:(5):3/8	TAPER PLUG:(5):3/8	2	2
A 66	X570003000ZZ	ロッカクプラグ::3/8	HEX. PLUG::3/8	2	2
A 67	Z565001700EE	マルパッキン::17	GASKET::17	2	2
A 68	Z540200054ZZ	フランジ:(2):0X54	FLANGE:(2):0X54	1	1
A 69	Z541204054AZ	パッキン:(2):40X54	GASKET:(2):40X54	1	1
A 70	X200012025ZZ	ボルト::M12X1.75X25	BOLT::M12X1.75X25	4	4
A 71	X200112025ZZ	ボルト:(1):M12X1.75X25	BOLT:(1):M12X1.75X25	4	4
A 72	Z321010050DZ	ハリガネ::1.0X500	WIRE::1.0X500	2	2
A 73	X210010040ZZ	スタッド::M10X1.5X40	STUD::M10X1.5X40	12	12
A 74	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	12	12
A 75	AG32001006A	メタルキャップ:211x234 (2):DRA-25F 1.23	METAL CAP:211x234 (2):DRA-25F 1.23	1	1
A 76	AG69001007A	カバー:ポンプトリツケフタ:DR-100F	COVER:PUMP FITTING COVER:DR-100F	1	1
A 77	X200010018ZZ	ボルト::M10X1.5X18	BOLT::M10X1.5X18	6	6
A 78	G003100470A	-:ポンプパツキン:	-:LO PUMP GASKET:	1	1
A 79	AG83001005A	フクロナット::M12x1.25x50	CAP NUT::M12x1.25x50	2	2
A 80	AG83001006A	ロッカクプラグ::G11/4xM12x1.25	HEX. PLUG::G11/4xM12x1.25	2	2
A 81	X251512070ZZ	トメネジ:(5):M12X1.25X70	SET SCREW:-5:M12X1.25X70	2	2
A 82	X220812000ZZ	ナット:(8):M12X1.25	NUT:(8):M12X1.25	2	2
A 83	Z565001300EE	マルパッキン::13	GASKET::13	4	4
A 84	AG72001003B	ベン:ユリョウキリカエ:DRA-30FS	VALVE:FLOW CHANGE:DRA-30FS	2	2
A 85	G889700700Z	-:バネ, ユリヨウキリカエ:	-:SPRING FLOW CHANGE VALVE:	2	2
A 86	Z565004200EE	マルパッキン::42	GASKET::42	2	2
A 87	AG89001007A	フタ:クラッチジクウシロ:DRA-25J	COVER:CLUTCH SHAFT REAR:DRA-25J	1	1
A 88	X210012046ZZ	スタッド::M12X1.75X46	STUD::M12X1.75X46	8	8
A 89	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	8	8

番号 No.	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	DRA-25JL
A 90	AG89001008A	ツリカナグ:フレノリンクボルト:A-16	HANGER:FRENO LINK BOLT:A-16	4	4
A 91	G005500110Z	-:ブリーザ:	-:BREEZER:	1	1
A 92	X200010020ZZ	ボルト::M10X1.5X20	BOLT::M10X1.5X20	2	2
A 93	X200012020ZZ	ボルト::M12X1.75X20	BOLT::M12X1.75X20	4	4
A 94	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	2	2
A 95	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	2	2
A 96	X200012020ZZ	ボルト::M12X1.75X20	BOLT::M12X1.75X20	4	4
A 97	Z541203044AZ	パッキン:(2):30X44	GASKET:(2):30X44	1	1
A 98	Z540200044ZZ	フランジ:(2):0X44	FLANGE:(2):0X44	1	1
A 99	Z310010000ZZ	バネザガネ::M10	SPRING WASHER::M10	4	4
A 100	X200010025ZZ	ボルト::M10X1.5X25	BOLT::M10X1.5X25	4	4
A 101	Z503303500ZZ	JISフランジトク:(3):35-00	JIS SPECIAL FLANGE:(3):35-00	1	1
A 102	Z501303500AZ	JISパッキン:(3):35	JIS GASKET:(3):35	1	1
A 103	X200010022ZZ	ボルト::M10X1.5X22	BOLT::M10X1.5X22	4	4
A 104	X200012025ZZ	ボルト::M12X1.75X25	BOLT::M12X1.75X25	3	3
B 1	AG89003001B	クラッチジク::DRA-25J	CLUTCH SHAFT::DRA-25J	2	2
B 2	AG89003003B	ニュウリョクジク::6DKM-20	INPUT SHAFT::6DKM-20	1	1
B 3	G253220050Z	-:シュツリヨクジク:	-:OUTPUT SHAFT:	1	1
B 4	G051700270Z	-:トクシュキクナツトM30X2.0:	-:SPECIAL CASTLE NUT M30X2.0:	16	16
B 5	G193220140Z	-:トメネジ PF3/8X1.5:	-:SET SCREW PF3/8X1.5:	2	2
B 6	G193220130Z	-:ヘイコウピン 30X50:	-:STRAIGT PIN 30X50:	4	4
B 7	G193220090Z	-:ボルト M24X2.0X95-6G:	-:BOLT M24X2.0X95-6G:	14	14
B 8	G253220170Z	-:ユアツリング 450X8:	-:OIL PRESS.RING 450X8:	2	2
B 9	Z321010200DZ	ハリガネ::1.0X2000	WIRE::1.0X2000	1	1
B 10	Z335610016ZZ	ヘイコウピン:(6):10X16	STRAIGHT PIN:(6):10X16	2	2
B 11	Z335610016ZZ	ヘイコウピン:(6):10X16	STRAIGHT PIN:(6):10X16	2	2
B 12	G253220100A	リーマボルト:ドウリョクデンタツ:32X164-M30X2.0	REAMER BOLT:POWER TRANSMISSION:32X164-M30X2.0	8	8
B 13	Z320006060ZZ	ワリピン::6X60	SPLIT PIN::6X60	16	16
B 14	AG72003005A	セッシュ:LOキュウユ:DRA-30FS	JOINT:LO SUPPLY:DRA-30FS	2	2
B 15	AG72003006A	ボルト:(1):M16x2.0x130	BOLT:-1:M16x2.0x130	4	4
B 16	G875800070Z	-:ボルト M16X2.0X162:	-:BOLT M16X2.0X162:	4	4
B 17	G010610260Z	-:ユアツリング130X4:	-:OIL PRESS.RING 130X4:	8	8
B 18	Z336016040ZZ	ダンツキピン::16X40	STEPPED PIN::16X40	3	3
B 19	Z321010200DZ	ハリガネ::1.0X2000	WIRE::1.0X2000	1	1
C 1	G253320010Z	-:メタル 250X260X130:	-:METAL 250X260X130:	1	1
C 2	G253320020Z	-:メタル 155X163X115:	-:METAL 155X163X115:	1	1
C 3	G253320030Z	-:ブシュ 190X211X90:	-:BUSH 190X211X90:	1	1
C 4	G253320040Z	-:ブシュ 190X211X135:	-:BUSH 190X211X135:	1	1
C 5	G193320050Z	-:ギヤブシュ 155X169X100:	-:GEAR BUSH 155X169X100:	4	4
C 6	G042200120Z	-:スラストメタル156X250X15:	-:THRUST METAL 156X250X15:	2	2
C 7	G193320080Z	-:スラストメタル 140X225X20:	-:THRUST METAL 140X225X20:	2	2
C 8	G193320090Z	-:スラストメタル 158X207X10:	-:THRUST METAL 158X207X10:	2	2
C 9	G193320130Z	-:ベアリングナツトAN28:	-:BEARING NUT AN28:	4	4
C 10	G193320150Z	-:ベアリングザガネAW28:	-:BEARING WASHER AW28:	4	4
C 11	G193320180Z	-:ストツプリング:	-:STOP RING:	2	2

番号 No.	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	数量 Qty. DRA-25JL
C 12	G253325010Z	ベアリング::22228CAME4S11	BEARING::22228CAME4S11	2	2
C 13	G253325020Z	ベアリング::23228CAME4S11	BEARING::23228CAME4S11	2	2
D 1	AG89006	1ダンギヤ:(1):	PRIMARY GEAR:(1):	1	1
D 2	AG89006	1ダンギヤ:(2):	PRIMARY GEAR:(2):	1	1
D 3	AG89006	2ダンピニオンギヤ::	SECONDARY PINION::	2	2
D 4	AG89006	2ダンギヤ::	SECONDARY GEAR::	1	1
D 5	AG89006	ギャクテンギヤ:(1):	ASTERN GEAR:(1):	1	1
D 6	AG89006	ギャクテンギヤ:(2):	ASTERN GEAR:(2):	1	1
E 1	AG89002001A	クラッチドラム:9クミ:DRA-25J	CLUTCH DRUM:9 SETS:DRA-25J	2	2
E 2	AG89002002A	クラッチピストン::DRA-25J	CLUTCH PISTON::DRA-25J	2	2
E 3	AG89002003A	クラッチウケイタ::DRA-25J	CLUTCH SUPPORT PLATE::DRA-25J	2	2
E 4	G253120230A	ライニングプレート:DR-25J:P186	LINING PLATE:DR-25J:P186	18	18
E 5	G253120110B	ボルト:ドラム:M20X2.0X177	BOLT:DRUM:M20X2.0X177	36	36
E 6	G011400300Z	-:ユアツリング220X7:	-:OIL PRESS.RING(2)220X7X7:	2	2
E 7	G014500300Z	-:ユアツリング 360X8:	-:OIL PRESS.RING:	2	2
E 8	G253120150Z	-:ダンツキピン 20X38:	-:STEPPED PIN 20X38:	8	8
E 9	G253120160Z	-:バネ(ピストン):	-:SPRING (PISTON):	12	12
E 10	Z321010300DZ	ハリガネ::1.0X3000	WIRE::1.0X3000	1	1
E 11	G886800040A	スチールプレート::DR-25J	STEEL PLATE::DR-25J	18	18
E 12	X251206008ZZ	トメネジ:(2):M6X1.0X8	:SET SCREW:(2)	12	12
F 1	G840900400A	ベアリング:6215:75X130X25	BEARING:6215:75X130X25	4	4
F 2	G859200090Z	-:スナツプリング(アナ)130:	-:SNAP RING (HOLE) 130:	2	2
F 3	NN00733	ギヤ:ポンプクドウ(1):	GEAR:PUMP DRIVE (1):	1	
F 4	NN00733	ギヤ:ポンプクドウ(2):	GEAR:PUMP DRIVE (2):	1	
F 5	NN00733	ギヤ:ポンプクドウ(2):	GEAR:PUMP DRIVE (2):	1	
F 6	NN00733	ギヤ:ポンプクドウ(1):	GEAR:PUMP DRIVE (1):		1
F 7	NN00733	ギヤ:ポンプクドウ(2):	GEAR:PUMP DRIVE (2):		1
F 8	NN00733	ギヤ:ポンプクドウ(2):	GEAR:PUMP DRIVE (2):		1
G 1	G005000020Z	-:パツド 240X442X35.9:	-:PAD 240X442X35.9:	20	20
G 2	G005000150Z	-:パツドピンブシユ:	-:PAD PIN BUSH:	20	20
H 1	G001701895A	-:マルベルト 7X555:	-:O BELT 7X555:	1	1
H 2	G001701973A	-:マルベルト 7X810:	-:O BELT 7X810:	1	1
J 1	A245073550Z	-:サーモセンサーザE-PF3/4X65:	-:BLOCK SENSOR E-PF3/4X65:	1	1
J 1 a	X200012095ZZ	ボルト::M12X1.75X95	BOLT::M12X1.75X95	4	4
J 1 b	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
J 1 c	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	2	2
J 2	Z580403000ZA	ドレンコック:(4):3/8	DRAIN COCK:(4):3/8	1	1
J 2 a	Z565001700EE	マルパッキン::17	GASKET::17	1	1
J 3	AG65024012A	フタ:コシアミトリツケフタ:60X60	COVER:STRAINER FITTING COVER:60X60	2	2
J 4	C007770080A	-:スイコミカントリツケフタパツキン-20:	-:GASKET-20 HOLDER:	3	3
J 4 a	X210010043ZZ	スタッド::M10X1.5X43	STUD::M10X1.5X43	6	6
J 4 b	X210010055ZZ	スタッド::M10X1.5X55	STUD::M10X1.5X55	6	6

番号 No.	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	数量 Qty. DRA-25JL
J 4 c	Z310010000ZZ	バネザガネ::M10	SPRING WASHER::M10	12	12
J 4 d	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	12	12
J 5	G875900050Z	ケース:コシアミ:50A	CASE:SUCTION FILTER:50A	2	2
J 6	AG83024001A	フィルター:コシアミ:50A 100メッシュ	FILTER:SUCTION:50A 100MESH	2	2
J 6 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	2	2
J 7	G001700030A	-:コシアミ:	-:L.O.STRAINER:	1	1
J 8	G001700700Z	-:ウケザラ:	-:SAUCER:	1	1
J 8 a	X200012055ZZ	ボルト::M12X1.75X55	BOLT::M12X1.75X55	4	4
J 9	C060700011Z	-:ロキ 50PC5L/1-X/KZ:	-:FILTER 50PC5L/1-X/KZ:	1	1
J 10	G001700660Z	-:ロキセツシユ(50X):	-:BLOCK L.O.FILTER(50X):	1	1
J 10 a	Z540200065ZZ	フランジ:(2):0X65	FLANGE:(2):0X65	4	4
J 10 b	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	4	4
J 10 c	X200012135ZZ	ボルト::M12X1.75X135	BOLT::M12X1.75X135	8	8
J 11	AG89024006A	ブラケット:ロキ:DRA-25J	BRACKET:FILTER:DRA-25J	1	1
J 12	G886900800Z	-:LOカン, ポンプデグチ:	-:L.O.PIPING PUMP OUTLET:	1	1
J 13	G002200230Z	-:メツキメクラフランジ45:	-:EYE FLANGE 45:	1	1
J 14	B056370020Z	-:メツキメクラフランジ:	-:EYE FLNGE:	1	1
J 15	AG83020034A	LOカン:チー:25A	L.O. PIPING:TEE:25A	2	2
J 15 a	X200010030ZZ	ボルト::M10X1.5X30	BOLT::M10X1.5X30	4	4
J 16	AG72020028A	LOカン:ロキデグチ L90:32A STPG370	L.O. PIPING:FILTER OUTLET L90:32A STPG370	1	1
J 16 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
J 17	G005500330Z	-:サシコミブロツク(2)43X65:	-:BLOCK(2)43X65:	1	1
J 18	G167120050Z	-:トクシユボルト3/8X3.0:	-:SPECIAL BOLT 3/8X3.0:	3	3
J 19	G001700730Z	-:トクシユボルトPF3/8:	-:SPECIAL BOLT PF3/8:	1	1
J 20	Y231004000ZC	オンドケイ::100CX1/2P	THERMOMETER::100CX1/2P	2	2
J 20 a	Z565002100EE	マルパッキン::21	GASKET::21	2	2
J 21	AG72024013A	カンセッシュ::G54x44x1/2	CONNECTOR::G54x44x1/2	1	1
J 21 a	X555002040ZZ	セッシュザ::1/4X1/2	JOINT SEAT::1/4X1/2	2	2
J 21 b	Z541203044AZ	パッキン:(2):30X44	GASKET:(2):30X44	1	1
J 21 c	Z541204054AZ	パッキン:(2):40X54	GASKET:(2):40X54	2	2
J 21 d	Z540200044ZZ	フランジ:(2):0X44	FLANGE:(2):0X44	1	1
J 21 e	Z540200054ZZ	フランジ:(2):0X54	FLANGE:(2):0X54	1	1
J 21 f	X200010028ZZ	ボルト::M10X1.5X28	BOLT::M10X1.5X28	4	4
J 21 g	Z565002100EE	マルパッキン::21	GASKET::21	4	4
J 21 h	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	12	12
J 21 i	X570004000ZZ	ロッカクプラグ::1/2	HEX. PLUG::1/2	2	2
J 22	AG89024001A	ブラケット:クーラ:8.28	BRACKET:COOLER:8.28	1	1
J 23	AG89024002A	パイプ:ケンユボウ:DRA-25J	PIPE:OIL LEVEL STICK:DRA-25J	1	1
J 24	AG20024001B	ケンユボウ::618	SOUNDING PIPE::618	1	1
J 24 a	Z310010000ZZ	バネザガネ::M10	SPRING WASHER::M10	4	4
J 24 b	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	4	4
J 24 c	X210010047ZZ	スタッド::M10X1.5X47	STUD::M10X1.5X47	4	4
J 25	AG89024003A	カンササエ:ケンユカン:DRA-25J	PIPE SUPPORTER:OIL LEVEL PIPE:DRA-25J	1	1
J 25 a	Z260102000ZZ	Uボルト:(1):20	U-BOLT:(1):20	1	1
J 25 b	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	2	2
J 25 c	X200010015ZZ	ボルト::M10X1.5X15	BOLT::M10X1.5X15	2	2

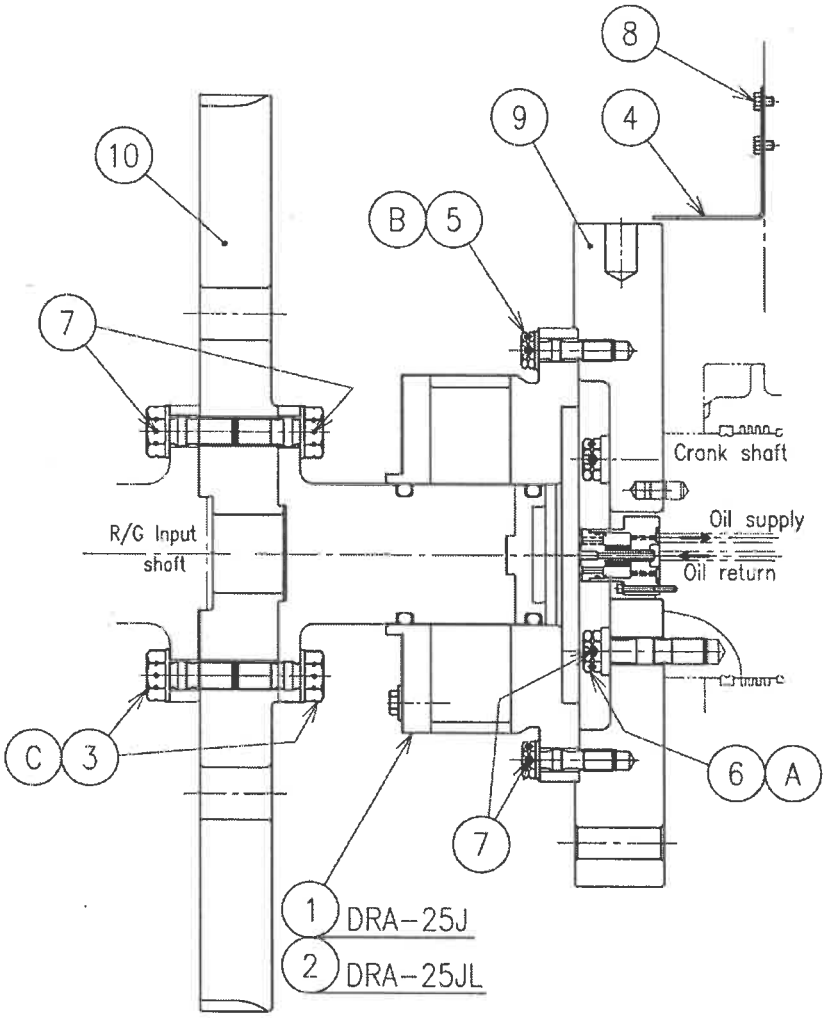
番号 No.	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	DRA-25JL
J 26	AG89024004B	ブラケット:コウアツチョウアツベン:DRA-25J	BRACKET:HRV:DRA-25J	1	1
J 26 a	X200012025ZZ	ボルト::M12X1.75X25	BOLT::M12X1.75X25	4	4
J 27	AG89024005A	ブラケット:テイアツチョウアツベン:DRA-25J	BRACKET:LRV:DRA-25J	1	1
J 28	AG89024007A	カンササエ:ポンプサクション:DRA-25J	PIPE SUPPORTER:PUMP SUCTION:DRA-25J	1	1
J 28 a	Z260105000ZZ	Uボルト:(1):50	U-BOLT:(1):50	2	2
J 28 b	Z260108000ZZ	Uボルト:(1):80	U-BOLT:(1):80	1	1
J 28 c	X200010015ZZ	ボルト::M10X1.5X15	BOLT::M10X1.5X15	2	2
J 28 d	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	4	4
J 28 e	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	2	2
J 29	Z540200465ZZ	フランジ:(2):1/2X65	FLANGE:(2):1/2X65	1	1
J 30	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
J 31	AG72020008A	LOカン:LRVイリグチ:50A	L.O. PIPING:LRV INLET:50A	1	1
J 32	G001700180Z	-:カンセツシユ S 60X65:	-:PIPE JOINT S 60X65:	1	1
J 32 a	Z540200065ZZ	フランジ:(2):0X65	FLANGE:(2):0X65	1	1
J 32 b	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
J 32 c	X210012140ZZ	スタッド::M12X1.75X140	STUD::M12X1.75X140	4	4
J 32 d	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
J 32 e	Z310012000ZZ	バネザガネ::M12	SPRING WASHER::M12	4	4
J 33	Z540100060ZZ	フランジ:(1):0X60	FLANGE:(1):0X60	1	1
J 34	Z541103560AZ	パッキン:(1):35X60	GASKET:(1):35X60	1	1
J 35	X200010045ZZ	ボルト::M10X1.5X45	BOLT::M10X1.5X45	2	2
J 36	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	2	2
J 37	Z540200065ZZ	フランジ:(2):0X65	FLANGE:(2):0X65	1	1
J 38	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
J 39	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
J 40	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
J 41	X570003000ZZ	ロッカクプラグ::3/8	HEX. PLUG::3/8	1	1
J 42	Z565001700EE	マルパッキン::17	GASKET::17	1	1
K 1	AG89020001A	カンビヒン:LOカン:EPサクション DRA-25J(L)	ASSY.:L.O. PIPING:EP SUCTION DRA-25J(L)	1	1
K 1 a	Z541209096AZ	パッキン:(2):90X96	GASKET:(2):90X96	1	1
K 1 b	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
K 1 c	X200016045ZZ	ボルト::M16X2.0X45	BOLT::M16X2.0X45	4	4
K 1 d	X220016000ZZ	ナット::M16X2.0	NUT::M16X2.0	4	4
K 1 e	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 2	AG89020002A	カンビヒン:LOカン:Hポンプサクション(1) DRA-25J(L)	ASSY.:L.O. PIPING:H PUMP SUCTION (1) DRA-25J(L)	1	1
K 2 a	Z541209096AZ	パッキン:(2):90X96	GASKET:(2):90X96	1	1
K 2 b	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
K 2 c	X200016045ZZ	ボルト::M16X2.0X45	BOLT::M16X2.0X45	4	4
K 2 d	X220016000ZZ	ナット::M16X2.0	NUT::M16X2.0	4	4
K 2 e	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 2 f	Z310012000ZZ	バネザガネ::M12	SPRING WASHER::M12	4	4
K 3	AG89020003A	カンビヒン:LOカン:Hポンプサクション(2) DRA-25J(L)	ASSY.:L.O. PIPING:H PUMP SUCTION (2) DRA-25J(L)	1	1
K 3 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
K 3 b	Z541204054AZ	パッキン:(2):40X54	GASKET:(2):40X54	1	1
K 3 c	X200012080ZZ	ボルト::M12X1.75X80	BOLT::M12X1.75X80	4	4

番号 No.	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	数量 Qty. DRA-25JL
K 3 d	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
K 3 e	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 4	AG89020004A	カンビヒン:LOカン:Lポンプサクション(1) DRA-25J(L)	ASSY.:L.O. PIPING:L PUMP SUCTION (1) DRA-25J(L)	1	1
K 4 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	2	2
K 4 b	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 4 c	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
K 4 d	Z310012000ZZ	バネザガネ::M12	SPRING WASHER::M12	4	4
K 4 e	X200012045ZZ	ボルト::M12X1.75X45	BOLT::M12X1.75X45	4	4
K 5	AG89020005A	カンビヒン:LOカン:Lポンプサクション(2) DRA-25J(L)	ASSY.:L.O. PIPING:L PUMP SUCTION (2) DRA-25J(L)	1	1
K 5 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	2	2
K 5 b	X200012045ZZ	ボルト::M12X1.75X45	BOLT::M12X1.75X45	4	4
K 5 c	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
K 6	AG89020006A	カンビヒン:LOカンHP:Hポンプデグチ DRA-25J(L)	ASSY.:L.O. PIPING HP:H PUMP OUTLET DRA-25J(L)	1	1
K 6 a	Z541204054AZ	パッキン:(2):40X54	GASKET:(2):40X54	2	2
K 6 b	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 6 c	X200012035ZZ	ボルト::M12X1.75X35	BOLT::M12X1.75X35	4	4
K 7	AG89020007A	カンビヒン:LOカン:Lポンプデグチ DRA-25J(L)	ASSY.:L.O. PIPING:L PUMP OUTLET DRA-25J(L)	1	1
K 7 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	2	2
K 7 b	X200012050ZZ	ボルト::M12X1.75X50	BOLT::M12X1.75X50	4	4
K 7 c	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 7 d	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
K 8	AG89020008A	カンビヒン:LOカン:HRVデグチ DRA-25J(L)	ASSY.:L.O. PIPING:HRV OUTLET DRA-25J(L)	1	1
K 8 a	Z541204054AZ	パッキン:(2):40X54	GASKET:(2):40X54	2	2
K 8 b	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 8 c	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
K 8 d	X200012045ZZ	ボルト::M12X1.75X45	BOLT::M12X1.75X45	4	4
K 9	AG89020009B	カンビヒン:LOカン:クーライリグチ DRA-25J(L)	ASSY.:L.O. PIPING:COOLER INLET DRA-25J(L)	1	1
K 9 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	2	2
K 9 b	X200012045ZZ	ボルト::M12X1.75X45	BOLT::M12X1.75X45	8	8
K 9 c	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	8	8
K 10	AG89020010A	カンビヒン:LOカン:クーラデグチ DRA-25J(L)	ASSY.:L.O. PIPING:COOLER OUTLET DRA-25J(L)	1	1
K 10 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	2	2
K 10 b	X200012045ZZ	ボルト::M12X1.75X45	BOLT::M12X1.75X45	8	8
K 10 c	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	8	8
K 11	AG89020011A	カンビヒン:LOカン:LRVドレン DRA-25J(L)	ASSY.:L.O. PIPING:LRV DRAIN DRA-25J(L)	1	1
K 11 a	Z541204054AZ	パッキン:(2):40X54	GASKET:(2):40X54	1	1
K 11 b	X200012075ZZ	ボルト::M12X1.75X75	BOLT::M12X1.75X75	4	4
K 11 c	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 12	AG89020012C	カンビヒン:LOカン:ギヤボックス(2)イリグチ DRA-25J(L)	ASSY.:L.O. PIPING:GEAR BOX (2) INLET DRA-25J(L)	1	1
K 12 a	Z541203044AZ	パッキン:(2):30X44	GASKET:(2):30X44	1	1
K 12 b	X550230000ZZ	ユニオンネジ::30	UNION SCREW::30	1	1
K 12 c	Z565003400EE	マルパッキン::34	GASKET::34	1	1
K 12 d	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 13	AG89020013C	カンビヒン:LOカン:ギヤボックス(1)イリグチ(1)	ASSY.:L.O. PIPING:GEAR BOX (1) INLET (1)	1	1
K 13 a	Z565002100EE	マルパッキン::21	GASKET::21	1	1
K 14	AG89020014A	カンビヒン:LOカン:ギヤボックス(1)イリグチ(2)	ASSY.:L.O. PIPING:GEAR BOX (1) INLET (2)	1	1

番号 No.	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	数量 Qty. DRA-25JL
K 14 a	X545210000ZZ	ツギボルト::1	BANJO PLUG::1	1	1
K 14 b	Z565003400EE	マルパッキン::34	GASKET::34	1	1
K 15	AG89020015A	カンビヒン:LOカン:シュツリョクジクウシロ DRA-25J(L)	ASSY.:L.O. PIPING:OUTPUT SHAFT REAR DRA-25J(L)	1	1
K 15 a	Z565001700EE	マルパッキン::17	GASKET::17	2	2
K 16	AG89020016A	カンビヒン:LOカン:メインスラストキュウユ DRA-25J(L)	ASSY.:L.O. PIPING:MAIN THRUST SUPPLY DRA-25J(L)	1	1
K 16 a	Z565001700EE	マルパッキン::17	GASKET::17	2	2
K 17	AG89020017A	カンビヒン:LOカン:HRVドレン DRA-25J(L)	ASSY.:L.O. PIPING:HRV DRAIN DRA-25J(L)	1	1
K 17 a	X545203000ZZ	ツギボルト::3/8	BANJO PLUG::3/8	1	1
K 17 b	Z565001700EE	マルパッキン::17	GASKET::17	2	2
K 18	AG89020018A	カンビヒン:LOカンHP:FCVイリグチ DRA-25J(L)	ASSY.:L.O. PIPING HP:FCV INLET DRA-25J(L)	1	1
K 18 a	Z541203044AZ	パッキン:(2):30X44	GASKET:(2):30X44	1	1
K 18 b	X550228000ZZ	ユニオンネジ::28	UNION SCREW::28	1	1
K 18 c	Z565003400EE	マルパッキン::34	GASKET::34	1	1
K 18 d	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 19	AG89020019A	カンビヒン:LOカンHP:FCVデグチ DRA-25J(L)	ASSY.:L.O. PIPING HP:FCV OUTLET DRA-25J(L)	1	1
K 19 a	X550228000ZZ	ユニオンネジ::28	UNION SCREW::28	2	2
K 19 b	Z565003400EE	マルパッキン::34	GASKET::34	2	2
K 20	AG89020020A	カンビヒン:LOカン HP:COVデグチR AHD DRA-25J	ASSY.:L.O. PIPING HP:COV OUTLET R AHD DRA-25J	1	
K 20 a	X550222000ZZ	ユニオンネジ::22	UNION SCREW::22	2	
K 20 b	Z565002700EE	マルパッキン::27	GASKET::27	2	
K 21	AG89020021A	カンビヒン:LOカン HP:COVデグチR AST DRA-25J	ASSY.:L.O. PIPING HP:COV OUTLET R AST DRA-25J	1	
K 21 a	X550222000ZZ	ユニオンネジ::22	UNION SCREW::22	2	
K 21 b	Z565002700EE	マルパッキン::27	GASKET::27	2	
K 22	AG89020047B	カンビヒン:LOカン:クラッチキュウユ(1) DRA-25J(L)	ASSY.:L.O. PIPING:CLUTCH SUPPLY (1) DRA-25J(L)	1	1
K 22 a	Z541204054AZ	パッキン:(2):40X54	GASKET:(2):40X54	1	1
K 22 b	Z541103560AZ	パッキン:(1):35X60	GASKET:(1):35X60	1	1
K 22 c	X200012050ZZ	ボルト::M12X1.75X50	BOLT::M12X1.75X50	2	2
K 22 d	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 22 e	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
K 23	AG89020048A	カンビヒン:LOカン:クラッチキュウユ(2) DRA-25J(L)	ASSY.:L.O. PIPING:CLUTCH SUPPLY (2) DRA-25J(L)	1	1
K 23 a	Z541103560AZ	パッキン:(1):35X60	GASKET:(1):35X60	2	2
K 23 b	X220010000ZZ	ナット::M10X1.5	NUT::M10X1.5	4	4
K 23 c	X200010050ZZ	ボルト::M10X1.5X50	BOLT::M10X1.5X50	4	4
K 24	Z540206065ZZ	フランジ:(2):60X65	FLANGE:(2):60X65	1	1
K 24 a	X200012080ZZ	ボルト::M12X1.75X80	BOLT::M12X1.75X80	4	4
K 24 b	X220012000ZZ	ナット::M12X1.75	NUT::M12X1.75	4	4
K 24 c	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
K 25	G886900560Z	-:LOカン, LOポンプイリグチ:	-:L.O.PIPING L.O.PUMP INLET:	1	1
K 25 a	Z541206065AZ	パッキン:(2):60X65	GASKET:(2):60X65	1	1
K 25 b	X200012030ZZ	ボルト::M12X1.75X30	BOLT::M12X1.75X30	4	4
K 26	AG89020022A	カンビヒン:LOカン HP:COVデグチL AHD DRA-25JL	ASSY.:L.O. PIPING HP:COV OUTLET L AHD DRA-25JL		1
K 26 a	X550222000ZZ	ユニオンネジ::22	UNION SCREW::22		2
K 26 b	Z565002700EE	マルパッキン::27	GASKET::27		2
K 27	AG89020023A	カンビヒン:LOカン HP:COVデグチL AST DRA-25JL	ASSY.:L.O. PIPING HP:COV OUTLET L AST DRA-25JL		1
K 27 a	X550222000ZZ	ユニオンネジ::22	UNION SCREW::22		2
K 27 b	Z565002700EE	マルパッキン::27	GASKET::27		2

第 5 図　　フレキシブルカップリング組立図

Fig. 5 Flexible coupling assembly



SYMBOL	SIZE	TIGHTENING TORQUE	WIDTH ACROSS FLATS
A	M30X2.0	1275 N·m	41 mm
B	M24X3.0	640 N·m	41 mm
C	M33X3.5	1470 N·m	50 mm

· APPLY MOLYKOTE1000 SPRAY TO THE THREADED PORTIONS AND THE CONTACT SURFACES.

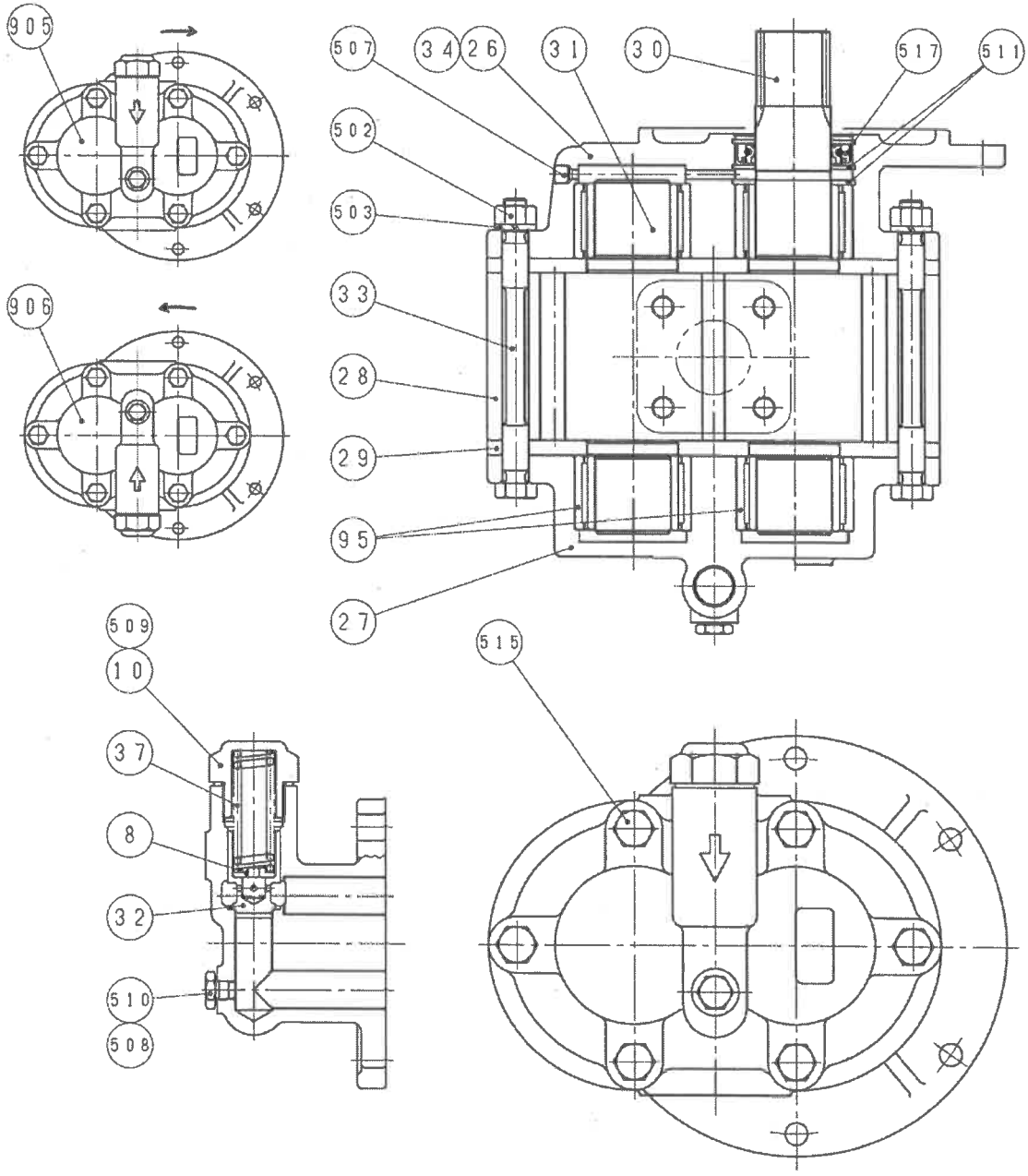
AG24005 § 09,10

第 5 図　フレキシブルカップリング組立図
Fig.　5 Flexible coupling assembly

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty. DRA-25J	数量 Qty. DRA-25JL
1	AG24005015B	カップリング:ガイスリンガー:BC41/10/120H/R	COUPLING:GEISLINGER:BC41/10/120H/R	1	
2	AG24005016A	カップリング:ガイスリンガー:BC41/10/120H/L	COUPLING:GEISLINGER:BC41/10/120H/L		1
3	AG39005004A	ボルト:ドウリョクデンタツ:M33X3.5X73	BOLT:POWER RANSMISSION:M33X3.5X73	16	16
4	AG24005021A	シシン:フライホイール760:6DKM-20	POINTER:FLYWHEEL760:6DKM-20	1	1
5	G885915310Z	ボルト:ドウリョクデンタツ:M24X3.0X85	BOLT:POWER TRANSMISSION:M24X3.0X85	8	8
6	G885900040Z	ボルト:ドウリョクデンタツ:M30X2.0X108-6g	BOLT:POWER TRANSMISSION:M30X2.0X108-6g	12	12
7	Z321010200DZ	ハリガネ::1.0X2000	WIRE::1.0X2000	4	4
8	X200010014ZZ	ボルト::M10X1.5X14	BOLT::M10X1.5X14	2	2
9	AG24005019A	フライホイール:6DKM-20 J28.8:BC41/10/120H	FLYWHEEL:6DKM-20 J28.8:BC41/10/120H	1	1
10	AG24005020A	カンセイエンバン:6DKM-20 J73.5:BC41/10/120H	INERTIA PLATE:6DKM-20 J73.5:BC41/10/120H	1	1

第　6－1　図　　潤滑油ポンプ組立図（２３２）

Fig.6-1　Lubricating oil pump assembly(232)



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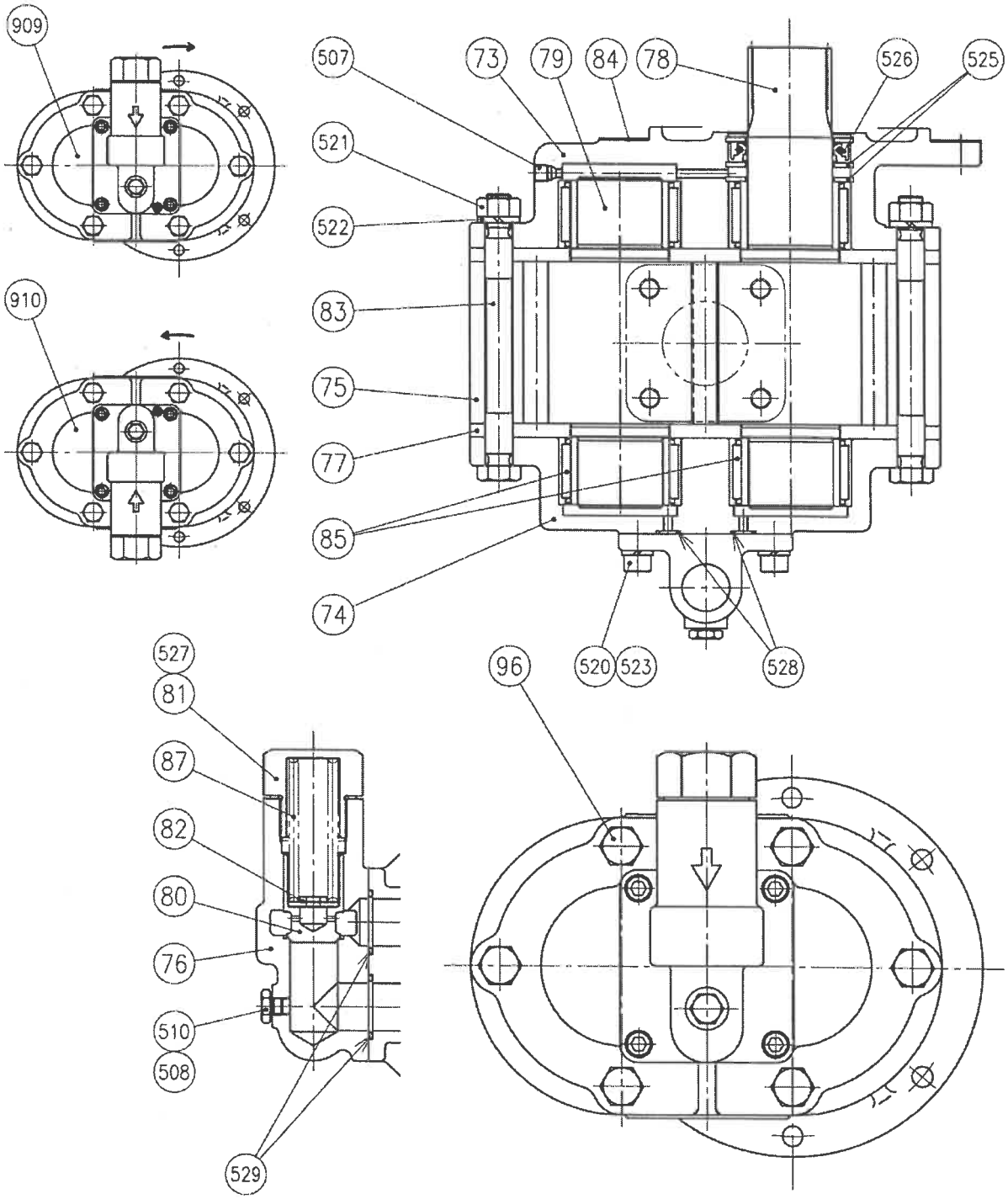
第6-1図　潤滑油ポンプ組立図 (232)
Fig.6-1　Lubricating oil pump assembly (232)

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty 232L
[906]	G811409060B	-:LOポンプASSY V232DL:	L.O.PUMP ASSY V232DL	1
[008]	C036340260Z	-:アンゼンベンバネウケ:	RETAINER: SAFETY VALVE SPRING	1
[010]	G003100370Z	-:アンゼンベンバネオサエ:	CAP: SAFETY VALVE SPRING	1
[026]	G003100020A	-:ジクウケ, ポンプ:	BEARING: PUMP	1
[027]	G003100060B	-:フタ, ポンプタイ:	COVER: PUMP BODY	1
[028]	G003100120A	-:タイ, ポンプ:	BODY:PUMP	1
[029]	G003100170B	-:イタ, シキリ:	PLATE:PARTITION	2
[030]	G811400300Z	-:ポンプモトギヤ:	L.O. PUMP DRIVE GEAR	1
[031]	G003100250Z	-:ポンプウケギヤ:	L.O.PUMP DRIVEN GEAR	1
[032]	G003100310Z	-:アンゼンベン:	SAFETY VALVE	1
[033]	G003100360Z	-:ポンプタイリーマボルト14X129:	REAMER BOLT 14X129	2
[034]	G003100460A	-:ポンプパツキン:	GASKET: L.O.PUMP	1
[037]	G811400361Z	-:バネ(アンゼンベン):	SPRING (SAFETY VALVE)	1
[095]	G811400950Z	-:ニードルベアリングNA6908:	BEARING NA6908	4
[502]	X220012000ZZ	ナット::M12X1.75	NUT M12X1.75	2
[503]	Z310012000ZZ	バネザガネ::M12	SPRING WASHERM 12	2
[507]	Z571501000ZZ	テーパプラグ:(5):1/8	TAPER PLUG (5) 1/8	1
[508]	Z665010000ZZ	マルパッキン::10	GASKET 10	1
[509]	Z665034000ZZ	マルパッキン::34	GASKET 34	1
[510]	X670010000ZZ	ロッカクプラグ::M10X1.25	HEX. PLUG M10X1.25	1
[511]	Z412006200ZZ	スナップリング:アナ:62	SNAP RING (HOLE) 62	3
[515]	X200012140ZZ	ボルト::M12X1.75X140	BOLT M12X1.75X140	4
[517]	Z461406212SB	オイルシール::SB406212	OIL SEAL SB406212	1
				DRA-25J(L)

第 6－2 図　　潤滑油ポンプ組立図（340）

Fig. 6–2　Lubricating oil pump assembly (340)

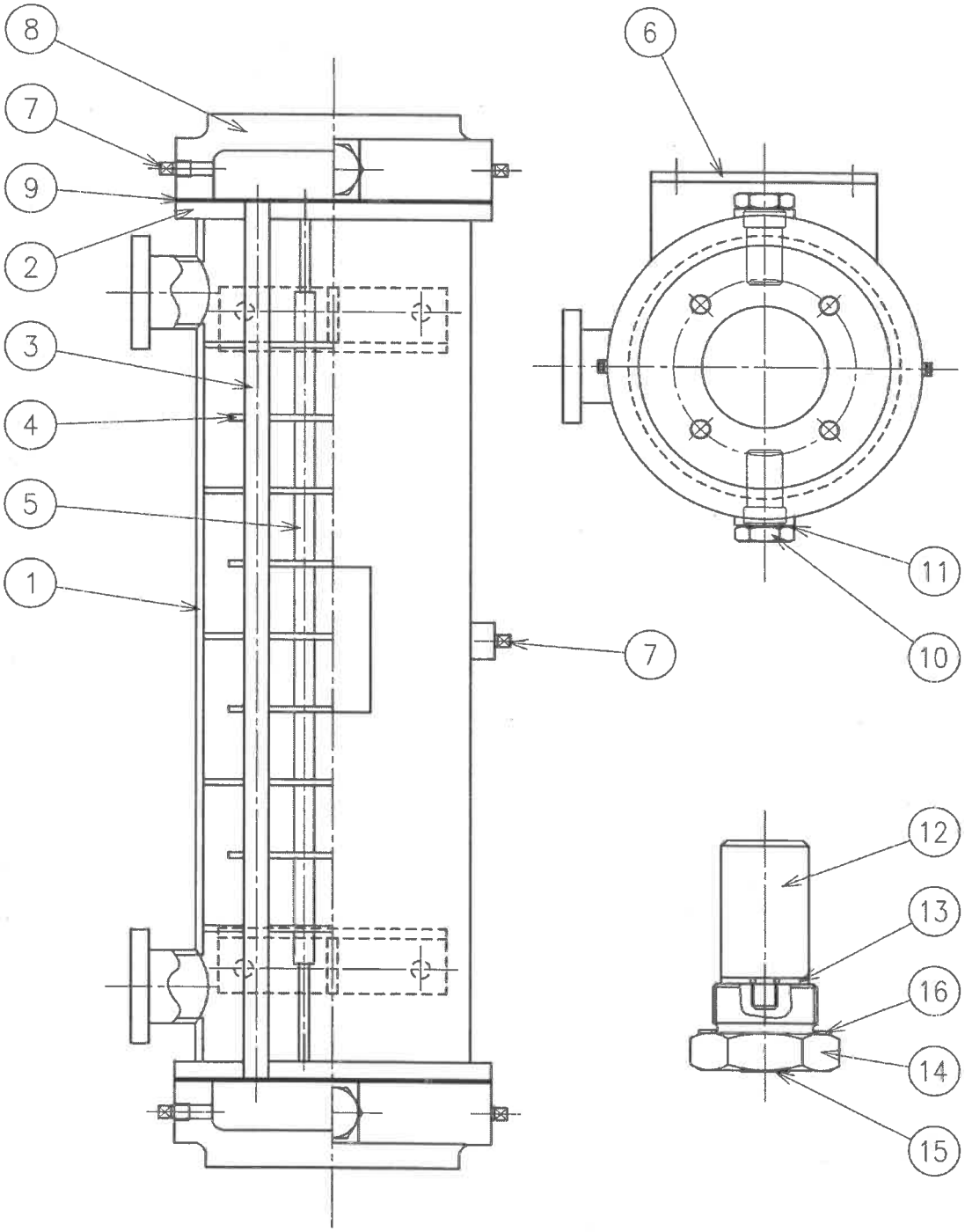


第6-2図　潤滑油ポンプ組立図 (340)

Fig.6-2　Lubricating oil pump assembly (340)

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty 340L
[910]	G811409100F	カンビヒン:LOポンプ V340DL:24	ASSY.L.O.PUMP V340DL24	1
[073]	G003100040B	-:ジクウケ, ポンプ:	BEARING PUMP	1
[074]	G003100070B	-:フタ, ポンプタイ:	COVER PUMP BODY	1
[075]	G003100130B	-:タイ, ポンプ:	BODY PUMP	1
[076]	G003100150B	-:タイ, アンゼンベン:	L.O. PUMPBODY SAFETY VALVE	1
[077]	G003100180C	-:イタ, シキリ:	PLATE PARTITION	2
[078]	G811400780Z	-:ポンプモトギヤ:	L.O.PUMP DRIVE GEAR	1
[079]	G003100270Z	-:ポンプウケギヤ:	L.O.PUMP DRIVEN GEAR	1
[080]	G003100280Z	-:アンゼンベン:	SAFETY VALVE	1
[081]	G003100290A	-:アンゼンベンバネオサエ:	CAP SAFETY VALVE SPRING	1
[082]	G003100380Z	-:アンゼンベンバネウケ:	CAP SAFETY VALVE SPRING	1
[083]	G003100410A	-:リーマボルト, ポンプ, 16X135:	REAMER BOLT PUMP 16X135	2
[084]	G003100470A	-:ポンプパツキン:	LO PUMP GASKET	1
[085]	G003100440A	ベアリング:NA6910X265:	BEARING NA6910X265	4
[087]	G811400861Z	-:バネ #1:	SPRING	1
[096]	G003100580Z	-:HTボルト, M14X2.0X150:	HT BOLT M14X2.0X150	4
[507]	Z571501000ZZ	テーパプラグ:(5):1/8	TAPER PLUG (5) 1/8	1
[508]	Z665010000ZZ	マルパッキン::10	GASKET 10	1
[510]	X670010000ZZ	ロッカクプラグ::M10X1.25	HEX. PLUG M10X1.25	1
[520]	Z212010022ZZ	アナツキボルト::M10X22	HEX. SOCKET BOLT M10X22	4
[521]	X220514000ZZ	ナット:(5):M14X1.5	NUT (5) M14X1.5	2
[522]	Z310014000ZZ	バネザガネ::M14	SPRING WASHER M14	2
[523]	Z310010000ZZ	バネザガネ::M10	SPRING WASHER M10	4
[525]	Z412007200ZZ	スナップリング:アナ:72	SNAP RING HOLE 72	3
[526]	Z461507212SB	オイルシール::SB507212	OIL SEAL SB507212	1
[527]	Z665040000ZZ	マルパッキン::40	GASKET 40	1
[528]	Z560101024AZ	Oリング::P10A	O-RING P10A	2
[529]	Z560103235AZ	Oリング::P32	O-RING P32	2
				DRA-25J(L)

第 7 図　　潤滑油冷却器組立図
Fig. 7 Lubricating oil cooler assembly



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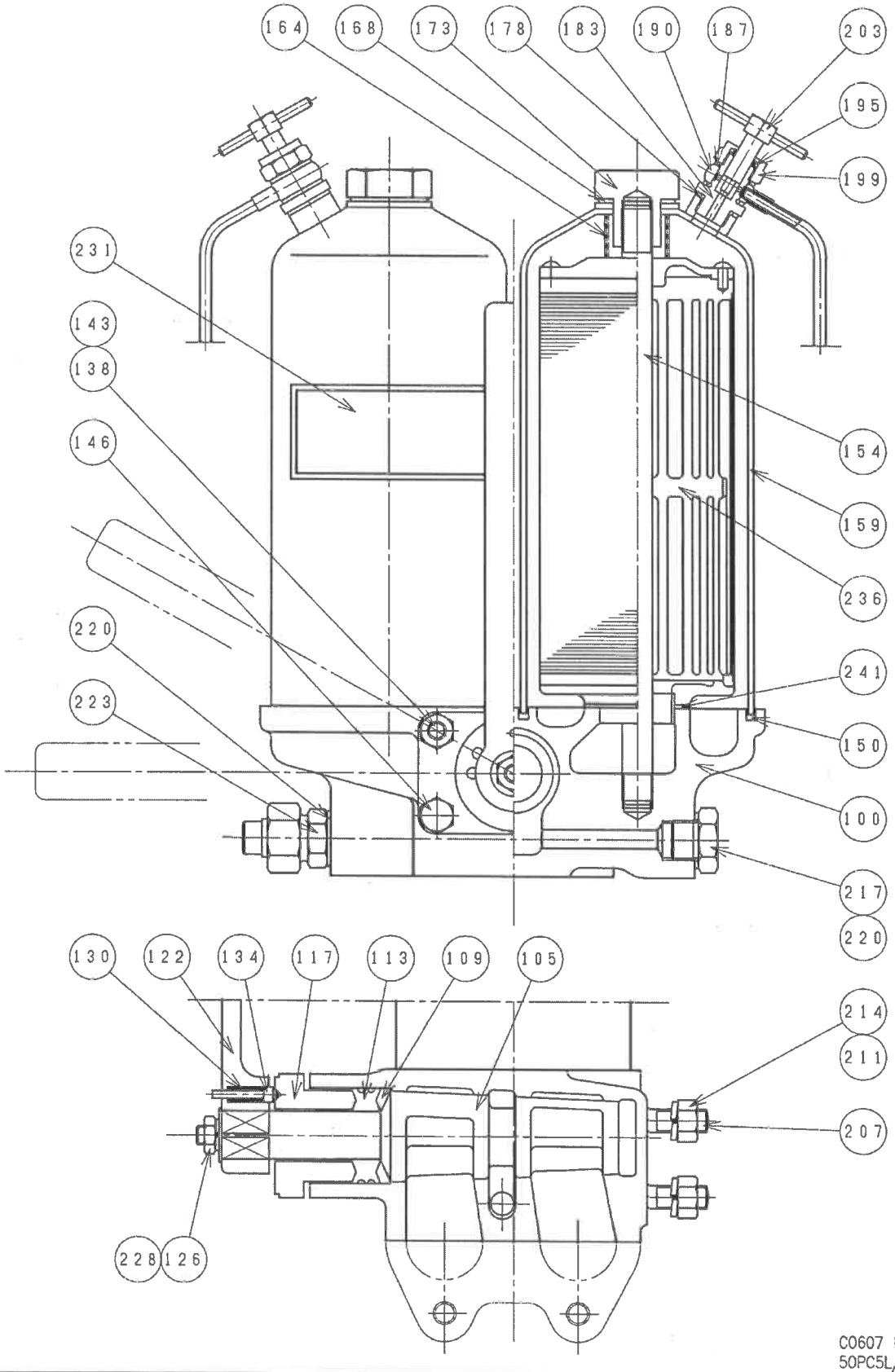
第7図　潤滑油冷却器組立図

Fig.7　Lubricating oil cooler assembly

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
Assy.	G004700260A	オイルクーラASSY. NL-083BX	OIL COOLER ASSY. NL-083BX	1
1	G004700260A-01	ドウタイ	BODY	1
2	G004700260A-02	カンイタ	HOLDER PLATE, COOLER PLATE	2
3	G004700260A-03	レイキャクカン	OIL COOLER PIPE	118
4	G004700260A-04	ジャマイタ	BAFFLE PLATE	1set
5	G004700260A-05	シジボウ、カンカクカン	HOLDER, DISTANCE PLATE	4set
6	G004700260A-06	アシ	FITT PLATE	2set
7	G004700260A-07	クウキドレンヌキプラグ	PLUG, AIR DRAIN	5
8	G004700510A	スイシツ, 200A	S.W. CASE, 200A	2
9	G004700520A	ガスケツト, 200A	GASKET 200A	2
10	X570010000JZ	ロツカクプラグ 1	HEX. PLUG 1	4
11	Z565003400EE	マルパツキン 34	GASKET 34	4
保護亜鉛付き　WITH PROTECTIVE ZINC				
12	G004800040Z	ホゴアエン 28X45	PROTECTIVE ZINC 28X45	4
13	C015170110Z	マルパツキン (2) 9X28	GASKET (2) 9X28	4
14	G004800100Z	ホゴアエンプラグ PF1	PLUG PF1, PROTECTIVE ZINC	4
15	A506470730Z	ホゴアエン メイバン	NAME PLATE, PROTECTIVE ZINC	4
16	Z565003400EE	マルパツキン 34	GASKET 34	4

第 8 図　潤滑油濾器組立図

Fig. 8 Lubricating oil filter assembly



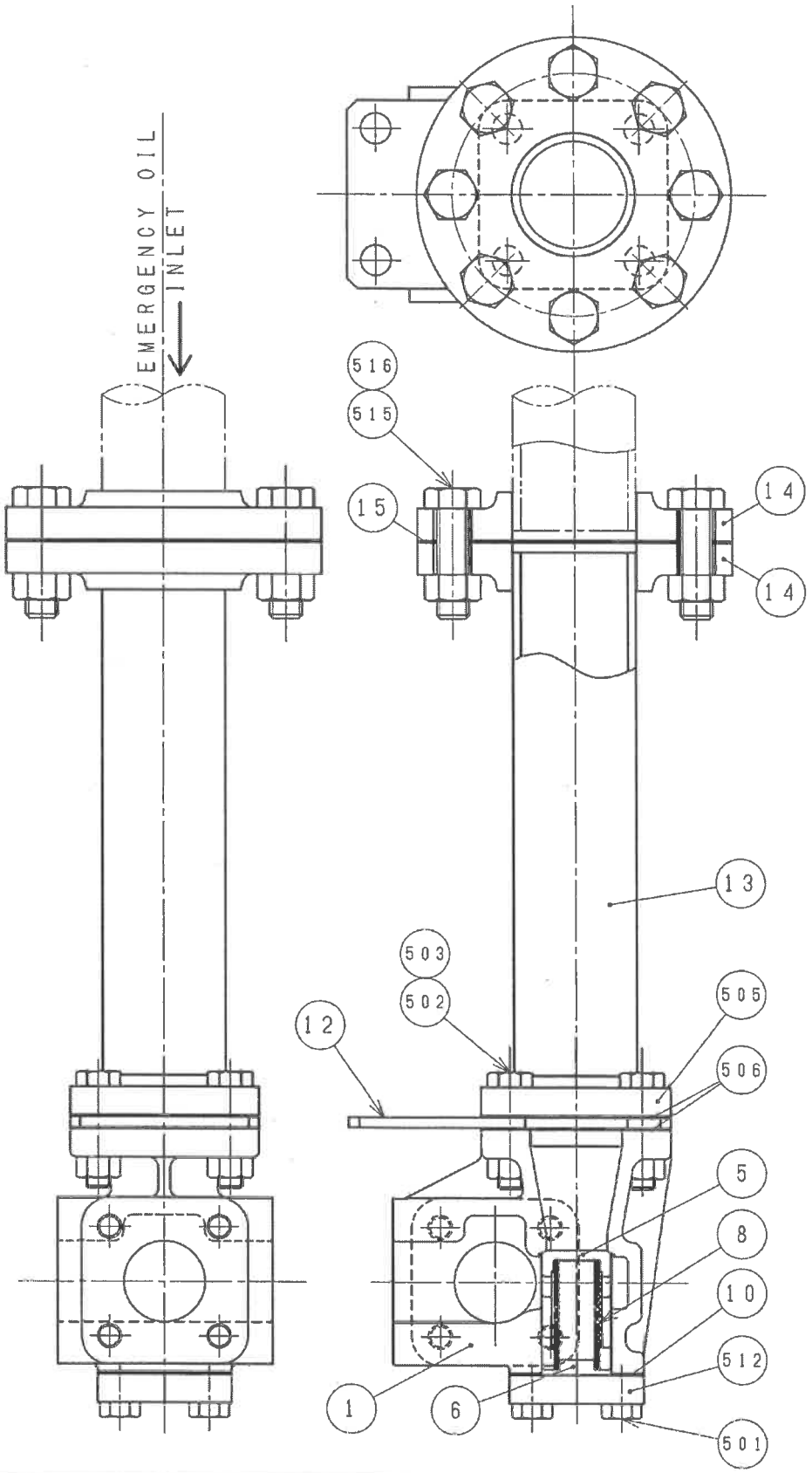
第8図　潤滑油濾器組立図

Fig.8 Lubricating oil filter assembly

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[002]	C060700011Z	ロキ 50PC5L/1-X/KZ	FILTER 50PC5L/1-X/KZ	1
[100]	Y529000005ZZ	ロキホンタイ	FILTER BODY	1
[105]	Y529000015ZZ	キリカエコツク	COCK	1
[109]	Y529000024ZZ	パツキンシート	PACKING SEAT	1
[113]	Y529000033ZZ	パツキン (コツク)	PACKING, COCK	1
[117]	Y529000047ZZ	オサエフタ	COCK COVER	1
[122]	Y529000058ZZ	ハンドル	HANDLE	1
[126]	Y529000065ZZ	ナツト (ハンドル)	NUT, HANDLE	1
[130]	Y529000072ZZ	バネ (トメセン)	SPRING, NOTCH BAR	1
[134]	Y529000079ZZ	トメセン	NOTCH BAR	1
[138]	Y529000086ZZ	スタツド(フタ)	STUD, COVER	2
[143]	Y529000093ZZ	ナツト (フタ)	NUT, COVER	2
[146]	Y529000099ZZ	ボルト (フタ)	BOLT, COVER	2
[150]	Y529000109ZZ	パツキン(ロキケース)	PACKING, CASING	2
[154]	Y529000124ZZ	ジク	SHAFT	2
[159]	Y529000142ZZ	ケース	CASE	2
[164]	Y529000149ZZ	バネ(ロキケース)	SPRING, CASING	2
[168]	Y529000155ZZ	パツキン (シメツケナツト)	PACKING, CAP NUT	2
[173]	Y529000164ZZ	シメツケナツト	CAP NUT	2
[178]	Y529000178ZZ	パツキン (ベンザ)	PACKING, VALVE SEAT	2
[183]	Y529000184ZZ	ベンザ	VALVE SEAT	2
[187]	Y529000189ZZ	パツキン (カンセツシユ)	PACKING, CONNECTOR	4
[190]	Y529000194ZZ	カンセツシユ	CONNECTOR	2
[195]	Y529000200ZZ	パツキン (フクロナツト)	PACKING, CAP NUT	2
[199]	Y529000207ZZ	フクロナツト	CAP NUT	2
[203]	Y529000211ZZ	クウキヌキセン	AIR PLUG	2
[207]	Y529000217ZZ	スタツド(ロキトリツケ)	STUD, FILTER FITTING	4
[211]	Y529000224ZZ	バネザガネ (ロキトリツケ)	SPRING WASHER,FILTER FITTING	4
[214]	Y529000229ZZ	ナツト (ロキトリツケ)	NUT, FILTER FITTING	4
[217]	Y529000233ZZ	プラグ (ドレン)	PLUG, DRAIN	1
[220]	Y529000239ZZ	パツキン	PACKING	2
[223]	Y529000244ZZ	ニツプル	NIPPLE	1
[228]	Y529000262ZZ	ザガネ (ハンドル)	WASHER, HANDLE	1
[231]	Y529000273ZZ	チユウイメイバン	NAME PLATE, CAUTION	1
[236]	Y529000296ZZ	エレメントASSY-100ME	ELEMENT ASSY. 100ME	2
[241]	Y529000310ZZ	エレメントパツキン	PACKING, ELEMENT	2

第 9 図　　逆 止 弁 組 立 図

Fig.9 Check valve assembly

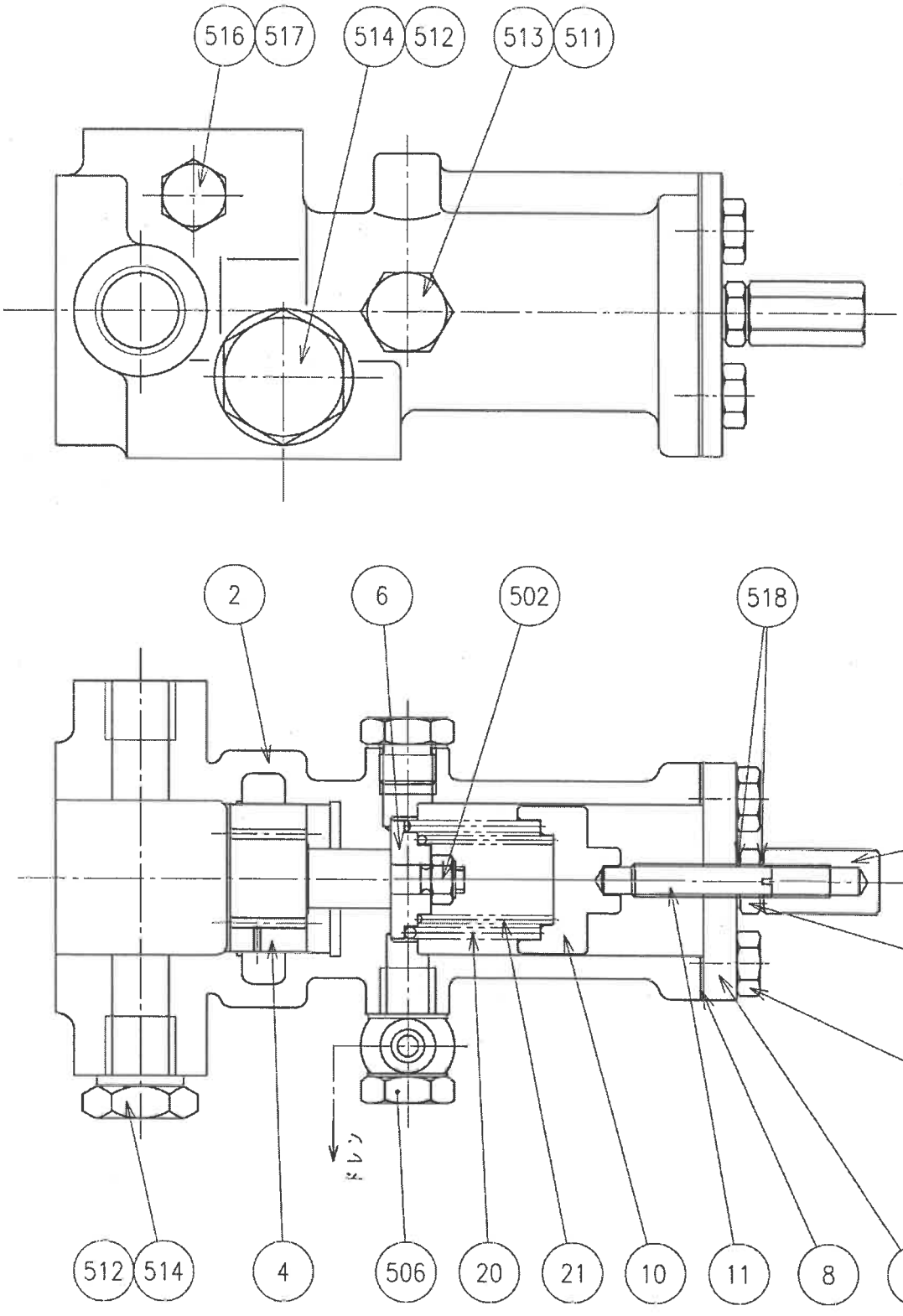


第9図　逆止弁組立図
Fig.9　Check valve assembly

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[001]	G005700010B	ギヤクシベンタイ	BODY CHECK VALVE	1
[005]	G005700050Z	ベン	VALVE	1
[006]	G005700060Z	バネウケ	SPRING SUPPORTER	1
[008]	G005700071Z	バネ	SPRING	1
[010]	G005700100A	パツキン 36X44	GASKET 36X44	1
[012]	B056370020Z	メツキメクラフランジ	EYE FLNGE	1
[013]	G005700130Z	パイプ，ギャクシベン	PIPE CHECK VALVE	1
[014]	G005700140Z	JIS-16K フランジ，50A	JIS-16K FLANGE 50A	2
[015]	G005700150Z	JIS-16K パッキン，50A	JIS-16K GASKET 50A	1
[501]	X200010025ZZ	ボルト M10X1.5X25	BOLTM10X1.5X25	4
[502]	X200012050ZZ	ボルト M12X1.75X50	BOLTM12X1.75X50	4
[503]	X220012000ZZ	ナツト M12X1.75	NUTM12X1.75	4
[505]	Z540206065ZZ	フランジ(2)60X65	FLANGE(2)60X65	1
[506]	Z541206065EE	パツキン(2)60X65	GASKET(2)60X65	2
[512]	Z540200044ZZ	フランジ(2)0X44	FLANGE(2)0X44	1
[515]	X200016055ZZ	ボルト M16X2.0X55	BOLTM16X2.0X55	4
[516]	X220016000ZZ	ナツト M16X2.0	NUTM16X2.0	4

第 10 図　　高圧調圧弁組立図

Fig. 10　High pressure relief valve assembly



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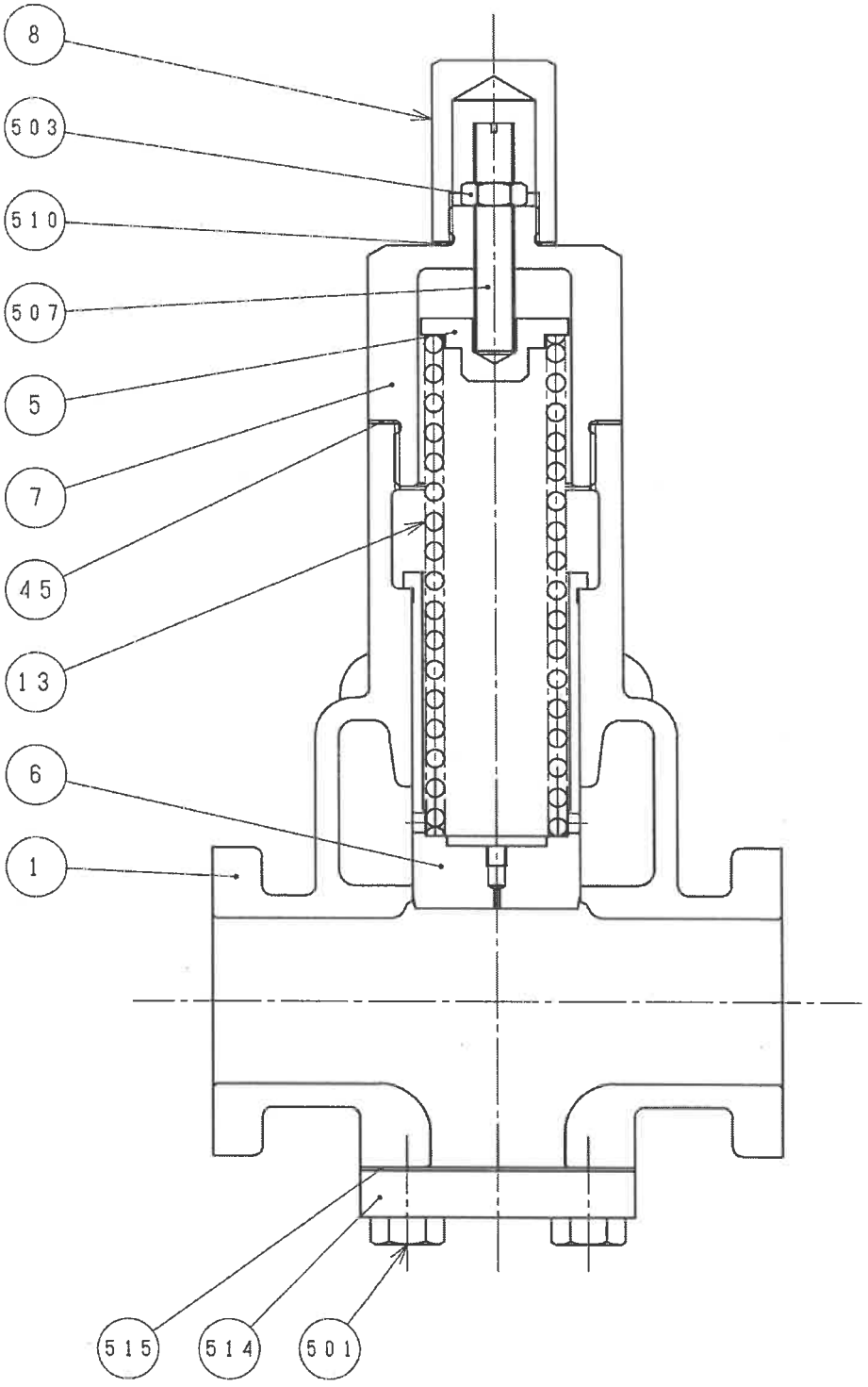
第10図　高圧調圧弁組立図
Fig.10 High pressure relief valve assy.

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[910]	G005309100D	-:チヨウアツベン(コウアツ)ASSY:	RELIEF VALVE(HIGH PRESS.)ASSY	1
[002]	G005300020Z	-:ベンタイ:	VALVE BODY	1
[004]	G005300040Z	-:ベン:	VALVE	1
[006]	G005300060Z	-:バネウケ(1):	SPRING RETAINER	1
[008]	G005300080A	-:ベンタイパツキン:	GASKET BODY	1
[010]	G005300100Z	-:バネウケ(2):	SPRING RETAINER	1
[011]	G005300110Z	-:ネジ, チヨウセイ, M10X1.5X63:	SCREW ADJUSTING M10X1.5X63	1
[012]	G005300120Z	-:ベンタイフタ:	VALVE BODY COVER	1
[014]	G005300140Z	-:フクロナツトM10X1.5:	CAP NUT M10X1.5	1
[020]	G005300191Z	-:バネ#1:	SPRING #1	1
[021]	G005300192Z	-:バネ#2:	SPRING #2	1
[501]	X200010022ZZ	ボルト::M10X1.5X22	BOLTM10X1.5X22	4
[502]	X227008000ZZ	Uナット::M8X1.25	U-NUTM8X1.25	1
[503]	X220310000ZZ	ナット:(3):M10X1.5	NUT(3)M10X1.5	1
[506]	X546110030ZZ	ツギワASSY.::10X3/8	BANJO ASSY.10X3/8	1
[511]	Z565001700EE	マルパッキン::17	GASKET17	2
[512]	Z565002100EE	マルパッキン::21	GASKET21	3
[513]	X570003000ZZ	ロッカクプラグ::3/8	HEX. PLUG3/8	1
[514]	X570004000ZZ	ロッカクプラグ::1/2	HEX. PLUG1/2	3
[516]	X570002000ZZ	ロッカクプラグ::1/4	HEX. PLUG1/4	2
[517]	Z565001300EE	マルパッキン::13	GASKET13	2
[518]	Z565001000EE	マルパッキン::10	GASKET10	2

第 11 図　　低圧調圧弁組立図

Fig. 11　Low pressure relief valve assembly



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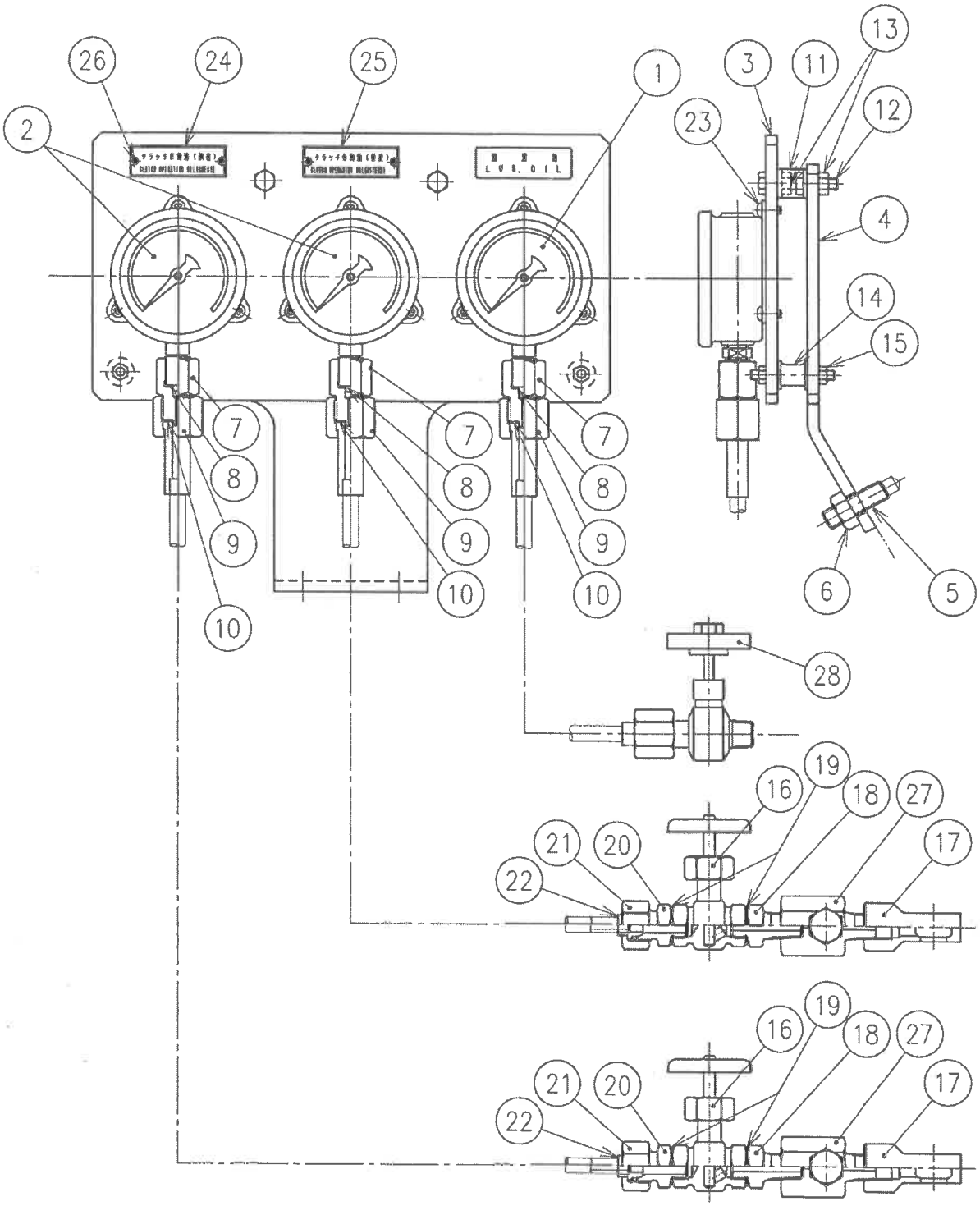
第11図　低圧調圧弁組立図

Fig.11　Low pressure relief valve assy.

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[901]	C037590010F	オイルチヨウアツベンASSY.1	OIL RELIEF VALVE ASSY.1	1
[001]	C037510010Z	チヨウアツベンタイ #1	RELIEF VALVE BODY #1	1
[005]	C037540050Z	バネウケ	RETAINER, VALVE SPRING	1
[006]	C037540060B	ベン	VALVE	1
[007]	C037540070A	バネケース	VALVE SPRING CASE	1
[008]	AE02077005A	キヤツプ: :	CAP	1
[013]	C037570151A	バネ 5.5X36.5X209	SPRING 5.5X36.5X209	1
[045]	C037570450Z	ガスケツト, 61-75	GASKET 61-75	1
[501]	X200012030ZZ	ボルト M12X1.75X30	BOLT M12X1.75X30	4
[503]	X222312000ZZ	コガタナツト(3)M12	NUT(3)M12	1
[507]	X251512070ZZ	トメネジ(5)M12X1.25X70	SET SCREW(5)M12X1.25X70	1
[510]	Z565002700EE	マルパツキン 27	GASKET 27	1
[514]	Z540200054ZZ	フランジ(2)0X54	FLANGE(2)0X54	1
[515]	Z541204054AZ	パツキン(2)40X54	GASKET(2)40X54	1

第 12 図 計器板組立図

Fig. 12 Instrument board assembly



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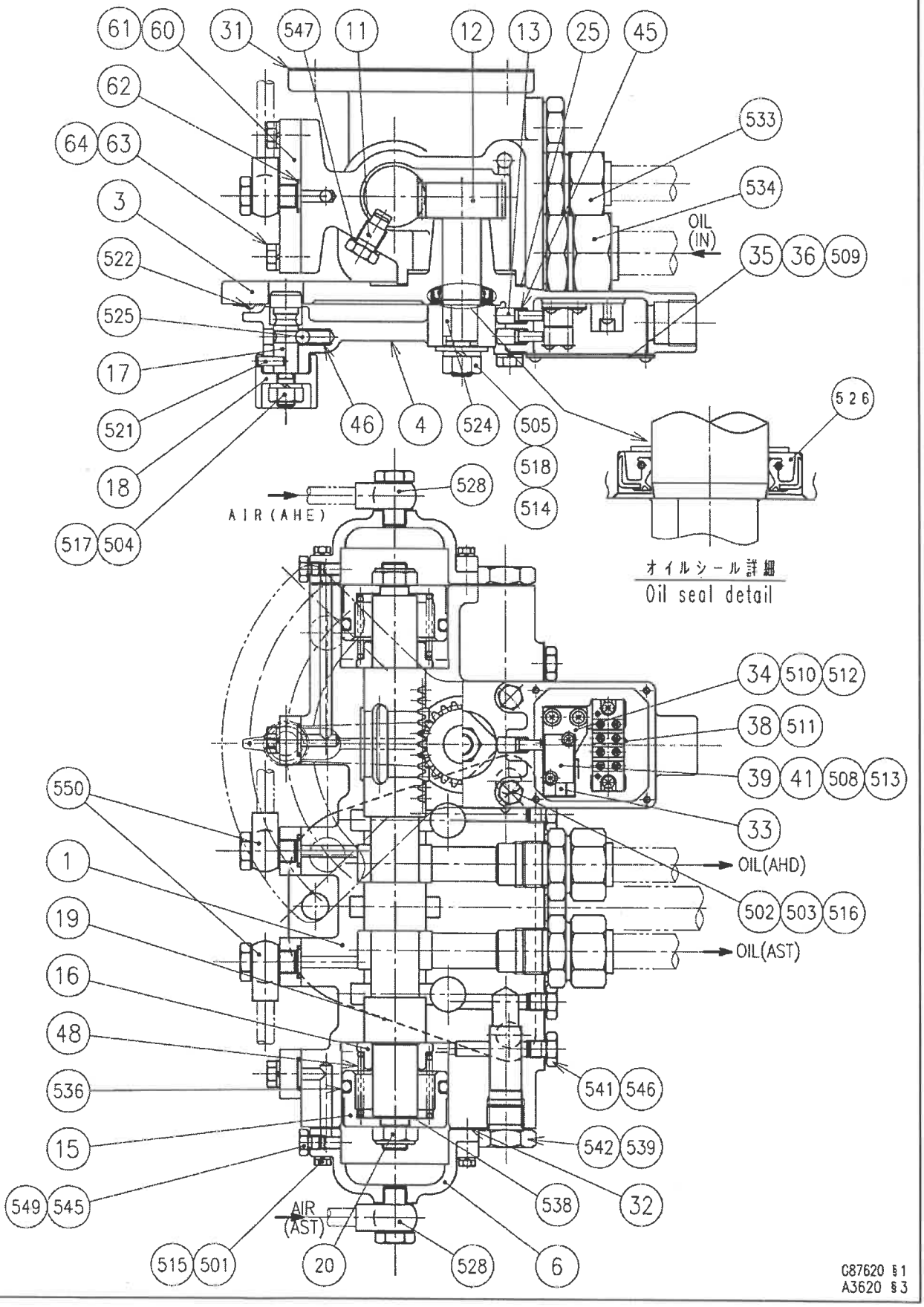
第12図　計器板組立図

Fig.12 Instrument board assembly

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[001]	NN00393002A	60Gゲージ:サイコウ(0.6MPa):アオマーク(0.25-0.4MPa)	60G PRESS.GAUGEMAX(0.6MPa) BLUE MARK(0.25-0.4MPa)	1
[002]	NN00393003A	60Gゲージ:サイコウ(3.5MPA):ブルーマーク(1.9-2.1MPa)	60G PRESS.GAUGEMAX(3.5MPa) BLUE MARK(1.9-2.1MPa)	2
[003]	NN00393005A	ケイキバン:60G:3	GAUGE BARD60G 3	1
[004]	G876100020A	-:ブラケツト, (4), ケイキバン:	-:BRAKECT (4) GAUGE BOARD:	1
[005]	X210010035ZZ	スタッド::M10X1.5X35	STUD M10X1.5X35	2
[006]	X220010000ZZ	ナット::M10X1.5	NUT M10X1.5	2
[007]	AE01130014A	セッシュ::G1/4XG3/8	JOINT G1/4XG3/8	3
[008]	Z565000600ZZ	マルパッキン::6	GASKET6	3
[009]	Y143080803ZZ	セッシュ::PF3/8X8	UNION PIPE JOINT PF3/8X8	3
[010]	Z565000800ZZ	マルパッキン::8	GASKET 8	3
[011]	AE01130007A	ボウシンゴム::1	ISOLATION RUBBER 1	2
[012]	X200006018ZZ	ボルト::M6X1.0X18	BOLT M6X1.0X18	6
[013]	X220006000ZZ	ナット::M6X1.0	NUTM6X1.0	6
[014]	G009200080Z	-:ボウシンゴムA5001:	CUSHION RUBBER A5001	2
[015]	X220005000ZZ	ナット::M5X0.8	NUT M5X0.8	4
[016]	Y024350085ZZ	ニードルバルブ::20KBS 8	NEEDLE VALVE 20KBS 8	2
[017]	NN00065001B	ツギワ:バルブ:1/4X1/4	BANJO VALVE 1/4X1/4	2
[018]	G009200060Z	-:セツシユザPF1/4XPT1/4:	JOINT SEAT PF1/4XPT1/4	2
[019]	Z565001300EE	マルパッキン::13	GASKET 13	4
[020]	G009200170Z	-:ユニオンネジ 8:	UNION SCREW 8	2
[021]	X550308000ZZ	ユニオンナット::8	UNION NUT 8	2
[022]	X550108000ZZ	ユニオンツバ::8	UNION NIPPLE 8	2
[023]	X245104010ZZ	プラスマルコネジ::M4X0.7X10	PLUS ROUND MACHINE SCREW M4X0.7X10	9
[024]	G832600161Z	-:クラツチユアツメイバン(1):	NAME PLATE(1)	1
[025]	G832600162Z	-:クラツチユアツメイバン(2):	NAME PLATE(2)	1
[026]	Z352025500ZZ	リベット::2.5X5	RIVET 2.5X5	4
[027]	Y142222200ZZ	ダンプナー::Rc1/4xR1/4-SUS	DAMPNER Rc1/4xR1/4-SUS	2
[028]	NN00141007C	ユニオンバルブ::10K SUS-8	UNION VALVE 10K SUS-8	1

第 13 図　クラッチ操作弁組立図

Fig.13　Clutch operating valve assembly



第13図　クラッチ操作弁組立図

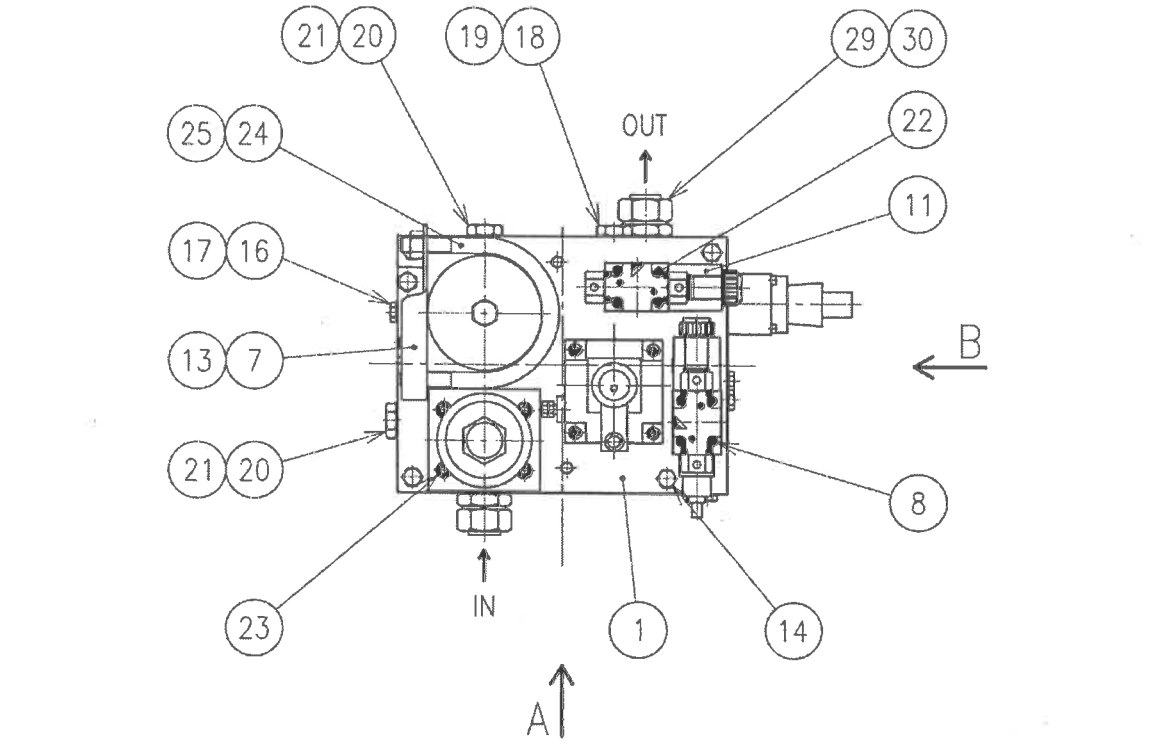
Fig.13 Clutch operating valve assembly

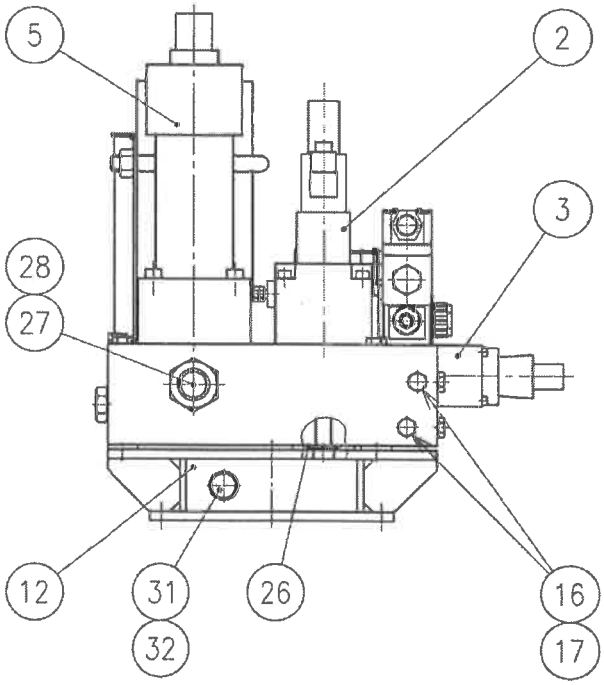
番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[903]	A362090030C	カンビヒン:クラッチソウサベン:3	ASSY.:CLUTCH OPERATING VALVE:3	1
[001]	A362010010Z	ベンタイ	VALVE BODY	1
[003]	A362010030A	シジバン	HOLDER PLATE	1
[004]	A362010040A	レバ (1)	LEVER (1)	1
[006]	A362010060Z	ベンタイフタ	COVER VALVE BODY	2
[011]	A362040010Z	マワリドメ	TAB WASHER	1
[012]	A362040020A	ジク:レバ:	SHAFT:LEVER:	1
[013]	A362040030Z	ロツド	ROD	2
[015]	A362040050Z	ピストン	PISTON	2
[016]	A362040060Z	バネウケ	SPRING RETAINER	2
[017]	A362040070Z	ストツパ	STOPPER	1
[018]	A362040080Z	ストツパハンドル	HANDLE LEVER STOPPER	1
[019]	A362040090Z	ベンジク	VALVE SHAFT	1
[020]	A362000200Z	Uナット M14X1.5	U-NUT M14X1.5	2
[025]	A362070010A	シジバンパツキン	PACKING HOLDER PLATE	1
[031]	A362070070A	ベンタイパツキン	GASKET VALVE BODY	1
[032]	A362070080A	フタパツキン	GASKET COVER	2
[033]	A362070090Z	ゼツエンイタ	INSULATION PLATE	2
[034]	A362070100Z	ブラケツト	BRACKET	1
[035]	A362070110Z	ケースフタ	CASE COVER	1
[036]	A362070120Z	ケースフタパツキン	GASKET CASE COVER	1
[038]	NN00104006A	タンシバン: :4P	4P	1
[039]	A001070161A	マイクロスイツチ V-15-1E5	MICRO SWITCH V-15-1E5	2
[041]	A362070170Z	ビニールコード 0.75X0.5M	CORD 0.75X0.5M	1
[045]	A362079801Z	ロツドバネ	ROD SPRING	2
[046]	A362079802Z	レバーバネ	SPRING LEVER	1
[048]	A362079807Z	ピストンバネ	SPRING PISTON	2
[060]	G001800510Z	ベンタイメクラフタ (2)	COVER, VALVE BODY (2)	2
[061]	G001800520Z	ベンタイメクラフタ	COVER, VALVE BODY	2
[062]	Z560101424ZZ	Oリング P14	O RING P14	4
[063]	X200008022ZZ	ボルト M8X1.25X22	BOLT M8X1.25X22	8
[064]	Z310008000ZZ	バネザガネ M8	SPRING WASHER M8	8
[501]	X200005025ZZ	ボルト M5X0.8X25	BOLT M5X0.8X25	12
[502]	X200008025ZZ	ボルト M8X1.25X25	BOLT M8X1.25X25	2

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[503]	X200008050ZZ	ボルト M8X1.25X50	BOLT M8X1.25X50	2
[504]	X220010000ZZ	ナット M10X1.5	NUT M10X1.5	1
[505]	X220012000ZZ	ナット M12X1.75	NUT M12X1.75	1
[508]	X245103025ZZ	プラスマルコネジ M3X0.5X25	PLUS ROUND MACHINE SCREW M3X0.5X25	2
[509]	X245104006ZZ	プラスマルコネジ M4X0.7X6	PLUS ROUND MACHINE SCREW M4X0.7X6	4
[510]	X245104010ZZ	プラスマルコネジ M4X0.7X10	PLUS ROUND MACHINE SCREW M4X0.7X10	2
[511]	X245104016ZZ	プラスマルコネジ M4X0.7X16	PLUS ROUND MACHINE SCREW M4X0.7X16	2
[512]	Z300004000ZZ	ヒラザガネ 4	FLAT WASHER 4	2
[513]	Z300103000ZZ	ヒラザガネ (1) 3	FLAT WASHER (1) 3	2
[514]	Z300014000ZZ	ヒラザガネ 14	FLAT WASHER 14	1
[515]	Z310005000ZZ	バネザガネ M5	SPRING WASHER M5	12
[516]	Z310008000ZZ	バネザガネ M8	SPRING WASHER M8	4
[517]	Z310010000ZZ	バネザガネ M10	SPRING WASHER M10	1
[518]	Z310012000ZZ	バネザガネ M12	SPRING WASHER M12	1
[521]	Z335605016ZZ	ヘイコウピン (6) 5X16	STRAIGHT PIN (6) 5X16	1
[522]	Z352020500ZZ	リベット 2X5	RIVET 2X5	3
[524]	Z400005018ZZ	キー 5X18	KEY 5X18	1
[525]	Z445002500ZZ	ボール 5/16	BALL 5/16	1
[526]	A362005010Z	オイルシール HMSA22357	OIL SEAL HMSA22357	1
[528]	X546110020JZ	ツギワ ASSY. 10X1/4	BANJO ASSY. 10X1/4	2
[533]	X550020060ZZ	ユニオンセツシユASSY. 20X3/4	UNION ASSY. 20X3/4	2
[534]	X550025060ZZ	ユニオンセツシユASSY. 25X3/4	UNION ASSY. 25X3/4	1
[536]	Z560104857ZZ	Oリング P48A	O-RING P48A	2
[538]	Z565001700EE	マルパツキン 17	GASKET 17	2
[539]	Z565002100EE	マルパツキン 21	GASKET 21	3
[541]	X570001000ZZ	ロツカクプラグ 1/8	HEX. PLUG 1/8	4
[542]	X570004000ZZ	ロツカクプラグ 1/2	HEX. PLUG 1/2	3
[545]	Z665008000ZZ	マルパツキン 8	GASKET 8	2
[546]	Z665010000ZZ	マルパツキン 10	GASKET 10	4
[547]	Z665014000ZZ	マルパツキン 14	GASKET 14	1
[549]	X670008000ZZ	ロツカクプラグ M8X1.25	HEX. PLUG M8X1.25	2
[550]	X546110020ZZ	ツギワ ASSY. 10X1/4	BANJO ASSY. 10X1/4	2

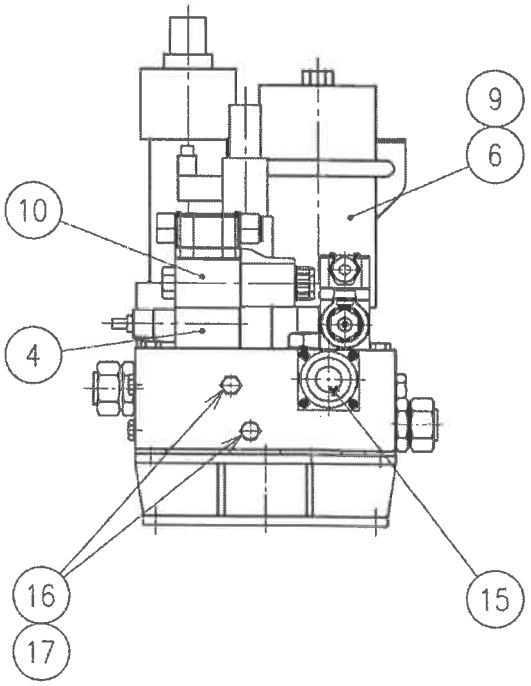
第 14 図 比例電磁弁ブロック組立図

Fig.14 Flow control valve block assembly





View "A"



View "B"

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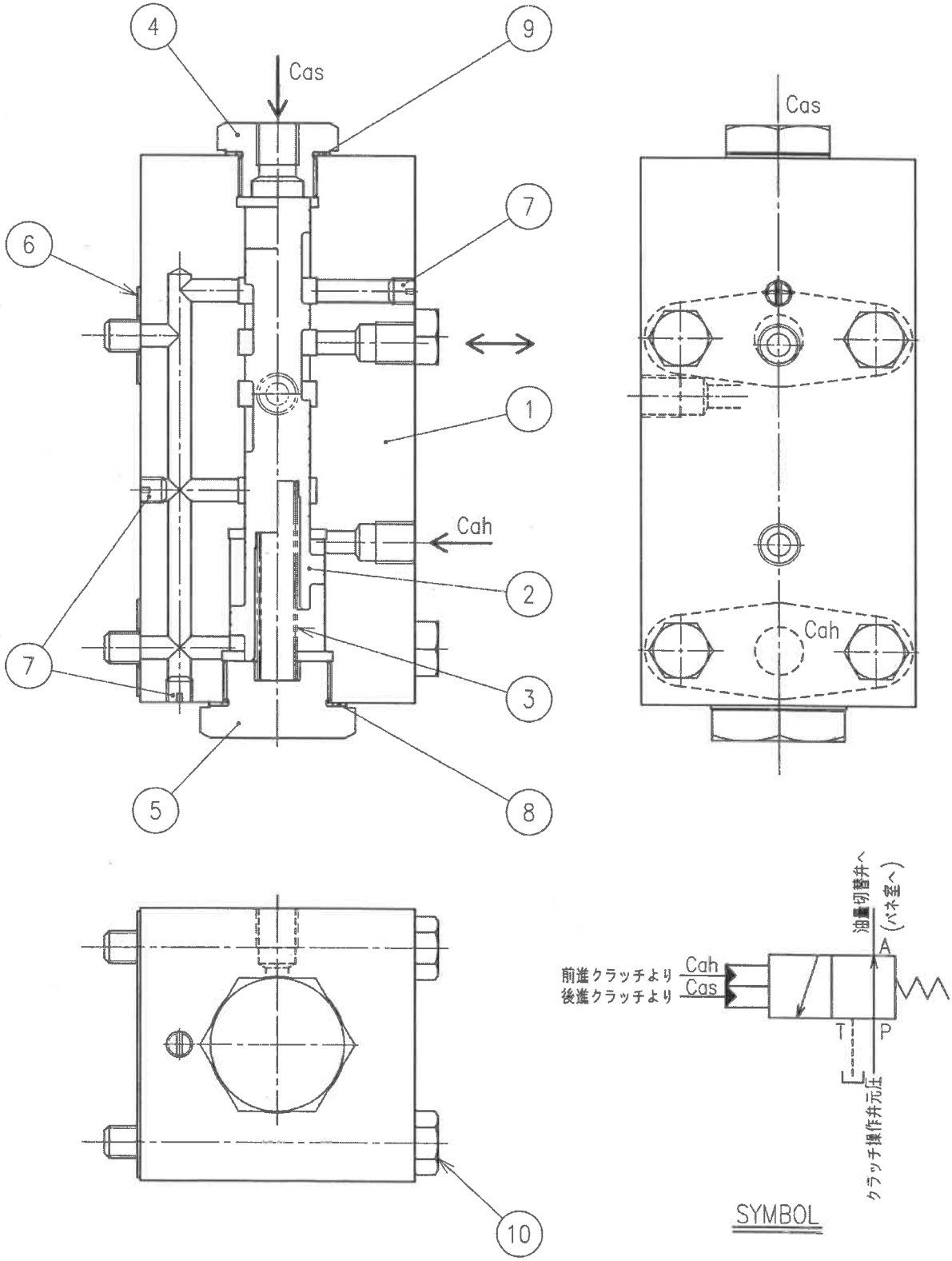
第14図　比例電磁弁ブロック組立図

Fig.14 Flow control valve block assembly

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[001]	NN00734001A	ブロック:ヒレイデンジベン:	BLOCK:FLOW CONTROL VALVE:	1
[002]	NN00734002A	デンジベン:ヒレイ:ES-G02-30-F-12	MAGNETIC VALVE:PROPORTIONAL:	1
[003]	NN00734014A	ベン:セイミツシボリベン: F10P-10-3/50L(KPM)	VALVE:-:F10P-10-3/50L(KPM)	1
[004]	NN00734004A	ベン: モジュラースタックガタシボリベン: MT-02Bi-55	VALVE:MT THROTTLE VALVE: MT-02Bi-55	1
[005]	NN00734015A	フィルター:マニホールドガタ: GF-A-06-3-40UW-IV	FILTER:-:GF-A-06-3-40UW-IV	1
[006]	NN00734006A	アキュムレータ:ミニレータ: MC70-1000-30/6(NOK)	ACCUMULATOR:MINILATOR: MC70-1000-30/6(NOK)	1
[007]	NN00734007A	ブラケット:アキュムレータ: MC70-1000-30/6	BRACKET:ACCUMULATOR: MC70-1000-30/6	1
[008]	NN00734008A	アナツキボルト:M5x0.8x85:	HEX. SOCKET BOLT:M5x0.8x85:	4
[009]	NN00734009A	セッシュ:アキュムレータ: MC70-1000(2000)-30/6	JOINT:ACCUMULATOR: MC70-1000(2000)-30/6	1
[010]	NN00734011A	デンジベン::KSO-G02-20BP-30	MAGNETIC VALVE:: KSO-G02-20BP-30	1
[011]	G875200150A	デンジベン:KSO-G02-2BP:	MAGNETIC VALVE:KSO-G02-2BP:	1
[012]	NN00734012A	ダイ:ヒレイデンジベンブロック:240X2	MOUNT:FLOW CTRL V.:240X240	1
[013]	X200010016ZZ	ボルト::M10X1.5X16	BOLT::M10X1.5X16	2
[014]	X200010110ZZ	ボルト::M10X1.5X110	BOLT::M10X1.5X110	4
[015]	X251901010ZZ	トメネジ:(9):1/8X10	SET SCREW:-9:1/8X10	1
[016]	X570001000ZZ	ロッカクプラグ::1/8	HEX. PLUG::42012	5
[017]	Z565001000EE	マルパッキン::10	GASKET::10	5
[018]	X570004000ZZ	ロッカクプラグ::1/2	HEX. PLUG::42006	1
[019]	Z565002100EE	マルパッキン::21	GASKET::21	1
[020]	X570006000ZZ	ロッカクプラグ::3/4	HEX. PLUG::42067	2
[021]	Z565002700EE	マルパッキン::27	GASKET::27	2
[022]	Z212005045ZZ	アナツキボルト::M5X45	HEX. SOCKET BOLT::M5X45	4
[023]	Z212010080ZZ	アナツキボルト::M10X80	HEX. SOCKET BOLT::M10X80	4
[024]	Z260110000ZZ	Uボルト:(1):100	U-BOLT:-1:100	1
[025]	X220016000ZZ	ナット::M16X2.0	NUT::M16X2.0	2
[026]	Z560102635ZZ	Oリング::P26	O-RING::P26	1
[027]	X550025100ZZ	ユニオンセッシュASSY.::25X1	UNION ASSY.::25X1	1
[028]	Z565003400EE	マルパッキン::34	GASKET::34	1
[029]	X550025060ZZ	ユニオンセッシュASSY.::25X3/4	UNION ASSY.::25X3/4	1
[030]	Z565002700EE	マルパッキン::27	GASKET::27	1
[031]	X570003000ZZ	ロッカクプラグ::3/8	HEX. PLUG::42071	2
[032]	Z565001700EE	マルパッキン::17	GASKET::17	2

第 15 図 油量切替指令弁組立図
Fig.15 Flow change control valve assembly



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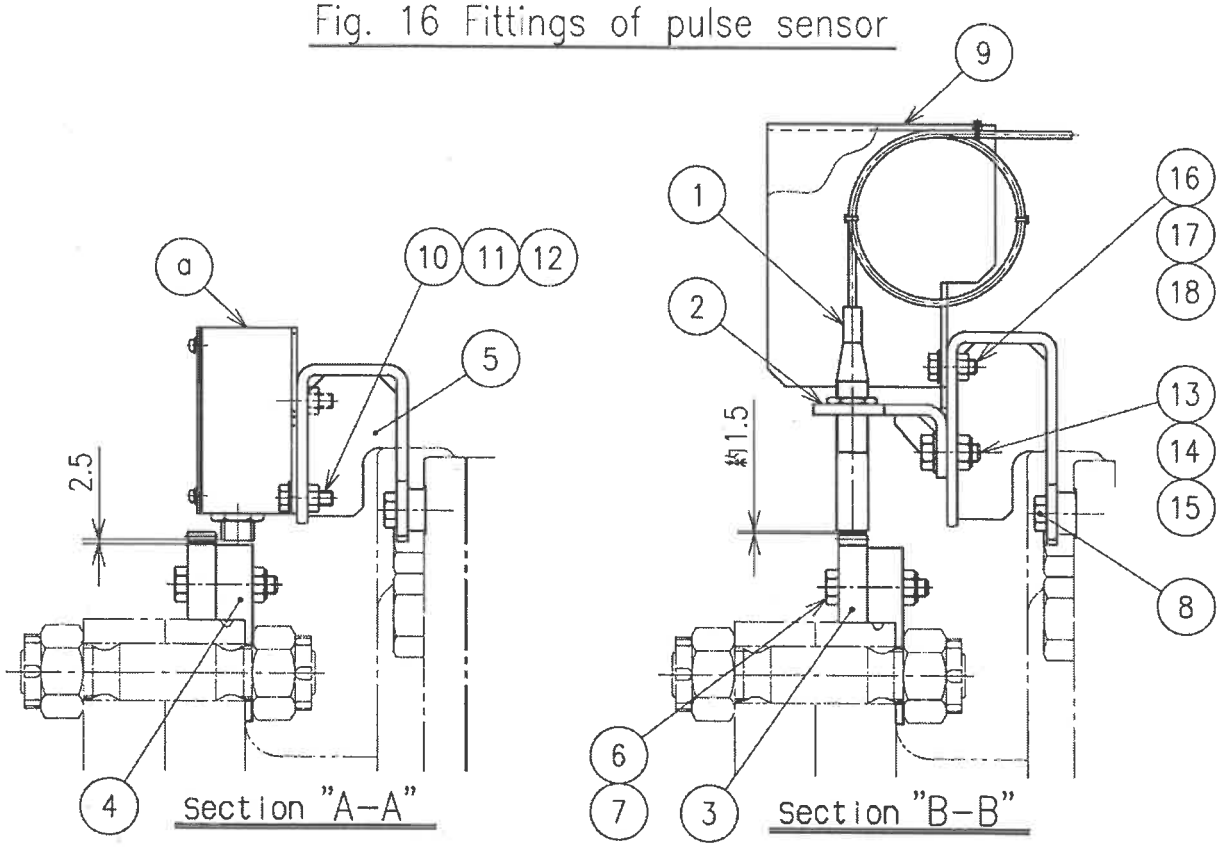
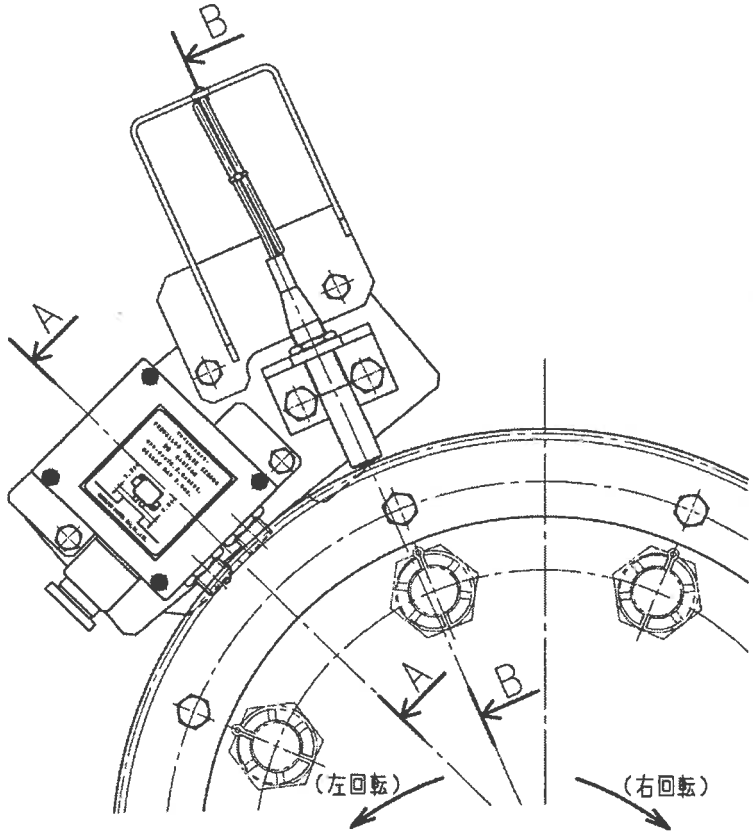
第15図　油量切替指令弁組立図
Fig.15 Flow change control valve assembly

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[001]	NN00738001A	タイ:ユリョウキリカエシレイベン:	BODY:FLOW CHANGE CONTROL VALVE:	1
[002]	NN00738002A	ベン:ユリョウキリカエシレイベン:	VALVE:FLOW CHANGE CONTROL VALVE:	1
[003]	NN00738003A	バネ:ユリョウキリカエシレイベン: 1.2X11.2X78XN27	SPRING:FLOW CHANGE CONTROL VALVE: 1.2X11.2X78XN27	1
[004]	G003400060Z	−:セツシユザ:	−:JOINT SEAT:	1
[005]	G003400080Z	−:トクシユプラグ M33X2:	−:SPECIAL PLUG M33X2:	1
[006]	G003400110A	−:ベンタイパツキン:	−:GASKET VALVE BODY:	2
[007]	X251208008ZZ	トメネジ:(2):M8X1.25X8	SET SCREW:−2:M8X1.25X8	3
[008]	Z565003400EE	マルパッキン::34	GASKET::34	1
[009]	Z565002500HZ	マルパッキン::25	GASKET::25	1
[010]	X200010095ZZ	ボルト::M10X1.5X95	BOLT::M10X1.5X95	4

第 16 図 パルスセンサ取付図

Fig. 16 Fittings of pulse sensor





・取付の詳細については、取扱説明書を参照のこと。

・For detail of fittings, refer to instruction manual.

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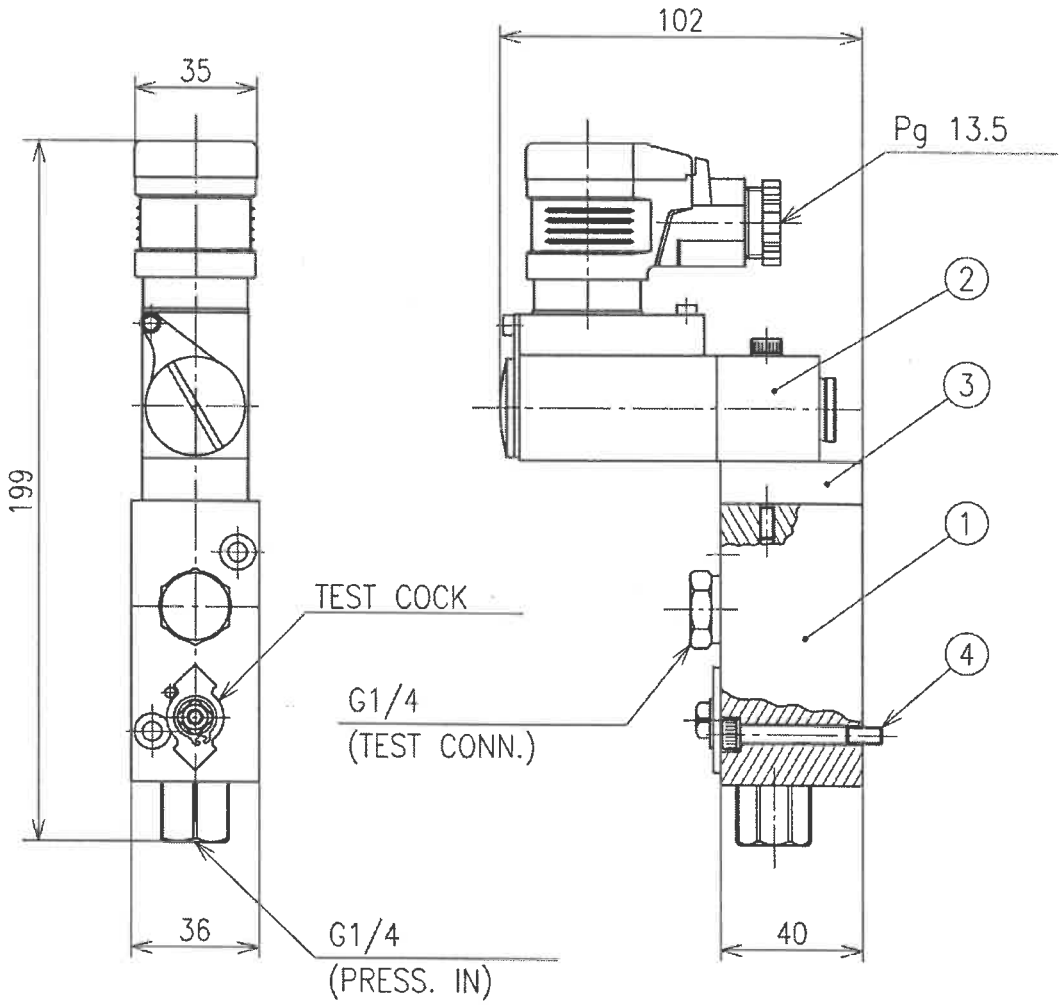
第 16 図　パルスセンサ取付図
Fig. 16　Fittings of pulse sensor

番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty
[001]	NN00736001A	パルスセンサ:ジャケット: DSD1810.11ATW	PULSE SENSOR: JAQUET:DSD1810.11ATW	1
[002]	NN00736004B	ブラケット:(2): パルスセンサ(DSD1810.11ATW)	BRACKET:(2): PULSE SENSOR(DSD1810.11ATW)	1
[003]	NN00736018A	ギヤ:カイテンケンシュツ: DRB-19J(L)S	GEAR:PULSE SENSOR: DRB-19J(L)S	1
[004]	NN00736019A	プレート:カイテンケンシュツ: DRB-19J(L)S	PLATE:PULSE SENSOR: DRB-19J(L)S	1
[005]	NN00736022B	ブラケット:(1): パルスセンサ DRA-25JS	BRACKET:(1): PULSE SENSOR DRA-25JS	1
[006]	X200010050ZZ	ボルト::M10X1.5X50	BOLT::M10X1.5X50	8
[007]	X227010000ZZ	Uナット::M10X1.5	U-NUT::M10X1.5	8
[008]	X200010028ZZ	ボルト::M10X1.5X28	BOLT::M10X1.5X28	2
[009]	NN00736021B	カバー: パルスセンサ(DSD1810.11ATW): DRB-19JS	COVER: PULSE SENSOR (DSD1810.11ATW): DRB-19JS	1
[010]	X200008025ZZ	ボルト::M8X1.25X25	BOLT::M8X1.25X25	2
[011]	X220008000ZZ	ナット::M8X1.25	NUT::M8X1.25	2
[012]	Z300008000ZZ	ヒラザガネ::8	FLAT WASHER::8	2
[013]	X200010025ZZ	ボルト::M10X1.5X25	BOLT::M10X1.5X25	2
[014]	X227010000ZZ	Uナット::M10X1.5	U-NUT::M10X1.5	2
[015]	Z300010000ZZ	ヒラザガネ::10	FLAT WASHER::10	2
[016]	X200008022ZZ	ボルト::M8X1.25X22	BOLT::M8X1.25X22	2
[017]	X220008000ZZ	ナット::M8X1.25	NUT::M8X1.25	2
[018]	Z300008000ZZ	ヒラザガネ::8	FLAT WASHER::8	2
[a]	L512009010A	カンビヒン:プロペラパルスセンサー(1):	PROPELLER PULSE SENSOR (1)	1

第 17 図　　圧力スイッチ組立図
Fig. 17　Pressure switch assembly



B10218 11

番号 Number	組立品番号 Assembly number	組立品名称	Name of assembly	数量 Qty.
	B102109110A	カンピセン：アツリョクSW:061B0102-1.0	ASSY.:PRESS.SWITCH:061B0102-1.0	1
番号 Number	部品番号 Parts number	部品名称	Name of parts	数量 Qty.
1	NN00008004A	バルブ:テスト061B7013	VALVE:TEST:061B7013	1
2	B102170038Z	アツリョクスイッチ::061B0102-1.0	PRESS.SWITCH::061B0102-1.0	1
3	NN00008039A	ダンパー:パルススナバ:061B7221	DAMPER::PULSE SNUBBER:061B7221	1
4	Z212005040ZZ	アナツキボルト::M5X40	BOLT::M5X40	2

Proportional Solenoid Valve Controller Manual

(Outline, Specifications, Controls, Maintenance and Inspection, and Failures and Corresponding Measures)

To: PHILIPPINE IRON CONSTRUCTION & MARINE WORKS,INC.(SNo.64)

Reduction Gear Unit Type DRA-25J(L)

DAIHATSU

DAIHATSU DIESEL MFG. CO., LTD.

Document No.	INB000126A

2017/05/09 TECHNICAL DEPARTMENT

1. Outline

(1) Outline

The unit aims to reduce the amount of black smoke exhausted and make main engine rotation drop less frequent with the following measures: a proportional solenoid valve controller is arranged in a clutch hydraulic oil line, a proportional solenoid valve is used to control the flow of clutch hydraulic oil when a clutch is engaged, the opening of the proportional solenoid valve is enlarged at the same time when the clutch is engaged so that the clutch will be filled with hydraulic oil swiftly, and during a process to engage the clutch, optimal slip control is adopted to implement clutch engagement without shock.

'In addition, propeller rotation speed and main engine rotation speed are taken into account for the unit's control, and the effects of oil temperature and loads are minimized.

Even if it has become impossible to detect propeller rotation speed, a direct compensation system is adopted which provides timers for respective operations and ensures that the clutch is engaged if an operation is continued beyond a prescribed time period.

The setup time and sensitivity of a clutch engagement operation are adjusted during pre-shipment factory operations. In addition, a numeric keypad in a control panel of the proportional solenoid valve can be used to change default values so that adjustment can be made to realize optimal operations.

(2) Controller's composition and control system

[Controller's Composition]

'Control equipment related to the proportional solenoid valve controller is arranged together on a block secured on a gear box's cover. The controller consists of the proportional solenoid valve, a drain valve, a drain solenoid valve, a bypass valve, a bypass solenoid valve, a filter, an accumulator to prevent pulsing motions and other equipment.

In addition, a propeller pulse sensor is attached to an output shaft and a controller (a PLC) is attached to a control panel.

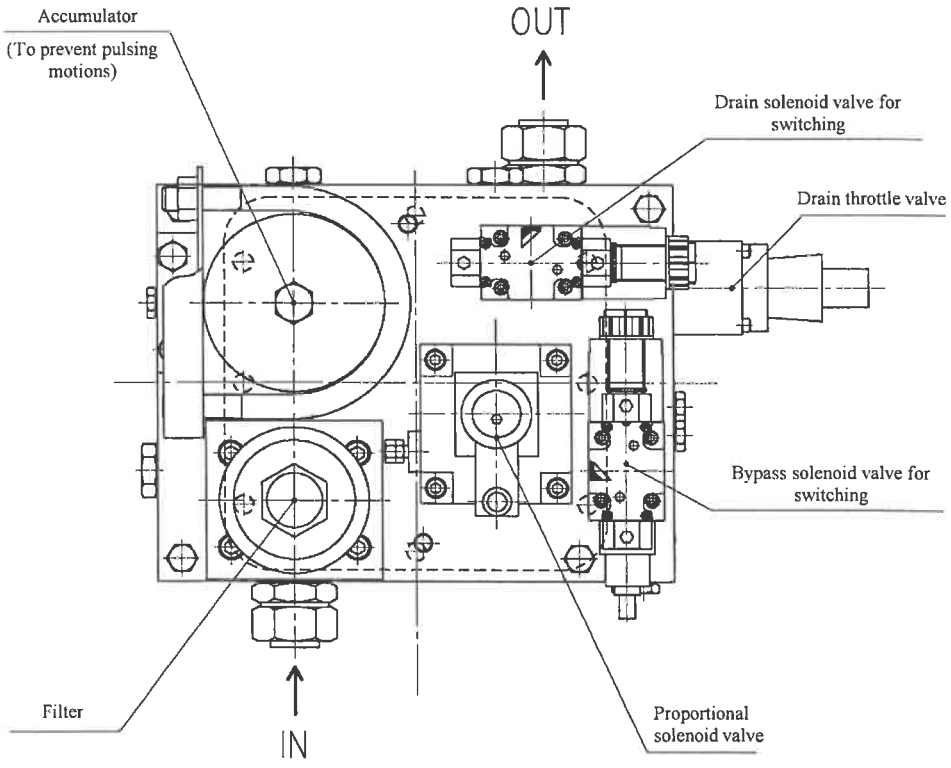


Fig. 1 Exterior view of proportional solenoid valve block

[Control system]

1) A control handle's handle position signal and the output shaft's actual rotation speed signal detected by the propeller pulse sensor are input to the controller embedded in the control panel so that the PLC can perform PID control, and a resulting control electric current signal is output to the proportional solenoid valve.

   The proportional solenoid valve changes the amount of its valve opening depending on a control electric current value to change slip hydraulic pressure and control propeller rotation speed within prescribed values.

2) A clutch control valve's ahead/astern signal and a main engine's rotation speed are input to the controller to realize improved response and control without shock during a clutch engagement process.
3) When hydraulic pressure is increasing after the clutch is engaged, the drain solenoid valve (a precision throttle valve), which is installed at the proportional solenoid valve's secondary side, drains part of clutch hydraulic oil to make slip control easier.
   When the clutch is engaged, the drain solenoid valve, which is installed on the line, is closed to keep the clutch's hydraulic pressure high.
   It is possible to use the drain solenoid valve's dial to change the amount of hydraulic oil to be drained.
   For your information, the opening amounts of the drain solenoid valve were adjusted during pre-shipment factory operations, and are locked.
4) A hydraulic pressure circuit which bypasses the proportional solenoid valve is provided, and a bypass solenoid valve and a variable throttle are installed there.
   When the clutch is engaged, the bypass solenoid valve is excited to also fill the clutch with hydraulic oil through the bypassing hydraulic pressure circuit so that a propeller can swiftly start rotating.

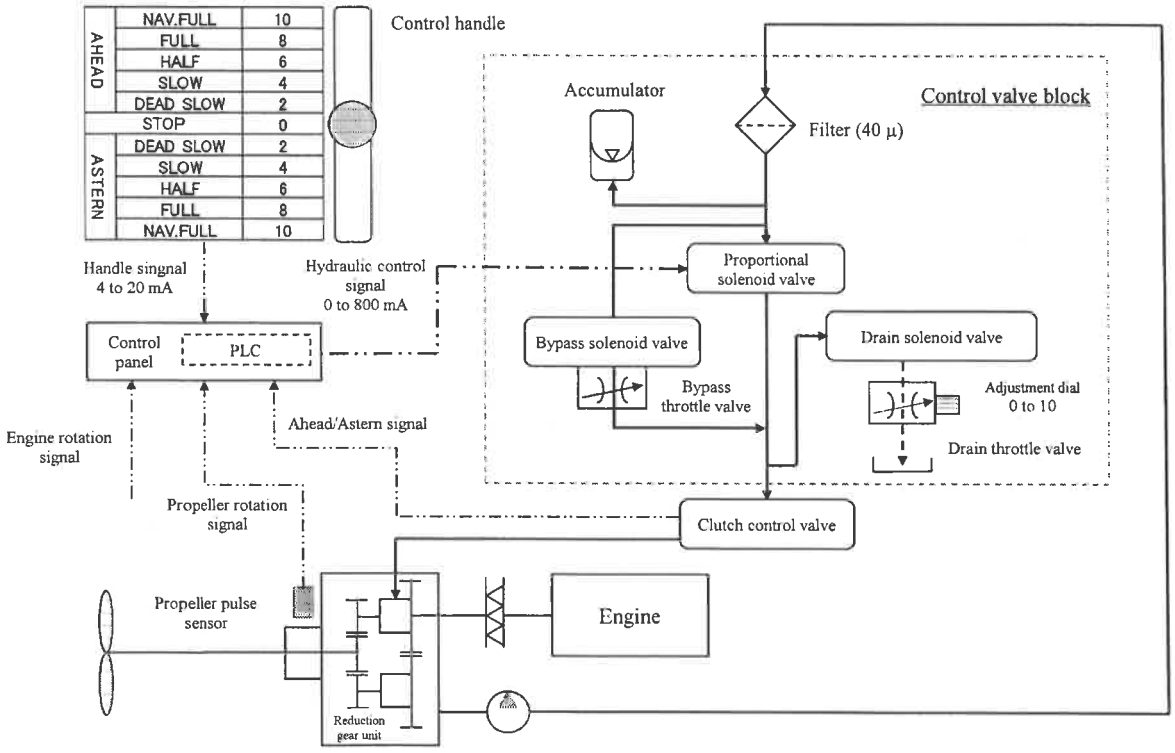


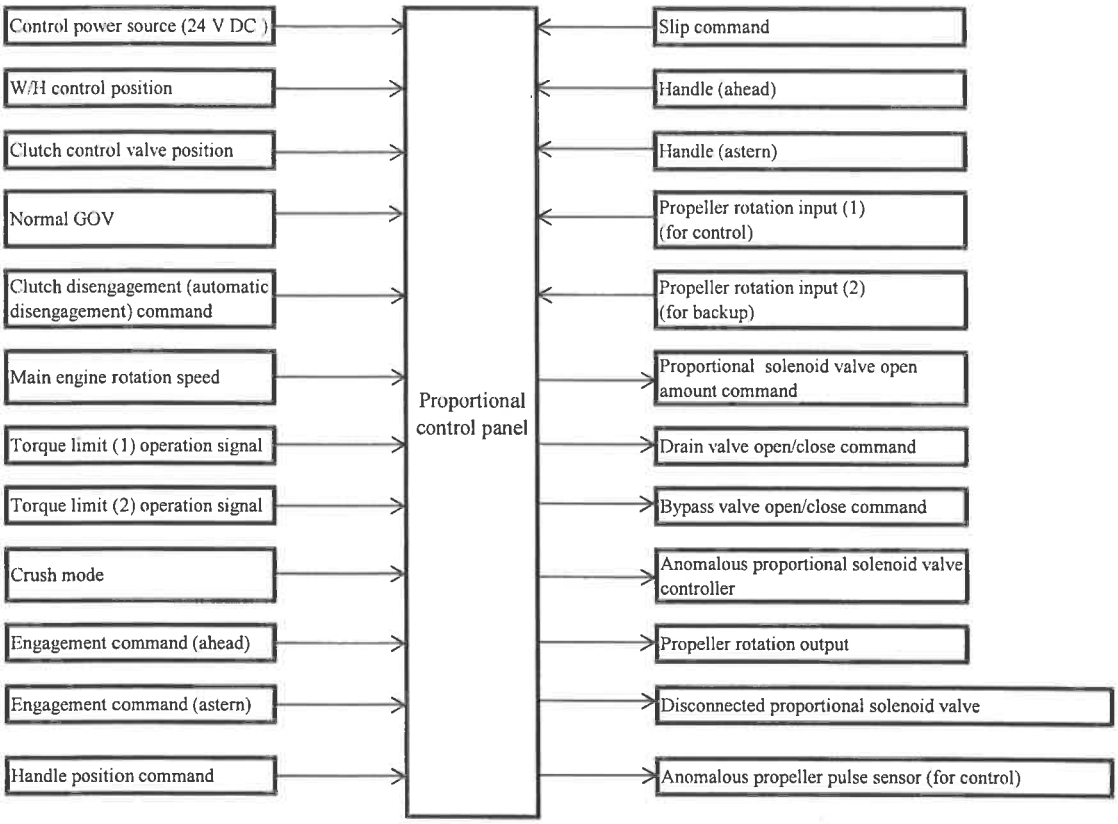
Fig. 2 Control system

2. Specifications

(1) Vessel specifications

Clutch engagement control

(2) Power source, and input and output signals



- Control power source: 24 V DC

  For a PLC inner power source, a 24 V DC / 24 V DC converter is used to make its control unit insensitive to the fluctuations of an external input power source's voltage.

  A contact input power source and an output relay operation power source operate with a control system's 24 V DC.

- Input signal

Handle position command	• • • •	ASmax to Neutral to AHmax = 4 to 12 to 20 mA
W/H control position	• • • •	Closed at the control position.
Clutch operating valve position	• • • •	Closed when the clutch operating valve's position is either AH or AS.
Clutch disengagement command	• • • •	Closed by clutch automatic disengagement operations.
Normal GOV	• • • •	Closed when GOV is normal.
Main engine rotation speed	• • • •	0 to full-scale (min^{-1}) = 4 to 20 mA
Torque limit (1) operation	• • • •	Closed during low torque limit operations.
Torque limit (2) operation	• • • •	Closed during high torque limit operations.
Propeller rotation input (1) (for control)	• • • •	Pulse signals (used for normal operations).
Propeller rotation input (2) (for backup)	• • • •	4 to 20 mA (used when the Propeller rotation signal (1) (for control) is not normal)

- Output signal

Solenoid valve open amount command	• • • •	For open amounts, 0 to 100% = 0 to 800 (100%) mA (a dither is available).
Drain valve open/close command	• • • •	Closed if commanded.
Bypass valve open/close command	• • • •	Closed if commanded.
Anomalous proportional solenoid valve controller	• • • •	Open if a PLC power source and PLC signals are not normal.
Disconnected proportional solenoid valve	• • • •	Open if the proportional solenoid valve is disconnected.
Propeller rotation output	• • • •	AS full-scale to 0 to AH full-scale (min^{-1}) = 4 to 12 to 20 mA.
Anomalous propeller pulse sensor (for control)	• • • •	Open if the propeller pulse sensor (for control) is not normal.

3. Controls

(1) Clutch engagement control

When the control handle is moved from a neutral position to an ahead position, the clutch control valve is switched to an ahead position and at the same time ejection output force acts on the proportional solenoid valve to increase its open amount, as well as the bypass solenoid valve is opened and additional hydraulic oil is also supplied from the bypass line. This will shorten the time required to fill the clutch with hydraulic oil.

If a propeller started rotating, and it reached to the rotation speed to release the ejection, after the open amount is decreased and the drain valve is opened to decrease clutch hydraulic pressure momentarily, the propeller's rotation speed is increased while the clutch is kept slipping and PID controlled with respect to a reference output so that it will conform with a prescribed propeller rotation speed increase operation.

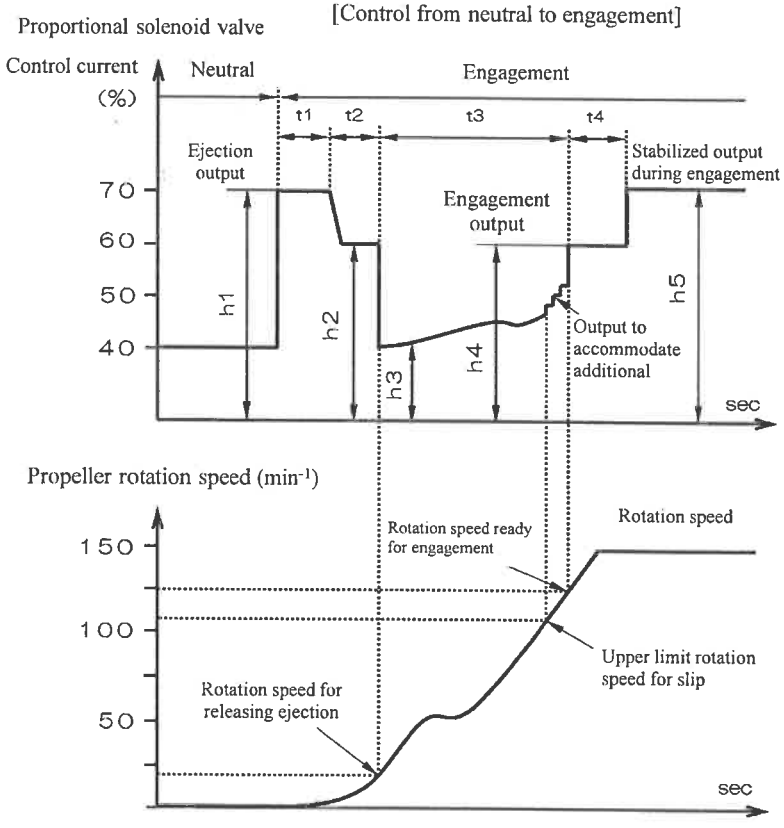
During the above process, if a signal indicating the decrease of engine rotation speed due to overloading is detected, the reference output will be lowered and the control to absorb shock due to the overloading is performed.

If propeller rotation speed reaches upper limit slip rotation speed, an output to accommodate additional shock absorption is applied, the open amount is increased stepwise, and the propeller rotation speed swiftly increases to one which allows the clutch to be engaged.

If propeller rotation speed becomes one which allows the clutch to be engaged, an engagement output works to close the drain valve and increase clutch hydraulic pressure to source pressure (2 MPa) after a prescribed time period is passed, and a clutch engagement state will be maintained.

When a time limit set by an engagement command compensation timer is passed after the control handle was operated to engage the clutch, the open amount of the proportional solenoid valve will be forced to be that of the clutch engagement, and the drain valve is closed to realize an engagement state.

If the control handle is operated for astern sailing from the neutral position, operations will be the same for ahead sailing.



Individual pre-set values of those marked with "t" and "h" are adjusted during pre-shipment factory operations, and their final adjustments are conducted during marine operations.
The individual pre-set values are documented in a pre-set value list.

Fig. 3 Clutch engagement control

PID operations

1) A proportional (P) operation is a control to output an operation amount proportional to a rotation speed deviation between a prescribed value and actual propeller rotation speed.
2) An integration (I) operation is to set 3 zones of large, medium and small for rotation speed deviations, and to change rotation speed, whose rotation speed deviation belongs to either one of the 3 zones, with integral sensitivity appropriate to diminish the rotation speed deviation within an appropriate operation time (sec). If a deviation is large, a corresponding integration action is performed swiftly, and if a deviation is small, a corresponding integration action is performed slowly.
3) A differentiation (D) operation is a control to adjust a response to an external disturbance. If a large D operation value is set, a large operation amount is applied to a rapid disturbance so that an original state can be achieved as soon as possible.

Please refer to a pre-set value list for respective pre-set values.

(2) Adjusting individual pre-set values for PLC control

1) Individual pre-set values are those adjusted during pre-shipment factory operations. If there is any difficulty to operate a vessel during an actual sailing or other problem, it can be adjusted by changing corresponding pre-set values.

If informed of a difficulty or other problem, our company's service engineer will solve it.

2) Individual pre-set values, which were set before shipment, are documented in a DM list.

If any pre-set values are changed after shipment, please record those changes in the DM list.

4. Controller and its operations

(1) Clutch control valve

When the control handle in a bridge is operated, control air acts on the clutch control valve and slides a spool valve to switch to neutral, ahead and astern modes.

A micro switch attached to the clutch control valve's handle transmits an ahead/astern clutch engagement signal. Please refer to the annex "Reduction Gear Unit Manual."

(2) Proportional solenoid valve

1) **Before starting an operation, loosen an air vent to remove air sufficiently**, and fill sufficient hydraulic oil to the proportional solenoid valve. After that, start an operation.
2) If no input current is supplied during an initial adjustment or due to an electric failure, a manual adjustment screw on top of the proportional solenoid valve can be tightened till the end so that the clutch will be engaged. **Make sure that the manual adjustment screw is turned back completely, and its lock nut is fastened during normal operations.**

(3) Drain throttle valve (precision throttle valve)

1) Drain throttle valve , by rotating the dial to alter the drainage quantity , and changes the clutch engaging time and the control oil pressure.

The key is kept by the person responsible, only when there is a need, please make adjustments by inserting the key.

If re-adjustment is required, please follow below steps;

[Dial adjustment]

1. Insert a key to a dial and turn the key to the left 90 degree angle.
2. Turn the dial onerevolution any direction and confirm snick. (The dial can be adjusted after confirming the snick.)
3. Adjust the dial position.

   Clockwise: Inrease drain. (Extend clutch engaging duration.)

   Counterclockwise: Decrease drain. (Shorten clutch engaging duration.)
4. Remove the key to finish.

※Keep the key by person in charge. Do not leave the key in the drain throttle valve.

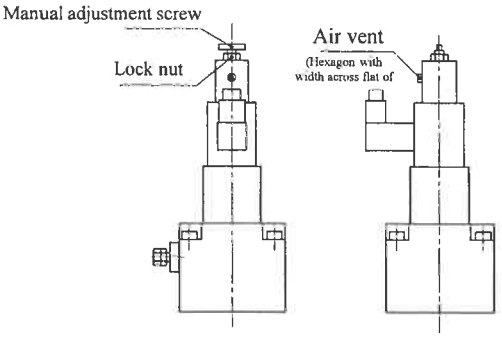


Fig. 4-1 Proportional solenoid valve

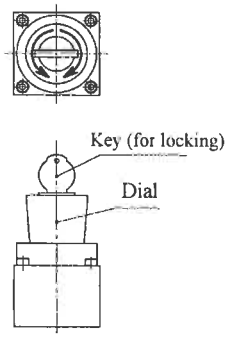


Fig. 4-2 Drain throttle valve

(4) Drain solenoid valve

1) The drain solenoid valve is used to drain part of hydraulic oil for slip control during a slip operation or when increasing hydraulic pressure after a clutch engagement operation.
If the drain solenoid valve is excited, a drain line is "opened," and if the drain solenoid valve is unexcited, the drain line is "closed."

2) It is possible to confirm whether the drain solenoid valve is excited by checking if the valve's lamp is turned on. **If the drain solenoid valve is not excited even when it is commanded to perform a slip operation, please measure the voltage values of terminal blocks No. 20 and No. 21 of the control panel's controller to confirm whether their output voltage values are 24 V DC .**

(5) Bypass solenoid valve

1) A bypass circuit with a proportional solenoid valve is provided for the clutch hydraulic oil line. When the clutch is engaged from its neutral position, this solenoid valve is excited to open the bypass circuit to swiftly fill the clutch hydraulic line with hydraulic oil.
A throttle valve is installed under this switchable solenoid valve, and the throttle valve is adjustable. However, the throttle valve is already adjusted during pre-shipment factory operations.

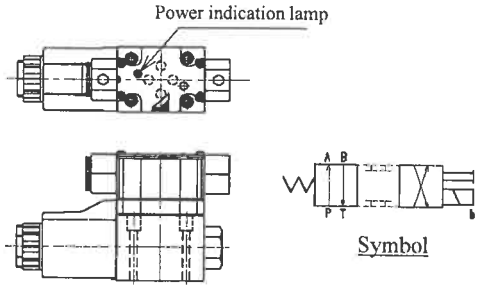


Fig. 5-1 Drain solenoid valve

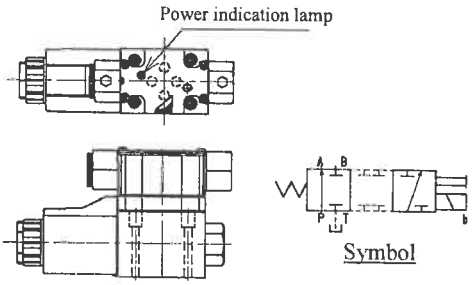


Fig. 5-2 Bypass solenoid valve

(6) Filter

1) A 40 μ filter is installed to prevent the clutch hydraulic oil line from clogging.
If differential pressure becomes greater than or equal to 0.3 MPa, a clogging indicator is activated, and if the differential pressure becomes greater than or equal to 0.35 MPa, the filter is relieved.

2) If the clogging indicator is activated, please inspect the filter, and if no anomaly is found, please clean filter elements and use the filter again. (The filter elements are made of metal.)

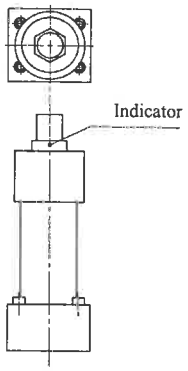


Fig. 6 Filter

(7) Propeller pulse sensor

1) There are 2 types of the propeller pulse sensor: one for control, and the other for backup. Propeller rotation speed is detected by a detection gear and a detection plate attached to an output shaft coupling.

2) Please refer to "5. How to install and adjust propeller pulse sensors."

5. How to install and adjust propeller pulse sensors

(1) Outline of how to install propeller pulse sensors

There are 2 types of the propeller pulse sensor: one for control, and the other for backup. Propeller rotation speed is detected by a rotation detection gear (for control) and a rotation detection plate (for backup) attached to the output shaft coupling.

The propeller pulse sensor is a proximity switch, and it can also distinguish ahead and astern rotational directions. This means that **if a propeller pulse sensor is not installed properly, the sensor may not be able to detect propeller rotation signal, and this may pose problems for clutch engagement and slip operations. Therefore, please make sure to install a propeller pulse sensor in accordance with the following instructions**.

* Note) A rotation detection gear is a single unit, and therefore it must be inserted to an output shaft before it is directly coupled to a middle shaft.

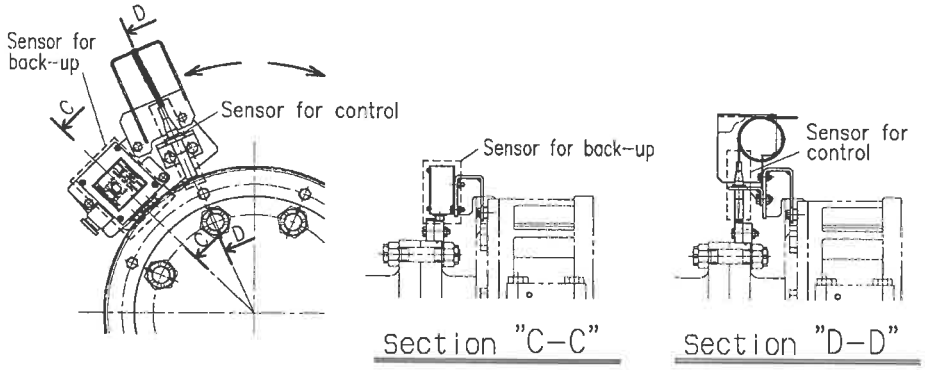


Fig. 7 Exterior view of propeller pulse sensor

(2) How to install propeller pulse sensors

1) How to install rotation detection gear (for control) and rotation detection plate (for back-up)

i) Set the rotation detection plate and the rotation detection gear to the output shaft coupling.

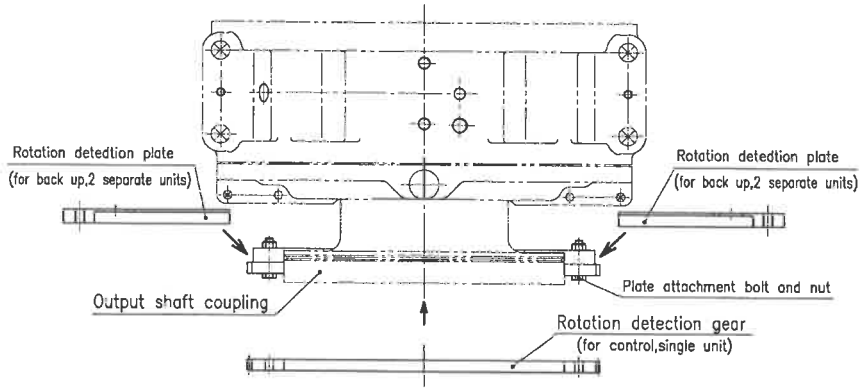


Fig. 8 Installing rotation detection gear/plate

ii) Adjust so that their clearances with the output shaft couplings in the places marked with "★" (10 places in total) will be uniform.

<Note> Standard radial clearance X,Y,Z:0.050~0.145

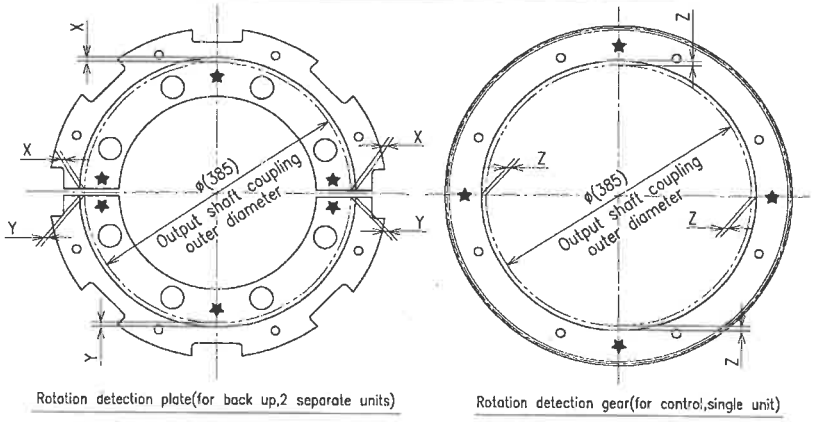


Fig. 9 Rotation detection gear/plate clearance adjustments

2) How to install propeller pulse sensor (for control)

i) Attach the sensor to a bracket so that its protrusion allowance is about 67 mm, and adjust the sensor's slit with the bracket's mark (a hole with the diameter of 2.5 mm). While keeping the sensor touching the rotation detection gear's tooth tip, fix a bracket (1) with a bolt.

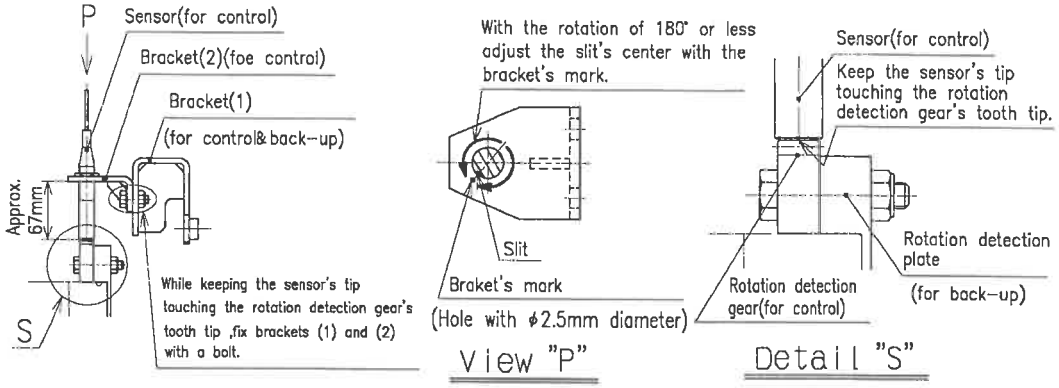


Fig. 10 Sensor position adjustments

ii) Rotate back the propeller pulse sensor (for control) one rotation, and adjust its clearance with its rotation detection gear to the value specified below.

A clearance between the propeller pulse sensor (for control) and its rotation detection gear must be 1.5

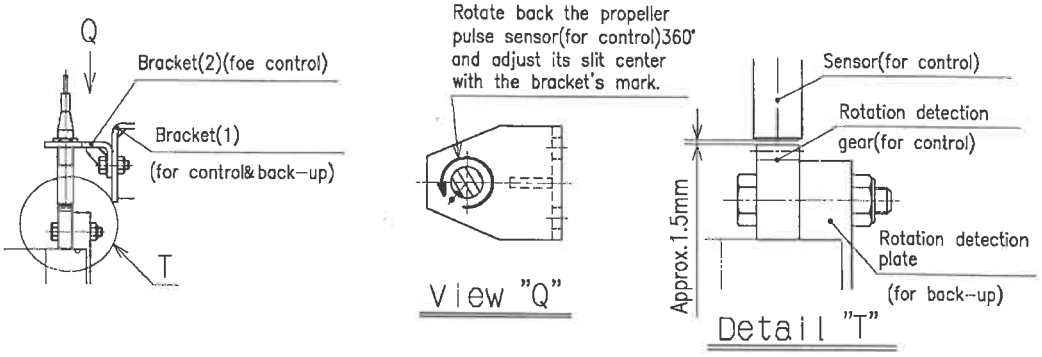
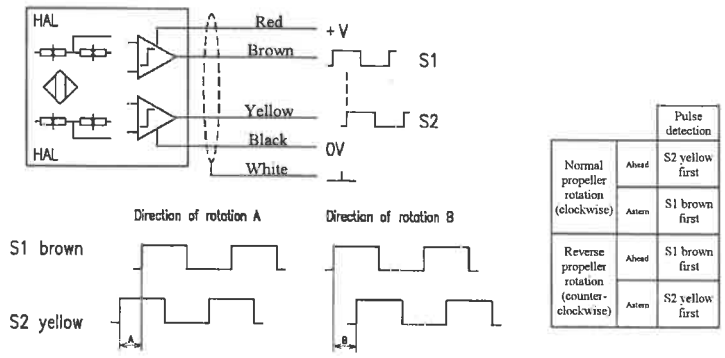


Fig. 11 Sensor clearance adjustment

iii) Connect wires in accordance with the following wire connection instructions.



		Pulse detection
Normal propeller rotation (clockwise)	Ahead	S2 yellow first
	Astern	S1 brown first
Reverse propeller rotation (counter-clockwise)	Ahead	S1 brown first
	Astern	S2 yellow first

Fig. 12 Wire connection instructions

iv) Points to be checked after installation

- Turn on an electronic controller.
- It can be confirmed whether the propeller pulse sensor (for control) is operating normally by measuring the voltage values of the control panel's terminal blocks No. 13 and No. 14.

3) How to install propeller pulse sensor (for backup)

The output signals of the propeller pulse sensor (for backup) are also square waves. It is used as a backup when the propeller pulse sensor (for control) fails.

2 sensors are installed in a sensor box to detect ahead and astern motions.

- It is a box construction. Therefore, use bracket attachment hole's looseness to adjust its clearance.

A clearance between the rotation detection plate and the propeller pulse sensor (for backup) must be 2.5 mm.

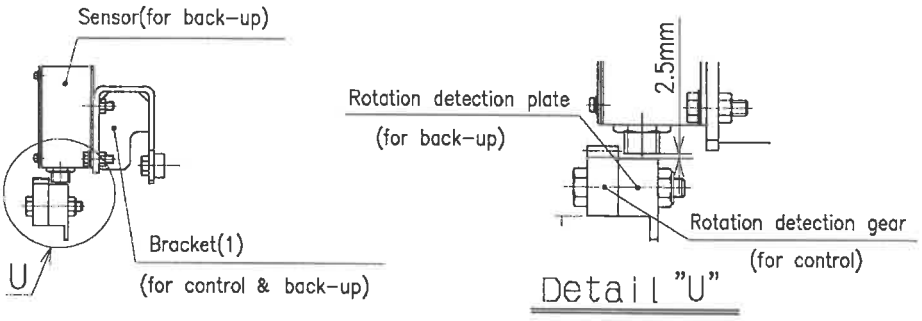


Fig. 13 Installing sensor for display

6. Emergency operations

(1) If the control panel for the proportional solenoid valve loses power or control signals to the proportional solenoid valve are lost, a "anomalous proportional solenoid valve controller (PLC)" alarm on a monitor panel is turned on. In that case,as back-up a prescribed current is separately sent from a main engine's panel to the proportional solenoid valve so that the proportional solenoid valve is opened by a prescribed open amount and clutch engagement operations can be performed by control handle operations.The prescribed current value is already

Although a dial provided on the main engine's control panel can be used to change the prescribed current's value, **In case that the engine is running under the situation of "anomalous proportional solenoid valve controller (PLC)", make sure the engaging clutch oil must be within green mark range.**

Basically customers do not have to adjust the prescribed current, the engaging clutch oil pressure by any possibility is below blue mark range,turn clockwise (increase the proportional solenoid valve open angle) the dial located near the control panel for the proportional solenoid valve in main engine control panel in order to increase the pressure within blue mark range.

[Alarm reset]

Turn off the circuit breaker in the the control panel for the proportional solenoid valve, and then turn on it again. If the alram is not reset by above precedure, PLC might be failure. Please contact our service engineer.

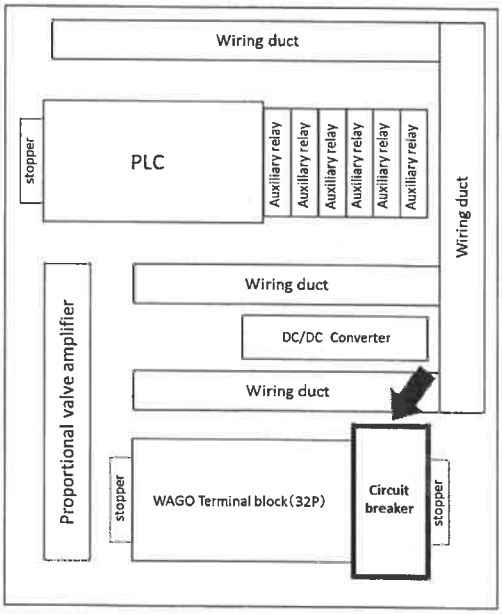


Fig. 14 The position of the circuit breaker in the the control

(2) If the proportional solenoid valve's control cable is disconnected or the proportional solenoid valve itself failed, a "disconnected" alarm on the monitor panel is turned on.
It is possible to force the proportional solenoid valve to open by screwing in the manual adjustment screw on top of the proportional solenoid valve (Refer to Fig.4-1) , so that clutch engagement can be performed.

(3) If the normal power supply or emergency is lost, the power of DC24V for remote-control is kept, therefore the panel can be operated as the clutch control panel. But, both of them are lost, the signal of the proportional solenoid valve is lost and clutch cannot be performed.
The solution of the situation is forced to engage the clutch. That method is same way write-down above section (2).

(4) If the signals of a main engine's governor are lost, a hold-as-is approach will be adopted.
That is, if the clutch is engaged, it will remain so.

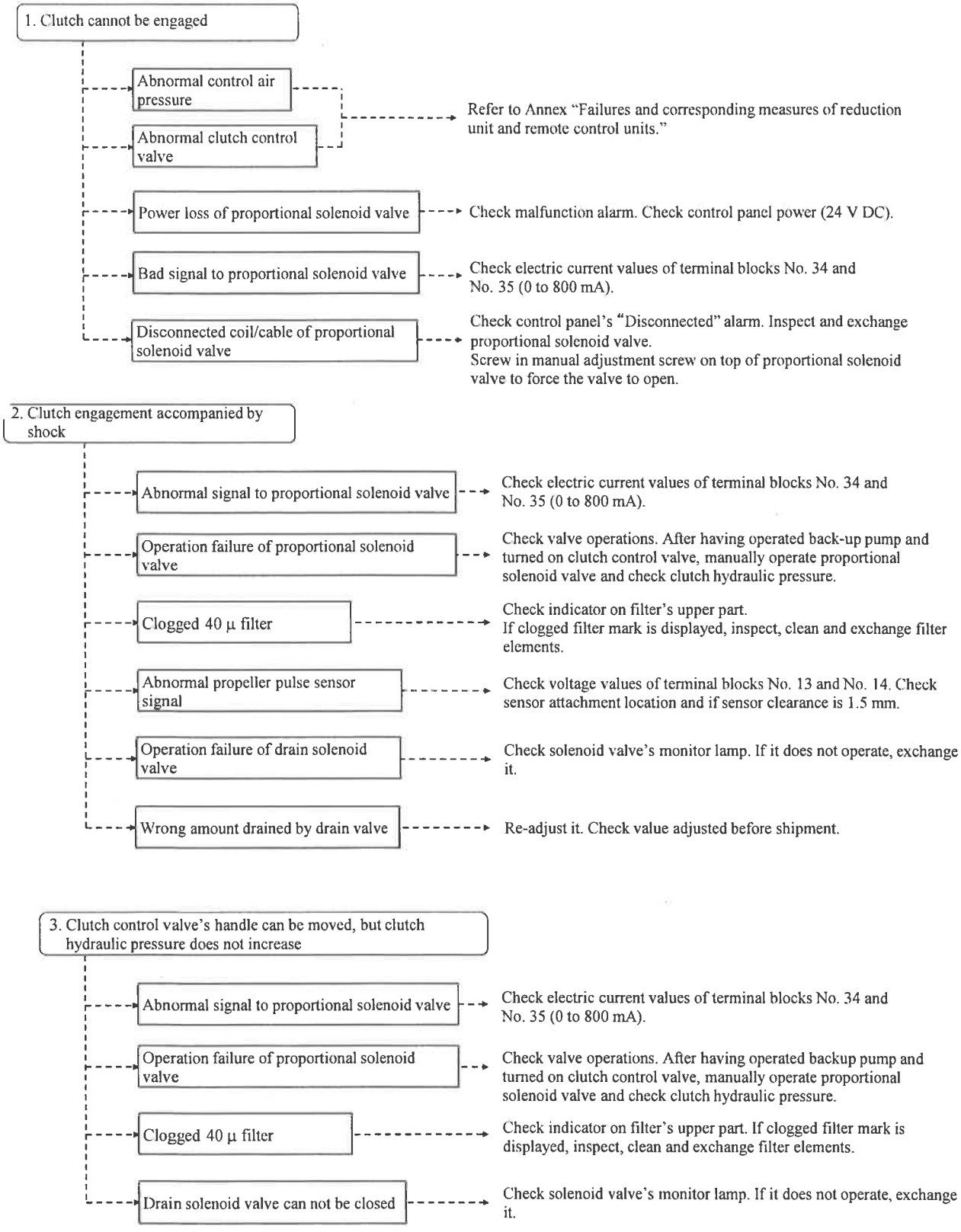
(5) If the signal of the propeller pulse sensor (for control) is lost, the "disconnected" alarm will be issued.
The loss of propeller signals for a short time will be detected, and monitoring operations performed by the propeller pulse sensor (for control) will be automatically switched to those performed by the propeller pulse sensor (for backup). Thus, continued vessel operations will be ensured.

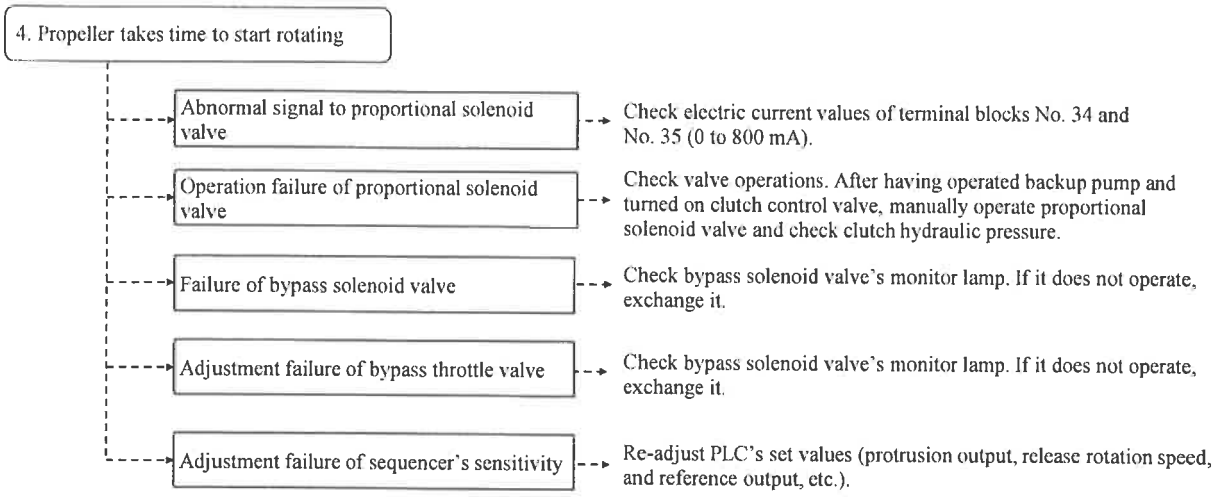
The disconnection alarm starts due to other reasons in some conditions. Please reset the disconnection alarm.
Two procedures of alarm reset are as follows;

[Alarm reset]
Procedure 1
Turn off the circuit breaker in the control panel for the proportional solenoid valve, and then turn on it again.
Procedure 2
Change DMO (pick up sensor failure reset) from 0 to 1 by PLC board.
To keep DMO at 1 makes abnormality detection not to work. To return DMO to 0 makes abnormality detection to work in normal condition.

If alarm cannot be reset by above two precedure, please contact our service engineer.
It is assumerd that the control propeller pulse sensor or wire might be failure.

7. Causes of proportional solenoid valve control system's troubles and measures to solve them




4. Propeller takes time to start rotating
Abnormal signal to proportional solenoid valve
Check electric current values of terminal blocks No. 34 and No. 35 (0 to 800 mA).
Operation failure of proportional solenoid valve
Check valve operations. After having operated backup pump and turned on clutch control valve, manually operate proportional solenoid valve and check clutch hydraulic pressure.
Failure of bypass solenoid valve
Check bypass solenoid valve's monitor lamp. If it does not operate, exchange it.
Adjustment failure of bypass throttle valve
Check bypass solenoid valve's monitor lamp. If it does not operate, exchange it.
Adjustment failure of sequencer's sensitivity
Re-adjust PLC's set values (protrusion output, release rotation speed, and reference output, etc.).

Proportional Solenoid Valve Controller Specifications

ON-OFF

Ver. 1.02

August 29, 2016

MIT ENGINEERING, INC.

History of Revision

Revision Number	Description of Revision	Date

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1. Parts List

No.	Product Name	Manufacturer	Type	Amperage Consumed	Quantity
1	PLC CPU unit	OMRON	CP1H-XA40DT1-D	0.15 A	1
2	PLC LCD option board	OMRON	CP1W-DAM01	0.02 A	1
3	PLC memory cassette	OMRON	CP1W-ME05M	-	1
4	DC/DC converter	ETA	SVB-24SC-24	MAX 2.5 A	1
5	Proportional valve amplifier	NACHI	EBA-PD1-NZ-D2-10	MAX 1.25 A	1
6	Auxiliary relay	OMRON	G2R-2-SND DC24V	21.8 mA	6
7	Circuit breaker	MITSUBISHI	CP30-BA 3A	-	1
8	Terminal block	WAGO	100X280-681	-	41
9	Resistance		10KΩ 1/4W	-	3
10	Capacitor		50V 22μF	-	1

2 Specification

1) Overview

The clutch operation in this application destination ship will be the "ON-OFF" clutch.

This device uses a proportional solenoid valve to control the clutch operation pressure at the time of the forward/reverse operation of the clutch in the reduction gears. The filling speed of the oil in the clutch cylinder of the reduction gears are gradually controlled to minimize the variance of the engine load at the time of clutch engaging. The black smoke in emission gas is also reduced and the decrease of the revolution speed of the engine is mitigated, which are the objectives of this system.

Detection of the engine and propeller revolution speeds are included in the control.

Any influence from oil temperature or external thrust is suppressed to the minimum.

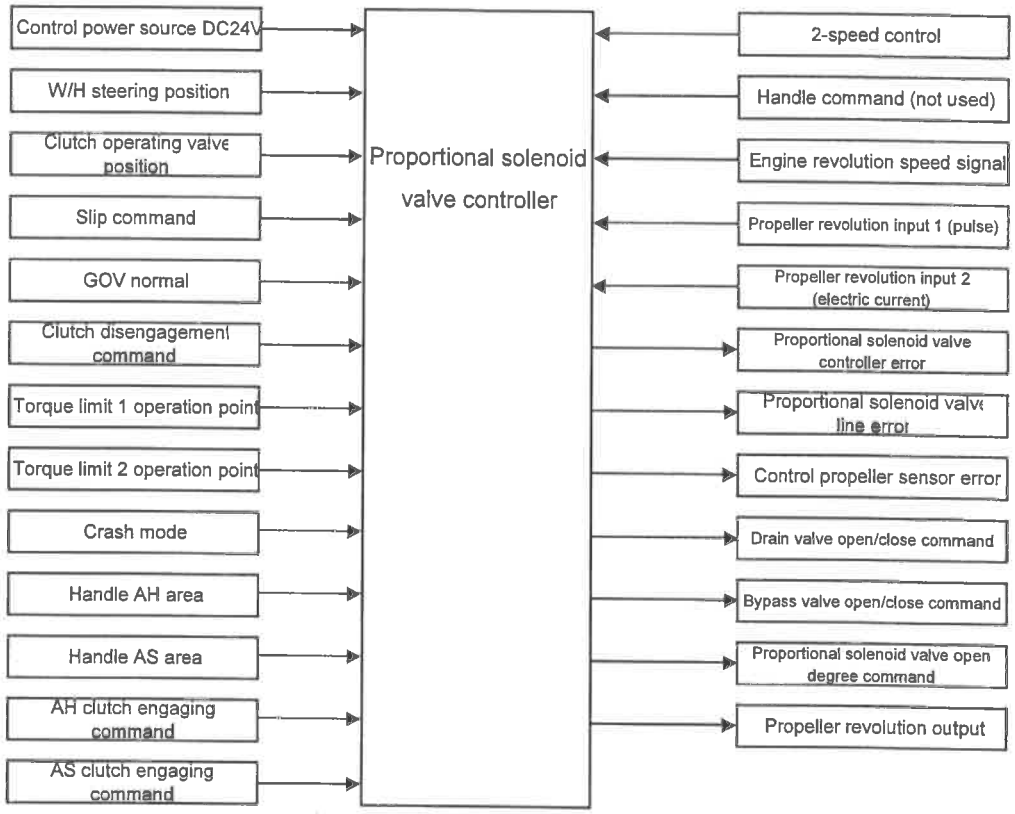
It will equalize the load transfer from the clutch to the main machine when the clutch is engaged, (to boost the clutch hydroulic pressure from the clutch-cylinder filling.)

If the revolution speed of the engine is reduced lower than the setting of the idle revolution speed while the clutch is engaging, the revolution speed of the propeller will be reduced synchronously to mitigate the decrease in the revolution speed of the engine.

*Even when it becomes impossible to detect the revolution speeds of the engine and the propeller, if the handle is operated to the clutch engaging position, and if the timer is set up for each operation to compensate the engaging of the clutch and the same operation is continued within the specified time, the operation of the timer is prioritized to conduct an accurate clutch engaging operation.

Each of the settings for specified revolution speeds, sensitivities, and engaging time for the clutch engaging operation and the slip operation can be changed to any setting on the PLC (setting board) in the clutch control panel.

2) System Configuration



・Specifications of the power source and I/O signals

Control power source DC24V

The internal power source for the PLC uses the DC24V/DC24V converter, so the voltage from the external input power, even when it fluctuates, does not easily exercise influence over the control part in this state.

Input signal ・Closed at the W/H operation position ・Closed at each of the AH and AS positions of the control valve

・Closed at the slip command ・Closed at the GOV normal

・Closed at the declutch command ・Closed at the torque limit 1 operation point

・Closed at the torque limit 2 operation point ・Closed at the crash mode

・Closed at the handle AH area ・Closed at the handle AS area

・Closed at the AH direct connection command ・Closed at the AS direct connection command

・Closed at the 2-speed control

・Handle command AS MAX - neutral - AH MAX = 4 - 12 - 20 mA (available depending on the DM97 setting)

・Engine revolution signal 0 - full scale min-1 = 4 - 20 mA

・Propeller revolution signal 1 pulse signal (for regular use)

・Propeller revolution signal 2 amperage signal 4 - 20 mA・Used at the time of the propeller revolution signal 1 error

Output signal ・Open at the proportional valve control device error・Open at the proportional valve line error

・Open at the control propeller sensor error ・Closed at the drain valve command

・Closed at the bypass valve command

・Proportional solenoid valve open degree output 0 - 100% = 0 - 800 mA (dither available)

・Propeller revolution output AS full scale - 0 - AH full scale min-1 = 4 - 12 - 20 mA

3) Control at Each Operation Position

W/H steering

·Waiting, jump up, slipping, buffering, or clutch engaging operation as well as an optimal operation of the bypass valve and the drain valve is conducted for neutral-engaging operation in the W/H handle operation position while the clutch disengagement command and the GOV error signal are not sent. The revolution of (the load on) the propeller is increased smoothly.

·Operation from the clutch engaging position to the neutral position

The proportional solenoid valve output turns to waiting output after the open degree becomes low, and the drain valve is excited (opened) for the specified time.

·If a clutch disengagement signal is input, the proportional solenoid valve output becomes the same operation as the command neutral operation.

·When a GOV error occurs

The setting operation closest to the load (propeller's revolution) status at the time a GOV error is detected will be used and maintained until the error is canceled.

At this time, the bypass valve and the drain valve also maintain the current status.

E/S steering

As the clutch engaging (Ahead, Astern) operation is made with the clutch disengaging (N) of the E/S steering, the proportional solenod valve performs control depending on the jump up, slipping, buffering, or clutch engaging operations based on the position information of the clutch operating valve.

At this time, the bypass valve and the drain valve make optimal operations based on the status of the clutch engaging operation.

*GOV error detectionis not conducted.

4) Controlling the Drain Solenoid Valve

At the time of W/H steering and any other control, the drain solenoid valve is excited (opened) until the handle neutral operation (until the clutch operating valve is set to the neutral position) and from the secondary jump up output operation to the clutch engaging output operation.

After the clutch engaging operation, the drain solenoid valve is demagnetized (closed) when the specified time passes.

*When the clutch operating valve is in the neutral position, demagnetization (closing) continues to reduce the time for filling the clutch-cylinder while the primary jump up is in operation.

From the time of clutch engaging to the time of neutral operation, excitation (opening) continues for a specified time to help reduce the hydraulic pressure of the clutch.

5) Controlling the Bypass Solenoid Valve

At the time of primary jump up operation, excitation (opening) will reduce the time for filling the clutch cylinder.

6) Crash Mode

When a crash mode signal is sent from an external part, the specified sensitivity in the PLC is changed to a suitable operation for crash operation.

In addition, a lower limit propeller revolution command is used to continue the command for an arbitrary duration of time.

7) 2-Speed Control

When a 2-speed control signal is sent from an external part, the propeller and engine revolution settings in the PLC are changed to suit 2-speed operation. The engaging/disengaging operations of the clutch are conducted.

8) Proportional Solenoid Valve Controller Error

An alarm is issued when the power source voltage to the PLC becomes low or if the proportional solenoid valve output reaches or becomes lower than the setting. At this time, the proportional solenoid valve loses its output, so the clutch is disengaged.

If this alarm is issued, the M/E remote control side switches the line of the proportional solenoid valve output to the emergency line. Clutch control is conducted without using the proportional solenoid valve controller. However, in this case, the slip control does not work.

To reset this error, turn off the power switch of the proportional solenoid valve controller.

9) Control Propeller Revolution Sensor Error

If the PLC detects a control propeller revolution sensor error, an error signal is output to the outside, and an emergency propeller revolution signal turns on for control.To reset this error, use the PLC setting board to change DM0 (pickup sensor error reset) from 0 to 1.

This resets the error and the error detection can be canceled by continuing "1".

If you change it back to "0", the status changes and the error detection is validated again.

You can also reset it by turning off the power switch of the proportional solenoid valve controller.

10) Proportional Solenoid Valve Line Error

An alarm will be issued when a proportional solenoid valve output signal is received by the PLC and it is judged that the electric signal from the proportional solenoid valve lines is broken or that a signal exceeding a planned setting value is being output.

This output is for alarms only and is not used for any control.

11) When Remote Control Power Source is Lost (1) (When the Regular Power Source or the Emergency Power Source is Lost)

Even if one of the remote-control power sources is lost, the clutch control panel can continue its regular operation since the DC24V in the remote control is secured.

12) When Remote Control Power Source is Lost (2) (When Both Regular Power Source and Emergency Power Source are Lost)

If the above-mentioned trouble occurs with the clutch operating valve in the engaging position, the proportional solenoid valve output is lost, so the clutch is disengaged.

To make the clutch engaging position again, tighten the T-screw part for the manual operation of the proportional solenoid valve. This enables the engaging operation of the clutch.

13) When Remote Control Power Source is Lost (3)

(When the Clutch Control Panel Power Source is Lost)

If the above-mentioned trouble occurs with the clutch operating valve in the clutch fitting position, the proportional solenoid valve loses its output and the clutch is disengaged.

The proportional circuit switches to the emergency circuit, so the clutch-engaging operation can be conducted. (Slipping operation is not available.)

3. Sensitivity Adjustment (Adjustment at the Time of Defective Operation) *See the operation flow for "a" and "b."

a It takes time to start rotating the propeller. Or, the startup is sudden and the revolution speed of the engine decreases markedly when it starts rotating.

(1) Electrical adjustment (PLC)

DM4 AH(ahead) primary jump up output DM11 AS(astern) primary jump up output

It takes a shorter time to start rotating when the setting value is larger.

It takes a longer time to start rotating when the setting value is smaller.

If the setting value is too large, the propeller starts up all of sudden, causing an overshoot.

When changing the setting, try changing it by 300 (3%) to 500 (5%).

*You can change the setting up to 10000 (100%), but the setting values exceeding the DM565 proportional solenoid valve upper limit will not be output.

(2) Mechanical adjustment (hydraulic equipment)

Adjusting the open degree of the bypass valve

When you turn the knob counterclockwise, the open degree increases and the amount of oil into the clutch cylinder increases.

This will result in a shorter time to start rotating. When you turn it clockwise, the open degree decreases and it will take longer to start operations.

Try changing it by half a turn to one full turn.

*The full open position is about 6.5 turns from the complete close.

b The startup takes time after the propeller starts rotating. Or, the startup is too fast.

(1) Electrical adjustment (PLC)

DM3 At crash, clutch engaged, AH addition output DM10 At crash, clutch engaged, AS addition output

An addition is made to the PID output during the slipping shown in the operation flow arrow "b."

As the setting value becomes larger, the startup time of the propeller revolution becomes shorter.

As the setting value becomes smaller, the startup takes longer.

When changing the setting, try changing it by 300 (3%) to 500 (5%).

(2) Mechanical adjustment (hydraulic equipment)

Adjusting the open degree of the drain valve

The amount of released oil (variable) is adjusted when the drain valve is in the open status in the operation flow.

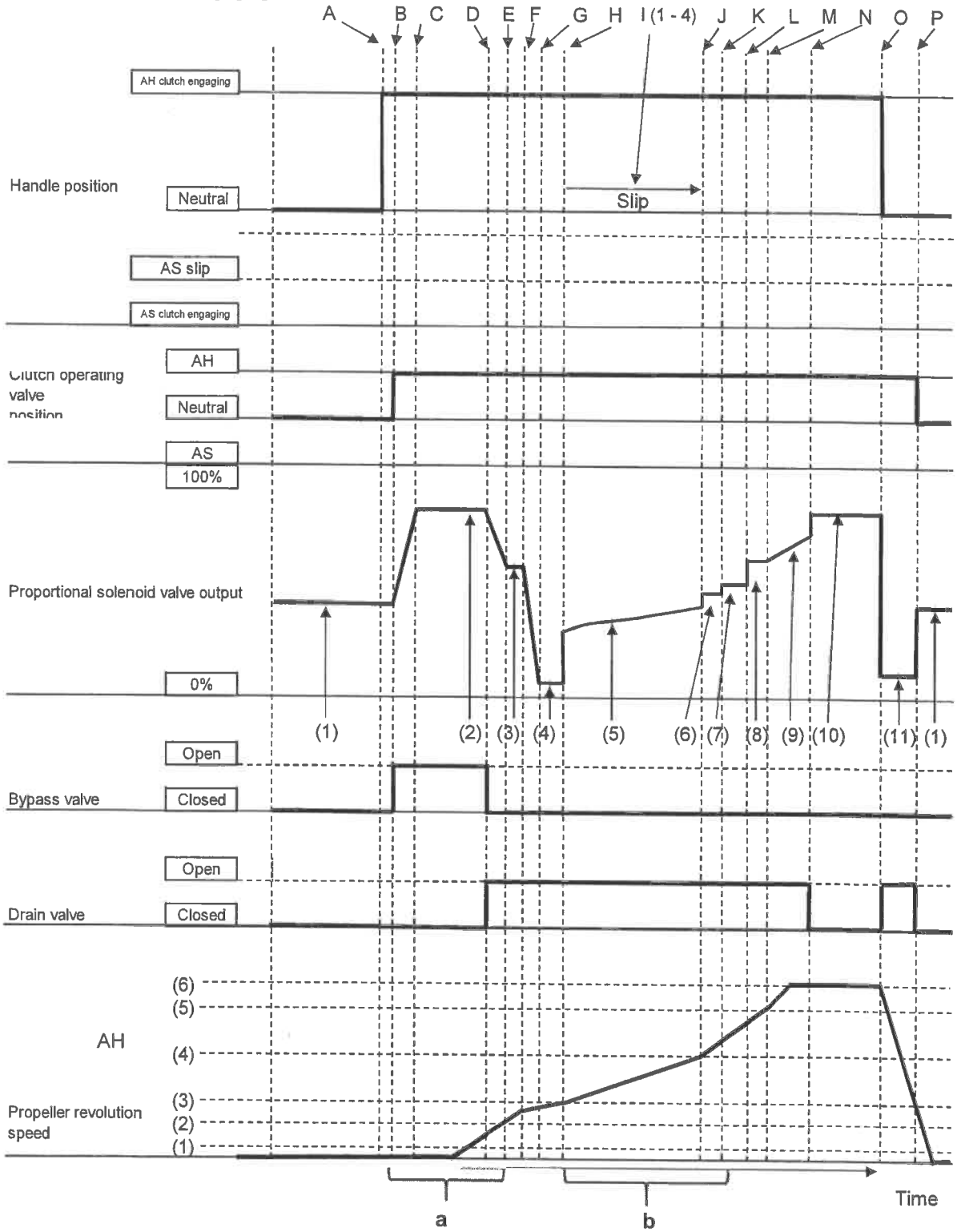
When the value of the valve graduation becomes smaller, the amount of released oil decreases and the startup time of the propeller revolution becomes shorter.

As the value becomes larger, the startup time becomes longer.

When changing the setting, try changing it by 0.5 to 1 at a time on the valve graduation scale.

4. ON-OFF Slip Operation Flow (For A-P and (1)-(11), see the description in the next section.)

Neutral ⇒ AH clutch engaging ⇒ Neutral



(1) Primary jump up release revolution speed
(2) Secondary jump up release revolution speed
(3) Slip command AH lower limit/1-point command revolution speed
(4) Slip command AH upper limit revolution speed
(5) Revolution (possible clutch engaging)
(6) Clutch direct connection revolution (idle)

5. Description of ON-OFF Slip Operation Flow (About Signs A to P)

A Issuance of handle command AH clutch engaging command

B Clutch operating valve AH position followed⇒ Proportional solenoid valve output AH primary jump up output operation + bypass valve open

AH primary jump up operation time count start

B - C Time changing from the wait output to the AH primary jump up output

C - D Status to maintain AH primary jump up

D Starting AH secondary jump up operation time + AH secondary jump up operation time count start

There are two types of release conditions in the operations of D.

D1: The retention time for AH primary jump up ends.

D2: The AH primary jump up release propeller revolution speed is detected.

D1 or D2, whichever is faster, is prioritized and the AH secondary jump up output starts.

At this time, AH primary jump up output release + bypass valve open + drain valve open

D - E The time to change from the AH primary jump up output to the AH secondary jump up output

F1 End of AH secondary jump up retention time⇒ AH secondary jump up output release

F2 AH secondary jump up release propeller revolution speed detection⇒ AH secondary jump up output release

F1: Operation time, F2: Release revolution speed detection. The one that starts operating earlier is prioritized.

F Overshoot prevention output operation + overshoot prevention operation time count start

F - G Time to change from the AH secondary jump up output to the overshoot prevention output

H End of overshoot prevention operation⇒ Slip operation start + Slip compensation time count at direct connection operation start

I1 Maintain slip command lower limit revolution speed (1-point slip command)

I2 Slip command lower limit revolution speed (1-point slip command) followed by propeller revolution speed

I3 Slip command increases the setting speed from the lower limit to the upper limit.

I4 Slip command upper limit revolution speed followed by propeller revolution speed⇒ Slip end

When slip compensate time at the time of the clutch engaging ends, the buffer starts regardless of the processes from I1 to I4.

J Slip end⇒ Buffer (1) operation (Buffer (1) operation time count start + Clutch engaging command compensate time count start)

K AH buffer (1) release⇒ AH buffer (2) operation (Buffer (2) operation time count start)

L AH buffer (2) release⇒ AH buffer (3) operation (Buffer (3) operation time count start)

M AH buffer (3) release⇒ Clutch engaging output operation (Clutch engaging settling operation time count start)

When the propeller revolution speed reaches or exceeds the revolution speed that enables a clutch engaging in each of the buffering processes from J to M, the clutch engaging output starts regardless of the ongoing buffering process.

The clutch engaging output starts even if the clutch engaging command compensate time ends.

N Clutch engaging output release⇒ Clutch engaging settling output operation + Drain valve closed

O Handle command AH clutch engaging⇒ Neutral From proportional solenoid valve output clutch engaging settling output to low level output + drain valve open

P Clutch operating valve neutral (following handle command)⇒ From proportional solemoid valve output low level to wait output + Drain valve closed

6. Operation Flow of DM Setting Values

The DM setting list of the operation flow proportional solenoid valve output descriptions (1) to (10)

*The operation flow is for AH operations, but the AS side is included for the following DM.

Proportional solenoid valve open degree output setting

(1)	DM25	Waiting output		
(2)	DM4	AH primary jump up time	DM11	AS primary jump up time
(3)	DM5	AH secondary jump up time	DM12	AS secondary jump up time
(4)	DM8	at crash, clutch engaging, AH overshoot prevention output	DM15	at crash, clutch engaging, AS overshoot prevention output
(5)	At AH	DM3 at crash, clutch engaged, AH addition output + DM110 after PID APRLMT output + DM111 reference output		
	At AS	DM10 at crash, clutch engaged, AS addition output + DM110 after PID APRLMT output + DM111 reference output		
(6)	DM16	AH clutch buffer adding time 1	DM19	AS clutch buffer adding time 1
(7)	DM17	AH clutch buffer adding time 2	DM20	AS clutch buffer adding time 2
(8)	DM18	AH clutch buffer adding time 3	DM21	AS clutch buffer adding time 3
(9)	DM22	Proportional solenoid valve output when the clutch is engaged.		
(10)	DM23	Proportional solenoid valve output directly after settling of engaged clutch		
(11)	DM24	Proportional solenoid valve output transition from handle neutral to clutch operating valve neutral		

Proportional solenoid valve output time setting

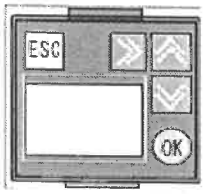
(2)	DM26	AH primary jump up time	DM32	AS primary jump up time
(3)	DM27	AH secondary jump up time	DM33	AS secondary jump up time
(4)	DM28	AH jump up output delay = overshoot prevention operating time	DM34	AS jump up output delay = overshoot prevention operating time
(5)	DM40	Slip compensate time at clutch engaging operation		
(6)	DM29	AH clutch buffered (1) adding time	DM35	AS clutch buffered (1) adding time
	DM41	Clutch engaging command compensate time		
(7)	DM30	AH clutch buffered (2) adding time	DM36	AS clutch buffered (2) adding time
(8)	DM31	AH clutch buffered (3) adding time	DM37	AS clutch buffered (3) adding time
(9)	DM42	Engaged settling operating time		

The DM setting list of the operation flow propeller revolution speed descriptions (1) to (5)

(1)	DM174	AH primary jump up release revolution speed	DM172	AS primary jump up release revolution speed
(2)	DM176	AH secondary jump up release revolution speed	DM170	AS secondary jump up release revolution speed
(3)	DM157	Slip command AH lower limit / one point command	DM156	Slip command AS lower limit / one point command
(4)	DM152	Slip command AH upper limit	DM134	Slip command AS upper limit
(5)	DM184	Revolution (possible AH clutch engaging)	DM162	Revolution (possible AS clutch engaging)

7. Changing DM Setting Values

PLC setting board



The DM setting value can be changed by operating the setting board of the PLC.

You need to carry out two steps to change the setting.

1. Change the DM setting value.

2. Transfer the data to the memory cassette.

Note

1. Transfer the data before turning off power.

If you do not transfer the data to the memory cassette after you change the DM setting value, the settings you have changed will become invalid when the power is turned off.
The values will return to the former setting values.
After changing the DM setting value, the data needs to be transferred to the memory cassette before the power is turned off.

2. Data should be transferred when the clutch is not in use or when the engine is not in operation.

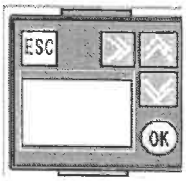
You need to change the mode of the PLC to the program mode when you transfer the changed settings to the memory cassette.
When the program mode is on, the PLC is not in operation and is not controllable.
Data should be transferred to the memory cassette when the clutch is not in use or when the engine is not in operation.
*Changing the DM setting value does not stop the PLC. The DM settings can be changed even when the clutch and engine are in use or in operation.

7-1. How to Change the DM Setting Values

The DM setting value can be changed by operating the setting board of the PLC.

Example of change: To change the value of DM00004 AH primary jump up output (set) from "6000" to "6500".

PLC setting board



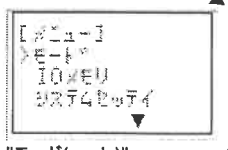
Initial screen of the PLC setting board

"メニュー(menu)" screen

Release the security setting and go to the menu screen.

On the initial screen or the Ver. display screen, press "ESC" and "OK" at the same time.

The security setting is released and the "メニュー(menu)" screen appears.

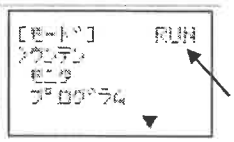


"モード(mode)" screen

Go to the "モード(mode)" screen in the menu.

Press "∧" and "∨". Move the "›" sign to "モード(mode)" and press "OK".

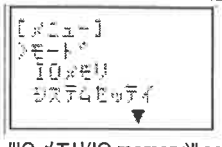
The "モード(mode)" screen appears.



"メニュー(menu)" screen

Switch the mode to the "モニタ(monitor)" mode.

Press "∧" and "∨"and move the "›" sign to "モニタ(monitor)" and press "OK". The upper right corner changes from "RUN" to "MON", indicating that the monitor has changed into the "モニタ(monitor)" mode. Press "ESC" to go back to the menu.



"IO メモリ(IO memory)" screen

The "IO メモリ" screen in the menu appears.

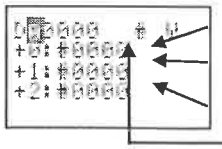
Press "∧" and "∨". Move the "›" sign to "IO メモリ(IO memory)" and press "OK".

The "IO メモリ(IO memory)" screen appears.

DM screen

The "DM" screen in the "IO メモリ(IO memory)" appears.

Press "∧" and "∨". Move the "›" sign to "DM" and press "OK". The DM screen appears.



rent value DM00000

rent value DM00001

rent value DM00002

Change the DM number to "D00004".

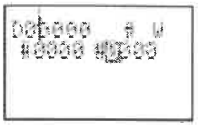
Use "≫" to move the blinking cursor to the row you want to change.

Press "∧" and "∨" and move to "D00004".

Change the data type from "#" to "+".

Use "≫" to move the blinking cursor to "#". Press "∧" and "∨" to change it to "+".

Setting value changing screen



Change the current setting value of "6000" to "6500".

Press "OK" while the display is showing "D00004 + W". The setting value changing screen appears. Use "≫" to move the blinking cursor to the row you want to change. Use "∧" and "∨" to change the setting value to "6500". Press "OK" and the setting value will change.

*If you want to cancel the setting value change, do not press "OK" but press "ESC" to escape from the change screen and go back to the DM screen. If you do not press "OK" after you enter a value to change the setting, the setting will not change.

After you finish all the settings and adjustments, press the ESC screen and escape from the menu screen to go back to the mode screen.
Change the mode from "モニタ(monitor)" to "ウンテン(run)". The process is complete after you go back to the initial screen from the ESC screen.

7-2. How to Transfer Data to a Memory Cassette

After you change the DM setting, it is necessary to transfer the data to the memory cassette before the PLC **power source turns off.**
If this is not conducted, the changed DM setting will not be reflected and the former setting value will be used when you turn on the power next time.

1. Switch to the program mode.

(The PLC does not work in the program mode. Follow these steps while the clutch is not being used and **the engine is not in operation. A PLC control device error alarm is also issued.**)

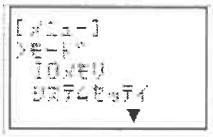
Initial screen of the PLC setting board

"メニュー(menu)" screen

Release the security setting and go to the "メニュー(menu)" screen.

On the initial screen or the Ver. display screen, press "ESC" and "OK" at the same time.

The security setting is released and the "メニュー(menu)" screen appears.

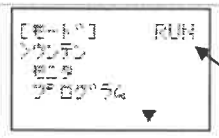


"モード(mode)" screen

Go to the "モード(mode)" screen in the menu.

Press "へ" and "﹀". Move the "›" sign to "モード(mode)" and press "OK".

The "モード(mode)" screen appears.



Switch the mode to the program mode.

Press "へ" and "﹀". Move the "›" sign to "プログラム(program)" and press "OK".

The upper right corner changes from "RUN" to "PRG". The mode changes into the "プログラム(program)" mode. Press ESC and go back to the menu screen.

2. Transfer the data to the memory cassette.

"メニュー(menu)" screen

"メモリカセット(memory cassette)" screen

The "メモリカセット(memory cassette)" screen in the menu appears.

Press "へ" and "﹀". Move the "›" sign to "メモリカセット" and press "OK".

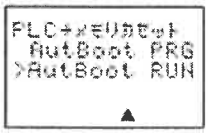
The "メモリカセット" screen appears.

Mode selection screen after data transfer

Select "PLC → メモリカセット(memory cassette)" in the memory cassette.

Press "へ" and "﹀". Move the "›" sign to "PLC → メモリカセット" and press "OK".

The mode selection screen after data transfer appears.



Confirmation screen

Select "AutBootRUN" in the mode after transfer.

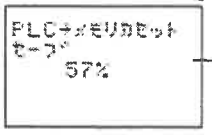
Press "へ" and "﹀". Move the "›" sign to "AutBootRUN" and press "OK".

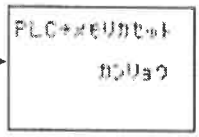
The confirmation screen appears.

To confirm transfer, select "OK".

Press "へ" and "﹀". Move the "›" sign to "OK" and press "OK".

Data will start being transferred from the PLC to the memory cassette.

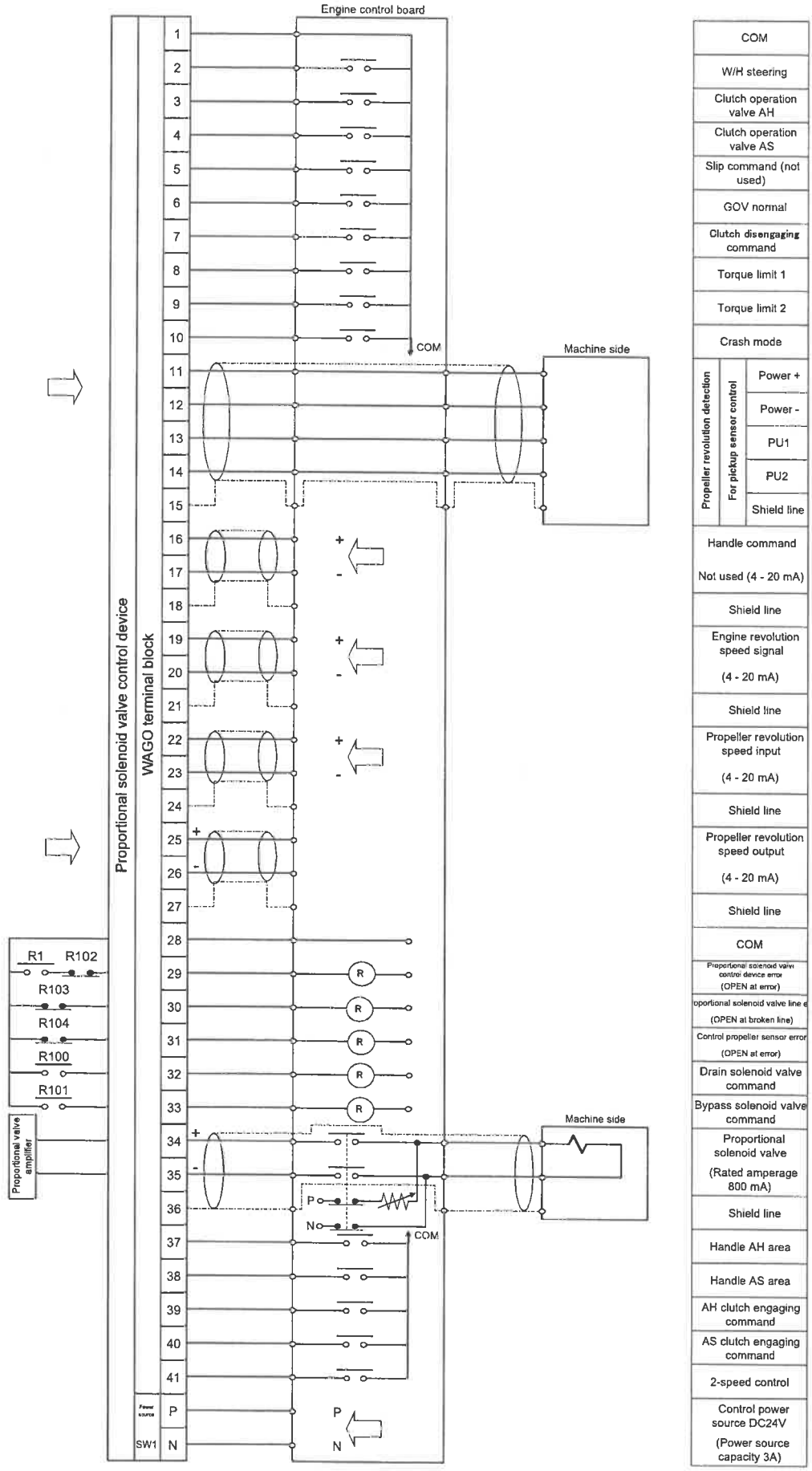




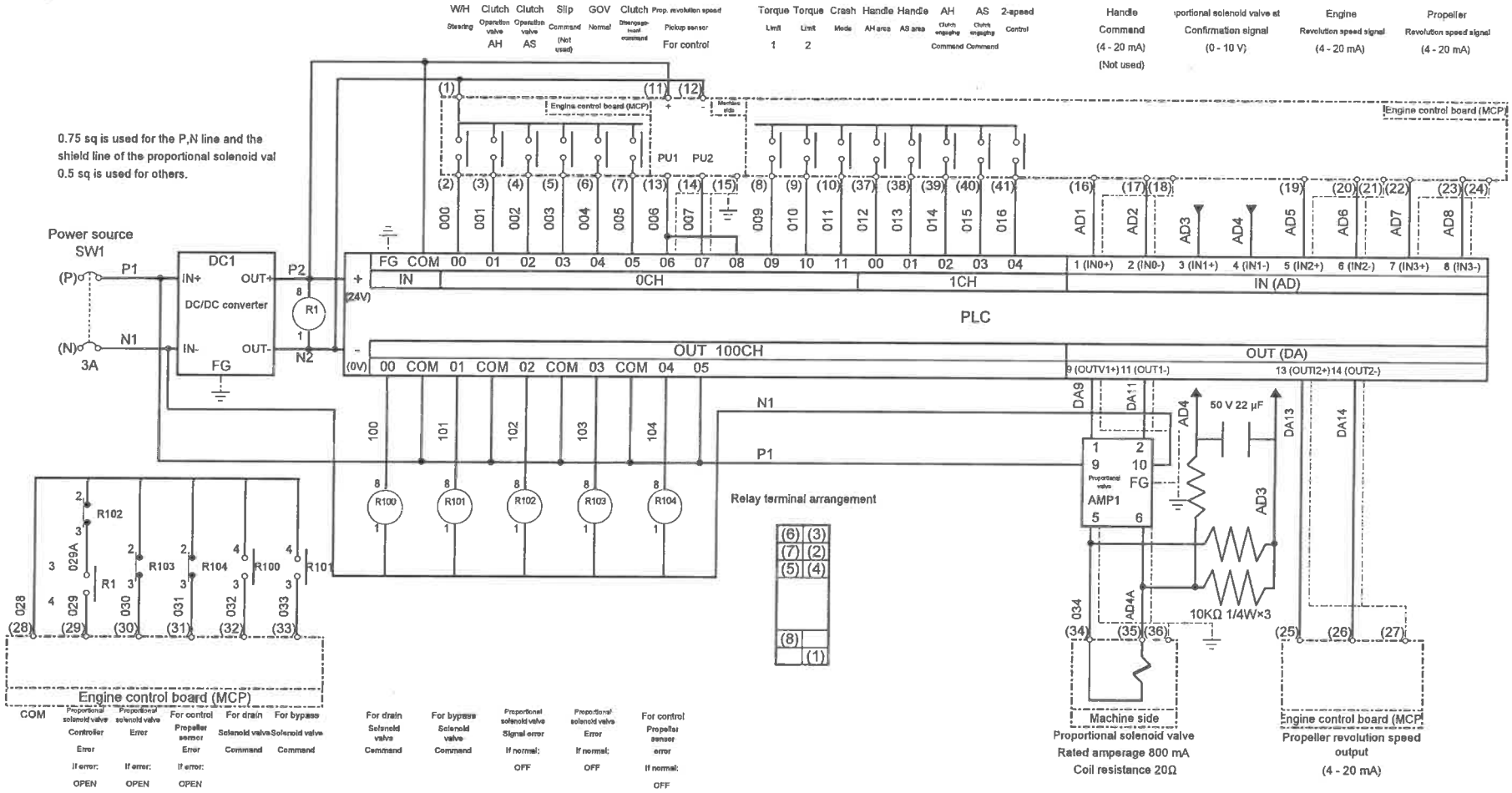
When the PLC finishes transferring the data to the memory cassette, the message "カンリョウ(finished)" will be displayed to indicate that the transfer has ended.

After the transfer ends, turn off the power or switch from the program mode to the operation mode.

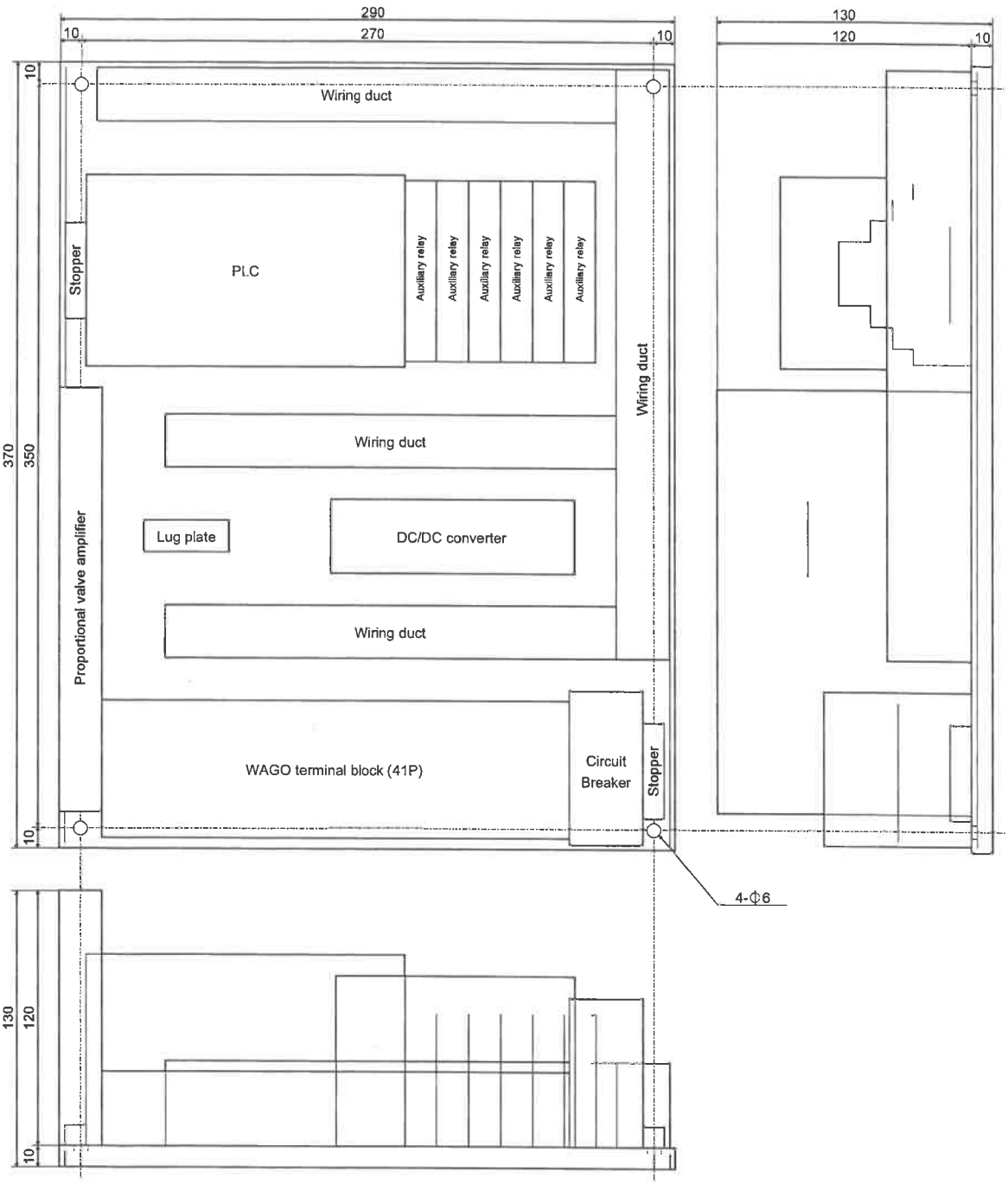
8. Terminal Connection Chart of Proportional solenoid Valve Control Device



9. Proportional Solenoid Valve Controller Circuit Chart



10. External View of the Proportional Solenoid Valve Control Panel

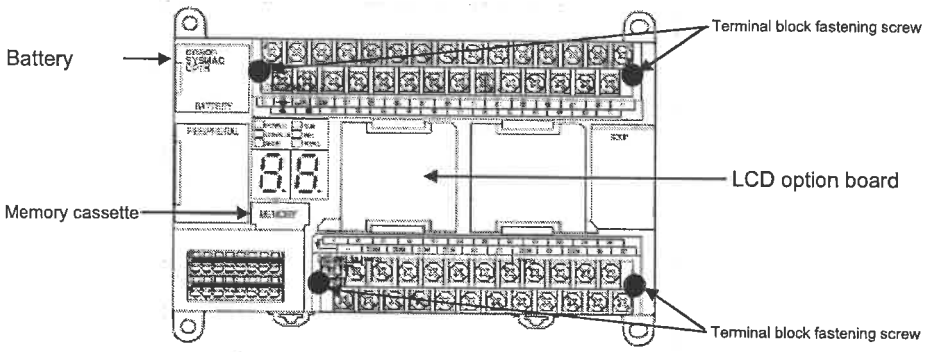


11. How to Replace the PLC Main Body and to Transfer Data by Replacing the Memory

1 Turn off the power of the proportional solenoid valve controller.

2 Remove the wiring from the PLC that will be replaced.

If you will keep using the same terminal block and the connector, remove the terminal block fastening screws and remove the entire terminal block.

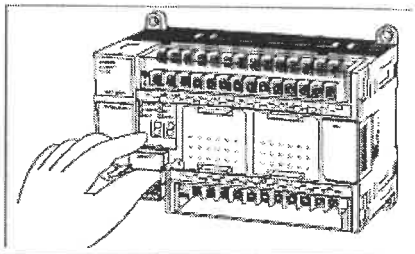


3 Remove the memory cassette and the LCD option board from the PLC that will be replaced. Mount them on the new PLC.

Memory cassette

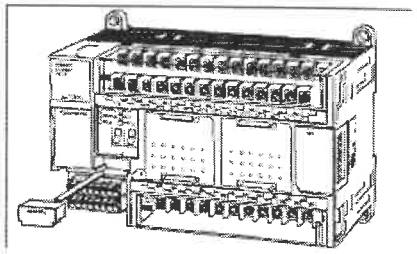
Removal

Place your thumbnail on the protruding part in the lower part of the upper surface of the memory cassette.
Support the memory cassette with other fingers and move it upwards to remove it.



Mounting

Hold the memory cassette with the name label side facing upwards.
Push it straight into the slot until it is fully inserted.



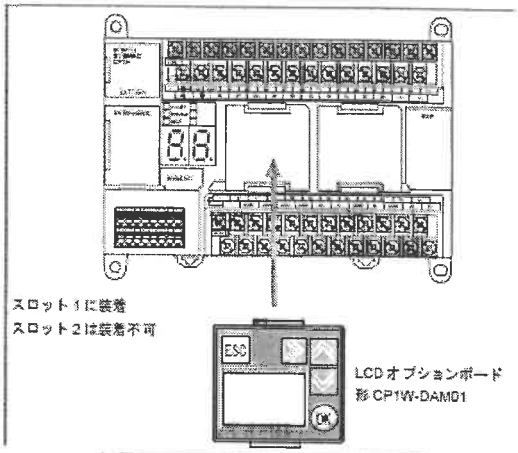
LCD option board

Removal

Hold the upper and lower lock levers on the option board at the same time.
The lock is released and you can pull out the board.

Mounting

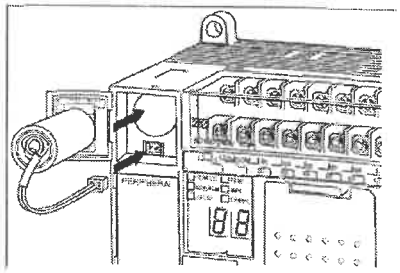
Remove the cover from option slot 1.
Make sure that the board is facing the correct direction for mounting. Push the board in firmly until you hear a click.



4 Remove the battery from the new PLC.

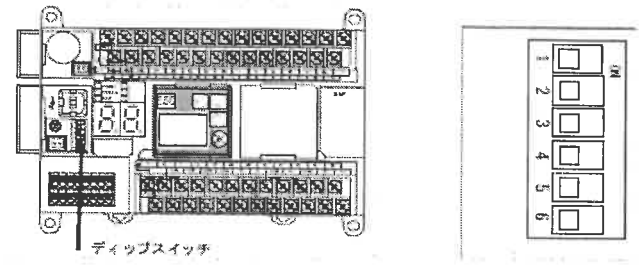
As the machine is supposed to operate without batteries, please remove if your new PLC has a battery.

Open the cover and remove and pull out the battery connector.

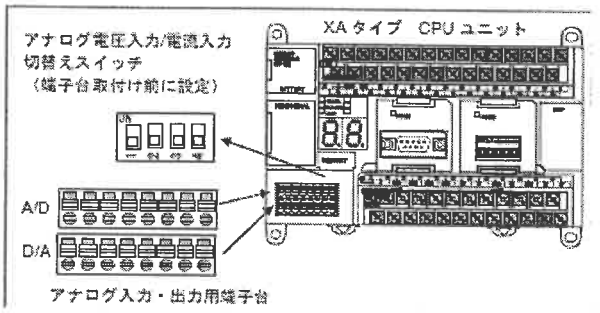


5 Set up the dipswitch of the new PLC.

Turn on SW2 and SW4 on the dipswitch as shown below.



Turn on SW1, SW3, and SW4 on the dipswitch as shown below.



6 Remove the old PLC from the panel, and mount the new PLC.

7 Connect the wires.

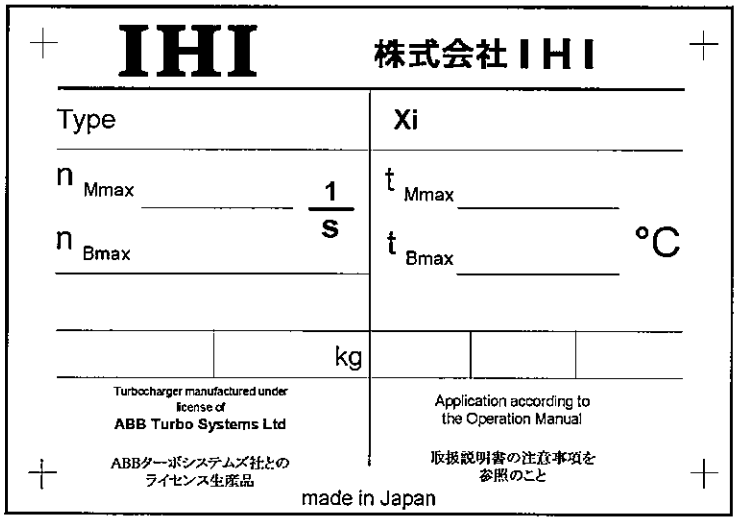
8 Turn on the power of the proportional solenoid valve controller.

When you turn on the power, each of the data on the old PLC is automatically transferred to the new PLC from the memory cassette.

Turbocharger
過給機

Operation Manual
取扱説明書

TPS 48 - 61F



IHI Corporation

IHI

NV50FPB-1.R2

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1

Preliminary remarks

予備知識

1 Preliminary remarks

1 予備知識

1.1 Purpose of the manual

1.1 取扱説明書について

Operating manual

This operation manual will help you to become familiar with your turbocharger and to use it to full effect in its intended application. You will find important instructions as to the safe, correct, and economical operation of the turbocharger.

取扱説明書

この取扱説明書は、過給機をよく理解していただき、それぞれのアプリケーションの中で最大限活用していただくためのものです。本書には安全で正しい過給機の使用方法および経済的な運用方法のための重要な事項が記述されています。

Target group

The content of this operation manual is intended for persons such as engineers, marine mechanics and service technicians who are responsible for the running of the engine and the turbocharger installed thereon.

本書の対象者

取扱説明書の内容は、エンジンならびにエンジンに搭載された過給機の運転に責任を持つエンジニア、すなわち船舶技術者およびサービス要員の方を対象にしています。

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Content of operating manual

This operation manual includes helpful information on the following topics:

- Information on the operation of the turbocharger and ways to prolong the turbocharger's service life.
- Early identification and avoidance of hazards.
- Minimizing repair costs and downtime.

取扱説明書の内容

取扱説明書には下記事項について役立つ情報も含まれています。

- 過給機の運転に関する情報およびサービス間隔を延長する方法
- 不具合の早期発見とその回避方法
- 修理費用および故障による休止時間を抑えるための方法

This operating manual complements and extends existing national regulations on protection against accidents and accident prevention.

本書は事故の予防および防止に関する各国の法規および関連規則に対応しています。

NOTE

An operation manual must be available at all times at the site where the turbocharger is used. It is essential that this operation manual is read before beginning work by all persons working with or on the turbocharger. Anyone who works only occasionally on the machine, e.g. during installation and maintenance, must also read and fully understand the operating manual before beginning work.

注記

過給機が使用される場所に本書を少なくとも1冊常備し、使用できるようにしてください。過給機を取扱う全ての人が必ず事前に本書を熟読し、本書の内容を完全に理解してください。また一時的な作業者（艤装、メインテナンスの場合等）であっても同様に作業前に本書を理解してください。

The instructions included in this operating manual must be followed without fail.

本書に記載されている全ての指示には確実に従ってください。

WARNING

This applies especially to the general and special safety instructions preceding the respective chapters and in them. Ignoring hazards and the safety instructions can lead to serious personal injury and damage to equipment.

警告

特にそれぞれの章の中の安全に対する一般的および特別な指示を確実に守ってください。

危険と安全に関する指示を無視した場合は、重大な人身事故および装置の損傷につながることがあります。

If in doubt, consult the safety officer responsible and the person accountable for protection against accidents in your domain.

疑問のある場合、プラントの安全および事故防止の管理責任者に相談してください。

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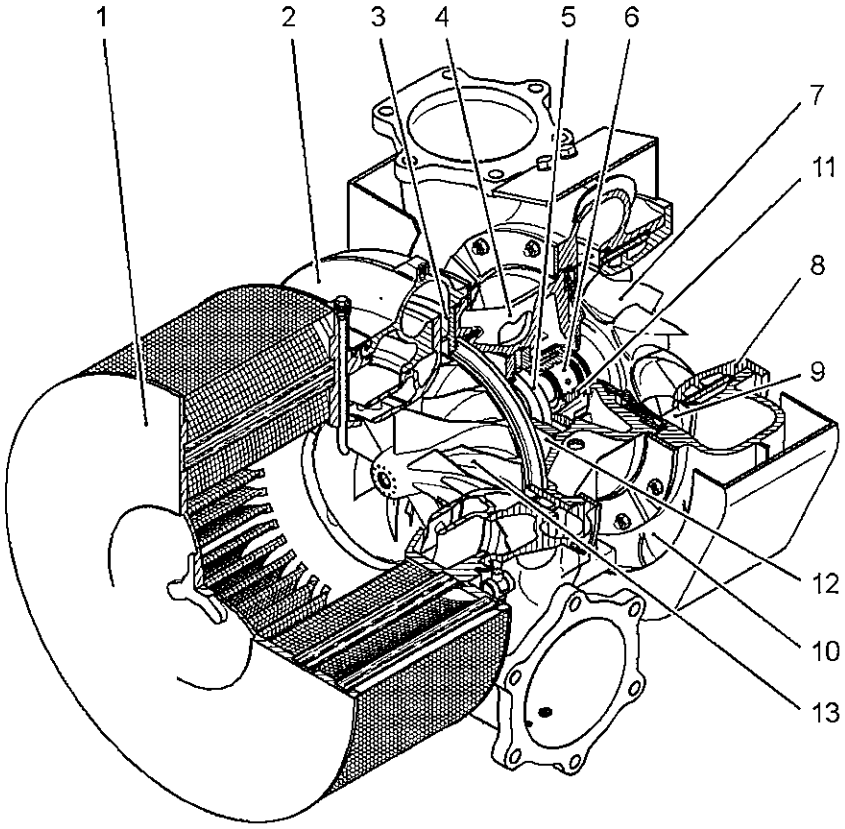
1.2 Layout and function 1.2 構造と機能



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1	Suction branch / Filter silencer 空気吸込囲 / 消音器	8	Gas outlet flange ガス出口フランジ
2	Compressor casing コンプレッサケーシング	9	Nozzle ring ノズルリング
3	Diffuser ディフューザ	10	Turbine casing タービンケーシング
4	Bearing casing 軸受ケーシング	11	Turbine-end bering flange タービン側軸受フランジ
5	Axial thrust bearing スラスト軸受	12	Compressor-end bearing flange コンプレッサ側軸受フランジ
6	Radial plain bearing(only visible turbine-end) ラジアル軸受 (タービン側)	13	Compressor wheel コンプレッサ
7	Turbine タービン		

Working principle

The turbocharger is a turbo-machine consisting of the following main components:

- Turbine
- Compressor

These are mounted on a common shaft.

Exhaust gases from the diesel or gas engine flow through the turbine casing (10) and nozzle ring (9) to impinge on the turbine (7).

The turbine (7) uses the energy contained in the exhaust gas to drive the compressor wheel (13). The compressor draws in fresh air, compresses it and then forces it into the engine's cylinders.

The exhaust gases escape through the exhaustgas duct which is connected to the gas outlet flange (8).

The air, which is necessary for operation of the diesel or gas engine and compressed in the turbocharger, is drawn into the compressor wheel (13) through the suction branch or filter silencer (1).It then passes through the diffuser (3) and leaves the turbocharger through the outlet on the compressor casing (2).

The rotor runs in two radial plain bearings (6), which are located in bearing flanges (11/12) between the compressor and the turbine. The axial thrust bearing (5) is located between the two radial plain bearings.

The bearings are connected to a central lubricating oil feed which is supplied by the engine's oil lubrication circuit. The oil outlet is always at the lowest point of the bearing casing (4).

機能の原理

過給機はターボ機械であり、下記の主要な要素(部品)から構成されています。

- タービン
- コンプレッサ

この 2 つの要素は同一軸上に配置されています。

ディーゼルまたはガスエンジンより出た排気ガスは、タービンケーシング(10)およびノズルリング(9)を通りタービン(7)に導かれます。

タービン(7)はコンプレッサ(13)を駆動するために排気ガスに含まれるエネルギーを使用します。それによってコンプレッサは新鮮な空気を吸込み、圧縮された空気をシリンダの中へ送り込みます。

タービンを通過した排気ガスはガス出口フランジ(8)に接続された排気管へ排出されます。

ディーゼルまたはガスエンジンが必要をする空気は、空気吸込囲または消音器(1)を通り、コンプレッサ(13)へ導かれ過給機の中で圧縮されます、その後空気はディフューザ(3)を抜け、コンプレッサケーシング(2)の外側を通って過給機より排出されます。

ロータはコンプレッサとタービンの間にある軸受フランジ(11/12)に配置された 2 つのラジアル軸受(6)により支持され回転します。スラスト軸受(5)は 2 つのラジアル軸受の間にあります。

軸受部は、エンジンの潤滑システムから潤滑油が供給される過給機中央の給油部につながっています。潤滑油は常に軸受ケーシング(4)の最下部から排出されます。

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1.3 Correct use of the turbocharger

NOTE

This turbocharger has been developed for use on diesel engines to generate the volume of air and the charging pressure required to operate the engine.

The engine manufacturer has provided ABB Turbo Systems or Turbo Systems United Co., Ltd with information regarding the intended purpose of the engine, from which the operating parameters specific to the turbocharger (such as rotating speeds, temperatures, replacement periods / exchange intervals) have been derived and are indicated on the rating plate.

If it is **used** in conjunction **with a gas engine,** the engine must **not** be installed in a potentially **explosive environment,** and precautionary measures must be taken to ensure that the machine room as a whole is classified as **not potentially explosive.**

Any other use will be regarded as a special application which must first be discussed with ABB Turbo Systems or Turbo Systems United Co., Ltd. The manufacturer accepts no liability for other applications. If it is used otherwise, we reserves the right to reject all claims under the warranty.

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Latest technology

This turbocharger was built according to state of the art technology and is operationally safe according to prevailing safety regulations.

1.3 過給機の正しい使用方法

注記

この過給機は、ディーゼルエンジン用としてその運転に必要とされる空気量と給気圧力を供給するために開発されたものです。

エンジン製造会社から提示されたエンジン用途に関する情報を基に規定された過給機の許容限界は、銘板上に記載されています。(回転数、温度、部品交換時間)

ガスエンジンに搭載する場合、エンジンは**爆発・引火の危険性がある環境**に設置することはできません。危険予防の対策として**防爆処置**を施したエンジン室を確保してください。

所定外の使用方法に関してはABB 社または TSU 社と十分協議してください。協議がない使用方法については責任を負うことはできません。もし誤った方法で使用された場合、保障期間内であっても保証できかねますので予めご確認ください。

製品の水準

この過給機は最新の技術と安全規則に沿って製造されており、運転取扱い上安全です。

WARNING

Improper operation and maintenance of the turbocharger can result in danger to life and limb of the user or third parties. In addition, improper use may cause damage to the machine. The machine may only be operated by trained personnel.

警告

過給機の不適切な運転やメインテナンスを行った場合は、使用者或いは第三者に対して重大または致命的な人身事故を引き起こすことがあります。加えて不適切な取扱いは、エンジンや過給器に損傷をもたらすことがあります。
訓練を受けた作業者だけが取扱を行ってください。

Correct use of the turbocharger also includes observation of the assembly, disassembly, operating, maintenance and repair conditions specified by the manufacturer. Disposal regulations set down by local authorities must be observed.

また過給機を正しく使用するため、指示された取付け・取外し・運転・メインテナンスおよび修理の条件を守ってください。過給機の廃棄処理は、当該管轄官庁の規定に従ってください。

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Perfect condition

万全な状態

The turbocharger must only be installed in a technically perfect condition, corresponding to its intended purpose and following the instructions in the engine manufacturer's manual, and must be operated and used in compliance with the operation manual.

過給機はその機能を果たすために、エンジン製造会社の取扱説明書に従って設置し、取扱説明書の記述内容を遵守して操作し、使用してください。

▶ Malfunctions / defects which could affect safety must be eliminated immediately.

▶ 安全に影響する機能不良や不具合は直ちに取除いてください。

The manufacturer shall not accept liability for damages resulting from unauthorised alterations to the turbocharger.

過給機を許可なく改造したり、または不適切な使用により生じた損害の責任は負いかねます。

1.4 Essential information

Organizational measures

In addition to this manual, the general, statutory regulations applicable in the respective country for the prevention of accidents and the protection of the environment must be observed.

This also applies to the provision and wearing of personnel protection equipment.

NOTE

The safety and risk consciousness of the personnel working on and with the turbocharger must be checked regularly with reference to this manual.

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▶ The turbocharger must be shut down immediately by stopping the engine in the event of alterations affecting safety or of corresponding operating behaviour. The fault must be reported to the person or department responsible.

NOTE

Additions and alterations to, and conversions of the turbocharger that could compromise safety require the prior approval of ABB Turbo Systems Ltd or Turbo Systems United Co., Ltd.

Original parts and safety

Original parts and accessories are especially designed for the turbocharger supplied by ABB Turbo Systems or Turbo Systems United Co., Ltd.

1.4 基本情報

組織的方策

この取扱説明書の他に、事故防止や環境保護のために各国の法令で定められた規則が適用されますので、従ってください。

また保護具の準備および着用も規定の中に定められています。

注記

過給機を扱う作業者の安全と危険意識を取扱説明書を参照し定期的にチェックしてください。

▶ 安全に影響する不具合または運転状態の変化が確認された場合は、エンジン停止により直ちに過給機を停止させてください。不具合は責任者または責任部門へ報告してください。

注記

またこの他に、安全を損なう可能性がある過給機の改造や変更については ABB 社または TSU 社の事前承認が必要になります。

純正部品および安全性

純正部品は、過給機専用に設計され ABB 社または TSU 社より供給されます。

NOTE

We herewith expressly state that parts and accessories not supplied by us have also not been tasted and approved by us.

注記

ABB 社ならびに TSU 社以外が供給した部品は、ABB 社によるテストおよび承認が成されておりません。

WARNING

If components and assemblies not manufactured and tested by ABB Turbo Systems are used, this may have a negative effect on the technical design specifications of the turbocharger.

Moreover, the active and/or passive safety of the turbocharger may be compromised by the use of non-original parts. Note that such improper use may endanger personnel and the machine.

警告

ABB 社によりテスト・製造されていない部品を使用した場合、過給機の機能に有害な影響を与えることがあります。

また、専用に設計されていない部品を使用することで、直接または間接的に過給機の安全を損なうことがあります。純正部品以外を使用した場合、人的・物的危険が生じることがありますのでご注意願います。

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ABB Turbo Systems and Turbo Systems United Co., Ltd shall not be liable for any damage caused by using non-original parts and accessories.

ABB 社および TSU 社は純正部品や付属品でないものを使用した場合の損害に対しては一切責任を負いかねます。

Qualification of personnel

作業者の資格

The turbocharger must only be operated and maintained by trained and authorised personnel.

過給機の取扱いおよびメインテナンスは訓練を受けた資格のある作業者だけが行ってください。

CAUTION

Work on mechanical components, such as bearings or rotors, may only be performed by qualified fitters from an official service station of ABB Turbo Systems.

注意

軸受またはロータのような構成部品の作業は、ABB 社サービスステーションの認定作業者のみが行うことができます。

It must be ensured that only authorised personnel work on the turbocharger.

必ず認定作業者のみが過給機を取扱ってください。

Design variants

This operating manual is valid for various turbocharger design variants.

This means that some sections and component descriptions in this operating manual do not apply to your turbocharger.

Our service stations and agencies will gladly answer your questions relating specifically to your model.

過給機のモデル

この取扱説明書は、オプションなどを含め過給機全般について記述しています。

このため、お求めいただいた過給機に対応していない記述もあります。

過給機に関するご質問につきましては ABB 社認定のサービスステーションおよび代理店にお問い合わせください。

Accuracy of illustrations

The graphics and images shown in this manual are of a general nature and are intended to assist understanding of the work steps and instructions. Differences may exist in certain details.

イラストの正確さ

本書のイラストは作業ステップや指示を理解しやすくするためのものです。細部では相違する点があります。

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Storage of new turbochargers

New turbochargers supplied by ABB Turbo Systems or Turbo Systems United Co., Ltd can be stored without additional safeguarding measures for a period of 6 months after the delivery date.

Suitable storage locations are dry rooms where the relative humidity is between 40-70% and no condensation can form.

- ▶ After expiry of the 6 months, all surfaces and accessible areas of the turbocharger should be sprayed with anticorrosive agent.
- ▶ To do this, the insulating materials must be removed.

NOTE

Repeat the described measures every 6 months to keep your turbocharger free from rust.

新しい過給機の保管

ABB 社または TSU 社から供給された過給機は、出荷から 6 ヶ月間は特別な処置がなくても保管することができます。

相対湿度 40～70%で結露の発生しない乾燥した場所に保管してください。

- ▶ 6ヵ月後、表面および全ての施工可能な箇所に防錆油をスプレーしてください。
- ▶ この作業をするときは、インシュレーションを取外してください。

注記

また 6 ヶ月ごとに、過給機に防錆処置を繰り返し行ってください。

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1.5 Abbreviations and symbols

The following abbreviations are used in this document:

▷ Prerequisite
▶ Work step
⇨ Result
✓ Positive checkpoint (activity that must be carried out without fail)
✕ Negative checkpoint (activity that must not be carried out under any circumstances)

1.5 略号とシンボルマーク

下記の略号が本書には使われています:

▷ 前提条件
▶ 作業ステップ
⇨ 結果
✓ 正のチェックポイント(間違いなく実施されなくてはならない事項)
✕ 負のチェックポイント(どのような状況下でも実施してはならない事項)

NV50F01-1.R1

Basic Instructions

The following symbol is used for notes in this manual:

NOTE

This note provides suggestions and tips to make work on the turbocharger easier.
The note provides further essential information.

CAUTION

Reference to safety instructions
The warnings and cautions are described in the chapter **Safety**.

基本的注意事項

本取扱説明書では本書中の注記箇所に下記のシンボルマークが使われています。

注記

過給機の取扱いを容易にするための提案および注意事項
基本的な情報

注意

安全指示について
警告と注意事項は「**安全**」の章で述べられています。

ABB Turbo Systems

ABB Turbo Systems Ltd is specified in this manual as **ABB Turbo Systems.**

TSU

Turbo Systems United Co., Ltd is specified in this manual as **TSU.**

Official ABB Turbo Systems service station

The term **Official ABB Turbo Systems service station** used in this manual signifies ABB Turbocharger service stations, that have been tested and certified by ABB Turbo Systems.

ABB

本書の中で ABB とは ABB ターボシステムズ社のことです。

TSU

本書の中で TSU とはターボシステムズユナイテッド社のことです。

ABB 社認定サービスステーション

本書で言及される ABB 社認定サービスステーションとは、ABB 社によって確認・認証をうけた ABB 社製過給機に関する認定サービスステーションです。

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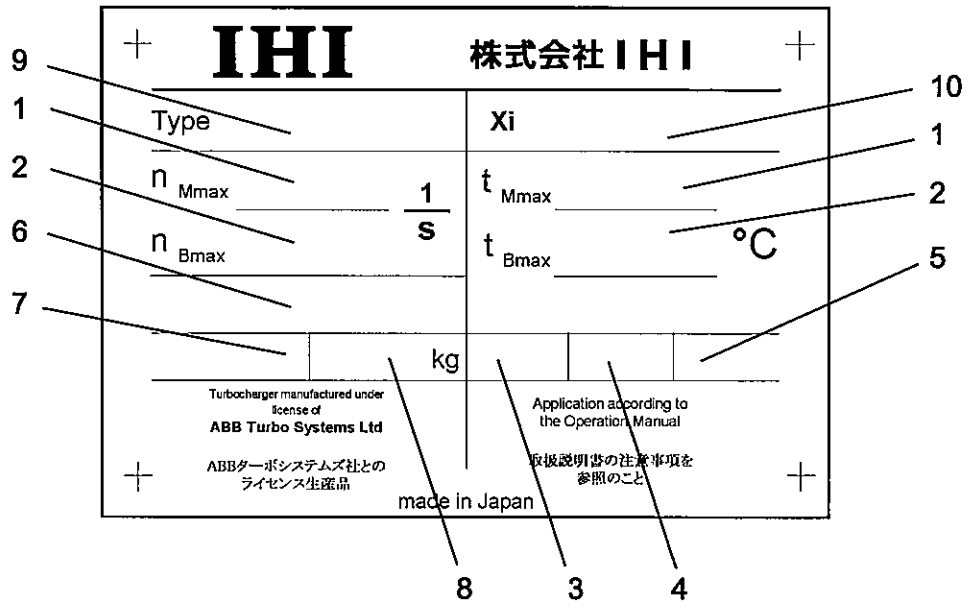
1.6 Turbocharger rating plate

1.6 過給機の銘板



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Operational limits

運転限界

1 Turbocharger operational limits at engine overload (110%). In test rig operation only, unless otherwise agreed with the engine manufacturer.

2 Turbocharger operational limits in service.

1 エンジン過負荷時(110%) の最高許容回転数およびガス温度
特に定められない限り、試験運転時のみ使用可能です。

2 エンジン常用負荷時の最高許容回転数およびガス温度

Recommended inspection / replacement intervals for turbocharger components

過給機部品の推奨交換時間

3 Inspection interval for plain bearings in 1000 h

4 Replacement interval for compressor in 1000 h

5 Replacement interval for turbine in 1000 h

3 平軸受点検間隔(x1000 時間)

4 コンプレッサ交換時間(x1000 時間)

5 タービン交換時間(x1000 時間)

Additional specifications:

追加事項：

6	Customer part number	6	客先部品番号
7	Field for special design designation	7	特殊仕様の識別記号
8	Mass of the turbocharger in kg	8	過給機の質量(単位:kg)
9	Turbocharger type	9	過給機形式
10	Series number	10	機械番号

Explanations of the rating plate

The conditions agreed with the engine manufacturer apply to the values indicated on the rating plate with regard to operational limits, inspection intervals and replacement intervals.

CAUTION

Operation at values higher than those given n_{Bmax}, t_{Bmax} might considerably reduce the recommended replacement intervals. In such cases, we recommend that you contact the nearest official service station of ABB Turbo Systems.

n_{Mmax}, t_{Mmax} normally apply only when running at overload (110%) during trials on the engine test bed. These limits can also be permitted during contiuous operation for special applications. Operation above n_{Mmax}, t_{Mmax} is not permitted.

Non-compliance with the recommended replacement intervals might increase the risk of unpredictable component failures.

過給機の銘板表示に関する説明

運転限界、点検間隔および交換時間に関する銘板上の値は、エンジン製造会社の同意している条件を基に刻印されています。

注意

n_{Bmax}, t_{Bmax} を超えた状態で運転した場合は、推奨交換時間が大幅に短縮します。そのような運転が確認された場合は、ABB 社認定のサービスステーションにご相談ください。

n_{Mmax}, t_{Mmax} の数値は通常、エンジンテストスタンドにおける過負荷(110%) 試運転時のみに適用されます。特殊な用途には上記数値での継続運転が許容されることがありますが、いかなる場合も n_{Mmax}, t_{Mmax} を超えることは認められません。

推奨交換時間を遵守しない場合、予想不可能な部品損傷が生じる危険性を高めます。

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The inspection intervals and replacement intervals for the turbocharger components specified must be observed without fail!

指定されている過給機部品の点検間隔および推奨交換時間は、確実に守ってください。

Turbocharger components whose permissible operating times have expired may not be used again under any circumstances!

交換時間の過ぎた部品は、どんな事情であれ再利用することはできません。

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2

Safety

安全

2 Safety

2.1 Basic safety instructions

The following symbols and terms are used for the safety instructions in this manual:

WARNING

Failure to follow work steps and operating instructions marked with this symbol completely and rigorously may result in serious personal injury or even fatal accidents.

Warning signs must be strictly observed at all times.

CAUTION

Serious damage to the machine or to other property may result if the working and operation instructions marked with this symbol are not, or only partly, observed.

Caution signs must be strictly observed at all times.

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2 安全

2.1 安全に関する基本的指示

下記のシンボルマークおよび用語は、本書中の安全指示箇所に使われています。

警告

このシンボルマークの付いている運転・取扱上の指示に従わない場合または部分的にのみ従う場合は、重大または致命的な人身事故につながることがあります。

いかなる場合も警告事項は厳守してください。

注意

このシンボルマークの付いている運転・取扱上の指示に従わない場合または部分的にのみ従う場合は、機械およびその他の機器に重大な損害が発生する可能性があります。

いかなる場合も注意事項は厳守してください。

2.2 General safety instructions

The instructions listed here are designed to ensure your safety and to promote safe handling of the turbocharger and its associated operating materials.

WARNING

The following safety instructions must be followed when operating and when working on the turbocharger.

- Do not work on the turbocharger if you are under physical or mental stress.
- Wear protective clothing at all times.
  - Helmet
  - Ear protection
  - Protective glasses
  - Protective suit
  - Oil and petrol resistant gloves
  - Protective shoes
- Attach individual parts and larger component modules carefully to suitable hoists/lifting devices which are in technically perfect condition and which have adequate load-bearing capacity.
- Do not stand under suspended loads.
- If welding work is to be carried out above the turbocharger, the air filter must be covered to prevent damage to the filter mat.
- When working on the turbocharger, be aware of the risk of it falling.
- For overhead assembly work, use the steps and platforms provided for this purpose.

2.2 安全に関する一般的指示

ここに挙げられている指示は使用者の安全を守ると同時に、過給機およびその関連機器を安全に取扱う上で役立ちます。

警告

過給機を運転または取扱う際は下記に示す安全指示に必ず従ってください。

- 心身共に安定した状態のときにのみ、過給機を取扱ってください。
- 常時保護具を着用してください。
  - ヘルメット
  - 耳栓
  - 保護めがね
  - 作業服
  - 耐油/耐ガソリンの手袋
  - 安全靴
- 個々の部品および大きな組立品を取扱う際は、欠陥がなく、所要吊上げ荷重能力を持ったホイスト/吊上要具を使用してください。
- 吊上げた物体の下には立たないでください。
- 溶接作業を過給機上で行う場合は、吸込フィルタの損傷を防ぐために覆いをしてください。
- 過給機上で作業する際には、落下の危険性に十分配慮してください。
- 高い位置での取付け作業時には専用の踏み台を使用してください。

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2.3 Special safety instructions

Safety when connecting

- Work only on those connections for which you are qualified.

Safety when commissioning and during operation

- Before starting work, make a visual inspection of the working area.
- Remove obstacles and any objects lying around from the work area.
- Before commissioning check the turbocharger for damage and leaks.
- Refrain from any activity which could compromise safety while working with the turbocharger.
- Inspect the turbocharger after about 12 hours of operation or at least once a day for visible damage and defects.
- Report any damage or changes in operational performance to the person/department responsible immediately.
- In the event of damage, shut down the turbocharger immediately and secure it against inadvertent or unauthorized use.

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2.3 安全に関する特別指示

接続作業時の安全

- 過給機取付け作業は資格所有者のみが行ってください。

作業時および運転中の安全

- 作業を開始する前に作業場を目視で点検してください。
- 障害物および不要な物は作業場外へ移動させてください。
- 運転前に過給機に損傷や漏れがないかチェックしてください。
- 過給機の取扱中は安全性を損なわせる行動は慎んでください。
- 約 12 時間運転毎または少なくとも 1 日 1 回、過給機に外部損傷や欠陥がないか点検してください。
- 損傷や性能上の変化があった場合は直ちに担当者または担当部門に報告してください。
- 損傷が発生した場合は過給機を直ちに停止し、不測の事態を生じさせないため適切な処置を実施してください。

Safety when cleaning

When handling detergents, solvents, acids and alkalis be sure to observe the respective safety directions on the manufacturers' labels.

Follow directions regarding the environmentally compatible collection, storage and disposal of these liquids.

WARNING

Explosion hazard

Cleaning work should be carried out outdoors due to explosion hazard. If this is not possible, ensure that there is adequate ventilation and air circulation. Before beginning cleaning work in confined spaces, extinguish naked flames (including cigarettes!) due to the risk of explosion.

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- Before cleaning, protect the floor against unintended leaks of oil and operation materials.
- You must wear protective clothing (see section **General safety instructions**).
- Handle operation materials and detergents with utmost caution.
- When handling solvents, wear protective clothing **and** observe the following rules:
  - Avoid skin contact with solvents.
  - Avoid inhaling vapours under all circumstances.
  - Do not allow solvents to soak protective clothing.
- After cleaning, check electric cables for signs of wear or damage.

清掃作業時の安全

洗浄剤、溶剤、酸およびアルカリ剤を使用する場合は、その製造会社のラベルに表示された安全に関する全ての指示を確実に守ってください。

また、これらの液体の回収、保管および廃棄に関しては環境汚染を防ぐための指示に従ってください。

警告

爆発の危険

清掃作業は爆発のおそれがあるため屋外で行ってください。もし屋外での作業ができない場合は、十分な換気および空気の循環を行ってください。狭い空間で清掃作業を始める前には、爆発の危険があるため火(タバコを含む!)を消してください。

- 潤滑油等の漏れがあっても良いように、清掃前に過給機周りの床を保護してください。
- 必ず保護具を着用してください。(「**安全に関する一般指示**」の項をご参照ください。)
- 潤滑油等・洗浄剤を取扱う際には十分注意してください。
- 溶剤を取扱うときは保護具を着用し、さらに下記の指示事項を厳守してください。
  - 溶剤が皮膚に触れないように注意してください。
  - いかなるときも気化した溶剤を吸引しないようにしてください。
  - 溶剤が作業服に吸収され、染み込まないようにしてください。
- 清掃後、電気ケーブルの損耗または損傷がないかチェックしてください。

Safety during disassembly, assembly, maintenance and troubleshooting

- Follow the prescribed maintenance and inspection schedules for all adjustment, maintenance and inspection work.
- Inform operating personnel about special work and repairs before starting.
- Ensure that absorbent materials are available to clean up any operation materials that escape accidentally.
- Ensure safe and environmentally-friendly drainage, collection and disposal of operating and auxiliary materials.
- Before opening a cover or removing a protective component on the turbocharger, the engine must have been switched off and must not be started up again until all parts have been properly re-assembled.

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CAUTION

Assembly and disassembly work must only be performed by trained personnel. Work on mechanical components, such as bearings or rotors, must only be performed by qualified fitters from an official service station of ABB Turbo Systems.

分解、組立、メインテナンスおよび不具合対処中の安全

- 全ての調整・メインテナンス・点検作業は本書に規定の間隔で実施してください。
- 特別な作業や修理を開始する場合は、事前に運転管理者に知らせてください。
- 潤滑油等の漏れがあっても良いように吸収材を準備してください。
- 潤滑油や洗浄剤等を完全に抜き取って回収し、安全および環境保護にかなった方法で確実に処理してください。
- 過給機のカバーや保護用部品を取外す前に、エンジンを停止させ、すべての部品が適切に組立てられるまで起動しないでください。

注意

組立および分解作業は、訓練を受けた資格所有者のみが行わなければなりません。軸受やロータのような機械構成部品の作業は、ABB 社サービスステーションの認定作業者のみに作業させるようにしてください。

Safety when taking out of operation or putting into storage

- Secure the rotor against turning.
- Clean the turbocharger before mothballing.
- Wear protective clothing when mothballing the turbocharger.
- Keep the work area clean and free of oil and operating materials.
- Remove any obstacles lying around on the floor.

過給機の使用停止または保管時の安全

- ロータを固定し回転しないようにしてください。
- 防錆処置を施す前に過給機を清掃してください。
- 過給機の防錆作業中は保護服を着用してください。
- 作業場付近をきれいにし、油および洗浄剤のない状態にしておいてください。
- 床にある障害物は移動してください。

2.4 Safety instructions and hazard protection

Associated hazards may arise during operation of and work on the turbocharger:

- from the turbocharger and its accessories.
- from the operating materials used.
- due to failure to comply with the safety instructions.
- from inadequate performance of maintenance and inspection work.

Mechanical influences

WARNING

Mechanical influences can cause serious injury or even fatal injury.

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Personal injury

Unconsciousness and injury due to:

- crushing
- shearing
- cutting
- winding
- smashing
- drawing in
- knocking
- stabbing
- rubbing
- escape of liquids under high pressure and / or at high temperature
- slipping
- tripping
- falling

2.4 安全指示と危険防止

過給機の取扱・運転中には危険が伴います。

- 過給機およびその付属品から
- 潤滑油など運転使用剤から
- 安全指示に従わなかった場合
- 不適切なメインテナンス・点検作業から

機械的要因

警告

機械的要因によって人体に重大もしくは致命的な障害を及ぼすことがあります。

人体への障害

下記のようなことで意識を失ったり、負傷したりすることがあります。

- つぶされる
- はさまれる
- 切られる
- 巻き込まれる
- たたかれる
- 引き込まれる
- ぶつかる
- 刺さる
- 擦る
- 高圧力および高温の液体の噴出
- すべる
- つまずく
- 転落

Causes

- sharp edges, pointed parts
- crush, shear and winding areas
- fragmentation / ejection of parts
- fracture or rupture of parts due to overload
- elastic elements (springs), liquids and / or gases under pressure/in a vacuum
- slippery, oily work area, obstacles on the floor

Protective measures

- ✓ Wear leather protective gloves
- ✓ Wear close-fitting protective clothing
- ✓ Tie up long hair and beards (wear hair protection)
- ✓ Wear face and eye protection
- ✓ Wear protective shoes
- ✓ Keep floor, equipment and the turbo-charger clean
- ✓ Ensure the availability of oil absorbing materials and have oil collector basins ready or in position
- ✓ Eliminate leaks

Operation and process materials

Operation and process materials refer to **chemical and hazardous substances** :

- Oil
- Grease
- Coolants
- Detergents and solvents
- Acids

CAUTION

Note that operation and process materials are harmful to the environment if they penetrate soil or water.

原因

- 鋭い縁や先端の尖った部品
- つぶされる、はさまれる、巻込まれるような場所
- 部品の破片の破砕/飛散
- 過負荷による部品の破損、破断
- 弾性体(ばね)、高圧下または負圧下の液体およびガス
- 滑りやすい、油で濡れた作業場、床にある障害物

防護方策

- ✓ 皮製の手袋を着用してください。
- ✓ ぴったりした作業服を着用してください。
- ✓ 長い髪やひげは結わいて保護具を着用してください。
- ✓ 顔や目の保護具を着用してください。
- ✓ 安全靴を着用してください。
- ✓ 床・設備・過給機は常にきれいにしておいてください。
- ✓ 油を吸着する素材または油を回収する容器を常に準備し、所定の場所に配置してください。
- ✓ 漏洩物を取除いてください。

運転および設備用各種使用剤

運転および設備用各種使用剤は、下記の**化学物質および有害物質**にあたります。

- 油脂
- グリース
- 冷却剤
- 洗浄剤および溶剤
- 酸

注意

運転および設備用各種使用剤を自然界へ漏洩させると環境汚染になりますので、注意してください。

NV50F02-1

Personal injury

The following injuries can be caused by operation and process materials:

- Allergies
- Skin disease
- Loss of consciousness
- Poisoning or nausea after inhalation

人体への傷害

運転および設備用各種使用剤は、下記の人体傷害の原因になることがあります。

- アレルギー
- 皮膚疾患
- 意識喪失
- 吸入による中毒または意識朦朧状態

Causes

- Inhalation of toxic gases, smoke and vapours
- Skin contact with aggressive fluids
- Clothing wet or soaked with hazardous substances
- Spilling and tipping over of vessels containing substances used during operation
- Escape due to leakage
- Tanks and collectors which are not gas-tight and which do not comply with the regulations on hazardous substances
- Burning cigarettes or open flame in the proximity of operation materials

原因

- 有毒ガス・煙・蒸気を吸入した場合
- 活性な液体に皮膚が触れた場合
- 服に危険物質が染み込んだ場合
- 運転使用剤をこぼしたり、容器をひっくり返した場合
- 漏洩物の流出
- タンクや容器が密閉されておらず、危険物質の取扱規程に沿っていない場合
- 運転使用剤の近くでタバコまたは火を扱った場合

NV50F02-1

Protective measures

- ✓ Utmost caution when handling process and operation materials
- ✓ Protective clothing must be worn
- ✓ Avoid skin contact and inhalation
- ✓ Ensure that the workroom is properly ventilated
- ✓ Observe the hazard indications on the containers for operation and process materials
- ✓ Seal the hazardous substance container tight immediately after use

防護方策

- ✓ 設備・運転使用剤を取扱うときは最大限の注意を払ってください。
- ✓ 必ず保護具を着用してください。
- ✓ 状況にかかわらず薬物に直接触れたり、吸込んだりすることは避けてください。
- ✓ 作業場の換気をしてください。
- ✓ 運転・設備使用剤用の容器上に示されている危険指示には従ってください。
- ✓ 危険な物質の入った容器は使用後直ちにしっかり密封してください。

- ✓ Used operation materials must be collected safely in accordance with the legal provisions, stored separately in suitable containers as prescribed, and disposed of as special waste in an environmentally compatible manner
- ✓ Ensure that containers with new / used operation materials are sealed tight
- ✓ In the event of leaks or spills, immediately apply a suitable absorbent material and dispose of this in a technically and environmentally compatible manner as special waste

- ✓ 使用した運転使用剤は法規に従い安全に回収し、決められた適切な容器に入れて保管してください。もしくは、特殊な廃棄物として環境を汚染しないよう技術的に配慮の上処理してください。
- ✓ 新しいあるいは使用済の運転使用剤の容器はしっかりと封をしてください。
- ✓ 漏れたり、こぼした場合は、直ちに適切な吸着材で吸い取り、特殊な廃棄物として環境を汚染しないように技術的に配慮の上処理してください。

Handling insulating materials

インシュレーションの断熱材の取扱い

- ✓ Ensure that the work area is well ventilated
- ✓ Keep the work area clean
- ✓ Avoid disturbing dust
- ✓ Dust-suppressing tools and processing steps
- ✓ Remove from packaging only in the work area
- ✓ Exercise particular care when removing old insulating materials
- ✓ Dispose of insulating materials in a technically and environmentally compatible manner
- ✓ If large quantities of dust are created, wear protective glasses
- ✓ Use half or quarter masks
- ✓ Wear suitable work clothing and gloves
- ✓ For sensitive skin, apply moisturising, protective barrier cream

- ✓ 作業場の換気を十分行ってください。
- ✓ 作業場のきれいな状態を保持してください。
- ✓ 粉じんを飛散させないでください。
- ✓ 粉じんを抑えるような要具・処理手順に配慮してください。
- ✓ 開梱は作業場のみで行ってください。
- ✓ 使用済みのインシュレーションの断熱材を除去する場合は特別な注意を払ってください。
- ✓ インシュレーションの断熱材は環境を汚染しないように技術的に配慮の上処理してください。
- ✓ 多量の粉じんが発生する場合は保護めがねを着用してください。
- ✓ マスクを着用し、口および鼻を覆ってください。
- ✓ 適切な作業服および手袋を着用してください。
- ✓ 肌が過敏な作業者は、湿気を与える保護用クリームを使用してください。

NV50F02-1

Noise

The effects of noise above a legally permitted level can cause personal injury.

Personal injury

- Loss of hearing
- Deafness
- Impaired hearing
- Health disorders such as loss of balance or of consciousness
- Cardiac and circulatory disturbances

Causes

- Machine noise level above 85 dB (A)

Protective measures

✓ Wear ear protection

騒音

規定上許容されるレベルを超える騒音は、人体に傷害を引き起こすことがあります。

人体への障害

- 難聴
- 聴覚喪失
- 聴力障害
- バランス感覚もしくは意識喪失などの健康障害
- 心臓病および循環器系障害

原因

- 機械の騒音レベルが 85dB(A)を超える場合

防護方策

✓ 騒音用耳栓を着用してください。

NV50F02-1

Heat hazard

When the turbocharger is operated, surfaces become very hot.

Personal injury

- Burns

Causes

- Missing or incorrectly fitted insulation.

Protective measures

✓ Wear leather protective gloves.

高温による危険

過給機の運転中、その表面は非常に高温になります。

人体への傷害

- 火傷

原因

- インシュレーションがないか、正しく取付けられていない場合

防護方策

✓ 皮製の手袋を着用してください。

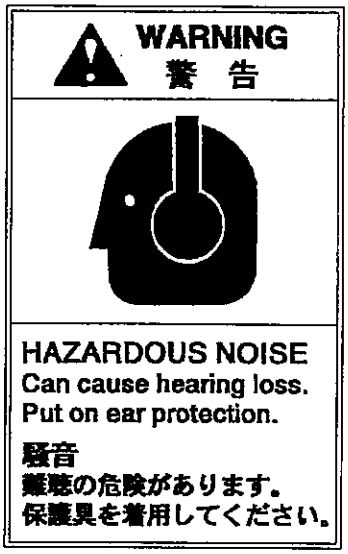
2.5 Warnings on the turbocharger

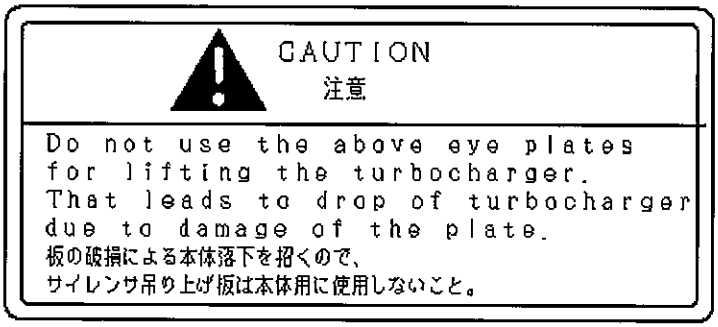
Warnings that have been applied to turbochargers must not be removed, painted over or obscured with other stickers. Missing or illegible warnings must be replaced by new warnings.

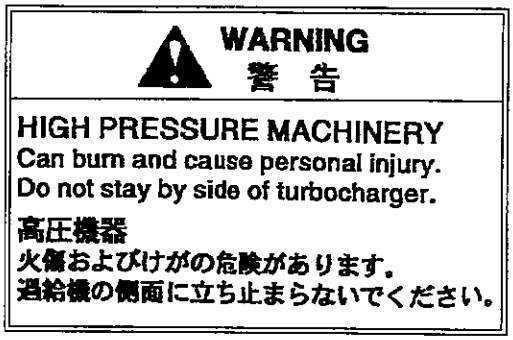
2.5 警告表示

過給機出荷時に取付けられている警告ラベルは取除いたり、その上に別の物を貼りつけたり塗装を上塗りしないでください。警告ラベルがない場合、または解読不能になった場合は新しいものに交換してください。

NV50F02-1







NOTE

Apply warnings so that they are clearly visible in the positions intended for that purpose. Please contact an official ABB Turbo Systems service station for further information.

注記

警告ラベルは、はっきり確認できる場所に取付けてください。更に詳細な情報に関しては ABB 社認定のサービスステーションへお問い合わせください。

2.6 Hot surfaces on uninsulated turbochargers

WARNING

Hot surfaces on uninsulated turbochargers

Uninsulated turbochargers can lead to serious personal injuries(burns).

The turbocharger is supplied by TSU without insulation depending on the order from the engine builder. If it is, the engine builder is responsible for insulating the turbocharger properly or for providing protection to prevent hot surfaces from being touched.

▶ The engine buiter's information and specifications about protection against hot turbocharger surfaces must be observed without fail.

2.6 高温部位

警告

過給機の高温部位

過給機にインシュレーションが装着されていない場合、重大な怪我(火傷)を負うことがあります。

エンジン製造会社の要求により、インシュレーションを装着せずに過給機を TSU 社より納入することがあります。その場合、エンジン製造会社は高温面に触れられないように正しく過給機を断熱する責任があります。

▶ エンジン製造会社の情報および過給機高温面の断熱仕様には、十分注意して確認してください。

NV50F02-1

3

Start-up

起動

3 Start-up

3.1 Oil supply

A carefully designed oil supply, which functions under all operating conditions, is an important prerequisite for trouble-free operation of the turbocharger.

Lubrication of the turbocharger is normally carried out with oil from the engine oil circuit.

If a separate lubrication system is used, emergency (standby) lubrication must also be provided.

NOTE

Observe the instructions of the engine manufacturer regarding lubrication oil selection and oil change intervals.

NV50F03-1

Oil filtration

CAUTION

Contaminants must be filtered out of the lubrication oil to prevent bearing parts from becoming dangerously worn. These impurities may be of different sizes depending on the turbine specification and the bearing used.

The standard oil filtration specification for the turbocharger is **34 µm.** This means that contaminants that are larger than **0.034 mm** must be filtered out of the oil with a separation efficiency **> 99%** .

3 起動

3.1 給油

過給機を支障なく運転するためには、全ての運転条件において機能する給油システムが必要です。

過給機への給油は通常エンジンの給油システムから供給されます。

もし独立給油システムを使用する場合は非常時用の給油システムが必要です。

注記

潤滑油の種類および交換時期については、エンジン製造会社の指示に従ってください。

潤滑油フィルタ

注意

軸受の異常磨耗を防ぐため潤滑油の不純物をろ過してください。これらの不純物はタービンおよび軸受仕様によりサイズが異なります。

過給機用の標準的な潤滑油フィルタは 34 µm です。これは、0.034 mm よりも大きい不純物を 99% 以上潤滑油からろ過するためのものです。

- If the oil is not filtered adequately by the engine filter, an additional filter must be fitted.
- When the engine is started from cold, and when dirt deposits in the additional filter cause the flow resistance to rise above **50 kPa**, a bypass must be activated to direct the flow of oil around the filter and ensure that the oil supply to the turbocharger is uninterrupted.
- Before putting into operation for the first time, check that the oil filters are clean.

NOTE

The engine manufacturer's instructions regarding filtration fineness and separation efficiency must also be followed.

- エンジン用潤滑油フィルタにて潤滑油が十分ろ過されない場合は、過給機用としてフィルタの追加が必要です。
- 冷態起動時または追加フィルタへのごみの堆積により流路抵抗が 50 kPa を超えるときは、過給機に確実に給油するために追加フィルタにバイパスを設けてください。
- 起動前にフィルタにつまりがないことを確認してください。

注記

フィルタのメッシュサイズやろ過能力についてはエンジン製造会社の指示に従ってください。

NV50F03-1

Lubricant

NOTE

All lubricating oils used for engines are admissible.

潤滑油

注記

全てのエンジン用潤滑油は使用可能です。

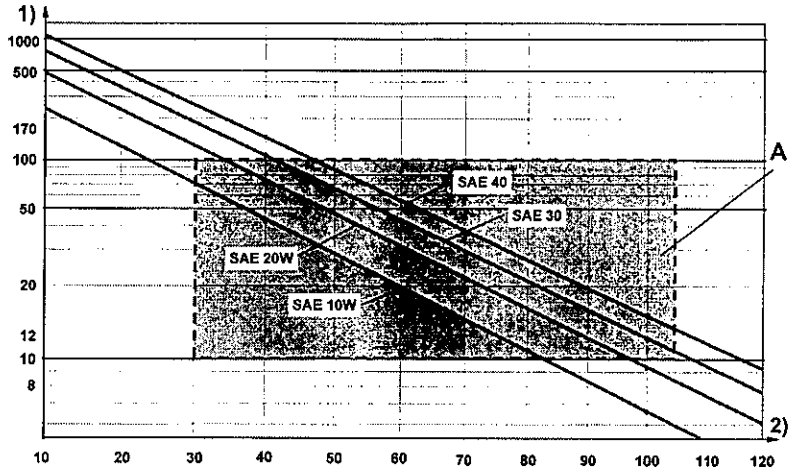
Oil inlet viscosity and temperature

The oil entry temperature must not exceed 105°C. Permissible oil entry viscosities and oil entry temperatures are shown in the following chart.

過給機入口での潤滑油粘度と温度

潤滑油入口温度は 105℃を超えないようにしてください。潤滑油入口での粘度と温度の許容値は下表の通りです。

NV50F03-1



S_00267

1)	Kinematic viscosity (mm^2/s = cSt)	動粘度 (mm^2/s = cSt)
2)	Oil entry temperature (°C)	潤滑油入口温度 (°C)
A	Permissible range	許容範囲

NOTE

The temperature may be allowed to fall below the lower limit briefly when the engine is started.

At this time, however, the oil entry temperature must not fall below 10 °C.

注記

起動時短時間ながら、潤滑油入口温度が下限値を下回る場合がありますが、その場合でも 10 °C 以下にならないようにしてください。

Oil pressure

The oil pressure in front of the bearings must be maintained precisely to ensure trouble-free operation.

When the engine is under load, the oil pressure (gauge pressure) before the turbocharger should be between **200 ... 450 kPa** and when idling at least **20 kPa** .

When the oil is cold, the oil pressure may be allowed to rise briefly, to **800 kPa.** The oil pressure must not exceed **100 kPa** if pre-or post-lubrication is used.

油圧

損傷のない安全な運転を行うためには軸受前の規定油圧を確実に保持することが必要です。

過給機前潤滑油圧(ゲージ圧)は負荷運転時では200～450 kPa に 、アイドリング時は少なくとも 20 kPa を確保するように調節してください。

起動時、潤滑油が低いため一時的に油圧が上昇する場合がありますが、最高 800 kPa まで許容可能です。エンジン起動前、停止後の潤滑油圧は100 kPa を超えないようにしてください。

Prelubrication

CAUTION

Turbochargers must be supplied with oil from the very first moment the engine is started.

▶ Actuate prelubrication device before starting engine each time.

エンジン起動前の潤滑油供給

注意

過給機はエンジン起動時にはすでに油が供給され潤滑状態でなければなりません。

▶ 必ずエンジン起動前に潤滑油装置を作動させてください。

NV50F03-1

3.2 Inspection work

This inspection work includes preventive visual controls, as well as monitoring and measurement to ensure the correct functioning of the turbocharger. These serve as an aid to detecting anomalies before and during commissioning, thus preventing possible damage to the machine.

- The safety instructions must be followed before and during all inspection work.

3.2.1 Inspection before commissioning

Monitoring equipment

▶ Check for correct functioning.

Oil fiters

▶ Check cleanliness before start-up.

Manometers in oil lines

▶ Check oil pressure in oil supply lines.

Air filter mat

▶ Inspect for damage.

CAUTION

We urgently recommend that you install a bypass pipe so that the entire lubrication system can be thoroughly flushed with warm, clean oil before the initial start-up and after any servicing work on the lubrication system.

Also, we strongly advise you to use a separate start-up filter when running in the engine and after any servicing work on the lubrication system.

3.2 点検作業

点検は、監視や計測だけでなく目視チェックによる予防作業により、過給機を正常に作動させるためのものです。これらは起動前および運転中の変化を見つけるのに役立ち、機械の損傷を防ぎます。

- 全ての点検作業は安全指示に従ってください。

3.2.1 起動前の点検

監視装置

▶ 機能点検を行ってください。

潤滑油フィルタ

▶ 起動前に汚れがないことを確認してください。

潤滑油ラインのマノメータ

▶ 給油ラインの油圧を点検してください。

フィルタマット

▶ 損傷がないことを確認してください。

注意

初めての起動前や潤滑システム分解開放後に運転を再開する前には、温かい油で潤滑システム全体をフラッシングしてください。フラッシングにはバイパス管を使用し、過給機をバイパスさせてください。

潤滑システムの分解開放後に初めてエンジンを運転する際には、その都度新しいフィルタを使用することを推奨いたします。

NV50F03-1

Prelubrication

CAUTION

Turbochargers must be supplied with oil from the very first moment the engine is started.

▶ Actuate prelubrication device before starting engine each time.

エンジン起動前の潤滑油供給

注意

過給機はエンジン起動時にはすでに油が供給され潤滑状態でなければなりません。

▶ 必ずエンジン起動前に潤滑油装置を作動させてください。

3.2.2 Inspection after commissioning

Turbocharger speed

▶ Measure. (This is optional as a turbocharger speed measuring device is not supplied as standard with all types of turbocharger.)

Charging pressure

▶ Measure.

Temperatures

▶ Measure in front of and behind turbine, compressor and oil feed at various engine outputs.

Gas, air, and oil lines

▶ After starting engine, check all gas, air and oil lines for leaks.

3.2.2 起動後の点検

過給機の回転数

▶ 計測(回転計はオプションであり、全ての過給機形式について標準で含まれていません。)

給気圧

▶ 計測

温度

▶ タービン・コンプレッサ出入口および潤滑油入口の温度をさまざまなエンジン出力で計測してください。

ガス、空気、潤滑油配管

▶ エンジン起動後、ガス、空気および潤滑油配管の全てに漏れがないか点検してください。

NV50F03-1

NOTE

In the first few houes after starting up the engine, lubricants and pastes used during turbocharger assembly liquefy or vaporize and might be given off as oily liquid or smoke. If further oily liquid continues to appear after this period, an oil leak must be suspected. Then, the first step is to check for leakage of the oil supply to the turbocharger. If something wrong is detected, contact an official ABB Turbo Systems service station.

注記

過給機の組立中に潤滑油やペーストは液化または気化し、起動後数時間の間に油状の液体として流れ出ることがあります。もし、この現象が起動後数時間経ってもおさまらない場合は、油漏れとして扱い原因を特定してください。まず最初に過給機への給油ラインの漏れをチェックしてください。漏れ等の不具合がある場合には、ABB 社認定のサービスステーションへお問い合わせください。

NV50F03-1

- ▶ Measure speed, oil pressure, charging pressure and temperatures in front of and behind turbine and compressor at various engine outputs.
- ▶ Compare measured values with values in inspection / commissioning report, taking into account the different operating conditions.

- ▶ 過給機回転数・油圧・給気圧およびタービン・コンプレッサ出入口の温度を様々なエンジン出力で計測してください。
- ▶ これらの計測結果を、運転状況の違いを考慮して、過去の運転データと比較してください。

3.2.3 Inspection after 100 service hours

Clean or replace lubricating oil filters after the first 100 service hours.

3.2.3 100 時間運転後の点検

使用後、100 時間経過したら、潤滑油フィルタを洗浄または交換してください。

3.3 Putting into operation after laying up

If they are present

- ▶ Remove the cover plates (blind flanges) between the compressor casing outlet and the charge-air receiver, upstream of the gas inlet, downstream of the gas outlet.
- ▶ Inspect the exhaust-gas duct before and after the turbine for any combustion residues, clean / remove foreign bodies or residual water.
- ▶ Inspect air supply line or filter silencer for any foreign bodies, clean / remove.
- ▶ Start up the turbocharger's engine end oil circulation system.
- ▶ Prepare the turbocharger for operation as instructed in the section **Inspection work before commissioning** in the chapter entitled **Commissioning.**
- ▶ Commission the turbocharger.

NV50F03-1

3.3 休止後の再起動

供給されている場合

- ▶ コンプレッサケーシング出口と給気管の間、およびガス出入口の応急フランジ(カバーフランジ)を取外してください。
- ▶ タービン前後の排気管に燃焼残留物や異物、水分残渣がないか点検し、必要に応じて清掃/取除いてください。
- ▶ 吸込空気配管或いは消音器に異物がないか点検し、必要に応じて清掃/取除いてください。
- ▶ エンジンから過給機へ供給される潤滑油の循環を開始してください。
- ▶ 「**起動**」章、「**起動前の点検**」の項を参照して過給機の運転準備をしてください。
- ▶ 過給機を稼動させてください。

4

Operation

運転

4 Operation

4 運転

4.1 Servicing work

Servicing work during operation includes visual checks, monitoring and measuring as well as inspection and function checks in order to ensure correct functioning of the turbocharger. It serves as an aid to detecting anomalies during operation, thus preventing damage to the machine. The inspection, measurement and servicing operations listed must be carried out at the intervals indicated.

4.1 取扱

運転中の取扱作業には、目視点検、監視、計測、過給機が正常に作動していることを確認する検査、機能点検が含まれます。これらの点検によって運転中の変化が確認でき、過給機の損傷を回避できます。リストアップされた点検、計測、取扱作業を指定通り定期的に実行してください。

WARNING

The pertinent safety instructions must be observed during all servicing work.

警告

全ての取扱作業中は安全指示を適切に守ってください。

CAUTION

Servicing work must be carried out at the prescribed intervals in order to avoid damage to and malfunctioning of the turbocharger.

注意

過給機の損傷ならびに運転停止を回避するため、取扱作業は指定通り定期的に実行してください。

NV50F04-1.R1

4.1.1 Servicing work every 25 to 50 hours

Operation data

- ▶ Record
- ▶ Record in the machine logbook
- ▶ In the event of significant deviations, establish the cause.

4.1.1 25～50 時間毎の取扱作業

運転データ

- ▶ 記録してください。
- ▶ 機関日誌に記録してください。
- ▶ もし運転データが過去の運転データから大きく変化している場合は、その原因を調査してください。

CAUTION

If you are unable to establish the cause of the anomalies, contact an official ABB Turbo Systems service station.

注意

変化の原因が特定できない場合は ABB 社サービスステーションへお問い合わせください。

- ▶ Visual inspection for air, exhaust gas, water and oil leaks.
- ▶ 空気、ガス、水、油等の漏れの目視チェックをしてください。

4.1.2 Servicing work according to the engine manufacturer's instructions

Oil filters

- With the engine at standstill clean or replace oil filters

4.1.2 エンジン製造会社指示間隔での取扱作業

潤滑油フィルタ

- エンジンが停止した状態で潤滑油フィルタの洗浄または交換を行ってください。

4.1.3 Servicing work every 8,000 -12,000 hours

- Dismantle the turbocharger.
- Measure clearances.
- Clean turbine and compressor casing and check for any cracks and erosion or corrosion.
- Clean bearing casing. Blow through oil ports.
- Clean nozzle ring and check for cracks and erosion.
- Inspect and assess rotor and bearing parts.

4.1.3 8,000～12,000 時間毎の取扱作業

- 過給機を分解してください。
- クリアランスを計測してください。
- タービンケーシングとコンプレッサケーシングを洗浄し、クラックおよび腐蝕、または侵食がないか確認してください。
- 軸受ケーシングを洗浄し、潤滑油通路をエアーブローしてください。
- ノズルリングを洗浄し、クラック、腐蝕がないか確認してください。
- ロータと軸受部品の点検および評価を行ってください。

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NOTE

Inspecting and assessing rotor

The first inspection and assessment of the bearing parts must be carried out in accordance with the instructions on the rating plate or between 8,000 and 12,000 hours.
The rotot and the bearing parts must be inspected and assessed by an official service station of ABB Turbo Systems.

注記

ロータの点検と評価

軸受部品の最初の点検および評価は過給機銘板に従い実施するか、8000～12000 時間運転の間に実施してください。
ロータおよび軸受部品の点検および評価は ABB 社認定のサービスステーションに依頼してください。

4.1.4 Entries in the machine logbook

Monitoring of the machine plant provides information about the performance of the turbocharger.

The following operation data and measurement values must be entered regularly in the engine manufacturer's machine log book:

- Output and speed of the engine
- Air intake temperature
- Exhaust temperature before and after the turbine
- Pressure of the charge-air
- Pressure drop in the charge-air cooler
- Lubricating oil pressure, lubricating oil temperature

If provided

- Air temperature after compressor and after the charge-air cooler
- Speed of the turbocharger
- Pressure loss in the air filter

4.1.4 機関日誌の記入要領

機関を監視することにより過給機性能についての情報を得ることができます。

下記の運転データおよび計測値をエンジン製造会社の機関日誌に定期的に正しく記入してください。

- エンジンの出力と回転数
- 吸込空気温度
- タービン出入口の排気ガス温度
- 給気圧
- 給気冷却器での圧力損失
- 潤滑油圧、潤滑油温度

供給されている場合

- コンプレッサおよび空気冷却器後の空気温度
- 過給機の回転数
- フィルタリングの圧力損失

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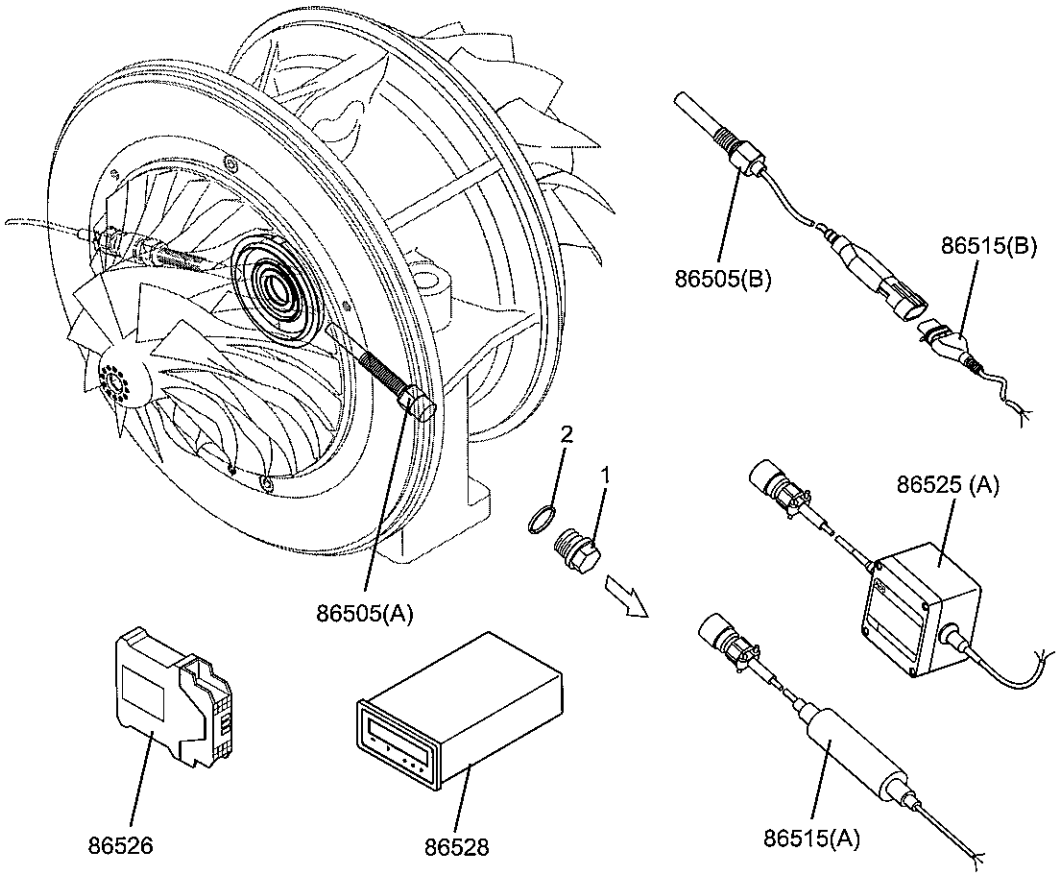
4.2 Measuring speed 4.2 回転数の計測



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S_01302

Part no. 部品番号	Description / 名称	Part no. 部品番号	Description / 名称
86505(A)	Speed sensor (cable not integreated) 回転計ピックアップ(ケーブル別体)	86526	F/I converter F/I コンバータ
86505(B)	Speed sensor(cable integreated) 回転計ピックアップ(ケーブル一体)	86528	Tachometer タコメータ
86515(A)	Ccble connector for 86505(A) 86505(A)用ケーブルコネクタ		
86515(B)	Ccble connector for 86505(B) 86505(B)用ケーブルコネクタ	1	Screw plug プラグ
86525	Power amplifier パワーアンプ	2	Gasket パッキン

Turbochargers can be equipped with a speed measuring system.

オプションとして回転数計測装置が装備できます。

This system permits continuous monitoring of the turbocharger speed which can be used as a suitable control parameter for continuous monitoring of the machine installation.

この計測装置により過給機回転数を連続して監視することができます。そしてそれはエンジンに搭載された連続監視システムでの運転状況把握に適当なパラメータとなります。

4.2.1 Assembly

If the speed sensor is not fitted on the turbocharger, the following procedure must be adopted to fit it:

- ▷ Engine and turbocharger are not running.
- ▷ The supply of lubricating oil to the turbocharger is switched off.
- ▶ Remove screw plug (1) with gasket (2) from bearing casing (left / right, depending on desired position of speed sensor).
- ▶ Screw in speed sensor (86505) as far as it will go and tighten to specified torque (see chapter Table of tightening torques).
- ▶ Connect speed sensor with cable connector (86515) or power amplifier (86525).

4.2.1 組付け

回転計ピックアップが過給機に組付けられていない場合は下記の手順に従い組付けてください。

- ▷ エンジンおよび過給機を停止してください。
- ▷ 過給機への給油装置を停止してください。
- ▶ プラグ(1)とパッキン(2)をベアリングケーシングから取外してください。(左右どちらか回転計ピックアップを取付ける方)
- ▶ 回転計ピックアップ(86505)を規定の締付けトルクで止まるまでしっかりと締付けてください。(「締付けトルク」の項をご参照ください)
- ▶ ケーブルコネクタ(86515)またはパワーアンプ(86525)を回転計ピックアップと接続してください。

4.2.2 Replacement

If the speed sensor has to be replaced, proceed as follows:

- ▷ Engine and turbocharger are not running.
- ▷ The supply of lubricating oil to the turbocharger is switched off.
- ▶ Disconnect cable connector (86515) or power amplifier (86525) from speed sensor (86505).
- ▶ Screw out defective speed sensor.
- ▶ Screw in new speed sensor as far as it will go and tighten to specified torque (see chapter Table of tightening torques).
- ▶ Connect new speed sensor with cable connector or power amplifier.

4.2.2 交換

回転計ピックアップを交換する場合は下記の手順で行ってください。

- ▷ エンジンおよび過給機を停止してください。
- ▷ 過給機への給油装置を停止してください。
- ▶ ケーブルコネクタ(86515)またはパワーアンプ(86525)を回転計ピックアップ(86505)から取外してください。
- ▶ 古い回転計ピックアップを取外してください。
- ▶ 新しい回転計ピックアップを規定の締付けトルクで止まるまでしっかりと締付けてください。(「締付けトルク」の項をご参照ください)
- ▶ ケーブルコネクタまたはパワーアンプを新しい回転計ピックアップと接続してください。

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NOTE

Replacement / fitting speed sensor

It is recommend that you have work on the speed sensor carried out by an official ABB Turbo Systems service station.

For work on the speed measuring system, the engine manufacturer's instructions must be followed.

注記

回転計ピックアップの交換 / 取付け

回転計ピックアップの作業は、ABB 社認定のサービスステーションへ依頼するよう推奨いたします。

回転計ピックアップの作業については、エンジン製造会社の指示に従ってください。

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4.2.3 Sources of error

A drop in turbocharger speed can signify:

- Damaged rotor components or bearings.
- Defective cylinders connected during pulse operation.
- Defective or leaking exhaust manifolds or charge-air ducts.

An increase in turbocharger speed can signify:

- A very dirty turbine

Possible reasons for failure of speed measuring system:

- Defective speed measuring system.
- Turbocharger damage.

Possible reasons for measuring errors of speed measuring system:

- Fitting error.
- Dirty sensor tip.
- Defective speed measuring system.

NOTE

Problems with speed measuring system

If problems arise with the speed measuring system, it is recommended that the engine output be reduced to idling speed and the engine stopped. The cause of the failure should then be determined.

If the cause of the failure or the measuring errors cannot be found, please contact an official ABB Turbo Systems service station.

4.2.3 エラーの原因

過給機回転数の低下

- ロータの部品または軸受の損傷
- 動圧過給システムでのエンジンシリンダの不具合
- 排気配管および給気配管の損傷もしくは漏れ

過給機回転数増大

- タービンのひどい汚れ

回転計系統の故障原因

- 回転数計測装置の不具合
- 過給機の故障

回転数計測装置の計測不良

- 組付け不良
- 回転計ピックアップ先端の汚れ
- 回転数計測装置の不具合

注記

回転数計測装置の問題

回転数計測装置に問題が起きた場合はエンジン出力をアイドリング状態に落とした後、停止させることを推奨いたします。停止後故障の原因を明らかにしてください。

故障または計測不良の原因が不明の場合は ABB 社認定のサービスステーションへ連絡してください。

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4.3 Surging of turbocharger

Surging of the turbocharger can occur with certain engine operating states, such as rapid load removal or while manoeuvring. The direction of flow in the compressor is then briefly reversed. However, sporadic surging of this kind does not generally affect the safe operation of the turbocharger.

CAUTION

Prolonged or periodic surging

If surging is prolonged or occurs periodically, the following components may be damaged:

- Compressor wheel (vibration fractures,...)
- Turbine blades (cracks)
- Bearing (overheating)
- Filter silencer

NOTE

Action to be taken:

The operating safety of the turbocharger is no longer ensured with continuous or periodically recurring surging. The operator must take immediate steps to correct the surging together with the closest official ABB Turbo Systems service station.

The components listed above must be inspected for damage by an official ABB Turbo Systems service station, and replaced in cases of doubt.

4.3 過給機のサージング

急速な負荷軽減等、特定の運転状況下で、過給機のサージングが生じる場合があります。その際、コンプレッサ内で一時的に逆流が発生します。単発的なサージングであれば過給機の安全運転に影響はありません。

注意

連続的または、周期的なサージング

連続的または周期的なサージングは、下記の構成部品を損傷させる場合があります。

- コンプレッサ（振動による疲労等）
- タービンブレード（亀裂）
- 軸受（過熱）
- 消音器

注記

取るべき方策：

連続的または周期的なサージングが生じた場合、過給機の運転上の安全は保障できません。最寄りの ABB 社認定のサービスステーションにご相談の上早急に解決策を実施してください。前述の構成部品の損傷点検は、ABB 社認定のサービスステーションに依頼し、懸念がある場合は交換してください。

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4.4 Touching of rotating parts

Very slight and uniform wear around the periphery of rotor components, caused by their slight local touching adjacent components, is not hazardous and is permissible. This somewhat shortens the length of the compressor and / or turbine blades. Certain tolerances must be observed to avoid a significant loss of efficiency. See section 8.4, **Radial clearances N and R.**

NOTE

Please note that the N and R clearances are only an initial estimate of the touching state. If there is any doubt about the extent of touching, an official ABB Turbo Systems service station must be contacted.

4.4 回転部の接触

ロータの外周部分の均一で軽い磨耗は、周囲の部品とわずかにこすれたのが原因であり、危険な現象の兆候ではなく許容される現象です。コンプレッサまたはタービンブレードが幾らか短くなります。大きな効率損失の原因となるのを回避するため、一定の許容値を保持してください。8.4 項、「**ラジアルクリアランス N および R**」をご参照ください。

注記

ラジアルクリアランスの N および R は接触状態判断の第一段階です。 接触程度に疑問がある場合は、必ず ABB 社認定のサービスステションに連絡してください。

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4.5 Shutting the engine down

CAUTION

Before shutting the engine down, it must be allowed to idle for **5** to **10 minutes** longer so that the circulating lubricating oil can dissipate the heat in the turbocharger bearings.

4.5 エンジンの停止

注意

潤滑油の循環により過給機の軸受を冷却するために、エンジン停止前 **5** 分～**10** 分間のアイドリング運転をしてください。

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5

Maintenance

メインテナンス

5 Maintenance

5.1 Foreword of Maintenance

Maintenance and servicing work involves regular visual checks and cleaning to ensure that the turbocharger and its attached units function trouble-free.

- The external condition and how dirty the cleaning points specified in this chapter are, must be established by visual checks at the specified intervals.
- The safety precautions must be observed during all maintenance and servicing work.

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The cleaning points described in the following are:

- Filter silencer
- Compressor
- Turbine and nozzle ring

5 メインテナンス

5.1 メインテナンスにあたって

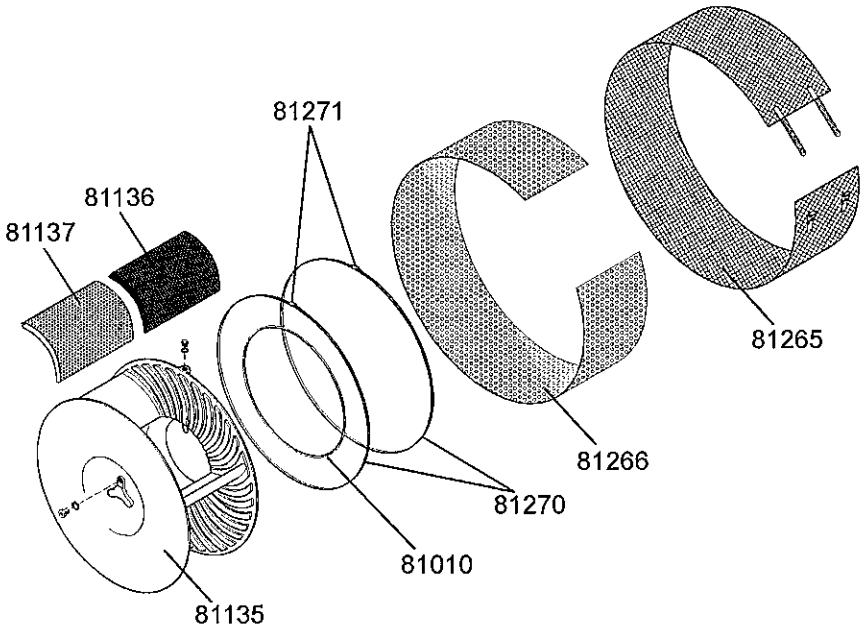
目視検査と洗浄作業からなる定期的なメインテナンスおよび補修作業は、過給機とその補助装置の正しい性能を確保するためのものです。

- 下記部品の外観と清掃ポイントの汚れ度合いを、目視検査にて指定された間隔でチェックしてください。
- 全てのメインテナンスおよび取扱作業で安全上の注意に従ってください。

下記の部品について清掃ポイントを記述します:

- 消音器
- コンプレッサ
- タービンおよびノズルリング

5.2 Cleaning filter silencer　5.2 消音器の洗浄



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If provided

The filter ring (81265) can be replaced or cleaned by washing.

NOTE

To guarantee perfect operation during washing, it is advisable to replace the filter ring with a second (filter ring available in pack of two).

CAUTION

Wash or replace the filter ring every 500 operating hours. The filter ring must not to be washed more than five times.

供給されている場合

フィルタリング(81265) は交換ないしは取外して水洗浄をすることができます。

注記

汚れたフィルタリングの洗浄中にも安全な運転を行うために、交換用の新しいフィルタリングの使用をお勧めします。(2 個セットのフィルタリングをご用意しています。)

注意

フィルタリングの洗浄または交換は 500 時間運転毎行なってください。フィルタリングは 5 回を超えて洗浄しないでください。

Washing the filter ring

- ▶ Rinse out filter ring using water (up to 40° C) containing light-duty detergent or, if heavily soiled, let it soak while squeezing carefully. Rinse in cold water. Avoid high mechanical stressing (wringing, powerful water jet).
- ▶ Let filter ring dry completely before fitting.

フィルタリングの洗浄

- ▶ 粉末洗浄剤を使って温水(40℃以下)で洗い落としてください。特に汚れがひどい場合には洗浄水に浸してください。その後、絞った上で冷水ですすいでください。なお、一連の洗浄中にはフィルタリングに無理な力がかからないように注意してください。(例:絞りすぎ、高圧水による洗浄)
- ▶ 組付ける前に乾燥させてください。

Cleaning silencer

- ▶ Release tension bands (81270) and take out cover grid (81266).
- ▶ Withdraw cover plates (81137)and bend upwards, then remove felt segments (81136).

CAUTION

When cleaning, take care that the felt segments do not get wet!

消音器の洗浄

- ▶ テンションバンド(81270)をゆるめ、カバーグリッド(81266)を取外してください。
- ▶ 仕切板金網(81137)を引抜き、仕切板(81136)を取外してください。

注意

洗浄中、仕切板を濡らさないよう注意してください!

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- ▶ Remove dirt with cloth, soft brush or a jet of compressed air.
- ▶ Have very dirty felt segments replaced by an official service station of ABB Turbo Systems Ltd.

- ▶ 柔らかいブラシまたは圧縮空気で汚れを取除いてください。
- ▶ ひどく汚れた仕切板はABB社サービスステーションに交換を依頼してください。

CAUTION

Damaged tension bands must be replaced with new ones.

注意

テンションバンドが損傷している場合は、新しい部品と交換してください。

5.3 Cleaning the compressor during operation

How dirty the compressor stage (compressor wheel and diffuser) becomes, depends on how clean the drawn-in air is.

Deposits of dirt can build up in the flow ducts if the following substances are in the intake air:

- Oil or saline mist
- Solid combustion residues
- Dust of various kinds

Dirty deposits on the compressor stage have a negative influence on the compressor efficiency and on the charging pressure.

This results in higher exhaust temperatures and higher fuel consumption by the engine. Dirty deposits on the compressor can also increases rotor unbalance.

Periodic cleaning of the compressor during operation prevents or delays a severe increase in dirty deposits. In no way, however, does it replace the regular service work during which the turbocharger is completely dismantled and the compressor cleaned mechanically.

NOTE

If the layer of dirt is very thick and hard, the compressor can be cleaned only manually when disassembled. This cleaning should be carried out by an official ABB Turbo Systems service station.

Interval between cleaning

The interval between periodic cleaning depends considerably on the operating conditions. In general, cleaning should be carried out every **25** to **100 operating hours.**

5.3 運転中のコンプレッサ洗浄

コンプレッサ側(コンプレッサおよびディフューザ)の汚れは、吸込空気の清浄度によります。

下記の物質が吸込空気中に存在する場合、ダクト中に汚れが堆積していきます。

- 油や塩分を含んだミスト
- 燃焼残渣物
- ちりやほこり

コンプレッサ側のひどい汚れはコンプレッサ効率や吐出圧力に悪影響を与えます。

その結果、エンジン排気温度の上昇や燃料消費量の増加をもたらします。また、コンプレッサの汚れはロータのアンバランス量を増大させます。

運転中のコンプレッサの定期的な洗浄は通常のオーバーホール時の清掃効果に及ぶものではありませんが、汚れがひどくなるのを防いだり遅らせたりする効果があります。

注記

汚れが非常に厚く硬く堆積している場合、コンプレッサは取外した状態での洗浄が必要です。この洗浄は、ABB社サービスステーションにご依頼ください。

洗浄間隔

定期的な洗浄間隔は運転状況に大きく左右されますが、通常 **25～100 運転時間毎**に洗浄してください。

NV50F05-1.R1

NOTE

Should the prescribed cleaning intervals prove incompatible with the operation of the engine, please contact an official ABB Turbo Systems or TSU.

注記

規定の洗浄間隔が運行と合わない場合には、ABB 社または TSU 社にお問い合わせください。

Cleaning method

The cleaning of the compressor in operation is carried out with the cleaning method described in the following:

- Wet cleaning

This cleaning method has been tested and approved by ABB Turbo Systems or TSU.

Principle of wet cleaning

To clean the compressor stage while in operation, water is injected in front of the compressor wheel from an injection pipe that is built in the filter silencer or suction branch.

The water does not act as a solvent during this process, but the doposit is removed by the mechanical action of the impacting droplets. This is a very suitable process, provided that the build-up of dirt deposits is not too advanced.

NV50F05-1.R1

洗浄方法

運転中のコンプレッサ洗浄は、下記に記載した洗浄方法で行ってください。

- 水洗浄

この洗浄方法は、ABB 社/TSU 社が確認し認定した方法です。

水洗浄の原理

消音器または空気吸込囲に内蔵された注水管からコンプレッサ前に洗浄水が噴射され、運転中のコンプレッサを洗浄します。

洗浄過程において洗浄水は溶剤として作用せず、衝突する水滴の機械的衝撃によって汚れを除去します。汚れがそれ程進行していない段階から洗浄を実施することにより、最適な洗浄効果が期待できます。

CAUTION

Corrosion and deposits during cleaning

Damage and impairment of turbocharger parts through salt water and cooling water additives.

▶ Always use **clean water only** for cleaning.

注意

洗浄中の腐食と堆積物

海水や冷却添加剤を含む洗浄水を使用すると過給機部品に損傷を与えます。

▶ 洗浄は常に**清水**を使用してください。

CAUTION

The injection pipe must on no account be connected directly via a cock to a water pipe or a dosing vessel larger than the one supplied. This prevents uncontrolled volumes of water entering the turbocharger and engine, which can lead to serious damage.

注意

注水パイプは、水道管ないしは供給品の容器よりも大きな容量の容器に直接継がないでください。これにより、過給機およびエンジンに予想外の水量が入ることを防ぎ、重大な損傷を防止することができます。

Cleaning more than one turbocharger compressor

If several turbochargers are installed on one engine, we recommend cleaning the turbocharger compressors one after the other.

複数台の過給機が搭載されている場合

1 台のエンジンに複数の過給機が搭載されている場合は、1 台ごとに洗浄を実施されることを推奨いたします。

V-engines

For V-engines with **several turbochargers** on each engine, we recommend parallel cleaning of the compressors. This cleaning process is faster and the risk of turbocharger surging is avoided.

V 型エンジン

複数の過給機が各バンクごとに搭載された V 型エンジンに関しては、各過給機のコンプレッサの洗浄を同時に行うことを推奨いたします。この洗浄工程は、より速く過給機サージングの危険を回避します。

Sequential charging

In the case of sequential charging, it must be ensured that both turbocharger compressors are cleaned, especially following periods of operation in the lower performance range.

シーケンシャル過給システム

シーケンシャル過給システムの場合、必ず双方の過給機のコンプレッサを洗浄してください。低負荷運転時は運転されていない過給機がある場合がありますので注意してください。

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Supplier of cleaning water

専用洗浄水供給業者

Turbo Systems United Co., Ltd.(TSU)

ターボシステムズユナイテッド株式会社(TSU)

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5.3.1 Wet cleaning compressor using external water pressure vessel (XC1)

NOTE

These guidelines for wet cleaning apply exclusively for **cleaning with clean water** and are subject to permission for the procedure being obtained from the **engine manufacturer.**

NOTE

The supply of water with the externally fitted dosing vessel is only suitable for applications **with underpressure** before the compressor wheel (not applicable for: Blower connected upstream of the compressor or high-pressure compressor stage in the case of two-stage charging).

5.3.1 高圧型洗浄装置 (XC1) によるコンプレッサ水洗浄

注記

コンプレッサ水洗浄に関する下記の指示事項は、エンジン製造会社が容認した場合の水洗浄のみに適用してください。

注記

高圧型洗浄装置は、コンプレッサ前が**負圧状態**である場合のみ使用できます。(コンプレッサの上流側にブロワが設置されている場合や二段過給の高圧段コンプレッサの場合は使用できません)

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Operating state prerequisite for cleaning compressor using XC1

NOTE

In order to carry out a successful cleaning procedure that has been tested and is recommended by ABB Turbo Systems, the following prerequisites must be met:

▷ Engine loading as high as possible

▷ High turbocharger speed

▶ Cleaning cycle started according to the following description in **Wet cleaning procedure for compressor using XC1.**

XC1 によるコンプレッサ洗浄時の運転条件

注記

ABB 社が確認し推奨する洗浄を確実に行う為には、以下の前提条件が満たされていなければなりません:

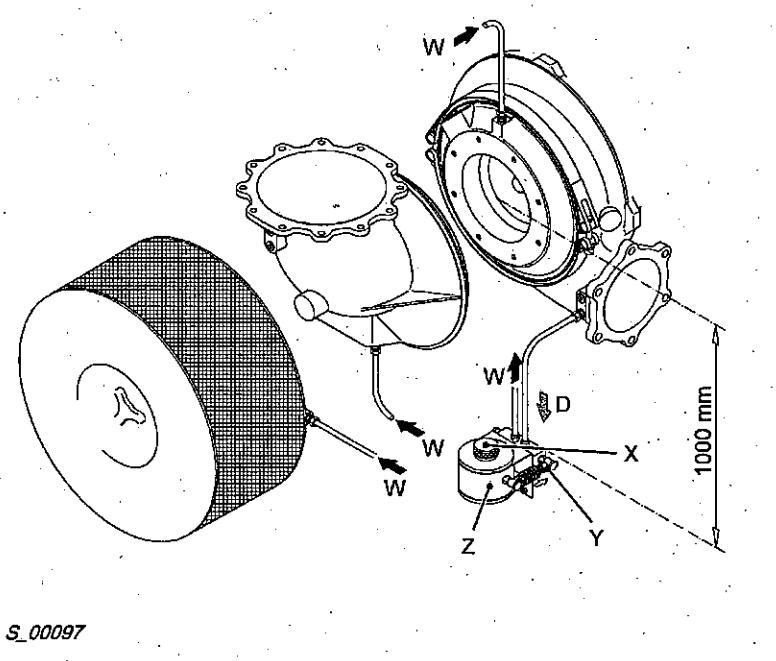
▷ エンジンはできる限り全負荷状態

▷ 過給機が高回転数であること

▶ 洗浄する場合は、後述の「**XC1によるコンプレッサの水洗浄手順**」をご参照ください。

Wet cleaning procedure for compressor using XC1

XC1 によるコンプレッサの水洗浄手順



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X	Screw plugs プラグ	D	Compressed air 圧縮空気
Y	Valve activator バルブレバー	W	Water 水
Z	Pressure vessel 圧縮容器		

Operating water pressure vessel

- ▶ Screw out screw plug(X).
- ▶ Fill vessel with clean water.
- ▶ Screw in screw plug.
- ▶ Push back valve activator(Y)against spring pressure and hold it for **10** to **15 seconds** until full volume of wate has been injected.
- ▶ Do not repeat cleaning cycle until a stabilization period of at least **10 minutes** has elapsed.

高圧洗浄装置の取扱い

- ▶ 閉塞プラグ(X)を取外してください。
- ▶ 容器に清水を注ぎ、満たしてください。
- ▶ 閉塞プラグを締付けてください。
- ▶ バルブレバー(Y)をスプリングと逆方向にスライドさせ、全容量を噴射するまで **10～15 秒間** 保持してください。
- ▶ 安定するまで少なくとも **10 分間**は再洗浄しないでください。

NOTE

Whether or not cleaning has been successful, can be seen from the charging or flushing pressure and also from the exhaust gas temperatures .

If the cleaning operaion was not successful, it can be repeated up to **2 times**.

If cleaning is still not successful **after three attempts** and the engine operating values are unsatisfactory, we recommend that you have the turbocharger checked and cleaned by an official ABB Turbo Systems service station.

注記

洗浄効果は給気圧と排気ガス温度により確認できます。

最初の洗浄が不十分な場合、追加 2 回まで繰り返して実施することができます。

洗浄を 3 回実施し、それでも効果が得られずエンジンの性能の回復が不十分な場合は、ABB 社認定のサービスステーションに点検と清掃を依頼されることを推奨いたします。

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Cleaning parameters per turbocharger compressor　コンプレッサ洗浄のパラメータ

Turbocharger type 過給機形式	Turbocharger speed 過給機回転数	Contents of the dosing vessel [dm³] 注入量(注入容器量) [dm³]	Water injection time t_1 [s] 水噴射時間 t_1 [s]
TPS	approx. n_{Bmax} 可能な限り n_{Bmax} に近づける	0.4	10

5.4 Turbine cleaning during operation

5.4 運転中のタービン洗浄

The combustion of fuels in diesel engines, such as marine diesel oil (**MDO**) and heavy fuel oil (**HFO**), produces dirty deposits on the turbocharger turbine components. In combination with a high exhaust gas temperature, poor quality fuel can lead to particularly hard deposits. Especially on the nozzle ring.

ディーゼルエンジンにおいて船舶用ディーゼルオイル(MDO)や重油(HFO)が燃焼することにより、過給機のタービンブレードに汚れが付着します。特に高い排気ガス温度と低質な燃料が組み合わされた時には、ノズルリングに非常に硬い汚れが付着します。

Consequences of dirty deposits:

- Poor turbine efficiency
- Elevated exhaust gas temperatures
- Higher charging and ignition pressures as the turbocharger speed rises
- Lower engine performance

汚れの影響

- タービン効率の低下
- 排気ガス温度の上昇
- 過給機回転数の上昇による給気圧、シリンダ内燃焼圧力の上昇
- エンジン性能の低下

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Experience with turbocharger operation shows that the intervals between overhauls can be extended if cleaning during operation is carried out periodically.

経験上、運転中の定期的な洗浄によりオーバーホールの間隔を延ばせることが確認されています。

Cleaning the turbine in the way described in the following can be used for 4-stroke applications when there are heavy deposits (for example MDO and HFO applications).

下記に述べられているタービン洗浄は、汚れの激しい(船舶用ディーゼルオイルや重油を使用する)4ストロークエンジンで実施してください。

Generally, no turbine cleaning is required with the other applications.

通常、その他の機関ではタービン洗浄は必要ありません。

NOTE

Rergular cleaning

Regular turbine cleaning during operation hinders or delays an excessive build-up of deposits.

注記

定期的な洗浄

運転中定期的にタービン洗浄を実施することにより、堆積物の形成を防ぐ、ないしは遅らせることができます。

This cleaning does not replace the normal maintenance work during which the turbocharger is completely disassembled.

しかしながら、水洗浄は定期的な過給機の分解整備に代わるものではありませんので、別途定期的な分解整備も行ってください。

NOTE

Very dirty turbocharger

Turbochargers with very heavy dirty deposits must no longer be cleaned this way. In this cace, the dirty components must be cleaned by an official ABB Turbo Systems service station.

注記

過給機の激しい汚れ

過給機の汚れが激しい場合にはこの洗浄方法を適用することはできません。その場合は、汚れた部品の洗浄を ABB 社サービスステーションに依頼してください。

Cleaning intervals

The interval between periodic cleaning greatly depends on the operating conditions (fuel quality, engine load and engine type). ABB Turbo Systems recommends the following intervals:

洗浄間隔

定期的な洗浄間隔は運転条件に大きく依存します（燃料品質、エンジン負荷およびエンジン型式）、ABB 社は下記の洗浄間隔を推奨します。

Cleaning interval in service hours [h] 運転時間の洗浄間隔 [時間]	50 ～ 200

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NOTE

Cleaning intervals

If the specified cleaning intervals are unsuitable for the operation of the engine, please contact ABB Turbo Systems or TSU.

注記

洗浄間隔

指定された洗浄間隔が運転に不適当な場合は ABB 社/TSU 社にお問い合わせください。

Cleaning method

The cleaning of the turbine components during operation is carried using the **wet cleaning methods** described in the following. These have been tested and are approved by ABB Turbo Systems or TSU.

The turbine casing design and the type of turbine casing sealing are decisive for the choice of cleaning method.

洗浄要領

運転中のタービン洗浄は下記に記載されている**水洗浄要領**により実施してください。この方法は ABB 社/TSU 社で確認され承認されたものです。

タービンケーシングの種類とシール方法により洗浄方法が選択されます。

NOTE

Approval by engine manufacturer

The prerequisite for wet cleaning is that the engine manufacturer approves the method and his instructions are followed.

注記

エンジン製造会社による承認

水洗浄に必要な条件はエンジン製造会社により承認されますので、エンジン製造会社の指示に従ってください。

5.4.1 Wet cleaning methods

Method with short water injection time (3 x 30 seconds cleaning)

The cleaning method with short water injection time is used for turbocharger designs **without** additional sealing by means of C-rings.

The method is based on the principle of thermal shock in combination with a subsequent flushing phase.

Method with long water injection time (10 minutes cleaning)

The cleaning method with long water injection time is used for turbocharger designs **with** additional sealing by means of C-rings.

Based on a 10-minute injection phase, this cleaning method utilises the principle of water solubility of the dirt deposits.

5.4.1 水洗浄要領

短時間注水要領(30 秒 x 3 回洗浄)

短時間注水による洗浄は C リングの**付かない**過給機で行います。

この方法は、サーマルショック(熱衝撃)の原理に加え引続き水が吹きつけられることにより洗浄するものです。

長時間注水要領(10 分間洗浄)

長時間注水による洗浄は C リング**付き**過給機で行います。

この洗浄方法は 10 分間注水し続けることにより、汚れの水溶性の原理を利用するものです。

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NOTE

Choice of cleaning method

If you need information about the choice of cleaning method(short / long), please contact the engine manufacturer or an official ABB Turbo Systems service station.

注記

洗浄方法の選択

洗浄方法(短時間/長時間)の選択についての情報が必要な場合はエンジン製造会社または ABB 社サービスステーションへお問い合わせください。

5.4.2 Prerequisites

CAUTION

Risk of corrosion and deposits

Impairment and damage of turbocharger by salt water and cooling water additives in flow ducts.

- ▶ Always use **clean water only** for cleaning.

Avoid corrosion on the casing inner surfaces.

- ▶ Continue to operate engine **for 10 minutes** after a wet cleaning operation.

CAUTION

Maximum temperature in front of turbine during cleaning

During cleaning, the temperature in front of the turbine will rise and can heavily stress the material of turbine components.

- ▶ Don't exceed maximum temperature in front of turbine during cleaning.

CAUTION

Impermissible thermal stressing and flooding of turbine

Smaller volumes of water can result in poor cleaning. Larger volumes of water lead to impermissible thermal stressing of the turbine components and can cause flooding of the turbine.

- ▶ Observe turbine cleaning parameters without fail.

5.4.2 前提条件

注意

腐蝕と堆積物の危険

塩水と冷却水添加物による過給機の損傷と障害

- ▶ 洗浄には常に**清水**を使用してください。

ケーシング内面での腐蝕を避ける。

- ▶ 水洗浄後 **10 分間**はエンジンを運転し続けてください。

注意

洗浄中のタービン前排気ガス最高温度

洗浄中はタービン前排気ガス温度の上昇やタービン材料に高い応力がかかることがあります。

- ▶ 洗浄中はタービン前排気ガスの最高許容温度を上回らないようにしてください。

注意

高い熱応力とタービンからの水漏れ

注水量が少ないと十分な洗浄効果が得られません。また注水量が多すぎるとタービンに高い熱応力が発生したり、タービンからの水漏れの原因となります。

- ▶ タービン洗浄のパラメータを観察してください。

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NOTE

V-engines

For V-engines with **several turbochargers** on each engine, we recommend parallel cleaning of all turbochargers. This cleaning process is quicker and avoids surging of the turbochargers.

注記

V 型エンジン

複数の過給機が各バンクごとに搭載された V 型エンジンに関しては、各過給機の洗浄を同時に行うことを推奨いたします。この洗浄により、短時間で過給機の洗浄を行うことができ、またサージングを回避することができます。

Recommended operating state for wet cleaning turbine

タービン水洗浄時の推奨運転状態

The following operating state has been tested and is approved by ABB Turbo Systems:

下記の表は ABB 社により確認および承認された運転状態です。

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Characteristic / component 特徴 / 部品	Conditions / 状態	
	short water injection time 短時間注水	long water injection time 長時間注水
Engine load (guide value) エンジン負荷(目安)	20...40 %	
Turbine inlet temperature T_{TE} before water injection (guide value) 注水前 タービン入口温度 T_{TE}（目安）	350...400°C	350...380°C
Water supply 注水	ensured 確実に行う	
Differential pressure: water to exhaust gas at inlet 注水圧:入口排気ガス圧	≥ 150kPa	

If necessary, the engine output might have to be reduced to be able to maintain these conditions.

必要ならば、これらの状態を維持するためにエンジン出力を下げてください。

The cleaning cycle can now be started based on the following descriptions.

洗浄は下記の手順にそって開始してください。

5.4.3 Procedure for method with short water injection time

▷ Prerequisites have been fulfilled.

▶ Open shut-off valve.

▶ Set required water flow rate V_W using flowmeter and inject for **30 seconds**.

▶ Pause for 3 or 5 minutes (see cleaning parameters).

▶ Repeat 30-second water injection two times.

▶ Close shut-off valve.

This completes the turbine cleaning operation.

5.4.3 短時間注水の手順

▷ 必要条件を満たしてください。

▶ 遮断弁を開けてください。

▶ 流量計を使って要求供給量 Vw にて **30 秒間**注水してください。

▶ 3～5 分間隔をあけてください(注水条件参照)。

▶ 繰り返し 30 秒の注水を 2 回行ってください。

▶ 遮断弁を閉じてください。

これでタービン洗浄は完了です。

Cleaning parameters　**注水条件**

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TPS	Maximum temperature in front of turbine during cleaning [°C] タービン洗浄中の ガス入口最高温度[℃]	Water flow rate [dm³/min] 水供給量 [dm³/min]	Injection time [min] 注水時間 [min]	Pause between inspections [min] 注水間隔 [min]
48	450	6	0.5 (3 x)	3
52		8		
57		11		
61		17		5

5.4.4 Procedure for method with long water injection time

▷ Prerequisites have been fulfilled.

▶ Wait 5 minutes as stabilizing phase after reducing engine load.

▶ Open shut-off valve.

▶ Set required water flow rate V_W using flowmeter and inject for **10 minutes**.

▶ Close shut-off valve.

▶ Wait 5 minutes as stabilizing phase before increasing engine load again.

This completes the turbine cleaning operation.

5.4.4 長時間注水の手順

▷ 必要条件を満たしてください。

▶ エンジン負荷を下げた後 5 分間静定してください。

▶ 遮断弁を開けてください。

▶ 流量計を使って要求供給量 Vw にて 10 分間注水してください。

▶ 遮断弁を閉じてください。

▶ 再びエンジン負荷を上げ 5 分間静定してください。

これでタービン洗浄は完了です。

Cleaning parameters　**注水条件**

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TPS	Maximum temperature in front of turbine during cleaning [°C] タービン洗浄中のガス入口最高温度[℃]	Water flow rate [dm³/min] 水供給量 [dm³/min]	Injection time [min] 注水時間 [min]	Stabilizing phase before / after water injection [min] 注水前後の静定間隔 [min]
48	450	4	10	5
52		5.5		
57		8.5		
61		12		

6

Troubleshooting

トラブルシューティング

6 Troubleshooting

6 トラブルシューティング

6.1 Possible remedies for failures

6.1 不具合への対処法

Exhaust temperature too high

排気ガス温度が高すぎる

Engine output and speed unchanged

エンジン出力および回転数が一定

Engine

エンジン

▷ Malfunction in the injection system
▶ Repair, or contact the manufacturer.

▷ 燃焼噴射システムの不具合
▶ 修理またはエンジン製造会社に相談する

Turbocharger

過給機

▷ Air starvation, e.g. air filter clogged with dirt
▶ Clean
▷ Compressor / turbine soiled
▶ Clean

▷ 空気不足、例えばエアフィルタが汚れて閉塞
▶ 洗浄する
▷ コンプレッサ / タービンの汚れ
▶ 洗浄する

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▷ Exhaust counterpressure too high
▶ Clean or repair boiler or exhaust silencer

▷ 排気ガスのタービン背圧が高すぎる
▶ ボイラまたは排気ガス消音器の洗浄または修理する

▷ Turbine damaged or worn
▶ Contact an official ABB Turbo Systems service station.

▷ タービンの損傷または摩損
▶ ABB 社認定のサービスステーションへ問い合わせる

Charge-air cooler

空気冷却器

▷ Cooler soiled
▶ Clean

▷ 空気冷却器の汚れ
▶ 洗浄する

▷ Insufficient coolant water quantity
▶ Replenish

▷ 冷却水が少なすぎる
▶ 補充する

▷ Coolant water inlet temperature too hot
▶ Clean/inspect cooling system

▷ 冷却水供給温度が高すぎる
▶ 冷却システムを点検 / 洗浄する

▷ Ventilation inadequate
▶ Improve ventilation

▷ 換気が十分でない
▶ 換気を良くする

Charge-air pressure too low

給気圧力が低すぎる

Engine output and speed unchanged

エンジン出力および回転数が一定

Intake condition normal

通常の吸込条件

Engine

エンジン

▷ Air receiver leaking
▶ Repair

▷ 給気レシーバに漏れ
▶ 修理する

▷ Gas line between engine and turbine leaking
▶ Repair

▷ エンジンとタービンの間の排気管からの漏れ
▶ 修理する

▷ Injection misadjusted
▶ Correct adjustment

▷ 燃料噴射タイミングの調整が不適切
▶ 正しく調整する

▷ Valve control misadjusted
▶ Correct adjustment

▷ バルブ制御の調整が不適切
▶ 正しく調整する

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Turbocharger

過給機

▷ Manometer display faulty
▶ Replace manometer

▷ 圧力計表示が正しくない
▶ 圧力計を交換する

▷ Leak in line to the manometer
▶ Repair leak

▷ 圧力計配管の漏れ
▶ 漏れを修理する

▷ Air filter dirty, causing excessive loss of pressure
▶ Clean

▷ エアフィルタの汚れによる圧力損失上昇
▶ 洗浄する

▷ Compressor / turbine soiled
▶ Clean

▷ コンプレッサ / タービンの汚れ
▶ 洗浄する

▷ Compressor / turbine damaged
▶ Contact an official ABB Turbo Systems service station.

▷ コンプレッサ / タービンの損傷
▶ ABB 社認定のサービスステーションへ問い合わせる

▷ Excessive exhaust counterpressure
▶ Clean boiler or exhaust silencer

▷ 排気ガスのタービン背圧が高すぎる
▶ ボイラまたは排気ガス消音器を洗浄する

Charge-air pressure too high

給気圧力が高すぎる

Engine output and speed unchanged

エンジン出力および回転数が一定

Intake condition normal

通常の吸込条件

Engine

エンジン

▷ Malfunction in the injection system
▶ Correct adjustment

▷ 燃料噴射システムの不具合
▶ 正しく調整する

▷ Engine output higher than expected
▶ Check engine output

▷ エンジン出力が計画より高い
▶ エンジン出力を確認する

▷ Injection misadjusted
▶ Correct adjustment

▷ 燃焼噴射タイミングの調整が不適切
▶ 正しく調整する

Turbocharger

過給機

▷ Manometer display faulty
▶ Replace manometer

▷ 圧力計表示が正しくない
▶ 圧力計を交換する

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Vibrations

振動

Turbocharger

過給機

▷ Rotor unbalance due to heavy fouling of compressor/turbine
▷ Turbine or compressor damaged
▷ Bearing defective
▶ Contact an official ABB Turbo Systems service station.

▷ コンプレッサ / タービンのひどい汚れによるロータのアンバランス
▷ タービンまたはコンプレッサの損傷
▷ 軸受の故障
▶ ABB 社認定のサービスステーションへ問い合わせる

Noises during run-out

エンジン停止から過給機停止までの異常騒音

Turbocharger

過給機

▷ Turbocharger dirty
▶ Clean

▷ 過給機が汚れている
▶ 洗浄する

▷ Bearing damaged
▷ Rotor grazing
▷ Foreign bodies in the turbocharger
▶ Contact an official ABB Turbo Systems service station.

▷ 軸受の損傷
▷ ロータの接触
▷ 過給機に異物が混入している
▶ ABB 社認定のサービスステーションへ問い合わせる

Run-out time too short

Turbocharger

▷ Turbocharger dirty
▶ Clean

▷ Bearing damaged
▷ Rotor grazing
▷ Foreign bodies in the turbocharger
▶ Contact an official ABB Turbo Systems service station.

エンジン停止から過給機停止までの所要時間が短すぎる

過給機

▷ 過給機が汚れている
▶ 洗浄する

▷ 軸受の損傷
▷ ロータの接触
▷ 過給機に異物が混入している
▶ ABB 社認定のサービスステーションへ問い合わせる

Sluggish start-up

Turbocharger

▷ Turbocharger dirty
▶ Clean

▷ Bearing damaged
▷ Rotor grazing
▷ Foreign bodies in the turbocharger
▶ Contact an official ABB Turbo Systems service station.

起動時の回転上昇が遅い

過給機

▷ 過給機が汚れている
▶ 洗浄する

▷ 軸受の損傷
▷ ロータの接触
▷ 過給機に異物が混入している
▶ ABB 社認定のサービスステーションへ問い合わせ

Lubricating oil pressure too low

Engine

▷ Oil filter heavily soiled
▶ Clean

▷ Oil pump in the lubricating system defective
▶ Inspect

▷ Manometer provides false reading
▶ Replace manometer

Turbocharger

▷ Axial clearance of the rotor too big
▶ Contact an official ABB Turbo Systems service station.

潤滑油圧が低すぎる

エンジン

▷ 潤滑油フィルタの極端な汚れ
▶ 洗浄する

▷ 潤滑油循環ポンプの故障
▶ 点検する

▷ 圧力計表示の故障
▶ 圧力計を交換する

過給機

▷ ロータのアキシャルクリアランスが過大
▶ ABB 社認定のサービスステーションへ問い合わせる

NV50F06-1

Constant surging of the turbocharger

Engine

- ▷ Exhaust pressure to the turbine elevated because boiler or exhaust silencer is dirty
- ▷ Trap dirty
- ▶ Clean

Turbocharger

- ▷ Charge-air filter or silencer dirty
- ▷ Heavy deposits of contamination in the turbine
- ▶ Clean

NOTE

If it is not possible to determine the reason for the surging, contact an official ABB Turbo Systems service station.

過給機が連続的にサージングする

エンジン

- ▷ ボイラまたは排気ガス消音器の汚れによるタービン背圧の圧力上昇
- ▷ キャッチグリッドの汚れ
- ▶ 洗浄する

過給機

- ▷ 空気冷却器または消音器の汚れ
- ▷ タービンの極端な汚れの堆積
- ▶ 洗浄する

注記

サージングの原因が特定できない場合は、ABB 社認定のサービスステーションへお問い合わせください。

NV50F06-1

7

Removal and installation

取外しと搭載

7 Removal and installation

7 取外しと搭載

7.1 Turbocharger mass

7.1 過給機の質量

WARNING

Carefully fasten individual parts and larger assemblies/modules to suitable hoists/lifting devices which are in technically perfect condition and have adequate load-bearing capacity.

Loads that are not suspended in accordance with regulations during disassembly and assembly might fall and cause serious or even fatal injury.

- ▶ Select a suitable rope taking into account mass of load.
- ▶ Hang rope crossed over on the crane hook.
- ▶ Don't stand under suspended loads.

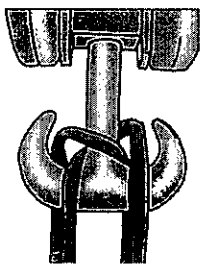
警告

個々のパーツおよび大きな組立ユニットは、適切な荷重能力を持つ万全な吊上装置に入念に固定してください。

過給機の取外しおよび搭載時に、不適切な吊上装置を使用した場合には、吊上げた部品の落下により、重大または致命的な人身事故につながることがあります。

- ▶ 吊上げる部品の質量を考慮して適切なワイヤロープを使用してください。
- ▶ ロープを交差させてクレーンのフックに吊下げてください。
- ▶ 吊上げた部品の下には立たないでください。

NV50F07-1.R1



I_01308

CAUTION

Work that exceeds the scope of the description in this chapter must only be performed by a trained fitter from an official ABB Turbo Systems service station.

注意

この章の記述を超える作業については、ABB 社サービスステーションの認定作業者のみに作業させてください。

7.1.1 Turbocharger mass TPS 48-61

7.1.1 過給機の質量 TPS48-61

Complete turbocharger without silencer / air suction branch and gas outlet elbow

過給機完備品(消音器/空気吸込囲およびガス出口エルボなし)

Mass [kg]
質量[kg]

TPS 48	TPS 52	TPS 57	TPS 61
110	160	260	450

NV50F07-1.R1

Complete turbocharger without gas outlet elbow and with silencer

過給機完備品(消音器付・ガス出口エルボなし)

Mass[kg]
質量[kg]

TPS 48	TPS 52	TPS 57	TPS 61
129	187	300	512

Complete turbocharger without gas outlet elbow and with air suction branch (radial)

過給機完備品(空気吸込囲付・ガス出口エルボなし)

Mass [kg]
質量[kg]

TPS 48	TPS 52	TPS 57	TPS 61
116	168	271	471

The value on the rating plate is the heaviest possibility in the turbocharger type depending on the specifications.

銘板には、同一形式の中で最も質量の大きい仕様の値が刻印されています。

7.2 Remove the turbocharger

▶ Disconnect all gas and air lines in accordance with the engine manufacturer's instructions.

If provided

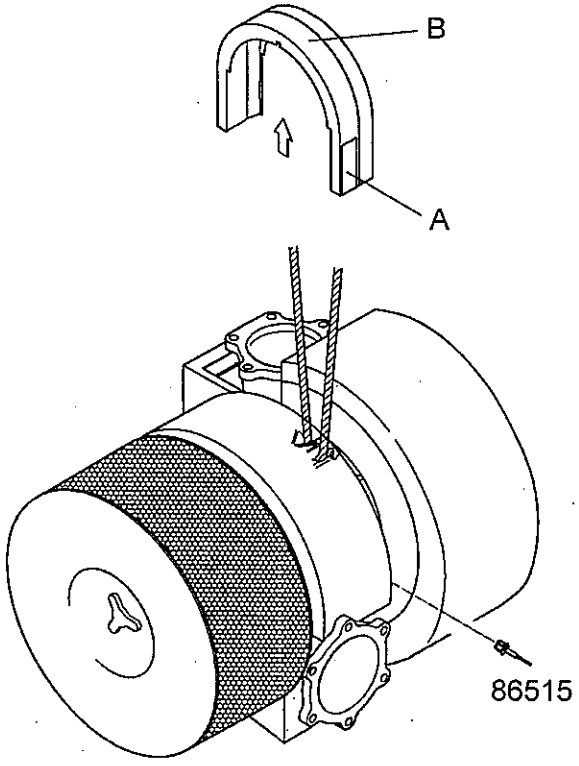
Remove hard insulation as follows :

7.2 過給機の取外し

▶ エンジン製造会社の取扱説明書に従って、ガスおよび空気配管を取外してください。

供給されている場合

次のようにインシュレーションを取外してください:



NV50F07-1.R1

S_00098

▶ Remove screws and sheet-metal covering (A) from bearing casing insulation.

▶ Remove remaining screws from bearing casing insulation (B).

▶ Pull bearing casing insulation up and off. Squeeze bearing casing insulation together for easier removal.

▶ Unplug cable connector (86515) of speed sensor.

▶ Check lifting equipment.

▶ 軸受ケーシング用インシュレーションからボルトとメタルカバー(A)を取外してください。

▶ 軸受ケーシング用インシュレーション(B)の残りのボルトを取外してください。

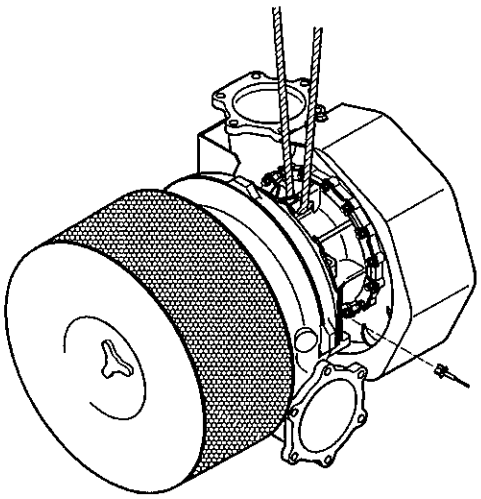
▶ 軸受ケーシング用インシュレーションを持ち上げて取外してください。押しながら持ち上げるとより簡単に取外せます。

▶ 回転計のケーブルコネクタ(86515)を取外してください。

▶ 吊上装置を点検してください。

Turbocharger without gas outlet elbow

ガス出口エルボなし過給機の場合



S_00099

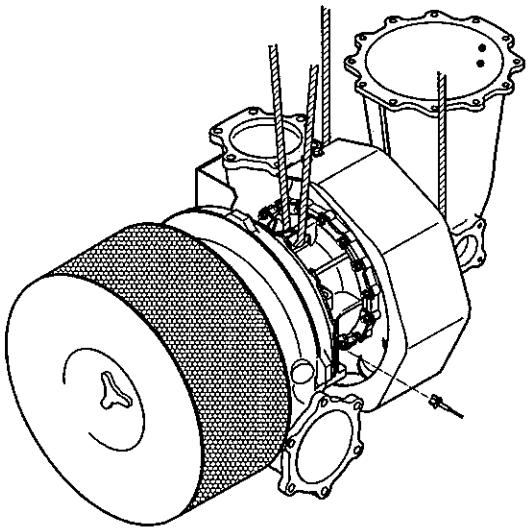
NV50F07-1.R1

▶ Fasten lifting equipment to bearing casing.

▶ 軸受ケーシングにワイヤロープを確実に取付けてください。

Turbocharger with gas outlet elbow without insulation:

ガス出口エルボ付、インシュレーションなし過給機の場合



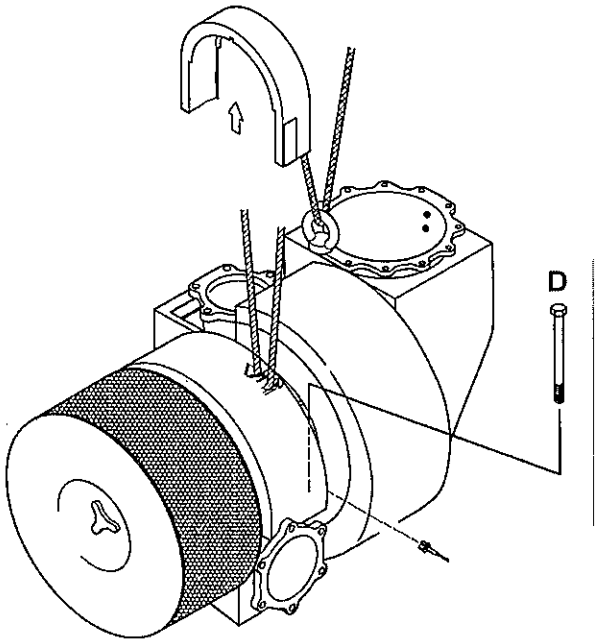
S_00100

▶ Loop lifting equipment around gas outlet elbow and fasten securely to bearing casing.

▶ ガス出口エルボの周りにワイヤロープを回し、また軸受ケーシングにワイヤロープを確実に取付けてください。

Turbocharger with gas outlet elbow and hard insulation:

ガス出口エルボ付、インシュレーション付過給機の場合



S_00101

NV50F07-1.R1

▶ Fasten lifting equipment to flange of gas outlet elbow and to bearing casing using a lifting eye bolt and nut.

▶ ガス出口エルボのフランジにアイボルトおよびナットを取付け、そのアイボルトと軸受ケーシングにワイヤロープを確実に取付けてください。

WARNING

During removal, a turbocharger that is not lifted in compliance with regulations could fall and cause serious or even fatal injury.

▶ Select a suitable rope taking into account the mass of turbocharger.

警告

取外し中に規定外の方法で過給機を吊上げた場合には、落下により重大または致命的な人身事故につながることがあります。

▶ 過給機の質量を考慮して適切なワイヤロープを使用してください。

▶ Loosen fixing screws (D) on bearing casing.

▶ Lift turbocharger off engine and put it down to one side.

▶ Cover oil connections.

▶ 軸受ケーシングのボルト(D)をゆるめてください。

▶ エンジンから過給機を取外し、適切な場所に置いてください。

▶ 潤滑油配管接続部に覆いをしてください。

7.3 Installing the turbocharger

WARNING

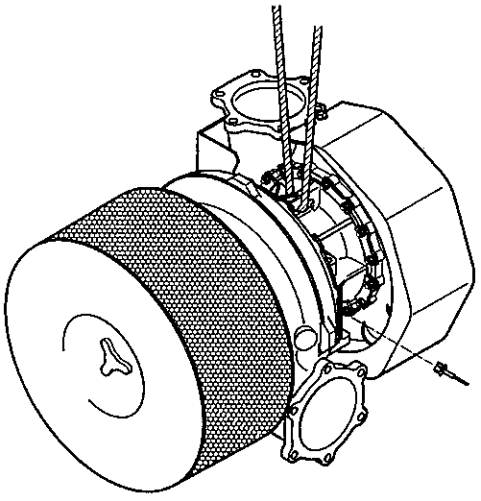
During installation work, a turbocharger that is not lifted in compliance with regulations could fall and cause a serious or even fatal injury.

▶ Select a suitable rope taking into account the mass of the turbocharger.

- ▶ Remove covers from the oil connections.
- ▶ Visually inspect O-ring seals of oil supply and drain pipe. (O-ring seals for engine are not included in scope of delivery from ABB Turbo Systems or TSU).
- ▶ Check lifting equipment.

NV50F07-1.R1

Turbocharger without gas outlet elbow



S_00099

- ▶ Fasten lifting equipment to bearing casing.

7.3 過給機の搭載

警告

組付け中に規定外の方法で過給機を吊上げた場合には、落下により重大または致命的な人身事故につながることがあります。

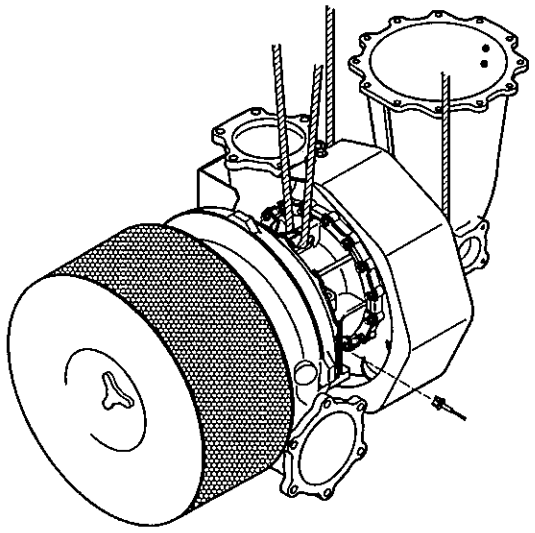
▶ 過給機の質量を考慮して適切なワイヤロープを使用してください。

- ▶ 潤滑油配管接続部の覆いを取外してください。
- ▶ 給排油配管の O リングを目視検査してください。(エンジン側 O リングは ABB 社または、TSU 社の供給範囲ではありません)
- ▶ 吊上装置を点検してください。

ガス出口エルボなし過給機の場合

- ▶ 軸受ケーシングにワイヤロープを確実に取付けてください。

Turbocharger with gas outlet elbow without insulation:　**ガス出口エルボ付、インシュレーションなし過給機の場合**



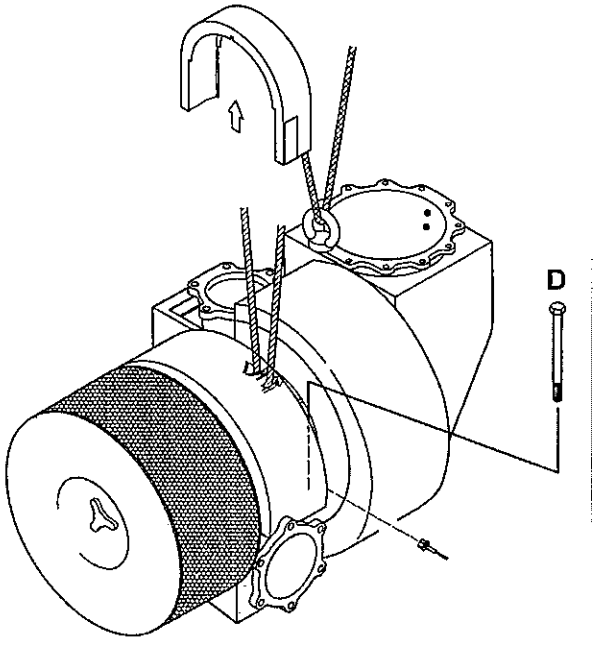
S_00100

NV50F07-1.R1

▶ Loop lifting equipment around gas outlet elbow and securely fasten to bearing casing.

▶ ガス出口エルボの周りにワイヤロープを回し、また軸受ケーシングにワイヤロープを確実に取付けてください。

Turbocharger with gas outlet elbow and hard insulation:　**ガス出口エルボ付、インシュレーション付過給機の場合**



S_00101

- ▶ Fasten lifting equipment to flange of gas outlet elbow and to bearing casing using a lifting eye bolt and nut.
- ▶ Adapt turbocharger to suit engine.
- ▶ Tighten fixing screws (D) on bearing casing.
- ▶ Plug in cable connector (86515) of speed sensor.

- ▶ ガス出口エルボのフランジにアイボルトおよびナットを取付け、そのアイボルトと軸受ケーシングにワイヤロープを確実に取付けてください。
- ▶ 過給機をエンジンに据えつけてください。
- ▶ 軸受ケーシングにボルト(D)を締めこんでください。
- ▶ 回転計のケーブルコネクタ(86515)を差し込んでください。

If provided

Fasten hard insulations as follows:

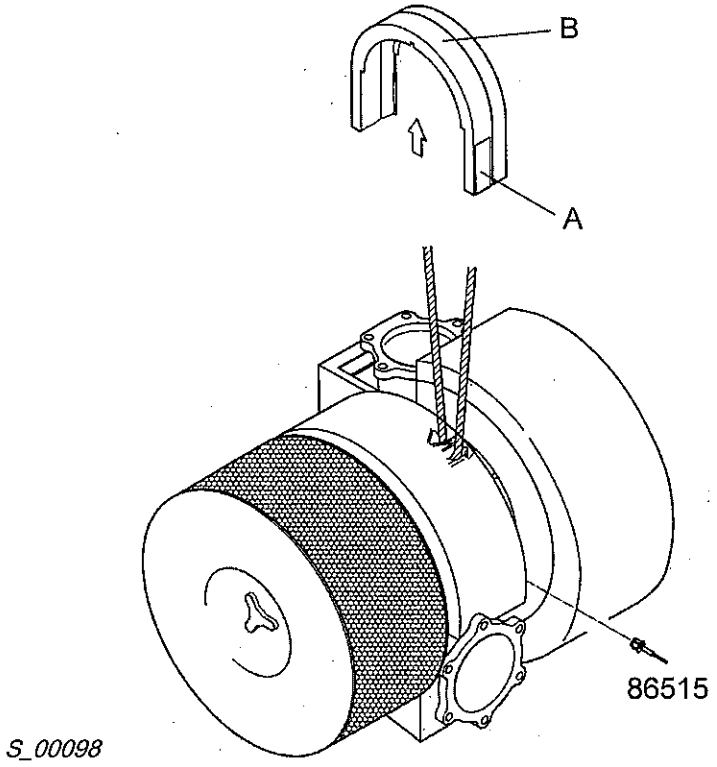
- ▶ Adapt bearing casing insulation to suit.

供給されている場合

次のようにインシュレーションをしっかりと取付けてください:

- ▶ 軸受ケーシングにインシュレーションを取付けてください。

NV50F07-1.R1



S_00098

- ▶ Tighten screws of bearing casing insulation (B).
- ▶ Adapt screws and sheet-metal covering (A) of bearing casing insulation to suit.
- ▶ Fasten all gas and air lines in accordance with engine manufacturer's instructions.

- ▶ 軸受ケーシング用インシュレーション (B) のボルトを締め付けてください。
- ▶ 軸受ケーシング用インシュレーションのメタルカバー(A) とボルトを取付けてください 。
- ▶ エンジン製造会社の取扱説明書に従って全てのガスおよび空気配管をしっかりと取付けてください。

8

Disassembly and assembly

分解と組立

8 Disassembly and assembly

8.1 Module mass

WARNING

Carefully fasten individual parts and larger assemblies/modules to suitable hoists/lifting devices which are in technically perfect condition and have adequate load-bearing capacity.

Loads that are not suspended in accordance with regulations during disassembly and assembly might fall and cause serious or even fatal injury.

- ▶ Select a suitable rope taking into account mass of load.
- ▶ Hang rope crossed over on crane hook.
- ▶ Don't stand under suspended loads.

NV50F08-1.R1

8 分解と組立

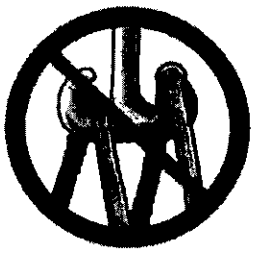
8.1 構成部品の質量

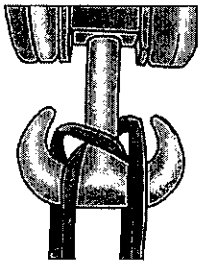
警告

個々のパーツおよび大きな組立ユニットは、適切な荷重能力を持つ万全な吊上装置に入念に固定してください。

過給機の取外しおよび搭載時に、不適切な吊上装置を使用した場合には、吊上げた部品の落下により、重大または致命的な人身事故につながることがあります。

- ▶ 吊上げる部品の質量を考慮して適切なワイヤロープを使用してください。
- ▶ ロープを交差させてクレーンのフックに吊下げてください。
- ▶ 吊上げた部品の下には立たないでください。





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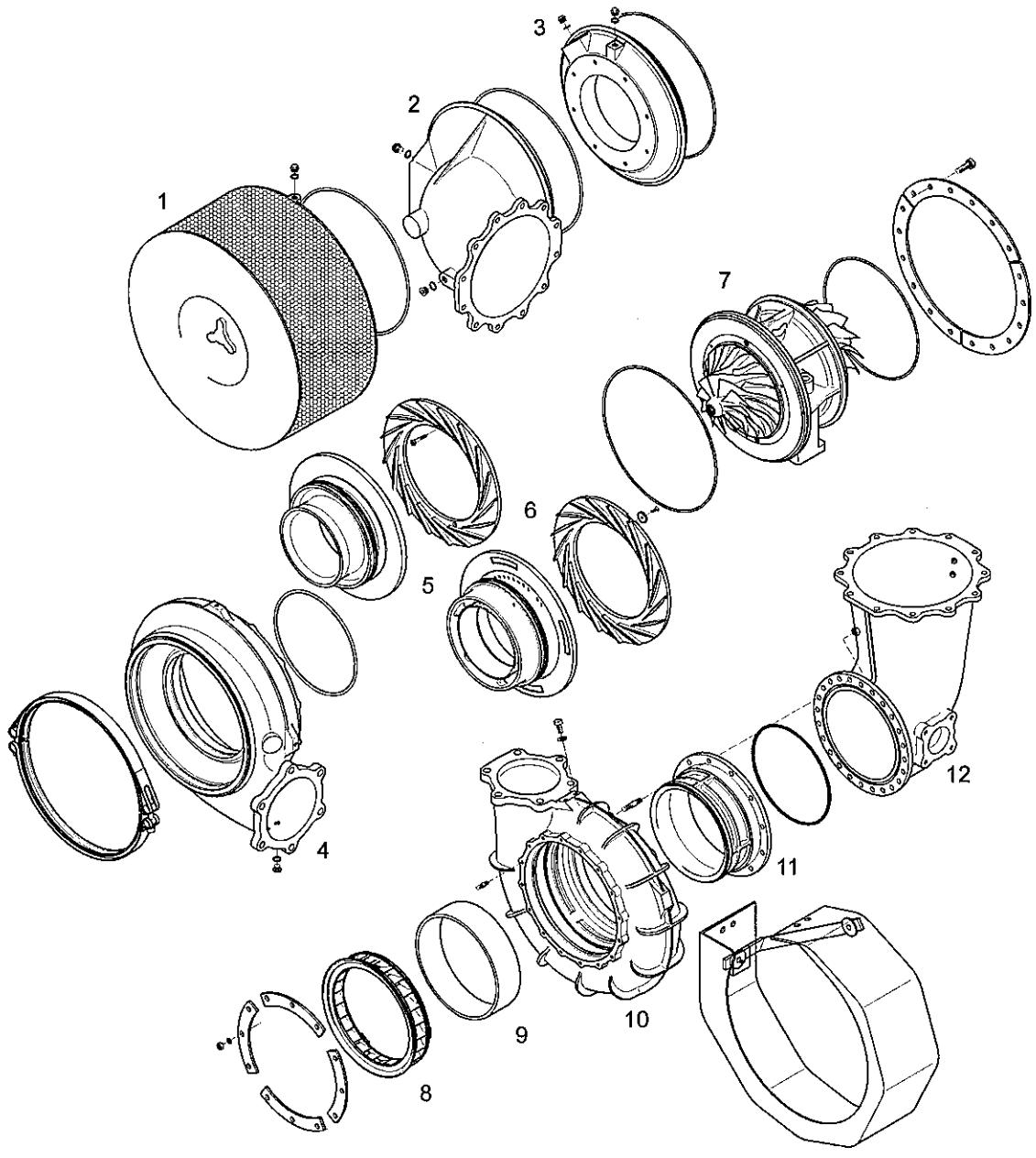
Modules

構成部品



S_00945

NV50F08-1.R1

Mass [kg]
質量[kg]

	Description / 名称		TPS 48	TPS 52	TPS 57	TPS 61
1	Silencer 消音器		19	27	40	62
2	Air suction branch, radial 空気吸込囲(径方向流入タイプ)		6	8	11	21
3	Air suction branch, axial 空気吸込囲(軸方向流入タイプ)		4	4	6	9
4	Compressor casing コンプレッサケーシング		24	35	51	89
5	Wall insert 扇車覆		7	10	13	30
6	Diffuser ディフューザ		2	3	6	10
7	Cartridge group センターカートリッジ部		32	48	83	136
8	Nozzle ring ノズルリング		1	2	3	6
9	Burst ring バーストリング		2	3	5	12
10	Turbine casing with external burst protection タービンケーシング(バーストプロテクション付)	1 inlet 1 口	32	47	77	130
		2 inlets 2 口	35	50	86	145
		3 inlets 3 口	-	53	93	162
		4 inlets 4 口	-	-	93	162
11	Gas outlet flange ガス出口フランジ		4	8	16	30
12	Gas outlet elbow ガス出口エルボ		18	25	40	56

NV50F08-1.R1

8.2 Disassembling and assembling turbocharger

8.2 過給機の分解と組立

NOTE

Not all tools are marked with a part number. The tool list ensure proper identification of the tools (list enclosed in the toolbox).

注記

要具には部品番号が記されていないものがあります。要具表を参照して要具を選択してください（表は要具箱に入っています。）

WARNING

Attach individual parts and larger component modules carefully to suitable hoists / lifting devices which are in technically perfect condition and which have adequate load-bearing capacity.

Turbocharger parts that are not suspended in accordance with the regulations during disassembly and assembly may fall and can cause serious or even fatal injury.

- ▶ Select a suitable rope taking into account the mass of the turbocharger parts.
- ▶ Do not stand under suspended loads.

警告

個々のパーツおよび大きな組立ユニットは、適切な荷重能力を持つ万全な吊上装置に入念に固定してください。

過給機の取外しおよび搭載時に、不適切な吊上装置を使用した場合には、過給機部品の落下により、重大または致命的な人身事故につながることがあります。

- ▶ 吊上げる部品の質量を考慮して適切なワイヤロープを使用してください。
- ▶ 吊上げた部品の下には立たないでください。

NV50F08-1.R1

Removing compressor casing

- ▶ Disconnect all air ducts in accordance with engine manufacturer's instructions.

If provided

- ▶ Remove bolts of compressor casing insulation and remove insulation.
- ▶ Loosen V-band (72020) and remove silencer (81000) or air suction branch (82000).

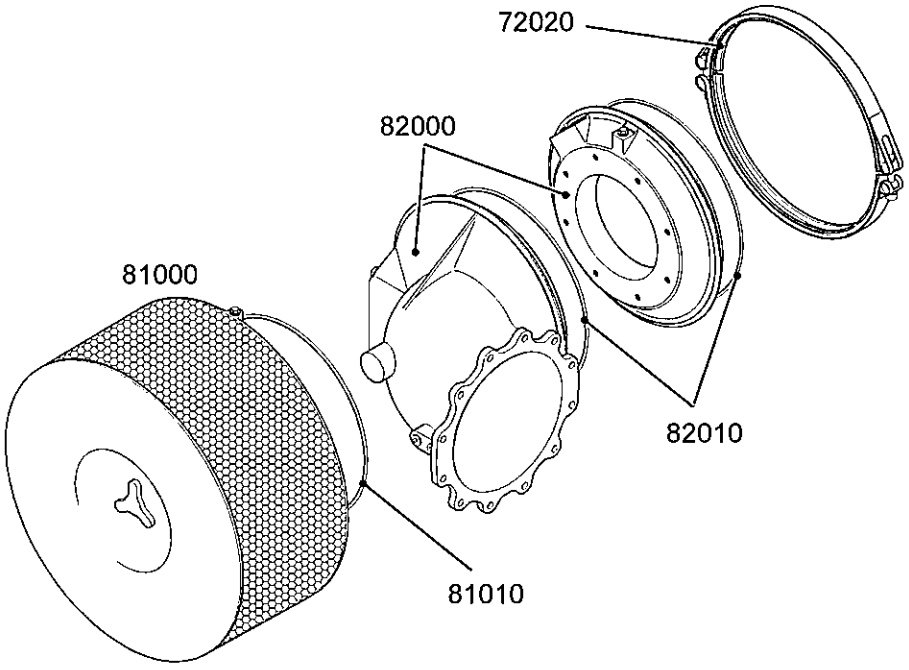
コンプレッサケーシングの取外し

- ▶ エンジン製造会社の取扱説明書に従って全ての空気配管を取外してください。

供給されている場合

- ▶ コンプレッサケーシング用インシュレーションのボルトを取外し、インシュレーションを取外してください。
- ▶ V 形クランプ(72020)をゆるめ、消音器(81000)または空気吸込囲(82000)を取外してください。

NV50F08-1.R1

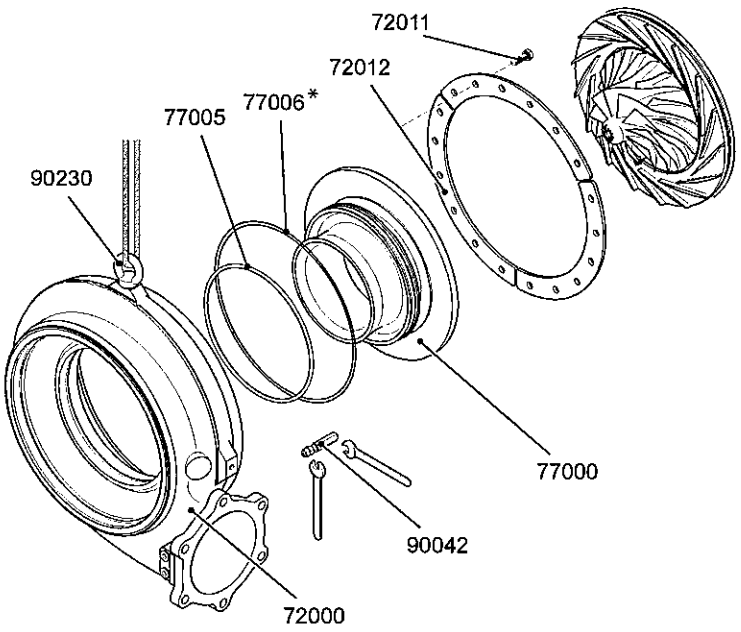


S_00103

81000	Silencer 消音器	**82000**	Air suction branch 空気吸込囲

▶ Slightly loosen screws (72011) and turn compressor casing (72000) until lifting eye bolt (90230) can be fastened to lifting equipment.

▶ ボルト(72011)を少しゆるめ、吊上用アイボルト(90230)に吊上装置がかけられる位置までコンプレッサケーシング(72000)を回転させてください。



S_00114

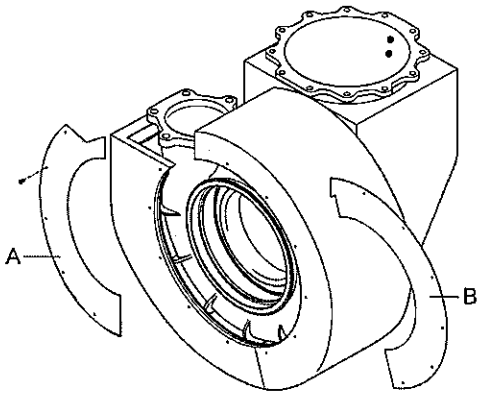
NV50F08-1.R1

(77006* only provided with wall insert with acceleration aid)

(77006* は加速補助機能付きのみ)

If provided

供給されている場合



NOTE

If the compressor casing cannot be removed easily, it can be pressed off using the press-off tool (90042) against the turbine casing. To do this, remove the two cover plates (A+B) from the hard insulation on the turbine casing.

注記

コンプレッサケーシングを簡単に取外せない場合、押上要具(90042)でタービンケーシングから押し出すことができます。その際は、タービンケーシング用インシュレーションの 2 枚のメタルカバー(A+B)を取外す必要があります。

CAUTION

Axial force

The press off tool can generate a high axial force and, if it is not used properly (too much pressure on one side), it can damage the rotor.

- ▶ It should thus be used alternately on both sides while avoiding excessive pressure on either side.

注意

軸方向の力

押上要具により軸方向に大きな力がかかることがあります。要具が適切に使用されていない場合(片側にかかる圧力が高すぎる)はロータが損傷する可能性があります。

- ▶ 従ってどちらか一方だけ圧力が高すぎるということがないように両側を交互に押し出してください。

NV50F08-1.R1

- ▶ Remove screws (72011) and clamping straps (72012).
- ▶ Move compressor casing (72000) with wall insert (77000) away carefully and horizontally.
- ▶ Tap wall insert (77000) out of compressor casing using nylon hammer and remove O-ring seal (77005).

- ▶ ボルト(72011)をゆるめ押さえ板(72012)と共に取外してください。
- ▶ 扇車覆(77000)とコンプレッサケーシング(72000)を慎重に水平移動してください。
- ▶ プラスチックハンマーで軽く叩きながらコンプレッサケーシングから扇車覆(77000)を取外してください。O リング(77005)も取外してください。

TPS61

Use lifting eye blots (90255) to lift wall insert (77000).

TPS61

リングボルト(90255)を使用し、扇車覆(77000)を持ち上げてください。

CAUTION

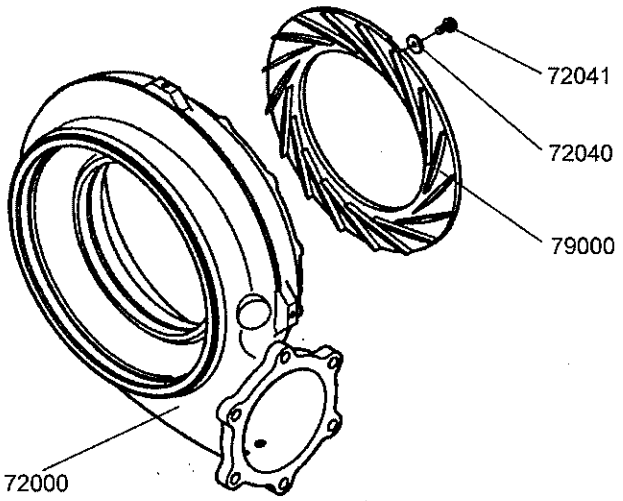
Work that exceeds the scope of the description in this chapter must only be performed by a trained fitter from an official ABB Turbo Systems service station.

注意

この章の記述を超える作業については、ABB 社サービスステーションの認定作業者のみに作業させてください。

Diffuser disassembly
TPS..-F31/32 / FV31/32

ディフューザの取外し
TPS..-F31/32 / FV31/32



S_00115

NV50F08-1.R1

▶ Unscrew bolts (72041) and remove them from compressor casing (72000) with their locking discs (72040) and diffuser (79000).

▶ ボルト(72041)をゆるめ、ガイドワッシャ(72040)とディフューザ(79000)をコンプレッサケーシング(72000)から取外してください。

Installing compressor casing

コンプレッサケーシングの取付け

▶ Re-fit wall insert (77000) and O-ring (77005) in reverse order.

▶ 扇車覆(77000)と O リング(77005)を逆の順序で取付けてください。

NOTE

Always replace O-ring seals with new ones.(See section **Reserve and spare parts**)

注記

O リングは常に新しいものと交換してください。(「**予備品**」の項をご参照ください)

Fitting the diffuser TPS..-F31/32/FV31/32

ディフューザの組付け TPS..-F31/32/FV31/32

▶ Fit diffuser in reverse order.

▶ ディユーザも逆の順序で取付けてください。

▶ Mount compressor casing in reverse order.

▶ コンプレッサケーシングも逆の順序で取付けてください。

▶ Thoroughly clean clamping straps(72012) before fitting them.

▶ 押さえ板(72012)は取付ける前に十分きれいにしてください。

Removing cartridge group

センターカートリッジ部の取外し

▶ Check lifting equipment and fasten to bearing casing(42001).

▶ 吊上装置を点検し、軸受ケーシング(42001)に取付けてください。

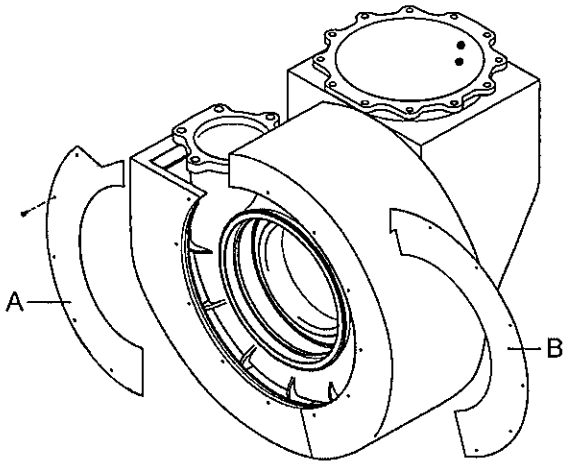
If provided

供給されている場合

▶ Disconnect all lines in accordance with engine manufacturer's instructions and remove insulation, turbocharger, compressor casing and filter silencer (see also previous chapter).

▶ エンジン製造会社の取扱説明書に従って各接続配管から切り離し、インシュレーションを取外した後、過給機をエンジンから取外し、消音器およびコンプレッサケーシングを取外してください。(前章参照)

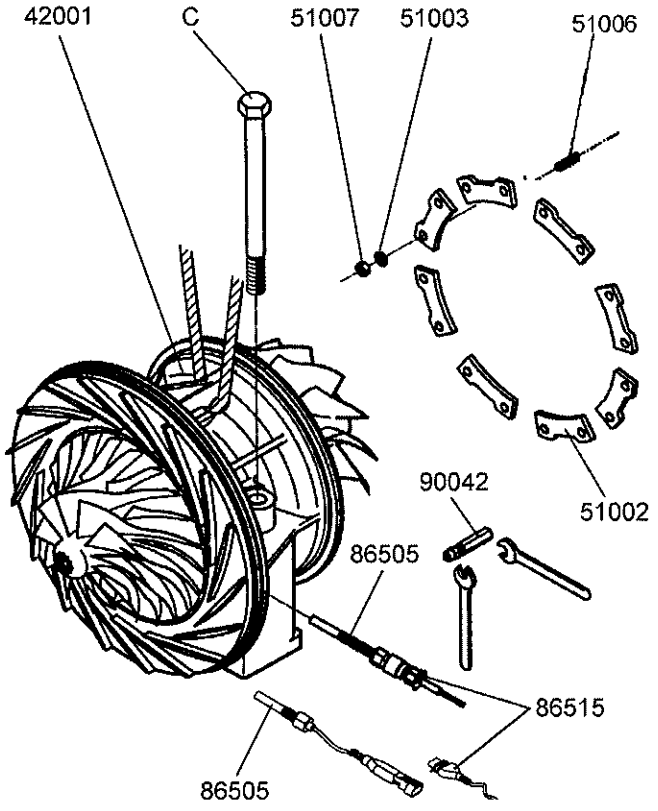
NV50F08-1.R1



S_00105

▶ Remove screws from plates (**A+B**) of hard insulation on turbine casing and take away panels.

▶ タービンケーシング用インシュレーションの 2 枚のメタルカバー(A+B)にあるボルトを取外し、板を取外してください。



S_00117

NV50F08-1.R1

If provided

- ▶ Unplug cable connector (86515) of speed sensor (86505).
- ▶ Apply penetrating oil to stud threads (51006) and allow to act.
- ▶ Unscrew nuts (51007) and remove Ver-busripp wahers (51003) with clamping straps (51002).
- ▶ Unscrew and remove bearing casing fixing screws (C) (42001).
- ▶ Remove cartridge group.

- ▷ Bearing casing is difficult to remove.
- ▶ Use press-off tool (90042) against turbine casing to press off bearing casing (see following illustration).

供給されている場合

- ▶ ケーブルコネクタ(86515)を回転計ピックアップ(86505)から抜いてください。
- ▶ スタッドねじ(51006)のねじ部に油(浸透効果のある)をさし、効果が現れるまで十分浸透させてください。
- ▶ ナット(51007)をゆるめ、特殊ワッシャ(51003)と押さえ板(51002)を取外してください。
- ▶ 軸受ケーシング(42001)の固定ボルト(C)をゆるめて取外してください。
- ▶ センターカートリッジ部を取外してください。

- ▷ 軸受ケーシングを簡単に取外すことができない場合。
- ▶ 押上要具(90042)でタービンケーシングから軸受ケーシングを押し出すことができます。(写真参照)

CAUTION

Axial force

The press-off tool can generate a high axial force and, if not used properly (too much pressure on one side), it can damage the rotor.

▶ In view of this, use it alternately on both sides, but don't exert excessive pressure on either side.

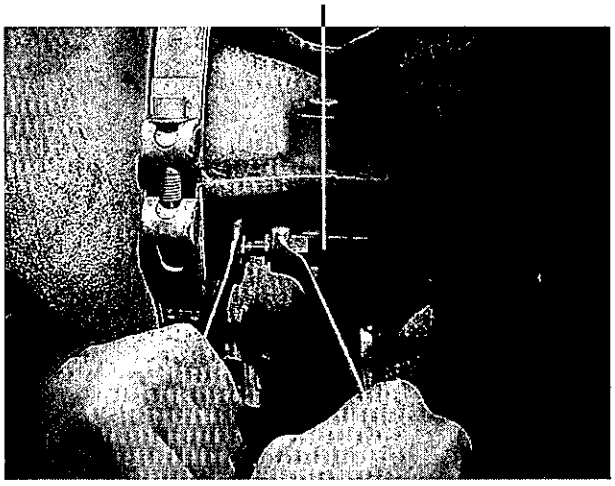
注意

軸方向の力

押上要具により軸方向に大きな力がかかることがあります。要具が適切に使用されていない場合(片側にかかる圧力が高すぎる)はロータが損傷する可能性があります。

▶ 従ってどちらか一方だけ圧力が高すぎるということがないように両側を交互に押し出してください。

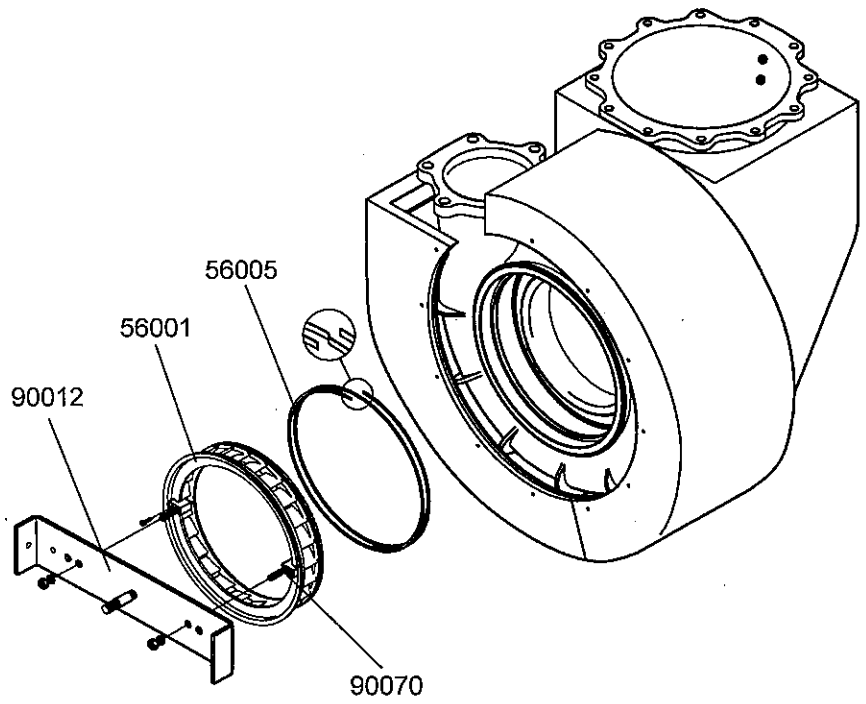
NV50F08-1.R1



S_00128

▶ Cover oil connections.

▶ 潤滑油配管接続部をカバーしてください。



S_00107

NV50F08-1.R1

▶ Withdraw nozzle ring (56001) using two extractors (90070) and bottom part of service support (90012).

▶ 2 つの引抜要具(90070)および支え板(90012)を用いノズルリング(56001)を引き抜いてください。

If provided

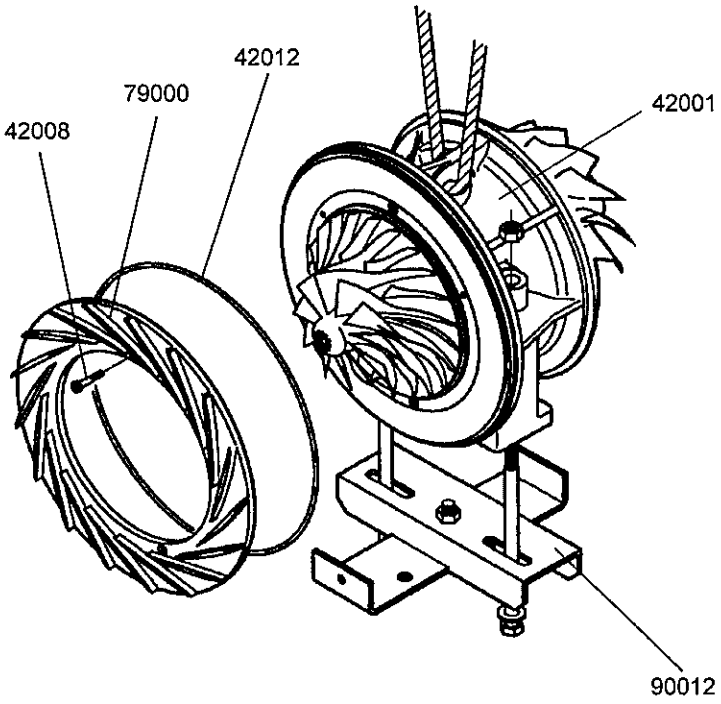
▶ Remove lamellar sealing ring (56005).

供給されている場合

▶ 特殊パッキン(56005)を取外してください。

Removing diffusor
TPS..-F33

ディフューザの取外し
TPS..-F33



S_00110

NV50F08-1.R1

- ▶ Mount cartridge group on service support (90012) and bolt firmly.
- ▶ Unscrew bolt (42008) and remove diffuser (79000) and O-ring seal (42012).
- ▶ Measure axial and radial clearances (see section **Axial and Radial Clearances**).

- ▶ 支え板(90012)にセンターカートリッジ部を置き、きつくボルトを締付けてください。
- ▶ ボルト(42008)を取外し、ディフューザ(79000)およびOリング(42012)を取外してください。
- ▶ アキシャルおよびラジアルクリアランスを計測してください。(「**アキシャルおよびラジアルクリアランス**」の項をご参照ください。)

Installing the cartridge group

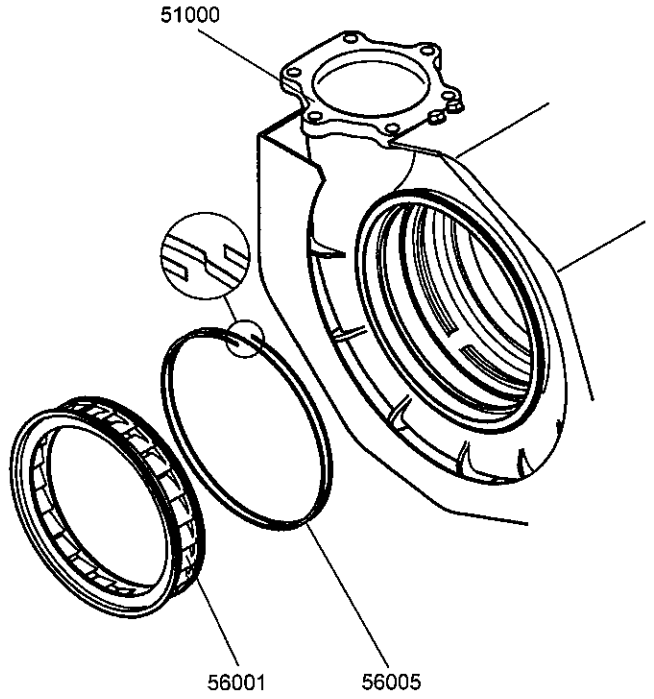
センターカートリッジの組付け

NOTE

Always replace O-ring seals with new ones. (See section **Reserve and spare parts**)

注記

Oリングは常に新しいものと交換してください。(「**予備品**」の項をご参照ください)



S_00109

NV50F08-1.R1

If provided

供給されている場合

▶ Fit lamellar gasket ring (56005).

▶ 特殊パッキン(56005)を取付けてください。

NOTE

Take care to coil the lamellar sealing ring (56005) correctly.

注記

特殊パッキン(56005)の巻き形状を確認し正しく組付けてください。

▶ Push nozzle ring (56001) into turbine casing (51000) as for as it will go.

▶ Make sure cams are in right position.

▶ タービンケーシング(51000)の中へノズルリング(56001)を挿入し、止まるまでスライドさせてください。

▶ ノズル突起部が正しい位置にあるか確認してください。

NOTE

To ensure the nozzle ring is fixed during operation, it must be clamped between the partition wall and the turbine casing.

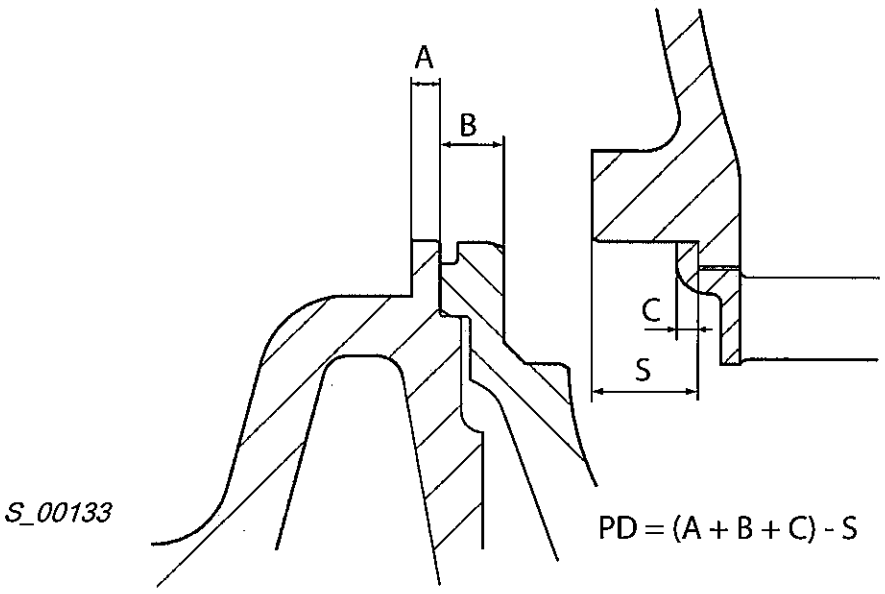
注記

ノズルリングは隔壁とタービンケーシングの間にはさまれて固定されます。

▶ Calculate press fit (PD), as shown.

▶ 図のように、圧縮幅(PD)を計算してください。

NV50F08-1.R1



NOTE

If the calculated value (PD) is less than **0.1 mm**, an official ABB Turbo Systems service station must be contacted.

注記

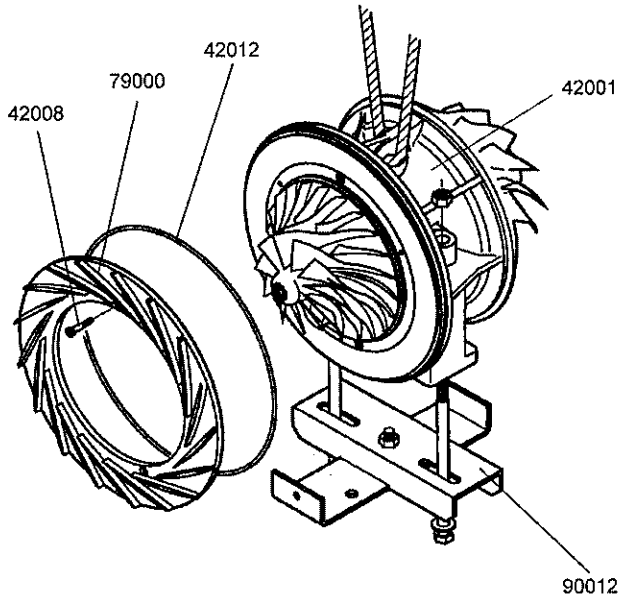
計算結果が 0.1mm よりも小さい場合は、ABB 社サービスステーションへ連絡してください。

▶ Measure axial and radial clearances (see chapter **Axial and radial clearances**).

▶ アキシャルおよびラジアルクリアランスを計算してください。(「**アキシャルおよびラジアルクリアランス**」の章をご参照ください。)

Fitting diffuser
TPS..-F33

ディフューザの取付け
TPS..-F33



S_00110

NV50F08-1.R1

▶ Secure diffuser (79000) and O-ring seal (42012) using bolt (42008).

▶ ディフューザ(79000)および O リング(42012)をボルト(42008)で固定してください。

NOTE

See section "Removing rotor".

注記

「ロータの取外し」の項をご参照ください。

▶ Check lifting equipment and fasten to bearing casing (42001).

▶ Unscrew cartridge group and lift it off service support (90012).

▶ Apply a layer of Antiscuff (or similar high-temperature lubricating paste) to self-centring mounts and screw threads.

▶ Remove temporary covers from oil connections.

▶ Fit cartridge group as far as possible by hand.

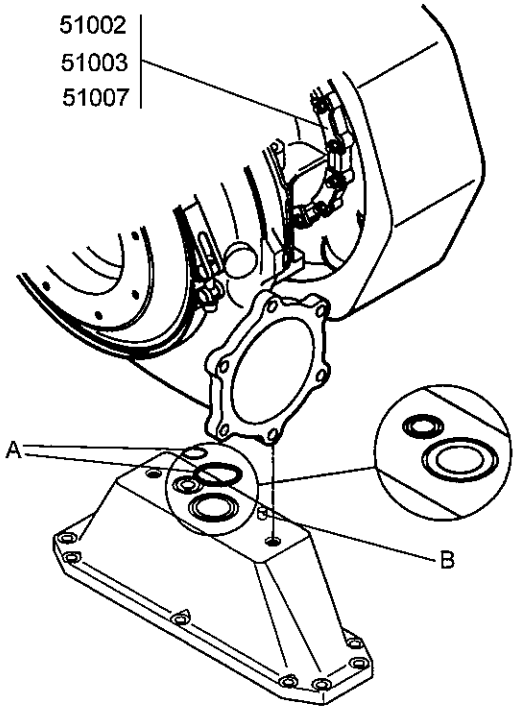
▶ 吊上装置を点検し、軸受ケーシング(42001)に取付けてください。

▶ 取付ボルトを取外しセンターカートリッジを支え板(90012)から持ち上げてください。

▶ ボルトのねじ部に焼付防止剤または耐熱性潤滑剤を塗布してください。

▶ 潤滑油配管接続部から、分解時に一時的に取付けたカバーを取外してください。

▶ センターカートリッジはできるだけ手作業で取付けてください。



S_00118

NV50F08-1.R1

CAUTION

Take care not to damage or move gasket rings (A) in the mounting support when fitting the cartridge group. Gasket rings (A) belong to the engine and are not included in the scope of delivery from ABB Turbo Systems or TSU.

注意

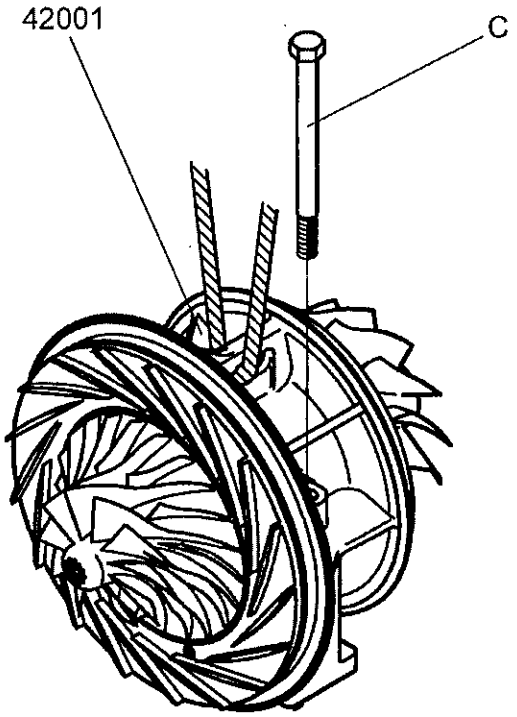
センターカートリッジを取付ける時に取付台上のパッキン(A)が損傷したり、動いたりしないように注意してください。パッキン(A)はエンジン側の供給品であり、ABB 社および TSU 社の供給範囲外です。

NOTE

If two turbochargers are mounted on one engine, one with a left-hand and one with a right-hand oil inlet, a dowel can be fitted, as a precaution, to ensure correct positioning in the support. This dowel (B) fits in the respective groove on the foot of the bearing casing.

注記

エンジン上に 2 台の過給機が搭載され、各々が左右の異なる油入口位置である場合、誤組立てを防ぐために取付台にピン(B)を立てることができます。このピン(B)が軸受ケーシングの取付脚部の相対する穴に合います。



S_00113

NV50F08-1.R1

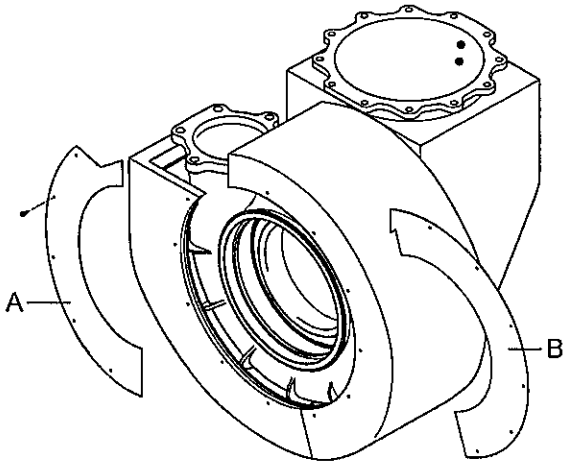
- Secure cartridge group using clamping straps (51002) and nuts (51007) with Verbusripp washers (51003) (see also chapter Tightening torques table).
- Secure bearing casing (42001) using fixing screws (C) .
- Tighten fixing screws (C) to tightening torques listed in following table.

- センターカートリッジ部が、押さえ板(51002)、ナット(51007)、特殊ワッシャ(51003)により、しっかりと取付けられていることを確認してください。(「締付けトルク」の章をご参照ください。)
- 軸受ケーシング(42001)を固定ボルト(C)で固定してください。
- 下表の締付けトルクでボルト(C)を締付けてください。

TPS	Fixing screws C [mm] 固定ボルト C [mm]	Tightening torques [Nm] 締付けトルク [Nm]
48	M16	230
52	M20	455
57	M20	455
61	M24	780

If provided

供給されている場合



S_00105

NV50F08-1.R1

▶ Plug in cable connector(86515) of speed sensor (86505) and fit insulation plates (A/B).

▶ ケーブルコネクタ(86515)を回転計ピックアップ(86505)に差し込み、インシュレーションメタルカバー(A/B)を取付けてください。

CAUTION

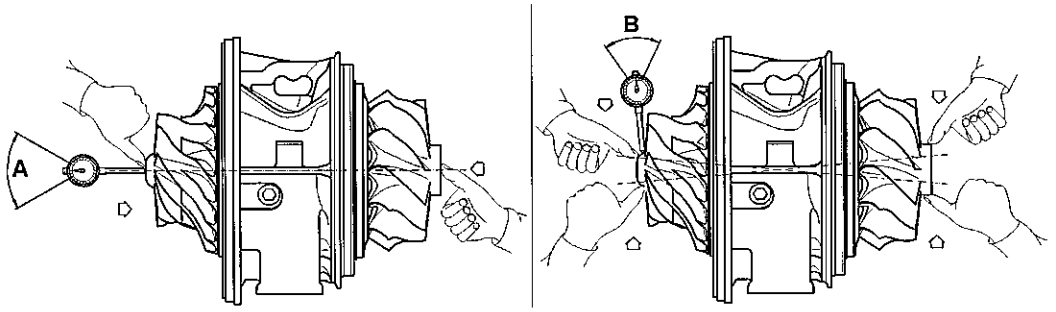
Work that exceeds the scope of the description in this chapter must only be performed by a trained fitter from an official ABB Turbo Systems service station.

注意

この章の記述を超える作業については、ABB 社サービスステーションの認定作業者のみに作業させてください。

8.3 Axial clearance A and possible radial movement B

8.3 ロータの軸方向遊び A と径方向遊び B



S_00119

NV50F08-1.R1

▶ After removal and before installation of cartridge group, measure and note axial clearance A and radial movement B.

▶ センターカートリッジ部を取外した後および取付ける前にロータの軸方向遊び A および径方向遊び B を計測し、記録してください。

NOTE

In order to measure axial clearance A correctly, the turbine must be raised slightly.

注記

ロータの軸方向遊び A を正しく計測する為に、タービン側を軽く持ち上げてください。

Axial and radial clearances[mm]

アキシャルおよびラジアルクリアランス[mm]

	TPS48	TPS52	TPS57	TPS61
A min. A 最小値	0.08	0.10	0.12	0.15
A max. A 最大値	0.16	0.18	0.21	0.25
B min. B 最小値	0.50	0.61	0.72	0.81
B max. B 最大値	0.99	1.15	1.31	1.55

CAUTION

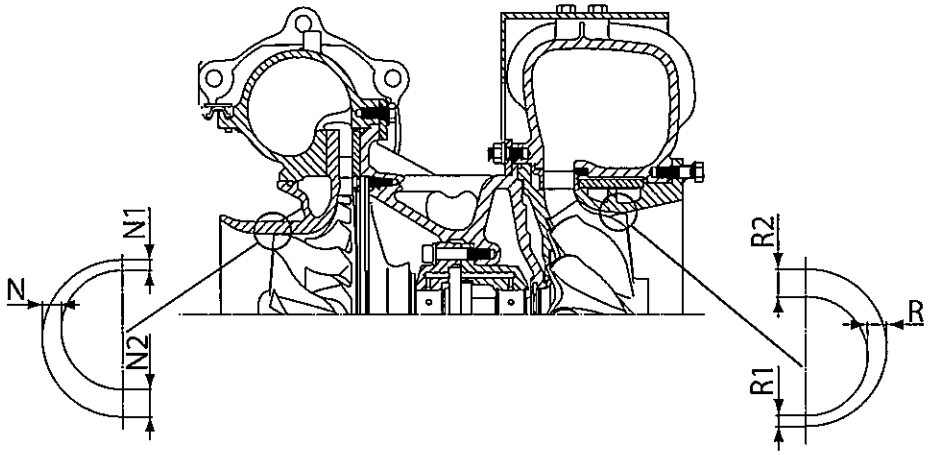
If the clearances are out of tolerance, an official ABB Turbo Systems service station must be contacted.

注意

クリアランスが許容範囲から外れる場合は、ABB 社認定のサービスステーションにお問い合わせください。

8.4 Radial clearances N and R

8.4 ラジアルクリアランス N および R



S_01184

NV50F08-1.R1

N and R

These are the theoretical radial clearance between compressor wheel and wall insert, or between turbine wheel and gas outlet flange.

The minimum and maximum clearances result from the manufacturing tolerances of the compressor and turbine wheel or the wall insert and gas outlet flange.

N1 and R1

Clearances in the bearings and casing centring components are distributed non-uniformly by the rotor mass so that minimum values result.

N2 and R2

Clearances in the bearings and casing centring components are distributed non-uniformly by the rotor mass so that maximum values result.

N および R

これらはコンプレッサホイールと扇車覆の間、またタービンホイールとガス出口フランジの間の理論上のラジアルクリアランスです。

最小と最大のクリアランスはコンプレッサホイール、タービンホイールおよび扇車覆、ガス出口フランジの製造公差により生じます。

N1 および R1

ロータ質量により、ロータは、ベアリングおよびケーシング内で偏心するため、クリアランスは均一ではありません。その最小値を表します。

N2 および R2

ロータ質量により、ロータは、ベアリングおよびケーシング内で偏心するため、クリアランスは均一ではありません。その最大値を表します。

Feeler gauge

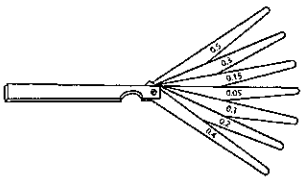
N1/N2 or R1/R2 are measured simultaneously each time using two feeler gauges.

The feeler gauges must not be pushed between wall insert / compressor wheel or gas outlet flange / turbine wheel resp.With excessive force.

すきまゲージ

N1/N2 および R1/R2 は2つのゲージを使用して、それぞれ同時に測定してください。

すきまゲージは扇車覆とコンプレッサホイールの間、ガス出口フランジとタービンホイールの間に適切な力で押し込んでください。



S_01182

NV50F08-1.R1

NOTE

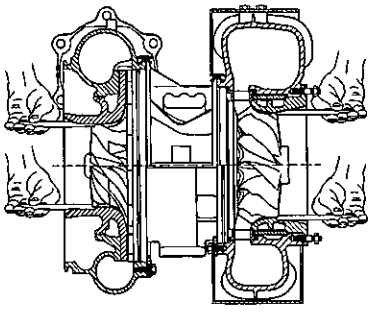
When measuring, make sure that commercially available feeler gauges (valve-setting gauges) with rounded ends are used.

注記

先端が丸い市販ゲージを使用し、計測してください。

Measuring N and R

NとRの測定



S_01181

$$\frac{N1 + N2}{2} = N \qquad \frac{R1 + R2}{2} = R$$

The measured mean values must be within the permissible N or R clearance.

測定値は N および R のクリアランス許容範囲内でなければなりません。

Disassembly and assembly	分解と組立	8	
Radial clearances N and R	ラジアルクリアランス N および R	8.4	Page23

Radial clearance N and R [mm]

ラジアルクリアランス N および R [mm]

	TPS 48	TPS 52	TPS 57	TPS 61
N min. N 最小値	0.22	0.30	0.40	0.50
N max. N 最大値	0.58	0.67	0.86	0.94
R min. R 最小値	0.47	0.59	0.75	0.90
R max. R 最大値	0.77	0.92	1.11	1.32

NV50F08-1.R1

CAUTION

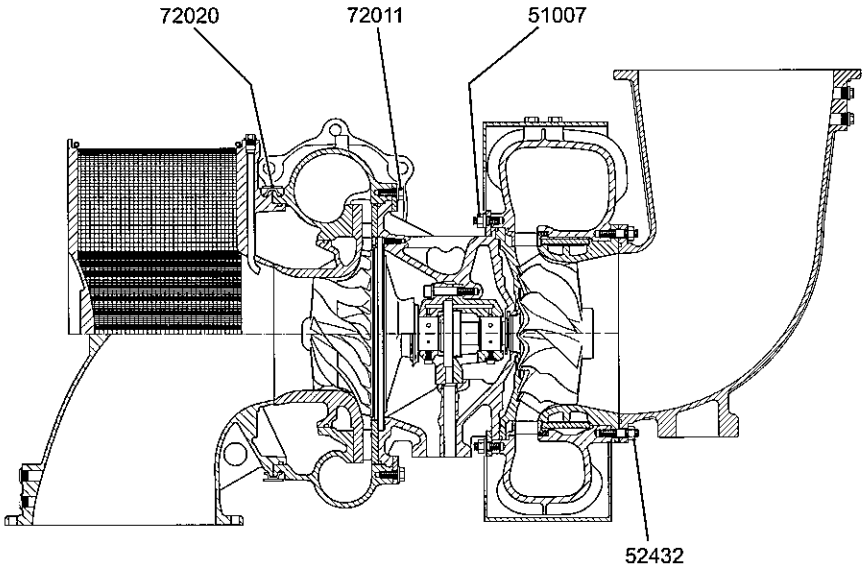
If the clearances are out of tolerance, an official ABB Turbo Systems service station must be contacted.

注意

クリアランスが許容範囲から外れる場合は、ABB 社認定のサービスステーションにお問い合わせください。

8.5 Table of tightening torques

8.5 締付けトルク一覧表



S_00120

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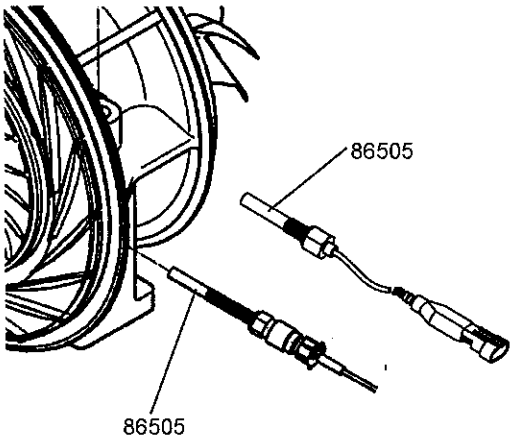
If provided

供給されている場合



S_00121

Disassembly and assembly	分解と組立	8	
Table of tightening torques	締付けトルク一覧表	8.5	Page25

The following tightening torques must be maintained for the specified screwed connections:

ボルトの締付けは下表のトルクを遵守してください。

Tightening torques [Nm]

締付けトルク [Nm]

Part no. 部品番号	TPS 48	TPS 52	TPS 57	TPS 61
51007	25	45	75	75
52432	20	40	65	65
72011	35	70	105	105
72020	60	60	60	60
86505	15	15	15	15

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9

Shutting down turbocharger

無過給運転

9 Shutting down turbocharger

9.1 Shutting down turbocharger

If the turbocharger is damaged and the engine can only be shut down for a short period for emergency repairs, proceed as follows:

▶ Fit cover plate.

CAUTION

The engine manufacturer's instructions for operation of the engine with a blocked / shut down turbocharger must be followed precisely in all cases!

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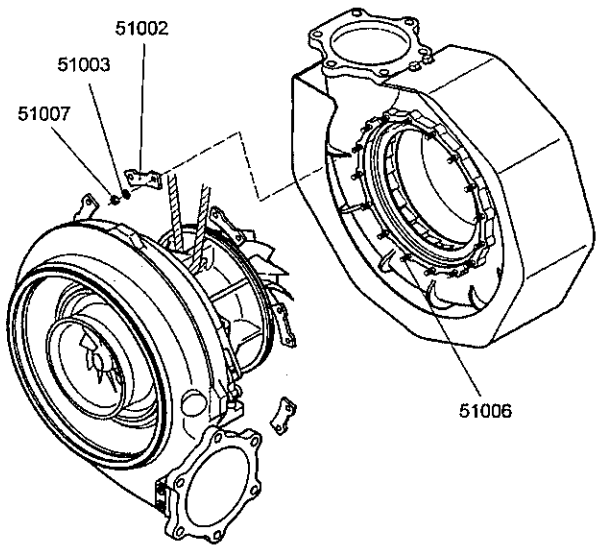
Disassembling turbocharger



S_00125

▶ Disassemble the turbocharger(see section **Disassembling and assembling turbocharger**).

9 無過給運転

9.1 無過給運転

過給機が損傷し、応急修理のためわずかな時間しかエンジンを停止できない場合には、次の方法で対応してください。

▶ 応急フランジの取付け。

注意

過給機の固定および切り離しによるいかなる運転についても、エンジン製造会社の指示に忠実に従ってください。

過給機の分解

▶ 過給機を分解してください(「**過給機の分解と組立**」の項をご参照ください。)

NOTE

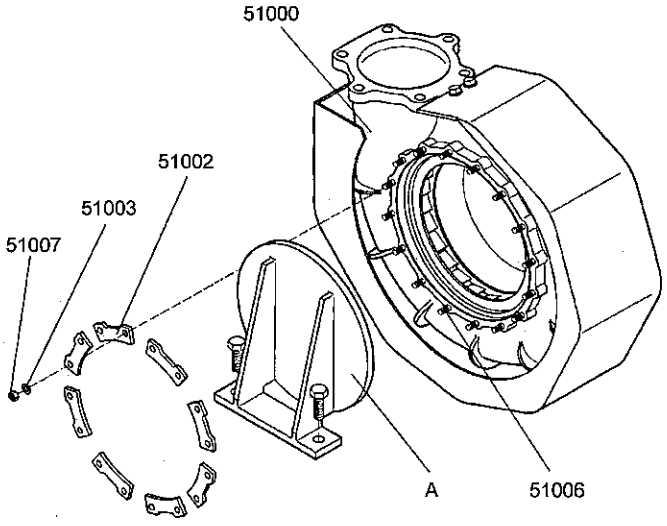
Do not remove the nozzle ring.

注記

ノズルリングは取外さないでください。

Fitting cover plate

応急フランジの取付け



S_00126

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- ▶ Close off the opening in turbine casing with cover plate(A).
- ▶ Fasten cover plate to turbine casing (51000) using clamping straps (51002), verbusripp washers(51003) and nuts (51007)and bolt tightly to support.

- ▶ 応急フランジ(A)でタービンケーシングの開口部を塞いでください。
- ▶ 応急フランジを押さえ板(51002)、特殊ワッシャ(51003)および六角ナット(51007)でガス入口ケーシング(51000)に取付け、エンジンの過給機支持台にボルトでしっかりと固定してください。

NOTE

Clean clamping straps (51002) thoroughly before fitting. Apply Antiscuff (or similar high-temperature lubricating paste) to threads of studs(51006).

注記

組付ける前に押さえ板 (51002)を十分に洗浄してください。またスタッド(51006)のねじ部に焼付防止剤(または耐熱性潤滑剤)を塗布してください。

Drawing of cover plate

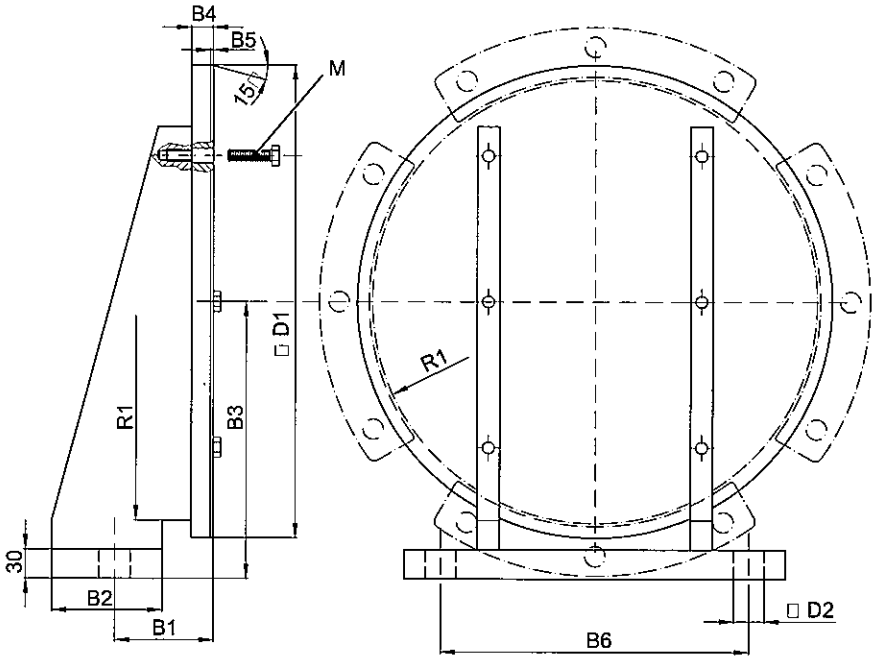
応急フランジの図面

NOTE

The cover plate (material: normal structural steel, in accordance with DIN EN 10025-2) must be made in-house according to the drawing.

注記

応急フランジ(材質： DIN EN 10025-2 に準拠した一般的建築鋼材)はこの図に従って製作してください。



S_00127

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Cover plate with strap connector

応急フランジ寸法表

TPS	B1 [mm]	B2 [mm]	B3 [mm]	B4 [mm]	B5 [mm]
48	65.7 ± 0.5	60	130	11.7 ± 0.2	1.5
52	79.6 ± 0.5	78	155	14.2 ± 0.2	1.5
57	98.2 ± 0.5	100	190	17.2± 0.2	2
61	116.8 ± 0.5	120	226	20.5 ± 0.2	2.4

TPS	B6 [mm]	ØD1 [mm]	ØD1* [mm]	ØD2 [mm]	R1 [mm]	M
48	150	222.7 ± 0.2	227.7 ± 0.2	17	maximum.150 最大	M8
52	180	265.7 ± 0.2	271.7 ± 0.2	21	maximum.125 最大	M8
57	220	325.7± 0.2	332.5 ± 0.2	21	maximum.153 最大	M10
61	260	387.7± 0.2	395.9 ± 0.2	25	maxmum.182 最大	M10

*) Diameter for turbine casing designs for optional sealing with C-ring

*) Cリング(オプション品)付きのタービンケーシング直径

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NOTE

If you require information about the turbine casing design (sealing with / without C-Ring), please contact an official ABB Turbo Systems service station.

注記

もし、タービンケーシング(Cリングあり/無し)についての情報が必要でしたら、ABB社サービスステーションにお問い合わせください。

10

Mothballing the turbocharger

長期休止時の防錆

10 Mothballing the turbocharger

10.1 Taking out of operation for up to 12 months

If a turbocharger put out of use for up to 12 months, any possible special turbocharger storage measures depend primarily on the condition of the lubricating oil before the shut-down.

If the Total Acid Number (TAN) is lower than 2 mg KOH/g no special measures need to be taken. If, however, the Total Acid Number is greater, the turbocharger must be dismantled after its shut-down, cleaned and bright parts oiled with protective oil.

If the turbocharger is left mounted on the engine and the engine oil is replaced by protective oil which is circulated by the pre-lubrication pump prior to the shut-down, no measures need to be taken with the turbocharger. The remains of the old engine oil will be washed away and the sensitive bearing parts will be largely protected against corrosion.

If the rotor turns due to a draught from the flue, fit a blind flange between the compressor casing and the charge-air duct.

10 長期休止時の防錆

10.1 休止期間が 12 ヶ月までの場合

過給機の休止期間が 12 ヶ月までの場合、過給機に特別な保護処置が必要かどうかは休止前の潤滑油の状態によります。

全酸値が 2mgKOH/gよりも低い場合は、特別な処置は必要ありません。しかしながらそれより高い場合は、休止後過給機を取外して洗浄し、防錆油を機械加工面へ塗布してください。

エンジン上に過給機を搭載したままでエンジン用潤滑油を防錆油に交換し、休止前にプライミングを実施した場合は、過給機に関して特別な処置は必要ありません。古いエンジン潤滑油の残りは洗い流し、特に軸受部品は十分な防錆処置を行ってください。

ロータが煙突からドラフトにより回転する場合はコンプレッサケーシングと給気管との間にカバーフランジを取付けてください。

NV50F10-1

10.2 Taking out of operation for more than 12 months

If the turbocharger is removed from the engine and stored for a prolonged period of time, its disassembly and the special storage measures must be carried out by an official ABB Turbo Systems service station.

Suitable storage locations are dry rooms where the relative humidity is between 40% and 70% and no condensation can form.

NOTE

Inspect stored turbochargers yearly for corrosion.

If there are signs of rust, clean parts and renew protection against corrosion.

10.2 休止期間が 12 ヶ月を超える場合

過給機をエンジンから取外して長期間使用しない場合は、過給機の分解、保管準備のための作業を ABB 社サービスステーションに依頼してください。

相対湿度 40～70%で結露の発生しない乾燥した場所に保管してください。

注記

保管した過給機は年に 1 回錆の発生状態をチェックしてください。

錆の兆候が見られる場合は、錆を落とし再度防錆処置を行ってください。。

NV50F10-1

11

Disposal

廃棄

11 Disposal

11.1 Environmentally compatible disposal of turbocharger components

NOTE

Environmentally compatible disposal of turbocharger components

Component disposal must be environmentally compatible, professional and in accordance with local regulations.

The turbocharger mainly consists of metal (cast-iron materials, steel, nickel-steel alloys, aluminium, bearing brass).

Other components are: non-metallic materials (filter components made of felt and polyethylene), lubricants (engine oil), electronic components (speed sensor and associated components), thermal insulation.

- ▶ Dispose of metals by melting down and recycling.
- ▶ Dispose of non-metallic materials as waste.
- ▶ Dispose of lubricant left-overs and residues as used oil.
- ▶ Dispose of electronic components as electrical / electronic scrap.

11 廃棄

11.1 環境に配慮した過給機構成部品の廃棄

注記

環境に配慮した過給機構成部品の廃棄

構成部品の廃棄は、環境に十分配慮し、地域の条例に従って行わなければなりません。

過給機は主に金属部品で構成されています。(鋳鉄、鋼、ニッケル合金、アルミ合金、銅合金など。)

他の構成部品:非金属材料(消音器の構成部品はフェルトやポリエチレン)、潤滑油(エンジンオイル)、電気部品(回転計ピックアップとその関連部品)、断熱材。

- ▶ 金属は、リサイクルしてください。
- ▶ 非金属材料は廃棄してください。
- ▶ 潤滑油の使い残し、および残留物は使用済み潤滑油と同様に処分してください。
- ▶ 電気部品は、電気スクラップとして処理してください。

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Disposal	廃棄	11	
Environmentally compatible disposal of turbocharger components	環境に配慮した過給機構成部品の廃棄	11.1	Page2

WARNING

Handling damaged thermal insulation

Damaged thermal insulation can burden the air with dust. The glass fibres can cause mechanical irritation of the eyes, skin and respiratory tracts.

- ▶ Prevent dust from being produced.
- ▶ Remove dust with a suitable vacuum cleaner.
- ▶ Wear respiratory mask as protection against particles (P1 or P2 mask).
- ▶ Wear leather protective gloves.
- ▶ Wear close-fitting safety goggles.

警告

損傷したインシュレーションの断熱材の取扱い

損傷したインシュレーションの断熱材は、空気中に粉じんを飛散させます。そのガラス繊維は目や皮膚、呼吸器官に炎症を起こします。

- ▶ 粉じんを飛散させないでください。
- ▶ 掃除機で粉じんを取除いてください。
- ▶ 粉じん対策としてマスクを着用してください。
- ▶ 皮手袋を着用してください。
- ▶ ぴったり合うゴーグルを着用してください。

NV50F11-1

- ▶ Dispose of thermal insulation as special waste.

- ▶ インシュレーションの断熱材は、特殊廃棄物として処理してください。

12

Reserve and spare parts

予備品

12 Reserve and spare parts

12 予備品

12.1 Ordering spare parts

12.1 予備品の注文

When making inquries or ordering spare parts, the following data should be specified:

- Turbocharger type
- Xi number
- Description and part number

- 予備品に関するお問い合わせおよびご注文の際には、下記の事項が必要となります。
- 過給機の形式
- 機番 Xi
- 部品名称および番号

Our service stations and agents will take orders for spare parts.

▶ If special variants / cases are not addressed in these general instructions, then contact an official ABB Turbo Systems service station or an ABB agent.

ABB 社認定のサービスステーションおよび代理店にて、予備品のご注文を承っております。

▶ また、取扱説明書の中にも記載されていないような個々のご相談にも対応いたしますので、ABB 社認定のサービスステーションまたは代理店へお問い合わせください。

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Spare parts set

予備品セット

NOTE

The spare parts set (97070) is required for the work described in this manual. The parts included in the spare parts set are available only as a whole set.

注記

予備品セット(97070)は本取扱説明書に記載されている作業に必要なものです。予備品はセットでのみ販売しております。

Spare parts set (97070)

予備品セット(97070)

Quantity 数量	Description 名称	Part number 部品番号
1	Hexagon socket head screw 六角ボルト	42008
1	O-ring O リング	42012
1	Gasket パッキン	52406
1	O-ring O リング	77005
2	Counter-sunk screw 皿ボルト	72041
1	O-ring O リング	81010/82010

CAUTION

All spare parts that were ordered with the turbocharger should be kept complete and serviceable.

▶ Rusty parts should be carefully cleaned and greased.

注意

過給機本体と一緒に納入された全ての予備品は、いつでも使用できる状態にしておいてください。

▶ 錆びた部品は錆を落としグリス等で防錆してください。

▶ Dispose of replaced, unusable parts in a technically and environmentally compatible way.

▶ 交換され使用できなくなった部品は、環境に配慮した方法で廃棄してください。

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12.2 Directory of part numbers

12.2 部品番号表

The following part numbers and descriptions must be used in ALL correspondence or orders:

お問い合せやご注文の際には下記の部品番号と部品名称をお申し付けください:

Turbocharger
過給機

Part no. / 部品番号	Description / 名称
42008 (in spare parts set) (予備品セットに含まれる)	Hexagon socket head screw / 六角ボルト
42012 (in spare parts set) (予備品セットに含まれる)	O-ring / O リング
51000	Turbine casing / タービンケーシング
51002	Clamping straps / 押さえ板
51100	Gas outlet elbow / ガス出口エルボ
52400	Gas outlet flange / ガス出口フランジ
52406 (in spare parts set) (予備品セットに含まれる)	Gasket / パッキン
56001	Nozzle ring / ノズルリング
56005	Lamellar gasket ring / 薄型パッキン
57200	Burst protection / バーストプロテクション
57210	Burst ring / バーストリング
72000	Compressor casing / コンプレッサケーシング
72011	Hexagon-head screw / 六角ボルト
72012	Clamping straps / 押さえ板
72020	V-Strip / V 形クランプ
72041(in spare parts set) (予備品セットに含まれる)	Counter-sunk screw 皿ボルト
77000	Wall insert / 扇車覆
77005 (in spare parts set) (予備品セットに含まれる)	O-ring / O リング
79000	Diffuser / ディフューザ
81000	Filter silencer / 消音器
81010 (in spare parts set) (予備品セットに含まれる)	O-ring / O リング
82000	Air suction branch, axial 空気吸込囲(軸方向流入)
82010 (in spare parts set) (予備品セットに含まれる)	O-ring / O リング

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Cartridge group
センターカートリッジ部

Part no. 部品番号	Description 名称
21000	Shaft タービン軸
21002 (in spare parts set) (予備品セットに含まれる)	Piston ring ピストンリング
25000	Compressor wheel コンプレッサ
32101 (in spare parts set) (予備品セットに含まれる)	Plain bearing bush 平軸受ブッシュ
32103 (in spare parts set) (予備品セットに含まれる)	Circlip スナップリング
32105 (in spare parts set) (予備品セットに含まれる)	Thrust ring スラストリング
32106 (in spare parts set) (予備品セットに含まれる)	Thrust bearing スラスト軸受
32108 (in spare parts set) (予備品セットに含まれる)	Piston ring ピストンリング
32109	Sealing disc シーリングディスク
32110	Compressor-end bearing flange コンプレッサ側軸受フランジ
32111 (in spare parts set) (予備品セットに含まれる)	Auxiliary bearing 反スラスト軸受
32112	Turbine-end bearing flange タービン側軸受フランジ
32113 (in spare parts set) (予備品セットに含まれる)	Hexagon socket head screw CE 六角ボルト
32114 (in spare parts set) (予備品セットに含まれる)	Hexagon socket head screw TE 六角ボルト
32221	Bearing cover 軸受蓋
32222 (in spare parts set) (予備品セットに含まれる)	O-ring O リング
42001	Bearing casing 軸受ケーシング
42002	Hexagon socket head screw 六角ボルト
42008 (in spare parts set) (予備品セットに含まれる)	Hexagon socket head screw 六角ボルト
43001	Partition wall 隔壁

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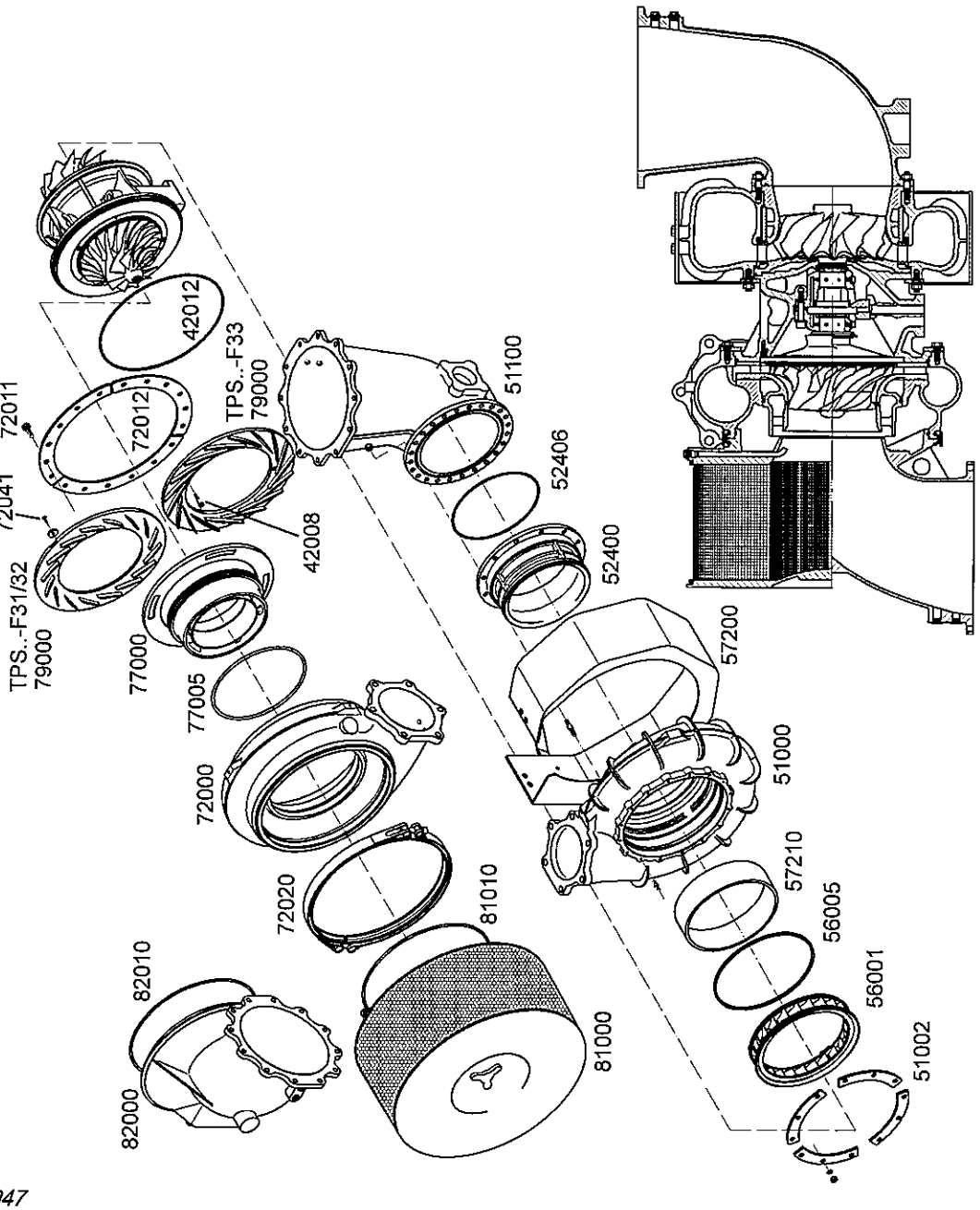
12.3 General view of turbocharger

12.3 過給機の全体図

NV50F12-1.R1



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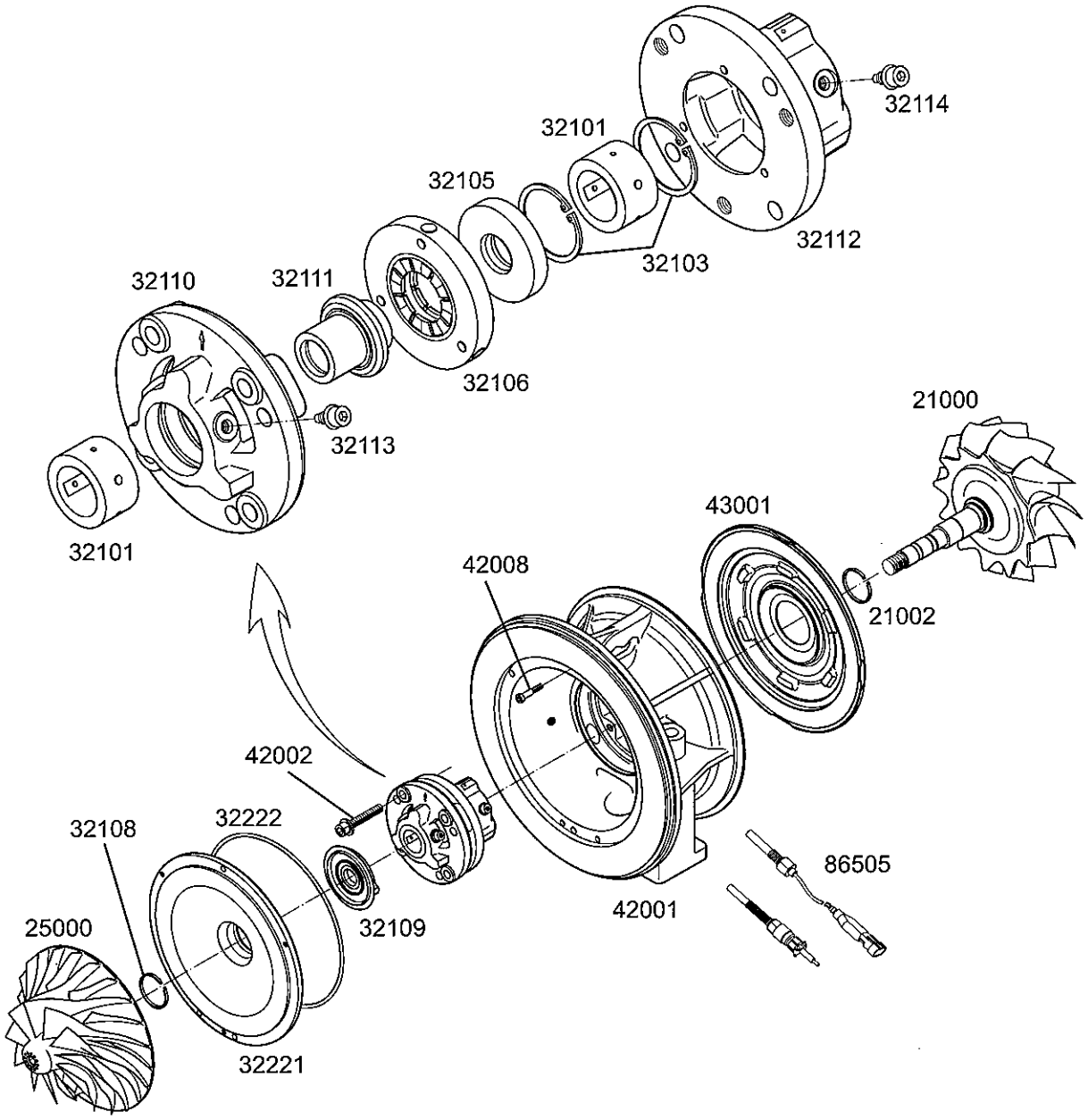
12.4 General view of cartridge group

12.4 センターカートリッジの全体図



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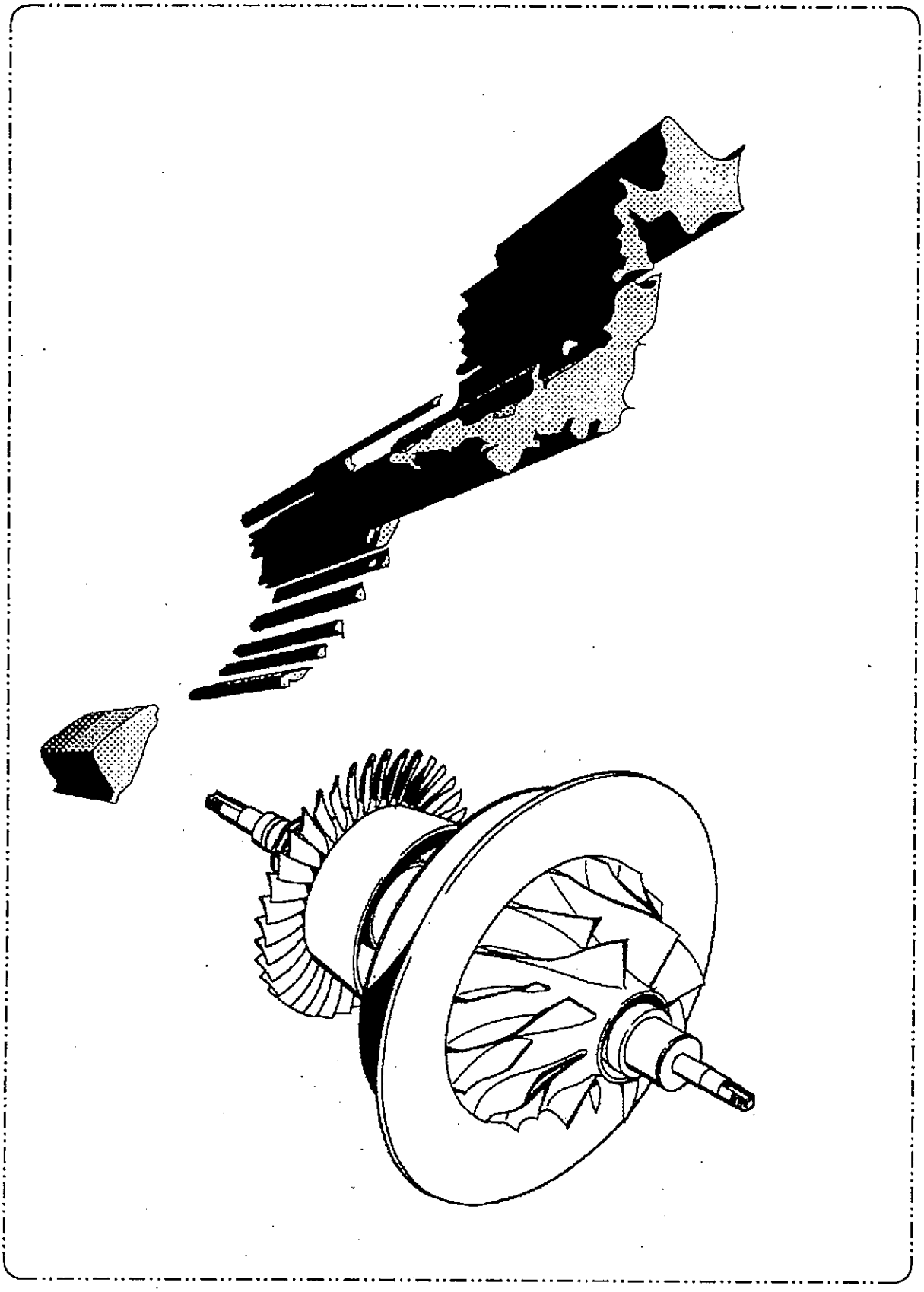
13

Guide to service stations

サービスステーション

1
TSU Turbocharger Service Division

1
TSU 過給機サービス事業部



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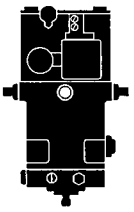
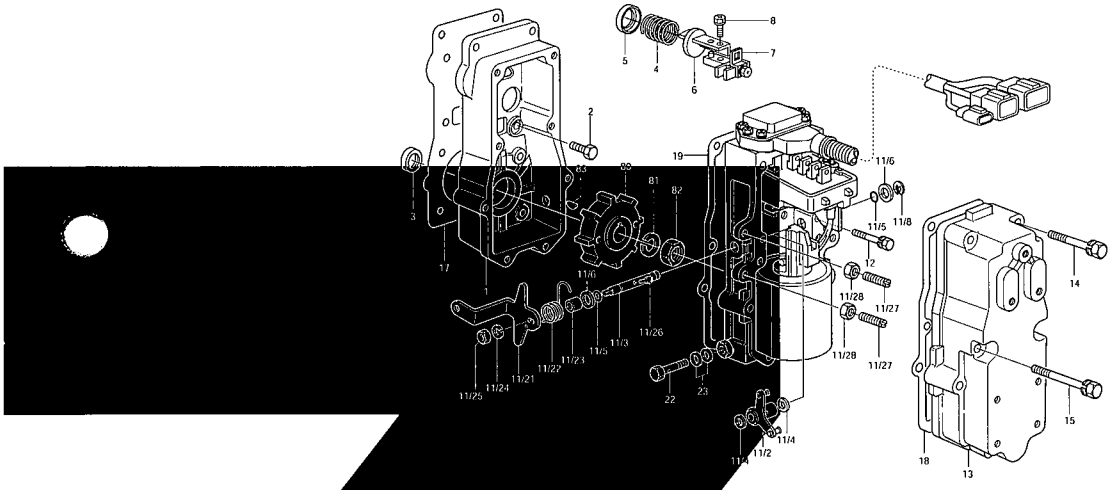
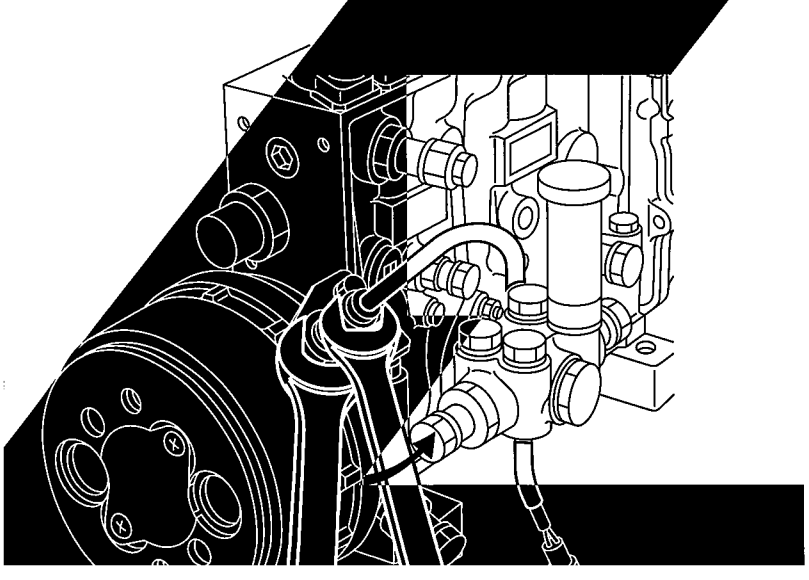


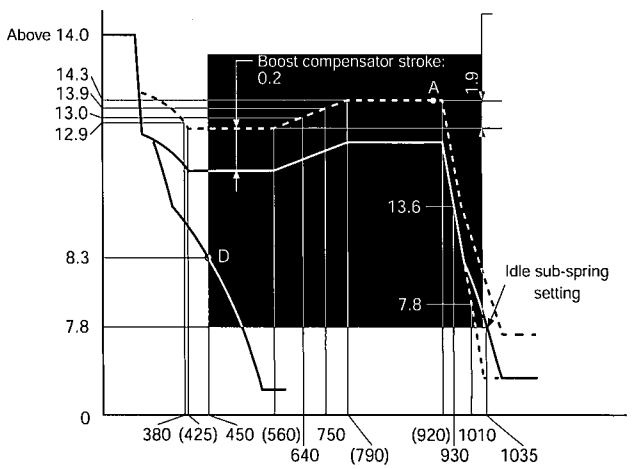
Service Manual - Operation & Maintenance

Excellence inside. ZEXEL.

RHD MODEL GOVERNOR (RHD6 & RHD10)







WARNINGS

The following definitions and warning signs are used in this service manual.
These are extremely important to safe operation.
Important points are described to prevent bodily injury and property damage.
They must be fully understood before beginning governor maintenance.

⚠CAUTION Improper maintenance can result in injury and property damage.

MEANINGS OF MARKS

The following marks are used in this service manual to facilitate correct governor maintenance.

Advice Procedures that must be performed to enable the best possible governor maintenance.

Note Information assisting in the best possible governor maintenance.

FOREWORD

The RHD hydraulic governor is a high performance variable speed governor for use with main and auxiliary marine engines, engines for electric generators and general power engines.
This service manual describes the construction and operation, repair and maintenance, and adjustment of the RHD6 and RHD10 hydraulic governors.

The contents of the manual, including illustrations, drawings and specifications were the latest available at the time of printing.
The right is reserved to make changes in specifications and procedures at any time without notice.

Bosch Corporation
Automotive Aftermarket Division

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1. The RHD governor has a large controlling force, despite its compact size.
2. As the hydraulic fluid is contained in the governor, piping and sub-tank installation are unnecessary when installing the governor on the engine.
3. As the power piston utilizes a differential system, the operating force is uniform. Moreover, it is not necessary to install a return spring in the fuel control linkage. When play in the linkage system is excessive, however, it is necessary to install a spring at the end of the system to absorb this play.
4. The governor is able to rotate in either direction, and therefore it can be used without modification on engines which rotate in the reverse direction.
5. Because the normal operating speed is low, governor endurance is high, and gearing is simple when installing the governor on low and intermediate speed engines.
6. The governor has a speed droop mechanism and a compensator mechanism, enabling it to be easily and freely adjusted.
   The adjustment range is very wide, enabling governor use on a wide range of engines. The governor can also be used as an isochronous governor with 0% speed droop.
7. The governor body's high tensile aluminum alloy casting makes it extremely lightweight.
8. The governor can be equipped with a governor motor, a pneumatic controller and a hydraulic controller.
9. RHD6 and RHD10 governors are interchangeable. For example, if a higher pressure injection system is required or if the governor's control force is insufficient to cope with the demand for increased engine output, decreased fuel consumption or the use of lower quality fuel oil, then the RHD10 can replace the RHD6.
10. The governor is equipped with a standardized piping connection to enable the installation of an auxiliary starting booster.
11. Of the RHD10 governor's main components, 80% are identical to those of the RHD6. Almost 100% of the remote control device's components are identical to those of the RHD6.

2 SPECIFCATIONS

	RHD6	RHD10
Control force	5.9 J {0.6 kgf·m}	9.8 J {1.0 kgf·m}
Output shaft torque	9.8 N·m {1.0 kgf·m}	15 N·m {1.5 kgf·m}
Output shaft operation angle	35°	40°
Power piston diameter	Small: ϕ17.7; Large: ϕ25	Small: ϕ18.6; Large: ϕ27
Controlled speed range	Standard rating: 600 ~ 2,000 r/min (Normal: 1,700 ~ 2,000 r/min) High speed rating: 800 ~ 2,400 r/min (Normal: 1,900 ~ 2,400 r/min)	
Maximum allowable speed	2,500 r/min	
Compensator	Needle valve type	
Speed droop adjustment range	0 ~ approx 10% (rated)	
Direction of rotation	Clockwise & counterclockwise	
Hydraulic fluid	SAE20W~40 (Mobiloil)	
Dry weight (lever type)	5.5 kg	5.7 kg
Governor drive power	Max approx 0.2 kW {0.3ps}	
Applicable engines	Low speed: up to approx 1,500 kW {2,000 ps} Intermediate speed: up to approx 1,850 kW {2,500 ps} High speed: up to approx 2,200 kW {3,000 ps}	Low speed: up to approx 2,200 kW {3,000 ps} - High speed: up to approx 2,950 kW {4,000 ps}
Applicable injection pumps	PE (all types) PF1Z x 18 cyl PF1C, -1CD x 16 (18) cyl PF1W x 12 (16) cyl PF1GD x 8 cyl PF1DD x 6 cyl	PE (all types) PF1CX x 18 cyl PF1WX x 16 cyl PF1GX x 12 cyl PF1DX x 8 cyl PF1EX x 6 cyl

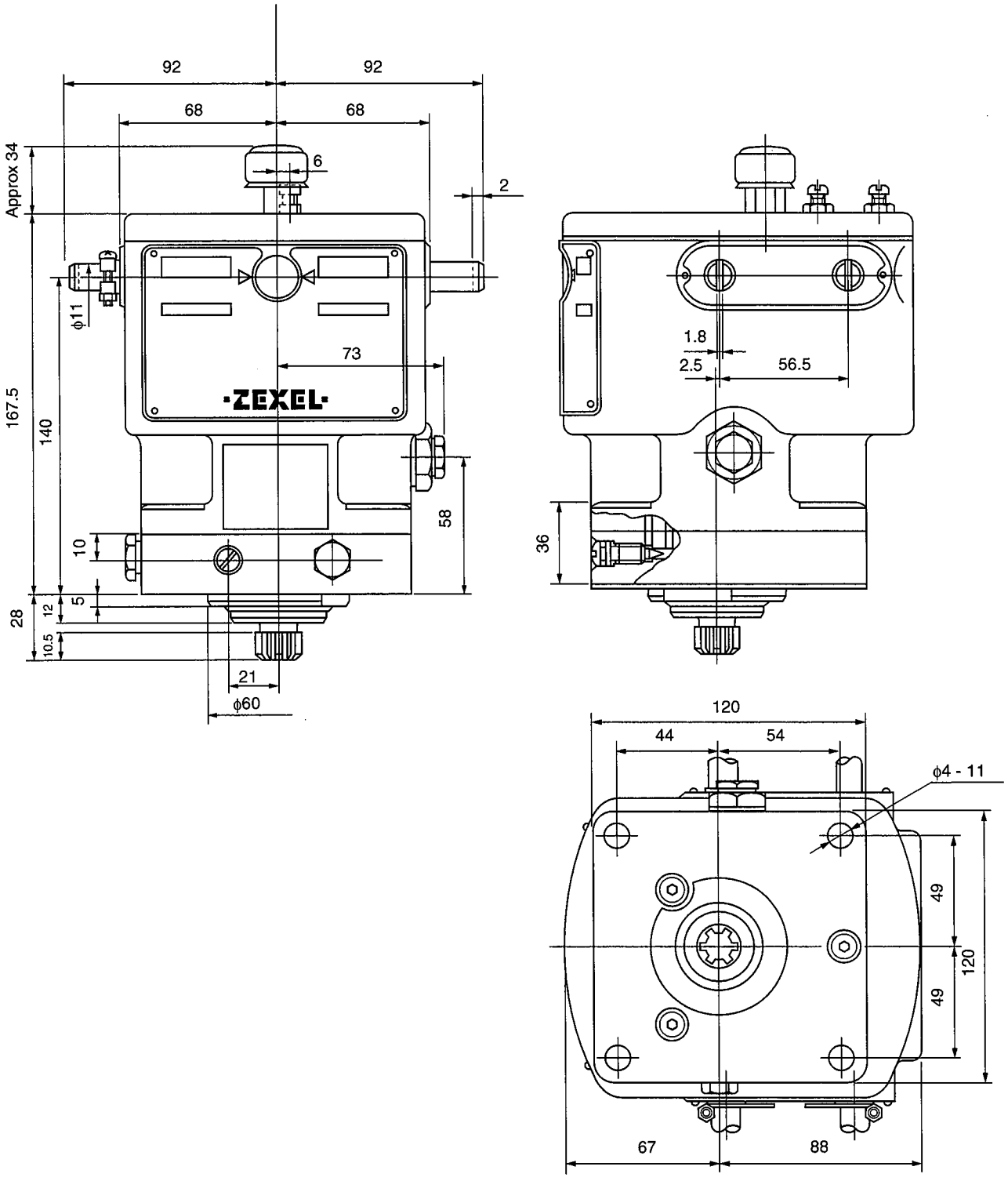
The main difference between the RHD6 and the RHD10 is the power piston. The parts adjacent to the power piston also differ.

Note:

1. The method of connecting fuel linkages and the amount of corresponding resistance differ depending on the size of the applicable engine and the size and number of the injection pumps.
   Consequently, engines with little friction loss can be used in high output engine applications and, conversely, engines with large friction loss can only be used in applications equal to or less than those specified above.
2. Simply because the governor has surplus control force does not necessarily mean that stable operation can be ensured.
   In order to ensure stable operation and to prevent the speed from exceeding the necessary speed, the inertia of the engine and the drive system should be as large as possible.

The figure below shows the RHD6 hydraulic governor (lever type). The RHD10 has a special plug (piping connection) for auxiliary starting booster installation located below the center of the name plate.

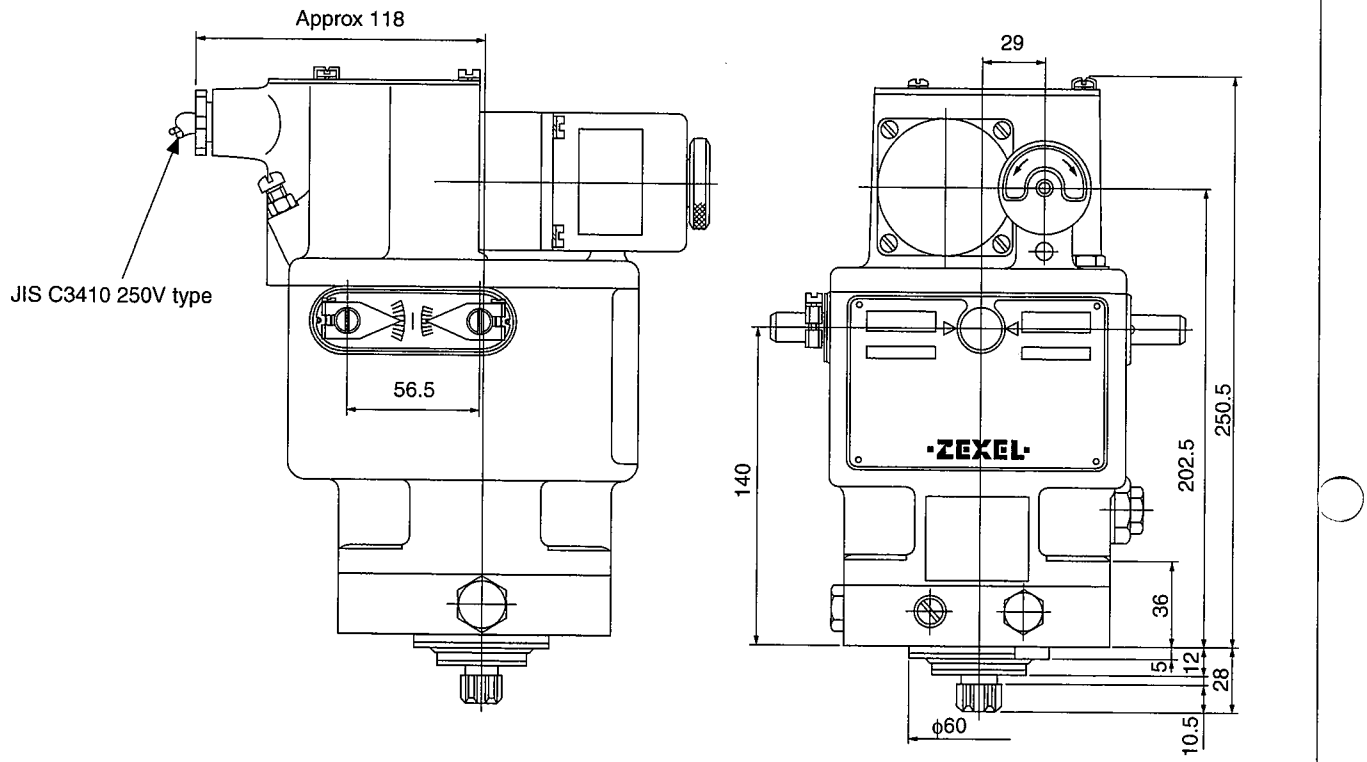
Lever type



P-RHDR- 001

RHD6 and RHD10-MC, -MCL type governor motors

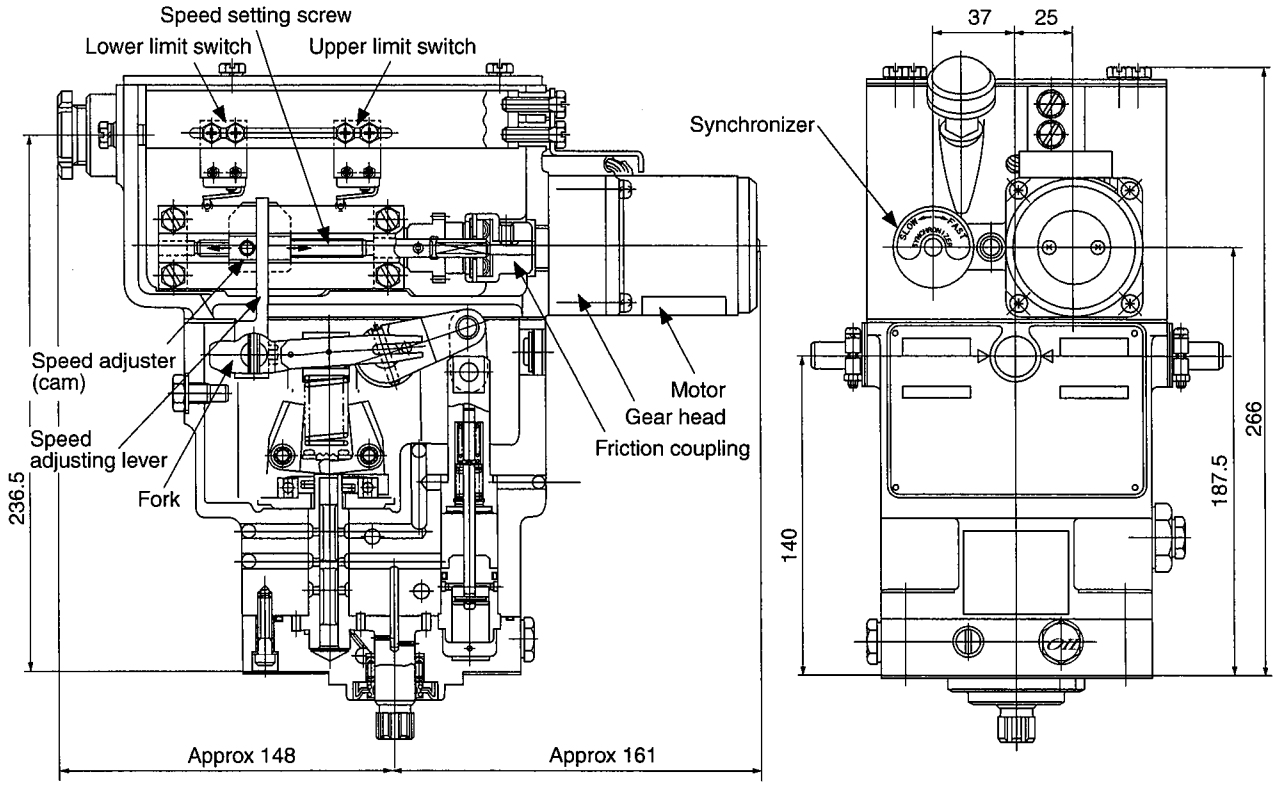
MC type with governor motor



P-RHDR- 002

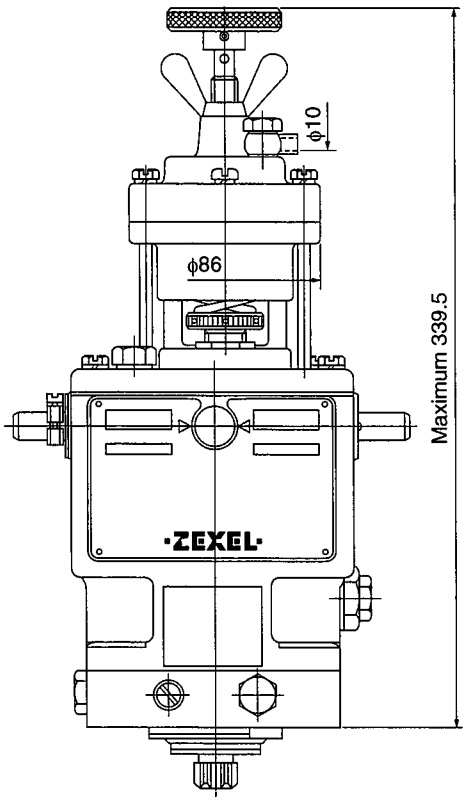
	Voltage	Cycle	Speed	Current consumption	Output
Motor	AC100V	50/60 Hz	1,200/1,450 r/min	20 W	4 W
	AC220V	50/60 Hz	1,200/1,450 r/min	20 W	4 W
	DC24V	-	2,200 r/min	20 W	3 W
	DC100V	-	2,200 r/min	20 W	3 W
Gear head (speed reduction ratio)	1/18, 1/30, 1/50, 1/75, 1/100, 1/150, 1/300, 1/375, 1/450, 1/600, 1/900, 1/1500				
Speed setting time	The speed setting time can be determined freely by choosing from any of the above gear heads. The standard specified time, however, is approximately that shown below. Main engines: 20 ~ 25 sec dead slow to rated speed Engines for generators: 7 ~ 9 sec/1 Hz				
Limit switch adjustment range (RHD6, 10 -MCL)	Standard specification Lower limit: 600 r/min Upper limit: 2,000 r/min Prior to shipment, the lower limit is set at 600 r/min, and the upper at 1,600 r/min. High speed specification Lower limit: 800 r/min Upper limit: 2,400 r/min Prior to shipment, the lower limit is set at 800 r/min, and the upper at 2,000 r/min.				

MCL type with governor motor

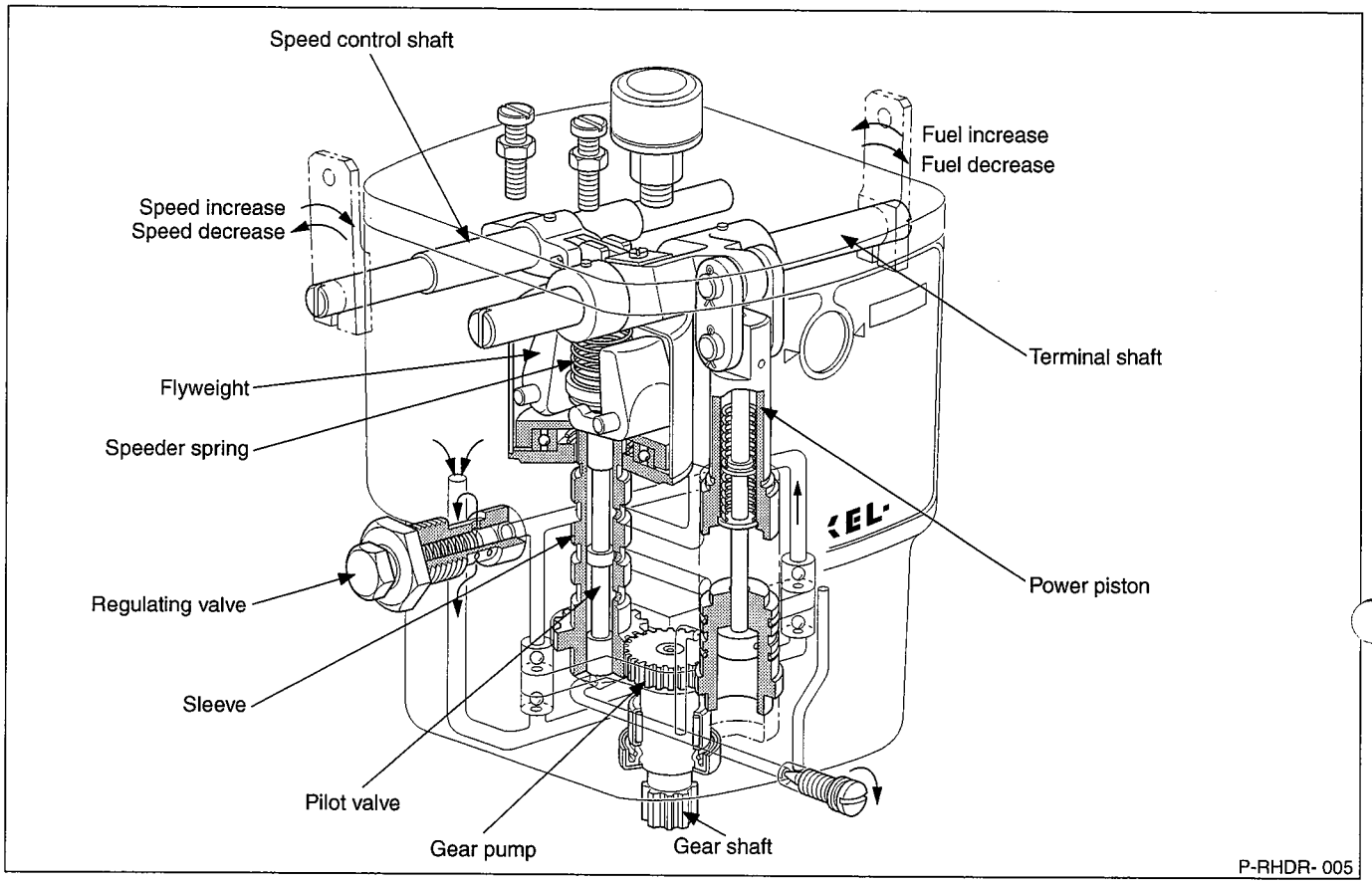


P-RHDR- 003

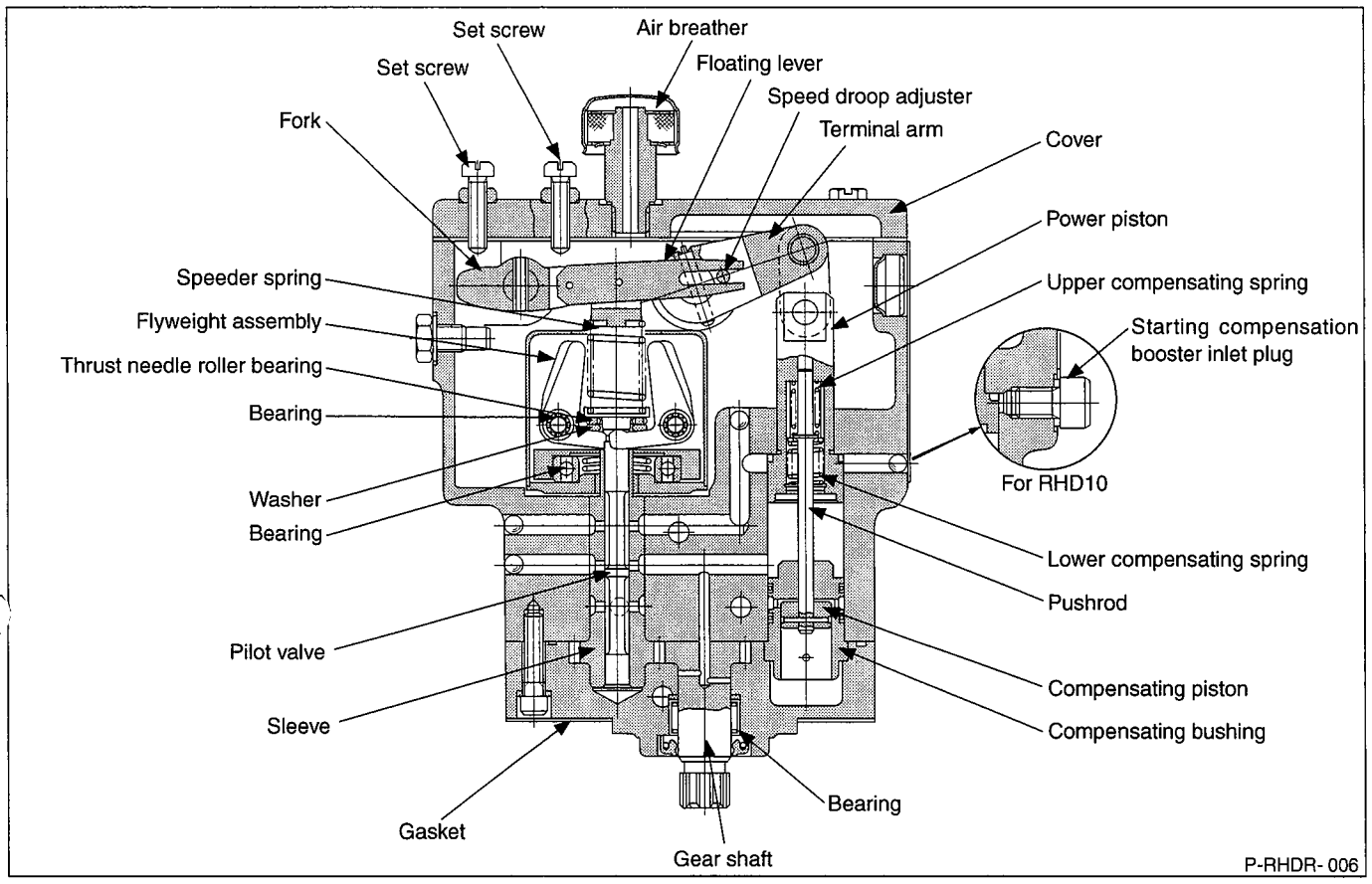
Pneumatic controller dimensions



P-RHDR- 004



The above figure shows a cutaway view of the RHD6 hydraulic governor.
The governor is installed directly on the engine.
Engine crankshaft speed is increased using a suitable gear ratio, and is then transmitted to the governor gear shaft. This rotation is transmitted to the hydraulic governor's gear pump, which pressurizes the hydraulic fluid.
The gear pump pumps hydraulic fluid from the governor's hydraulic fluid tank in through the low pressure side of the regulating valve, and delivers the high pressure hydraulic fluid to the pilot valve. The regulating valve also regulates the pressure of the fluid pressurized by the gear pump (RHD6: 1.18 MPa {12 kfg/cm^2}, RHD10: 1.47 MPa {15 kgf/cm^2}) and returns excess fluid to the gear pump's intake side.
The gear pump has four check valves so that governor performance does not vary, even when the governor rotates in the reverse direction. The gear pump's driven gear and the sleeve are unified. One end of the sleeve is connected to the governor flyweight. The pilot valve inside the sleeve is moved up and down in response to the flyweight's centrifugal force to control the flow of hydraulic fluid to the power piston, reacting promptly to variations in engine load to rotate the terminal shaft.



The flyweight's centrifugal force is transmitted to the pilot valve by the thrust needle roller bearing. The speeder spring's force is always acting against the flyweight's centrifugal force. The set force of the speeder spring can be changed by moving the control lever.

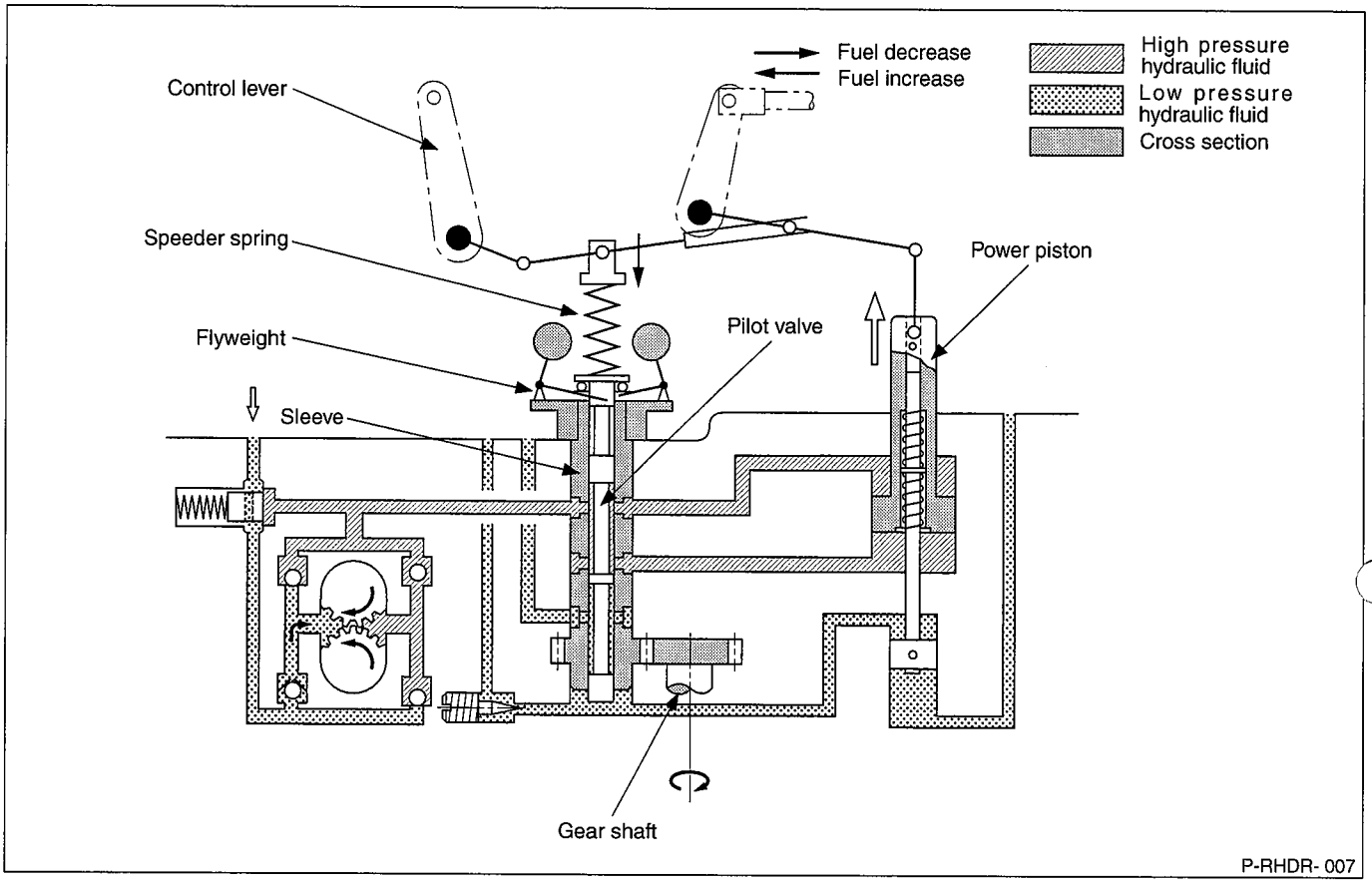
The flyweight assembly is an oil damper type assembly. When sleeve rotation is transmitted to the flyweight, the oil acts to absorb unnecessary high frequency rotation fluctuations and enable stable output.

The power piston, which controls the fuel injection quantity to the engine, is a simple mechanism which enables high stabilized output.

The power piston is operated by the hydraulic fluid to move in both the fuel increase and fuel decrease directions. The power piston rotates the terminal shaft via the guide lever and the terminal arm to directly control fuel injection quantity.

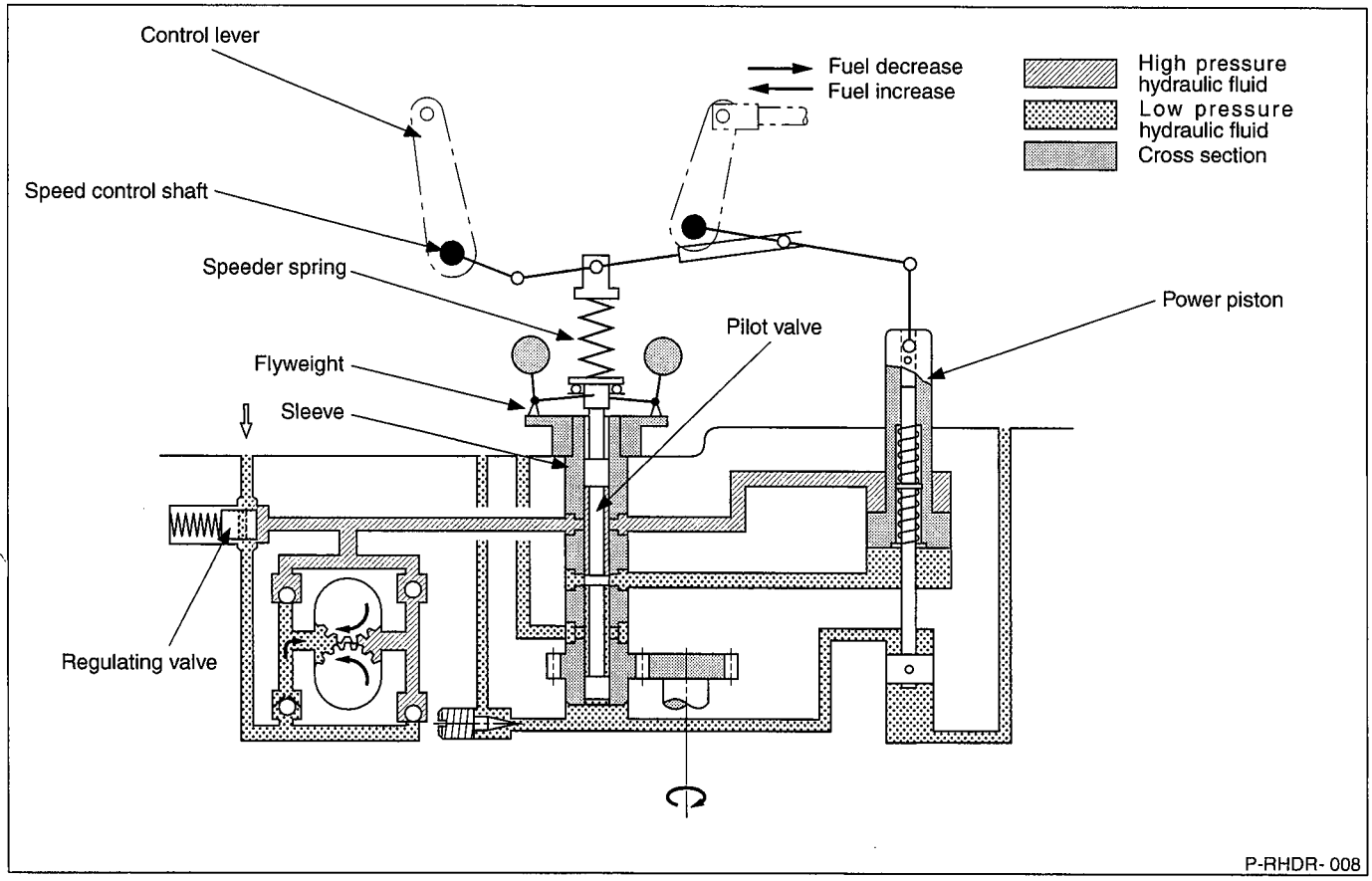
In addition to having a larger diameter power piston, the RHD10 is equipped with a piping connection to enable the connection of an auxiliary starting booster to the top of the power piston.

ENGINE STARTING



When the engine is stopped, the flyweight is pushed down and closed by the force of the speeder spring. Consequently, the pilot valve is in its lowermost position.
When the control lever is then moved in the fuel increase direction, the engine is rotated by compressed air, etc, the gear pump is operated through the gear shaft and the hydraulic fluid is pressurized.
Because the flyweight is pushed down by the speeder spring and the pilot valve does not move, the pressurized hydraulic fluid passage and the passage to the power piston are open, and the hydraulic fluid pressurized by the gear pump is delivered through the pilot valve to the top and bottom of the power piston.
Because the ratio of the areas of the power piston subject to hydraulic pressure is 1:2, the power piston is immediately moved up (in the fuel increase direction) to move the output shaft side lever in the fuel increase direction through the system of links and facilitate engine starting.

NORMAL OPERATION

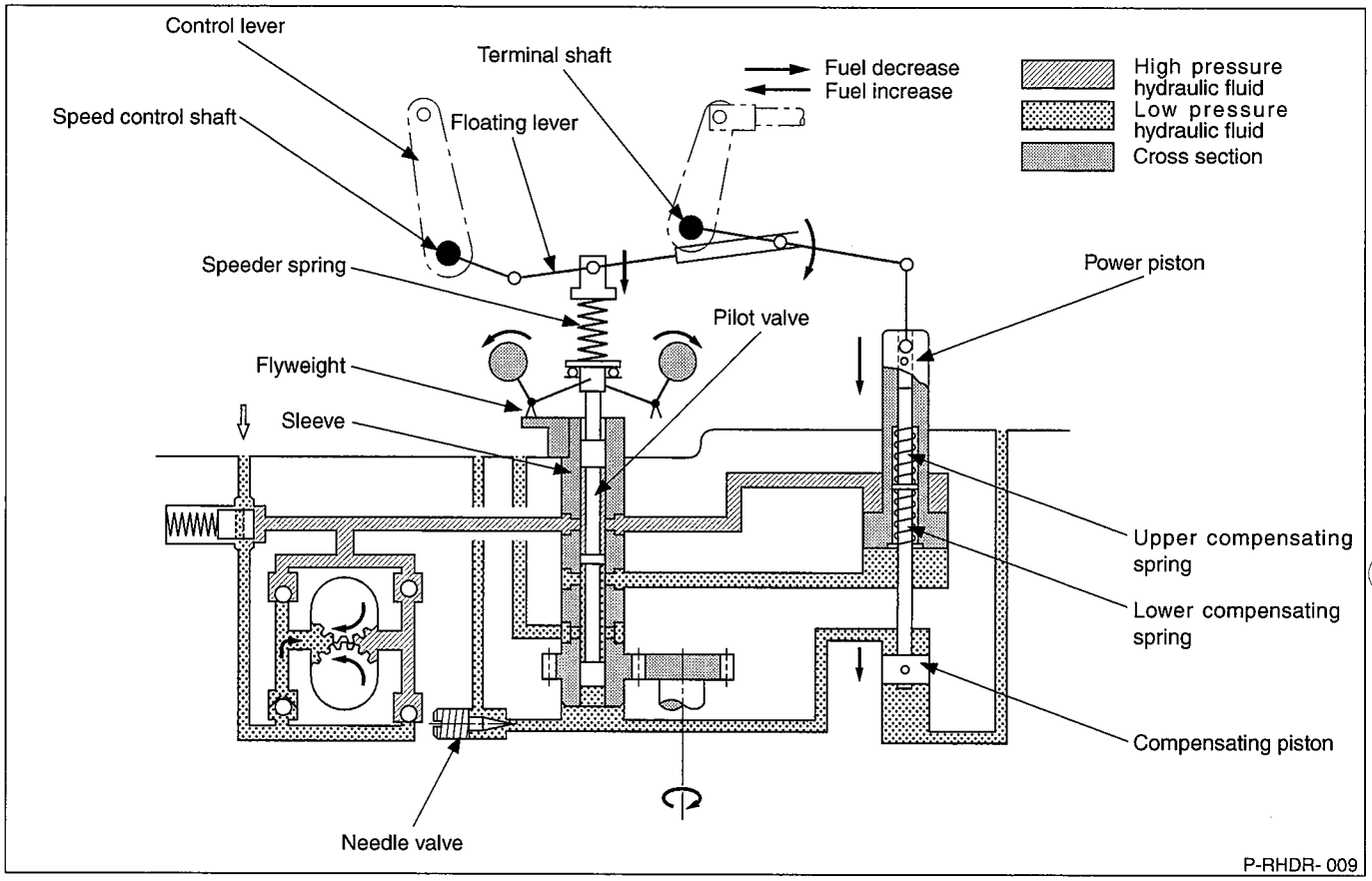


When the control lever is moved in the speed increase direction, the speeder spring is compressed by the speed control shaft. When the speeder spring force exceeds the flyweight' s centrifugal force, the pilot valve is moved down. Because of this, the high pressure hydraulic fluid passage and the top and bottom power piston passages open, and the high pressure hydraulic fluid pressure moves the power piston in the fuel increase direction to increase engine speed. As the engine speed increases, the flyweight's centrifugal force increases and the pilot valve is pulled up.

At the position where the speeder spring force and the flyweight's centrifugal force balance, the high pressure hydraulic fluid passage and the bottom power piston passage are closed by the pilot valve land. Because of this, the power piston maintains this position and a constant stable speed is maintained.

At this time, excess hydraulic fluid overflows through the regulating valve and is returned to the gear pump's intake side to maintain a stable hydraulic fluid pressure to the pilot valve.

LOAD DECREASE



When engine load decreases, and governor rotation has increased, the flyweight's centrifugal force increases and moves the pilot valve up. This opens the center pilot port so that the hydraulic fluid in the bottom of the power piston returns to the governor's hydraulic fluid tank. Consequently, the power piston is moved down (in the fuel decrease direction) by the high pressure hydraulic fluid in the top of the cylinder and the speed is decreased to the previous balanced condition.

When the power piston moves down, it compresses the speeder spring through the speed droop mechanism's floating lever, the pilot valve moves down, and the land moves to a position where it closes the pilot port.

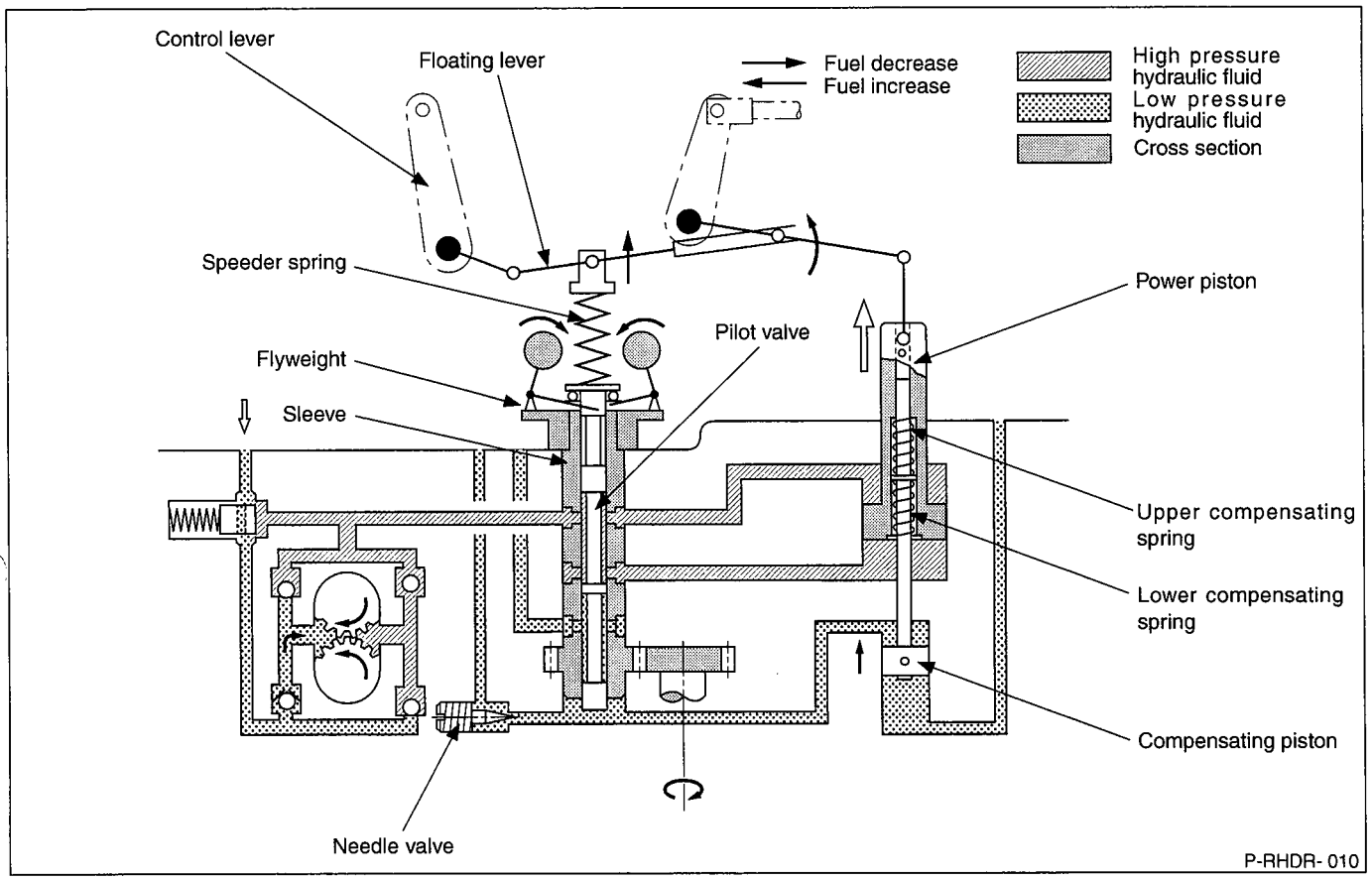
When the power piston moves down, its downward movement is also transmitted to the compensating pushrod and piston through the compensating spring in the power piston.

Because of this, the negative pressure generated in the compensator chamber relieves the upward movement of the pilot valve, momentarily applying a force in the downward direction.

The compensating effect ends when hydraulic fluid from the governor chamber flows in through the needle valve for a fixed period to restore the negative pressure in the compensator chamber to atmospheric pressure.

Consequently, when the pilot port is closed by the speed droop mechanism and the compensator mechanism, the power piston and the terminal shaft stop at a new position where the fuel necessary to operate the engine at normal no-load speed is supplied when the engine load is cut.

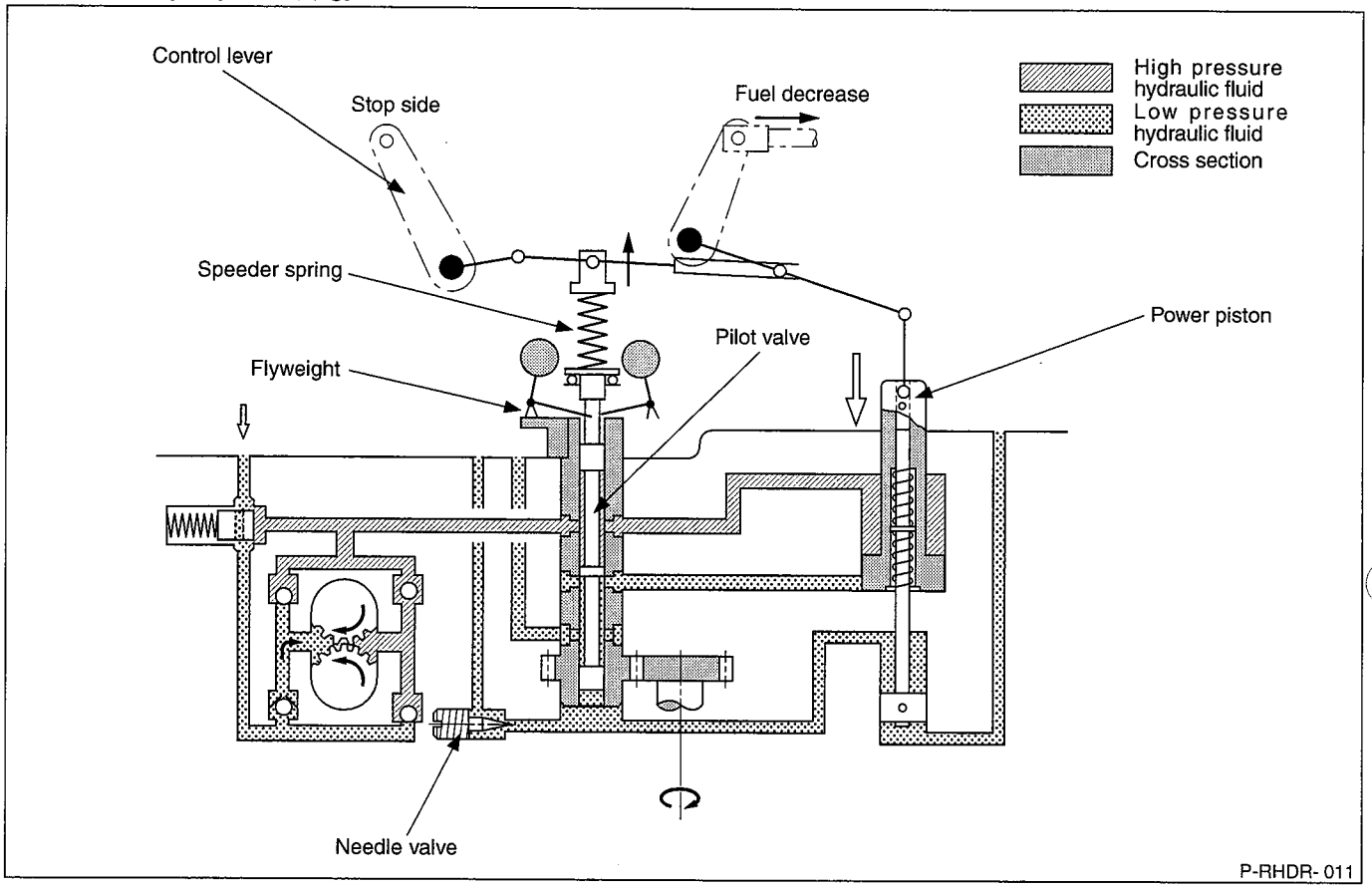
LOAD INCREASE



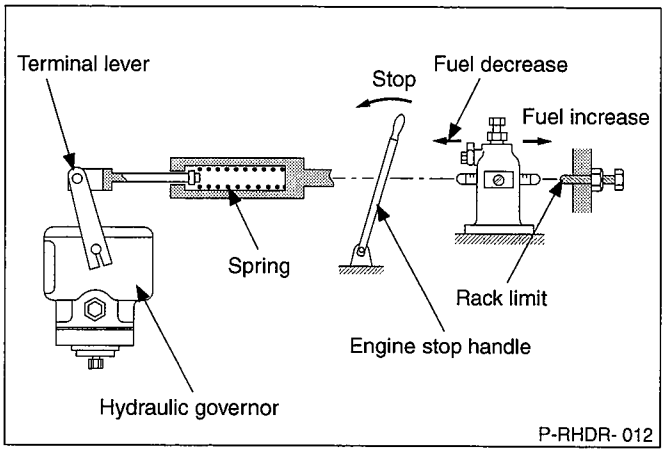
When the engine load increases, engine speed decreases and the governor's operation is exactly opposite to that at load decrease.
Thus, the pilot valve moves down, high pressure hydraulic fluid flows into the chamber below the power piston, and the power piston moves upward in the fuel increase direction to increase the engine speed and return it to the previous balanced condition.
When the power piston moves up, the speeder spring is moved upward by the speed droop mechanism through the floating lever. Consequently, the pilot valve also moves upward until the pilot valve land closes the pilot port.

As the power piston moves up, the compensating piston is drawn up by the lower compensating spring, and the compensator chamber pressure becomes positive.
Because of this, the speed at which the pilot valve moves down is relieved as this pressure is applied in an upward direction.
The compensating effect ends when the hydraulic fluid flows from the governor chamber in through the needle valve for a fixed period to restore the pressure in the compensator chamber to atmospheric pressure.
In this way, the pilot valve returns to its regular position and the engine is maintained at the new power piston position.

ENGINE STOPPING



Because the governor's power piston is moved in the fuel decrease direction by the high pressure hydraulic fluid above the power piston, when the control lever is moved to the stop side the pilot valve is moved up by the speeder spring.



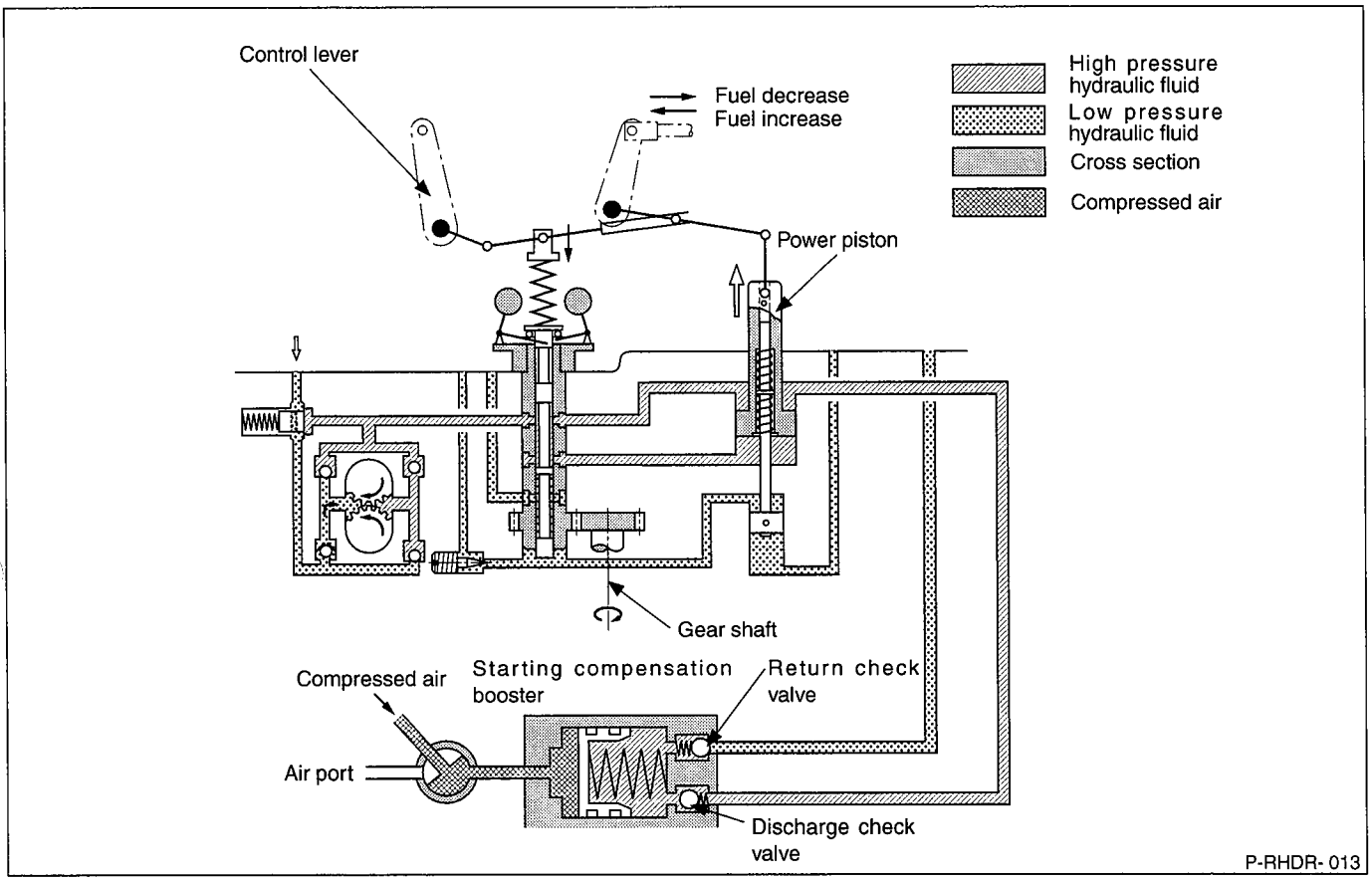
The high pressure hydraulic fluid in the bottom of the power piston is then returned to the governor's hydraulic fluid tank, the power piston moves down (in the fuel decrease direction) and the engine can be forcibly stopped.

The engine can also be stopped by remote control by a governor motor or pneumatic controller. These remote control devices are also installed with a synchronizer to enable manual engine stopping.

When the engine is not stopped on the governor side as described above, but is instead stopped by a handle which mechanically sets the control rack to the non-injection position, it is necessary to install a spring in the link system to prevent excessive force being applied to the governor.

As this governor does not have a load limit device, it is also necessary to install a spring as above when it is necessary to mechanically limit the load.

AUXILIARY STARTING BOOSTER



An auxiliary starting booster can be connected to the RHD10 governor when necessary. The following explains engine operation at starting when a booster has been connected to the governor.

When compressed air is supplied to the booster piston to facilitate starting, the piston compresses the spring and the hydraulic fluid in the chamber opposite the compressed air is guided through the discharge side check valve to the chamber above the power piston.

As the governor at this time is in the engine starting status, as explained previously, the hydraulic fluid also flows through the pilot valve to the chamber below the power piston.

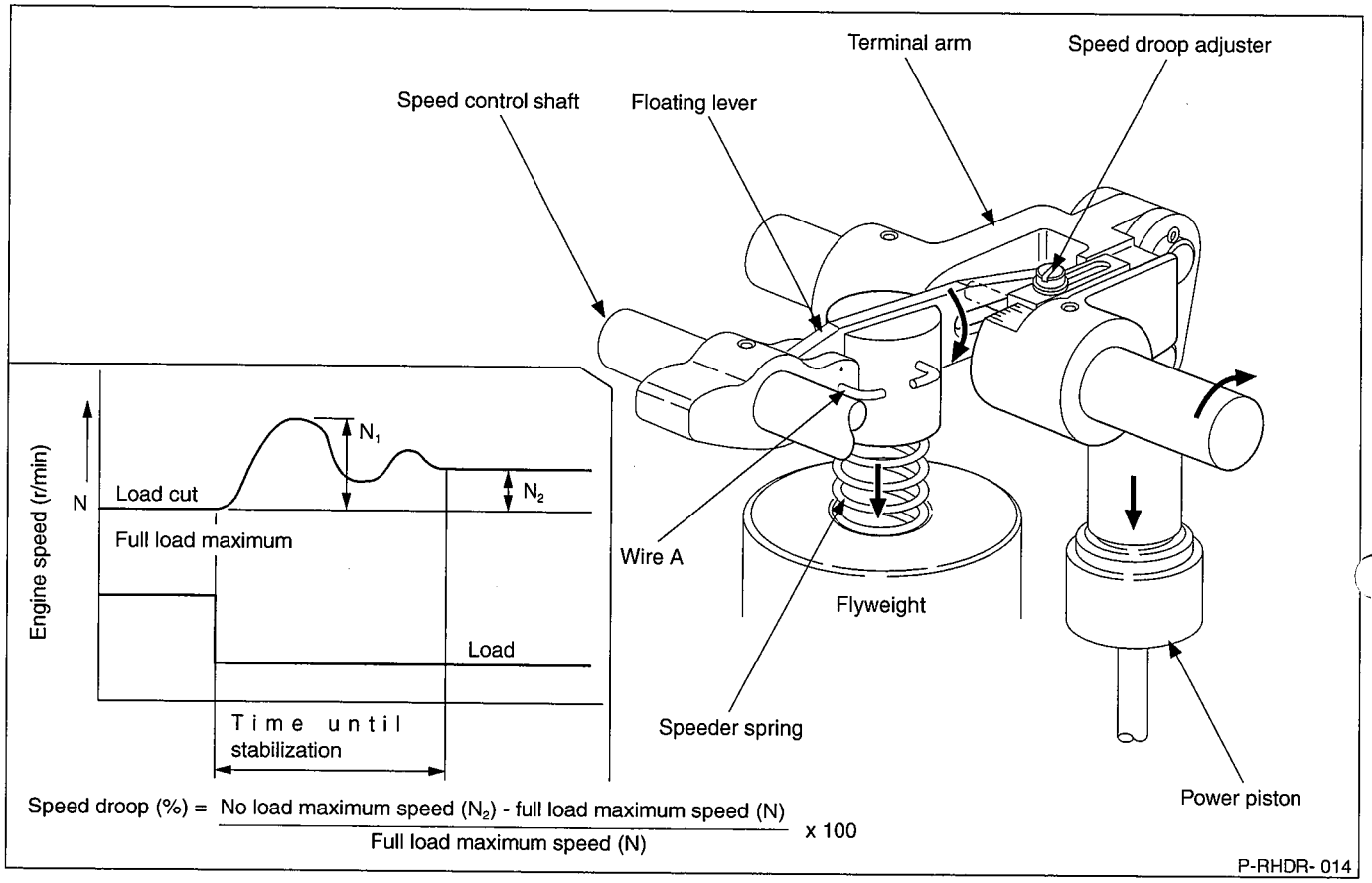
As the governor is simultaneously being turned, high pressure hydraulic fluid from the gear pump is also acting on the top and bottom chambers of the power piston.

Consequently, hydraulic fluid pressurized by the booster moves the power piston in the fuel increase direction faster than normal, and it arrives at the fuel position necessary for starting faster than normal.

The booster is an effective means of assisting starting for engines such as those for emergency power generators, where mis-starting must be avoided, and high output engines where the governor's surplus output is small.

Boosters are manufactured especially for the RHD6 and RHD10 governors.

SPEED DROOP MECHANISM



With the speed droop mechanism, speed droop can be varied freely from 0% ~ 10% by adjusting the speed droop adjuster.
When the speed control shaft position is fixed using the control lever, and the engine speed is maintained at a speed suitable for a fixed load, should the load decrease, the power piston is moved in the fuel decrease direction (ie, down) to prevent an excessive speed increase.
With this power piston movement, the speed droop adjuster attached to the terminal arm turns and pivots the floating lever in a clockwise direction around the wire A to compress the speeder spring and apply a new set load.
Because the speeder spring's set load is increased, the pilot valve is again moved down to a position where it closes the pilot port, the flyweight's centrifugal force and the speeder spring are balanced at an engine speed a little higher than the initial balanced condition, and this engine speed is maintained.

Conversely, when the load increases, power piston movement in the fuel increase direction extends the speeder spring so that the engine speed is maintained at a speed a little lower than that when the engine speed was originally balanced.
Consequently, by adjusting the speed droop adjuster scale, the link lever ratio is changed and, as the speeder spring's set force can be changed, speed droop can be set freely.

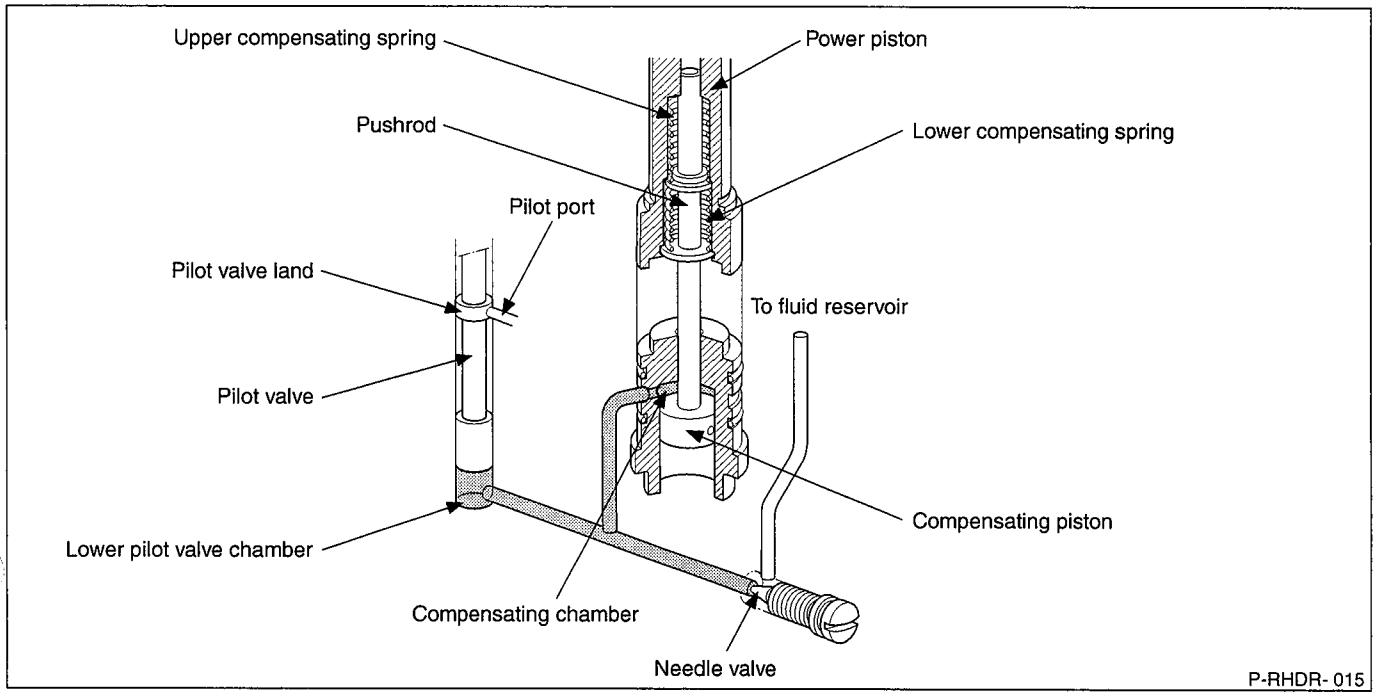
- Speed droop (stabilizing droop)

When load is cut at full-load maximum speed (N), the speed will increase and then stabilize at no-load maximum speed (N_2).
Speed droop is the no-load maximum speed (N_2) expressed as a percentage of the full-load maximum speed (N).

- Momentary droop

When the load is cut at full-load maximum speed (N), the maximum speed (N_1) attained while the speed is stabilizing expressed as a percentage of the full-load maximum speed (N) is referred to as the momentary droop.

COMPENSATOR MECHANISM



The compensator mechanism is used to prevent hunting, which generally more easily occurs when speed droop is decreased.

The compensator mechanism's upper compensating spring and lower compensating spring are both assembled with a set load.

When engine speed increases and the power piston has moved in the fuel decrease direction (ie, down), the power piston and pushrod and the compensating piston first move down together. Because of this, the pressure in the compensating chamber becomes negative. When this negative pressure falls below a certain value, as it overcomes the upper spring's set load and compresses the spring, the amount of compensating spring movement becomes relatively less than the amount of power piston movement (the rate that negative pressure is generated becomes less than power piston movement). As the compensating chamber is connected to the lower pilot valve chamber, the lower pilot valve chamber pressure also becomes negative, and the pilot valve works to move down against the flyweight's centrifugal force and close the previously open pilot port. Because of this, power piston movement in response to engine fluctuations can be slowed to a certain extent and hunting prevented.

In other words, providing the compensating spring with a set load and varying the rate at which negative pressure is generated by power piston movement can improve stability by preventing hunting, and improve response (ie, the speed at which the power piston moves in the fuel increase direction) by minimizing momentary droop.

Conversely, when engine speed decreases, the power piston moves up. As the lower compensating spring has a set force, however, the compensating piston also initially moves up, and the compensating chamber pressure becomes positive. When this positive pressure exceeds a certain level, the spring is compressed and the rate at which the positive pressure is generated is slowed. When the compensating chamber pressure becomes positive, the lower pilot valve chamber pressure also becomes positive, and the pilot valve is pushed up to close the pilot port.

Because of this, excessive power piston movement and therefore hunting can be prevented.

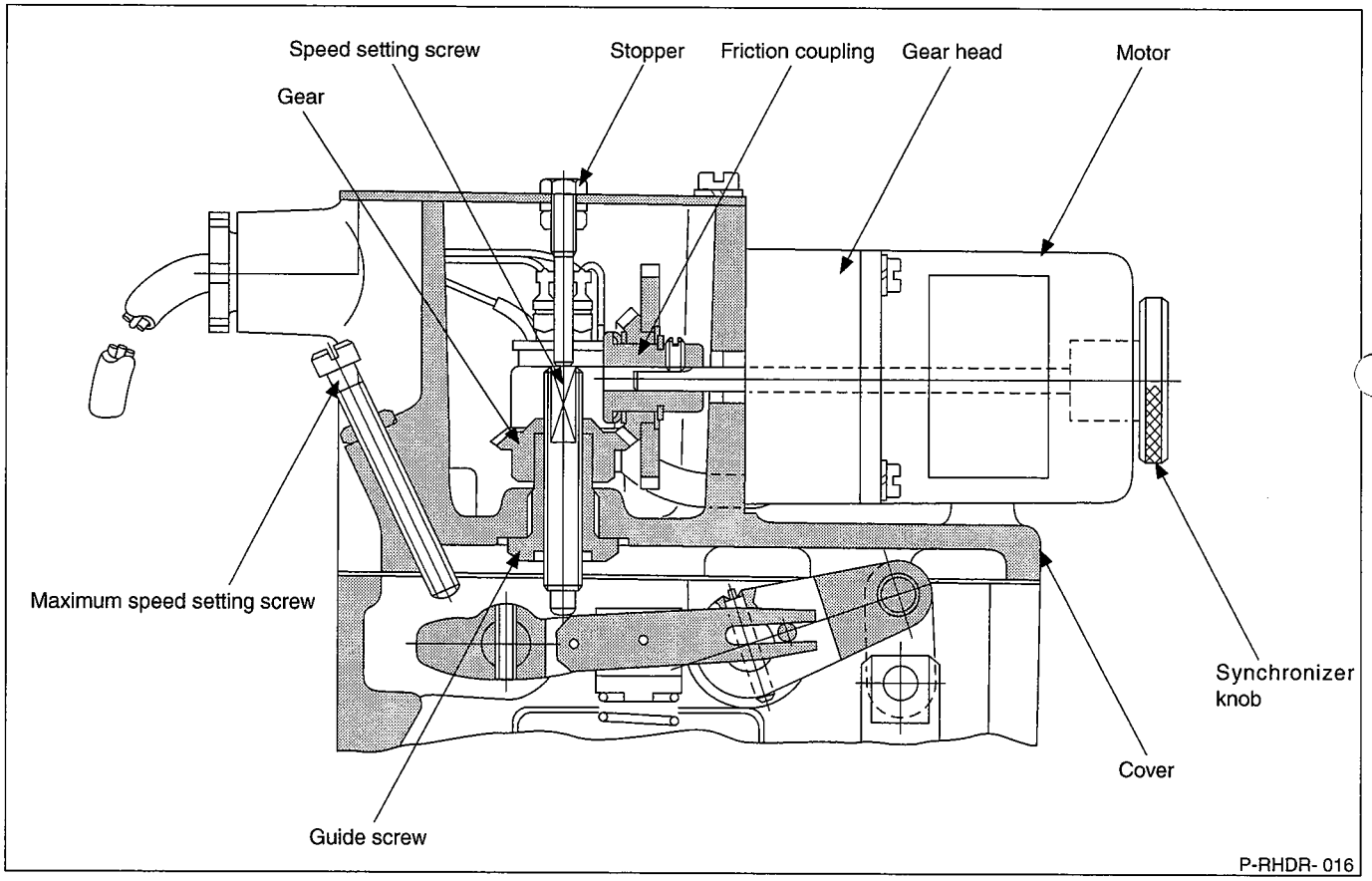
Closing the needle valve increases the pressure generated in the compensating chamber, and thus increases the compensator effect, so that response can be further slowed.

The governor can be equipped with a governor motor and a pneumatic controller. These facilitate remote engine control from a central control room.

Their construction and operation are explained below.

GOVERNOR MOTOR

RHD6 and RHD10 - MC type governor motor



The governor motor operates when the engine speed set switch in the control room is turned ON.

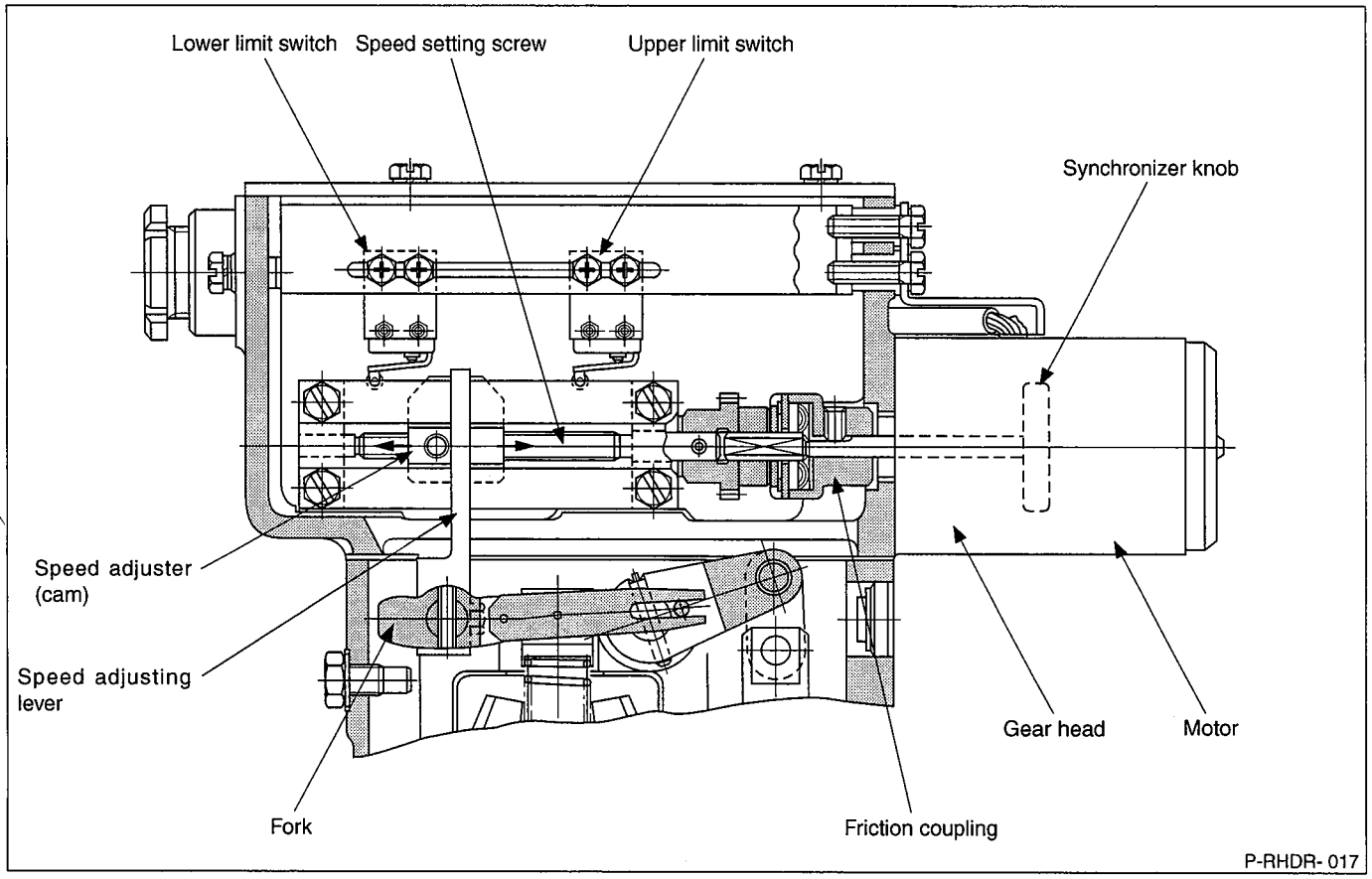
When the motor operates, the speed setting screw is rotated by the friction coupling and gear.

As the speed setting screw is screwed into the guide screw, this rotation results in vertical movement.

Because the speed setting screw acts on the speed adjusting mechanism within the governor, speed setting screw movement changes the set load of the speeder spring to vary engine speed.

The speed can also be changed manually in the same way by turning the synchronizer knob.

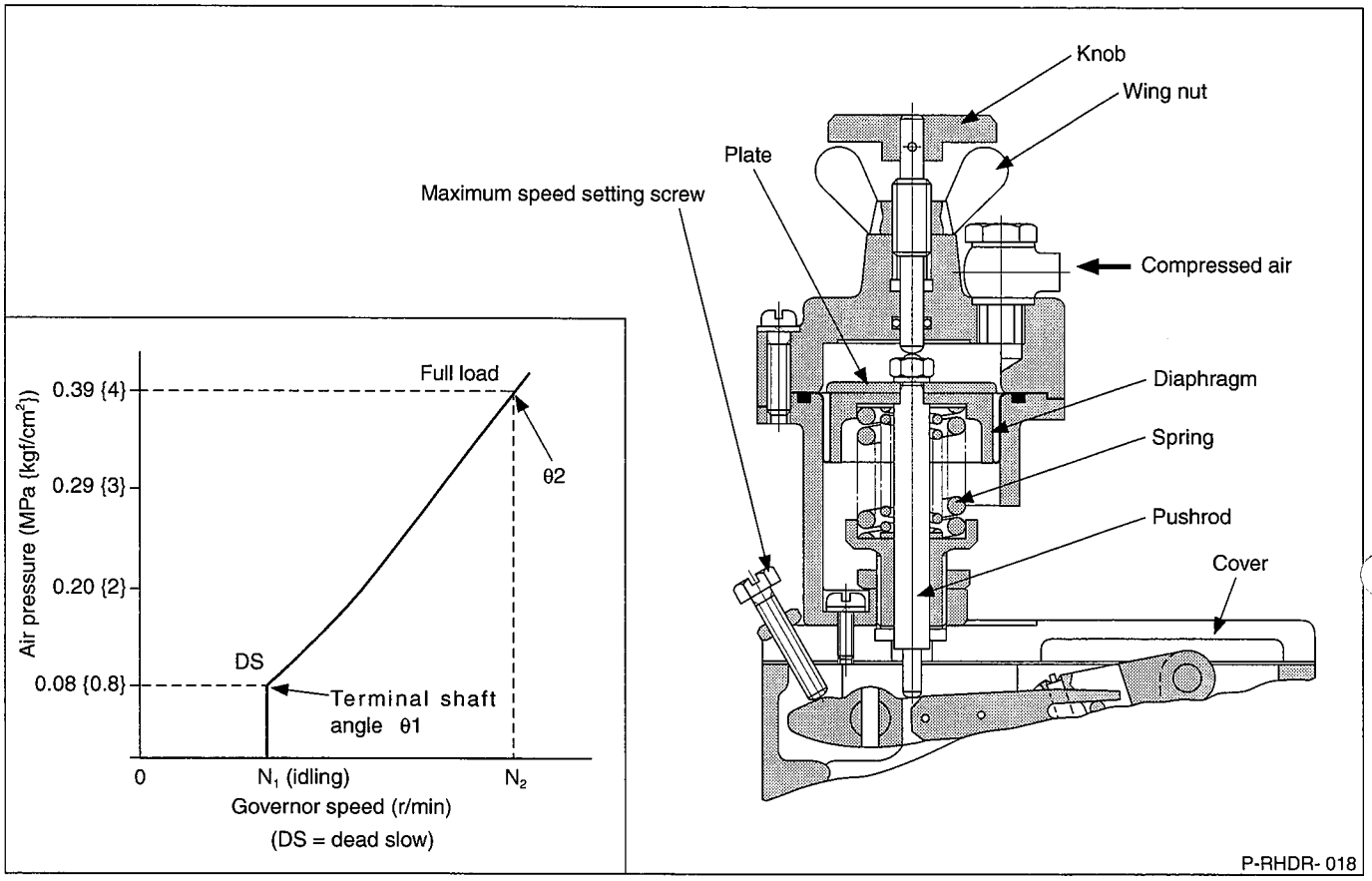
RHD6 and RHD10 - MCL type governor motor (with limit switches)



This governor motor enables electrical control of the set governor speed's upper and lower limits through limit switches.
Three types are available for use with ships' main engines, pumps, and generators. The control ranges and speed setting times differ depending on the type used.
The motor is decelerated by the gear head which, through the friction coupling, turns the speed setting screw to slide the speed adjuster in an axial direction. A pin protruding from the speed adjuster turns the speed adjusting lever fixed to the speed control shaft to vary the set speed.

The upper part of the speed adjuster also serves as a cam to operate the limit switches and move them within the range determined by the slot to enable adjustment of the upper and lower set speeds.
When setting the speed manually, the motor side synchronizer can be used to freely vary the speed from the minimum to the maximum irrespective of limit switch control.
Also, when an attempt is made to simultaneously change the set speed by remote control, the friction coupling ensures that the speed is first changed on the engine side to prevent any adverse effect on the motor.

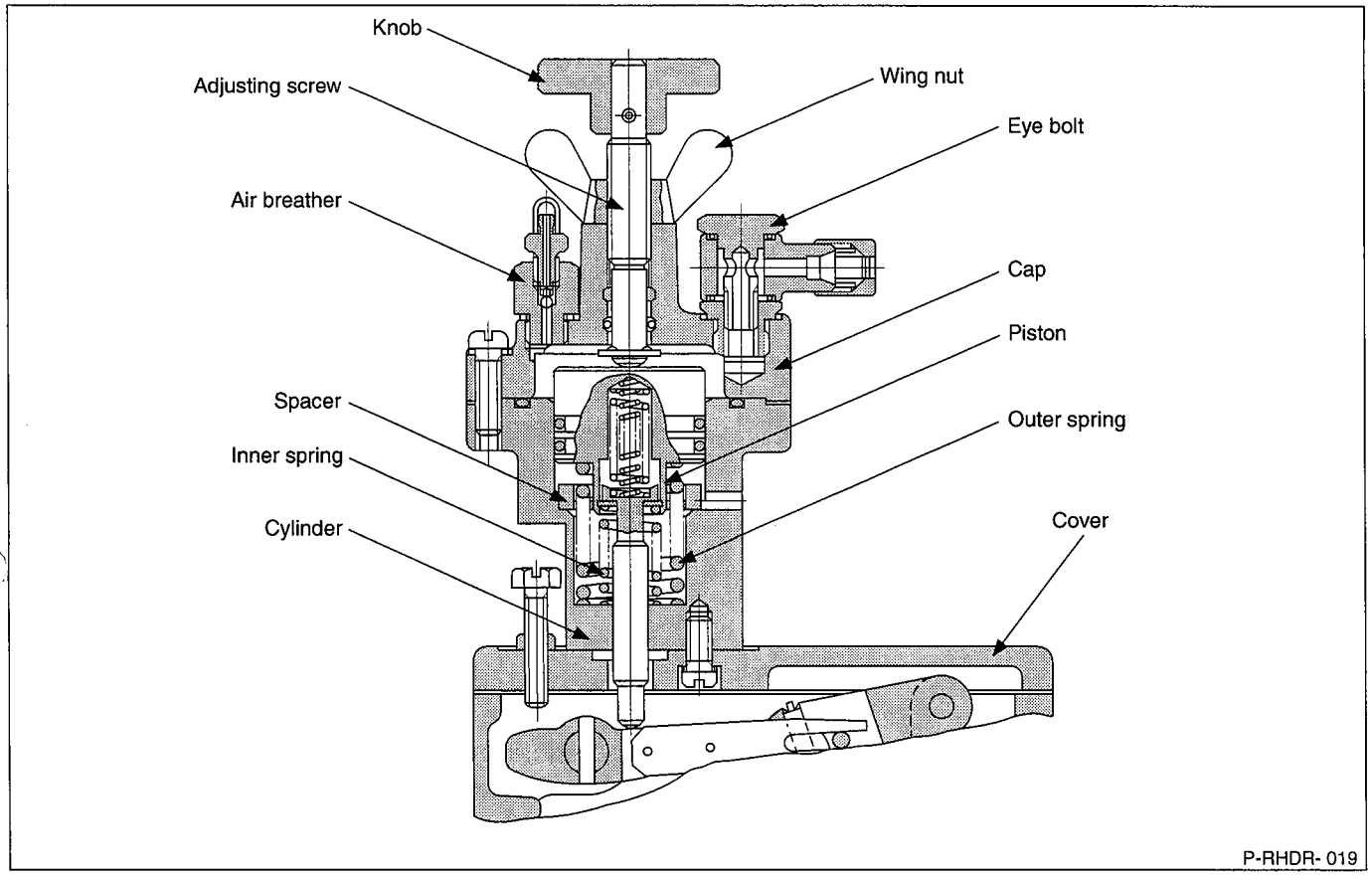
PNEUMATIC CONTROLLER



The pneumatic controller is used to control engine speed.
When air at a specified pressure is supplied to the pneumatic controller, the air pressure acts on the diaphragm and plate, compressing the spring and pushing the pushrod down.

The pushrod acts on the speed adjusting link mechanism in the governor to change the set load of the speeder spring.
The engine speed can also be freely changed manually in the same way by turning the knob on the top of the controller.

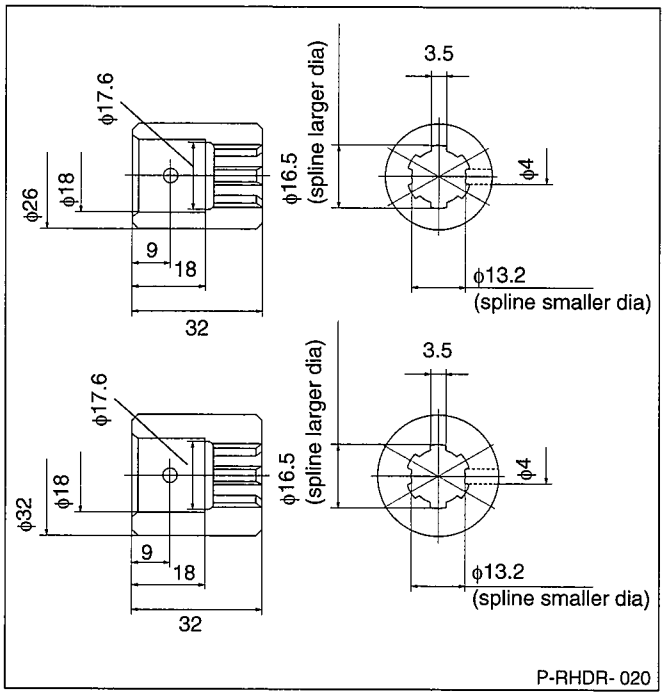
HYDRAULIC CONTROLLER



The hydraulic controller is used to control engine speed, but utilizes hydraulic pressure instead of air pressure, and a piston instead of a diaphragm.

The hydraulic controller's piston displacement is 12 cm^3.

GOVERNOR INSTALLATION



1. Install the governor vertically to the governor drive case. Use the accompanying gasket at the installation surface.
2. Ensure the governor gear shaft is not subjected to a direct radial load, and that eccentricity and acceleration-gear play are as small as possible.
   Gear shaft spline coupling dimensions are shown at left. These are the same for RHD6 and RHD10 governors.

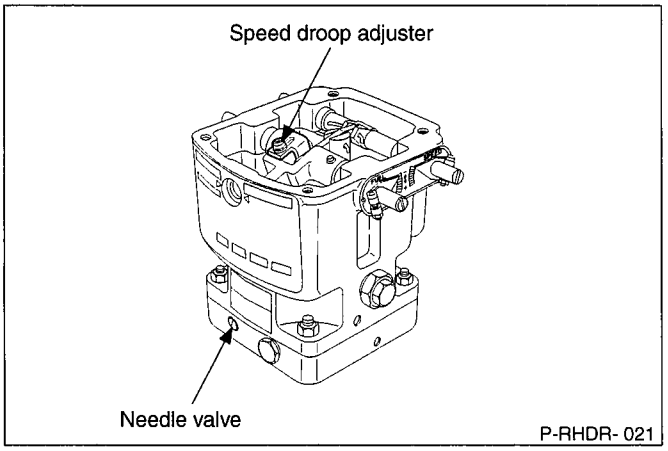
⚠ CAUTION

When handling the governor, ensure the end of the governor's gear shaft is not subject to force or impact.

Note:
Although the spline is a standard JIS square-shaped spline (6 x 13 x 16), the dimensions shown at left are recommended to avoid even minute eccentricity when installing the coupling. The spline couplings shown at left are standard Bosch Corporation spline couplings.

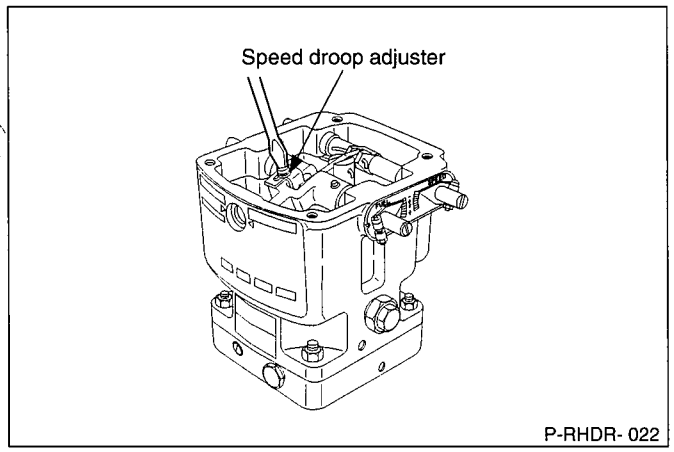
HYDRAULIC FLUID

Hydraulic fluid type	Fluid charge	Inspection	Replacement
The type of hydraulic fluid used varies depending on the governor oil temperature during normal operation. Less than 50°C: SAE20 50°C ~ 70°C: SAE30 Over 70°C: SAE40	Approx 1,300 cm^3. With the governor horizontal, fill the governor with fluid until the level is slightly above the middle of the governor case window. If the fluid charge is insufficient, bubbles will form, performance will decrease, and fluid deterioration will be accelerated.	Check the hydraulic fluid daily. If the charge is less than that specified, check for leaks. If repair is not possible, contact an authorized Bosch Corporation service station.	The fluid must initially be replaced after 1,000 hours or 3 months. Thereafter, the fluid must then be replaced every 3,000 hours or 6 months.

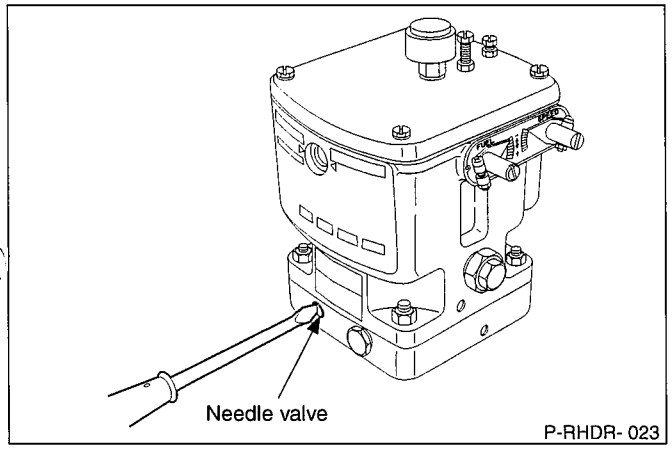


AIR BLEEDING AND COMPENSATION ADJUSTMENT

1. Set the speed droop at the smallest value possible using the speed droop adjuster (0 ~ 2 on the adjuster scale) and fully open the compensating needle valve (3 turns from the fully closed position).
2. Operate the engine at low speed and cause the engine to 'hunt' or 'surge' for approximately 30 seconds to bleed the air.



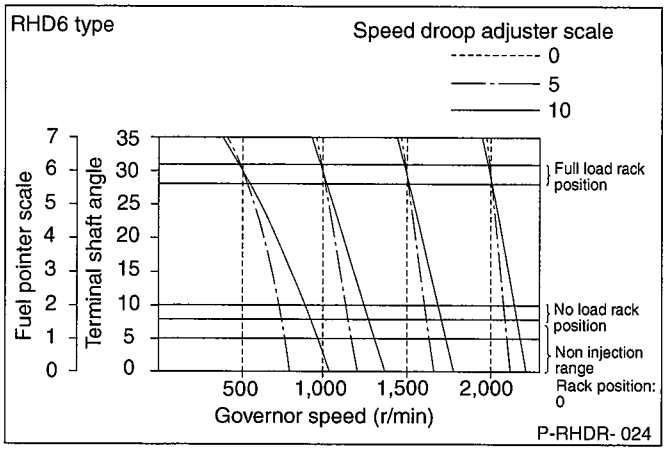
3. Stop the engine, set the speed droop to the setting estimated for the particular application (determined from the performance diagram) and then restart the engine.



4. Close the needle valve until 'hunting' stops.

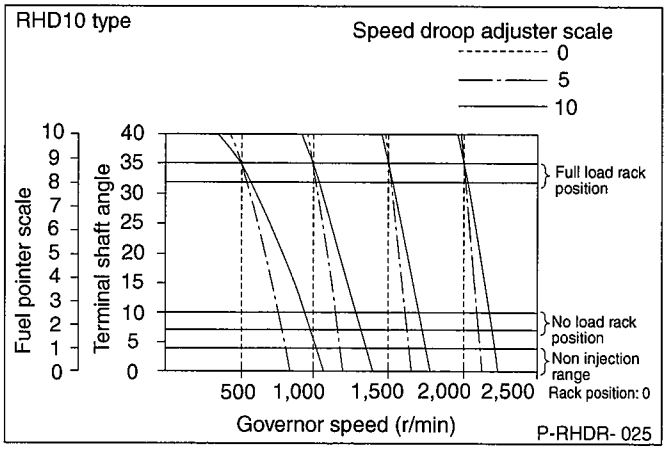
Advice

The above completes air bleeding and compensation adjustment. If, however, the speed droop adjustment setting is changed during engine adjustment, readjust the needle valve to as open a position as possible at which stabilized engine operation is obtained.



LINKAGE CONNECTIONS

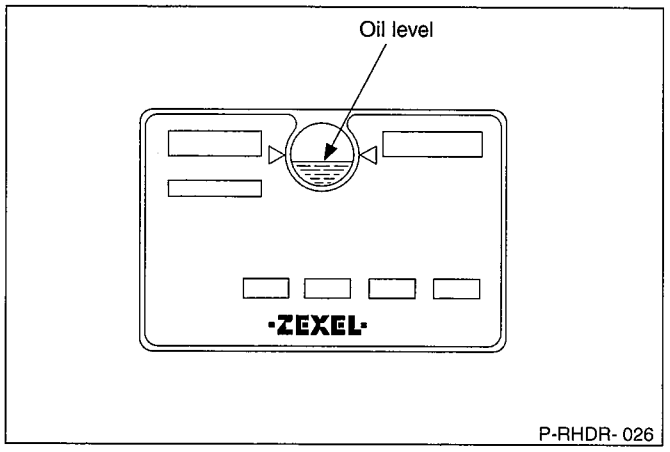
1. Use a split tightening type lever. After determining the correct position, secure it using a knock pin, etc.
2. The linkage must be as simple and direct as possible. Minimize resistance and the actual mass of the linkage.



3. The figures at left show the recommended relationships for connections between the terminal shaft angle and the injection pump for a static governor.

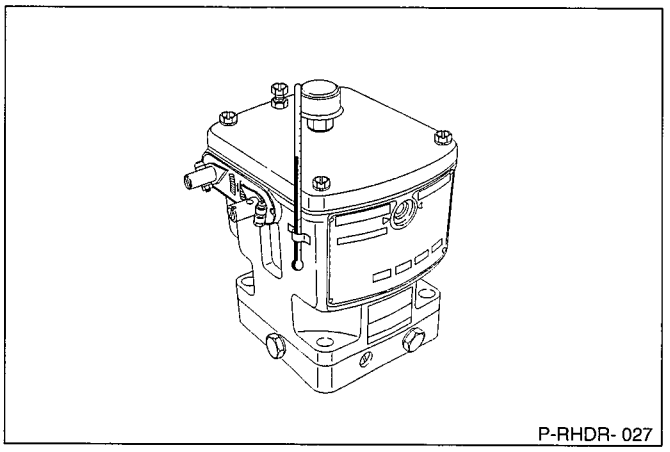
Advice

As play in the link system adversely affects performance, ensure play is as small as possible.



DAILY INSPECTION

1. Check the fluid level daily before beginning operation.
   If the level is below the middle of the gauge, add fluid to the governor.
2. Check that the governor mounting bolts are not loose and that there are no faults in the linkage connections from the governor terminal shaft to the injection pumps' control rack.



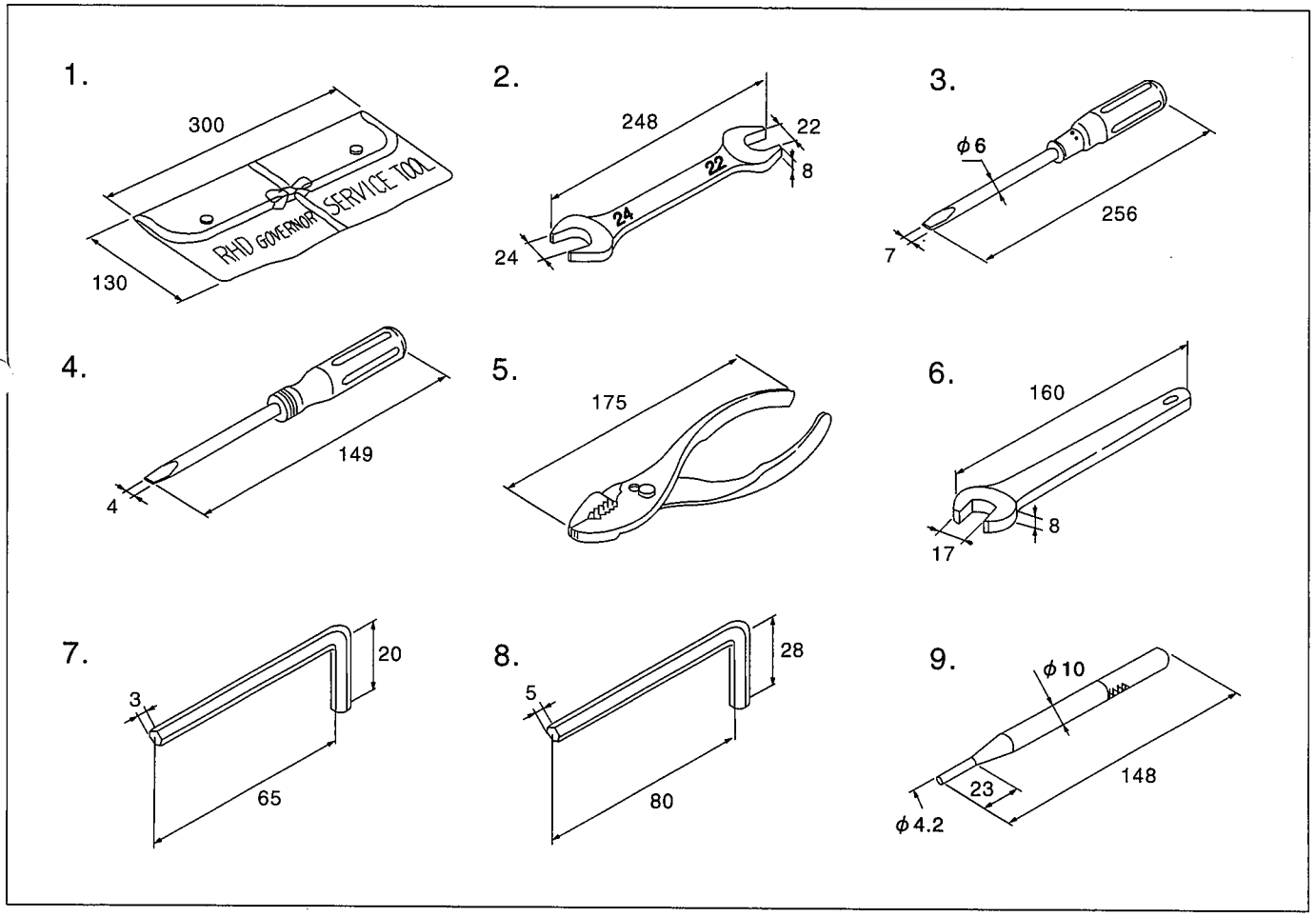
IN-SERVICE INSPECTION

1. Check the governor temperature. The temperature should generally increase to approx 60°C after 30 ~ 40 min of high speed engine operation. If the ambient temperature is high, the temperature may increase to 80°C after continued high speed operation (over 1,800 r/min).
   Temperatures over 100°C are abnormal.
2. Check for abnormal noises.
3. Wipe any fluid from the outside of the governor and check that no fluid leaks during operation.

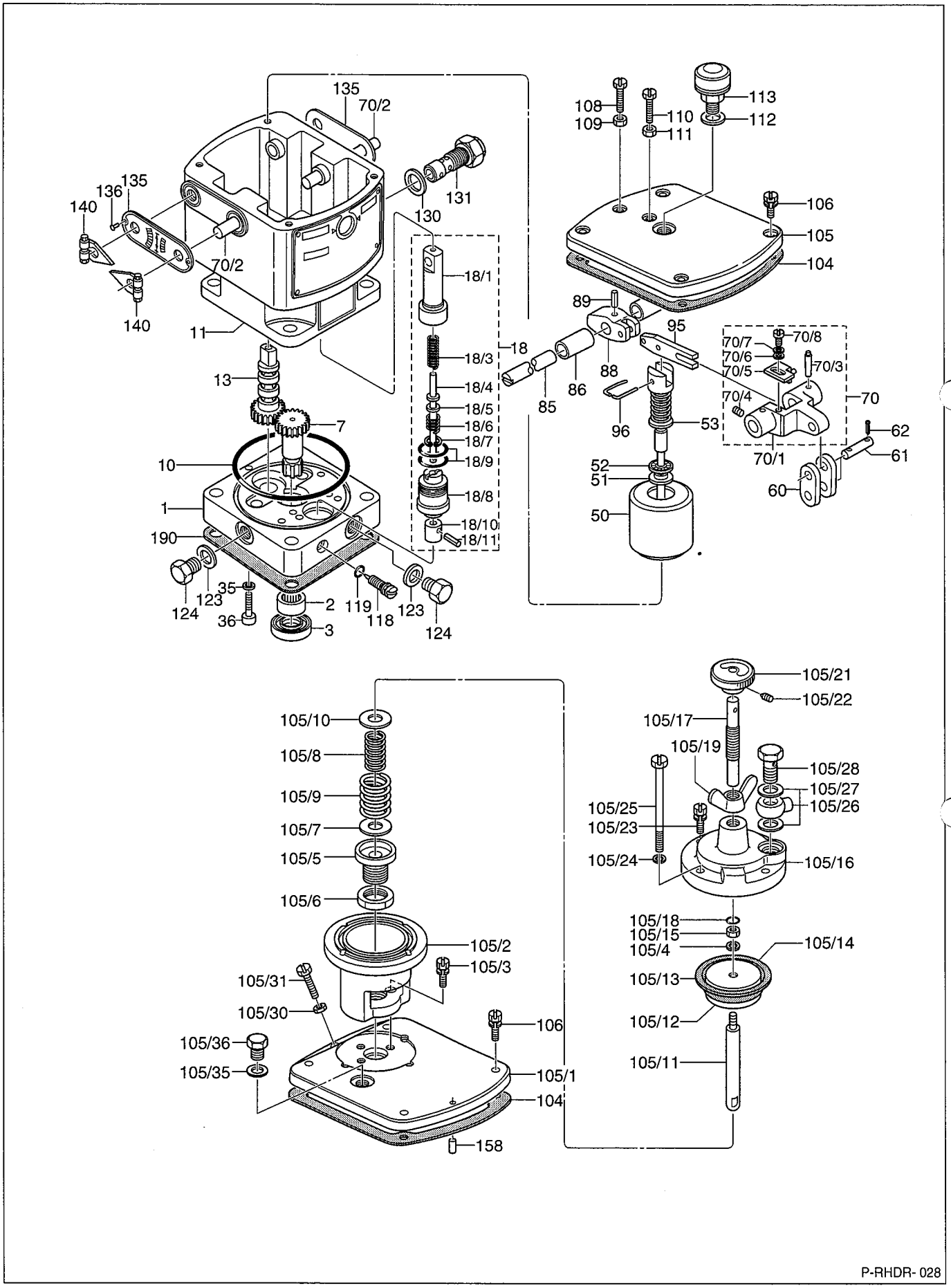
SPECIAL TOOLS

The following special tools (in addition to general tools) are required for disassembly and reassembly of the RHD hydraulic governor.

Key no.	Part name	Zexel part no.	Bosch part no.	Qty	Application
1	Case	376100-2000	9 421 622 974	1	For tools
2	Spanner	376100-2100	9 421 622 811	1	For removing/installing regulating valve (SW22mm, 24mm)
3	Screwdriver	376100-2200	9 421 622 812	1	For removing/installing cover bolts
4	Screwdriver	376100-2300	9 421 622 813	1	For speed droop adjuster adjustment
5	Pliers	376100-2400	9 421 622 814	1	For removing split pins
6	Spanner	376100-2500	9 421 622 815	1	For removing/installing plug (SW17mm)
7	Allan wrench	376100-2600	9 421 622 816	1	SW3mm
8	Allan wrench	376100-2700	9 421 622 817	1	SW5mm
9	Extractor	376100-2800	9 421 622 818	1	For removing pin



EXPLODED VIEW



P-RHDR- 028

PART LIST

RHD-LC type hydraulic governor

Key no.	Part name	Key no.	Part name	Key no.	Part name
1	Base	36	Bolt	89	Roll pin
2	Needle roller bearing	50	Flyweight	95	Floating lever
		51	Washer	96	Wire
3	Oil seal	52	Thrust needle roller bearing	104	Gasket
7	Gear shaft			105	Cover
10	O-ring	53	Pilot valve assembly	106	Bolt
11	Housing			108	Maximum speed setting screw
13	Sleeve	60	Guide lever		
18	Compensator assembly	61	Pin	109	Nut
		62	Split pin	110	Minimum speed setting screw
18/1	Power piston	70	Terminal arm assembly		
18/3	Upper compensating spring			111	Nut
		70/1	Terminal arm	112	Gasket
18/4	Pushrod	70/2	Terminal shaft	113	Air breather
18/5	Washer	70/3	Taper pin	118	Needle valve
18/6	Lower compensating spring	70/4	Set screw	119	O-ring
		70/5	Speed droop adjuster	123	Gasket
18/7	Snapring			124	Screw plug
18/8	Compensator bushing	70/6	Washer	130	Gasket
		70/7	Lock washer	131	Regulating valve
18/9	O-ring	70/8	Screw	135	Indication plate
18/10	Compensating piston	85	Speed control shaft	136	Nail
18/11	Roll pin	86	Collar	137	Indication plate
35	Gasket washer	88	Fork	140	Pointer

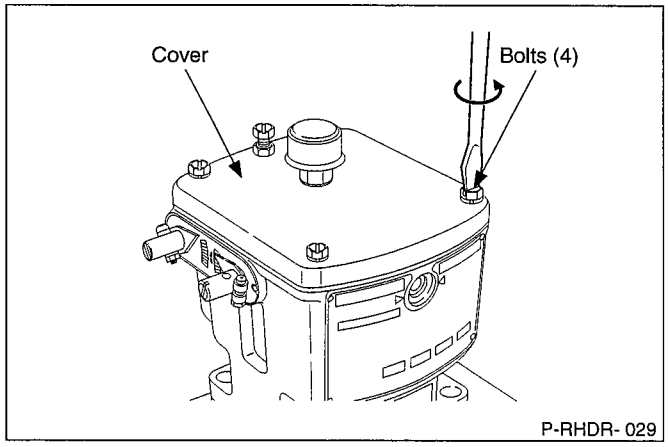
Pneumatic controller

Key no.	Part name	Key no.	Part name	Key no.	Part name
105	Pneumatic controller	105/12	Piston	105/25	Bolt
105/1	Cover	105/13	Diaphragm	105/26	Eye
105/2	Cylinder	105/14	Plate	105/27	Gasket
105/3	Screw	105/15	Nut	105/28	Eye bolt
105/4	Spring washer	105/16	Cover	105/30	Nut
105/5	Spring seat	105/17	Screw	105/31	Bolt (set screw)
105/6	Nut	105/18	O-ring	105/35	Gasket
105/7	Washer	105/19	Wing nut	105/36	Bolt
105/8	Spring	105/21	Knob	(158)	Pin
105/9	Spring	105/22	Set screw		
105/10	Washer	105/23	Screw		
105/11	Pushrod	105/24	Spring washer		

PREPARATION

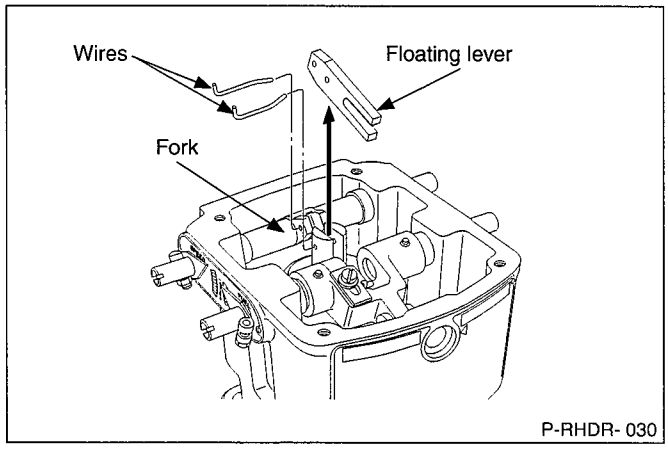
Keep the work bench clean and tidy.
Before starting disassembly, record performance data and the positions of adjustable parts for later reference. This data will facilitate the detection and diagnosis of any governor malfunctions and defects.

During disassembly, put the disassembled parts neatly and sequentially on the work bench, labelling them if necessary to facilitate later reassembly.
Clean the outside of the governor before disassembly.

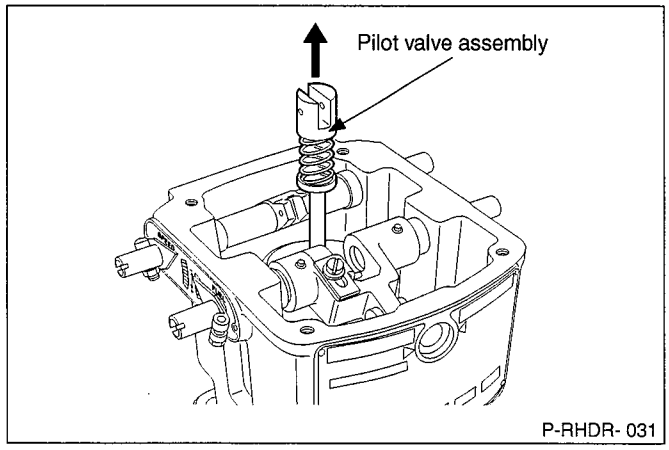


DISASSEMBLY

1. Remove the four bolts using a screwdriver and then remove the cover together with the gasket.
2. Turn the governor upside down and drain the fluid.

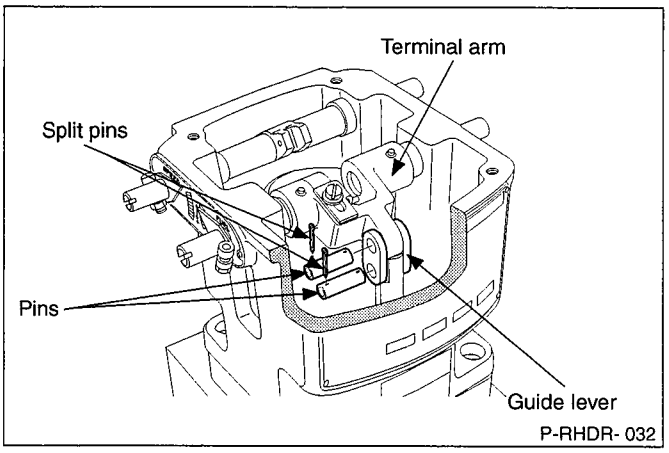


3. Remove the two wires connecting the fork to the floating lever and then remove the floating lever.

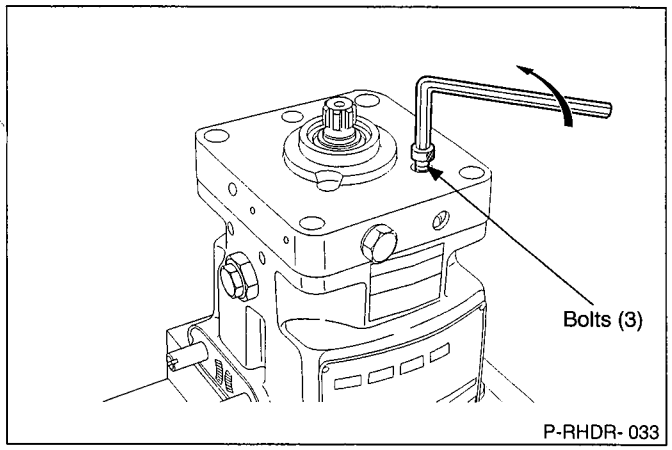


4. Remove the pilot valve assembly.

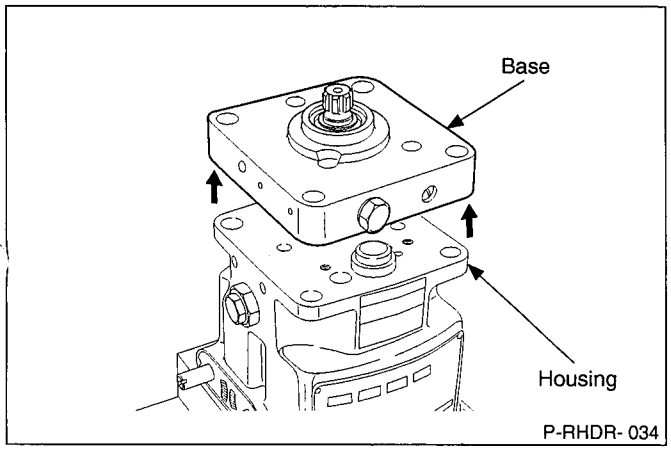
Advice
Put the pilot valve in clean light oil.



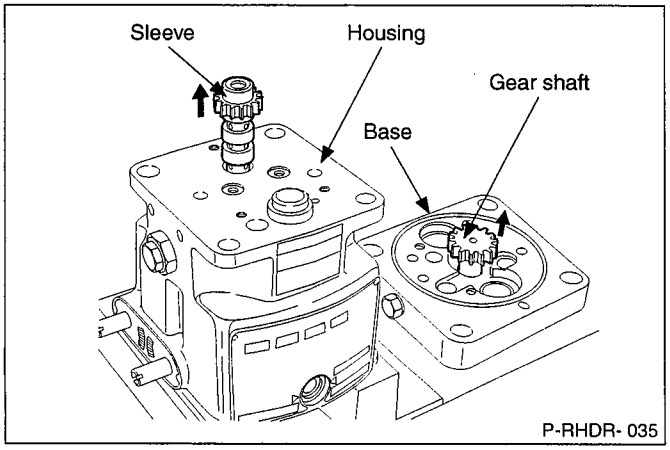
5. Remove the two split pins and then remove the two pins connecting the terminal arm and the guide lever to the power piston.



6. Remove the bolts using an allan wrench (SW 5 mm).



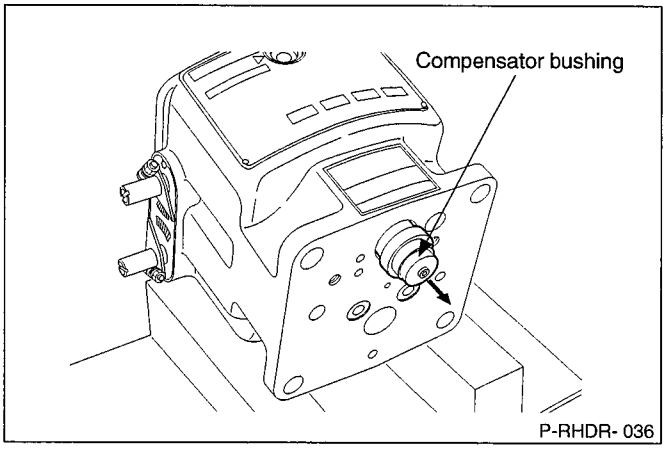
7. Tap the base lightly with a plastic hammer to separate it from the housing.



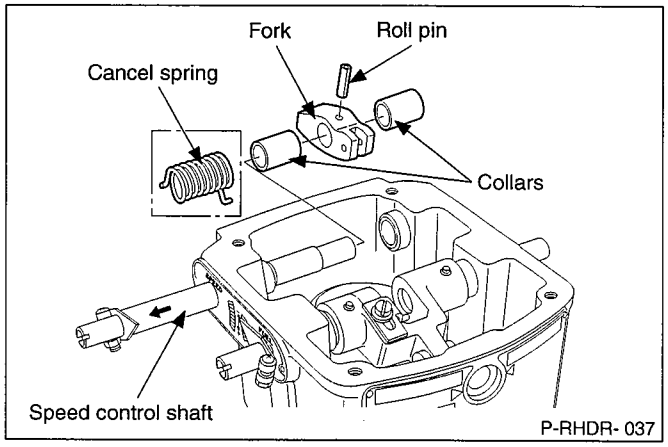
8. Remove the sleeve from the housing and the gear shaft from the base.

Advice

Put the sleeve in clean light oil.

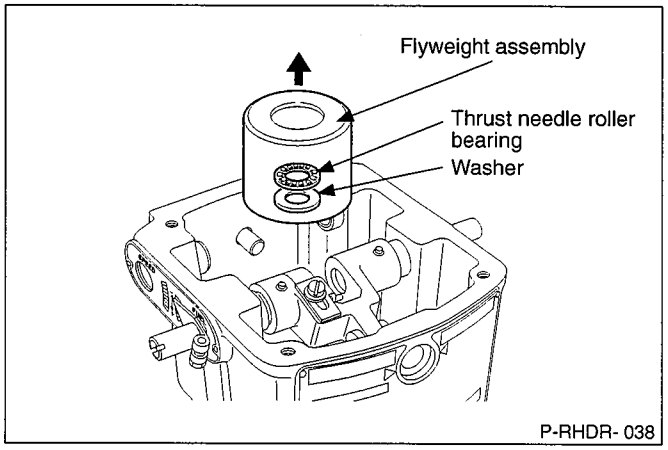


9. Push the power piston down from the top of the governor and then remove the compensator bushing and the compensator assembly.

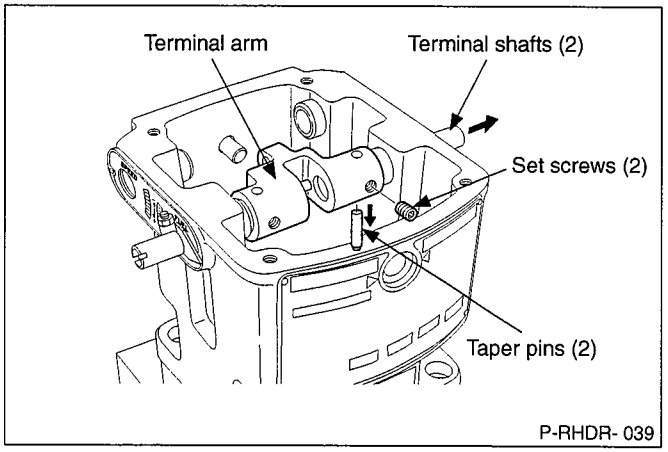


10. Remove the roll pin securing the fork to the speed control shaft, and then remove the speed control shaft, the collars and the fork.

Note:
PC, MC, and MCL type governors are equipped with a cancel spring on the speed control shaft. Remove the cancel spring when removing the speed control shaft.



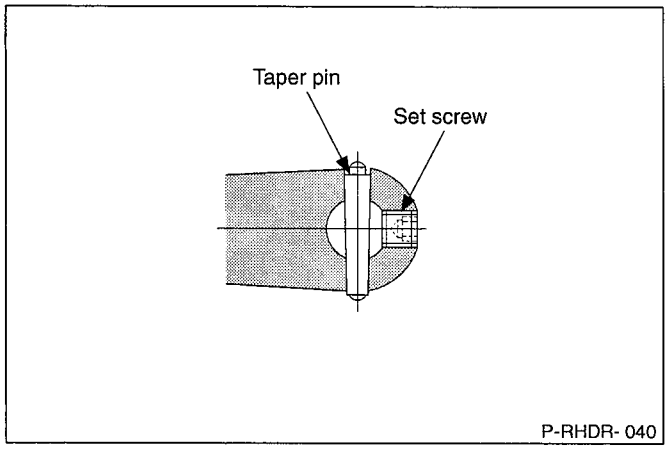
11. Remove the flyweight assembly together with the thrust needle roller bearing and the washer from the housing.

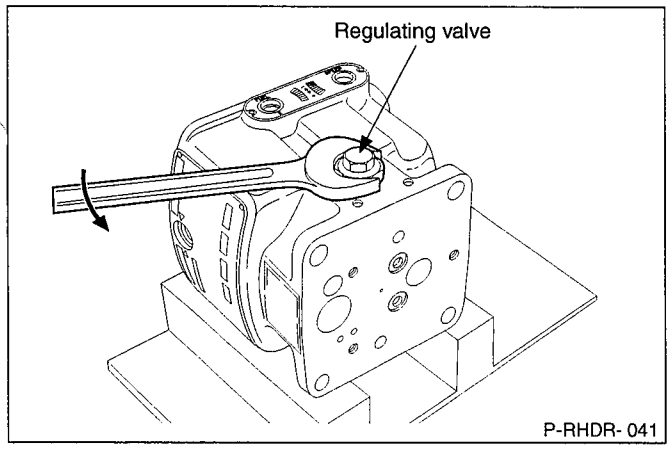


12. Turn one terminal shaft one half turn and remove the terminal shafts' taper pins. Remove the set screws using an allan wrench (SW 3 mm), then remove the terminal shafts and arm.

Advice

1. **Do not remove the terminal arm assembly unless it is not operating smoothly or it is necessary to replace parts.**
2. **When replacing the terminal shafts and arm, they must be replaced as an assembly.**





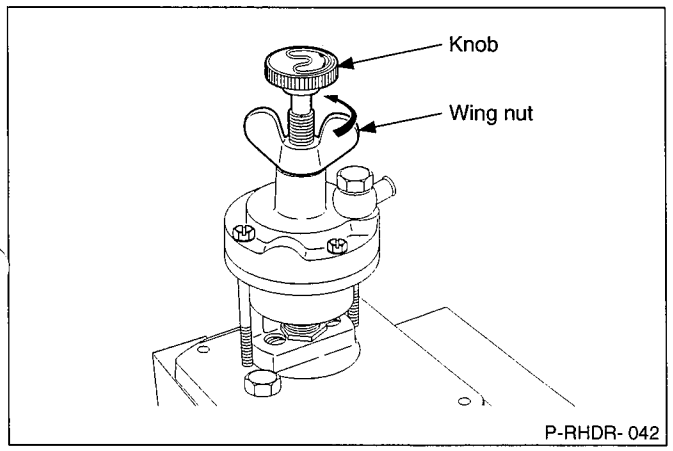
13. Remove the needle valve and regulating valve (hexagon head, SW 24 mm) and any other external parts.

The above completes disassembly of main RHD governor components.

Disassemble other components when necessary.

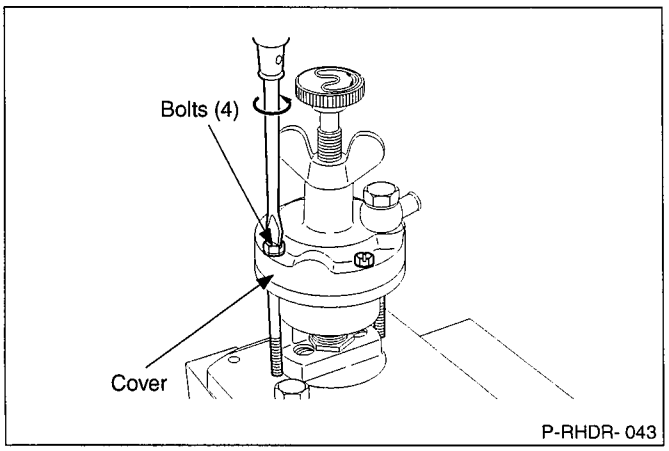
The pneumatic controller is the most commonly equipped additional device.

Disassembly is described below.

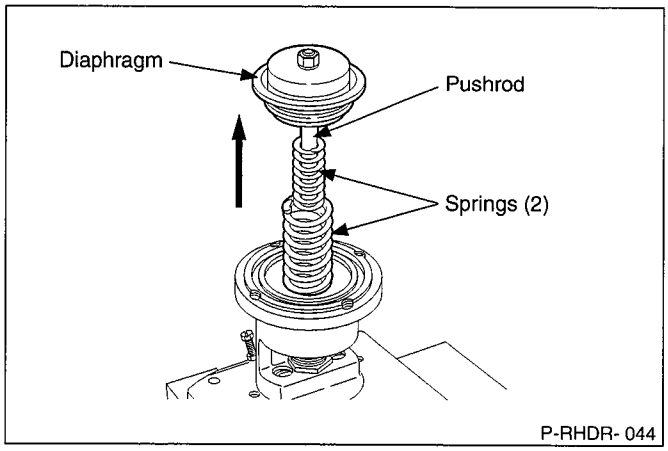


PNEUMATIC CONTROLLER DISASSEMBLY

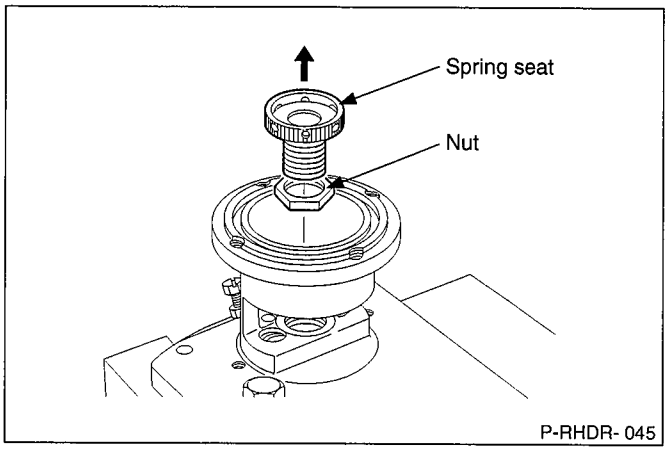
1. Loosen the wing nut. Then, loosen the knob until it can be easily turned.



2. Remove the four bolts using a screwdriver and then remove the cover.



3. Remove the diaphragm together with the pushrod, and then remove the two springs.



4. Loosen the nut and then remove the spring seat.

The above completes disassembly of main pneumatic controller parts.
Disassemble other parts when necessary.

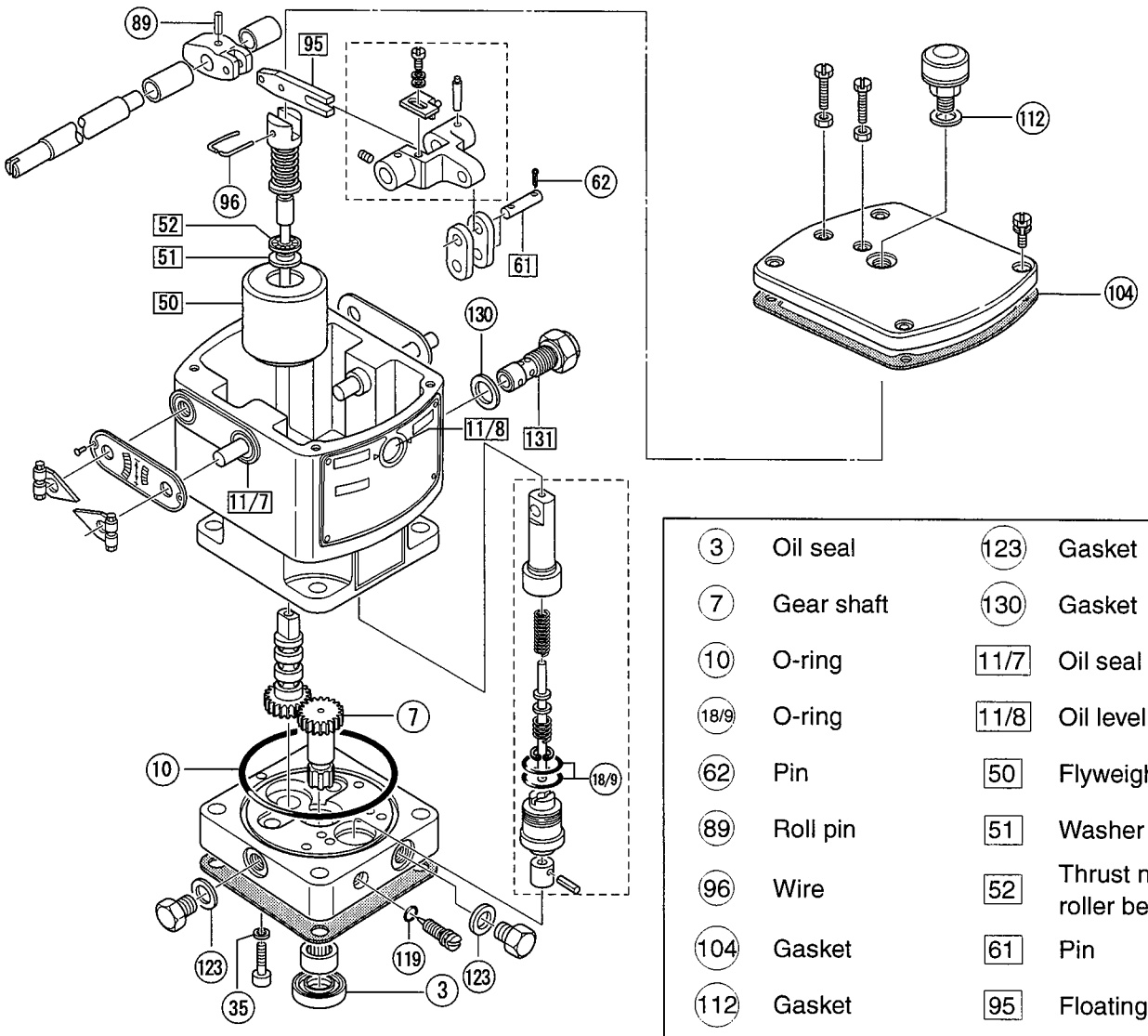
Record the details of all inspections and repairs.
With parts such as those listed below, first check their external appearance to determine whether further disassembly is necessary.

- Terminal arm assembly
- Pilot valve assembly
- Housing assembly
- Flyweight assembly
- Governor motor assembly

Wash all parts thoroughly in clean light oil, and check for wear, damage and scratches. Replace any parts that cannot be reused with new parts.

REPLACEMENT STANDARDS

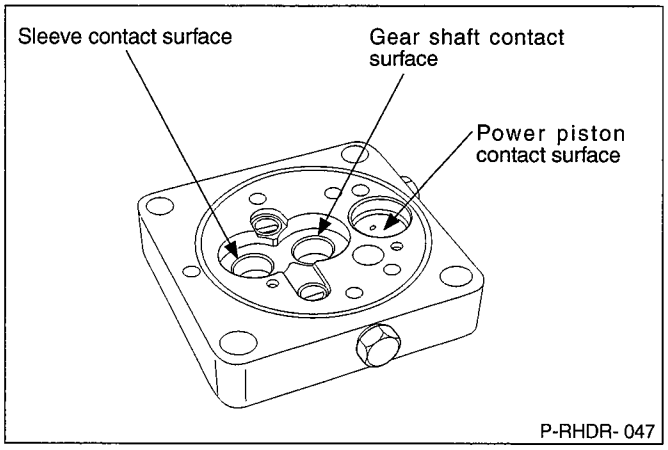
Parts with key numbers marked ○ :
Replace at disassembly or every 2 years
Parts with key numbers marked □ :
Replace every 4 years or 20,000 hours
In principle, assemblies must be replaced when any of their component parts have been used for 8 years or 40,000 hours.



P-RHDR- 046

3	Oil seal	123	Gasket
7	Gear shaft	130	Gasket
10	O-ring	11/7	Oil seal
18/9	O-ring	11/8	Oil level gauge
62	Pin	50	Flyweight
89	Roll pin	51	Washer
96	Wire	52	Thrust needle roller bearing
104	Gasket	61	Pin
112	Gasket	95	Floating lever
119	O-ring	131	Regulating valve

If parts other than the above are worn or damaged, replace them with new ones.

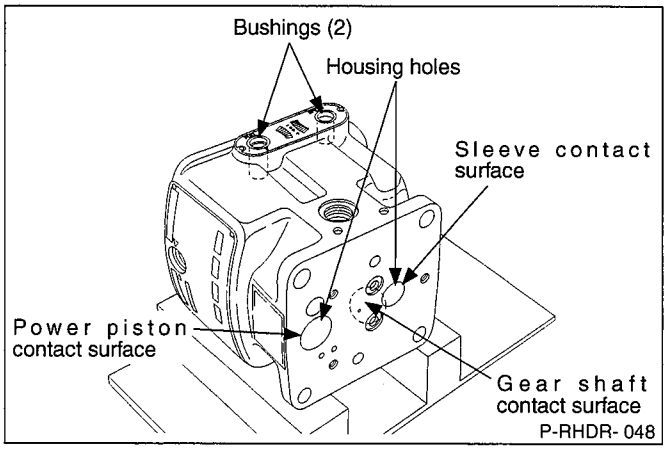


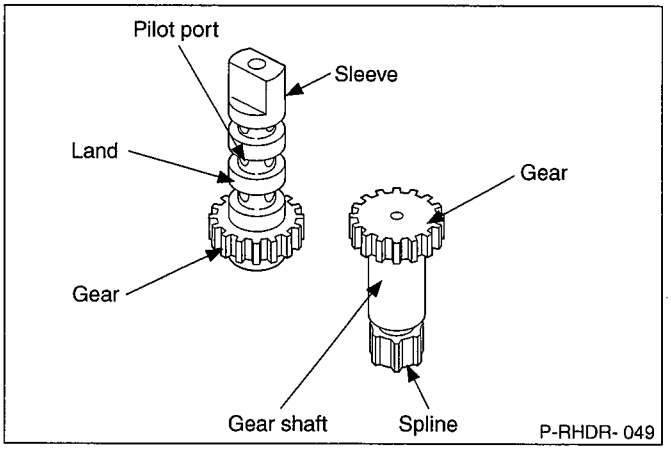
Base assembly and housing

Replace the base if the power piston, gear shaft or sleeve contact surfaces are worn or damaged.

Replace the housing if the power piston, gear shaft or sleeve contact surfaces, or the housing holes, are worn or damaged.

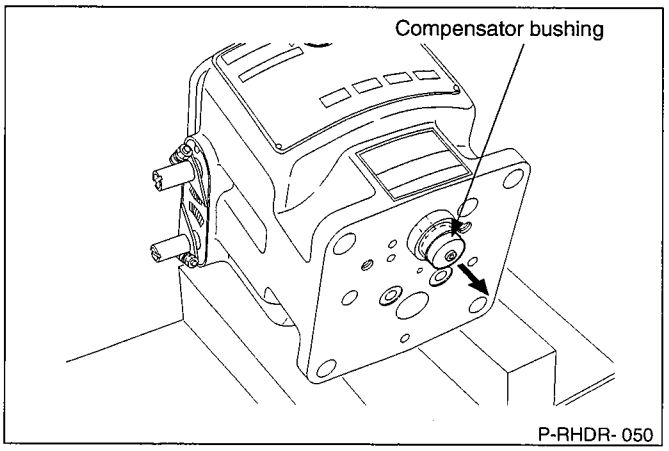
Replace the housing if the pressfitted terminal arm bushings are worn or loose.





Gear shaft and sleeve

Replace the gear shaft if the outside, spline, oil seal contact surface or gear is worn or damaged. Replace the sleeve if the gear, pilot ports or lands are worn or damaged.



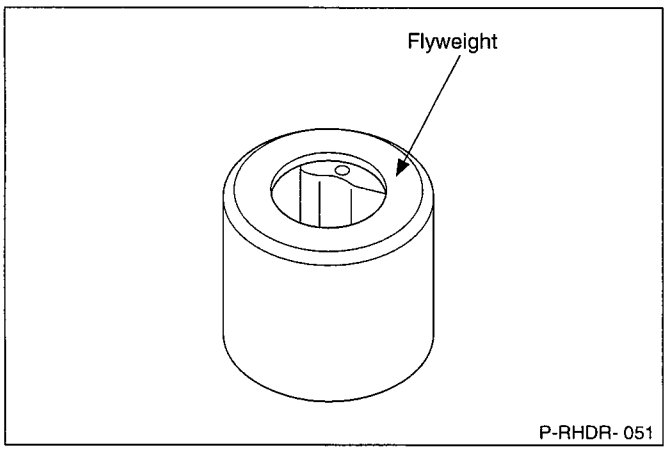
Compensator assembly

Assemble the compensator assembly's power piston, pushrod and compensator bushing in the housing and check that they move smoothly up and down under their own weight when the housing is moved.

Repair or replace them if they do not move smoothly.

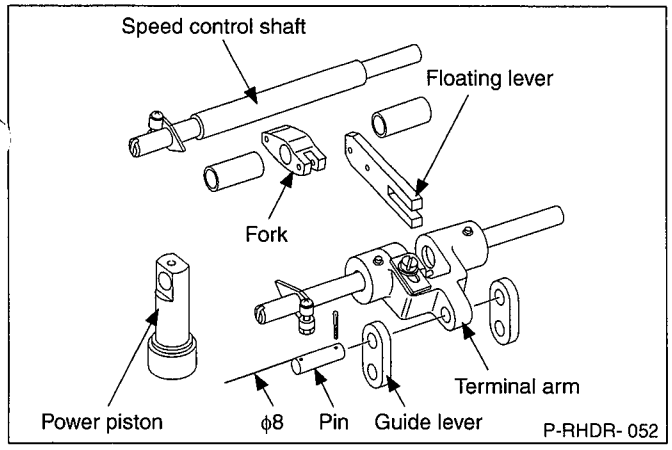
Note:

If the power piston does not move smoothly, hunting or excessive speed droop will prevent engine adjustment.



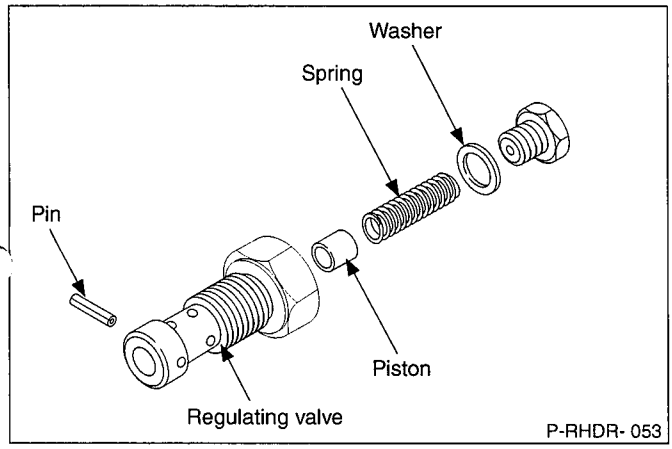
Flyweight assembly

Foreign matter in the flyweight will prevent smooth flyweight movement.
Check that the flyweight moves smoothly.
Wash the flyweight thoroughly.



Power piston, guide lever and terminal arm

- Replace the guide lever if the pin holes are worn.
  Replace the pins if they are worn.
  Replace the power piston if the inside is worn.
- Replace the speed control shaft if it is bent or the shaft journals are worn.

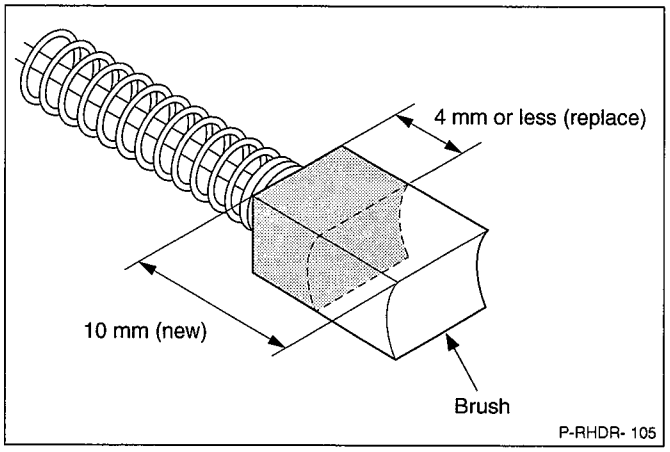


Regulating valve

Check that the piston moves smoothly.
Repair or replace the piston if it is worn or does not move smoothly.

Other parts

- Check all other parts for cracks, damage, damaged threads and rust.
  Repair or replace them if necessary.
- Replace all O-rings and gaskets.
- Replace springs that are damaged, bent or rusted.



GOVERNOR MOTOR BRUSH INSPECTION (MC, MCL TYPES)

Advice

Inspection applies only to direct current (DC) type motors. The following is not necessary for alternating current (AC) motors.

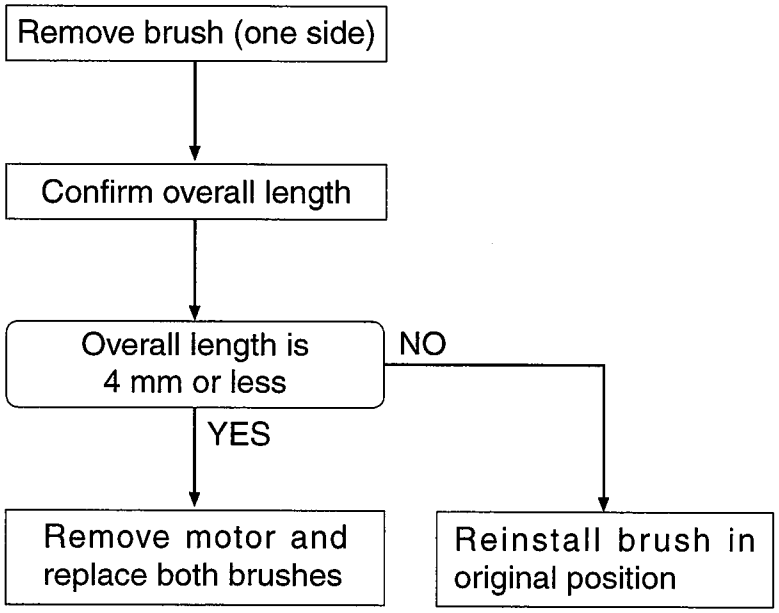
Brush inspection intervals

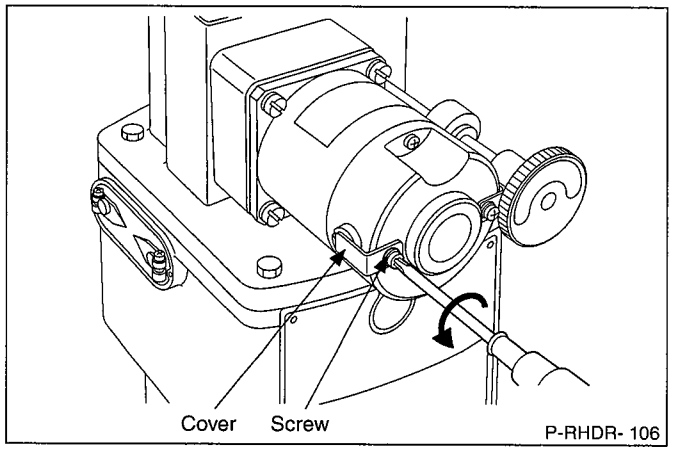
Brush inspection intervals are shown below.

Part name	Inspection interval	Remarks
Brush	Monthly	When overall length is 4 mm or less

Note:
Overall length of new brushes: 10 mm

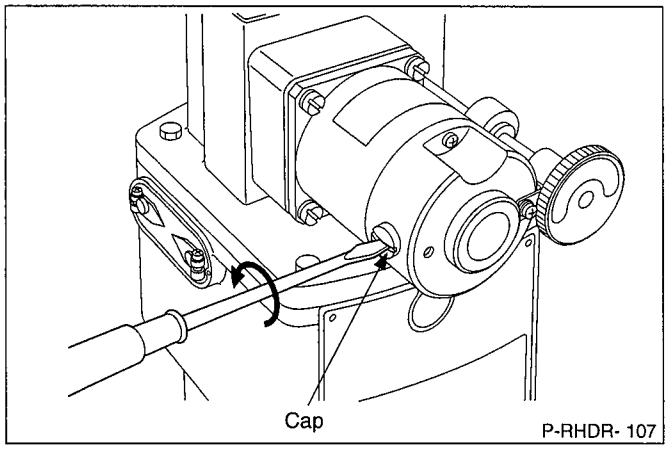
Brush inspection



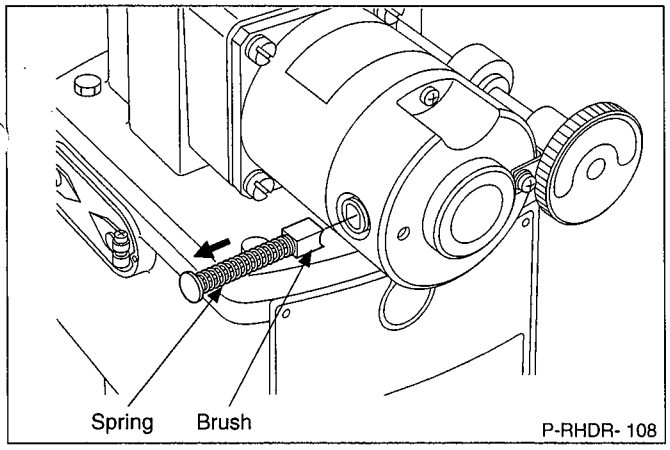


Brush removal

1. Remove the screw and then remove the cover.



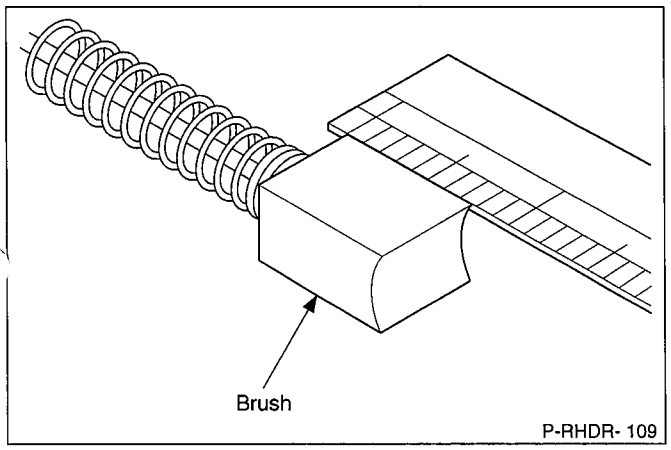
2. Loosen and then remove the cap.



3. Remove the brush from the motor.

Advice

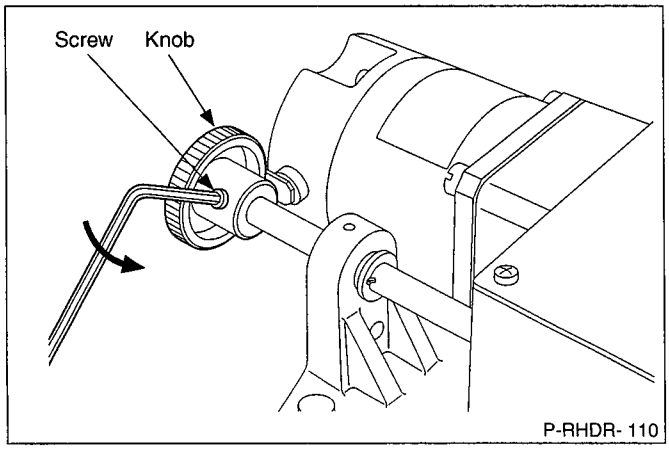
Do not stretch the brush spring during removal.



4. Measure the overall length of the brush.

Advice

- **When overall brush length still exceeds the wear limit of 4 mm, reverse the removal procedure to reinstall the brush. (Inspection is complete.)**
- **When overall brush length is 4 mm or less, continue with the following steps. (Replace both brushes at the same time.)**

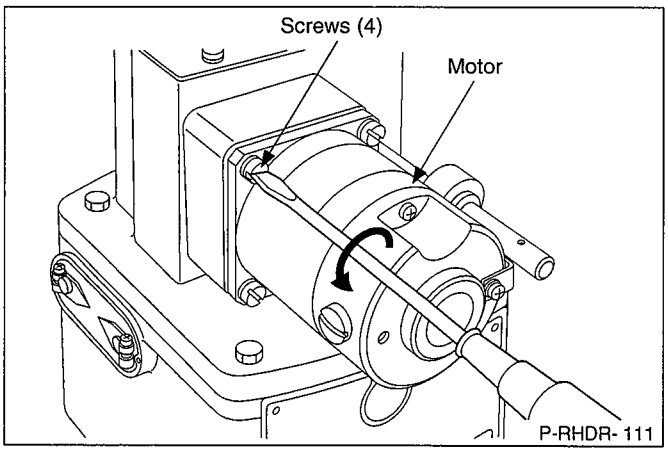


Motor removal

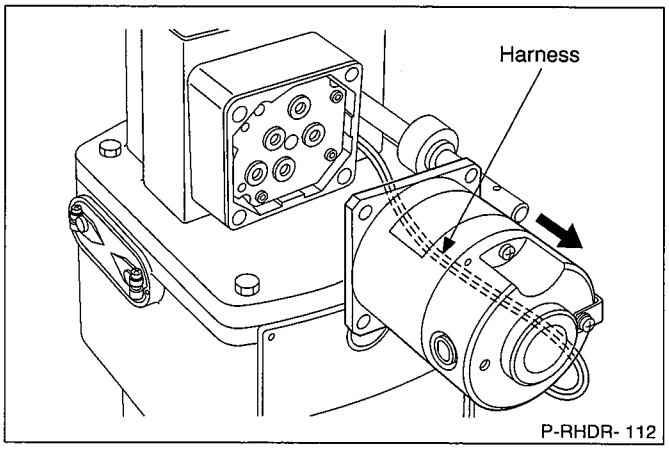
1. Loosen the screw and then remove the knob.

Note:

Screw: SW3

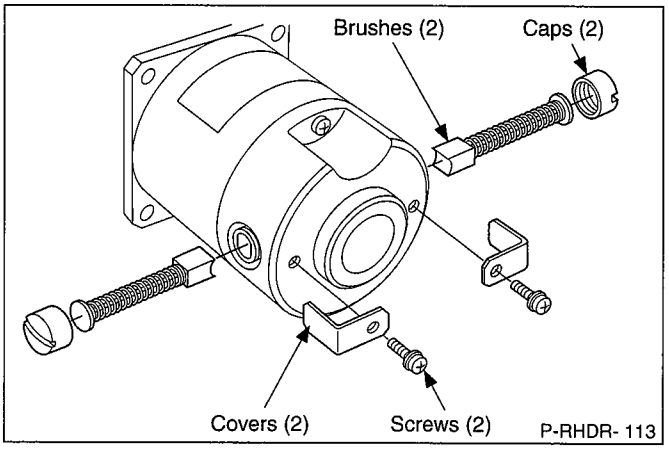


2. Remove the screws and then remove the motor horizontally.



Advice

The motor is connected to a harness. Do not pull the harness excessively.



Brush replacement

1. Remove the knob-side brush using the same procedure as above.
2. Replace both brushes with new ones and then reinstall the motor and knob by reversing the removal procedure.

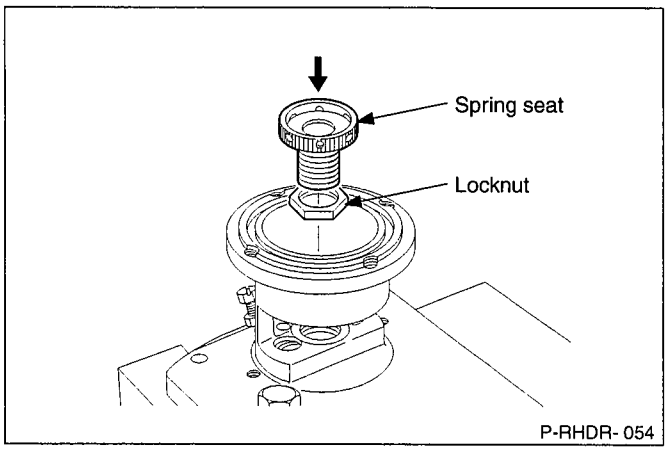
Advice

- **When reinstalling the motor, install the motor so that the side with the nameplate is facing up.**
- **Tighten diagonally opposed motor fixing screws (4) gradually and evenly.**

Tightening torque: 2 ~ 2.2 N·m
{0.2 ~ 0.22 kgf·m}

Brush part numbers

Part name	Zexel part no. (Bosch part no.)	Remarks
Brush	158901-4500 (9 421 622 899)	For DC24V
	158901-4600 (9 421 622 900)	For DC100V

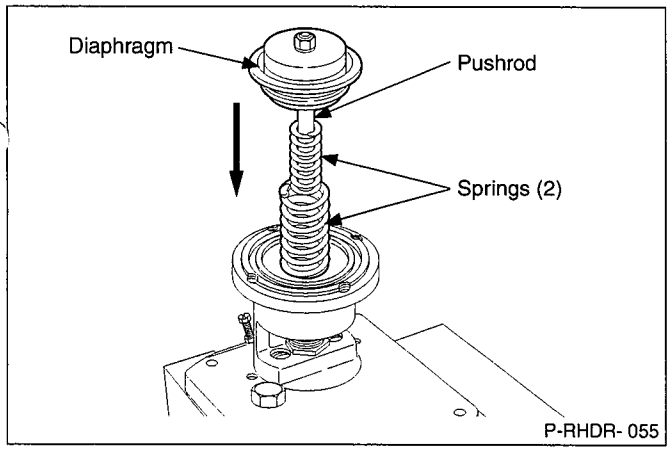


Governor reassembly is described below.

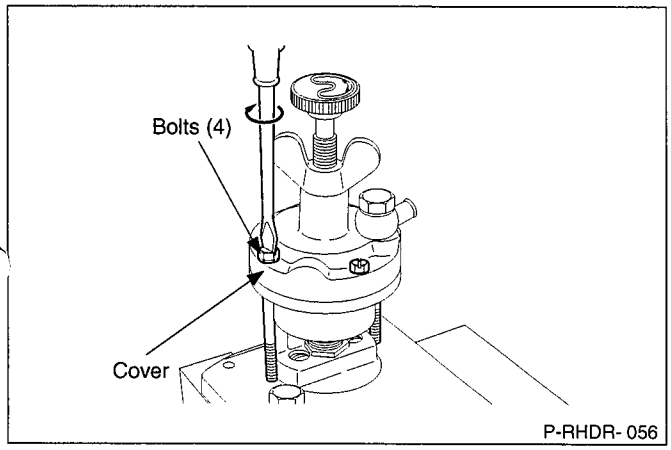
PNEUMATIC CONTROLLER

(if installed)

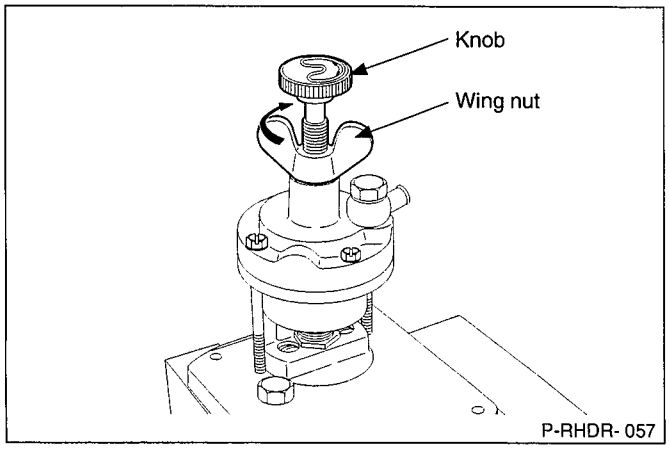
1. Screw the locknut onto the spring seat, then screw the spring seat into the cylinder.



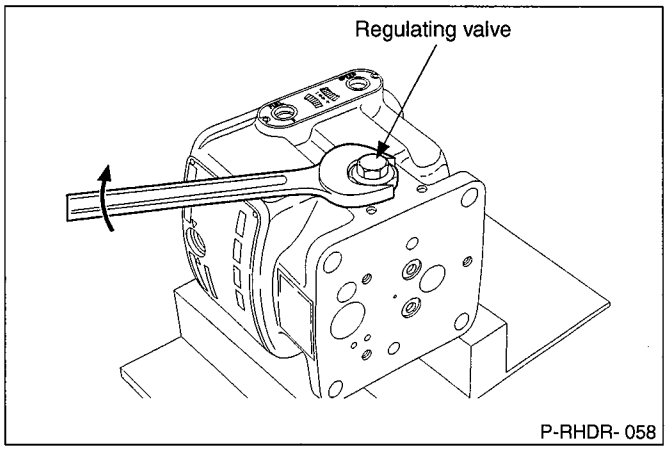
2. Install the diaphragm and pushrod together with the two springs.



3. Install the cover and secure it using the four bolts.



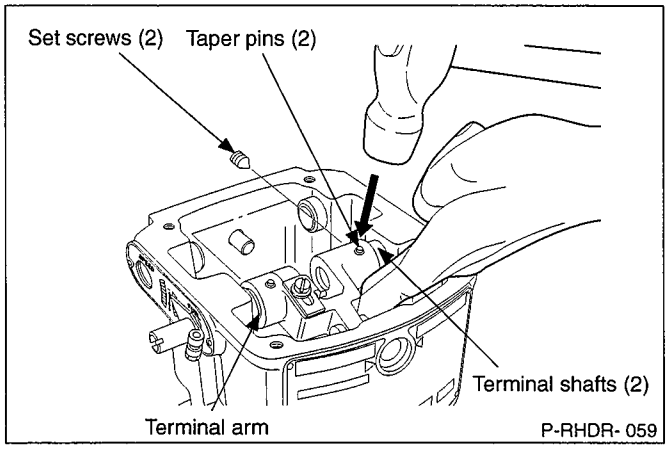
4. Install the knob and wing nut.



GOVERNOR ASSEMBLY

1. Install the regulating valve.

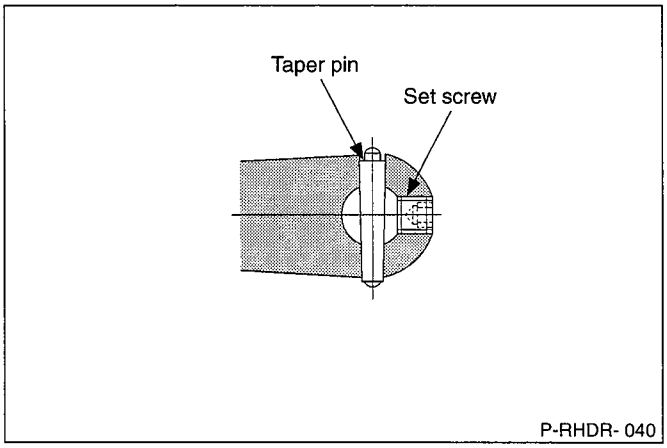
Tightening torque: 25 ~ 29 N·m {2.5 ~ 3.0 kgf·m}



2. Install the terminal shafts and arm. Align each terminal shaft's and terminal arm's taper hole.
   Coat the taper pins with an adhesive and install the taper pins.

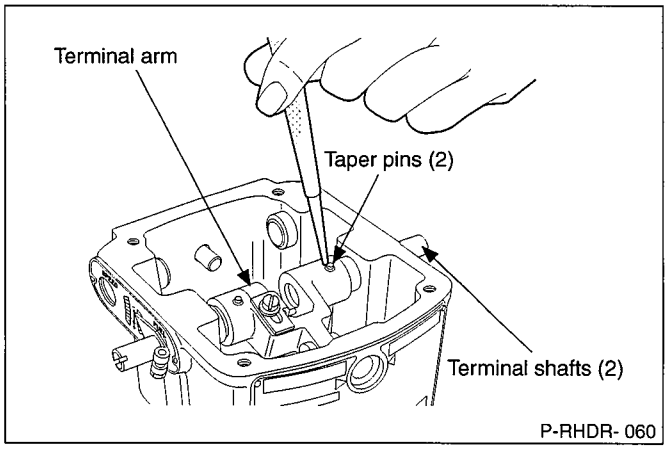
⚠ CAUTION

When installing the two taper pins in the terminal shafts, support the terminal arm and terminal shafts to avoid damaging the bushings pressfitted to the housing.

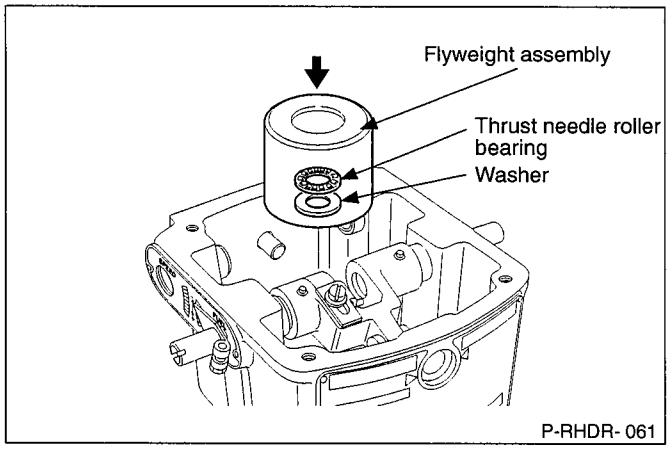


3. Coat the set screws with an adhesive and install the two set screws.

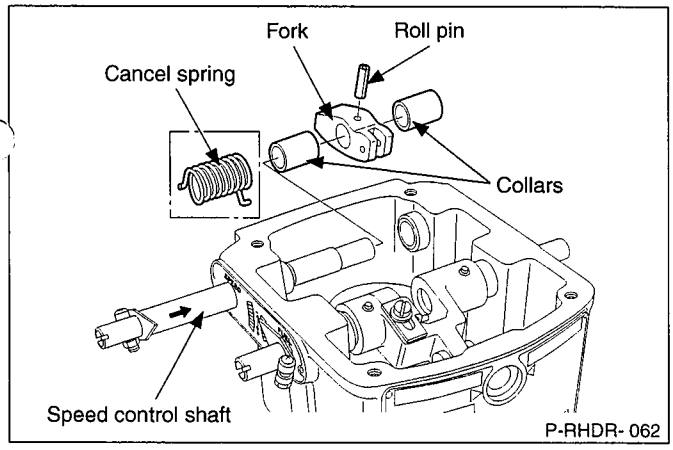
Tightening torque: 3.9 ~ 4.9 N·m {0.4 ~ 0.5 kgf·m}



4. Caulk the two taper pins after final tightening.



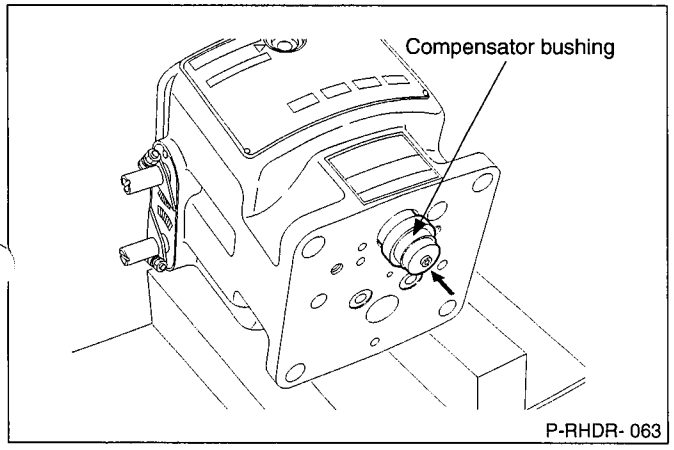
5. Install the flyweight assembly, thrust needle roller bearing and washer in the housing.



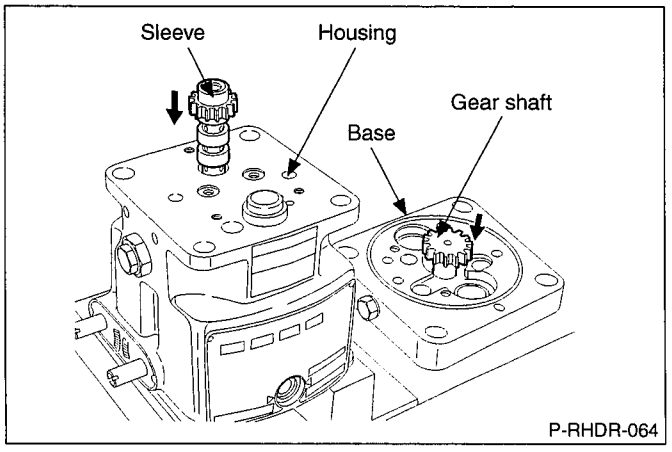
6. Install the speed control shaft, collars and fork, and then install the roll pin.

Advice

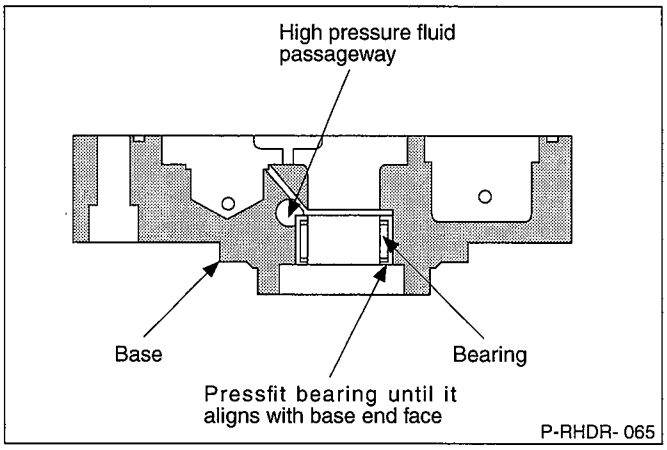
PC, MC, and MCL governors are equipped with a cancel spring. Install the cancel spring when installing the speed control shaft.



7. Install the power piston, the compensator assembly and the compensator bushing.

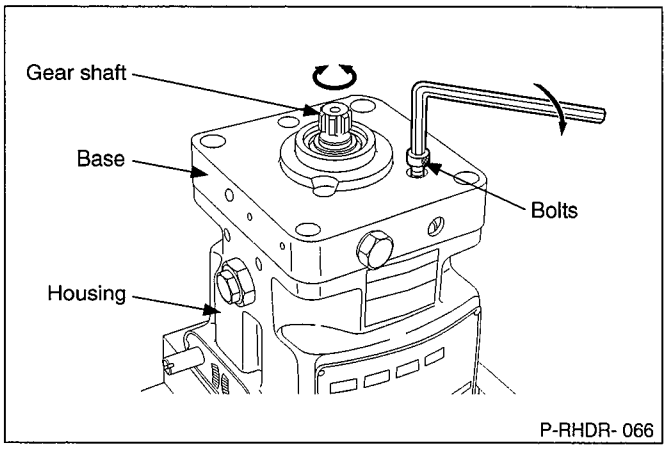


8. Install the gear shaft in the base, and the sleeve in the housing.



Advice

When pressfitting the gear shaft bearing into the base, pressfit the bearing to align with the base face, as shown. If the bearing is pressfitted too deep, the bearing will block the high pressure fluid passageway.

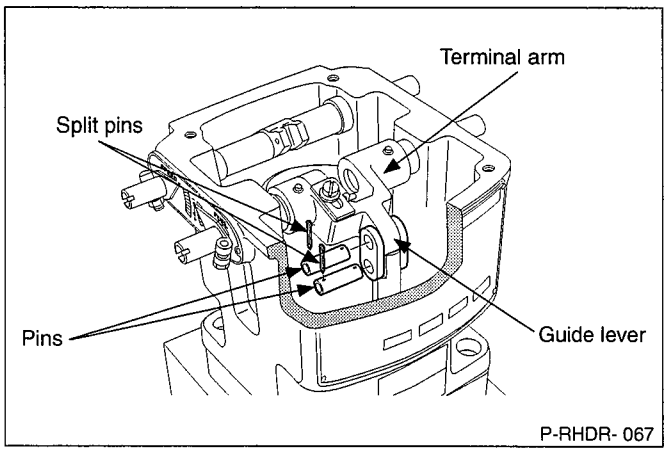


9. Install the base to the housing using the bolts and gasket washers.

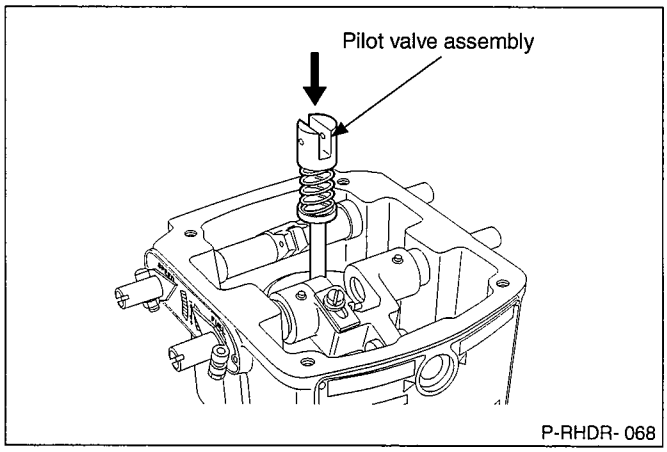
Tightening torque: 10 ~ 15 N·m {1.0 ~ 1.5 kgf·m}

⚠ CAUTION

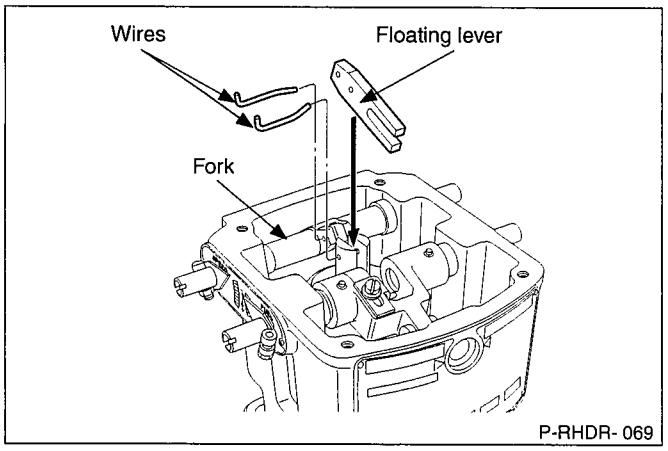
While securing the base to the housing, check that the gear shaft turns smoothly. If the gear shaft does not turn smoothly, the oil delivery gear may be damaged.



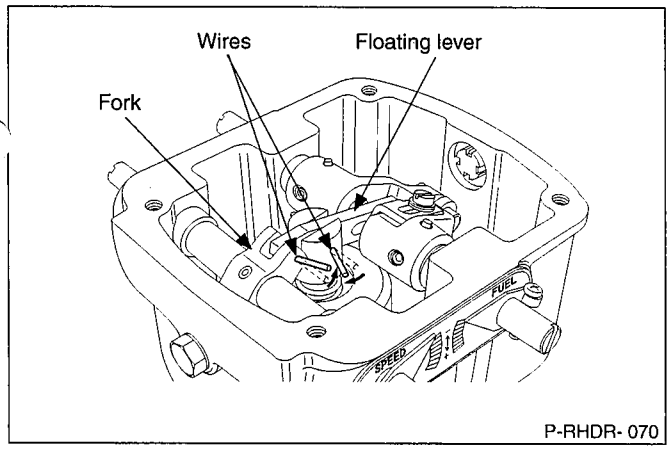
10. Attach the guide lever to the terminal arm and the power piston using the two pins and two split pins.
Then, bend the split pins over.



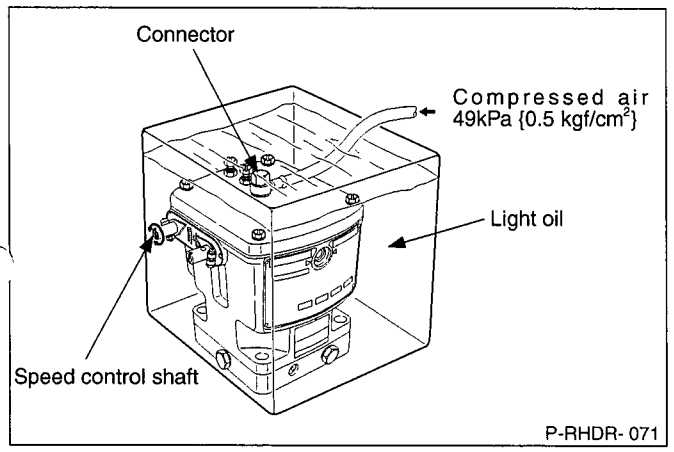
11. Install the pilot valve assembly.



12. Install the floating lever and secure it to the fork and pilot valve using the two wires.



Advice
Bend the ends of the wires to prevent them from loosening.



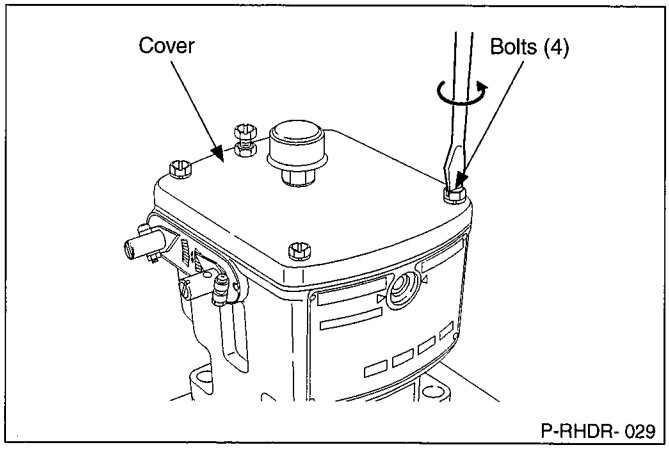
13. After assembling the governor's inner components, thoroughly wash the inside and then install the cover using the bolts.

Air tightness test

-1 Before filling the governor with hydraulic fluid, remove the air breather and install the connector, gaskets and eye bolt. Then, supply compressed air at 49 kPa {0.5 kgf/cm^2} to the governor through the connector, and immerse the governor in light oil.

-2 Operate the speed control shaft and check that no air leaks from the governor. If air leaks, repair the governor.

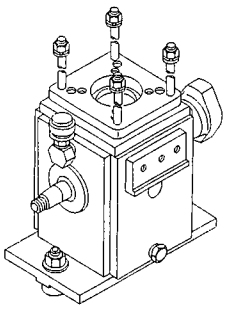
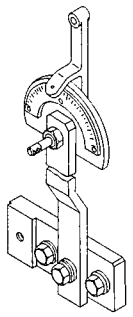
Advice
If the governor is equipped with an overflow valve, seal it using a blind plug.
If the governor is equipped with booster connections, remove them and seal the openings with blind plugs.
Remove the blind plugs and reinstall all components after adjustment.

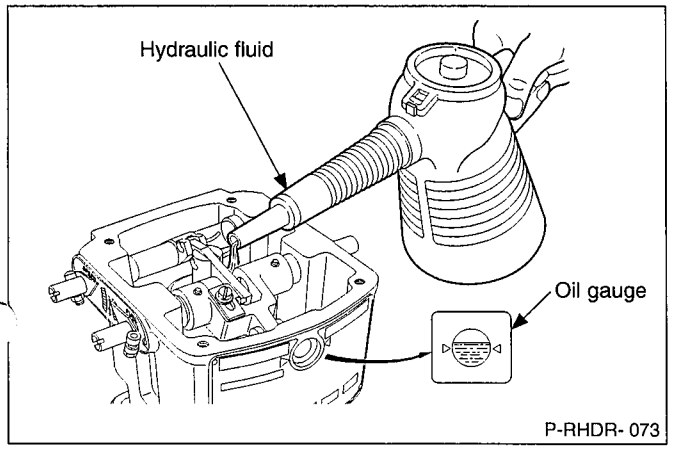


14. Remove the connector, gaskets and eye bolt.
Reinstall the air breather, then remove the cover.

In addition to general tools, the following special tools are necessary for governor adjustment.

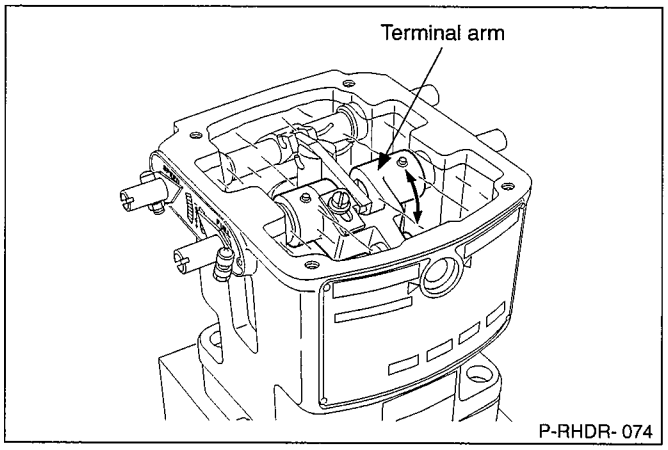
SPECIAL TOOLS

Key no.	Part name	Part no.	Shape	Remarks
1	Driving stand	307610-0050		For driving RHD governor
2	Adjusting device	307610-1010		For measuring lever angle

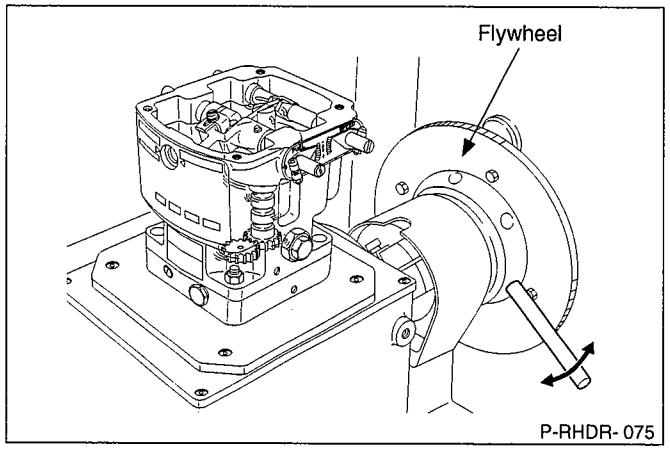


PREPARATION

1. Attach the driving stand to the pump tester and install the governor on the driving stand.
   Fill the governor with the specified amount of hydraulic fluid (approx 1,300 cm^3; the fluid should be a little over the middle of the gauge).



2. Move the terminal arm by hand until it becomes difficult to move to bleed the air from the governor.

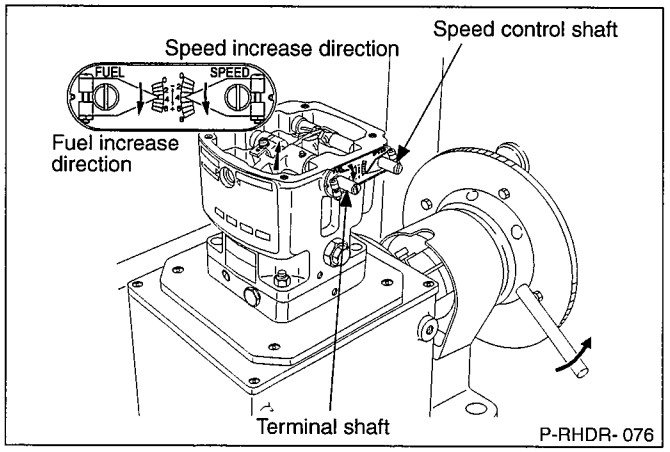


3. Turn the flywheel by hand and check that the gear shaft can be turned easily, and that the flyweight turns together.

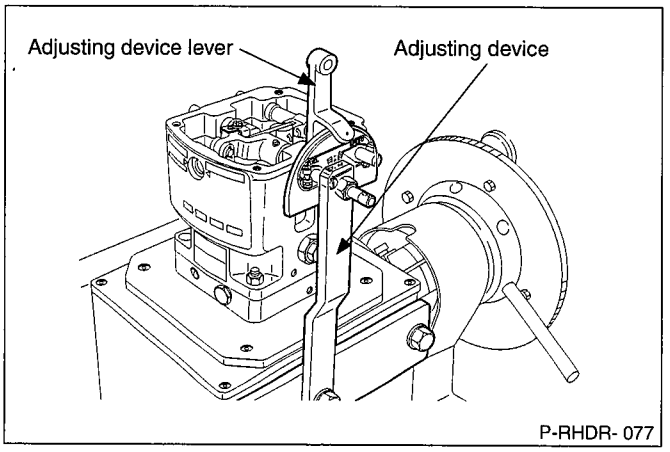
Note:
If the gear shaft does not turn easily, it will heat up during operation and adversely affect endurance.

⚠ CAUTION

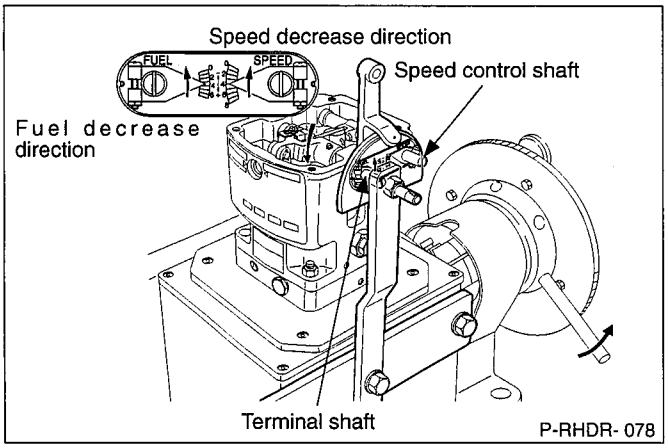
Remove the lever after operating the flywheel.
Operating the test bench without removing the lever can cause serious injury and damage the test bench.



4. Set the speed control shaft in the speed increase position, rotate the gear shaft and check that the terminal shaft moves in the fuel increase direction.



5. Attach the adjusting device to the driving stand and then attach the adjusting device lever to the terminal shaft.

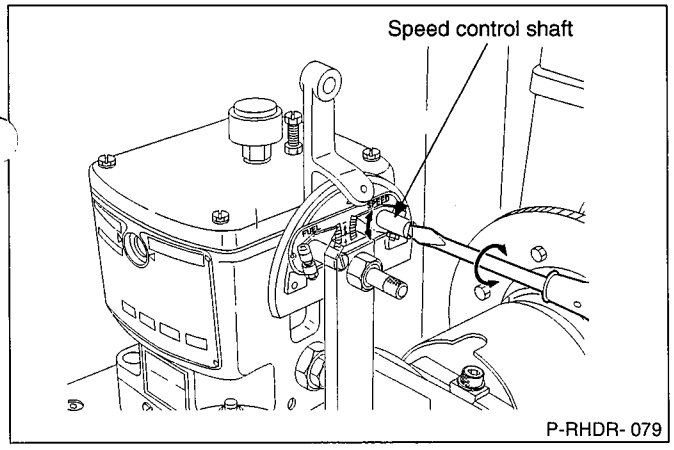


6. Set the speed control shaft in the speed decrease position, rotate the gear shaft and check that the terminal shaft moves in the fuel decrease direction.

Note:

If the cover is installed, the terminal shaft may not return in the fuel decrease direction, depending on the set position of the speed control shaft.

7. Install the governor cover.



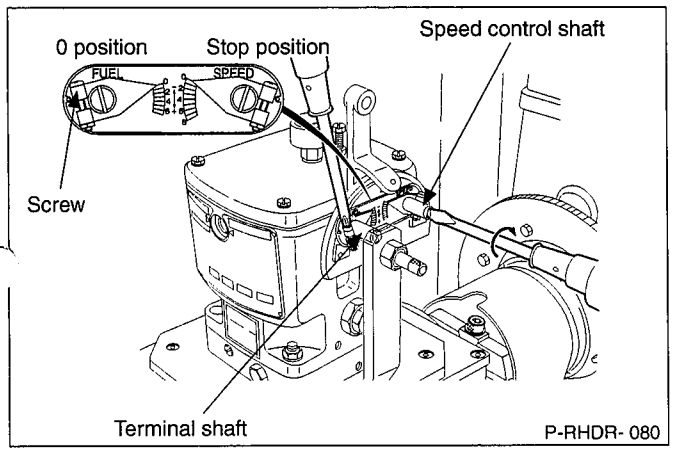
Warming-up operation

Run the governor at 1,000 r/min for 20 mins and check that no fluid leaks.

Air bleeding

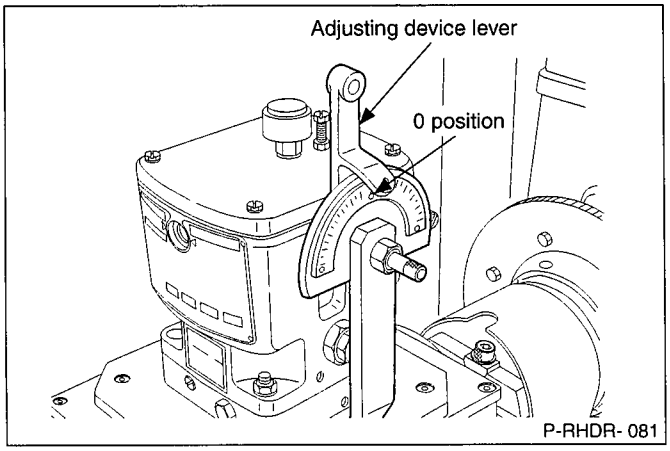
Fully open the needle valve (approx 3 turns from the fully closed position) and run the governor at 1,000 r/min. Operate the speed control shaft so that the power piston moves through its full stroke.

Do this at least 10 times to bleed all air from the governor's hydraulic system.

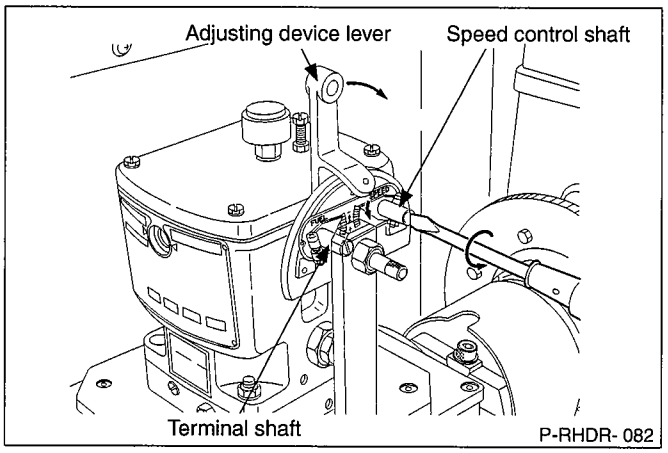


Terminal shaft pointer adjustment

1. Run the governor at approx 1,000 r/min and move the speed control shaft to the stop position.
2. In the above condition, align the terminal shaft side pointer with the indication plate's (fuel side) 0 position and fix it using the screw.



3. This position is the terminal shaft angle's 0 position. Align the adjusting device's 0 position with this position.



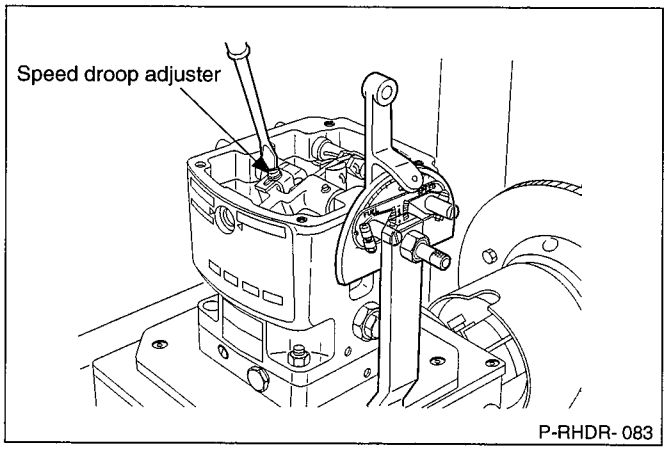
4. Run the governor at approx 600 r/min (with the needle valve fully open, ie, returned 3 turns from the fully closed position) and turn the speed control shaft. Check that the terminal shaft moves easily from 0° ~ 35° (RHD10: 0° ~ 40°) when the power piston moves through its full stroke.
   At this time, check that the terminal shaft operating torque is 9.8 N·m {1 kgf·m} for the RHD6 or 14.7 N·m {1.5 kgf·m} for the RHD10 using a spring balance attached to the adjusting device.

OPERATION TEST

The operation test is the same for LC, PC, MC, and MCL type governors.
Perform the operation test with the needle valve fully open.

Speed control shaft pointer angle adjustment

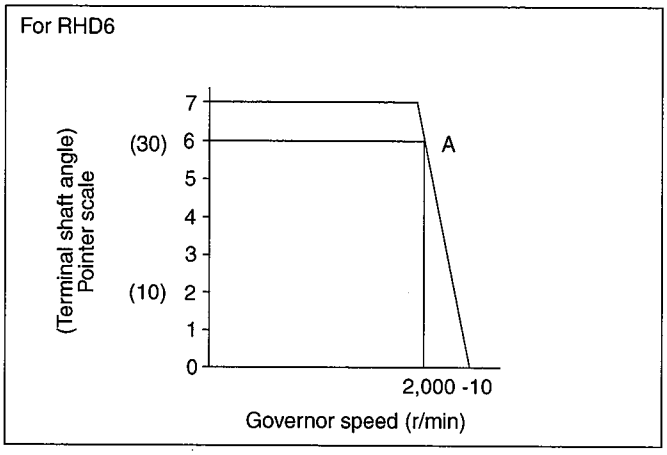
1. Remove the governor cover and set the droop adjuster pointer at the specified value.



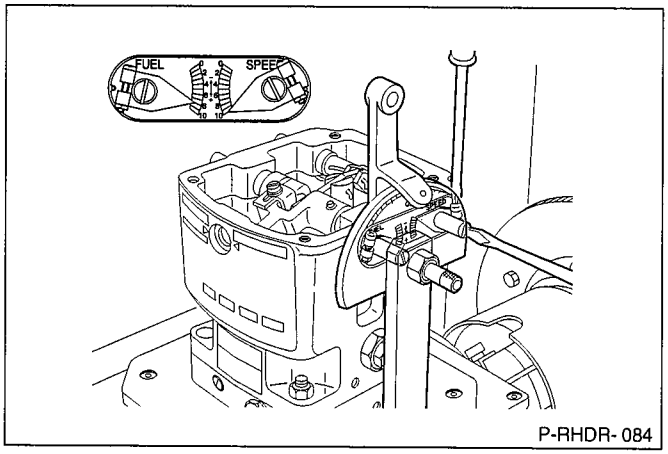
2. Run the governor at 2,000-10 r/min and secure the speed control shaft at the position (A) where the terminal shaft angle is 30° (ie, 6 on the scale). Then, align the speed control shaft pointer at 8 (on scales graduated from 0 ~ 8) and tighten the screw.

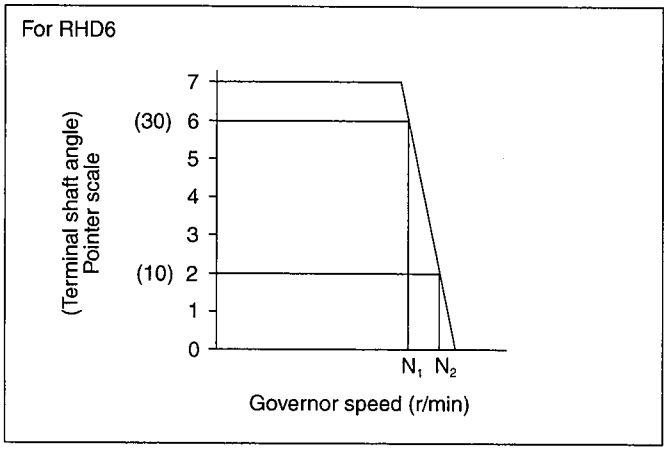
Note:

-1 On PC, MC and MCL governors, the terminal shaft is fixed (at A) by a knob.



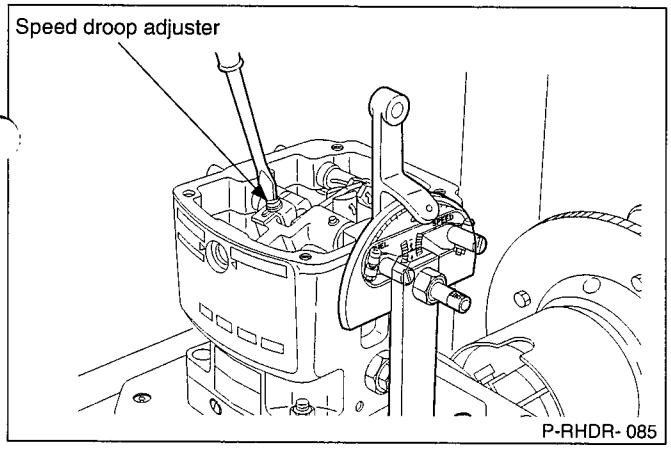
-2 On RHD10 governors, when the terminal shaft angle is 32° (ie, 8 on the scale), align the speed control shaft pointer at 10 using the screw.





Speed droop adjustment

1. Temporarily fix the droop adjuster pointer at the specified position.
2. Operate the governor at N_1 r/min and then fix the speed control shaft when the terminal shaft angle is 30° for the RHD6 or 32° for the RHD10.



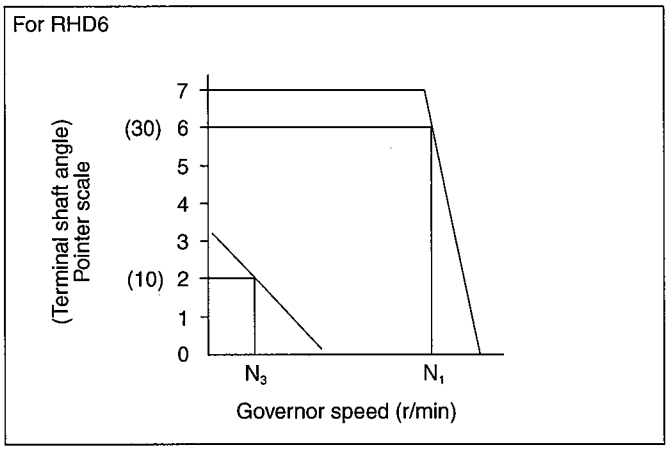
3. Gradually increase governor speed and further adjust the speed droop adjuster so that governor speed is N_2 r/min when the terminal shaft angle is returned to 10° for the RHD6 or 8° for the RHD10.

Note:
When the speed droop adjuster pointer is moved toward 0, speed droop decreases.
When the speed droop adjuster is moved toward 10, speed droop increases.

Advice

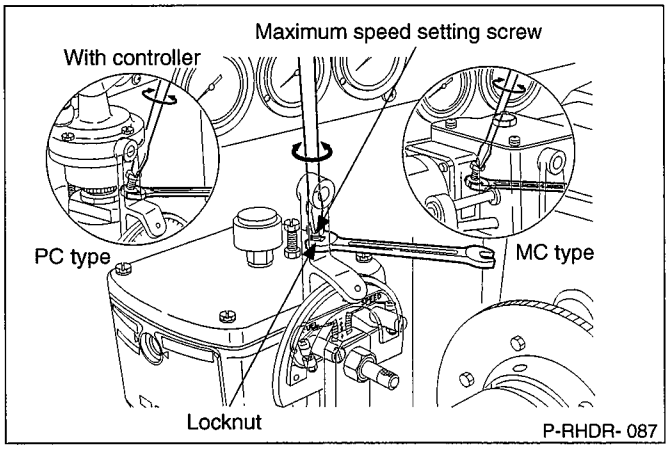
Confirm that N_2 is within the allowable speed when the speed droop adjuster is exactly as specified.

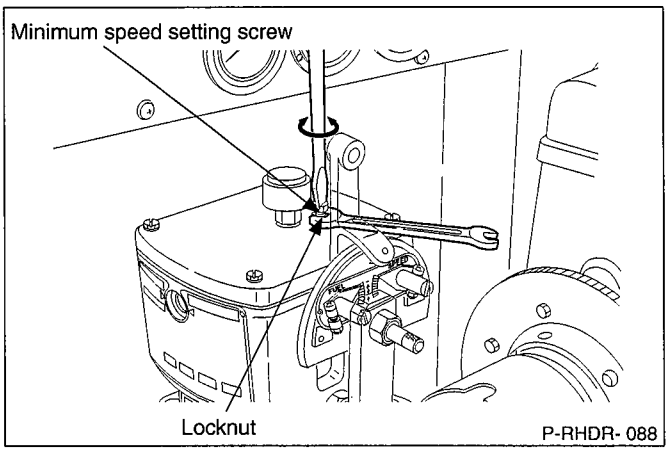
Note:
If the lever attached to the terminal shaft deviates more than ±1° during high speed operation (ie, at approx 1,800 r/min), check as follows.
Regular deviation: Check compensator related parts (housing, base, assembly, regulating valve). Replace them if necessary.
Irregular deviation: Check the flyweight assembly. Replace it if necessary.
Perform the following adjustments with the governor adjusted as described above.



Maximum speed setting screw adjustment
(not necessary on MCL type governors)

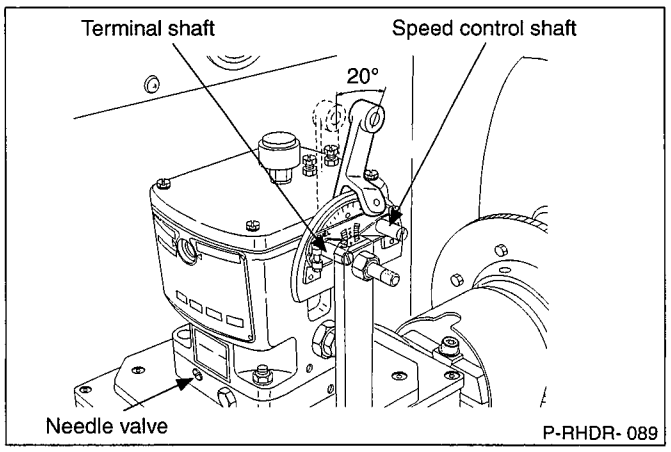
1. Install the governor cover.
   Operate the governor at N_1 r/min, adjust the maximum speed setting screw until the terminal shaft angle is 30° for the RHD6 or 32° for the RHD10, and then fix the screw using the locknut.





Minimum speed setting screw adjustment
(not necessary on PC, MC or MCL type governors)

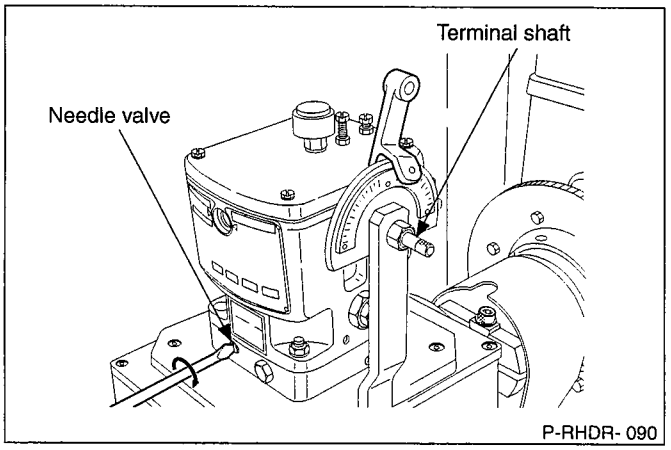
1. With the governor stopped, move the speed control shaft as far as possible to the low speed side (0 or less on the scale), and then temporarily set the minimum speed setting screw.
2. Operate the governor at N_3 r/min and adjust the minimum speed setting screw so that the terminal shaft angle does not exceed 10° (RHD6) or 8° (RHD10). Then, fix the screw using the locknut.



Compensator chamber high pressure hydraulic fluid flow

The flow of a large quantity of hydraulic fluid to the compensator is not desirable.
To check for this, perform the following test.

1. Fully open the needle valve (approx 3 turns from the fully closed position) and maintain governor speed at approx 1,000 r/min.
2. Fix the speed control shaft so that the terminal shaft angle is approx 20° (4 on the scale).

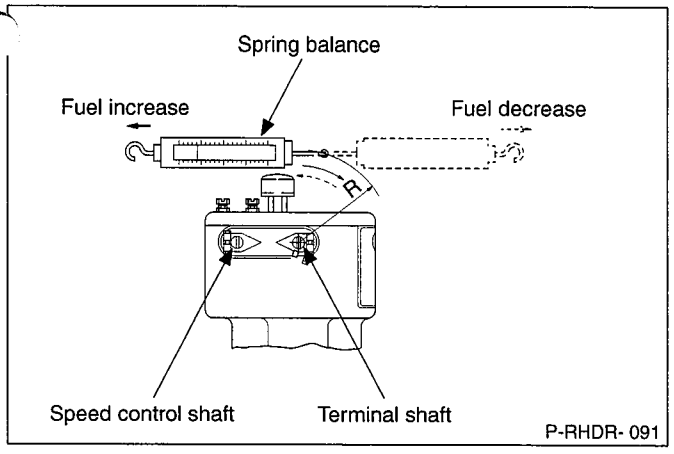


3. Fully close the needle valve and check that the terminal shaft does not move more than 1° in the 0 direction.

Advice

If the terminal shaft moves more than 1° in the 0 direction, check for the following.

- **Clearance between the sleeve and base.**
- **The base-housing contact surfaces may not be even, or may be scratched.**
- **The needle valve escape ports (ϕ 0.4 and ϕ 0.8) and the base escape ports (ϕ 1.5) may be blocked.**
- **The compensator bushing O-ring may be damaged.**



Governor control confirmation

Under the following conditions, operate the speed control shaft to move the terminal shaft in the fuel increase direction (RHD6: 0° → 35°; RHD10: 0° → 40°) and in the fuel decrease direction (RHD6: 35° → 0°; RHD10: 40° → 0°) and check that the lever resistance is as specified.

Measurement conditions

Governor speed:	600 r/min
Needle valve position:	fully open
R (see fig at left):	100 mm

Note :

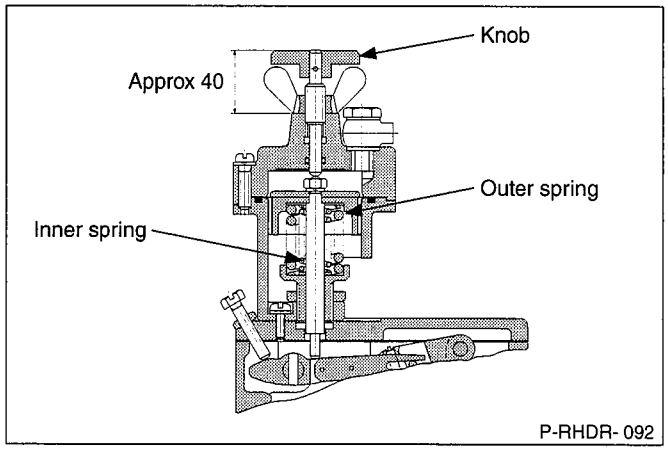
If the lever resistance is not as specified, replace the following parts.

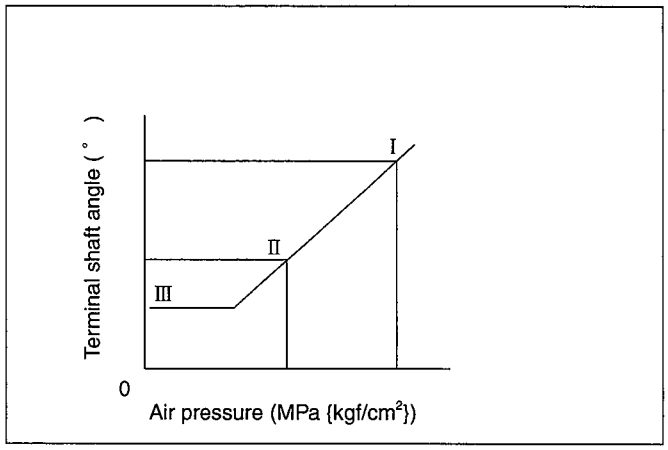
- Regulating valve assembly
- Check for openings or leaks in the gear pump or hydraulic fluid system and replace faulty or damaged parts.

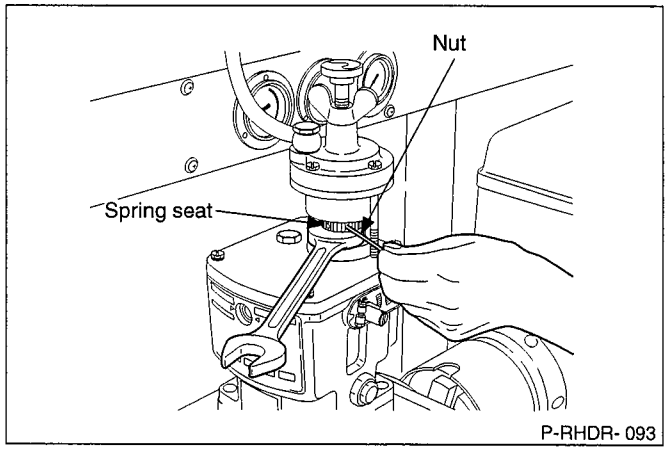
Reverse operation

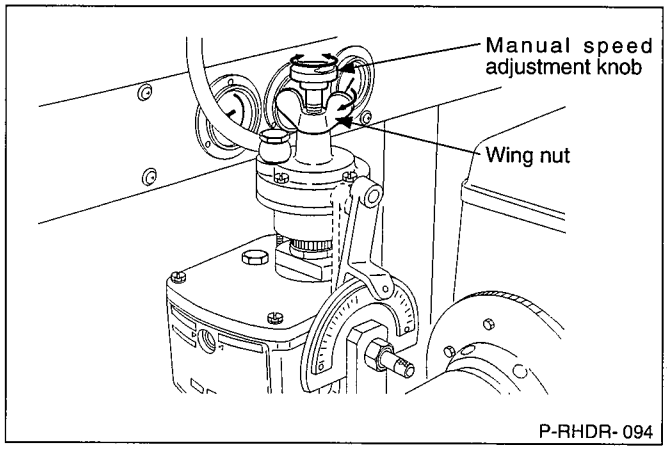
The RHD governor is constructed so that identical performance is obtained whether it is operating clockwise or counterclockwise.

To confirm this, operate the governor in the reverse direction and check governor control as described above under the exact same conditions and procedures, and check that performance is as specified.









ADDITIONAL DEVICES

Pneumatic controller adjustment

1. Before adjustment, check that the outer and inner springs' distinguishing colors are as specified.
2. Temporarily position the manual knob as shown at left and connect a compressed air source capable of supplying 0 ~ 0.49 MPa {0 ~ 5 kgf/cm^2 } to the eye bolt.

Note :

Perform the following adjustments and confirmations based on the individual test standards.

3. Point I adjustment

Example:

1,680 ~	**0.37 MPa ~ {3.8 kgf/cm^2}**	**30±0.5**
(speed)	**(air pressure)**	**(terminal shaft angle)**

Operate the governor at 1,680 r/min and adjust the spring seat so that the terminal shaft angle is 30±0.5° when when 0.37 MPa {3.8 kgf/cm^2} air pressure is supplied. Then, secure the spring seat using the nut.

Advice

When adjusting point I, return the air pressure to 0 and the manual speed adjusting knob to its original position to prevent diaphragm damage.

4. Point II adjustment

Example:

1,055 ~	**0.16 MPa ~ {1.6 kgf/cm^2}**	**(Approx 16.5)**
(speed)	**(air pressure)**	**(terminal shaft angle)**

Operate the governor at 1,055 r/min and confirm that the terminal shaft angle is 16.5° when 0.16 MPa {1.6 kgf/cm^2} air pressure is supplied.

5. Point III adjustment

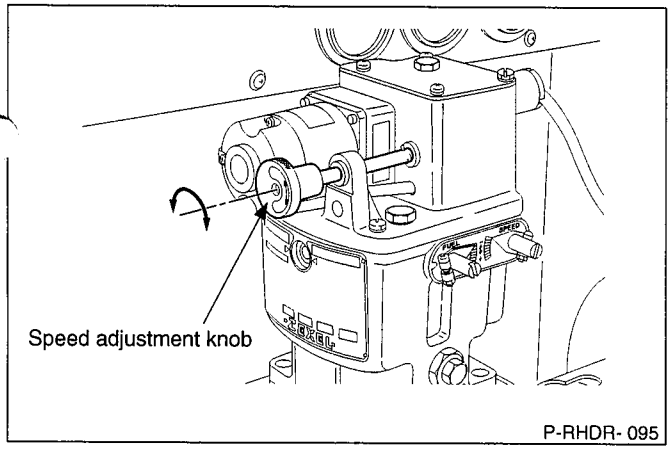
Example:

700 ~	**0 ~**	**8.5+1.5**
(speed)	**(air pressure)**	**(terminal shaft angle)**

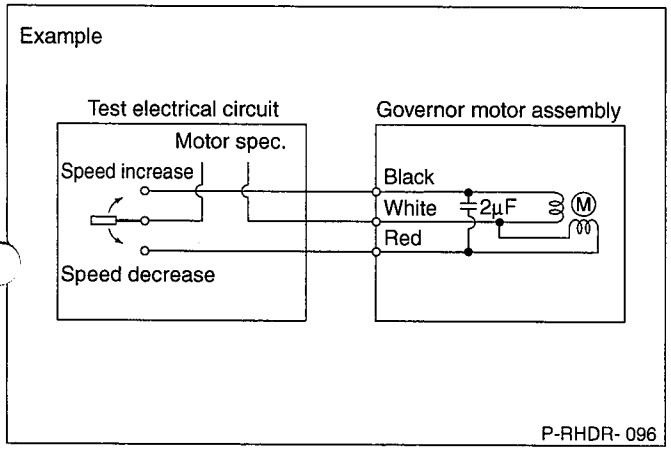
Operate the governor at 700 r/min and adjust the manual speed adjustment knob so that the terminal shaft angle is 8+1.5° at an air pressure of 0. Then, secure the knob using the wing nut.

Governor motor operation confirmation (MC type)

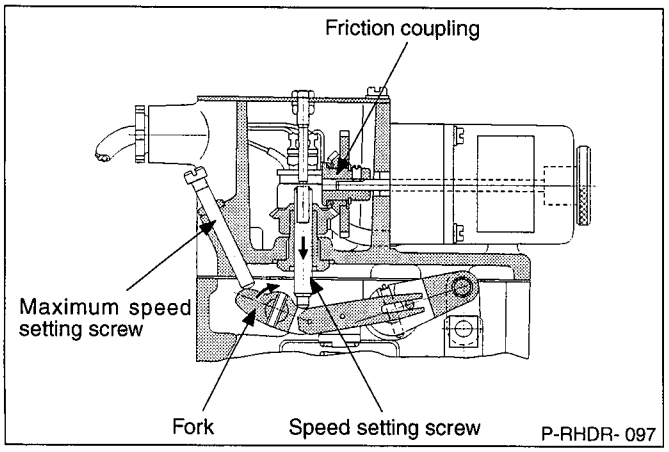
1. Specification confirmation
   Confirm from the individual standards that the specifications of individual parts are correct.
- Motor specification (voltage)
- Condenser capacity
- Speed reduction ratio



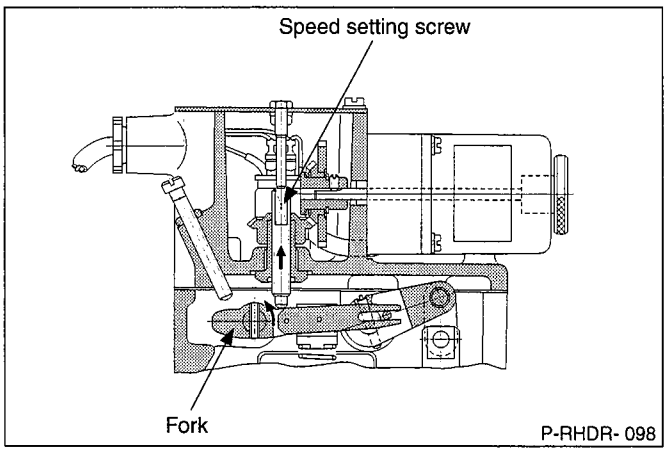
2. Operation confirmation (using speed adjustment knob)
   Turn the knob manually, and confirm that the speed setting screw turns smoothly over the whole range by observing speed control shaft pointer movement.
   If it does not move smoothly over the whole range, loosen the friction coupling set screw, adjust the positions of the bevel gear teeth and recheck screw movement.



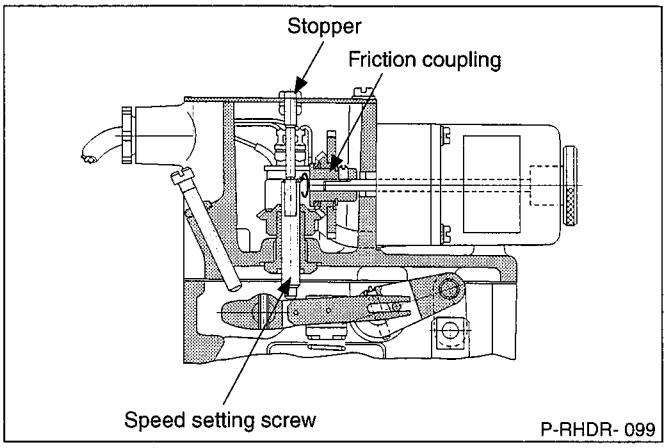
3. Operation confirmation (using the motor)
   Connect the governor motor assembly to the electrical test circuit in accordance with the particular motor's specifications, as shown at left.



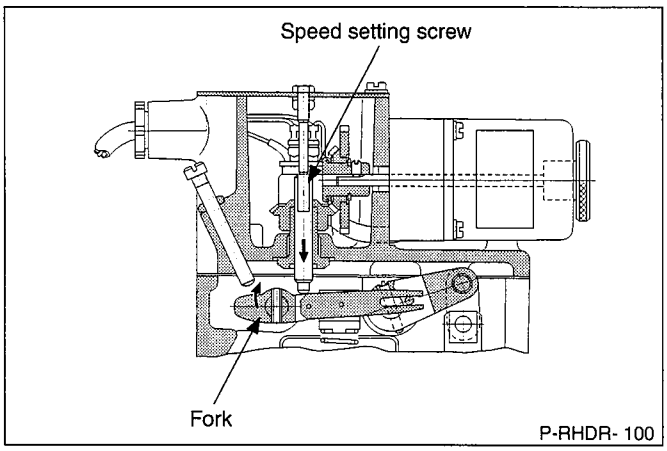
4. Operation test
- Turn the governor motor's speed switch ON in the speed increase direction to operate the motor.
  Then, operate the motor until the fork contacts the maximum speed setting screw adjusted above and the friction coupling is rotating freely.



- Then, turn the switch OFF and then ON in the speed decrease direction, and confirm that the speed setting screw begins moving smoothly.



- Operate the governor motor until the speed setting screw contacts the stopper and the friction coupling is rotating freely.



- Then, turn the switch OFF and then ON in the speed increase direction and confirm that the speed setting screw moves smoothly.

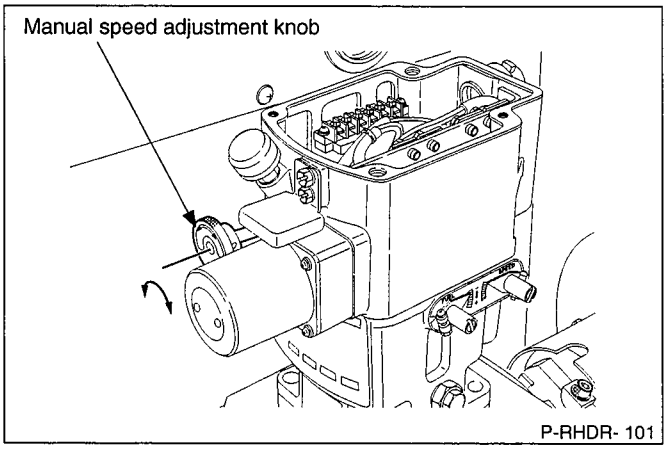
Advice

During the above operation test, if the friction coupling rotates freely in other than the following two cases, check the alignment of the bevel gear teeth.

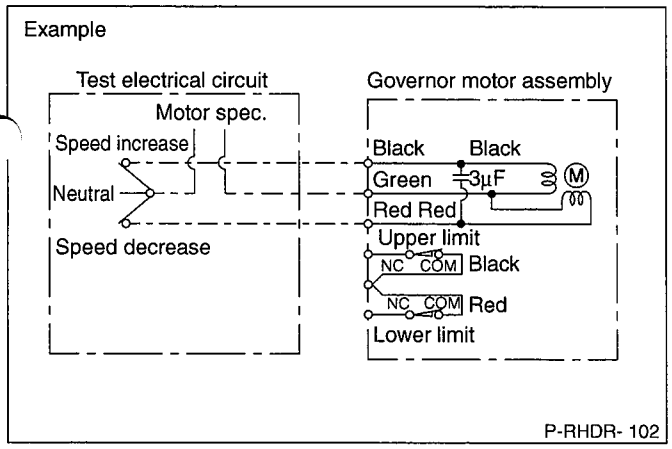
- **When the maximum speed setting screw is contacting the fork**
- **When the speed setting screw is contacting the stopper**

Governor motor operation test and limit switch adjustment (MCL type)

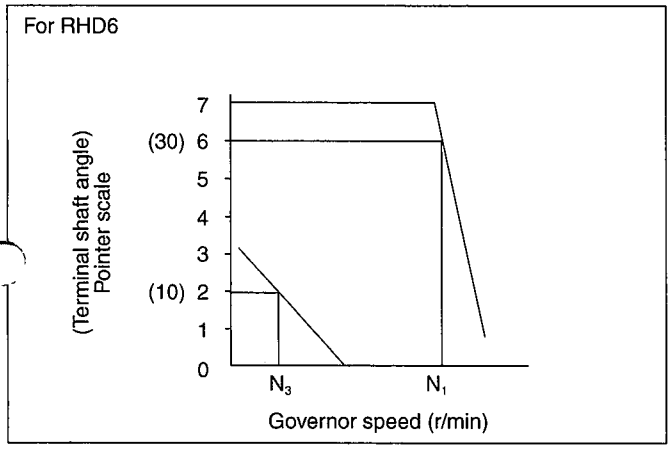
1. Specification confirmation
   Confirm from the data sheet that the specifications of individual parts are correct.
- Motor specification (voltage)
- Condenser capacity
- Speed reduction ratio



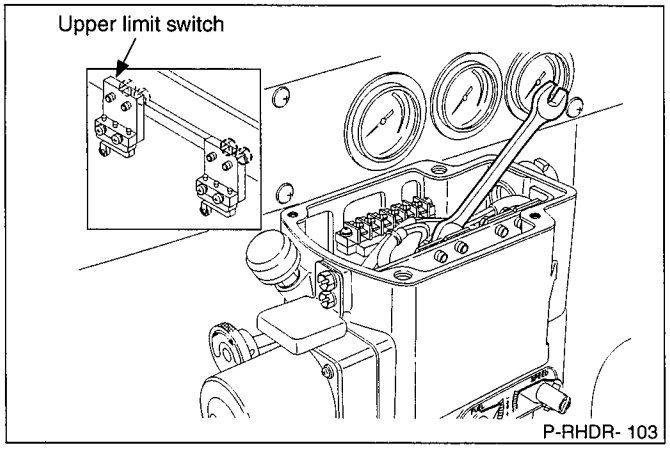
2. Operation confirmation
(using the manual speed adjustment knob)
Confirm operation as described below before adjusting the terminal shaft side pointer.
Operate the knob manually and confirm that the speed setting screw rotates smoothly over the full range.
If it does not move smoothly in some places, check the installation of the holder or the speed setting screw.

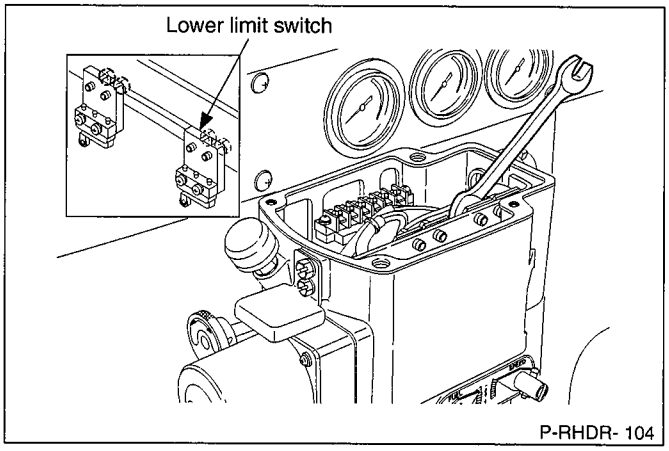


3. Governor motor operation test
(limit switch adjustment)
Connect the governor motor assembly to the electrical test circuit in accordance with the particular motor's specifications, as shown at left.
Then, adjust each limit switch.



- Maximum speed limit switch adjustment
Operate the governor at the specified speed (N_1 r/min), and the governor motor in the speed increase direction.
Adjust the limit switch's upper limit position so that the governor motor stops at the terminal shaft angle of 30±3° for the RHD6 or 32±3° for the RHD10.





- Minimum speed limit switch adjustment
  Operate the governor at the specified speed (N_3 r/min), and the governor motor in the speed decrease direction.
  Adjust the limit switch's lower limit position so that the governor motor stops at the terminal shaft angle of 10±3° for the RHD6 or 8±3° for the RHD10.
  Set the governor speed at 1,500 r/min and confirm that the governor motor operates correctly between the upper and lower operation limits and that the friction coupling does not rotate freely.

TROUBLESHOOTING

Of the phenomena listed below, some originate solely in the governor, and some solely in the engine.

- Faulty starting
- Hunting
- Insufficient engine output
- Engine will not operate at maximum speed
- Excessive momentary speed droop, or inability to control speed

First, check the engine for the following faults. If engine faults are found, perform repairs in accordance with the engine maker's instructions.

- Compression pressure of each cylinder is extremely uneven
  Excessive play in fuel control and speed adjustment linkage
- Excessive resistance in injection pump control rack and fuel control linkage
- Excessive play because of worn governor drive system gear or rubber damper
- Relative positions of linkages incorrect
- Dashpot spring in fuel control linkage bent or damaged

If no engine faults are found, check the governor.

As the governor uses many precision parts, there is a danger of the engine overrunning if these parts are incorrectly assembled.

⚠CAUTION

Never attempt to repair the governor on-site.

Always consult your nearest Bosch Corporation representative.

12 PART NUMBER EXPLANATION

MODEL NUMBER

Example: RHD 6 L C
(1) (2) (3) (4)

(1) Independent type hydraulic governor, manufactured by Bosch Corporation
(2) Control force (work load)
Indicates 10 times the controlling force (work load) in kgf·m.
(3) L: lever type
M: with governor motor
P: with pneumatic controller
(4) C: with compensator

Note:
No symbol 'C' indicates that a compensator is not used.

CODE NUMBER

Example: 10 58 5 6 - 0 000
(1) (2) (3) (4) (5) (6)

(1) 10: Assembly part number
(2) 58: Hydraulic governor manufactured by Bosch Corporation
(3) Type
5: RHD6 type
6: RHD10 type
(4) Direction of pump rotation and governor installation position

	Direction of pump rotation	Governor installation position	Remarks
0	Clockwise	Right side	For in-line pumps
1	Clockwise	Left side	
2	Counterclockwise	Right side	
3	Counterclockwise	Left side	
4	Clockwise	-	
5	Counterclockwise	-	
6	Reversible	-	

Note:
With in-line pumps, the direction of pump rotation is viewed from the pump's drive side.
Governor installation position is viewed from the pump's cover plate side.

(5) Remote control device installation
3: LC type without remote control device
4: PC type
5: MC type
6: MC type
7: MCL type
8: MCL type
9: HC type, others
} with remote control device

(6) Characteristic number

Pub. No: EE74E-11016

GOVERNOR MODEL RHD (RHD6 & RHD10)
Service Manual
Operation & Maintenance
Printing: August 2007
Published by:
Bosch Corporation
Automotive Aftermarket Division

Printed in Japan
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EE74E-11016
JA/ASA5
8.2007
Printed in Japan

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DAIHATSU

GEISLINGER COUPLING
INSTRUCTION BOOK

ダイハツディーゼル株式会社

DAIHATSU DIESEL MFG. CO., LTD

List No.INB000081A

2017/05/09 TECHNICAL DEPARTMENT

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1. INTRODUCTION

Manufacturer :	Geislinger GmbH
Contact & Service Centres	: www.geislinger.com
Manual No.; Revision No.; Language	: 0010 03 EN (English)
Date of issue	: 2002-02-01
Date of last change	: 2010-10-19

GENERAL INFORMATION ON THE OPERATING INSTRUCTIONS

These operating instructions contain important information on the use of the product. Carry out these instructions carefully and precisely. This helps to avoid hazards, downtime and repair costs whilst also increasing the reliability and service life of the product.

These operating instructions are part of the product.

Keep these instructions in a safe place for the entire life of the product.

If the product is transferred to a new owner, these instructions and all associated documents are to be passed on to the new owner.

The meaning and appearance of safety symbols used in these operating instructions are as follows.

WARNING!	The text after the signal word "**WARNING!**" contains important information, which must be followed to protect the user from injury.

CAUTION!	The text after the signal word "**CAUTION!**" contains information, which must be followed to protect the product and the system from damage.

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2. SAFETY INSTRUCTIONS

WARNING!

ANY WORK CARRIED OUT ON AND IN CLOSE PROXIMITY TO THE SYSTEM SHALL ONLY BE UNDERTAKEN

→ By trained personnel

→ With the system shut down and the supply of power switched off

→ In compliance with the regulations that apply in relation to

- Health and safety at work
- and accident prevention in agreement with international, national and local requirements and regulations

→ In compliance with the safety information and data provided in these operating instructions

→ In compliance with national installation and erection regulations

→ In compliance with the generally accepted rules of engineering practice

THE FOLLOWING PERSONAL PROTECTIVE EQUIPMENT IS TO BE USED FOR ALL WORK ON AND IN THE IMMEDIATE VICINITY OF THE PRODUCT.

SAFETY HELMET

SAFETY FOOTWEAR

PROTECTIVE GLOVES

PROTECTIVE GOGGLES

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3. APPLICATION OF THE COUPLING

This product is an elastic damping coupling manufactured by Geislinger.

This coupling is a special product designed for a designated system.

The coupling will only function properly if it has been installed in this system correctly.

CAUTION!	Installation of the coupling in equipment other than the designated system may cause damage to the system and to the coupling.

4. TECHNICAL DATA

Technical data, dimensions of the components and weights: See attached assembly drawings and the arrangement drawings (if available)

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5. PRODUCT DESCRIPTION

The Geislinger coupling is an elastic steel coupling with an internal hydraulic damping system.

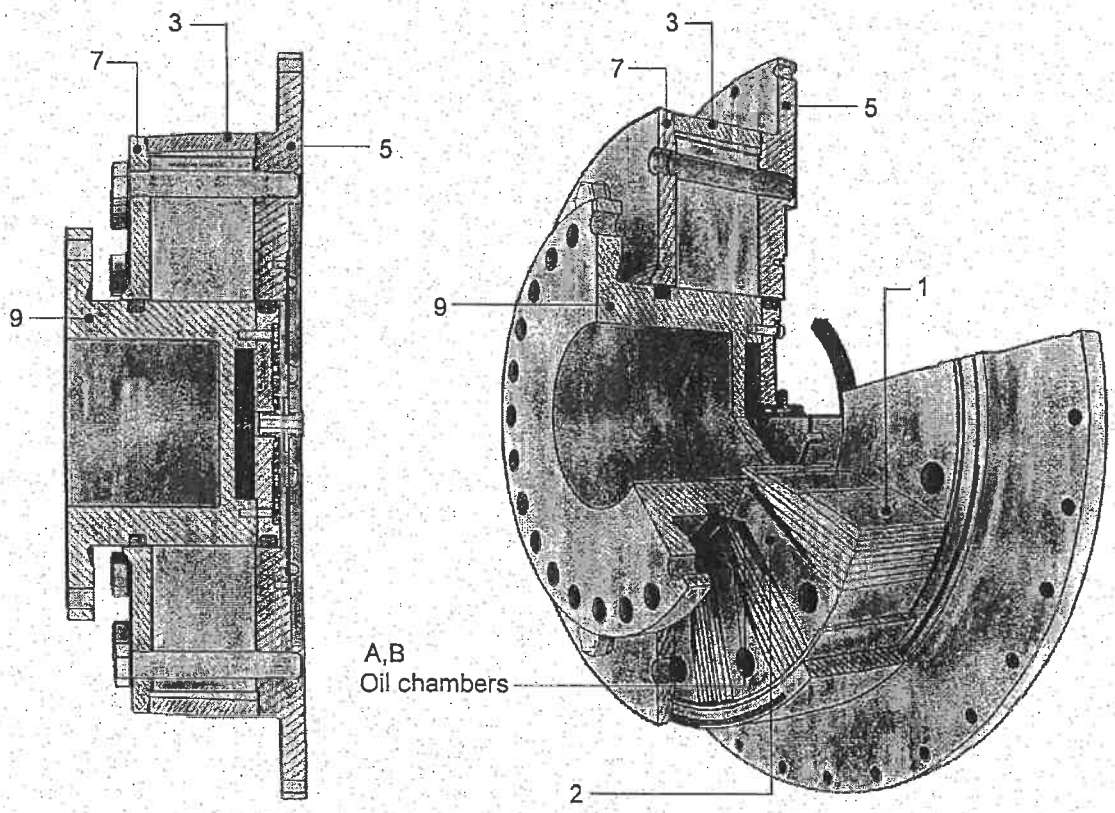
Radially arranged spring blades transmit the elastic torque. Torsional vibrations are damped by oil displacement from one chamber to the adjoining one.

The spring packs (1) and intermediate pieces (2) are clamped together using a clamping ring (3). The inner ends of the spring packs (1) reach into the axial grooves of the inner star (12). The chambers (A,B), which are formed by the intermediate pieces (2) and the spring packs (1), are filled with oil. If torque is applied to the coupling, the two parts of the coupling (external and internal) will be twisted against each other. The spring blades (1) will be deflected during this process. This changes the volume of the oil in chambers A and B. The oil will be displaced from one chamber into the adjoining one through a narrow gap. The resultant oil flow resistance delays the relative movement of both parts of the coupling (outer and inner) and dampens the torsional vibrations.

The intermediate pieces (2) also act as a stop, therefore providing overload protection for the deflected spring blades.

The main coupling bolts connect the middle piece of the coupling (1,2,3) to the flange (5) by means of friction locking. The side plate (7) seals the front of the coupling and seals off the inner part of the coupling from the outer part.

Sketch illustrating the principle



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6. TRANSPORT AND LIFTING INSTRUCTIONS

WARNING!

The product may only be lifted and handled by personnel who

→ Have the required licence to

- Work with cranes
- Drive industrial trucks

→ And are also familiar with transport and lifting instructions in

- The operating instructions
- and the assembly diagram

Supports must always be provided for overhanging components.

6.1. Transport instructions

→ Only move the product in a horizontal position.

→ Do not tilt the shipping crate or pallet

→ Avoid hard impacts during transportation.

→ Max. lifting speed 50 m/min.

→ Do not damage the packaging and preserving agents.

TRANSPORTING THE COUPLING WHEN MOUNTED

If the coupling is mounted on the engine flange and then transported together with the engine (e.g. from the engine manufacturer to the place where it will be used), projecting and suspended parts of the coupling must always be supported. This relates to the installation of the coupling inner star (9) on the engine flange (coupling outer part suspended) as well as the installation of the coupling outer part on the engine flange (inner star (9) suspended) → See transport sketch.

CAUTION!	If the suspended or projecting parts are not supported, the following problems may occur: → The seals may be damaged (leaks) → The system may possibly be damaged → And the service life of the coupling may be reduced.

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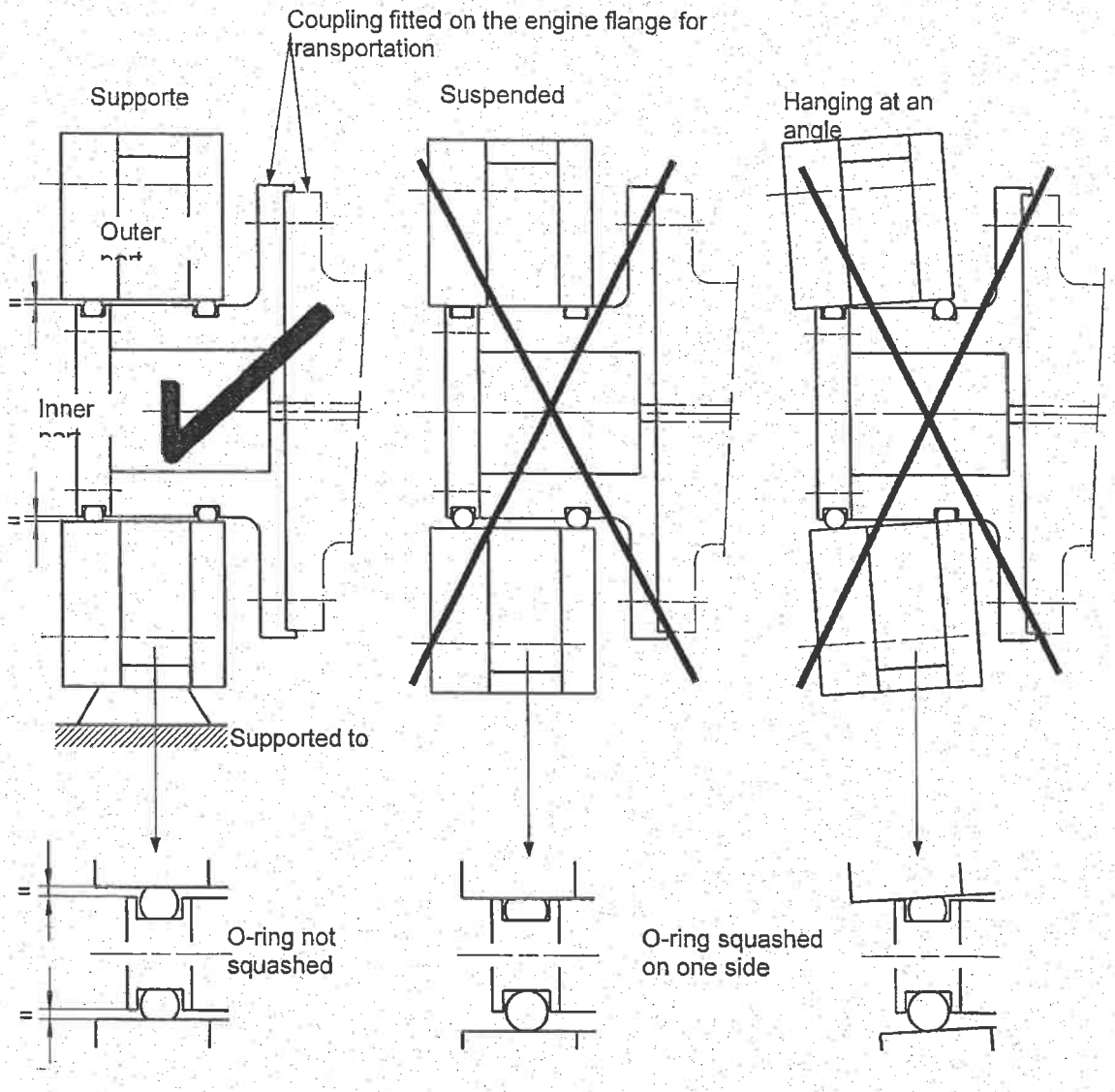
TRANSPORT SKETCH



→ For transportation the coupling is to be arranged in such a way that the outer part of the coupling is aligned centrally with the inner part of the coupling. The distance between the outer part and the inner part of the coupling must be identical at the top and the bottom.

→ If the transportation safety device for the coupling is already provided and fitted (see assembly diagram) the transport support does not need to be mounted.

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6.2. Lifting tackle

WARNING!	ONLY USE HIGH-STRENGTH LIFTING TACKLE for lifting. The nominal carrying capacity of a SINGLE lifting device must be greater than the total weight of the product or the component to be carried!

LIFTING TACKLE SUPPLIED WITH THE ORDER

These are

→ loadable from all sides

→ and to be mounted and used in accordance with the instructions in

- the lifting tackle manufacturer's operating instructions
- the assembly diagram
- and the lifting sketch (see Chapter 6.4).

LIFTING TACKLE PROVIDED BY THE CUSTOMER

(Lifting tackle not in the scope of delivery)

WARNING!	ONLY HIGH-STRENGTH LIFTING TACKLE is permitted to lift the product. The nominal carrying capacity of a SINGLE lifting device must be greater than the total weight of the product or the component to be carried!

In addition, the following regulations must be observed

→ It is only permissible to use lifting tackle approved in the country where it will be used.

→ The lifting tackle must comply with the relevant national and international safety specifications.

→ The lifting tackle must not be damaged or corroded.

→ The lifting tackle must be 100% crack-tested.

→ It must be possible to screw together the lifting tackle to make a complete unit.

→ The tapped lifting holes on the product must not be damaged when doing this.

→ The lifting tackle must be mounted and used in accordance with the instructions in

- the lifting tackle manufacturer's operating instructions
- the assembly diagram
- and the lifting sketch (see Chapter 6.4).

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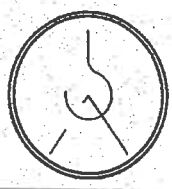
6.3. Tapped holes for lifting purposes

In the assembly diagram these are indicated with 

WARNING!	Only use tapped holes marked in this way for lifting.

CHECKING THE TAPPED LIFTING HOLES BEFORE ASSEMBLING THE LIFTING TACKLE

Before assembling the lifting tackle check the relevant tapped lifting hole to make sure that it is suitable to use.

WARNING!	The tapped lifting hole cannot be used if → the thread is damaged, chipped, crossed or distorted → the thread is corroded → if there are signs of reworking such as Helicoil screw thread inserts

MULTI-PURPOSE TAPPED HOLES

These are both tapped holes for screws used for other purposes and tapped lifting holes.

Use:

→ Remove bolt from multi-purpose tapped hole.

→ Fit lifting tackle as specified.

→ After the lifting operation has been completed, remove the lifting tackle and, complying with the

- tightening torque
- sealing specifications
- and safety measures

refit bolt as per the assembly diagram.

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6.4. Lifting instructions

Only lift the product

→ Using the lifting tackle provided → See Chapter 6.2.

→ With suitable carrying straps such as ropes, chains or belts.

→ After lifting tackle and carrying straps have been fitted in accordance with the regulations.

→ Using the tapped lifting holes indicated in the assembly diagram → See Chapter 6.3

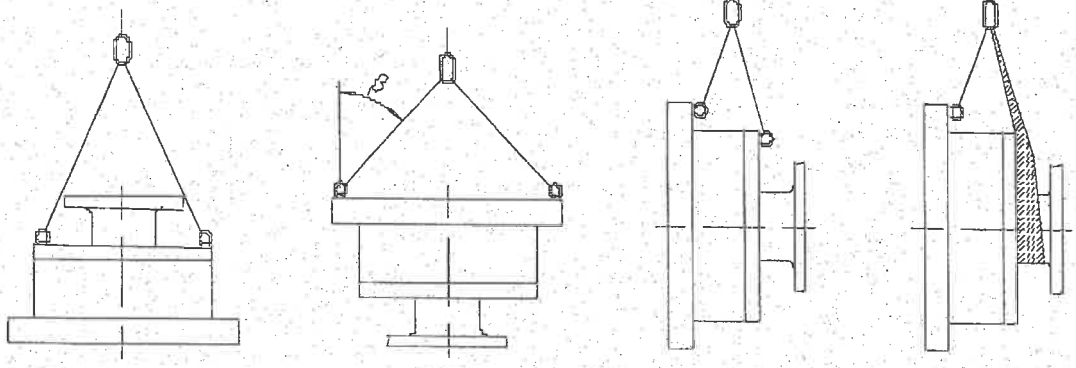
→ Attached to at least two lifting points simultaneously.

→ With the load distributed as equally as possible.

CAUTION!

Never lift the product using only one connection point for the lifting tackle. $\beta_{maximum} = 45°$

Lifting examples



WARNING!

STANDING UNDER A SUSPENDED LOAD IS PROHIBITED!

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7. P ACKAGING AND STORAGE

STANDARD PACKAGING

All metal parts are protected against corrosion and sealed in polythene film.

All parts are packed in crates made of soft wood or are placed on wooden pallets.

NOTE:
Special packaging can also be used for the couplings in line with the customer's requirements.

CAUTION!	If the packaging is damaged, the product must be → Installed in the system within 100 hours → Or the preservatives must be reapplied and the packaging restored to its original condition.

DISPOSAL OF STANDARD PACKAGING

The PE film can be recycled.

NOTE:
Contact Geislinger or a Geislinger service centre to ask for data sheets on packaging/special packaging.

STORAGE INSTRUCTIONS

→ Do not damage the packaging and preserving agents.

→ Do not stack the products or place heavy parts on top of them.

→ Store the product in dry conditions: max. permissible relative humidity 85%.

→ Avoid variations in temperature of more than 25° Celsius.

→ Make sure that the permissible temperature application range for seals as per the attached data sheet is adhered to during storage.

CAUTION!	If the temperature rises above or falls below the permissible temperature application range for seals during storage, it may damage the seals and then subsequently reduce the service life of the coupling.

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8. UNP ACKING

GENERAL

The coupling, coupling components and connecting parts may be delivered separately or packed individually. Each package may contain separately packed tools or small components (e.g. screws, sealing rings, lifting tackle, etc.). Do not throw these away with the packaging.

→ Take the coupling out of the wooden crate.

→ Remove the packing film and rust-inhibiting paper (if provided) from the coupling.
Check the coupling for possible damage that may have occurred during transportation.

→ Fit the lifting tackle (730) (if supplied), as appropriate. Clean the surfaces underneath prior to mounting.

→ First bring the coupling into an upright position, with the main axis in a horizontal position.

WARNING!	Secure the coupling against falling over and rolling away.

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9. COUPLING INSTALLATION INSTRUCTIONS

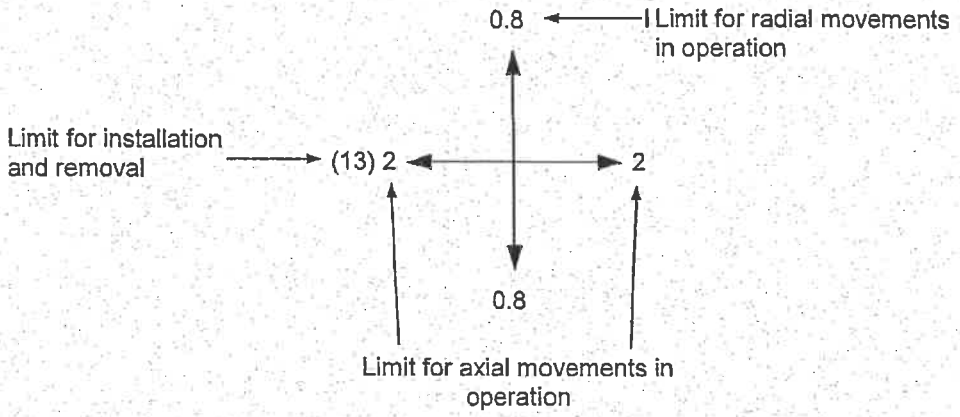
9.1. General

CAUTION!	Does not apply to bearing mounted couplings.

The distance by which the coupling may be pushed together axially (limit for removal and installation) is indicated in the displacement cross.

The displacement cross is shown in the area of the spring leaves (1).

The values shown are only intended as an example. **For the relevant values please see the appropriate Geislinger assembly drawing for the coupling concerned.**



9.2. Aligning shafts

→ Align shafts that connect to the coupling so that the radial, axial or angular displacements that occur do not exceed the permissible values for the coupling in any operating modes.

→ The values for parallel (Kr) and angular (Kw) errors and their combination are stated in a displacement diagram on the assembly drawing.

→ Radial movements of the parts to be connected (e.g. engine and gear casings) due to heat expansion are to be considered.

→ Try to achieve the best possible shaft alignment at operating temperature or in a condition in which the installation is mostly operated.

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→ The tolerance for the axial installation dimension of the coupling is preferably the "central position" plus / minus 0.5 mm. The nominal dimension for the installation length in the central position is shown on the assembly drawing.

CAUTION!	The displacement capacity of a coupling must not be seen as a wide tolerance allowance for the static shaft setting. It is mainly meant to allow a maximum of movements during dynamic service conditions. **Bad shaft setting reduces the displacement capacity available for dynamic operation.**

9.3. Installing the coupling

PREPARATION

The coupling must be fully assembled.

Check the oil feed and return bores to make sure that they are clean and oil can flow through. Remove any masking tape.

Remove the rust inhibitor (e.g. Anticorit) from all flange-mounting surfaces and centring shoulders.

CAUTION!	Do not scratch or damage the surfaces during cleaning.

O-rings on the flanges (for oil supply and venting) must be well greased and undamaged.

INSTALLATION

Install the coupling in the system in accordance with the

→ Installation specifications in the assembly diagram and the

→ Safety regulations and transport and lifting instructions given in the operating instructions.

CAUTION!	Do not damage the centring shoulders, seals and sealing surfaces during the mounting operation.

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ACTIVITIES AFTER INSTALLATION

Measure the coupling length again.

If the installation length is not correct as per the assembly diagram

→ Slacken off the connection bolts.

→ Realign the coupling and install again.

The measured length should not exceed the installation length as stated on the assembly diagram plus/minus 0.5 mm tolerance.

CAUTION!	Remove lifting tackle and ancillary installation tools (if present) from the coupling and keep them for use later.

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10. STARTING UP THE COUPLING

10.1. Venting the coupling

COUPLINGS WITH VENTING SCREW 756

→ Rotate the coupling until the venting screws (756) are in top position.

→ Remove the venting screw.

→ Supply hydraulic oil to the coupling with a pre-lubrication pump (engine or gear circuit) until oil escapes from the venting hole.

→ Using a screw and disc spring, seal the venting hole until it is oil tight.

COUPLINGS WITH CONTINUOUS VENTING

→ Supply the coupling with a pre-lubrication pump (engine or gear circuit) with hydraulic oil for at least 2 minutes.

The coupling is vented by a hole in the centre of the coupling or by nozzles (714). Oil entrained through the venting hole is fed back into the oil system through return bores.

10.2. Under load

→ When the coupling is used for the first time or after the spring packs have been changed (1), at first only increase the operating load slowly.

→ The coupling should not be operated at full load for approx. half an hour after it has been started up.

10.3. Checks after the first test run

→ Signs of oil loss, e.g. oily marks (radial) on the side plate (7) or on the flange (5) indicate that the O-ring (850 or 853) is leaking.

→ Check screws for leaks (visual check).

→ Measure the shaft offset of the connecting shafts at operating temperature.

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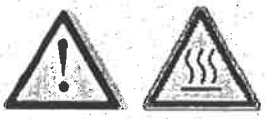
11. REMOVING THE COUPLING

 WARNING!	**COUPLING MAY BE HOT! RISK OF BURNING!** Allow the coupling to cool off to room temperature before removing it.

→ Remove the coupling from the system, following the safety instructions as well as the transport and lifting instructions indicated in the operating instructions and the assembly diagram.

CAUTION!	Do not damage the centring shoulders, seals and sealing surfaces during removal.

WARNING!	If the coupling is placed in an upright position (the main axis is horizontal), secure the coupling against falling over and rolling away.

→ Use a suitable container to catch the oil remaining in the coupling.

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12. STRIPPING DOWN AND ASSEMBLING THE COUPLING

12.1. Separating the inner part from the outer part

→ Position the coupling on the flange of the inner star (9).

→ Put wooden blocks of suitable thickness between the inner star flange (9) and the side plate (7). This prevents the outer part of the coupling from subsiding after the cover (12) has been removed.

→ Slacken off the cover bolts (758) and remove the cover (12).

→ Slacken off the screws (757) and remove the end cover (11).

→ Fit the lifting device (730) on the outer part of the coupling.

→ Lift the outer part of the coupling and pull it off the inner star (9).

12.2. Stripping down the outer part

CAUTION!	Only open the main bolts (750) if it is absolutely necessary (e.g. for a complete service). If there is no centring ring between the middle piece (1,2,3) and flange (5) or the side plate (7), the concentricity setting of the coupling outer part will be lost. Make sure how the outer part is to be reassembled → see Chapter 12.4 – Assembling outer part. If the components are to be reassembled using depth gauge measurements, it is essential to mark the position of the side plate (7), middle piece group (1,2,3), the flange (5) and the inner star (9) in relationship to each other beforehand.→ See Chapter 12.4 – Assembling outer part.

→ Place the coupling on the flange plate (5). (The coupling shaft is vertical.)

→ Open and remove the main bolts (750).

→ Take off the side plate (7).

→ Remove the middle piece of the coupling (1, 2, 3).

→ Remove the screws (768) and retainer ring (21) from the side plate (7).
This gives access to the L-rings (20) and the O-rings (853) (if they are present).

A full inspection can be carried out on the coupling in this condition.
If the outer part needs to be dismantled further (middle piece 1, 2, 3), e.g. to exchange the spring packs (1), this can only be carried out in the manufacturer's plant or by a Geislinger service centre.

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12.3. Prepare prior to reassembly

→ Clean every single part carefully before assembly.

CAUTION!	Venting holes as well as oil supply and return holes must be free of dirt and contamination. Thoroughly remove any carbon deposits that may be present in the spring chambers.

→ Renew all seals (O-rings, etc.).

→ Carefully grease all

- O-rings/O-ring grooves
- Bearings and running surfaces of bearings (if present)

with KLÜBER PETAMO GHY 133N paste or an equivalent product.

→ Fit O-rings in their relevant grooves, ensuring that they are not damaged.

→ Fit new disc springs.

→ Renew all sealing rings (copper rings, etc.).

→ Renew all bolts up to and including thread size M20.

→ Bolts larger than M20 may only be reused if the following criteria are met.

- The bolt head must not be damaged.
- The bolt thread must not be damaged or stretched.
- The surface on which the bolt head rests must not be damaged.
- The bolt must be cleaned impeccably. Only use LOCTITE CLEANER to remove deposits of LOCTITE.

CAUTION!	Any damaged or distorted bolts must always be replaced by new ones.

CAUTION!	Only use LOCTITE CLEANER to remove any deposits of LOCTITE.

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12.4. Assembling the outer part

ROUNDNESS OF OUTER PART

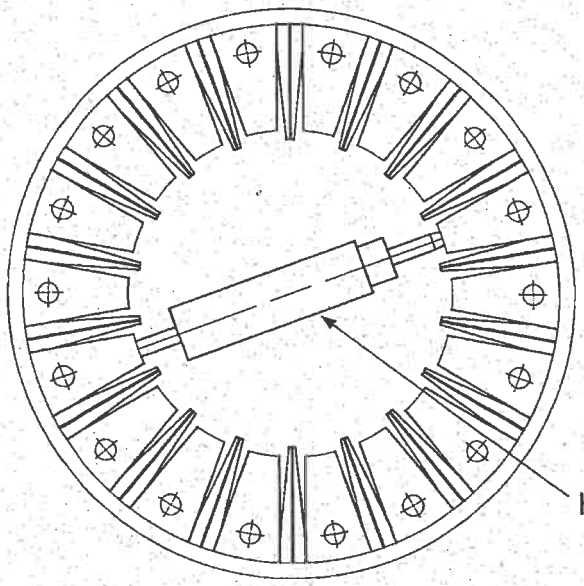
→ Check the divergence from concentricity at the inside diameter of all intermediate pieces (2).

→ Improve the concentricity to the values listed below as a minimum. Then fit the middle piece.

- Large couplings require the use of a hydraulic jack to adjust concentricity. (→ See diagram)
- Small couplings can be corrected with radial strikes on to the outside of the clamping ring (3), using a soft-faced hammer.

→ Take readings at top and bottom of opposite pairs of intermediate pieces (2).

The concentricity of the inner diameter of all intermediate pieces (2) is to be within the following values:



Up to size 56	Max. 0.1 mm
63 to 140	Max. 0.2 mm
Greater than 140	Max. 0.3 mm

CAUTION!	When adjusting concentricity do not damage the inner surfaces of the intermediate pieces.

CAUTION!	If the stated values for roundness are not achieved, contact Geislinger or a Geislinger service centre.

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ASSEMBLING THE OUTER PART

The process is the same as described for dismantling the outer part but in reverse order.

CAUTION!	Ensure that the right direction of rotation is used for the coupling. In order to ensure that the original direction of rotation is used, fit the spring packs (1) as they are shown on the assembly diagram.

NOTE:
If the coupling is equipped with centring shoulders on the flange (5) and side plate (7) to the clamping ring (3), the centring of the components in accordance with Points A or B is not required.

→ Position the flange plate (5) on the work top. (O-rings are visible.)

→ Place the middle piece (1, 2, 3) of the coupling on the flange plate (5).

→ Position the side plate (7) on the middle piece of the coupling.

→ Coat the disc springs (810) on both sides with an adequate amount of LOCTITE. (See assembly diagram for type.)

CAUTION!	The disc springs and the matching contact surfaces must be free of grease and dry.

→ Treat the threads of the main bolts (750) in accordance with the mounting instructions on the assembly diagram.

→ Fit the main bolts (750) according to the installation instructions given in the assembly diagram.

A) Assembling outer part with the help of a centring stage (preferred variant)

→ Rest the flange plate (5) on the centring stage. (O-rings are visible.)
As an alternative to the centring table, a horizontal lathe can also be used for centring the components.

→ Place the middle piece (1, 2, 3) of the coupling on the flange plate (5).

→ Position the side plate (7) on the middle piece of the coupling.

→ Centre the flange (5), coupling middle piece (1,2,3) and side plate (7) with a dial gauge in accordance with the concentricity tolerances provided in the assembly drawing.

→ Coat the disc springs (810) on both sides with an adequate amount of LOCTITE. (See assembly diagram for type.)

CAUTION!	The disc springs and the matching contact surfaces must be free of grease and dry.

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→ Treat the threads of the main bolts (750) in accordance with the mounting instructions on the assembly diagram.

→ Fit the main bolts (750) according to the installation instructions given in the assembly diagram.

→ Recheck the concentricity tolerances given in the assembly diagram to ensure that they are correct.

CAUTION!	A deviation of 30% in concentricity tolerances is permissible in the case of repairs, providing that there are no balancing problems. If the tolerance deviation is more than 30%, the outer part of the coupling must be dismantled again and reassembled.

B) Assembling the outer part with the aid of depth gauge measurements

→ Before dismantling measure the position of the side plate (7), middle piece group (1,2,3), flange (5) and inner star (9) in relation to each other. →See the sketch below.

→ The depth gauge measurements must be taken at four points, each offset at an angle of 90° to each other.
The surfaces at these points must be cleaned thoroughly beforehand. Depth gauges with display accuracy of 0.01 mm must be used for measuring purposes.

→ The measuring positions must be marked with a scriber and numbered 1 – 4.

→ The coupling can now be dismantled.

→ When the coupling is reassembled, the components must be aligned to the same depth measurement.

→ Reassemble the flange (5), middle piece group, inner star (9) and side plate (7), observing the marking. Roughly align the parts with each other beforehand. Use an aluminium hammer to help with this task. It is an advantage to use an assembly lever (see sketches for design and how to use) which can be used to adjust the parts in the hole for the main bolt.

→ Fit four main bolts in the holes adjacent to the measuring points in each case, without using Loctite. (See ** in the assembly sketch below.)

→ Tighten the four main bolts uniformly and in sequence in small steps until the stop is reached. The disc springs must lie flat and fit tightly. In order to check that the parts have not slipped while the bolts are being tightened, it is a good idea to fit four depth gauges (using magnetic feet), each opposite each other,
on the flange (5) and side plate (7).

→ Then slacken off each of the four marked main bolts by a quarter of turn.

→ Adjust the parts in relation to each other to the measured depth.
Tolerance = +/-0.02 mm for bearing-mounted couplings (bronze bearing (15) provided)
Tolerance = +/-0.04 mm for couplings without bearings.

→ Tighten the four main bolts again in small stages until they fit tightly and mark them.

→ Coat the disc springs (810) on both sides with an adequate amount of LOCTITE. (See assembly diagram for type.)

CAUTION!	The disc springs and the matching contact surfaces must be free of grease and dry.

→ Treat the threads of the main bolts (750) in accordance with the mounting instructions on the assembly diagram.

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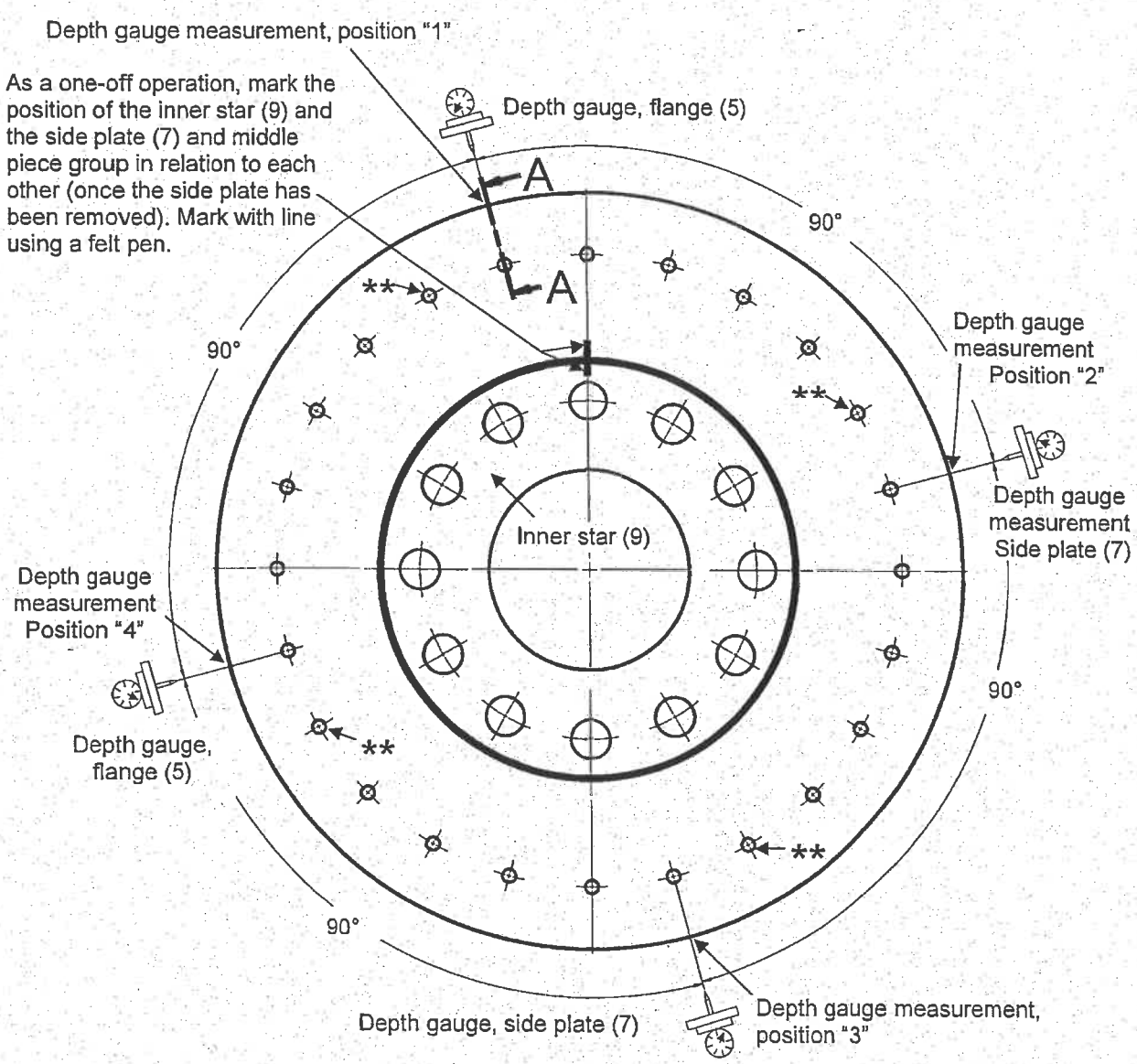
→ Fit all remaining main bolts (750) according to the installation instructions given in the assembly diagram.

→ Slacken off the four main bolts and then fit them in accordance with the assembly instructions.

→ Recheck the concentricity tolerances and then the depth measurements given in the assembly diagram to ensure that they are correct.

CAUTION!	A deviation of 30% in concentricity tolerances is permissible in the case of repairs, providing that there are no balancing problems. If the tolerance deviation is more than 30%, the outer part of the coupling must be dismantled again and reassembled.



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Sketch diagram of A-A cross section

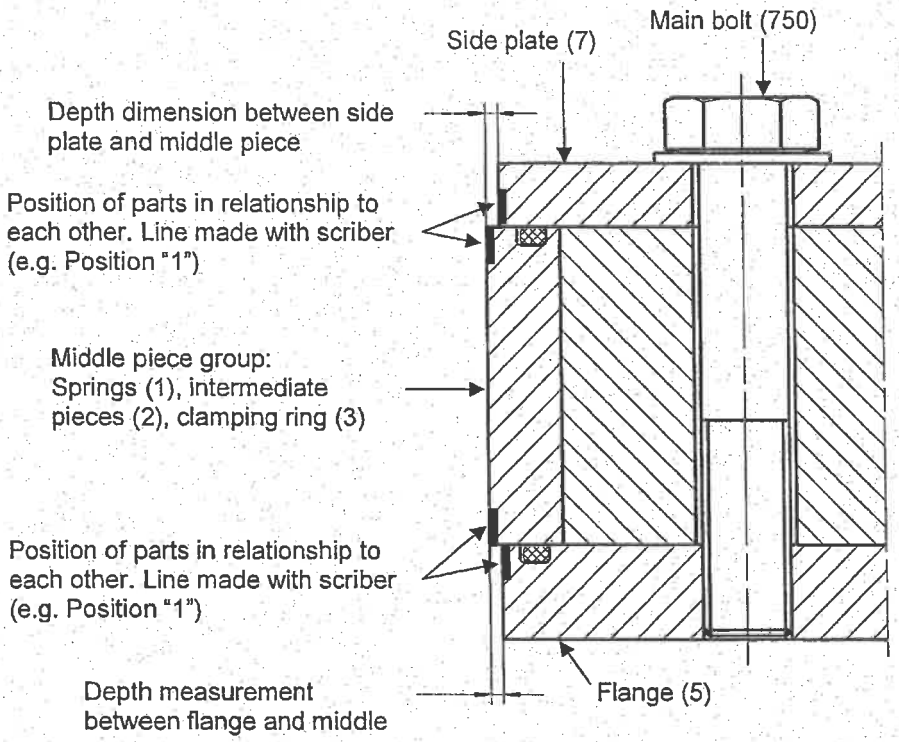
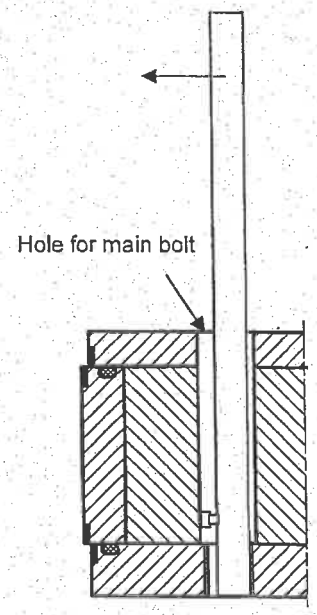
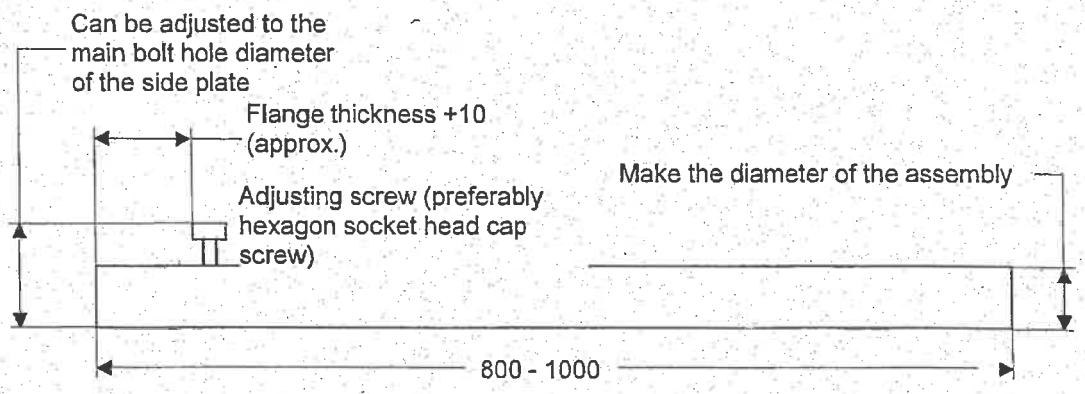


Diagram showing how to use the assembly lever



Suggestion for assembly



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COMPLETING THE OUTER PART ASSEMBLY

If present

L-ring (20) assembly

→ Position L-ring on side plate
→ Fit O-rings 853 and 854 in the correct grooves.
→ Position supporting ring (21) on the L-ring.
→ Fit the bolts (768) according to the installation instructions given in the assembly diagram.
→ Fit the L-ring on the flange as for the side plate (if present)

Fitting the oil supply line (55) in the cover (12)

CAUTION!	Venting holes as well as oil supply and return holes must be free of dirt and contamination.

CAUTION!	Only use LOCTITE CLEANER to remove any deposits of LOCTITE.

→ Check whether the O-rings are fitted in the correct grooves.
→ Fit the oil supply line (55) in the cover (12).
→ Use a retaining ring (56).
→ Fit the bolts (759) according to the installation instructions given in the assembly diagram.

12.5. Assembling the outer part – inner part

PREPARING THE ASSEMBLY

CAUTION!	All oil supply and return holes must be free of dirt and contamination.

CAUTION!	Only use LOCTITE CLEANER to remove any deposits of LOCTITE.

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→ Carefully grease bearing/bearing running surfaces (if present) with KLÜBER PETAMO GHY 133N paste or an equivalent product.

→ Apply a thin coat of TECTYL oil or a comparable rust inhibitor to all external surfaces of the inner star (9) including the grooves.

→ Fit O-rings in their relevant grooves, ensuring that they are not damaged.

ASSEMBLY

→ Position the inner star (9) on its flange.

→ Fit the lifting device on the outer part of the coupling and lift it above the inner star (9).

→ Fit the outer part of the coupling with the inner ends of the springs in the inner star grooves.

→ Push the outer part of the coupling along the inner star grooves until the end cover (11) can be fitted.

CAUTION!	If the outer part of the coupling is pushed too far into the inner star, the O-ring (850) of the inner star may be damaged. Therefore, use the same spacers between the outer part of the coupling and the inner star as when stripping the coupling.

→ Fit end cover (11).

CAUTION!	Any oil return holes in the inner star (9) and the end cover (11) must be aligned.

→ Fit the sealing screws (757) according to the installation instructions given in the assembly diagram.

→ Fit the cover (12) with the oil supply (55, if present) in the end cover (11) and flange (5).

→ Fit the bolts (758) for the cover according to the installation instructions given in the assembly diagram.

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13. INSTALLATION OF THE COUPLING AFTER SERVICING

→ Align the connecting shafts as described in Chapter 9.2.

→ Fit the coupling back into the system as described in Chapter 9.3.

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14. SERVICING

14.1. Visual check

Check coupling for oil leaks whenever possible but at least once a month.

14.2. Lubricating oil quality

The quality of the lubricating oil should be checked at regular intervals.
If the quality of the lubricating oil is poor or if it contains too much water, it may damage the coupling.
This reduces the service life of the coupling.
See Chapter 16 on what to do if the oil quality is poor.

14.3. Comprehensive servicing (Stripping down the coupling)

The coupling is to be given a comprehensive service in accordance with the stated servicing frequencies and servicing instructions.

SERVICING INTERVALS AND INSTRUCTIONS
Intervals in hours worked during normal operation

18,000 – 20,000	40,000 – 60,000
Type of service	Type of service
Dismantle coupling Separate inner part of coupling from outer part (do not strip down outer part of coupling) Clean all parts Treat or exchange all parts that can be reached in accordance with Chapter 12.3 Check permissible wear as per Chapter 15. Repair worn parts or renew them, if necessary Dispose of worn parts as per the instructions in Chapter 18.	Carry out service as for 20,000 hours of operation **If necessary, also:** Strip down the outer part of coupling Exchange all spring packs (1) and the inner star (9) **CAUTION!** The middle piece of the coupling (1, 2, 3) may only be stripped down by Geislinger or a Geislinger service centre.

If permissible wear between servicing intervals has been exceeded by more than 50%, subsequent servicing intervals are to be shortened accordingly. For this, it is assumed that wear is at a linear rate.

Servicing frequencies also depend on the specific conditions of use in each case.

NOTE

If the coupling is serviced and monitored on a regular basis, it facilitates long-term planning and spare parts can be ordered in good time.

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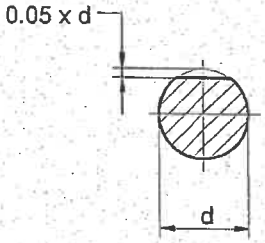
15. PERMISSIBLE WEAR

15.1. O-rings, items 850 and 853

→ Total wear must not exceed 5% of the ring diameter "d".

→ Total wear is the sum of
O-ring wear
+ depth of lead-in grooves on the running surfaces.



Running surfaces are:
For item 850, the O-ring groove bottom on the inner star (9) and end cover (11) as well as the inside diameter of the side plates (5 and 7) or at the inside diameter of the L-ring (20) (if present)
at item 853 - if present – the side surfaces to the L-ring (20) and the retainer ring (21)

→ You can obtain the O-ring nominal diameter at Geislinger or a Geislinger service centre

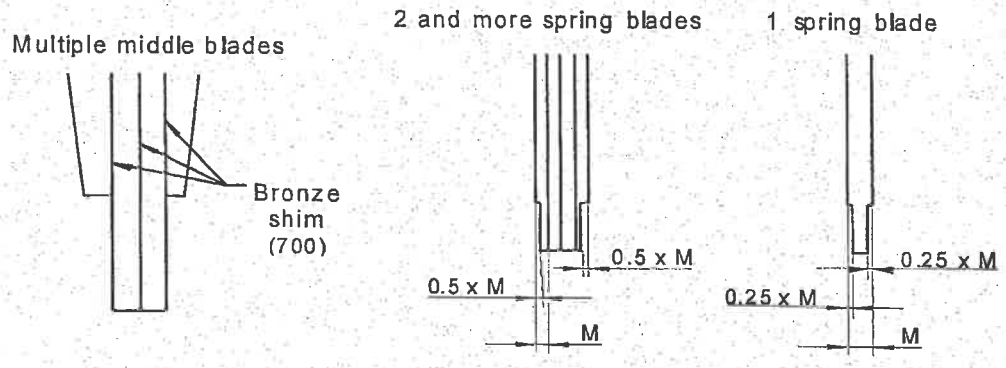
15.2. Wear to the spring blades

The calculated dimension "**X**" specifies the max. permissible wear at the inner ends of the spring blades. The dimension "**M**" is the thickness of one single spring blade. The dimension "**M**" can either be provided by GEISLINGER upon request or can be measured at the tips of the spring blades when the coupling is opened.

Measuring the permissible wear per insertion side "**X**"

Spring packs with two or more spring blades $X = 0.5 \times M$

Spring packs with one individual spring blade $X = 0.25 \times M$



Wear or deformation of the bronze shims (700) (in the case of multiple blades), which are inserted between the spring leaves, is normal and therefore no reason to service or exchange the spring packs.

CAUTION!	Bad oil quality or an inadequate supply of oil at the spring ends leads to severe wear very quickly.

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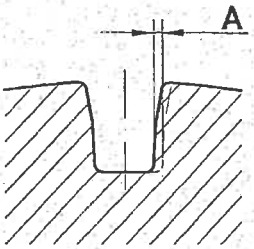
15.3. Inner star

→ Wear to the groove flanks must not exceed the value "A" given in the figure.



Coupling size	"A" [mm]
< 140	1.0
≥ 140 < 200	1.5
≥ 200	2.0

15.4. Side plate – Flange

→ The total wear at the sealing surfaces of the O-rings (850) must not exceed 5% of the nominal O-ring diameter ("d") (→ See Chapter 15.1)

Such sealing surfaces are either

- the inside diameter of the side plate (7),
- of the flange (5)
- bushes inserted in these (if present)
- or the inside diameter of an L-ring (20) (if present)

15.5. Bearings (if present)

See the assembly diagram for permissible values for axial and radial bearing wear.

Note:
If no max. permissible figures for wear or bearing clearances are given on the assembly diagram, contact Geislinger or a Geislinger service centre.

15.6. Other wear/damage

If other damage or wear occurs, these are generally to be clarified with Geislinger or a Geislinger service centre.

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16. REPAIR INFORMATION

INSPECTED CONDITION	CORRECTIVE MEASURE
"O-rings" sweat during operation; oil traces are present radially on the flange and side plate; "no drips are formed".	No effect on the function of the coupling; no repair required.
O-rings leaking; drips are formed.	Remove and strip down coupling; renew O-rings. Check lead-in grooves and renew worn parts, if applicable (see below) Check system alignment
Traces of oil in a radial position on the flange or side plate; main bolts (750) leaking.	Open leaking main bolts, seal them anew and fit again.
Traces of oil in a radial position on the side plate; bolts (756) leaking.	Open leaking bolts, seal them anew and fit again.
Total wear of O-rings 850 and 853 (if present) less than permissible wear dimension.	Renew O-ring(s); sand and polish lead-in grooves.
Total wear of O-rings 850 and 853 (if present) more than permissible wear dimension.	Renew O-ring(s) and worn part(s). (Repair by means of rebushing possible; clarify with Geislinger or Geislinger service centre.)
Wear at the spring blades greater than the permissible wear dimension "**X**".	Renew all spring packs (1) (by Geislinger or a Geislinger service centre).
Wear or deformation of the bronze shims (700) in the case of multiple blades.	No influence on the function of the coupling; first exchanged when the spring packs (1) were completely renewed.
Wear at the groove flanks greater than the permissible wear dimension "**A**".	Renew inner star (9) or rework groove flank; clarify with Geislinger or a Geislinger service centre.
Signs of wear on the outer diameter of the inner star (9) in the area of the intermediate pieces (2) of the flange (5) or the side plate (7).	Check alignment of shaft and correct. Renew damaged parts, if necessary. (Clarify with Geislinger or a Geislinger service centre.)
Bearing play more than permissible; see assembly diagram for permissible values.	Repair or renew worn parts.
Signs of abrasion marks (fretting) on the running surfaces of bearings.	Contact Geislinger or a Geislinger service centre; renew damaged/worn parts.
Signs of oil carbon deposits near the spring chambers	No appreciable impairment of the coupling function. In spite of this, any oil carbon deposits present are carefully removed during each service. Clean all oil ducts thoroughly

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INSPECTED CONDITION	CORRECTIVE MEASURE
Poor oil quality or higher water content in the engine oil than permitted by the engine manufacturer. No damage to engine bearings.	Remove venting screws (756). Drain coupling completely. Bleed coupling as described in Chapter 10.1 and supply with fresh oil.
Poor oil quality or higher water content in the engine oil than permitted by the engine manufacturer. Bearing positions in engine already damaged as a result of this.	18,000 – 20,000 hours: Service coupling as described in Chapter 14.3.
Other damage or wear that has occurred but which was not described earlier.	Contact Geislinger or a Geislinger service centre.

CAUTION!	The coupling is to be assembled in accordance with the instructions provided in Chapter 12.

WARNING!	Dispose of worn parts as instructed in Chapter 18.

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17. OPERATING CONDITIONS

Permissible ambient temperature maximum 100° Celsius (unless other instructions are given).

Oil temperature at the coupling inlet maximum 60° Celsius.

Permissible temperature ranges of the sealing elements: See data sheet provided.

CAUTION!	If the temperature rises above or falls below the permissible temperature application ranges for seals during operation, it may damage the seals and then consequently damage the coupling and the system.

Required oil inlet pressure of the coupling

→ For the value required, see the assembly diagram (technical data field).

→ If no value is provided, the required oil pressure is to be calculated using the following formula:

$$p \geq \frac{T_d}{T_{d,p}}$$

p = Required minimum oil pressure in the coupling [bar] (absolute value)

T_d = The calculated damping torque in the coupling as a synthesis of all orders [kNm].

$T_{d.}$ = Permissible damping torque per 1 bar oil pressure [kNm/bar]. The permissible value is given in the technical data field of the assembly drawing.

CAUTION!	DO NOT EXCEED THE MAXIMUM PERMISSIBLE SHAFT MISALIGNMENT STATED IN THE ASSEMBLY DIAGRAM IN ANY OPERATING CONDITION.

WARNING!	FREELY ROTATING PARTS ARE TO BE COVERED BY A GUARD TO PREVENT ACCIDENTS.

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18. WITHDRAWING THE COUPLING FROM SERVICE & DISPOSING OF USED PARTS

After withdrawing the coupling from service or after exchanging individual components –

→ All metal parts can be recycled (scrap metal).

→ Oil still in the coupling or remaining oil must be disposed of as used oil.

→ All old seals (O-rings, etc.) must be disposed of as oily consumables (special waste).

WARNING!	National and local regulations must always be observed when disposing of the coupling or parts thereof.

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19. ORDERING SPARE PARTS

CAUTION!	USE ORIGINAL GEISLINGER SPARE PARTS ONLY

Please provide the following information so that the product can be accurately identified if enquiries are sent:

- → Type designation of coupling (→ See the stamped designation on the outer part)
- → Coupling serial number (→ See rating plate)
- → Part number of the spare part (→ See assembly diagram/parts list)
- → Designation of the spare part
- → Quantity of the parts required

EXAMPLE

Designation	C 80 / 17.5 / 62H / 10
Serial number	10106273/2/10
Part number of the spare part	850
Quantity of the parts	2
Designation of the spare part	O-ring

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20. APPENDIX

These operating instructions contain the following supplements:

→ PERMISSIBLE TEMPERATURE APPLICATION RANGES FOR O-RINGS AND SEALS
→ MOUNTING INSTRUCTIONS FOR SCREWS
→ INSTALLATION DECLARATION (INSIDE EUROPEAN UNION)
→ ASSEMBLY DIAGRAM(S) WITH PARTS LIST
→ WARRANTY
→ USE OF A WEAR GAUGE (OPTIONAL)
→ CODE BOOK NUMBER CODING (OPTIONAL)
→ OPERATING INSTRUCTIONS FOR BOUGHT-IN PARTS (OPTIONAL)

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Permissible temperature operating range for sealing elements (O-rings and so on)

Permissible temperature operating range at standstill, transport and storage of the product	
from (minimum)	
- 10° C (+ 14° F)	

Permissible temperature operating range before start-up and during operation	
from (minimum)	
- 10° C (+ 14° F)	

ATTENTION!

The exceeding respectively the dropping below of the permissible temperature operating range of the sealing elements can lead to damages on the sealing elements and in further consequence to damages on the product and the machinery!

General assembly instructions for screws and screw/nut connections

These instructions are intended to supplement the assembly instruction for screws given on the assembly drawing

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2. APPLYING ASSEMBLY AIDS 4

3. ASSEMBLY 6

4. AFTER ASSEMBLY 7

5. RE-USE 8

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1. CLEANING THE THREADED PARTS

Before handling the threaded parts using the assembly tools, the contact surfaces must be cleaned carefully and thoroughly.

These include the contact areas

- the screws on the head support and on the thread ► see Figure 1
- the nuts (if present) on both support surfaces and on the thread ► see Figure 2
- both sides of the washers ► see Figure 3
- the threaded holes (if present) ► see Figure 4
- and the contact surfaces on the parts being connected ► see Figure 5

Fig. 1

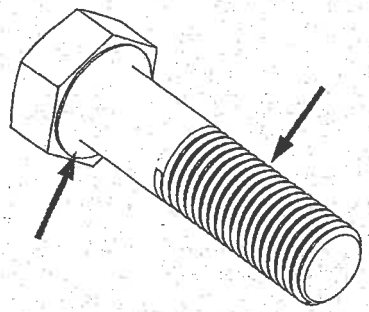


Fig. 2

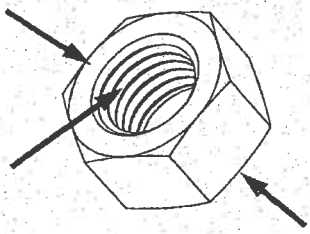


Fig. 3

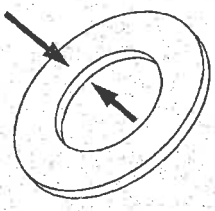


Fig. 4

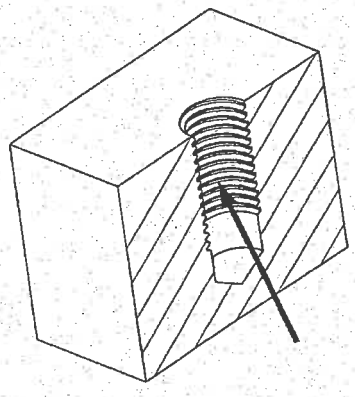
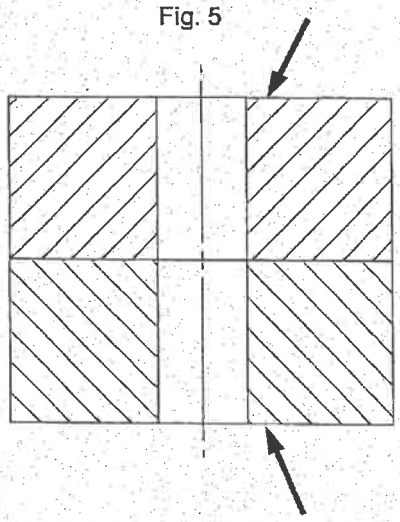


Fig. 5



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Oxide layers

If the contact surfaces have oxidised, first carefully remove the oxide layers mechanically, for example using a wire brush.

Dirt and grease

After removing the oxide layers (if present), all contact surfaces must be cleaned and degreased thoroughly.
Isoparaffin or acetone can be used as agents for cleaning and degreasing.

If Loctite is used being for the assembly, degreased, cleaned contact surfaces must not afterwards be touched with bare hands.
If dirty or if coming into contact with grease from hands, the relevant contact surfaces must be cleaned **again**!

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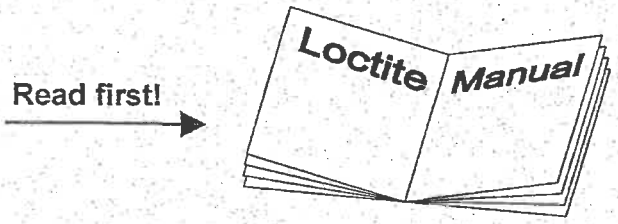
2. APPLYING ASSEMBLY AIDS

For assembly, the contact surfaces and thread must be treated with seal and/or screw securing agents or with lubricants.
Normally used as screw securing or sealants are Loctite products and engine oil or Molykote paste as lubricants.
The precise types and application of the assembly aids can be found on the relevant assembly drawings.

Before applying the assembly aids, the contact surfaces must be dried thoroughly!

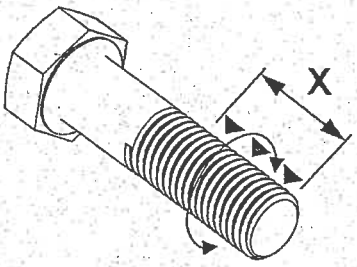
Note on Loctite products

Loctite products should always be used within the processing time specified by Loctite.
These processing times can be found in the instructions for use for the relevant Loctite products being used. Do not exceed the maximum processing times permitted! Try and keep the processing times as short as possible.



Application in threaded parts

Before applying the assembly aid, the bearing threaded length "X" (see diagram) should be calculated for the parts being screwed together. This threaded length should then be wetted with the assembly aid across the entire area. In terms of the quantity, use only so much that only small quantities of excess assembly aid overflows after assembly.
A test assembly is recommended to be able to estimate the amount required.



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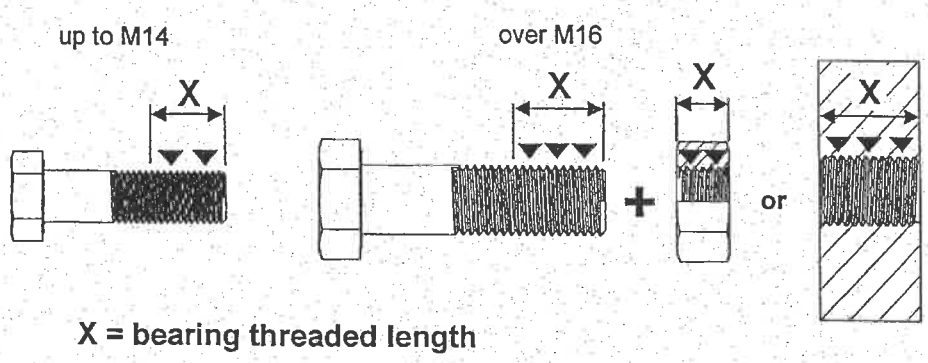
With sealing unions in particular, ensure that the threaded gap is completely filled with sealant. For thread size M16 and over, the threaded hole or the nut thread should also normally be coated with the assembly aid.

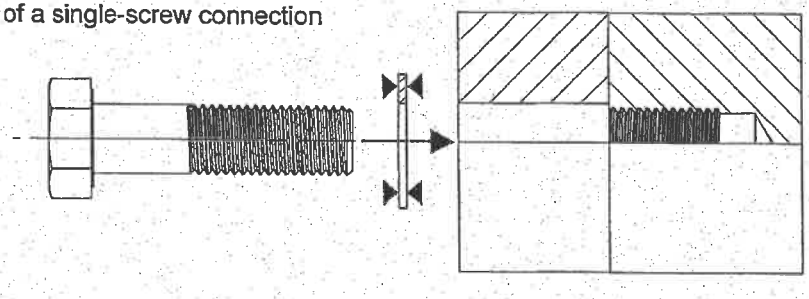


X = bearing threaded length

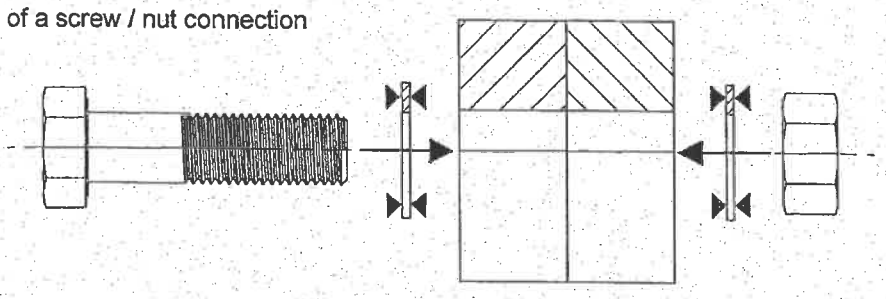
Application under screw heads and on washers

When using the assembly aids under washers or screw heads, wet the surface so that after the assembly process is complete for the screw, the contact surfaces (see diagram) is completely covered with the assembly aid.
When using on washers, apply to both the front and the back.
In terms of the quantity, use only so much that only small quantities of excess assembly aid overflows after assembly.
A test assembly is recommended to be able to estimate the amount required.

Example of a single-screw connection



Example of a screw / nut connection



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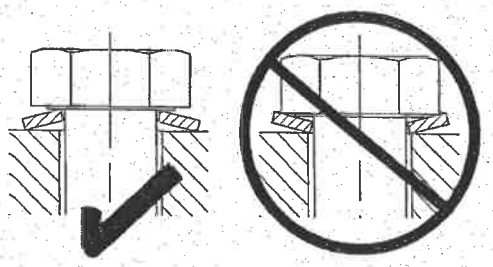
3. ASSEMBLY

The screws should only be tightened at the torques and in accordance with the guidelines given on the assembly diagram.

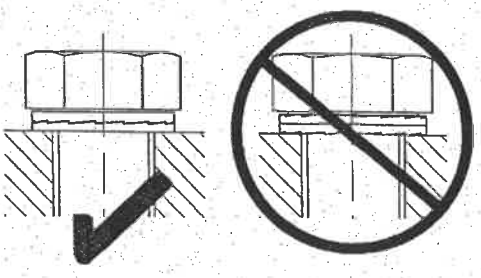
Note on washers

Disc springs and sometimes NORD-LOCK are normally only used as washers for securing screws where applicable. These should be assembled as follows.

Disc spring usage



NORD-LOCK washer usage

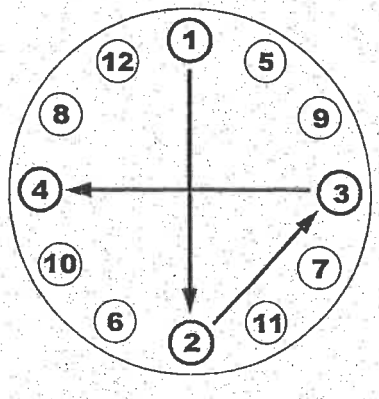


Tightening process

The screws should normally be tightened crosswise, i.e. opposite pairs of screws should be tightened immediately after one another.
The assembly should always be carried out according to the same diagram, as described in the example below for a twelve-point division.
The torque values, the tightening of the screw and the application area for the assembly aid are described in the assembly guidelines in the assembly diagram.

TWELVE DIVISION EXAMPLE

a) Mark screws 1 - 4
Pre-assemble screws 1 - 4 **without** Loctite assembly aid to approx. 30% of the torque value given on the assembly diagram and in the order given



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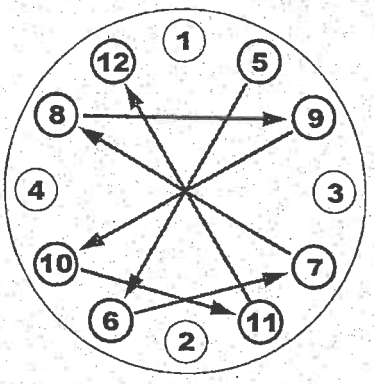
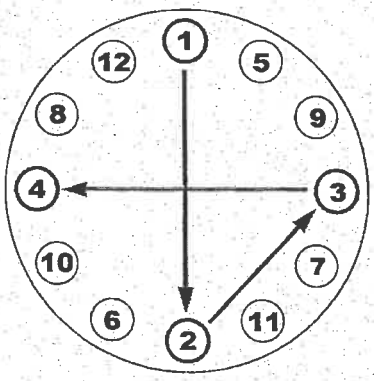
b) Final assembly of screws 5 – 12 in the order given



c) Remove screws 1 - 4.
Final assembly of screws 1 - 4 according to the assembly guidelines in the assembly diagram and in the order given.



4. AFTER ASSEMBLY

Immediately after the screw assembly is complete, the excess residue of the assembly aids, especially Loctite residue, should be removed

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5. RE-USE

If screw connections secured or sealed using Loctite are unfastened, the any old Loctite reside remaining on the screws should be completely removed mechanically, for example using a wire brush.
With threaded connections and internal threads (e.g. with nuts), old Loctite residue should also be removed completely.

All washers from unfastened screw connections should normally be replaced by new washers!

Afterwards, all parts should be prepared for re-assembly (see chapters 1 to 4).

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Orderer		Recipient	

Speed Switch Unit (-D)
MDP-REV-D
Instruction Manual

Article

(30 pages including the front page)

DAIHATSU DIESEL MFG. CO., LTD.
ENGINE DESIGN SECTION
TECHNICAL DEPARTMENT

		DRAWN BY	T.NAKAMITSU
TYPE		**CHECKED BY**	M.M
LIST No.	INB000104A		
DATE	27. MAY. 2015	**APPROVED BY**	M. MOCHIZUKI
REVISION	27,May,2015 : This book is what of Q7LT333140FD the p13 hysteresis value definition and is reviewed, and replaced with the KIDS number.		

	#

Speed Switch Unit-D Safety Precautions

This manual describes precautions for safety installation and operation of the speed switch unit-D. Operate equipment following these precautions.

Hazards and damage likely to result from incorrect wrong use or, neglecting of the indicated precautions, is explained below in this Manual using the following symbol marks shown below. The symbols:

... CAUTION = State likely to cause damage of light or intermediate damage.

1. Caution against electric shock

(1) Voltage remains in the internal terminal block even after POWER is turned OFF. See page 24(Fig.17).

(2) When using 100 or 200 VAC for signal output contacts, be aware of possible electrical shock from the terminal block.

2. Caution against current leakage

(1) To prevent current leakage, be sure to ground equipment properly. See page 24(Fig.17).

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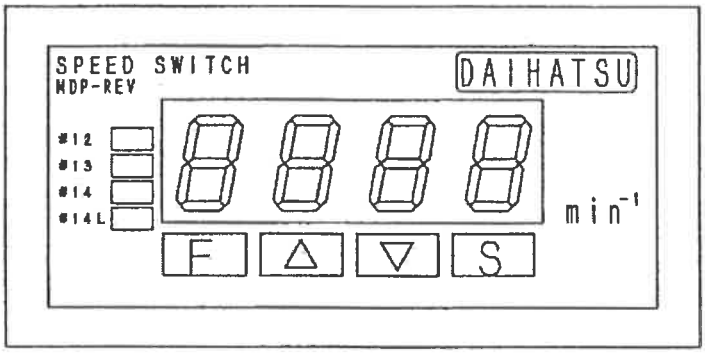


Fig. 1 Speed switch unit–D panel display

1 Overview

This product is a revolution detector available with a digital display function. It is a new type of speed switch unit, compact in size and highly functional, that indicates measurements digitally by a micro computer, ensuring high precision, flexible setting that meet a wide range of applications.

2 Features and functions

- High measurement precision by digital measurement (0.05% (at F.S.) ± 1 digit)

- Revolution speed displayed digitally to a maximum of 4 digits.

- Three-point revolution detection and one-point contact for start-up interlock.

- High withstand voltage semiconductor contact outputs
(100VAC, 0.1A, withstand voltage 1500V)

- 4 ~ 20mA analogue output

- Revolution speed frequency output (1 Hz per revolution)

- Input specification and revolution range can be set at any value.

- Detachable terminal block is used.

3 Specifications

Item		
Speed detection	Number of detection points	3 speeds (#12, #13, #14) (#14L : contact for start-up interlock)
	Detection speed range	100～9990m in^{-1} (#14L :1～999m in^{-1})
	Measurement precision	0.5m in^{-1} when it is 1000m in^{-1} (0.05% (at F.S.) ± 1digit)
	Setting method	Detection point is set by key operation
	Response time	Contact output response 0.3 seconds
	Hysterisis	Standard 3.0% (0.0 ～ 1.0% at F.S. setting changeable)
Display	Display range	0～ 9999
	Display	4-digit red digital display (brightness adjustable)
	Display update	Approx. 0.5 sec.
Pulse input	Puls input range	2～6kHz
	Setting method	Set number of pulses and reduction gear ratio per revolution
	Input pulse	Dedicated pulse sensor (AC 2Vpp ～ 288Vpp (AC 0.7 ～ 100Vrms))
Frequency output (pulse output)	Output signal	1Hz, 0/15V (5kΩ) per revolution Allowable load resistance 5 KΩ or more, non-insulated with the power supply
	Output potential range	1～9999 Hz (1～9999m in^{-1})
	Output error	0.1% (2000Hz or higher)
Analogue output	Output signal	Current signal 1 output 4-20mA load resistance 500Ω
	Output precision	±0.5%(FS)
	Full scale	100～9990m in^{-1}
Contact point output	Output contact	3 contact (#12 : Overspeed, #13 regular speed, #14, low speed) 1 contact (#14L : for start-up interlock)
	Contact structure	form a
	Contact capacity	No-voltage semi-conductor contact, AC/DC 100V 0.1A, withstand voltage 1500V
Power supply	Voltage	24VDC -25～+25% (18.0V～30.0V) Ripple 10%or less.
	Power consumption	2.5W
Environmental conditions	Working temperature	5～ 55℃ (Max. 95% RH, with no dew condensation)
	Storage temperature	-20～65℃ (MAX 95%RH, with no dew condensation)
	Ambience	No corrosive gas
	Vibration/impact	5 ～ 13.2Hz:± 1mm, 13.2 ～ 100Hz:0.7G Impact: max. 10G
Case	Dimension	72×36×137
	Weight	0.2Kg
	Color	Black

4 Outline

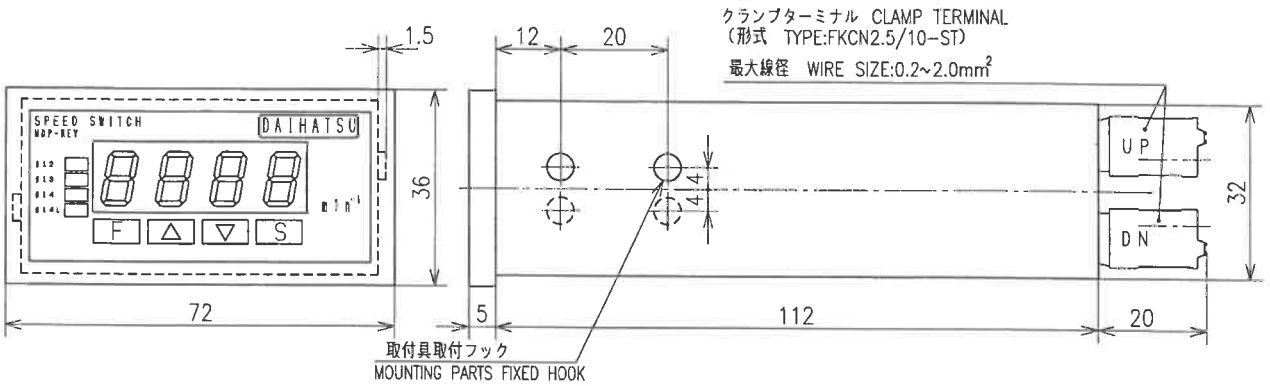


Fig. 2 Outline drawing

5 Configuration

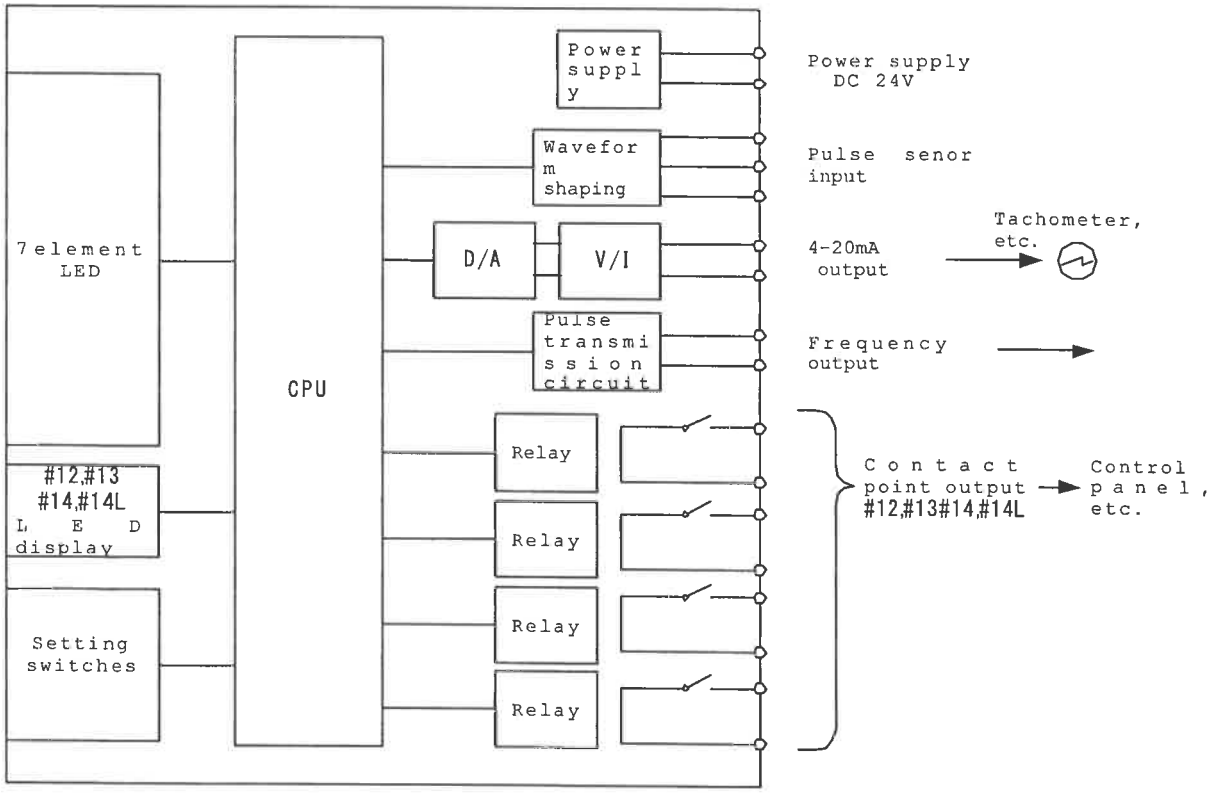


Fig. 3 Internal block diagram

6 Operation and modes

6.1 Display and operation

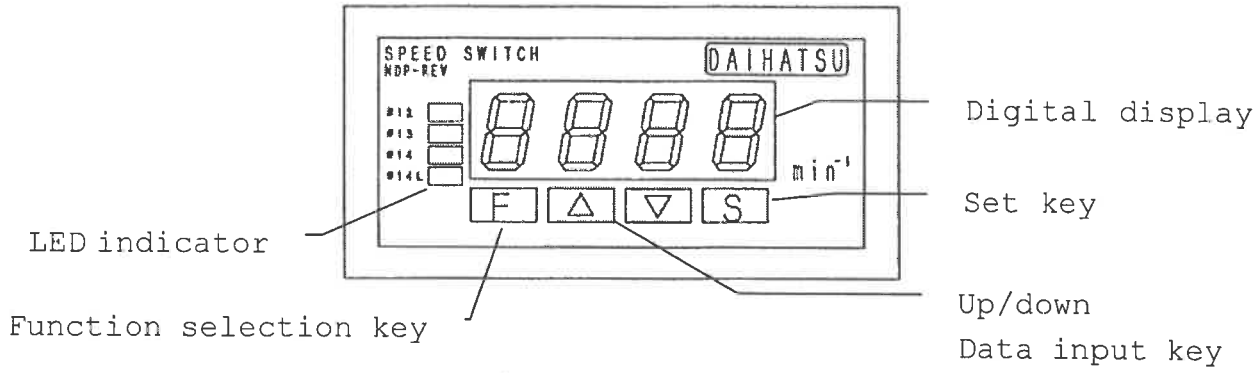


Fig. 4 Panel display

<Display function>

Digital display: Engine revolution speed is indicated digitally on the 4-digit digital LED display.
Set values, etc. are displayed, during various setting operations.

<LED indicator>

#12 : Means overspeed detection and indicates that the engine is rotating faster that the rated speed.

#13 : Means rated speed detection and indicates that the engine is running at the rated speed.

#12 : Means Low-speed detection and indicates that the engine speed has declined below the low-speed set value.

#14L : Indicates that the conditions for start-up interlock have been met. It is applicable in start-up interlock circuits as it can detect complete stop of the engine.

<Input>

[F] Function selection key:

Switch for changing the mode and items. Pressing this key during normal mode changes the mode to setting display/setting mode.
Pressing and holding this key for more than 3 seconds changes the mode to initial setting display/setting mode.

[△][▽] Data input key:

Allows you to increase and decrease the set value.

[S] Set key:

Starts the setting of items and also enters and saves the set value.

6.2 Actions when the power supply is turned on

When the power supply is turned on, all LEDs turn on for about 1 second and normal mode starts. During the above action, revolution detection, output and error detection functions are suspended whether or not input signal exists.

Note) Before using the speed switch unit for the first time, various settings must be made to enable normal function, as initially set values are only temporary values set for shipment.
Please set each parameter correctly reading this instruction manual to enable correct use of the unit.

6.3 Normal state/display (normal mode)

During normal mode, the revolution speed presently detected is displayed. When the revolution speed exceeds the detection set values, the relevant contact output operates and at the same time the relevant LED indicator lights. This mode is displayed except when the setting change mode is selected.

During normal mode, the display brightness setting can be changed by the [▲][▼] data entering keys.(The initial setting value is set at the maximum brightness of 31, and normally it is not particularly necessary to change it.)

The next time the power supply is turned, the brightness is displayed at the value preciously saved.

▲	Increase the brightness.
▼	Decrease the brightness
S	Pressing this button for more than 3 seconds saves the present brightness.

6.4 Initial set values

Note) To use this product, it is necessary to pre-set various set values correctly.
Using this unit without setting these values may cause critical problem in engine operation and control.

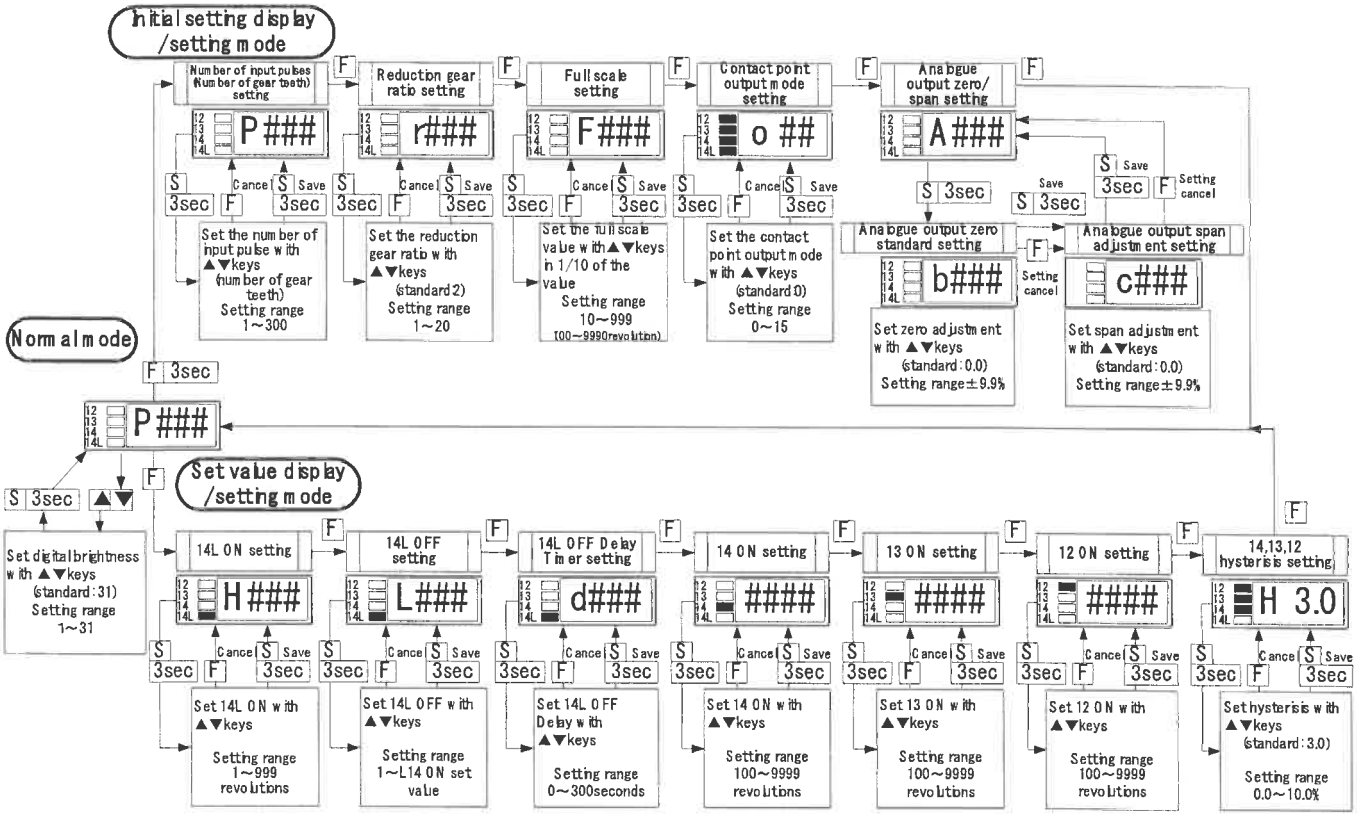
For the initial set values the following values are entered before shipment.
To encourage setting by the users, the values set in the initial setting are normally impractical values.
Refer to the table below and set items that have no ※ mark attached.

※ marked items can be used without changing the initial set value.

Section	Set Items		Settable range	Initial Set Value	Set Value
8. 2	Number of gear teeth		1～300	1	
8. 2	Reduction gear ratio		1～20	2	
8. 3	Analogue value full scale		10～999	10	
8. 4	Output mode		0～15	0	※
7. 2	#14L	ON revolution speed	1～999	1	
7. 2		OFF revolution speed	1～999	1	
7. 2		Off delay timer	0～300	0	
7. 3	#14 ON revolution speed		100～9999	100	
7. 4	#13 ON revolution speed		100～9999	100	
7. 5	#12 ON revolution speed		100～9999	100	
7. 6	#14, 13, 12 contact hysterisis		0～10.0	3	※
8. 5	Analogue output zero setting		-9.9～0～9.9	0	※
8. 5	Analogue output span setting		-9.9～0～9.9	0	※
6. 3	Display LED, brightness adjustme		1～31	31	※

※ marked items basically require no change

Set value display/setting mode and initial set value display/ setting mode



F : Press the "[F]Function selection key" once. S : Press the "[S]set key" once.

F 3sec : Press and hold (3seconds) the "[F]Function selection key" . S 3sec : Press and hold (3seconds) the "[S]set key".

Fig. 5 Setting procedure list

To change the set values, it is necessary to perform setting change work using the "Set value display/setting mode" and "Initial setting value display/setting mode".

"Set value display/setting mode":

A mode for setting mainly items concerning speed detection.

"Initial setting value display/setting mode":

A mode for setting constants that should be pre-set such as number of gear teeth and full scale value, etc.

To change from the "Normal mode" to "Set value display/setting mode" or "Initial setting value display/setting mode", press the "[F] function selection key" at normal mode.
To change to each mode, press the "[F] function selection key" as follows. Please refer to Fig. 5 Setting procedure list.

① To change to the "Set value display/setting mode", press the "[F] function selection key" once.
② To change to the "Initial setting value display/setting mode" press and hold (3 seconds) the "[F] function selection key".

To increase or decrease the set values by pressing the [▲][▼] keys, pressing the key once increases/decreases the value by ± 1 step, but pressing and holding the same key changes the mode to repeat mode and the values can be entered continuously. Keep pressing even further, changes the mode to fast forward and values can be changed in 10-fold, 50-fold steps.

7 Set value display/setting mode

7.1 Mode and setting items

Pressing the "[F] function selection key" from "Normal mode", changes the mode to "Set value display/setting mode".

In this mode, each time the "[F] function selection key" is pressed, the table below is displayed sequentially and pressing the "[F] function selection key" once again at the last item, returns the mode to "Normal mode". (Refer to Fig. 5)

The set value display/setting mode consists of the following setting items.

① Setting #14L ON:
Revolution detection for start-up interlock, ON point revolution speed
② Setting #14L OFF:
Revolution detection for start-up interlock, OFF point revolution speed
③ Setting #14L OFF Delay:
Contact start-up interlock, OFF DELAY TIMER (second)
④ Setting #14 ON:
Low speed detection ON point revolution speed
⑤ Setting #13 ON:
Rated revolution speed detection ON point revolution speed
⑥ Setting #12 ON:
Overspeed detection ON point revolution speed
⑦ Hysteresis (#14, #13, #12):
ON-OFF width of #14, #13, #12 (% at FS)

Pressing and holding the "[S] set key" during the item display changes the mode to the set value change mode of the applicable item.

When it enters into the set value setting mode, measurement is suspended and external output and external contact are frozen temporarily.
(When it returns to the normal mode again, it restarts but with a delay of about 1 to 2 seconds)

To add, when no operation is made for 3 minutes when any set value is being displayed or set, it will automatically return to normal mode.

When the brightness degree had been decreased in normal mode, the maximum brightness will be displayed temporarily in set value display mode but will display the original brightness again when returned to normal mode.

7.2 Start-up interlock contact (#14L)

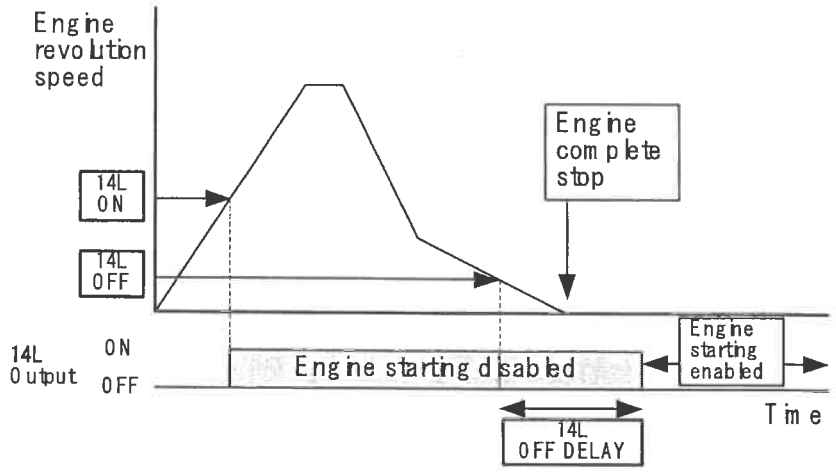


Fig. 6 14L OFF DELAY TIMER contact operation outline

Internal combustion engine employs fly wheel (inertial mass) at the output axis to reduce output fluctuation. Therefore, when the engine is stopped, revolution remains longer than expected due to inertia of fly wheel, and even when the users stop the machine, the output axis may not be stopped completely. When start-up command is given to the engine under such situation, it may trigger unexpected accidents.

Contact for start-up interlock outputs a condition that can block restarting of engine when the output axis is rotating, with the following 3 conditions combined, so as to enable secure detection of engine stop (refer to Fig. 6 "Engine Activation Disabled" part).

(1) "#14L ON detected revolution speed" as engine operating condition,
(2) "#14L OFF detected revolution speed" as engine condition immediately before stopping,
(3) Time counter timer from (2) to complete stop "#14L DELAY TIMER"

For using the start-up interlock contact (#14L), the following 3 values must be set.

① #14L ON setting
H = 1 ~ 999
② #14L OFF setting
L = 1 ~ "#14 ON set value"
③ #14L OFF Delay setting
d = 0 ~ 300 sec.

Description:

① is an operation point setting for start-up interlock contact and when the engine revolution speed exceeds this set value, #14L contact output operates.

② #14L starts the OFF DELAY timer when the engine revolution speed becomes below this set value.

③ is a set value of #14L OFF DELAY timer and sets the delay time in seconds. #14L contact output turns off when the time set by this timer is up.

7.3 Low speed detection contact (#14)

To use the low speed detection contact (#14), the following value must be set.
When the engine revolution speed increases above this set value, #14 contact output operates.

④ #14 ON setting
#14 = 100 ~ 9999

7.4 Rated rotation detection contact (#13)

To use the rated rotation detection contact (#13), the following value must be set.
When the engine revolution speed increases above this set value, #13 contact output operates.

⑨ #13 ON setting
#13 = 100 ~ 9999

7.5 Overspeed detection contact (#12)

To use the overspeed detection contact (#12), the following value must be set.
When the engine revolution speed increases above this set value, #12 contact output operates.

⑩ #12 ON setting
#12 = 100 ~ 9999

7.6 Hysteresis (#14, #13, and #12)

Hysteresis is provided to secure stability in detection operation even when the revolution speed sways. At the point of time the revolution speed exceeds the set value (detected revolution speed), the contact turns on and at the point of time the revolution speed become less than set value (return revolution speed) for the amount of hysteresis, the contact turns off.

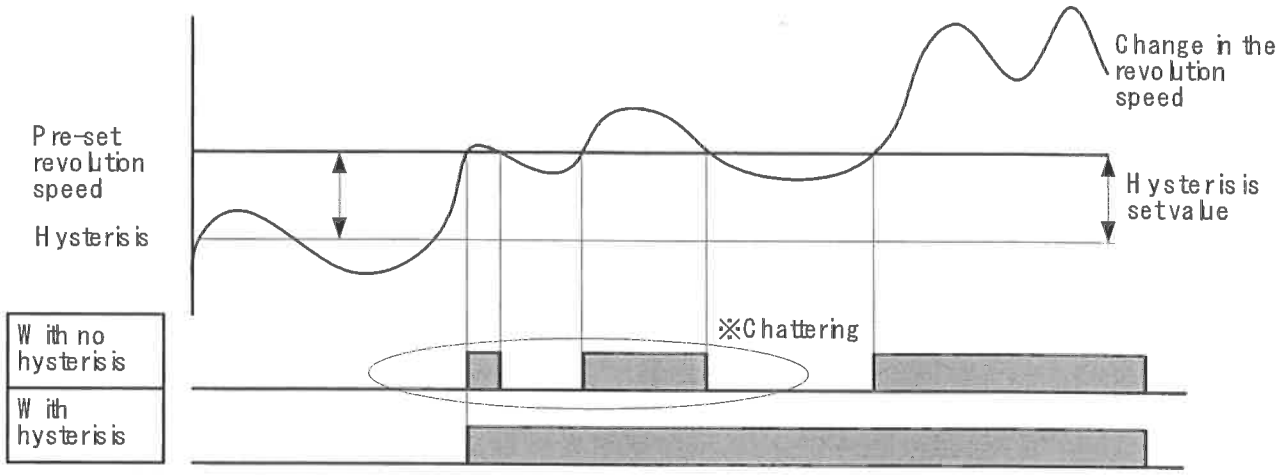


Fig. 7 Hysteresis

※Chattering may generally cause relay contact depositing by contacts repeating ON/OFF in a short time.

Detection revolution speeds for #14, #13 and #12 are set individually, but for return revolution speed, this hysteresis value will be applied to all.

⑪ Hysteresis (#14, #13, and #12)
0.0 ~ 10.0% (at FS)

※ 3.0% is registered as initial set value.
※ It is not particularly necessary to change.

Hysteresis means the range between (detection revolution speed) – (return revolution speed), and The percent value sets it for each rotating speeds.

[Ex. 4] When it is #13 = 1800, H = 3.0,
(#13: rated rotation detection set value, H: hysterisis set value)

Hysteresis = $1800 \times 3.0/100 = 54\text{min}^{-1}$
#13 ON point (detection revolution speed) ----- 1800min^{-1}
#13 OFF point (return revolution speed) ----- 1746min^{-1}

8 Initial setting display/setting mode

8.1 Mode and setting items

Pressing and holding (keep pressing for 3 seconds) the "[F] function selection key" from "Normal mode" or "Set value display/setting mode" changes the mode to "Initial setting display/setting mode".

In this mode, each time the "[F] function selection key" is pressed, the following display will be displayed sequentially and pressing the "[F] function selection key" at the last item returns the mode to normal mode. (Refer to Fig. 5)

Initial set value display/setting mode consists of the following setting items.

① Pulse setting:
Pulse number per revolution (number of teeth)
② Reduction gear ratio setting:
Reduction gear ratio from the crank shaft
③ Full scale setting:
Scale range of analogue output
④ Output mode setting:
N/C, N/O modes of output contacts
⑤ Analogue output zero/span adjustment:
For analogue output range adjustment

Pressing and holding the "[S] set key" during display of each item changes the mode to set value changing mode of the relevant item.

When it changes to the initial setting mode, measurement is suspended and external output and external contacts are frozen temporarily.
(When it returns to the normal mode again, it restarts but with a delay of 1 to 2 seconds).

To add, when either of the set value displayed or set is left for 3 minutes without any operation, automatically returns to normal mode.

When the brightness degree had been decreased in normal mode, the maximum brightness will be displayed temporarily in set value display mode but will display the original brightness again when returned to normal mode.

8.2 Pulse setting/reduction gear ratio set values

Engine revolution speed (or revolution speed of rotor) is calculated from the frequency of pulse signal impressed on the sensor input terminal based on the pulse specification set and is displayed.

For the calculation of revolution speed, it is necessary to set the following pulse specifications.

① Number of pulses per revolution (number of gear teeth)
P = 1 ~ 3 0 0
② Reduction gear ratio
r = 1 ~ 2 0

① is the number of pulses generated per revolution of the rotation axis (gear) installed with a pulse sensor.
② is the reduction gear ratio to set when the engine revolution speed and the rotation axis installed with a pulse sensor are not the same.

For instance, when the revolution of the engine output axis is detected by pulse sensor in the original condition, the reduction gear ratio of ② becomes r = 1.

[Ex. 1] Engine output axis direct measurement: when the number of gear teeth is 142.
P = 1 4 2, r = 1

When the pulse sensor is attached to the gear that interlocks with the cam shaft of 4 cycle engine, r becomes 2. (Cam shaft only rotates 1/2 revolutions per engine revolution, so the reduction gear ratio becomes 2).

[Ex. 2] Cam shaft measurement: number of gear teeth 75
P = 7 5, r = 2

Also when the number of pulses per revolution is not an integral number, set a value 10-times ① and at the same time set a value 10 times large also for ②.

[Ex. 3] Cam shaft measurement: tacho generator 7.5 pulse
P = 7 5, r = 2 0

(Reference)
The pulse frequency to be impressed is expressed by the following formula.

Frequency = Revolution speed × number of pulses
÷ reduction gear ratio ÷ 60

On the contrary, when calculating the revolution speed from the frequency, it is expressed by the following formula.

Revolution speed = Frequency × reduction gear ratio × 60
÷ number of pulses

8.3 Full scale set value

The revolution speed can be output by the analogue signal of DC 4 ~ 20mA.
It is necessary to set the full scale value so it meets the specification of the equipment to which this output is connected.

③ Analogue full scale setting range
F = 1 0 0 ~ 9 9 9 0 min^{-1}

The set value is determined by defining the revolution speed necessary at the time of 20mA output to meet the specification of the equipment to be connect to and the value dividing the revolution speed by 10 shall be the set value. 4mA is equivalent to 0 revolution speed.

[Ex. 4] Full scale = 2 5 0 0 min^{-1}
F = 2 5 0

8.4 Contact output mode

Contacts that operate when the contact output and the conditions of (start-up interlock contact (#14), low-speed detection contact (#14), rated rotation detection contact (#13) and overspeed detection contact (#12)) are met can be specified whether it should be ON during operation or OFF by the following setting.

④ Output mode setting range
o = 0 ~ 1 5

Note) As a standard setting, it is set at [0] (all contacts are A contact operation) at the time of shipment from the plant.

This value is expressed by the mode numbers from 0 to 15 and is applied to each contact, #14L, #14, #13 and #12, as follows:

No.	#14L	#14	#13	#12
0	A	A	A	A
1	A	A	A	B
2	A	A	B	A
3	A	A	B	B
4	A	B	A	A
5	A	B	A	B
6	A	B	B	A
7	A	B	B	B
8	B	A	A	A
9	B	A	A	B
1 0	B	A	B	A
1 1	B	A	B	B
1 2	B	B	A	A
1 3	B	B	A	B
1 4	B	B	B	A
1 5	B	B	B	B

A : N/O (on when error occurs) mode
B : N/C (on when normal) mode

8.5 Analogue output zero/span adjustment

When there is a slight deflection in the analogue output range (4 ~ 20mA), this range can be corrected by the adjustment of zero output (4mA) and full output (20mA).

Note 1)

The speed switch unit has been adjusted at the time of shipment to ensure that correct analogue output can be made.
Normally, it is not necessary to change this setting.

Note 2)

Do not attempt to adjust the deflection in the revolution speed caused by the error in the equipment (for instance tachometer) connected to the analogue output by this function of speed switch unit.

⑤ Analogue zero output/full output adjustment setting range
－９．９～０．０～＋９．９％ (no correction when it is 0.0%)

When you enter into this mode, "A" is displayed on the leftmost side of the numerical value display and the present output on the right side.(Continuous lighting)

Under this condition the following keys are accepted.

F	changes to normal mode
S	Pressing and holding this for 3 seconds changes to "Analogue output adjustment"

[Analogue zero output adjustment]

When you enter into this mode, "b" is displayed on the leftmost side of the numerical value display in continuous display and the present zero correction value on the right side by flashing. The analogue output will be fixed at "4.00mA".
The correction value is the % value added to the standard state.

Here, with the following keys, fine adjustment of output value (4mA) will be made.

F	Cancels the setting and changes to "full output adjustment".
▲	Raises the correction value (output rises at the same time)
▼	Lowers the correction value (output declines at the same time)
S	Pressing and holding for 3 seconds sets and saves the present indicated value.

To add, the setting range will be ± 9.9% and no correction value when it is 0.0%.

[Analogue full output adjustment]

When you enter into this mode, "c" is displayed on the leftmost side of the numerical value display in continuous display and the present zero correction value on the right side by flashing. The analogue output will be fixed at "20.00mA".
The correction value is the % value added to the standard state.

Here, with the following keys, fine adjustment of output value (20.00mA) will be made.

F	Cancels the setting and changes to "full output adjustment".
▲	Raises the correction value (output rises at the same time)
▼	Lowers the correction value (output declines at the same time)
S	Pressing and holding for 3 seconds sets and saves the present indicated value.

To add, the setting range will be ± 9.9% and no correction value when it is 0.0%.

9 Installation

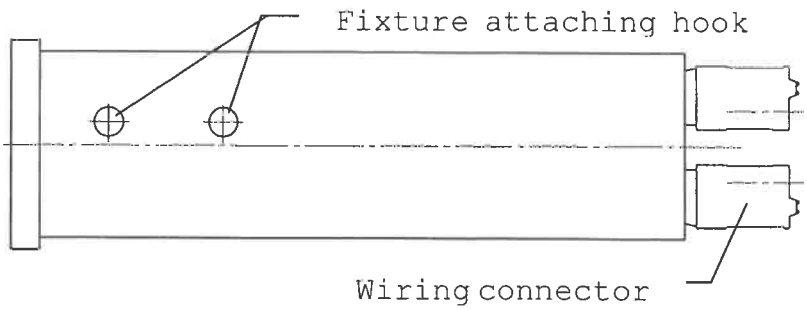


Fig. 8 Speed switch unit–D side face

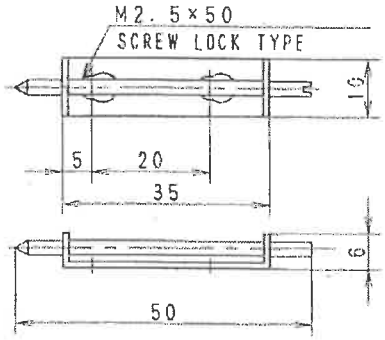


Fig. 9 Fixture (accessory)

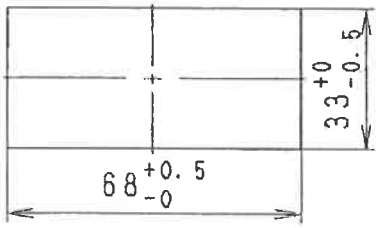


Fig. 10 Panel cut dimension

① Cut the panel of the place to install into in the panel cut size (Fig. 10).
② Insert speed switch unit–D from the front side of the panel cut hole. (Fig. 11)
③ After inserting, hook the attached fixture to the fixture installing hook and screw in metal fitting screw with a slotted screwdriver to securely fasten to the panel side. (Fig. 12)

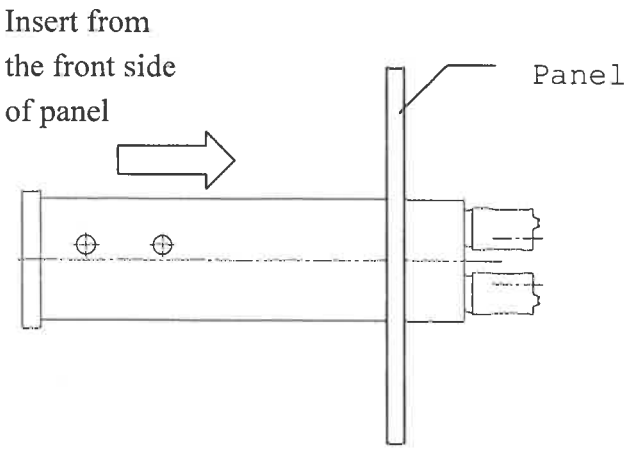


Fig. 11 Installation (insertion)

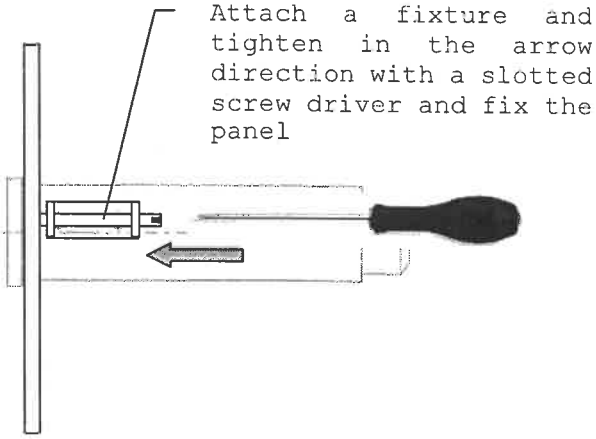


Fig. 12 Installation (fixing)

① When there is vibration on the installation panel face, you can use "bracket: panel installation (NN00262007A)" (option) which can be installed more securely.

② When converting the conventional speed switch unit and speed switch unit 2 to speed switch unit–D during maintenance, etc., you can use the speed switch unit–D available with a compatible case attached with a bracket that fit each unit,

10 Wiring

10.1 Input/output connector terminal configuration

Speed switch unit–D has double decked input/output connector on the backside of the main body. The terminal configuration is described on the nameplate seal on the upper side of the main body in the same way as below.
Please refer to the drawing for correct wiring.

The connector configuration and terminals are as described below.

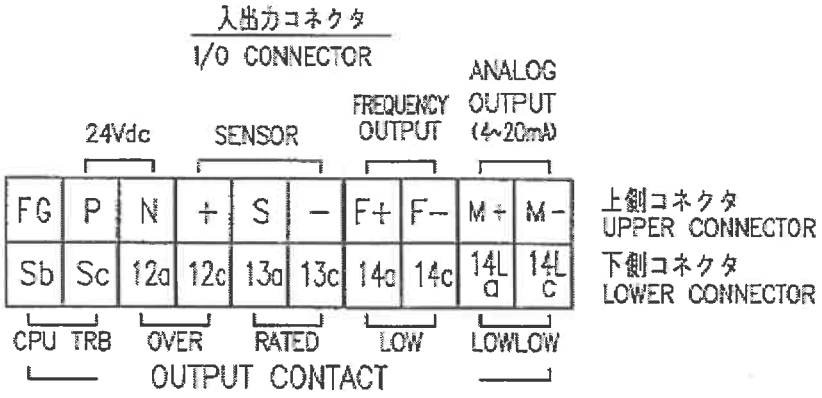


Fig. 13 Terminal configuration

Upper deck terminal			
		FG	No connect.
	24Vdc	P	Connect the power supply. Connect + 24V to [P] and 0V to [N]. Power supply specification, power voltage: DC 24V, allowable voltage range: − 25 ~ + 25%, power consumption: 2.5W
		N	
	SENSOR	+	Connect the sensor output. [+] and [-] are 24vdc power supplies. [S] is for the pulse sensor output which is input in pulse.
		S	
		−	
	FREQUENCY OUTPUT	F+	Outputs the pulse of detected revolution speed. (1Hz per rotation) 0/15V, allowable load resistance is 5KΩ or more and non-insulated from the power supply.
		F-	
	ANALOG OUTPUT	M+	Outputs current at 4 - 20mA by scaling determined by the full scale setting. Load resistance 500Ω or below.
		M-	
Lower deck connector			
OUTPUT CONTACT	CPU TROUBLE	Sb	Turn off (Open) the contact when speed SW is normal.
		Sc	Turn on (Close) the contact when speed SW has cpu trouble or power supply failure.
	OVER	12a	Turns on (off) the contact when it reaches the pre-set speed limit. No-voltage semiconductor contact, contact rating: AC/DC 100V 0.1A, withstand voltage: 1500V
		12c	
	RATED	13a	Turns on (off) the contact when it reaches the pre-set rated speed. No-voltage semiconductor contact, contact rating: AC/DC 100V 0.1A, withstand voltage: 1500V
		13c	
	LOW	14a	Turns on (off) the contact when it reaches the pre-set low speed. No-voltage semiconductor contact, contact rating: AC/DC 100V 0.1A, withstand voltage: 1500V
		14c	
	LOWLOW	14La	Contact attached with a start-up interlock off-delay timer when the engine stops. No-voltage semiconductor contact, contact rating: AC/DC 100V 0.1A, withstand voltage: 1500V
		14Lc	

10.2 Wiring installation to I/O connector

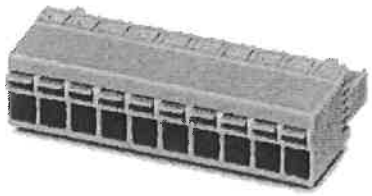


Fig. 14 Connector

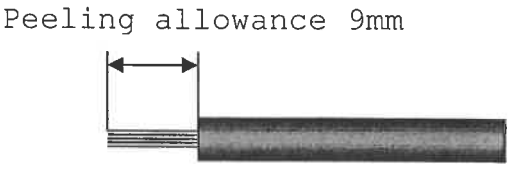


Fig. 15 Peeling allowance of wiring

1) Electric wires to connect to the connector must satisfy the following specification.

Recommended: Single wire: φ 1.0mm (AWG 18)
Twisted wire: 0.75mm^2 (AWG 20)
Strand diameter: Min. φ 0.18mm

Usable electric wire range
Single wire: φ 0.4 ~ 1.0mm (AWGF 26 ~ 18)
Twisted wire: 0.3 ~ 0.75mm^2 (AWG 22 ~ 20)
Strand diameter: Min. φ 0.18mm

2) Secure 9mm for peeling allowance of wiring. Twist it lightly so it will not spread out. Solderless terminal is not necessary.

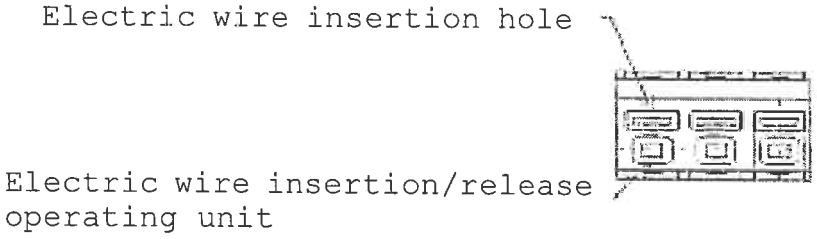


Fig. 16 Connector ele

3) Insert and release wiring to the connector by pressing the wire insertion/release operation unit located at the side of the wiring insertion hole, with a slotted driver or the like. Insert it fully and release the driver from the electric wire insertion/release operation unit. Wiring is grasped inside the connector and fixed securely.

4) Connector can be inserted, only one line to one pin. Should it become necessary to insert multiple wirings into one pin, a separate terminal block will be necessary to bundle the wiring into one before connecting to the wiring.

5) Connector itself can be removed easily by pulling it out. If it is difficult to install wiring, remove the connector to make installation easier. Also at the time of speed switch unit–D replacement, wirings can be removed in a lump at the connector, and there is no need to remove wirings one by one.

6) After completing installation, be sure to check that the wiring is correct and that no short circuiting has occurred due to spreading of wiring. Also pull the wirings to see that they do not come off.

10.3 External wiring

External wiring example of speed switch unit-D is given below.

1) Standard connection

A combination of one unit of pulse sensor and one unit of speed switch.

There are one analogue output system, three contact output systems, frequency (pulse) output and start-up interlock contact output. The contact signal is connected to the engine control panel for control.

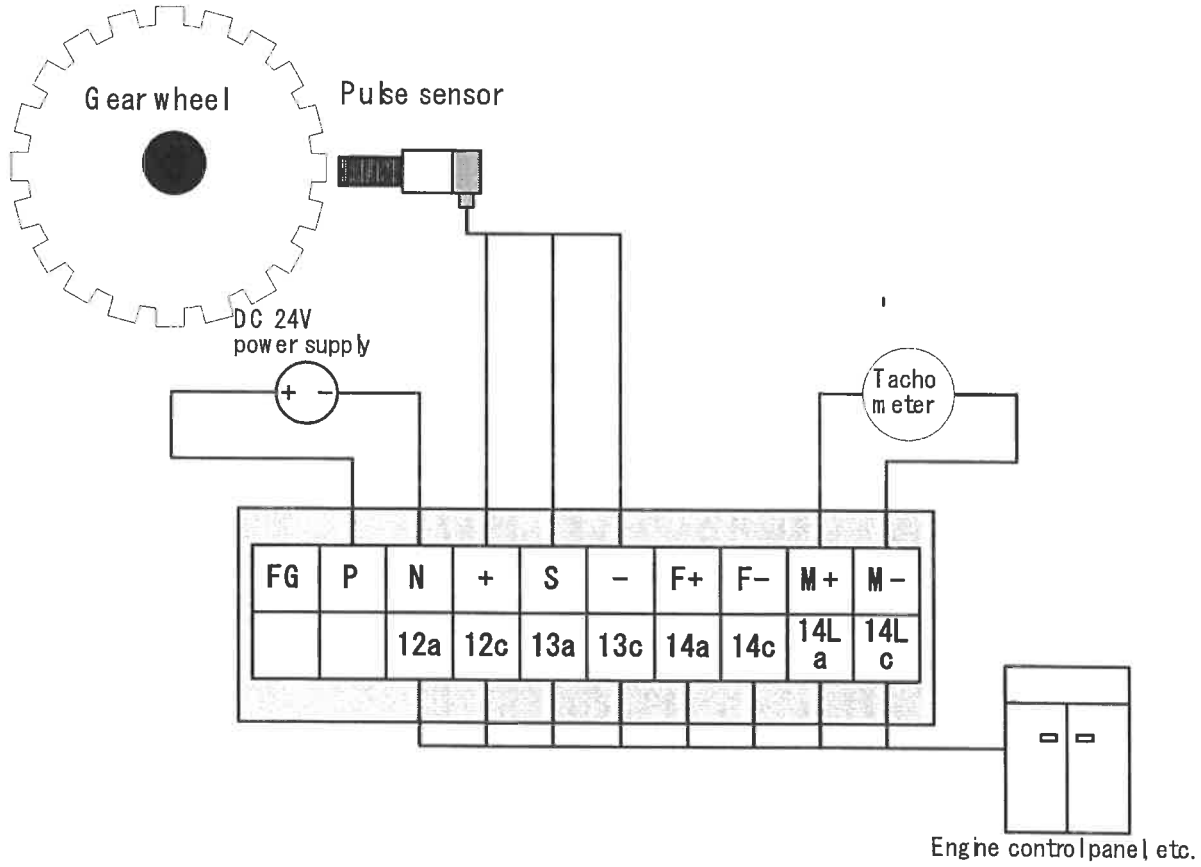


Fig. 17 Standard connection with one unit of pulse sensor and one unit of speed switch

2) Driving two units of speed switch

One unit of pulse sensor can drive up to three units of speed switches.

By parallel driving, the number of outputs can be expanded to 3-fold. (3 systems of analogue output, 9 systems of contact output).

When you perform the parallel drive, connect the wiring each the speed switches as the following figure. (Fig. 18)

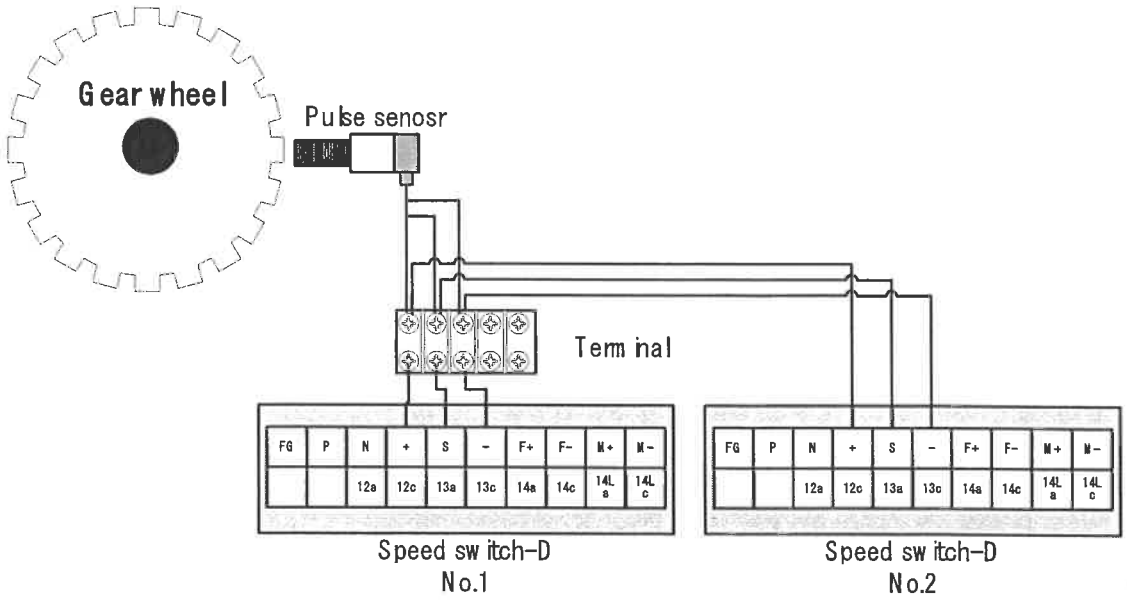


Fig. 18 Parallel driving connection by one unit of pulse sensor and two units of speed switches.

10.4 Connection with the pulse sensor

1) When connecting the pulse sensor NN00006014A (new type)

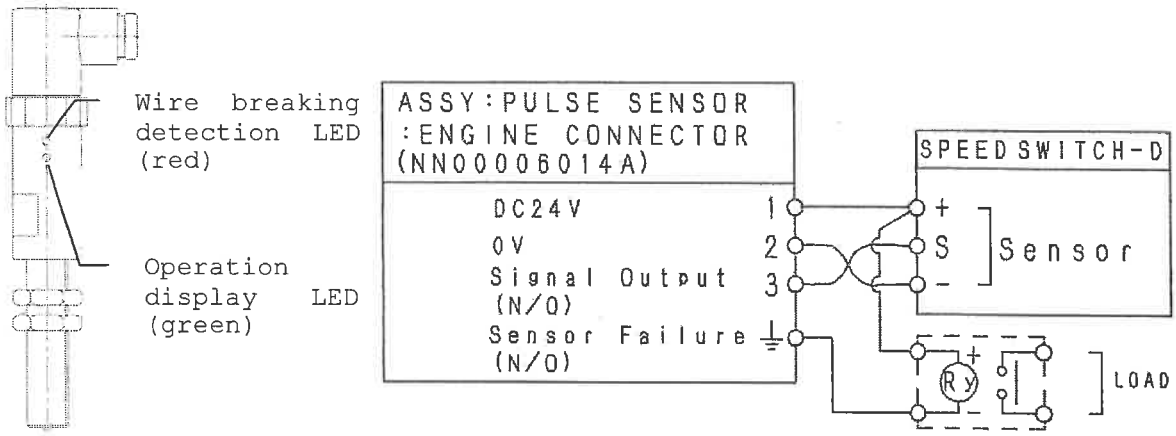


Fig. 19 Pulse sensor NN00006014A

Fig. 20 Pulse sensor speed switch connection diagram

When connecting this pulse sensor to the speed switch unit –D, refer to the connection diagram in Figure 20.

Operation display LED is linked with rotation of gear, and blinks.

The LED put out at the time of the stop.

This pulse sensor is provided with a wire breaking detection function for coil and the red LED at the upper part of pulse sensor turns on when the coil wire breaks, to display the state of wire breakage and at the same time to output it outside. When using the coil breakage detection, prepare a separate relay as in the drawing above and connect it.

For the relay to be used in wire breaking detection, select the one with a coil rated current 30mA or below during operation.(Omron: MY 4-D equivalent)

This pulse sensor is compatible with the (2) pulse sensor (conventional type) in terms of the dimension at the installation section and electrically, it is possible to use it as an alternative model.

2) When connecting pulse sensor 50A, 100, 200 (L51131) (conventional type)

Fig. 21 Pulse sensor (L51131)

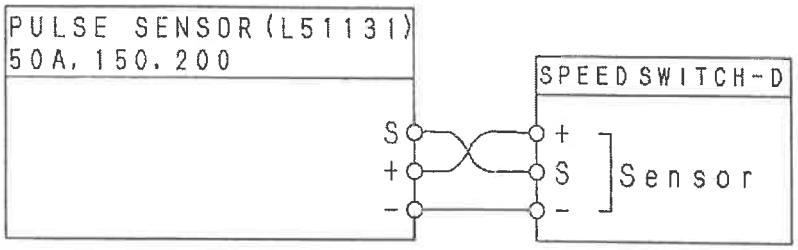


Fig. 22 Pulse sensor (conventional) – speed switch connection diagram

When connecting this pulse sensor to the speed switch unit–D, refer to the connection diagram in Figure 22.

This pulse sensor has no display/output function for coil breaking detection.

Note)

At the wiring work of a pulse sensor, disconnect the power supply to the speed switch.

After the wiring work and check are finished, turn on the power supply to the speed switch.

When do you mistake the wiring that connect reversely to the pulse sensor, the speed switch may broken by the differ polarity.

10.5 Connection and branching of analogue output

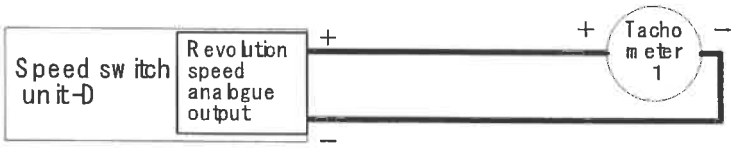


Fig. 23 Analogue output connection (one system)

The speed switch unit –D has one output of analogue output signal (4-20mA signal) that outputs the rotational speed. For the connection diagram, refer to Figure 23. The allowable load resistance of connecting instrument is 500Ω maximum.

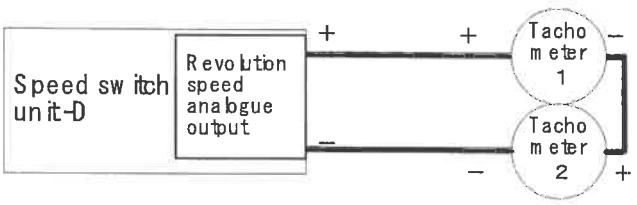


Fig. 24 Analogue output connection (two systems) part 1

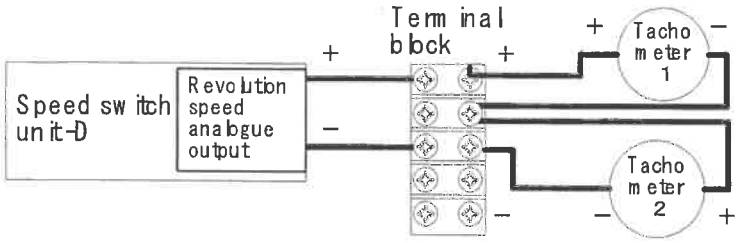


Fig. 25 Analogue output connection (two systems) part 2

When outputs of two systems are necessary for sending signals to the meter and data logger, connect the instrument in serial referring to Figure 24 and 25.

However, in the following cases, use an isolator and divide the analogue signal into two.

① When load resistance is set to over 500 Ω in total.

② When floating of the input signal is not carried out in either instrument.

Note)

① When connecting to an instrument with a load resistance exceeding 500Ω, nonconformity may occur such as discrepancy between the actual revolution speed and the revolution speed indicated on the instrument.

② When negative terminal sides of two units of connecting instruments are grounded, current wraparound occurs through grounding and as a result, the input of equipment on the downstream side bypasses, causing disability in signal input to this instrument.
These incidents can be avoided by inserting an isolator and insulating the signal by direct current.
For the isolator, select the insulator type with two-outputs.
(For example) Maker: M system: W2YV-AAA-R2 equivalent

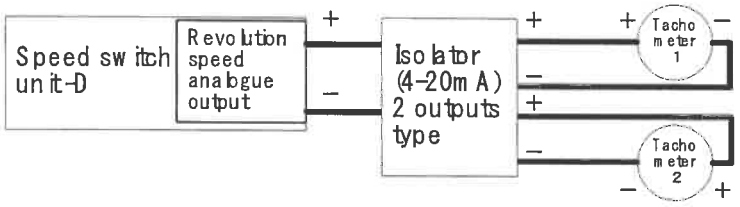


Fig. 26 Analogue output connection
(Isolator used for dividing into 2)

10.6 Precautions in wiring handling

1) Wiring and shield handling

Wiring to the speed switch unit must be installed taking note of the following points.

① For power supply wirings, pair negative and positive wirings and twist them together.
② For the pulse sensor wiring, use three-wire shield wire.
③ For analogue output, use two--wire shield wire.
④ For frequency output, use two-wire shield wire.

⑤ Use the shield wire on the basis of single-point grounding. Make sure to connect grounding.

⑥ Keep wirings connected to the speed switch unit away from the power line and large size relays.

2) Twisted pair wiring

Twisted pair wiring means twisting 2 wirings. This is generally done on electric wires to improve the resistance to noise easily and reduce the effect of electromagnetic induction to the outside.

Fig. 27 Twisted pair wiring

Fig. 28 Normal parallel wiring

3) Grounding of shield wire (single-point grounding)

Grounding of three-core shield wire between pulse sensor speed switch units.

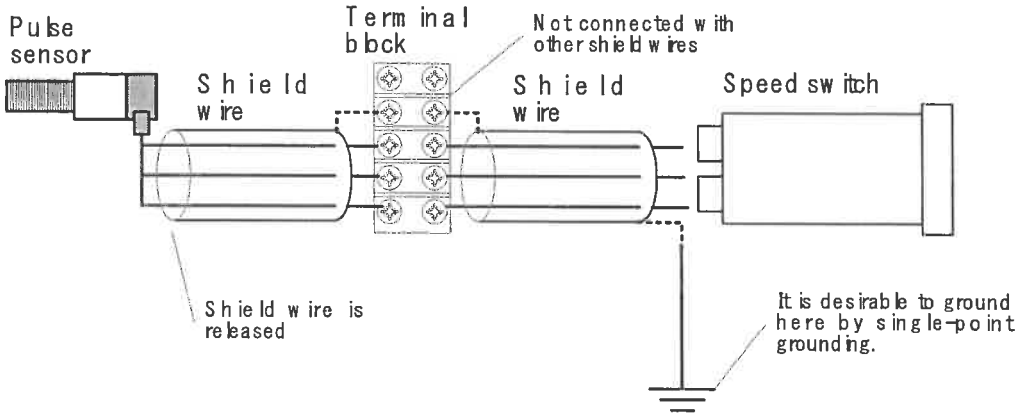


Fig. 29 Shield wire grounding (single-point grounding)

Single-point grounding is an effective connecting method for strengthening the noise resistance of wiring, using the weak signal and pulse, by grounding the shield wire at only single point as shown in the above diagram.

If the sensor side is also connected to the ground, two sides, the sensor side and speed switch side would be grounded. This would cause potential difference between the two points that would provide a good condition for the noise to adhere, it is normally not done.

But if neither grounding is available, there would be no place for noise ingredient to escape to when external noise ingredient adheres to the shield wire, making it possible for the noise to adhere to the signal line, which otherwise should be protected, impeding the effective use of shield wire.

The shield wire must be connected based on single-point grounding.

10.7 The cautions about a power supply

The speed switch is an electronic device.

The power line connected to the speed switch should use the rectified"24VDC" by the DC/DC converter or same quality power supply.

When used with the power supply with high voltage fluctuation and ripple rate exceeding the specification limits, it may not operate normally.

10.8 The cautions about the insulated check which used the insulation resistance meter

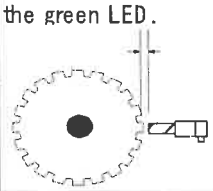
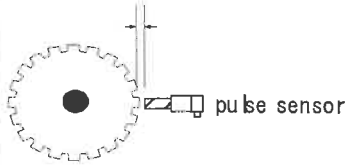
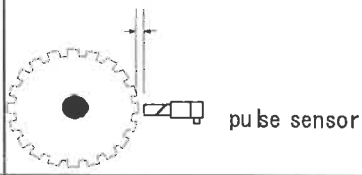
When you check the circuit insulation, must not supply the high voltage on the power supply line of a speed switch by an insulation resistance meter or same equipments.

The inside circuit of the speed switch may damaged by the high voltage.

When you use insulation resistance meter, please disconnect the external wiring connector of a speed switch. It can remove easily.

11 Troubleshooting

11.1 Troubleshooting table

	Phenomenon	Cause	Countermeasure
1	Display/LED does not light	The power is not supplied	Turn on the power supply
		Power supply mis-wired.	Redo wiring work.
		Mis-wired to pulse sensor	Redo wiring work.
2	Contact signal does not output The meter does not work	Wrong set value	Redo the setting work.
		Wrong wiring	Redo wiring work.
		Pulse sensor attaching gap inappropriate	Fix within the range of 1 to 2 turn return from the rotor. It can be checked if it is detected during operation of pulse sensor by the green LED. 
		This unit or pulse sensor failure	Check if either the unit or sensor is defective and replace with anew one if any defect is found.
3	Meter indication swings during constant rotation. Contact output turns off sometimes.	Pulse sensor attaching gap inappropriate	Fix within the range of 1 to 2 turn return from the rotor. It can be checked if it is detected during operation of pulse sensor by the green LED. 
		Looseness in pulse sensor lock nut	Retighten
		Pulse sensor failure	Replace the pulse sensor with a new one.
		Noise influence	Contact the person in charge of engineering
		Looseness, omission of wiring	Redo wiring work.
4	Meter reading differs from the actual measurement Contact output differs from the revolution speed detected by the contact output.	Incorrect speed switch set value	Redo the setting work
		Anabgue adjustment defect (Only for the phenomenon of meter reading being different from the actual measurement)	Make adjustment by anabgue setting (Correctable up to ± 9.9%)
		Pulse sensor attaching gap inappropriate	Fix within the range of 1 to 2 turn return from the rotor. It can be checked if it is detected during operation of pulse sensor by 
		Anabgue output bad resistance exceeds 500Ω	Lower the bad resistance

MCY Filter

INSUTRUCTION MANUAL

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E-MAIL info@ameroid.co.jp URL http://www.ameroid.co.jp/

1. 『Safety Precautions』

1-1. Before use, please use it correctly in accordance with the instructions on this "Safety Precautions" carefully read.

1-2. Matter that was listed in the "WARNING" is, might lead to serious injury or death, serious, such as vessels of damage result depending on the situation. Please be sure to observe because the instruction manual lists the important information about safety.

1-3. The matters described in the column of "CAUTION" might lead to failure or damage to the equipment.

1-4. After you have finished reading, please be sure to store it in a place where people where you intend to use is seen at any time.

1-5. If you did not correctly use, failure of any product, for the accident, the Company does not assume the responsibility therefore please acknowledge.

⚠ WARNING

(1) Scattering of liquid might occur during and at the time of cleaning the use of this equipment.
If the fluid is harmful to the human body, safety glasses, face protection, please wear the appropriate safety equipment for gloves, etc. liquid permeability.
(2) You can ride on top of the device, please do not place any object. There is a risk of deformation or damage.
(3) Do not use in applications other than those listed in the product specifications.
(4) Dangerous liquid (strong acid or alkali, toxic, etc.) Please do not use.
(5) Please keep from discharging all the liquid in the case for a long time is not used.
If you do not use a long time without the discharge of the liquid, it may be a liquid (by freezing or heating) expansion by the surrounding environment, leakage of corruption and liquid products occurs.
(6) Please do not modify this product.
(7) Transportation, maintenance, installation, during the inspection, etc., prior to turn off the power source of the device, please do not operate in accidental. In addition, it should be performed by those who are expert knowledge.
(8) Please do not use compressed air to, drain in the filter device.
(9) If you have any questions In use of this product, please contact your dealer or our sales office of purchase.

⚠ CAUTION

(1) Doorway hose, piping connection, please conducted by those who are expert knowledge.
(2) Components wear by operating time, there is damage.
Please conduct regular maintenance and inspection.
(3) Repair, disassembly, inspection, etc., please go people always do some specialized knowledge.

1. Mounting

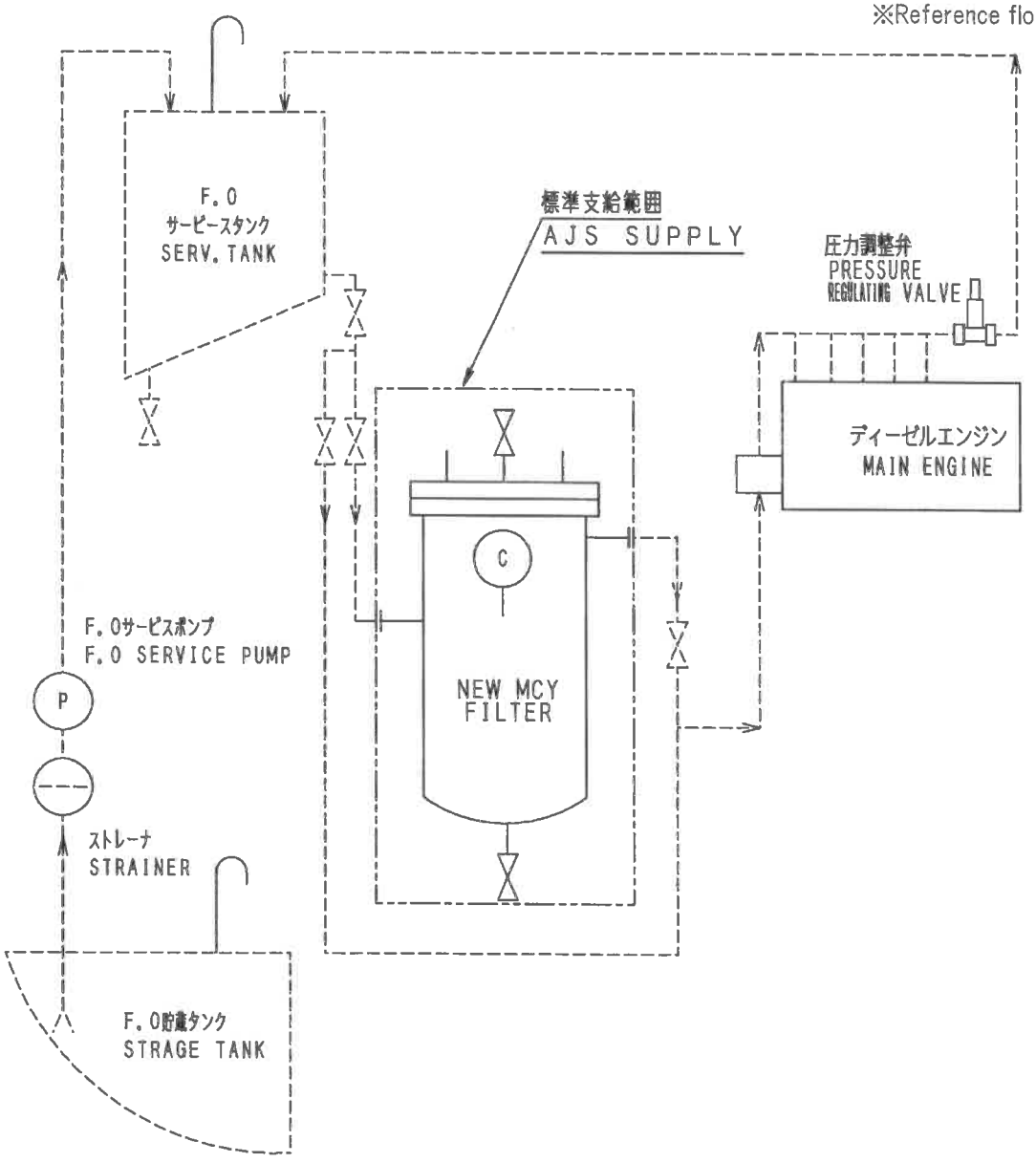
1) This unit Please be installed in a flat location.
   Also, please provide a margin in the upper space for element replacement.

2) Filter in the fuel oil pipe system between the service tank and the engine, please be attached to the lower than the service tank position.

3) Piping steel pipe, use the copper tube, the extreme rise in the shortest distance, please take into account so as not to fall Piping.

2. Piping

※Reference flow diagram



1) Please provided a filter bypass piping as able to respond at the time of unexpected failure of the machine.

2) Inlet and outlet of the machine, please provide a valve to bypass piping.

3. Oil passing, driving

3-1. After installation, the first time through oil.

1) If there is any mistake in the path of the pipes, forget tightening such as a flange connections, please be inspected if there is any piping or the like in the forgotten Remove dust protection tape.
2) Please be in the filter inlet valve "open".
3) Filter outlet valve, please to the drain valve "closed".
4) Open the valve taken out of service tank, please supply the oil to the filter.
5) Please be in the filter air vent valve "open". Please execute the air vent until the oil is blown. Please to "closed" after the balloon.
6) The filter outlet valve in the "open", please supply the oil to the engine.
7) Please be inspected or other abnormal leaks, etc..

3-2. In-use maintenance

1) While before the engine start-up and engine operation, sludge from once filter bottom of the drain valve to 1-2 times, please drain the water and the like.
   Although it is not a problem in the discharge of about 200cc as a guide, if the water is further Dell, please be carried to the discharge of the water is finished.
2) Even when supplemented with fuel oil to the ship, please discharged from the filter bottom of the drain valve with the drainage of the service tank bottom.
3) If the engine is in operation by might drain it does not come out.
   In that case, please draining to "slightly open" the air vent valve to the bypass valve "half open" is.
   After drainage, the bypass valve is set to "closed" to confirm that the oil was blowing from the air vent valve please to "closed".

4. Usage of maintenance, inspection

1) While before the engine start-up and engine operation, sludge from the filter bottom of the drain valve once in 1 to 2, please drain the water and the like.
2) Open the drain valve of the service tank, sludge once in the tank to 1 to 2, please drain the water and the like.
3) Record the indicated value of the filter coupled meter, please make sure that it is within -0.04MPa.

5. Exchange of elements, cleaning

Indicated value of the coupling meter is taken out of the element After shows more than -0.04MPa, please do the replacement or cleaning.
(MS type washing type, KF type is exchange type)
Please element replacement is carried out in the following procedure.
The filter inlet and outlet valve "closed", it can be supplied to the engine without passing through the oil filter by to the bypass valve "open".

1) To perform the engine is stopped, close the service tank source valve, a filter inlet valve "open", the filter outlet valve, please carry out 2) or later after the "closed".
If you do not want to stop the engine, the filter inlet and outlet valve "closed", and from then on the bypass valve "open" Please implement 2) or later.

2) Filter the airvent valve "open", please the oil in the vessel in the filter drain valve "open" to all emissions.

3) Please remove Loosen the body cover mounting bolts and nuts.

4) Please remove Loosen the element retainer mounting bolts.

5) Please remove the element pack using the eyebolt.
Please be careful so as not to scratch the element in the casing or the like.

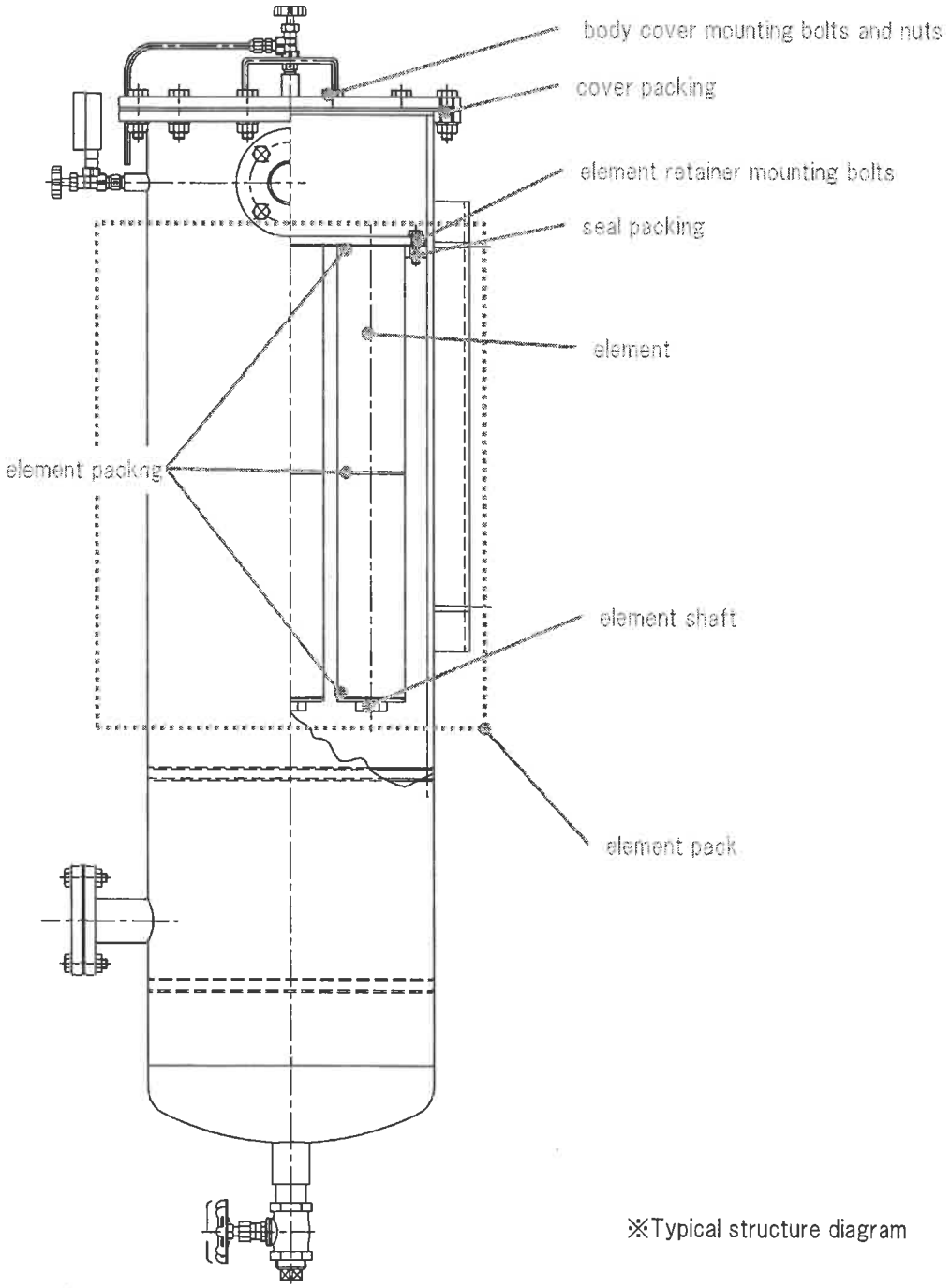
6) Please remove the element Loosen the element shaft.

7) In the case of exchange-type elements (KF type)
Element is the type of 1-month top, 2 months of the bottom, element 2 months type of top, middle, there is the element packing each in the bottom so please be replaced at the same time.
In the case of-cleaning element (MS type)
Element is the type of 1-month top, 2 months of the bottom, element 2 months type of top, middle, there is the element packing each in the bottom so please be replaced at the same time.
Element is attached to the inside of the white kerosene, please remove the sludge and the like to wash attached with a clean sponge or the like.

8) Is loaded elements, please assemble the element pack.

9) Seal packing, please replace the cover packing.

10) 3) Follow steps to 5) in reverse, please restoration.



※Typical structure diagram

6. Specification

Model		MCY-800MS	MCY-2000MS	MCY-2000KF	MCY-4000MS II
Use oil		A Fuel Oil			
Operating temperature		Normal temperature			
Filtration accuracy		10 μ (nominal)			
Water separation efficiency		More than 90%			
Processing flow rate (maximum)		300 ℓ/h	800 ℓ/h		2000 ℓ/h
Element type		Washing type	Washing type	Exchange type	Washing type
Element material		Micro screen	Micro screen	KaneFeel	Micro screen
Element quantity		1	3	6	12
Vessel in the amount of oil		7 ℓ	25 ℓ	30 ℓ	80 ℓ
Port size		JIS 5K-32A	JIS 5K-32A		JIS 5K-40A
Size		280W×230D×760H	360W×340D×830H (1100H/KF)		400W×390D×1420H
Weight	Dry	23kg	55kg (60kg/KF)		130kg
	Full	30kg	80kg (90kg/KF)		210kg
Accessories		Compound gauge × 1 Air vent valve × 1 Drain valve × 1			

～ M e m o ～

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INSTRUCTION MANUAL

FOR

MOTOR-DRIVEN CENTRIFUGAL PUMPS

NANIWA PUMP MFG.CO.,LTD.

1-4A

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SAFETY INSTRUCTIONS

CAUTIONS FOR YOUR SAFETY

Before using this pump, read the INSTRUCTION MANUAL (S). Follow WARNING LABEL(S), INSTRUCTION (S) and CAUTION PLATE (S) in order to use it correctly. It is also highly recommended to ALWAYS KEEP the INSTRUCTION MANUAL (S) at the SAME PLACE for easy access.
TRANSPORTATION, INSTALLATION, PIPING, WIRING, OPERATION INSPECTION and MAINTENANCE WORK must be done by ONLY a LICENSED and / or AUTHORIZED PERSON(S) who has enough knowledge on health and safety rules and regulations as well as on his or her profession.
In any respect, we will NOT GUARANTEE any DEATH, INJURIES, DAMAGES AND LOSSES which are result in modification without our written authorization or using and assembling unauthorized parts. or imitation parts. Genuine parts must be used or assembled for replacement.
When DISPOSING a pump, any accessories, used parts and oil, they should be treated as a general INDUSTRIAL WASTE.
WARNING LABEL(S) and SIGN in the instruction manual(S) are classified into WARNING, CAUTION and NOTICE as described below.

WARNING : indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION : indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury

NOTICE : indicates a potentially hazardous situation which, if not avoided, may result in damaging or defecting a product. It will be only used for protecting the property, but not for personal safety.
Footnote : ANSI Z 535

WARNING

TRANSPORTATION and INSTALLATION

* TRANSPORTATION work must be carried by only a LICENSED or AUTHORIZED PERSON(S) who has enough knowledge on his or her profession : special attention and caution are required when fitting a HANGING WIRE in relation to its weight and gravity center.
  DROPPING or FALLING → DEATH or SERIOUS INJURY.

OPERATION and MANIPULATION

* NEVER ALLOW an UNAUTHORIZED PERSON to operate the pump. DO NOT TOUCH or contact to ROTATING parts or portion(s).
  ROTATING → ROLLED IN, BIT, PINCHED and SPILLED (contacted things)

CAUTION

OPERATION and MANIPULATION

* DO NOT TOUCH or contact to SEALING part or HOT parts while the pump is IN OPERATION.
  HOT → BURNED and INJURED

INSPECTION and MAINTENANCE

* When OVERHAULING the pump, carefully handle the HEAVY weighted parts ; especially, fitting a HANGING WIRE.
  DROPPING or FALLING → INJURY.
* Before STARTING a MAINTENANCE work, clearly SING IN MAINTENANCE WORK and CUT ELECTRIC SUPPLY.
  ELECTRIC SHOCK → DEATH
  UNINTENDED ROTATION → ROLLED IN, BIT and PINCHED.

NOTICE

OPERATION and MANIPULATION

* Without a specific purpose, do not manipulate the valves and cocks attached to or supplied for the pump.
  WRONG MANIPULATION → DAMAGING or DEFECTING the pump.

1. Purpose

This manual has been prepared for the standard motor-driven centrifugal pumps used on commercial vessels for the convenience of persons concerned with handling pumps who belong to shipbuilders and shipowners.

2. Scope of Application

This manual is applicable to A.C. motor-driven centrifugal pumps in general such as cooling water pumps, sanitary pumps, fresh water pumps, circulating pumps, fire & bilge pumps. (hereinafter called pump.)

3. Transport

In transport it is necessary to proceed as follows.

a. When lifting, pay attention to the weight of the units lifted and the method of using wires.
b. Be careful to avoid damage to accessories such as piping, valves, cocks, etc.

4. Prevention of Vibration

In order to minimize the vibration of pump and piping, pay sufficient attention to the reinforcement of pump foundation and piping supports.

5. Installation

In installation, pay attention as follows

a. Pump foundation must be as rigid and strong as practical.
b. Considering operation and overhaul, give as ample space as possible.
c. Installation face must be flat.

6. Piping

Sufficient attention must be given to piping since it has a great effect on pump performance.

a. Suction pipe must be as short as possible.
b. Suction pipe must be free from air pocket or air invasion.
c. Valves on the suction side must be guarded against air invasion through the gland.
d. Flanges connecting to the suction and discharge nozzles should be placed correctly in order to avoid disturbance of pump alignment when flange bolts are unduly tightened.
e. Precautions must be taken so that expansion and contraction due to temperature and weight of piping and valves may not abnormally affect the pump.
f. Interior of the piping must be as clean as possible.

7. Connection

On completing installation, the alignment must be checked and care must be taken so that no misalignment will occur after connecting with piping. Since alignment is duly carried out in the factory, unreasonable readjustment should be avoided.
Even though misalignment occurs, readjustment must be made carefully by loosening both of the bolts on the suction and discharge flanges and the foundation.

Smooth movement should be ensured by rotating by hand
Case where alignment is checked with motor and pump shaft coupled on the surface plate, permissible amount of misalignment must comply with the value in Table 1.

Table.1 Permissible amount of misalignment

OUT PUT (kW)	Alignment for Motor and Impeller (mm)	Alignment for Impeller and Casing (mm)
37 and less	0.03 and less	0.03 and less
More Than 40	0.05 and less	0.05 and lese

(Reference)
Take either of the two measurement methods of alignment, which are described below.

a) After coupling motor and pump shaft, measure the value of the shaft end periphery, a dial indicator is fixed, as shown in Fig 1 (a) , on the motor flange, and the pump shaft is given one complete turning, and in this case the half value of reading is taken as the measurement value, which must comply with the value in Table 1.

b) After coupling motor and pump, measure the value of casing ring inner side, a dial indicator is fixed, as shown in fig 1 (b), on the shaft, and the pump shaft is given one complete turning, and in this case the half value of reading is taken as the measurement value, which must comply with the value in Table 1.

(a) (b)

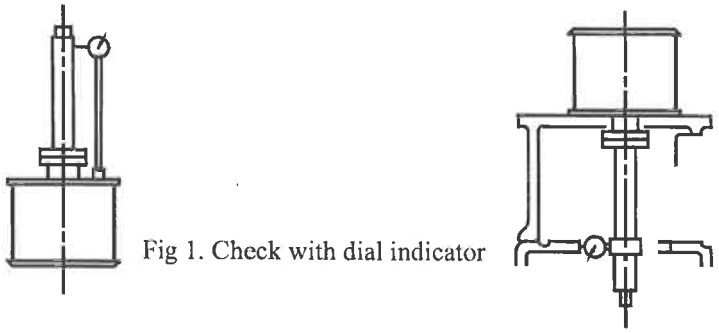

Fig 1. Check with dial indicator

Note : A thickness gauge should be adopted for alignment only at the times of repair, intermediate inspection and periodical inspection. It is preferable to use a dial indicator for checkup after initial installation or piping arrangement.

8. Operation

Operation after installation or reassembly must be carried out in the following order.

8-1 Preparation before Operation

a. Open the air vent valve on top of the casing and fill the casing with water until it comes out of the vent.
b. In the case of reservoir-type self-priming pumps, fill the casing with water after removing the plug on top of the casing. However, after the second starting there is no need of priming if water remains in the casing.
c. In the case of self-priming pump provided with a vacuum pump, fill the replenishing water tank.
d. Open fully the valve on the suction side and close completely that on the discharge side.
e. Check if the bearings are lubricated.
f. Turn the pump shaft by hand to see whether it turns smoothly.

8-2 Operation

a. Start the motor, but at first repeat "ON" and "OFF" operations once or twice and enter into operation on confirming that there is no abnormal condition.
b. When pump revolutions and pressure have risen, open gradually the valve on the discharge side until the pressure drops down to the specified pressure.
c. Avoid absolutely dry operation, especially by careful when a mechanical seal is used. Even though a vacuum pump is provided, stop the pump if priming is not effected within 5 minutes.

8-3 Stopping

a. Close the valve on the discharge side.
b. Stop the motor.
c. When the pump stops suddenly during operation, switch off first of all and then close the valve on the discharge side.

9. Maintenance

In order to maintain an efficient operation over a long period, the utmost care must be taken.

9-1 Cautions during Operation

a. Be careful to operate as near the design point as practical.
b. Check the vibration and noise. If abnormal condition is surveyed, stop the pump immediately. Particularly, for the pump provided with a mechanical seal, protect against vibration in view of the service life of the mechanical seal.
c. Pay attention to bearing temperature rise. If the bearing housing can be felt by hand safely, there is no fear, otherwise, measure the temperature with a thermometer. Keep the permissible bearing temperature at "ambient temperature plus 40℃" or "liquid temperature plus 20℃", but in any case keep it below 75℃.
d. Filling too much grease will cause overheating, so when replacing grease, pour it with the drain plug removed during operation so that the old grease can be drawn out.

e. Be sure to keep small amount of continuous leakage through the gland, which must be in the from of drops. Too hard tightening of the packing will cause premature wear of the shaft sleeve and overheating.
f. When a mechanical seal is used, there is almost no leakage, so if continuous leakage is observed check the seal.
g. Never throttle the valve on the suction side. Adjust the flow by means of the valve on the discharge side.
h. Never operate for more than 10 minutes with the valve closed on the discharge side, otherwise a temperature rise of the liquid in the casing will cause seizure.
i. See whether the standby pump is reversing due to the leakage from its non return valve.
j. In the case of a pump with automatic starting and stopping device such as hydrophore system, check the pressure at the start and stop of pump as well as the interval of operation, and if the interval is too short, regulate the air amount, water level, etc.
k. In the case of automatic operation under central control system, take care of the indication of gauge on the control panel board.

9-2 Cautions during Standstill

a. Keep the valves closed on the discharge side.
b. Keep the valves closed on the suction side at standstill over a long period.
c. In cold regions, keep the pumps at standstill and standby completely drained.

10. Inspection

10-1 Periodical Inspection

Excepting the case of abnormal conditions, it is desirable to carry out inspection periodically in the following way, but since it varies according to the method of mounting, place of installation, liquid handled, etc. It is preferable that the operator should make a final plan of inspection in accordance with conditions.

Item	Action to be taken	Period		
		Every 3 months	Every 12 months or 8,000 hrs.	Every 4 years or 20,000 hrs.
Ball bearing	Pour oil	○		
	Check up inner and outer races as well balls for exfoliation, and if its trace is found, renew it.			○
Submerged line bearing	Check up sliding surfaces for conditions of contact and flaw. Measure inside diameter and take necessary steps according to manufacturer's standard.		○	
Gland packing	Check up conditions of fitting and amount of leakage. Renew it if leakage is too much.	○		

Item	Action to be taken	Period		
		Every 3 months	Every 12 months or 8,000 hrs.	Every 4 years or 20,000 hrs.
Mechanical seal	Check up conditions of mating faces.		○	
Shaft sleeve	Check up contact points and flaw in portions facing packing and bearing. Measure outside diameter and take necessary steps according to manufacturer's standard.		○	
Shaft	Check up conditions of contact, flaw in sliding faces, and bend of shaft.			○
"O" ring	Check up surface flaw and condition of Deterioration. Renew it if deformed.		○	
Casing wearing ring	Check up conditions of contact, wear and corrosion in sliding faces. Measure inside diameter and take necessary steps according to manufacture's standard.		Ⓢ	Ⓕ
Impeller	Check up conditions of wear and corrosion of all parts. Measure sliding face in contact with casing wearing ring and take necessary steps according to manufacture's standard.		Ⓢ	Ⓕ
Shaft coupling bolts and rubber rings	Check up condition of wear and renew them if abnormal points are found.		○	

Note : 1: The hours quoted refer to the operating hours.
2: Ⓢ refer to sea water pump and Ⓕ a fresh water pump.
3: When a sea water pump casing is made of cast iron, its interior must be painted at the time of overhaul.

10-2 Overhaul

When overhauling the pump, attention must be paid as follows.

a. Understand the construction well by referring to the assembly drawing and make no mistake in the order of overhaul.
b. When separating fit and flange faces, use jack bolts and wooden hammers, and never apply force with chisels or drivers.
c. When removing the rotating element, take care to avoid flaw on sliding faces and machined surfaces. Particularly, never damage the mechanical seal mating faces.
d. When removing rotating parts from the shaft, draw off each one carefully after removing the locking device.
e. Handle the longsized-parts such as shaft carefully so that it may not bend.
f. Handle the parts carefully, by arranging them on sheets of paper or cloth in good order.
g. At overhauling, put suitable match marks as many as possible to avoid mistakes when reassembling.

10-3 Assembly

Carry out assembly, by reversing the order of disassembly and paying attention as follows.

a. Remove dust and stain from each part by washing it thoroughly with kerosene. Repair it if flaw is found.
b. Fit the locking device perfectly in each rotating part if necessary.
c. When fitting the parts with match marks, be sure no follow them.
d. Install the mechanical seal carefully and confirm its movement by hand after installation.
e. Insert each packing ring in good order softly one by one from the bottom, staggering each joint by 90° or 180° .
f. Check up alignment as mentioned in Page. 7.
g. Turn the shaft by hand to see whether it turns smoothly.

11. Troubles and Remedies

Should troubles occur, their causes must be traced and necessary remedies must be carried out. The following, for instance, can be conceivable as troubles.

Troubles	Causes	Remedies
Pump does not start.	• Motor is in trouble. • Pump seizes. • No power source. • Wiring is broken, or relay, etc. are in trouble. • Lack in voltage.	• Repair motor. • Repair pump. • Check up electric system. • Repair • Check up electric system
Pump starts, but does not discharge water.	• Pump is not primed or in-sufficiently primed. • Valves are not open. • Valves will not open. • Pump is sucking air. • Suction pipe or strainer is clogged. • Impeller is clogged. • Vacuum pump is in trouble. • Motor rotating direction is wrong.	• Prime pump once more. • Open valves. • Repair valves. • Check up suction system. • Clean suction pipe or strainer. • Clean impeller. • Repair vacuum pump. • Change wiring.
Pump starts, but specified discharge pressure is not reached.	• Pump is sucking air. • Pump speed are too low. • Discharge pressure is too high. • Suction pipe or strainer is clogged. • Impeller is clogged. • Casing wearing ring is worn. • Cavitation exists. • Gauges are wrong. • Impeller is worn.	• Check up suction system. • Correct electric source. • Check up discharge piping. • Clean suction pipe or strainer. • Clean impeller. • Renew casing wearing ring. • Throttle discharge valve. • Replace gauges by new ones. • Replace impeller by new one.

Troubles	Causes	Remedies
Pump starts and discharges water, but soon ceases to discharge water.	• Priming is insufficient. • Air pockets exist in suction line. • Pump is sucking air. • Pump is sucking air through stuffing box.	• Prime pump. • Correct piping. • Check up suction system. • Check up sealing pipe • Adjust mechanical seal. • Adjust packing.
Pump starts, but motor gets overloaded.	• Abnormal metal contact exists in the rotating parts. • Connection is wrong. • Packing is too tightened. • Pump shaft is bent. • Pump casing is deformed. • Pump discharges too much. • Pump bearing are wrong.	• Remove contact. • Check up alignment. • Loosen gland or otherwise renew packing • Renew shaft. • Check up foundation and piping. • Throttle discharge valve. • Renew pump bearings.
Pump starts, but bearings get over-heated.	• Lubricant is in shortage. • Lubricant is too much. • Lubricant is unsuitable or incorrect. • Connection is wrong. • Bearings are wrong. • Shaft is bent. • Thrust has increased. • Impeller balance is wrong. • Bearing assembly is wrong.	• Supply lubricant. • Draw out lubricant. • Replace lubricant by specified one. • Check up alignment. • Renew bearings. • Renew shaft. • Check up whether excessive wear exists or impeller balance holes are clogged, and reassemble if necessary. • Repair or renew impeller. • Readjust bearings.
Pump starts, but vibration takes place.	• Impeller balance is wrong. • Connecting is wrong. • Shaft is bent. • Installation is wrong. • Foundation is weak. • Other vibration is transmitted. • Cavitation exists. • Bearing are worn.	• Repair or renew impeller. • Check up alignment. • Renew shaft. • Correct installation condition. • Reinforce foundation. • Reinforce piping. • Throttle valve on discharge side. • Renew ball bearing or submerged bearing.

In case troubles cannot be remedied in spite of the above countermeasures, the causes may be in the design conditions of the pump, so it is preferable to consult the shipyard or manufacturer.

INSTRUCTION MANUAL

FOR

MOTOR-DRIVEN GEAR PUMPS

naniwa pump
Manufacturing Co.,Ltd.

NANIWA PUMP MFG.CO.,LTD.

No. 3-1A

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  Footnote : ANSI Z 535

WARNING

TRANSPORTATION and INSTALLATION

* TRANSPORTATION work must be carried by only a LICENSED or AUTHORIZED PERSON(S) who has enough knowledge on his or her profession : special attention and caution are required when fitting a HANGING WIRE in relation to its weight and gravity center.
  DROPPING or FALLING → DEATH or SERIOUS INJURY.

OPERATION and MANIPULATION

* NEVER ALLOW an UNAUTHORIZED PERSON to operate the pump. DO NOT TOUCH or contact to ROTATING parts or portion(s).
  ROTATING → ROLLED IN, BIT, PINCHED and SPILLED (contacted things)

CAUTION

OPERATION and MANIPULATION

* DO NOT TOUCH or contact to SEALING part or HOT parts while the pump is IN OPERATION.
  HOT → BURNED and INJURED

INSPECTION and MAINTENANCE

* When OVERHAULING the pump, carefully handle the HEAVY weighted parts ; especially, fitting a HANGING WIRE.
  DROPPING or FALLING → INJURY.
* Before STARTING a MAINTENANCE work, clearly SING IN MAINTENANCE WORK and CUT ELECTRIC SUPPLY.
  ELECTRIC SHOCK → DEATH
  UNINTENDED ROTATION → ROLLED IN, BIT and PINCHED.

NOTICE

OPERATION and MANIPULATION

* Without a specific purpose, do not manipulate the valves and cocks attached to or supplied for the pump.
  WRONG MANIPULATION → DAMAGING or DEFECTING the pump.

1. Purpose

This manual has been prepared for the standard motor-driven gear pumps used on commercial vessels for the convenience of persons concerned with handling pumps who belong to shipbuilders and shipowners.

2. Scope of Application

This manual is applicable to A. C. motor-driven ordinary pumps such as lubricating oil, fuel oil and fuel valve cooling oil pumps. (hereinafter called pumps.)

3. Outline of Construction

This pump has in its casing one set of gear and shaft supported with bearings and liquid is contained between the teeth. With the rotation of the gear, liquid is continuously discharged around the casing inside. (See the reference figure)
The bearings located inside the seal section are self-lubricated with the liquid.
The lubrication liquid is supplied to oil passage through oil openings provided on the discharge side and is returned to the suction side after lubrication of the bearings.
The safety valve protects the pump or the motor by opening the port and by-passing the liquid to the suction side when the discharge pressure rises abnormally higher than the set pressure.

Reference : The procedure for controlling the set pressure is that after the adjust screw of the safety valve is fully tightened with discharge suction valves fully open, these valves are gradually closed to set the discharge pressure at the set pressure and the suction pressure at -380mmHg. Then the adjust screw is slowly loosed and locked at a point where the discharge pressure or the suction pressure fluctuates very slowly. When control is over pump operation is started with discharge suction valves open.

4. Transport

In transport it is necessary to proceed as follows.

a. When lifting, pay attention to safety in respect to the weight of the units lifted and the method of using wires.
b. Be careful to avoid damage to accessories such as piping, valves, cocks, etc.

5. Prevention of Vibration

In order to minimize the vibration of pump and piping, pay sufficient attention to the reinforcement of pump foundation and piping supports.

6. Installation

In installation, pay attention as follows:

a. Pump foundation must be as rigid and strong as practical.
b. Considering operation and overhaul, give as ample space as possible.
c. Installation face must be flat.
d. Location of installation must be determined in consideration of pump suction head.

7. Piping

Sufficient attention must be given to piping since it has a great effect on pump performance.

a. Suction pipe must be as short as possible, and bends must be as few as possible to minimize friction loss.
b. Suction pipe must be free from air pocket or air invasion.
c. Valves on the suction side must be guarded against air invasion through the gland.
d. Filter must be provided on the suction side for pump protection.
e. Flanges connecting to the suction and discharge nozzles should be placed correctly in order to avoid disturbance of pump alignment when flange bolts are unduly tightened.
f. Precautions must be taken so that expansion and contraction due to temperature and weight of piping and valves may not abnormally affect the pump. And at the same time, precautions must be taken to avoid misalignment of the flange caused by pipe vibration.
g. Interior of the piping must be as clean as possible.

8. Connection

8-1 Alignment

On completing installation, the alignment must be checked and care must be taken so that no misalignment will occur after connecting with piping. Since alignment is duly carried out in the factory, unreasonable readjustment should be avoided.
Even though misalignment occurs, readjustment must be made carefully by loosening both of the bolts on the suction and discharge flanges and the foundation. After the alignment is made a knock pin shall be inserted.

For Horizontal pump, it is desirable that adjustment be made by inserting a liner between common bedplate and foundation, but if necessary, shims can be placed between common bedplate and motor base.

8-2. Checking of alignment

a. Untighten the bolts of the shaft coupling between pump and motor and check the alignment at four points, 90° apart, on the coupling periphery.
b. For comparing the distance between the faces, fix a dial indicator on the shaft coupling of the motor side as shown in Fig.1 (a) and give the motor coupling one complete rotation by hand. Then half of the variation in readings is taken as the value of measurement, which must comply with the value in Table 1.

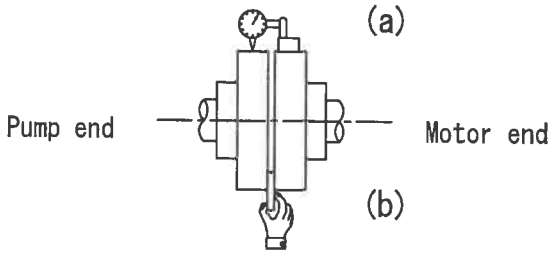


Fig. 1

Table .1

No. of motor frame	Coupling periphery (mm)
M180L(ML5-180L) and less	0.05 and less
M200M(ML5-200M) and more	0.08 and less

Table .2

No. of motor frame	Coupling end face (mm)
M132M(ML5-132M) and less	0.10 and less
M160M(ML5-160M) and more	0.18 and less

c. For measuring the clearance between coupling faces, insert a thickness gauge at four points of equal interval as shown in Fig.1 (b). Then the value of measurement, which must comply with the value in Table 2.
d. Ensure the rotating direction of the motor.
e. Tighten the coupling bolts.
f. Rotate the pump by hand to see whether it rotates smoothly.

Note : A thickness gauge should be adopted for alignment only at the times of repair, intermediate inspection and periodical inspection. It is preferable to use a dial indicator for checkup after initial installation or piping arrangement.

【 FOR SURE FLEX COUPLING 】

* Use straight edge or a caliper (pass) as shown in Fig.1 or Fig.2.
* Adjust alignment to comply with the values shown in table 1.
* Confirm motor rotating direction and that pump turns smoothly by hands.

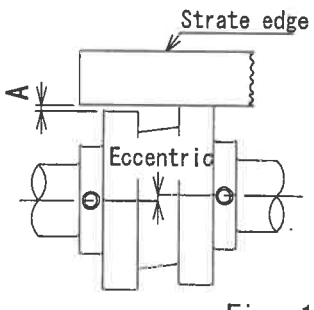


Fig. 1.

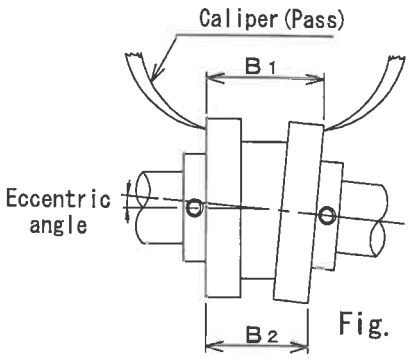


Fig. 2.

COUPLING SIZE	MOTOR FRAME NO.	ERRORS OF ECCENTRICITY A mm	ERRORS OF ANGULARITY B1-B2 mm
4S	80 90L	0.25	0.55
5S	100L 112M	0.38	0.7
6S	132S	0.38	0.9
7S	132M 160M	0.51	1.0

9. Operation

Operation after installation or reassemble must be carried out in the following order.

9-1 Preparation before Operation

a. Fully open the valves of both suction and discharge sides.
b. Open the air vent valve (or cock) on the casing top to fill the casing with liquid.
c. If the pump is located above the suction level, pull the plug out of the casing top and fill the casing with oil while rotating the pump by hand in order to fully spread oil over the teeth and the bearings. Keep the air vent plug (or cock) on the discharge side open.
d. See that the cock (or plug) of the pressure gauge and compound gauge is open.
e. Rotate the pump by hand to see whether it turns smoothly.

9-2 Operation

a. Start the motor, but at first repeat "ON" and "OFF" operations once or twice and enter into operation on confirming that there is no abnormal condition with respect to rotating direction, noise, vibration, starting current, etc.
b. Tighten the air vent plug (or cock) after speed has risen, and air is discharged from the air vent and liquid is pumped up in some time. If liquid is not pumped up in about 30 seconds after start-up, stop the motor and check the cause.
c. Avoid dry operation absolutely as the sliding contact surfaces of the bearings, the teeth and the mechanical seal are of self-lubrication construction.
d. Pay a careful attention in the winter season because there may be a fear of excessive torque of motor and abnormal noise and vibration of pump caused by oil viscosity increase. Some treatment such as heating is preferable.

9-3 Stopping

a. Stop the motor.
b. After stopping the pump, close the suction valve and then close the discharge valve.

10. Maintenance

In order to maintain an efficient operation over a long period, the utmost care must be taken.

10-1 Cautions during Operation

a. Try to operate the pump within the range of the design condition.

b. Check the vibration, noise, pressure gauge and ammeter. If abnormal condition is surveyed stop the pump immediately. Particularly, for the pump provided with a mechanical seal, protect against vibration in view of the service life of the mechanical seal.

c. Pay attention to bearing temperature rise. If the bearing housing can be felt by hand safely, there is no fear, otherwise, measure the temperature with a thermometer. Keep the permissible bearing temperature at "ambient temperature plus 40℃" or "liquid temperature plus 20℃". If low viscosity oil outside the range of design condition, be careful as the bearings and the gear may be seized.

d. Keep a small amount of continuous leak from the gland. Too hard tightening of the packing will cause premature wear of the shaft and the sleeve, overheating and overload. Especially pay a careful attention to a case where oil of high temperature or low viscosity is used.

e. When a mechanical seal is used, oil leak is generally very small. If oil leaks in drips, check the seal.

f. Never throttle the valve on the suction and discharge sides. Adjust the flow by means of the by-pass valve on the discharge side.

g. Never operate with the valve closed on the discharge side, otherwise a temperature rise of the liquid in the casing will cause seizure.

h. See whether the stand by pump is reversing due to the leakage from its non return valve.

i. In the case of a pump with automatic starting and stopping device, check the pressure at the start and stop of pump as well as the interval of operation, and if the interval is too short, check the system.

j. In the case of automatic operation under central control system, take care of the indication of gauge on the control panel board.

k. Keep the strainer clean to avoid clogging.

10-2 Cautions during Standstill

a. Keep closed the discharge and suction valves. But such valves must be kept open in a case where the pump is of auto start type with a foot valve or a check valve, where the pump is under positive suction head or where oil is accumulated in the pump casing in some way or other.

b. When the pump is shut down over a long period, the pump must be rotated by hand or motor periodically. (once a week or so)

11. Inspection

11-1 Periodical Inspection

Excepting the case of abnormal conditions, it is desirable to carry out inspection periodically in the following way, but since it varies according to the method of mounting, place of installation, liquid handled, etc. It is preferable that the operator should make a final plan of inspection in accordance with conditions.

Item	Action to be taken	Period		
		Every 3 months	Every 12 months or 8,000 hrs.	Every 4 years or 20,000 hrs.
Ball bearing	Check up inner and outer races as well as balls for exfoliation, and if its trance is found, renew it.			○
Line bearing	Check up sliding surfaces for conditions of contact and flaw. Measure inside diameter and take necessary steps according to manufacture's recommendation.		○	
Gland Packing	Check up condition of fitting and amount of leakage. Renew it if leakage is too much.	○		
Mechanical seal	Check up condition of mating faces.		○	
	Renew it.			○
Shaft	Check up conditions of contact, flaw in sliding faces. Measure the outside diameter of journal and treat it according to manufacturer's recommendation.			○
"O" ring	Check up surface flaw and condition of deterioration. Renew it if deformed.		○	

Item	Action to be taken	Period		
		Every 3 months	Every 12 months or 8,000 hrs.	Every 4 years or 20,000 hrs.
Safety Valve	Throttle the discharge valve to raise discharge pressure and confirm operation of the safety valve		○	
Gear	Check up tooth surfaces and condition of its setting to the shaft. Measure the outside diameter and width and treat it according to manufacturer's recommendation			○
Shaft coupling bolts and rubber rings	Check up conditions of wear and renew them if abnormal points are found.		○	
Foundation bolts	See to it that the foundation bolts are tightened enough.		○	

Note : The hours quoted refer to the operating hours.

11-2 Overhaul

When overhauling the pump, attention must be paid as follows.

a. Understand the construction well by referring to the assembly drawing and make no mistake in the order of overhaul.
b. When separating fit and flange faces, use jack bolts and wooden hammers, and never apply force with chisels or drivers.
c. When removing the rotating element, take care to avoid flaw on sliding faces and machined surfaces. It is preferable not to overhaul the mechanical seal unless there is anything abnormal.
d. When removing rotating parts from the shaft, draw off each one carefully after removing the locking device.
e. Handle the longsized-parts such as shaft carefully so that it may not bend.
f. Handle the parts carefully, by arranging them on sheets of paper or cloth in good order.
g. At overhauling, put suitable match marks as many as possible to avoid mistakes when reassembling.

11-3 Assembly

Carry out assembly, by reversing the order of disassembly and paying attention as follows.

a. Remove dust and stain from each part by washing it thoroughly with kerosene. Repair it if flaw is found.

b. Fit the locking device perfectly in each rotating part if necessary.

c. When fitting the parts with match marks such as fittings and gear, be sure to follow them.

d. Before assembling the sliding contact faces of the bearings, tooth faces and mechanical seal, apply enough clean lube oil to them.

e. When fitting the bearings and side cover, take into account the position of oil channel for bearing lube oil and return oil and the direction of pump discharge and suction.

f. Insert each packing in good order softly one by one from the bottom, staggering each joint by 90° or 180° to a fixed position and support it lightly with the packing gland.

g. Carefully fit the mechanical seal and then confirm its movement by hand.

h. Tighten the bolts with an equal force.

i. Check up the alignment as mentioned in Section 8.

j. Rotate the pump by hand to see that it turns smoothly.

12. Troubles and Remedies

Should troubles occur, their causes must be traced and necessary remedies must be carried out. The following, for instance, can be conceivable as troubles.

Troubles	Causes	Remedies
Pump does not start.	• Motor is in trouble. • Pump seizes. • No power source. • Wiring is broken, or relay, etc. are in trouble. • Foreign matters exist in the pump.	• Repair motor. • Repair pump. • Check up electric system. • Repair • Overhaul the pump and remove them.
Pump starts, but does not discharge liquid.	• Pump is not well filled with oil. • Air is not vented enough. • Valves are not open. • Valves will not open. • Pump is sucking air. • Suction pipe or strainer is clogged. • Motor rotating direction is wrong.	• Fill it again. • Open air plug (or cock) • Open valves. • Repair valves. • Check up suction system and gland packing • Clean suction pipe or strainer. • Change wiring.
Pump starts, but specified capacity or discharge pressure are not reached.	• Pump is sucking air. • Pump speed are too low. • Safety valve is open. • Suction pipe or strainer is clogged. • Liquid viscosity is too low. • Gear shaft is worn. • Suction pressure is too high. • Instruments are wrong.	• Check up suction system and gland packing • Correct electric source. • Adjust its setting. • Clean it. • Check design specification. • Renew it. • Check viscosity. • Suction Pressure shall be within -380mmHg. • Confirm full opening of suction valve. • Replace them with new ones.
Pump starts and discharges liquid but soon ceases to discharge liquid.	• Air pockets exist in suction line. • Pump is sucking air. • Pump is sucking air through stuffing box.	• Correct piping. • Check up suction system. • Check up sealing pipe • Adjust mechanical seal. • Adjust packing.

Troubles	Causes	Remedies
Pump starts, but motor gets overloaded.	• Abnormal metal contact exists in the rotating parts. • Connection is wrong. • Packing is too tightened. • Pump shaft is bent. • Motor is in trouble. • Liquid viscosity is too high. • Foreign matters exist in the pump. • Discharge pressure is too high. • Bearings are wrong.	• Check clearance. • Check up alignment. • Loosen gland or otherwise renew packing • Renew shaft. • Repair it. • Conform it to design condition. • Remove them. • Conform it to design condition. • Renew them.
Pump starts, but bearings get overheated.	• Lubricant is in shortage. • Connection is wrong. • Bearings are wrong. • Shaft is bent. • Thrust has increased. • Bearing assembly is wrong.	• Check oil supply to bearings and return oil system. • Check up alignment. • Renew them. • Renew it. • Check whether excessive wear exists on bearing side or whether pump is in abnormal suction pressure condition. • Readjust bearings.
Pump starts, but vibration and accidental noise takes place.	• Connecting is wrong. • Shaft is bent. • Installation is wrong. • Foundation is weak. • Other vibration is transmitted. • Cavitation exists. • Bearings are worn. • Safety valve is chattering. • Foreign matters exist in the pump. • Tooth contact is wrong.	• Check up alignment. • Renew shaft. • Correct installation condition. • Reinforce foundation. • Reinforce piping. • Conform suction pressure and viscosity to design condition. • Renew them. • Repair it • Overhaul the pump to remove them. • Renew the gear.

In case troubles cannot be remedied in spite of the above counter-measures, the causes may be in the design conditions of the pump, so it is preferable to consult the shipyard or manufacturer.

TP WAX-TYPE AUTOMATIC TEMPERATURE CONTROL VALVE

INSTRUCTION MANUAL

TPR GROUP TEIPI THERMAL ENGINEERING CO.,LTD.

Structure and Operation

Fig-1* 20A～32A

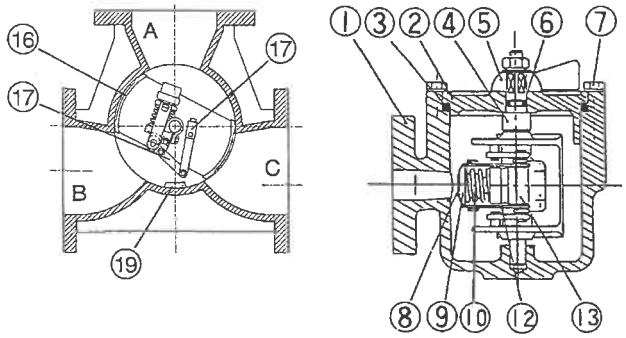


Fig-2* 40A～80A

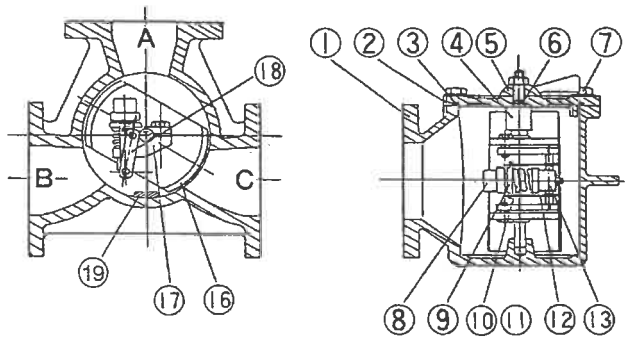


Fig-3* 100A～150A

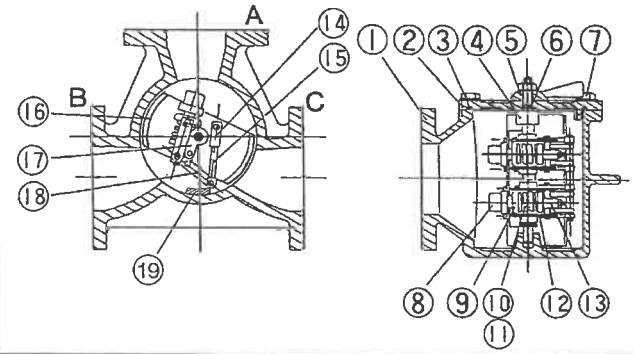
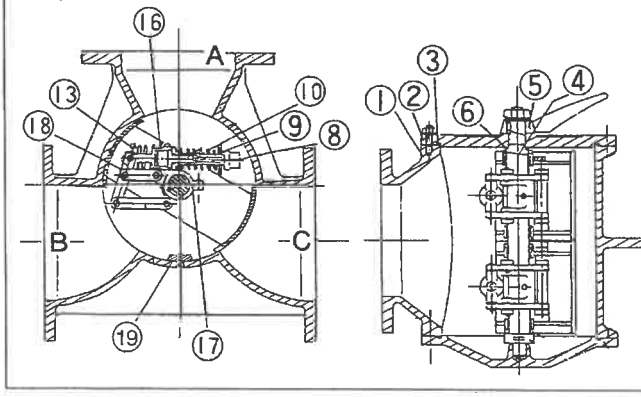


Fig-4* 200A/250A



The Parts

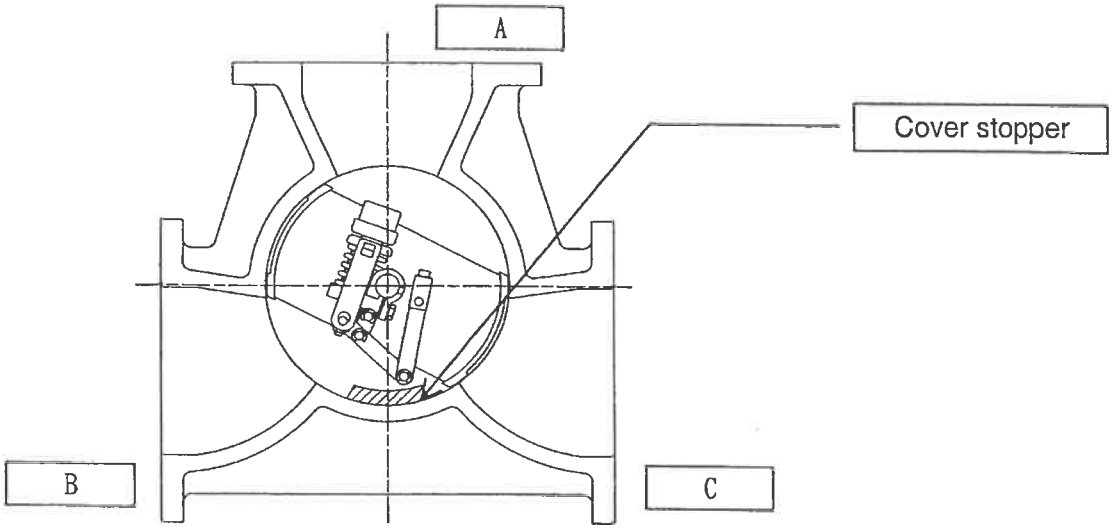
No.	Part
1	Body
2	Top cover
3	Cover seal
4	Shaft
5	Lever
6	Shaft seal
7	Cover bolt
8	Wax element
9	Spring seat
10	Element spring (inner)
11	Element spring (outer)
12	Strap
13	Housing
14	Cross piece
15	Rod
16	Rotor
17	Element bracket
18	Lever link
19	Cover stopper

Operation

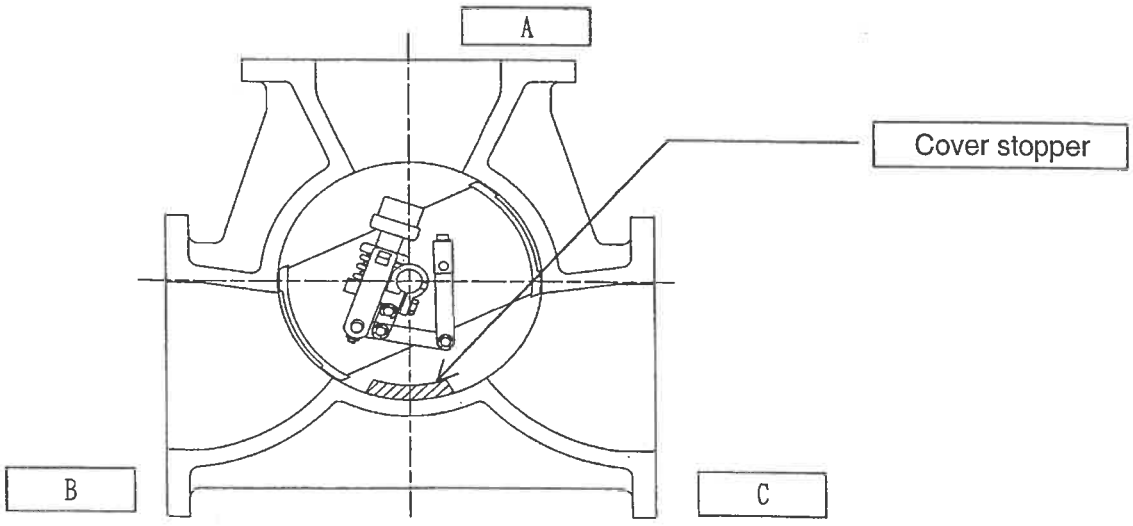
※Outline of Operation

When the temperature of the fluid passing through the valves reaches the wax element's lower limit temperature, the wax starts to expand.
The expanding force is transmitted to the plunger, which pushes out the housing.
The force from this is transmitted to the link connected to the rotor, turning the rotor around the shaft and gradually closing Port B and opening Port C.
The degree of openness of the ports is controlled in response to the fluid's changing temperature, altering the flows through the ports in a mutually reciprocal manner.
When the fluid's temperature reaches the wax element's upper limit temperature, Port B is fully closed and Port C is fully opened.
Conversely, when the temperature drops, the element spring, compressed when the wax element expanded, extends as the wax element contracts, the force from the spring moving the rotor in the opposite direction from when the temperature was rising.

Fluid temperature is below wax element's lower limit temperature.



Fluid temperature reaches wax element's upper limit temperature.



※The structure is such that the cover stopper and the rotor do not come in contact when the fluid's temperature reaches the wax element's upper limit temperature.

Applications

Standard Configurations

Configuration for controlling cooling water and lubricant temperatures in engines

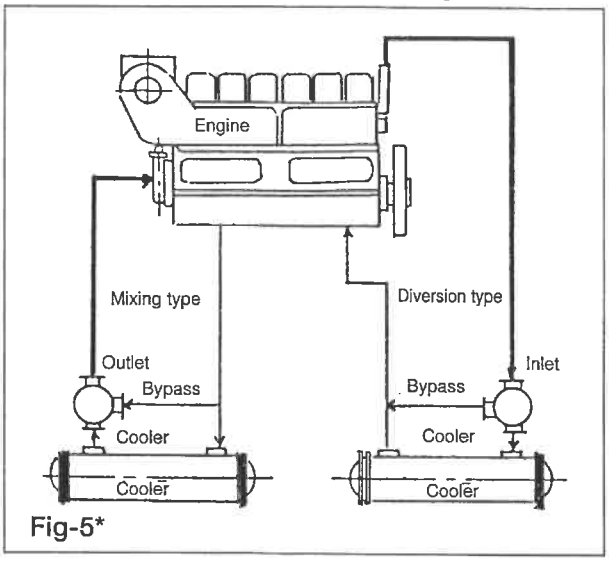
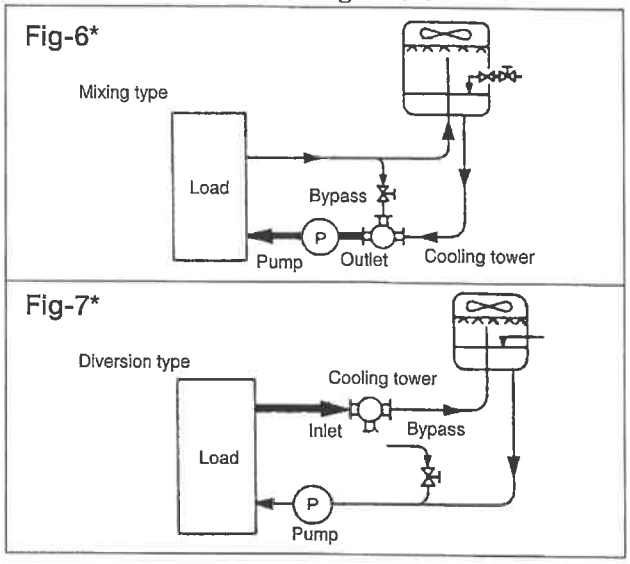


Fig-5*

Configuration for controlling condenser cooling water in refrigerators



Examples of Piping

The TP Wax-Type Automatic Temperature Control Value is a three-way valve. One of its ports is always open; the other two are opened and closed according to the temperature of the fluid. Understanding how the ports (A, B and C) operate will allow you to use this product in a wide range of applications.

(For the ports, select the optimal positions from the six available settings: LI, LB, LC, RI, RB, and RC).

(※) Operation of ports

Port A: This port is always open, regardless of changes in the fluid's temperature.

Port B: This port is open when the fluid's temperature is below the wax element's lower limit temperature, and gradually closes as the fluid's temperature rises.

Port C: This port is closed when the fluid's temperature is below the wax element's lower limit temperature, and gradually opens as the fluid's temperature rises.

Temperature control for heated pools, floor heating, public baths

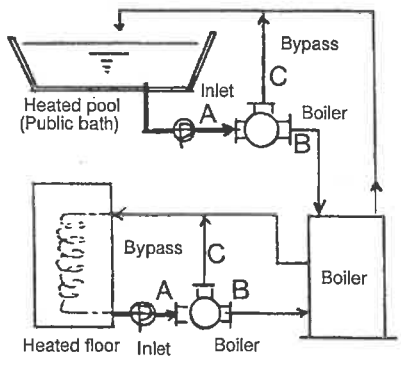


Fig-8*

Temperature control for hot water supply systems and shower systems

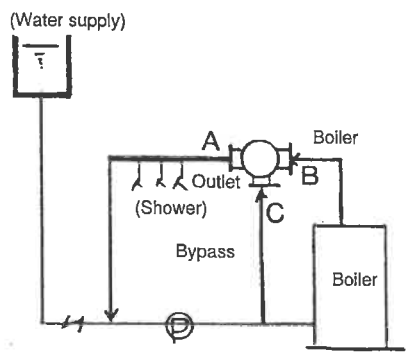


Fig-9*

Temperature control for solar heating systems

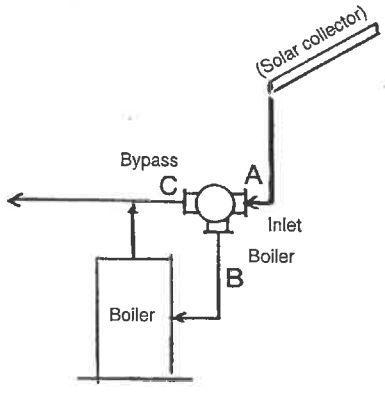


Fig-10*

Installation

1. **Make sure that there is enough space in front of the Temperature Control Valve's side with the lever so that the internal mechanism can be removed.**

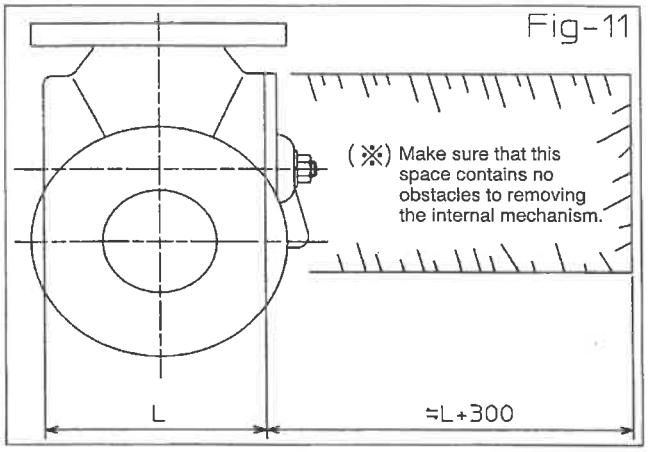


Fig-11

2. **Be careful that the body doesn't get deformed.**

When tightening the flange bolts, be careful not to deform the body of the Temperature Control Valve.

To determine whether there is deformation, loosen the lever nut, then move the lever back and forth several times between AUTO and the stopper pin at E, checking whether the lever moves smoothly. If it doesn't move smoothly, redo the piping correctly.

3. **Don't let welding waste or other foreign matter get into the Temperature Control Valve.**

If the lever does not move smoothly, there may be welding waste or other foreign matter inside the Temperature Control Valve. In such a case, remove the internal mechanism and clean out the Temperature Control Valve's interior.

To prevent welding waste and other foreign matter from getting into the Temperature Control Valve, we recommend flushing out the pipes. To prevent flushing from having an adverse effect on the Temperature Control Valve, do it with the internal mechanism removed.

4. **Pressure differences among the three valves will prevent proper temperature control.**

There will be some pressure difference, depending on the sizes of the port apertures. However, if the pressure difference is 0.1 MPa or greater, adjust it by placing a gate valve on the Port B line or the Port C line.

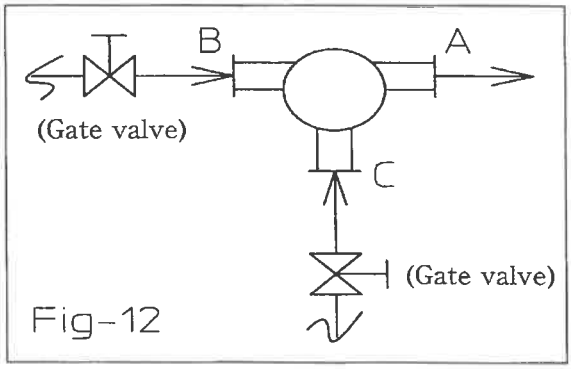


Fig-12

5. **Examples of good piping and undesirable piping**

5-1. Piping for direct cooling of seawater for ships

(1). If the Temperature Control Valve is installed above the waterline, air will get into it when it is not operating and it will gradually stop operating normally. Therefore, install it below the waterline. (See Fig. 13.)

(2). Arrange the bypass line so that there is no negative pressure around the Temperature Control Valve. If there is negative pressure around it, air will get sucked into it from the outboard discharge side. (See Fig. 14.)

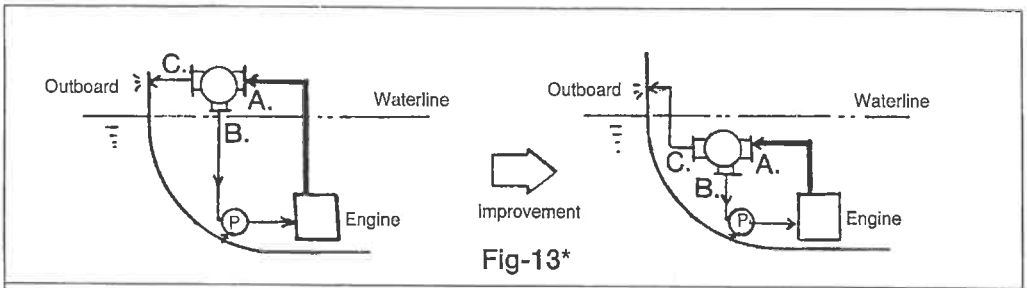


Fig-13*

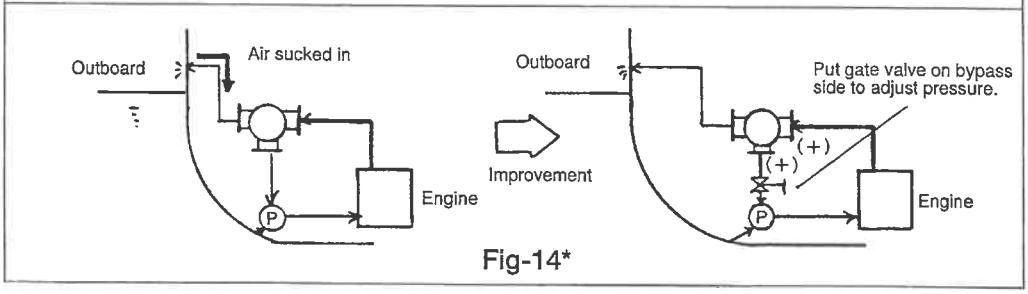


Fig-14*

5-2. Piping for refrigerators (condenser cooling water)

(1). If the Temperature Control Valve is installed above the water surface of the cooling tower, air will get into it when it is not operating and it will gradually stop operating normally. Therefore, install it below the water surface. Position the bypass division point below the water surface as well. (See Fig. 15 for an example of undesirable piping and Fig. 16 for an example of good piping.)

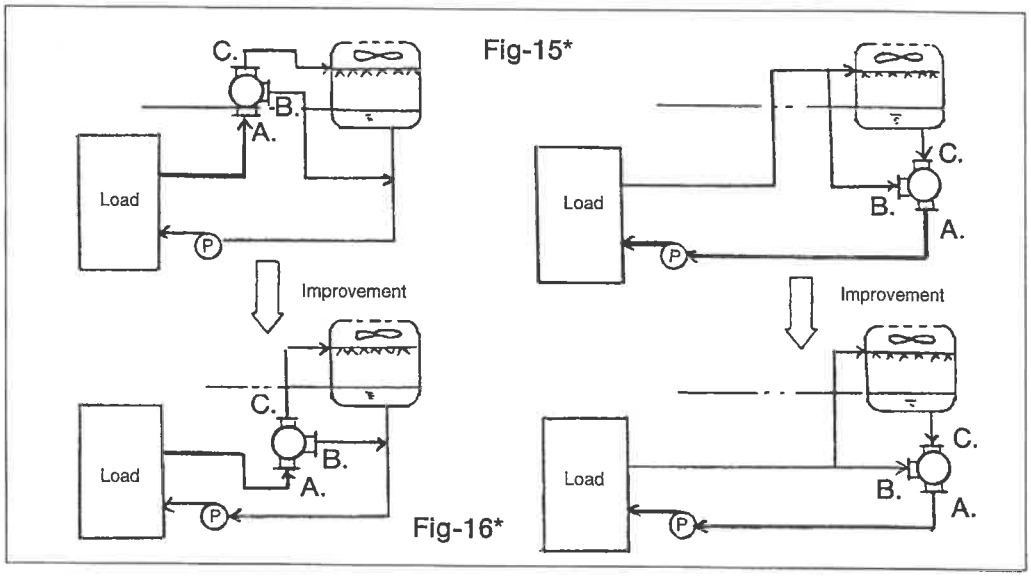


Fig-15*

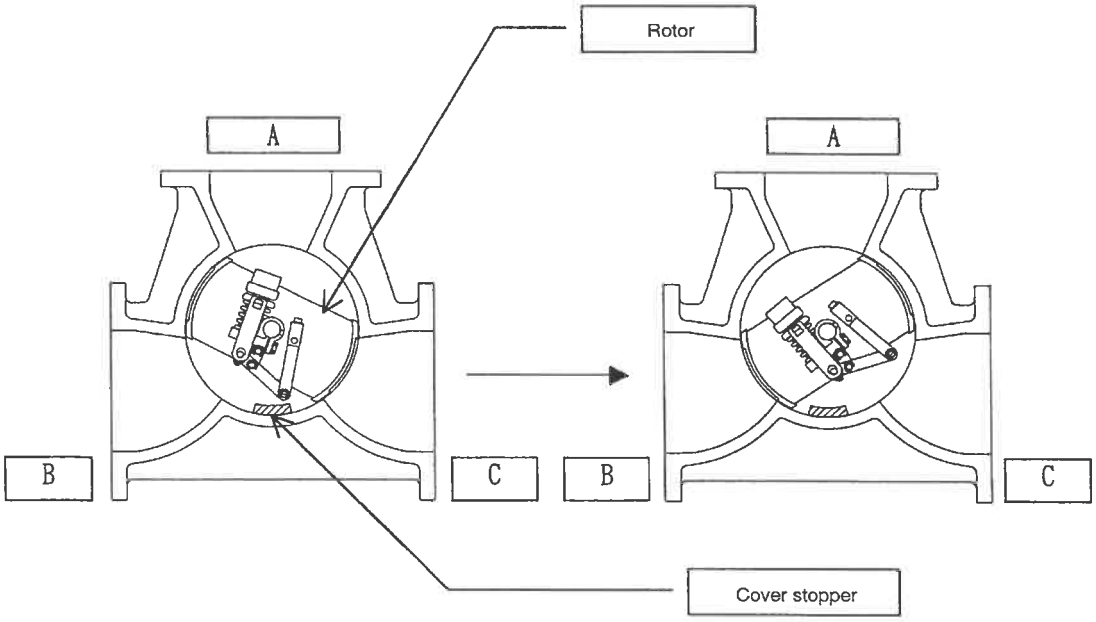
Fig-16*

(2). Install the Temperature Control Valve near the cooling tower. Especially when the cooling tower is installed in a building, a head difference can develop between it and the condenser; and if the Temperature Control Valve is installed near the condenser, the pressure difference between the bypass and the cooling tower will become large, making it impossible for the Temperature Control Valve to operate normally. To avoid this problem, install the Temperature Control Valve as close as possible to the cooling tower.

Things to Check Before the Trial Run

1. Has the piping been properly done?
   (The piping for Ports A, B and C; the selection of other materials than wax; etc.)
2. Have the installation precautions been heeded?
3. Does the lever move smoothly when operated? (See "Inspection of lever operation.")
4. While the Temperature Control Valve is operating, keep the lever set in the AUTO position except when there is an emergency (e.g., an abnormal rise in fluid temperature).

   If the Temperature Control Valve is operated with the lever set in some other position than AUTO, it's possible that, as the fluid's temperature rises, the rotor will come in contact with the cover stopper, damaging the wax element. If the lever is in the AUTO position, the rotor will not come in contact the cover stopper even when the fluid's temperature reaches the wax element's upper limit temperature.



Inspections and Adjustments When Irregularities Occur

1.Pulsation occurs in the pump.

This happens when air suction occurs or the pressure balance is not normal. In such a situation, check the following.

(1). Is the Temperature Control Valve installed in a position higher than the water level, as shown in Fig. 13 and Fig. 15? If so, redo the piping correctly so that it is as shown in Fig. 14 or Fig. 16.

(2). Has negative pressure occurred near the Temperature Control Valve? (See Fig. 14.) If so, adjust the gate valve installed on Port B (the bypass side) and eliminate the negative pressure.

(3). Is any of the piping system's other equipment (for example, the strainer) clogged or damaged?

2.The temperature rises abnormally during operation.

(1). Move the lever towards "E" (emergency) and set it at the position of the desired temperature. The closer the lever is to "E," the lower the temperature that can be obtained. (However, if the wax element inside the Temperature Control Valve senses the temperature, it may not be possible to move the lever towards "E." In such a case, do not force the lever.)

(2). If the temperature does not go down even after the lever is moved towards "E," check for dirt in the cooler or whether the cleaning tower's inlet and outlet temperatures are abnormal.

(3). If there are no abnormalities with the cooler or the cleaning tower, perform an overhaul inspection (see the "Overhaul inspection guidelines" on the following page) and check whether there are any abnormalities with the Temperature Control Valve.

3.The temperature becomes abnormally low during operation.

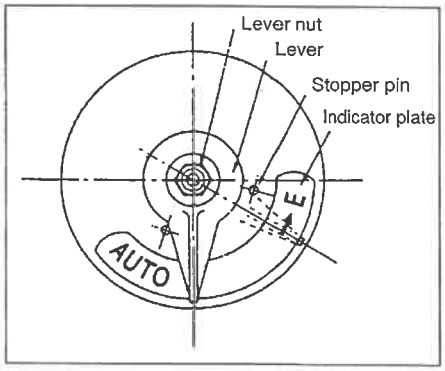
(1). Operate the lever and check that it moves smoothly. Then restart operation.

(2). If the temperature is low even after operation is restarted, perform an overhaul inspection (see the "Overhaul inspection guidelines" on the following page) and check whether there are any abnormalities with the Temperature Control Valve.

Guidelines for Periodic Inspections

1. Inspection of lever operation

To prevent the rotor from sticking due to foreign matter, scales, etc., check, once a week, that the lever moves smoothly. (Checking is not necessary if the fluid is oil, etc., since adhesions of foreign matter, scales, etc., do not occur with such fluids.) Before performing this inspection, stop the pump. If the pump cannot be stopped, grab the lever with a water pump pliers, etc., and slowly move it. (This is to prevent pulsation in the pump.)



(※) How to operate the lever

① Loosen the lever nut.

② Move the lever back and forth several times between AUTO and the stopper pin at E.

③ Check that the lever moves smoothly. Return it to its original position and tighten the lever nut.

※When moving the lever back and forth, do not hit the stopper pin too hard, as that could cause it to come out.

2. Overhaul inspection guidelines (See Figs. 1-4.)

(1). Prior to this inspection, eliminate the internal pressure.

(2). Remove all of the tightening bolts (indicated by (7) in Fig. 1) from the top cover.

(3). Remove the conjoined top cover and internal mechanism from the body and then separate them.

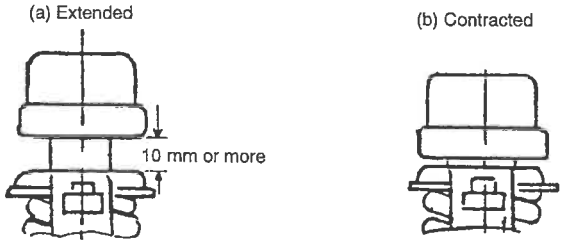
(4). Check the interior of the body for foreign matter, rust, etc. If any is found, remove it.

(5). Check the internal mechanism for scales and other adhesions. If any are found, remove them by immersing the internal mechanism in 5% Hcl (hydrochloric acid) for 30 minutes. After removing the scales, rinse the internal mechanism with water.

(6). Check the wax element for abnormalities according to the following procedure.

(a). Pour hot water, with a temperature 2℃ above the upper limit temperature, on the wax element for about 1 minute and confirm that the wax element extends.

(b). Pour hot water, with a temperature below the lower limit temperature, on the wax element for about 1 minute and confirm that the wax element contracts to its original state.



If any abnormality is found, replace the wax element.
(See "Replacement of wax element" on the following page.)

(7). Replace the internal mechanism in the body. Make sure that the O-ring and other parts are in their prescribed places.

(8). Replace the top cover and secure its tightening bolts.

(9). Operate the lever and make sure that it moves smoothly.

That completes the overhaul inspection. Fill the piping with water and restart operation.

3. Replacement of wax element (See Fig. 2 "40A - 80A.")

(1). Remove all (7) tightening bolts from the top cover.

(2). Remove the conjoined top cover and internal mechanism from the body and then separate them.

(3). Place the internal mechanism as shown in Fig. 18. With a hammer, lightly tap the head of the wax element (indicated by ▼ in the drawing) in order to break the bond of the "Loctite" adhesive on the threads of the wax element. With a water pump pliers, grab the neck of the wax element and, turning it counterclockwise, remove the wax element. (So that the neck of the wax element can be grabbed with the pliers, turn the rotor by hand as shown in Fig. 19.)

※When removing the wax element, place the rotor in the position shown in Fig. 18. Otherwise, the spring might come off.

Fig-18*

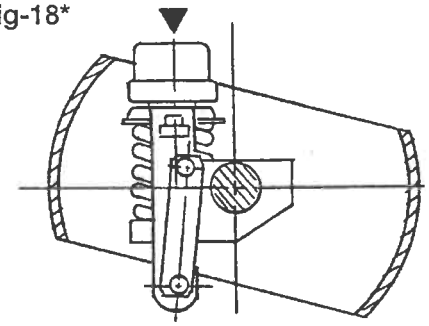
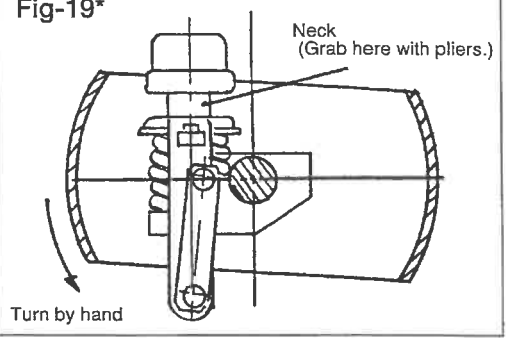


Fig-19*



(4). Apply "Loctite" to the threads of the new wax element, then screw the element into bracket (17) (without using too much force).

(5). Replace the internal mechanism in the body. Make sure that the O-ring and other parts are in their prescribed places.

(6). Replace the top cover and secure its tightening bolts.

(7). Operate the lever and make sure that it moves smoothly.

That completes the replacement of the wax element. Fill the piping with water and restart operation.

Note: The above procedure applies to port apertures 40A-80A (and also to 100A-150A). For apertures 32A and below and 200A and above, please request the corresponding instruction manual.

4. Measures for periods of non-operation

The following measures are to be taken if the cooler's operation is stopped during winter or some other period.

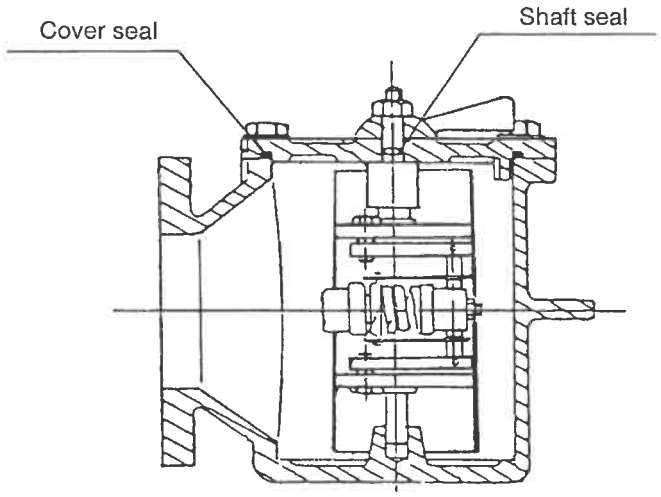
(1). If the period of non-operation is going to be long (2-3 months or more), drain the water from the piping, remove the internal mechanism, and cover the body.

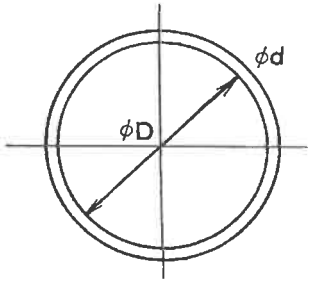
(2). Draining the water is particularly important in cold regions. (If there is also concern about the water freezing during operation, wrap the Temperature Control Valve with heat insulating material.)

(3). If the period of non-operation is half a month to a month, do not drain the water in the piping (provided that there is no chance of its freezing), as the water serves to inhibit rusting and scale adhesion.

(4). When resuming operation after a period of non-operation, first confirm that the internal mechanism is working properly.

Table of Wax-Type Automatic Temperature Control Valve Seals





Cover Seals (Material: NBR)

Aperture	Internal diameter (D)	Diameter (d)	Regulation (AN)	Regulation (JIS)
20-25A	79.4	3.1	–	B-2401 G80
32A	104.4	3.1	–	B-2401 G105
40A	132.94	3.53	AN6230-30	1517-30
50A	〃	〃	〃	〃
65A	〃	〃	〃	〃
80A	〃	〃	〃	〃
100A	151.99	〃	AN6230-36	1517-36
125A	177.39	〃	AN6230-40	1517-40
150A	202.56	6.98	AN6227-72	1516-72
200A	266.06	6.98	AN6227-77	1516-77
250A	〃	〃	〃	〃
300A	354.96	6.98	AN6227-84	1516-84
350A	405.26	6.98	AS568-461	ARP568-461

Shaft Seals (Material: NBR)

Internal diameter (D)	Diameter (d)	Regulation (JIS)
8.8	1.9	B-2401 P9
〃	〃	〃
11.8	2.4	B-2401 P12
〃	〃	〃
〃	〃	〃
〃	〃	〃
〃	〃	〃
〃	〃	〃
〃	〃	〃
31.34	3.53	AN6227-23
〃	〃	〃
50.16	5.33	AN6227-32
〃	〃	〃

M E M O

M E M O

Manufacturer

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Current as of April 2005.
The contents of this instruction manual are subject to change without notice.

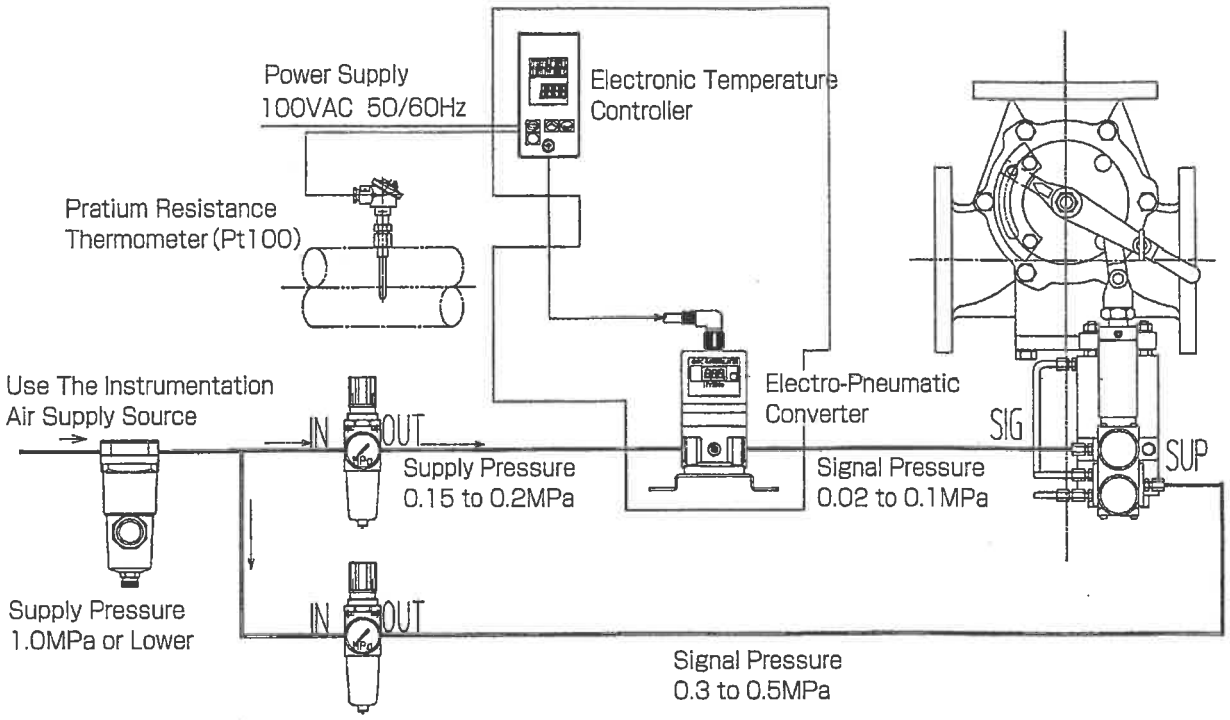
ELECTRO-PNEUMATIC AUTOMATIC TEMPERATURE REGULATOR

INSTRUCTION MANUAL

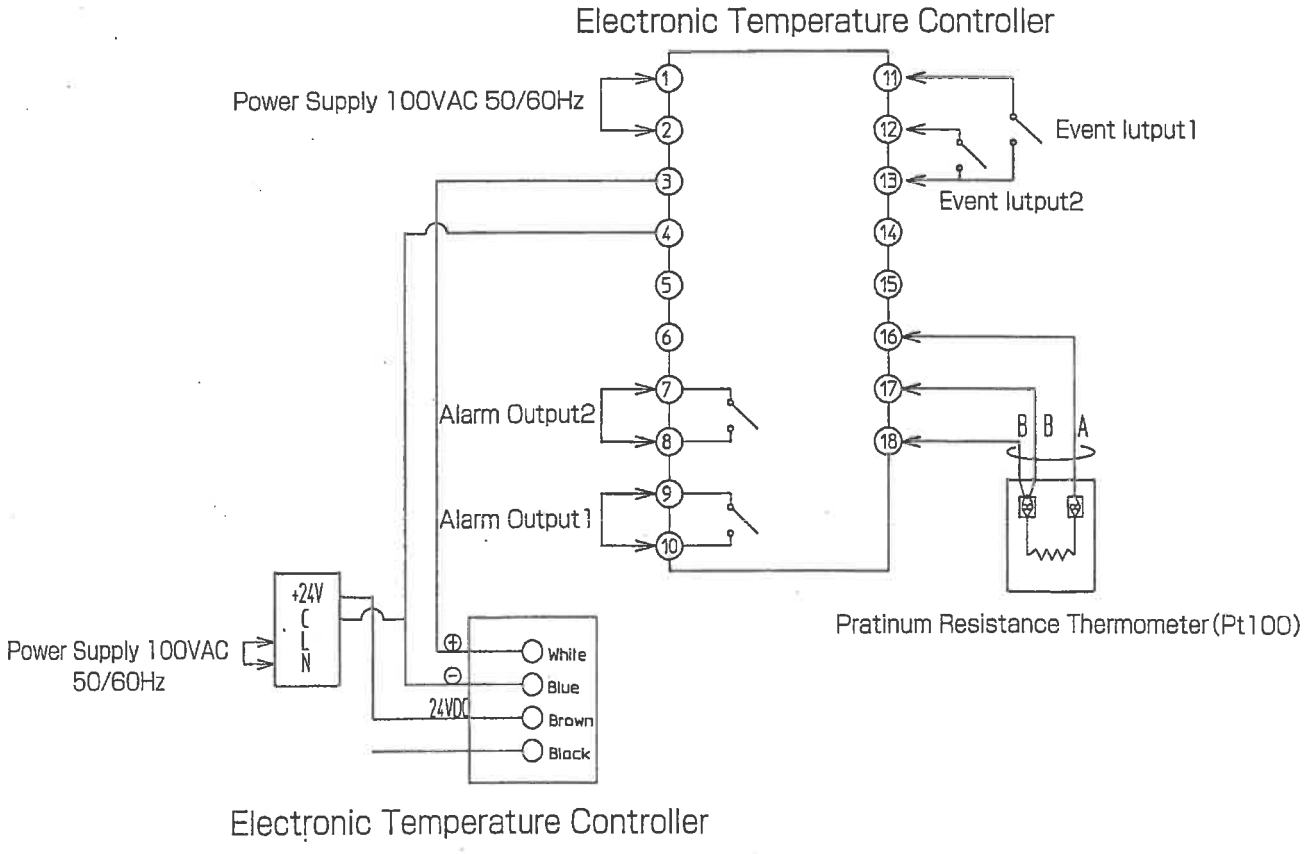
TP TEIPI THERMAL ENGINEERING CO.,LTD.

Structure and Function

1. Instrumentation Arrangement by User



2. Electrical Wiring Diagram



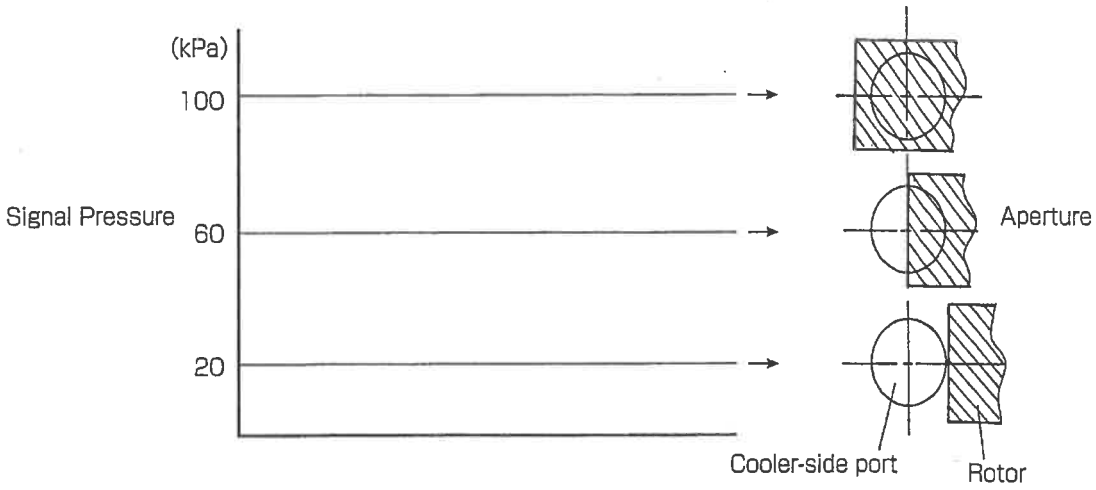
Operation Mechanism

The sensor (Platinum Resistance Thermometer) installed to the particular position for the desired control in the piping detects the temperature of the applied fluid to generate the resistance value which corresponds to each temperature level and sends it to the electronic temperature controller. The resistance value thus reaching the temperature controller causes the output current 4mA to 20mA to be delivered to the electropneumatic converter, thus actuating the solenoid valve so that the supplied air pressure 200kPa is converted to the output signal pressure 20 to 100kPa. This signal pressure is then input to the positioner at the main body driving section.

The rotor of the driving section is rotated in proportion to the signal pressure (20 to 100kPa), so that the flow rates on both side B and C are changed by the aperture adjustment at the said two sides made every moment in accordance with the temperature variation.

As for the control by the temperature controller, advanced PID control with two degrees of freedom is employed. The controller needs the shortest rise time, does not allow overshooting, and resumes a stabilized condition very fast in the event of disturbance.

※Relatoin Between Signal Pressure and Aperture (In Case of Reverse Action)



Temperature Set Value

The previously speciffied set value (T0℃) is set upon shipment from the factory.

How to change Temperature Set Value

Upon delivery, the set value is 'T0℃'. In order to change the set value, follow the steps described below.

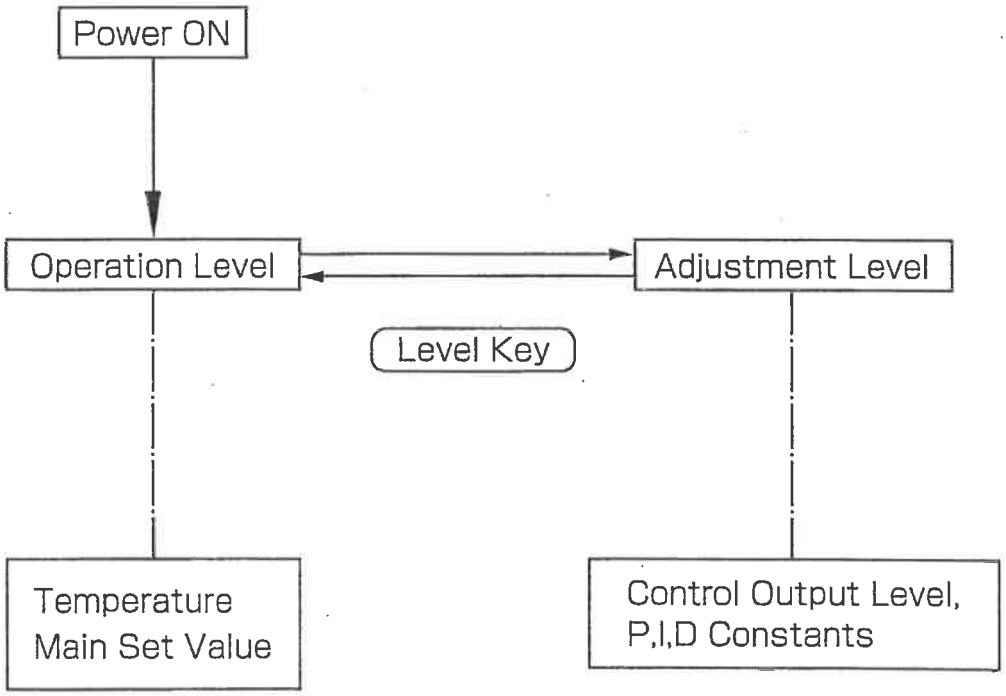
Ex: Changing from T0℃ to T1℃

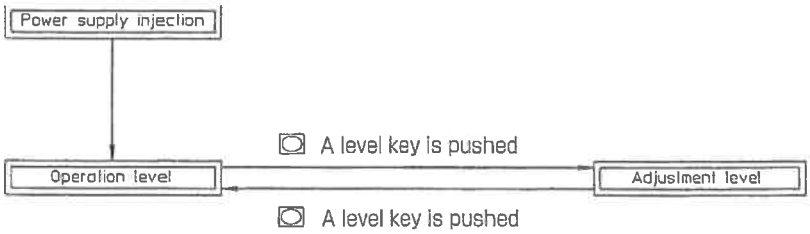
Press the setting key (UP) and get T1℃ displayed. (In order to lower the set value, press the setting key (DOWN) and get the desired temperature value displayed.)

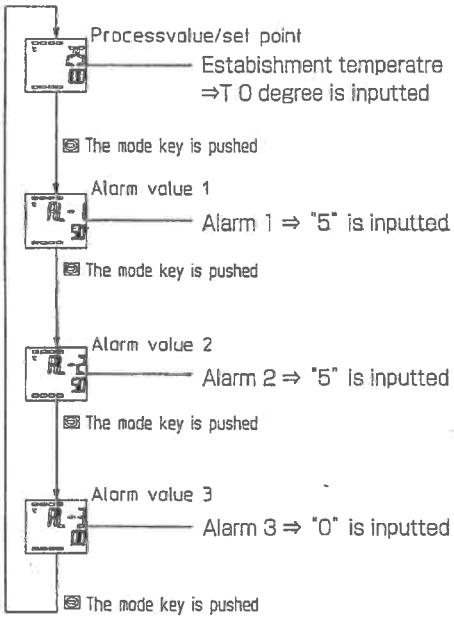
Temperrature control is carried out for the value newly set as above.

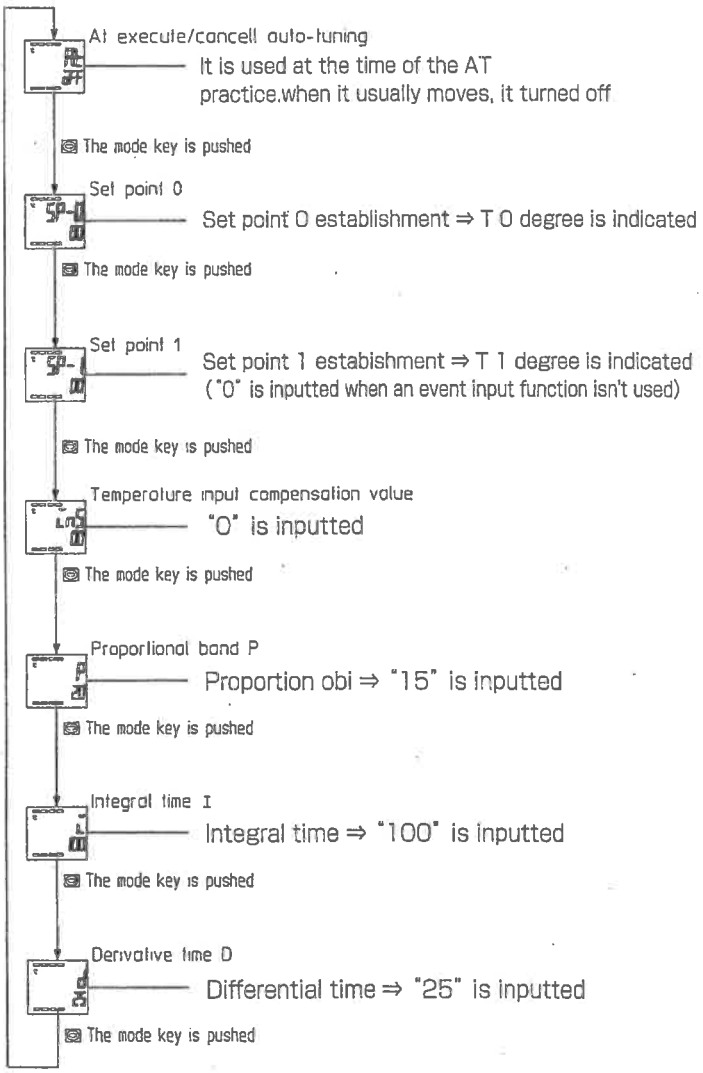
Temperature Setting Method (as set by maker upon shipment)

Indication Level Flow (Specifications prior to Drive)









*The numerical value of P.I.D after
auto-tuning is different from the above numercial value

Automatic Tuning

PID control with inadequate constants provieded may result in unstable control, occurrence of hunting, or extreme delay of adjustment which further leads to excdssive consumption of the time required for restoration of the deviation generated by disturbance.

For the optimum selection of PID constants, the automatic tuning function is employed. (The temperature controller for this model employs the 'Limit Cycle Method'.)

Automatic tuning (AT) allows the preparation of parameters by automatic internal computation to realize the optimum control for the system.

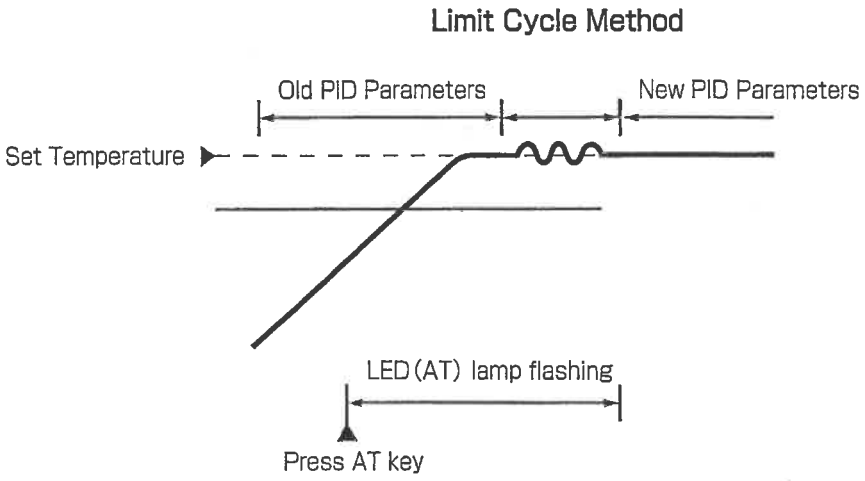
*AT Method :

If you press the Display key and Level key at the same time for one second or longer, automatic tuning starts with Display (SV) flashing.

Temperature control is carried out based on the respective set PID parameters (P=15℃ I=100sec D=25sec).

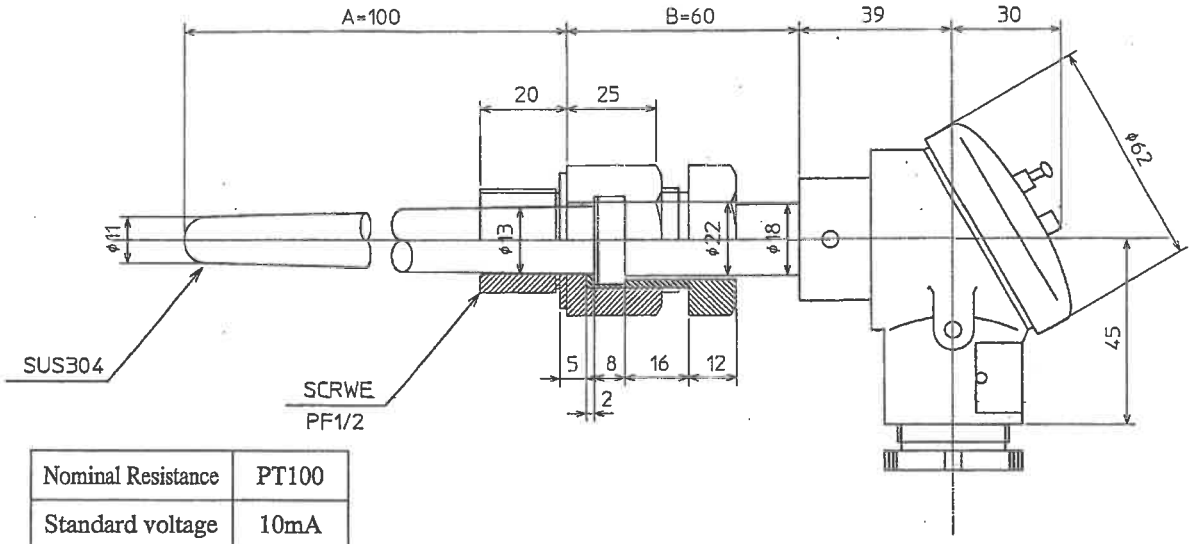
When the temperature of the controlled system reaches the set temperature level, AT is executed by the limit cycle method.

If the Display(SV) goes off, AT is completed and the parameters are replaced by new values produced by the said computation.



Specifications of Main Devices

Platinum Resistance Thermometer (PT100)



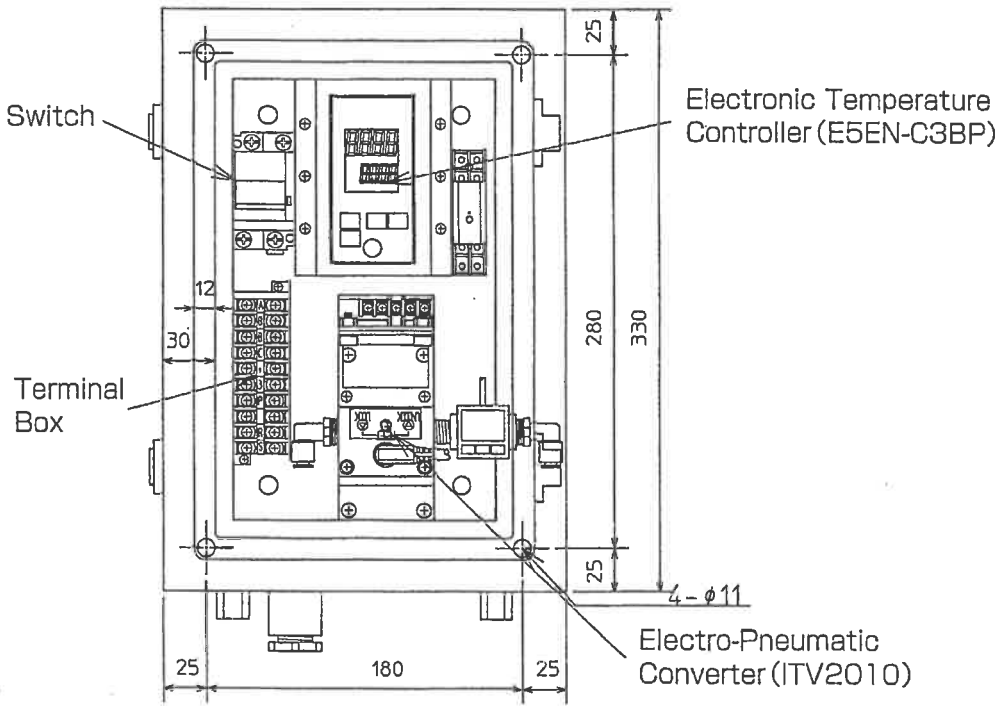
Nominal Resistance	PT100
Standard voltage	10mA
Class	0.5
Temperature range	0 to 200℃

*Conforms with JIS C 1604

Control Box

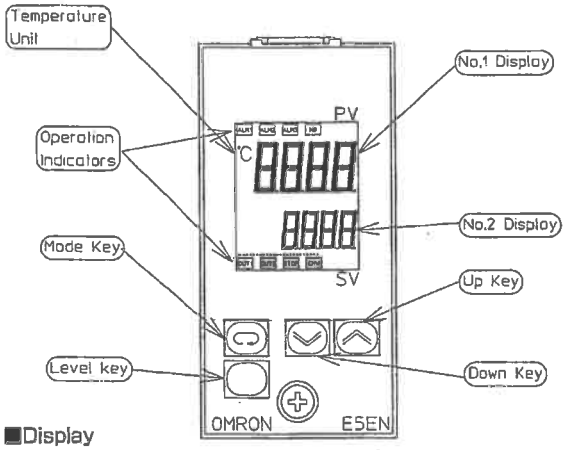
For free and easy settinng up, the temperature controller and the electro-Pneumatic converter are built into the control box.

Two types of control boxes, indoor-use and drip-proof types, are available for your particular purpose.



Electronic Temperature Controller

■Front panel



■Display

●No.1 display

Displays the process value or parameter type.
Lights for approximately one second during startup.
Displays the set point, parameter operation read value or the manipulated variable.
Lights for approximately one second during startup.

(1) ALM1 (alarm 1)
Lights when alarm 1 is ON.
ALM2 (alarm 2)
Lights when alarm 2 is ON.
ALM3 (alarm 3)
Lights when alarm 3 is ON.

(2) HB (heater burnout alarm display)
Lights when a heater burnout is detected.

(3) OUT1,OUT2 (control output 1, control output 2)
Lights when control output 1 and/or control output 2 are ON.However, whenever control output is the current output, OUT1 stays off.

(4) STOP (stop)
Lights when operation is stopped.
During operation, this indicator lights when an event or the run/stop function is stopped.

(5) CMW (communications writing control)
Lights when communications writing is "enabled" and is out when it is "disabled."

●Temperature unit

The temperature unit is displayed when the display unit parameter is set to a temperature, indication is determined by the currently selected "temperature unit" parameter set value. When this parameter is set to "℃","℉" is displayed, and when set to "℉","℉" is displayed.Flashes during ST operation.

■How to use keys

The following describes the basic functions of the front panel keys.

- **(level)key** — Press this key to select the setting leaves. The setting level is selected in order "operation level" ⟷ "adjustment level", "intial setting level" ⟷ "communications setting level".
- **(mode)key** — Press this key to select parameters within each level.
- **(up)key** — Each press of this key increases value displayed on the No.2 display.Holding down this key speeds up the incrementation.
- **(down)key** — Each press of this key decreases value displayed on the No.2 display.Holding down this key speeds up the incrementation.
- **+ key combination** — EThis key combination sets the E5EN to the "protect level." For details on. the protect level, see Chapter 5 Parameters.

Patings

Supply Voltage	100 to 240 VAC 50-60/Hz
Operating voltage range	85 to 110%of rated supply voltage
Power consumption	9VA
Sensor input	Platinum resistance thermometer : Pt100.jpt100
Control output	Current output : DC4-20mA load600 max.resolution approx2,600
Alarm output	SPST-NO.250VAC,3A (resistive load) electrical life : 100,000 operations min applicate load 1V 1mA
Control method	2-PID or ON/OFFcontrol
Setting method	Digital setting using front panel keys
Indication method	7-segment digital display and single-lighting indicator
Ambient temperature	-10 to 55℃ 1 with no condensation or icing 1
Ambient humidity	Relative humidity 25 to 85%
Storange temperature	-25 to 65℃ (with no condensation or icing 1)
Altitude	2,000 m or less
Recommended fuse	T2A.250VAC time lag low shut-off capacity
installation environment	installation category2.pollution class2 (IEC61010-1 comptiant)

Characteristics

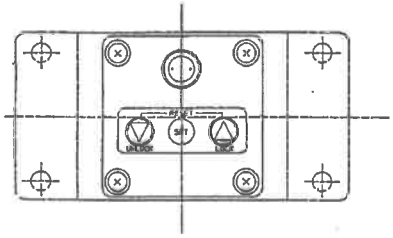
Supply Voltage	Thermocouple : (±0.5% of indication value or ±1℃.whichever is greater)±1digit max. Platinum resistance thermometer : (±0.5% of indication value or ±1℃.whichever is greater)±1digit max. Analog input : ±5%FS±1 digit max. CT input : ±5%FS±1 digit max.
Hysteresis	0.1 to 999.9EU (in units of 0.1EU)
Proportional bond (P)	0.1 to 999.9EU (in units of 0.1EU)
integral time (I)	0 to 3999s (in units of 1second)
Derivative time (D)	0 to 3999s (in units of 1second)
Control period	1 to 99 (in units of 1second)
Manual reset value	0.01 to 100.0% (in units of 0.1%)
alarm seting range	-1999 to 9999 (decimal point position dependent on input type)
sampling penod	500ms
insutation resistance	200MΩmin. (by 500VDC megger)
Dielectric strength	200VAC 50/60Hz 1 min
Vibration resistance	300 10 to 55 Hz 20m/s2 for 2 hrs, each in X,Y and Z derections
Shock resistance	300 m/s2 max,3 times each in 3 axes,6 directions (reloy :100m/s2)
Weight	Approx 260g
Protective structure	Front panel : NEMA4X for indoor use (equivalent IP66) Rear case : IP20.terminals : IP00
Memory protection	EEPROM (non-valatile memory) (number of writes :100,000)

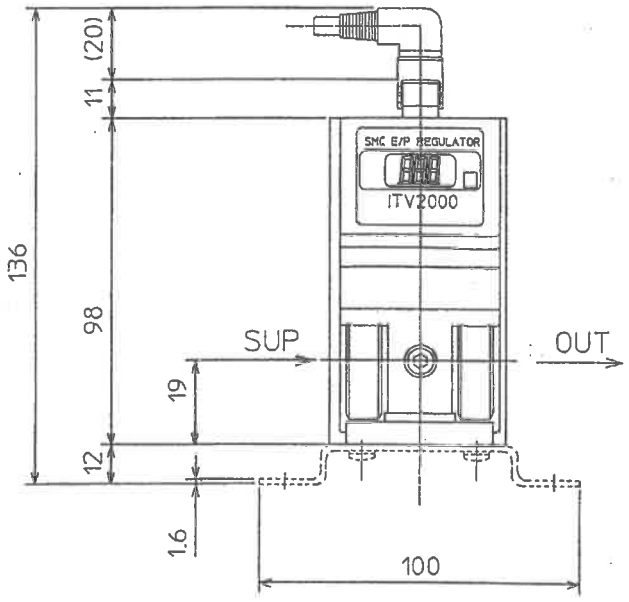
Electronic-Pneumatic Converter (ITV2000)

The electro-pneumatic converter receives the input electrical signal (4mA to 20 mA DC) to convert the supplied air pressured at 200 kPa to the signal pressure of 20 to 100 kPa.

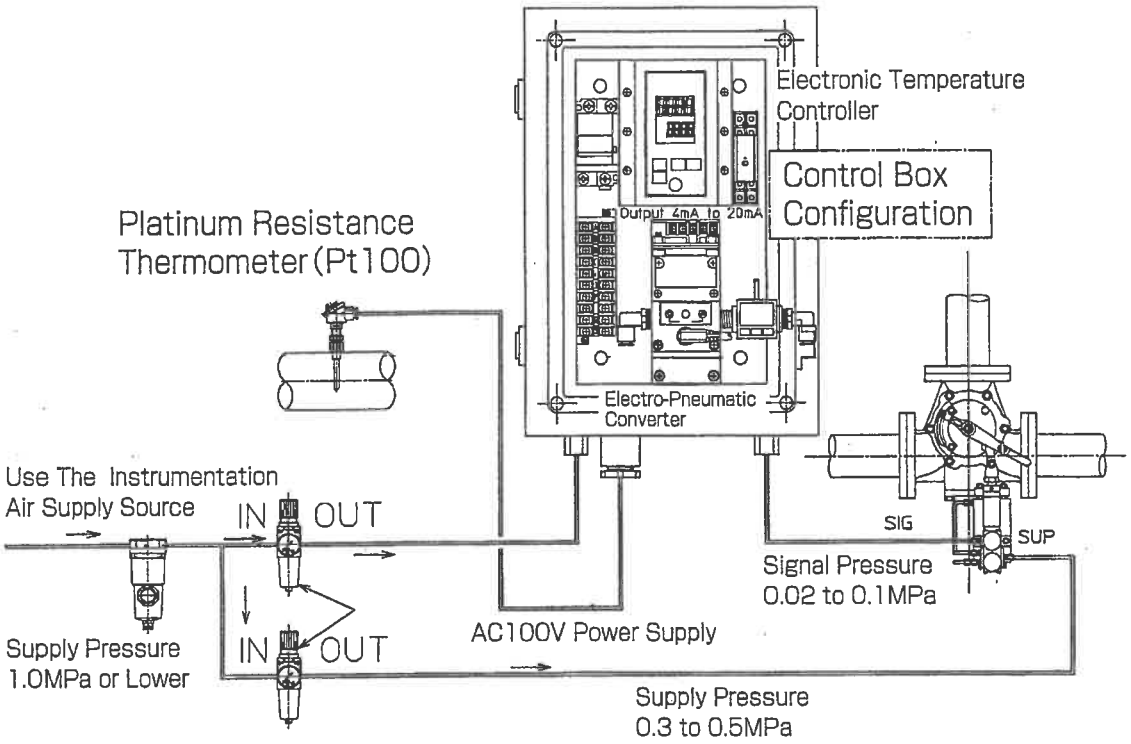
Supply pressure	From set pressure +0.1MPa to Max. 1MPa Note1
Output pressure	0.005~0.1MPa (ITV201*-*1)
Max. flow rate	1500 l/min (ANR) (SUP : at 0.7MPa)
Power voltage Note2)	24VDC±10% (ITV20*0*-*1) 12~15VDC (ITV20*1*-*1)
Current consumption	Supply Voltage 24VDC type : 0.12A or less Supply Voltage 12VDC type : 0.18A or less
Input signal Note2)	4~20mADC (ITV20**-01)
Input impedance	250Ω or less (ITV20**-01,ITV20**-11) Approx. 6.5kΩ (ITV20**-21,ITV20**-31)
Linearity	±1.0% F.S. or less
Hysteresis	0.5% F.S. or less
Repeatability	±0.5% F.S. or less
Additional functions	Min.pressure adjustment : Adjusting range 0~50% of rated value Max.pressure adjustment : Adjusting range 10~100% of rated value Key locking function : It is possible to lock the keys by pushing up key for 2 sec. or more and push SET key.
Display Accuracy	±3% F.S. or less
Monitor output	Analog output 1~5DVC (Input Impedance : 1kΩ or more) Output Accuracy : ±6% F.S. or less
Temperature charecteristics	±0.12% F.S. or less/ ℃
Operating temperature	0~50℃ (Without condensation)
Protective function for input signal	When a signal exceeding 150% of rated value is input, an error ( ) is indicated on LED and functions are stopped. In the case of current input type, protective circuit is operated at 40mADC or more.
Withstand voltage	Between a batch of external terminals and case :1500VAC 50/60Hz For 1 minute
Insulation resistance	Between a batch of external terminals and case :100MΩ or more (by 500VDC mega-meter)
Vibration proof	10~500Hz at either of smaller value of amplitude of 1.5mm or acceleration of 10G in X, Y and Z directions for 2 hours each.
Impact resistance	30G in X, Y and Z directions for 3 times each
Enclosure	Main unit : Equivalent to IP65. Cable connector : IP67
Weight	Approx. 350g (No accessory)

Note1) Up to 0.2MPa for the specification of 0.1MPa output pressure.
Note2) Take care that operation out of the specification range may cause damage.





Airless/Electroless Condition of Electro-Pneumatic (PID Control) Temperature Regulator

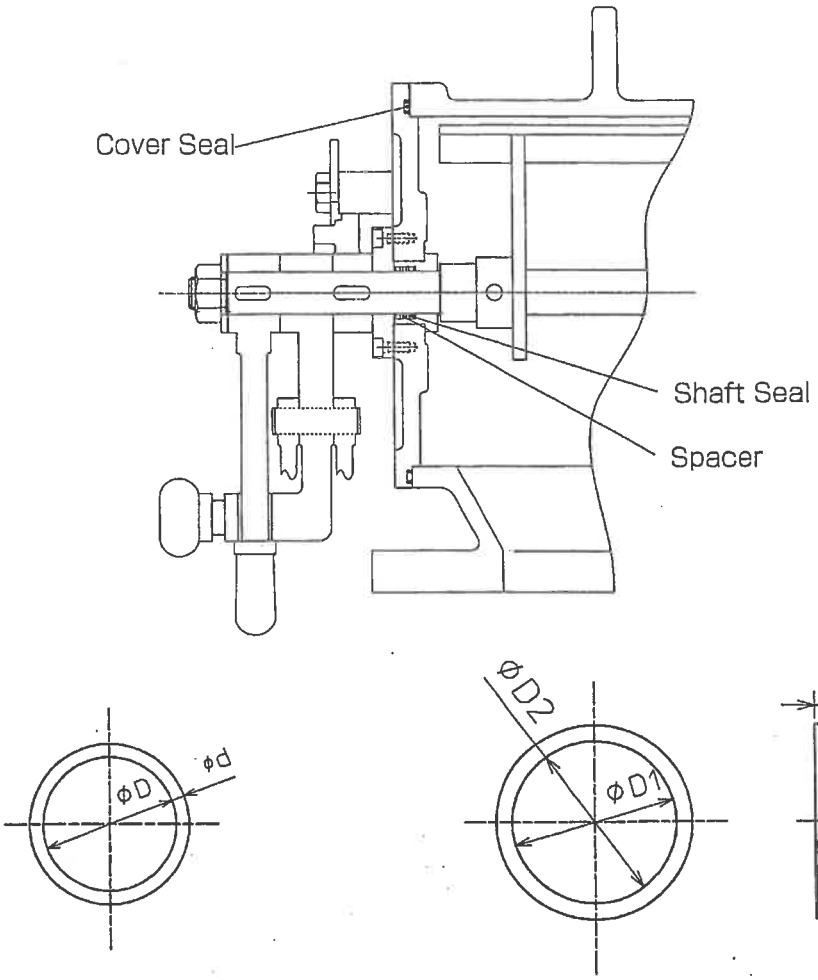


Condition		Standard Temperature Regulator	Temperature Regulator with Built-in Spring
Airless	Operation pressure 300 to 700 kPa is cut off	Valve stops at the position before the air supply is cut	Valve stops when fully opend at C side
	Signal pressure 20 to 100 kPa is cut off	Valve stops when fully opend at C side	Same as above
	Main air supply is cut off	Valve stops at the position before the air supply is cut	Same as above
Condition		Valve stops when fully opend at C side	Same as above

List of Seals

List of Seals Used in Regulator Driving Section

Applicable to Air Type, Electro-Pneumatic Type, and Electro-Pneumatic PID Type



Cover Seal

Type	Bore Diameter (D)	Diameter (d)	Standard (JIS)
40A	132.94	3.53	1517-30
50A	132.94	3.53	1517-30
65A	132.94	3.53	1517-30
80A	132.94	3.53	1517-30
100A	152.41	3.53	1517-36
125A	177.8	3.53	1517-40
150A	202.56	6.98	1516-72
200A	266.07	6.98	1516-77
250A	266.07	6.98	1516-77
300A	354.96	6.98	1516-84
350A	406.4	6.98	ARP-461

Shaft Seal

Bore Diameter (D)	Diameter (d)	Standard (JIS)
15.8	2.4	P-16
15.8	2.4	P-16
15.8	2.4	P-16
15.8	2.4	P-16
15.8	2.4	P-16
19.8	2.4	P-20
19.8	2.4	P-20
29.7	3.5	P-30
29.7	3.5	P-30
49.7	3.5	P-50
49.7	3.5	P-50

Spacer

Bore Diameter (D)	Diameter (d)	Standard (JIS)
16.0	20.0	1.5
16.0	20.0	1.5
16.0	20.0	1.5
16.0	20.0	1.5
16.0	20.0	1.5
20.0	24.0	1.5
20.0	24.0	1.5
30.0	36.0	1.5
30.0	36.0	1.5
50.0	56.0	1.5
50.0	56.0	1.5

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